Four-dimensional quantum Yang–Mills theory: the proof, outlined (detailed notes)

A natural-language account of the Lean proof (OpenAI-YM-20261028, commit 3456e5e)

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Contents

1 The proof in one reading **2**

2 The architecture **22**

3 The units **26**

3.1 I.1. Lattice, symmetries, gauge groups and the Wilson measure 26
3.2 II.1. Field labels and their lattice representatives 42
3.3 II.2. Marker algebra, polymer combinatorics and Gaussian estimates 63
3.4 III.1. The block-spin step and the symmetric gauge fixing 83
3.5 III.2. Choosing the constants: the order of choice, the level schedule and the shooting argument 101
3.6 III.3. Free lattice kernels and covariances 118
3.7 III.4. Gaussian integration: change of Gaussian sector, determinants, window law . 135
3.8 III.5. Cluster expansion of the polymer gas 156
3.9 III.6. The stationary branch: background field of the effective action 178
3.10 III.7. Matching the Wilson energy to fourth order 195
3.11 III.8. Connected activities and backward dependence cones 215
3.12 III.9. The marginal flow: the running coupling and its reset 229
3.13 III.10. One full nonlinear renormalization step: bounds on irrelevant terms 249
3.14 III.11. Comparison with the free Gaussian measure: interacting rank and determinant budgets 267
3.15 III.12. Bounds uniform in the volume and convergence as the spacing tends to zero 284
3.16 III.13. Extension to all labels: one common Wilson family 299
3.17 III.14. The construction run: flat representations and the ultraviolet rate 318
3.18 III.15. Cost accounting for observables and the remainder comparison 332
3.19 III.16. Analytic toolkit 350
3.20 IV.1. Infinite volume: extraction of a subsequence and the Euclidean axioms of the limit 364
3.21 V.1. Decomposition of the limit into phases 383
3.22 V.2. Osterwalder–Schrader reconstruction of a Wightman theory 401
3.23 V.3. Reconstruction in every phase 423

3.24 VI.1. Ward identities, the rotation action and the conserved stress tensor 439
3.25 VI.2. The free Maxwell stress tensor and the local Ward charge 461
3.26 VII.1. Short distances: convergence to free Maxwell fields 477
3.27 VII.2. The trace of the stress tensor and the trace anomaly 492
3.28 VII.3. The theory is not free: Gram positivity and exclusion of free-field profiles . 509
3.29 VII.4. Operator product expansion and contact extensions 527
3.30 VII.5. Asymptotic freedom of the running coupling 548
3.31 VII.6. Scaling variants and fixed-field estimates 561
3.32 VIII.1. Assembly of the main theorem 577

1 The proof in one reading

This is a substantial overview, about 10,000 words. The contents links locate individual topics; the exact statement fixes the scope of every claim.

1.0.1 The theorem

Let $G \subset U(m)$ be a closed subgroup whose Lie algebra $\mathfrak{g}$ is simple and nonabelian, for instance $SU(N)$ with $N \geq 2$; G need not be connected. The main theorem is `YangMills.jointExistence`; its precise statement is in statement.md. It asserts that there is one family of Wilson lattice gauge theories for G on four-dimensional tori, consisting of:

- spacings $a_N = 1/D_N \downarrow 0$ and physical side lengths $L_p \uparrow \infty$;
- bare couplings $\beta_{N,p}$ with $\beta_{N,p}/\log(1/a_N) \to b_* > 0$ uniformly in p;
- insertions $I_{N,p}(v, y)$ indexed by tokens v, each a continuous gauge-invariant function of the links in a stencil of fixed size in lattice units around the site y.

The family has the following property. For every strictly increasing sequence $p(k)$ of volume indices there are a further subsequence $p(q(k))$ and a continuum hierarchy satisfying (1)–(5) below.

(1) Lattice origin. Let $\mathrm{Conf}_n \subset (\mathbb{R}^4)^n$ be the open set of configurations of distinct points. For every n, every list of tokens v and every $f \in C_c^\infty(\mathrm{Conf}_n)$,

$$\lim_{k\to\infty} \lim_{N\to\infty} a_N^{4n} \sum_y f(\hat{y}) \left\langle \prod_{i=1}^n I_{N,p(q(k))}(v_i, y_i) \right\rangle_{N,p(q(k))} = \int f(x)\, S^{\mathrm{mix}}(\pi v; x)\, dx.$$

Here the sum runs over n-tuples of sites, $\hat{y}$ are their physical positions, and $\langle\cdot\rangle_{N,p}$ is the Wilson measure. - The inner limit runs over the whole sequence of spacings, at every volume whose fundamental box contains the support of f; this holds for all large k. - The test functions are joint functions of the n points and vanish near all coincidences. - One subsequence serves all n, v and f.

(2) Phases. On Conf_n, $S^{\mathrm{mix}} = \int_X S^\xi \, d\nu(\xi)$. Here X is a compact metric space, ν is a probability measure of full support, and $\xi \mapsto S^\xi(a; x)$ is continuous. All correlations S^s, $s \in X \sqcup \{\mathrm{mix}\}$, are smooth and permutation symmetric on Conf_n.

(3) Reconstruction. A selected phase ξ_* carries a Wightman theory: - a strongly continuous unitary representation of the proper orthochronous Poincaré group; - spectrum in the closed forward cone; - a vacuum that is unique up to scalars among translation-invariant vectors; -

fields that are operator-valued tempered distributions on the invariant domain spanned by field monomials applied to the vacuum; - local commutativity and faithful field labels.

Its Wightman functions are boundary values, on product test functions, of the continuation of S^{ξ_*} that is holomorphic in the ordered complex times.

(4) Operator product expansion. One system of coefficients, with finitely many local fields of each degree, satisfies finite-order OPE bounds in the mixed state and in every phase.

(5) Nonfreeness. Let e_4 be the time direction. A field Θ of degree 4 with vanishing vacuum expectation satisfies

$$\lim_{r\downarrow 0} r^8 \log^2(1/r)\, G_\Theta(r) = \frac{3\dim_{\mathbb{R}} \mathfrak{g}}{\pi^4}, \qquad G_\Theta(r) = S^{\xi_*}(\Theta(-\tfrac{r}{2}e_4)\,\Theta(\tfrac{r}{2}e_4)).$$

Moreover, G_Θ agrees on no interval $(0,\delta)$ with a finite Wick profile, that is, a finite sum of products of free mass-shell kernels $\int E^q h(\mathbf{p})(2E)^{-1}e^{-rE}\,d^3\mathbf{p}$.

The lattice data are fixed before the volume sequence. The refinement and all continuum objects may depend on that sequence.

The stronger statement. The theorem is a projection of a longer statement, `YangMills.constructed`. That statement holds for every value $b > 0$ of the penalty parameter of the block-spin kernel; the public theorem takes $b = 1$. The construction supplies the following facts; the slope value is established in the proof of regularity, rather than stated as a separate conjunct of `YangMills.constructed`: - The slope is $b_* = \kappa_G = 11C_{\mathfrak{g}}/(24\pi^2)$. - A token is a pair (d, c), with c in the space V_d of gauge-invariant polynomials of weight at most d in the curvature and its covariant derivatives (curvature has weight 2, and each derivative adds 1). - The field spaces are $F_d = V_d/N_d$, where N_d consists of the labels whose mixed correlations with all separated spectators vanish. The label space of the relativistic theory is the directed union of the F_d, so there is no finite field multiplet. - Θ is the centred trace of a constructed conserved stress tensor. - Every phase carries an irreducible Wightman theory, with low-energy bounds that are uniform in the phase. - The correlations extend to distributions across coincident points; uniqueness is not asserted. - After finite-dimensional changes of frame, the rescaled correlations converge at short distance to those of $\dim \mathfrak{g}$ free Maxwell fields. - Nested and integrated forms of the OPE hold, as do further nonfreeness tests.

This overview describes the proof of the stronger statement.

What is not claimed. - No mass gap is proved; the available spectral bounds in every phase are polynomial. - The infinite-volume limit is not shown to be unique or independent of the subsequence. - The phases are not shown to agree, and X is not shown to be a point. - The mixed state is not shown to cluster. - The limit is not shown to be independent of the lattice data. - The two limits are taken in a fixed order. No diagonal limit, exchange of limits or rate of convergence is asserted. - No phase is shown to be a limit of lattice states; the lattice limit is the mixture. - The public theorem asserts nothing at coincident points and no Euclidean axioms. The construction proves those axioms internally. - There is no continuum measure on connections, no Wilson-loop observable or area law, and no particle or scattering theory. - Only finite Wick profiles are excluded. Generalized free fields with a continuous mass distribution, infinite sums and general composites of Gaussian fields are not excluded. - The OPE is a finite-order bound and is not compared with perturbation theory.

Normalizations. On $\mathfrak{g}$, put $\langle X, Y\rangle_{\mathfrak{g}} = -\frac{1}{m}\mathrm{Re}\,\mathrm{tr}(XY)$, and define $C_{\mathfrak{g}} > 0$ by $\mathrm{tr}(\mathrm{ad}X\,\mathrm{ad}Y) = -C_{\mathfrak{g}}\langle X, Y\rangle_{\mathfrak{g}}$. Two facts fix the coupling convention: - $1 - \frac{1}{m}\mathrm{Re}\,\mathrm{tr}\,e^X = \frac{1}{2}|X|^2_{\mathfrak{g}} + O(|X|^4)$; - each unoriented plaquette is counted once in the Wilson action.

Hence $\beta = 1/g_0^2$ for the continuum action $\frac{1}{2g_0^2}\int \sum_{\mu<\nu} |F_{\mu\nu}|^2_{\mathfrak{g}}$. The constructed couplings satisfy, uniformly in p,

$$\frac{\beta_{N,p}}{\log(1/a_N)} \longrightarrow \kappa_G = \frac{11C_{\mathfrak{g}}}{24\pi^2} = 2b_0, \qquad b_0 = \frac{11C_{\mathfrak{g}}}{48\pi^2}.$$

For $SU(N)$ in the defining representation, $C_{\mathfrak{g}} = 2N^2$ and $\kappa_G = 11N^2/(12\pi^2)$. In that case β is the usual Wilson coupling $2N/g^2$, and the law above is the one-loop running $1/g^2 \simeq (11N/24\pi^2)\log(1/a)$ of Gross–Wilczek and Politzer.

1.0.2 The strategy

Order of limits. One lattice family is fixed first. Correlations of renormalized insertions are taken to the limit $a_N \to 0$ at fixed volume, along the whole sequence of spacings. The volume then tends to infinity along a subsequence of the prescribed one. The renormalization group (RG) is used only for the first limit. Its levels run from the Wilson lattice (level N) to the torus of side L_p at unit spacing (level 0). The bare coupling is tuned so that the effective coupling at level 0 is one fixed number $\bar{\beta}$, the same for all N and p. No RG step is taken above the unit scale; infinite volume is reached by compactness, from local bounds that are uniform in the volume.

Ultraviolet construction (Parts I–III). At each level the density is written as two pieces: - an explicit gauge-invariant action, whose normalized curvature-squared coefficient is the running coupling $\beta_{N,p,m}$; - polymer activities: regular ones, controlled where the curvature is small, and forced ones, which carry the large-field regions.

One block-spin step integrates the fine links against an exactly normalized, gauge-covariant probability kernel: - In the small-field region this is a Gaussian integral around a stationary background field, organized by a cluster expansion. - Large-field regions are kept and paid for by a lower bound on the quadratic energy of the step. - The marginal term is absorbed into the coupling. The coupling changes by the one-loop amount $\kappa_G \log L_m$ up to an error, and block ratios that grow slowly along the levels make the accumulated error $o(\log(1/a_m))$.

The same steps transport inserted observables. This second layer produces what later parts use: - chessboard bounds on renormalized insertions and on their difference quotients, uniform in the volume; - convergence at fixed volume, obtained from fixed-level convergence and uniform observable tails. The construction also supplies analytic estimates in a tuning parameter.

Euclidean limit (Part IV). - Chessboard bounds make the correlations locally bounded, uniformly in the volume. - A transfer-matrix representation and complex interpolation make them equicontinuous, and Arzelà–Ascoli gives the subsequence. - Difference quotients of insertions are close to insertions of higher degree. This yields derivative maps on the field spaces and kernels that are smooth on Conf_n. - The limit inherits translations, the signed permutations of the axes and reflection positivity. It does not inherit continuous rotations or clustering.

Symmetry (Part VI). - Lattice integration by parts gives shell identities for a conserved tensor. Its normalization is fixed by short-distance comparison with free Maxwell theory. - Reflection

positivity controls the flux at infinity, and analyticity extends the Ward identities to all separated configurations. - The infinitesimal rotations integrate to an $O(4)$ action, again by comparison with free Maxwell theory.

Phases and reconstruction (Part V). - The zero-energy subspace of the Osterwalder–Schrader (OS) space of the mixed state carries a commutative C^*-algebra; its proof of commutativity uses the $O(4)$ action. The characters of this algebra are the phases. - Each phase is reconstructed as a Wightman theory, from bounds on operators inserted in thin time slabs. Lorentz covariance comes from the Ward identities. - A low-energy spectral bound, uniform in the phase and derived from locality, upgrades vacuum uniqueness from almost every phase to every phase.

Identification (Part VII). - Along sequences of scales tending to zero, the theory converges, after changes of frame, to free Maxwell theory. - The one-loop coefficient produces an inverse-logarithmic law for the centered stress-tensor trace two-point function, and that law excludes the specified finite Wick profiles. - The RG also yields the OPE.

The logical order is not the order of the parts. Part VI uses the OS spaces of the mixed state from Part V and the free comparison from Part VII. Part V uses the $O(4)$ action from Part VI. Part VIII assembles everything for one family and one limit table.

1.0.3 Part I. The lattice theory

Unit: I.1 — Lattice, symmetries, gauge groups and the Wilson measure.

On $\Lambda_\ell = (\mathbb{Z}/\ell\mathbb{Z})^4$, a configuration assigns $U(y,\mu) \in G$ to each positively oriented link. With h the normalized Haar measure, the Wilson measure is

$$d\mu_{\ell,\beta}(U) = Z^{-1} \exp\Big[-\beta \sum_{y} \sum_{\mu<\nu} (1 - \tfrac{1}{m} \mathrm{Re}\,\mathrm{tr}\, U_{y;\mu\nu}) \Big] \prod_{(y,\mu)} dh(U(y,\mu)).$$

The energy lies in $[0, 12\ell^4]$, so this is a probability measure for every real β. It is invariant under gauge transformations, translations and the signed permutations of the axes.

Lattice schedule. - Block ratios are odd integers $L_m \geq 3$, with $D_{m+1} = D_m L_m$ and $a_m = 1/D_m$. - The level-m torus at volume p has $L_p D_m$ sites per side, with $4 \mid L_p$. - Odd ratios make the coarse lattice compatible with the signed permutations. - Supports are measured by cover numbers by balls of radius B, which obey a coarsening inequality uniform in the volume.

Group facts. On G the unit proves: - a real-analytic logarithm near 1; - an equivariant nearest-point retraction onto G, whose derivative is the orthogonal projection onto $\mathfrak{g}$; - a curvature map $F(u) \in \mathfrak{g}$ with $|F(u)| \leq \|u-1\|/\sqrt{m}$ and $F(vuv^{-1}) = \mathrm{Ad}(v)F(u)$.

These are qualitative statements about single group elements; quantitative control comes in Part II.

Role of simplicity. Simplicity enters through one fact: every Ad-invariant symmetric bilinear form on $\mathfrak{g}$ is a multiple of $\langle \cdot, \cdot \rangle_{\mathfrak{g}}$. Together with lattice symmetry, this leaves a single marginal coupling. The same fact is used in the one-loop computation and in the uniqueness of the stress tensor.

Reflection positivity. The Wilson measure is reflection positive for every $\beta \geq 0$, for reflections through planes of sites and through planes between sites (the argument of Osterwalder and Seiler).

This is proved in Part IV for the original lattice measure, not for an effective action, and it is used only through the chessboard inequality.

1.0.4 Part II. Fields, insertions and polymer activities

Units: II.1 — Field labels and their lattice representatives; II.2 — Polymer activities, norms, the Wilson family, the construction record, reflection-positive cells.

Labels. In radial gauge, a connection near a point is described by its Taylor coefficients, and the coefficient of order j has weight $j+2$. V_d is the space of $\mathrm{Ad}(G)$-invariant complex polynomials of weight at most d in these coefficients. Equivalently, it consists of the gauge-invariant polynomials of weight at most d in the curvature and its covariant derivatives. It is finite-dimensional, and $1 \in V_d \subset V_{d+1}$.

Lattice representatives. The basic lattice variables are "slots": plaquette curvatures transported to a base point along walks. - For each d, finitely many slots have linearizations with no common kernel on the jets of weight at most d. - The slot map then has an analytic Ad-equivariant left inverse near 0; it is chosen non-constructively. - Compose a basis of V_d with the order-d Taylor polynomial of this inverse, and average over the signed permutations. For each label c this gives a gauge-invariant polynomial Φ_c in the link entries of a fixed box stencil.

Two properties follow: - On links obtained by parallel transport of polynomial radial-gauge potentials at spacing h, Φ_c reproduces the classical label to order h^d. - Suppose the plaquettes near the stencil have chord at most t. Then a component with $f \geq 1$ curvature factors, and its derivatives, are $O(t^f)$. Conversely, a function with such bounds determines a label with uniformly bounded coefficients.

Renormalized insertions. The insertion of $v \in V_d$ at a site o is

$$O_{N,p}(v,o) = a_N^{-d}\, \Phi^{(d)}_{S_{1/\theta_N}\tau^{-1}v}(o).$$

- S_t multiplies the coefficient of a component with f curvature factors by t^f.
- θ_N is the small-field threshold of the finest level.
- τ is a linear automorphism of V_d. It commutes with conjugation and with the signed permutations, and it fixes the unit.

The RG defines τ. The running label coordinates of an insertion obey a transfer relation from level N down to level m, and τ is their value at a reference level $r(d)$ fixed in advance. So each insertion is normalized to have running coordinates exactly v at level $r(d)$. All labels of weight at most d, including the unit, may mix; no triangular structure is imposed. The large-field part of the inserted density goes into reserved forced polymers, so the construction never divides by the vacuum density.

Equation relations. Haar invariance gives the lattice Schwinger–Dyson identity $\langle\langle A, \nabla_e S_W\rangle\, S\rangle_\beta = 0$ for S independent of the link e. Products of such forces with slot polynomials span the linearized free-Maxwell equation ideal, up to terms of higher weight. They give lattice observables whose correlations with distant spectators vanish exactly. Two facts are established later, through the comparison with free theory: these relations become null in the limit, and they exhaust N_d.

Polymers, budgets and marks. - A polymer activity has a finite support of sites. - Regular activities are anchored at a site and measured by suprema of derivatives over small-field configurations. - Forced activities carry the large-field regions, together with an inventory of flagged sites. - Activities are controlled by budgets summed through each site, $\sum_{X\ni x} e^{a\,\mathrm{cov}_B(X)}\|K_X\| \le \text{budget}$. - Insertions carry marks: commuting variables λ with $\lambda^2 = 0$, one for each label or site that may occur. Exponentials of marked quantities are finite sums, and logarithms are finite recursions in the number of marks. - The weighted norm $\|a\|_w = \sum_x \prod_{\lambda\in x} w_\lambda\,|a(x)|$ is submultiplicative, so bounds do not grow with the number of marks. - A tree-graph bound without factorials controls connected clusters of overlapping supports.

1.0.5 Part III. The ultraviolet construction

1.0.5.1 The block-spin step and the background field

Units: III.1 — The block-spin step and the symmetric gauge fixing; III.6 — The stationary branch: background field of the effective action; III.3 — Free lattice kernels and covariances.

At level m, relative to product Haar measure, the density is

$$\rho_m(U) = \exp\big(v|\Lambda_m| - \beta\,\mathcal{K}(U) - \mathcal{L}(U) - \mathcal{E}(U)\big)\,\mathcal{G}(U).$$

- $\mathcal{K}$ is built from lattice paths: a plaquette term, smeared curvature squares, and cubic and quartic polynomials in transported curvatures. It is averaged over the signed permutations and normalized so that its quadratic symbol at zero momentum is the identity; this fixes the meaning of β.
- $\mathcal{L}$ is a loop term independent of β.
- $\mathcal{E}$ is the sum of the regular activities times small-field indicators.
- $\mathcal{G}$ is the gas of forced activities.
- At the finest level, $\rho_N = e^{-\beta_{N,p,N} S_W}$.

The kernel. One step is $\rho_m(V) = \int k_\beta(V,U)\,\rho_{m+1}(U)\,dU$. Given the fine configuration U, a finite tag t is drawn with probability $\lambda(U,t)$. Each coarse link is then drawn independently, with Haar density

$$Z_\beta^{-1}\exp\Big(-\beta\kappa(1-\tfrac{1}{m}\mathrm{Re}\,\mathrm{tr}\,V_b Q_b(U,t)^{-1})\Big),$$

where $\kappa > 0$ is the penalty. The predictor Q_b is built in two steps: 1. On a patch of radius $(J+3)L$ around each coarse site, a gauge frame is chosen. - If all route holonomies on the patch are close to 1, the frame is the unique minimizer of an averaged holonomy discrepancy. - Otherwise it is a uniformly drawn eligible axial gauge; such gauges exist for every configuration. 2. Fine transports of length L, read in these frames, are combined by a nonlinear (minimizing) average with a J-fold convolution weight, with a random fallback.

Properties of the kernel: - Haar invariance gives $\int k_\beta(V,U)\,dV = 1$ exactly. - The kernel is exactly covariant under fine gauge transformations and the lattice symmetries. - Hence $\int \rho_m$ equals the Wilson partition function at every level, and the step is defined without any small-field assumption.

Background field. For the integration, the links of a spanning forest are set to 1, and each remaining link is $\exp(a)$ times a fixed lift of the coarse field x. The background is the critical branch $a = \rho(x)$ of the classical Taylor model of the action (quadratic, cubic and quartic terms)

plus the blocking penalty, at fixed x. The steps are: - The Hessian at the flat field is a positive "bed" form plus the penalty. - Each non-tree link is bounded by the curvature in its block plus the response of the predictor. With this, the Hessian is uniformly positive in the vertical directions, with a constant independent of the volume and the forest. - The analytic implicit function theorem then gives ρ as a locally unique analytic germ with explicit linear part. Statements with a uniform radius and localized jets give control independent of the volume. - The branch is not shown to be a global minimizer, and this is not used. - Changing the forest multiplies the Hessian determinant by the square of a Jacobian and the Haar measure by the Jacobian itself. So the critical value and the one-loop term do not depend on the forest.

Free kernels. The free covariances are written in Fourier space through the actual averaging filters, which are entire and multiplicative under composition of blockings. Aliased momenta carry extra powers of $1/L$. - All constants (decay rates, strip widths, lower bounds) are fixed before the block ratio L. - The quadratic forms decay exponentially and are bounded below uniformly in the volume. - An exact identity on each pair of tori splits a quadratic form into a coarse part and a fluctuation part (a Schur complement).

1.0.5.2 Gaussian integration and the cluster expansion

Units: III.4 — Gaussian integration: change of Gaussian sector, determinants, window law; III.5 — Cluster expansion of the polymer gas; III.8 — Connected activities and backward dependence cones.

Change of variables. Active links are written in exponential coordinates around the background. Inactive links stay in Haar measure. - The change of variables to a Gaussian integral with covariance C and mean $m(Y)$ is an exact identity. It holds for any positive-definite C, keeps all Jacobians, and uses no smallness assumption. - The covariance actually used is the inverse of the gauge-fixed Gram form on the active links, not the inverse of the Wilson quadratic form. Its precision is bounded below uniformly in the volume. - Window law: the global inverse is approximated by inverses on windows of radius r, with error Ce^{-vr} uniform in the volume. - $\log\det$ separates into local terms and far-field terms with summable errors.

Cluster expansion. The gas is $\Xi(w) = \sum_s \prod_{i\in s} w_i$, summed over families of polymers with pairwise disjoint supports and real weights. With $B = \lceil (\log\widehat{\beta})^2 \rceil$, the smallness condition is

$$\sum_{i:\, x\in X_i} |w_i|\, e^{(a+\delta)\operatorname{cov}_B(X_i)} \le \gamma \quad \text{for every site } x, \qquad 4(2B+1)^4\gamma/\delta \le q < 1.$$

Under this condition: - $\log\Xi$ is an absolutely convergent sum of connected clusters, bounded through spanning trees (Penrose's partition scheme); - removing every polymer that meets a set Y changes $\log\Xi$ by at most $|Y|\gamma/(1-q)$, uniformly in the volume.

So the vacuum logarithm is extensive, but a local change costs only its size. A second, graph-free argument based on a deletion recursion gives bounds of the same type.

Connected activities and dependence cones. - Fixed-order connected coefficients come from the heat equation $\partial_t W = \frac{1}{2}(\Gamma : \nabla^2 W + \Gamma : \nabla W \nabla W)$ for $W_t = \log \mathbb{E}_{t\Gamma} e^V$. A rooted ℓ^1 norm with weight $e^{\sigma\,\mathrm{diam}}$ makes contraction and joining lossless, so these coefficients are polynomially bounded uniformly in the volume. There is no resummation. - Pulling a coarse observable back through the block kernel gives a function of the fine links in a window of fixed radius around its support.

The backward cone therefore grows by a bounded physical width per step. - For disjoint supports, the pull-back factorizes exactly.

1.0.5.3 Large fields and one nonlinear step

Unit: III.10 — One full nonlinear renormalization step: bounds on irrelevant terms.

At reference coupling $\hat{\beta}$ the small-field threshold is

$$t = \hat{\beta}^{-1/2} P(\hat{\beta}), \qquad P(\hat{\beta}) = \exp\left((\log\log\hat{\beta})^2\right),$$

well above the Gaussian fluctuation scale $\hat{\beta}^{-1/2}$.

Seeds. A seed is one of: - a fine plaquette with chord at least $t/4$; - a coarse link on which the coarse field departs from the predictor by more than a fixed multiple of t; - a flagged coarse site.

Energy lower bound. For every store A of seeds,

$$\kappa\, t^2\, \sharp A \;\le\; \text{quadratic energy of the step integrand, after subtracting the stationary predictor.}$$

The energy is a sum over five channels, one of which is the blocking penalty. The proof assigns charges to fine cells with bounded multiplicity, and controls the linearization defects on cells without seeds.

Why this suffices. Since $\beta t^2 \simeq P^2$, each seed gains a factor e^{-cP^2}. P^2 grows faster than any power of $\log\hat{\beta}$, so this gain beats: - the polylogarithmic number of connected seed sets through a site; - any per-seed loss of a fixed power of $\hat{\beta}$, or of order P^z with $z < 2$.

The decomposition into large-field sectors is exact: large-field sectors are integrated, not discarded.

One step. The step takes inputs at level $m+1$: regular norms at most $\hat{\beta}^{-11/10}$, forced norms at most $e^{-P^{1/5}}$, and the coupling in the band $[\hat{\beta}/2, 2\hat{\beta}]$. For L large, and then $\hat{\beta}$ large, it produces: - site-summable budgets for the new regular activities; - the bound $\exp\left((C\,\mathrm{bill}^5 - \beta\mu t^2)\sharp A\right)$ for each large-field sector, where $\mathrm{bill} = 1 + B + (\log\hat{\beta})^2$; - bounds on the probabilities of the remaining catastrophic events.

The marginal and vacuum terms are extracted, and the level-m bounds restored, by the reset below.

1.0.5.4 The marginal coupling and the schedule

Units: III.7 — Matching the Wilson energy to fourth order; III.9 — The marginal flow: the running coupling and its reset; III.2 — Choosing the constants: the order of choice and the RG schedule.

Matching and the one-loop coefficient. - At the finest level, a path representation is constructed whose jets of order at most 4 agree with those of the Wilson energy in exponential coordinates. - At later levels, the cubic and quartic parts are represented in curvature variables by quasi-local lifts (a cost-controlled Poincaré lemma). - In each step, the coefficient $\alpha_{N,m}$ of curvature squared produced by the one-loop term satisfies $|\alpha_{N,m} + \kappa_G \log L_m| \le D$, with D fixed before L.

The value of κ_G comes from an explicit momentum integral. The comparison symbol is

$$h(k) = 4C_{\mathfrak{g}}\Big(\frac{k_1^2}{|k|^6} - \frac{4k_1^2k_0^2}{|k|^8} - \frac{1}{|k|^4}\Big).$$

Its angular average is $4C_{\mathfrak{g}}(\frac{1}{4} - \frac{1}{6} - 1)|k|^{-4} = -\frac{11}{3}C_{\mathfrak{g}}|k|^{-4}$, and $\int_{1/L<|k|\le 1} |k|^{-4}\, d^4k = 2\pi^2 \log L$. Hence $(2\pi)^{-4}\int_{1/L<|k|\le 1} h = -\kappa_G \log L$. Comparing the step's Hessian with h up to $O(|k|^{-3})$ gives the bound on $\alpha_{N,m}$.

The reset. After the step, the blocked density is rewritten exactly, at every configuration including large fields: - a new coupling $\beta_{N,p,m} = \beta_{N,p,m+1} + \alpha_{N,m} + \delta$; - a new vacuum constant; - reorganized activities: flat values and isotropic quadratic parts are subtracted, compensating path terms are added, and mismatches of the small-field masks go into the forced gas.

The correction δ is the isotropic Hessian coefficient of the regular activities: - By simplicity and hyperoctahedral symmetry, their quadratic part at a point is δ times the identity on plaquette orientations. - $|\delta| \le 1$, and δ is fixed by a scalar equation with a unique root.

Altogether, with ω_j a drift allowance attached to regularity index j,

$$\big|\beta_{N,p,m+1} - \beta_{N,p,m} - \kappa_G \log L_m\big| \le \omega_{J_{m+1}}.$$

Endpoint. For real $|b| < r_*$ there is a trajectory with level-0 coupling $\bar\beta + b$. Its bare coupling is unique within the construction, and the trajectory continues analytically to complex $|b| < r_*$, with r_* independent of N and p. The family uses $b = 0$, so $\beta_{N,p,0} = \bar\beta$ for all N and p. An intermediate-value shooting argument also produces such trajectories; the analytic version is the one used for convergence at fixed volume below.

Running. Put $s_m = \log(1/a_m)$ and $\hat\beta_m = \bar\beta + \kappa_G s_m$. Induction from level 0 gives $|\beta_{N,p,m} - \hat\beta_m| \le \sum_{k\le m} \omega_{J_k}$, and the schedule makes this $o(s_m)$:

$$\beta_{N,p,m} = \bar\beta + \kappa_G \log(1/a_m) + o(\log(1/a_m)) \qquad \text{uniformly in } N \ge m \text{ and } p.$$

At $m = N$ this is the slope law with $b_* = \kappa_G$. Since $\bar\beta$ is chosen large, every level stays in the band $[\hat\beta_m/2, 2\hat\beta_m]$. The statement is leading-logarithmic only: there is no two-loop term.

Why the block ratios grow. The one-loop error per step is bounded but does not decrease with L. For the accumulated error to be $o(s_m)$, the allowance per unit of logarithmic scale must tend to zero. The schedule therefore imposes two conditions: - $\omega_{J_{m+1}} \le J_{m+1}^{-2} \log L_m$, which forces $\log L_m \to \infty$; - $\log L_m / s_m \to 0$, which keeps each step small compared with the total scale.

The ratios are constant while the regularity index is fixed, $L_m = \ell_{J_{m+1}}$. The regularity index J is nondecreasing, rises by at most one per level, tends to infinity toward the fine end, and stays constant on long windows of levels. One sequence (L_m) serves every cutoff N.

Order of choices. The constants are chosen in a fixed order: 1. the kernel with its penalty, and for every index j its regularity profile (derivative order, spatial power, support exponent, widths, path counts, drift allowance), together with the one-loop error constant D; 2. all block ratios ℓ_j, so that the allowances and the logarithms of the combinatorial costs are at most $j^{-2}\log \ell_j$; 3. the

smallness exponents; 4. the entry couplings and the volume conditions; 5. $\bar{\beta}$; 6. the assignment of regularity profiles to levels.

Each degree d has a reference level $r(d)$ with regularity index at least $d+3$. Beyond $r(d)$ the derivative order exceeds d and the spatial power is at least d. All choices precede any observable or extraction order, so every degree is served by the same family.

1.0.5.5 Renormalized observables, uniformly in the volume

Units: III.11 — Comparison with the free Gaussian measure: interacting rank and determinant budgets; III.12 — Bounds uniform in the volume and convergence as the spacing tends to zero; III.15 — Cost accounting for observables and the remainder comparison; III.16 — Analytic toolkit.

The chessboard root. The quantity propagated for observables is the chessboard root. For a function F of the links in one cell of a tiling of the torus by cells of side a_m,

$$\|F\|_T = \Big| \mathbb{E}_{\beta_{N,p}} \Big[\prod_{\text{cells}} \text{reflected copies of } F \Big] \Big|^{1/\sharp\text{cells}}.$$

The main bound reads as follows. For every degree d, radius R and $\varepsilon > 0$ there is A such that, for every $m \geq r(d)$, there is N_0 independent of p with

$$\|O_{N,p}(v,o)\|_{T_m(o)} \leq A\, a_m^{-d-\varepsilon} \qquad (N \geq N_0,\ \text{all } p,\ \|v\| \leq R).$$

It is proved by running the insertion through the same RG steps; ε absorbs losses smaller than any power. Reflection positivity enters only afterwards, through the chessboard inequality. That inequality makes the root a seminorm and bounds correlations of separated observables by products of roots.

No loss of $\log\beta$ per site. A root over c copies tolerates a fixed cost per copy, but not a factor $\log\beta$ per site. The determinant budget avoids this by a gauge count on a terminal torus with $c = L^4$ sites: - A tree gauge leaves $4c-(c-1) = 3c+1$ free links. - In the numerator, $3c-3$ of them are integrated one at a time, each being the only new link of a chosen plaquette. Each integral gives at most $b\,E^{\dim G}$, with $E \propto g = \beta^{-1/2}$. The four links that wrap around the torus, one per axis, are left over. - The normalization is bounded below by confining all $3c+1$ free links to Haar balls of radius $g/4$. - The ratio is therefore at most $\exp(c\,\pi - 4\dim G\,\log g)$, where the cost π per site is independent of g. - Since the number of copies is c, the root carries only $g^{-4\dim G/c}$.

Field content. Free Gaussian detectors give uniform rank detection: $\|\pi x\| \leq Z\|\text{probe}(\pi x)\|$, where π is the projection onto quotient labels. The limit field spaces are isomorphic, degree by degree, to those of free Maxwell theory, compatibly with conjugation and reflections.

Difference quotients. - A normalized lattice difference of an insertion of degree d equals an insertion $O(x)$ of degree $d+|l|$, up to a term whose root tends to zero. - Write $x = \pi x + (x - \pi x)$. The part $x - \pi x$ lies in the Maxwell equation ideal and has small root by the lattice equations of motion. The part πx is bounded by the rank detector. - So difference quotients satisfy root bounds of the same form. By the chessboard inequality, lattice correlations with any number of differences are bounded at r-separated points, uniformly in p.

These bounds are propagated through the RG; they do not come from differentiating a sup norm.

A rank-free trace bound. Let $X \geq 0$ be a matrix and let the insertions b_a be separated by gaps of more than $2r$ factors of X. Moving $m \leq r$ factors from one gap to another changes $\operatorname{tr} \prod_a b_a X^{Y_a}$ by at most $(\mathrm{cost}/r)\, m$. The cost is a product of trace norms with no dimension factor. The proof is a Hölder inequality for complex exponents, obtained from Hadamard's three-lines theorem, followed by a Cauchy estimate. By quadrature it extends to continuous positive kernels. Part IV uses it for equicontinuity.

Bookkeeping. The numerical conditions of all steps can be met in one fixed order: constants before L, then L, then the later constants, then β large. This holds uniformly across regularity profiles.

1.0.5.6 Convergence at fixed volume and the construction run

Units: III.12; III.13 — Extension to all labels: one common Wilson family; III.14 — The construction run: flat representations and the ultraviolet rate.

The obstacle. At fixed volume, two cutoffs N and N' have different bare couplings, each fixed by the condition $\beta_{N,p,0} = \bar{\beta}$. Continuity in a fixed bare coupling therefore compares nothing.

Fixed-level convergence and tails. The construction has a one-parameter family of trajectories: - The level-0 coupling is $\bar{\beta} + ws$, with a real tuning parameter $s \in [-1, 1]$. For each s the bare coupling is unique within the recursion. - The bare coupling and effective data have analytic dependence on the tuning parameter. The estimates on that dependence are separate from the Cauchy argument below. - For fixed j, convergence of the trajectory and stability of the observable transfers give convergence of the effective insertions and their polymer parts. Uniform convergence passes through the Haar integrals, and a positive lower bound on the normalized partition function gives convergent correlation approximations. - The omitted part is bounded, uniformly in the cutoff, by $2F_0 e^{-B_j/64}$, where $B_j = \lceil (\log \hat{\beta}_j)^2 \rceil \to \infty$. - Choose j to make this tail small, then take the cutoffs large enough that their level-j approximations are close. This proves the Cauchy property first at lattice-fitted configurations; equicontinuity extends it to all configurations distinct modulo the period. See III.12, §§4.5–4.6.

Combined with the difference bounds, this gives at each fixed volume: - limit kernels smooth away from periodic coincidences; - convergence of difference quotients to derivatives; - convergence of smeared moments along the whole sequence of spacings.

The family used by the public theorem. VIII.1 assembles the construction for one family, using the bounds and convergence results described in III.9–III.12. III.13 describes an alternative coordinated family with root bounds, an analytic shooting family and Maxwell rank comparisons. Its existence statements do not identify that family with the one used by `YangMills.constructed`; properties from the two constructions should not be combined without such an identification.

The construction run (III.14). III.14 packages the construction run. It also proves quantitative convergence of the order-zero quadratic part of the effective action (its position-space Hessian kernel) at each fixed level i, at rate $(a_N/a_i)^c$. - The difference between two cutoffs is written in curvature variables as $d^{\mathsf{T}} A d$, with A defined modulo kernels annihilated by this map. - So the curvature precision, which has zero modes, is never inverted. - This rate concerns the quadratic germ only, not the full effective action or the final limits.

1.0.6 Part IV. Infinite volume

Unit: IV.1 — Infinite volume: extraction of a subsequence and the Euclidean axioms of the limit.

Local bounds. - Reflection positivity gives the chessboard inequality $|\langle\prod_c F_c\rangle| \le \prod_c \|F_c\|_T$ for every $\beta \ge 0$. Log-convexity of roots under translation handles cells not centred at the insertion. - Put each insertion in its own cell of side $a_m < r$. Then $|C_{N,p}(v;x)| \le A^n$ whenever each pair of points has some coordinate at distance at least r, uniformly in p and for large N. - Near collisions, the same argument gives $|S(v;x)| \le Cr^{-4n(1+\max d_i)}$ when all points are at mutual distance at least r.

Equicontinuity. Let Gen_n be the set of configurations whose points have pairwise distinct coordinates on every axis. 1. Integrating out the links along one axis gives a positive transfer kernel K. A correlation becomes $\mathrm{Tr}\prod_k B_k K^{Y_k}/\mathrm{Tr}\,K^M$, with slab kernels B_k and gaps Y_k. 2. Moving one point shifts units between two gaps. The trace is entire in the size of the shift. 3. The rank-free Hölder bound controls the trace on a disc, and a Cauchy estimate gives a Lipschitz bound. 4. The trace norms of the slab kernels are themselves chessboard roots of $2q$ alternating copies, so the bound is uniform in the volume.

The result is $|C(x) - C(y)| \le 16nA^n|x-y|/R$ on compact subsets of Gen_n whose coordinate gaps exceed R. No derivative bound is used.

Extraction. Arzelà–Ascoli is applied twice, over countably many degrees, basis labels and compact sets: first to extract a spacing subsequence that works for all volumes, then a volume subsequence. This gives limit kernels on Gen_n. The complement $\mathrm{Conf}_n \setminus \mathrm{Gen}_n$ is a null set, and the kernels are locally bounded on Conf_n, so dominated convergence gives the smeared limits.

Derivatives and smoothness. - Exact summation by parts in the lattice positions moves a difference quotient onto the test function. - The ultraviolet estimates show that the lattice difference quotient of an insertion is chessboard-close to an insertion of one degree higher. - A dimension count, from the rank comparison, identifies the kernel of the limiting projections with the null labels. - Finite-dimensional compactness then gives fixed derivative maps $\partial_a : F_d \to F_{d+1}$ with

$$\partial_{x_i^a} S(\dots, v_i, \dots; x) = S(\dots, \partial_a v_i, \dots; x) \qquad \text{in } \mathcal{D}'(\mathrm{Conf}_n).$$

Iteration and a Fourier argument give $S \in C^\infty(\mathrm{Conf}_n)$, with derivatives polynomially bounded near collisions.

What the limit inherits. The limit inherits: - translations; - the group of signed permutations of the axes, acting on points and labels; - permutation symmetry, reality and the unit label; - reflection positivity at coordinate planes, for point and smeared tests.

It does not inherit continuous rotations, clustering or uniqueness.

Removing the spacing refinement. The extraction refines both the spacing and the volume sequences. At each admissible fixed volume, the smeared limit along the whole spacing sequence exists (Part III) and agrees with the subsequential one. This removes the spacing refinement from the final statement.

1.0.7 Part V. Phases and relativistic reconstruction

Units: V.1 — Decomposition of the limit into phases; V.2 — Osterwalder–Schrader reconstruction of a Wightman theory; V.3 — Reconstruction for every phase and the reconstructed theory.

The OS space of the mixture. Positive-time field words, modulo the null space of $\langle \theta w, w' \rangle_S$ and completed, form a Hilbert space. It carries: - a contraction semigroup e^{-tH} with $H \geq 0$; - a unitary representation of the spatial translations.

The joint spectrum lies in a half-space. No positive spectral gap is asserted. The zero-energy space may be large. Its projection Π is the strong limit of e^{-tH} as $t \to \infty$.

Thin slabs. A word w placed in a thin time slab defines an operator B on the OS space. To bound it, consider the train $v_{n+1} = B^* B v_n$, which contains n reflected copies of w and an arbitrary exterior word. - The lattice bounds give $\|v_n\|^2 \leq B_v A^{4n}$. Here A is polynomial in the word data and in the inverse buffer width, and B_v depends on the exterior word but not on n. - A power lemma removes B_v: if $\|(BA)^n v\| \leq K_v D^{2n}$ for all n, then $\|Av\| \leq D\|v\|$.

So the exterior word is paid once and disappears under the n-th root.

Phases. - The compressions $M(w) = \Pi\, B_w|_{\ker H}$ are independent of the width and placement of the slab. - $\langle \Omega, M(w_1) \cdots M(w_k) \Omega \rangle$ is the limit of S as the clusters are moved apart in time order.

The compressions commute: 1. Place four clusters on the rays $n(ue_4 + ve_1)$, with $(u, v) = (0,0), (1,2), (2,1), (3,3)$. 2. Exchanging the time axis with a spatial axis reverses the order of the middle pair. 3. $O(4)$ covariance and permutation symmetry of S then give $\langle \Omega, M_x M_y M_w M_z \Omega \rangle = \langle \Omega, M_x M_w M_y M_z \Omega \rangle$. 4. By density, $[M_y, M_w] = 0$.

The resulting phase decomposition: - The $M(w)$ generate a commutative unital C^*-algebra. Its Gelfand spectrum is X, which is compact and metrizable. - ν represents the vacuum state, and it has full support because the vacuum is separating. - The phase correlations are $S^\xi(w) = \xi(M(w))$. They are continuous in ξ, and $\int S^\xi \, d\nu = S^{\mathrm{mix}}$. - Reflection positivity, $O(4)$ invariance, the derivative and stress identities and uniform growth bounds hold for every ξ. They are proved as operator identities before any character is applied. - Every phase has the same null spaces as the mixture. - A variance argument over a countable total family of words gives a one-dimensional zero-energy space for ν-almost every phase. - The selected phase ξ_* is one such phase, chosen by this existence argument; no further condition is imposed. The trace law of Part VII is proved for the phase so chosen.

Reconstruction. For each phase, slab operators bounded by $C p^{-A}$ (with p the slab half-width) define

$$F_n(z; \mathbf{x}) = \big\langle \Omega,\, B(v_1)\, e^{i(z_1 - z_2)H} U(\mathbf{x}_1 - \mathbf{x}_2)\, B(v_2) \cdots \Omega \big\rangle.$$

This function is holomorphic for $\operatorname{Im} z_1 < \cdots < \operatorname{Im} z_n$, and it equals the Euclidean correlation at imaginary times. The Wightman theory is obtained in five steps: 1. **Boundary values.** Bounds of order k^{-m}, when all height gaps exceed k, give tempered boundary values W_n. They converge against every joint Schwartz test as the vector of heights tends to zero through all ordered vectors. 2. **Positivity and fields.** This mode of convergence allows two independently regularized vectors to be compared. So W is a limit of Gram matrices of OS vectors: it is positive, and the Wightman space embeds isometrically into the OS space. Fields act on finite words modulo the null space by

prepending a letter. This gives the invariant domain, the formal adjoints and the cyclic vacuum. 3. **Covariance and spectrum.** The rotation Ward identities of Part VI, in planes containing the time axis, continue to infinitesimal boost identities. These integrate to Lorentz covariance without any spectral assumption. The spectral measure of a Wightman vector equals that of its OS image, so it is supported in $p^0 \geq 0$; covariance then moves the support into the forward cone. 4. **Locality.** Only adjacent transpositions are needed. For a spacelike adjacent pair, both orderings have holomorphic germs, constructed with complex boosts. The germs agree to all orders on the Euclidean slice, because the Euclidean kernel is symmetric and smooth away from coincidences. 5. **Faithfulness.** A label whose Wightman functions vanish has a vanishing tube function, hence a vanishing Euclidean kernel.

Every phase. The low-energy bound uses only locality and the spectrum condition. Take A a finite sum of field words with compactly supported tests. - The commutator of time-translated copies of A vanishes for times at most 1 and spatial separations beyond a radius R independent of ξ. - Average over unit time, pair with a spatial-momentum bump of radius $2u$ (whose Fourier transform is $O(u^3)$ in three space dimensions), and use $|\mathbf{p}| \leq E$ on the forward cone. This bounds the spectral weight of each shell $u/2 < E \leq u$ by $O(u^2)$, and dyadic summation gives the first bound below.

$$\mu_{A\Omega_\xi}(0 < E \leq u) \leq M_A u^2 \quad (0 < u \leq 1), \qquad 0 \leq \langle A\Omega_\xi, (e^{-sH_\xi} - P_{0,\xi})A\Omega_\xi \rangle \leq M_A s^{-2} \quad (s \geq 1).$$

The constant M_A is independent of ξ, by compactness of X.

Vacuum uniqueness then extends from almost every phase to every phase: - The function $\xi \mapsto \langle A\Omega_\xi, e^{-sH_\xi} A\Omega_\xi \rangle - |\langle \Omega_\xi, A\Omega_\xi \rangle|^2$ is continuous. - For fixed A and s, the set of phases where this function lies in $[0, M_A s^{-2}]$ is closed by continuity. It contains the full-measure set of phases with unique vacuum; full support of ν makes this closed set all of X. - Letting $s \to \infty$ gives a unique vacuum in every phase.

The comparison map from the Wightman space to the OS space is surjective, proved independently by pushing time-ordered vectors apart. This transfers vacuum uniqueness to the OS space. Each phase theory is irreducible. Its Wightman distributions are continuous in ξ, and their ν-average is the boundary value of the continuation of S^{mix}. The bounds allow spectrum down to zero and give only polynomial decay s^{-2} in Euclidean time, not exponential clustering.

1.0.8 Part VI. Ward identities, rotations and the stress tensor

Units: VI.1 — Ward identities, the rotation action and the conserved stress tensor; VI.2 — The free Maxwell stress tensor and the local Ward charge.

The lattice has only the hypercubic symmetries. The proof obtains the generators of rotations from stress fluxes, and the group action by comparison with free Maxwell theory. It uses the OS spaces of the mixed state in the coordinate directions and the free comparison of Part VII. It uses no decay of correlations.

Free Maxwell theory. Consider $\dim \mathfrak{g}$ free Maxwell fields, with field spaces the G-invariant polynomials in on-shell curvature jets. - The centred, real, symmetric tensors of degree 4 that are covariant under signed permutations form a three-dimensional space. - For simple $\mathfrak{g}$, the divergence maps this space onto a two-dimensional space. So the conserved tensors form a line, spanned by

the Maxwell stress tensor. - For every Killing field v, the flux of $\sum_b v_b T_{ab}$ through a shell around an insertion equals the Lie derivative of the insertion. For elementary fields this follows from $-\Delta G_0 = \delta$; for polynomials, from the Leibniz rule and Wick contraction. - An explicit three-point function fixes the sign and normalization of the Ward constant.

Conserved tensor. There are scaling sequences $r_k \downarrow 0$ with frames U_k, isomorphisms for large k. Along them the rescaled interacting divergence converges to the free one. - Surjectivity is an open condition, so the interacting divergence is also onto, and its kernel is one-dimensional. - A nonzero conserved J in this kernel converges, after rescaling, to a multiple of the free stress tensor.

Shell identities. Lattice integration by parts in the link variables, with ultraviolet control of the error terms, gives shell identities. Take χ equal to 1 near x_0 and vanishing near the spectators:

$$-\sum_a \int \partial_a \chi(y)\, S(J_{ab}(y)\, B(x_0) \cdots)\, dy = S((p_b B)(x_0) \cdots).$$

With weight $K(y - x_0)$, for a rotation generator K, the same construction gives maps Y_K. Here p_b and Y_K are linear maps on labels, not yet identified with derivatives and rotations. Moving the centre of the shell gives $\partial p = p\, \partial$ and $\partial_b Y_K + \sum_c K_{cb}\, p_c = Y_K \partial_b$.

Flux at infinity. - For rotations, cut off the rotation field so that it is compactly supported and equal to the exact field on a box far larger than the configuration. - Conservation and symmetry cancel the cross terms. - The remaining exterior contribution is bounded through spectral measures of the OS semigroups along and transverse to a coordinate line, and it tends to zero. - This gives the angular identity for configurations on one coordinate line, together with a line-sum identity, and hence transversality of the label maps.

Normalization. In free theory, the finitely many relations satisfied by p and ∂ at a given degree have a one-dimensional solution space. Injectivity is an open condition, so the same holds along the scaling sequence. Hence $p_b = c\, \partial_b$, with c independent of the degree and $c \neq 0$. The tensor $T = J/c$ satisfies

$$-\sum_a \int \partial_a \chi(y)\, S^s(T_{ab}(y)\, B(x_0) \cdots)\, dy = \frac{\partial}{\partial x_0^b}\, S^s(B(x_0) \cdots)$$

in the mixed state. By the operator identities of Part V, it holds in every phase. Since $\partial\chi$ vanishes near every insertion, no contact terms arise.

All separated configurations. Positivity of Gram functions, together with the Ward values already obtained, gives analytic-vector bounds $\|f^{(m)}\| \leq C(\rho/2)^{-m} m!$. The angular Ward functional is therefore real-analytic along walks that move one point at a time. Starting from configurations on a line, it vanishes on all configurations whose points have pairwise distinct first coordinates, and by continuity on all of Conf_n.

Integration to $O(4)$. - The bracket defect $[Y'_u, Y'_v] - Y'_{[u,v]}$ (with $Y' = Y/c$) satisfies a summed identity. A rigidity statement, proved in free theory and transferred by the comparison, forces the defect to vanish. - The infinitesimal action converges to the Maxwell one: $U_k^{-1} Y'_K U_k \to \mathrm{Lie}_K^{\mathrm{Max}}$. - A perturbation of spectral projections of the Casimir operators gives exact intertwiners V_d. The action on F_d is $R_d(k) = V_d R_d^{\mathrm{Max}}(k) V_d^{-1}$. - Since $t \mapsto S(R(e^{tK})v; e^{tK}x)$ has zero derivative, and reflections act through the real structure, $S(R(k)v; kx) = S(v; x)$ for all $k \in O(4)$. - Averaging J/c

over $O(4)$ gives a conserved, real, symmetric, covariant tensor that satisfies both the translation and the rotation Ward identities.

1.0.9 Part VII. Identification as Yang–Mills theory

1.0.9.1 Free Maxwell fields at short distance

Units: VII.1 — Free Maxwell comparison theory; VI.2 — Interacting stress comparison; VII.6 — Scaling variants and fixed-field estimates. The comparison is assembled in VIII.1.

The reference theory. It consists of $\dim \mathfrak{g}$ free Maxwell fields built from $\Gamma(x) = 1/(4\pi^2|x|^2)$. - On the lattice, corrected covariances of finite-difference jets converge to $\partial_a \partial_b \Gamma$ away from coincidences. - Wick moments satisfy the same Gaussian recursion on the lattice and in the continuum, so correlations of quotient labels converge. - A triangular basis built from the Kelvin transform gives strict reflected Gram positivity for every label outside the ideal I_d generated by the free Maxwell equations and the Bianchi identities. - So the free null space is exactly I_d, and the free field space $M_d = P_d/I_d$, where P_d is the corresponding space of polynomials in free curvature jets, is detected by finitely many correlations.

The comparison. There are scales $r_k \downarrow 0$ and linear frames U_k on the field spaces with the following property. The correlations of the mixed state, rescaled by r_k and transformed by U_k, converge to the free Maxwell correlations: - with all derivatives; - uniformly on compact subsets of Conf_n; - for all degrees along one sequence.

The stress tensor converges to the Maxwell stress tensor. An upper bound on $\dim F_d$ makes the frames isomorphisms for large k, so $F_d \cong M_d$.

The frames are necessary: the claim is not that a fixed normalization of each composite field has a limit along every sequence of radii. The schedule contains a real parameter that moves the scale at which observables are read, and the estimates are uniform in it. Tuning this parameter, every sequence of radii tending to zero has a subsequence along which the comparison holds.

1.0.9.2 Running coupling

Unit: VII.5 — Asymptotic freedom of the running coupling.

With $g_{N,p}(m) = \beta_{N,p,m}^{-1/2}$, Part III gives

$$g_{N,p}(m)\,\sqrt{\log(1/a_m)} \longrightarrow \frac{1}{\sqrt{2b_0}} \qquad (m \to \infty),$$

uniformly in $N \geq m$ and p. This is the leading-logarithmic running of the coupling of the construction's effective actions. No two-loop term, no Λ parameter and no comparison with another renormalization scheme is claimed. For $SU(N)$, the conventionally normalized coupling $g_{\mathrm{std}} = \sqrt{2N}\,g$ satisfies $g_{\mathrm{std}}\sqrt{\log(1/a_m)} \to (11N/24\pi^2)^{-1/2}$.

1.0.9.3 The trace law and nonfreeness

Units: VII.2 — The trace of the stress tensor and the trace anomaly; VII.3 — The theory is not free: Gram positivity and exclusion of free-field profiles.

Leading form of the trace. The classical Maxwell and Yang–Mills stress tensors are traceless. The quantum trace comes from the scale dependence of the coefficient of curvature squared. - The proof pulls the trace of the lattice stress observable back through the RG steps. - Fourth-order

matching identifies the leading part of the trace as the derivative of the effective action with respect to scale. - The one-loop coefficient enters through a momentum-shell estimate of the kind used in Part III.

Along the scaling sequences, with β_k the running coupling at the RG level that reads scale r_k, and O_{F^2} the normalized curvature-squared label,

$$\beta_k\, r_k^4\, U_k^{-1}\Theta \longrightarrow 4d_2\, O_{F^2}, \qquad d_2 = -\frac{\kappa_G}{4}, \qquad \frac{\log(1/r_k)}{\beta_k} \longrightarrow \frac{1}{\kappa_G}.$$

In perturbative terms, this is the familiar $\Theta \simeq \frac{\beta(g)}{2g^3}F^2$, with $\beta(g)/(2g^3) = -b_0/2 = d_2$ at one loop. The trace is a curvature-squared field suppressed by $g^2(r) \simeq 1/(\kappa_G \log(1/r))$.

The two-point law. Here $\Theta = \sum_\mu T_{\mu\mu} - S^{\xi_*}(\sum_\mu T_{\mu\mu})\mathbf{1}$, in the selected phase, which has a unique vacuum. 1. The free two-point function of O_{F^2} is $3\dim\mathfrak{g}/(\pi^4|x|^8)$. 2. So at lattice-fitted configurations along the scaling sequence, $r_k^8 \log^2(1/r_k)\langle\Theta\Theta\rangle \to (4d_2/\kappa_G)^2 \cdot 3\dim\mathfrak{g}/\pi^4 = 3\dim\mathfrak{g}/\pi^4$. 3. Rotation invariance moves the pair onto the time axis. 4. The spectral representation in the phase gives $G_\Theta(r) = \int e^{-rE}\, d\mu(E)$ with a positive measure. So G_Θ is nonincreasing, which sandwiches $G_\Theta(r)$ between its values at fitted points. 5. If the limit failed along some sequence of radii, the flexible scaling above would give a subsequence along which it holds, a contradiction.

Hence the limit exists over all radii:

$$\lim_{r\downarrow 0} r^8 \log^2(1/r)\, G_\Theta(r) = \frac{3\dim_{\mathbb{R}}\mathfrak{g}}{\pi^4}.$$

The amplitude does not depend on the value of the one-loop coefficient. κ_G enters once through the trace and once through the running, and $(4d_2/\kappa_G)^2 = 1$ identically. So the test detects the inverse-logarithmic suppression and the free normalization of F^2, not the value of b_0; that value is established by the running law of Part III.

Exclusion of free profiles. Consider a finite Wick profile $W(r) = \sum_\alpha c_\alpha \prod_\beta K_{\alpha\beta}(r)$. Each factor $K(r) = \int E^q\, h(\mathbf{p})(2E)^{-1}e^{-rE}\, d^3\mathbf{p}$ has $E = \sqrt{\mathsf{m}^2 + |\mathbf{p}|^2}$ and h homogeneous. - W has an asymptotic expansion at $r = 0$ in the functions $r^a(\log r)^k$, with $a \in \mathbb{Z}$ and $k \geq 0$. - Along $r = 2^{-n}$, $r^8 \log^2(1/r)\, W(r)$ is then a finite sum of terms $c\,\rho^n n^{k+2}$, which can converge only to 0. - Since the limit for G_Θ is positive, G_Θ differs from W at points arbitrarily close to 0.

This excludes exactly the finite class of profiles in the statement, and it concerns only G_Θ on the time axis.

The stronger statement adds further tests: - a Wightman-side Wick test; - Maxwell–Fock tests. Consider any isometric embedding of the selected phase's Hilbert space into a Fock space of transverse photons that preserves the vacuum and translations and has transverse symmetric image. No such embedding makes Θ local with respect to the Weyl operators.

1.0.9.4 Operator product expansion and contact extensions

Unit: VII.4 — Operator product expansion and contact extensions.

The lattice statement. The product of insertions in one cell equals a single insertion of the same total degree, plus a remainder with small chessboard root. The proof is an induction over

the RG levels between the cluster scale and the observation scale: the part of degree at most D is kept, and the rest is paid into the remainder.

The continuum coefficients. They are read off rather than given by a formula. - A prescription fixes, degree by degree, biorthogonal systems of fields and detectors. The detectors are finite combinations of mixed-state correlations of the cluster with fixed, separated exterior words. - The prescription is averaged over $O(4)$ and symmetrized for reality. - The coefficient $c_{j,\kappa}(a;x,o)$ is a coordinate of the cluster's new component of degree j in a real basis.

Consider an (r,M,η)-cluster: points within Mr of o and at mutual distance at least ηr, carrying fields of total degree w. Take spectators at distance at least c from o and from each other. Then, for all sufficiently small r,

$$|c_{j,\kappa}(a;x,o)| \le r^{\,j-w-\varepsilon}, \qquad \Big|S^s(a,x;b,y) - \sum_{j\le D}\sum_{\kappa} c_{j,\kappa}(a;x,o)\, S^s(O_{j,\kappa},o;b,y)\Big| \le r^{\,D+1-w-\varepsilon}.$$

Uniformity and limits of the statement. - The coefficients do not depend on the spectators b and y, on the state s, or on D. - The threshold for r depends on the fields, on M,η,c,ε,D, and in the first bound on the channel, but it is uniform in s, x, o and y. - Uniformity in the spectators comes from the train argument of Part V, in which the exterior word is paid once. - Uniformity in the state comes from bounding the error in the zero-energy algebra, whose norm dominates every character and the mixture. - The loss ε absorbs logarithms. There is no logarithmic refinement, so the bound is weaker than the perturbative OPE. - Uniqueness of the coefficients is not asserted. Neither convergence as $D\to\infty$ nor agreement with perturbation theory is claimed. - Nested versions over finite trees of clusters, and integrated versions against measures of bounded variation, also hold.

Contact extensions. These use Taylor subtractions over forests of collapsing clusters on dyadic shells. The OPE remainder makes the shell sums converge. For every state and every list of labels this gives a distribution of finite order on $(\mathbb{R}^4)^n$ that agrees with the correlation on Conf_n. Not established: uniqueness, reflection positivity on overlapping supports, contact Ward identities, real-time products, and convergence of the lattice correlations on the diagonals.

1.0.10 Part VIII. Assembly

Unit: VIII.1 — Assembly of the main theorem.

The construction run depends only on G and the penalty b. It is chosen before any extraction order or observable. For each extraction order (a spacing sequence and a volume sequence), the assembly composes further refinements: - one refinement for the limits of the running coupling; - one that produces the limit table, with the Euclidean hierarchy, stress tensor, phases and reconstruction of Parts IV–VI.

To these it adds the reconstructions of all phases, the OPE prescription, the nested and contact results and the Maxwell comparison. Everything is transported to a common refinement, so all objects refer to one table.

The public theorem is the case $b=1$, with the full spacing sequence and the given volume sequence; its q is the volume part of the refinement.

No new estimate enters, but identifying all objects with one table is a real constraint. Separate existence results for a lattice limit, a Wightman theory, an OPE and a trace law would not show that they concern the same correlation functions. The formal construction supplies the hypotheses under which the limit theory is first proved. The outlines have not explained every supplier in full; the assembly unit records the limits of its source reading.

1.0.11 Relation to earlier work, and where the difficulty lies

Classical sources. - Balaban's program is the model for the ultraviolet analysis: block-spin transformations, gauge fixing around a background field, fluctuation covariances built from averaging operators, the small-field/large-field decomposition, and renormalization of the marginal coupling. - The classical cluster expansion (Glimm–Jaffe, Brydges, Kotecký–Preiss) organizes the polymer activities. - Reflection positivity (Osterwalder–Seiler) and chessboard estimates (Glimm–Jaffe–Spencer, Fröhlich–Israel–Lieb–Simon) connect lattice estimates with bounds on observables. - Osterwalder–Schrader and Glimm–Jaffe reconstruction turns Euclidean correlations into Wightman fields. - The slope is the one-loop coefficient of Gross–Wilczek and Politzer.

Departures from Balaban's scheme. - The block kernel is an exactly normalized, gauge-covariant Markov kernel with random tags, not a linear averaging operator in a fixed background gauge. - Integration uses a spanning-forest gauge. - The block ratios grow along the levels. - The threshold carries the factor $e^{(\log\log\widehat{\beta})^2}$. - The construction includes an analytic family in a tuning parameter. The unit outlines do not establish a comprehensive comparison with Balaban's papers.

Beyond the ultraviolet analysis, the proof does the following: - One schedule and one family serve composite fields of every degree. Each field has a fixed normalization, root bounds uniform in the volume, and derivatives controlled by rank detection, with no coupling-dependent loss per site. - Convergence at fixed volume is proved despite tuning that depends on the cutoff. - Equicontinuity in infinite volume comes from a rank-free trace inequality, without derivative bounds. - Commutativity of the zero-energy algebra is proved, not assumed. - Vacuum uniqueness in every phase follows from a uniform low-energy bound that replaces a mass gap or a cluster argument. - Lorentz covariance comes from Ward identities. - Rotation invariance is restored without decay of correlations, and the group action is obtained by comparison with free Maxwell theory. - The trace law holds as a limit over all radii. - The OPE is derived from the RG, with one system of coefficients for the mixture and every phase.

Where the difficulty lies. 1. **Gauge degeneracy and large fields** must be controlled with a bounded cost per site and per seed. This is done by the gauge count of the determinant budget, the energy lower bound for seeds, and the exactness of the kernel. 2. **Effective-action estimates must become estimates for observables**: for every composite field, its derivatives and its fixed-volume limit, inside one family. 3. **The passage from a lattice mixture to relativistic phases** has no continuous symmetry and no clustering to start from. It is handled by Ward identities with free comparison, the commutative zero-energy algebra, and spectral bounds based on locality.

The order of choice of constants and the transport of estimates through the construction account for a substantial part of the formal development. Its length alone does not measure mathematical novelty.

1.0.12 Steps whose informal account is incomplete

The theorem is proved in Lean, and the conditional statements of the limit theory are discharged within its proof. The unit outlines, however, describe the following steps only at the level of statements, or with readings marked tentative. A written proof will have to supply them.

- **One-loop identification (III.7, VII.2).** The slope bound is first proved for an auxiliary functional and then transferred to the completed Hessian of the actual step. The revised III.7 explains the determinant factorization and the transfer of the coefficient; several subsidiary real-quotient, complex-strip, tadpole, bubble and alias estimates remain inputs at statement level. The trace amplitude does not test this value.
- **Covariance and Hessian (III.4).** The Gaussian covariance is the inverse of the gauge-fixed Gram form on the active links. Whether it equals the inverse Hessian of the effective action at the background is left open.
- **Nonlinear step.**
  - The correspondence between the outputs of the one-step bounds (III.10) and the norms restored by the reset (III.9) is not written out.
  - Several hypotheses of the step were read only as statements.
  - The cluster expansion is proved for real weights (III.5), while the fixed-volume argument uses complex tuning.
- **Fixed-volume convergence (III.12, IV.1).** III.12 now explains fixed-level convergence, the uniform tail bound and the Cauchy argument; the underlying trajectory-convergence and multiscale estimates remain imported inputs. IV.1 locates the upgrade to the full spacing sequence, while leaving the upstream construction of its fixed-volume kernels outside that unit's review.
- **Shell identities and the flux at infinity (VI.1, VI.2).** The lattice derivation of the shell identities is given only at statement level. The exterior-flux estimate was not read. The descent of the rotation action to exact intertwiners is marked tentative.
- **Reconstruction (V.2).** The boundary values, the spectrum condition and the locality argument are described from statements; their mechanisms were confirmed in the Lean but not written out in the unit outlines.
- **Contact extensions and nested OPE (VII.4).** The shell estimates and the nested rates are described from docstrings.
- **Field relations (II.1).** It is not shown directly that the lattice equation relations exhaust the null space. The identification $F_d \cong M_d$ comes from the comparison with free theory.

1.0.13 Reading guide

The exact statement separates the public theorem from the stronger construction. The architecture describes what each part hands to the next. For the stated inputs and remaining gaps in this account, follow the relevant unit's "What it uses" and "What remains unexplained" sections. The assembly unit describes how the construction is put together.

Question	Units to read
What are the fields, and how are their normalizations fixed?	II.1, III.12, IV.1, VII.1
How does the ultraviolet analysis relate to Balaban's?	III.1, III.3, III.4, III.6, III.5, III.10

Question	Units to read
Where does the one-loop coefficient enter?	III.7, III.9, VII.5, VII.2
Why does one family serve every observable and every limit?	III.2, II.1, III.12, VIII.1; III.13 treats an alternative family
How are bounds made uniform in the volume?	III.11, III.12, III.16
How is infinite volume reached without an infrared RG?	III.16, IV.1
Why are there phases, and why does every phase reconstruct?	V.1, V.2, V.3; VII.4 for the train bounds
How are rotations and the stress tensor recovered?	VI.2, VI.1, with VII.1 and VII.6 for the free comparison
What makes the theory nonfree?	VII.2, VII.3, with VII.6 for arbitrary short-distance scales
What does the OPE say, and what is known at coincident points?	VII.4, then VIII.1 for the assembly

2 The architecture

This page describes how OpenAI's Lean proof of the construction of four-dimensional Yang–Mills theory is organised: its eight parts, what each part hands to the next, and the units that make up each part (detailed map).

It is read from the Lean statements and documentation and from the dependency graph. The mathematical framing follows the Yang–Mills group's assessment. Lean names link to the fixed source commit `3456e5e`.

2.0.1 The order of limits

The construction uses Wilson lattice theories on tori of side L_p, with lattice spacings a_N and couplings $\beta_{N,p}$ satisfying $\beta_{N,p}/\log(1/a_N) \to 11C_G/(24\pi^2)$. The public theorem leaves the positive limiting constant unspecified; see the exact statement.

Correlations of renormalized local observables are first taken to the limit $N \to \infty$ **at fixed volume**, along the whole sequence of spacings. They are then taken to the limit $p \to \infty$, along a subsequence that may be extracted from any prescribed sequence of volumes. The public form of this statement is `WilsonFamily.Converges`.

The renormalization group is used only for the first limit. The second is obtained by compactness.

2.0.2 The eight parts

part	name	units
I	Setting	I.1
II	Data of the construction	II.1, II.2
III	Ultraviolet construction	III.1–III.16
IV	Infinite volume	IV.1
V	Phases and reconstruction	V.1, V.2, V.3

part	name	units
VI	Symmetry	VI.1, VI.2
VII	Identification as Yang–Mills	VII.1, VII.2, VII.3, VII.4, VII.5, VII.6
VIII	Assembly	VIII.1

2.0.2.1 Ultraviolet units

- Blocking, constants and Gaussian estimates: III.1, III.2, III.3, III.4.
- Cluster expansion, stationary field, matching and dependence: III.5, III.6, III.7, III.8.
- Coupling reset, nonlinear step and observable limits: III.9, III.10, III.11, III.12.
- Alternative coordinated family, construction data and analytic tools: III.13, III.14, III.15, III.16.

The limit theory (Parts IV–VI) is written in the code around numbered work items; the units of those parts follow them.

2.0.3 What each part hands to the next

2.0.3.1 Ultraviolet → infinite volume Let G be a compact matrix group with simple nonabelian Lie algebra, and $b > 0$ a parameter of the block-spin kernel. The ultraviolet construction provides a Wilson family with the following properties.

1. **Specification.** The family satisfies its normalization, covariance and budget conditions, and its couplings obey the asymptotic-freedom law above.
2. **Chessboard bound, uniform in the volume.** For each degree d and $\varepsilon > 0$, every renormalized local observable at block scale m satisfies a chessboard bound of order $a_m^{-d-\varepsilon}$, uniformly in the volume, once the lattice is fine enough. A chessboard bound is a bound on the root of a product of reflected copies.
3. **Differences.** The same holds for finite-difference quotients, so correlation kernels at r-separated points are bounded uniformly in the volume.
4. **Fixed volume.** At each fixed volume the correlation kernels converge along the whole sequence of spacings, to smooth limits.
5. **Corrected fields.** The same estimates hold for the corrected fields.

For the subsequential extraction, the infinite-volume part uses 1–3 and 5, together with the reflection-positivity inequality on cells, which holds for every $\beta \geq 0$. Its conclusion: for every increasing sequence of volumes there is a subsequence along which all smeared correlations converge to kernels that are smooth away from coincident points. The fixed-volume convergence in item 4 then removes the spacing refinement; see IV.1. These two limits correspond to Theorems 4.1 and 5.1 of the group's assessment.

A caution about the code. About three quarters of the code filed under "full volume" (folder `YM/FullVolume`) belongs to the ultraviolet construction: it proves the volume-uniform bounds. The genuinely infinite-volume argument is short, about 7,000 lines.

2.0.3.2 Infinite volume → phases, reconstruction and symmetry The limit satisfies a Euclidean *hierarchy*:

- translation homogeneity;
- reflection positivity, including for smeared joint test functions;
- smoothness on separated configurations, with controlled derivatives;
- a compatible calculus of derivatives.

Reconstruction also needs a uniform bound on products of fields in thin time slabs.

2.0.3.3 Reconstruction → identification For every phase, a completed Wightman theory: unique vacuum, a light-cone bound, heat-kernel decay, and a trivial weak commutant. The heat bound is polynomial; no mass gap follows.

2.0.3.4 Symmetry → identification A conserved stress tensor and its Ward identities. Both are compared, along sequences of scales tending to zero, with the corresponding objects for free Maxwell fields.

2.0.3.5 Into the assembly A single Wilson family serves every part of the public theorem, from extraction to the trace test; VIII.1 describes the assembly. The construction run exists for every allowed G and every $b > 0$. III.13 describes an alternative coordinated family; its properties require a separate identification before they can be used for this family.

2.0.4 The work items of the limit theory

Parts IV–VI are written in the code as about seventy numbered work items. Each states one obligation, a "proof request" in the code's vocabulary, and proves it in its own file.

family	items	part	content
D	6	IV	the limit passes to derivatives of the correlation kernels; the Euclidean hierarchy
B	18	V	time-slab operators, decomposition into phases, reflection positivity and smoothness for each phase
C	29	V	Osterwalder–Schrader → Wightman: continuation in time, boost covariance, gluing of tubes, locality

family	items	part	content
A	17	VI	so(4) brackets, integration to the rotation action, the averaged conserved stress tensor, boosts, comparison with free Maxwell fields

These families interleave. D supplies the limit hierarchy; B begins with its OS spaces and slab operators. A uses those spaces and the free Maxwell comparison to obtain rotations and Ward identities. Those symmetry results enter both the phase decomposition in B and the boost and covariance arguments in C. The part numbers organize the subjects; they do not give a linear dependency order. See the strategy and the work-item introductions.

The same mechanism appears in the ultraviolet construction: families E–I cover the energy chain of one renormalization step, its stationary point and Hessian, label polynomials, and support bookkeeping.

See work-items.md for the complete list.

2.0.5 What does not follow the order of the parts

The Lean is not written in this order everywhere. Some definitions are needed before the part that explains their properties:

- the reflection-positive cells and chessboard roots;
- the record of the whole construction run;
- the basic types of the renormalization group and of the labels.

Some sections are filed far from their mathematical home. Two examples:

- the rotation-invariance estimate, in folder `Next/ROTATIONS`;
- some renormalization-group pieces filed with the trace-anomaly code.

None of this involves new mathematics. A written account should simply state these definitions early.

2.0.6 How reliable is this

- Reachable declaration sets for seventeen named results were compared with OpenAI's compiled record. This comparison does not verify every edge or the mathematical interpretation of the interfaces (method).
- The plain-language interfaces summarize statements and documentation. The unit index gives each outline's review status; its local limitations describe which arguments remain at statement level. The exact statement distinguishes the public theorem from additional construction conclusions.
- Report a correction with the page, heading and source reference.

3 The units

3.1 I.1. Lattice, symmetries, gauge groups and the Wilson measure

Part: I, Setting. **Lean:** `YM/Lattice` (1,215 lines), `YM/Gauge` (1,131), `YM/Space` (430), `YM/Corona` (405), `YM/Mink` (363), `YM/Wilson` (157). **Size:** 3,701 lines in 595 declarations used by the main theorem.

3.1.1 1. What this unit proves

The unit fixes the setting of the construction:

- the periodic lattices and their symmetry group;
- parallel transport and gauge transformations;
- the subdivision of a torus into blocks and the schedule of lattice spacings;
- a combinatorial measure of the size of a lattice set;
- the gauge group in its defining representation, with its Lie algebra, Casimir constant, local logarithm and nearest-point retraction;
- the Wilson measure and its invariance;
- the continuum symmetry groups of $\mathbb{R}^4$.

The results require no input theorem from another unit. Simplicity of $\mathfrak{g}$ is assumed only in Proposition I.1.6(d)–(f); the other hypotheses are stated locally.

Conventions. G is a closed subgroup of $U(N)$, with $N \geq 1$ fixed. For an integer $\ell \geq 1$, write $\Lambda_\ell = (\mathbb{Z}/\ell\mathbb{Z})^4$. Axes $0, 1, 2$ are spatial and axis 3 is time; e_i are the unit vectors. A link (x, i) joins x to $x + e_i$. A configuration is a map $U : \Lambda_\ell \times \{0, 1, 2, 3\} \to G$; the same transport and gauge definitions apply on the cover $\mathbb{Z}^4$. The norm $\|\cdot\|_F$ is the Frobenius norm on $M_N(\mathbb{C})$; norms on real coordinate spaces are Euclidean unless marked otherwise.

Proposition I.1.1 (torus geometry). For $y \in \mathbb{Z}^4$ let $[y]_\ell \in \Lambda_\ell$ be its reduction mod ℓ. For $x \in \Lambda_\ell$ let $r_\ell(x) \in \mathbb{Z}^4$ be the representative with coordinates in $(-\ell/2, \ell/2]$ (the centred lift; the tie $x_i = \ell/2$ lifts to $+\ell/2$). Then:

- $[r_\ell(x)]_\ell = x$, and $r_\ell(x)$ is the unique y with $[y]_\ell = x$ and $-\ell < 2y_i \leq \ell$ for all i;
- reduction is injective on this centred box, and $|\Lambda_\ell| = \ell^4$.

Put $|x|_\ell = \max_i |r_\ell(x)_i|$. The separation $|x - y|_\ell$ is a metric on the torus:

- it is symmetric and is zero exactly when $x = y$;
- it satisfies the triangle inequality;
- it is invariant under translations and under the hyperoctahedral group B_4 of Proposition I.1.2;
- reduction does not increase distances: $|[y]_\ell|_\ell \leq \max_i |y_i|$.

Status: theorem. *Lean:* `Lattice.Site`, `wrap`, `lift`, `lift_eq_iff`, `wrap_injOn`, `site_card`, `separation`, `separation_comm`, `separation_triangle`, `separation_translate`, `separation_signed`, `Corona.lap`.

Proposition I.1.2 (lattice symmetries, transport, gauge transformations). Let B_4 be the group of signed permutations of the four axes; it has order $2^4 \cdot 4! = 384$. An element $\rho = (\pi, \epsilon)$, with $\epsilon_i \in \{\pm 1\}$, acts on $\mathbb{Z}^4$ and on Λ_ℓ by $(\rho x)_{\pi(i)} = \epsilon_i x_i$. On either lattice Λ, an affine map is $A = (\rho, b)$, $Ax = \rho x + b$, with $b \in \Lambda$.

A walk w is a finite word of steps $\pm e_i$. The transport $U(x;w) \in G$ is the ordered product along w from x: a forward step along the link (y,i) contributes $U(y,i)$, and a backward step from y contributes $U(y-e_i,i)^{-1}$.

The pullback A^*U is

$$(A^*U)(x,i) = \begin{cases} U(Ax, \pi(i)), & \epsilon_i = +1, \\ U(Ax - e_{\pi(i)}, \pi(i))^{-1}, & \epsilon_i = -1, \end{cases}$$

that is, the transport of U along the image of the link (x,i). Then:

- (a) $1^* = \mathrm{id}$ and $B^*A^* = (A \circ B)^*$, i.e. $(\rho', b')^*(\rho, b)^* = (\rho\rho', \rho b' + b)^*$. For $A = (1, b)$ the pullback is the translation $(\tau_b U)(x,i) = U(x+b,i)$.
- (b) $(A^*U)(x;w) = U(Ax; \rho w)$, where ρw is the image word.
- (c) For $g : \Lambda \to G$ put $U^g(x,i) = g(x)U(x,i)g(x+e_i)^{-1}$. Then $U^g(x;w) = g(x)\, U(x;w)\, g(x_{\mathrm{end}})^{-1}$; for a closed walk this is conjugation by $g(x)$.
- (d) (locality) If U and V agree on every link whose two endpoints are visited by w from x, then $U(x;w) = V(x;w)$.
- (e) (plaquettes) A plaquette is $p = (x; i < j)$ with holonomy $U_p = U(x; e_i, e_j, -e_i, -e_j)$. Each A induces an injective map \$A_\#\$ of plaquettes, bijective on Λ_ℓ. It sends the axis pair to $\{\pi(i), \pi(j)\}$ in increasing order and the base point to $Ax - \sum_{k \in \{i,j\}:\, \epsilon_k = -1} e_{\pi(k)}$. For every set M and every $\phi : G \to M$ satisfying $\phi(u^{-1}) = \phi(u)$ and $\phi(vuv^{-1}) = \phi(u)$ for all $u, v \in G$, one has $\phi((A^*U)_p) = \phi(U_{A_\# p})$.
- (f) (periodic extension) For $b \in \Lambda_\ell$ let $(E_b U)(y,i) = U([y]_\ell + b, i)$ on $\mathbb{Z}^4$. Then:
  - $(E_b U)(y;w) = U([y]_\ell + b; w)$;
  - $E_b((\rho,a)^*U) = (\rho, 0)^*(E_{\rho b + a} U)$ for $a \in \Lambda_\ell$;
  - $E_b(U^g) = (E_b U)^{g([\cdot]_\ell + b)}$;
  - if F depends only on links inside $X \subset \mathbb{Z}^4$, then $F \circ E_b$ depends only on links inside $[X]_\ell + b$.

A function F of configurations is *gauge invariant* if $F(U^g) = F(U)$ for all g. It is *local on* X if $F(U) = F(V)$ whenever U, V agree on all links with both endpoints in X. These are the notions of (Reg3)–(Reg4) in the main statement. *Status:* theorem. *Lean:* `Signed` with its group law, `Signed.act`, `travel`, `pull`, `pull_pull`, `pull_one`, `travel_pull`, `gauge`, `travel_gauge`, `GaugeInvariant`, `Agrees`, `LocalOn`, `travel_local`, `Face.send`, `Face.injective_send`, `Face.send_price`, `expandAt`, `expandAt_pull`, `travel_at`, `expandAt_gauge`, `LocalOn.expandAt`.

Proposition I.1.3 (schedule of spacings; subdivision and blocks).

(a) *Schedule.* A schedule consists of odd integers $L_m \geq 3$ and a strictly increasing sequence of positive integers P_p with $4 \mid P_p$, indexed by $m, p \in \mathbb{N}$. Put

$$D_0 = 1, \quad D_{m+1} = D_m L_m, \quad a_m = 1/D_m, \quad \ell(m,p) = P_p D_m, \quad s_m = \log(1/a_m).$$

Then:

- D_m is odd and strictly increasing;

- $a_0 = 1$, $a_{m+1} = a_m/L_m$, and $a_m \downarrow 0$ strictly;
- $s_0 = 0$, $s_{m+1} = s_m + \log L_m$, and $s_m \to \infty$;
- $4 \mid \ell(m,p)$ and $a_m \ell(m,p) = P_p$;
- the placement $x \mapsto a_m\, r_{\ell(m,p)}(x)$ maps $\Lambda_{\ell(m,p)}$ into $(-P_p/2, P_p/2]^4$.

For $m \le m'$, $\Lambda_{\ell(m,p)} \to \Lambda_{\ell(m',p)}$ is a subdivision of odd ratio $L_m \cdots L_{m'-1}$ that preserves placements.

(b) *Mesh inequality.* If $\log L_m/s_m \to 0$ as $m \to \infty$ (the schedule is *gentle*), then there is $c > 0$, depending on the schedule, with $a_{m+1} \ge c\, a_m^2$ for all $m \in \mathbb{N}$. The ratio defining gentleness is considered for $m \ge 1$, when $s_m > 0$.

(c) *Subdivision.* Let $L \ge 1$ be an integer and $\ell' = \ell L$. Put $\iota(x) = [L\, r_\ell(x)]_{\ell'}$. Then:

- $r_{\ell'}(\iota x) = L\, r_\ell(x)$;
- $\iota : \Lambda_\ell \to \Lambda_{\ell'}$ is an injective group homomorphism commuting with B_4.

The residue map $\mathrm{res} : \Lambda_{\ell'} \to \Lambda_L$ (reduction mod L) is additive with $\mathrm{res} \circ \iota = 0$. The map $(x, y) \mapsto \iota(x) + [r_L(y)]_{\ell'}$ is a bijection $\Lambda_\ell \times \Lambda_L \to \Lambda_{\ell'}$, with inverse $z \mapsto (\mathrm{par}\, z, \mathrm{res}\, z)$; moreover $\mathrm{par} \circ \iota = \mathrm{id}$.

(d) *Odd ratio.* If L is odd, then:

- r_L commutes with B_4 and $\mathrm{par}(\rho z) = \rho\, \mathrm{par}(z)$;
- the block $\mathrm{par}^{-1}(x)$ is $\iota(x) + [\{-\frac{L-1}{2}, \dots, \frac{L-1}{2}\}^4]_{\ell'}$;
- $\mathrm{par}(z + e_i) = \mathrm{par}(z) + e_i$ if $r_L(\mathrm{res}\, z)_i = \frac{L-1}{2}$, and $\mathrm{par}(z + e_i) = \mathrm{par}(z)$ otherwise.

(e) *Metric comparison.* For every integer $L \ge 1$, all $x, x' \in \Lambda_\ell$ and all $z, z' \in \Lambda_{\ell'}$:

- $|z - \iota\, \mathrm{par}\, z|_{\ell'} \le L/2$;
- $L\, |x - x'|_\ell \le |\iota x - \iota x'|_{\ell'}$;
- $L\, |\mathrm{par}\, z - \mathrm{par}\, z'|_\ell \le |z - z'|_{\ell'} + L$ (foreshortening).

Status: (a) the schedule is a definition (data with conditions); the schedule of the construction is chosen in III.2. The consequences are theorems. *Lean:* `Schedule`, `denominator`, `side`, `side_four`, `a`, `strictAnti`, `shrinks`, `scale`, `divergent`, `span_side`, `place`, `places_period`, `toDepth`, `toDepth_odd`, `embeds_place`, `Gentle`, `mesh_inequality`, `Ratio`, `Ratio.embed`, `Ratio.root`, `Ratio.signed`, `Ratio.residue`, `Ratio.assemble_bijective`, `Ratio.parent`, `Ratio.parent_signed`, `Ratio.half_box`, `Ratio.parents_hop`, `Ratio.embed_parent`, `Corona.near`, `Corona.below`, `Corona.foreshorten`.

Proposition I.1.4 (cover numbers of lattice sets). For $X \subset \Lambda_\ell$ and $M \in \mathbb{N}$ let $\mathrm{halo}_M(X) = \{y : \exists x \in X,\ |x - y|_\ell \le M\}$. Use the graph joining sites at sup-distance at most 1; diagonal moves are allowed. Connectivity means that every two vertices can be joined, so the empty set is allowed. A set D *covers* X *at scale* M if $X \subset \mathrm{halo}_M(D)$ and $\mathrm{halo}_M(D)$ is connected in this graph. Let $\mathrm{cov}_M(X)$ be the least $|D|$ over such D; it is finite since $D = \Lambda_\ell$ covers. For all $X, Y, D \subset \Lambda_\ell$ and $M \in \mathbb{N}$, the following bounds hold independently of ℓ:

- (a) $X \subset Y$ implies $\mathrm{cov}_M(X) \le \mathrm{cov}_M(Y)$, and $\mathrm{cov}_M(X \cup Y) \le \mathrm{cov}_M(X) + \mathrm{cov}_M(Y)$ if $X \cap Y \ne \emptyset$.
- (b) $|\mathrm{halo}_M(D)| \le (2M+1)^4 |D|$ and $|X| \le (2M+1)^4\, \mathrm{cov}_M(X)$.
- (c) $\mathrm{halo}_M(D)$ is connected in this graph if and only if any two points of D are joined by a chain in D with steps of sup-distance at most $2M+1$.
- (d) (diameter) For $x, y \in X$: $|x - y|_\ell \le (4M+2)\, \mathrm{cov}_M(X) + 2M$.

- (e) (change of scale) For $b, k, m \in \mathbb{N}$ with $b \geq 1$ and $x \in \Lambda_\ell$, $\mathrm{cov}_b(\mathrm{halo}_{kb}\{x\}) \leq (2k+1)^4$. If D covers X at scale mb, then $\mathrm{cov}_b(X) \leq (2m+3)^4|D|$.
- (f) (coarsening) Let $\mathrm{par} : \Lambda_{\ell L} \to \Lambda_\ell$ be the parent map of a subdivision of integer ratio $L \geq 1$. Let $X \subset \Lambda_{\ell L}$ and $M, B, s \in \mathbb{N}$ with $B \geq 1$ and $M \leq 2B$. Then

$$L \cdot \mathrm{cov}_B(\mathrm{halo}_{sB}(\mathrm{par}\, X)) \leq (L + 10\,\mathrm{cov}_M(X))\,(2s+13)^4,$$

  with the left side computed on the coarse torus. Thus a fine set of cover number k has coarse cover number $\leq (2s+13)^4(1+10k/L)$.

Status: theorem. *Lean:* `halo`, `covers`, `coverNumber`, `Corona.cover_mono`, `Corona.meets_sum`, `Corona.halo_size`, `Corona.pays`, `Corona.ship_halo`, `Corona.halo_ship`, `Corona.extent`, `Corona.tile`, `Corona.tiling`, `Corona.collapse`.

Proposition I.1.5 (gauge group, plaquette cost, chord, Haar measure). G is compact and second countable, with its Borel structure. For $u \in G$ put

$$\sigma(u) = 1 - \tfrac{1}{N}\mathrm{Re}\,\mathrm{tr}\,u, \qquad \chi(u) = \|u-1\|_F/\sqrt{N}.$$

Then:

- $\chi^2 = 2\sigma$, $0 \leq \chi \leq 2$, $0 \leq \sigma \leq 2$, and $\sigma(1) = \chi(1) = 0$;
- $\chi(uv) \leq \chi(u) + \chi(v)$;
- σ and χ are continuous and invariant under $u \mapsto u^{-1}$ and $u \mapsto vuv^{-1}$.

The Frobenius norm is invariant under left and right multiplication by unitaries, and $\|u\|_F = \sqrt{N}$. The normalized Haar measure μ_G is invariant under left and right translations and under inversion; hence $u \mapsto vuw$ preserves μ_G. *Status:* theorem. *Lean:* `Gauge`, `compactGaugeU`, `price`, `chord`, `chord_price`, `chord_le_two`, `price_range`, `chord_triangle`, `price_inv`, `price_conj`, `chord_inv`, `chord_conj`, `mul_frob_eq`, `probHaar`, `probHaar_rightInv`, `preserve_inv`, `preserve_two_sided`.

Proposition I.1.6 (Lie algebra, colours, Casimir constant, slope constant).

(a) $\mathfrak{g} = \{X \in M_N(\mathbb{C}) : e^{tX} \in G \ \forall t \in \mathbb{R}\}$ is a closed real subspace of skew-Hermitian matrices. It is closed under $[X,Y] = XY - YX$ and under $X \mapsto uXu^{-1}$, $u \in G$. *Tangent criterion:* if $f : \mathbb{R} \to M_N(\mathbb{C})$ takes values in G, $f(0) = 1$, and f is differentiable at 0 with $f'(0) = X$, then $X \in \mathfrak{g}$.

(b) *Colours.* $\langle X, Y\rangle_{\mathfrak{g}} = -\frac{1}{N}\mathrm{tr}(XY)$ is real on $\mathfrak{g}$ and is an $\mathrm{Ad}(G)$-invariant inner product. Fix an orthonormal basis and write $\mathfrak{g} \cong \mathbb{R}^d$, $c \mapsto X_c$, $d = \dim \mathfrak{g}$. Then:

- $\|X_c\|_F = \sqrt{N}\,|c|$;
- if $P_{\mathfrak{g}}$ is the orthogonal projection onto $\mathfrak{g}$ for the real Frobenius inner product $\mathrm{Re}\,\mathrm{tr}(XY^*)$, define $\Pi : M_N(\mathbb{C}) \to \mathbb{R}^d$ by $X_{\Pi A} = P_{\mathfrak{g}}A$. Then $\Pi X_c = c$ and $|\Pi A| \leq \|A\|_F/\sqrt{N}$;
- $\mathrm{Ad}(u)c = \Pi(uX_cu^{-1})$ is an orthogonal representation of G;
- $\mathrm{ad}(c)c' = \Pi[X_c, X_{c'}]$ satisfies $X_{\mathrm{ad}(c)c'} = [X_c, X_{c'}]$;
- $c \mapsto \exp c := e^{X_c} \in G$ is continuous.

(c) G is *simple* if two conditions hold: there are $X, Y \in \mathfrak{g}$ with $[X,Y] \neq 0$, and every real subspace $I \subset \mathfrak{g}$ with $[\mathfrak{g}, I] \subset I$ is 0 or $\mathfrak{g}$. This agrees with the simplicity hypothesis of the main theorem: its Lie algebra is characterized by the same exponential condition (`GaugeGroup.coreSimple`, unit VIII.1).

If G is simple, then:

- (d) for every symmetric real bilinear form B on $\mathbb{R}^d$ satisfying $B(\mathrm{ad}(x)y,z) = -B(y,\mathrm{ad}(x)z)$ for all x,y,z, there is $c \in \mathbb{R}$ such that $B(x,y) = c\langle x,y\rangle$ for all x,y;
- (e) $-\mathrm{tr}_{\mathbb{R}^d}(\mathrm{ad}(x)\mathrm{ad}(y)) = C_{\mathfrak{g}}\langle x,y\rangle$ for all $x,y \in \mathbb{R}^d$, where $C_{\mathfrak{g}} := -\frac{1}{d}\sum_{i=1}^{d} \mathrm{tr}_{\mathbb{R}^d}(\mathrm{ad}(b_i)^2) > 0$ and $(b_i)_{i=1}^d$ is the standard orthonormal basis;
- (f) the *slope constant* $\kappa := 11C_{\mathfrak{g}}/(24\pi^2)$ is positive.

Status: theorem; $C_{\mathfrak{g}}$ and κ are definitions. *Lean:* `lie`, `closed_lie`, `generates_skew`, `tangent_generates`, `bracket_mem`, `conjugate_mem`, `Simple`, `colorIso`, `colorMat`, `trace_color`, `trace_color_real`, `colorMat_norm`, `projectColor`, `colorMat_project`, `project_colorMat`, `project_norm`, `adjoint`, `adjoint_mul`, `norm_adjoint`, `ad`, `colorMat_ad`, `lieExp`, `lieExp_cont`, `casimir`, `slopeConstant`, `invariant_scalar`, `trace_adjoint`, `casimir_pos`, `slopeConstant_pos`.

Proposition I.1.7 (Haar mass of small balls around 1). For $0 < r \le 1$, $\mu_G\{u : \chi(u) \le r\} \ge (r/4)^{2N^2}$. The exponent is the real dimension of $M_N(\mathbb{C})$, not $\dim G$. *Status:* theorem (not sharp). *Lean:* `shell`, `shell_ge`.

Proposition I.1.8 (local logarithm, nearest point, curvature of a group element). Let $\mathfrak{g}^\perp$ be the orthogonal complement of $\mathfrak{g}$ in $M_N(\mathbb{C})$ for $\mathrm{Re}\,\mathrm{tr}(XY^*)$.

(a) There are a map $\log_G : M_N(\mathbb{C}) \to \mathbb{R}^d$, an open neighbourhood V of 1 in $M_N(\mathbb{C})$ and an open neighbourhood W of 0 in $\mathbb{R}^d$ such that:

- $\log_G$ is real-analytic on V and $\log_G(1) = 0$;
- $\log_G(e^{X_c}) = c$ for all $c \in W$;
- $\exp(\log_G u) = u$ for all $u \in G \cap V$.

It is the first component of the local inverse of $(c,\nu) \mapsto e^{X_c}e^{\nu}$ on $\mathbb{R}^d \times \mathfrak{g}^\perp$. Connectedness of G is not assumed, and nothing is claimed away from 1.

(b) $u \in G$ is a *nearest point* to $X \in M_N(\mathbb{C})$ if $\|X-u\|_F \le \|X-v\|_F$ for all $v \in G$. Then:

- nearest points exist;
- for every $v \in G$, u is nearest to X iff vu is nearest to vX, iff uv is nearest to Xv;
- if u is nearest to X, then $u^*X - 1 \in \mathfrak{g}^\perp$;
- there is $\delta > 0$, depending only on G, such that X has exactly one nearest point whenever $\|X-v\|_F < \delta$ for some $v \in G$.

On a neighbourhood of 1 the nearest point is $\mathrm{foc}(X) = \exp(c)$, where (c,ν) is the local inverse of $\Psi(c,\nu) = e^{X_c}(1+\nu)$ at X. As a matrix-valued map, $X \mapsto \mathrm{foc}(X)$ is real-analytic at 1, with strict derivative the orthogonal projection $P_{\mathfrak{g}}$ onto $\mathfrak{g}$. Consequently, for every $\varepsilon > 0$ there is $\delta > 0$ such that, for all $X, Y \in M_N(\mathbb{C})$ and $u, v \in G$, the conditions $\|X-1\|_F, \|Y-1\|_F < \delta$ and u, v nearest to X, Y imply

$$\|u - v - P_{\mathfrak{g}}(X-Y)\|_F \le \varepsilon \|X-Y\|_F.$$

(c) The curvature of $u \in G$ is $F(u) = \Pi(\frac{1}{2}(u-u^{-1})) \in \mathbb{R}^d$, the projected anti-Hermitian part. It is continuous and satisfies:

- $F(1) = 0$ and $F(u^{-1}) = -F(u)$;
- $F(vuv^{-1}) = \mathrm{Ad}(v)F(u)$;
- $|F(u)| \le \chi(u)$;

- $\frac{d}{dt}F(\exp(tc))\big|_{t=0} = c$.

Status: theorem. The neighbourhoods and δ come from the inverse function theorem and compactness and are not explicit. *Lean:* `plog`, `splitNormal`, `plog_diff`, `plog_left`, `plog_right`, `Nearest`, `nearest_exists`, `nearest_left`, `nearest_right`, `nearest_normal`, `focus`, `nearest_is_focus`, `focus_diff`, `nearest_tube`, `focus_deriv`, `nearest_approximate`, `curvature`, `curvature_inv`, `curvature_conjugate`, `curvature_norm`, `leading_curvature`.

Proposition I.1.9 (the Wilson measure on the torus). Let $\mu_\ell = \mu_G^{\otimes(\Lambda_\ell\times\{0,\dots,3\})}$ be the product Haar measure on configurations. Let

$$E_\ell(U) = \sum_{x\in\Lambda_\ell}\sum_{i<j}\sigma(U_{(x;i<j)})$$

be the Wilson energy: $6\ell^4$ terms, each unoriented plaquette counted once. For every $\beta\in\mathbb{R}$ put

$$Z_\ell(\beta) = \int e^{-\beta E_\ell}\,d\mu_\ell, \qquad d\mathbb{P}_{\ell,\beta} = Z_\ell(\beta)^{-1}e^{-\beta E_\ell}\,d\mu_\ell.$$

For integrable complex F, write $\langle F\rangle_{\ell,\beta} = \int F d\mathbb{P}_{\ell,\beta}$. Then:

- E_ℓ is continuous, with $0\le E_\ell\le 12\ell^4$;
- $Z_\ell(\beta)>0$ and $\mathbb{P}_{\ell,\beta}$ is a probability measure;
- every continuous complex function, and every bounded measurable complex function, is $\mathbb{P}_{\ell,\beta}$-integrable;
- for integrable complex F, $\langle F\rangle_{\ell,\beta} = Z_\ell(\beta)^{-1}\int e^{-\beta E_\ell}F d\mu_\ell$.

These are the energy and measure of Definitions 1.3–1.4 of the main statement, with $h=\mu_G$; the identification is proved in VIII.1 (`energy_eq`, `law_eq`). *Status:* definition, with theorems. *Lean:* `Wilson.Config`, `energy`, `energy_nonneg`, `energy_cont`, `prior`, `bare`, `partition`, `partition_pos`, `law`, `moment`, `continuous_integrable`, `bounded_integrable`, `moment_formula`.

Proposition I.1.10 (invariance of the Wilson measure). For every $\beta\in\mathbb{R}$, every lattice affine map $A=(\rho,b)$ and every $g:\Lambda_\ell\to G$, the maps $U\mapsto A^*U$ and $U\mapsto U^g$ preserve μ_ℓ, E_ℓ and $\mathbb{P}_{\ell,\beta}$.

This covers translations, axis permutations, and reflections $x_a\mapsto b_a-x_a$. When ℓ is even, the reflection planes pass through sites for even b_a and bisect links for odd b_a. Hence $\langle F\circ A^*\rangle_{\ell,\beta}=\langle F\rangle_{\ell,\beta}$ for every integrable complex F. More generally, any measurable map preserving μ_ℓ and E_ℓ preserves $\mathbb{P}_{\ell,\beta}$. Reflection positivity is a separate assertion. *Status:* theorem. *Lean:* `energy_pull`, `energy_gauge`, `prior_pull`, `prior_gauge`, `law_preserved`, `law_pull`, `law_gauge`, `law_translate`, `moment_pull`.

Proposition I.1.11 (continuum symmetry groups). $\mathbb{R}^4$ carries the Euclidean norm and Lebesgue measure. Write $x=(\vec{x},x_t)$, with time the last coordinate. Let $\mathsf{m}(x)=x_t^2-|\vec{x}|^2$, $\overline{V}_+=\{|\vec{x}|\le x_t\}$ and $V_+=\{|\vec{x}|<x_t\}$.

(a) *Euclidean.* There is a homomorphism $B_4\to O(4)$ whose image acts on coordinates as in Proposition I.1.2. The mirror in axis a maps to the reflection $x_a\mapsto -x_a$. The real skew matrices form $\mathfrak{so}(4)$, with plane generators $\Omega_{ij}=E_{ij}-E_{ji}$ for $i<j$, and $t\mapsto e^{t\Omega}\in O(4)$.

(b) *Lorentz group.* $\mathcal{L}=\{g\in GL(\mathbb{R}^4):\det g=1,\ \mathsf{m}\circ g=\mathsf{m},\ g\overline{V}_+=\overline{V}_+\}$; this is $SO^+(1,3)$. It is a topological group of Lebesgue-preserving maps that preserves V_+, and it contains every

element of $SO(4)$ fixing the time coordinate. For a unit spatial vector e, the boost

$$B_e(r):\ x_t \mapsto \cosh r\, x_t + \sinh r\, (\vec{x}\cdot e), \quad \vec{x} \mapsto \vec{x} + \big(\sinh r\, x_t + (\cosh r - 1)(\vec{x}\cdot e)\big)e$$

is a continuous one-parameter subgroup of $\mathcal{L}$. Its generator at $r = 0$ is $K_e : (\vec{x}, x_t) \mapsto (x_t\, e,\ \vec{x}\cdot e)$.

(c) *Dual action and cone.* With $R(\vec{x}, x_t) = (-\vec{x}, x_t)$, put $g^\vee = RgR$. Then $g^\vee \in \mathcal{L}$, $(g^\vee)^\vee = g$, $g^\vee = (g^{-1})^T$, and $\langle gx, g^\vee p\rangle = \langle x, p\rangle$ for the Euclidean pairing. Moreover:

- $x \in \overline{V}_+$ iff $\langle x, p\rangle \geq 0$ for all $p \in \overline{V}_+$;
- if $x \in V_+$ and $p \in \overline{V}_+ \setminus 0$, then $\langle x, p\rangle > 0$;
- $x \in \overline{V}_+$ iff $(g^\vee x)_t \geq 0$ for every $g \in \mathcal{L}$.

(d) *Poincaré group.* $\mathcal{P} = \mathcal{L} \ltimes \mathbb{R}^4$ consists of pairs (Λ, b) and acts by $x \mapsto \Lambda x + b$, with $(\Lambda, b)(\Lambda', b') = (\Lambda\Lambda', \Lambda b' + b)$. It is a topological group, determined by its action on points, and preserves Lebesgue measure on $\mathbb{R}^4$ and, diagonally, on $(\mathbb{R}^4)^n$ for every $n \in \mathbb{N}$.

(e) *Configurations.* For $n \in \mathbb{N}$, $r \in \mathbb{R}$ and $x \in (\mathbb{R}^4)^n$, call x:

- *separated* ($x \in \mathrm{Conf}_n$) if the x_i are pairwise distinct; Conf_n is open;
- *generic* if for every axis the n coordinates on that axis are pairwise distinct (generic implies separated);
- *r-spaced* if $|x_i - x_j|_\infty \geq r$ for $i \neq j$ ($r > 0$ implies separated).

Lebesgue measure on $(\mathbb{R}^4)^n$ is the product of the four-dimensional ones. *Status:* theorem, apart from the definitions. *Lean:* `Signed.ortho`, `Signed.ortho_move`, `Signed.inclusion0`, `Signed.mirrored`, `Ortho.skewSpace`, `Ortho.oneParameter`, `Ortho.plane`, `Mink.Proper`, `Lorentz.preserves`, `Lorentz.stays`, `Lorentz.room`, `Ray.boost`, `Ray.generator`, `Lorentz.dual`, `Lorentz.dual_sent`, `Lorentz.dual_adjoint`, `cone_test`, `strict_pair`, `Lorentz.from_half`, `Poincare`, `Poincare.by_placement`, `Poincare.together`, `conf`, `generic`, `spaced`, `conf_open`, `Points.volume_product`.

Proposition I.1.12 (boxes, axis routes and slabs on the cover). For $r \in \mathbb{N}$ let $B_r = \{-r, \dots, r\}^4 \subset \mathbb{Z}^4$; then $|B_r| = (2r+1)^4$ and $\rho B_r = B_r$ for $\rho \in B_4$. For $x, y \in \mathbb{Z}^4$ and an ordering o of the four axes, the *axis route* $\mathrm{route}_o(x, y)$ goes from x in straight legs along the axes in the order o, adjusting one coordinate at a time. It ends at y, and:

- reversing the order gives the reverse of $\mathrm{route}_o(y, x)$;
- the image of $\mathrm{route}_o(x, y)$ under $A = (\rho, b)$ is $\mathrm{route}_{\pi\circ o}(Ax, Ay)$.

For an anchor $a \in \mathbb{Z}^4$ and a set J of axes, define $\mathrm{slab}(r, a, J) = \{y \in B_r : y_i = a_i \text{ for } i \notin J\}$. If $i \notin J$, then, for any function f with values in a commutative additive monoid,

$$\sum_{y \in \mathrm{slab}(r,a,J\cup\{i\})} f(y) = \sum_{z=-r}^{r} \sum_{y \in \mathrm{slab}(r, a[i\leftarrow z], J)} f(y).$$

For $z, w \in \{-r, \dots, r\}$, translation by $(w - z)e_i$ maps the layer with pinned coordinate z bijectively onto the layer with pinned coordinate w. Write $\mathrm{straight}_j(d)$ for the signed leg of displacement de_j, $d \in \mathbb{Z}$. For every cover configuration U,

$$U(x; \mathrm{straight}_j(d+1)) = U(x; \mathrm{straight}_j(d))\, U(x + de_j, j).$$

Status: theorem. *Lean:* `box`, `box_covariant`, `Corona.box_size`, `route`, `route_end`, `back_route`, `send_route`, `slab`, `slab_sum`, `slab_update_image`, `carry_succ`.

3.1.2 2. What it uses

No result from another unit is needed. The definitions and estimates are used throughout the proof. Main consumers include:

- the nearest-point retraction (I.1.8(b)): gauge fixing and alignment in III.1;
- the invariant-form lemma (I.1.6(d)): the one-loop computation in III.7 and the stress-tensor forms in VI.1–VI.2;
- the coarsening estimate (I.1.4(f)): III.5 and III.10;
- the mesh inequality: V.1;
- invariance of the law: IV.1, III.13 and VII.2;
- positivity of $C_{\mathfrak{g}}$: VIII.1 (the slope in (Reg1)) and VII.2;
- the Haar-ball bound (I.1.7): a product lower bound for configurations whose links are near 1 (`YM.RG.space`), a lower bound for the block-kernel noise normalizer (`YM.SectorNext.Vacuum.shell_pZ`), and positivity of the ball mass used in corrected-field tests (`YM.Push.Corrected.Tests.Offer.stamp`).

Standard mathematics. The underlying results from Mathlib, together with references for the mathematical context, are:

- Existence and uniqueness of Haar measure on a compact group, and its right and inversion invariance (Halmos, *Measure Theory*, §§58–60; Folland, *A Course in Abstract Harmonic Analysis*, Ch. 2).
- Product measures and invariance of product measures under coordinate permutations; change of variables for measure-preserving maps; Bochner integration.
- Matrix exponential; the inverse function theorem in the real-analytic class.
- The local structure of a closed matrix group near 1: the Lie algebra as the set of generators of one-parameter subgroups, the Lie–Trotter limit, and von Neumann's argument that a closed subgroup meets $\exp(\mathfrak{g}^{\perp})$ only at 1 near the identity (B. C. Hall, *Lie Groups, Lie Algebras, and Representations*, Ch. 2–3). The local structure is derived here from the exponential definition of the Lie algebra.
- Spectral theorem for symmetric operators on a finite-dimensional Euclidean space; the Schur-type uniqueness of invariant forms on a simple real Lie algebra (Knapp, *Lie Groups Beyond an Introduction*, Ch. I).
- Scaling of Lebesgue measure of balls in a finite-dimensional space.
- Lattice gauge theory: Wilson's action (K. Wilson, Phys. Rev. D 10 (1974) 2445); gauge-invariant lattice measures and their symmetries (Osterwalder–Seiler, Ann. Phys. 110 (1978) 440; Seiler, Lecture Notes in Physics 159).
- Minkowski geometry and the restricted Lorentz and Poincaré groups (Streater–Wightman, *PCT, Spin and Statistics, and All That*, Ch. 1).

Imported constants and assumptions. The group G is an arbitrary closed subgroup of $U(N)$, possibly disconnected. Simplicity is assumed only in I.1.6(d)–(f), and agrees with the main theorem's hypothesis by the bridge cited in I.1.6(c). The local radii depend on G; the mesh constant also depends on the schedule.

3.1.3 3. Strategy

The torus metric comes from the absolute values of centred representatives. Odd subdivision ratios give symmetric residue windows, so the parent map commutes with signed permutations even when the coarse torus has even side. The schedule separates the physical period from the mesh; gentle growth of the block factors gives the mesh inequality. Cover numbers measure the size of lattice sets by connected unions of balls. Sampling a path through a fine cover at intervals of LB yields the coarsening bound $O(1 + k/L)$ for a set of fine cover number k, with constants independent of the period.

Affine pullback assigns to each link the transport along its image. Reflections reverse links and shift their base points. The pullback therefore permutes link variables and inverts some of them, preserving product Haar measure. Image plaquette holonomies differ by conjugation and possibly inversion; the cost σ is invariant under both. Summing over the plaquette bijection proves invariance of the energy and hence of the Wilson measure.

For the closed matrix group, the tangent criterion follows by approximating e^X with $f(1/n)^n$. It gives the linear and bracket closure of $\mathfrak{g}$. The inverse function theorem, together with isolation of $\exp(\mathfrak{g}^\perp)$ from G near 1, gives a local logarithm. Applying the same theorem to $e^{X_c}(1 + \nu)$ identifies the nearest-point retraction and its derivative. Finally, the spectral theorem reduces an invariant symmetric form to an eigenspace; real simplicity forces that eigenspace to be all of $\mathfrak{g}$. Applied to the negative Killing form, this gives $C_{\mathfrak{g}} > 0$.

3.1.4 4. Steps

3.1.4.1 4.1 Torus, subdivision, schedule

Lemma I.1.a (centred lift and separation). Proposition I.1.1. *Proof sketch.* Coordinatewise, r_ℓ is the representative of minimal absolute value, with the tie sent to $+\ell/2$. This gives uniqueness, $|r_\ell(a+b)_i| \le |r_\ell(a)_i| + |r_\ell(b)_i|$ (hence the triangle inequality for the maximum), and $|r_\ell([y]_\ell)_i| \le |y_i|$. A zero lift reduces to the zero site, so zero separation implies equality. Signed permutations permute coordinates and change signs, and the centred absolute value is even, including at the tie. Thus $|\rho x|_\ell = |x|_\ell$. There are ℓ choices in each coordinate, giving ℓ^4 sites. *Lean:* `lift_eq_iff`, `pdist_triangle`, `pdist_invariant`, `Corona.lap`.

Lemma I.1.b (subdivision). Proposition I.1.3(c). *Proof sketch.*

- $L\, r_\ell(x)$ lies in the centred window of $\Lambda_{\ell L}$, so $r_{\ell L}(\iota x) = L\, r_\ell(x)$, and ι is injective.
- For every $y \in \mathbb{Z}^4$, $\iota([y]_\ell) = [Ly]_{\ell L}$, because changing y by ℓz changes Ly by $\ell L z$. This gives additivity and B_4-covariance.
- Applying res to an equality of assembled points gives equality of their residue coordinates. Injectivity of ι then gives equality of the coarse coordinates. The count $\ell^4 L^4 = (\ell L)^4$ makes the assembly map bijective. Since ιx is assembled from $(x, 0)$, its parent is x.

Lean: `Ratio.lift_embed`, `Ratio.embed_injective`, `Ratio.root`, `Ratio.signed`, `Ratio.residue_embed`, `Ratio.assemble_bijective`, `Ratio.unpack_at`.

Lemma I.1.c (odd blocks and their symmetry). Proposition I.1.3(d). *Proof sketch.* For odd L, the window $(-L/2, L/2] \cap \mathbb{Z} = [-\frac{L-1}{2}, \frac{L-1}{2}] \cap \mathbb{Z}$ is symmetric. Hence $r_L(\rho y) = \rho\, r_L(y)$, the assembly map intertwines (ρ, ρ) with ρ, and par commutes with ρ. Every residue has exactly one representative in $B_{(L-1)/2}$, so the assembly bijection identifies each parent fibre with the stated translated box. For the crossing rule, write $z = \iota(\mathrm{par}\, z) + [q]$ with $q = r_L(\mathrm{res}\, z) \in B_{(L-1)/2}$.

- If $q_i < \frac{L-1}{2}$, then $q + e_i$ stays in the block.
- If $q_i = \frac{L-1}{2}$, then $z + e_i = \iota(\operatorname{par} z + e_i) + [q - (L-1)e_i]$, because $\iota(x + e_i) = \iota(x) + Le_i$.

Lean: `Ratio.lift_signed_of_odd`, `Ratio.assemble_signed`, `Ratio.parent_signed`, `Ratio.half_box`, `Ratio.half_lift`, `Ratio.parents_hop`, `Ratio.internal`.

Lemma I.1.d (metric comparison across scales). Proposition I.1.3(e). *Proof sketch.*

- The vector $r_L(\operatorname{res} z)$ has coordinates bounded by $L/2$. Its reduction is $z - \iota(\operatorname{par} z)$, and reduction cannot increase the torus norm.
- The lift of $\iota(x - x')$ is $L\, r_\ell(x - x')$, which gives the second inequality.
- Foreshortening follows from the second inequality and two applications of the first.

Lean: `Corona.near`, `Corona.below`, `Corona.foreshorten`.

Lemma I.1.e (schedule and mesh inequality). Proposition I.1.3(a),(b). *Proof sketch.* The product D_m is odd, strictly increasing and at least 3^m, which gives $a_m \to 0$ and $s_m \to \infty$. Placement follows by multiplying $-\ell < 2r_\ell \le \ell$ by a_m and using $a_m \ell(m,p) = P_p$. The scaled-lift identity gives compatibility under subdivision. Gentleness gives $\log L_m < s_m$ for all $m \ge m_0$, hence $a_{m+1} > a_m^2$. Take c to be the minimum of the finite nonempty set $\{1\} \cup \{a_{m+1}/a_m^2 : m < m_0\}$; every member is positive. *Lean:* `den_strict`, `a_step`, `scale_succ`, `embeds_scale`, `mesh_inequality`.

3.1.4.2 4.2 Cover numbers Lemma I.1.f (elementary properties). Proposition I.1.4(a),(b). *Proof sketch.* A cover of a set covers its subsets. For the union, take optimal covers A of X and B of Y. The halo of $A \cup B$ is the union of the two haloes, which is connected through a common point of $X \cap Y$. Counting: $\operatorname{halo}_M(D)$ is the image of $D \times B_M$ under $(x, q) \mapsto x + [q]$, and $X \subset \operatorname{halo}_M(D)$ for an optimal D. *Lean:* `Corona.cover_mono`, `Corona.meets_sum`, `Corona.halo_size`, `Corona.pays`.

Lemma I.1.g (connectivity of haloes). Proposition I.1.4(c). *Proof sketch.* Use lattice segments $t \mapsto x + [\operatorname{clamp}_{[-t,t]} r_\ell(y - x)]$, which move by ≤ 1 per unit time and reach y at $t = |x - y|_\ell$. If $|x - y|_\ell \le 2M + 1$, the segment stays in $\operatorname{halo}_M\{x, y\}$. Conversely, a path with steps of sup-distance at most 1 in $\operatorname{halo}_M(D)$ yields a chain of centres with consecutive distances $\le 2M + 1$, by the triangle inequality. *Lean:* `Corona.ship_halo`, `Corona.halo_ship`.

Lemma I.1.h (a path through a chained set; diameter). Let $D \subset \Lambda_\ell$ and $b \in \mathbb{N}$. Suppose any two points of D can be joined in D by a chain with steps of sup-distance at most b. Then there are $n \le 2b|D|$ and sites $R_0, \dots, R_n$ visiting every point of D, with $|R_{k+1} - R_k|_\ell \le 1$ for $k < n$. Consequently Proposition I.1.4(d) holds. *Proof sketch.* For nonempty D, start at one point. The chain condition gives a point of D outside the visited set within distance b of a visited point. Insert a detour out and back along a coordinatewise-clamped segment, costing at most $2b$ steps. For $D = \emptyset$, take a constant path of length 0. A cover has chain bound $b = 2M + 1$, so the resulting path has length at most $(4M + 2)\operatorname{cov}_M(X)$. Distances along it are bounded by its length, and the two halo radii add $2M$. *Lean:* `Corona.ring_path`, `Corona.path_span`, `Corona.extent`.

Lemma I.1.i (change of scale). Proposition I.1.4(e). *Proof sketch.* The grid $x + [b\, q]$, $q \in B_k$, covers $\operatorname{halo}_{kb}\{x\}$ at scale b:

- division with remainder puts every point within b of a grid point;
- consecutive grid points along coordinatewise-clamped paths are at most b apart on the torus, so the grid is b-chained.

For a general D covering X at scale mb, I.1.g gives chains of centres with steps at most $2mb+1 \le 2(m+1)b$. The balls $\mathrm{halo}_{(m+1)b}\{x\}$ along such a chain overlap. For each centre, the grid above gives a cover of its ball whose b-halo is connected and contains that ball. These connected haloes meet successively, so their union covers X and is connected. The union of their centres has at most $(2m+3)^4|D|$ points. *Lean:* `Corona.tile`, `Corona.tiling`.

Lemma I.1.j (coarsening). Proposition I.1.4(f). *Proof sketch.*

1. Take an optimal cover A of X at scale M. By I.1.h there is a fine path $R_0, \dots, R_n$ visiting A, with steps of sup-distance at most 1 and $n \le (4M+2)|A|$.
2. Sample its parents every LB steps: $D' = \{\mathrm{par}\, R_{jLB} : 0 \le j \le \lfloor n/(LB) \rfloor\}$. By foreshortening, parents of path points at most LB steps apart are within $2B$ on the coarse torus. Thus D' is $2B$-chained, every $\mathrm{par}\, R_k$ with $0 \le k \le n$ lies in $\mathrm{halo}_{2B}(D')$, and $LB|D'| \le n + LB$.
3. A point of $\mathrm{halo}_{sB}(\mathrm{par}\, X)$ lies within $sB + 3B + 2B$ of D'. The middle term uses $L|\mathrm{par}\, c - \mathrm{par}\, v|_\ell \le M + L \le 3BL$ for $c \in A$ within M of $v \in X$. The $(s+5)B$-halo of D' is connected by I.1.g, since D' is $2B$-chained, so D' is a cover at that scale.
4. Apply I.1.i with $m = s{+}5$, giving a factor $(2s{+}13)^4$. Since $M \le 2B$ and $B \ge 1$, $4M{+}2 \le 10B$. Thus $LB|D'| \le LB + 10B|A|$, and division by B gives $L|D'| \le L + 10|A|$.

Lean: `Corona.fuel`, `Corona.collapse`.

3.1.4.3 4.3 Links, transport, plaquettes

Lemma I.1.k (pullback is a right action). Proposition I.1.2(a). *Proof sketch.* Evaluate at a link. The image step of $\pm e_i$ under ρ' and then ρ is its image under $\rho\rho'$, and the affine maps compose as $(\rho, b) \circ (\rho', b') = (\rho\rho', \rho b' + b)$. *Lean:* `Step.rotate_mul`, `pull_id`, `pull_pull`, `pull_one`.

Lemma I.1.l (transport under pullback and gauge; locality). Proposition I.1.2(b)–(d). *Proof sketch.* Induction on the walk. One step: the variable that A^*U assigns to a step equals the variable that U assigns to the image step at the image point. For a backward step, this is the inverse of the forward variable at the shifted base point, which is why base points move under reflections. Gauge: the product telescopes, $\prod_k g(x_k)U_k g(x_{k+1})^{-1} = g(x_0)(\prod_k U_k)g(x_n)^{-1}$. Locality: each factor reads one link whose endpoints are visited. *Lean:* `Step.pull_read`, `travel_pull`, `Step.gauge_read`, `travel_gauge`, `travel_local`.

Lemma I.1.m (plaquettes under symmetries). Proposition I.1.2(e). *Proof sketch.*

- By I.1.l, $(A^*U)_p$ is the transport of U around the image square, read from the image of the base point along the image steps.
- Reversing the orientation of a square inverts its holonomy, and starting it at the next corner conjugates its holonomy.
- Hence ϕ of the holonomy equals ϕ of the holonomy of the positive-axis square based at $Ax - \sum_{k\in\{i,j\}:\,\epsilon_k=-1} e_{\pi(k)}$, with axes in increasing order.
- Injectivity of \$A_\sharp\$: the axis pair is permuted bijectively, and the shifted base point and the axis pair determine Ax. On the finite torus, injective implies bijective.

Lean: `oriented_swap`, `oriented_next`, `price_flip`, `price_shift_pos`, `Face.lower`, `Face.injective_send`, `Face.send_price`.

Lemma I.1.n (periodic extension). Proposition I.1.2(f). *Proof sketch.* $E_b U$ evaluates U at the reduced, shifted point. Reduction is additive, commutes with B_4 and sends unit steps to unit steps. So transport, pullback and gauge action commute with E_b step by step, and an

agreement set on the cover maps to its image. *Lean:* `expandAt_pull`, `travel_at`, `expandAt_gauge`, `LocalOn.expandAt`.

Lemma I.1.o (boxes, routes, slabs). Proposition I.1.12. *Proof sketch.* The endpoint of a route is computed axis by axis: each leg changes one coordinate by $y_a - x_a$, and the axes are distinct. Reversal and covariance hold leg by leg, since the image of a straight leg along a is a straight leg along $\pi(a)$ of signed length $\epsilon_a(y_a - x_a)$. When $i \notin J$, the slab sum is a disjoint decomposition by y_i; translating a pinned layer by $(w - z)e_i$ changes only that coordinate. The transport identity follows by appending a forward step for $d \geq 0$ and cancelling a final backward step for $d < 0$. Box covariance follows coordinatewise from the symmetric interval $[-r, r]$, and its cardinality is the product of the four interval cardinalities. *Lean:* `route_end`, `back_route`, `send_route`, `slab_sum`, `carry_succ`.

3.1.4.4 4.4 Gauge group

Lemma I.1.p (cost, chord, Haar). Proposition I.1.5. *Proof sketch.*

- $\|u - 1\|_F^2 = \operatorname{tr}((u-1)(u-1)^*) = 2N - 2\operatorname{Re}\operatorname{tr} u$, using $uu^* = 1$.
- $\|u - 1\|_F \leq \|u\|_F + \|1\|_F = 2\sqrt{N}$.
- $uv - 1 = (u - 1) + u(v - 1)$ and $\|u(v-1)\|_F = \|v - 1\|_F$ give the triangle inequality.
- The trace is conjugation invariant, and $\operatorname{tr} u^{-1} = \overline{\operatorname{tr} u}$.
- A right translate of μ_G is again a left Haar probability measure, hence equals μ_G by uniqueness. Right invariance then makes the image under inversion a left Haar probability measure as well. Continuity of σ and χ follows from continuity of the trace and norm.

Lean: `frob_sq`, `unitary_frob`, `chord_price`, `chord_triangle`, `price_inv`, `probHaar_rightInv`, `probHaar_invInv`.

Lemma I.1.q (the Lie algebra). Proposition I.1.6(a). *Proof sketch.*

- Skewness: differentiate $f(t)f(t)^* = 1$ at 0.
- Tangent criterion: if f takes values in G with $f(0) = 1$ and $f'(0) = X$, then X is skew and

$$\|f(1/n)^n - e^X\|_F \leq n\,\|f(1/n) - e^{X/n}\|_F = o(1),$$

  by the telescoping $\|A^n - B^n\|_F \leq n\|A - B\|_F$ for unitary A, B. Unitary multiplication is isometric, so the bound has no exponential factor. Closedness of G gives $e^X \in G$; apply this to $f(t\,\cdot)$.
- Additivity: apply the criterion to $e^{tX}e^{tY}$.
- Scalar multiplication follows by rescaling the one-parameter subgroup. The resulting real subspace is closed because it is finite dimensional. Conjugation preserves it since $e^{s(uXu^{-1})} = ue^{sX}u^{-1}$ for $u \in G$.
- Brackets: $e^{tX}Ye^{-tX} \in \mathfrak{g}$ and the closed subspace $\mathfrak{g}$ contains the limit $[X, Y]$ of its difference quotients.

Lean: `tangent_skew`, `frob_pow_difference`, `tangent_approx`, `tangent_generates`, `generates_add`, `conjugate_mem`, `bracket_mem`.

Lemma I.1.r (colours, invariant forms, Casimir). Proposition I.1.6(b)–(f). *Proof sketch.*

- The colour frame is a Frobenius-orthonormal basis of $\mathfrak{g}$ scaled by $\sqrt{N}$. For skew X, Y, $\operatorname{Re}\operatorname{tr}(XY^*) = -\operatorname{tr}(XY)$ and $\operatorname{tr}(XY)$ is real. Orthogonal projection followed by the inverse colour frame gives Π and its norm bound. Unitary conjugation preserves $\mathfrak{g}$ and the Frobenius inner product, so it induces the orthogonal adjoint representation. The exponential is continuous by the matrix exponential and the induced topology on G.
- (d): let A be the symmetric operator with $\langle Ax, y\rangle = B(x, y)$. Invariance of B and skewness of ad give $A\,\mathrm{ad}_x = \mathrm{ad}_x A$. Nonabelianity gives $d > 0$. Choose a real eigenvalue c of A; its nonzero eigenspace is an ad-invariant subspace, hence an ideal. Simplicity makes it the whole space, so $A = c \cdot \mathrm{id}$.
- (e): $K(x, y) = -\operatorname{tr}(\mathrm{ad}_x \mathrm{ad}_y)$ is symmetric and invariant, by $\mathrm{ad}_{[x,y]} = [\mathrm{ad}_x, \mathrm{ad}_y]$ and cyclicity, so $K = c\langle\cdot,\cdot\rangle$. Taking the trace over an orthonormal basis gives $c = C_{\mathfrak{g}}$.
- Positivity: $K(x, x) = \sum_i |\mathrm{ad}_x b_i|^2$, which is > 0 for some x because $\mathfrak{g}$ is nonabelian. This forces $C_{\mathfrak{g}} > 0$, and the positive factor $11/(24\pi^2)$ gives $\kappa > 0$.

Lean: `inner_frob_color`, `Simple.ideals`, `ad_jacobi`, `ad_skew`, `minusKilling_invariant`, `invariant_scalar`, `minusKilling_sq`, `trace_adjoint`, `casimir_pos`.

Lemma I.1.s (Haar mass of small balls). Proposition I.1.7. *Proof sketch.* Points of G lie on the Frobenius sphere of radius $\sqrt{N}$ in $M_N(\mathbb{C}) \cong \mathbb{R}^{2N^2}$. Take a finite set $p \subset G$ whose distinct points have Frobenius distance strictly greater than $r\sqrt{N}$:

- the balls of radius $r\sqrt{N}/2$ around its points are disjoint and lie in the ball of radius $2\sqrt{N}$, so $|p|(r/4)^{2N^2} \le 1$; in particular a maximal such set exists;
- if p is maximal, then $G = \bigcup_{i\in p} i \cdot \{\chi \le r\}$, and left invariance gives $1 \le |p|\,\mu_G\{\chi \le r\}$.

Lean: `shell_ge`.

Lemma I.1.t (local logarithm). Proposition I.1.8(a). *Proof sketch.*

- $\Phi(c, \nu) = e^{X_c} e^{\nu}$ on $\mathbb{R}^d \times \mathfrak{g}^{\perp}$ is analytic, with $D\Phi(0)(c, \nu) = X_c + \nu$ invertible, so it has a local analytic inverse.
- *Isolation:* near 0, $e^{\nu} \in G$ with $\nu \in \mathfrak{g}^{\perp}$ forces $\nu = 0$. Otherwise take $\nu_k \to 0$, $\nu_k \neq 0$, $e^{\nu_k} \in G$, with $\nu_k/\|\nu_k\|_F \to Y$ along a subsequence. For $t \ge 0$, $e^{tY} = \lim (e^{\nu_k})^{\lfloor t/\|\nu_k\|_F \rfloor} \in G$ by closedness; inverses handle $t < 0$. So $Y \in \mathfrak{g} \cap \mathfrak{g}^{\perp} = 0$, contradicting $\|Y\|_F = 1$.
- For $u \in G$ near 1, write $u = \Phi(c, \nu)$; then $e^{\nu} = e^{-X_c} u \in G$ forces $\nu = 0$.

Lean: `direction_in_lie`, `isolates`, `splitDeriv`, `plog_diff`, `plog_left`, `plog_right`.

Lemma I.1.u (nearest point). Proposition I.1.8(b). *Proof sketch.*

- Existence by compactness; equivariance because left and right unitary multiplication are Frobenius isometries.
- First-order condition: if 1 is nearest to X, then $t \mapsto \|e^{tY} - X\|_F^2$ is minimal at 0, so $\langle X - 1, Y\rangle = 0$ for $Y \in \mathfrak{g}$. The general case follows by left translation.
- Identification near 1: for any choice $b(X)$ of nearest points, $\|b(X) - 1\|_F \le 2\|X - 1\|_F$. Hence $\nu(X) = b(X)^* X - 1 \in \mathfrak{g}^{\perp}$ tends to 0, and $X = \Psi(\log_G b(X), \nu(X))$ with both arguments near 0. Injectivity of the local diffeomorphism Ψ gives $b(X) = \mathrm{foc}(X)$. To obtain one neighbourhood valid for every nearest point, choose $b(X)$ to differ from $\mathrm{foc}(X)$ whenever such a nearest point exists. The same argument forces this selection to equal foc on a neighbourhood, excluding all such exceptions there.
- Uniqueness in a tube: translate by v^{-1}.

- The derivative of foc at 1 is $X_{(\cdot)} \circ \mathrm{pr}_1 \circ D\Psi(0)^{-1} = P_{\mathfrak{g}}$, and the approximation bound is the strict differentiability of foc at 1.

Lean: `nearest_exists`, `nearest_left`, `nearest_normal`, `tubular`, `tubularDeriv`, `nearest_is_focus`, `nearest_tube`, `focus_deriv`, `nearest_approximate`.

Lemma I.1.v (curvature of a group element). Proposition I.1.8(c). *Proof sketch.* Conjugation by $v \in G$ is a Frobenius isometry preserving $\mathfrak{g}$, hence also $\mathfrak{g}^\perp$, so $\Pi(vAv^{-1}) = \mathrm{Ad}(v)\Pi A$. For the bound:

$$|F(u)| \le \tfrac{1}{2\sqrt{N}}\|u - u^{-1}\|_F \le \tfrac{1}{2}(\chi(u) + \chi(u^{-1})) = \chi(u).$$

The derivative of $\frac{1}{2}(e^{tX_c} - e^{-tX_c})$ at 0 is X_c, and $\Pi X_c = c$. *Lean:* `project_conjugate`, `curvature_conjugate`, `curvature_norm`, `leading_curvature`.

3.1.4.5 4.5 The Wilson measure **Lemma I.1.w (the law).** Proposition I.1.9. *Proof sketch.* E_ℓ is a finite sum of continuous functions on a compact space. There are $6\ell^4$ terms, each in $[0, 2]$, giving the energy bounds. The weight $e^{-\beta E_\ell}$ is continuous and strictly positive, so it is integrable and $Z_\ell > 0$. The density $e^{-\beta E_\ell}/Z_\ell$ has integral 1. Continuous complex functions are bounded by compactness, and bounded measurable functions are integrable against a probability measure. The moment formula follows by integration against the density. *Lean:* `bare_integrable`, `partition_pos`, `moment_formula`.

Lemma I.1.x (invariance of Haar measure, energy and law). Proposition I.1.10. *Proof sketch.*

- *Haar.* A^* is the composite of two maps. The first is the relabelling $U \mapsto U \circ e_A$, where e_A maps the link (x, i) to the positively oriented link carrying its image; this is injective, hence a permutation of the finite link set. The second inverts the variables on links with $\epsilon_i = -1$. Permutations of factors preserve a product measure, and inversion preserves μ_G. A gauge transformation acts on each factor by $u \mapsto g(x)ug(x + e_i)^{-1}$, which preserves μ_G.
- *Energy.* Gauge: plaquette holonomies are conjugated. Pullback: by I.1.m the energy is re-summed over the bijection \$A_\sharp\$.
- *Law.* A map preserving μ_ℓ and E_ℓ preserves the density, hence the law.

Lean: `sendEdge`, `sendEdge_injective`, `prior_pull`, `prior_gauge`, `face_gauge`, `energy_pull`, `law_preserved`, `moment_pull`.

3.1.4.6 4.6 Continuum symmetry groups **Lemma I.1.y (Euclidean group and configurations).** Proposition I.1.11(a),(e). *Proof sketch.* The matrix of ρ has entry ϵ_a at $(\pi(a), a)$; it is orthogonal and multiplies like B_4. The exponential of a skew matrix is orthogonal. The set Conf_n is a finite intersection of open sets $\{x_i \neq x_j\}$. Distinct time coordinates imply distinct points, as does a positive lower bound on pairwise sup-distances. Splitting off one Euclidean four-dimensional factor at a time identifies Lebesgue measure on $(\mathbb{R}^4)^n$ with the product measure. *Lean:* `Signed.ortho_move`, `Signed.inclusion0`, `generic_conf`, `spaced_conf`, `conf_open`.

Lemma I.1.z (Lorentz group, boosts, cone, Poincaré group). Proposition I.1.11(b)–(d). *Proof sketch.*

- The three defining conditions of $\mathcal{L}$ are stable under products and inverses, and $\det = 1$ gives Lebesgue invariance.

- Boosts: the addition law $B_e(r)B_e(r') = B_e(r+r')$ follows from the addition formulas for cosh, sinh, and m is preserved by $\cosh^2 - \sinh^2 = 1$. For the determinant: $RB_e(r)R = B_e(-r)$ gives $\det B_e(r) = \det B_e(-r) = \det B_e(r)^{-1}$, and $\det B_e(r) = \det B_e(r/2)^2 > 0$, so $\det = 1$. The cone is preserved by $|\sinh r| \le \cosh r$. Differentiating the boost formula at $r = 0$ gives K_e; continuity follows from cosh and sinh.
- Dual action: the identity $\langle gx, g^\vee p\rangle = \langle x, p\rangle$ is polarization of $\mathsf{m}(x) = \langle x, Rx\rangle$.
- Cone duality: Cauchy–Schwarz gives $\langle x, p\rangle \ge x_t p_t - |\vec{x}|\,|\vec{p}| \ge 0$ for $x, p \in \overline{V}_+$. Conversely, test against $p = (-\vec{x}/|\vec{x}|, 1)$ when $\vec{x} \ne 0$, and against $(0, 1)$ otherwise. If $x \in V_+$ and $p \in \overline{V}_+ \setminus \{0\}$, then $p_t > 0$ makes the bound strict.
- Half-space characterization: for $x \in \overline{V}_+$, every $g^\vee x$ remains in that cone and has nonnegative time coordinate. If $x_t < 0$, take $g = 1$. Otherwise, if $x \notin \overline{V}_+$, put $e = \vec{x}/|\vec{x}|$, choose $x_t/|\vec{x}| < s < 1$, and set $r = \operatorname{artanh} s > 0$ and $g = B_e(r)$. Then $g^\vee = B_e(-r)$ and $(g^\vee x)_t = \cosh r\,(x_t - s|\vec{x}|) < 0$.
- The open cone is characterized by $x_t > 0$ and $\mathsf{m}(x) > 0$, and is preserved by the cone and quadratic-form identities. A Euclidean rotation fixing time preserves $|\vec{x}|$ and hence both cones and m; determinant 1 places it in $\mathcal{L}$. Affine composition gives the Poincaré group law. Evaluation at 0 recovers the translation, and subtracting it recovers the Lorentz map. Joint continuity follows from linear evaluation; the diagonal measure statement follows from preservation of each factor.

Lean: `Ray.adds`, `Ray.positive`, `Ray.staysM`, `Lorentz.dual_sent`, `cone_test`, `Lorentz.from_half`, `Poincare.preserving`, `Poincare.together`.

3.1.5 5. Where the difficulty lies

The local logarithm and nearest-point retraction require a closed-subgroup argument that also handles disconnected groups. The coarsening estimate must control arbitrary periodic sets uniformly in the torus side. The invariant-form argument uses simplicity over $\mathbb{R}$; irreducibility of the defining representation is not assumed. Invariance of the Wilson law reduces to Haar invariance, but reflections require the correct shifts of link and plaquette base points.

The representation $G \subset U(N)$ fixes the cost and the colour inner product, and hence $C_{\mathfrak{g}}$ and κ. Each positive-axis plaquette is counted once. The later running-coupling estimate says that the ratio of the level-m coupling to s_m tends to κ, uniformly in the ultraviolet cutoff $N' \ge m$ and the volume index p, for a family satisfying the construction conditions (`zeroUp_bound`). Its normalization is $\kappa = 2b_0$, where $b_0 = 11C_{\mathfrak{g}}/(48\pi^2)$ (`b0_normalizes`).

Even torus sides create tie points in the centred lift. Absolute values, and hence separation, remain invariant under reflections. Placements commute with signed permutations on the strict interior of the fundamental box. Odd subdivision ratios remove this ambiguity from the residue window, which gives equivariance of the parent map.

3.1.6 6. Notation

object	code name	proposed notation
torus, side, cover	`Site`, `Side`, `Cover`	$\Lambda_\ell = (\mathbb{Z}/\ell\mathbb{Z})^4$, ℓ, $\mathbb{Z}^4$
reduction, centred lift	`wrap`, `lift`	$[y]_\ell$, $r_\ell(x)$
torus norm, separation	`pdist`, `separation`	$\lvert x\rvert_\ell$, $\lvert x-y\rvert_\ell$

object	code name	proposed notation
signed permutations, affine maps	`Signed`, `affine`	$\rho = (\pi, \epsilon) \in B_4$, $A = (\rho, b)$
step, walk, transport	`Step`, `Walk`, `travel`	$\pm e_i$, w, $U(x; w)$
configuration	`Links`, `Wilson.Config`	U
pullback, translation	`pull`, `translate`	A^*U, $\tau_b U$
gauge transformation; invariance; locality	`gauge`, `GaugeInvariant`, `LocalOn`	U^g
plaquette, holonomy, image plaquette	`Face`, `faceValue`, `Face.send`	$p = (x; i < j)$, U_p, $A_\sharp p$
periodic extension	`expandAt`	$E_b U$
box, axis route, slab	`box`, `route`, `slab`	B_r, $\mathrm{route}_o(x, y)$, $\mathrm{slab}(r, a, J)$
schedule: block factors, periods	`Schedule`	(L_m), (P_p); the statement writes L_p for P_p and b_k for L_k
denominator, spacing, side, log-scale	`denominator`, `a`, `side`, `scale`	D_m, a_m, $\ell(m, p)$, $s_m = \log(1/a_m)$
placement	`place`	$a_m\, r_\ell(x)$
subdivision	`Ratio`, `embed`, `residue`, `parent`	ratio L, ι, res, par
halo, cover number	`halo`, `coverNumber`	$\mathrm{halo}_M(X)$, $\mathrm{cov}_M(X)$ (as in III.5)
gauge group	`Gauge`, `El`, `mat`	$G \subset U(N)$, $N = n + 1$ (the statement writes m)
plaquette cost, chord	`price`, `chord`	$\sigma(u) = 1 - \frac{1}{N}\mathrm{Re}\,\mathrm{tr}\,u$, $\chi(u)$
Haar probability	`probHaar`	μ_G
Lie algebra, colours, projection	`lie`, `Color`, `colorMat`, `projectColor`	$\mathfrak{g} \cong \mathbb{R}^d$, $c \mapsto X_c$, Π; $X_{\Pi A} = P_{\mathfrak{g}} A$
adjoint action, bracket, exponential	`adjoint`, `ad`, `lieExp`	$\mathrm{Ad}(u)$, $\mathrm{ad}(c)$, $\exp c$
simplicity	`Simple`	G simple
Casimir, slope constant	`casimir`, `slopeConstant`	$C_{\mathfrak{g}}$, $\kappa = 11 C_{\mathfrak{g}}/(24\pi^2)$ ($= b_* = 2b_0$ of the statement)
local logarithm, nearest point	`plog`, `Nearest`, `focus`	$\log_G$, nearest point, $\mathrm{foc}(X)$
curvature of a group element	`curvature`	$F(u)$
Wilson energy, weight, partition function	`energy`, `bare`, `partition`	$E_\ell(U)$, $e^{-\beta E_\ell}$, $Z_\ell(\beta)$
product Haar, Wilson law, expectation	`prior`, `law`, `moment`	μ_ℓ, $\mathbb{P}_{\ell,\beta}$, $\langle\cdot\rangle_{\ell,\beta}$
spacetime, time, space, sup-distance	`E4`, `time`, `spatial`, `cubedist`	$\mathbb{R}^4$, x_t, $\vec{x}$, $\lvert x - y\rvert_\infty$
n-point configurations	`Points`, `conf`, `generic`, `spaced`	$(\mathbb{R}^4)^n$, Conf_n, generic, r-spaced
orthogonal group, rotation generators	`Ortho`, `skewSpace`, `plane`	$O(4)$, $\mathfrak{so}(4)$, Ω_{ij}
Minkowski square, cones	`m`, `cone`, `inside`	$\mathsf{m}(x)$, $\overline{V}_+$, V_+

object	code name	proposed notation
Lorentz group, dual, boost, Poincaré group	`Mink.Lorentz`, `dual`, `Ray.boost`, `Poincare`	$\mathcal{L} = SO^+(1,3)$, $g^\vee$, $B_e(r)$, $\mathcal{P}$

3.1.7 7. What remains unexplained

1. **Additional helper lemmas.** The assignment of declarations to I.1 includes helpers outside the six principal folders, including in `Cells`, `Extra`, `Further` and `Next`. Their statements and proofs are not covered here; their elementary character should not be assumed.
2. **Coarsening along the selected schedule.** Direct consumers retain $M \le 2B$ explicitly, for example `YM.Next.Scheduling.Buy.footage`, or include it in a conjunction, as in `YM.Extra.Iteration.Shapes.raw`. The derivation of this condition for the particular consecutive cover scales used in III.5 and III.10 remains to be traced through their schedule hypotheses.
3. **Bibliographic precision.** The references in §2 give context. Exact theorem locations and agreement of their conventions with the statements here remain to be checked.

3.2 II.1. Field labels and their lattice representatives

Part: II, Data of the construction. **Lean:** work items G01–G07: `YM/Labels/`, `YM/Source/`, `YM/Work/` (G01–G07), `YM/Next/FOUNDATIONS/`, `YM/Push/RELAY_MARK/`, `YM/Extra/SOURCE_HISTORIES/`, `YM/Push/LOCAL_SPLITTING/`, `YM/Push/RELAY_DIFFERENCE/`, `YM/Planned/ATOM_GUARDED_LAYOUT/`, `YM/Further/LOCAL_LOG_JETS/`, `YM/LocalH/`. **Size:** 10,805 lines in the main-theorem closure:

package	lines
equation sections (RELAY_MARK)	2,336
source histories (SOURCE_HISTORIES)	1,801
local splitting (LOCAL_SPLITTING)	1,485
primary sections and readout (FOUNDATIONS)	1,196
comparison estimates (RELAY_DIFFERENCE)	858
record, sections, transfers (Source)	594
work items G05–G07	934
guarded layout, Haar chart, log jets	749
other files (labels, G01–G04)	852

Conventions.

- **Gauge group.** $G \subset U(n)$ is a compact matrix group, with $n \ge 1$; nothing in this unit uses simplicity. Use the Ad-invariant inner product $\langle X, Y\rangle = -\mathrm{Re}\,\mathrm{tr}(XY)/n$ on its Lie algebra $\mathfrak{g}$, and the Frobenius norm on matrices. For a plaquette P, $F_P(U) \in \mathfrak{g}$ is the orthogonal projection of $\frac{1}{2}(U_P - U_P^{-1})$, and $\vartheta(U_P) = \|U_P - 1\|/\sqrt{n}$ is its chord. $S_W(U) = \sum_P (1 - \mathrm{Re}\,\mathrm{tr}\,U_P/n)$ is the Wilson energy, summed once over each site and axis pair $\mu < \nu$, and $\langle\cdot\rangle_\beta$ is the Wilson expectation on a torus $\Lambda_\ell = (\mathbb{Z}/\ell)^4$, where $\ell \ge 1$.

- **Cover and placement.** Many objects live on the cover $\mathbb{Z}^4$. A function f of finitely many cover links is *placed* at a site $o \in \Lambda_\ell$ by lifting the torus periodically around o; the result is written $f(o;U)$.
- **Small fields.** $\mathrm{Hy}_t(X)$ is the set of configurations in which no plaquette touching X has chord $> t$.
- **Levels.** $m = 0$ is the coarsest level and $m = N$ the Wilson lattice. L_m is the odd block factor of the step $m+1 \to m$, the spacing satisfies $a_m = L_m a_{m+1}$, and $s_m = \log(1/a_m)$ is the scale.
- **Couplings and level constants (III.9).** $\hat\beta_m = \bar\beta + b_0 s_m$ with $b_0 = 11C_A/(24\pi^2)$. The small-field threshold is $\theta_m = \hat\beta_m^{-1/2}\exp((\log\log\hat\beta_m)^2)$ and the *forced allowance* is $w_m = \exp(-\exp((\log\log\hat\beta_m)^2/5))$. $J(m)$ is the nondecreasing stage index, $\bar\kappa_j \geq 1$ the stage-j constant, $\kappa_m = \bar\kappa_{J(m)}$ the level-m constant, and b_m its budget point. $\rho_m = \rho_{N,p,m} = e^{\mathcal{V}_m}\,\mathrm{gas}_m$ is the polymer representation of the level-m density, with vacuum exponent $\mathcal{V}_m$ (II.2). $T_\beta^{(m)}$ is the blocking push-forward from level $m+1$ to level m (III.1).
- **Budget norms (II.2).** A budget point b has order $k_b \geq 6$, power $P_b \geq 2$, threshold $t_b \in (0,1]$, a support exponent and a cover size. f is *b-regular on X* if it is local on links with both endpoints in X, measurable and gauge invariant, and $A \mapsto f(e^A U)$ is C^{k_b} at $A = 0$ for every $U \in \mathrm{Hy}_{t_b}(X)$. All ordered left derivatives through order k_b must exist there. Write $\|f\|_{b;o,X} \leq c$ when $c \geq 0$ bounds every such derivative, including the value, in every list of directions satisfying $|A_e| \leq t_b(1+\mathrm{dist}(e_-,o))^{P_b}$. Here e_- is the initial vertex of the positive link e, and dist is the periodic ℓ^∞ distance. All bounds are pointwise, including at the small-field boundary.

What the unit does. It defines the field labels and their lattice representatives, and specifies the renormalized observables $O_{N,p}$.

1. **Sections.** For each label it constructs gauge-invariant polynomials in the link matrices on a fixed stencil that reproduce the label to Taylor order d in the spacing, with symmetry and small-field bounds (§4.1–4.3).
2. **Readout.** It proves the converse: every regular lattice function has a label, and the label's size is bounded by the function's budget norm uniformly in the lattice side, support and norm parameters (§4.4).
3. **Equation sections.** From the lattice Schwinger–Dyson identity it builds a second kind of section on which every label of the free-Maxwell equation ideal becomes orthogonal to remote spectators (§4.5).
4. **Observables.** It specifies the record carrying a label through the RG history, proves the exact birth formula at the Wilson end, and collects the numerical conditions and comparison estimates used downstream (§4.6–4.8).

3.2.1 1. What this unit proves

Proposition II.1.1 (label spaces and frames). For $j \geq 0$ let Tay_j be the real space of 4-tuples $A = (A_\mu)$ of symmetric $(j+1)$-linear maps $(\mathbb{R}^4)^{j+1} \to \mathfrak{g}$ satisfying the radial-gauge condition $\sum_\mu z_\mu A_\mu(z,\dots,z) = 0$. Its elements are the Taylor coefficients of weight $j+2$ of a connection in radial gauge. A *jet* is a sequence $J = (J_j)_{j\geq 0}$. Fix a real basis of each Tay_j, and give its coordinates weight $j+2$. Let V_d be the space of complex polynomials in these coordinates of total weight $\leq d$ that are invariant under $J \mapsto \mathrm{Ad}(g)J$ for all $g \in G$. Then:

(a) V_d is finite-dimensional, $1 \in V_d \subset V_{d+1}$, and an element of V_d is determined by its values on real jets.

(b) $V_d = \bigoplus_{w,f} V_d^{(w,f)}$, where $(w, f) =$ (weight, number of factors). $V_d^{(0,0)} = \mathbb{C}1$, and $V_d^{(w,0)} = 0$ for $w > 0$. Complex conjugation $p \mapsto \bar{p}$ and the orthogonal group $O(4)$, acting by rotating the jet, preserve the grades.

(c) There is a coherent system of *frames*. For each d there is a basis $b_0 = 1, b_1, \dots, b_{K_d}$ of V_d with these properties:

- each b_i is real ($\bar{b}_i = b_i$) and of pure grade (w_i, f_i), with $(w_0, f_0) = (0, 0)$, $w_i \le d$ and $f_i \ge 1$ for $i \neq 0$;
- the basis of V_d is contained in that of V_{d+1}.

In coordinates $c \in \mathbb{C}^{K_d+1}$ we write $(S_t c)_i = t^{f_i} c_i$, $\|c\|_1 = \sum_i |c_i|$, $c^\circ = c - c_0 e_0$, and $s \cdot c$ for the $O(4)$ action, defined by $p_{s\cdot c}(J) = p_c(s^{-1}J)$. *Status:* definitions; (a)–(c) are theorems. *Lean:* `Labels.Taylor`, `radial`, `limited`, `invariantSpace`, `Label`, `cap_finite`, `eval_ext`, `Grade`, `decompose`, `Frame`, `Frames`, `frames`, `Frame.scaled`, `Frame.act`, `Frame.core`, `Frame.coeffSize`. The field spaces F_d of the main theorem are quotients of V_d by the null labels. That quotient is formed in V.1/V.2, not here.

Proposition II.1.2 (test holonomies and sound sections).

(a) Fix d, a jet J, $h \in \mathbb{R}$ and a cover edge (x, μ), and let $A_\mu^{(d)}(z) = \sum_{j+2\le d} J_{j,\mu}(z, \dots, z)$, with an empty sum when $d < 2$. Let $u\colon \mathbb{R} \to G$ solve $u'(t) = u(t)\, hA_\mu^{(d)}(h(x + te_\mu))$ with $u(0) = 1$. The solution exists and is unique, and we put $U_{J,h}^{(d)}(x, \mu) = u(1)$. Moreover $U_{J,h}^{(d)} = U_{\mathrm{sc}_h J,1}^{(d)}$, where $\mathrm{sc}_h J = (h^{j+2} J_j)_j$, and $U_{J,0}^{(d)} \equiv 1$. A *holonomy datum* is such a choice of solutions; by uniqueness its links do not depend on the choice.

(b) A *section* on a frame of V_d is a pair (S, Φ): a finite stencil $S \subset \mathbb{Z}^4$ and a $\mathbb{C}$-linear map $c \mapsto \Phi_c$ into polynomials in the entries, and their conjugates, of the link matrices on S.

(c) The section is *sound* if:

- $0 \in S$ and S is invariant under signed permutations;
- each Φ_c is gauge invariant, $\Phi_{e_0} = 1$, and $\Phi_{\bar{c}} = \overline{\Phi_c}$;
- $\Phi_c(s^* U) = \Phi_{s\cdot c}(U)$ for every signed permutation s, where s^* is the pull-back under $y \mapsto sy$;
- (*projection*) for every c and every jet J, the functions $h \mapsto \Phi_c(U_{J,h}^{(d)})$ and $h \mapsto p_c(\mathrm{sc}_h J)$, with $p_c = \sum_i c_i b_i$, are C^d at 0 and have equal derivatives of order $\le d$ there.

Status: definitions; the ODE facts are theorems. *Lean:* `Source.Hol`, `partConnection`, `SolvesLink.unique`, `Hol.exists_ode`, `scaleJet`, `Projects`, `SectionData`, `SectionData.Sound`.

Proposition II.1.3 (primary sections). For every gauge group G, every holonomy datum and every coherent system of frames, there are sections $\Sigma_d = (S_d, \Phi^{(d)})$ on the frames of V_d such that, for every d:

1. Σ_d is sound.
2. (*slot structure*) Each column is $\Phi_{e_i}^{(d)}(U) = P_i(\sigma_1(U), \dots, \sigma_m(U))$. Here the σ_l are slots with stencils in S_d, P_i is a complex polynomial in their real components, and every monomial of

P_i has degree $\geq f_i$. A *slot* $\sigma = (\omega, P)$ is a lattice walk ω from 0 followed by a plaquette P at its end, with $\sigma(U) = \mathrm{Ad}(U_\omega)F_P(U) \in \mathfrak{g}$.

3. (*cost*) For all $k, P \in \mathbb{N}$ there is $A \geq 0$, depending on the chosen section and on k, P, with the following property. For every budget point b of order k and power P (any threshold and support exponent), every side ℓ, site o and index i, the placed column $\Phi^{(d)}_{e_i}(o;\cdot)$ is b-regular on its footprint $Y_o = o + S_d$ with sup bound $A\, t_b^{f_i}$.

For the fixed primary sections used in the final family, two further facts hold:

4. one nondecreasing bound $\pi(D) \geq 1$ serves in item 3 for all $d, k, P \leq D$;
5. there is $c_d \geq 1$ with $|\Phi^{(d)}_{e_i}| \leq c_d$ at every configuration and every placement.

Status: theorem, assembled from G01–G07; no simplicity assumption on G. *Lean:* `Integrated.source_sections`, `Bid.teaching`, `Bid.Learns`, `Bid.pop`, `Bid.Credit`, `Foundations.hol`, `localSection`, `sound`, `valuation`, `price`, `priced`, `full_value_bound`. These are the sections of the final Wilson family (Prop. II.1.6).

Proposition II.1.4 (readout of a lattice function). Fix a holonomy datum, d, and a frame of V_d.

(a) Let b be a budget point of order $\geq d$, let $X \subset \Lambda_\ell$, $o \in \Lambda_\ell$, and let f be b-regular on X. There is a unique $p \in V_d$ that *reads* f at o:

- if X does not wrap around o ($10\,\mathrm{dist}(x, o) < \ell$ for all $x \in X$), then for every jet J the function $h \mapsto f(U^{(d),o}_{J,h})$ is C^d at 0 and has the same derivatives of order $\leq d$ there as $h \mapsto p(\mathrm{sc}_h J)$;
- if X wraps, $p = f(1) \cdot 1$ by convention.

Here $U^{(d),o}_{J,h}(x, \mu) = U^{(d)}_{J,h}(\widetilde{x - o}, \mu)$, where the tilde denotes the centred periodic lift. These are torus test links, defined by lifting each initial vertex; they need not be periodic samples of a connection across the seam.

(b) There is $C \geq 1$, depending only on the holonomy datum and the frame, with the following property. For all ℓ, o, X, all budget points b with $k_b \geq d$ and $P_b \geq d$, all f that are b-regular on X with sup bound c, and all p reading f:

$$\|S_{t_b}\,\mathrm{coord}(p)\|_1 \leq C\, c.$$

(c) If Σ is sound and $20(|x_\mu| + 1) \leq \ell$ for every $x \in S$ and every axis μ, then the placed function $\Phi_c(o;\cdot)$ reads p_c at o.

Status: theorem. *Lean:* `exists_read`, `read_projection`, `Reads`, `price_of_readout`, `placed_reads`. This bounds the local-splitting projection and the remainder comparisons in III.12–III.13 and VIII.1.

Proposition II.1.5 (equation sections). Let $\mathrm{Sh}_j \subset \mathrm{Tay}_j$ be the radial potentials whose curvature is a degree-j homogeneous Taylor coefficient of a $\mathfrak{g}$-valued solution of the *free* Maxwell equations. That is: F is antisymmetric, satisfies the Bianchi identity, and $\sum_a \partial_a F_{ab} = 0$. The *equation ideal* is

$$\mathcal{E}_d = \{p \in V_d : \; p(J) = 0 \text{ for every jet with } J_j \in \mathrm{Sh}_j \text{ for all } j\}.$$

This is the weight-$\leq d$ part of the ideal of the linearized field equations, imposed degree by degree. For every holonomy datum, every d and every frame of V_d, there are a section Σ and a $\mathbb{C}$-linear automorphism R of V_d (the *weave*) such that:

- Σ is sound and has the slot structure and cost of Prop. II.1.3;
- R commutes with the signed-permutation action and with conjugation;
- for every $p \in \mathcal{E}_d$, the cover function $\Phi_{\mathrm{coord}(Rp)}$ is *micro*.

A function f is *micro* if it is continuous and there are $r, M \in \mathbb{N}$ with the following property. For every side $\ell \geq M$, site o, coupling $\beta \neq 0$ and bounded measurable S depending only on links with both endpoints at distance $> r$ from o,

$$\langle f(o;\cdot)\, S\rangle_\beta = 0.$$

The same holds simultaneously for all d along a fixed system of frames. The constructed R is unitriangular: if p has pure grade (w, f), every nonzero component of $Rp - p$ has grade (w', f') with $(f', w') > (f, w)$ in lexicographic order. The existence statement does not require this extra property. *Status:* theorem, unconditional. *Lean:* `Maxwell.equationIdeal`, `ideal_meaning`, `Maxwell.onShell`, `gaugeJet`, `Micro`, `MicroAt`, `Portfolio`, `Portfolio.build`, `commissioned`, `regrade_tri`. In the main chain these sections enter through the auxiliary-section data in `Run.Ties` (III.12), as a second family of sections on the same history, together with `Hearthside`. `Memento` and `awarded` would make them the family's own sections, but they are outside the main-theorem closure.

Proposition II.1.6 (the construction record and the renormalized observables). A *record* C consists of:

- an RG history H (III.9) and a holonomy datum;
- coherent frames, and sections Σ_d on them;
- reference levels $r(d)$;
- for all d, N, p, a transfer family $\mathcal{T}_{d,N,p} = ((E_m)_m, \tau)$ (Prop. II.1.7), with $\tau = \tau_{d,N,p}$ a linear automorphism of $\mathbb{C}^{K_d+1}$.

The renormalized observable of $v \in \mathbb{C}^{K_d+1}$ at $o \in \Lambda_{N,p}$ is

$$O_{N,p}(v, o; U) = a_N^{-d}\, \Phi^{(d)}_{S_{1/\theta_N}\tau^{-1}v}(o; U).$$

The *consistency predicate* Makes(C) states:

- (i) H is selected;
- (ii) every Σ_d is sound;
- (iii) $d + 3 \leq J(r(d))$, and for $m \geq r(d)$ the level-m budget has order $\geq d + 1$ and power $\geq d$;
- (iv) if $r(d) \leq N$, then $20(|x_\mu| + 1) \leq \ell_{N,p}$ for every $x \in S_d$ and every axis μ;
- (v) if $r(d) \leq N$, each placed column $\Phi^{(d)}_{e_i}(o;\cdot)$ is b_N-regular on Y_o with sup bound $\kappa_N \theta_N^{f_i}$;
- (vi) if $r(d) \leq N$, then $\mathcal{T}_{d,N,p}$ is based for $(\Sigma_d, r(d))$.

Then:

(a) Without any hypothesis: $v \mapsto O_{N,p}(v, o)$ is linear, $O_{N,p}(v, o)$ is continuous and depends only on links in Y_o, and $O_{N,p}(v, o + b; U) = O_{N,p}(v, o; U(\cdot + b))$ (literal translation covariance).

(b) If the sections are sound, $O_{N,p}(v, o)$ is gauge invariant for every N.

(c) Under Makes and $r(d) \le N$:

- reality: $O_{N,p}(\bar{v}, o) = \overline{O_{N,p}(v, o)}$;
- hypercubic covariance: $O_{N,p}(s \cdot v, A_{s,b} o; U) = O_{N,p}(v, o; A_{s,b}^* U)$ for the lattice affine map $A_{s,b} y = sy + b$;
- normalization of the unit: $O_{N,p}(a_{r(d)}^d e_0, o) \equiv 1$;
- normalized readout: the level-$r(d)$ coordinates of the transfer of $\tau^{-1} v$ equal v.

The renormalization map of the main theorem is $B_{N,p,d} = S_{1/\theta_N} \tau^{-1}$: writing $P_d[c] = \Phi_c^{(d)}$, its insertion formula is $I_{N,p} = a_N^{-d} P_d[B_{N,p,d} c]$. *Status:* (a)–(c) are theorems. Makes is proved for the final family in `Bandage.squad` (VIII.1) and included in the construction record `Run` (III.12). That family uses the fixed coherent frames, holonomies and primary sections of Prop. II.1.3. Clauses (ii)–(v) come from Prop. II.1.9(c) of this unit and clause (vi) from G08 (III.10). *Lean:* `Family`, `Family.O`, `initial`, `Family.Makes`, `linear0`, `continuous0`, `local0`, `translation`, `invariant`, `reality`, `rotated`, `constant`, `normalized`.

Proposition II.1.7 (based transfer families). *Source coefficients.* A source coefficient at level m is $E = (x, (X_i), (g_i), (h_i))_{i \in \mathbb{N}}$. Here $x \in \mathbb{C}^{K_d + 1}$ are the running label coordinates, the X_i are supports, the g_i are *regular* remainders, and the h_i are *forced* functions attached to the forced-polymer addresses i of ρ_m, whose activities are K_i.

Density. Its density at o is the λ-coefficient, at $\lambda^2 = 0$, of the density with one inserted source:

$$\mathcal{D}_m(E)(U) = \rho_m(U) \Big[x_0 + \mathbf{1}_{\mathrm{Hy}_{\theta_m}(Y_o)} \Phi_{S_{1/\theta_m} x^\circ}(o; U) + \sum_i \mathbf{1}_{\mathrm{Hy}_{\theta_m}(X_i)} g_i(U) \Big] + e^{\mathcal{V}_m(U)} \sum_{b \text{ adm}} \sum_{i \in b} h_i(U) \prod_{j \in b \setminus \{i\}} K_j(U).$$

Format at o. The g_i are b_m-regular on $X_i \ni o$, with $g_i(1) = 0$. When X_i does not wrap around o, $h \mapsto g_i(U_{J,h}^{(d),o})$ is C^d at 0 and all its derivatives of order $\le d$ vanish there, for every J. If $k_{b_m} \ge d$, this is exactly zero readout in Prop. II.1.4(a). Each h_i is gauge invariant and measurable, vanishes unless polymer i is eligible and carries o, and depends only on the links and flags of the support of polymer i.

Remainder bound (κ, c). Require $\kappa > 0$ and nonnegative a, b with $a + b \le c$, together with nonnegative weights e_i, e_i' with $\sum_i \mathrm{mass}(X_i, e_i) \le a$, $\|g_i\|_{b_m; o, X_i} \le e_i$, $\sum_i \mathrm{mass}(\mathrm{supp} K_i, e_i') \le \kappa b$ and $|h_i| \le e_i'$ at every configuration. Each weighted sum is required to converge; $\mathrm{supp} K_i$ denotes the prescribed support of forced polymer i.

Based family. A family $(E_m(v))_m$ with terminal automorphism τ is *based* for (Σ, r) if $r \le N$ and, for all v and $r \le m \le N$:

- (1) $E_m(v)$ has Format at 0 and remainder bound $(\sqrt{w_m},\ \kappa_m \|x_m(v)^\circ\|_1)$;
- (2) x_m, g and h are linear in v, and the supports do not depend on v;
- (3) (*bare end*) $x_N(v) = v$, $g \equiv 0$, and $\mathcal{D}_N(E_N(v)) = \rho_N \Phi_{S_{1/\theta_N} v}(0; \cdot)$;
- (4) (*transfer*) $\mathcal{D}_m(E_m(v)) = L_m^d T_{\beta_{N,p,m+1}}^{(m)} \mathcal{D}_{m+1}(E_{m+1}(v))$ for $m < N$;
- (5) $\|x_{m+1}\|_1 \le \kappa_{m+1} \|x_m\|_1$ and $\|x_m\|_1 \le \kappa_{m+1} L_m^d \|x_{m+1}\|_1$;
- (6) $x_r(v) = \tau v$, τ commutes with conjugation and with the signed-permutation action, and $x_m(a_N^d e_0) = a_m^d e_0$.

Status: definition (specification). G08 (III.10) produces such families from the per-step source estimates, together with the room and bank inequalities of Prop. II.1.9. A coefficient with Format

and a remainder bound has bounded measurable density whenever the underlying representation is lawful (Lemma 4.26). For a based family on a selected history, and $r \le m \le N$,

$$\int \mathcal{D}_m(E_m(\tau^{-1}v))\,dU = a_m^d\, Z_{N,p}\,\langle O_{N,p}(v,0)\rangle_{\beta_{N,p}}.$$

Lean: `Coefficient`, `Coefficient.density`, `exponentialEntry`, `Format`, `remBound`, `TransferData`, `TransferData.Based`, `integral_marked` (not used by the main theorem), `Bid.G08`.

Proposition II.1.8 (birth at the Wilson end). Let ρ be a lawful polymer representation on Λ_ℓ with parameter $w > 0$ (II.2) and threshold θ. Let $o \in \Lambda_\ell$, and let Σ be a sound section whose placed columns satisfy $|\Phi_{e_i}(o;\cdot)| \le B$ with $B \ge 1$. Put $Y = Y_o$ and $f_{\max} = \max_i f_i$, and assume the *fee condition*

$$\mathrm{fee} := B\,\theta^{-f_{\max}}\,\mathrm{mass}(Y,1)\,(e^{|Y|w} - 1) \le \sqrt{w}.$$

Assume a *shelf* of reserved forced addresses: for every pair of finite supports $\emptyset \ne I \subseteq X$ and every auxiliary index in $\mathbb{N}$, there is a distinct forced address with support X, flagged inventory I, identically zero activity, and an auxiliary eligibility test satisfied at every configuration. For every fixed reservation index there is a coefficient $E(v)$ with the following properties:

- $x = v$ and $g \equiv 0$;
- the forced part is the large-field *supplement* $\Phi_{S_{1/\theta}v}(o;\cdot) - v_0 - \mathbf{1}_{\mathrm{Hy}_\theta(Y)}\Phi_{S_{1/\theta}v^\circ}(o;\cdot)$, spliced into the reserved addresses;
- the components x, g, h depend linearly on v, the supports are independent of v, and $E(v)$ has Format at o and remainder bound $(\sqrt{w}, \|v^\circ\|_1)$;
- for every configuration U,

$$\mathcal{D}(E(v))(U) = \rho(U)\,\Phi_{S_{1/\theta}v}(o;U).$$

Status: theorem. The main chain uses `Guard.summon` at level N, with $w = w_N$. It requires a selected history and reserved addresses whose auxiliary tests need hold only when the support has exactly the prescribed flagged inventory, together with a separate zero-activity assumption. The section bound and fee condition are unchanged. The address conditions are `Keeper` and `Keeper.blank`. *Lean:* `launched`, `born`, `supplement`, `fullCost`, `fee`.

Proposition II.1.9 (numerical conditions for the sections and transfers).

(a) *Plans.* Let P be a stage profile, $C_j \ge 1$, and let (K_j) be nondecreasing with $\mathsf{K}_j \ge 1$. For the fixed degree-d frame put $\Omega_d = 1 + 2\sum_{s,i}\|s \cdot e_i\|_1$, where s ranges over signed permutations and i over frame indices; the factor 2 accounts for conjugation. This constant bounds the finite symmetry average. Assume P satisfies the following, writing $\bar\kappa_j$ for the constant of stage j:

- $j \le$ order and $j \le$ power of stage j;
- $\pi(\max(\mathrm{order}_j, \mathrm{power}_j)) \le \bar\kappa_j$;
- $(C_j + 1)\,\Omega_d \le \mathsf{K}_j$ for $d \le j$;
- $64\mathsf{K}_{j+1}^2 \le \bar\kappa_j$.

Then there are lower bounds n_j for the block factors and coupling thresholds β_j^* such that every admissible level system with profile P, block factors at least n_j at stage j, and $\hat\beta_m \ge \beta^*_{J(m+1)}$ satisfies the following numerical conditions:

- $J(m) \le$ order and power at level m;
- $\pi(\max(\text{order}, \text{power})) \le \kappa_m$;
- the latch of (b) for all $d \le J(m)$;
- room: $64\mathsf{K}^2_{J(m)+1} \le \kappa_m$;
- bank: $10^5\mathsf{K}^6_{j+1}(1+\kappa_{m+1})^2 \le L_m^{1/8}$, with $j = J(m+1)$;
- for every $d \le J(m)$, with $j = J(m+1)$ and $\eta = L_m^{-2/5}$,

$$\eta(1+\kappa_{m+1})\Omega_d \le L_m^{-3/8}, \qquad (C_j + \eta\kappa_{m+1})\Omega_d \le \mathsf{K}_j.$$

If a profile already satisfies $j \le \text{order}_j, \text{power}_j$, then, for any fixed $C_j \ge 1$, one can choose (K_j) and enlarge only its constants $\bar\kappa_j$ to obtain the four stage conditions above.

(b) *Latch.* At level m, Latch(d, m) holds if, for every p, $\ell_{m,p}$ is at least the minimal side of Σ_d, and the fee condition of Prop. II.1.8 holds with $B = c_d$, $\theta = \theta_m$, $w = w_m$, at every placement. For each d and support exponent, Latch(d, m) holds once $\widehat\beta_m$ is large enough.

(c) *Section clauses of Makes.* If the numerical conditions of (a) hold on a history and $d + 3 \le J(r(d))$ for all d, then the primary sections satisfy clauses (ii)–(v) of Makes.

Status: theorems. The two per-step inequalities are `Buys`; the finite symmetry constant is `orbitCost`. *Lean:* `Doors.Plans`, `Doors.Budget`, `Doors.afford`, `alone`, `Latch`, `latch_cost`, `entrances`, `Issue.refer`, `Foundations.refLi`, `Bid.RefLi`.

Proposition II.1.10 (chessboard comparison at all offsets). Let $\mathcal{C}_k(x)$ denote the tiling of $\Lambda_{N,p}$ by link cells of odd side $D_{k,N} = a_k/a_N$, for $k \le N$, centred at x. Write rt_β for the chessboard root test of a tiling, and say that F is *interior* to a cell if it is measurable in that cell with a uniform bound.

(a) *Centred symmetry.* Suppose F is interior to the central cell of $\mathcal{C}_k(0)$. For every diagonal sign change s and every x, the function $F \circ A^*_{s,x}$ is interior to the central cell of $\mathcal{C}_k(x)$ and has the same root test as F in $\mathcal{C}_k(0)$. The same holds after complex conjugation. Diagonal sign changes do not permute the axes.

(b) *All offsets from one centred bound.* Let $F_o = F_0(\cdot + o)$, and assume:

- F_0 is interior to the central cell of $\mathcal{C}_k(0)$ with root test $\le A$;
- T is a tiling whose side is a multiple of the level-k cell side and which satisfies the reflection-positivity seam inequality at β.

Then for every o in the inner margin of cell 0 of T (distance $\ge 3\times$ the level-k side from its boundary), F_o is interior to that cell and $\mathrm{rt}^T_\beta(F_o) \le A$.

(c) *Combinations.* Assume Makes, fix $\gamma > 0$ and an extraction order f, and let $Q_i(N, p, o)$ be finitely many families. Each family is eventually a literal translate of its value at 0. For each i and every w, assume there is $k \ge w$ such that eventually $k \le N$ and $Q_i(N, p, 0)$ is interior to the central cell of $\mathcal{C}_k(0)$, with root test at most $e^{-\gamma s_N}$. Suppose also that, for every $\varepsilon > 0$, eventually $|r_i(N, p)| \le e^{\varepsilon s_N}$. The coefficients are independent of o.

"Eventually" along f means: for all sufficiently large volume indices, then for all sufficiently large mesh indices; this is the fused order of IV.1. Put $Q = \sum_i r_i Q_i$ and $y(N, p) = e^{-\gamma s_N/2}$. Then $y \to 0$ along this order and Q is *close*: for every $j, m \in \mathbb{N}$ there is $k \ge m$ such that eventually

$k \le N$ and, for every link tiling T of level-k side and every $o \in T.\mathrm{margin}(0, \lfloor T.\mathrm{side}/(8(j+1))\rfloor)$, $Q(N,p,o)$ is interior to cell 0 and its root test is at most $y(N,p)$.

Status: theorems. *Lean:* `center_price`, `corner_root`, `offset`, `gather_close`, `Bid.close`, `roof_of_tabs` (a bookkeeping corollary). Consumers: III.12, III.13, VII.4, VIII.1.

3.2.2 2. What it uses

From other units.

- I.1 (lattice and Wilson measure): tori and the cover $\mathbb{Z}^4$, periodic placement, gauge transformations, signed permutations and their pull-backs, translations, Haar probability measure, Wilson energy and expectation (`Wilson.moment`).
- II.2 (polymer activities, norms, record, cells):
  - budget points, regularity and sup bounds (`BudgetPoint`, `Regular`, `supBounds`);
  - healthy sets and masks (`Healthy`, `masked`);
  - slots (`Slot`);
  - polymer representations and lawfulness (`Rep`, `Rep.Lawful`), together with mass, admissibility and the activity bound;
  - tilings, root test, interior and seam inequality (used in Prop. II.1.10).
- III.1 (block-spin step): the blocking push-forward $T_\beta^{(m)}$ and the fact that it preserves total mass (used in Lemma 4.26).
- III.9 (marginal flow): the history, its selection, $\hat{\beta}_m$, θ_m, w_m, the stage index $J(m)$ and the stage constants κ_m.
- III.2 (order of constants): sheets, profiles and the level systems quantified over in Prop. II.1.9.
- III.10: G08 turns per-step estimates into based families; III.11 supplies the per-step estimates used in VIII.1. Both use the formats fixed here.
- III.16 (toolkit): block Taylor polynomials and jets of compositions, the Faà di Bruno formula, and the Maxwell exterior calculus (curl, recovery of a radial potential from its curvature). The radial-gauge identity identifies a homogeneous potential with its closed curvature jet.
- IV.1: extraction orders and the fused order (used in Prop. II.1.10(c)).

Standard mathematics.

- Linear ODEs on matrix groups: existence by an exponential chart and left translation; uniqueness through $\frac{d}{dt}(uv^*) = 0$.
- Analytic dependence on parameters through the Volterra equation and the Neumann series ([Dieudonné, *Foundations of Modern Analysis*, ch. X]).
- The analytic inverse function theorem in finite dimensions.
- Reynolds averaging over a finite group, and averaging over a compact group with Haar measure.
- Polarization of symmetric multilinear forms, and norm equivalence on finite-dimensional spaces.
- Radial (Fock–Schwinger) gauge and its Taylor expansion [Novikov–Shifman–Vainshtein–Zakharov, Fortschr. Phys. 32 (1984) 585].
- Integration by parts for Haar measure, i.e. the lattice Schwinger–Dyson identity [Makeenko–Migdal, Phys. Lett. B 88 (1979) 135; Chatterjee, Comm. Math. Phys. 366 (2019) 203].

- Chessboard estimates from reflection positivity [Fröhlich–Israel–Lieb–Simon, Comm. Math. Phys. 62 (1978) 1].

Imported constants and assumptions.

- In Prop. II.1.6 the record and Makes are hypotheses of (c). They are discharged for the final family in VIII.1, using Prop. II.1.9(c) and G08.
- In Prop. II.1.8 lawfulness and the fee condition are hypotheses. The fee condition at level N is the latch of Prop. II.1.9(b). Coupling thresholds fixed before the block sizes ensure it (III.2), since w_m decays faster than every inverse power of $\hat{\beta}_m$.
- The bounds $\pi(D)$ and c_d depend on G and the fixed frames and sections, including the finite slot family chosen in G03. They are fixed before the block factors; the stage constants must satisfy $\pi(\max(\mathrm{order}_j, \mathrm{power}_j)) \le \bar{\kappa}_j$.

3.2.3 3. Strategy

Labels are invariant polynomials in radial-gauge Taylor coefficients. Test holonomies $U_{J,h}$ compare them with functions of lattice links by matching derivatives in h through order d. Finitely many transported plaquette curvatures give an analytic map of the truncated jet whose derivative is injective. An equivariant left inverse, followed by Taylor expansion, produces slot polynomials representing every label. Averaging over signed permutations gives covariance. Every slot and each of its admissible left derivatives costs one threshold factor, so a column of factor degree f has norm $O(\theta^f)$. Conversely, Taylor expansion of a regular lattice function gives a unique read label. Testing on jets of amplitude θ bounds its scaled coordinates uniformly in the lattice side and support.

The equation sections use Haar integration by parts in one link. For $\ell \ge 2$, $\beta \ne 0$ and a bounded measurable multiplier independent of that link, the expectation of the corresponding Wilson force is zero. Transported forces are linear combinations of slots; their leading continuum symbols are the free Maxwell divergence. Every equation polynomial is a sum of products of a linear Maxwell equation functional and other linear jet functionals. Represent the first factor by a force and the remaining factors by slot polynomials avoiding its link. Each product is orthogonal to remote spectators and reads the prescribed generator modulo higher grade. A unitriangular change of labels absorbs those higher terms. The resulting equation section describes vanishing at separated lattice sites; its transfer to null labels of the limit requires the later comparison arguments.

At the Wilson end the observable is inserted to first order in a parameter λ with $\lambda^2 = 0$. The density coefficient consists of the vacuum density times a masked local section and regular remainders, together with a gas in which one forced polymer is marked. Splicing the section's large-field complement into reserved forced addresses gives the exact bare-end identity. This uses no division by the vacuum density. Along the RG history the running coordinates describe mixing of local observables; their values at the reference level define τ, hence the renormalization map $S_{1/\theta_N}\tau^{-1}$. This unit fixes the transfer specification and proves the birth identity; III.10–III.11 supply the per-step estimates.

3.2.4 4. Steps

3.2.4.1 4.1 Label spaces and frames Lemma 4.1 (finiteness and evaluation). V_d is finite-dimensional, and a polynomial in the jet coordinates that vanishes on all real jets is zero. *Sketch:* every coordinate has weight ≥ 2. So a monomial of weight $\le d$ has exponents $\le d$ and

involves only Tay_j with $j \le d-2$, and there are finitely many such monomials. A polynomial vanishing on $\mathbb{R}^n \subset \mathbb{C}^n$ is zero, because each variable ranges over an infinite set. *Lean:* `cap_finite`, `eval_ext`.

Lemma 4.2 (real graded frames). Every linearly independent set of real pure-grade labels in V_d containing 1 extends to a frame. Consequently frames exist, and there is a coherent sequence of them. *Sketch:*

- V_d is spanned by the elements $q + \bar{q}$ and $i(q - \bar{q})$, where q runs over the graded components; these elements are real and of pure grade.
- Extend the given independent set inside this spanning set, and move 1 to index 0.
- Recursion in d gives coherence, since $V_d \subset V_{d+1}$ preserves reality and grades.
- $f_i > 0$ for $i \neq 0$, since the only nonzero zero-factor grade is $\mathbb{C}1$.

Lean: `pure_real_span`, `extend_subset`, `frame_exists`, `extend_frame`, `frames`.

3.2.4.2 4.2 Test holonomies and the slot map (G01–G04)

Lemma 4.3 (transport ODE; covariance, G01).

- Existence and uniqueness hold as stated in Prop. II.1.2(a).
- Scaling: $U_{J,h} = U_{\mathrm{sc}_h J,1}$, with only the components J_j, $j \le d-2$, entering.
- Gauge covariance: the links of $\mathrm{Ad}(g)J$ are the constant gauge transform by g of the links of J.
- Hypercubic covariance: $s^* U^{(d)}_{J,h} = U^{(d)}_{s^{-1}J,h}$, where $s^{-1}J$ is the jet rotated by $s^{-1} \in O(4)$.
- In particular each slot of $\mathrm{Ad}(g)J$ is $\mathrm{Ad}(g)$ applied to the slot of J.

Sketch:

- Uniqueness: if u, v solve the same equation, then uv^* has zero derivative.
- Existence: take a local solution in an exponential chart, left-translate it, and glue the pieces.
- Scaling and covariance: the transformed function solves the transformed equation, so uniqueness concludes; a reflected axis reverses the parameter of the ODE.

Lean: `SolvesLink.unique`, `Hol.exists_ode`, `Bid.G01`, `Work.G01.answer`, `links_colored`.

Lemma 4.4 (analytic germ, G02).

(a) Let x range over a finite-dimensional real vector space, let $A_x(t)$ depend linearly on x and continuously on t, and let u_x solve $\dot{u} = uA_x$ with $u_x(0) = 1$. Then $x \mapsto u_x(1)$ is analytic at 0, equals 1 there, and its derivative along every ray is $\int_0^1 A_x$.

(b) Consequently each link, each walk and each slot is an analytic function of the truncated jet near 0. The derivative of the slot (ω, P) at 0 is the circulation of the edge-integrated potential around P. The transport along ω does not contribute at linear order. *Sketch:* write the ODE as $F = 1 + T(x)F$ on $C([0,1])$, where $T(x)$ is the Volterra operator, linear in x, with $\|T(x)\| < 1$ near 0. Then $F = (1 - T(x))^{-1}1$, which is analytic since inversion is analytic near 1 in a Banach algebra. Differentiate the fixed-point equation at 0. The bounds are link by link; no analytic radius uniform over an infinite stencil is claimed. *Lean:* `endpointGerm`, `linkGerm`, `walkGerm`, `slotGerm`, `Drops.Peg`.

Lemma 4.5 (finitely many slots detect the jet, G03). For each d there are finitely many slots whose linearizations have no common kernel on the truncated jet space $\bigoplus_{j \le d-2} \mathrm{Tay}_j$. *Sketch:*

- *All slots detect the jet.* Suppose all linearized slots vanish on A. Write $N_l(x,t)$ for the difference of the two iterated averages of partial derivatives over a square of side t based at x. The unit-square circulation is $\sum_l N_l(x,1)$. Homogeneity gives $N_l(tx,t) = t^l N_l(x,1)$, hence $\sum_{l\le k} t^{k-l} N_l(tx,t) = 0$ when A_k is the top component and $x \in \mathbb{Z}^4$.
- *Mesh to zero.* Letting $t \to 0$ with $tx \to y$ gives $\mathrm{curl}(A_k)(y) = 0$ for the top component. In radial gauge the potential is recovered from its curvature, so $A_k = 0$. Descend in k.
- *Finitely many suffice.* Among finite families choose one whose common kernel has minimal dimension. Adding any slot cannot lower the dimension, so this kernel lies in the common kernel of all slots, which is 0.

The choice is non-constructive. *Lean:* `Bid.G03`, `top_term`, `zero_of_zero_partials`, `Work.G03.answer`.

Lemma 4.6 (equivariant left inverse, G04). The slot map $\mathfrak{S}$ (finitely many slots) is analytic at 0. There is a linear C with $C \circ D\mathfrak{S}(0) = \mathrm{id}$ and $C \circ \mathrm{Ad}_g = \mathrm{Ad}_g \circ C$. *Sketch:* $D\mathfrak{S}(0)$ is injective (Lemma 4.5) and equivariant (it is the derivative of an equivariant map). Its range and orthogonal complement are Ad-invariant, so the inverse composed with the orthogonal projection onto the range is equivariant. No Haar averaging is needed. *Lean:* `Bid.G04`, `Work.G04.answer`.

Lemma 4.7 (local analytic equivariant inverse). There is f, analytic at 0, with $f(0) = 0$, $f \circ \mathfrak{S} = \mathrm{id}$ near 0, and $f \circ \mathrm{Ad}_g = \mathrm{Ad}_g \circ f$ near 0. *Sketch:* apply the analytic inverse function theorem to $C \circ \mathfrak{S}$, whose derivative at 0 is the identity, and put $f = (C \circ \mathfrak{S})^{-1} \circ C$. Equivariance follows from that of C and $\mathfrak{S}$, together with local injectivity. *Lean:* `Drops.insight`, `Vision`.

Lemma 4.8 (Haar measure in exponential coordinates). There are $r, c > 0$ and a measurable function J such that exp is injective on $B_r(0) \subset \mathfrak{g}$, J is analytic and positive there, $J(0) = 1$, and $\exp_*(cJ(A)\mathbf{1}_{B_r(0)}\,dA)$ is normalized Haar measure restricted to $\exp(B_r(0))$, an open set. *Sketch:* take J to be the determinant of the derivative of $B \mapsto \log(e^{-A}e^{B})$ at $B = A$. It is analytic and equals 1 at $A = 0$, so it is positive on a small ball. The chain rule gives its transformation under local left translation. Change of variables then proves the corresponding local measure identity; averaging over the compact group identifies the scalar c with probability Haar normalization. *Lean:* `LocalH.actual`, `Foundations.chart_ok`. This supplies the measure normalization in the stationary-point analysis (III.6) and the step estimates.

3.2.4.3 4.3 Primary sections (G05–G07)

Lemma 4.9 (invariant slot polynomials, G05). Let f be as in Lemma 4.7, let $T_d f$ be its block Taylor polynomial of order d at 0, and put $P_i := b_i \circ T_d f$. Write $\sigma(U)$ for the chosen finite tuple of slot values. Then:

- $P_0 = 1$, each P_i is Ad-invariant and real-valued on real slot values, and each monomial of P_i has degree $\ge f_i$;
- $P_i(\sigma(U_{J,h}))$ and $b_i(\mathrm{sc}_h J)$ have the same d-jet in h.

Sketch:

- $T_d f$ is equivariant because f is equivariant and Ad is linear.
- $T_d f(0) = f(0) = 0$, so every coordinate of $T_d f$ lacks a constant term; since b_i is homogeneous of degree f_i, every monomial of P_i has degree $\ge f_i$.
- Along $X(h) = \mathrm{trunc}(\mathrm{sc}_h J)$, which is smooth with $X(0) = 0$, the slots of $U_{J,h}$ equal $\mathfrak{S}(X(h))$.
- Since $f \circ \mathfrak{S} = \mathrm{id}$ near 0, and the d-jet of a composition depends only on the d-jets of its factors, $T_d f \circ \mathfrak{S} \circ X$ and X have the same d-jet.

- Labels of weight $\le d$ see only the components of weight $\le d$.

All Taylor terms of order $\le d$ are kept, so P_i may have degree $> d$ in the slots. *Lean:* `Work.G05.answer`, `recovery_projects`, `blockTaylor_jets`, `Bid.Cumuli`.

Lemma 4.10 (covariant section, G06). From the P_i and hypercubic covariance of the links one obtains a sound section with the slot structure, whose stencil is a box $[-r, r]^4$ containing all slot stencils. *Sketch:*

- Write $\Psi_c(U) = \sum_i c_i P_i(\sigma(U))$ before averaging, and let $\mathcal{H}$ be the signed-permutation group. Set $\Phi_c(U) = |\mathcal{H}|^{-1} \sum_{s\in\mathcal{H}} \Psi_{s^{-1}\cdot c}(s^*U)$. A transformed slot is a slot multiplied by a sign. The coefficient action preserves grades, so the average remains a slot polynomial with the same lower bound on degrees.
- Gauge invariance: slots transform by Ad of the gauge transformation at the base point, and the P_i are Ad-invariant.
- The projection property survives because jets and labels transform compatibly.
- Each slot component is a polynomial in link entries and their conjugates, since $U^{-1} = U^*$.

Lean: `Work.G06.answer`, `enclose`, `realize`, `entryPoly_slot`.

Lemma 4.11 (cost, G07). For all (k, P) there is A such that the section has the cost property of Prop. II.1.3, item 3. *Sketch:*

- *One slot costs t.* On $\mathrm{Hy}_t(S)$ a slot has norm $O(t)$. Each left derivative in an admissible direction A, with $|A_e| \le t(1 + \mathrm{dist})^P$, is also $O(t)$, with constants depending on the stencil.
- *Algebra.* These bounds are stable under sums and products, with exponents of t adding, and $t \le 1$ lowers exponents.
- *Columns.* Every monomial of column i has $\ge f_i$ slot factors, which gives $c_i t^{f_i}$; take $A = \sum_i c_i$.

A is not uniform in d, k, P. *Lean:* `Work.G07.answer`, `paid_lumin`, `paid_slot`, `raw_slot`.

Lemma 4.12 (assembly = Prop. II.1.3). Lemmas 4.3–4.11 combine into sections for all d.

- The uniform price is $\pi(D) = 1 + \sum_{d,k,P\le D} \max(0, A_{d,k,P})$, which is nondecreasing in D.
- The value bound c_d holds because each column is a continuous function of finitely many variables in the compact group G.

Lean: `Bid.teaching`, `Integrated.source_sections`, `priced`, `full_value_bound`.

3.2.4.4 4.4 Readout

Lemma 4.13 (logarithmic chart of test holonomies). Fix d, a torus, a site o and a jet J. Near $h = 0$, the torus test configuration $U^{(d),o}_{J,h}$ is the exponential of an analytic function of h vanishing at 0. Thus $h \mapsto f(U^{(d),o}_{J,h})$ is C^d at 0 whenever $A \mapsto f(e^A)$ is C^d at $A = 0$. *Sketch:* apply the local group logarithm to each test link, use Lemma 4.4 and the polynomial scaling of the jet, and intersect the finitely many neighbourhoods on the torus. The logarithm inverts the exponential there, so the assertion follows by composition. *Lean:* `testLog_diff`, `increment_testLog`.

Lemma 4.14 (existence and uniqueness of the read label = Prop. II.1.4(a)). Fix d, a torus and o. If f is gauge invariant and $A \mapsto f(e^A)$ is C^d at $A = 0$, there is a unique $p \in V_d$ such that, for every J, $f(U^{(d),o}_{J,h})$ and $p(\mathrm{sc}_h J)$ have equal d-jets at $h = 0$. *Sketch:*

- *Polynomial in the jet.* Take the block Taylor polynomial of order d at 0 of the map $X \mapsto f(\text{test links of } X)$ on the truncated jet space. It is a polynomial in the jet coordinates; truncate it to weight $\le d$.
- *Jets agree.* Since $\mathrm{trunc}(\mathrm{sc}_h J)$ has components $h^{j+2}J_j$, truncation does not change h-derivatives of order $\le d$.
- *Invariance.* Gauge invariance of f and covariance of the test links (Lemma 4.3) make each weight component of p invariant.
- *Uniqueness.* The h-derivatives at 0 determine the weight components, and then Lemma 4.1 applies.

For regular f with order $\ge d$ the hypothesis holds at the identity configuration, which is healthy; wrapping supports use the constant convention. *Lean:* `read_projection`, `exists_read`.

Lemma 4.15 (uniform bounds along scaled test jets).

(a) Fix d. For each jet J and $k \in \mathbb{N}$ there is $C > 0$ such that, for all torus sides, sites o, positive links e and $\theta \in [0,1]$,

$$\left|\partial_h^k \log U^{(d),o}_{\theta J,h}(e)\Big|_{h=0}\right| \le C\,\theta\,(1+\mathrm{dist}(e_-,o))^{\max(k-1,0)}.$$

(b) For $k \le d$ there is $C_{d,k,J} \ge 0$ such that $|\partial_h^k f(U^{(d),o}_{t_b J,h})|_{h=0}| \le C_{d,k,J}c$ whenever $d \le \min(k_b, P_b)$ and f is b-regular with sup bound c. The bound is uniform in side, placement, support and b. *Sketch:*

- (a): set $r = (1+\mathrm{dist}(e_-,o))^{-1}$. The rescaled initial position lies in a fixed compact ball. Express the link at parameter hr by a polynomial-in-time ODE profile of amplitude θr. Its logarithmic endpoint is analytic near the zero profile. The k-th h-derivative at 0 varies smoothly over the compact set of normalized positions, lengths and amplitudes, and vanishes at amplitude 0. A uniform Lipschitz bound in the amplitude gives $C\theta r$. Undoing the rescaling gives $C\theta r^{1-k}$ for $k \ge 1$; the order-zero logarithm is 0.
- (b): apply Faà di Bruno to $(f \circ \exp) \circ \log U$ at the identity. Its terms contain symmetric Fréchet derivatives of $A \mapsto f(e^A)$, evaluated on the logarithmic derivatives from (a). Pull each tensor back to the span of these finitely many directions, and bound it by its diagonal values through polarization. Repeated left derivatives in one direction equal derivatives along the corresponding exponential ray. This avoids a factor depending on the number of lattice links; it does not assume that ordered left derivatives commute.

Lean: `testLog_bound`, `test_bound`, `mixed_budget`.

Lemma 4.16 (readout estimate = Prop. II.1.4(b)). *Sketch:*

- *Functionals.* For a jet J and $k \le d$ let $\lambda_{J,k}(c) := \partial_h^k\, p_c(\mathrm{sc}_h J)|_{h=0}$.
- *Scaling.* Since $p_{S_\theta c}(K) = p_c(\theta K)$, one has $\lambda_{J,k}(S_\theta\,\mathrm{coord}(p)) = \partial_h^k p(\mathrm{sc}_h(\theta J))|_{h=0} = \partial_h^k f(U^{(d),o}_{\theta J,h})|_{h=0}$.
- *Bound.* By Lemma 4.15(b) this is bounded by $C_{J,k}c$.
- *Norm equivalence.* The $\lambda_{J,k}$ separate points (Lemma 4.14), and on a finite-dimensional space a separating family of functionals controls the norm.
- *Wrapping case.* $p = f(1)\cdot 1$, and $|f(1)| \le c$ because the identity configuration is healthy.

The constant depends on the holonomy datum and the frame only. *Lean:* `coeff_bound`, `budget_probes`, `read_frame_scaled`, `price_of_readout`.

Lemma 4.17 (placed sections read their labels = Prop. II.1.4(c)). Let Σ be sound. If $20(|x_\mu|+1) \le \ell$ on the stencil, the footprint does not wrap around o, the periodic lift of the stencil is injective, and the placed function along the test holonomies is the cover function along $U_{J,h}$. Soundness (projection property) then concludes. *Lean:* `nonwrapping_placement`, `placed_reads`.

3.2.4.5 4.5 Equation sections Lemma 4.18 (lattice Schwinger–Dyson identity). Assume $\ell \ge 2$. *Gradient.* For a link e and $A \in \mathfrak{g}$, $\frac{d}{dt} S_W(e^{tA} U_e)|_0 = \langle A, \nabla_e S_W(U)\rangle$. With $F_{ab}(x)$ based at x, put $\Delta_c F_{ab}(x) = F_{ab}(x) - \mathrm{Ad}(U^{-1}_{x-e_c,c}) F_{ab}(x-e_c)$. Then

$$\nabla_{(x,\nu)} S_W = \sum_{\nu<b} \Delta_b F_{\nu b}(x) - \sum_{a<\nu} \Delta_a F_{a\nu}(x).$$

This local formula also defines the force on cover configurations. *Integration by parts.* For continuous W, W' with W' the derivative of W along $e^{tA}U_e$, and bounded measurable S invariant under that motion,

$$\big\langle (W' - \beta \langle A, \nabla_e S_W\rangle W) S \big\rangle_\beta = 0.$$

For $\beta \ne 0$ and $W = 1$ this gives $\langle \langle A, \nabla_e S_W\rangle S\rangle_\beta = 0$. *Sketch:* the motion $U_e \mapsto e^{tA}U_e$ preserves the product Haar measure, and the Boltzmann factor is differentiated under the integral. *Lean:* `field_deriv`, `turn_preserving`, `ibp`, `equation_silent`.

Lemma 4.19 (transported forces are micro). Let b be a positive cover link, let ω be a walk from 0 avoiding b, and let g be continuous. Assume there is M such that, on every torus of side $\ell \ge M$ and at every placement o, the placed g is invariant under left increments of the placed link b. Then $U \mapsto \langle X, \mathrm{Ad}(U_\omega)\nabla_b S_W(U)\rangle\, g(U)$ is micro, for every $X \in \mathfrak{g}$. *Sketch:*

- Write $\langle X, \mathrm{Ad}(U_\omega)Y\rangle = \sum_a \langle \mathrm{Ad}(U_\omega)^{-1}X, E_a\rangle\langle E_a, Y\rangle$ in an orthonormal basis.
- Once the torus is large enough, the placed link b lies within a fixed radius of o.
- Then every factor other than $\langle E_a, \nabla_b S_W\rangle$ (namely the transport, g, and a spectator supported beyond that radius) is invariant under moving U_b. Apply Lemma 4.18.

The periodic independence hypothesis holds, in particular, if g depends on finitely many cover links, all distinct from b. No continuity or gauge invariance of the spectator is needed, and β may be negative. *Lean:* `micro_force`, `force`.

Lemma 4.20 (forces span the equation functionals).

(a) Let J be truncated through J_j, with $j \ge 1$, and write $\mathfrak{F}_\nu(J;x)$ for the linearization at the zero jet of the transported force at (x,ν). It is independent of the transport path. For every $y \in \mathbb{R}^4$ and $x_t \in \mathbb{Z}^4$ with $t x_t \to y$ as $t \downarrow 0$,

$$\lim_{t\downarrow 0} t^{j-1} \mathfrak{F}_\nu(J; x_t) = -\sum_a \partial_a (\mathrm{curl}\, J_j)_{a\nu}(y).$$

For $j = 0$ the linearized force vanishes. At finite mesh the force is a transverse difference of plaquette averages.

(b) For jets truncated through weight $j+2$, if every linearized transported force on a walk ending at the initial vertex of its link and avoiding that link vanishes, then the component J_j lies in Sh_j.

(c) Dually, for every real linear functional Q on Tay_j vanishing on Sh_j, the functional $Q \circ \mathrm{proj}_{\mathrm{top}}$ is a finite combination of linearized transported forces. *Sketch:* express each transverse

difference as an additional integral of a derivative of the plaquette average. Homogeneity and continuity of these averages give (a). If every force vanishes, every lattice point can be reached by a walk avoiding the varied link; (a) then makes the top curvature divergence-free. Its Bianchi identity follows because it is a curl, proving (b). In finite dimensions, the annihilator of the common force kernel is the span of the force functionals, giving (c). *Lean:* `continuum_current`, `top_shell`, `force_kernel_last`, `force_span`.

Lemma 4.21 (generators of the equation ideal). $p \in \mathcal{E}_d$ if and only if $p \circ \Pi = 0$, where Π is the projection of each Tay_j onto Sh_j. Every $p \in \mathcal{E}_d$ of pure grade m is a combination of products $\ell \cdot M$, of total grade m, of a linear functional ℓ on Tay_j vanishing on Sh_j and a product M of real linear jet functionals, with complex scalar coefficients. Grade components of elements of $\mathcal{E}_d$ lie in $\mathcal{E}_d$, and $\mathcal{E}_d$ contains no nonzero constant. *Sketch:* write $p = p - p \circ \Pi$. For a monomial $M = \prod_r \ell_r$, the difference $M - M \circ \Pi$ telescopes into $\sum_r (\ell_r - \ell_r \circ \Pi) \cdot (\cdots)$, and each $\ell_r - \ell_r \circ \Pi$ vanishes on Sh. *Lean:* `erase_shell`, `multiplied`, `generation_ready`.

Lemma 4.22 (micro lifts at pure grade). For $p \in \mathcal{E}_d$ of grade m (with $m_1 \le d$) there are a gauge-invariant lattice function f and a label q with:

- f represents q to order d;
- $q - p$ has only components of strictly higher grade;
- f is a slot polynomial with $\ge m_2$ slot factors;
- f is micro.

Sketch:

- *Generators.* For a generator $\ell \cdot M$, write $\ell \circ \mathrm{proj}_{\mathrm{top}}$ as a combination of linearized forces (Lemma 4.20).
- *Lift of a force.* A transported lattice force component is a degree-one slot polynomial. Its readout is the full linearized force plus terms with at least two jet factors.
- *Lift of the monomial.* Slots avoiding one specified link still detect every truncated jet: excluding that link removes only finitely many plaquette positions, which does not change the mesh-to-zero argument away from 0; continuity gives the value at 0. Thus every factor of M can be represented, through all retained weights, by a slot polynomial independent of that link after periodic placement.
- *Product.* The force combination matches ℓ through its own weight $j+2$; extending it to cutoff d adds only higher-weight one-factor terms. Nonlinear readout errors have more factors. Multiplication by the lifted M therefore gives ℓM modulo higher grade. The product is micro by Lemma 4.19. Combine linearly.
- *Gauge invariance.* Average over constant gauge transformations with Haar measure. The orbit spans a finite-dimensional space, so the average stays a slot polynomial and the relation above persists.

Lean: `stanza`, `coherent_verse`, `mounted_read`, `averaged`, `positioned`, `Cone`.

Lemma 4.23 (a covariant lifting map). There is a linear map $F : V_d \to V_d \times \{\text{lattice functions}\}$, with $F(p) = (q, f)$, $F(1) = (1, 1)$, such that for p of grade m:

- f is gauge invariant and represents q;
- $q - p$ has only components of higher grade;
- f has $\ge m_2$ slot factors;
- f is micro whenever $p \in \mathcal{E}_d$.

F commutes with signed permutations and conjugation. *Sketch:*

- *One grade.* Choose a linear projection $P_m : V_d \to V_d^{(m)} \cap \mathcal{E}_d$. Apply the micro lift to $P_m p$, and the primary-section lift $q \mapsto (q, \Phi_{\mathrm{coord}(q)})$ to $p - P_m p$; call the sum $X_m(p)$. On pure grade m both summands have the required grade and slot bounds. On $V_d^{(m)} \cap \mathcal{E}_d$ the second summand vanishes.
- *All grades.* Sum $X_m \circ$ (grade-m part) over m.
- *Symmetrization.* Average over the signed-permutation action, and with the conjugate. These operations preserve all properties above.

Lean: `shelf`, `Offered`, `Payment`, `Campaign`, `unsym`, `impression`.

Lemma 4.24 (weave and section = Prop. II.1.5). *Weave.* Let R be the first component of F. It is unitriangular for the order of grades, hence an automorphism. *Section.* The inverse of R also does not lower grades, so $R^{-1}b_i$ has no component preceding the grade of b_i. Hence the map $b_i \mapsto f(R^{-1}b_i)$ sends basis vectors to slot polynomials with $\geq f_i$ factors. Enclose all their stencils in one box and rewrite them as entry polynomials; this gives a section Σ with $\Phi_{\mathrm{coord}(Rp)} = f(p)$. *Properties.*

- Soundness: gauge invariance, unit, covariance and reality are inherited from F; the projection property holds because $f(p)$ represents Rp.
- Cost: G07 (Lemma 4.11).
- Microness: for $p \in \mathcal{E}_d$, $f(p)$ is the sum of the micro grade components of $F(p)$.

Choose these sections and automorphisms for every d. *Lean:* `twist`, `regrade_tri`, `paid_reel`, `weaveBuild`, `Portfolio.build`, `commissioned`.

3.2.4.6 4.6 Source histories **Lemma 4.25 (splicing a large-field multiplier).** Let f be bounded and vanish on $\mathrm{Hy}_\theta(Y)$, let ρ be lawful with forced-activity parameter w, and let a shelf as in Prop. II.1.8 be given. Then for every configuration U,

$$\sum_b \sum_i \mathrm{line}_i(f)(U)\, \mathbf{1}_{\{i \in b,\ b \text{ adm}\}} \prod_{j \in b \setminus \{i\}} K_j(U) = f(U)\, \mathrm{gas}(U),$$

as an identity on all configurations. The *line* coefficients satisfy $\sum_i \mathrm{mass}(X_i, p_i) \leq \sup|f| \cdot \mathrm{mass}(Y, 1)(e^{|Y|w} - 1)$, where $p_i \geq \sup_U |\mathrm{line}_i(f)(U)|$. *Sketch:*

- On a large-field configuration every admissible polymer set meets Y, because the forced polymers cover the flagged sites.
- For each nonempty compatible set c of old polymers meeting Y, reserve one address with support $Y \cup \bigcup_{i \in c} X_i$ and inventory $\bigcup_{i \in c} I_i$. Its coefficient is $f \prod_{i \in c} K_i$ when the flagged sites in that support equal the inventory, and 0 otherwise.
- Replace all polymers meeting Y by that address and leave the remaining polymers unchanged. This is a bijection between old admissible sets meeting Y and admissible sets with a marked reserved address; its inverse expands the marked address. Paired summands agree. The forced-activity bounds give absolute convergence and the displayed weighted estimate, so the double sum can be reindexed.

Lean: `Attach.paid`, `Attach.routes`, `Attach.line_budget`.

Lemma 4.26 (birth = Prop. II.1.8; summability; moment identity). *Birth.* The supplement vanishes on $\mathrm{Hy}_\theta(Y)$, because $\Phi_{e_0} = 1$. It is gauge invariant, measurable, determined

by the links and flags of Y, and bounded by $B\theta^{-f_{\max}}\|v^\circ\|_1$. Splice it by Lemma 4.25. The local part of the density is then the masked healthy part and the marked series is supplement × gas; their sum is $\rho\,\Phi_{S_{1/\theta}v}$. The forced budget is $\mathrm{fee}\cdot\|v^\circ\|_1 \le \sqrt{w}\,\|v^\circ\|_1$. *Guarded birth.* Use the same reserved addresses and coefficients after temporarily replacing the forced-polymer auxiliary tests by the always-true test. The vacuum activities, supports and inventories stay fixed, so the density and bounds are unchanged. Every nonzero inserted coefficient enforces the correct inventory and therefore satisfies the original auxiliary test. Restoring those tests gives the birth identity on the original selected history.

Summability. A remainder bound dominates the masked regular series by a summable sequence. If the representation is lawful, combine summable majorants for the activities and forced coefficients into $p_l \ge 0$; the marked summand for a set b is bounded by $\prod_{l\in b} 2p_l$, summable over finite sets. Format supplies measurability of the coefficients. The lawful vacuum representation and the finite section then give bounded measurable density. *Moment identity.* For a selected history, each transfer multiplies the total integral by L_m^d (the push-forward preserves mass) and $a_m = L_m a_{m+1}$. This gives the identity $\int \mathcal{D}_m = a_m^d Z\langle O\rangle$ of Prop. II.1.7. *Lean:* `launched`, `healthy_supplement`, `supplement_bound`, `regular_series`, `marked_series`, `envelope`, `integral_marked`, `onTop`. *(proofs read)*

Lemma 4.27 (numerical conditions = Prop. II.1.9). *Latch.* $\theta_m^{-1} \le \hat\beta_m$. Once the cover size, which grows like $(\log\hat\beta_m)^2$, exceeds the minimal side, $\mathrm{mass}(Y,1) \le e^{\text{support exponent}}$. Also $e^{|Y|w} - 1 \le |Y| w e^{|Y|}$. Hence $\mathrm{fee} \le C\hat\beta_m^{f_{\max}} w_m$. Since $w_m = \exp(-\exp((\log\log\hat\beta_m)^2/5))$ decays faster than every inverse power, $\hat\beta^{n+1} w \le \sqrt{w}$ eventually. Thresholds for all $d \le J(m)$ are combined by summing finitely many thresholds. *Numerical conditions.*

- bank: $L^{1/8} \to \infty$;
- per-step prices: $L^{-2/5} s \le 1$ and $L^{-2/5}(1+s)T \le L^{-3/8}$ for large L;
- block lower bounds: maximum of finitely many requirements per stage;
- Enlarge each stage constant $\bar\kappa_j$ to $\max(\bar\kappa_j, \pi, 64\mathsf{K}_{j+1}^2)$, with $\mathsf{K}_k = 1 + \sum_{j\le k}\sum_{d\le j}(C_j+1)\,\Omega_d$.

Section clauses. For $m \ge r(d)$, $d+1 \le d+3 \le J(m) \le \mathrm{order}_m$ (and likewise for the power). Latch gives the minimal side. $\pi(\max(\mathrm{order},\mathrm{power})) \le \kappa_N$ turns the bound of Prop. II.1.3, item 4, into clause (v) of Makes. *Merger delay.* For each section and support-exponent ceiling u there is a threshold q such that, whenever the level-m support exponent is at most u and $\hat\beta_m \ge q$, $(2B_m+1)^4\sqrt{w_m} \le 1$, where $B_m = \lceil(\log\hat\beta_m)^2\rceil$ is the cover size. The same threshold ensures the bare-section bounds needed for local merging. *Lean:* `latch_cost`, `entrances`, `afford`, `alone`, `refLi`, `Issue.refer`, `Push.Local.delay`.

Lemma 4.28 (local-merger identities and differentiation).

(a) Fix a valid forced-polymer family, a budget point, and a family f_i of measurable gauge-invariant functions. Each f_i is determined by links and flags on X_i and vanishes on $\mathrm{Hy}_\theta(X_i)$. For every i and nonempty compatible bundle b of polymers meeting X_i, assume a distinct reserved address with support $X_i \cup \bigcup_{j\in b} X_j$, inventory $\bigcup_{j\in b} I_j$, and auxiliary test satisfied everywhere. Pack $f_i \prod_{j\in b} K_j$ into that address, multiplied by its eligibility indicator. If $\sum_i |f_i(U)| < \infty$, the marked-gas series converges at U and equals $(\sum_i f_i(U))\mathrm{gas}(U)$. For a fixed admissible set, the marked summand is the sum over its possible marked addresses, with all other activities multiplied unchanged.

(b) Suppose f is C^k in exponential coordinates at U. Take a list of $r \le k$ pairs (A_j, B_j) of left

and right directions. Assume that at each link either all A_j vanish or all B_j vanish. The ordered mixed derivative is the sum of 2^r ordered left derivatives: for each pair choose its left direction or the direction $\mathrm{Ad}(U)B_j$; put the chosen left directions first in their original order, followed by the chosen right directions in reverse order.

(c) Let W be an open set of configurations. Suppose every f_i is C^k in exponential coordinates at every $U \in W$. For every fixed ordered list l of at most k left directions, including the empty list, assume a summable majorant $d_i(l)$ for $|D_l f_i(U)|$, uniform in $U \in W$. Then $\sum_i f_i$ is C^k in those coordinates throughout W, and $D_l \sum_i f_i = \sum_i D_l f_i$ there.

Sketch: in (a), replace the polymers meeting X_i by their reserved marked address. Expanding that address is the inverse correspondence, and the activity bound and summability of $f_i(U)$ justify reindexing. In (b), move the right derivatives past the left derivatives on disjoint link sets; replacing right increments by left increments conjugates their directions by U and reverses their order. For (c), work in one exponential chart contained in W. Bounds on finitely many basis derivatives give a summable bound on the Fréchet derivative; the chart change is uniformly bounded on a smaller neighbourhood. Differentiate the series there and induct on k, retaining the order of each left-derivative list.

Lean: `gasMark_eq`, `Slots.recover`, `mix_convert`, `smooth_exchange`. These identities enter the conversion of local functions into source coefficients and the differentiation of unmasked series in the per-step estimates.

3.2.4.7 4.7 Properties of the observables Lemma 4.29 (= Prop. II.1.6). *Sketch:*

- Translation is literal, since the placement commutes with translations.
- Linearity, continuity and locality are properties of Φ and of the linear maps S_{1/θ_N} and τ^{-1}.
- Gauge invariance follows from soundness of the section.
- Reality and covariance use three facts: τ commutes with conjugation and with the signed-permutation action (Prop. II.1.7(6)); S_t commutes with the $O(4)$ action because the grades are preserved; and the section is sound.
- Unit: Prop. II.1.7(6) gives $\tau(a_N^d e_0) = x_r(a_N^d e_0) = a_r^d e_0$, hence $\tau^{-1}(a_r^d e_0) = a_N^d e_0$. Since $S_t e_0 = e_0$ and $\Phi_{e_0} = 1$, the observable is $a_N^{-d} a_N^d = 1$.
- Normalization is $x_r = \tau$.

Lean: see Prop. II.1.6 and `origin_unit`.

3.2.4.8 4.8 Comparison estimates Lemma 4.30 (= Prop. II.1.10). *Sketch:*

- (a) Translations and diagonal axis reflections carry the centred tiling and all its cell copies to the corresponding centred tiling at the new anchor. Reindex the cells and use invariance of the Wilson measure. Conjugating every copy conjugates the expectation, leaving its absolute value unchanged.
- (b) At offsets o that are centres of the finer centred grid, apply the chessboard inequality for T. Each cell copy of F_o is a reflected and translated copy of F_0 planted in a distinct finer cell, so the product of roots is $\le A^{\sharp\mathrm{cells}}$ by (a).
- (b), general offsets: bracket each coordinate between aligned corners, retaining the $2\times$-side margin for all intermediate positions. For literal translates F_u along an axis the seam inequality gives $r_{u+1}^2 \le r_u r_{u+2}$, where r_u is the root test. A nonnegative sequence satisfying this inequality cannot exceed both endpoint bounds. Iterate over the four coordinates.

- (c) The root test is subadditive under finite sums. Slowly varying coefficients cost $e^{\varepsilon s_N}$ with $\varepsilon < \gamma/2$, which leaves the rate $e^{-\gamma s_N/2}$.

Lean: `center_price`, `corner_root`, `offset`, `gather_close`.

3.2.5 5. Where the difficulty lies

The main analytic estimate is the volume-independent readout bound. Polarization in the full lattice direction space could introduce constants depending on the volume. Pulling each tensor back to the finitely many test directions avoids this, while rescaling each link position gives the required distance weights. The analytic radii of individual links need not be uniform.

The equation lift must serve two purposes at once: reproduce a prescribed jet and permit integration by parts without a derivative of its multiplier. Slot sampling away from the varied link supplies this independence. The triangular argument orders grades first by factor number, then by weight; higher-factor corrections need not have higher engineering weight. The final family retains its primary sections, and the equation sections form an auxiliary family whose comparison requires later estimates.

The birth identity requires reserved addresses in the same forced-polymer alphabet. Lawfulness alone does not supply them. The guarded construction weakens the auxiliary-test requirement while retaining exact eligibility whenever an inserted coefficient is nonzero. Its numerical bound is possible because the forced allowance decays faster than every inverse coupling power.

The finite slot family is chosen nonconstructively. Consequently the stencils and constants are finite but not explicit. Their dependence on degree and derivative parameters must be included before the block factors are chosen.

3.2.6 6. Notation

Written proof	Lean (glossary)	Meaning
Tay_j, jet J	`Labels.Taylor`, `Jet`	radial-gauge Taylor coefficient of weight $j+2$; sequence of them
V_d	`Labels.Label`	invariant polynomials of weight $\le d$ (labels)
$p(J)$	`read`	value of a label on a jet
(w, f), $V_d^{(w,f)}$	`Grade`	weight and number of factors
frame (b_i), $c \in \mathbb{C}^{K_d+1}$	`Frame`, `Frame.Coeff`	real graded basis starting with 1; coordinates
S_t, c°, $\|c\|_1$	`Frame.scaled`, `core`, `coeffSize`	$c_i \mapsto t^{f_i} c_i$; drop the unit coordinate; ℓ^1 norm
$s \cdot c$, $\bar{c}$	`Frame.act`, `realBar`	$O(4)$ action; conjugation
$U_{J,h}^{(d)}$	`Hol`, `Hol.links`	test holonomies
$\mathrm{sc}_h J$	`scaleJet`	$(h^{j+2} J_j)_j$
slot $\sigma = (\omega, P)$	`RG.Slot`	$\mathrm{Ad}(U_\omega) F_P(U)$
slot map $\mathfrak{S}$, truncated jets	`bring`, `Snap`	finitely many slots as a function of $(J_j)_{j \le d-2}$

Written proof	Lean (glossary)	Meaning
section (S, Φ); $\Phi_c(o; U)$	`SectionData`, `places`	stencil and linear map to entry polynomials; placement
sound	`Sound`	symmetries, unit, projection property
slot structure; cost	`Bid.pop`; `Bid.Credit`	columns are slot polynomials with $\geq f_i$ factors; t^{f_i} budget bound
$\pi(D)$, c_d	`price`, `cameras`	uniform price; value bound of the columns
p reads f at o	`Reads`	readout (constant convention when wrapping)
Sh_j, $\mathcal{E}_d$	`Maxwell.onShell`, `gaugeJet`, `equationIdeal`	free-Maxwell components; free-Maxwell ideal
$\nabla_e S_W$, transported force	`field`, `force`	link gradient of the Wilson energy; its transport to 0
micro	`Micro`	orthogonal to remote bounded spectators
micro lift; lifting map F	`Cone`; `Campaign`	lift agreeing modulo higher grade; all grades at once
weave R; equation section	`twist`; `Portfolio`	unitriangular regrading; section with micro equation part
record C; Makes	`Family`; `Family.Makes`	Wilson family; consistency predicate
$O_{N,p}(v, o)$	`Family.O`	renormalized observable $a_N^{-d}\Phi_{S_{1/\theta_N}\tau^{-1}v}(o; \cdot)$
$r(d)$, $a_r^d e_0$	`Family.ref`, `Family.refUnit`	reference level; physical unit vector
$E_m(v)$, $\mathcal{D}_m$	`Coefficient`, `density`	source coefficient; first-order inserted density
τ; based	`TransferData.terminal`; `Based`	terminal map; transfer specification
supplement; fee	`supplement`; `fee`	large-field part of the bare observable; birth condition
latch; Budget; Plans	`Latch`; `Budget`; `Plans`	birth condition per level; level inequalities; stage-wise numerical plan
price list (K_j)	`Bid.Worth`	nondecreasing constants ≥ 1
close	`Bid.close`	root-test bounds tend to 0 in all prescribed link tilings and margins

3.2.7 7. What remains unexplained

1. **Identification in the limit (tentative).** The auxiliary equation sections enter the volume-uniform comparison through `Run.Ties` and `Hearthside`. This unit does not trace their regrading and transfer normalizations into the null-label quotient, or prove the converse identification with the free-Maxwell quotient. A literal equality $N_d = \mathcal{E}_d$ in the original primary coordinates has not been verified.
2. **Later applications.** The section constants and reference-level conditions are explicit here. Their simultaneous compatibility with every later per-step estimate, and the full use of the local-merger identities, belong to III.2 and III.10–III.15 and are not verified by these local results.

3.3 II.2. Marker algebra, polymer combinatorics and Gaussian estimates

Part: II, Data of the construction. **Lean:** `YM/Marker/`, `YM/Ac/`, `YM/Ovens/Coffee.lean`, `YM/Further/SHELL_BIRTH_49/Fusing/Coal.lean`, `YM/Extra/MULTISOURCE_TRAJECTORIES/`, `YM/Push/GRID_APPROXIMATION/Nil.lean`, `YM/MergeProof/`, `YM/Next/RANK/`, `YM/Vac/Treasury.lean`. **Size:** 3,691 lines in 793 declarations used by the main theorem.

This unit supplies the algebraic, combinatorial and numerical estimates used to multiply and sum polymer activities in Part III, especially in the Gaussian comparison of III.11. The activity format is defined in III.1; the Wilson family and construction record in II.1; reflection-positive cells in IV.1.

The results fall into five groups: marker algebra (II.2.1–II.2.4), polymer combinatorics and forced-activity fusion (II.2.5–II.2.8), switching (II.2.9), Gaussian polynomial calculus (II.2.10–II.2.11), and numerical scale comparisons (II.2.12–II.2.13). They are used in III.8–III.13, III.15, VII.2, VII.4 and VII.6; the vacuum-energy estimate also enters V.2.

3.3.1 1. What this unit proves

Conventions.

- Λ is a four-dimensional periodic lattice.
- A *support* is a finite subset of Λ, and dist is the lattice separation.
- For $M \in \mathbb{N}$ and a support X, the **cover number** $\mathrm{cov}_M(X)$ is the least cardinality of a set $D \subseteq \Lambda$ such that the union $B_M(D)$ of the balls of radius M around D contains X and is connected for nearest-neighbour steps (`coverNumber`, `covers`, I.1).
- The parameters of the polymer norms are $P = (r, q, e_P, B, t_P)$, with $r, q, B \in \mathbb{N}$, $r \geq 6$, $q \geq 2$, $B \geq 1$, $e_P > 0$ and $0 < t_P \leq 1$: smoothness order, power, support exponent, cover size and threshold. The mass of an envelope c on support X is $e^{e_P \mathrm{cov}_B(X)} c$.
- Two facts from I.1 are used:
  - $|X| \leq (2M+1)^4 \,\mathrm{cov}_M(X)$ (`Corona.pays`);
  - $\mathrm{cov}_M(X \cup Y) \leq \mathrm{cov}_M(X) + \mathrm{cov}_M(Y)$ if $X \cap Y \neq \emptyset$ (`Corona.meets_sum`).

3.3.1.1 A. Marker algebra

Proposition II.2.1 (marker algebra). Let V be a finite set and R a *census* on V: a finite family of finite subsets of V that contains $\emptyset$ and all singletons and is closed under taking subsets. Let $\mathcal{A}_R$ be the space of functions $a : R \to \mathbb{C}$, written $a = \sum_{x \in R} a(x)\delta_x$,

with product

$$(ab)(z) = \sum_{x,y\in R:\ x\cap y=\emptyset,\ x\cup y=z} a(x)\,b(y) \qquad (z\in R).$$

Then:

1. $\mathcal{A}_R$ is a commutative unital $\mathbb{C}$-algebra with unit $\delta_\emptyset$.
2. $\delta_x\delta_y = \delta_{x\cup y}$ if $x\cap y=\emptyset$ and $x\cup y\in R$, and $\delta_x\delta_y = 0$ otherwise. So every marker $\delta_{\{v\}}$ squares to zero, and products that leave R vanish.
3. $(ab)(\emptyset) = a(\emptyset)\,b(\emptyset)$.
4. If $a(\emptyset)=0$, then $a^k(x)=0$ whenever $|x|<k$; in particular, $a^k=0$ for $k>|V|$.

Status: theorem. *Lean:* `Marker.Census`, `Poly`, `Poly.prod`, ring structure, `Poly.quiet_mul`, `Poly.high`, `Poly.stops`.

Proposition II.2.2 (weighted ℓ^1 norm). For $w : V\to(0,\infty)$ put $w^x=\prod_{v\in x} w_v$ and $\|a\|_w=\sum_{x\in R} w^x|a(x)|$. Then $\|1\|_w=1$, $\|c\,\delta_x\|_w = w^x|c|$, and

$$\|ab\|_w\le\|a\|_w\|b\|_w, \qquad \Big\|\prod_{i\in T} a_i\Big\|_w \le \prod_{i\in T}\|a_i\|_w, \qquad \|a^k\|_w\le\|a\|_w^k.$$

Set $\operatorname{Exp}(a)=\sum_{k=0}^{|V|} a^k/k!$.

- $\|\operatorname{Exp}(a)\|_w\le e^{\|a\|_w}$ for every a.
- If a countable family satisfies $\|f_i\|_w\le b_i$ with $\sum_i b_i<\infty$, then $\sum_i f_i$ converges in $\mathcal{A}_R$, $\|\sum_i f_i\|_w\le\sum_i b_i$, and coefficients of the sum are sums of coefficients.
- $\max_x|a(x)|\le(\min_{x\in R} w^x)^{-1}\|a\|_w$.

The multiplicative and exponential bounds have constant 1, independently of R and $|V|$. The norm-comparison constant is used to establish convergence.

Status: theorem. *Lean:* `Poly.toll`, `Poly.price`, `multiplication`, `multiple`, `powers`, `Poly.pass`, `pass_budget`, `budgeted`, `share`, `equivalent_norm`.

Proposition II.2.3 (exponential and connected polynomial). Use the marker algebra and exponential of II.2.1–II.2.2.

(a) If $a(\emptyset)=b(\emptyset)=0$, then $\operatorname{Exp}(a+b)=\operatorname{Exp}(a)\operatorname{Exp}(b)$ and $\operatorname{Exp}(-a)\operatorname{Exp}(a)=1$. More generally, $\operatorname{Exp}(\sum_{i\in s} a_i)=\prod_{i\in s}\operatorname{Exp}(a_i)$ for a finite family with all $a_i(\emptyset)=0$. The zero-constant-term hypotheses are necessary for these identities in general.

(b) If $x\neq\emptyset$, then $\operatorname{Exp}(c\,\delta_x)=1+c\,\delta_x$. Hence, for a finite family of nonempty $x_i\in R$ (not necessarily distinct),

$$\operatorname{Exp}\Big(\sum_{i\in s} c_i\delta_{x_i}\Big)=\prod_{i\in s}(1+c_i\delta_{x_i}).$$

(c) If $F(\emptyset)=1$, there is exactly one $C\in\mathcal{A}_R$ with $C(\emptyset)=0$ and $\operatorname{Exp}(C)=F$: the **connected polynomial** of F. It is obtained by iteration:

- $C_0=0$ and $C_{k+1}=C_k+F-\operatorname{Exp}(C_k)$;
- $F-\operatorname{Exp}(C_k)$ vanishes on all marker sets with at most k elements, so $C=C_{|V|}$;
- each C_k is a polynomial, hence continuous, function of F.

(d) If $C(\emptyset)=0$ and $\operatorname{Exp}(C)=F$, then $F\operatorname{Exp}(-C)\,P=P$ for every P.

Status: theorem. *Lean:* `Exponential.pass_add`, `pass_neg`, `pass_sum`, `Trajectories.pass_send`, `Trajectories.pass_sum`, `Expands`, `roll`, `rolls`, `roll_cont`, `Expands.uniq`, `connected_spec`, `correct`.

Proposition II.2.4 (tagged products; repacking). Use positive weights w as in II.2.2.

(a) Let $\tau : \iota \to R$ and let $s \subseteq \iota$ be finite. Say that s **fits** if the sets $\tau(i)$, $i \in s$, are pairwise disjoint and their union lies in R.

- If s fits, then $\prod_{i\in s} c_i\delta_{\tau(i)} = (\prod_{i\in s} c_i)\,\delta_{\bigcup_{i\in s}\tau(i)}$ and $w^{\bigcup_{i\in s}\tau(i)} = \prod_{i\in s} w^{\tau(i)}$.
- Conversely, if $\prod_{i\in s} c_i\delta_{\tau(i)} \neq 0$, then s fits.

(b) Let:

- $\sigma : \iota \to \kappa$ assign a *slot* to each entry;
- "admissible" be any property of finite sets of slots;
- $F : \iota \to \mathcal{A}_R$ satisfy $\sum_i \|F_i\|_w < \infty$.

For finite $s \subseteq \iota$ put $\mathrm{val}(s) = \prod_{i\in s} F_i$ if σ is injective on s and $\sigma(s)$ is admissible, and $\mathrm{val}(s) = 0$ otherwise. Then val is summable over finite subsets of ι, and

$$\sum_s \mathrm{val}(s) = \sum_{S\subseteq\kappa \text{ finite, admissible}} \prod_{k\in S} \sum_{i:\,\sigma(i)=k} F_i.$$

Status: theorem. *Lean:* `Trajectories.Fits`, `tagged_prod`, `fits_of_tagged_prod`, `toll_union`, `Packets.summable_value`, `Packets.repack`.

3.3.1.2 B. Polymer combinatorics **Proposition II.2.5 (connected components).** Let $X : \iota \to \{\text{supports}\}$ and let $s \subseteq \iota$ be finite.

- Two labels of s are *chained in* s if they are joined by a sequence of labels of s whose successive supports intersect.
- A **spark** is a nonempty s in which all labels are chained.
- $\mathrm{Parts}(s)$ is the set of chain classes of s, and $X(p) = \bigcup_{i\in p} X_i$.

Then:

1. The parts of s are sparks, pairwise disjoint, with union s and with pairwise disjoint supports $X(p)$.
2. If S is a finite set of sparks with pairwise disjoint supports, then $\mathrm{Parts}(\bigcup S) = S$.
3. If s is a spark and $i \in s$, every label of $s \setminus \{i\}$ is chained in $s \setminus \{i\}$ to a label whose support meets X_i.
4. If every label of s is chained in s to a label meeting a support Y (call s a **pack for** Y), then every part of s contains a label meeting Y.

The same component statements hold when the supports are finite subsets of any set.

Status: theorem. *Lean:* `Ac.Spark`, `Parts`, `reconstruct`, `separate`, `separates`, `excise`, `packHits`, `Collide.uncluttered`.

Proposition II.2.6 (tree bound in cover-number weights). Let:

- ι be any set of labels and $X : \iota \to \{\text{supports}\}$;
- $M \in \mathbb{N}$, $f : \iota \to [0,\infty)$, $a \geq 0$ and $\gamma \geq 0$ with $(2M+1)^4\gamma \leq 1$.

Assume the **hit condition with exponent** $a+1$: for every site x the family $\big(\mathbf{1}[x \in X_i]\, f_i\, e^{(a+1)\mathrm{cov}_M(X_i)}\big)_i$ is summable, with sum at most γ. Then for every site x,

$$\sum_{s \text{ spark},\ x\in X(s)} e^{a\,\mathrm{cov}_M(X(s))} \prod_{i\in s} f_i \ \le\ e^{\gamma}-1.$$

The sum runs over finite sets of labels, so each label is used at most once, and the family is summable.

Status: theorem. Used by III.9–III.13, VII.2 and VII.6. *Lean:* `Ac.Hit`, `Ac.roots`, `Ac.forest`.

Proposition II.2.7 (unique recovery for two kinds of overlap). A **kiln** on a set I of labels consists of:

- finite sets $\mathrm{men}(i) \subseteq V$ ("mentions");
- finite sets $\mathrm{wall}(i) \subseteq W$ (W another set);
- a set of *dirty* labels.

For finite $s \subseteq I$:

- a label $i \in s$ is **hot in** s if it is joined to a dirty label of s by a chain in s of overlapping mentions;
- the **fuel** of i in s is the subset of $V \sqcup W$ consisting of its mentions in the V component and, only when i is hot, its walls in the W component;
- s is a **mole** if it is a spark for fuel overlap;
- the **slabs** of s are its parts for fuel overlap.

Then:

1. Every slab p of s is a mole, and the fuel of its labels computed in p equals the fuel computed in s. Distinct slabs have disjoint mention-unions. Distinct slabs that contain a dirty label have disjoint wall-unions. (In a mole, containing a dirty label is equivalent to all labels being hot.)
2. Conversely, let S be a finite set of moles with pairwise disjoint mention-unions, and with pairwise disjoint wall-unions among the members that contain a dirty label. Then the slabs of $\bigcup S$ are exactly the members of S.
3. By 1 and 2, $s \mapsto \mathrm{slabs}(s)$ and $S \mapsto \bigcup S$ are inverse bijections between finite sets of labels and such finite sets of moles.
4. Suppose the walls are supports and, inside a mole s, overlapping mentions imply overlapping walls. Then s is a spark for the walls. Thus a family of moles can be estimated by II.2.6 whenever its wall supports and weights satisfy that proposition's hypotheses.

Status: theorem. Used by III.11 and VII.4. *Lean:* `Kiln`, `Kiln.Hot`, `fuel`, `Mole`, `good`, `fuel_comp`, `clubs_away`, `walls_away`, `takes_mole`, `Kiln.undo`, `Kiln.attends`.

Proposition II.2.8 (fused gas of forced activities: definitions). Fix a gauge group G and a periodic lattice Λ.

Data.

- **Pipes**: regular addresses (reg_i); forced addresses (ker_i), each with a support, a nonempty inventory contained in it, and an extra test on configurations; nonempty "softer" supports (Y_i), for $i \in \mathbb{N}$.

- **Tap** d: three families of real-valued functions of the configuration, namely regular coefficients E_i, forced coefficients K_i and logarithmic coefficients ℓ_i.
- **Labels** are $\mathbb{N} \sqcup \mathbb{N}$: *left* labels are forced kernels, *right* labels are logarithms.
  - The *patch* of a left label i is $\operatorname{supp}(\ker_i)$; that of a right label i is Y_i.
  - The *reach* of a left label is $K_i(U)$; that of a right label is $e^{\ell_i(U)} - 1$.
- A **pellet** is a finite set of labels that contains at least one left label, is a spark for patch overlap, and whose left labels have pairwise disjoint kernel supports. Pellets form a countably infinite set.

Fused activities. Fix a activity parameter point P and a bijective enumeration F of the pellets. Eligibility is always tested at the threshold of P. The fused activities keep the regular part (reg_i, E_i). Their forced part is indexed by pellets:

- address of $F(i)$: support = union of the patches, inventory = union of the left inventories, extra test = all left kernels eligible;
- value
$$\mathrm{solid}_i(U) = \mathbf{1}[\text{the address of } F(i) \text{ is eligible at } U] \prod_{j \in F(i)} \mathrm{reach}_j(U).$$

The unit also defines the hypotheses under which the fused gas is used. Below, B is the cover size of P.

- **Digest condition at** (d, P, δ, b):
  1. the unfused activities $(\mathrm{reg}, \ker, E, K)$ are valid at P with regular and forced bounds δ, b;
  2. $\ell_i(U) \neq 0$ only if Y_i meets the large-field set of U;
  3. $|\ell_i(U)| \le B_i$ for every i, U, with $\sum_i B_i < \infty$;
  4. each ℓ_i is gauge invariant and measurable, and has the same value on two configurations that agree on links internal to Y_i and have the same large-field flags on Y_i;
  5. lattice affine maps permute the ℓ_i covariantly.
- **Fusion bounds at** (d, P, κ, γ):
  1. $\gamma, \kappa \ge 0$ and $(2B+1)^4 \gamma \le 1$;
  2. there is a label weight $f \ge 0$, independent of U, with $|\mathrm{reach}| \le e^{-\kappa} f$ on left labels and $|\mathrm{reach}| \le f$ on right labels;
  3. f satisfies the hit condition of II.2.6 for the patches, with exponent (support exponent of P) $+1$ and bound γ.
- For a parameter set $T \subseteq \mathbb{R}$ and taps d_a, $a \in T$, the parametric conditions mean:
  1. for each label j, $a \mapsto \mathrm{reach}_j(d_a)$ is continuous in the supremum over configurations, relative to T;
  2. one common choice of κ, γ, f satisfies the reach and hit bounds for every $a \in T$;
  3. the fused values are continuous in the norm $\sum_i e^{e_P \mathrm{cov}_B(S_i)} \sup_U |K_i(U)|$, where e_P is the support exponent and S_i the fused support.

Status: definitions, plus one theorem (an enumeration exists). The properties of the fused gas are the work items H01–H04, stated in `YM/Bid/Kitchen.lean` and proved in III.10:

- **H01** (under the digest condition and fusion bounds): $\mathrm{gas}_{\mathrm{fused}}(U) = \mathrm{gas}_{\mathrm{unfused}}(U) \exp \sum_i \ell_i(U)$ for every configuration U.
- **H02** (under the digest condition and fusion bounds): the fused forced activities satisfy the K-bound v whenever $e^{-\kappa}(e^{\gamma} - 1) \le v$.
- **H03** (under the digest condition): they form a forced family with the lattice moves.

- **H04**: if the unfused representations form a continuous parameter curve, conditions 1 and 2 above imply that the fused representations form such a curve. The path data, cutoff and parameters P are fixed along the curve.

At startup in III.9, the digest and fusion bounds follow from clip compatibility, threshold and coupling bounds, and the packet estimates in the cited results (`Start.digests`, `Loan.buys`). *Lean:* `Ovens.Pipes`, `Tap`, `reach`, `contains`, `addr`, `solid`, `fused`, `Digests`, `Buys`, `invariantJet`, `little`, `retakes`, `encoding`; the obligations `Bid.H01`–`H04`, proved by `Work.H01.answer`, `H02.answer`, `H03.answer`, `H04.answer` (III.10).

3.3.1.3 C. Switching Proposition II.2.9 (switching).

Let ι be finite. For $x \in \mathbb{R}$ and $g : 2^\iota \to \mathbb{R}$ put $\mathrm{val}_x(g)(r) = \sum_{i \subseteq r} x^{|i|} g(i)$. This is the value at $x\mathbf{1}_r \in \{0, x\}^\iota$ of the multilinear polynomial $\sum_i g(i) \prod_{v \in i} X_v$.

(a) For $x > 0$, val_x is a linear bijection of $\mathbb{R}^{2^\iota}$.

(b) Let $(f_x)_{x \in [0,1]}$ be functions on 2^ι. Say (g_x) **models** f if $\mathrm{val}_x(g_x) = f_x$ for $x \in [0,1]$ and $g_x \to g_0$ as $x \downarrow 0$. In that case $\mathrm{jet}(f) := \lim_{x \downarrow 0} \mathrm{val}_x^{-1}(f_x)$ exists and equals g_0.

(c) A family $(f^u)_u$ is **paid at cost** c if there is $b \geq 0$ on 2^ι with $\sum_i b(i) \leq c$ such that, for each u, some model g^u of f^u satisfies $|g^u_x(i)| \leq b(i)$ for all $x \in [0,1]$. Then:

- $|f^u_x(r)| \leq c$ for every u, r and $x \in [0,1]$;
- paid families are closed under finite sums (costs add), finite products (costs multiply), scalar multiples (cost multiplied by the absolute value), and restriction by zero outside a set of parameters. Families of paid functions with summable costs may also be added, and their jets sum to the jet of the sum. Products of models obey the rule $X_v^2 = xX_v$: val_x turns the product $(p *_x q)(k) = \sum_{i \cup j = k} x^{|i \cap j|} p(i) q(j)$ into the pointwise product;
- a monomial $u \mapsto c(u)\, \mathbf{1}[i \subseteq r]\, x^{|i|}$ with $|c| \leq b$ is paid at cost b.

(d) If f has a model and $f_x(r) = f_x(r \setminus \{v\})$ for every r and $x \in [0,1]$, then $\mathrm{jet}(f)(i) = 0$ whenever $v \in i$. For a paid family on a measurable parameter space, if $u \mapsto f^u_x(r)$ is measurable for each fixed $x \in [0,1]$ and r, then $u \mapsto \mathrm{jet}(f^u)(i)$ is measurable for every i.

(e) For $\rho : \iota \to \mathcal{A}_R$ put $\mathrm{snd}_\rho(f) = \sum_{i \subseteq \iota} \mathrm{jet}(f)(i) \prod_{v \in i} \rho_v$.

- If (f^u) is paid at cost d and $\|\rho_v\|_w \leq 1$ for all v, then the coefficients of $\mathrm{snd}_\rho(f^u)$ are dominated, uniformly in u, by a nonnegative array of w-norm at most d.
- If $\rho_v(\emptyset) = 0$ for all v, the coefficient of $\mathrm{snd}_\rho(f)$ on a singleton $\{p\}$ is $\sum_v \mathrm{jet}(f)(\{v\})\, \rho_v(\{p\})$.

Status: theorem. For square-zero ρ_v, snd_ρ is multiplicative on functions with models. For paid families with summable costs, it commutes with sums; for paid f, it sends e^f to the exponential series of $\mathrm{snd}_\rho(f)$ (`mul_sounds`, `soundSum`, `soundExp`). *Lean:* `Switch.value`, `entire`, `jet`, `Models`, `Models.exact`, `Paid`, `Paid.tasting`, `Paid.add`, `Paid.sum`, `paidClick`, `comb`, `Paid.mul`, `Paid.prod`, `jet_erase`, `Paid.jet_meas`, `mapScores`, `sound`, `Accent.crystal`, `Accent.quality`.

3.3.1.4 D. Gaussian polynomial calculus Conventions.

- $\mathbb{C}[X] = \mathbb{C}[X_v : v \in V]$. The variable set V need not be finite; each polynomial uses finitely many variables.
- $\|p\|_1$ is the sum of the absolute values of the coefficients of p.

- For $b \in \mathbb{C}^V$, ∂_b is the derivation with $\partial_b X_v = b_v$.
- A *covariance* is a symmetric $Q : V \times V \to \mathbb{C}$; no positivity is assumed.
- $\pi(B, d) = (2B(d+1))^d$.

Proposition II.2.10 (algebraic Gaussian expectations). Let Q be a covariance. The operators $X_v + \partial_{Q_{v\cdot}}$ on $\mathbb{C}[X]$ commute; define $\mathcal{G}_Q p = p(X + \partial_Q)\,1$.

(a) $\mathcal{G}_Q$ satisfies:

- $\mathcal{G}_Q 1 = 1$ and $\mathcal{G}_Q(X_v p) = X_v\,\mathcal{G}_Q p + \partial_{Q_{v\cdot}}\mathcal{G}_Q p$;
- $\mathcal{G}_0 = \mathrm{id}$ and $\mathcal{G}_P\mathcal{G}_Q = \mathcal{G}_{P+Q}$;
- $\partial_b \mathcal{G}_Q = \mathcal{G}_Q \partial_b$.

(Equivalently $\mathcal{G}_Q = \exp(\frac{1}{2}\sum_{v,w} Q_{vw}\partial_v\partial_w)$, so $\mathcal{G}_{-M}$ is Wick ordering with respect to M. On each polynomial the derivative sum and exponential series have only finitely many nonzero terms.)

(b) For $m \in \mathbb{C}^V$ put $\mathbb{E}_{Q,m}(p) = (\mathcal{G}_Q p)(m)$. This is a linear functional with $\mathbb{E}_{Q,m}(1) = 1$, and it satisfies, for every finite set s and choice of variables z_j, the Wick recursion

$$\mathbb{E}_{Q,m}\Big(X_v \prod_{j\in s} X_{z_j}\Big) = m_v\,\mathbb{E}_{Q,m}\Big(\prod_{j\in s} X_{z_j}\Big) + \sum_{j\in s} Q_{vz_j}\,\mathbb{E}_{Q,m}\Big(\prod_{k\in s\setminus\{j\}} X_{z_k}\Big).$$

(c) *Change of variables.* For a matrix $A = (A_{vz})$ with finitely supported rows, define the linear substitution $p \mapsto p \circ A$ by $X_v \mapsto \sum_z A_{vz} X_z$. If Q is a covariance and m a mean on the new variables, then:

- $\mathcal{G}_Q(p \circ A) = (\mathcal{G}_{AQA^{\mathsf{T}}} p) \circ A$;
- $\mathbb{E}_{Q,m}(p \circ A) = \mathbb{E}_{AQA^{\mathsf{T}}, Am}(p)$.

(d) *Identification.* Let ξ be a random vector in $\mathbb{R}^V$ under a probability measure. Assume all polynomial moments of ξ are finite and $\mathbb{E}[\xi_v\, q(\xi)] = \mathbb{E}[(\partial_{Q_{v\cdot}} q)(\xi)]$ for every polynomial q (Gaussian integration by parts). Then $\mathbb{E}[q(m + \xi)] = \mathbb{E}_{Q,m}(q)$ for every $m \in \mathbb{R}^V$.

(e) *Bounds.* Let $B \geq 1$, $|m_v| \leq B$, $|Q_{vw}| \leq B$ and $\deg p \leq d$. Then

$$|\mathbb{E}_{Q,m}(p)| \leq \|p\|_1\,\pi(B, d).$$

If (Q', m') obeys the same bounds and each corresponding entry of the means and covariances differs by at most $E \geq 0$, then $|\mathbb{E}_{Q,m}(p) - \mathbb{E}_{Q',m'}(p)| \leq \|p\|_1\, E\,\pi(B, d)$. In the first bound, and in the second when $m = m'$, it suffices to assume the hypotheses for entries indexed by the variables of p.

Status: theorem. Used by III.11 and VII.6. *Lean:* `Normal.Symmetric`, `slit`, `cream_comm`, `milk`, `feeds`, `skims`, `froze`, `warms`, `blend`, `spring`, `prepared`, `wets`, `VMeasure`, `payment`, `Common.priced`, `Common.width`, `Common.differs`, `Tapes.bound_v`, `Tapes.differently`.

Proposition II.2.11 (estimates for Gaussian polynomial steps). All polynomial degrees below are total degrees.

(a) *Weighted coefficient norms.* For $w : V \to \mathbb{N}$ and $r \geq 0$, the quantity $\|p\|_{w,r} = \sum_\alpha r^{w\cdot\alpha}|c_\alpha|$ is submultiplicative, and $\|\cdot\|_1 = \|\cdot\|_{0,1}$. It is a norm when $r > 0$ and may be a seminorm at $r = 0$. For a linear substitution $X_v \mapsto e_v$ with $\|e_v\|_{w,r} \leq c$ and $c \geq 1$,

$$\|p \circ e\|_{w,r} \leq \|p\|_1\, c^{\deg p}.$$

(b) *Smoothing near the identity.* If $0 \le e \le 1$, $|Q_{vw}| \le e$ and $\deg p \le D$, then $\|\mathcal{G}_Q p - p\|_1 \le \|p\|_1 \, e\,(2(D+2))^D$ and $\deg \mathcal{G}_Q p \le D$.

(c) *One step.* Let $T_{H,Q} p = (\mathcal{G}_Q p) \circ H$, where $H = (H_v)$ are linear forms in new variables and Q is a covariance.

- If $C \ge 1$, $\|H_v\|_1 \le C$, $|Q_{vw}| \le C$ and $\deg p \le d$, then $\|T_{H,Q} p\|_1 \le \|p\|_1 (2(d+2)C)^d$ and $\deg T_{H,Q} p \le d$.
- For two such pairs, assume $\varepsilon \ge 0$, $\|H_v - H'_v\|_1 \le \varepsilon$ and $|Q_{vw} - Q'_{vw}| \le \varepsilon$. The difference bound is $\|p\|_1 \, \varepsilon \, (2(d+2)C)^d$ (`crossing`).
- If Q vanishes between the variables of P and those of R, then $T_{H,Q}(PR) = T_{H,Q}(P)\, T_{H,Q}(R)$.

(d) *Linearization of a rescaled substitution.* Let:

- f_v be linear forms and A_v polynomials with no terms of degree < 2, with $\|f_v\|_1, \|A_v\|_1 \le B$ and $B \ge 1$;
- $0 < t \le 1$, and Φ_t the algebra map $X_v \mapsto f_v + t^{-1} A_v(tX)$;
- K is a covariance, $0 \le \varepsilon \le 1$, $|K_{vw}| \le \varepsilon$ and $\deg P \le n$.

Then

$$\big\|\Phi_t(P) - (\mathcal{G}_K P) \circ f\big\|_1 \le \|P\|_1 (n\, t B (2B)^n + \varepsilon (2(n+2))^n B^n).$$

(e) *Guards and second moments.* Call g an **η-guard** for μ if g is measurable, $0 \le g \le 1$, and $\int (1-g)\, d\mu \le \eta$.

- For a finite measure μ, the product of an η-guard and an ε-guard is an $(\eta+\varepsilon)$-guard.
- For a probability measure, measurable f with integrable $|f|^2$ and $\int |f|^2 d\mu \le B$, and an η-guard g: $\big|\int g f\, d\mu - \int f\, d\mu\big| \le \varepsilon' B + \eta/\varepsilon'$ for every $\varepsilon' > 0$.
- Under II.2.10(d), if $\deg p \le d$, $B \ge 1$ and $|m_v|, |Q_{vw}| \le B$ on the variables of p, then $|p(m+\xi)|^2$ is integrable and $\int |p(m+\xi)|^2 \le \|p\|_1^2 \, \pi(B, 2d)$.

(f) *Coefficients from values.*

- For finite V, $d \in \mathbb{N}$ and $\varepsilon > 0$, there are a finite nonempty set $T \subseteq [-\varepsilon, \varepsilon]^V$ and $C \ge 0$ such that $\|p\|_1 \le C \max_{x \in T} |p(x)|$ for all p of degree $\le d$.
- A polynomial is determined by its values on a real cube of positive side length.
- If $\deg p \le D$, $B \ge 1$, $v, w \in \mathbb{R}^V$, $\delta \ge 0$, $|v_i|, |w_i| \le B$ and $|v_i - w_i| \le \delta$, then $|p(v) - p(w)| \le \|p\|_1 \, D B^D \delta$.

Status: theorem. Used by III.11 and VII.6. *Lean:* `Rise.cents`, `Rise.funds`, `Rise.rise`, `LiftMark.bounds`, `Normal.pipe`, `Tapes.Valve`, `Valve.Fair`, `crest`, `crux`, `bridge`, `LiftMark.garnish`, `LiftMark.stock`, `Guarded`, `guards_mul`, `escaped`, `squareFits`, `Spindle.molds`, `Spindle.listen`, `Spindle.amove`.

3.3.1.5 E. Numbers

Proposition II.2.12 (scale functions). For $s > e$ put

$$y = \log\log s, \quad \Pi = e^{y^2}, \quad \theta = s^{-1/2}\Pi, \quad h = e^{y^{3/2}}, \quad j = h^{1/5}, \quad \omega = e^{-\frac{7}{10} j}, \quad M = \lceil (\log s)^2 \rceil, \quad W = M^2,$$

all as functions of s. Here θ is the original small-field threshold. Then:

1. For every $b \in \mathbb{R}$ and $p > 0$, eventually $\Pi(s)^b \le s^p$.
2. Eventually $s\theta^2 = \Pi^2$, $s\theta^3 \le 1$, $s\theta^4 \le 1$, and $2\Pi^v \theta^2 \le 1$ for every $v \le 1$.
3. For all $a > 0$, $k \in \mathbb{N}$ and $c \in \mathbb{R}$, eventually:

- $W(s)^k \le \Pi(s)^a$ and $h(s) \le \Pi(s)^a$;
- $c \log W(s) \le a\, y(s)^2$;
- $y(s)^2 \le a\, W(s/2)$ and $y(s)^2 \le a\, j(s/2)$;
- $j(s) \le h(s/2)$ and $y(s) \le y(s/2) + 1$.

4. For every $c \in \mathbb{R}$ and $a < b$, $c\,\Pi^a \le \Pi^b$ eventually.

Here $f \ll g$ means $f/g \to 0$ as $s \to \infty$. For every fixed $a, p > 0$, $y^2 \ll j \ll h \ll \Pi^a \ll s^p$; also $W(s) \sim (\log s)^4$.

Status: theorem. Item 2 is used in III.10, III.11 and V.2 (the per-site vacuum-energy bound `treasury_stage`); item 3 is used in III.11. *Lean:* `Numbers.monitor`, `Hat`, `P`, `theta`, `cost`, `uniform_loop`, `Perch.bill`, `walls`, `hot`, `join`, `w`, `logspan`, `rate_wall`, `rate_inverse`, `rate_fire`, `fire_stretch`, `Spindle.hotter`, `Paint.q_power`, `Glimmer.transports`.

Proposition II.2.13 (numerical conditions for the rank comparison). Use the scales of II.2.12, and write $\Pi = \Pi(s)$, $y = y(s)$ in formulas involving both s and u.

(a) *Shelf conditions.* Fix $m, Q \in \mathbb{N}$, $e, U, V \ge 0$ and $a > 0$. Put $A_Q = (2Q+1)^4$ and $G(u) = e^{-a h(u)^2}$. For all sufficiently large s, the following ten inequalities hold simultaneously for every $u \in [s/2, s]$:

$$\begin{aligned}
8 &\le W(u),\\
W(u)^2 + 2QW(u) &\le M(s)^{10},\\
64(mU+1)y^2 &\le W(u),\\
64(mU + 4\cdot 2^m)y^2 &\le W(u),\\
2^m\Pi^{-25/32} + \Pi^{-1} + \Pi^{-2} &\le \Pi^{-3/4},\\
e^{mUy^2 + 4\cdot 2^m y^2 + 2j(u)} G(u) + 2^m\Pi^{-3} &\le \Pi^{-1},\\
(2W(u)+1)^4\Pi^{-1} &\le 1,\\
e^{eA_Q}\Pi^{1/8}\Pi^{-31/32} &\le \Pi^{-25/32},\\
e^{(e+1)A_Q}\Pi^{1/8}\Pi^{1/4} &\le \Pi,\\
e^{-2j(u)}(e^{\Pi^{-1}} - 1) &\le \Pi^{-3/4}\omega(u).
\end{aligned}$$

The fifth line uses $e^{mVy^2}\Pi^{-mV-2} = \Pi^{-2}$. The seventh is the geometric hypothesis of II.2.6 with cover size $W(u)$ and hit bound Π^{-1}.

(b) *Tally conditions.* For the same data, fix $\alpha \ge 1$ and $L > 1$, and put

$$n(L,s) = \left\lceil \frac{10 \log M(s)}{\log L} \right\rceil.$$

For all sufficiently large s,

$$(\alpha+1)^{n(L,s)}\Pi(s)^{-3/4} \le \Pi(s)^{-1/2},$$

and (a) holds for every $u \in [s/2, s]$. Moreover,

$$G(u) \le \Pi(s)^{-3/4}\omega(v) \qquad (u, v \in [s/2, s]).$$

(c) *Gaussian prices.* For $k, n \in \mathbb{N}$, set $\phi = 1/(4(nk+k+1))$. For every fixed $A \geq 1$, eventually in s,

$$n\Pi^{-2}(A+\Pi^{-2})^n \pi(\Pi^\phi, nk) + A^n \Pi^{-2} \pi(\Pi^\phi, nk) \leq \Pi^{-3/2}.$$

Thus the error parameters $\xi = \Pi^{-2}$, covariance error Π^{-2}, mean and covariance bound Π^ϕ, and reference norm bound A satisfy the polynomial comparison condition with error $\Pi^{-3/2}$.

(d) *Iteration parameters.* A parameter record consists of an era (Q, e, U, V, a, α) as above, integers $t, g, H_0 \geq 0$, and real numbers b, c, f, F, K satisfying

$$c \geq 1, \quad K \geq 1, \quad F \geq 0, \quad f + c \leq -8, \quad g \geq 6, \quad U = V, \quad f = \tfrac{1}{16} - U, \quad 2V + 2 \leq g - 1.$$

Here t is the Taylor order, g the target order, H_0 a support parameter, b an exponent, c a coefficient exponent, and F an error exponent. Write $C_d = (2(d+2)K)^d$. At scale s and integer height H, the iteration conditions are

$$(\alpha+2)^H \leq \Pi, \qquad C_t^H \leq \Pi, \qquad C_{2t}^H \leq \Pi, \qquad \Pi^{2c+1} + \Pi^{2c+2} \leq \Pi^{2c+3}, \qquad H \leq \Pi.$$

They imply, for every integer $0 \leq r \leq H$,

$$(\alpha+2)^r \Pi^{-g} \leq \Pi^{1-g} \leq \Pi^{-4}.$$

For $b \geq 0$, the block-size conditions

$$L > 1, \qquad \log L \geq 2(b+2)\log q \quad (q \in \{\alpha+2, C_t, C_{2t}\})$$

hold for all sufficiently large L. For any such fixed L, the iteration conditions hold for all sufficiently large s at height

$$H(L,s) = \left\lceil \frac{(b+2)y(s)^2}{\log L} \right\rceil$$

(racetrack).

(e) *Moment and fusion errors.* Define $R_Q(z) = e^{zA_Q}$ and the following polynomial error function. For $B, \varepsilon, w \geq 0$ and $D \in \mathbb{N}$, put

$$\begin{aligned} q_0 &= w\pi(B,D), & d_0 &= wDB^D\varepsilon, \\ q_1 &= d_0 + \varepsilon w^2 \pi(B,2D) + \varepsilon + w\varepsilon\pi(B,D), \\ q_2 &= 2q_0 d_0 + \varepsilon w^4 \pi(B,4D) + 2\varepsilon + w^2\varepsilon\pi(B,2D), \\ \mathcal{Q}(B,\varepsilon,w,D) &= q_1 + q_2 + (q_0+q_1)q_1 + q_1 q_0. \end{aligned}$$

For a parameter record in (d), set $\tau(s) = 3h(s)$. The remaining numerical conditions are

$$\begin{aligned} 1 \leq \tau(s) \leq \Pi - 1, \quad \pi(\tau(s), 2t) &\leq \Pi, \\ K\pi(\tau(s),1) \leq \Pi, \qquad K + \Pi^{-F} &\leq \Pi, \\ 4R_Q(e+2)\Pi^{-6} &\leq \Pi^{-4}, \\ 4R_Q(e+2)\mathcal{Q}(\Pi, \Pi^{-F}, \Pi^{2c+3}, 2t) &\leq \Pi^{-g}. \end{aligned}$$

For every $u \in [s/2, s]$ they also require

$$\begin{aligned} \tau(u) &\leq \tau(s), \\ G(u) + 2\Pi^{-2}R_Q(e+3) &\leq \Pi^{-1}, \\ (2W(u)+1)^4\Pi^{-1} \leq 1, \qquad e^{\Pi^{-1}} - 1 &\leq 1. \end{aligned}$$

In the second line, $G(u) = e^{ah(u)^2} G(u)^2$, as in the source definition. All these conditions hold for sufficiently large s whenever

$$g + 10Z + 10 \le F, \qquad Z = 2c + 8t + 13$$

(`funding`).

Status: theorems and numerical definitions. The constants entering the block-size inequalities in (d) are fixed before L. The lower bound on s may depend on all the fixed parameters and on L. *Lean:* `Perch.Demand`, `Shelf`, `mint`, `Era`, `Tally`, `on_time`, `Glimmer.fuel`, `collected`, `shed`, `Tapes.Purse`, `Purse.budget`, `Racer`, `Racer.pos`, `Faring`, `torque`, `Digits`, `pulse_ok`, `Panel`, `rentQ`, `load`, `capZ`, `defect`, `grant₁`, `grant₂`, `grant`, `Paint.trib`, `Station.more`, `Station.bump`.

3.3.2 2. What it uses

- **III.1, activity format.**
  - `Activities` consists of regular addresses (a site belonging to the support), forced addresses (support, inventory, extra test) and coefficient functions.
  - Eligibility of a forced address: support ∩ large-field set = inventory, and the extra test holds.
  - `Valid` requires regularity and vanishing conditions for the regular coefficients, gauge invariance, measurability, final-field dependence and eligibility for the forced coefficients, affine covariance, and the prescribed regular and forced bounds. The K-bound v requires nonnegative envelopes $b_i \ge \sup_U |K_i(U)|$ whose masses have a summable hit bound v at each site. The gas sums products of forced coefficients over finite selections with disjoint supports and inventories whose union is exactly the large-field set.
  - `SumBudget` (nonnegative, summable, sum ≤ bound).
  - `BudgetPoint` (order, power, support exponent, cover size B, threshold).
  - The large-field set `flags`, `Healthy`, `UsesFinal`, `Dominated`.
  - II.2.8 uses this format; II.2.6 uses its notion of a nonnegative summable bound.
- **I.1, lattice.** Sites, supports, separation, `halo`, cover numbers, and the cardinality and intersecting-union bounds quoted at the head of §1. The digest condition also uses gauge invariance and lattice affine maps.
- **Standard mathematics.**
  - Polymer gases and the Mayer expansion; the Kotecký–Preiss criterion and its inductive proof (R. Kotecký, D. Preiss, *Comm. Math. Phys.* 103 (1986); R. L. Dobrushin, *AMS Transl.* 177 (1996)); D. Brydges, *A short course on cluster expansions* (Les Houches 1984); Glimm–Jaffe, *Quantum Physics*, cluster expansions.
  - Cover-number (connected block-count) weights for polymers in the style of Bałaban's renormalization-group papers.
  - Exponential and logarithm on the nilpotent ideal of a commutative algebra. Connected parts as logarithms of generating functions; Möbius inversion on the Boolean lattice (G.-C. Rota 1964) for the triangular inversion in II.2.9.
  - Gaussian integration by parts, Wick recursion and Wick ordering (Glimm–Jaffe; B. Simon, *The $P(\varphi)_2$ Euclidean (Quantum) Field Theory*, Ch. I). Gaussian convolution $e^{\frac{1}{2}\Delta_C}$ on polynomials in the renormalization group (D. Brydges, Park City lectures 2009; Brydges–Slade, *J. Stat. Phys.* 159 (2015)).
  - Equivalence of norms on finite-dimensional spaces (II.2.11(f)).
- **Imported bounds and assumptions.**

- The tree bound needs $(2M+1)^4\gamma \le 1$ and the exponent $a+1$ in the hit condition. These are hypotheses of each application. In the rank comparison they come from II.2.13(a), (e) with $M = W(u)$ and $\gamma = \Pi(s)^{-1}$; in the fused gas, from the fusion bounds with $M = B$ and exponent $e_P + 1$.
- Fusion requires the digest and hit bounds in II.2.8. Their startup hypotheses are stated in III.9; their verification in every subsequent application belongs to the corresponding RG step.
- Given one compatible set of stage parameters for each index, a sequence of numerical records can be chosen with common exponent $b \ge d+6$, common Taylor order $t \ge d$, common f, and coefficient exponents $c_i \ge bd + d + 2$. This is the parameter selection used in III.4 (`preSheet`).

3.3.3 3. Strategy

Products of local activities are represented in a finite commutative algebra of square-zero markers. The allowed marker sets determine which products vanish. The exponential converts sums of nonconstant monomials into products, and a triangular correction constructs the unique connected polynomial whose exponential is a given polynomial with constant term 1. The weighted ℓ^1 norm controls these operations without a loss depending on the number of markers. Switching connects scalar functions of activity parameters to this algebra: for positive switch size x, coefficients are recovered from values at $0/x$ configurations. A convergent model supplies their limit as $x \downarrow 0$, while a uniform majorant controls that limit. The relation $X_v^2 = xX_v$ then becomes the square-zero relation.

Connected-component decompositions reduce polymer sums to sums over packs attached to a support. A Kotecký–Preiss induction bounds the latter by $e^{\mathrm{cov}_M(Y)}$. Attaching X_i costs $e^{\mathrm{cov}_M(X_i)}$, paid by the extra unit in the hit exponent $a+1$. The geometric input is $|Y| \le (2M+1)^4 \mathrm{cov}_M(Y)$. For two interactions, mentions always connect, while walls connect only hot labels. Hotness is unchanged on restriction to a component, so the component decomposition can be recovered uniquely. The forced-activity fusion uses ordinary patch components; its density identity and bounds are proved in III.10.

The Gaussian comparison in III.11 uses algebraic convolution on polynomials with complex symmetric covariances. Wick recursion gives coefficient-norm and moment bounds depending on the degree and entrywise bounds, independently of the number of variables. Gaussian integration by parts identifies this algebraic functional with an actual expectation when a measure is supplied. The numerical inequalities then absorb the errors and iteration losses. There are two iteration lengths: $n(L,s) = O(y(s)/\log L)$ and $H(L,s) = O(y(s)^2/\log L)$ at fixed L. The first gives losses $\Pi(s)^{o(1)}$; the second requires the explicit lower bounds on L in II.2.13(d). The constants in those block-size inequalities are chosen before L; all remaining constants must be fixed before the lower bound on s is chosen.

3.3.4 4. Steps

3.3.4.1 4.1 Marker algebra Lemma 4.1 (associativity). $\mathcal{A}_R$ is a commutative unital ring. *Sketch:*

1. On basis elements, $(\delta_a\delta_b)\delta_c \ne 0$ iff a, b, c are pairwise disjoint and $a \cup b \cup c \in R$.

2. This condition is symmetric in a, b, c. Downward closure ensures that both intermediate unions are allowed whenever the total union is allowed.
3. Extend by bilinearity.

Lean: `Console.lean`.

Lemma 4.2 (norm). $\|ab\|_w \le \|a\|_w \|b\|_w$. *Sketch:* $\|c\delta_x \cdot c'\delta_y\|_w \le w^x|c|\, w^y|c'|$, with equality when the product is nonzero, because $w^{x\cup y} = w^x w^y$ on disjoint unions. Expand a, b in the basis and use the triangle inequality. *Lean:* `binary`, `multiplication`.

Lemma 4.3 (nilpotency; norm of Exp**).** If $a(\emptyset) = 0$, then a^k vanishes on sets with fewer than k elements. For arbitrary a, $\|\mathrm{Exp}(a)\|_w \le \sum_{k\le|V|} \|a\|_w^k/k! \le e^{\|a\|_w}$. *Sketch:* in $a^{k+1} = a^k \cdot a$, each factor a contributes a nonempty disjoint marker set (induction on k). The norm bound follows from Lemma 4.2. Summable families are handled by comparing with the sup norm (equivalent norms) for convergence, and by passing the triangle inequality to the limit for the bound. *Lean:* `high`, `pass_budget`, `budgeted`.

Lemma 4.4 (additivity of Exp**).** II.2.3(a)–(b). *Sketch:*

1. Apply the binomial theorem to $(a+b)^n/n!$.
2. Terms $a^i b^j$ with $i + j > |V|$ vanish by Lemma 4.3, so the truncation at $|V|$ is consistent on both sides.
3. For (b): $(c\delta_x)^2 = 0$ when $x \neq \emptyset$, and induction over s.

A second proof, via the exponential of commuting nilpotent elements in Mathlib, is in `Trajectories.pass_add`. *Lean:* `Exponential.pass_add`, `Trajectories.pass_sum`.

Lemma 4.5 (logarithm by iteration). II.2.3(c)–(d). *Sketch:* say u vanishes below order m if $u(x) = 0$ for $|x| < m$.

1. **Key estimate.** If $c(\emptyset) = 0$ and u vanishes below order $m \ge 1$, then $\mathrm{Exp}(c+u) - \mathrm{Exp}(c) - u = (\mathrm{Exp}(c) - 1)\, u + \mathrm{Exp}(c)\,(\mathrm{Exp}(u) - 1 - u)$ vanishes below order $m + 1$.
2. **Existence.** With $u = F - \mathrm{Exp}(C_k)$, the key estimate gives that $F - \mathrm{Exp}(C_{k+1})$ vanishes below order $k + 2$. After $|V|$ steps it vanishes identically.
3. **Uniqueness.** For two solutions C, D, additivity gives $\mathrm{Exp}(C - D) = 1$. If $\mathrm{Exp}(p) = 1$ and $p(\emptyset) = 0$, then $\mathrm{Exp}(p)$ agrees with $1 + p$ one order beyond the vanishing order of p. So p vanishes to every order, i.e. $p = 0$.
4. **(d)** follows from $\mathrm{Exp}(-C)\,\mathrm{Exp}(C) = 1$.

Lean: `roll`, `rolls`, `roll_cont`, `Expands.uniq`, `correct`.

Lemma 4.6 (majorant arrays; first-order coefficients).

- For nonnegative arrays A, B on R define the product $(A * B)(z) = \sum_{x\sqcup y=z} A(x)B(y)$. Then $|a| \le A$ and $|b| \le B$ coefficientwise imply $|ab| \le A * B$, and $\sum_z w^z (A * B)(z) \le (\sum w^x A(x))(\sum w^y B(y))$. These are the uniform-in-configuration majorants used in III.8.
- Every p decomposes as $p(\emptyset)1 + \sum_v p(\{v\})\delta_{\{v\}} +$ (terms of order ≥ 2).
- $\mathrm{Exp}(F)$ and F have the same singleton coefficients when $F(\emptyset) = 0$.
- For a finite family, singleton coefficients satisfy

$$(\prod_{i\in s} a_i)(\{v\}) = \sum_{i\in s} a_i(\{v\}) \prod_{j\in s\setminus\{i\}} a_j(\emptyset).$$

Sketch: apply the triangle inequality to each disjoint-union coefficient sum. The weights factor on each surviving product. Splitting coefficients by marker-set size gives the decomposition. A singleton can receive a nonempty marker set from only one factor; powers of order at least two therefore make no singleton contribution to the exponential.

Lean: `Nil.combine`, `merged_price`, `Accent.ballots`, `whisper`, `shockProd`.

Lemma 4.7 (repacking). II.2.4(b). *Sketch:*

1. Sum val fibrewise over the slot set $S = \sigma(s)$.
2. For admissible S, the selected sets in the fibre are in bijection with sections $S \ni k \mapsto i_k \in \sigma^{-1}(k)$.
3. The sum over sections of $\prod_k F_{i_k}$ is the limit over finite truncations of $\prod_{k\in S}\sum_{i\in\sigma^{-1}(k)} F_i$.
4. The majorant $\prod_{i\in s}\|F_i\|_w$ is summable over finite sets s because $\sum_i \|F_i\|_w < \infty$. II.2.2 then gives summability of the algebra-valued terms.

Lean: `Packets.repack`.

3.3.4.2 4.2 Components and the tree bound **Lemma 4.8 (components).** II.2.5. *Sketch:* "chained in s" is the reflexive–transitive closure of "supports meet" restricted to s. Its classes are the parts. A chain cannot leave a member of a family with disjoint supports, which gives item 2. For item 3, follow a chain from j to i and stop immediately before the first visit to i. *Lean:* `separates`, `excise`.

Lemma 4.9 (root decomposition). For finite $D \subseteq \iota$ and a support Y, let $\mathrm{film}(D, Y) = \sum\prod_{i\in s} f_i$, the sum over packs $s \subseteq D$ for Y (including $s = \emptyset$). For $f \geq 0$,

$$\mathrm{film}(D, Y) \leq \prod_{i\in D,\ X_i\cap Y\neq\emptyset} (1 + f_i\,\mathrm{film}(D\setminus\{i\}, X_i)).$$

Sketch:

1. Each part of a pack contains a label meeting Y (II.2.5, item 4). Fix one such label for each part, its *root*; distinct parts have distinct roots.
2. Removing the root from its part leaves a pack for X_{root} inside $D\setminus\{\mathrm{root}\}$ (II.2.5, item 3).
3. The map $s \mapsto$ (for each candidate root: nothing, or the remainder of the part it roots) is injective, and the weight factorizes.
4. The right-hand side sums over all such assignments with nonnegative terms.

No sum over cuts or orderings occurs, and no label is repeated inside a part. *Lean:* `Ac.pouch`, `film`, `lookup`.

Lemma 4.10 (cover number of a spark). If s is a spark, then $\mathrm{cov}_M(X(s)) \leq \sum_{i\in s}\mathrm{cov}_M(X_i)$. *Sketch:* enumerate s so that each new label meets the union of the previous ones (possible since s is connected), and use the intersecting-union bound from I.1 at each step. Subadditivity is not claimed for disjoint unions. *Lean:* `Ac.cover`.

Lemma 4.11 (Kotecký–Preiss induction). Under the hypotheses of II.2.6, put $p_i = f_i e^{a\,\mathrm{cov}_M(X_i)}$. Then for every finite D and every support Y,

$$\mathrm{film}_p(D, Y) \leq e^{\mathrm{cov}_M(Y)}.$$

Sketch: strong induction on D. By Lemma 4.9 and the induction hypothesis for $D \setminus \{i\}$,

$$\mathrm{film}_p(D, Y) \le \prod_{i \in D,\ X_i \cap Y \neq \emptyset} (1 + f_i e^{(a+1)\mathrm{cov}_M(X_i)}) \le \exp\Big(\sum_{i \in D,\ X_i \cap Y \neq \emptyset} f_i e^{(a+1)\mathrm{cov}_M(X_i)} \Big) \le e^{|Y|\gamma}.$$

The last step sums the hit condition over the sites of Y. Finally $|Y|\gamma \le (2M+1)^4 \gamma \mathrm{cov}_M(Y) \le \mathrm{cov}_M(Y)$. *Lean:* `demand`, `Tree.lean` (`step`, `all_pack`).

Lemma 4.12 (tree bound). II.2.6. *Sketch:*

1. Take a finite family of sparks whose supports contain x, and let A be the union of their labels.
2. Each such spark is a nonempty pack for $\{x\}$ inside A.
3. By Lemma 4.10 and $a \ge 0$, $e^{a\,\mathrm{cov}_M(X(s))} \prod_{i \in s} f_i \le \prod_{i \in s} p_i$.
4. The bound $e^{|Y|\gamma}$ in the proof of Lemma 4.11 with $Y = \{x\}$ (so $|Y| = 1$) gives $\mathrm{film}_p(A, \{x\}) \le e^{\gamma}$.
5. Subtract the empty pack. Uniform bounds on finite partial sums give summability.

Lean: `Ac.forest`.

3.3.4.3 4.3 Two kinds of overlap Lemma 4.13 (hotness is local).

- Let p be a slab of s and $i \in p$. Then i is hot in p iff it is hot in s, and its fuel in p equals its fuel in s.
- In a mole, containing a dirty label is equivalent to all labels being hot.
- Hotness spreads along fuel chains.

Sketch: a mention chain is a fuel chain, so it stays inside the slab. A wall link joins only hot labels. Moles are also invariant under label permutations compatible with bijections of mentions and walls that preserve dirtiness (`Kilner.molecule`). *Lean:* `hot_comp`, `fuel_comp`, `takes_mole`.

Lemma 4.14 (recovery). II.2.7, items 1–3. *Sketch:* let $s = \bigcup S$.

1. A mention chain in s cannot leave a member of S, since mention-unions are disjoint. So hotness in s equals hotness in the member, and fuels agree.
2. The fuel-unions of distinct members are disjoint: the mention components by hypothesis; the wall components because only members containing a dirty label have hot labels, and their wall-unions are disjoint.
3. Apply II.2.5, item 2 to the fuel overlap.
4. For slabs of a given set, use Lemma 4.13 to identify their internal fuels. Disjoint fuel-unions imply disjoint mention-unions; when both slabs contain a dirty label, all their labels are hot, so their wall-unions are disjoint as well.

Lean: `Kiln.undo`, `clubs_away`, `walls_away`.

Lemma 4.15 (moles are wall-sparks). II.2.7, item 4. *Sketch:* a fuel overlap is either a mention overlap, which implies a wall overlap by hypothesis, or a wall overlap between hot labels. *Lean:* `Kiln.attends`, `Collide.spark_ac`.

Lemma 4.16 (pellets). Pellets are countably infinite. *Sketch:* every singleton {left i} is a pellet, and finite subsets of $\mathbb{N} \sqcup \mathbb{N}$ form a countable set. *Lean:* `Pipes.encoding`.

3.3.4.4 4.4 Switching Lemma 4.17 (triangular inversion). For $x > 0$, val_x is injective. *Sketch:* strong induction on r: $\mathrm{val}_x(g)(r) = x^{|r|}g(r) + \sum_{i \subsetneq r} x^{|i|}g(i)$ and $x^{|r|} \neq 0$. This is Möbius inversion on the Boolean lattice, scaled by $x^{|i|}$. Hence val_x is a linear bijection of a finite-dimensional space. *Lean:* `entire`, `Digits.lean` (`vane`, `decodes`).

Lemma 4.18 (jets; products).

- If g models f, then $\mathrm{val}_x^{-1}(f_x) = g_x$ for $0 < x \leq 1$, so the jet is g_0.
- If g, g' model f, f', then $x \mapsto g_x *_x g'_x$ models ff', since $\mathrm{val}_x(p *_x q) = \mathrm{val}_x(p)\,\mathrm{val}_x(q)$.
- Bounds: $|p *_x q| \leq |p| *_1 |q|$ for $x \in [0,1]$, and $\sum(b *_1 b') = \sum b \cdot \sum b'$.

Lean: `Models.exact`, `Paid.mul`; the model product `value_comb`, `Models.mul`.

Lemma 4.19 (absent markers; measurability). II.2.9(d). *Sketch:* $\mathrm{val}_x(g)(r) - \mathrm{val}_x(g)(r \setminus \{v\}) = \sum_{v \in i \subseteq r} x^{|i|}g(i)$. If f does not see v, this vanishes for all r, so $g_x(i) = 0$ for $v \in i$ by induction, for every $0 < x \leq 1$. Let $x \downarrow 0$. Measurability: the jet is the pointwise limit of $\mathrm{val}_x^{-1}(f_x)$, and val_x^{-1} is a continuous linear map. *Lean:* `jet_erase`, `Paid.jet_meas`.

Lemma 4.20 (transfer bound). II.2.9(e). *Sketch:*

1. The coefficient of $\mathrm{snd}_\rho(f^u)$ at z is bounded by $\sum_i b(i)\,|(\prod_{v \in i} \rho_v)(z)|$, using $|g_0(i)| \leq b(i)$.
2. The w-norm of this majorant array is at most $\sum_i b(i)\,\|\prod_{v \in i} \rho_v\|_w \leq \sum_i b(i) \leq d$, by II.2.2.

Lean: `Accent.crystal`.

3.3.4.5 4.5 Gaussian calculus Lemma 4.21 (construction of $\mathcal{G}_Q$). II.2.10(a)–(b). *Sketch:*

1. $[X_v + \partial_{Q_v}, X_w + \partial_{Q_w}] = Q_{vw} - Q_{wv} = 0$ by symmetry, and derivations commute.
2. The commutative subalgebra they generate receives an algebra map from $\mathbb{C}[X]$; apply it to 1.
3. The semigroup law and commutation with ∂_b follow by induction on polynomials from the recursion. Applying evaluation at m gives the moment functional and its Wick recursion. The terminating series $e^{\Delta_Q/2}$ satisfies the same polynomial recursion, giving the equivalent heat-operator formula.

Lean: `cream_comm`, `make`, `feeds`, `warms`, `spring`.

Lemma 4.22 (size of moments). II.2.10(e). *Sketch:*

1. By linearity it suffices to treat monomials, each written as $\prod_{j \in s} X_{z_j}$ with $|s| \leq d$. The coefficient sum gives the factor $\|p\|_1$.
2. Let c_d bound the moments of degree $\leq d$. The Wick recursion gives $c_{d+1} \leq B\,c_d + d\,B\,c_d$, and $c_d \leq (2B(d+1))^d$ by induction.
3. For differences, apply $|xv - yw| \leq |x||v - w| + |x - y||w|$ to each term of the recursion.
4. Localization: restrict Q and m to the variables of p by a change of variables.

Lean: `Common.lean`, `width`, `differs`, `Guard.lean` (`pinned`).

Lemma 4.23 (change of variables; factorization). II.2.10(c) and II.2.11(c), last clause. *Sketch:* both sides of II.2.10(c) satisfy the same recursion in X_v. One uses $\partial_m(p \circ A) = (\partial_{Am}p) \circ A$ and the fact that $\mathcal{G}_Q$ applied to (A-linear form) $\cdot\, q$ obeys the Wick rule with covariance AQA^T. If Q vanishes between the variables of P and R, then $\partial_{Q_v \cdot}\mathcal{G}_Q R = 0$ for every variable v of P. The Leibniz rule then gives $\mathcal{G}_Q(PR) = (\mathcal{G}_Q P)(\mathcal{G}_Q R)$; linear substitution preserves this identity. Related results:

- if Q is block diagonal, Gaussian expectations of products over blocks factorize (`bridge`; `Particle.tide` for marginals);
- applying Wick ordering separately at each vertex subtracts its chosen covariance from the diagonal block (`lodge`), and two steps compose as $T_{P,R}T_{H,Q} = T_{HP,\,HRH^{T}+Q}$ (`piped`).

Lean: `prepared`, `wets`.

Lemma 4.24 (identification with a true expectation). II.2.10(d). *Sketch:*

1. Substituting $X \mapsto X + m$ turns integration by parts into the recursion $\mathbb{E}[(m+\xi)_v q(m+\xi)] = m_v \mathbb{E}[q(m+\xi)] + \mathbb{E}[(\partial_{Q_v} q)(m+\xi)]$.
2. Both $q \mapsto \mathbb{E}[q(m+\xi)]$ and $\mathbb{E}_{Q,m}$ are linear, equal 1 on 1, and satisfy the Wick recursion on monomials.
3. Induction on the degree.

Lean: `payment`.

Lemma 4.25 (step bounds). II.2.11(a)–(d). *Sketch:*

- **(a)**: expand in monomials. Submultiplicativity bounds the image of a degree-k monomial by $c^k \le c^{\deg p}$.
- **(b)**: induction on the degree, using $\mathcal{G}_Q(X_v q) - X_v q = X_v(\mathcal{G}_Q q - q) + \sum_j Q_{vj}\,\mathcal{G}_Q(\partial_j q)$.
- **(c)**: the same recursion with X_v replaced by the linear form H_v; each step multiplies the bound by at most $C(1+d) \le 2(d+2)C$. For the difference bound, subtract the two recursions and split each product difference into the change in one factor at a time.
- **(d)**: write $\Phi_t(P) - (\mathcal{G}_K P) \circ f = [\Phi_t(P) - P \circ f] + [(P - \mathcal{G}_K P) \circ f]$.
  - First term: $\|t^{-1} A_v(tX)\|_1 \le tB$, since A_v has no terms of degree < 2, and $\|U(p) - T(p)\|_1 \le \|p\|_1\, d\, B' A^d$ for algebra maps that are B'-close on generators (bounded by A).
  - Second term: (b) followed by (a).

Lean: `Rise.rise`, `LiftMark.puff`, `LiftMark.bounds`, `crux`, `LiftMark.message`, `LiftMark.stock`.

Lemma 4.26 (guards; second moment). II.2.11(e). *Sketch:*

- $1 - gz \le (1-g) + (1-z)$ for $g, z \in [0,1]$.
- $|gf - f| = (1-g)|f| \le \varepsilon' |f|^2 + (1-g)/\varepsilon'$, using $0 \le 1 - g \le 1$; integrate.
- Second moment: $|p(m+\xi)|^2 = (p\bar{p})(m+\xi)$, where $\bar{p}$ conjugates the coefficients. Apply II.2.10(d)–(e) to $p\bar{p}$, which has degree at most $2d$ and $\|p\bar{p}\|_1 \le \|p\|_1^2$.

Lean: `guards_mul`, `escaped`, `squareFits`.

Lemma 4.27 (reading coefficients). II.2.11(f). *Sketch:*

1. Evaluation at all points of the cube is injective on polynomials of degree $\le d$, since a polynomial vanishing on a product of infinite sets is zero.
2. In finite dimension, a finite subfamily of evaluations is already injective: minimize the kernel dimension.
3. Take a left inverse and bound its ℓ^1 operator norm.

The Lipschitz bound follows by telescoping over monomials. A version with explicit dimension dependence and constant $(|V|+1)^d T_{d,\varepsilon}$ is proved in III.11 (`ball_count`). For polynomials of degree at most n, an $o(t^n)$ difference at 0 forces equality; for C^n functions it forces equality of their Taylor polynomials through order n (`Spindle.decipher`, `trill`). *Lean:* `listen`, `molds`, `amove`.

3.3.4.6 4.6 Numbers Lemma 4.28 (loop bounds). II.2.12, items 1–2. *Sketch:*

1. $(\log\log s)^2 = o(\log s)$, so $b\,y^2 \le p \log s$ eventually; this is item 1.
2. Then $s\theta^2 = \Pi^2$ exactly.
3. $s\theta^3 = \Pi^3 s^{-1/2} \le 1$ by item 1 with $b = 3$, $p = \frac{1}{2}$.
4. Item 1 with $b = 1$, $p = 1/2$ gives $\theta \le 1$ eventually, hence $s\theta^4 \le s\theta^3$.
5. $s \cdot 2\Pi^v \theta^2 \le 2\Pi^3 \le 2\sqrt{s} \le s$ for $s \ge 4$.

Lean: `cost`, `uniform_loop`.

Lemma 4.29 (comparison of scales). II.2.12, items 3–4. *Sketch:*

- $\log W(s) \le 6y(s)$ eventually, so fixed powers of W are bounded by Π^a. The ceiling inequalities $(\log s)^2 \le M(s) < (\log s)^2 + 1$ also give $W(s) \sim (\log s)^4$. Also $\log h = y^{3/2} = o(y^2)$.
- If $a < b$, then $c\Pi^{a-b} \to 0$, proving item 4.
- $y(s) \le y(s/2) + 1$, since $\log s = \log(s/2) + \log 2$.
- $y^2 e^{-y} \to 0$ handles $y(s)^2 \le a\,W(s/2)$ and $y(s)^2 \le a\,j(s/2)$; for the latter use $j(s/2) = e^{y(s/2)^{3/2}/5} \ge e^{y(s/2)}$ once $y(s/2) \ge 25$.
- Eventually $y(s) \le 2y(s/2)$; then $2^{3/2} \le 5$ gives $j(s) \le h(s/2)$.

Lean: `rate_wall`, `rate_inverse`, `rate_fire`, `fire_stretch`.

Lemma 4.30 (feasibility of the shelf and tally conditions). II.2.13(a)–(b). *Sketch:*

1. Use $y(s)^2 \ll W(s/2)$ for the two linear lower bounds on $W(u)$. Since $W(u) \le W(s) = M(s)^2$, the support-size inequality follows once $M(s)$ is large compared with Q. Fixed powers of W are $o(\Pi)$, giving the geometric condition.
2. The purely exponential terms reduce to fixed constants times powers of Π with strictly smaller exponents than the target; for example, $1/8 - 31/32 = -27/32 < -25/32$. For the sixth shelf inequality, uniformly in $u \in [s/2, s]$,

$$(mU + 4 \cdot 2^m + 2)y(s)^2 + 2j(s) \le ah(u)^2$$

   eventually. Hence its first term is at most Π^{-2}. The reserve inequality follows from $j(s/2) \gg y(s)^2$ and $e^{\Pi^{-1}} - 1 \le \exp(1)$.
3. The ceiling formula gives $n(L,s)\log(\alpha+1) = O(y(s)) \le y(s)^2/4$ eventually. The same uniform tail estimate, with coefficients $3/4$ and $7/10$, gives $G(u) \le \Pi^{-3/4}\omega(v)$.
4. Uniformity uses $W(s/2) \le W(u) \le W(s)$, $h(u) \ge h(s/2)$ and $j(u) \le j(s)$. Intersect the finitely many eventual conditions.

Lean: `mint`, `force`, `on_time`.

Lemma 4.31 (numerical parameter bounds). II.2.13(c)–(e). *Sketch:*

- The block-size conditions are finitely many lower bounds on $\log L$. At height $H(L,s)$, each logarithmic loss is at most $y(s)^2$ for large s.
- For $r \le H$, $(\alpha+2)^r \le (\alpha+2)^H \le \Pi$; combine this with $g \ge 6$.
- For (c), $\pi(\Pi^\phi, nk) \le (2(nk+1))^{nk}\Pi^{1/4}$ and $\Pi^{-2} \le 1$. The left side is therefore at most a fixed constant times $\Pi^{-7/4}$, which is eventually at most $\Pi^{-3/2}$.
- For (e), fixed-degree moments of $\tau = 3h$ are bounded by arbitrarily small positive powers of Π. With $Z = 2c + 8t + 13$, the error function in its sixth inequality is $O(\Pi^{-F+6Z})$. The condition $g + 10Z + 10 \le F$ makes this at most Π^{-g} after multiplication by the fixed factor.

The remaining tail bound uses $G(u) \le \Pi^{-2}$ uniformly on $[s/2, s]$, and the geometric bound follows from $W(s)^4 \ll \Pi$.

Lean: `pulse_ok`, `Racer.pos`, `collected`, `shed`.

3.3.5 5. Where the difficulty lies

The tree estimate must preserve the connected cover weight. Cover numbers are subadditive along intersecting unions, whereas a disconnected union may require an additional connecting region. The induction therefore uses connected components and spends exactly one extra cover exponent on each attachment. It counts finite sets of labels; it does not estimate Mayer coefficients with arbitrary repeated labels.

For two interactions, the graph itself depends on the selected set, through hotness. Unique recovery requires both locality of hotness and the asymmetric separation conditions: all components separate through mentions, while only components containing a dirty label separate through walls.

Switching requires convergence as well as a bound. Bounded coefficients alone need not have a limit as $x \downarrow 0$. The model supplies convergence, and the common majorant allows sums and coefficient estimates to pass to the limit. The Gaussian bounds allow complex covariances without positivity; their constants grow like d^d up to exponential factors at fixed entry bounds. The degree is fixed before L and s.

The numerical conditions establish feasibility for fixed parameters. They do not supply the analytic estimates to which those numbers are applied. The two iteration lengths in §3 must remain distinct.

The source contains duplicate exponential and component constructions. Its second trajectory logarithm and `Candle.small` are absent from the main theorem's dependency set. The latter bounds a restarted threshold by $3h(\widehat{\beta})\beta^{-1/2}$.

3.3.6 6. Notation

Written proof	Lean	Meaning
R, census	`Marker.Census`	finite downward-closed family of allowed marker sets
$\mathcal{A}_R$	`Marker.Poly`	marker algebra
$c\,\delta_x$	`Poly.send`	monomial with marker set x
w^x, $\|a\|_w$	`toll`, `price`	weight of a marker set; weighted ℓ^1 norm
Exp	`Poly.pass`	exponential truncated at order $\|V\|$
connected polynomial Log F	`Exponential.connected`, `Expands`	C with $C(\emptyset) = 0$, $\mathrm{Exp}(C) = F$
majorant array	`Nil.Sheet`	nonnegative coefficient bound
$\mathrm{cov}_M(X)$	`coverNumber`	connected cover number (I.1)
spark, Parts(s)	`Ac.Spark`, `Ac.Parts`	connected label set; components
pack for Y, film(D, Y)	`Ac.Pack`, `Ac.film`	label set attached to Y; its weighted count

Written proof	Lean	Meaning
hit condition	`Ac.Hit`	site-wise summability of $f e^{a\,\mathrm{cov}}$
kiln, hot, fuel, mole, slab	`Kiln`, `Hot`, `fuel`, `Mole`, `slabs`	two-interaction polymer data and components
pipes, tap, pellet, reach	`Ovens.Pipes`, `Tap`, `contains`, `reach`	data and pieces of the fused gas
val_x, model, jet, paid	`Switch.value`, `Models`, `jet`, `Paid`	switching objects
snd_ρ	`Switch.sound`	jets inserted into the marker algebra
$\|p\|_1$, $\|p\|_{w,r}$	`Rise.coin`, `Rise.cents`	coefficient norm; weighted coefficient seminorm
$\mathcal{G}_Q$, $\mathbb{E}_{Q,m}$	`Normal.milk`, `Normal.blend`	Gaussian smoothing; algebraic expectation
∂_b	`Normal.slit`	derivation $X_v \mapsto b_v$
$p \circ A$	`Normal.exchange`	linear substitution
$T_{H,Q}$	`Normal.pipe`, `Valve.read`	one Gaussian polynomial step
$\pi(B,d)$	`Common.priced`	$(2B(d+1))^d$
η-guard	`Guarded`	$[0,1]$-valued cutoff of small defect
$\Pi(s)$, $\Pi(s)^a$	`Numbers.P`, `Perch.bill`	$e^{(\log\log s)^2}$ and its powers
$\theta(s)$	`Numbers.theta`	$s^{-1/2}\Pi(s)$, small-field threshold
$h,\ j,\ \omega,\ W$	`hot`, `join`, `w`, `walls`	$e^{y^{3/2}}$, $h^{1/5}$, $e^{-0.7j}$, $\lceil(\log s)^2\rceil^2$
parameter record $(t, g, H_0, b, c, f, F, K)$	`Tapes.Purse`	`taylor`, `goal`, `horizon`, `band`, `credit`, `f`, `foil`, `L`
iteration conditions at height H	`Racer`	five inequalities in II.2.13(d)
moment and fusion conditions	`Faring`	inequalities in II.2.13(e)

3.3.7 7. What remains unexplained

1. **Restarted thresholds.** The original threshold is $s^{-1/2}\Pi(s)$; the restarted Gaussian comparison uses $s^{-1/2}h(s)$ and cover size $W(s)$ (`Perch.Candle`, III.11). Their ratio $h(s)/\Pi(s)$ tends to zero. The numerical estimates here do not establish the existence of restarted representations satisfying the required density and trajectory identities. That construction must be explained in III.11.
2. **Analytic inputs to the numerical conditions.** The shelf, tally and moment conditions are explicit in II.2.13. Their use still requires the one-step estimates with the stated supports, covariances and constants. The verification of those analytic hypotheses along the actual RG trajectories, including every later application of the digest and fusion bounds, is outside this unit.
3. **Auxiliary declarations.** The additional regrouping results in `EventBale` and the cost estimates in `MergeProof/West`, `North` and `East` are not formulated above, apart from the cited permutation result. Their full role in III.11 remains to be traced.

3.4 III.1. The block-spin step and the symmetric gauge fixing

Part: III, Ultraviolet construction (renormalization group). **Lean:** YM/RG/ (Block, Patch, Measurable, Push, Masks, Activities, PathActions, Bounded, History), YM/Branch/, YM/Pressure/, YM/PREC/, YM/PCalc/, YM/Tree/, YM/Exterior/, and the small packages YM/RG17/, YM/Push/CORRECTED_COLUMNS/, YM/Planned/CALIBRATED_MIXED_HISTORY/, YM/UVNext/ANNULUS_COMPARISON/, YM/FullVolume/Train/, YM/SectorTools/ACTUAL_SCALAR_LEDGER/. **Size:** 4,219 lines (693 declarations) in the main-theorem closure.

Conventions. $N \geq 1$ and $G \subset U(N)$ is a closed subgroup with probability Haar measure dg. Its Lie algebra $\mathfrak{g} \subset M_N(\mathbb{C})$ carries the real inner product $\operatorname{Re}\operatorname{tr}(X^\dagger Y)$, and $\Pi_{\mathfrak{g}}$ denotes the orthogonal projection onto $\mathfrak{g}$. Matrix norms below are Frobenius norms. In path actions, Lie-algebra fields, Hessians and the six works, the physical inner product is $N^{-1}\operatorname{Re}\operatorname{tr}(X^\dagger Y)$; write $|X| = \|X\|/\sqrt{N}$ for its norm. This is the coordinate normalization of Gauge.inner_frob_color. The curvature field is $\operatorname{curv}(u) = \Pi_{\mathfrak{g}}(u - u^{-1})/2$, with $|\operatorname{curv}(u)| \leq \operatorname{ch}(u)$. The *chord* is $\operatorname{ch}(u) = \|u-1\|/\sqrt{N}$ and the *price* is $\operatorname{pr}(u) = 1 - \operatorname{Re}\operatorname{tr} u/N$. They satisfy $\operatorname{ch}^2 = 2\operatorname{pr}$, both are class functions, $\operatorname{ch}(u^{-1}) = \operatorname{ch}(u) \leq 2$, $\operatorname{ch}(uv) \leq \operatorname{ch}(u) + \operatorname{ch}(v)$, and $\|u - v\|^2 = 2N\operatorname{pr}(uv^{-1})$. The Wilson energy is $\sum_p \operatorname{pr}(U_p)$.

A step goes from the fine torus $\Lambda_f = (\mathbb{Z}/f)^4$ to the coarse torus $\Lambda_c = (\mathbb{Z}/c)^4$, with $c \geq 1$, $f = cL$ and $L \geq 3$ odd. The coarse site x has root Lx and block $Lx + B_{(L-1)/2}$, where $B_r = \{-r, \dots, r\}^4 \subset \mathbb{Z}^4$. Configurations are $U \in G^{\text{links}}$, and μ_f is the product of the probability Haar measures. The gauge action is $(k \cdot U)_{(y,i)} = k_y U_{(y,i)} k_{y+e_i}^{-1}$. $T_U(y; w)$ is the parallel transport along a walk w. $T_U(y \to z; o)$ is the transport along the coordinate route that changes the coordinates in the order given by $o \in S_4$. For a coarse site x, $U^{(x)}$ is the periodic lift of U to $\mathbb{Z}^4$, translated so that Lx sits at the origin. All patch constructions take place on $\mathbb{Z}^4$, so a patch may be larger than the torus. For finite $X \subset \mathbb{Z}^4$, $E_X(U)$ is the sum of the chords of the links with both ends in X, and $F_X(U)$ is the sum of the chords of the plaquettes with all four corners in X; F_X is gauge invariant. $\mathfrak{B}_4$ is the group of signed permutations of the axes (384 elements). For $s \in \mathfrak{B}_4$ and a point b, $s_b U$ is the pullback of U under $z \mapsto b + s(z)$; a link reversed by s carries the inverse variable. Further, $\kappa_{\text{slope}} = 11C_A/(24\pi^2)$, where C_A is the adjoint Casimir, and $\operatorname{temp}(x) = x^{-1/2} e^{(\log\log x)^2}$ for $x > 1$.

Definition (the block kernel). A *patch* is $P = (L, J)$ with $L \geq 3$ odd and $J \geq 4$. Its radius is $R = (J+3)L$ and its points are those of B_R, with centre 0. The *kernel constants* are $K = (\kappa, \delta, \varepsilon, a)$, all positive: penalty, frame cutoff, predictor cutoff, axial factor. For a configuration W on $\mathbb{Z}^4$ the following objects are defined.

1. *Frame tags and axial frames.* A frame tag is $t = (r, o) \in B_R \times S_4$. The axial frame of t is $a_y^t = T_W(r \to y; o)$.
2. *Minimizing branch.* $W \in M_\delta$ if $\operatorname{ch}(a_y^{(0,o)}\, T_W(y \to z; o')\, (a_z^{(0,o)})^{-1}) < \delta$ for all $o, o' \in S_4$ and $y, z \in B_R$. In words: in each of the 24 axial gauges from the centre, every route transport in the patch is δ-close to 1.
3. *Score.* For $h : B_R \to G$, $S_W(h) = \operatorname{avg}_{y,z \in B_R} \operatorname{avg}_{o \in S_4} \operatorname{pr}(h_y T_W(y \to z; o) h_z^{-1})$.
4. *Eligible tags.* $\mathcal{E}(W) = \{t : E_{B_R}(a^t \cdot W) \leq a(2R+1) F_{B_R}(W)\}$.
5. *Frame and tag mass.* If $W \in M_\delta$, the frame h^W is a minimizer of S_W subject to $h_0 = 1$, and the tag mass is uniform, $1/(24|B_R|)$. Otherwise $h_y^{W,t} = (a_0^t)^{-1} a_y^t$ (the axial gauge from r, normalized at the centre), with mass $\mathbf{1}_{\mathcal{E}(W)}(t)/|\mathcal{E}(W)|$.
6. *Weight.* w is the law on B_R of $\xi_1 + \cdots + \xi_J$, where the ξ_k are i.i.d. uniform on $B_{(L-1)/2}$. It is the J-fold convolution of the uniform weight on a centred block; it is $\mathfrak{B}_4$-invariant and

supported in $B_{J(L-1)/2}$.

7. *Connector.* For a coarse link $b = (x, i)$ and frame tags $(t_x)_{x\in\Lambda_c}$, set $Y_y = h_y^{(x)}\, T_{U^{(x)}}(y;\ L \text{ steps along } e_i)\, (h_y^{(x+e_i)})^{-1}$ for $y \in B_R$. Here $h^{(x)}$ is the frame of $U^{(x)}$ with tag t_x. This is the fine transport from $Lx + y$ to $L(x + e_i) + y$, written in the frames of the two patches.
8. *Prediction.* Let $\Phi_Y(u) = \sum_y w_y \operatorname{pr}(Y_y u^{-1})$. If $\min \Phi_Y < \varepsilon$, then Q_b is a chosen minimizer, equivalently a nearest point of G to the matrix mean $\sum_y w_y Y_y$ (an extrinsic, chordal mean); the bond tag then has uniform mass $1/|B_R|$. Otherwise $Q_b = Y_{r_b}$, with the bond tag r_b drawn with probability w_{r_b}.
9. *Kernel.* The tag law is $\lambda(U, t) = \prod_x m^{\mathrm{fr}}(U^{(x)}, t_x) \prod_b m^{\mathrm{pr}}(Y_b, r_b)$, and

$$\rho_\beta(U, V) = \sum_t \lambda(U, t) \prod_b Z_\beta^{-1} e^{-\beta\kappa \operatorname{pr}(V_b Q_b(U,t)^{-1})}, \qquad Z_\beta = \int_G e^{-\beta\kappa \operatorname{pr}(g)}\, dg.$$

The *branch conditions* for (P, K, G) are:

- (B1) for $W \in M_\delta$ the constrained minimizer in item 5 is unique;
- (B2) whenever $\min \Phi_Y < \varepsilon$, the minimizer of Φ_Y is unique;
- (B3) $\mathcal{E}(W) \neq \emptyset$ for every W.

Compactness gives the chosen minimizers; (B1)–(B2) make the choices unique on the minimizing branches. Lean: `Patch`, `KernelChoices`, `usesMin`, `score`, `eligible`, `frameMass`, `frame`, `weight`, `bondCost`, `predict`, `predictorMass`, `Branches`, `StepData`, `connector`, `prediction`, `likelihood`, `density`.

3.4.1 1. What this unit proves

The block-spin kernel is normalized, Borel and covariant under gauge and lattice symmetries. Its branch cutoffs can be chosen independently of the patch size. The geometric estimates synchronize neighbouring frames and bound the predicted coarse curvature by L^{-2} times the fine curvature in a surrounding box. Near the trivial configuration the prediction is analytic, with linearization $\mathcal{M} + d_0\mathcal{F}$. Tree gauge reduces the step integral to the free links. The unit also defines effective densities, RG histories and the requirements used in the later energy and stationary-branch estimates.

Proposition III.1.1 (the block kernel is a Markov kernel). Assume (B1)–(B3). For every $\beta \in \mathbb{R}$:

- $\rho_\beta \geq 0$;
- $(U, V) \mapsto \rho_\beta(U, V)$ is Borel;
- $\sup \rho_\beta < \infty$;
- $\int \rho_\beta(U, V)\, d\mu_c(V) = 1$ for every U.

Consequently, for every Banach-space-valued $F \in L^1(\mu_f)$ the push $(\mathcal{P}_\beta F)(V) = \int \rho_\beta(U, V) F(U)\, d\mu_f(U)$ is defined for every V. Moreover $\mathcal{P}_\beta F \in L^1(\mu_c)$, $\mathcal{P}_\beta$ is linear, and $\int \mathcal{P}_\beta F d\mu_c = \int F d\mu_f$. This is the operation that takes level $m + 1$ to level m. *Status:* theorem. Positivity and integrability in V hold without (B1)–(B3). *Lean:* `IsKernel`, `isKernel`, `push`, `push_integrable`, `push_integral`, `push_add`, `push_smul`.

Proposition III.1.2 (exact covariance). Assume (B1)–(B3). For all $\beta \in \mathbb{R}$, U, V:

- (a) $\rho_\beta(k\cdot U,(k\circ\mathrm{embed})\cdot V)=\rho_\beta(U,V)$ for every $k:\Lambda_f\to G$, where $\mathrm{embed}(x)=Lx$;
- (b) $\rho_\beta(s_{Lb}U,s_bV)=\rho_\beta(U,V)$ for every $s\in\mathfrak{B}_4$ and $b\in\Lambda_c$;
- (c) if the kernel constants of a calendar satisfy (B1)–(B3) for every patch, then every step of the calendar has *block control* (branch conditions together with (a) and (b)), and is a Markov kernel by III.1.1.

Status: theorem. *Lean:* `covary`, `channel`, `Calendar.blocks_go`, `Calendar.BlockControl`, `Levels.transition_mass`.

Proposition III.1.3 (branch constants depending only on G). Let $d=d(G)$ be the constant of III.1.5. Let $r_G>0$ be such that every $X\in M_N(\mathbb{C})$ at distance $<r_G$ from G has a unique nearest point in G. Put $\delta_0=d/\sqrt{N}$ and $\varepsilon_0=r_G^2/(2N)$. Then for every $\kappa>0$, $\delta\in(0,\delta_0]$, $\varepsilon\in(0,\varepsilon_0]$ and $a\ge 16$, the constants $(\kappa,\delta,\varepsilon,a)$ satisfy (B1)–(B3) for every patch, that is, for every odd $L\ge 3$ and every $J\ge 4$. Condition (B3) holds in a stronger form: for every W and every ordering o there is a root r with (r,o) eligible. *Status:* theorem. The existence theorem records the case $(\delta_0,\varepsilon_0,16)$ for a given penalty κ. *Lean:* `Axial.viable`, `RG.frame_unique`, `RG.predictor_unique`, `Axial.eligible`.

Proposition III.1.4 (axial gauge controlled by curvature). Let l be a list of distinct axes, $r\in\mathbb{N}$, $o\in B_r$, and U any configuration on $\mathbb{Z}^4$. Let $S=\{y\in B_r:\ y_j=o_j \text{ for } j\notin l\}$ be the slab. Then there is a root $x\in S$ such that the axial gauge $g_y=T_U(x\to y;\text{ axes in the order }l)$ satisfies

$$E_S(g\cdot U)\ \le\ 4\,|l|\,(2r+1)\,F_S(U).$$

Status: theorem. *Lean:* `Axial.existence`.

Proposition III.1.5 (least-squares synchronization: uniqueness up to a global left multiple). There is $d=d(G)\in(0,1/4]$ with the following property. Let n,Λ be finite nonempty sets. Let $W:n\times n\times\Lambda\to G$ satisfy $W_{xxk}=1$ and $W_{yx\sigma(k)}=W_{xyk}^{-1}$ for some permutation σ of Λ, and $\|W_{xyk}-1\|\le d$ for all indices. Then any two global minimizers h,h' of $\mathrm{cost}(h)=\mathrm{avg}_{x,y,k}\,\mathrm{pr}(h_xW_{xyk}h_y^{-1})$ over G^n satisfy $h'=c\,h$ for a single $c\in G$. The constant d does not depend on $|n|$ or $|\Lambda|$. *Status:* theorem. *Lean:* `Branch.align`, with `Pairs`.

Proposition III.1.6 (synchronization of the realized frame with any gauge). There is $A=A(G)>0$ with the following property. Let P be any patch and K any kernel constants; no branch condition is needed. Let W be a configuration on $\mathbb{Z}^4$, t a frame tag with $m^{\mathrm{fr}}(W,t)>0$, h the realized frame, and $H:\mathbb{Z}^4\to G$ any gauge. Then there is $c\in G$ with

$$\sum_{y\in B_R}\mathrm{ch}(c^{-1}h_yH_y^{-1})\ \le\ A\,(2R+1)\Big(E_{B_R}(H\cdot W)+a\,(2R+1)\,F_{B_R}(W)\Big).$$

Status: theorem. *Lean:* `Pressure.synchronized`.

Proposition III.1.7 (one gauge for a neighbourhood of blocks). Fix G, K and $j\ge 4$, and put $\ell_j=2j+15$. Then $D=16\,\ell_j$ and $C=A\,\ell_j(D+a\,\ell_j)$, with A from III.1.6, have the following property. Let S be any step with kernel constants K and $J=j$ (any L, any c). Let U be fine, x a coarse site, and t a tag that is *usable* at x, meaning that $m^{\mathrm{fr}}(U^{(x+y)},t_{x+y})>0$ for all $y\in B_2$. Then there are a gauge $H:\mathbb{Z}^4\to G$ and pads $p_y\in G$ $(y\in B_2)$ with

$$E_{B_{R'}}(H \cdot U^{(x)}) \le D\,L\,\mathrm{port}(x,U), \qquad \sum_{i \in B_R} \mathrm{ch}(p_y^{-1} h_i^{(x+y)}\, H_{i+Ly}^{-1}) \le C\,L^2\,\mathrm{port}(x,U) \quad (y \in B_2).$$

Here $R' = R + 4L = (J+7)L$ and $\mathrm{port}(x,U) = F_{B_{R'}}(U^{(x)})$ is the fine curvature in the macro-box around x. Every tag with $\lambda(U,t) > 0$ is usable at every coarse site. *Status:* theorem. *Lean:* `Pressure.planting`, `usable`, `wraps`, `port`, `macroR`.

Proposition III.1.8 (predicted versus actual coarse curvature). For G, K, j let $C_8 = 8\max(1,\varepsilon^{-1})(2C+D)$, with C, D from III.1.7. Let S be any step with constants K and $J = j$. Then for all U and V, every coarse site x, every tag t usable at x and every plane p:

$$\mathrm{ch}(Q(U,t)_{(x,p)}) \le \frac{C_8}{L^2}\,\mathrm{port}(x,U), \qquad \mathrm{ch}(V_{(x,p)}) \le \mathrm{fuel}(Q,V)_{(x,p)} + \frac{C_8}{L^2}\,\mathrm{port}(x,U).$$

Here $Q_{(x,p)}$ denotes the plaquette holonomy, and $\mathrm{fuel}(Q,V)_{(x,p)} = \sum_{e \in \partial(x,p)} \mathrm{ch}(V_e Q_e^{-1})$ is the sum of the four *wires* of the plaquette, counted with multiplicity even when edges coincide on a short torus. The macro-box contains $6(2R')^2(2R'+1)^2$ internal plaquettes. Thus L^{-4}port differs from their average chord by a factor bounded above and below in terms of J only. The bound therefore says that the coarse predicted plaquette is at most $C(G,K,J)\,L^2$ times the average fine curvature: the area scaling of the non-abelian Stokes formula, with no further loss in L. *Status:* theorem. *Lean:* `Pressure.qdc`, `fuel`, `wire`.

Proposition III.1.9 (effective densities and RG histories). The following definitions carry four elementary consequences.

(a) *Budget points, flags, masks.* A budget point is a tuple (integer power ≥ 2, integer order ≥ 6, support exponent > 0, cover size $B \ge 1$, threshold $t \in (0,1]$). The flagged sites $\mathrm{flags}_t(U)$ are the sites touched by a plaquette of chord $> t$. U is *healthy* on X if $X \cap \mathrm{flags}_t(U) = \emptyset$, and the masked function is $\mathbf{1}_{\{\text{healthy on } X\}}F$. A function is *regular* on X if it depends only on links in X, is Borel and gauge invariant, and at each configuration healthy on X is C^q in exponential left increments, where q is the order; its ordered left derivatives through order q must exist. Its *sup bounds* control derivatives of orders $0, \dots, q$ in directions with $|A_e| \le t(1 + \mathrm{dist}(e^-, o))^{\mathrm{power}}$, where e^- is the base of the positive link. *Theorem:* a function that reads the links in X and the flags in X depends only on the links in $\mathrm{inspected}(X)$, the union of X and the corners of all plaquettes meeting X.

(b) *Activities.* There are regular activities E_i, with an address (a site o_i in a support X_i), and forced activities K_i, with a support Y_i, an inventory $\emptyset \ne I_i \subset Y_i$ and an extra test. *Validity* at bounds $\delta_E, w_K \ge 0$ means the following:

- E_i is regular on X_i and $E_i(1) = 0$;
- if $10\,\mathrm{dist}(y, o_i) < n$ for every $y \in X_i$, where n is the torus side, the Hessian of E_i at the trivial configuration vanishes on affine directions $A_{(y,\mu)} = a_\mu + \sum_\nu \widetilde{(y - o_i)}_\nu d_{\mu\nu}$, using the canonical centred lift $\widetilde{(y - o_i)}$; equivalently, $D^2[E_i(\exp A)]_{A=0}[A,B] = 0$ for every two such directions A, B;
- K_i and its extra test are gauge invariant, Borel, and depend only on links and flags in Y_i; K_i vanishes unless $Y_i \cap \mathrm{flags}_t = I_i$ and the test holds;

- covariance under every element of $\mathfrak{B}_4$ and every base point (index permutations transporting addresses, inventories, tests and values);
- there are nonnegative majorants e_i, k_i, with e_i bounding the regular derivatives and $|K_i| \le k_i$, such that, for every site x,

$$\sum_{i:o_i=x} e^{\sigma \operatorname{cov}_B(X_i)} e_i \le \delta_E, \qquad \sum_{i:x\in Y_i} e^{\sigma \operatorname{cov}_B(Y_i)} k_i \le w_K.$$

Here σ is the support exponent and $\operatorname{cov}_B(X)$ is the least number of radius-B balls with connected union covering X; both series are required to converge.

The *gas* is $\sum_b \prod_{i\in b} K_i(U)$, summed over finite families b with pairwise disjoint supports whose inventories cover $\operatorname{flags}_t(U)$ exactly. *Theorem:* the gas equals 1 when $\operatorname{flags}_t(U) = \emptyset$.

(c) *Leading action.* A clip has $c_{\mathrm{dir}} > 16\, c_{\mathrm{lin}} > 0$ and a relaxation length. Path data consist of row series $\ell_p(U) = \sum_k r_{p,k} \operatorname{slot}_k(U)$ (weighted parallel-transported curvatures) and loop, cubic and quartic atoms. Their bounds require exponentially weighted summability, invariance, the normalization $c_{\mathrm{dir}}\delta_{pq} + \sum_k \hat{s}_k(p)\hat{s}_k(q) = \delta_{pq}$ of the zero-momentum symbols (the sums of row coefficients at each plaquette orientation), and vanishing zeroth and first moments of the loop coefficients. The row term at a plaquette based at x is kept exactly when $\operatorname{ch}(U_q) \le t \exp(\operatorname{dist}(x, q^-)/r_{\mathrm{rel}})$ for every plaquette q, where $r_{\mathrm{rel}} > 0$ is the relaxation length and q^- its base. The per-plaquette term is $\operatorname{clip}_t(\operatorname{ch} U_p) + \mathbf{1}_{\mathrm{kept}} \frac{1}{2} |\ell_p(U)|^2$, where $\operatorname{clip}_t(s) = c_{\mathrm{dir}} s^2/2$ for $s \le t$ and $c_{\mathrm{lin}} t s$ for $s > t$. The atoms are masked by health on their path and plaquette footprints. The leading action $\mathcal{S}$ is the $\mathfrak{B}_4$-average of $\beta[\sum_p(\cdot) + \text{cubic} + \text{quartic}] + \text{loop}$. A representation $\mathcal{R} = (\text{path data, clip, activities}, p, \beta, v)$ evaluates to $\operatorname{eval}_{\mathcal{R}} = \exp\left(v|\Lambda| - \mathcal{S} - \sum_i \mathbf{1}_{\mathrm{healthy}} E_i\right) \cdot \text{gas}$. *Theorem:* if $\mathcal{R}$ is lawful (path bounds and valid activities), then $\operatorname{eval}_{\mathcal{R}}$ is Borel and bounded on that finite torus.

(d) *Histories.*

- A *calendar* holds torus sides $n_{m,p}$ and factors L_m, with $n_{m+1,p} = L_m n_{m,p}$ and $\operatorname{scale}(m) = \sum_{k<m} \log L_k$, the kernel constants, the barrier $\bar{\beta} > 0$, stage constants, orders $J_m \ge 4$ and exponents θ. It sets $\hat{\beta}_m = \bar{\beta} + \kappa_{\mathrm{slope}} \operatorname{scale}(m)$, $t_m = \operatorname{temp}(\hat{\beta}_m)$ and cover size $\lceil (\log \hat{\beta}_m)^2 \rceil$.
- Step m goes from level $m+1$ to level m, with patch (L_m, J_{m+1}).
- The *fits* conditions include: J_m nondecreasing with $J_{m+1} \le J_m + 1$ and $J_m \to \infty$; $\hat{\beta}_m > 1$; $t_m \le 1$; the level-m torus side at least $100(\text{cover size}_m + 1)$ and the fine side of step m at least $100(R_m + 1)$; drift and logarithmic-cost bounds.
- *Levels* means a calendar with its fits conditions and block control at every step.
- A history assigns a representation $\mathcal{R}_{N,p,m}$ to each mesh index N, volume p and level $m \le N$. It is *selected* if: the path data do not depend on p; each $\mathcal{R}_{N,p,m}$ has the assigned level-m clip and budget point and is lawful with the level-m constants and $\beta_{N,p,m} \in [\hat{\beta}_m/2, 2\hat{\beta}_m]$; $v_{N,p,N} = 0$ and $\operatorname{eval}_{N,p,N}$ is the bare Wilson density; $\operatorname{eval}_{N,p,m} = \mathcal{P}_{\beta_{N,p,m+1}} \operatorname{eval}_{N,p,m+1}$ with $|\beta_{m+1} - \beta_m - \kappa_{\mathrm{slope}} \log L_m| \le \operatorname{drift}_{m+1}$; $\beta_{N,p,0} = \bar{\beta}$; and $|\beta_{N,p,m} - \hat{\beta}_m| = o(\operatorname{scale}(m))$ uniformly in N, p.

The lawful bounds at level m use the stage path width and row and higher-atom bound, the loop bound $2\operatorname{monitor}_m^{\theta(J_m)}$, and activity bounds $\delta_E = \hat{\beta}_m^{-11/10}$ and $w_K = \exp[-\operatorname{monitor}_m^{1/5}]$, where $\operatorname{monitor}_m = \exp[(\log\log \hat{\beta}_m)^2]$.

The remaining fits conditions are: $\log L_m/\mathrm{scale}(m) \to 0$; L_m is the factor assigned to order J_{m+1}, and the assigned factors are nondecreasing; the derivative orders, powers and support exponents are nondecreasing along the levels; θ is antitone with $0 < \theta(J_m) \le J_m^{-1}$; and

$$\frac{\mathrm{drift}_{m+1}}{\log L_m} \le J_{m+1}^{-2}, \qquad \frac{\log(1 + C_{m+1} + \sigma_{m+1}^{-1} + C_{m+1}^{\mathrm{col}} + r_{m+1} + s_{m+1})}{\log L_m} \le J_{m+1}^{-2}.$$

Here $C, \sigma, C^{\mathrm{col}}, r, s$ are the stage path bound, path width, column bound, relaxation length and support exponent.

Theorem: for a selected history, $\int \mathrm{eval}_{N,p,m}\, d\mu = Z_{N,p}(\beta_{N,p,N})$, the Wilson partition function, for every $m \le N$.

Status: definitions. The four theorems are proved. The quantitative path and activity bounds needed to construct such histories belong to III.3–III.10. *Lean:* `BudgetPoint`, `BudgetPoint.Regular`, `flags`, `UsesFinal.allReads`, `Activities`, `RegularFamily`, `ForcedFamily`, `Moves`, `Valid`, `admissible`, `gas`, `gas_one`, `Clip`, `PathData`, `PathData.Bounds`, `baseTerm`, `principal`, `inChannels`, `Rep`, `Rep.eval`, `Rep.envelope`, `Calendar`, `Calendar.Fits`, `Levels`, `Levels.budget`, `History`, `History.Selected`, `History.partition_eq`.

Proposition III.1.10 (the tube identity). A *tube* for a step is a tower of transfer kernels together with path data and clips at the fine level (up) and at the coarse level (down). Fix a path-length cutoff, a fine threshold t and a coarse threshold s. From V one builds:

- the *vision* w (the thinned coarse curvature and row data);
- linear maps $\Phi, \Phi', \mathcal{N}$, assembled from the adjoint action of parallel transports of V along cut-off routes;
- the fine data $X_f(U)$, the coarse data $X_n(V)$, and the noise field $Z_b = \sqrt{\kappa}\,\mathrm{curv}(V_b Q_b^{-1})$.

Use counting inner products and norms on these finite arrays. Let C_f and C_c be the sums of the fine and coarse per-plaquette crowns, and let $P = \sum_b \frac{\kappa}{2}\,\mathrm{ch}(V_b Q_b^{-1})^2$. A crown is the clipped direct term plus half the kept row's squared physical norm, with the row truncated at the path cutoff. The totals of the six local works are

$$\begin{aligned} W_F &= C_f + \tfrac{1}{2}\|\Phi w\|^2 - \langle \Phi w, X_f\rangle, & W_I &= P + \tfrac{1}{2}\|\Phi' w\|^2 - \langle \Phi' w, Z\rangle, \\ W_C &= \langle w, X_n\rangle - \tfrac{1}{2}\|w\|^2 - C_c, & W_L &= \langle w, \Phi^\dagger X_f + \Phi'^\dagger Z - X_n\rangle, \\ W_N &= \tfrac{1}{2}\|\mathcal{N} w\|^2, & W_D &= \tfrac{1}{2}\langle w, (1 - \Phi^\dagger\Phi - \Phi'^\dagger\Phi' - \mathcal{N}^\dagger\mathcal{N}) w\rangle. \end{aligned}$$

For every path cutoff, thresholds $t, s \in \mathbb{R}$ and configurations U, Q, V,

$$W_F + W_I + W_C + W_N + W_L + W_D = C_f + P - C_c.$$

The fine and coarse data have two components per plaquette: $\sqrt{c_{\mathrm{dir}}}\,\mathrm{curv}(U_p)$ and the truncated row, using the respective clip and configuration. The fine transfer uses parallel transports of the blockwise copy of V; the other two maps use V on the coarse torus. The vision caps the direct component by $\min(1, s/\mathrm{ch}(V_p))$, taking this factor to be 1 at zero chord, and masks the row by the kept condition. For the local works, assign each square or inner-product summand to its array coordinate. The two components at each plaquette are summed together: W_F is indexed by

fine plaquettes, W_I by coarse links, W_C, W_L, W_D by coarse plaquettes, and W_N by coarse gain coordinates. The crowns and penalties are charged at the same indices.

Status: theorem (an algebraic identity). *Lean:* `Tube`, `Tube.locates`, `workF`, `workD`.

Proposition III.1.11 (energy per seed: the requirement, and its transfer to a history).

The requirement. Write $\widehat{\beta}$ for the coarse reference coupling, $t = \mathrm{temp}(\widehat{\beta} + \kappa_{\mathrm{slope}} \log L)$ for the fine-level threshold and $\varrho = \lceil (\log \widehat{\beta})^2 \rceil$ for the cluster radius.

- A *store* A is a finite set of fine plaquettes (costs), coarse links (aisles) and coarse sites (flags). Its size $|A|$ is the sum of the three cardinalities. Its locations are the parent coarse sites of all corners of its fine plaquettes, both endpoints of its coarse links, and its coarse sites. Write $D_r(A)$ for their radius-r neighbourhood, using the maximum-norm distance on the coarse torus.
- (A, U, k, V) is a *plot* at $(\widehat{\beta}, W)$ if the following conditions hold:
  - every fine plaquette whose parent-corner set meets $D_{5\varrho}(A)$ has chord $\geq t/4$ if it belongs to A, and $\leq t/2$ otherwise;
  - every coarse link with an endpoint in $D_{5\varrho}(A)$ has wire $\mathrm{ch}(V_b Q_b(U,k)^{-1}) \geq Wt$ if it belongs to A, and $\leq 2Wt$ otherwise;
  - on $D_{5\varrho}(A)$, the coarse sites in A are exactly $\mathrm{flags}_{\mathrm{temp}(\widehat{\beta})}(V)$;
  - every site in $D_{5\varrho}(A)$ has positive frame mass under k.
- The *estimator data* fix kernel constants, two stage records, an order j, a decay width $\eta > 0$ and a bound C. A tube is *compatible* with these data if the kernel and order agree, the branch conditions hold, and its transfer kernels have decay bounds through $j-2$ forward fine-coordinate differences: the fine transfer row has bound C/L^2, and differences of order q of the background and curvature kernels have bounds C/L^{1+q} and C/L^{2+q}. Its other two rows have bound C. The clips, path widths and row and higher-atom bounds are those of the two stages, with some finite loop bounds; the path rows must also be attached to quadratic metrics satisfying the tower's transfer identities (III.3).
- The *energy bound* at (L, W, κ_0) holds if for all sufficiently large $\widehat{\beta}$ the following inequality is true. The threshold for $\widehat{\beta}$ depends on the estimator data, L, W and κ_0, not on the torus. The inequality is required for every step with factor L and coarse side $c \geq 100(1+\varrho)$, every compatible tube, every forest, and every plot with $U \equiv 1$ on its trunks:

$$\kappa_0 \, t^2 \, |A| \;\leq\; \mathrm{Work}(A).$$

  Here $\mathrm{Work}(A)$ sums the six local works of III.1.10 with path cutoff $\lfloor \varrho/10^5 \rfloor$: a fine plaquette is included when its base's parent lies in $D_{2\varrho}(A)$, and a coarse plaquette, link or gain coordinate is included when its base lies there.
- The estimator data satisfy the *energy-per-seed requirement* if there is L_0 such that for every $L \geq L_0$ and every sufficiently large W there is $\kappa_0 > 0$ for which the energy bound holds.

The transfer (theorem). Let B be a directed history and suppose the *ringing* condition holds: for each pair of orders $j_{\min} \leq i \leq j \leq i+1$, the inequality holds with some $\mu > 0$ for every coarse reference coupling above the assigned entry threshold and wire parameter $F(j,i) > 0$. Here

$j_{\min} = J_0$ is the first admissible order. Then for each such pair there is $\mu > 0$ with the following property. For all N, all $m < N$ with $J_{m+1} = j$ and $J_m = i$, all p, and every plot at $(\hat{\beta}_m, F(j,i))$ with U trivial on the trunks,

$$\mu\, t_{m+1}^2\, |A| \le \mathrm{Work}(A).$$

This uses $\hat{\beta}_{m+1} = \hat{\beta}_m + \kappa_{\mathrm{slope}} \log L_m$, so that $t = \mathrm{temp}(\hat{\beta}_{m+1}) = t_{m+1}$.

Status: the requirement is a definition (a hypothesis of other units). It is proved for every choice of estimator data in III.10 (`Next.Fluctuations.grants`). Its proof applies `Integrated.pressure_grants`, which follows from the work items E01–E09 and from the coarse-row bound `EBare`, itself proved by `Fluctuations.bare`. The ringing condition is derived from these grants by raising the β-entry thresholds (`Bid.pricing_order`, `pressure_pricing`, III.10). The transfer is a theorem of this unit; its constant is uniform in N, m, p for the fixed pair (j,i). *Lean:* `Stores`, `Stores.Plot`, `Stores.assessed`, `Teller`, `Teller.Funds`, `Teller.live`, `Teller.cash`, `Teller.Grants`, `Testing.Stamp`, `Testing.Ringing`, `Testing.notation_ticket`.

Proposition III.1.12 (kinetic specification of a directed history). Fix levels, a horizon of quadratic kernels, a local exponential-chart density and positive stage constants. Kinetic data assign, for each N, m, a shift $\alpha_{N,m}$ and path data, and, for each volume and forest F, a map ψ from coarse Lie-algebra fields to Lie-algebra fields on the free fine links.

To state the stationary requirements, let $\mathcal{A}_f^\circ$ be the unmasked, unclipped fine action with coupling 1 and without loop terms, averaged over signed symmetries. Write $\mathcal{L}_f^\circ$ for the corresponding loop action and $\mathcal{Q}_c^\circ$ for the coarse quadratic part. For a coarse field u and a free-link field b, let $V = \exp u$ and form $U_F(u,b)$ by putting 1 on trunks and $\exp(b_e)(\iota V)_e$ on free links, where ιV is 1 on internal links and copies V on links crossing blocks. For each tag t, set

$$\mathcal{J}_t(u,b) = \mathcal{A}_f^\circ(U_F(u,b)) + \kappa \sum_e \mathrm{pr}(V_e Q_e(U_F(u,b),t)^{-1}).$$

The kinetic requirements are:

- (i) the stages and block factors agree with the fixed choices;
- (ii) for every $m \le N$, the path rows are attached to the horizon's quadratic kernels and satisfy the stage bounds of III.1.9(c);
- (iii) at level N, the unmasked, unclipped action is analytic and has the same derivatives of orders $0, \dots, 4$ at 0 as the Wilson energy in exponential coordinates, and every loop coefficient vanishes;
- (iv) for $m < N$, with the prescribed positive stage constant η, every volume and every forest satisfy

$$D_b^2 \mathcal{J}_t(0,0)[v,v] \ge \eta \sum_{e\ \mathrm{free}} |v_e|^2$$

for every t, v. The map ψ is analytic at 0, with $\psi(0) = 0$; each $\mathcal{J}_t$ is analytic at $(0,0)$, and $D_b \mathcal{J}_t(u, \psi(u)) = 0$ near 0. The derivative of ψ is the prescribed linear background, adjusted to tree gauge, minus the linearized copy $\iota_1 u$, restricted to free links. Explicitly, if $\mathcal{B}u$ is the horizon's linear background, put

$$\chi_u(y) = \int_{y \to \mathrm{root}(y)} \mathcal{B}u - (\mathcal{F}\mathcal{B}u)(\mathrm{parent}(y)), \qquad D\psi(0)u = (\mathcal{B}u + d_0\chi_u - \iota_1 u)|_{\mathrm{free}},$$

where the integral follows the forest and $\mathcal{F}$ is defined in III.1.14. Finally the unmasked coarse action $\mathcal{A}_c^\circ(\exp u)$ is analytic at 0 and has the same derivatives as $\mathcal{I}_t(u,\psi(u))$ of every order through 4, for every t;

- (v) for $m < N$,
$$|\alpha_{N,m} + \kappa_{\text{slope}} \log L_m| + 1 \le \text{drift}_{m+1};$$
- (vi) for $m < N$ and every volume and forest, let $H_t(u)$ be the Hessian of $\mathcal{I}_t(u,\cdot)$ at $\psi(u)$, in the counting basis of the physical Lie-algebra coordinates, and define
$$T_t(u) = \mathcal{L}_f^\circ(U_F(u,\psi(u))) + \tfrac{1}{2}\log\det H_t(u) - \log \prod_{e \text{ free}} J(\psi(u)_e), \qquad D(u) = \alpha_{N,m}\mathcal{Q}_c^\circ(\exp u) + \mathcal{L}_c^\circ(\exp u).$$
For every tag t, $\det H_t > 0$ near 0, both T_t and D are analytic at 0, and
$$|D^2 T_t(0)[a,b] - D^2 D(0)[a,b]| \le |\Lambda_c|\,\Theta\Lambda e^{-\gamma c}$$
for any two real plane waves a, b. Each wave is the real part of a lattice Fourier character times a complex amplitude whose every direction and colour component has modulus at most 1. Here $\Theta, \gamma > 0$ are the assigned stage constants, Λ is the fine loop bound, and c is the coarse torus side.

A *directed history* consists of paid levels (III.2), a horizon satisfying the decay, positivity and transfer identities of III.3, kinetic data satisfying (i)–(vi), and a recipe with the same path data. It also carries a local Haar-chart formula: on some ball $B_r \subset \mathfrak{g}$, the exponential is injective with open image, J is positive and analytic, $J(0) = 1$, and the pushforward of $c_J J(X)\,dX$ is Haar measure restricted to that image, for some $c_J > 0$; J is Borel globally.

A *move* between a fine representation $\mathcal{R}_{m+1}$ and a coarse representation $\mathcal{R}_m$ requires the exact push identity, the recipe's continuation and retention conditions, and agreement of the coarse path data, clip, norm parameters and lawful bounds with level m. It also requires a forced-activity bound $\exp[-g_j\,\text{monitor}_{m+1}^2]$, where $g_j > 0$ is prescribed and $j = J_{m+1}$, and
$$|\beta_m - \beta_{m+1} - \alpha_{N,m}| \le 1.$$
Together with (v), this implies the drift bound of III.1.9(d).

Status: definitions of requirements. For simple nonabelian $\mathfrak{g}$, the conditional construction theorems `Roost.companyK` and `FundedCompanySelection.company` supply directed histories with prescribed paid levels as part of the assembly in III.2. The kinetic requirements themselves do not assert a coupling reset.

Lean: `PriceGap`, `Critical`, `Guessed`, `ClassicalMatch`, `extracted`, `Horizon.Suffices`, `Kin`, `Kin.Planned`, `FiniteBound`, `thermal`, `hessian`, `LocalRatio`, `Directed`, `Directed.Move`.

Proposition III.1.13 (tree gauge on blocks). Functions in (c)–(d) take values in a real Banach space. A *forest* assigns to every non-root fine site y (one with $y \neq L \cdot \text{parent}(y)$) a unit step $\text{up}(y)$ and a height $\le 4L$, such that the height decreases along up and the parent block does not change. The *trunks* are the links used by these steps; the others are *free*.

- (a) For odd L a forest exists: choose an axis on which the offset from the block root is nonzero and step towards the root, with the ℓ^1 norm of the offset as height.
- (b) Gauging by $\text{fold}_y = T_U(y \to \text{root along the tree})^{-1}$ makes every trunk link equal to 1. Moreover fold is 1 at the roots and depends only on the trunk links.

- (c) If $F \in L^1(\mu_f)$ is invariant under gauge transformations equal to 1 at the block roots, then $\int F d\mu_f = \int F(\mathrm{drape}\, s)\, d\mu_{\mathrm{free}}(s)$. Here $\mathrm{drape}\, s$ has trunk links 1 and free links s, and μ_{free} is product Haar measure on the free links.
- (d) Hence, if ρ_β is a kernel satisfying III.1.2(a) and F is gauge invariant and integrable, then $(\mathcal{P}_\beta F)(V) = \int \rho_\beta(\mathrm{drape}\, s, V)\, F(\mathrm{drape}\, s)\, d\mu_{\mathrm{free}}(s)$.

Status: theorem. *Lean:* `Tree.Forest`, `Tree.select`, `Forest.compress`, `Forest.names`, `Forest.pack`, `Forest.pushing`.

Proposition III.1.14 (prediction near the trivial configuration). Assume (B1)–(B3). For a $\mathfrak{g}$-valued 1-form v on Λ_f, let $U_v = (\exp v_e)_e$. Then for every tag t and coarse link $b = (x, i)$:

- (a) $Q_b(U_0, t) = 1$;
- (b) $v \mapsto Q_b(U_v, t)$ is real analytic near $v = 0$; in a sufficiently small neighbourhood all patch frames and bond predictors use their minimizing branches;
- (c) the derivatives at $v = 0$ are, componentwise in $\mathfrak{g}$,

$$\tfrac{d}{ds} h_y^{(x)}(U_{sv})\big|_0 = \tau_x(y) - \tau_x(0), \qquad \tfrac{d}{ds} Q_b(U_{sv}, t)\big|_0 = (\mathcal{M}v + d_0 \mathcal{F} v)(b).$$

Here $d_0\varphi(x,i) = \varphi(x + e_i) - \varphi(x)$ and $\tau_x(y) = -\mathrm{avg}_{z \in B_R} \mathrm{avg}_o \int_{\mathrm{route}(Lx+y \to Lx+z;\, o)} v$, $(\mathcal{M}v)(x,i) = \sum_y w_y \int_{Lx+y}^{Lx+y+Le_i} v$ (a straight line integral), and $(\mathcal{F}v)(x) = \tau_x(0) - \sum_y w_y \tau_x(y)$. - (d) The linearization $\mathcal{Q} = \mathcal{M} + d_0\mathcal{F}$ is independent of t and maps fine pure gauge to coarse pure gauge: $\mathcal{Q}\, d_0\varphi = d_0(\varphi \circ \mathrm{embed})$.

Status: theorem, with the neighbourhood allowed to depend on the fixed step. Since both tag laws are locally uniform, the kernel itself is analytic in fine and coarse exponential coordinates near $(1, 1)$ for each fixed β. *Lean:* `Push.settled`, `Push.prediction_cont`, `Push.derivative`, `Push.prices`, `Precision.probe`, `Precision.root_gradient`.

3.4.2 2. What it uses

From other units.

- I.1 (lattice, gauge groups). This unit uses:
  - the torus and its cover $\mathbb{Z}^4$, the block subdivision (root, parent), coordinate routes, slabs and layers, parallel transport and its gauge rule, the signed symmetries, and the lattice forms with $d \circ d = 0$;
  - the gauge group data: chord, price, conjugation invariance, probability Haar measure and its two-sided invariance, the exponential map;
  - the nearest-point map π of $G \subset M_N(\mathbb{C})$: uniqueness of the nearest point within a tubular radius r_G (`nearest_tube`); analyticity of π near 1 with $D\pi(1) = \Pi_{\mathfrak{g}}$ (`focus_deriv`); the strict-differentiability bound $\|\pi(X) - \pi(Y) - \Pi_{\mathfrak{g}}(X - Y)\| \le \eta \|X - Y\|$ for X, Y near 1 (`nearest_approximate`); and the first-order condition that 1 nearest to X implies $X - 1 \perp \mathfrak{g}$ (`nearest_normal_one`);
  - the Wilson energy, the prior and the partition function; and C_A (`casimir`).
- III.16 (analytic tools): the ℓ^1 size of lattice offsets used for forest heights (`Laurent.size`).
- III.2 (constants and schedules): the staged choices of kernel and analytic constants, block sizes, entry thresholds, barrier and windows, with their compatibility conditions (`Paid`). They enter the statements of III.1.11–III.1.12.

- III.10: the proof of the energy-per-seed requirement (see III.1.11). III.2 supplies directed histories (see III.1.12).

Standard mathematics.

- Block-spin renormalization for lattice gauge fields, in the background-field and axial-gauge setting: T. Bałaban, *Propagators and renormalization transformations for lattice gauge theories I, II*, Comm. Math. Phys. 95 (1984) 17–40, 96 (1984) 223–250; *Averaging operations for lattice gauge theories*, CMP 98 (1985) 17–51; *Spaces of regular gauge field configurations on a lattice and gauge fixing conditions*, CMP 99 (1985) 75–102; *Renormalization group approach to lattice gauge field theories I, II*, CMP 109 (1987) 249–301, 116 (1988) 1–22.
- Axial and tree gauges, and the absence of a Faddeev–Popov factor for compact groups: E. Seiler, *Gauge theories as a problem of constructive quantum field theory and statistical mechanics*, Lecture Notes in Physics 159 (1982). Curvature controls the connection in a suitable gauge (continuum analogue): K. Uhlenbeck, CMP 83 (1982) 31–42.
- Least-squares group synchronization: A. Singer, *Angular synchronization by eigenvectors and semidefinite programming*, Appl. Comput. Harmon. Anal. 30 (2011) 20–36. Extrinsic means on embedded manifolds: R. Bhattacharya, V. Patrangenaru, Ann. Statist. 31 (2003) 1–29.
- The tubular neighbourhood theorem and the nearest-point projection; the real-analytic implicit function theorem (Krantz–Parks, *The Implicit Function Theorem*, Birkhäuser 2002), as in `Push.authored`.
- Continuity of a unique minimizer depending on a parameter, when the argmin relation is closed and the target compact (`minArg_cont`). Fubini's theorem, Jensen's inequality, and invariance of Haar measure.

Imported constants and assumptions.

- The constants $d(G)$, r_G and $A(G)$, and the clustering constant of Lemma 4.14, depend only on G, through π. The kernel and geometric theorems require no simplicity assumption. The construction of directed histories cited in III.1.12 assumes that the Lie algebra of G is simple and nonabelian.
- The kernel constants are chosen before the block factors ("pre-L"). L odd ≥ 3 and $J \geq 4$ are part of the patch; in a calendar, J_m and L_m come from III.2.
- $\kappa_{\mathrm{slope}} = 11C_A/(24\pi^2)$ enters only through $\hat{\beta}_m$ and the identity $\hat{\beta}_{m+1} = \hat{\beta}_m + \kappa_{\mathrm{slope}} \log L_m$.
- The energy-per-seed requirement (III.1.11) and the kinetic data (III.1.12) are hypotheses here, discharged in III.10 and III.2 respectively.

3.4.3 3. Strategy

The step follows the block-spin method: fix a frame around each coarse site, transport across coarse links, average the connectors, and add independent Haar-normalized Wilson noise. Two rules define the frames on all configurations. When route holonomies in axial gauges are small, least-squares synchronization gives a unique frame normalized at the centre. Otherwise an axial frame is chosen uniformly among tags satisfying the curvature bound. The predictor likewise uses a unique nearest point to the weighted matrix mean when the minimum price is small, and a connector sampled with weight w otherwise. Uniqueness and symmetric sampling make these rules covariant. The tag probabilities sum to 1, and the noise has total mass 1, so the full kernel is normalized.

The branch cutoffs are independent of the number of patch sites. For frame uniqueness, normalize each minimizer by a nearest point to its centroid. Centroid differences then have zero tangential projection, making the feedback comparison a contraction. Predictor uniqueness follows from the tubular neighbourhood of G. Eligible axial frames exist by induction over coordinate slabs: choose a layer of least curvature, fix its gauge, and compare the other layers by plaquette bounds.

For quantitative control, one axial gauge on the macro-box synchronizes the surrounding patch frames. Each frame defect costs $O(L^2)$ times the curvature sum; the averaging weight contributes L^{-4}. Fine transports cost another L times a link sum of order L, giving the same L^{-2} coarse-curvature estimate. Near the trivial configuration, the analytic implicit function theorem gives the frame and predictor derivatives. Tree gauge then reduces integration to free links. These geometric and analytic results enter the later energy estimates and stationary-branch analysis.

3.4.4 4. Steps

3.4.4.1 4.1 The kernel and its normalization

Lemma 4.1 (the tag law is a probability). Under (B3), $\sum_t \lambda(U,t) = 1$. *Sketch:* first sum over the bond tags with the frame tags fixed. The connectors are then fixed, and each bond factor sums to 1: the mass is uniform on the minimizing branch, and it is w, with $\sum w = 1$, otherwise. The remaining sum factorizes over coarse sites, and each frame factor sums to 1: the mass is uniform on all tags, or on the nonempty eligible set. *Lean:* `StepData.chosen`, `frameMass_sum`, `predictorMass_sum`.

Lemma 4.2 (Borel measurability). Under (B1)–(B2), for each tag the frames, connectors, predictions and $\lambda(U,t)$ are Borel functions of U. *Sketch:* M_δ and $\{\min \Phi_Y < \varepsilon\}$ are open. The argmin relations are closed and the targets compact, so a unique minimizer depends continuously on the parameter. Hence the selected frame is continuous on M_δ and the minimizing predictor is continuous where $\min \Phi_Y < \varepsilon$. For a fixed tag, fallback frames are continuous in U. Eligibility is a closed condition. Piecewise definitions over Borel sets are Borel. *Lean:* `minArg_cont`, `selected_cont`, `best_cont`, `min_open`.

Lemma 4.3 (Proposition III.1.1). *Sketch:* for each link, right invariance of Haar measure moves Q_b to 1, so $\int Z_\beta^{-1} e^{-\beta\kappa\,\mathrm{pr}(V_b Q_b^{-1})} dV_b = 1$. Fubini over the coarse links then gives $\int \rho_\beta\, dV = \sum_t \lambda = 1$. The noise is continuous on a compact space, hence bounded by some A_β, and $\rho_\beta \le A_\beta \sum_t \lambda = A_\beta$. The properties of the push follow from Fubini for the bounded kernel times an integrable F. *Lean:* `noise_mass`, `density_total`, `density_bound`, `push_slice`, `push_integral`.

3.4.4.2 4.2 Covariance

Lemma 4.4 (gauge covariance of the patch data). Assume (B1)–(B2). For every $k : \mathbb{Z}^4 \to G$, M_δ, the eligible sets and frame masses are unchanged under $W \mapsto k \cdot W$, and the frame becomes $h_y \mapsto k_0 h_y k_y^{-1}$. For a gauge $k : \Lambda_f \to G$, the connector of (x,i) becomes $k_{Lx}\, Y_y\, k_{L(x+e_i)}^{-1}$. The prediction transforms in the same way, with unchanged bond mass. *Sketch:* $T_{k\cdot W}(y \to z; o) = k_y T_W(y \to z; o) k_z^{-1}$, and ch, pr are class functions. Then $S_{k\cdot W}(k_0 h k^{-1}) = S_W(h)$, the normalization $h_0 = 1$ is preserved, and (B1) identifies the minimizer. The axial costs are conjugated, hence unchanged. $\Phi_{uYv}(uqv) = \Phi_Y(q)$, and (B2) identifies the minimizer. *Lean:* `frame_gauged`, `min_gauged`, `updateGauged`, `bond_gauge`, `predicts`.

Lemma 4.5 (lattice-symmetry covariance of the patch data). Assume (B1)–(B2). Let $s \in \mathfrak{B}_4$ act about the centre of a patch. Then:

- M_δ is invariant; this is why all 24 orderings enter its definition;

- frame tags are permuted by $(r, o) \mapsto (sr, \bar{s} \circ o)$, where $\bar{s} \in S_4$ is the underlying axis permutation, and eligibility and masses correspond;
- the frame at tag t for $s_0 W$ equals the frame at the transformed tag for W, composed with s;
- w is invariant;
- after reindexing site tags, the connector for the transformed field equals the original connector at the image link, composed with s, and inverted when s reverses that link;
- the prediction commutes with both operations.

Sketch: routes and axial gauges are mapped to routes and axial gauges, and the score is invariant; (B1) gives the minimizer. Moreover $\mathrm{pr}(k^{-1}u) = \mathrm{pr}(ku^{-1})$, so $\Phi_{Y^{-1}}(u^{-1}) = \Phi_Y(u)$, and (B2) applies. *Lean:* `frames_channel`, `axial_channel`, `min_channel`, `good_channel`, `weight_invariant`, `bond_pull`, `bond_inv`, `StepData.positive`, `StepData.negative`. *(proof read, including `Axial.channels`)*

Lemma 4.6 (Proposition III.1.2). *Sketch:* (b) The bijection of tag configurations that relocates frame tags to image sites, maps them by s, and maps bond tags by s gives $\lambda(s_{Lb}U, t) = \lambda(U, t')$ and $Q(s_{Lb}U, t) = s_b Q(U, t')$. The noise is invariant under the simultaneous symmetry, including inversion of both arguments. Reindex the sum over tags. (a) λ is invariant and $Q \mapsto (k \circ \mathrm{embed}) \cdot Q$; the noise is invariant because $\mathrm{pr}(kVk'^{-1}(kQk'^{-1})^{-1}) = \mathrm{pr}(VQ^{-1})$. (c) is immediate. *Lean:* `mass_channel`, `prediction_channel`, `noise_channel`, `withinChoice`, `channel`, `covary`.

3.4.4.3 4.3 Axial gauges and the branch constants

Lemma 4.7 (comparison of parallel layers). Fix r, o, l as in III.1.4 and an axis $j \notin l$. Let S' be the slab for $l + [j]$, and suppose all j-links inside S' equal 1. Let S_z be the layer of S' with j-th coordinate z. Then $|E_{S_z} - E_{S_u}| \le 4F_{S'}$ for every $z, u \in \{-r, \dots, r\}$. *Sketch:* with trivial j-links, the (i, j)-plaquette at y has holonomy $U_{(y,i)}U^{-1}_{(y+e_j,i)}$. The inequality $|\mathrm{ch}(a) - \mathrm{ch}(b)| \le \mathrm{ch}(ab^{-1})$ is a discrete Stokes inequality. Summing it over the four axes i and over y in a layer bounds $|E_{S_{z+1}} - E_{S_z}|$ by $4\times$ the plaquette sum at the sites of S_z. Telescoping uses each layer once. *Lean:* `variation`, `squares`, `slide_le`, `window_le`.

Lemma 4.8 (Proposition III.1.4). *Sketch:* induction on $|l|$; the case $l = \emptyset$ is trivial. For $l' = l + [j]$, the slab S' is the disjoint union of $2r + 1$ layers S_z, each a slab for l. Choose the layer z_0 with the least curvature, so that $(2r + 1)F_{S_{z_0}} \le F_{S'}$. By induction there is a root $x \in S_{z_0}$ with $E_{S_{z_0}} \le 4|l|(2r + 1)F_{S_{z_0}} \le 4|l|F_{S'}$ in its axial gauge along l. Extend that gauge by moving along j last. Then all j-links in S' are trivial, and the gauge on S_{z_0} is unchanged. By Lemma 4.7,

$$E_{S'} = \sum_z E_{S_z} \le (2r + 1)(4|l| + 4)F_{S'} = 4|l'|(2r + 1)F_{S'}.$$

The choice of the cheapest layer is what closes the induction. *Lean:* `existence`, `total_slab`, `vertical`, `faces_pins`.

Lemma 4.9 ((B3) for $a \ge 16$). For every W and every ordering o there is a root r with $E_{B_R}(a^{(r,o)} \cdot W) \le 16(2R + 1)F_{B_R}(W)$. *Sketch:* apply III.1.4 to the whole box with the four axes in the order o; then $4 \cdot 4 = 16$. *Lean:* `Axial.eligible`, `edge_cost`.

Lemma 4.10 (structure of the synchronization problem). Let W be as in III.1.5, with centroid $m(h) = \mathrm{avg}_x h_x$ and *feedback* $F_x(h) = \mathrm{avg}_{y,k} h_y W^{-1}_{xyk}$, both computed in $M_N(\mathbb{C})$.

- (a) u minimizes $\sum_i w_i \,\mathrm{pr}(f_i u^{-1})$ (with $w_i \ge 0$, $\sum w_i = 1$) if and only if u is a nearest point of G to $\sum_i w_i f_i$.

- (b) The cost is invariant under $h \mapsto ch$. For every global minimizer, h_x is a nearest point of G to $F_x(h)$ for every x; this is the self-consistency condition. Changing only h_x combines the two off-diagonal contributions by reflection symmetry. The original diagonal contribution is zero; the added diagonal term for the competitor is nonnegative. Thus the comparison also holds with the self term included in $F_x(h)$.
- (c) If h is a global minimizer and $\|W - 1\| \le d$, then $\mathrm{cost}(h) \le \mathrm{cost}(1) \le d^2/(2N)$, so $\mathrm{avg}_{x,y}\|h_x - h_y\|^2 \le 4d^2$. If moreover 1 is a nearest point to $m(h)$, then $\|m(h) - 1\| \le 2d$ and $\|F_x(h) - 1\| \le 3d$.
- (d) If $\max_x \|p_x - q_x\| \le u$, then $\|F_x(p) - F_x(q)\| \le u$ and $\|F_x(p) - F_x(q) - (m(p) - m(q))\| \le ud$.

Lean: `min_iff`, `price_diff`, `min_cov`, `self_consistent`, `compare_energy`, `cluster`, `closeness`, `changes`, `weak_changes`.

Lemma 4.11 (Proposition III.1.5). *Sketch:* let δ_1 be the radius for which the strict-differentiability bound of π at 1 holds with $\eta = 1/4$, and take $d = \min(\delta_1/4, 1/4)$, so that $3d < \delta_1$. Left-multiply each minimizer by the inverse of a nearest point to its centroid. Then 1 is nearest to both centroids, and by 4.10(b) both are still minimizers. Let p, q be the normalized minimizers, $u = \max_x \|p_x - q_x\|$ attained at x, and $B = F_x(p) - F_x(q)$. By 4.10(c) both feedbacks lie within $3d < \delta_1$ of 1, and self-consistency makes p_x, q_x nearest points to these feedbacks. Hence $\|p_x - q_x - \Pi_{\mathfrak{g}} B\| \le \frac{1}{4}\|B\| \le \frac{u}{4}$. Since $m(p) - 1$ and $m(q) - 1$ are normal to $\mathfrak{g}$, $\Pi_{\mathfrak{g}}(m(p) - m(q)) = 0$, and 4.10(d) gives $\|\Pi_{\mathfrak{g}} B\| \le ud$. So $u \le ud + u/4 \le u/2$, hence $u = 0$ and $p = q$. Undoing the normalization gives $h' = ch$. *Lean:* `align`.

Lemma 4.12 ((B1) and (B2) on patches). (B1) holds if $\delta \le d/\sqrt{N}$, and (B2) holds if $\varepsilon \le r_G^2/(2N)$, for every patch. *Sketch:* (B1). Take $n = B_R$, $\Lambda = S_4$ and $W_{yzo} = a_y T(y \to z; o) a_z^{-1}$, where a is the axial frame from the centre with the identity ordering. The involution σ reverses the ordering, since the reversed route is the route with the reversed ordering. On M_δ, $\|W - 1\| = \sqrt{N}\,\mathrm{ch}(W) < d$. Since $S_W(h) = \mathrm{cost}(ha^{-1})$ and the cost is left invariant, a constrained minimizer is a global minimizer. By III.1.5 two of them differ by a constant, which equals 1 at the centre. (B2). If $\Phi_Y(u) < r_G^2/(2N)$, Jensen's inequality gives $\|\sum w_y Y_y - u\|^2 \le \sum w_y \|Y_y - u\|^2 = 2N\Phi_Y(u) < r_G^2$. So the mean has a unique nearest point, which by 4.10(a) is the unique minimizer. *Lean:* `frame_unique`, `predictor_unique`, `weighted_unique`.

Lemma 4.13 (Proposition III.1.3). Combine 4.9 and 4.12. *Lean:* `viable`.

3.4.4.4 4.4 Synchronization and the L^{-2} estimate **Lemma 4.14 (clustering of minimizers for arbitrary data).** There is $C_G > 0$ such that for all finite nonempty n, Λ and all W satisfying the diagonal and reflection conditions of III.1.5, every global minimizer h admits $y \in G$ with $\mathrm{avg}_x \mathrm{ch}(h_x y^{-1}) \le C_G \,\mathrm{avg}_{x,z,k} \mathrm{ch}(W_{xzk})$. *Sketch:* write $\mathrm{inc}_x = \mathrm{avg}_{z,k} \mathrm{ch}(W_{xzk})$ and $\bar{\mu} = \mathrm{avg}_x \mathrm{inc}_x$. Since $\mathrm{ch}^2 \le 2\,\mathrm{ch}$, $\mathrm{cost}(h) \le \mathrm{cost}(1) \le \bar{\mu}$, so $\mathrm{avg}\|h_x - h_z\|^2 \le 8N\bar{\mu}$. Let δ_1 be the radius for which the strict-differentiability bound of π at 1 holds with $\eta = 1$.

- If $\bar{\mu} < \sigma = \delta_1^2/(32N)$: normalize so that 1 is nearest to the centroid, which then lies within $\delta_1/2$ of 1. Also $\|F_x - m\| \le \sqrt{N}\,\mathrm{inc}_x$. For x with $\mathrm{inc}_x < \delta_1/(2\sqrt{N})$, the derivative bound gives $\|h_x - 1\| \le 2\|F_x - m\|$, that is $\mathrm{ch}(h_x) \le 2\,\mathrm{inc}_x$. For the other x, use $\mathrm{ch} \le 2$.
- If $\bar{\mu} \ge \sigma$: use $\mathrm{ch} \le 2 \le (2/\sigma)\bar{\mu}$.

Lean: `Pressure.cluster`, `price_square`, `income_bounds`.

Lemma 4.15 (route counting). For every ordering o: $\sum_{y,z \in B_r} \mathrm{ch}\, T_U(y \to z; o) \le 4\,|B_r|\,(2r +$

1) $E_{B_r}(U)$. *Sketch:* induction over the axes of the route. The last segment, along the axis j, starts at the point $\zeta(y,z)$ that has the already-processed coordinates of z and the other coordinates of y. Its chord is at most the sum of the link chords on the whole j-line through $\zeta(y,z)$. The map $(y,z)\mapsto\zeta(y,z)$ is $|B_r|$-to-one onto B_r, and each j-link lies on one line containing $2r+1$ points. *Lean:* `route_sum`, `marginal`, `slidings`, `Road.revenue`.

Lemma 4.16 (Proposition III.1.6). *Sketch: Minimizing branch.* Since $T_{H\cdot W}=HT_WH^{-1}$, the family $y\mapsto h_yH_y^{-1}$ is a global minimizer of the synchronization cost for the data $T_{H\cdot W}(y\to z;o)$. By 4.14 and 4.15 there is c with $\sum_y \mathrm{ch}(h_yH_y^{-1}c^{-1})\le 4C_G(2R+1)E_{B_R}(H\cdot W)$. The cutoff δ plays no role here. *Fallback branch.* Here h is the normalized axial frame of an eligible tag, so $E_{B_R}(h\cdot W)\le a(2R+1)F_{B_R}(W)$. Put $g=hH^{-1}$; the pure-gauge configuration $g\cdot 1$ satisfies $E(g\cdot 1)\le E(h\cdot W)+E(H\cdot W)$ by the triangle inequality. Apply 4.15 to $g\cdot 1$, whose route transports are $g_yg_z^{-1}$, and choose v with $\sum_y \mathrm{ch}(g_yg_v^{-1})$ at most its average over v. Then $c=g_v$. In both cases $A=\max(4C_G,4)$, and $\mathrm{ch}(c^{-1}x)=\mathrm{ch}(xc^{-1})$. *Lean:* `synchronized`, `Road.strips`, `risks`, `fallback`.

Lemma 4.17 (elementary bounds on the step geometry).

- (a) If $X+v\subset Y$, then the link and plaquette sums over X of the configuration translated by v are at most those over Y.
- (b) $w_y\,L^4\le 1$ for all y. One summand ξ_1 is uniform on L^4 points and independent of the others; only $J\ge 1$ is used.
- (c) $\sum_{y\in B_R}\mathrm{ch}\,T_U(y+v\to y+v+\ell e_j)\le \ell\,E_X(U)$, provided all these segments lie in X.
- (d) For all $\xi,\eta\in G$ and every bond tag, $\mathrm{ch}(\xi^{-1}Q_b\eta)\le 2\max(1,\varepsilon^{-1})\sum_y w_y\,\mathrm{ch}(\xi^{-1}Y_y\eta)$. On the minimizing branch Q_b is nearest to the mean $\bar Y=\sum w_yY_y$, so $\|Q_b-\xi\eta^{-1}\|\le 2\|\bar Y-\xi\eta^{-1}\|$. On the fallback branch, $\varepsilon\le\Phi_Y(\xi\eta^{-1})\le\sum w_y\mathrm{ch}(\xi^{-1}Y_y\eta)$ because $\mathrm{pr}\le\mathrm{ch}$, while $\mathrm{ch}\le 2$.

Lean: `spread`, `light`, `traverse`, `price_query`.

Lemma 4.18 (Proposition III.1.7). *Sketch:* apply III.1.4 once, on $B_{R'}$ around Lx with all four axes. This gives one gauge H with $E_{B_{R'}}(H\cdot U^{(x)})\le 16(2R'+1)\,\mathrm{port}\le 16\,\ell_jL\,\mathrm{port}$. For $y\in B_2$, the patch of $x+y$ is the translate of B_R by Ly and lies inside $B_{R'}$. Apply III.1.6 to it, with the translated gauge $H(\cdot+Ly)$; this gives p_y. Bound its edge and face sums by those of $B_{R'}$ using 4.17(a), and use $2R+1\le\ell_jL$. Every tag with $\lambda>0$ is usable, because a vanishing frame mass would make λ vanish. *Lean:* `planting`, `wraps`.

Lemma 4.19 (link comparison). In the setting of III.1.7, for $y,y+e_i\in B_2$ with pads $\xi=p_y$ and $\eta=p_{y+e_i}$:

$$L^2\,\mathrm{ch}(\xi^{-1}Q_{(x+y,i)}\eta)\le 2\max(1,\varepsilon^{-1})(2C+D)\,\mathrm{port}(x,U).$$

Sketch: write $\xi^{-1}Y_k\eta=f_k\cdot T_{H\cdot U^{(x)}}(k+Ly\to k+Ly+Le_i)\cdot g_k^{-1}$, where f_k,g_k are the frame defects of III.1.7. Then $\mathrm{ch}(\xi^{-1}Y_k\eta)\le\mathrm{ch}f_k+\mathrm{ch}T+\mathrm{ch}g_k$. Multiply by $w_kL^4\le 1$ and sum: the defects contribute $\le 2CL^2$ port, and by 4.17(c) the transports contribute $\le L\,E_{B_{R'}}\le DL^2$ port. Conclude with 4.17(d). *Lean:* `indicated`.

Lemma 4.20 (Proposition III.1.8). *Sketch:* conjugate the plaquette holonomy of Q at x by p_0. It factors into four padded links $p_y^{-1}Q\,p_{y'}$, with $y,y'\in\{0,e_i,e_k,e_i+e_k\}\subset B_1$, and the intermediate pads cancel. The triangle inequality and 4.19 give $\mathrm{ch}\le 4\cdot 2\max(1,\varepsilon^{-1})(2C+D)L^{-2}$ port. For V, $\mathrm{ch}(V_{(x,p)}Q_{(x,p)}^{-1})\le$ fuel by a telescoping product estimate. *Lean:* `qdc`, `comp`.

3.4.4.5 4.5 Near the trivial configuration Lemma 4.21 (Proposition III.1.14). *Sketch:*

1. U_0 lies in M_δ, which is open, and at $v = 0$ the frame is $h \equiv 1$ by (B1).
2. By 4.10(b) the frame solves $h_0 = 1$ and $h_y = \pi(\mathrm{avg}_{z,o} h_z T_{U_v}(y \to z; o)^{-1})$ for $y \neq 0$. This system is real analytic near $(0, 1)$. A variation u in the kernel of its partial derivative satisfies $u_0 = 0$ and $u_y = \Pi_{\mathfrak{g}}(\mathrm{avg}\, u)$ for $y \neq 0$. If $n = |B_R|$ and $A = \Pi_{\mathfrak{g}}(\mathrm{avg}\, u)$, then $A = (n-1)A/n$, so $u = 0$. Finite dimensionality makes the derivative invertible. Continuity of the selected minimizer identifies it with the analytic implicit solution.
3. Differentiate the fixed-point equation, using $D\pi(1) = \Pi_{\mathfrak{g}}$ and the fact that the derivative of a transport along w is the line integral of v along w. This gives $h'_y = \Pi_{\mathfrak{g}}(\mathrm{avg}\, h') + \tau(y)$; with $h'_0 = 0$ this is $h'_y = \tau(y) - \tau(0)$.
4. Near 0, $\min \Phi_Y < \varepsilon$ (it is 0 at $v = 0$), so $Q_b = \pi(\sum w_y Y_y)$, which is analytic. Its derivative is $\sum_y w_y Y'_y = (\mathcal{M}v + d_0 \mathcal{F} v)(b)$.
5. For an exact form $d_0\varphi$, the line integrals telescope; this gives (d).

Lean: `hats_zero`, `stars`, `scheme`, `focused`, `derivative`, `lips`, `prices`, `root_gradient`, `Exterior.curvature_linear`.

3.4.4.6 4.6 Tree gauge Lemma 4.22 (Proposition III.1.13). *Sketch:*

(a) Because L is odd, blocks are centred at their roots and the offset q lies in $B_{(L-1)/2}$. A step towards 0 along an axis with $q_a \neq 0$ decreases $|q|_1 \leq 4L$ and stays in the block. (b) $\mathrm{fold}_y U_{\mathrm{up}(y)} = \mathrm{fold}_{y+\mathrm{up}(y)}$, by induction on the height. (c) Split the links into trunks and free links; this split preserves the measure. For fixed trunk values, gauging by fold (trunks set to 1) sends the trunk values to 1. On the free links it acts as $s_e \mapsto k_{e^-} s_e k_{e^+}^{-1}$, which preserves product Haar measure, and F is unchanged because $k = 1$ at the roots. (d) $U \mapsto \rho_\beta(U, V) F(U)$ is invariant under gauge transformations that are trivial at the roots, by III.1.2(a). *Lean:* `select`, `compress`, `names`, `pack`, `pushing`.

3.4.4.7 4.7 Formats and energy accounting Lemma 4.23 (the theorems of III.1.9).

- Gas without flags: an admissible family must cover the empty flag set by nonempty inventories, so only the empty family occurs.
- Locality: flags on X depend only on the plaquettes meeting X.
- Envelope: summing the per-site bounds on the finite torus gives summable majorants for the regular and forced activities. The gas is dominated by the convergent sum of products of the forced majorants. Row and atom series have summable majorants as well; the Borel masks preserve these bounds. The exponent and gas are therefore Borel and bounded.
- Partition function: decreasing induction on m, using $\mathrm{eval}_m = \mathcal{P}\, \mathrm{eval}_{m+1}$, the identity $\int \mathcal{P} F = \int F$ (III.1.1, with the kernel property at each level), and the initial condition $\mathrm{eval}_N = e^{-\beta \sum \mathrm{pr}}$.

Lean: `gas_one`, `UsesFinal.allReads`, `Rep.envelope`, `partition_eq`.

Lemma 4.24 (Proposition III.1.10). *Sketch:* expand the six works. The adjointness $\langle \Phi q, p \rangle = \langle q, \Phi^\dagger p \rangle$ matches the cross terms of W_F, W_I with W_L, and the squares $\frac{1}{2}\|\Phi w\|^2$, $\frac{1}{2}\|\Phi' w\|^2$, $\frac{1}{2}\|\mathcal{N} w\|^2$, $\frac{1}{2}\|w\|^2$ cancel against W_D, W_N, W_C. What remains is the right-hand side. III.1.11 requires a lower bound for the corresponding local sum near the seeds; the identity alone supplies no such bound. *Lean:* `Tube.locates`, `arity`.

Lemma 4.25 (transfer of the energy per seed, III.1.11). Assume the ringing condition and $j_{\min} \leq i \leq j \leq i + 1$. *Sketch:* the stamp for the pair (j, i) gives μ and a β-entry threshold. In a

directed history:

- the tube of step m is compatible with the estimator data;
- the side condition $c \geq 100(1 + \varrho)$ is part of the fits conditions;
- $\widehat{\beta}_m$ exceeds the entry threshold by the price conditions;
- $\widehat{\beta}_{m+1} = \widehat{\beta}_m + \kappa_{\text{slope}} \log L_m$ turns $\operatorname{temp}(\widehat{\beta}_m + \kappa_{\text{slope}} \log L)$ into t_{m+1}.

Lean: `notation_ticket`.

3.4.5 5. Where the difficulty lies

The synchronization cutoff must be independent of the patch size. The feedback map alone has Lipschitz constant 1; normalizing the centroids removes its tangential average and gives the contraction factor $d + \frac{1}{4} \leq \frac{1}{2}$.

The curvature estimate must hold on both branches. It combines clustering of arbitrary minimizers, route multiplicities and curvature-controlled axial gauges. Losing a power of L in any of these comparisons would spoil the area scaling in III.1.8.

Normalization and tree-gauge integration use finite sums and Haar invariance. Global regularity is more limited: the kernel is Borel, while analyticity is proved near the trivial configuration. The energy bound and stationary requirements in III.1.11–III.1.12 need the later units.

3.4.6 6. Notation

Written proof	Lean	Meaning
$\rho_\beta(U, V)$, $\mathcal{P}_\beta$	`StepData.density`, `push`	block kernel and push
$P = (L, J)$, $R = (J + 3)L$	`Patch`, `radius`	block factor, order, patch radius
$K = (\kappa, \delta, \varepsilon, a)$	`KernelChoices`	penalty, frame cutoff, predictor cutoff, axial factor
(B1)–(B3)	`Patch.Branches`	branch conditions
M_δ	`usesMin`	minimizing branch
$S_W(h)$	`score`	synchronization cost on a patch
frame h, tag $t = (r, o)$	`frame`, `FrameTag`	gauge fixing on a patch; root and axis ordering
$\mathcal{E}(W)$	`eligible`	eligible axial tags
w	`weight`	J-fold convolution of the block-uniform weight
Y, Φ_Y, Q	`connector`, `bondCost`, `prediction`	connector, bond cost, prediction
$\lambda(U, t)$	`likelihood`	tag law
ch, pr	`Gauge.chord`, `Gauge.price`	chord distance to 1; Wilson plaquette density
E_X, F_X	`Axial.edges`, `Axial.faces`	link and plaquette chord sums in X
$m(h)$, $F_x(h)$	`Pairs.centroid`, `Pairs.feedback`	centroid and feedback of the synchronization problem

Written proof	Lean	Meaning
π	`Gauge.focus`	nearest point of G (near 1)
port, R'	`port`, `macroR`	fine curvature in the macro-box of radius $(J+7)L$
usable	`usable`	positive frame masses on the 5^4 neighbours
toll, wire, fuel	`toll`, `wire`, `fuel`	plaquette chord; chord of $V_b Q_b^{-1}$; sum of four wires
$\mathcal{M} + d_0\mathcal{F}$, τ	`probe`, `mean`, `fixRoot`, `tau`	linearized prediction (averaging operator)
forest, trunks, fold	`Tree.Forest`, `trunks`, `foldAt`	tree gauge in blocks
budget point, flags	`BudgetPoint`, `flags`	norm parameters; large-field sites
$\mathcal{R}$, eval	`Rep`, `Rep.eval`	effective density: exponential of the vacuum term minus the action and regular activities, times the gas
$\hat\beta_m$, t_m	`Calendar.hatBeta`, `Calendar.threshold`	reference coupling; small-field threshold temp$(\hat\beta_m)$
tube, works	`Tube`, `Tube.vision`	transfer tower with fine and coarse path data; six local works
store, plot, Work	`Stores`, `Stores.Plot`, `assessed`	seeds; their large-field characterization; local work
estimator data, energy-per-seed requirement	`Teller`, `Teller.Grants`	parameters of one energy estimate; energy-per-seed requirement
$t = \mathrm{temp}(\hat\beta + \kappa_{\mathrm{slope}} \log L)$	`fiery`, `temp`, `betaF`	fine-level threshold
ϱ	`radiusB`	cluster radius $\lceil (\log \hat\beta)^2 \rceil$
kinetic data, directed history	`Kin.Planned`, `Directed`	one-loop shift and stationary-branch requirements

3.4.7 7. What remains unexplained

1. **Energy estimates.** III.1.10 is an identity for the clipped direct term and truncated rows. The estimates converting it into III.1.11, including bounds on W_D and the truncation remainders, belong to III.10. No sign is asserted here for $1 - \Phi^\dagger\Phi - \Phi'^\dagger\Phi' - \mathcal{N}^\dagger\mathcal{N}$ at arbitrary V.
2. **Construction of the kinetic data.** The stationary conditions and the outputs of the two cited construction theorems are identified in III.1.12. Their full construction from the estimates of III.3, III.6–III.7 and III.9 is outside this unit. The alternative constructor `Extra.Iteration.planned` has the required output but is outside the main-theorem closure.
3. **Choice of neighbourhood and smoothing order.** The curvature proof uses pads at plaquette corners in B_1, whereas the stated synchronization result supplies them throughout B_2. The need for the larger neighbourhood in later estimates is not determined here. Likewise,

$w \le L^{-4}$ needs only $J \ge 1$; the later use of the stronger patch condition $J \ge 4$ remains to be specified.

4. **Additional files.** The small packages listed in the header are not described by the numbered results. Their contributions, and any further results from the exterior calculus or quadratic-kernel files beyond those cited here, remain to be incorporated into the relevant later units.

3.5 III.2. Choosing the constants: the order of choice, the level schedule and the shooting argument

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/Price/` (the constants and the level schedule, 406 lines), `YM/Shoot/` (shooting, 228 lines), `YM/Integration/RG05Preconstruction/`, `YM/Integration/NativeSelection/` and other files of `YM/Integration/`, `YM/RG08/Calendar/`, and parts of `YM/RG08/`, `YM/RG12/`, `YM/FullVolume/` (in particular `Roll/`) and `YM/TraceExtension/`. **Size:** 18,156 lines in the main-theorem closure:

part	lines
proof of the plentiful construction (III.2.5)	about 6,250
proof of the joint same-factor purchase (III.2.4)	about 4,700
proof of the final run (III.2.7)	about 4,000
proof of the sheet and radar purchase	about 2,400
the two directories that carry the mathematics of the schedule (`YM/Price/`, `YM/Shoot/`)	634

Conventions. These are those of III.9.

- **Levels.** Levels $m = 0, 1, \dots, N$ run from the coarsest ($m = 0$, unit spacing) to the finest ($m = N$, the Wilson lattice).
- **Steps and scales.** The step $m + 1 \to m$ blocks by an odd factor $L_m \ge 3$. The scale is $s_m = \sum_{i<m} \log L_i$. The level-$m$ torus at volume index p has side $n_p e^{s_m}$, where n_p is the side of the level-0 torus. III.9 writes L_p for n_p; the letter L is kept here for block factors.
- **Slope.** $\kappa = 11C_G/(24\pi^2)$ (III.9.1), and $\kappa > 0$ for simple G.
- **Reference quantities (III.9.2).** $\hat\beta_m = \bar\beta + \kappa s_m$, $M_m = e^{(\log\log\hat\beta_m)^2}$, the threshold $t_m = \hat\beta_m^{-1/2} M_m$, the cover size $B_m = \lceil (\log\hat\beta_m)^2 \rceil$ and the band $\mathfrak{B}_m = [\hat\beta_m/2,\, 2\hat\beta_m]$.
- **Patch radius.** $R_m = (J_{m+1} + 3)L_m$ (`Patch.radius`).

The main results concern numerical choices and a conditional shooting construction:

1. A long list of requirements can be met simultaneously by choosing the constants in one fixed order. Each requirement has the form "this inequality holds once the block factor, the coupling or the lattice is large enough, given the constants chosen before".
2. There is a level schedule meeting every entrance condition.
3. Given the one-step construction as a hypothesis, there is a trajectory of effective actions with $\beta_{N,p,0} = \bar\beta$ (shooting).

The terminology of sheets, prices and purchases records numerical bounds and their order of choice (glossary, family 1). The unit also contains a group of analytic lemmas of the Gaussian step, filed here by the dependency assignment (III.2.9).

3.5.1 1. What this unit proves

Proposition III.2.1 (the constants, in the order in which they are chosen). Let G be simple. The construction fixes the following data, in this order.

1. *Sheet (data chosen before the block factors).* It consists of the following.
   - A kernel choice K with branches on every patch.
   - For every stage $j \in \mathbb{N}$, a stage record X_j with:
     - a clip with relaxation r_j;
     - derivative order $k_j \geq 6$ and spatial power $q_j \geq 2$;
     - support exponent $a_j > 0$, width $w_j > 0$ and path count $\varpi_j > 0$;
     - drift allowance $\omega_j \geq 0$ and column bound $\gamma_j \geq 1$;
     - the constraint $r_j w_j \geq 12$.
   - A first stage $j_0 \geq 4$.
   - Floors $f_j \in \mathbb{N}$.

   The sequences k_j, q_j, a_j are nondecreasing.

2. *Stage block factors ℓ_j.* They are odd, satisfy $\ell_j \geq \max(3, f_j)$, are nondecreasing in j, and for $j \geq j_0$

$$\frac{\omega_j}{\log \ell_j} \leq \min\Big(\frac{\kappa}{4},\ \frac{1}{j^2}\Big), \qquad \frac{\log \mathcal{C}_j}{\log \ell_j} \leq \frac{1}{j^2}, \qquad \mathcal{C}_j := 1 + \varpi_j + w_j^{-1} + \gamma_j + r_j + a_j.$$

3. *Smallness exponents $\theta_j > 0$.* They are nonincreasing, with $\theta_j \leq 1/j$ for $j \geq j_0$.

4. *Delays.* These are the following.
   - Entry couplings $\beta_j^{\mathrm{ent}} \in \mathbb{R}$.
   - Entry scales $\sigma_j^{\mathrm{ent}}(\bar\beta)$. These are functions of the stage *and of the barrier* still to be chosen.
   - Volume shells $S_j \in \mathbb{N}$ and growth exponents $g_j \in \mathbb{N}$.

   Delays are ordered componentwise.

5. *Barrier $\bar\beta > 1$.* It satisfies $\beta_{j_0}^{\mathrm{ent}} \leq \bar\beta$ and $x^{-1/2} e^{(\log\log x)^2} \leq 1$ for all $x \geq \bar\beta$.

6. *Wait.* This consists of:
   - readiness scales σ_j^{rdy};
   - windows $h_j : \mathbb{R} \to \mathbb{N}$ with $h_j(s)/s \to 0$ as $s \to \infty$ (for each fixed j, not uniformly);
   - a minimal period $n_{\min}$.

   Waits are ordered componentwise by readiness scales, windows and minimal period.

Items 2, 3 and 5 always exist:

- item 2 exists for every sheet;
- in item 3, θ can be chosen below any prescribed positive sequence;
- in item 5, $\bar\beta$ can be chosen above any prescribed real number.

Item 4 has no further conditions. In item 6, only the stated sublinearity is required; the readiness scales are real and $n_{\min} \in \mathbb{N}$.

Status: items 1–6 are definitions (specifications). There are three existence theorems. The existence of item 2 uses $\kappa > 0$, hence the simplicity of G. The later existence theorems III.2.4–III.2.7 meet specified requirements in this order. *Lean:* `Sheet`, `Stage`, `Large`, `Rolls`, `Rolls.exists_rolls`, `Protection`, `Protection.exists_margin`, `Delays`, `DelayLe`, `Bar`, `Bar.exists_above`, `Wait`.

Proposition III.2.2 (a level schedule meeting all entrance conditions). Let G be simple and let data 1–6 of III.2.1 be given. Then there is a level system with the following data:

- a stage index J_m;
- step block factors $L_m = \ell_{J_{m+1}}$;
- level-0 sides $n_p = 4(M + p + 1)$ for some $M \in \mathbb{N}$;
- reference couplings $\hat{\beta}_m = \bar{\beta} + \kappa s_m$.

It satisfies:

- (a) *identification:* the kernel, stage records, ℓ, θ and $\bar{\beta}$ are those of III.2.1, and $J_0 = J_1 = j_0$;
- (b) *entry:* $\beta^{\mathrm{ent}}_{J_{m+1}} \le \hat{\beta}_m$ for all m, and $\sigma^{\mathrm{ent}}_{J_{m+1}}(\bar{\beta}) \le s_m$ whenever $J_{m+1} > j_0$;
- (c) *volume:* $n_p \ge n_{\min}$, and $S_j(1 + B_m + B_{m+1})^{g_j} \le n_p e^{s_m}$ for all m, p, with $j = J_{m+1}$;
- (d) *windows:* for every $j \ge j_0$ there is a level k with $\sigma^{\mathrm{rdy}}_j \le s_k$, $h_j(s_k) \le k$, and $J_i = j$ for $|i - k| \le h_j(s_k)$;
- (e) *the scale conditions of III.9.2:*
  - J is nondecreasing, $J_{m+1} \le J_m + 1$ and $J_m \to \infty$;
  - k_{J_m}, q_{J_m}, a_{J_m} are nondecreasing in m;
  - $\hat{\beta}_m > 1$ and $t_m \le 1$;
  - $100(B_m + 1) \le n_p e^{s_m}$ and $100(R_m + 1) \le n_p e^{s_{m+1}}$;
  - $0 < \theta_{J_m} \le 1/J_m$, and θ is nonincreasing;
  - $\omega_{J_{m+1}} / \log L_m \le J_{m+1}^{-2}$ and $\log \mathcal{C}_{J_{m+1}} / \log L_m \le J_{m+1}^{-2}$;
  - $\log L_m / s_m \to 0$;
- (f) *block control:* at every step the kernel is well posed, symmetric under the lattice symmetries and gauge covariant.

The same construction also ensures that every positive integer divides n_p for infinitely many p.

Status: theorem. The divisibility variant is filed in III.10 and is the one used in the final assembly (III.2.7). Note that (d) asserts the existence of one level k per stage. Taking σ^{rdy}_j large forces k to be large. *Lean:* `Wait.ready`, `Paid`, `Calendar.Fits`, `Levels`, `Schedule.Gentle`, `Calendar.blocks_go`; variant `Wait.engine`, `AsDivisors`.

Proposition III.2.3 (shooting: a selected history from a recipe). Let G be simple. Fix a level system with

$$\omega_{J_{m+1}} \le \tfrac{\kappa}{4} \log L_m \qquad \text{for all } m.$$

This holds for every level system of III.2.2, by III.2.1(2).

Recipe. Let a recipe be given: path data $P^{\mathcal{R}}_{N,m}$ and three predicates (a start test, a retention test at each level, a transition test for consecutive levels).

Admissible representations. A level-m representation ρ (III.9: path data, clip, activities, budget point, β, v) is *admissible for* (N, p) if:

- its path data are $P^{\mathcal{R}}_{N,m}$;

- its budget point, clip and polymer bounds are those of level m: width w_{J_m}, path count ϖ_{J_m}, loop cap $2M_m^{\theta_{J_m}}$, allowances $\widehat{\beta}_m^{-11/10}$ and $e^{-M_m^{1/5}}$;
- it passes the retention test.

Continuous families. A family $x \mapsto \rho_x$ on an interval is *continuous* if:

- β and v are continuous in x;
- the address sets of the regular and forced activities do not depend on x;
- differences of regular activities tend to zero in the sum of budget masses of their derivative bounds, up to the budget order on the healthy domain; differences of forced activities tend to zero in the sum of budget masses of their uniform bounds over all configurations.

Hypotheses. Assume that for every N and p:

- **(start)** There is a continuous family $\beta \mapsto \rho_\beta$, $\beta \in \mathfrak{B}_N$, of admissible level-N representations. Each ρ_β has coupling β, $v = 0$, density equal to the Wilson weight at β, and passes the start test.
- **(advance)** Let $m < N$ and $l \le u$, and let $x \mapsto \rho'_x$, $x \in [l, u]$, be a continuous family of admissible level-$(m+1)$ representations with $\beta(\rho'_x) \in \mathfrak{B}_{m+1}$ for all x. Then there is a continuous family $x \mapsto \rho_x$ of admissible level-m representations such that, for every x:
  - the density of ρ_x is the block-spin push-forward of the density of ρ'_x at coupling $\beta(\rho'_x)$;
  - $|\beta(\rho'_x) - \beta(\rho_x) - \kappa \log L_m| \le \omega_{J_{m+1}}$;
  - the transition test holds.

Conclusion. There are representations $\rho_{N,p,m}$, $0 \le m \le N$, for all N, p, such that:

- every $\rho_{N,p,m}$ is admissible with $\beta_{N,p,m} \in \mathfrak{B}_m$;
- $\rho_{N,p,N}$ is the Wilson weight with $v = 0$;
- consecutive levels are related as in (advance);
- $\beta_{N,p,0} = \bar{\beta}$;
- the path data at (N, m) do not depend on p.

The resulting history is *selected* in the sense of III.9.3. In particular it obeys the running law: for every $\varepsilon > 0$ there is m_0 such that $|\beta_{N,p,m} - \widehat{\beta}_m| \le \varepsilon s_m$ for all $m \ge m_0$, $N \ge m$ and all p.

Status: theorem, conditional on (start) and (advance).

- In the main chain both are verified in III.10 (`Graces.winnow`):
  - (start) from the Wilson start-up curves of III.9 (`hops`);
  - (advance) from the one-step construction with coupling reset (`Mallow.faces`).
- The drift hypothesis is derived from III.2.2 by `Drifting.pay`.
- A second route, through the step program of III.1/III.9 (`Directed.Program`), is not used by the main theorem.

Lean: `Recipe.races`, `Recipe`, `Lives`, `Curve`, `Recipe.Relays`, `Recipe.Starter`, `Recipe.Issues`, `Drifting`, `History.Selected`.

Proposition III.2.4 (the joint purchase on one emitted factor).

Hypotheses. Assume:

- G is simple;
- K is a kernel choice with branches at every patch;

- order data v are given:
  - a first stage ≥ 4;
  - derivative orders $k_j \geq \max(6, j)$ and powers $q_j \geq \max(2, j)$, both nondecreasing;
  - support exponents a_j, nondecreasing and ≥ 100;
- $Q : \mathbb{N} \to \mathbb{N}$ is nondecreasing with $Q \geq 1$;
- the records Ω (`Contemple`) and Δ (`Decal`) are given, together with parameters $(\xi, \omega_v, \omega_t)$ and a degree-300 interpolation tableau θ_1 at parameter ξ (`Tableau`), and the contraction condition `sphere` at these parameters; in particular $\xi \geq 2$, $0 < \omega_v \leq 1$ and $\omega_t > 0$;
- π is a predicate on emitted factors (a *prediction*) that is prepaid at every sheet and price map.

Conclusion. There exist:

- a sheet s with kernel K and the first stage, orders, powers and support exponents prescribed by v;
- a price map T;
- a real sequence Y and a nondecreasing real sequence $K' \geq 1$;
- a floor $\tau \geq f^s$.

These are followed by the alternation below:

> for every floor sequence $k \geq \tau$ (block factors exist for the sheet s_k, which is s with floors k), and every choice of stage block factors ℓ for s_k, there is a wage threshold U such that for every wage $w \geq U$ and every prescribed postage rate and exchange datum, there are post-L cost constants F such that for every positive sequence θ^0 there is a smallness exponent $\theta \leq \theta^0$ such that for every prescribed menu, auxiliary record and delay T_0 there is a delay $V \geq T_0$ such that for every delay $V' \geq V$, every barrier $\bar{\beta}$ for V' and every wait W, there is a wait $Z \geq W$ such that for every wait $W' \geq Z$ and every level system a that meets the entrance conditions of W' (III.2.2(a)–(d), with (e)–(f)) and whose periods have the divisibility property, there exist a directed datum B on a (kinetic data and a recipe on the level system a) and an emitted factor X such that: - X records all these choices; - $\pi(X)$ holds; - all requirements listed below hold.

The requirements (with the unit that defines each predicate) are:

requirement	owner
wages, exchanges, menus and auxiliary records equal the prescribed data; postage rates do not exceed the prescribed rates	III.10 (`Factor`)
`Degree`, `Savvy`, `Roofing`, `PaysW`, `Spire`, `Wild`, tower bound `Tower.Sav` with `Canyon`, `Cadre`	III.12
`AccordV` (with the sequence Q)	III.11
kinetic data at the top level (`AtTop`)	III.9
amplification (`Amplify`)	III.1
support exponents ≥ 100 at every level	this unit
$\exp((a_j + 2)(2\,\mathrm{span}_{ij} + 13)^4) \leq \ell_i$ for all $i, j \in \mathbb{N}$ with $j \leq i \leq j + 1$	this unit
a bandage b (`Bandage`, III.10) with the extras (Ω, Δ, b) (`Extras`)	III.15

Here $\mathrm{span}_{ij} \in \mathbb{N}$ is the factor's transition support parameter; the exponent is given by `base` and `toll`.

The bandage also supplies the conditional family package of `Unlatched` with a vacuous additional conclusion; the family hypotheses are those stated in III.2.5.

Status: theorem. The estimates hold on the same emitted factor, with the displayed order of choice. The hypotheses Ω, Δ, θ_1, $(\xi, \omega_v, \omega_t)$ and v are constructed in the final assembly III.2.7, or supplied there by III.14. The version without a prediction is `richFares`. *Lean:* `richFaresWithPrediction`, built on `joinRadarsShipWithPrediction` and `joinExpensesShipWithPrediction`.

Proposition III.2.5 (the plentiful construction). Assume the hypotheses of III.2.4 and $Q_i \geq 2$. Then its conclusion holds with the following additional clause. Consider any trajectory compatible with the emitted factor, any reference levels $t(d)$ with $d+3 \leq J_{t(d)}$, any section-transfer data and any gauge-door data between consecutive levels. Assume that the resulting family satisfies the family-consistency conditions, that its gauge doors satisfy the step compatibility conditions, and that its section transfers satisfy the prescribed bounds from level $t(d)$ whenever $t(d) \leq N$. Then, for every extraction order f, the family satisfies the complete weak-continuum requirement (`CompleteWeak`, III.11) and the trace gallery (`Gallery`, VII.2). These are the hypotheses and additional conclusion of `Unlatched`. The floors, delays and waits may increase, and the prediction is retained. *Status:* theorem. This is where the requirements of the trace test (Part VII) are bought on the same constants as the ultraviolet estimates. *Lean:* `plentifulWithPrediction`, `plentiful` (vacuous prediction).

Proposition III.2.6 (a prediction on the same factor). Let G be simple, let s carry the estimate package of Lemma 4.16 for fixed tariff and scope, and let $\tau > 0$. Let π be a prediction that is *prepaid* at the price map of s, in the following sense.

- For every choice of targets with the canonical chart, expenses with prescribed deficit $-7/16$, programs, initial wages and baton, there is a length floor u.
- For every floor $l \geq u$, every choice of block factors, wages, duration bounds, register, postage, scouts, protection, step records and post-L costs, and every delay V, there is a delay $W \geq V$. The step records must have $\varepsilon \leq 1/4$ on transitions whose output stage is at least the first stage.
- For every delay $W' \geq W$, every barrier, wait, pricing record and directed datum, and every choice of the remaining factor data, π holds on the assembled factor if its shelter and step conditions hold and its periods have the divisibility property.

Then the complete native purchase of Lemma 4.17 can be carried out so that, on level systems with divisible periods, π holds on the factor it emits. The original floor is replaced by its maximum with u. The delay W is chosen before the original booth and final delay. *Status:* theorem. In the main-theorem closure it is used only with the vacuous prediction (`Always`): the only caller of III.2.7 passes it. *Lean:* `company`, `Prepaid`, `Prediction`, `AllPrices`.

Proposition III.2.7 (the complete run).

Inputs. Let G be simple and $b > 0$ (the kernel penalty). Let the following records be given: Ω, Δ and a custody request $\mathcal{H}$. Let Λ be a property of level systems that is *eventually paid* (III.2.8(a)). Let a *demand* be given:

- for every sheet s and price map q, a nonempty set of choices c;

- for each choice c, a floor $e_c : \mathbb{N} \to \mathbb{N}$;
- a real array $F_c(n, i)$;
- a real array $F'_c[\ell](https : //github.com/scottnarmstrong/yang - mills - informalization/blob/main/outline/units/n, i)$, which may depend on the block-factor sequence ℓ.

Conclusion. There is a run $\mathcal{R}$ (the record of the whole construction, III.12) with a campaign such that:

- (a) the kernel penalty is b;
- (b) *preselection:* there are s, q, c and a floor $n \geq \max(f^s, e_c)$ such that:
  - $\mathcal{R}$ uses the sheet s with floors n and the price map derived from q;
  - both arrays are paid in the sense of III.2.8(b) on the level system of $\mathcal{R}$, with F' evaluated at the actual block factors $\ell^{\mathcal{R}}$;
  - $F'_c[\ell^{\mathcal{R}}](https : //github.com/scottnarmstrong/yang - mills - informalization/blob/main/outline/units/0, J_{m+1}) \leq \hat{\beta}_m$ for all m;
- (c) $k_j \geq j$ and $q_j \geq j$ at every stage;
- (d) Λ holds;
- (e) the treasury of III.2.8(d) is funded;
- (f) the common-period purchase is funded;
- (g) the campaign has pillars, the funded, news and read-out records hold, and the custody and sky clauses hold.

There is also a version in which any prediction prepaid at every sheet and price map holds on the factor of the run; the intermediate factor theorem below supplies it. *Status:* theorem. Its only caller in the main closure is III.14 (`constructionRun`), which passes:

- $\Lambda \equiv$ true;
- the demand of its native component recipe: a floor and the pre-L and post-L arrays;
- concrete $\Omega, \mathcal{H}, \Delta$;
- the vacuous prediction.

Lean: `selectRunWithTreasuryPeriod_preselected_quote_orders_postnumeric`, `PostnumericPreselection`, `allFactorWithTreasuryPeriod_preselected_quote_orders_postnumeric_of_pred`

Proposition III.2.8 (purchases that persist under all later choices).

(a) Eventually paid properties. A property U of level systems is *eventually paid* if the following holds. For every sheet, block factors, smallness exponent and delay V there is a delay $X \geq V$ such that every level system meeting the entrance conditions of any delay $W \geq X$ (with any barrier and wait) satisfies U. A conjunction of two eventually paid properties is eventually paid.

(b) Diagonal payment of constants. For every real array $F(n, i)$, the following property is eventually paid. At every level m, $1 \leq 1 + \log\log\hat{\beta}_m$. For every n there is m_n such that

$$|F(n, i)| \leq 1 + \log\log\hat{\beta}_m \qquad \text{for all } m \geq m_n \text{ and all } i \leq J_{m+1}.$$

Likewise, for every $F : \mathbb{N} \to \mathbb{R}$, the property "$F(J_{m+1}) \leq \hat{\beta}_m$ for all m" is eventually paid.

(c) Countably many step requests. Fix a countable index set with an encoding in $\mathbb{N}$, and let (w_α) be a family of requests indexed by it. Each request consists of a floor $\mathrm{pre}_\alpha : \mathbb{N} \to \mathbb{N}$ and a cost function of a stage transition (j, i), its block geometry and a stage record; at step m the record is

X_{J_m}. Then there is a single floor pre, depending only on the family, with the following property. If $\ell_j \geq \mathrm{pre}(j)$ for all j, then every delay can be raised so that every later level system satisfying the entrance conditions for the enlarged delay, with any barrier and wait, satisfies, for all α and n with $\mathrm{code}(\alpha) \leq J_n$ and all $m \geq n$:

- $\mathrm{pre}_\alpha(J_{m+1}) \leq L_m$;
- the cost of step m is at most $1 + \log(2 + \log \hat{\beta}_m)$.

(d) The treasury. For simple G and every kernel choice there is a *plan*. A plan fixes a naming of charges and, for every request, a dictionary of finite words and spectra. If the block factors exceed the plan's floor, then every delay can be raised so that every later level system satisfying its entrance conditions is *treasury funded*: it meets the step requests of the plan (as in (c)) and the inverse charge from level 0 on.

Status: theorems. Dictionaries are chosen once, before the block factors, and never reselected. *Lean:* `EventuallyPaid`, `postnumeric_quote_diagonal_pair_requests`, `eventuallyPaid`, `PaidRequests`, `StagePaid`, `minDelay`, `Need`, `Affords`, `Plan`, `plan_exists`, `buy`, `TreasuryFunded`.

Proposition III.2.9 (analytic estimates used by the purchases). The row and cross norms are componentwise exponentially weighted ℓ^1 bounds on displacement kernels. Multilinear kernels use the corresponding sum over displacements and colour indices. The colour spaces are fixed and finite; constants may depend on them. A cross maps coarse fields to fine fields at block size L. All vanishing orders refer to the Fourier symbol at zero momentum.

- *Pullback gain.* For $\sigma, \tau > 0$, $C \in \mathbb{R}$ and $d \in \mathbb{N}$, there are $0 < \eta \leq \sigma$ and $K \geq 0$ such that, for every positive odd integer L with $\tau \leq L\eta$, the following holds. If $\|A\|_\tau \leq C/L$, $\|\nabla^s B\|_\tau \leq C/L^{1+|s|}$ for every list of coordinate differences with $|s| \leq d$, and R has $\|R\|_\sigma \leq M$ with $M \geq 0$ and vanishes to order d, then $\|A^{\mathsf{T}} R B\|_\tau \leq K M L^{2-d}$. `pullback_gain`

- *Contraction of Hessian differences.* For $\sigma, \tau > 0$ and $C \in \mathbb{R}$, there is $N \in \mathbb{N}$ such that, for every positive odd $L \geq N$, crosses A, B obeying the preceding bounds through order 3, and a row R on one-form components with $M \geq 0$, $\|R\|_\sigma \leq M$ and vanishing order 3, one has $\|A^{\mathsf{T}} R B\|_\tau \leq M L^{-1/2}$. `large_contraction`

- *Soft transfer of multilinear kernels.* For $r, d \in \mathbb{N}$, $\sigma, w > 0$ and $C \geq 0$, there are $0 < \eta \leq \sigma$ and $K \geq 0$, fixed before L, with the following property. Let F be a real $(r+1)$-linear coefficient kernel with width σ and bound B, supported on tuples whose first site is the origin, and vanishing to order d in the independent momenta. Let L be positive and odd with $(r+2)w \leq L\eta$, and let Q satisfy $\|\nabla^i Q\|_{(r+2)w} \leq C/L^{1+|i|}$ for every list $|i| \leq d$. The transfer through $r+1$ copies of Q, followed by symmetrization and recentering, has width w and bound KBL^4/L^{r+1+d}. `flow_soft`

- *Division by differences.* Fix a finite coordinate set I, $n \in \mathbb{N}$ and $\sigma > 0$. There are $0 < \eta \leq \sigma$ and $K \geq 0$ such that, for every $A \geq 0$ and real kernel $f : \mathbb{Z}^I \to \mathbb{R}$ with $\sum_x e^{\sigma|x|_1} |f(x)| \leq A$ whose symbol vanishes to order n, there are real kernels g_b, indexed by $b : \{1, \dots, n\} \to I$, with $\sum_x e^{\eta|x|_1} |g_b(x)| \leq KA$ and

$$\hat{f}(p) = \sum_b \left(\prod_{a=1}^{n} (e^{i p_{b(a)}} - 1) \right) \hat{g}_b(p)$$

  for every real momentum p. `real_difference_division`

- *Odd-part cancellation.* Fix a sequence of step scenes whose input coupling tends to infinity, leaves M_i, a cutoff parameter χ, chart radii satisfying the logarithm-chart conditions, and fluctuation data z_i. Let $H_i = \frac{1}{2}\log \beta_i^{\mathrm{out}}$ and $\eta_{*,i} = e^{-H_i}$. Assume that the fluctuation bounds, the leaf weights and the third-derivative bounds D_i have upper exponential rate 0 relative to H_i; that eventually each leaf is a logarithmic term given by a masked regular activity whose expression in logarithmic coordinates, multiplied by the square of its threshold, has zero first derivative at 0 and derivatives through order 3 bounded by D_i on the unit ball; that both z_i and $-z_i$ lie in the healthy domain; and that both cutoffs equal 1 near the origin in the two deformation parameters. If $P_i(z_i; y)$ denotes the paved expansion, then, for every $\varepsilon > 0$, eventually
$$\left\| D_y^2 \left(P_i(z_i; y) + P_i(-z_i; y) \right) \big|_{y=0} \right\| \le \eta_{*,i}^{2-\varepsilon}.$$
Here an upper rate a means an eventual bound e^{rH_i} for every $r > a$. `mirror_price`

- *Uniqueness of local read-out.* Fix a representation R, a degree-d frame and a section readable at the origin for the budget of R. Let P, Q be coefficient presentations in the prescribed format for the same R: their regular terms have supports containing the origin, vanish at the identity and have zero local degree-d projection on nonwrapping supports; their nonzero forced terms have supports containing the origin and satisfy the eligibility, support, gauge-invariance and measurability conditions. Suppose they have equal densities for this section and remainder bounds a, b with the same positive forced-term parameter. For every $m \in \mathbb{N}$ with $10(m(4B+2)+2B) < n$, their coefficient vectors satisfy
$$\|x_P - x_Q\|_{\mathrm{coeff}} \le K_{F,H} e^{-a_* m}(a+b),$$
where n is the torus side, B and a_* are the cover size and support exponent of R, and $K_{F,H}$ depends on the frame and holonomy tests. `unique_part`

Further linked estimates concern a post-block Hessian bound (`design`), extraction of cubic and quartic local coefficients (`tanker`), and asymptotic bounds for the trace insertion (`cloche`). *Status:* theorems, with the hypotheses above. The detailed scene, coefficient-format and norm interfaces are listed among the open points in section 7.

3.5.2 2. What it uses

From other units.

- I.1: $\kappa > 0$ for simple G (`slopeConstant_pos`).
- III.1 (interfaces):
  - the scale sequence (`Schedule`), the calendar, its conditions and level systems (`Calendar`, `Fits`, `Levels`);
  - histories, transfer and selection (`History`, `Transfer`, `Selected`);
  - block control from the kernel's branches (`blocks_go`);
  - the directed datum (`Directed`).
- III.9: the reference couplings, threshold and band (III.9.2); the meaning of a selected history (III.9.3); the Wilson start-up.
- III.10:
  - the verification of the shooting hypotheses (`Graces.winnow`);
  - the divisible-period schedule (`Wait.engine`);
  - the emitted factor and bandage records.

- III.9, III.11, III.12, III.15, VII.2: the predicates bought in III.2.4–III.2.5 (table in III.2.4) and their suppliers. This unit proves only that they hold on one common choice of constants. Their analytic content belongs to those units.
- III.12: the run record (`Run`). III.14: the demand passed to III.2.7.
- III.3, III.6 (probable): the free-kernel and stationary-branch estimates from which the sheet is built. The two entry points, `reapsY` and `verdant`, are themselves filed in this unit.
- III.16: strip and Laurent norms, Taylor jets (for III.2.9).

Standard mathematics.

- The intermediate value theorem, in a refined form (Lemma 4.11).
- Elementary asymptotics: $(\log x)^2 = o(x)$; polynomials in $\log s$ are $o(e^s)$.
- Diagonal enumeration of countably many requests.
- For background on the order of choice and bare-coupling tuning in constructive RG:
  - Choosing constants in a fixed order: first the constants of the step, then the block factor L large depending on them, then the coupling large depending on L. See Bałaban's papers on lattice gauge theories (Comm. Math. Phys. 1983–1989) and Bauerschmidt–Brydges–Slade, *Introduction to a renormalisation group method* (LNM 2242, 2019).
  - Tuning the bare coupling as a function of the cutoff so that the effective coupling at unit scale is fixed. Compare Gawędzki–Kupiainen on the Gross–Neveu and ϕ^4_4 models (Comm. Math. Phys. 99, 102, 1985).

  Here the tuning is a nested-interval argument on continuous families (§4.3).

Imported constants and hypotheses.

- κ enters only through $\kappa > 0$ and through the form $\hat{\beta}_m = \bar{\beta} + \kappa s_m$. The shift of β by $\kappa \log L$ up to error ω is part of the (advance) hypothesis of III.2.3, supplied in III.10.
- The hypotheses (start) and (advance) of III.2.3.
- In III.2.7, the records Ω, $\mathcal{H}$, Δ and the demand come from III.14. Inside the assembly, the kernel with penalty b comes from III.1 (`Axial.viable`). The tableau and sphere data are constructed in the assembly.

3.5.3 3. Strategy

Choose the stage records and floors first, then the block factors ℓ_j, post-L costs and smallness exponents, delays, the barrier $\bar{\beta}$, and finally the wait. Increasing the appropriate threshold meets each requirement: the block factors dominate the stage constants, entry couplings pay later costs, and the wait provides comparison windows and large tori. Persistent requirements combine by taking maxima of floors and chaining delay and wait increases. This gives common constants for III.2.4–III.2.7.

The stage index J_m advances by at most one per level. It enters a stage only after its entry thresholds are met and stays there long enough for one prescribed comparison window. At each fixed stage the volume bill is bounded by a polynomial in $\log s$, hence eventually by e^s. Since $s_m \geq m \log 3$, this pays every later volume demand. One period offset pays the finitely many steps of the first plateau.

For fixed N, p, start with the Wilson family over $[\bar{\beta} + (\kappa/2)s_N, \bar{\beta} + 2\kappa s_N] \subset \mathfrak{B}_N$. At each step the drift bound sends the endpoint couplings strictly beyond the endpoints of the next window. A

refined intermediate value argument selects a smaller parameter interval on which the new family stays in that window, so the next step applies. At level 0 the window is $\{\bar\beta\}$. Summing the drifts upward from there and using $\omega_{J_{m+1}} \le \log L_m / J_{m+1}^2$ gives the running law uniformly in depth and volume.

3.5.4 4. Steps

3.5.4.1 4.1 Numerical existence lemmas In sections 4.1–4.2, G is simple and the data are those of III.2.1.

Lemma 4.1 (large block factors). For each stage i there is M_i such that every odd $n \ge M_i$ satisfies the conditions of III.2.1(2) at stage i. Hence the stage block factors $\ell_i = 2\max_{j\le i} M_j + 1$ exist. *Sketch:* below the first stage only oddness and the size floors are required. At later stages, the remaining conditions have the form $c/\log n \le \delta$ with $\delta \in \{\kappa/4, i^{-2}\}$ and $\delta > 0$. Each holds for large n because $\log n \to \infty$. The running maximum makes ℓ nondecreasing. *Lean:* `Rolls.eventually_large`, `Rolls.exists_rolls`.

Lemma 4.2 (smallness exponents). For every positive sequence α, the running minimum $\theta_j = \min_{i\le j} \min\left((i+1)^{-1}, \alpha_i\right)$ is positive, nonincreasing, and satisfies $\theta_j \le \min(\alpha_j, 1/(j+1))$. *Lean:* `Protection.exists_margin`.

Lemma 4.3 (barrier). For every b, take $\bar\beta = \max(2, x_*, b, \beta_{j_0}^{\mathrm{ent}})$, where $x^{-1/2} e^{(\log\log x)^2} \le 1$ for all $x \ge x_*$. This is a barrier. *Sketch:* $(\log y)^2 \le \frac{1}{2} y$ for large y, applied with $y = \log x$. *Lean:* `Bar.exists_above`.

Lemma 4.4 (the volume bill). Define

$$\mathrm{bill}_j(s) = S_j(1 + c + c')^{q_j} + 100(c+1) + 100((j+3)\ell_j + 1),$$

where $c = \lceil \log^2(\bar\beta + \kappa s)\rceil$ and $c' = \lceil \log^2(\bar\beta + \kappa(s + \log \ell_j))\rceil$. For each fixed j, $\mathrm{bill}_j(s) \le e^s$ for all sufficiently large s. At $s = s_m$ with $j = J_{m+1}$ we have $c = B_m$, $c' = B_{m+1}$ and $(j+3)\ell_j = R_m$. So the bill is the sum of the three volume demands of III.2.2(c),(e). *Sketch:* $c, c' = O((\log s)^2)$, so the bill is bounded by a polynomial in $\log s$. *Lean:* `Bar.bill`, `Bar.allocated`.

Lemma 4.5 (entering a stage). For each $j > j_0$ there is n_j with the following property. For every level $m \ge n_j$ and every $s \ge m \log 3$:

- $\beta_j^{\mathrm{ent}} \le \bar\beta + \kappa s$;
- $\sigma_j^{\mathrm{ent}}(\bar\beta) \le s$;
- $s > 0$;
- $\log \ell_j \le s/j$;
- $\mathrm{bill}_j(s) \le e^s$.

Sketch: each condition holds for all large s. Take the largest of the thresholds, and n_j with $n_j \log 3$ above it. *Lean:* `Bar.enters`.

Lemma 4.6 (windows). For each j there is c_j such that, for all $k \ge c_j$ and all s with $k\log 3 \le s \le k \log \ell_j$, we have $\sigma_j^{\mathrm{rdy}} \le s$ and $h_j(s) \le k/2$. *Sketch:* for large s, $h_j(s) \le s/(2\log \ell_j)$ by sublinearity, and $s \le k \log \ell_j$. *Lean:* `Wait.lead`.

3.5.4.2 4.2 The level schedule Lemma 4.7 (exhaustion). For any $b, c : \mathbb{N} \to \mathbb{N}$ there is $J' : \mathbb{N} \to \mathbb{N}$ with the following properties:

- $J'_0 = J'_1 = 0$, J' is nondecreasing, $J'_{i+1} \le J'_i + 1$ and $J' \to \infty$;
- $b(J'_{i+1}) \le i$ whenever $J'_{i+1} > 0$;
- for every i there is $k \in \mathbb{N}$, $k \ge c(i)$ with $J' \equiv i$ on $[k - \lfloor k/2 \rfloor, k + \lfloor k/2 \rfloor]$.

Sketch: set $a_0 = 0$ and $a_{i+1} = a_i + b(i+1) + 2k_i + 1$, where $k_i = c(i) + 2a_i + 1$. Put $J' = i$ on $[a_i, a_{i+1})$. Stage $i+1$ starts at level $a_{i+1} > b(i+1)$. The window around k_i lies in $[a_i, a_{i+1})$ because $k_i/2 \ge a_i$ and $\frac{3}{2}k_i < a_{i+1}$. *Lean:* `Program.exhaust`.

Lemma 4.8 (the schedule meets all entrance conditions; proof of III.2.2).

Construction.

1. Apply Lemma 4.7 with $b(0) = 0$ and $b(i) = n_{j_0+i}$ for $i > 0$ (Lemma 4.5) and $c(i) = c_{j_0+i}$ (Lemma 4.6), and set $J_m = j_0 + J'_m$, $L_m = \ell_{J_{m+1}}$ and $s_m = \sum_{i<m} \log L_i$. Then $m \log 3 \le s_m \le m \log \ell_{J_m}$, using that ℓ and J are monotone.
2. Let m_* be such that $J'_m \ge 1$ for $m \ge m_*$. Put $M = n_{\min} + \max_{i \le m_*} \mathrm{bill}_{j_0}(s_i)$ and $n_p = 4(M + p + 1)$.

Verification.

- *Entry.* If $J_{m+1} = j_0$, then $\beta^{\mathrm{ent}}_{j_0} \le \bar\beta \le \hat\beta_m$. Otherwise $m \ge n_{J_{m+1}}$ and $s_m \ge m \log 3$, so Lemma 4.5 applies.
- *Volume off the first plateau.* There $\mathrm{bill}_{J_{m+1}}(s_m) \le e^{s_m} \le n_p e^{s_m}$. This gives (c), the cover condition, and the patch condition, the latter because sides increase with m.
- *Volume on the first plateau.* There $\mathrm{bill}_{j_0}(s_m) \le M < n_p$.
- *Mesh.* For $m \ge m_*$, Lemma 4.5 gives $\log L_m / s_m \le 1/J_{m+1} \to 0$.
- *Windows.* At the level k given by Lemma 4.7 for stage $j = j_0 + i$, s_k lies in the range of Lemma 4.6. So $h_j(s_k) \le k/2$, and $J \equiv j$ on $[k - h_j(s_k), k + h_j(s_k)]$.
- *Remaining conditions.* Monotonicity of $k_{J_m}, q_{J_m}, a_{J_m}$ comes from the sheet. The drift and log-cost bounds come from Lemma 4.1, and the bounds on θ from Lemma 4.2. Block control comes from the kernel's branches.
- *Divisibility variant.* Given $c \ge 1$ and z, the index $p = c(M + z + 1) - (M + 1) \ge z$ gives $n_p = 4c(M + z + 1)$.

Lean: `Wait.ready`, `Program.book`, `scaled_lower`, `scaled_upper`, `Wait.engine`.

3.5.4.3 4.3 Shooting Lemma 4.9 (running law from the transfer inequality). Consider a history on a level system of III.2.2 with $\beta_{N,p,0} = \bar\beta$ and $|\beta_{N,p,i+1} - \beta_{N,p,i} - \kappa \log L_i| \le \omega_{J_{i+1}}$ for all $i < N$. Then it obeys the running law of III.2.3. *Sketch:*

1. Since $\hat\beta_{i+1} - \hat\beta_i = \kappa \log L_i$ and $\beta_{N,p,0} = \hat\beta_0$, induction upward from level 0 gives $|\beta_{N,p,i} - \hat\beta_i| \le D_i := \sum_{k=1}^{i} \omega_{J_k}$.
2. Given $\varepsilon > 0$, choose i_0 with $J_{i+1}^{-2} \le \varepsilon/2$ for $i \ge i_0$. Then $\omega_{J_{i+1}} \le \frac{\varepsilon}{2} \log L_i$, so $D_i \le D_{i_0} + \frac{\varepsilon}{2}(s_i - s_{i_0})$.
3. Finally $D_{i_0} \le \frac{\varepsilon}{2} s_i$ for large i, because $s_i \to \infty$.

The bound is uniform in $N \ge i$ and p. *Lean:* `trajectory`.

Lemma 4.10 (gates). Put $\beta_m^- = \bar{\beta} + \frac{\kappa}{2} s_m$ and $\beta_m^+ = \bar{\beta} + 2\kappa s_m$. Then $[\beta_m^-, \beta_m^+] \subset \mathfrak{B}_m$, because $\bar{\beta} > 0$. Suppose a level-m representation ρ relays a level-$(m+1)$ representation ρ', and $\omega_{J_{m+1}} \le \frac{\kappa}{4} \log L_m$. Then:

- if $\beta(\rho') = \beta_{m+1}^-$, then $\beta(\rho) \le \beta_m^- - \frac{\kappa}{4} \log L_m < \beta_m^-$;
- if $\beta(\rho') = \beta_{m+1}^+$, then $\beta(\rho) \ge \beta_m^+ + \frac{3\kappa}{4} \log L_m > \beta_m^+$.

The drift hypothesis follows from the entrance conditions and III.2.1(2), since $L_m = \ell_{J_{m+1}}$. *Lean:* `floorBeta`, `ceilBeta`, `limits`, `gates`, `Drifting.pay`.

Lemma 4.11 (narrowing). Let $x \le y$ and let f be continuous on $[x, y]$ with $f(x) \le d \le u \le f(y)$. Then there are $x \le z \le w \le y$ with $f(z) = d$, $f(w) = u$ and $f([z, w]) \subset [d, u]$. *Sketch:* let w be the least point of $f^{-1}(u) \cap [x, y]$ and z the greatest point of $f^{-1}(d) \cap [x, w]$; both sets are nonempty by the intermediate value theorem and compact by continuity. Suppose some $c \in [z, w]$ has $f(c) < d$. Then the intermediate value theorem on $[c, w]$ gives a point of $f^{-1}(d)$ after z, a contradiction. Suppose instead $f(c) > u$. Then it gives a point of $f^{-1}(u)$ before w, again a contradiction. *Lean:* `narrow`.

Lemma 4.12 (nested intervals). Fix N, p and assume (start), (advance) and the drift bound. One can choose intervals $I_m = [l_m, u_m]$, $0 \le m \le N$, with $I_N = [\beta_N^-, \beta_N^+]$ and $I_m \subset I_{m+1}$ for $m < N$, and partial trajectories $(\rho_i^x)_{m \le i \le N}$ for $x \in [l_m, u_m]$, such that:

- ρ_N^x is the start representation at bare coupling x;
- every ρ_i^x is admissible with $\beta(\rho_i^x) \in \mathfrak{B}_i$;
- consecutive levels relay;
- $x \mapsto \rho_m^x$ is continuous;
- $\beta(\rho_m^{l_m}) = \beta_m^-$ and $\beta(\rho_m^{u_m}) = \beta_m^+$.

Sketch: downward induction.

- For $m = N$, restrict the start family to $[\beta_N^-, \beta_N^+]$.
- For $m + 1 \to m$, the family at level $m + 1$ has all couplings in $\mathfrak{B}_{m+1}$, so (advance) yields a continuous family ρ_m^x. By Lemma 4.10, $f(x) = \beta(\rho_m^x)$ satisfies $f(l_{m+1}) < \beta_m^-$ and $f(u_{m+1}) > \beta_m^+$.
- Lemma 4.11 gives $[l_m, u_m]$ on which f maps onto $[\beta_m^-, \beta_m^+] \subset \mathfrak{B}_m$ with the right endpoint values.

The restriction is needed: (advance) applies only to families whose couplings lie in the band at every point. Only the continuity of $x \mapsto \beta(\rho_m^x)$ enters the intermediate value argument. The other continuity clauses are carried because (advance) requires them at the next step. *Lean:* `steps`.

Lemma 4.13 (endpoint and assembly; proof of III.2.3). Since $s_0 = 0$, we have $\beta_0^- = \beta_0^+ = \bar{\beta}$. So $x = l_0$ gives a trajectory from level N to level 0 with $\beta_{N,p,0} = \bar{\beta}$. Choose one such trajectory for every (N, p). Then:

- the path data are $P_{N,m}^{\mathcal{R}}$, independent of p;
- admissibility gives the budget, clip, polymer bounds and band membership;
- (start) gives the Wilson weight with $v = 0$;
- the relays give the transfer identity and the drift inequality;
- Lemma 4.9 gives the running law.

Hence the history is selected. *Lean:* `successful`, `Competes`, `Recipe.races`.

3.5.4.4 4.4 Persistent purchases and their composition

Lemma 4.14 (composition rules). Persistent numerical requirements can be combined in the following forms:

- (i) a floor f such that the requirement holds for every later block factor;
- (ii) "for every delay V there is $V' \geq V$ such that every later delay, every barrier and wait, and every level system meeting those entrance conditions satisfy P";
- (iii) the same with waits instead of delays.

Floors combine by pointwise maximum. Suppliers of forms (ii) and (iii) combine by chaining, because the orders on delays and waits are transitive and the conclusions persist under raising. *Lean:* `postnumeric_quote_diagonal_pair_requests`, and the chains in `joinExpensesShipWithPrediction`.

Lemma 4.15 (diagonal payments; proof of III.2.8).

- For (b), raise β_j^{ent} to $\max\left(\beta_j^{\mathrm{ent}}, \exp\exp(1 + \sum_{n,i\leq j} |F(n,i)|)\right)$. The entrance condition $\beta_{J_{m+1}}^{\mathrm{ent}} \leq \hat{\beta}_m$ then bounds the partial sums at stage J_{m+1} by $\log\log\hat{\beta}_m$. Since $J \to \infty$, each fixed n satisfies $n \leq J_{m+1}$ from some level on.
- For the single-array form, raise β_j^{ent} to $\max(\beta_j^{\mathrm{ent}}, F(j))$.
- For (c), encode the requests. Take $\mathrm{pre}(j) = \max_{\mathrm{code}(\alpha)\leq j} \mathrm{pre}_\alpha(j)$. Raise β_j^{ent} until the finitely many costs with $\mathrm{code}(\alpha) \leq j$, evaluated on the stage-j geometry, are below $1 + \log(2 + \log x)$ for $x \geq \beta_j^{\mathrm{ent}}$.
- For (d), fix the plan before the block factors, apply (c) to its requests, and raise the delay once more for the inverse charge, whose block floor is zero.

Lean: `eventuallyPaid`, `minDelay`, `buy`.

Lemma 4.16 (the sheet). Let G be simple, let K have branches at every patch, let the order data v satisfy the order conditions of III.2.4, and fix a sequence $C_j \geq 1$. There are a sheet s with kernel K and the prescribed first stage, orders, powers and support exponents; two price maps O, T; real sequences λ, γ with $\gamma_j > 0$; and a nondecreasing sequence $K'_j \geq 1$. For every tariff and scope there is a floor $f \geq f^s$ with the following properties.

The sheet satisfies the stationary-path suitability conditions at stages $j \geq j_0$ and the budget plans for C, K'. For every floor $n \geq f$ and every admissible block-factor sequence ℓ, there is a sequence η satisfying the stationary budget-gap condition. For every post-L cost record F with $\eta_F = \eta$, $\Gamma_F \geq 1$ and $c_F \leq \gamma$, every protection, additional tariff, nonnegative lower bound z and delay A, there are a delay $D \geq A$, a transition-bound array bounded below by z, and two further transition arrays. Every later delay $E \geq D$ satisfies the pricing, ringing and comparison conditions. For every barrier, wait and level system meeting those entrance conditions, the budget and multisource conditions hold, and there are a Gaussian horizon and kinetic data with planned stationary paths, the Wilson top path, the cadre and amplification conditions, and the retained free and stationary data specified by the perceived-estimate package.

Sketch: the free-kernel and stationary suppliers first fix widths and preliminary bounds. The sheet construction fixes clips and the prescribed orders; the startup suppliers enlarge path bounds and floors. The final profile construction raises drift allowances to dominate the stationary supplier's bounds and raises column bounds without changing widths, paths or the prescribed orders. After choosing ℓ, the stationary and multisource suppliers raise the delay and construct the kinetic data on each admitted level system. The retained data identify these kinetic paths with the revised stationary paths. *Lean:* `sailsW`, `Perceived`. *(proof read at the level of the composition)*

Lemma 4.17 (the native purchase and prediction). Let G be simple, fix the perceived-estimate package of Lemma 4.16 at a floor v, and let $\tau > 0$. There are a floor $j \geq v$, programs, initial wages, targets, expenses and a baton, chosen before the block factors. The targets have the canonical chart and halo 500; the expenses have the prescribed earnings, deficit $-7/16$ and degree charge 0.

For every floor $l \geq j$, admissible block factors exist. For every such choice there is a wage threshold. For every wage above it there are duration bounds, a register and auxiliary size bounds. For every postage cap there are postage rates at most that cap and scouts satisfying the register condition. For every exchange datum and tariff there are reset prices, reset bounds and post-L costs. For every positive sequence θ^0 there are protection $\theta \leq \theta^0$ and step records with $\varepsilon \leq 1/4$ on transitions whose output stage is at least the first stage. For every menu and initial delay there are booths; for every auxiliary record there is an ordered delay above the initial delay.

Every later delay, barrier, wait and admitted level system satisfies the budget and multisource conditions and carries a directed datum on that same system. Its shelter conditions hold at the two pairs of tolerances $(1/16, 1/16)$ and (τ, τ), with the third tolerance $1/8$ in both cases. It also has the top kinetic data, cadre, amplification, step conditions, origins and rank-covariance data. The offering condition additionally assumes divisible periods.

Sketch: the shelter supplier gives the initial floor and data. A further floor pays the register's length requirement; a delay supplier supplies the step conditions before the booths are chosen. The shelter output then supplies the remaining data. For III.2.6, combine the prediction's floor u with j by maximum. Insert its delay before the booth delay; persistence then gives the prediction on the same emitted factor whenever the divisibility condition holds. *Lean:* `companyK` *(proof read at the level of the composition)*; `company`.

Lemma 4.18 (proof of III.2.4 and III.2.5). The proofs proceed in four steps.

1. `joinRadarsShipWithPrediction` takes the sheet from Lemma 4.16 and calls `joinExpensesShipWithPrediction`. That theorem starts from Lemma 4.17 and threads about thirty further suppliers through it:

   - floor suppliers, joined into one floor;
   - delay suppliers $V \leq V_1 \leq \ldots \leq V_7 \leq \ldots \leq w_6 \leq \ldots$, chained.

2. `richFaresWithPrediction` adds:

   - additional costs and floors before choosing the block factors;
   - a cross-term floor, so the final floor is $(t \vee L \vee \mu) \vee$ cross-floor;
   - shells raised to $\max(S_j, \cdot)$, the mesh and installment delays, and a wait enlargement.

3. `plentifulWithPrediction` adds:

   - trace floors and the fixed trace-control floor;
   - trace delays, including a raise of the entry couplings;
   - two wait enlargements.

   It concludes with `Trace.Over.gang`.

4. Floors are combined by maximum and delay and wait bounds by transitivity, as in Lemma 4.14. The final family conclusion uses the hypotheses of III.2.5.

(proof read at the level of the composition; the suppliers were not read)

Lemma 4.19 (the final assembly; proof of III.2.7). `allFactor…` makes the choices in order:

1. *Pre-L data.*
   - Order data: $j_0 = 4$, $k_j = 6 + j$, $q_j = 2 + j$, $a_j = 100$; and $Q_i = i + 2$.
   - A kernel with penalty b.
   - A tableau at a parameter $\xi \geq 2$ and sphere data with $\omega_v = 1/1024$ and $0 < \omega_t \leq 1$. The contraction supplier chooses ω_t; its auxiliary constant $u \geq 0$ satisfies $6u\omega_t \leq \omega_v$.
2. *The plentiful construction.* Apply III.2.5. Pick a choice c of the demand.
3. *Block floor and block factors.* The block floor is the maximum of five floors: the plentiful floor, a density floor, the treasury floor, the period-plan floor and e_c. Fix block factors, wages and post-L costs.
4. *Smallness exponent and delays.* Take $\theta^0 \equiv 1$ and initial data $\beta^{\mathrm{ent}} = \sigma^{\mathrm{ent}} = 0$, $S_j = g_j = 1$. Raise delays in this order: halo, step, thermal requests, Λ and both demand arrays together with the all-stage payment of the zeroth post-L row, treasury, period plan, auxiliary budget, silk, custody, the delay of III.2.5, the quote bounds, and the header bounds.
5. *Barrier.* Take $\bar{\beta} \geq 2$ (Lemma 4.3).
6. *Wait.* First apply the wait supplier of III.2.5 to a zero-window wait. Separately enlarge a slim wait for density bounds. Choose a common enlargement satisfying both waits and the required coverage bounds.
7. *Level system.* Take the level system of the divisibility variant of III.2.2, and the factor from III.2.5.

Finally `engageWithCustodyFactor` builds the run from the factor. *(proof read at the level of the composition)*

3.5.4.5 4.5 The analytic lemmas

Lemma 4.20 (mechanisms behind III.2.9). Under the hypotheses of III.2.9, the estimates follow as follows.

- *Division:* analytic division first gives complex coefficient packets on a smaller strip. Taking their real parts and using conjugation symmetry preserves the identity at real momenta and gives the stated real kernels.
- *Pullback gain:* divide R into terms with d forward differences and move those differences onto B. The two cross bounds contribute L^{-2}, the sum over fine sites contributes L^4, and the differences contribute L^{-d}, giving L^{2-d}.
- *Contraction:* apply the gain with $d = 3$ and choose $N > \max(\tau/\eta, K^2)$. Then $K/L \leq L^{-1/2}$ for $L \geq N$.
- *Multilinear transfer:* divide in the independent momenta and distribute the d differences among the $r + 1$ crosses. Their bounds contribute $L^{-(r+1+d)}$; summing the free translation gives L^4. The width reserve pays for symmetrization and recentering.
- *Odd-part cancellation:* reflection cancels the first odd contribution. The rate calculus controls the remaining second derivative with every exponent strictly below 2. The second derivative of a function composed with $y \mapsto -y$ equals its original second derivative at 0.
- *Local read-out:* subtract the presentations. Equality of densities expresses the coefficient difference as the difference of the regular remainders on the healthy domain. The local projection kills nonwrapping supports. A wrapping support has an exponentially small contribution, bounded using the side condition; finitely many test functionals then control the coefficient norm.

(proofs read at the level of the displayed declarations; the analytic division, transfer and local-projection inputs are not independently audited)

3.5.5 5. Where the difficulty lies

- **Order and persistence.** Each floor must be fixed before the block factors that pay it. Costs depending on the chosen factors can be paid by later entry thresholds. Delay and wait suppliers must remain valid under all subsequent increases; otherwise chaining them does not give a common set of constants.
- **Uniformity in depth and volume.** The constants and schedule are common to every N, p. The finite first plateau is handled by one period offset M, giving $n_p = 4(M + p + 1)$. Later volume conditions follow from the exponential growth of the side length.
- **Continuity during shooting.** Each new interval must keep every member of the family inside the next coupling band. The nested-interval argument preserves this condition while fixing the endpoint couplings. It requires continuity of the full representation family at each new step.
- **Two drift bounds.** The bound $\omega_j \le (\kappa/4)\log \ell_j$ gives the shooting gates. The bound $\omega_j \le \log \ell_j / j^2$ gives the running law. Both are paid after the stage constants are fixed.
- **Eventual payment.** The entry thresholds can absorb any fixed real array, including one evaluated after the block factors are chosen. Each fixed row is paid eventually; a single level need not pay every row. The stronger all-stage condition for the zeroth post-L row is purchased separately.

3.5.6 6. Notation

Written proof	Lean	Meaning
sheet; stage record X_j; first stage j_0; floors f_j	`Sheet`, `Stage`	data fixed before the block factors
$k_j, q_j, a_j, w_j, \varpi_j, \omega_j, \gamma_j, r_j$	`order`, `power`, `supportExponent`, `width`, `paths`, `drift`, `columns`, `clip.relaxation`	fields of the stage record
$\mathcal{C}_j$	(in `Large`)	$1 + \varpi_j + w_j^{-1} + \gamma_j + r_j + a_j$
ℓ_j; $L_m = \ell_{J_{m+1}}$	`Rolls`; `Schedule.block`	stage and step block factors
θ_j	`Protection`	smallness exponent of the loop cap
$\beta_j^{\mathrm{ent}}, \sigma_j^{\mathrm{ent}}(\bar\beta), S_j, g_j$	`Delays`	entry thresholds, shells, growth
$\bar\beta$	`Bar`	barrier: coupling at level 0
$\sigma_j^{\mathrm{rdy}}, h_j, n_{\min}$	`Wait`	readiness, windows, minimal period
meets the entrance conditions	`Paid`	III.2.2(a)–(d)
$\mathrm{bill}_j(s)$	`Bar.bill`	required sites per unit length
J', $J_m = j_0 + J'_m$	`Program`	stage index
$n_p = 4(M + p + 1)$	`Program.page`	level-0 side, with $M \in \mathbb{N}$
recipe $\mathcal{R}$	`Recipe`	path data and three tests
admissible; relays; start	`Lives`, `Relays`, `Starter`	
continuous family	`Curve`, `deviation`	

Written proof	Lean	Meaning
(start)+(advance)	`Recipe.Issues`	shooting hypotheses
$\mathfrak{B}_m$; $[\beta_m^-, \beta_m^+]$	`band`; `floorBeta`, `ceilBeta`	band; shooting window
eventually paid	`EventuallyPaid`	III.2.8(a)
plan; treasury funded	`Plan`, `TreasuryFunded`	III.2.8(d)
emitted factor; prediction	`Factor`, `Prediction`	record of all choices; predicate on it
sheet with floors n	`raise`, `bountyV`	
order data v	`Orders`, `Draft`, `WithStage`	

3.5.7 7. What remains unexplained

1. **Analytic content of the purchased conditions.** The common-choice statements III.2.4–III.2.7 and their quantifier order are verified against the displayed Lean interfaces. The predicates in the requirement table, the conditional family package, and the campaign's pillars, funding, news, custody and sky conditions are not fully expanded into mathematical estimates here. Their equivalence to the requirements stated in the owning units remains to be checked.
2. **Underlying suppliers.** The proofs composing the sheet, native, joint, plentiful and final-run constructions have been checked at their call sites. The proofs of the underlying analytic suppliers, including those used by the roughly thirty additions in the joint construction, are not independently verified here. The same limitation applies to the plan's dictionaries and inverse-charge supplier, and to the analytic division and projection results used in III.2.9.
3. **Auxiliary parameters.** K' is a nondecreasing sequence bounded below by 1; the tableau is a finite interpolation rule of degree 300; the sphere condition is a contraction estimate. The detailed analytic roles of Y, Q, wages, postage, exchanges, menus and the remaining auxiliary records still require identification in the step estimates.
4. **Drift normalization.** The sheet construction raises the drift allowance to dominate the stationary supplier's sequence. Its identification with the bound $1 + D$ used in III.9, including the relation between the constants attributed there to III.7 and III.13, remains unverified.
5. **Analytic interfaces.** The weighted summability and vanishing-order definitions, the first-site support condition in multilinear transfer, and the rate interpretation of odd-part cancellation have been checked. The full step-scene hypotheses of odd-part cancellation and the coefficient-format and readability conditions of local read-out still need an independent mathematical account. Only the statements of the supplementary Hessian, cubic/quartic extraction and trace-insertion estimates linked after III.2.9 were checked; their proofs remain outside this review.
6. **Scope of the source review.** The arithmetic construction, the divisibility argument, and the shooting proof are verified against the source. Other uses of the non-divisible period construction, the allocation of the supplementary analytic estimates to outline units, and the line-count breakdown given at the outset have not been independently checked.

3.6 III.3. Free lattice kernels and covariances

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/Next/FREE_KERNELS/` (5,450 lines), `YM/MultiNext/SIGNED_SOURCE_CONTINUATION/` (1,674), `YM/Further/FREE_COVARIANCE_KP/` (924), `YM/UVNext/PRICED_KERNELS/` (533),

YM/Further/CORRECTED_LIFT_COEFFICIENTS/ (248), YM/Push/CUTOFF_PROJECTORS/ (139), YM/ProofPorts/GRID_RENEWED_RESERVES/ (30). **Size:** 8,998 lines in the main-theorem closure.

Conventions.

- **Tori and forms.** $\Lambda_n = (\mathbb{Z}/n\mathbb{Z})^4$, $n \geq 1$. $\Omega^r(\Lambda_n)$ is the space of real r-cochains, with the counting inner product. $d_r : \Omega^r \to \Omega^{r+1}$ is the exterior derivative, and $d = d_1$. X^* denotes the transpose on real cochains and the Hermitian adjoint of a symbol at real momentum. Displacements use $|y|_1$; real momenta use the Euclidean norm $|p|$.
- **Levels.** In mesh N the levels are $i = N, \dots, 0$. Level N is the bare lattice. Step $i < N$ maps level $i+1$ (fine torus Λ_f) to level i (coarse torus Λ_c), with $f = cL_i$ and $L_i \geq 3$ odd. Level i has a stage index $J(i) \geq 4$, and $J(i) \leq J(i+1) \leq J(i) + 1$.
- **Kernels.** A *kernel* is a map $R : \mathbb{Z}^4 \to \mathrm{Hom}(\mathbb{R}^\beta, \mathbb{R}^\alpha)$, where α, β are sets of face types. It acts on every torus by periodisation: $(R_n\omega)(x) = \sum_{y \in \mathbb{Z}^4} R(y)\,\omega(x+y)$. Its symbol is $\hat{R}(p) = \sum_y R(y) e^{ip\cdot y}$.
- **Cross kernels.** A *cross kernel* of block size L also carries a fine residue $r \in (\mathbb{Z}/L)^4$ in its row index. It maps cochains on Λ_c to cochains on Λ_{cL}, for every c. Its displacements are measured on the coarse lattice.
- **Strip bound.** $R \in \mathcal{S}(\alpha, \Omega)$ means $\sum_y e^{\alpha|y|_1} |R_{ab}(y)| \leq \Omega$ for every entry. This implies that $\hat{R}$ is analytic and bounded on $|\mathrm{Im}\, p_a| < \alpha$. Conversely, a bounded periodic analytic symbol has such a weighted coefficient bound on each narrower strip.
- **Aliases and lattice momenta.** For a coarse momentum p, the fine aliases are $q_\ell = (p + 2\pi\ell)/L$, $\ell \in (\mathbb{Z}/L)^4$ (centred representatives). $\tau(p) = (e^{ip_a} - 1)_{a=1}^4$ is the lattice momentum, and $\sigma_r(p)$ is the symbol of d_r, so $\sigma_0 = \tau$.
- **Complex cubes.** Write $\mathcal{C}_\theta = \{p \in \mathbb{C}^4 : |\mathrm{Re}\, p_a| < \pi + \theta,\ |\mathrm{Im}\, p_a| < \theta\}$ and $\mathcal{C}_\theta^{\mathbb{R}} = \mathcal{C}_\theta \cap \mathbb{R}^4$. Absolute squares in the real-momentum formulas are continued holomorphically as $g(p)g(-p)$ for real-coefficient filters, and $|\tau(p)|^2$ as $\sum_a (e^{ip_a} - 1)(e^{-ip_a} - 1)$.
- **Constants from the kernel choice.** $\kappa > 0$ is the penalty of the kernel choice K (the number b of the main theorem). $c_i > 0$ is the direct coefficient of the clip of level i, that is, the coefficient of the bare plaquette term in the quadratic part of the action.

3.6.1 1. What this unit proves

The unit computes the Gaussian part of the block-spin step of III.1 in closed form, for every history of steps. It proves three things about it:

- exact algebraic identities on every torus, giving an orthogonal splitting of the quadratic form;
- exponential decay, uniform in the block sizes, history length and torus, with constants fixed by the kernel choice and stage index, and, after repair, by the sheet's direct clip coefficients;
- uniform ellipticity and hypercubic symmetry of the quadratic forms.

The constants are fixed before the block sizes (Proposition III.3.1). Positivity of the rolled kernel requires the lower bound $L \geq b_j$.

The package also contains:

- the free covariance of the curvature and its comparison with the continuum (III.3.6, used in III.11);
- calculus lemmas used in the nonlinear step (III.3.7, used in III.10 and III.2).

The identities of one step. Fix a step with tori Λ_f, Λ_c, $f = cL$. Let $Q : \Omega^1(\Lambda_f) \to \Omega^1(\Lambda_c)$ be the linearised block average of III.1: a weighted average, over a J-fold smoothing patch, of line integrals of the fine 1-form along L-link paths.

Metric data on a level is a quintuple $\mathcal{B} = (D, D^{-1}, H, H^{-1}, R)$ of kernels on 2-forms: a diagonal multiplier, its inverse, a metric, its inverse, and a root. With the direct coefficient c of the level's clip, set

$$V = (\sqrt{c}\,1,\ R) : \Omega^2 \to \Omega^2 \oplus \Omega^2.$$

Then $V^*V = c\,1 + R^*R = H$ and $\langle dA, H\,dA\rangle = |V dA|^2$.

Transfer kernels are $\mathcal{U} = (W, T, M_f, M_n, O)$ with:

- $W : \Omega^1(\Lambda_c) \to \Omega^1(\Lambda_f)$ and $T : \Omega^2(\Lambda_c) \to \Omega^2(\Lambda_f)$ (cross kernels);
- $M_f : (\Omega^2)^{\oplus 2}(\Lambda_c) \to (\Omega^2)^{\oplus 2}(\Lambda_f)$ (cross kernel);
- $M_n : (\Omega^2)^{\oplus 2}(\Lambda_c) \to \Omega^1(\Lambda_c)$;
- $O : (\Omega^2)^{\oplus 2}(\Lambda_c) \to ((\Omega^2)^{\oplus 2} \oplus \Omega^3)(\Lambda_c)$.

With fine data $(\mathcal{B}, V)$ and coarse data $(\mathcal{B}', V')$, the identities are:

- (D1) $d^*H\,dW = Q^*d^*H'd$
- (D2) $QW = 1 - \kappa^{-1}d^*H'd$
- (D3) $T^*d = dQ$
- (D4) $TD'd = D\,dW$
- (R1) $M_f^*M_f + M_n^*M_n + O^*O = 1$
- (R2) $\sqrt{\kappa}\,M_n^* = V'd$
- (R3) $M_fV'd = Vd\,W$
- (R4) $M_nV'd = \kappa^{-1/2}d^*H'd$
- (R5) $OV'd = 0$
- (R6) $M_f^*Vd = \sqrt{\kappa}\,M_n^*Q$
- (Δ) $H'(H'^{-1} - T^*H^{-1}T - \kappa^{-1}d\,d^*)H' = d_2^*Pd_2$, for a 3-form kernel P on the coarse level.

What the identities mean:

- **(D1)–(D2).** These say that $A = WB$ is a critical point, hence a minimiser, of $A \mapsto \langle dA, H\,dA\rangle + \kappa|QA - B|^2$, and that the minimum is $\langle dB, H'dB\rangle$. So W is Balaban's background-field (minimiser) map, and H' is the effective metric of the coarse level. The value identity is III.1's `action`.
- **(D3)–(D4).** (D3) makes T^* a 2-form average compatible with Q. (D4) says that T carries the weighted coarse curvature $D'dB$ to the weighted fine curvature $D\,dWB$ of the background field.
- **(R1)–(R6): orthogonal splitting.** For $A \in \Omega^1(\Lambda_f)$, $B \in \Omega^1(\Lambda_c)$ and $Z = A - WB$, (R3), (R4) and (D2) give

  $$(Vd A,\ \sqrt{\kappa}(B - QA)) = (M_f, M_n)V'dB + (VdZ,\ -\sqrt{\kappa}\,QZ).$$

  The two summands are orthogonal by (R6), and $|(M_f, M_n)V'dB| = |V'dB|$ by (R1) and (R5). Hence

  $$\langle dA, H\,dA\rangle + \kappa|QA - B|^2 = \langle dB, H'dB\rangle + \langle dZ, H\,dZ\rangle + \kappa|QZ|^2.$$

- **(Δ).** This is the same splitting for inverse metrics. It says $H'^{-1} = T^*H^{-1}T + \kappa^{-1}dd^* + H'^{-1}d_2^*Pd_2H'^{-1}$, and the last term sees only $d_2H'^{-1}\omega$.

Proposition III.3.1 (catalogue fixed before the block sizes). Let K be a kernel choice. There are numbers $\alpha_j > 0$, $\eta_j > 0$, $\Omega_j \in \mathbb{R}$, $\lambda_j \geq 0$ (amendments), $\gamma_j > 0$ (clip bounds) and $b_j \in \mathbb{N}$ (block thresholds), $j \in \mathbb{N}$, depending only on K, with the following properties.

(a) *Metric data of a history.* Let $j \geq 4$, let the clip have direct coefficient $c \leq \gamma_j$, and let v be a history of stage index j. Then there is metric data $(D, D^{-1}, H, H^{-1}, R)$ such that:

- D and D^{-1} are the explicit kernels of v with penalty κ, and $H^{-1} = D^{-1} + \lambda_j (d_2 D^{-1})^* (d_2 D^{-1})$ (Lemmas 4.4–4.7);
- all five kernels lie in $\mathcal{S}(\alpha_j, \Omega_j)$;
- on every torus: $DD^{-1} = HH^{-1} = 1$, $c\,1 + R^*R = H$, $\hat{H}(0) = 1$, $Hd = Dd$, and H and D are symmetric;
- on every torus: $\eta_j |\omega|^2 \leq \langle \omega, H\omega \rangle \leq \Omega_j |\omega|^2$, and the same for D;
- for every gauge group G, the form $\langle A, d^* H d\, A \rangle$ on $\mathfrak{g}$-valued 1-forms is invariant under the signed lattice symmetries;
- $D = 1$ if v is the empty history.

(b) *One step.* Let $j, t \geq 4$ with $t \in \{j, j-1\}$, and let P be a flight of index (j, t): odd block size $L \geq 3$ and fine history of index j. Then:

- the explicit transfer kernels W, T of P satisfy $\nabla_u W \in \mathcal{S}(\alpha_j, \Omega_j L^{-1-|u|})$ and $\nabla_u T \in \mathcal{S}(\alpha_j, \Omega_j L^{-2-|u|})$, for every word u of at most $j-2$ fine forward differences;
- the rolled 3-form kernel $\mathcal{P}$ of P, built with fine amendment λ_j and coarse amendment λ_t (Lemma 4.12), lies in $\mathcal{S}(\alpha_j, \Omega_j)$;
- if $L \geq b_j$, then at every real momentum $\hat{\mathcal{P}}(p)$ is Hermitian and $\eta_j \leq \hat{\mathcal{P}}(p) \leq \Omega_j$.

Status: theorem, with no hypothesis beyond the kernel choice. It uses III.6's square-root theorem to construct R. The catalogue does not depend on G. *Lean:* `Catalog`, `pledges`, `BottomQuote`, `HopQuote`, `Mag`, `Defined`, `Floor`, `Donate`, `Funds`.

Proposition III.3.2 (free kernels along an admissible schedule). Let G be a gauge group and K a kernel choice. There is a catalogue $(\alpha, \Omega, \eta, \lambda, \gamma, b)$ as in III.3.1 with the following property. For every sheet $\mathfrak{S}$ with kernel K (the pre-L constants, III.2) there is a second table $(\alpha'_j, \Omega'_j, \eta_j)$ such that the conclusion below holds.

Let A be an RG trajectory with kernel K and the stage profile of $\mathfrak{S}$, satisfying the *credit condition*:

- (C1) $c_i \leq \gamma_{J(i)}$ for every level i;
- (C2) $b_{J(i+1)} \leq L_i$ for every step i.

Then, simultaneously for all $N \geq 0$ and $0 \leq i \leq N$, there are two families of metric data and transfer kernels: a pre-repair family $\mathsf{M} = (\mathcal{B}^{N,i}, \mathcal{U}^{N,i})$ and an output family $\mathsf{H} = (\mathcal{B}^{N,i}, \tilde{\mathcal{U}}^{N,i})$. They have the same metric data and the same W, T, and satisfy:

1. *Explicit formulas.* $\mathcal{B}^{N,i}$ is the metric data of III.3.1(a) for the history of level i in mesh N, with amendment $\lambda_{J(i)}$. That history is the composite of the stage hops of steps $N-1, \dots, i$, starting from the empty history at level N. $W^{N,i}, T^{N,i}$ are the explicit transfer kernels of the flight of step i, of index $(J(i+1), J(i))$.
2. *Pre-repair conditions (M).*
   - $\mathcal{B}^{N,i} \in \mathcal{S}(\alpha_{J(i)}, \Omega_{J(i)})$, with lower and upper bounds $\eta_{J(i)}, \Omega_{J(i)}$ on every torus for the clip of level i, and the symmetry;

- $D^{N,N} = 1$;
- W, T have the derivative bounds of III.3.1(b) at stage $J(i+1)$, with $J(i+1) - 2$ differences;
- (D1)–(D4) hold on every schedule torus pair of step i;
- there is a 3-form kernel $P \in \mathcal{S}(\alpha_{J(i+1)}, \Omega_{J(i+1)})$, Hermitian at real momenta, with $\langle v, \hat{P}(p)v\rangle \geq \eta_{J(i+1)}|v|^2$ for real p and v in the range of $\sigma_2(p)$, such that (Δ) holds on every schedule torus pair.

3. *Output conditions (H), with the second table.*
   - the metric bounds of III.3.1(a), with strip constants α'_j, Ω'_j and the same lower bounds η_j;
   - $\tilde{M}_n, \tilde{O} \in \mathcal{S}(\alpha'_{J(i+1)}, \Omega'_{J(i+1)})$ and $\tilde{M}_f \in \mathcal{S}(\alpha'_{J(i+1)}, \Omega'_{J(i+1)} L_i^{-2})$, and the derivative bounds on W, T with α'_j, Ω'_j;
   - (D1)–(D4) and (R1)–(R6) on every schedule torus pair, with the clips of levels $i+1, i$;
   - $D^{N,N} = 1$, and the symmetry of every $\mathcal{B}^{N,i}$.

Status: theorem. The inputs F06–F09 (III.6) are discharged inside the proof. For trajectories with the stated kernel and profile, the additional hypothesis is the credit condition. The variant `before_paid` (assigned to III.4) replaces it by:

- two conditions on the sheet: clip coefficients $\leq \gamma_j$, and block-size floors $\geq b_j$, for all stages from the first one on;
- the schedule's payment record (III.2), which gives $L_i \geq$ floor.

III.9 builds the sheet so that these hold: its floors are exactly b and its clip coefficients are $\leq \gamma$ (`housing`). The variant also lets the second table be computed from a preliminary sheet $\mathfrak{S}$ and used for any later sheet $\mathfrak{S}'$, provided $\mathfrak{S}'$ has the same direct clip coefficients, the same kernel, a drift at least as large, and logarithmic parameter size at least as large.

Lean: `retained_support`; conditions `Present`, `Craft.Rack`, `Bid.Retained`, `Horizon.Suffices`, `Bid.Sworn`, `Credit`; discharge `credit_from_sheet`, `Willing`.

Proposition III.3.3 (identities on every torus). Let a family satisfy the output conditions of III.3.2(3) for some table. For every N, $m < N$ and positive sides c, f with $f = cL_m$, (D1)–(D4) and (R1)–(R6) hold, with the same kernels and the clips of levels $m+1, m$. The diagonal identities follow by rigidity (III.10, `panes`). They also follow from the explicit formulas when the family realises them (`universal_diagonal`). Given the diagonal identities, the roll identities follow from the output conditions by the same rigidity argument. *Status:* theorem. *Lean:* `complete_roll`, `Towers.Diagonal`, `Towers.Roll`, `Inputs`. Used in III.10.

Proposition III.3.4 (positivity of the rolled 3-form kernel). Fix a kernel choice K and $j, t \geq 4$. There are $C_0, B > 0$, depending only on (K, j, t), with the following property. For every coarse amendment $\lambda \geq C_0$ and fine amendment $\mu \geq 0$ there are $\eta, \sigma, \Omega > 0$ such that, for every flight P of index (j, t):

- $\mathcal{P}_{\mu,\lambda}(P) \in \mathcal{S}(\sigma, \Omega)$;
- if $\mu B \leq L^2$, then at every real p, $\hat{\mathcal{P}}_{\mu,\lambda}(p)$ is Hermitian and $\eta \leq \hat{\mathcal{P}}_{\mu,\lambda}(p) \leq \Omega$.

Status: theorem. *Lean:* `Flight.frontier`, `Flight.absorb_before`.

Proposition III.3.5 (line and box filters). For positive integers L, M, define the uncentred

line average below; for odd L, M, also define the centred line and box averages:

$$\mathrm{plank}_M(z) = M^{-1} \sum_{|s|\le (M-1)/2} e^{isz}, \qquad \mathrm{rail}_M(z) = M^{-1} \sum_{s=0}^{M-1} e^{isz}, \qquad a_L(p) = \prod_a \mathrm{plank}_L(p_a)$$

(centred line average, uncentred line average, box average). Then:

(a) $\mathrm{rail}_L(Mz)\,\mathrm{rail}_M(z) = \mathrm{rail}_{LM}(z)$ for all positive L, M, and $\mathrm{plank}_L(Mz)\,\mathrm{plank}_M(z) = \mathrm{plank}_{LM}(z)$ for odd L, M. For odd M, plank_M is even, 2π-periodic and real on $\mathbb{R}$.

(b) $|\mathrm{plank}_M(z)|, |\mathrm{rail}_M(z)| \le e^{M|\mathrm{Im}\, z|}$.

(c) There are absolute constants $c_*, c_1 > 0$ such that $|\mathrm{rail}_M(z)| \ge c_*$ for complex z with $M|z| \le 3\pi/2$, and $|\mathrm{plank}_M(z)| \ge e^{-c_1 M/L}$ for real z, $M \le L$ and $L|z| \le 3\pi/2$.

(d) There is an absolute B such that $|\mathrm{rail}_L((q_\ell)_a)|, |\mathrm{plank}_L((q_\ell)_a)| \le B/(1+|\ell_a|)$. This holds for odd L, every p in the complex cube $\{|\mathrm{Re}\, p_a| < \frac{5\pi}{4},\ |\mathrm{Im}\, p_a| < \frac{\pi}{4}\}$, every alias ℓ and every a.

Status: theorem. *Lean:* `plank`, `rail`, `area`, `plank_bind`, `rail_bind`, `plank_height`, `rail_lower`, `principal_plank`, `alias_rail`.

Proposition III.3.6 (free covariance of the curvature). Call a real 2-form kernel f *firm* with constants (α, Ω, η) if $f \in \mathcal{S}(\alpha, \Omega)$, $\hat{f}(0) = 1$, and $\eta|\omega|^2 \le \langle \omega, f_n \omega\rangle \le \Omega|\omega|^2$ for every real cochain on every torus. Symmetry is not required. Fix $\alpha, \eta > 0$ and $\Omega \in \mathbb{R}$, and let $U = \{p \in \mathbb{R}^4 \setminus \{0\} : |p_a| < 3\pi/2\}$. For $p \in U$ define:

- $\Phi_f(p) = \sigma_1(p)^* \hat{f}(p) \sigma_1(p) + \sigma_0(p)\sigma_0(p)^*$, the symbol of $d_1^* f d_1 + d_0 d_0^*$ on 1-forms;
- $G_f(p) = \sigma_1(p)\Phi_f(p)^{-1}\sigma_1(p)^*$, the curvature covariance symbol when f is symmetric and the gauge-fixed potential has precision $d_1^* f d_1 + d_0 d_0^*$;
- $G_\infty(p) = s(p)|p|^{-2} s(p)^*$, where $s(p)$ is wedging with ip on 1-forms: the continuum Feynman-gauge curvature symbol.

(a) There is $b > 0$, depending only on α, Ω, η, such that $\mathrm{Re}\,\langle x, \Phi_f(p) x\rangle \ge b|p|^2|x|^2$ for every firm f, $p \in U$ and complex 1-form x.

(b) For every $n \in \mathbb{N}$ there is $C_n \ge 0$, depending only on n, α, Ω, η, such that, for all firm f and $p \in U$, $\|D^n G_f(p)\| \le C_n |p|^{-n}$ and $\|D^n (G_f - G_\infty)(p)\| \le C_n |p|^{1-n}$.

(c) If f is a scalar kernel with $\sum_x e^{\alpha|x|_1}|f(x)| \le \Omega$, then $\hat{f}$ is C^∞ on $\mathbb{R}^4$ and $\|D^n \hat{f}(p)\| \le A_n \Omega$ for every p, where $A_0 = 1$ and $A_n = (n/\alpha)^n$ for $n \ge 1$. Here D^n is the full nth Fréchet derivative, with its multilinear operator norm; this notation also applies to (b).

Status: theorems. Part (b) is used in III.11. *Lean:* `tank_full`, `gem_quote`, `radial_prices`, `Firm`, `fuel`, `garnet`, `pearl`.

Proposition III.3.7 (calculus lemmas for the nonlinear step). These do not concern free kernels; they are consumed in III.10 and III.2.

(a) *Discrete Taylor expansion with scaled bounds.* Let $b, k \in \mathbb{N}$ with $k \le b$. There is $C = C(b,k) \ge 1$ with the following property. Let $L \ge 1$, $M \ge 0$, and let $p : \mathbb{Z}^4 \to E$ (E a real normed vector space) satisfy $\|\nabla_u p(x)\| \le M L^{-1-|u|}(1 + L^{-1}|x|_1)^b$ for all words u of forward differences with $|u| \le k$. Then the Newton pieces $P_i(x) = \sum_{|\nu|=i} \binom{x}{\nu} \nabla^\nu p(0)$, of degree at most i, satisfy $\|P_i(x)\| \le C M L^{-1-i}(1+|x|_1)^b$ for $i < k$, and the remainder satisfies $\|p(x) - \sum_{i<k} P_i(x)\| \le C M L^{-1-k}(1+|x|_1)^b$.

(b) *Weighted Taylor coefficients.* Let $d \in \mathbb{N}$ and let $f : E \to F$ be C^d at 0 between real normed vector spaces. Let $p_v : E \to E$ ($v \in V$ finite) be continuous linear maps with $\sum_v p_v = 1$, and let $Y_v \geq 1$ be integers. Suppose that for every z the function $t \mapsto f(\sum_v t^{Y_v} p_v z)$ has vanishing derivatives of orders $0, \dots, d$ at 0. Then $D^m f(0)(x_1, \dots, x_m) = 0$ whenever there are integers $k_i \geq 1$ with $\sum_i k_i \leq d$ and $p_v(x_i) = 0$ for every v with $Y_v > k_i$.

(c) *Moments of local gauge-invariant functions.* Fix $b, d \in \mathbb{N}$ with $b \leq d$, a torus Λ_n with $n \geq 2$, an anchor o, and a finite support X satisfying $10\,\mathrm{dist}(x, o) < n$ for every $x \in X$, where dist is the torus ℓ^∞ distance. Let $u : \mathrm{Config}(G, \Lambda_n) \to \mathbb{C}$ be gauge invariant and depend only on links whose two endpoints lie in X, with $u \circ \exp$ of class C^b at 0. For every radial polynomial connection jet J through weight d, let $U_{J,o}(t)$ be its link holonomies at scale t, based at o: a homogeneous connection component of degree $r - 1$ has weight r, $2 \leq r \leq d$, and is multiplied by t^r. Assume $\frac{d^k}{dt^k} u(U_{J,o}(t))|_{t=0} = 0$ for every J and $0 \leq k \leq b$. For any $m \in \mathbb{N}$, $p_1, \dots, p_m \geq 1$ with $\sum_i p_i \leq b$, and link fields w_i whose values on those links are polynomials of degree at most $p_i - 1$ in the centred lift of the initial site relative to o, one has $D^m(u \circ \exp)(0)(w_1, \dots, w_m) = 0$. Values of w_i on other links are unrestricted.

(d) *Radial decomposition.* Let $j \in \mathbb{N}$ and let $Q_a : \mathbb{R}^4 \to \mathfrak{g}$ ($a = 1, \dots, 4$) be homogeneous polynomials of degree $j + 1$. Then $Q_a = K_a + \partial_a r$, where $r = (j + 2)^{-1} \sum_a x_a Q_a$ is homogeneous of degree $j + 2$, and K is homogeneous of degree $j + 1$ with $\sum_a x_a K_a = 0$ (the Fock–Schwinger condition).

(e) *Small-field condition along rays.* There is $C = C(G) > 0$ with the following property. Let $0 \leq b \leq 1$, $\tau \geq Cb^2$, and let v be Lie-algebra link variables with $|v_x| \leq b$ on all links within distance 5 of Y. If every plaquette touching Y has holonomy within chord distance τ of 1 for the configuration $\exp(v)$, then the same plaquette bounds hold for $\exp(sv)$, $0 \leq s \leq 1$.

Status: theorems. *Lean:* `capital`, `cancel`, `converse`, `wiped`, `locale`, `strings`.

3.6.2 2. What it uses

From other units.

- **I.1.** The lattice, the signed (hypercubic) symmetries, and the gauge group. In the free-kernel construction the gauge group enters only the symmetry statement; III.3.7 also uses its exponential and holonomy maps.
- **III.1.** The block-spin data:
  - the block ratio, the patch (odd $L \geq 3$, $J \geq 4$), the kernel choices `KernelChoices` and the clips `Clip`;
  - the linearised average Q (`mean`) and the lattice exterior derivative;
  - the trajectory record `Levels`: block sizes, stage indices with $J(i) \leq J(i+1) \leq J(i) + 1$, and schedule tori whose sides tend to infinity.

  III.1 also defines the output objects:
  - metric data `Bed`, `Bed.full`, `Floor`;
  - transfer kernels `Towers` and their bounds `Towers.Deep`;
  - the identities (D1)–(D4), (R1)–(R6), with $d^* H d$ as `K`;
  - kernel families `Horizon`, the output conditions `Horizon.Suffices`, and price tables `MapPrices`.
- **III.2.** The sheet `Sheet`, the block sizes `Rolls`, and the payment record `Paid`.

- **III.6.** The four linear-algebra theorems `F06`, `F07`, `F08`, `F09`. Their statements are in Lemma 4.16. F07 is also used in Lemma 4.8.
- **III.10.**
  - The interface conditions `Craft.Rack`, `Bleeps`, `Reaches`, `Funds`, `Donate`, `Replaces`, `Bid.Retained`, `Bid.Sworn`.
  - The repair theorem `Bid.pinning` (Lemma 4.16), and `panes` for III.3.3.
- **III.16.** Kernels and their action on tori: `Row`, `Cross`, `installed`. The weighted norms `Row.strip` and `Laurent.Width`, where the size of a displacement is its ℓ^1 norm. The alias-resolved Fourier transform on Λ_f, which is unitary up to f^4.

Standard mathematics.

- Fourier analysis on $\mathbb{Z}^4$ and on periodic lattices. Bloch (alias) decomposition of operators that commute with coarse translations. Periodic strip analyticity gives exponential coefficient bounds on narrower strips.
- Gaussian block-spin transformations computed in Fourier space: K. Gawędzki, A. Kupiainen, *Massless lattice φ^4_4 theory*, CMP 99 (1985). Balaban's averaging and background-field operators for lattice gauge fields: T. Balaban, CMP 95 and 96 (1984), CMP 98 (1985). Exposition: J. Dimock, *The renormalization group according to Balaban I*, Rev. Math. Phys. 25 (2013).
- Lagrange's identity. The Schur-complement (orthogonal) splitting of a quadratic form.
- Exactness of the lattice de Rham (Koszul) complex for exponentially decaying translation-invariant kernels.
- Newton forward-difference interpolation. Weighted-homogeneous Taylor expansions. The Fock–Schwinger (radial) gauge.

Imported constants and assumptions.

- The kernel choice K; only the penalty κ enters the free kernels.
- The analyticity half-width $\pi/4$ (`zone`) and the radius $3\pi/2$ (`domainRadius`). The proved compactness bounds $c_-|w| \le |e^{iw} - 1| \le c_+|w|$ on $|w| \le 3\pi/2$ (`meter`).
- The credit condition, a hypothesis of III.3.2. It is discharged through `before_paid`, the construction of the sheet in III.9, and the payment record of III.2.
- F06–F09 are proved in III.6 and plugged in inside `retained_support`.

3.6.3 3. Strategy

The Gaussian block-spin problem is solved in Fourier space. Iterating the block average and its independent noise gives a diagonal 1-form covariance symbol. Minimising its precision over gauge directions gives the effective form d^*Dd by Lagrange's identity. The background-field map W and curvature transfer T are then explicit alias sums satisfying (D1)–(D4).

Uniform decay follows from the filters. A stage-J box filter gives $\prod_a (1 + |\ell_a|)^{-J}$ alias decay, leaving $J - 2$ fine differences summable. Along a history, successive dilation ratios decrease at least geometrically by $1/3$, so the extra filter products and accumulated noise stay uniformly bounded. Before estimating inverse symbols, the principal-alias poles are cancelled. The remaining denominators are bounded below at real momenta and hence on a thinner complex cube. These constants depend on the kernel choice and stage, uniformly in history length and block sizes.

The inverse-metric defect factors through d_2 as $X'^* C X'$, where $X' = d_2 D'^{-1}$ and C need not be

positive. Amending the fine and coarse inverse metrics leaves their quadratic forms on potentials unchanged and replaces C by $C+\lambda_t-\lambda_j S^*S$. The squared 3-form transfer satisfies $S^*S=O(L^{-2})$, including the sum over fine residues. Choose the coarse amendments first, then require $\lambda_j B/L^2\le 1$. The resulting positive defect yields M_f, M_n, O and (R1)–(R6) through III.6 and the repair theorem of III.10. The repair constants also depend on the sheet's direct clip coefficients. Rigidity extends the schedule identities to all tori.

3.6.4 4. Steps

3.6.4.1 4.1 Filters and histories Lemma 4.1 (filter algebra). This is III.3.5(a). *Sketch:* $M\,\mathrm{rail}_M(z)(e^{iz}-1)=e^{iMz}-1$; apply it twice and cancel $e^{iz}-1$ when nonzero. If $e^{iz}=1$, every uncentred average is 1. For the centred filter, the centring phases add up, since $M\frac{L-1}{2}+\frac{M-1}{2}=\frac{LM-1}{2}$. *Lean:* `rail_bind`, `plank_bind`.

Lemma 4.2 (size of filters). This is III.3.5(b)–(d). *Sketch:*

- (b) Triangle inequality.
- (c) The compactness constants and Lemma 4.1 give $|\mathrm{rail}_M|\ge c_-/c_+$. On the principal alias, averaging $|e^{isz}-1|\le c_+|sz|$ gives $|\mathrm{plank}_M(z)-1|\le c_+\frac{3\pi}{2}\frac{M}{L}$, so $|\mathrm{plank}_M|\ge\max(c_*,1-kx)\ge e^{-c_1x}$ with $x=M/L$. The exponent is linear in x with no prefactor, which is what makes products along a history converge.
- (d) $L\,\mathrm{rail}_L(z)(e^{iz}-1)=e^{ip_a}-1$ at $z=(q_\ell)_a$, and $L|z|\gtrsim 1+|\ell_a|$ when $\ell_a\neq 0$.

Lean: `plank_height`, `rail_height`, `rail_lower`, `plank_lower`, `principal_plank`, `alias_rail`.

Lemma 4.3 (histories). A history of stage index j is either empty, or a hop (L,J) with $j\le J\le j+1$ and odd $L\ge 3$, applied after a history of index J. Its dilation M_v is the product of its block sizes. Its cumulative filter is defined by $s_\emptyset=1$ and $s_{(L,J)\circ v}(p)=a_L(M_vp)^J s_v(p)$. Equivalently, $s_v=a^j_{M_v}e_v$, where e_v is a product of box filters at the hops where the index drops. Then:

- s_v is entire, $2\pi\mathbb{Z}^4$-periodic and invariant under signed permutations;
- on $\mathcal{C}_{\pi/4}$, $|s_v(q_\ell)|\le C_j\prod_a(1+|\ell_a|)^{-j}$ at the aliases $q_\ell=(p+2\pi\ell)/M_v$;
- for $p\in\mathcal{C}^{\mathbb{R}}_{\pi/4}$, $|s_v(p/M_v)|\ge c_j>0$.

Here C_j, c_j do not depend on v, that is, on the number of hops or the block sizes. *Sketch:* After k further hops, a hop of dilation M' is evaluated at the momentum scaled by $M'/M_v\le 3^{-k}$. Lemma 4.2 gives factors $e^{\pm CM'/M_v}$, and these products converge. The factor $a^j_{M_v}$ gives the alias decay by 4.2(d); the geometric product bounds the extra factors. *Lean:* `Train`, `Train.smooth`, `Train.bind`, `Train.smooth_upper`, `Train.smooth_lower`.

3.6.4.2 4.2 Explicit metric data Lemma 4.4 (auxiliary 1-form covariance symbols). Fix $\kappa>0$ and a history v of dilation M. For a direction a, let $\chi_{v,a}(p)=s_v(p)\,\mathrm{rail}_M(p_a)$. For $p\in\mathcal{C}^{\mathbb{R}}_{\pi/4}\setminus\{0\}$ and $q_\ell=(p+2\pi\ell)/M$, define

$$D_a(p)=M^{-2}\sum_\ell\frac{|\chi_{v,a}(q_\ell)|^2}{|\tau(q_\ell)|^2}+\kappa^{-1}\sum_{v'}M_{v'}^{-2}\sum_{\ell'}|\chi_{v',a}(q'_{\ell'})|^2.$$

The second sum runs over the proper final segments v' of v (the last k hops, $0\le k<\sharp v$), with $q'_{\ell'}=(p+2\pi\ell')/M_{v'}$. For the empty history, $D_a=|\tau|^{-2}$. The formula obeys the exact

one-step recursion $D'_a(p) = L^{-2}\sum_\ell |a_L(q_\ell)^j \mathrm{rail}_L((q_\ell)_a)|^2 D_a(q_\ell) + \kappa^{-1}$, which is the covariance recursion for an averaged field with independent noise. These singular symbols are used only away from zero; Lemma 4.6 constructs the regular 2-form symbols. *Lean:* `Train.levelD`, `D_feedback`, `Train.levelNoise`, `feed`. *(definitions and covariance recursion proof read)*

Lemma 4.5 (the diagonal). At the nonzero real momenta of Lemma 4.4, let $\rho = \sum_a |\tau_a|^2/D_a$ and $D_{\mu\nu} = (\rho\, D_\mu D_\nu)^{-1}$; this is a multiplier diagonal in the plaquette orientation. Then, with $F = \mathrm{diag}(D_a)$,

$$\sigma_1^* D\, \sigma_1 = F^{-1} - \rho^{-1} F^{-1}\tau\tau^* F^{-1}.$$

So d^*Dd is the precision F^{-1} minimised over gauge directions. For the empty history, $D = 1$ (the Wilson quadratic form). *Sketch:* Lagrange's identity $\sum_{\mu<\nu} w_\mu w_\nu |\tau_\mu x_\nu - \tau_\nu x_\mu|^2 = (\sum_a w_a|\tau_a|^2)(\sum_a w_a|x_a|^2) - |\sum_a w_a \bar\tau_a x_a|^2$, with $w_a = D_a^{-1}$. The weights and ρ are positive. *Lean:* `rotor_law`, `pulse`, `vault`, `pink`, `blue`, `top_metric`.

Lemma 4.6 (filled symbols, uniform analyticity). Fix $\kappa > 0$. Away from $p = 0$ in the real cube, the singular formulas become regular combinations. With $|\tau_M|^2 = \sum_a M^2 |e^{ip_a/M} - 1|^2$:

$$D_a = \beta_a/|\tau_M|^2, \qquad \rho = |\tau_M|^4 h, \qquad D_{\mu\nu} = (h\,\beta_\mu\beta_\nu)^{-1}.$$

Here β_a is the principal term $|\chi_{v,a}(p/M)|^2$ plus $|\tau_M|^2$ times the non-principal and noise terms, and h is a rational combination of filters whose denominators stay nonzero on a sufficiently thin cube. The regular formulas give $\beta_a(0) = h(0) = 1$. For each $j \ge 4$ there are $\theta, C, c > 0$, depending only on κ, j, such that, for all histories of index j, β_a and h are analytic and bounded by C on $\mathcal{C}_\theta$, and $|\beta_a|, |h| \ge c$ on $\mathcal{C}_\theta^{\mathbb{R}}$. Hence D and D^{-1} lie in a common $\mathcal{S}(\alpha, \Omega)$, uniformly over histories of index j. *Sketch:*

- Upper bounds: Lemma 4.3 summed over the aliases.
- Lower bound on β_a: its principal term (4.3 and 4.2(c)).
- Lower bound on h: away from zero use $|\tau_M|^2 h = \sum_a |\tau_a|^2/\beta_a$, $\beta_a \le C$ and $|\tau|^2 \asymp |\tau_M|^2 \asymp |p|^2$; at zero use $h(0) = 1$.
- Inverses: a lower bound at real momenta plus an analytic bound gives a lower bound on a thinner complex cube, so the inverses stay analytic.

Lean: `Train.bulb`, `Train.bend`, `Train.pole_split`, `Train.lift_bend`, `Train.recover_rho`, `Train.bulb_firm`, `Train.bend_firm`, `heart_banked`, `Moderate`.

Lemma 4.7 (amended metric). Fix $\kappa > 0$, $j \ge 4$ and $\lambda \ge 0$. For each history of index j, let

$$H^{-1} = D^{-1} + \lambda\, X^* X, \quad X = d_2 D^{-1}, \qquad \mathcal{T}_\lambda = (1 + \lambda\, X d_2^*)^{-1} \text{ on 3-forms.}$$

Then on every torus: $HH^{-1} = 1$, $Hd = Dd$, $XH = \mathcal{T}_\lambda d_2$, and $H, \mathcal{T}_\lambda$ are symmetric. Moreover $\widehat{H}(0) = 1$, and $H, H^{-1}, \mathcal{T}_\lambda, \mathcal{T}_\lambda^{-1}$ have strip bounds and two-sided bounds at real momenta, with constants depending only on κ, j, λ. The amendment changes H only away from exact 2-forms, and leaves $d^*Hd = d^*Dd$ unchanged. *Sketch:*

- $XDd = d_2 d = 0$, which gives $Hd = Dd$.
- $(1 + \lambda X d_2^*)X = d_2 H^{-1}$, which gives $XH = \mathcal{T}_\lambda d_2$.
- Uniform bounds $a\,1 \le D^{-1} \le A\,1$ and $\|X\| \le B$ give $a\,1 \le H^{-1} \le (A + \lambda B^2)\,1$ and hence two-sided bounds for H. The analogous argument applies to $1 + \lambda d_2 D^{-1} d_2^*$.

Lean: `pledge`, `amend`, `bail`, `treble`, `obligations`, `pledge_warm`, `insurance`, `forced_amend`.

Lemma 4.8 (metric data of every history; III.3.1(a)). Let $\kappa > 0$, $\lambda \geq 0$, $j \geq 4$. There are $\gamma, \sigma, \eta, C > 0$ with the following property. For every clip with $c \leq \gamma$ and every history of index j, the explicit D, D^{-1}, H, H^{-1}, together with a positive symmetric square root R of $H - c\,1$, satisfy III.3.1(a) with amendment λ and bounds σ, η, C. These constants depend only on κ, j, λ. *Sketch:*

- γ is half the real-momentum lower bound of H, so $H - c\,1$ has two-sided bounds.
- III.6's square-root theorem gives $R \in \mathcal{S}$ with $R^*R = H - c\,1$ on every torus.
- Torus bounds follow from symbol bounds, since Λ_n is diagonalised by the momenta $2\pi\mathbb{Z}^4/n$.
- The symmetry follows from Lemma 4.3.
- $D = 1$ for the empty history by Lemma 4.5.

Lean: `bid_bed`, `Defined`, `full_cover`, `low_ceiling`, `sworn_metric`.

3.6.4.3 4.3 Transfer kernels of one flight A flight of index (j, t) consists of an odd block size $L \geq 3$, a fine history v of index j, and the coarse history (hop (L, j) after v) of index $t \in \{j-1, j\}$.

Lemma 4.9 (alias symbols and derivative bounds). Fix $\kappa > 0$ and $j, t \geq 4$. There are $\theta, \alpha, C > 0$, depending only on κ, j, t, such that the following holds for every flight of index (j, t). Its explicit W and T are cross kernels; at this stage $M_f = M_n = O = 0$. Their fibre at coarse momentum p and fine residue r is

$$L^{-e} \sum_{\ell} e^{iq_\ell \cdot r} w_e(p, \ell), \qquad e = 1 \ (W), \quad e = 2 \ (T),$$

with explicit per-alias symbols built from the filters and from β_a, h of both histories. At the harmonic point $w_e(0,0) = 1$ and $w_e(0,\ell) = 0$ for $\ell \neq 0$. Each $w_e(\cdot, \ell)$ is analytic on $\mathcal{C}_\theta$ and bounded by $C \prod_a (1+|\ell_a|)^{-j}$, uniformly over flights. Hence, for words u with $|u| \leq j-2$, $\nabla_u W \in \mathcal{S}(\alpha, CL^{-1-|u|})$ and $\nabla_u T \in \mathcal{S}(\alpha, CL^{-2-|u|})$. *Sketch:* A fine difference in direction a multiplies the alias term by $e^{i(q_\ell)_a} - 1 = O((1+|\ell|)/L)$. The sum $\sum_\ell (1+|\ell|)^{|u|} \prod_a (1+|\ell_a|)^{-j}$ converges for $|u| \leq j-2$. In other words, the order j of the smoothing pays for $j-2$ derivatives. Decay is measured on the coarse lattice, so the fine-lattice rate is α/L. *Lean:* `Flight.duo`, `Flight.pilot_cost`, `Flight.ahead_cost`, `Flight.wire_budget`, `secure_both`, `Secured.propels`.

Lemma 4.10 (exact identities for one flight). Fix a flight of index (j, t) with $t \geq 4$, a kernel choice K, and any pair of tori with $f = cL$. Then (D1)–(D4) hold for:

- W, T of the flight and Q of the step;
- D, D' the diagonals of the fine and coarse histories;
- H, H' the amended metrics for any amendments ≥ 0.

Sketch:

- All operators commute with coarse translations. On the Bloch fibre over p, the identities become identities between per-alias symbols, for each alias $q = q_\ell$:
  - $(\sigma_1^* D \sigma_1)(q)\, W_\ell = L^{-3} h_1(q)^* (\sigma_1^* D' \sigma_1)(p)$, giving (D1);
  - $\sigma_1(q)^* T_\ell = L^{-3} h_1(q)^* \sigma_1(p)^*$, giving (D3);
  - $T_\ell D'(p) \sigma_1(p) = D(q) \sigma_1(q) W_\ell$, giving (D4);
  - the alias sum $\sum_\ell L h_1(q_\ell) W_\ell = 1 - \kappa^{-1} (\sigma_1^* D' \sigma_1)(p)$, giving (D2).
- Here $W_\ell = L^{-1} w_1(p, \ell)$, $T_\ell = L^{-2} w_2(p, \ell)$, and $h_r(q) = a_L(q)^j \prod_{a \in \mathrm{face}} \mathrm{rail}_L(q_a)$ is the alias symbol of the r-form average. The forward block average has alias row $L h_1(q)$,

and its adjoint has alias column $L^{-3}h_1(q)^*$ because the fine-to-coarse volume ratio is L^4. The form averages intertwine the exterior derivatives: $h_{r+1}(q)\,L\sigma_r(q) = \sigma_r(p)\,h_r(q)$, from $L\,\mathrm{rail}_L(q)(e^{iq}-1) = e^{iLq}-1$.
- The point $p = 0$ is treated separately: the box filter vanishes at the non-principal aliases of 0.
- The identities pass from the fibres to the tori through the alias-resolved Fourier transform.
- Replacing D by the amended H in (D1)–(D2) changes nothing, since $d^*Hd = d^*Dd$.

Lean: `Flight.tangent`, `diagonal`, `Flight.scaled_tri`, `hinge_transport`, `ZeroAlias.lean`, `pilot`, `wire`, `script_main`, `script_cross`, `script_dual`. *(proof read, including the nonzero-alias algebra and harmonic point)*

3.6.4.4 4.4 The defect and the amendment Lemma 4.11 (unamended defect). In the setting of 4.10, let $\mathcal{V} = D'^{-1} - T^*D^{-1}T - \kappa^{-1}dd^*$ on $\Omega^2(\Lambda_c)$. Then $D'\mathcal{V}D'$ is symmetric, and $D'\mathcal{V}D'd = 0$ on every torus. Consequently there is an explicit symmetric 3-form kernel C, decaying uniformly over flights of index (j,t), with

$$\mathcal{V} = (d_2D'^{-1})^*\,C\,(d_2D'^{-1}) \quad \text{on every torus.}$$

Sketch:

- By (D4), (D3), (D2): $T^*D^{-1}TD'd = T^*dW = dQW = d - \kappa^{-1}dd^*D'd$. Hence $\mathcal{V}D'd = 0$.
- A symmetric decaying 2-form kernel A with $Ad = 0$ on every torus equals $d_2^*Bd_2$, for an explicit decaying 3-form kernel B. This is a lattice Poincaré lemma for translation-invariant kernels, proved by an axis-by-axis contracting homotopy of the Koszul complex of $(e^{ip_a}-1)_a$.
- The hypothesis is needed on all tori, not on one: at a single period the harmonic (zero-momentum) forms obstruct the factorisation.

Lean: `Flight.gap`, `Flight.dep_zero`, `Double.duplex`, `Flight.fund`, `Flight.banked_dollars`. Quantitative control: `duplex_stock`.

Lemma 4.12 (amended defect: identity (Δ)). In the setting of Lemma 4.10, take fine amendment $\mu \geq 0$ and coarse amendment $\lambda \geq 0$. Then (Δ) holds on every pair of tori, with $P = \mathcal{P}_{\mu,\lambda}$ where

$$\mathcal{P}_{\mu,\lambda} = \mathcal{T}_\lambda^*(C + \lambda\,1 - \mu\,S^*S)\mathcal{T}_\lambda.$$

Here S is the 3-form transfer kernel of the flight, with $S\,d_2D'^{-1} = d_2D^{-1}T$, and S^*S has strip bounds $O(L^{-2})$. *Sketch:* Write $X = d_2D^{-1}$ and $X' = d_2D'^{-1}$. Then

$$H'^{-1} - T^*H^{-1}T - \kappa^{-1}dd^* = \mathcal{V} + \lambda X'^*X' - \mu\,T^*X^*XT.$$

Since $XT = SX'$, the last term is $-X'^*(\mu S^*S)X'$; with Lemma 4.11 the whole expression is $X'^*(C + \lambda - \mu S^*S)X'$. Conjugate by H' and use $X'H' = \mathcal{T}_\lambda d_2$ (Lemma 4.7). *Lean:* `Flight.rebased`, `Flight.rollFunds`, `Flight.render`, `Flight.intake`, `Flight.sale`, `Flight.netted`, `check_bleep`.

Lemma 4.13 (positivity; III.3.4). The constants and bounds of III.3.4 hold uniformly over flights of the fixed index (j,t). *Sketch:*

- The strip bound on C bounds its entries at real momenta, so $C \geq -C_1$.
- $\mu S^*S \leq \mu B/L^2 \leq 1$.
- Hence $C + \lambda - \mu S^*S \geq 1$ for $\lambda \geq C_0 = C_1 + 2$.

- $\mathcal{T}_\lambda^{-1} = 1+\lambda X' d_2^*$, with $X' = d_2 D'^{-1}$, has bounded entries, so $|x|^2 \le W|\mathcal{T}_\lambda x|^2$, and $\widehat{\mathcal{P}} \ge W^{-1}$.
- The upper bound and the strip bound follow from products of strip-bounded kernels.

Lean: `Flight.absorb_before`, `Flight.frontier`.

3.6.4.5 4.5 The catalogue Lemma 4.14 (III.3.1). For every kernel choice K, a catalogue satisfying III.3.1 exists. *Sketch:*

- *One hop.* For each (j,t) there is $\lambda_0(j,t)$ with the following property: for every coarse amendment $\lambda \ge \lambda_0$ and fine amendment $\mu \ge 0$, there are rates, constants and an integer $b > \mu B(j,t)$ for which III.3.1(b) holds. The proof combines Lemma 4.13 with Lemma 4.9 at $e = 1, 2$.
- *Assembly.* Set $\lambda_j = \max(\lambda_0(j,j), \lambda_0(j+1,j))$: each stage's amendment acts as the coarse amendment of the hops (j,j) and $(j+1,j)$, and as the fine amendment of (j,j) and $(j,j-1)$. Then:
  - α_j is the minimum of the rates of the bottom quote (Lemma 4.8 with λ_j) and of the hops (j,j), $(j,j-1)$;
  - Ω_j is the corresponding maximum, and η_j the corresponding minimum;
  - $b_j = \max(b(j,j), b(j,j-1))$;
  - weaker constants are always admissible.
- The block size enters only through b_j, which guarantees $\lambda_j B \le L^2$. The stronger choice $L \ge b_j > \lambda_j B(j,t)$ suffices for each admissible t.

Lean: `buys`, `pledges`.

3.6.4.6 4.6 Along a schedule: pre-repair, repair, output Lemma 4.15 (pre-repair family). Fix a catalogue of III.3.1 and a trajectory with its kernel, satisfying the credit condition. Define:

- $\mathcal{B}^{N,i}$: the metric data of Lemma 4.8 for the history of level i in mesh N, with the clip of level i (allowed by (C1));
- $W^{N,i}, T^{N,i}$: those of the flight of step i, of index $(J(i+1), J(i))$ (allowed since $J(i) \le J(i+1) \le J(i)+1$).

This family satisfies the pre-repair conditions of III.3.2(2):

- the bounds, from III.3.1;
- (D1)–(D4), from Lemma 4.10;
- (Δ), with P the rolled kernel;
- the positivity of P, from III.3.1(b) and (C2).

Lean: `shelves`, `stageHop`, `location`, `destination`.

Lemma 4.16 (repair; III.10 with III.6's theorems). For every sheet and every table X there is a table X' with the following property. Every pre-repair family for X (for a trajectory with the sheet's profile and kernel) has a repaired family with the same metric data and the same W, T, which satisfies the output conditions for X'. For step i the construction is:

1. **F06** applied to P from (Δ): a kernel P_1 with two-sided bounds at real momenta, a strip bound, and $d_2^* P_1 d_2 = d_2^* P d_2$ on every torus.
2. **F07**: a symmetric decaying square root, $P_1 = P_2^* P_2$.

3. **F08**: kernels with the same W, T and

$$M_f = VH^{-1}TV'^*, \qquad M_n = \kappa^{-1/2}d^*V'^*, \qquad O = (1-\Pi,\ P_2\, d_2 H'^{-1}V'^*), \qquad \Pi = V'H'^{-1}V'^*,$$

with the strip and derivative bounds of III.3.2(3). The constants depend only on X, κ and the sheet's direct clip coefficients.
4. **F09** (algebra) gives (R1)–(R6). Since $V'^*V' = H'$, Π is the orthogonal projection onto the range of V'. Then:
   - (R1): $M_f^*M_f + M_n^*M_n + O^*O = V'(T^*H^{-1}T + \kappa^{-1}dd^* + H'^{-1}d_2^*P_1 d_2 H'^{-1})V'^* + 1 - \Pi = 1$, by (Δ);
   - (R2) and (R4) follow from $V'^*V' = H'$;
   - (R3) follows from (D4) and $H'd = D'd$, $Hd = Dd$;
   - (R5) follows from $(1-\Pi)V' = 0$ and $d_2 d = 0$;
   - (R6) follows from (D3).

The second table depends on the sheet only through the direct clip coefficients, which enter V, V'. It can be chosen with $\alpha'_j \le \alpha_j$, $\Omega'_j \ge \Omega_j$ and unchanged η_j. *Lean:* `Bid.pinning`, `F06`, `F07`, `F08`, `F09`, `Replaces`. *(repair proof and F08–F09 proofs read)*

Lemma 4.17 (III.3.2 and its variant). The catalogue and repair construction give the families and constants asserted in III.3.2, including its preliminary-sheet variant. *Sketch:* Take the catalogue (4.14), then the second table from the repair theorem with III.6's theorems (4.16), then the pre-repair family (4.15), then its repair. The explicit formulas and the symmetry carry over because the metric data and W, T are unchanged. In the variant, the payment record of the schedule turns sheet-level bounds into (C1)–(C2). A preliminary sheet is handled by re-running the construction for a copy of the trajectory with the preliminary profile; this preserves the histories and the direct clip coefficients. *Lean:* `retained_support`, `retained_presents`, `before_paid`, `credit_from_sheet`.

3.6.4.7 4.7 All tori Lemma 4.18 (rigidity of decaying kernels). Suppose two finite sums of products of decaying kernels, or of cross kernels of the same block size, define equal matrices on tori Λ_{c_k} (respectively on pairs $(c_k, c_k L)$) with $c_k \to \infty$. Then they are equal on every torus (pair). *Sketch:* Sums and products of decaying kernels are decaying kernels, so each side is a single kernel. A summable kernel is the limit of its periodisations as the period tends to infinity. Dominated convergence justifies this coefficient limit; the schedule sides of each fixed step tend to infinity. *Lean:* `entry_limit`, `Echo.all`, `Echo.allAcross`, `Echo.Sound`, `Echo.schedule_sides`.

Lemma 4.19 (III.3.3). Every output family of III.3.2 satisfies (D1)–(D4) and (R1)–(R6) on every torus pair of the same block ratio. *Sketch:*

- (R1)–(R5) are identities between finite sums of products of kernels, with M_f and W as cross kernels. Apply 4.18 on the schedule tori.
- (R6) contains Q. Given (D3) and (R2) at (c, f), (R6) is equivalent to $(Vd)^*M_f = d^*(TV'^*)$, which is again an identity between kernels.
- (D1)–(D4) at arbitrary (c, f) follow by the same argument, because Q is itself given by a cross kernel (III.10). They also follow directly from Lemma 4.10 for any family realising the explicit formulas.

Lean: `complete_roll`, `orthogonal_iff`, `panes`, `universal_diagonal`.

3.6.4.8 4.8 Free covariance of the curvature (III.3.6) Lemma 4.20 (lower bound). The coercivity bound of III.3.6(a) holds uniformly over firm kernels. *Sketch:*

- $\sigma_1^*\sigma_1 + \sigma_0\sigma_0^* = |\tau|^2\,1$ on 1-form symbols; this is $d^*d + dd^* = -\Delta$ in Fourier space.
- A firm f satisfies $\operatorname{Re}\langle x, \hat{f}(p)x\rangle \geq \eta|x|^2$ at every real p: the torus bounds give it at the frequencies $2\pi\mathbb{Z}^4/n$, and these are dense; conclude by continuity.
- Hence $\operatorname{Re}\langle x, \Phi_f(p)x\rangle \geq \min(\eta, 1)|\tau(p)|^2|x|^2$.
- $|\tau(p)|^2 \geq c_0|p|^2$ for $|p_a| \leq 3\pi/2$, since $e^{ir} \neq 1$ for $0 < |r| \leq 3\pi/2$ (compactness).

Lean: `blade_id`, `ion`, `real_price`, `tank_full`.

Lemma 4.21 (smooth symbols). This is III.3.6(c). *Sketch:* Differentiate the Fourier series termwise and use $|y|_1^n \leq A_n e^{\alpha|y|_1}$, with A_n as in III.3.6(c). *Lean:* `radial_prices`.

Lemma 4.22 (comparison with the continuum). This is III.3.6(b). Let $S^k(U)$ be the class of families with $\|D^n g\| \leq C_n|p|^{k-n}$ on U, uniformly over the family. Sums preserve a fixed order, products add orders, and inversion reverses the order when the rescaled matrix has a uniform coercivity bound. *Sketch:*

- $\tau - ip \in S^2$ (Taylor at 0), and $\hat{f} - 1 \in S^1$ (since $\hat{f}(0) = 1$, and by Lemma 4.21).
- Hence $\Phi_f \in S^2$ and $\Phi_f - |p|^2\,1 \in S^3$.
- By (a) and the resolvent identity, $\Phi_f^{-1} \in S^{-2}$ and $\Phi_f^{-1} - |p|^{-2}\,1 \in S^{-1}$.
- Hence $G_f \in S^0$ and $G_f - G_\infty \in S^1$.

Lean: `gem_quote`, `Weight`, `Value`. *(proof read, including the symbol-class product and inverse estimates)*

3.6.4.9 4.9 Calculus lemmas (III.3.7) Lemma 4.23 (nonlinear calculus). The statements of III.3.7 hold under their respective hypotheses. *Sketch:*

- **(a)** Split at $|x|_1 = L$.
  - For $|x|_1 \leq L$: localised Newton remainder estimate, $\|p(x) - \sum_{i<k} P_i(x)\| \leq M'(1+|x|_1)^k$ with M' the bound on k-th differences.
  - For $|x|_1 > L$: bound p and the pieces directly, using $L^{-1-i}(1+|x|)^i \leq L^{-1-k}(1+|x|)^b$ when $L^{-1}(1+|x|) \geq 1$.
- **(b)** Substitute $z \mapsto \sum_v t^{Y_v} p_v z$ in the Taylor polynomial of f and read off each coefficient of t^r for $r \leq d$. Rescaling z by an independent scalar separates the ordinary Taylor degrees, and polarization recovers the multilinear coefficients. All weighted components of weight $\leq d$ of every Taylor coefficient vanish. A component of weight $> d$ vanishes on the given inputs by the support condition.
- **(c)** Locality restricts the logarithmic link fields to links with both endpoints in X. Gauge invariance identifies u near the identity with its pullback by the map taking a site gauge and a radial connection jet to gauged logarithmic holonomies. Part (b) gives weighted vanishing for this pullback. The derivative of that map covers every polynomial link field of degree at most $p-1$: part (d) supplies the highest degree, line integration changes it only by lower degrees, and induction removes the error. Its higher derivatives preserve the total weight. Faà di Bruno and induction on the number of inputs therefore transfer vanishing to $u \circ \exp$. Non-wrapping and $n \geq 2$ ensure that the centred lifts of adjacent sites differ by a unit vector.
- **(d)** $r = (j+2)^{-1}\sum_a x_a Q_a$ is homogeneous of degree $j+2$, and Euler's identity gives $\sum_a x_a(Q_a - \partial_a r) = 0$.

- **(e)** On the unit ball, $\|g(sv) - g(0)\| \le s\|g(v) - g(0)\| + Q(1-s)\|v\|^2$ for the matrix g of the plaquette holonomy of four link variables (second-order Taylor bound). Hence the chord distance at sv is at most $s\tau + (1-s)\tau$.

Lean: see III.3.7; `Maps`, `covers`, `attenuation`, `testLinks`. *(proof read for (a)–(e), including the moment-space definition and transfer argument)*

3.6.4.10 4.10 Auxiliary results Three auxiliary packages concern local observables and equation spaces:

- **Corrected observables.** An exact identity recovering holonomy-jet probes from the coefficients of corrected observables (`Screen.reproduces`). Used by the rank-covariance part of III.11, by VII.2 and by VII.4.
- **Equation projections.** Orthogonal projections onto a finite-dimensional equation subspace and its complement (`drop`, `retain`). The additional criterion `spectral_bank` gives spectral separation under construction and comparison hypotheses, exact top-jet tests, and approximation by a fixed triangular model on finitely many probes.
- **Local jet vanishing.** Vanishing of test-jet derivatives below the engineering degree for non-unit basis observables, assuming their local section is readable (`Speeds.lean`, `canon_first`). Used in the nonlinear step of III.10.

(statements read; the coefficient identity and jet-vanishing proofs read)

3.6.5 5. Where the difficulty lies

- **Uniformity.** The algebraic identities must be accompanied by bounds uniform over histories of any length and over all block sizes, with the kernel choice and stage index fixed. The mechanism is in Lemmas 4.2–4.3:
  - geometric dilation along the history;
  - a lower bound on the principal alias, with an exponent linear in M/L;
  - J-fold smoothing, which gives $\prod_a (1+|\ell_a|)^{-J}$ alias decay and therefore $J-2$ derivatives on W, T.
- **Positivity is proved only at real momenta.** $|\tau|^{-2}$ is never estimated on a strip. Every symbol is put in a filled form (Lemma 4.6) whose denominators are bounded below at real momenta, and inverses are taken on a thinner complex cube. The estimates use the Banach algebra of exponentially decaying kernels and its analytic symbols.
- **Unusual features.**
  - The amendment λ_j (Lemma 4.7) changes the inverse metric. It modifies the metric only off exact 2-forms, so $d^* H d$ is unchanged, and it exists only to make the defect's 3-form factor positive.
  - L enters the catalogue only through $\lambda_j B \le L^2$.
  - The second table depends on the sheet through its direct clip coefficients.
  - The factorisation of the defect (Lemma 4.11) needs the identities on all tori, because of the harmonic forms.
  - For explicit families, `universal_diagonal` gives the diagonal identities directly. The output conditions alone give them by rigidity.
- **Units.** Decay is in the ℓ^1 norm, on the lattice where the kernel acts. For W, T, M_f that is the coarse lattice, so the fine-lattice rate is α/L, with amplitudes L^{-1}, L^{-2} and an extra L^{-1} per fine difference. These are the scalings of Balaban's background field: smooth on scale L.

- **Statement strength.** III.3.2 is an existence statement whose output is a package of bounds and identities. Later units use (R1)–(R6), the floors and the strip bounds. The repaired family retains the pre-repair metric data and W, T, so the defect identity and positivity remain available, although they are not fields of the output conditions.

3.6.6 6. Notation

Written proof	Lean (glossary)	Meaning
R, $\hat{R}$, R_n; $\mathcal{S}(\alpha, \Omega)$	`Row`, `Row.strip`	translation-invariant kernel, symbol, action on Λ_n; strip bound
cross kernel	`Cross`, `installed`	coarse-to-fine kernel and its action
$\mathrm{plank}_M, \mathrm{rail}_M, a_L, a_L^J$	`plank`, `rail`, `area`, `massFilter`	line, box and J-fold box filters
history v, dilation M_v, filter s_v	`Train`, `Train.smooth`	finite history of block averages
flight of index (j, t)	`Flight`, `Flight.land`	one step: block size, fine history, coarse history
D_a; ρ; β_a, h	`Train.levelD`, `D_feedback`; `Train.rho`; `Train.bulb`, `Train.bend`	1-form covariance; gauge factor; filled symbols
D, D^{-1}	`pink`, `blue`	diagonal 2-form multiplier, inverse
H^{-1}, H; $\mathcal{T}_\lambda$; λ	`pledge`, `amend`; `treble`	amended inverse metric, metric; 3-form factor; amendment
metric data $\mathcal{B}$, V	`Bed`, `Bed.full`, `Floor`	$(D, D^{-1}, H, H^{-1}, R)$; $V = (\sqrt{c}\,1, R)$
transfer kernels (W, T, M_f, M_n, O)	`Towers` (`A0`, `T`, `mf`, `mn`, `o`), `Flight.duo`	background-field map, curvature transport, isometry blocks
(D1)–(D4); (R1)–(R6); (Δ)	`Towers.Diagonal`, `Inputs`; `Towers.Roll`; `Bleeps`	identities of one step
C, S, $\mathcal{P}_{\mu,\lambda}$	`Flight.fund`, `Flight.sale`, `Flight.rollFunds`	unamended 3-form defect, 3-form transport, rolled 3-form kernel
derivative bounds	`Donate`, `Towers.Deep`	∇_u bounds with $L^{-1-\lvert u\rvert}$, $L^{-2-\lvert u\rvert}$
positive on $\operatorname{ran}\sigma_2$; two-sided at real momenta	`Reaches`, `Funds`	positivity conditions on 3-form kernels
kernel family; pre-repair; output conditions	`Horizon`; `Craft.Rack`; `Horizon.Suffices`	metric data and transfer kernels for all (N, i)
explicit formulas; credit condition	`Present`; `Credit`	(III.3.2(1)); (C1)–(C2)
catalogue; price table	`Catalog`; `MapPrices`	constants fixed before L
hypercubic symmetry of d^*Hd	`Bid.Sworn`	invariance under signed permutations

Written proof	Lean (glossary)	Meaning
firm kernel; Φ_f; G_f; G_∞	`Firm`; `fuel`; `garnet`; `pearl`	curvature covariance, lattice and continuum

3.6.7 7. What remains unexplained

1. **Gaussian interpretation and zero modes.** Lemma 4.4 verifies the covariance recursion away from zero momentum. A measure-level derivation requires a stated treatment of gauge directions and harmonic modes; neither the auxiliary singular symbols nor III.1's `action` alone specifies that Gaussian measure. The identification with an iterated Feynman-gauge covariance remains probable.
2. **Change of Gaussian variables.** The quadratic splitting in §1 follows from the identities. Its use as a change of integration variables, including null directions and determinants, belongs to III.4 and III.10 and is not established by this unit alone.
3. **Auxiliary spectral criterion.** The finite-test hypotheses of the cited cutoff-projector criterion must be supplied at its point of use. Its statement does not assert that a trajectory satisfies them.
4. **Further proof detail.** The analytic coefficient-extraction machinery underlying the strip bounds and the deeper positivity and square-root proofs F06–F07 are inputs here; their full proofs remain outside this account. The symbol-class calculations in `Gem.lean` and `Weight.lean` support the covariance comparison, but no position-space continuum limit or contact-term statement is asserted in III.3.6.

3.7 III.4. Gaussian integration: change of Gaussian sector, determinants, window law

Part: III, Ultraviolet construction (renormalization group). **Lean:** about 40 work packages. The main ones are `YM/SectorNext/GAUSSIAN_SECTOR_CHANGE/` (exact change of variables, gap, windows), `YM/AnalyticBridge/RELATIVE_LOGDET_TRACE/` and `YM/PhysicalNext/AFTERNOON_INTERACTING_RANK_COMPARISON/` (determinants and the polymer form of a sector), `YM/AnalyticBridge/ANNULUS_RATIO_INSERTIONS/`, `YM/SectorTools/PHYSICAL_GAUSSIAN_INSERTION/` and `YM/Primitives/CONNECTED_GAUSSIAN_CUMULANTS/` (Gaussian moments), `SECTOR_AFFINE_JETS/`, `SECTOR_COVARIANCE_JETS/` and `SELECTED_ORDERED_KERNELS/` (jets), `LOCAL_GAUSSIAN_LAWS/`, `GAUSSIAN_FIELD_ALIGNMENT/`, `YM/Push/FREE_RESOLVENT/` and `YM/RG08/Gaussian/` (locality, infinite-volume kernels), `YM/SectorNext/LARGE_FIELD_REMAINDERS/`, and the constant-producing statements in `Next/STATIONARY`, `MergeProof` and `PhysicalNext/V9_JOINT_ENDPOINT_LOAD`. **Size:** 37,956 lines in 4,737 declarations of the main-theorem closure.

Conventions. The parameter x is the coarse inverse-coupling parameter: in applications $x = \hat{\beta}_m$. Throughout:

- $R(x) = \lceil (\log x)^2 \rceil$ is the cluster radius (`radiusB`).
- $t(x) = x^{-1/2} e^{(\log\log x)^2}$ is the small-field threshold (`temp`).
- $\beta_F(x, L) = x + s_G \log L$ is the coupling one level finer.
- $t_L(x) = t(\beta_F(x, L))$ is the fine-level threshold (`fiery`).
- $g = \beta^{-1/2}$.

- "For large x" means: there is x_0, depending only on the constants fixed before it, such that the statement holds for $x \geq x_0$.
- Write $\|A\|_s$ for the maximum of the row and column sums $\sum_j e^{s\,\mathrm{dist}(i,j)}|A_{ij}|$. Positions lie on the coarse torus; several coordinates may have the same position. Matrix inequalities refer to quadratic forms. Put $r_x = \lfloor R(x)/60 \rfloor$.

3.7.1 1. What this unit proves

Setting. One block-spin step S maps a fine torus Λ_f (side $f = Lc$, block size L odd) to a coarse torus Λ_c. The push-forward of the density at fixed coarse field V is decomposed (III.1, III.9) into sectors. A sector is given by:

- a store A: the large-field data, made of cost cells, aisles and flagged sites, with a partition of unity bill_A given by an allocation D;
- a choice k of gauge-fixing tags;
- a spanning forest F (tree gauge).

The integration variables are the free links of F, with product Haar measure. In each sector the unit splits the free links as follows (`chill`):

- **frozen links** $\mathcal{F}_A$: a parent block of an endpoint lies within distance $10R(x)$ of A. These are kept as Haar variables Y.
- **active links** $\mathcal{A}_A$: all others. These are written $u_e = \exp(a_e)\,\nu_e$, with $a_e \in \mathfrak{g}$ and $\nu = \iota V$ the copy of V on the fine lattice (`replica`).

We write $n_a = |\mathcal{A}_A| \dim \mathfrak{g}$.

On selected stencils, the step energy contains a sum of affine squares $Q_Y(a) = \frac{1}{2}|Ma + s(Y)|^2$ (`guide`):

- M is a linear map from active coordinates to rows of linearised plaquette curvatures and block constraints;
- $s(Y)$ depends on the frozen links.

From these the unit defines three objects:

- the **precision** $W = M^\top M$ (`Rows.gram`);
- the **covariance** $C = W_{\mathcal{W}}$, the symmetrised windowed inverse of W, with windows of coarse radius $R(x)/20$ (`fog`);
- the **mean** $m(Y) = -(W^{\mathrm{per}})^\top M^\top s(Y)$ (`meanLog`). This is a window approximation to the minimiser $-W^{-1}M^\top s(Y)$; its gradient need not vanish.

The unit proves four things:

- an *exact* Gaussian change of variables in every sector of every RG step;
- the gap and locality of W and C, uniformly in volume, store and field;
- local expansions of the log-determinants, and a polymer form of each sector with large-field suppression;
- Gaussian moment bounds, jet bounds, and the production of all post-L constants.

Proposition III.4.1 (exact Gaussian change of variables in one sector). Fix the following data:

- a step S and a tube T satisfying the funding conditions of a teller: the prescribed kernel and stage, the multiscale metric bounds, the prescribed fine and coarse cutoffs, bounds on both path families, and compatible free metrics and towers;
- a representation R of the fine density whose paths satisfy their bounds and whose activities satisfy the validity bounds, with $\beta = \beta_R$;
- x, V, k, A, F, an allocation D and a cutoff profile χ;
- a chart ratio J, meaning:
  - $J_{\mathrm{sc}} > 0$ and $\rho_J > 0$, and J is measurable;
  - J is real-analytic and positive on a ball $B_{\rho_J} \subset \mathfrak{g}$;
  - $J(0) = 1$;
  - exp is injective on B_{ρ_J};
  - $J_{\mathrm{sc}}\, J(a)\, da$ is mapped by exp onto normalised Haar measure restricted to $\exp B_{\rho_J}$;
  - $\exp B_{\rho_J}$ is open;
- a number $d > 0$.

Let C be a matrix on the active coordinates and m a map from frozen arrays to active coordinates. Assume they are *permitted* (`Permitted`):

- $C \succ 0$, $\beta > 0$, m measurable, and $4d < \rho_J$;
- the *standing condition* (`Standing`): whenever $\mathrm{bill}_A \neq 0$ and the gauge tags are eligible, every active link equals $e^{a_e}\nu_e$ with $|a_e| \le d$.

Let $\Phi_{A,k} = \mathrm{bill}_A \cdot \mathrm{exposed}_R \cdot \mathrm{enters}_{T,\beta,k,V}$ be the sector integrand. Here $\mathrm{exposed}_R$ is the interacting density of R, and enters is the tagged block-spin kernel times $e^{-\beta\cdot\text{gap energy}}$. Then the following hold (`Licenses`):

- (a) $\Phi_{A,k}$ is integrable on the slice;
- (b) for every Y, $\Psi(Y,\cdot) \in L^1(\gamma_C)$, and $\Psi \in L^1(\mathrm{Haar}^{\mathcal{F}_A} \otimes \gamma_C)$;
- (c) the identity

$$\int \Phi_{A,k}\, d\mathrm{Haar}^F = \mathrm{pref}_S(\beta)\, J_{\mathrm{sc}}^{|\mathcal{A}_A|}\, (2\pi g^2)^{n_a/2} (\det C)^{1/2} \int \mathrm{Haar}^{\mathcal{F}_A}(dY) \int \gamma_C(dz)\, \Psi(Y,z).$$

Here $\gamma_C = N(0,C)$. Set $a = m(Y) + gz$, and let U be the configuration with frozen links Y, active links $e^{a_e}\nu_e$ and tree links 1. Then (`dyed`)

$$\Psi(Y,z) = \tilde{J}_d(a)\, \chi_d(a)\, \mathrm{bill}_A\, \mathrm{lik}_k(U)\, \mathrm{exposed}_R(U)\, \exp\Big(-\beta\, \overline{\mathrm{res}}(Y,U,a) - \beta Q_Y(m(Y)) + \eta\cdot z + \tfrac{1}{2} z\cdot(C^{-1} - W)z\Big).$$

In this formula:

- $\eta = -g^{-1}(Wm(Y) + M^{\top} s(Y))$;
- $\overline{\mathrm{res}}$ is the orientation-averaged step energy minus the affine squares;
- $\tilde{J}_d(a)$ is the product of the extended link Jacobians, each equal to $J(a_e)$ for $|a_e| \le 2d$; χ_d is the product cutoff, equal to 1 when every $|a_e| \le d$.

Equivalently, $\mathrm{pref} \cdot \Psi(Y,z) = e^{\frac{1}{2} z\cdot C^{-1} z}\, r(Y, m(Y) + gz)$, where r is the integrand in Lebesgue log-coordinates (`recolored`). The identity holds for *every* permitted (C,m). No quantitative locality assumption is needed.

Status: theorem.

Lean: `authorized_change`.

Proposition III.4.2 (licence of every committed RG step; the transfer identity). Let $\mathcal{R}$ be a directed RG run (III.9) whose delays satisfy the rate conditions and the per-stage chart conditions (`Starcash`). Let $m < N$, let p be a period, and let R be a representation at level $m+1$ on its selected history with $\beta_R \in [\hat{\beta}_{m+1}/2,\, 2\hat{\beta}_{m+1}]$. Write:

- S for the step from level $m+1$ to level m;
- $x = \hat{\beta}_m$ and $\beta = \beta_R$;
- $j = J(m+1)$ and $i = J(m)$ for the stage indices;
- $d = a_{ji}\, t(\hat{\beta}_{m+1})$, and w for the allocation width at this stage pair.

Then:

- $w > 0$, $d > 0$, $\beta > 0$, $E_{ji} > 0$, $2d < \rho_{\text{first}}$ and $4d < \rho_J$;
- $|\log J(t)| \le E_{ji}|t|^2$ for $|t| \le 4d$;
- there is $\kappa > 0$ such that for all allocations D of width w, forests H and profiles χ:
  - (i) writing I_A for the signed sector column, its contribution to the push-forward, and L_A for the nonnegative column with absolute fine density, after removing the selected packet energy and kernel prefactor, both have bounded Borel envelopes and $|I_A(V)| \le \mathrm{pref}(\beta) e^{-\beta\kappa t_L(x)^2|A|} L_A(V)$ (`PortionCost`);
  - (ii) for all A, k, V, the chart is admitted (`Admitted`); when $\mathrm{flags}(A) = \mathrm{flags}_{t(x)}(V)$, also $C = W_{\mathcal{W}} \succ 0$, $W \succ 0$ and the selected predictor rows have analytic derivatives with a common bound; and the conclusions of III.4.1 hold with $C = W_{\mathcal{W}}$ and the windowed mean for every store (for wrong flags trivially, since $\mathrm{bill}_A \equiv 0$);
  - (iii) for every V:

$$\mathcal{S}_\beta(\rho_R)(V) = \sum_A \mathrm{sieve}(V) \sum_k \mathrm{pref}\, J_{\mathrm{sc}}^{|\mathcal{A}_A|} (2\pi g^2)^{n_a/2} (\det C_{A,k,V})^{1/2} \iint \Psi \, d\gamma_C \, d\mathrm{Haar}.$$

Here $\mathcal{S}_\beta$ is the block-spin push-forward, ρ_R is the fine density, and $\mathrm{sieve}(V) = \exp(-\beta \cdot$ orientation-averaged coarse energy of $V)$.

Status: theorem, conditional on the run, history and band hypotheses. III.4.12 supplies the rate and chart conditions; III.9 supplies the selected history.

Lean: `committed`, `displaced_wrong`.

Proposition III.4.3 (gap and locality of the actual precision and covariance). Fix G, a patch $P = (L, J)$ and a teller whose kernel satisfies the branch condition. There are $v, \eta, b, E > 0$ and $K \ge 0$ (post-L constants) with the following property. For large x, take any step with patch P and coarse side $c \ge 100(1 + R(x))$, any funded tube, and any A, F, k, V with $\mathrm{flags}(A) = \mathrm{flags}_{t(x)}(V)$. Then:

- (i) $\eta|a|^2 \le a \cdot Wa$ for all active a (`Gap.not_flat`);
- (ii) W has exponential off-diagonal decay in the Schur norm: $\sup_i \sum_j e^{v\,\mathrm{dist}(i,j)} |W_{ij}| \le K$, and the same for columns, where dist is the coarse distance between the parent blocks of the coordinates;
- (iii) $\|C - W^{-1}\|_0 \le E\, e^{-b r_x}$, and $C \succ 0$;
- (iv) $C \le K'$ for some post-L constant K' (`open_price`).

The constants do not depend on the volume, the store, the forest or the field.

Status: theorem.

Lean: `opaqueCost`.

Proposition III.4.4 (window law). Let $\sigma, \eta > 0$, $K \geq 0$, and fix a nonnegative function $B : \mathbb{R} \to \mathbb{R}$. There are $u, E, g > 0$ such that for every sufficiently large integer r the following holds, uniformly in the torus and the finite coordinate set. Let the coordinate positions p_j satisfy $\sum_j e^{-v \operatorname{dist}(i,p_j)} \leq B(v)$ for every coarse site i and every $v > 0$ (`Revenue`). Let A be a finite symmetric matrix with:

- Schur decay $\|e^{\sigma \operatorname{dist}} A\|_{\mathrm{Schur}} \leq K$;
- gap η;
- windows $\mathcal{W}_i$ that contain the r-ball around i.

Let L be the windowed inverse, $L_{\cdot i} = (A|_{\mathcal{W}_i})^{-1} e_i$ extended by zero, and $A_{\mathcal{W}} = \frac{1}{2}(L + L^{\top})$ (`clouded`). Then:

- $\|A_{\mathcal{W}} - A^{-1}\|_{\mathrm{Schur}} \leq E e^{-ur}$;
- $A_{\mathcal{W}} \geq g = \eta/(2(K+1)^2)$ and $A_{\mathcal{W}} \succ 0$.

A strengthened form (`same_purchase`) gives a rate $s \in (0, \sigma]$ and a constant K_1 such that A, A^{-1}, L and $(A_{\mathcal{W}})^{-1}$ all have s-weighted Schur norm $\leq K_1$, and $\|e^{s \operatorname{dist}}(L - A^{-1})\|_{\mathrm{Schur}} \leq K_1 e^{-sr}$.

Status: theorem. The constants depend only on (σ, η, K, B).

Lean: `Schur.windows`.

Proposition III.4.5 (log-determinants as local path series). Let $A \succ 0$ with windows $\mathcal{W}$ ($i \in \mathcal{W}_i$). Let P_u be the column-wise windowed inverse of $A + uI$, and $\Delta_u = (A + uI)P_u - I$. Assume the following bounds (`Priced`), at a fixed nonnegative Schur rate, with $C \geq 1$ and $0 \leq q < 1$, for every $u \geq 0$:

- $\|A\| \leq C$;
- $\|(A+u)^{-1}\|$, $\|P_u\| \leq C/(1+u)$;
- $\|\Delta_u\| \leq q/(1+u)$;
- $\|S\| \leq q$, where $S = AA_{\mathcal{W}} - 1$.

Assume also $A_{\mathcal{W}} \succ 0$. Then

$$\log\det A = \sum_i \Big(\ell_i + \sum_{n \geq 0} r_{i,n}\Big), \qquad \log\det(1+S) = \sum_i \sum_{n \geq 0} \frac{(-1)^n (S^{n+1})_{ii}}{n+1}, \qquad \log\det A_{\mathcal{W}} = -\log\det A + \log\det(1$$

with:

- $\ell_i = \int_0^\infty [(1+u)^{-1} - (P_u)_{ii}]\, du = (\log A|_{\mathcal{W}_i})_{ii}$, the single-window letter;
- $r_{i,n} = -\int_0^\infty (P_u(-\Delta_u)^{n+1})_{ii}\, du$ and $\int |\cdot| \leq C q^{n+1}$.

All series converge absolutely. Moreover (`labeled`), the matrix products expand into closed index paths. A term $r_{i,n}$ has one P_u factor and $n+1$ defect factors, hence $n+2$ jumps; the corresponding trace-series term has $n+1$ jumps. The absolute values, summed over the intermediate indices, are bounded by Cq^{n+1} and $q^{n+1}/(n+1)$, respectively.

Status: theorem.

Lean: `Priced.chains`.

For the relative expansions, the sector regularity conditions are: a lawful representation; matching fine paths and cutoff; a local chart with $d > 0$, $4d < \rho_J$ and $2d < \rho_{\text{first}}$; positive coupling and sector width; $0 < t_L(x) \le 1$; output threshold $t(x)$; and the tube's geometric, local-readability, predictor-derivative, covariance-positivity and gap conditions (`Chips`, `Boosted`).

Proposition III.4.6 (relative normalisation of a sector and its localisation). Let K_F be the flat kinetic precision: the Gram matrix of the move at $V = 1$ with empty store, on all free links of F.

(i) If the fine path family satisfies its path bounds and the kernel satisfies the patch's branch condition, $\det K_F$ does not depend on the forest or on the gauge tags (`Native.shared`).

(ii) Assume:

- the tube is funded, and its fine path family and cutoff are those of R;
- the precision certificate of Lemma 4.21 holds;
- J is a local chart ratio, $\beta > 0$, and the threshold of the output budget point is $t(x)$.

Then on every sector point $p = (A, F, k, V, Y, z)$ with matching flags, the positive density factor, with the sector masks and polymer product kept outside it, factorises as

$$\text{freight}(p) = \text{loading}(V)\, e^{\Lambda(p)}, \qquad \text{loading}(V) = \text{banker}(V)\, \text{pref}(\beta)\, e^{v_R|\Lambda_f|}\, \text{spare}(\beta)^{|F^{\text{free}}|} (\det K_F)^{-1/2} > 0.$$

Here $\text{freight} = J_{\text{sc}}^{|\mathcal{A}|}(2\pi g^2)^{n_a/2}(\det C)^{1/2} e^{\frac{1}{2} z\cdot C^{-1} z}\, \tilde{J} \cdot (\text{noise}) \cdot e^{\text{exponent}_R}$, and $\text{spare}(\beta) = (2\pi g^2)^{\dim \mathfrak{g}/2} J_{\text{sc}}$ is the Gaussian normalisation per link. The exponent is

$$\Lambda(p) = -\beta\, \mathcal{E}(p) + \eta \cdot z + \tfrac{1}{2} z \cdot (C^{-1} - W) z + \kappa(p) + \sum_e \log \tilde{J}(a_e) + (\text{tails}),$$

$$\kappa(p) = -|\mathcal{F}_A| \log \text{spare}(\beta) - \tfrac{1}{2}(\log\det W - \log\det K_F) + \tfrac{1}{2}\log\det(WC).$$

Here $\mathcal{E}$ is the orientation-averaged row energy, including the subtraction of the affine energy at a and its addition at m:

$$\mathcal{E} = E_{\text{noise}}(U, V) + E^{\text{cut}}_{\text{fine}}(U) - E^{\text{cut}}_{\text{coarse}}(V) - Q_Y(a) + Q_Y(m).$$

The tails are the remaining fine exponent after removing its volume constant and restoring its cut energy, together with the coarse cutoff correction. Thus $\mathcal{E}$ is not merely a sum of affine squares.

(iii) Assume in addition the sector regularity conditions below, a nonnegative polymer exponent, and both path-counting bounds of Lemma 4.21. Then $\kappa(p) = -|\mathcal{F}_A| \log \text{spare} + \sum_e \text{sect}_e + I + \text{Sh}$ (`book_price`), where:

- $\text{sect}_e = -\frac{1}{2} \sum_{\text{colours}} [\ell_e(\text{actual}) - \ell_e(\text{flat})]$, the difference of the single-window letters of III.4.5 for the identity-completed full precision at (A, V) and at $(\emptyset, 1)$;
- I is the corresponding difference of path letters;
- $\text{Sh} = \frac{1}{2}\log\det(1 + S)$.

(iv) Regrouping (`rate_pass`) gives $\kappa = \sum_o \text{rent}_o + \text{far} + I + \text{Sh}$:

- o runs over the cores of A (the components formed by overlap of the $500R(x)$-halos of its seeds);
- rent_o is the sum of $\text{sect}_e - \log \text{spare} \cdot 1_{e \in \mathcal{F}_A}$ over links within $200R(x)$ of o;
- far is the same sum over links farther than $200R(x)$ from A.

Status: theorem.

Lean: `Dyed.cardinal_spin`, `Dyed.keptEarly`, `Dyed.spoolLift`.

Proposition III.4.7 (polymer form of a sector, with large-field suppression). Fix:

- a teller and a patch D satisfying the branch condition;
- a determinant-jet fund and a local chart ratio J;
- a polymer exponent $e \geq 0$, a chart-width multiplier $h \geq 0$, constants $a, \alpha, \tau > 0$ with $2\alpha \leq a/2$, and certificate constants $C_0 \geq 0$, $g_0, b_0, d_0 > 0$, $\eta_0 \in \mathbb{R}$.

Then for large x the following holds. Let $y = \mathrm{retime}(x)$, $z = \beta_F(x, D.L)$ and $\mathrm{hot}(s) = e^{(\log\log s)^{3/2}}$. Eventually $x \leq y \leq x^2$ and $t(y) = x^{-1/2}\mathrm{hot}(x)$. Take a step with patch D, a sector datum with chart J, and positive cutoff parameter W_0, subject to the following assumptions:

- the representation is lawful; the tube has the same fine paths and cutoff as the representation and satisfies the funded, geometric and chart-admission conditions; the chart radius $d = h\, t_{D.L}(y)$ satisfies $d > 0$, $4d < \rho_J$ and $2d < \rho_{\mathrm{first}}$; the output threshold is $t(y)$;
- the step satisfies the free-metric size condition and $c \geq 100(1 + R(y))$, and its fine and coarse path families have a common path bound;
- $\beta \in [z/2, 2z]$; the fine threshold $s = z^{-1/2}\mathrm{hot}(z)$ also lies in $[t_{D.L}(y)/2, 2t_{D.L}(y)]$;
- on each eligible packet the row energy $\mathcal{E}$ is at least as^2 times its size, whenever the sector cutoffs are nonzero;
- the support exponents satisfy the reserve inequality with parameters (e, τ); the budget radius B_0 satisfies $25R(y) + 30 \leq B_0 \leq \mathrm{walls}(y)$, and the fine budget radius is at most $2B_0$;
- the lawful-representation bound can be chosen with activity constant $\Gamma \leq \beta_F(y, D.L)^{1/8}$ and smallness parameter $\delta \leq y^{-1/2}$, and the fine paths satisfy the required row-counting bound;
- the determinant-jet certificate at halo 20, exponential order 2 and derivative order 1, and the precision and two path-counting certificates of Lemma 4.21 hold at y, with constants $(g_0, C_0, \eta_0, b_0, d_0)$;
- uniformly over flag-matched sectors and frozen arrays, $\|C\|_0$ is bounded by a common constant, every component of m and of $Mm + s(Y)$ has absolute value at most 1, and $|\eta_i| \leq y^{-4}$; on the chart-cutoff support, $|z_i| \leq y^2$.

For such data there is a decomposition (`Plate`)

$$\Lambda(p) = \sum_{o \in \mathrm{cores}(A)} \mathrm{base}_o(p) + \sum_X \mathrm{aux}_X(p)$$

whenever the flags match and the product of the fine-field, block, gauge-tag and chart cutoffs is nonzero. The bounds below hold for every flag-matched p, including points outside that support. In it:

- base_o depends only on the data in the $500R(y)$-neighbourhood of o;
- aux_X depends only on the data near the local support X;
- both are gauge-invariant and lattice-covariant;
- $\mathrm{base}_o \leq -\alpha\,\mathrm{hot}(z)^2|o|$, where $|o|$ counts cost cells, aisles and flags;
- $|e^{\mathrm{aux}_X} - 1| \leq f_X$, with $f \geq 0$ satisfying the polymer bound $\sum_{X \ni y'} f_X e^{e\, N_{B_0}(X)} \leq y^{-1/48}$ at every site y', where N_{B_0} is the cover number at the output budget radius (`Hit`).

Status: theorem.

Lean: `foreshadows`, with hypotheses `Helps`.

Proposition III.4.8 (moments under the sector law). Fix G. For every covariance cap $L > 0$ and $k \in \mathbb{N}$ there is $D \geq 1$ with the following property. Take:

- any finite index set I and any probability space (Ω, μ);
- any $C \succeq 0$ on $\mathbb{R}^{I \times \dim \mathfrak{g}}$ with $C \leq L$;
- any finite family $(r_i)_{i \in Y}$ of measurable insertions with $w_i, d_i \geq 0$, $|r_i(\omega, z)| \leq w_i e^{d_i(1+|z_{a_i}|^2+|z_{b_i}|^2)}$, and $\sum_{i:\, x \in \{a_i, b_i\}} d_i \leq c = 1/(24L)$ at every coordinate site x;
- any $\rho, H \geq 0$ and sets $P, J \subset I$;
- any measurable v with $|v| \leq H(1 + \sum_{i \in P} |z_i|^2)^k$ that vanishes unless $|z_i| \geq \rho$ for all $i \in J$.

Then $f = v \prod_{i \in Y} r_i \in L^1(\mu \otimes \gamma_C)$ and

$$\int |f| \, d(\mu \otimes \gamma_C) \leq H\, D\, (1 + |P| \dim \mathfrak{g})^{2k+1}\, e^{-c|J|\rho^2}\, 2^{\dim \mathfrak{g}\, |J|} \prod_{i \in Y} (e^c\, 4^{\dim \mathfrak{g}} w_i).$$

In a sector, take $\mu = \mathrm{Haar}^{\mathcal{F}_A}$ and $C = W_{\mathcal{W}}$; III.4.3 supplies a uniform covariance cap. Moreover:

- (`r4`) For measurable q_i on the same product space, let $|q_i| \leq x_i^2(1 + |z_{a_i}|^2 + |z_{b_i}|^2)$ with $0 \leq x_i \leq 1$ and $\sum_{i \ni \mathrm{site}} 8x_i \leq c$. Then $\int (\prod_i (e^{q_i} - 1))^4 \leq (\prod_i M x_i)^4$, with $M = M(G, L)$.
- (`oasis`) Fix a patch, a teller satisfying its branch condition and a covariance cap $L > 0$. For large x, every funded step with that patch and $c \geq 100(1 + R(x))$, matching flags, $0 < d \leq 1$, $2d < \rho_{\mathrm{first}}$ and $\beta \in [\beta_F(x, L_{\mathrm{block}})/2, 2\beta_F(x, L_{\mathrm{block}})]$ has the fourth-moment insertion bounds above. The single-site exponents come from $\eta \cdot z$ and the pair exponents from $\frac{1}{2} z \cdot (C^{-1} - W) z$. Their weights are $De^{-uR(x)/480}$ for sites and exponentially decaying in distance for pairs; the constants are fixed before the threshold in x, which may depend on the covariance cap.

Status: theorem.

Lean: `same_law`.

Proposition III.4.9 (Gaussian insertions supported on selected coordinates). Fix $p \geq 1$, a finite jet order, G, a patch and a teller satisfying the branch condition. There are post-L constants $K, M, s, b > 0$ and $B \geq 0$ such that, for large x, the following holds. Take a funded step with that patch, $c \geq 100(1 + R(x))$ and matching flags. Fix any $g, d, \theta > 0$, frozen array and slot set U. Restrict the linear and quadratic correction to U, obtaining $t \cdot z_U + \frac{1}{2} z_U \cdot \mathcal{J} z_U$, where $t = \eta$ and $\mathcal{J} = C^{-1} - W$. Its tilted law is $N(\mathcal{A} t_U, \mathcal{A})$, with $\mathcal{A} = (C^{-1} - \mathcal{J}_U)^{-1} \succ 0$. The likelihood ratio of the U-marginals satisfies

$$\mathbb{E}_{\gamma_{C,U}}[(dN_U / d\gamma_{C,U})^p] \leq \exp\Big(2p(|U|\ell^2/(4\theta) + 2K|U|(\theta + \varepsilon/2))\Big)$$

whenever $Kp(\theta + \varepsilon/2) \leq 1/4$. The parameters are:

- $\varepsilon = Ke^{-bR(x)}$;
- $\ell = g^{-1} M^2 e^{-s r_x} B\, (t(x) + d)$.

The support restriction concerns the Gaussian variable z; the coefficients may depend on all the background and frozen data. The bound retains its dependence on g, d and $|U|$.

Status: theorem.

Lean: `Physical.blocks`, built on `insert_letter`.

Proposition III.4.10 (jets of the block coefficients and of the covariance). Fix G, a patch P, a teller satisfying the branch condition, and an order n. There are $r, d, v, \eta, \lambda_0, C_* > 0$ such that for large x the following holds. Take every funded step with patch P and $c \geq 100(1 + R(x))$, every A, F, V with matching flags, every k, and every logarithmic perturbation B of the background with $\|B\| \leq r$. The store, forest and tags remain fixed as B varies. Then:

- the move matrix $M(B)$ is real-analytic;
- M, $W = M^\top M$ and C satisfy weighted Schur bounds up to order n. For every list ℓ of at most n coordinate directions, with coarse positions p_b, and output and input positions p_i, p_j,

$$\prod_{b \in \ell} e^{d\,\mathrm{dist}(p_b, p_i)} \sum_j e^{d\,\mathrm{dist}(p_i, p_j)} |\partial_\ell X_{ij}(B)| \leq C_*,$$

  and the analogous column bound holds with the external weights anchored at p_j;
- $W(B) \geq \eta I$, $C(B) \geq \lambda_0 I$ and $C(B) \succ 0$;
- for every $u \geq 0$, the full, principal and windowed resolvents have the same marked jet bounds with cap $C_*/(1+u)$ at order zero and $C_*/(1+u)^2$ at positive orders;
- the window defect has marked jet cap $C_* e^{-vR(x)}/(1+u)$, and the difference between the windowed and full resolvents has cap $C_* e^{-vR(x)}/(1+u)^2$.

The selected predictor rows are real-analytic, with k-jets bounded by constants depending only on (G, P, K, k), on a ball of fixed radius (`sounds`, `chosen_branch`). The precision W is also real-analytic (Lemma 4.27); the covariance and resolvent conclusions used here are finite-order jet bounds.

Status: theorem.

Lean: `assembly`, `fromTeller`, `coverall`, `stores`.

Proposition III.4.11 (affine squares in the large-field annulus). Fix a teller, a block size L, $W_0 > 0$, $K \in \mathbb{R}$ and an integer halo $D \geq 0$. There are $C \geq 1$ and an eventually nonnegative function $e(x)$ with $(1+R(x))^n \beta_F(x, L) e(x) \to 0$ for every $n \in \mathbb{N}$, such that for large x the following holds. Take a funded step of block size L with $c \geq 100(1 + R(x))$, stores O, A, a forest F, fields U, V and tags k. Assume U is in tree gauge. In the $(D+5)R(x)$-halo of O, assume the following local sector conditions: fine plaquettes whose support meets that halo and which belong to A have toll at least $t_L(x)/4$, those outside A have toll at most $t_L(x)/2$; block bonds whose support meets that halo and which belong to A have discrepancy at least $W_0 t_L(x)$, those outside have discrepancy at most $2W_0 t_L(x)$; the coarse flags match V; and the chosen frames have positive mass. Then there is a direction a, vanishing on the tree links, such that for every pair of windows H, Q with $H_{\mathrm{face}} \subseteq Q_{\mathrm{face}}$ and $H_{\mathrm{noise}} \subseteq Q_{\mathrm{noise}}$, and with the face, noise and coarse rows of Q in $\{p \notin A^{(2R)},\ p \in O^{(DR)}\}$:

- Q is charted by a with constant C;
- the affine series on H converge, the required curvature and predictor derivatives exist with the stated linearisation, and $\mathrm{work}_Q \geq \mathrm{squares}_H(a) - e(x)|O|$;
- for every other a' charting H with constant K, the same convergence, derivative statements and bound hold with a'.

Here $|O|$ is the number of cost cells, aisles and flags. The comparison with a' uses uniqueness of the logarithm on the selected stencils.

Status: theorem.

Lean: `annulus_pay`, with `cleans_at` and `reacts`.

Proposition III.4.12 (production of the step constants). Assume:

- G is simple;
- the kernel choices K satisfy the branch condition for every patch;
- the initial stage is at least 4; the jet orders, position orders and support exponents are nondecreasing, the support exponents are positive, and at stage i the two orders are at least $\max(6, i)$ and $\max(2, i)$, respectively;
- costs $C_i \geq 1$ are given.

Then there are pre-L data with kernel K and the prescribed orders, two families of path prices, free-field and slope bounds, and a cost bound, with the following property. For every tariff and scope there is a sequence of block-size floors. After any increase of those floors and any choice of block sizes above them, there is a stagewise Hessian gap η. For every compatible post-L cost choice (gap η, activity cap at least 1, and slope bound below the chosen cap), protection and initial delays, the delays can be enlarged so that the rate and chart conditions hold for all further enlargements. On every paid run with these data there are stationary path families, including the full stationary estimates in the retained family. For each adjacent stage pair (j, i) above the initial stage, the per-step chart conditions hold with a_{ji}, E_{ji} chosen after L_j (`sponsored_permit`). `issue_cosmos` replaces each entry value by the maximum of its old value and the thresholds for (j, j) and $(j, j-1)$, preserving the rate conditions.

Status: theorem under these kernel and order assumptions.

Lean: `sailsV`, using `geared`.

3.7.2 2. What it uses

- **III.1 (block-spin step and gauge fixing):** fine and coarse tori with $f = Lc$, tree-gauge slices, gauge tags, predictor and push-forward (`StepData`). The branch condition requires unique frame and predictor minimisers on the selected supports and existence of an eligible axial gauge (`Branches`).
- **II.2, III.5, III.8 (representations and activities):** path and activity bounds for the fine density (`Rep.Lawful`), large-field allocations, and the polymer summability norm of III.4.7.
- **III.10 (step estimator):** the tube's fine, coarse and noise work rows, together with their path bounds, cutoff compatibility and free-metric data (`Funds`). These determine M and $s(Y)$.
- **III.6–III.7 (stationary branch and fourth-order matching):** stationary path families and their finite-volume estimates (`FullPaths`). Their production in III.4.12 uses the free-field estimates of III.3.
- **III.2 and III.9 (schedule and marginal flow):** ordered constants, selected histories, the running inverse coupling and its band. III.4.2 applies to these data through its directed-run hypotheses.
- **I.1 and III.16 (analytic toolkit):** the matrix gauge group, normalised Haar measure, exponential charts, analytic jets and finite-dimensional Gaussian measures.

The analytic tools are Gaussian affine changes of variables, the Haar Jacobian in exponential coordinates, Combes–Thomas estimates, the resolvent integral for log, Neumann series, Gaussian

integration by parts and Fourier decay from analytic strips. General references are Helgason, *Differential Geometry, Lie Groups and Symmetric Spaces*; Glimm–Jaffe, *Quantum Physics*; and Katznelson, *An Introduction to Harmonic Analysis*. Balaban's fluctuation-integral papers and Dimock's *The renormalization group according to Balaban* provide related RG constructions; precise bibliographic locations remain open (§7).

The order of constants is pre-L data, block sizes, post-L bounds, then thresholds in x. Statements with additional parameters retain those dependencies. The precision and path-counting certificates come from `late_price`, and the determinant-jet fund from Lemma 4.21. The packet-energy, chart, noise and representation hypotheses in III.4.7 are separate assumptions. The branch and order assumptions of III.4.12, and the rate assumptions of the scheduling statements in Lemma 4.33, remain explicit.

3.7.3 3. Strategy

Keep links within $10R(x)$ of the large-field store as Haar variables. Write the others in exponential coordinates about the coarse-field copy ιV, and separate the affine squares $Q_Y(a) = \frac{1}{2}|Ma + s(Y)|^2$ from the remaining energy. The substitution $a = m(Y) + gz$ and division by an arbitrary Gaussian density give the exact identity of III.4.1. Choose the symmetrised windowed inverse for C and the windowed mean for m. The endpoint windows give finite range, at most $\lfloor R(x)/20 \rfloor$; the linear correction and $C^{-1} - W$ remain in the integrand.

A local Poincaré inequality in tree gauge gives the gap of W. Predictor-derivative comparison permits absorption of the gauge errors at small threshold. Path decay gives a uniform weighted Schur bound; Combes–Thomas and the window law then give $C = W^{-1} + O(e^{-cR(x)})$ and positivity. The resolvent integral for log, followed by a Neumann expansion, expresses the relative Gaussian normalisation as single-window and path terms. Regroup these near the large-field cores. Their logarithmic cost is absorbed by the packet energy $\beta s^2 \gtrsim \mathrm{hot}(z)^2$, leaving exponentially suppressed core factors and a small polymer remainder on the cutoff support.

Gaussian moment bounds control the exponential corrections, marked jets control background derivatives, and the annulus estimates recover affine squares away from the store. The construction follows the fluctuation-integral method, with window errors retained explicitly. III.4.12 chooses the constants in the order needed along the run.

3.7.4 4. Steps

3.7.4.1 4.1 Exact change of variables

Lemma 4.1 (Gaussian density and scaling). For $C \succ 0$, $g > 0$, $m \in \mathbb{R}^n$, and any Banach-valued f on $\mathbb{R}^n$, $f \in L^1$ iff $z \mapsto e^{\frac{1}{2} z \cdot C^{-1} z} f(m + gz) \in L^1(\gamma_C)$, and

$$\int f = g^n (2\pi)^{n/2} (\det C)^{1/2} \int e^{\frac{1}{2} z \cdot C^{-1} z} f(m + gz)\, d\gamma_C.$$

Lean's Bochner integral assigns value zero to a nonintegrable function, so the identity also holds under that convention without an integrability premise. *Sketch:* the density of γ_C follows by pushing $N(0,1)^{\otimes n}$ forward by $C^{1/2}$ (`cloud_formula`); then translate and dilate.

Lean: `Gaussian.gauged`, `at_price`.

Lemma 4.2 (exponential chart on links). For a local chart ratio J and base links ν, the map $a \mapsto (e^{a_e} \nu_e)_e$ sends $\prod_e J_{\mathrm{sc}} J(a_e)\, da_e$ on $B_{\rho_J}^{\mathcal{A}}$ onto product Haar measure restricted to the image.

Hence $\int g\,d\mathrm{Haar} = \int_{B^{\mathcal{A}}_{\rho_J}} \prod_e J_{\mathrm{sc}} J(a_e)\, g(e^a \nu)\, da$ for strongly measurable g vanishing off the image, with equivalence of integrability. In the iterated form on the slice (frozen links Haar, active links in the chart), assume also that g is bounded; then every chart section and the outer integral are integrable.

Lean: `log_transport`, `release_integral`, `thaw_integral`.

Lemma 4.3 (regularised integrand). Assume the branch condition, a local chart ratio, the standing condition, $d > 0$ and $4d < \rho_J$. For a multiplier with a bounded Borel envelope, the chart integrand equals $J_{\mathrm{sc}}^{|\mathcal{A}|} \tilde{J}_d(a) \chi_d(a)$ times the sector multiplier everywhere. Its sector integral equals $J_{\mathrm{sc}}^{|\mathcal{A}|} \iint r\, da\, d\mathrm{Haar}(Y)$, with section-wise, outer and joint integrability. *Sketch:* on the support, $|a_e| \le d$ forces $\chi_d = 1$ and each extended Jacobian to equal J; off the support both sides vanish (`protected_pay`). The envelope supplies boundedness and measurability.

Lean: `sheltered`, `regularized_price`.

Lemma 4.4 (affine-square splitting). Under the fine path bounds, the averaged step energy equals $\overline{\mathrm{res}} + Q_Y(a)$ (`compose_residual`). For any C, m, z and $g > 0$, with $\beta = g^{-2}$ and $\eta = -g^{-1}(Wm + M^\top s(Y))$:

$$-\beta Q_Y(m) + \eta \cdot z + \tfrac{1}{2} z \cdot (C^{-1} - W) z = \tfrac{1}{2} z \cdot C^{-1} z - \beta Q_Y(m + gz).$$

Sketch: expand $\frac{1}{2}|M(m + gz) + s|^2$ and use $g^{-2} = \beta$.

Lean: `Rows.adjusted`, `recolored`.

Lemma 4.5 (proof of III.4.1). Apply Lemma 4.1 to $r(Y, \cdot)$ for each Y, then Lemma 4.4 to identify the Gaussian integrand with $\mathrm{pref} \cdot \Psi(Y, \cdot)$. The same change applied to norms multiplies their Gaussian integral by the positive constant $\mathrm{pref}\,(2\pi g^2)^{n_a/2} (\det C)^{1/2}$. Lemma 4.3 and the Fubini criterion therefore give joint integrability. Inserting the section identity into the outer Haar integral proves the formula.

Lean: `authorized_change`.

Lemma 4.6 (admission of the chart). Fix a teller, local chart ratio, block size L and allocation width $w > 0$. There are $D, E > 0$ such that for large x, with $d = D\, t_L(x)$:

- $0 < d$, $4d < \rho_J$ and $2d < \rho_{\mathrm{first}}$;
- $|\log J(a)| \le E|a|^2$ for $|a| \le 4d$;
- for every funded step of block size L with $c \ge 100(1 + R(x))$ and stencil span at most $R(x)/50$, and every store, forest, allocation of width w and V, each active link (and each frozen link read on a selected stencil) in the support of $\mathrm{bill} \cdot \mathrm{lik}$ equals $e^a \nu_e$ with $|a| \le d$.

Sketch: the allocation gives toll $\le t_L(x)$ off the cost cells and block discrepancy $\le 2w\, t_L(x)$ off the aisles. The axial-gauge estimate bounds the chord distance to ν_e by $m\, t_L(x)$. The logarithmic chart then gives $|a| \le Cm\, t_L(x)$.

Lean: `matinee`.

Lemma 4.7 (permits and the licence; proof of III.4.2). Under the fixed patch, branch, funding and volume hypotheses, combine Lemma 4.6, III.4.3 and continuity of the windowed mean. For large x, matching flags and every $\beta > 0$, $(W_{\mathcal{W}}, m)$ is permitted, $W \succ 0$, and the selected predictor derivatives are bounded. Enlarge the entry thresholds to cover every admissible stage pair. Then:

- apply III.4.1 to stores with matching flags;
- wrong flags force $\mathrm{bill}_A \equiv 0$, so all integrals vanish;
- insert both identities into the sector decomposition of the push-forward, using tagged kernel times density = sieve(V) · exposed · enters.

The step size is $d = a_{ji} t(\hat{\beta}_{m+1})$, because $t_L(\hat{\beta}_m) = t(\hat{\beta}_{m+1})$.

Lean: `sponsored_permit`, `committed`.

Lemma 4.8 (identity completion). Assume $W \geq \eta I$ with $\eta > 0$ and $R(x) \geq 120$. Complete W by the identity on the inactive positions (frozen and tree links), with the corresponding endpoint windows. The windowed inverse and its symmetrisation are the identity completions of W^{per} and $W_{\mathcal{W}}$; the mean is extended by zero. The determinant is unchanged. This puts the actual and flat log-determinants on the same full link set.

Lean: `Completion.stocked`.

3.7.4.2 4.2 Gap, decay and windows **Lemma 4.9 (gap of the active precision).** Under the patch, branch, funding and volume hypotheses of III.4.3, there is $\eta = \eta(G, P, \mathrm{teller}) > 0$ with $\eta|a|^2 \leq a \cdot Wa$ for large x, every flag-matched sector and every active vector a. *Sketch:*

- Each active link is bounded by a constant times a local sum of plaquette jets plus a block jet, computed in a gauge chosen per link. The comparison of the predictor's derivative with its value at the flat field (`block_comparison`) makes the gauge-rotation errors $O(\varepsilon)$ once $t(x) \leq \delta$.
- Summing gives $|a|^2 \leq c_1 \sum |\mathrm{plaq}|^2 + c_2 \sum |\mathrm{block}|^2 + O(\varepsilon^2)|a|^2$.
- The last term is absorbed, and the energy is bounded by the Gram form.

Lean: `Gap.not_flat`.

the derivative-comparison theorem itself states two things. For links within the predictor's reach that are δ-close to 1, the derivative of $a \mapsto$ curvature of $\mathrm{pred}(e^a U)\mathrm{pred}(U)^{-1}$ at 0 is $\leq B\|a\|_{\ell^1(\mathrm{stencil})}$. It also differs from its value at $U = 1$ by $\leq \varepsilon\|a\|_{\ell^1}$. The constant $\delta(\varepsilon)$ comes from continuity and is not explicit.

Lemma 4.10 (Combes–Thomas). Given $\sigma, \varepsilon > 0$ and $M \geq 0$, there are $v \in (0, \sigma]$ and $C = 2/\varepsilon$ such that every symmetric A with $\|e^{\sigma\,\mathrm{dist}} A\|_{\mathrm{Schur}} \leq M$ and gap ε satisfies $e^{v\,\mathrm{dist}(i,j)}|A^{-1}_{ij}| \leq C$, in any volume. The same holds for all principal submatrices, since they inherit the gap. *Sketch:*

- Conjugate A by $e^{v\,\mathrm{dist}(\cdot,i)}$ with $v = \min(1, \varepsilon/(2(M+1)))\sigma$.
- The conjugate differs from A by $\leq \varepsilon/2$ in Schur norm, so it keeps gap $\varepsilon/2$.
- Test it on the weighted i-th column of A^{-1}.

Lean: `Schur.inverse_tilt`, `Schur.provides`.

Lemma 4.11 (proof of the window law, III.4.4).

1. $(AL)_{ji} = \delta_{ji}$ for $j \in \mathcal{W}_i$, because column i of L solves the window system. So $AL - 1$ is supported at distance $> r$ and has unweighted Schur norm $\leq e^{-ur}\cdot$const.
2. Hence $L - A^{-1} = A^{-1}(AL - 1)$ has the same size, and so does its symmetrisation.
3. The gap of A^{-1} is at least $\eta/(K+1)^2$, because $\|A\| \leq K + 1$. For r large the error is below half of it.

Lean: `Schur.windows`.

Lemma 4.12 (proof of III.4.3).

1. Lemma 4.9 gives the gap.
2. The path-decay and predictor-derivative bounds give weighted row and column bounds for M; multiplying $M^\top M$ gives the weighted Schur bound on W.
3. The counting function is $B(v) = \dim \mathfrak{g} \cdot L^4 \cdot 4 \cdot \sum_{z \in \mathbb{Z}^4} e^{-v|z|_1}$ for $v > 0$: each coarse site carries at most $4L^4 \dim \mathfrak{g}$ coordinates.
4. The endpoint windows contain r_x-balls.
5. III.4.4 gives (iii) and $C \succ 0$.

Lean: `opaqueCost`. The decay certificate `holds` adds bounds for C^{-1} and $\|C^{-1} - W\| \le Me^{-aR(x)}$.

Lemma 4.13 (perturbation of inverses). Let $s \ge 0$ and $A, D \succ 0$ with $\|D^{-1}\|_s \le K$, $\|A - D\|_s \le \delta$ and $K\delta \le \frac{1}{2}$. Then $\|A^{-1}\|_s \le 2K$. *Sketch:* from $A^{-1} = D^{-1} - D^{-1}(A - D)A^{-1}$, a bound t on A^{-1} improves to $K + t/2$; iterate from any finite-volume bound and pass to the limit.

Lean: `inverse_perturb`.

Lemma 4.14 (infinite-volume kernels).

- Let $r > 0$, $M \ge 0$. A family of periodic kernels compatible under coverings and bounded pointwise by $Me^{-r\,\mathrm{dist}}$ is, for every $\gamma < r$, exactly the periodisation of a kernel on $\mathbb{Z}^4$ with strip norm at rate γ bounded by $M \sum_{z \in \mathbb{Z}^4} e^{-(r-\gamma)|z|_1}$. The proof extracts a pointwise limit along the periods $k!$ and passes compatibility to the limit by dominated convergence.
- Given a strip bound with positive rate and a uniform positive gap in every period, each nonnegative-shift resolvent is the periodisation of a strip-decaying kernel at any rate below the resulting inverse-decay margin (`strip_resolvent`).
- Under the analytic-strip inverse bound and real-symmetry condition on the symbol, Fourier inversion gives a real inverse kernel with strip decay at rate $\eta/4$ (`freezer_row`). Consequently, for each stage $J \ge 4$ there is a rate $\sigma > 0$, chosen before the patch, such that every patch at stage J satisfying the branch condition has a strip-decaying inverse of its flat gauge-completed Hessian. The cap may depend on the patch; both rate and cap are uniform in the permitted kinetic profiles, gauge tags and periods (`joint_universal`).

(proof read for periodic extraction and the uniform flat inverse; statement only for the other assertions)

Lean: `publish`.

Lemma 4.15 (tree inverse and the free covariance).

- Gauge-fixing completion: for a symmetric free metric and a step satisfying the branch condition, if the gauge-completed joint matrix $\mathcal{H}$ satisfies $\mathcal{H}N = I$, then $ZNZ^\top$ inverts the tree matrix and the projected N is the tree covariance (`projecting`, pure algebra, proof read).
- Fix a patch, kernel satisfying the branch condition, and positive strip and gap bounds. Each free metric satisfying these bounds has a strip-decaying operator representing its plane tree-inverse covariance at every coarse period at least 3. The rate and cap are uniform over these metrics. With odd L and the fine and coarse free-field lower bounds and diagonal tower compatibility, the same operator represents the conditional covariances of every admissible

step with that patch. Its range-d truncation has strip norm at rate $t/2$ at most $Ae^{-td/2}$ (`earned`, `onePlateau`, proof read).

- For a large step, a tube satisfying the fine path bounds, a sector with no frozen free links, and an attached, normalised free metric whose full kernel has Gram matrix equal to that metric and satisfies the free-field symmetry conditions, the windowed resolvent at flat background differs from the free tree resolvent by at most $Ce^{-tR(x)}(1+u)^{-2}$ in a positive-rate Schur norm, for every $u \geq 0$. The statement applies to the geometric windows used in this comparison. At $u = 0$ it also gives precision and log-determinant comparison bounds (`headComparison`, `freeHead`, statement only).

Lemma 4.16 (covariance and locality of entries).

- The windowed inverse commutes with site bijections combined with orthogonal colour rotations, for every u (`natural`).
- If the precision, source and windows are transported together by such a map, so are the ordered inverse, windowed covariance and mean; the covariance determinant is preserved (`moves`).
- Under the branch condition, $R(x) \geq 120$ and stencil span at most $R(x)/50$, agreement of the configurations, tags, trees and stores on a set containing the $14R(x)$-ball about the row position makes the corresponding full precision rows equal (`same_row`).

3.7.4.3 4.3 Determinants Lemma 4.17 (proof of III.4.5).

1. On the spectrum of $A \succ 0$, $\log \lambda = \int_0^\infty [(1+u)^{-1} - (\lambda+u)^{-1}]du$, so $\log\det A = \sum_i \int_0^\infty [(1+u)^{-1} - (A+u)^{-1}_{ii}]du$.
2. Split $(A+uI)^{-1} = P_u + ((A+uI)^{-1} - P_u)$.
3. The first part gives ℓ_i, bounded by $(C+1)C$ through $\int (1+u)^{-2} = 1$.
4. Expand the second part as $P_u \sum_{n\geq 1} (-\Delta_u)^n$. Its negative enters the logarithm. The nth correction in III.4.5 has absolute integral at most Cq^{n+1}.
5. For $\log\det(1+S)$ use the trace series; $1 + S = AA_{\mathcal{W}}$.

Lean: `Priced.chains`, `letterBase`.

Lemma 4.18 (proof of III.4.6 (ii)–(iv)).

- *(ii)* Use $J_{\mathrm{sc}}^{|\mathcal{A}|}(2\pi g^2)^{n_a/2} = \mathrm{spare}^{|F^{\mathrm{free}}|}\mathrm{spare}^{-|\mathcal{F}_A|}$ and $\det C = \det W^{-1}\det(WC)$, with $\det K_F$ independent of the forest. The quadratic part follows from Lemma 4.4, and the coarse factors are collected in the coarse normalisation.
- *(iii)* Apply the path-series expansion both to the identity-completed full precision at (A, V) ("panel") and at $(\emptyset, 1)$ ("frame"), and subtract letter by letter.
- *(iv)* Sum $\mathrm{sect}_e - \log\mathrm{spare} \cdot 1_{\mathcal{F}_A}(e)$ over links, split by distance to the cores $(200R(x))$.

Lean: `book_price`, `rate_pass`.

Lemma 4.19 (local bounds on the pieces). Call a function ε-*small* if it is a sum of local terms ϕ_X on flag-matched sectors where the cutoff product is nonzero, with $|\phi_X| \leq b_X$ on every flag-matched sector and $\sum_{X \ni y'} b_X e^{eN_B(X)} \leq \varepsilon$ at every site y'. Here B is the specified cover radius. Under the corresponding hypotheses of III.4.7, for large x:

- (a) At the retimed scale $y = \mathrm{retime}(x)$, with $z = \beta_F(x, L)$ and $\beta \in [z/2, 2z]$, $|\mathrm{rent}_o| \leq \delta\,\mathrm{hot}(z)^2|o|$ for any fixed $\delta > 0$ (`wages`). Each link contributes $O(\log y)$, the $200R(y)$-

halo of a core contains $O((1+R(y))^4|o|)$ links, and $\mathrm{hot}(z)^2$ dominates this logarithmic cost.

- (b) The far-field term is $x^{-1/8}$-small (`bills`).
- (c) I and Sh are $e^{-dR(x)}$-small (`billIt`).
- (d) The chart terms $\sum_e \log \tilde{J}(a_e)$ are $x^{-1/8}$-small, since $|\log \tilde{J}| \le u(4d)^2$, $d \le ht_L(x)$ and $t_L(x)^2 \le x^{-1/4}$ eventually. The linear term $\eta \cdot z$ has the same polymer bound: the noise assumptions give $|\eta_i z_i| \le x^{-2}$ on the cutoff support (`bolsters`).
- (e) The remaining path and interaction correction term (path and cubic/quartic corrections) is $x^{-1/24}$-small (`Ancestor.abound`).

(proof read for (a), (b), (d); statement and outline for (c), (e))

Lemma 4.20 (proof of III.4.7).

1. On flag-matched sectors, $\Lambda = \sum_o (\mathrm{center}_o + \mathrm{rent}_o) + \mathrm{tenders}$ (`issued`). The center is $-\beta$ times the row energy of III.4.6 summed over row positions within $200R(y)$ of o (`center`). The remainder consists of the terms of Lemma 4.19 (b)–(e), the row energy outside these halos, and the quadratic window correction.
2. On the nonzero cutoff support, the packet-energy hypothesis gives $\mathrm{center}_o \le -\beta a s^2 |o|$ for each core of the store, with $s = z^{-1/2}\mathrm{hot}(z)$. Since $zs^2 = \mathrm{hot}(z)^2$ and $\beta \ge z/2$, the condition $2\alpha \le a/2$ gives $2\alpha\,\mathrm{hot}(z)^2 \le \beta a s^2$. The bound is `ranked`. Apply Lemma 4.19 (a) with $\delta = \alpha$ to obtain $\mathrm{center}_o + \mathrm{rent}_o \le -\alpha\,\mathrm{hot}(z)^2|o|$.
3. `paltry` uses the lower bound `minimum` to clamp the core term between fixed bounds, preserving its value on the nonzero cutoff support and its upper bound on all matching sectors.
4. The tenders are ℓ-small with $\ell = 4q + y^4\nu + 2\nu + a'$, where $q = y^{-1/8}$, $\nu = y^{-5}$ (from $e^{-cR(y)} \le y^{-5}$) and $a' = y^{-1/24}$. Then $2\ell \le 16y^{-1/24} \le y^{-1/48}$ for large y (the power comparison).
5. `issue_invoice` symmetrises over lattice motions and converts the small sum into $|e^{\mathrm{aux}_X} - 1| \le f_X$.

Lemma 4.21 (precision, counting and jet certificates). Fix G, a patch and a teller satisfying the branch condition, a halo multiplier $l \ge 1$ and a polymer exponent $e \ge 0$.

- `late_price` gives the precision and path-counting certificates, eventually in x:
  - positive constants v, η, b, d and $C \ge 1$, chosen before the threshold in x, giving the gap, positivity and resolvent bounds of III.4.5 for every funded step with that patch and $c \ge 100(1+R(x))$, uniformly in matching sectors and every window system containing the r_x-balls, with error factor $e^{-bR(x)}$;
  - cluster counting with factor $e^{-dR(x)}$.
- For each pair of nonnegative integer orders (n,k) and $l \ge 20$, `stores` chooses a fund whose covariance-jet and determinant-letter bounds hold eventually in x. The certificate also includes stencil-span and exponential cost bounds. These two results supply the corresponding certificates, rather than all the hypotheses of III.4.7.

3.7.4.4 4.4 Gaussian moments and insertions Lemma 4.22 (exponentials of quadratic forms). Let $x \sim N(0, I)$ and $\mathcal{Q} = \sum_i w_i \langle q_i, x\rangle^2$ with $w_i \ge 0$ and $\sum_j w_j |\langle q_i, q_j\rangle| \le 1/8$ for each i. Then $\mathbb{E}e^{\mathcal{Q}} \le e^{2\sum_i w_i |q_i|^2}$. *Sketch:* Gaussian integration by parts gives $F'(t) = \mathbb{E}[\mathcal{Q}e^{t\mathcal{Q}}] \le TF + \frac{1}{2}F'$, so $(e^{-2Tt}F)' \le 0$.

Lean: `authentic`, `amount`.

Lemma 4.23 (polynomial moments). For $L > 0$ and $k \in \mathbb{N}$ there is $D(L,k) > 0$, independent of n, such that every $0 \preceq C \le LI$ on $\mathbb{R}^n$ satisfies integrability and $\mathbb{E}_{\gamma_C}(1+|z|^2)^k \le D(L,k)(1+n)^{k+1}$. *Sketch:* dominate a power by an exponential with $t = 1/(4L)$, bound the sum of squared coordinates by n times its largest term, and use $\mathbb{E}e^{tz_x^2} \le 2$.

Lean: `many`.

Lemma 4.24 (proof of III.4.8).

1. On $\{v \neq 0\}$, $1 \le e^{-c|J|\rho^2} e^{c\sum_J |z_i|^2}$.
2. The site budget gives $\sum d_i(1 + |z_{a_i}|^2 + |z_{b_i}|^2) \le |Y|c + 2c\sum_X |z|^2$.
3. Use $ab \le a^2 + b^2$ to bound the product of the polynomial score and $e^{3c\sum_U |z|^2}$. The exponential square has expectation at most $2^{\dim \mathfrak{g}|U|}$, since $6c = 1/(4L)$ and $|U| \le |J| + 2|Y|$; Lemma 4.23 bounds the squared polynomial score.

For the fourth-moment bound: $|e^q - 1| \le xe^{2x(1+|z_a|^2+|z_b|^2)}$, then apply III.4.8 with $k = 0$. For the coupling-band estimate: the mean-shift equation writes η through $W^{\mathrm{per}} - W^{-1} = O(e^{-uR(x)/60})$, and $C^{-1} - W$ is controlled by the inverse-window estimate.

Lean: `same_law`, `r4`, `oasis`.

Lemma 4.25 (Gaussian likelihood ratios).

- For $A, D \succ 0$, any centres a, d and any real p, the p-th power of $dN(a,A)/dN(d,D)$ is $N(d,D)$-integrable iff $pA^{-1} + (1-p)D^{-1} \succ 0$ (`power_iff`).
- For $A, D \succ 0$, if $D^{-1} - A^{-1}$ is supported in $S \times S$ and $A^{-1}(d-a)$ on S, the ratio equals the ratio of the S-marginals at x_S (`likelihood_local`).
- Let $A, D \succ 0$, $p \ge 1$, $K, w_i \ge 0$ and $D \le KI$. If the change in the quadratic exponent about the reference centre has absolute value at most $b + \sum_i w_i z_i^2$, and $Kpw_i \le 1/4$ for every i, then the p-th likelihood-ratio moment is at most $e^{2p(b+2K\sum_i w_i)}$. Cauchy–Schwarz controls the normaliser (`weighted`).
- Let $0 \prec C \le KI$ with $K > 0$, $\mathcal{J}$ symmetric, $\varepsilon \ge 0$, $p \ge 1$ and $\theta > 0$. For $|z \cdot \mathcal{J}z| \le \varepsilon|z|^2$, $|t_i| \le \ell$ on S and $Kp(\theta + \varepsilon/2) \le 1/4$, the inserted covariance is positive definite and the moment bound of III.4.9 holds with $U = S$. Positivity follows from $C^{-1} \ge K^{-1}I > \varepsilon I$; the linear part contributes $e^{p|S|\ell^2/(2\theta)}$ by AM–GM (`insert_letter`).
- For symmetric A, integrability of $e^{-\frac{1}{2}z\cdot Az - b\cdot z + c}$ forces $A \succ 0$ (`posDef_of_integrable_quadratic`).
- Let $A, D \succ 0$, $\|A^{-1}\|_s \le M$, $\|A^{-1} - D^{-1}\|_s \le \delta$, $s \ge 0$, and $|\log\det A - \log\det D| \le n\delta$. The ratio $T(z)$ of the A-Gaussian density at $z + hu$ to the D-density at z obeys

$$|\log T(z)| \le \tfrac{\delta}{2}(n + |z|^2) + \tfrac{M}{2}(h^2|u|^2 + |h|(|u|^2 + |z|^2)).$$

  Expand the shifted quadratic form and apply the Schur bounds (`priceShift`).

(proof read for the weighted moment and insertion bounds; statement only for the other assertions)

3.7.4.5 4.5 Jets Lemma 4.26 (analytic predictor rows). Fix G, a patch P, a kernel K satisfying the branch condition and an order k. There are $C \ge 0$ and $r, \delta > 0$ such that the following holds. Take a step with that patch and kernel, matching flags, stencil span at most

$R(x)/50$, $\mathrm{allot}(P)+1 \le 9R(x)$, $2(\mathrm{allot}(P)+1) < c$, and $\mathrm{permit}(P)t(x) \le \delta$. On every selected block stencil, the predicted group element and its relative curvature are real-analytic on the r-ball of mixed coarse and fine logarithmic perturbations; the curvature has k-jets bounded by C. Here allot and permit are fixed patch-dependent geometric constants. *Sketch:*

- Transplant the stencil into a fixed reference torus (`impression`).
- Use a local trimming gauge in which the links have chord $\le \delta$.
- Write the prediction as $u \cdot \mathrm{paste}(...) \cdot v^{-1}$ with maps of norm ≤ 1 (`freshQ`).
- For large x, all gauge-tag choices give the same predictor in a neighbourhood of each $(B, 0)$ with $\|B\| \le r$ on the selected stencils; the common predictor has uniform jets.

Lean: `sounds`, `chosen_branch`.

Lemma 4.27 (precision near the background). Under the fixed patch, branch, funding, volume and flag hypotheses of III.4.10, constants $r, s, C_*, \eta > 0$ precede the threshold in x. On $\|B\| \le r$:

- M and W are real-analytic, with weighted jets $\le C_*$;
- the gap at $B = 0$ comes from Lemma 4.9;
- a Lipschitz perturbation keeps half of it.

For $d \ge 0$ with $(1+2k)d \le \mathrm{width}/2$, the move-matrix entries have marked k-jets with the corresponding off-diagonal exponential weight. *(proof read for the precision bound; statement only for the entry bound)*

Lean: `precision`, `entries`.

Lemma 4.28 (transport along walks). Fix the jet order and a bound on the marked jets of the link field. For each $\kappa > 0$ and endpoint jet cap $M > 0$, there are a positive mark radius and $N > 0$, uniform in the lattice, walk and scaling parameter $s \in [0, 1]$, such that adjoint parallel transport moves a marked jet bound from the end of a walk of length ℓ to its start, at base point 0, with cap multiplied by $Ne^{\kappa\ell}$ (`carried`). For a strip-decaying kernel of rate $\gamma > 0$, walks with length at most $B(1+|z|_1)$, $B > 0$, finite sets of walks below every length cutoff, and the required endpoint bounds, choosing $\kappa = \gamma/B$ gives a uniform marked bound on the truncated sum (`follows`). If the path family has strip bound (σ, M) with $M \ge 0$, $d \ge 0$, $\gamma > 0$ and $2\gamma + 2kd \le \sigma/2$, selecting only paths of length at least R gives marked k-jets bounded by a constant times $Me^{-\sigma R/4}$ (`transport_rows`).

Lemma 4.29 (cut and window errors). Assume fine path bounds with positive rate σ and nonnegative amplitude, and the branch condition. Near the flat background $B = 0$, the actual sector Gram matrix equals the reindexed path-cut affine block. If no free links are frozen, the tree-coordinate identification also identifies their geometric windowed resolvents (`notarized`). With a large step, $R(x) \ge 120$ and the marked gap and counting bounds needed for the inverses, the following bounds hold at $B = 0$ for $u \ge 0$:

- cutting the affine inverse germ at range R changes its k-jet by $O(4^k C_0^2 e^{-\sigma R/4})$, with $C_0 \propto (1+u)^{-1}$ (`shortening`);
- the ordered windowed resolvent differs from the uncut one by a cap proportional to $(1+u)^{-2}(e^{-(k+1)\gamma r_x} + e^{-\sigma\lfloor R(x)/10^5\rfloor/4})$; symmetrisation costs at most a factor 2 (`total`);
- given the marked bounds and positive gaps for both W and C, and the two successive admissible inverse-decay rates, $C^{-1} - W$ has jets through order q bounded by a constant times $e^{-(q+1)dr_x}$ in the reduced-rate marked norm (`residual`).

3.7.4.6 4.6 Large fields and constants Lemma 4.30 (clean fields).

- Under the funding, block-size, volume, tree-gauge and local sector hypotheses of III.4.11, there is $C \geq 1$, independent of the integer halo, with the following property. The field $a_e = \operatorname{plog}(U_e \nu_e^{-1})$ vanishes on the tree links. On the swept annulus it gives exponential coordinates: within $\lfloor 2R(x)/5 \rfloor$ of each swept site, $|a_e|$ and the linking are at most $C t_L(x)$. A coarse gauge and its pullback to the fine lattice make the corresponding coarse and fine chords at most $C(1 + R(x))^5 t_L(x)$ (`cleans_at`).
- Given a clean-field constant $X \geq 1$, there are $C > 0$ and $M \geq X$ such that, for large x, the local plaquette curvatures and block rows have first-order expansions in a, with linear term at most Cw and remainder at most Cw^2, where $w = M(1 + R(x))^5 t_L(x)$. Moreover $(1 + R(x))^n w \to 0$ for every $n \in \mathbb{N}$; in terms of x, $w = x^{-1/2+o(1)}$ (`reacts`).

(proof read for the response estimate; statement only for the clean-field construction)

Lemma 4.31 (affine squares along paths). Assume fine path bounds with amplitude C and cutoff direct coefficient d_0. Fix a path cutoff $R_0 \in \mathbb{N}$, thresholds t, s, $B > 0$ and $h \geq 0$. Suppose the face under consideration has toll at most t and satisfies the path-retention condition, its two coarse path rows have norm at most Bh, and its first-order curvature and curvature error are at most Bh and Bh^2. Require the same two curvature bounds at every face reached by a retained path of length at most R_0, with chord deviation from the reference field at most h along every such path. Then the affine path series converges and work minus the extracted square is at least $-4(mB)^2(1 + 2c_G R_0)h^3$, where $m = \sqrt{d_0} + C + 1$ (`fine_squares`). Under the clean-field and annulus hypotheses of Lemma 4.30, each fine-face, noise and coarse row at a swept site has error at least $-e(x)$, where e is eventually nonnegative and $(1 + R(x))^n \beta_F(x, L) e(x) \to 0$ for every n (`coat`).

Lemma 4.32 (proof of III.4.11).

1. Sum the row bounds of Lemma 4.31 over Q. The counted face, noise and coarse rows number at most $X(1 + R(x))^4 |O|$, so the total error is $e'(x)|O|$, where $e'(x) = X(1 + R(x))^4 e(x)$ still satisfies $(1 + R(x))^n \beta_F(x, L) e'(x) \to 0$ for every n.
2. Retain the affine squares selected by H and discard the other squares by positivity. The additional coarse rows in the work are nonnegative.
3. The extracted squares are unique, because the chart logarithm is unique once $\max(K, C) t_L(x)$ is below the chart radius.

Lean: `annulus_pay`.

Lemma 4.33 (proof of III.4.12).

1. Choose the free-field bounds.
2. Choose the stationary width caps and path amplitudes, build a sheet within them, then choose drift bounds, slope caps and block-size floors. The stationary theorem gives a Hessian gap and full stationary path estimates only after the free-field, retained-family, paid-run, chart and pre-L compatibility hypotheses have been supplied. Its construction combines the one-loop analytic estimate with downward induction over levels.
3. Build the startup and free-field data for that sheet, preserving the required orders and bounds.
4. For each L-sequence the delays are enlarged and the chart-threshold enlargement is applied.
5. Assemble the stationary path families and their estimates.

Related constant-producing statements:

- The scheduling theorems produce funded delays under their transition-rate and compatibility hypotheses;
- The Gaussian comparison theorem gives, for each fixed real moment parameter and patch satisfying the branch condition, uniform reverse column-switch bounds on sufficiently large funded steps with matching flags and the required size condition. It also bounds the source $M^\top s(Y)$ by $C(t(x)+d)$ for every $d>0$;
- The counting theorem gives a polymer budget $e^{-cR(x)}$ for fixed patch, nonnegative polymer exponent and prescribed integer orders; $c>0$ is fixed by the input bounds.

(proof read for the main constant selection and stationary theorem; statement only for the ancillary scheduling and comparison results)

Lean: `sailsV`, `geared`, `book`, `endures`, `post`, `takings`.

3.7.5 5. Where the difficulty lies

The gap is a gauge-theoretic local Poincaré estimate: the predictor derivative must be compared with its flat-field value after a different gauge choice for each active link. Its smallness radius comes from continuity rather than an explicit formula. Inverse estimates then spend decay rate, while every bound must remain uniform in the volume, store, forest, tags and field.

The determinant comparison must accommodate different active dimensions, hence identity completion and the frozen-link normalisation. The resulting core costs grow polynomially in $\log x$; the retimed large-field threshold supplies the stronger suppression $e^{-\alpha \mathrm{hot}(z)^2|o|}$. The final powers $x^{-1/8}$, $x^{-1/24}$ and $y^{-1/48}$ allow the separate polymer budgets to be added. Gaussian transport and the affine-square identity are finite-dimensional; their application still requires the chart support and integrability conditions.

3.7.6 6. Notation

Written proof	Lean (glossary)	Meaning
γ_C	`Gaussian.cloud`	centred Gaussian with covariance C
$z\cdot C^{-1}z$	`force`	precision quadratic form
$(2\pi g^2)^{n_a/2}(\det C)^{1/2}$	`chartCost`	Gaussian normalisation
$g=\beta^{-1/2}$	`coupling`	coupling of the representation
J, J_{sc}, ρ_J	`ChartRatio`, `LocalRatio`	normalised Haar Jacobian of exp, scale, radius
$\mathcal{F}_A$, $\mathcal{A}_A$	`chill` (`Frozen`, `Active`)	free links within $10R(x)$ of A / the others
ιV	`replica`	coarse field copied to the fine lattice
$Q_Y(a)$, W	`Rows.guide`, `Rows.gram`	affine squares; precision $M^\top M$
$C=W_{\mathcal{W}}$, $m(Y)$	`fog`, `meanLog`	windowed covariance; windowed mean
L, $A_{\mathcal{W}}$	`Schur.per`, `Schur.clouded`	column-wise window inverse; its symmetrisation

Written proof	Lean (glossary)	Meaning
Schur decay, gap	`Pours`, `Gapped`	weighted row and column sums; $a \cdot Aa \geq \eta\|a\|^2$
Ψ	`dyed`	sector integrand in Gaussian coordinates
permitted, licence	`Permitted`, `Licenses`	hypotheses and conclusions of III.4.1
ℓ_i, path letters, S	`letterBase`, `page`, `errata`, `excess`	window log letter; path terms; $AA_{\mathcal{W}} - 1$
$\kappa(p)$, $\Lambda(p)$	`keptEarly`, `spoolLift`	relative log-normalisation; full log-integrand
spare(β)	`spare`	$(2\pi g^2)^{\dim \mathfrak{g}/2} J_{\mathrm{sc}}$
rent$_o$, far term, I, Sh	`rents`, `nightly`, `introTax`, `showTax`	pieces of κ
ε-small	`Petty`	sum of local terms with polymer norm $\leq \varepsilon$
$y^{-1/48}$	`garlic`	final polymer budget
$R(x)$, $t(x)$, $t_L(x)$	`radiusB`, `temp`, `fiery`	cluster radius, small-field threshold, fine-level threshold
M, $s(Y)$	`move`, `source`	linear row map and frozen-link source
$\mathcal{E}$, center$_o$	`Milk.levy`, `Coffer.center`	row energy remainder; its $-\beta$-weighted core sum
sector support	`Dyed.Playing`	nonzero product of field, block, tag and chart cutoffs
matching flags	`Matched`	$\mathrm{flags}(A) = \mathrm{flags}_{t(x)}(V)$
branch condition	`Patch.Branches`	uniqueness of selected minimisers and eligible axial gauge
sector regularity, noise bounds	`Chips`, `noises`	assumptions in III.4.6–III.4.7
c_G	`Gauge.colorScale`	matrix norm factor in the path estimate
price, bill, tax, rent, teller	numerical bounds; `Teller`	the step-estimator interface

3.7.7 7. What remains unexplained

1. **Proof depth.** The supporting proofs marked "statement only" remain incomplete readings: the strip-resolvent and Fourier-inverse results in Lemma 4.14; the flat comparison in 4.15; 4.16; the path-counting and interaction estimates in 4.19(c),(e); 4.21; parts of 4.25; the entry bound in 4.27; 4.28–4.29; the clean-field construction in 4.30; 4.31; and the ancillary scheduling statements in 4.33. Their statements and the stated mechanisms have been compared with the cited source; these labels do not certify their dependency proofs.
2. **Run-level hypotheses.** The packet-energy and support-exponent assumptions of III.4.7 are substantive inputs. Their simultaneous supply on the selected run, and the transfer from stationary path families to the tube's work rows, still need a complete account. The precision, determinant-jet and noise bounds have local suppliers, but their use does not

establish these other premises.

3. **Free-kernel comparison.** Lemma 4.15 compares the flat sector under the no-frozen-link and free-metric compatibility conditions. A direct statement comparing an arbitrary interacting sector covariance with the infinite-volume free operator was not located. The full expansion of the free-metric compatibility conditions and the nested marked inverse bounds in Lemma 4.29 remains to be written.
4. **Consumers and coverage.** The downstream reset and repeatable-step statements have not been checked here. The additional assigned packages for Wilson bulk terms, vacuum stability, proof interfaces, multisource terms and flat RG data are not covered by these steps. The closure-size figures in the opening metadata were retained without recounting.
5. **References.** The precise results and bibliographic locations in Helgason, Glimm–Jaffe, Katznelson, Balaban and Dimock remain to be selected and verified. The cited Lean declarations are the evidence for the statements above.

3.8 III.5. Cluster expansion of the polymer gas

Part: III, Ultraviolet construction (renormalization group). **Lean:** work packages `YM/Push/ABSOLUTE_SUMS/` (3,624 lines), `YM/AnalyticNext/CONNECTED_ACTIVITY_BOUNDS/` (3,589), `YM/Push/SECTOR_MEASURE/` (1,153), `YM/SectorNext/VACUUM_NORMALIZATION/` (935), `YM/Progress/STATIONARY_SCENES/` (445). **Size:** 9,746 lines in 1,097 declarations used by the main theorem.

Conventions.

- $\Lambda_n = (\mathbb{Z}/n\mathbb{Z})^4$, with $n \geq 1$, is the periodic lattice with the maximum distance d. A *support* is a subset of Λ_n.
- $\mathrm{cov}_B(X)$ is the least number of d-balls of radius B whose union is connected and covers X (`coverNumber`). Two properties are used throughout (I.1):
  - $|X| \leq \mathrm{ball}_B \, \mathrm{cov}_B(X)$, with $\mathrm{ball}_B = (2B+1)^4$;
  - cov_B is subadditive on unions of overlapping sets.
- A family of supports is *connected* if it is nonempty and its overlap graph is connected.
- For $f \geq 0$ on an arbitrary index set, $\mathrm{SB}(f; C)$ means: f is summable and $\sum f \leq C$ (`SumBudget`).
- For polymers $(X_i, w_i)_{i \in \iota}$ with $w \geq 0$, the *site-sum condition* $\mathrm{Hit}_B(w; a, \gamma)$ means $\mathrm{SB}(i \mapsto \mathbf{1}[x \in X_i]\, w_i \, e^{a \, \mathrm{cov}_B(X_i)}; \gamma)$ for every site x (`Ac.Hit`, II.2).
- All weights are real. An upper bound on an infinite sum of nonnegative terms includes summability, as in SB.
- For a relation R that is not symmetric, the graph used below has an edge between distinct occurrences a, b exactly when both $R(z_a, z_b)$ and $R(z_b, z_a)$ hold. The coefficient of a one-vertex tuple is 1.

3.8.1 1. What this unit proves

The unit consists of five packages.

- **(A) Mayer expansion.**
  - The Mayer expansion of a hard-core polymer gas, with absolute bounds, and its re-exponentiation after forced polymers (III.5.1–III.5.6).

  - The verification of the smallness condition for gases of large-field seeds of a block-spin step (III.5.7–III.5.8).
- **(B) Vacuum normalization.** A second, graph-free construction of the logarithm by deletion identities (III.5.9), and a trace-log formula (III.5.10).
- **(C) Sector decomposition.** The decomposition of one block-spin integral into sectors labelled by packets of seeds, with a Gaussian gain per seed (III.5.11).
- **(D) Inverse decay.** A Combes–Thomas bound (III.5.12).
- **(E) Gaussian bounds.** Windowed inverses (III.5.13), and a uniform spectral gap and uniform decay for the precision matrix of the small-field Gaussian of a step (III.5.14).

III.5.1–III.5.7, III.5.9, III.5.10, III.5.12 and III.5.13 are general implications. III.5.8(c), III.5.11 and III.5.14 concern the block-spin step under the funding hypotheses of III.1; III.5.14 also fixes a patch satisfying its branch conditions.

Proposition III.5.1 (Mayer identity for a finite gas).

- *Data.* Let ι be finite, R a reflexive symmetric relation on ι ("incompatible"), and $w : \iota \to \mathbb{R}$. Let $\Xi(w) = \sum_s \prod_{i\in s} w_i$, the sum over subsets $s \subseteq \iota$ of pairwise compatible labels.
- *Ursell coefficient.* For $z = (z_0, \dots, z_m) \in \iota^{m+1}$, let $G_R(z)$ be the graph on $\{0, \dots, m\}$ with an edge $\{a, b\}$, $a \neq b$, whenever $R(z_a, z_b)$. Put $\phi_R^T(z) = \sum_G (-1)^{|G|}$, the sum over connected spanning subgraphs $G \subseteq G_R(z)$.
- *Claim.* Let $\ell_m = \frac{1}{(m+1)!} \sum_{z\in\iota^{m+1}} \phi_R^T(z) \prod_a w_{z_a}$. If $|\ell_m| \leq Cq^m$ for all m, with $C \geq 0$ and $0 < q < 1$, then $\exp\big(\sum_{m\geq 0} \ell_m\big) = \Xi(w)$.

Status: theorem. *Lean:* `Gas.exact_polynomial`; coefficient identities `Coef.recurrence`, `Partition.partition`.

Proposition III.5.2 (tree-graph bound). Let T be a graph on a finite vertex set with a distinguished vertex. Then $\big|\sum_G (-1)^{|G|}\big|$, summed over the connected spanning subgraphs $G \subseteq T$, is at most the number of spanning trees of T. In particular $|\phi_R^T(z)|$ is at most the number of spanning trees of $G_R(z)$. *Status:* theorem. Spanning trees are counted as acyclic parent maps towards the distinguished vertex. *Lean:* `Graph.pay`.

Proposition III.5.3 (pinned cluster bounds).

- *Data.* ι is any set; $X : \iota \to$ supports of Λ_n; $B \in \mathbb{N}$; $w : \iota \to [0, \infty)$; $a \geq 0$; R is a relation on ι with $R(i, j) \Rightarrow X_i \cap X_j \neq \emptyset$. R need not be symmetric.
- *Smallness condition.*

$$\mathrm{Ready}(w; a, \gamma, \delta, q) : \qquad \gamma > 0, \quad \delta > 0, \quad 4\,\mathrm{ball}_B\,\gamma/\delta \leq q < 1, \quad \mathrm{Hit}_B(w;\, a + \delta,\, \gamma).$$

- *Notation.* $X(z) = \bigcup_a X_{z_a}$ for a tuple z.

Then, for every site x and every integer $m \geq 0$:

- (a) $$\sum_{m'\geq m} \sum_{z\in\iota^{m'+1}:\ x\in X(z)} \frac{|\phi_R^T(z)|}{(m'+1)!} \prod_a w_{z_a}\, e^{a\,\mathrm{cov}_B(X(z))} \;\leq\; \frac{\gamma\, q^m}{1-q}.$$
- (b) The same bound holds for $\sum_s \prod_{i\in s} \big(w_i e^{a\,\mathrm{cov}_B(X_i)}\big)$. The sum runs over finite sets $s \subseteq \iota$ with $|s| \geq m + 1$ whose supports form a connected family containing x.

- (c) (Marked form.) Suppose $Q \subseteq \iota$ and $\mathrm{SB}(i \mapsto \mathbf{1}[i \in Q] w_i e^{(a+\delta)\mathrm{cov}_B(X_i)}; b)$. Then the sum over $z \in \iota^{m+1}$ with some $z_a \in Q$ of $|\phi_R^T(z)| \prod_{j=0}^{m} (w_{z_j} e^{a\,\mathrm{cov}_B(X_{z_j})})/(m+1)!$ is at most bq^m.

Each displayed bound is a summability statement of the form SB. *Status:* theorem. Form (a) is used in III.2, III.10, III.11, III.12, III.13 and III.15. *Lean:* `Ursell.log_hit_tail`, `Trees.touch_tail`, `Ursell.cluster_at`, `Ursell.cluster_tail`; hypothesis `Trees.Ready`.

Proposition III.5.4 (convergence and logarithm of the gas). Let X, B be as in III.5.3, R reflexive and symmetric with $R(i,j) \Rightarrow X_i \cap X_j \neq \emptyset$, $w : \iota \to \mathbb{R}$ and $a \geq 0$. Assume $\mathrm{Ready}(|w|; a, \gamma, \delta, q)$. Then:

- $\Xi(w) = \sum_s \prod_{i \in s} w_i$, over finite pairwise compatible $s \subseteq \iota$, converges absolutely, with $\sum_s \prod_{i \in s} |w_i| \leq e^{n^4 \gamma}$.
- The cluster series $\sum_{m \geq 0} \sum_{z \in \iota^{m+1}} \phi_R^T(z) \prod_a w_{z_a}/(m+1)!$ converges absolutely. Its absolute sum is at most $n^4\gamma/(1-q)$, and at most $n^4 \gamma q^m$ over the clusters of $m+1$ labels.
- $\Xi(w) > 0$, and $\log \Xi(w)$ equals the cluster series.

Status: theorem. Used in III.10. *Lean:* `Gas.price_log`, `Gas.absolute`, `Gas.absolute_at`.

Proposition III.5.5 (removing the polymers that meet a region). In III.5.4, take for R the overlap relation ($R(i,j)$ iff $i = j$ or $X_i \cap X_j \neq \emptyset$), with nonempty supports. For $Y \subseteq \Lambda_n$ let $w_i^Y = w_i\, \mathbf{1}[X_i \cap Y = \emptyset]$. Then

$$\Xi(w^Y) = \Xi(w)\, e^{-p}, \qquad |p| \leq \frac{|Y|\,\gamma}{1-q},$$

where p is the sum of the cluster terms with $X(z) \cap Y \neq \emptyset$.

- Hence $e^{-C|Y|}\Xi(w) \leq \Xi(w^Y) \leq e^{C|Y|}\Xi(w)$ with $C = \gamma/(1-q)$, uniformly in n.
- The clusters of at least $m+1$ labels that meet Y contribute at most $|Y|\gamma q^m/(1-q)$ in absolute value.

Status: theorem. Used in III.10. *Lean:* `Gas.restores_log`, `Gas.missing_tail`.

Proposition III.5.6 (re-exponentiation with forced polymers). Fix $B \in \mathbb{N}$ and a common lattice Λ_n.

- *Data.* "Unforced" polymers $(X_i, u_i)_{i \in \iota}$ and "forced" polymers $(Y_k, d_k)_{k \in \kappa}$, all with nonempty supports, $u, d \geq 0$; R a relation on ι contained in the overlap relation.
- *Atoms.* The atoms are of two kinds:
  - the forced labels k, with support Y_k and weight d_k;
  - the tuples $z \in \iota^{m+1}$, with support $X(z)$ and weight $|\phi_R^T(z)| \prod_a u_{z_a}/(m+1)!$.
- *Constants.* $A \geq 0$, $\delta' > 0$, $t \geq 1$, $0 \leq \sigma < 1$, $\mu \geq 0$, $\eta > 0$, $q' < 1$.
- *Hypotheses.* $\mathrm{Ready}(u; A + \delta', \gamma, \delta, q)$, $\mathrm{Hit}_B(d; A + \delta', c)$, and

$$\frac{t\, e^{\mu} c}{1-\sigma} + \frac{t\,\gamma}{(1-\sigma)(1-q)} \;\leq\; \eta \;\leq\; \sigma, \qquad 4\,\mathrm{ball}_B\, \eta/\delta' \leq q'.$$

Fix a site x and $m \geq 0$. Consider finite sets s of atoms with multiplicities $n_\alpha \geq 1$ such that:

- s contains at least one forced label;
- $\sum_\alpha n_\alpha \geq m$;
- the supports of the atoms in s form a connected family containing x.

Summed over all such pairs $(s, (n_\alpha)_{\alpha\in s})$, the quantity

$$e^{A\,\mathrm{cov}_B(\bigcup_{\alpha\in s}\operatorname{supp}\alpha)}\ \prod_{\alpha\in s}\frac{\mathrm{wt}(\alpha)^{n_\alpha}}{n_\alpha!}$$

is at most $t^{-m}e^{-\mu}\,\eta/(1-q')$.

Status: theorem. The general form, with an arbitrary selection predicate, is `Multiple.estimates`. Used in III.2, III.10, III.11 and III.13. *Lean:* `OmitClusters.restore`.

Proposition III.5.7 (gases of seed packets).

Setting. Let S be a block-spin step (III.1) from Λ_f to Λ_c with block side L, and let $l, B \in \mathbb{N}$.

- A *seed* is a fine cell, a coarse bond or a coarse site. Its location is the projection of the cell's vertices, the bond's two endpoints, or the site itself. Each location lies within distance 1 of its anchor.
- A *packet* A (`Stores`) is a finite set of seeds; $|A|$ is its number of seeds.
- Let $D_0(A)$ be the union of its seed locations and $D_r(A)$ its radius-r neighbourhood. The *skins* are the radius-lB neighbourhoods of the individual seed locations; their union is the *dome* $D(A) = D_{lB}(A)$. The packet is *connected* if its skins form a connected family. The sector and matrix estimates below use $D_r(A)$ directly, rather than neighbourhoods of $D(A)$.
- Put

$$\tau_L = L^4\,\mathrm{card}\{\text{plaquette orientations}\}{+}5, \qquad K_{\mathrm{hit}} = (2lB{+}3)^4\,\tau_L, \qquad K_{\mathrm{step}} = (4lB{+}5)^4\,\tau_L.$$

(a) Packet sums. Assume $a, C \ge 0$, $0 \le p_A \le Ce^{(s-g)|A|}$ for every connected A, and $\rho := K_{\mathrm{step}}^2\,e^{\,s-g+a(2l+3)^4} \le q < 1$. Then for every coarse site x and every $m \ge 0$,

$$\sum_{A\ \text{connected},\ x\in D(A),\ |A|\ge m} p_A\,e^{a\,\mathrm{cov}_B(D(A))}\ \le\ K_{\mathrm{hit}}\,C\,\frac{q^{\max(1,m)}}{1-q}.$$

The same bound holds for an arbitrary family of variants with nonnegative weights, each nonzero variant assigned to a connected packet and supported in its dome, provided the sum of the variant weights of each packet is at most $Ce^{(s-g)|A|}$. For variants, the sum is pinned by the actual support and uses its covering number.

(b) Mixed words.

- *Alphabet.* Add optional polymers $(X_i, w_i)_{i\in\iota}$ on Λ_c with $w \ge 0$ and lengths $\lambda_i \ge 0$. A packet has footprint $D(A)$ and length $e_s|A|$.
- *Hypotheses* (with $\varepsilon, \delta, \gamma_{\mathrm{root}} > 0$ and $a, C, e_s, \mu_s \ge 0$):
  - $0 \le p_A \le Ce^{(s-g)|A|}$ on connected packets;
  - (a) holds with $a + 2\varepsilon + \delta$ in place of a, $s + 2\varepsilon e_s + \mu_s$ in place of s, and some $q_p < 1$;
  - $\mathrm{Hit}_B(w\,e^{2\varepsilon\lambda};\ a + 2\varepsilon + \delta,\ \gamma_{\mathrm{opt}})$;
  - $K_{\mathrm{hit}}Cq_p/(1-q_p) + \gamma_{\mathrm{opt}} \le \gamma_{\mathrm{root}}$;
  - $4\,\mathrm{ball}_B\,\gamma_{\mathrm{root}}/\delta \le q < 1$.
- *Notation.* For a finite set s of packets and optional labels, $D(s)$ is the union of the footprints, and $\mathrm{ch}(s) = \sum_{\alpha\in s}(\mathrm{cov}_B(\text{footprint of }\alpha) + \text{length of }\alpha)$.
- *Claim.* For all $m, k, r \in \mathbb{N}$, $T \in \mathbb{R}$ and every site x,

$$\sum_s \left(\prod_{\alpha\in s}\mathrm{wt}(\alpha)\right)\big(1+\mathrm{ch}(s)\big)^k\,e^{a\,\mathrm{cov}_B(D(s))}\ \le\ \frac{k!\,e^{\varepsilon}}{\varepsilon^k}\,e^{-\varepsilon T-\mu_s r}\,\frac{\gamma_{\mathrm{root}}\,q^m}{1-q},$$

the sum over connected s with $x \in D(s)$, $|s| \geq m+1$, at least r seeds in total, and $\mathrm{ch}(s) \geq T$. Packets that are not connected have weight 0.

Status: theorem. The hypotheses of (b) are the records `Mixed.Costs.Buy` and `Mixed.Accounts`. (b) is used in III.2 and III.10–III.13. *Lean:* `Packets.packet_income`, `Packets.packet_variants_tail`, `Mixed.mixed_hit`, `Mixed.ready_vertices`.

Proposition III.5.8 (the numerical conditions hold at large coupling).

Notation. Fix the gauge group. In (a) and (b), also fix the block side L and $l \in \mathbb{N}$. For the coarse coupling x put:

- $\beta_F = x + b \log L$, with b a constant of the group;
- $B = \lceil (\log x)^2 \rceil$;
- $P(s) = e^{(\log\log s)^2}$;
- $t = \beta_F^{-1/2} P(\beta_F)$, the small-field threshold;
- $\mathrm{bill}(x) = 1 + B + (\log \beta_F)^2$;
- for $\mu, u > 0$, $M = \mu u\, P(\beta_F)^2$.

(a) Let $a, D \geq 0$, $d \in \mathbb{N}$ and $z < 2$. For all sufficiently large x the following holds. Whenever $s \leq D\,\mathrm{bill}^d P(\beta_F)^z$, $g \geq M$ and $0 \leq C \leq \exp(D\,\mathrm{bill}^d P(\beta_F)^z)$:

- the numerical condition $\rho \leq q < 1$ of III.5.7(a) holds with $q = e^{-M/2}$;
- $K_{\mathrm{hit}} C q^{\max(1,m)}/(1-q) \leq e^{-M\max(1,m)/4}$ for every m.

If $\beta \geq u\beta_F$, then $\beta\mu t^2 \geq M$.

(b) Let in addition $\varepsilon, \delta > 0$. For all large x, all $\beta \geq u\beta_F$, and all $C, e_s \geq 0$, $s \in \mathbb{R}$, and γ_{opt} with

$$C \leq e^{\mathrm{fee}}, \qquad s + 2\varepsilon e_s \leq \mathrm{fee} := D\,\mathrm{bill}^d P(\beta_F)^z, \qquad 0 \leq \gamma_{\mathrm{opt}} \leq \delta/(16\,\mathrm{ball}_B),$$

the numerical inequalities of III.5.7(b) hold with:

- gain $g = \beta\mu t^2$;
- mark $\mu_s = \mu u P(\beta_F)^2/2$ (half of the guaranteed gain is extracted per seed);
- $q = 1/2$;
- $\gamma_{\mathrm{root}} = e^{-\mu u P(\beta_F)^2/8} + \gamma_{\mathrm{opt}}$.

The packet envelope and optional site-sum estimate in III.5.7(b) remain hypotheses on the actual weights.

(c) For a step-estimator $\mathcal{T}$ (III.1), there is n such that for every $L \geq n$ and all sufficiently large W there is $\mu > 0$, chosen before l, d, u, a, D, z, with the following property for every choice of l, d, u, a, D, z satisfying (a). For all sufficiently large x, every funded step with block side L and coarse side $\geq 100(1+B)$ satisfies the point charge of III.5.11 with constant μ. For every $\beta \geq u\beta_F$, the numerical conclusions of (a) hold with $g = \beta\mu t^2$. The threshold in x may depend on the fixed numerical parameters but not on the volume.

Status: theorem. (c) takes the energy grant per seed from III.10. Used in III.10. *Lean:* `Delay.fits`, `Delay.comparison`, `Buy.quoted`, `Grant.order`.

Proposition III.5.9 (logarithm by deletion).

(a) Abstract form.

- *Data.* J finite; $X : \iota \to$ nonempty subsets of J (ι any set); $v : \iota \to [0,\infty)$; a weight w on subsets of J with $w \geq 1$ and $w(P \cup P') \leq w(P)w(P')$ whenever $P \cap P' \neq \emptyset$; $a \geq 0$; $0 \leq \delta \leq 1 - e^{-a}$.
- *Site condition.* For every $x \in J$,

$$\sum_{i:\, x \in X_i} w(X_i)\, v_i\, e^{a|X_i|} \leq \delta.$$

- *Gas and its pieces.* For $u : \iota \to \mathbb{R}$ with $|u_i| \leq v_i$, let $Z_u(B) = \sum_Q \prod_{i \in Q} u_i$ ($B \subseteq J$), the sum over finite families Q of labels with pairwise disjoint supports contained in B. Let λ_u be the Möbius pieces of $-\log Z_u$.
- *Claim.*
  - $Z_u(B) > 0$ and $Z_u(B) = \exp\big(-\sum_{A \subseteq B} \lambda_u(A)\big)$ for every B.
  - $|\lambda_u(A)| \leq \lambda_{-v}(A)$, and $\sum_{A \ni x} w(A)\, \lambda_{-v}(A) \leq a$ for every x.
  - For every $Q \subseteq J$, $Z_u(J \setminus Q) = Z_u(J)\, e^{l}$, with $l = \sum_{A \cap Q \neq \emptyset} \lambda_u(A)$ and $|l| \leq a|Q|$.

(b) At a coarse field.

- *Data.* Let ι be countable, with nonempty supports X_i. Let $E_i(V)$ ($i \in \iota$) be functions of the gauge field on Λ_n. Each is local in X_i, gauge invariant and Borel, with $|E_i(V)| \leq v_i$ whenever no site of X_i is flagged. Let $\tilde{E}_i$ be E_i set to 0 when X_i contains a flagged site.
- *Hypotheses.* $M \in \mathbb{N}$, $A, a \geq 0$, $v_i \geq 0$, $a(2M+1)^4 \leq 1$, $\delta \leq 1 - e^{-a}$, and $\sum_{i:\, x \in X_i} e^{(A+1)\mathrm{cov}_M(X_i)} v_i \leq \delta$ for every x.
- *Claim.* The gas Z_V of $\tilde{E}(V)$ satisfies, for every V and every Q:
  - $Z_V(Q) > 0$ and $Z_V(Q) = \exp\big(-\sum_{P \subseteq Q} \lambda_P(V)\big)$;
  - $|\lambda_P(V)| \leq g(P)$, with g independent of V, and $\sum_{P \ni x} e^{A\,\mathrm{cov}_M(P)} g(P) \leq a$;
  - $Z_V(\Lambda_n \setminus Q) = Z_V(\Lambda_n) e^{l}$ with $l = \sum_{P \cap Q \neq \emptyset} \lambda_P(V)$ and $|l| \leq a|Q|$;
  - there are gauge-invariant Borel functions λ_P^{raw}, local in the links with both endpoints in P, such that $\lambda_P(V) = \mathbf{1}[P \text{ contains no flagged site}]\lambda_P^{\mathrm{raw}}(V)$. Thus the masked coefficients also use the flag tests on P.

Status: theorem. (b) is used in III.11. *Lean:* `ReLog.catalog`, `ReLog.omitted`, `ReLog.swept`, `ReLog.intake`, `ReLog.retest`, `ReLog.retain_mask`.

Proposition III.5.10 (trace-log of a near-identity product). Let A be positive definite and C symmetric (real matrices), and $S = AC - 1$ with $0 \leq q < 1$ and $\sum_j |S_{ij}| \leq q$ for every i. Then:

- $\det C > 0$ and $\det(1+S) > 0$;
- $\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \operatorname{tr} S^n$ converges absolutely, and its sum is $\log\det(1+S)$.

Status: theorem. Used in III.4. *Lean:* `Round.exact_trace`, `Round.twin_log`.

Proposition III.5.11 (sector decomposition of a block-spin step; gain per seed).

Setting. Fix a step S with block side L, a step-estimator $\mathcal{T}$, a tube T with Funds(S,T) (III.1), a coarse coupling x with threshold t, and constants $W, \kappa > 0$.

Sector of a packet. Say (U,k,V) satisfies the sector prerequisites of A when:

- each fine cell whose projected vertices meet $D_{5B}(A)$ has plaquette chord $\geq t/4$ if listed in A, and $\leq t/2$ otherwise;
- each coarse bond with an endpoint in $D_{5B}(A)$ has predicted wire chord $\geq Wt$ if listed in A, and $\leq 2Wt$ otherwise;
- on $D_{5B}(A)$, the sites listed in A are exactly the sites flagged by V at the coarse threshold $t_C = x^{-1/2}e^{(\log\log x)^2}$;
- the frame tag has positive frame mass at each site of $D_{5B}(A)$.

Point charge. For every tree-gauge forest F, every fine field U equal to 1 on the trunks of F, every frame tag k, every coarse field V and every packet A such that (U, k, V) satisfies these prerequisites:

$$\mathcal{E}_A(U, V) \;\geq\; \kappa\, t^2\, |A|.$$

Here $\mathcal{E}_A$ is the *assessed energy* of A: the sum of the fine, block-constraint, coarse and cross terms of III.1 with anchors in $D_{2B}(A)$, using walks truncated at $\lfloor B/100000 \rfloor$.

Claim. Let R be a lawful representation of the fine density (III.1) adapted to T: its path data and clip are those of the upper stage of T, and its threshold is t. Let its coupling satisfy $\beta \geq 0$. Let F be any tree-gauge forest and $D = (D_A)_A$ any allocation: $D_A(k, U, V) \geq 0$ is Borel, $\sum_A D_A = 1$, and $D_A \neq 0$ implies the two seed tests above for every fine cell and coarse bond, together with equality of the flag sets on the whole coarse lattice. Positive likelihood then implies the local sector prerequisites. This global support condition is needed to apply the point charge in every lattice orientation. Then for every coarse field V

$$(\mathrm{push}_\beta\, \rho_R)(V) = \sum_A c_A(V), \qquad |c_A(V)| \;\leq\; \pi_\beta\, e^{-\beta\kappa t^2|A|}\, \ell_A(V),$$

where:

- c_A is the A-th sector output;
- $\pi_\beta = Z_{\mathrm{bond}}(\beta)^{-|\mathrm{Bond}(\Lambda_c)|}$, where $Z_{\mathrm{bond}}(\beta)$ is the single-bond penalty integrated against probability Haar measure;
- $\ell_A(V)$ is the remaining sector integral, including the coarse factor sieve(V), with the orientation-averaged assessed energy removed and absolute values on the exposed density;
- c_A and ℓ_A are bounded Borel functions.

Uniformity.

- At a committed pair of scales of the run, a single $\kappa > 0$ serves all volumes, all steps of that pair, and all lawful representations produced by the run whose coupling lies in the admissible band.
- For a fixed step-estimator, choose L sufficiently large, then W sufficiently large, then $\kappa > 0$. The point charge holds for all sufficiently large x and all funded steps of coarse side $\geq 100(1 + B)$, uniformly in their volumes.

Status: theorem, conditional on Funds and the point charge. The point charge is discharged in two ways:

- in general, from the Teller's energy grant (III.10);
- at committed scales, from the committed-scale energy hypothesis of the run, through `notation_ticket` (III.1).

Used in III.4 and III.10. *Lean:* `Presented.portionCost` (conclusion `PortionCost`), `Consumer.bought`, `points_grant`; definitions `Seeded`, `Alive`, `PointCharge`, `earn`, `Allocation`.

Proposition III.5.12 (Combes–Thomas bound). Let M be a real symmetric matrix on a finite index set, with positions $a_i \in \Lambda_n$. Assume $M \geq r > 0$, $\eta \geq 0$, and

$$\sup_i \sum_j e^{\eta d(a_i,a_j)} |M_{ij}| \leq C, \qquad \sup_j \sum_i e^{\eta d(a_i,a_j)} |M_{ij}| \leq C.$$

If $0 \leq \theta \leq 1$ and $\theta C \leq r/2$, then for every $u \geq 0$ the matrix $M + uI$ is invertible and

$$e^{\theta\eta\, d(a_i,a_j)} \left|(M+uI)^{-1}_{ij}\right| \leq \left(\min(1, r/2)\,(1+u)\right)^{-1} \quad \text{for all } i, j.$$

The hypotheses pass to principal submatrices. *Status:* theorem. *Lean:* `Green.inverse`, `Green.Bottom.restrict`, `Green.Rows.restrict`.

Proposition III.5.13 (windowed inverses).

Setting.

- A finite index set, positions $d_j \in \Lambda_n$, and $U \geq 0$ such that $\mathrm{card}\{j : d(y, d_j) \leq \rho\} \leq U(1+\rho)^4$ for every y and every $\rho \in \mathbb{N}$.
- A, r, C, η, θ as in III.5.12, with $C \geq 0$ and $\eta, \theta > 0$; and $0 \leq \mu < \nu < \theta\eta$.
- A radius $R \in \mathbb{N}$ and windows $P(j) \ni j$ such that $d(d_i, d_j) \leq R \Rightarrow i \in P(j)$.

Objects.

- $\mathrm{within}(A)_{ij} = \left((A|_{P(j)})^{-1}\right)_{ij}$ for $i \in P(j)$, and 0 otherwise.
- $E = A\,\mathrm{within}(A) - 1$ and $\mathrm{cov} = \frac{1}{2}(\mathrm{within} + \mathrm{within}^T)$.
- $\Pi_U(\eta) = U\,4!\,(2/\eta)^4\, e^{\eta/2}/(1-e^{-\eta/2})$, $Y = \Pi_U(\theta\eta-\nu)/\min(1, r/2)$, and $s = e^{-(\nu-\mu)R}(CY+1)$.

(a) Weighted row and column sums at rate μ satisfy:

- A^{-1} and $\mathrm{within}(A)$: at most Y;
- E: at most s;
- $\mathrm{within}(A) - A^{-1}$: at most Ys.

At any rate $0 \leq \xi \leq \eta$, if $\mathrm{within} - A^{-1}$ has row and column bounds $s' \geq 0$ and within has bounds $Y' \geq 0$, then cov, $\mathrm{cov} - A^{-1}$ and $A\,\mathrm{cov} - 1$ have bounds Y', s' and Cs', respectively. If also $C, Y' > 0$ and $s' \leq 1/(2C)$, then $\mathrm{cov} \geq 1/(2C)$ and $\mathrm{cov}^{-1} \geq (Y')^{-1}$.

(b) Fix positive bounds on the weighted row and column sums, spectral gap, rate α, and counting constant. There are $0 < e \leq \alpha$ and $K, c, c_1, c_2 > 0$, depending only on these data, such that for all sufficiently large integer R the following holds uniformly for all such matrices and windows with room R. At rate e, the full and windowed inverses of $tA + (1-t)I$ have bounds K, and their residual and inverse difference have bounds $Ke^{-cR} < 1$, for every $t \in [0, 1]$. At $t = 1$, the covariance has bound K, its difference from A^{-1} and the residual $A\,\mathrm{cov} - I$ have bounds Ke^{-cR}, and $\mathrm{cov} \geq c_1$, $\mathrm{cov}^{-1} \geq c_2$.

Status: theorem. *Lean:* `Win.Windows.window_price`, `covariance_price`, `covariance_floor`, `Win.Promise.receive`.

Proposition III.5.14 (precision matrix of the small-field Gaussian of a step).

Setting. Fix a step-estimator $\mathcal{T}$ and a patch $\mathcal{D}$ (block side L, order J) satisfying the branch conditions for its kernel and gauge group. For a step with patch $\mathcal{D}$, tube T, coupling x, tree-gauge forest F, packet A and coarse field V, define D as follows:

- *Rows.* Use the flat plaquette-curl and upper path-sum rows, with walks truncated at $\lfloor B/100000 \rfloor$, and the flat linearized block-constraint rows.
- *Transport.* Transport the input Lie-algebra field by the replica of V and a fixed path gauge centred at each row, in every lattice orientation.
- *Restriction.* Keep rows whose oriented centres lie outside $D_{8B}(A)$, and columns on *active* fine bonds: bonds outside the trunks of F whose projected endpoints lie outside $D_{10B}(A)$. Set all other entries to zero.
- *Normalization.* Write Lie-algebra coordinates and multiply each row by the inverse square root of the number of lattice orientations.

This is a transported flat linearization. Its equality to the derivative of the nonlinear cut rows at V is not asserted.

Let gram $= D^T D + \operatorname{diag}(\mathbf{1}_{\text{inactive}})$, with positions the coarse parents of the bonds.

Claim. There are constants $r, K, U, \alpha > 0$, with α half the decay rate of the upper path data, such that the following holds for all large x. For every step with patch $\mathcal{D}$ and Funds, coarse side $\geq 100(1+B)$, every F, and every A, V for which the flags of A are the flagged sites of V:

- gram is symmetric and gram $\geq r$;
- at most $U(1+\rho)^4$ indices lie within distance ρ of any site;
- its weighted row and column sums at rate α are at most K;
- $D_{ij} = 0$ when the row centre and column position are separated by more than $\lfloor B/100000 \rfloor + R_{\text{pred}}(\mathcal{D}) + 2$, where R_{pred} is the patch's predictor range. The range grows with B; the weighted bounds are uniform in x.

Consequently, for large x, III.5.13(b) applies to the following windows: a fine bond belongs to the window of an addressed bond exactly when at least one of its projected endpoints is within distance $\lfloor B/4 \rfloor$ of at least one projected endpoint of the addressed bond. Include every color coordinate of each such bond. Their room is $\lfloor B/4 \rfloor$, and the error is $K_{\text{win}} e^{-c_{\text{win}} \lfloor B/4 \rfloor} < 1$.

Status: theorem, conditional on Funds and Branches. Used in III.4 and III.11 through the window estimates below. *Lean:* `Aff.bedrock`, `Aff.surplus`, `Aff.gram_budget`, `Aff.install_promises`; definitions `Aff.Dmat`, `Aff.gram`, `Promise.Served`.

Further outputs.

- A calculus of budgets SB (`Basic.lean`): monotonicity, scaling, pull-back, sums over fibres, geometric tails, `budget_count_tail`. It is used by about 125 modules elsewhere.
- Site-sum bounds for bundles of walks on the fine lattice, with a tail in the true length of the walks (`Paths.bundle_hit`).
- A weighted row bound for the tree-gauge response kernel (`Response.response_price`).
- Auxiliary Gaussian estimates, listed in Lemma 4.36.

3.8.2 2. What it uses

From other units.

- **I.1:** the torus Λ_n; the distance d (`separation`); $|X| \le \mathrm{ball}_M \mathrm{cov}_M(X)$ (`Corona.pays`); subadditivity of cov_M on overlapping unions (`Corona.meets_sum`).
- **II.2:** the site-sum condition `Ac.Hit`; connected families `Ac.Spark`; cov_M of a connected union is at most the sum (`Ac.cover`).
- **III.1:**
  - SB;
  - the step data: patch, kernel, frame choices, the pushforward `StepData.push`, `Branches`;
  - lawful representations `Rep.Lawful`;
  - packets, domes, sectors and the assessed energy (`Stores`, `Plot`, `assessed`);
  - the step-estimator interface `Teller` and `Teller.Funds`, with Funds supplied for the run by `Testing.proceeds`;
  - tubes, tree-gauge forests (`Tree.Forest`) and flags;
  - the constants B, t, β_F (`radiusB`, `fiery`, `betaF`);
  - the point charge at committed scales (`notation_ticket`).
- **III.10:**
  - the step-estimator's energy grant per seed (`Next.Fluctuations.grants`), used for III.5.8(c) and III.5.11;
  - the sector integrals and their reduction (`Sector.peel`, `Sector.prefactor`);
  - the linearized block constraint `zLin`;
  - the growth of β_F and the uniform predictor bound used in the spectral gap estimate.
- **III.6:** a gauge-fixed Poincaré-type inequality (`Work.F01.link_bound`). A field vanishing on the trunks is bounded on each fine bond by its plaquette curls in a box of fixed radius around the block, plus its block average. This is the source of the spectral gap r in III.5.14.
- **III.2:** committed pairs of scales, the construction run and its energy hypothesis entering the uniform sector estimate.

Standard mathematics. The formal proofs establish the estimates directly. References for the methods are:

- Mayer expansion, Ursell functions and the Penrose tree partition: R. Fernández and A. Procacci, "Cluster expansion for abstract polymer models. New bounds from an old approach", *Commun. Math. Phys.* **274** (2007), 123–140.
- Polymer convergence criteria: R. Kotecký and D. Preiss, "Cluster expansion for abstract polymer models", *Commun. Math. Phys.* **103** (1986), 491–498; C. Gruber and H. Kunz, "General properties of polymer systems", *Commun. Math. Phys.* **22** (1971), 133–161.
- Lattice-animal counting by spanning walks; the Schur test for matrix norms; Möbius inversion on subsets.
- Exponential conjugation for inverse decay: J.-M. Combes and L. Thomas, "Asymptotic behaviour of eigenfunctions for multiparticle Schrödinger operators", *Commun. Math. Phys.* **34** (1973), 251–270.
- Large-field sectors and localized propagators belong to Bałaban's block-spin method. Here they are implemented through the explicit seed allocations and principal inverses stated above.

Imported constants and assumptions.

- The constants $B = \lceil (\log x)^2 \rceil$, $t = \beta_F^{-1/2} e^{(\log\log \beta_F)^2}$ and $\beta_F = x + b \log L$ (III.1).
- The halo factor l is arbitrary in the packet estimates; a particular choice belongs to their application.

- The block side $L \geq n$ and the constant W, chosen large (III.2).
- Assumed hypotheses:
  - Funds and Branches (III.1);
  - lawful representations (III.1);
  - the point charge (discharged as in III.5.11);
  - in the auxiliary Gaussian estimates, the coarse-side bound and flag equality, together with the compatible quadratic forms specified in Lemma 4.36.

3.8.3 3. Strategy

For a finite gas, inclusion–exclusion writes the partition function as a sum over tuples. Splitting off the connected component of a distinguished occurrence gives the coefficient recurrence $(m+1)g_{m+1} = \sum_{k=0}^{m}(k+1)\ell_k g_{m-k}$. For the gas polynomial $P(t) = \sum_n g_n t^n$ and cluster series $L(t) = \sum_m \ell_m t^{m+1}$, this becomes $P' = L'P$. A geometric bound on the cluster coefficients justifies differentiation on an interval containing $[0,1]$; then $Pe^{-L} = 1$. Absolute bounds allow passage to arbitrary label sets through the directed family of finite subsets.

The bounds combine the tree-graph inequality with a sum over trees. A parent support Y meets labels of total inflated weight at most $|Y|\gamma \leq \mathrm{ball}_B \gamma \,\mathrm{cov}_B(Y)$. The reserve exponent δ absorbs each parent's number of children through $x^k \leq k! e^x$. Substochastic kernels sum the labels; the sum of the resulting child factorials over parent maps is at most $4^m m!$. After cancellation of the occurrence factorial, the ratio per added vertex is $4\,\mathrm{ball}_B \gamma/\delta$. Subadditivity of the covering number transfers these bounds to cluster unions. The resulting estimates retain site pins, tails, deleted regions and repeated forced polymers.

For seed packets, a spanning-walk count gives an entropy factor K_{step}^2 per seed. The point charge supplies an energy gain $\beta\kappa t^2$ per seed. Since $t = \beta_F^{-1/2} P(\beta_F)$ and $\beta \geq u\beta_F$, the guaranteed gain is $\kappa u P(\beta_F)^2$. At fixed L, this dominates every cost $\mathrm{bill}^d P(\beta_F)^z$ with $z < 2$. Half the gain can be retained as a mark for later estimates. The partition of unity over global seed records then splits the block-spin pushforward while preserving a common field, tag array and tree-gauge slice.

The deletion construction obtains another logarithm from the region recurrence. Its negative-logarithm pieces form a union-convolution series pinned at a chosen site. A weighted ℓ^1 bound closes induction on the region and compares signed activities with the gas of negative envelope activities. The final matrix estimates use a transported flat linearization: a gauge-fixed Poincaré inequality and a small-background comparison give a uniform spectral gap, while path bounds give weighted row and column estimates. Exponential conjugation yields inverse decay, and principal window inverses approximate the full inverse with error $O(e^{-cB})$.

3.8.4 4. Steps

3.8.4.1 4.1 Algebra of the finite gas In Lemmas 4.1–4.4 the label set ι is finite, the weights are real, and R is reflexive and symmetric. All vertex sets are finite.

Lemma 4.1 (gas as a sum over tuples). Let R be reflexive and symmetric, and $g_n = \frac{1}{n!}\sum_{z\in\iota^n} \mathrm{all}(z) \prod_a w_{z_a}$, where $\mathrm{all}(z) = \sum_{G\subseteq G_R(z)} (-1)^{|G|}$. Then $g_n = \sum_{|s|=n,\ s \text{ compatible}} \prod_{i\in s} w_i$, and $\sum_n g_n = \Xi(w)$. *Sketch:*

- $\mathrm{all}(z) = \mathbf{1}[G_R(z) = \emptyset]$, by the binomial identity.

- Since R is reflexive, an edgeless tuple has distinct entries, and so orders a compatible set; each set arises $n!$ times.

Lean: `Partition.allMass_eq`, `Gas.row`, `Gas.sums`.

Lemma 4.2 (splitting off a component). For any relation R, any tuple z on a finite vertex set α, and $o \in \alpha$:

$$\mathrm{all}(z) = \sum_{s \ni o} \phi_R^T(z|_s)\, \mathrm{all}(z|_{\alpha \setminus s}).$$

Sketch: a subgraph is the same thing as three pieces of data:

- the vertex set s of the component of o;
- a connected spanning graph on s;
- an arbitrary subgraph on the complement, with no edges between s and its complement.

The signs multiply. *Lean:* `Partition.partition`. *(proof read; the bijection lemmas statement only)*

Lemma 4.3 (coefficient recurrence). $(m+1)\, g_{m+1} = \sum_{k=0}^{m} (k+1)\, \ell_k\, g_{m-k}$. *Sketch:*

- Sum Lemma 4.2 over all tuples on $\{o\} \sqcup \{1, \dots, m\}$; the tuple sums over s and over its complement factor.
- There are $\binom{m}{k}$ sets s of size $k+1$ containing o.
- Relabelling invariance of ϕ^T finishes the count.

Lean: `Coef.unbased`, `Coef.discrete`, `Coef.recurrence`.

Lemma 4.4 (III.5.1). Under the geometric coefficient bound of III.5.1, $\exp(\sum_m \ell_m) = \Xi(w)$. *Sketch:*

- Let $u = (1+q^{-1})/2 > 1$. On $(-u, u)$, the series $L(t) = \sum \ell_m t^{m+1}$ can be differentiated term by term, because $|(m+1)\ell_m t^m| \le C(m+1)(qu)^m$ is summable.
- $P(t) = \sum g_n t^n$ is a polynomial, since $g_n = 0$ for $n > |\iota|$.
- Lemma 4.3 and the Cauchy product give $P' = L'P$. Hence $H = Pe^{-L}$ has $H' = 0$ and $H(0) = g_0 = 1$.
- Evaluate at $t = 1$.

Lean: `Gas.exact_polynomial`.

3.8.4.2 4.2 Tree-graph bound

Lemma 4.5 (Penrose partition; III.5.2). The absolute connected-graph sum is bounded by the number of spanning trees. *Sketch:*

- To each connected spanning G attach its breadth-first tree from the distinguished vertex, breaking ties by a fixed enumeration.
- For a fixed tree p, the graphs with tree p form a Boolean interval $[E(p), E(p) \cup F(p)]$. Here $E(p)$ is the edge set of p, and $F(p)$ consists of the edges of T that neither change the breadth-first layers nor offer an earlier parent.
- The alternating sum over an interval vanishes unless the interval is a point: flip one free edge.
- Each fibre therefore contributes at most 1, and only trees of T contribute.

The bound follows without identifying which trees have nonzero fibres. *Lean:* `Graph.pay`; breadth-first construction in `Breadth.lean`. *(proof read; breadth-first construction statement only)*

3.8.4.3 4.3 Sums over trees Lemma 4.6 (substochastic sums along a forest). Let α, β be finite, $p : \alpha \to \alpha \sqcup \{o\}$ an acyclic parent map, Y a fixed root state, and X_j the state associated with $j \in \beta$. If $K(Y', j) \geq 0$ and $\sum_j K(Y', j) \leq 1$ for every state Y', then $\sum_{z:\alpha\to\beta} \prod_i K(X_{z_{p(i)}}, z_i) \leq 1$, with the first argument interpreted as Y when $p(i) = o$. *Sketch:* induction on the set of summed vertices. Sum out a vertex of maximal rank first: it is a leaf, and its row sum is ≤ 1. *Lean:* `Trees.substochastic`.

Lemma 4.7 (fan-out factorials). $\sum_{p:[m]\to[m]\sqcup\{o\}} \prod_v (\mathrm{card}\{i : p(i) = v\})! = \binom{2m}{m} m! \leq 4^m m!$. *Sketch:* with $m+1$ possible parents fixed, extending a map on n vertices multiplies the total factorial weight by $n+m+1$. Iterating from $n = 0$ to m gives $(2m)!/m!$, and $\binom{2m}{m} \leq 4^m$. *Lean:* `Trees.parent_fanout`, `fanout_bound`.

Lemma 4.8 (trees rooted at a support). Assume $w \geq 0$, $\gamma, \delta > 0$, $a \in \mathbb{R}$, and $\mathrm{Hit}_B(w; a+\delta, \gamma)$. For any support Y and any finite $D \subseteq \iota$,

$$\sum_p \sum_{z \in D^m} \mathbf{1}[X_{z_i} \cap X_{\mathrm{parent}(i)} \neq \emptyset \ \forall i] \ \frac{1}{m!} \prod_i \left(w_{z_i} e^{a\,\mathrm{cov}_B(X_{z_i})}\right) \ \leq \ e^{\delta\,\mathrm{cov}_B(Y)} \left(4\,\mathrm{ball}_B \gamma/\delta\right)^m,$$

where p runs over acyclic parent maps on the m labelled vertices with external root of support Y. *Sketch:*

- Define $K(Y', i) = \mathbf{1}[X_i \cap Y' \neq \emptyset]\, w_i e^{(a+\delta)\mathrm{cov}_B(X_i)} \big/ \left(\tau \max(\delta\,\mathrm{cov}_B(Y'), 1)\right)$ with $\tau = \mathrm{ball}_B \gamma/\delta$. The numerator summed over i is at most $|Y'|\gamma \leq \tau\delta\,\mathrm{cov}_B(Y') \leq \tau \max(\delta\,\mathrm{cov}_B(Y'), 1)$, so $\sum_i K(Y', i) \leq 1$, also for $Y' = \emptyset$.
- Rewrite each weight $w_i e^{a\,\mathrm{cov}}$ as $\tau \max(\delta\,\mathrm{cov}(\mathrm{parent}), 1)\, K\, e^{-\delta\,\mathrm{cov}(X_i)}$.
- A parent with k children contributes $\max(\delta\,\mathrm{cov}, 1)^k \leq k!\, e^{\delta\,\mathrm{cov}}$. The exponential at each internal vertex cancels its own $e^{-\delta\,\mathrm{cov}}$, leaving $e^{\delta\,\mathrm{cov}_B(Y)}$ at the root.
- Conclude with Lemmas 4.6 and 4.7.

Lean: `Trees.trees_afford`, `Trees.forest_at`.

3.8.4.4 4.4 Cluster sums and the logarithm Lemma 4.9 (rooted and marked cluster sums; III.5.3(c)). Under III.5.3's hypotheses, the marked sum at order m is at most bq^m. *Sketch:*

- $\frac{1}{(m+1)!} = \frac{1}{m+1} \cdot \frac{1}{m!}$: write the tuple sum as the average over the choice of root vertex. A transposition moves the root to position 0, and ϕ^T is invariant under relabelling.
- At root 0, Lemma 4.5 bounds $|\phi^T(z)|$ by the trees of $G_R(z)$. Since R is contained in the overlap relation, these are trees of supports rooted at X_{z_0} in the sense of Lemma 4.8.
- The root label is summed against its own weight $w_i e^{(a+\delta)\mathrm{cov}(X_i)}$, which absorbs the factor $e^{\delta\,\mathrm{cov}}$. This gives bq^m.

Lean: `Ursell.cluster_at`.

Lemma 4.10 (pinned bounds; III.5.3(a),(b)). Under III.5.3's hypotheses, both pinned tails are at most $\gamma q^m/(1-q)$. *Sketch:*

- If $\phi^T_R(z) \neq 0$, then $G_R(z)$ is connected, so $\mathrm{cov}_B(X(z)) \leq \sum_a \mathrm{cov}_B(X_{z_a})$.
- Apply Lemma 4.9 with $Q = \{i : x \in X_i\}$, where $b = \gamma$ by the site condition at x, and sum the geometric tail.
- The set version (b) follows in the same way from a count of distinct-label forests.

Lean: `Ursell.connected_pay`, `Ursell.log_hit_tail`, `Trees.touch_tail`. *(proof read; distinct-label forest count statement only)*

Lemma 4.11 (from finite to arbitrary label sets; III.5.4). Under III.5.4's hypotheses, the gas and cluster series converge absolutely, and $\Xi = \exp(\text{cluster series}) > 0$. *Sketch:*

- Every support is nonempty, so $\sum_i |w_i| e^{(a+\delta)\mathrm{cov}} \le n^4\gamma$. Hence $\sum_s \prod_{i\in s} |w_i| \le e^{n^4\gamma}$, and the cluster series has absolute sum $\le n^4\gamma/(1-q)$.
- On a finite $D \subseteq \iota$ the hypotheses restrict, and III.5.1 applies with $C = n^4\gamma$.
- Let the finite subsets $D \subseteq \iota$ increase by inclusion and use dominated convergence for both series; the dominating budgets do not depend on D.

Lean: `Gas.finite_log`, `Gas.price_log`, `Product.everyone_ne`, `Product.set_products`.

3.8.4.5 4.5 Deleted regions and forced polymers **Lemma 4.12 (III.5.5).** Under III.5.5's hypotheses, deleting polymers meeting Y multiplies the gas by e^{-p}, with $|p| \le |Y|\gamma/(1-q)$ and the stated cluster tails. *Sketch:*

- The cluster terms of w^Y are those of w minus the terms whose union meets Y.
- The removed terms are bounded by Lemma 4.10 with $m = 0$, summed over $x \in Y$.
- Apply III.5.4 to both gases; $|w^Y| \le |w|$, so w^Y satisfies the same hypotheses.

Lean: `Gas.restores_log`, `Gas.missing_tail`.

Lemma 4.13 (multiplicities on a fixed set). For a finite set s, if $0 \le \sigma < 1$ and $0 \le w_i \le \sigma$ for $i \in s$, then $\sum_{(n_i)\in\mathbb{N}^s_{\ge 1}} \prod_{i\in s} w_i^{n_i}/n_i! \le \prod_{i\in s} (w_i/(1-\sigma))$. *Sketch:* bound $w_i^{n_i}/n_i!$ by $w_i\sigma^{n_i-1}$ and multiply the geometric sums. *Lean:* `Multiple.buy_at`.

Lemma 4.14 (multiset sums). Let X_i be nonempty, $w_i \ge 0$, $a \ge 0$, $t \ge 1$, $0 \le \sigma < 1$, and $e_i \in \mathbb{R}$. Suppose $\mathrm{Ready}(t\,w\,e^e/(1-\sigma); a, \gamma, \delta, q)$ and $\gamma \le \sigma$. Then the sum over multisets with connected support through x, satisfying a predicate Q on which $\sum n_i e_i \ge v$, with total multiplicity $\ge m$, of $e^{a\,\mathrm{cov}_B(\cup)} \prod w_i^{n_i}/n_i!$, is at most $t^{-m}e^{-v}\gamma/(1-q)$. *Sketch:*

- Each weight $t w_i e^{e_i} e^{a\,\mathrm{cov}_B(X_i)}$ is at most σ, by the site condition at a point of its nonempty support.
- Connectedness bounds the union cover by the sum of the individual covers; multiplicities are at least one. Lemma 4.13 then sums the multiplicities at cost $1/(1-\sigma)$ per label, and III.5.3(b) sums the connected sets.
- The factor $t^{\sum n} e^{\sum ne} \ge t^m e^v$ is extracted on the selected multisets.

Lean: `Multiple.estimates`.

Lemma 4.15 (III.5.6). Under III.5.6's hypotheses, the pinned multiset sum containing a forced label is at most $t^{-m}e^{-\mu}\eta/(1-q')$. *Sketch:*

- The first-level clusters satisfy a site condition with constant $\gamma/(1-q)$ at exponent $A+\delta'$, by III.5.3(a) with $m = 0$; the forced polymers satisfy one with constant c.
- Join the two alphabets, scaling the weights by $te^{\mu}/(1-\sigma)$ and $t/(1-\sigma)$ respectively; the resulting constant is at most η.
- Apply Lemma 4.14 with $e = \mu$ on forced labels; the presence of a forced label gives $v = \mu$.

Lean: `OmitClusters.restore`, `Labels.together`.

3.8.4.6 4.6 Packets and the large-coupling numerics Lemma 4.16 (animals and covers). In the setting of III.5.7:

- For every $k \in \mathbb{N}$, the number of connected packets with k seeds whose dome contains x is at most $K_{\mathrm{hit}}(K_{\mathrm{step}}^2)^k$.
- $\mathrm{cov}_B(D(A)) \le |A|\,(2l+3)^4$ for connected A.

Sketch:

- A connected packet is the vertex set of a walk of length $2(k-1)$. The walk runs in the graph of seeds whose skins can meet, which has degree $\le K_{\mathrm{step}}$. It starts at one of at most K_{hit} seeds whose skin contains x.
- For $B \ge 1$, a skin lies in a ball of radius $lB+1 \le (l+1)B$, covered by $(2l+3)^4$ balls of radius B. For $B=0$, its connected location set lies in the radius-1 ball and costs at most $3^4 \le (2l+3)^4$ singleton balls. A connected union costs at most the sum.

Lean: `Packets.stores_count`, `Animals.animals`, `Packets.cover_support`. *(statement only for the animal count; covering proof read)*

Lemma 4.17 (III.5.7(a)). Under III.5.7(a)'s hypotheses, the packet tail is at most $K_{\mathrm{hit}} C q^{\max(1,m)}/(1-q)$. *Sketch:*

- On a connected packet, $p_A e^{a\,\mathrm{cov}} \le C\big(e^{s-g+a(2l+3)^4}\big)^{|A|}$.
- With the count of Lemma 4.16, sum the geometric series (`budget_count_tail`).
- Connected packets have at least one seed; hence $\max(1,m)$.

Lean: `Packets.packet_income`, `Packets.cover_income`.

Lemma 4.18 (III.5.7(b)). Under III.5.7(b)'s hypotheses, the mixed-word sum has the bound stated there. *Sketch:*

- *Ready.* Inflate the weights to $p_A e^{2\varepsilon e_s|A|+\mu_s|A|}$ and $w_i e^{2\varepsilon\lambda_i}$. The joint alphabet then satisfies Ready at exponent $a+2\varepsilon$: packets by Lemma 4.17, optional labels by hypothesis, and the constants add up to γ_{root}.
- *Connected sets.* III.5.3(b) sums the connected sets.
- *Extractions.*
  - $(1+\mathrm{ch})^k \le \varepsilon^{-k} k!\, e^{\varepsilon} e^{\varepsilon\,\mathrm{ch}}$;
  - one of the two factors $e^{\varepsilon\,\mathrm{ch}}$ of the inflation gives $e^{\varepsilon T}$;
  - the marks give $e^{\mu_s r}$.

Lean: `Mixed.ready_vertices`, `Mixed.mixed_hit`, `Reserves.sets`, `Reserves.reserve_tail`.

Lemma 4.19 (III.5.8). For fixed numerical parameters satisfying III.5.8, all its packet and tree conditions hold for sufficiently large x, uniformly in volume. *Sketch:*

- *Domination.* $D\,\mathrm{bill}^d \le P(\beta_F)^e$ eventually, for every $e>0$, since $(\log\beta)^{2d} = e^{2d\log\log\beta}$ is eventually below $e^{e(\log\log\beta)^2}$. Also $K_{\mathrm{hit}}, K_{\mathrm{step}}^2 \le e^{M/16}$, because $K_{\mathrm{step}} \le \mathrm{const}\cdot\mathrm{bill}^4$.
- *Ratio.* $\rho \le e^{M/16} e^{M/16-M+M/16} \le e^{-M/2}$.
- *Gain.* $\beta \ge u\beta_F$ gives $\beta\mu t^2 \ge \mu u P(\beta_F)^2$, because $\beta_F t^2 = P(\beta_F)^2$.
- *Assembly.* For (b), subtract the mark $M/2$ from the gain and apply (a) with $\mu/2$; the remaining gain is at least $M/2$. For (c), use the step-estimator's grant before applying the numerical estimates.

Lean: `Delay.monitor_pays`, `Delay.step_delay`, `Delay.fits`, `Buy.quoted`, `Grant.order`. *(proof read; the grant of III.10 statement only)*

3.8.4.7 4.7 The logarithm by deletion **Lemma 4.20 (deletion identity).** For the gas Z_u of III.5.9(a) and $x \in B$:

$$Z_u(B) = Z_u(B \setminus \{x\}) + \sum_{P \subseteq B,\ P \ni x} \varphi_u(P)\, Z_u(B \setminus P), \qquad \varphi_u(P) = \sum_{i:\, X_i = P} u_i,$$

and $Z_u(\emptyset) = 1$. *Sketch:*

- At most one member of a disjoint family contains x; removing it removes its support.
- The identity holds exactly for finite truncations of ι and passes to the absolutely convergent limit.

Lean: `ReLog.stock_deletion`, `ReLog.shelf`.

Lemma 4.21 (the incidence algebra). Let J be finite and let w satisfy the weight assumptions of III.5.9(a). For set functions on J define:

- $(f * g)(T) = \sum_{P \cup Q = T} f(P) g(Q)$;
- $\hat{f}(B) = \sum_{A \subseteq B} f(A)$, so that $\widehat{f * g} = \hat{f}\hat{g}$;
- $\|f\|_w = \sum_A w(A)|f(A)|$.

Then:

- $\|\cdot\|_w$ is submultiplicative on functions supported on sets containing a common point x.
- For such f with $\|f\|_w < 1$, $\lambda(f) = \sum_{n \geq 0} f^{*(n+1)}/(n+1)$ satisfies $\widehat{\lambda(f)} = -\log(1 - \hat{f})$ and $\|\lambda(f)\|_w \leq -\log(1 - \|f\|_w)$.
- If f and F are supported on sets containing x, $|f| \leq F$ pointwise and $\|F\|_w < 1$, then $|\lambda(f)| \leq \lambda(F)$.

Sketch: cumulative sums turn union convolution into multiplication. Two nonzero factors pinned at x have intersecting supports, so the weight inequality gives submultiplicativity. Apply it to convolution powers and sum $\sum_{n \geq 1} q^n/n = -\log(1-q)$ for $q < 1$; termwise absolute domination gives the last assertion.

Lean: `ReLog.crop_glue`, `ReLog.lamp_shed`, `ReLog.lamp_cropped`, `ReLog.lamp_mono`.

Lemma 4.22 (the induction). Let J be finite, w satisfy III.5.9(a)'s weight assumptions, and $a \geq 0$. Let U, z be functions of supports with $U \geq 0$, $|z| \leq U$, and $\sum_{P \ni x} w(P) U(P) e^{a|P|} \leq 1 - e^{-a}$ for every x. Suppose Z, Y satisfy the deletion identities with activities $-U, -z$, respectively, and $Z(\emptyset) = Y(\emptyset) = 1$. Then for every B:

- $Z(B) > 0$ and $Y(B) > 0$;
- $|\lambda_Y(B)| \leq \lambda_Z(B)$;
- $\sum_{A \subseteq B,\ A \ni x} w(A) \lambda_Z(A) \leq a$ for $x \in B$.

Sketch: strong induction on B. The empty region is immediate. For nonempty B, pick $x \in B$.

- Define f_x on subsets of B, extended by zero on other supports, by Möbius inversion from

$$\hat{f}_x(A) = \sum_{P \subseteq A,\ P \ni x} U(P) \exp\left(\sum_{T \subseteq A \setminus \{x\},\ T \cap P \neq \emptyset} \lambda_Z(T) \right), \qquad A \subseteq B.$$

The conditions $x \in P \subseteq A$ make f_x supported on sets containing x. The deletion identity gives $Z(A) = Z(A \setminus \{x\})(1 - \hat{f}_x(A))$.

- By submultiplicativity of w and the induction hypothesis at the points of P, $\|f_x\|_w \leq \sum_{P \ni x} w(P)U(P)e^{a|P|} \leq 1 - e^{-a}$. Hence $Z(B) > 0$.
- The pieces of $-\log Z$ on sets containing x are $\lambda(f_x)$, with $\|\lambda(f_x)\|_w \leq -\log e^{-a} = a$.
- The same computation for Y, with $|f_x^Y| \leq f_x$, gives the domination.

Lean: `ReLog.leash`, `ReLog.matures`, `ReLog.swept`, `ReLog.catalog`.

Lemma 4.23 (deleting sites). Under III.5.9(a)'s hypotheses, with $Z = Z_u$ and $\lambda = \lambda_u$, $-\log Z(J \setminus Q) = \sum_{A \cap Q = \emptyset} \lambda(A)$, hence $l = \sum_{A \cap Q \neq \emptyset} \lambda(A)$, and $|l| \leq \sum_{x \in Q} \sum_{A \ni x} w(A)\lambda_{-v}(A) \leq a|Q|$ because $w \geq 1$. *Lean:* `ReLog.omitted`.

Lemma 4.24 (at a coarse field; III.5.9(b)). Under III.5.9(b)'s hypotheses, its masked logarithm coefficients satisfy the common envelope and deletion-ratio bounds. *Sketch:*

- $e^{A \operatorname{cov}_M}$ is an admissible weight, since cov_M is subadditive on overlapping unions.
- $a(2M+1)^4 \leq 1$ gives $a|X| \leq \operatorname{cov}_M(X)$, which converts the condition into III.5.9(a).
- *Masking.* If P contains a flagged site y, the masked gas does not see y, so its Möbius piece at P vanishes.
- The raw pieces depend only on labels supported in P, so they are local in P and gauge invariant. Countability makes the gas sums Borel; finite Möbius inversion preserves this. Multiplying by the healthy-region indicator gives the masked coefficients, including the external flag tests.

Lean: `ReLog.intake`, `ReLog.native_rate`, `ReLog.permits`, `ReLog.retain_mask`, `ReLog.retest_meas`.

Lemma 4.25 (trace-log; III.5.10). Under III.5.10's hypotheses, the trace-log series converges absolutely to $\log\det(I+S)$, and both determinants are positive. *Sketch:*

- $AC = A^{1/2}(A^{1/2}CA^{1/2})A^{-1/2}$ is similar to a symmetric matrix.
- Its eigenvalues λ are those of $1+S$, and $|\lambda - 1| \leq \|S\|_{\infty\to\infty} \leq q$.
- Hence $\log\det(1+S) = \sum \log\lambda = \sum_n (-1)^{n+1} \operatorname{tr} S^n/n$, and $\det C = \det(1+S)/\det A > 0$.

Lean: `Round.twins`, `Round.twin_log`.

3.8.4.8 4.8 The sector decomposition Lemma 4.26 (allocations and columns). An allocation is the partition of unity with global seed tests specified in III.5.11. It satisfies $0 \leq D_A \leq 1$; where the likelihood is positive, $D_A \neq 0$ implies the local sector prerequisites.

- The hard allocation, the indicator of the actual booking, is one example.
- A smooth decreasing cutoff $\chi : \mathbb{R} \to [0,1]$, equal to 1 on $(-\infty, 1]$ and 0 on $[4, \infty)$, gives another allocation when $W, t > 0$.
- For every coarse V and $\beta \in \mathbb{R}$, if the step satisfies its branch conditions and g is integrable and gauge invariant, then $\sum_A \operatorname{out}_A(V) = \operatorname{push}_\beta g(V)$.

Sketch: sum the allocation before integrating, then apply the tree-gauge integration identity for g and the gauge-covariant block kernel.

Lean: `Allocation`, `Allocation.hard`, `Chi.canonical`, `push_columns`.

Lemma 4.27 (cut-out identity). For a representation adapted to the tube as in III.5.11, let $\Delta_{\mathrm{cut}}(U,V)$ be the orientation-averaged fine cut-row action minus its coarse counterpart. Then

$$\text{kernel} \times \rho_R = \text{sieve}(V) \cdot \text{exposed}(U) \cdot \text{kernel} \cdot e^{-\beta\,\Delta_{\mathrm{cut}}(U,V)}.$$

- In the input action, split the cut rows of the upper stage, averaged over the lattice symmetries, into two parts:
  - their coarse counterpart at V, which gives the factor $\text{sieve}(V)$, taken out of the integral;
  - the difference $\Delta_{\mathrm{cut}}(U,V)$, which is kept with the block kernel.
- The rest of the action and the polymer gas form $\text{exposed}(U)$.

Lean: `Presented.expel`, `Presented.use_gap`.

Lemma 4.28 (energy factorization). For a step satisfying its branch conditions,

$$\text{kernel} \cdot e^{-\beta\,\Delta_{\mathrm{cut}}} = \pi_\beta\, e^{-\beta\,\overline{\mathcal{E}}_A} \cdot (\text{likelihood})\, e^{-\beta\,\overline{\mathcal{E}}_A^{\mathrm{rest}}},$$

where bars denote averages over the lattice symmetries, applied to the rotated packet. On the tree-gauge slice, the global seed tests and positive likelihood imply the local prerequisites in every orientation. The point charge therefore gives $\overline{\mathcal{E}}_A \geq \kappa t^2 |A|$ wherever $D_A \neq 0$ and the likelihood is positive; zero-likelihood terms vanish. *Lean:* `Oriented.enters_eq`, `Oriented.mean_pays`.

Lemma 4.29 (III.5.11). Under III.5.11's funding, adaptation and point-charge assumptions, the sector outputs sum to the pushforward and satisfy its per-seed bound. *Sketch:*

- Express each signed sector output using Lemma 4.27; its sum is the pushforward by Lemma 4.26.
- Bound its absolute value by the integral of the absolute exposed density. Apply Lemma 4.28 with $\beta \geq 0$ to extract $\pi_\beta e^{-\beta\kappa t^2|A|}$.
- Bounded Borel factors give bounded Borel sector integrals at each fixed volume.

Lean: `Oriented.turn_charge`, `Presented.portionCost`.

Lemma 4.30 (one κ). The uniform point-charge conclusions of III.5.11 hold. *Sketch:* the general conclusion uses the step-estimator's energy grant (III.10). At a committed pair of scales, use the run's energy hypothesis (III.1); its compatibility conditions identify the representation's paths, clip and threshold with those of the tube. The constant is chosen before the volume, step and representation.

Lean: `points_grant`, `Consumer.bought`, `charged_at`. *(proof read; the inputs from III.1 and III.10 statement only)*

3.8.4.9 4.9 Matrices Lemma 4.31 (III.5.12). Under III.5.12's hypotheses, the shifted inverse has the stated pointwise exponential bound. *Sketch:* fix j and put $s_k = e^{\theta\eta d(a_k,a_j)}$ and $w = s \cdot (M+uI)^{-1} e_j$.

- Then $(M+uI+F)w = e_j$, with $F_{kl} = M_{kl}(s_k/s_l - 1)$.
- By convexity of exp, $|s_k/s_l - 1| \leq \theta e^{\eta d(a_k,a_l)}$. The Schur test then gives $|\langle w, Fw\rangle| \leq \theta C|w|^2 \leq \frac{r}{2}|w|^2$.
- Hence $\min(1, r/2)(1+u)|w|^2 \leq w_j \leq |w|$, and $|w_i| \leq |w|$ gives the claim.

Lean: `Green.inverse`, `Green.diminishing`, `Green.schur`.

Lemma 4.32 (III.5.13). Under III.5.13's hypotheses, the window residual, inverse difference and covariance satisfy its weighted bounds. *Sketch:*

- $E_{ij} = 0$ for $i \in P(j)$, because column j of within(A) solves the system on the window. So E lives at distances $> R$.
- Pointwise decay at rate $\theta\eta$, from III.5.12 applied to A and to its principal submatrices, becomes row-sum decay at rate ν by the counting bound. This costs $\Pi_U(\theta\eta - \nu)$.
- The far entries of E at the slower rate μ pay $e^{-(\nu-\mu)R}$.
- Finally within$-A^{-1} = A^{-1}E$, and the weighted Schur classes are closed under products.
- Part (b) applies (a) uniformly along $tA + (1-t)$.

Lean: `Windows.stitched`, `Windows.inverse_price`, `Windows.window_price`, `Win.planePrice`.

Lemma 4.33 (spectral gap). Fix $\mathcal{T}, \mathcal{D}$ as in III.5.14. There is $0 < r \le 1$ such that, for all sufficiently large x, all funded steps with this patch, coarse side $\ge 100(1+B)$ and the stated flag equality satisfy $r|v|^2 \le |Dv|^2$ for every v supported on active coordinates. *Sketch:*

- Take an active bond. Its projected endpoints lie outside $D_{10B}(A)$, so in an axial gauge the background V is within $\delta(x) \to 0$ of 1 near the bond.
- The Poincaré-type inequality of III.6 bounds the bond variable by flat curls and block averages near the block. These are the flat rows.
- In this gauge, the transported flat rows differ from the untransported flat rows by $O(\delta)|v|_{\mathrm{loc}}$.
- Square and sum with bounded overlap: $|v|^2 \le \mathcal{V}|Dv|^2 + o(1)|v|^2$, which gives $r = (2(1+\mathcal{V}))^{-1}$.
- Then $\langle v, \mathrm{gram}\, v\rangle = |Dv|^2 + |v_{\mathrm{inactive}}|^2 \ge r|v|^2$.

Lean: `Aff.inflow`, `Aff.surplus`, `Aff.decomposes`. *(proof read; per-bond comparison and III.6 inequality statement only)*

Lemma 4.34 (decay). Fix $\mathcal{T}, \mathcal{D}$ as in III.5.14. There is $K > 0$, uniform over all funded steps with that patch, all V, A, F and every real x, such that D and its unmasked version have weighted row and column bounds K at rate α equal to half the upper path-decay rate, and gram has bounds $K^2 + 1$. Moreover $\mathrm{gram}_{ij} = 0$ at distances greater than $2(\lfloor B/100000 \rfloor + R_{\mathrm{pred}}(\mathcal{D}) + 2)$. *Sketch:*

- Pointwise exponential decay of the row coefficients comes from the path-sum bounds of the upper stage and from the block-constraint rows.
- The counting bound converts it into row sums.
- Products and sums give gram.

Lean: `Aff.gram_budget`, `Aff.stencil`, `Aff.received`.

Lemma 4.35 (III.5.14). The uniform matrix and window conclusions of III.5.14 hold. *Sketch:* combine Lemmas 4.33 and 4.34 with the index-counting bound, then apply III.5.13(b). The windows have room $\lfloor B/4 \rfloor \to \infty$ as $x \to \infty$.

Lean: `Aff.bedrock`, `Aff.install_promises`.

Lemma 4.36 (auxiliary Gaussian estimates). Fix the step-estimator and patch of III.5.14. The auxiliary Gaussian (`Mint`) uses the completed precision $G = D^T D + I_{\mathrm{inactive}}$, the affine constant h (including the clipped frozen coordinates), source $D^T h$, and selected mean $\mu = -\mathrm{within}(G)^T D^T h$.

For the field estimates below, assume a funded step with this patch, coarse side $\ge 100(1+B)$, and

the flag equality of III.5.14. Write $H_{n,p} = B^p t^n$. A *reserve* $C_{\mathrm{res}} > 0$ supplies uniform bounds on the upper and lower row sums and the coarse predicted fields. A *compatible pair of quadratic forms* means upper and lower forms attached to the tube, satisfying its diagonal and renormalization identities, the two quadratic bounds and orientation covariance. The shift and plateau estimates assume such a pair and the window bounds of III.5.13(b). Each large-x threshold is uniform in the remaining volume and field data.

- `Aff.escape`: With a compatible pair, for every coarse Lie-algebra field b and every tree-gauge forest, the stationary linear extension solves the untruncated flat fine-row and block-constraint equations with the predicted coarse linear rows as right-hand sides, in every orientation. This is an identity for those linear operators; it does not assert that $D\mu + h = 0$.
- `Aff.velocity`, `Aff.shifted`: Assume a reserve. Let $o \notin D_{20B}(A)$ and let a gauge transform of V agree, on bonds anchored within $6B$ of o, with $\exp b$, where $\|b\|_\infty \le H_{1,7}$. Let v be the inverse-gauge transport of the stationary extension minus the replica extension, restricted to active coordinates. Then $|\mu_i - v_i| \le H_{2,27}$ whenever the column position is within $2B$ of o. At every row centred outside $D_{20B}(A)$, $|(D\mu + h)_i| \le H_{2,29}$.
- `Jewels.night`: Assume a reserve C_{res}, $W > 0$, and the two numerical margins $C_{\mathrm{res}}/L^2 \le s_0$ and $C_{\mathrm{res}} + s_0 \le (W/4)\sqrt{p_0}$, where $s_0 = \sqrt{c_{\mathrm{direct}}}/64$, c_{direct} is the upper direct quadratic coefficient, and $p_0 > 0$ is the kernel penalty. There is $\varepsilon > 0$ such that, for all sufficiently large x, the following holds when the frozen cutoff scale equals Bt. Reconstruct the fine field by inserting $\exp(\mu + z)$ relative to the replica of V on active bonds and keeping the fixed tree-gauge slice field on the other bonds. For any fluctuation z, a fine cell whose projected anchor lies outside $D_{20B}(A)$ has plaquette chord $< t/4$ if $|z_i| \le \varepsilon t$ on coordinates anchored within B of that anchor. The same test at the anchor of a coarse bond outside $D_{20B}(A)$ gives wire chord $< Wt$ and makes both likelihood cutoffs equal to 1.
- `Aff.tap_open`: Fix an order $p \in \mathbb{N}$ and $\delta > 0$. For all sufficiently large x, if every fine bond anchored within $B/2$ of a given coarse-bond anchor has chord at most $H_{1,p}$, then the frame and connector use their minimizing branches and the predicted coarse-bond chord is $< \delta$, for every tag. The uniqueness of those minimizers is part of the branch hypotheses.
- `Aff.perturb`: There is $K > 0$, depending only on the step-estimator and patch, such that for every truncation radius R and every unmasked transported row $L_{R,V}$, with $s_R = R + R_{\mathrm{pred}}(\mathcal{D}) + 2$, $\|(L_{R,V} - L_{R,1})u\| \le K(4s_R + 1)\delta C_u$ if $C_u, \delta \ge 0$, $\|u_j\| \le C_u$ on fine bonds anchored within $s_R + 1$ of that row, and all coarse-bond chords anchored within $4s_R + 1$ are at most δ. This compares with the flat background; it is not a Lipschitz estimate between arbitrary backgrounds.
- `Aff.arc_bill`: Assume a reserve and fix $p \ge 7$. For all sufficiently large x, take a row centred outside $D_{2B}(A)$ and a fine Lie-algebra field u bounded by $H_{1,p}$ on bonds anchored within B. If U agrees there with $\exp(u)\,\mathrm{replica}(V)$, its nonlinear row value $F(U,V)$ satisfies $\|F(U,V)\| \le H_{1,p+4}$ and $\|F(U,V) - F(\mathrm{replica}(V),V) - L_V u\| \le H_{2,2p+6}$, where L_V is the transported flat row at the prescribed truncation. This is the error estimate used in place of identifying L_V with a derivative at V.
- `Aff.reserves`: For every step-estimator there exists a reserve C_{res}. It is chosen before L; for each fixed L, its row and coarse-field bounds hold for all sufficiently large x and all funded steps of coarse side $\ge 100(1 + B)$.

3.8.5 5. Where the difficulty lies

The tree-graph argument is standard. The estimates must also retain arbitrary label sets, repeated occurrences, site pins and covering weights at the growing scale $B = \lceil (\log x)^2 \rceil$. The entropy factor $4(2B+1)^4$ is sufficient because it grows only polynomially in $\log x$.

The gain scale is $P(\beta_F)^2 = e^{2(\log\log\beta_F)^2}$. It absorbs $\mathrm{bill}^d P(\beta_F)^z$ for fixed d and $z < 2$, but grows more slowly than every positive power of β_F. The deletion route gives support-indexed logarithm coefficients and a common field-independent envelope; the graph route also gives occurrence tails and multiset bounds. Both give deletion ratios uniform in volume, although the full logarithm is extensive.

The sector estimate needs global seed records to preserve the local point-charge prerequisites under lattice orientations. Its remainder ℓ_A has no volume-uniform bound here. For the Gaussian estimates, the substantive inputs are the gauge-fixed Poincaré inequality and control of the transported flat rows away from the flagged region. Finite range grows with B; uniform exponential row bounds provide the useful control.

3.8.6 6. Notation

Written proof	Lean	Meaning
$\mathrm{SB}(f; C)$	`SumBudget`	$f \geq 0$ summable, $\sum f \leq C$
$\mathrm{Hit}_B(w; a, \gamma)$	`Ac.Hit`	site-sum condition
cov_B, ball_B	`coverNumber`, `ballCost`	connected covering number; $(2B+1)^4$
$\mathrm{Ready}(w; a, \gamma, \delta, q)$	`Trees.Ready`, `degreePrice`	smallness of the cluster expansion
$R \subseteq$ overlap	`Ursell.Tests`, `blocked`	incompatibility implies overlap; the overlap relation
$G_R(z)$, $\phi^T_R(z)$	`Ursell.pot`, `Ursell.coefficient`	incompatibility graph of a tuple; Ursell coefficient
$\Xi(w)$; cluster term; log series	`Gas.total`, `Gas.term`, `Gas.logSum`	gas, its cluster terms and their sum
g_n, ℓ_m	`Coef.gasRow`, `Coef.logRow`	coefficients of Ξ and of the log series
w^Y, p	`Gas.cut`, `Gas.missing`	weights off Y; clusters meeting Y
multisets, $\prod w^n/n!$	`Multiple.Reps`, `manyW`	labels with multiplicities
packet A, number of seeds, $D_r(A)$	`Stores`, `shares`, `dome`	finite set of typed seeds; radius-r neighbourhood of their locations
connected packet, K_{hit}, K_{step}, τ_L	`Packets.connected`, `hitCost`, `stepCost`, `typesAt`	animal-counting constants
ρ, packet tail	`ratioCost`, `fit`, `totalCharge`	per-seed ratio; tail of packet sums
hypotheses of III.5.7(b)	`Mixed.Costs.Buy`, `Mixed.Accounts`	numerical list; weight bounds

Written proof	Lean	Meaning
B, β_F, t, P	`radiusB`, `betaF`, `fiery`, `Vac.Numbers.P`	$\lceil(\log x)^2\rceil$; $x + b\log L$; $\beta_F^{-1/2}P(\beta_F)$; $e^{(\log\log s)^2}$
M, gain $\beta\mu t^2$	`hatWin`, `Grant.credit`	guaranteed and actual gain per seed
$Z(B)$, $\lambda(A)$	`fumes`, `locker`, `pieces`	disjoint-family gas; Möbius pieces of $-\log Z$
$f * g$, $\widehat{f}$, $\|f\|_w$, $\lambda(f)$	`glue`, `shed`, `crop`, `lamp`	union convolution, cumulative sum, weighted norm, log series
hypotheses of III.5.9	`Rate`, `Deletion`, `Receipt`, `Sufficient`, `Fund`, `Tags`	weight, deletion identity, site conditions, coefficient properties
allocation D; point charge	`Allocation`, `PointCharge`	partition of unity over packets; $\mathcal{E}_A \geq \kappa t^2$ per seed
$\mathcal{E}_A$; c_A; ℓ_A; $\pi_\beta e^{-\beta\kappa t^2 k}$	`assessed`, `cutEntry`, `leftOver`, `earn`	sector energy; sector output; remainder; gain
weighted Schur classes; $M \geq r$	`Green.Rows`, `Green.Both`, `Green.Bottom`	row/column sums at rate η; lower bound
windows, within, E, cov, Π_U	`Win.Windows`, `within`, `error`, `cov`, `planePrice`	localized inverse and covariance
uniform matrix data; window constants	`Win.Promise`, `Turns`, `Turns.Takes`	(mass, bottom, rate, counting); bounds of III.5.13(b)
D, gram	`Aff.Dmat`, `Aff.gram`	transported flat row matrix; completed precision matrix
transported flat derivative	`Aff.derivative`	row operator used to define D
auxiliary Gaussian assumptions	`Aff.Reserve`, `Aff.Alongside`, `Aff.cert`	reserve bounds; compatible quadratic forms; volume and flag tests
$H_{n,p}$; comparison field v	`Rates.allowance`, `Aff.unit`	$B^p t^n$; active stationary extension minus replica extension, transported back by the local gauge

3.8.7 7. What remains unexplained

1. **Imported analytic inputs.** The step-estimator's energy grant in III.10 and the Poincaré inequality in III.6 are used as stated. Their proofs, the full per-bond spectral comparison and the origin of the run's committed-scale energy hypothesis have not been audited here.
2. **Later applications.** The dependency database identifies consumers in the units listed above. Their verification of the site-sum, per-packet and multiset hypotheses remains outside this unit's audit. In particular, the bounds here do not construct renormalization outputs or bound the sector remainders uniformly in volume.
3. **Unread supporting proofs.** The breadth-first construction, the distinct-label forest count, the spanning-walk construction for animals and the integration of allocations restricted to sectors were used through their statements. In the auxiliary Gaussian chain, the statements and the displayed estimates of Lemma 4.36 were checked; its supporting local chart and

quadratic-form comparison lemmas were not audited throughout. These remain proof inputs, rather than additional conclusions of the present sketches.

4. **Flag locality.** In III.5.9(b), locality on P applies to the raw coefficients. Their masked versions also inspect flags on P; those flags can depend on plaquettes extending outside P. A written proof must retain this flag dependence or enlarge the physical support before claiming ordinary link locality for the masked coefficients.
5. **Verification scope.** This is a source correspondence check. No Lean build, axiom computation or rendered-page check was run.

3.9 III.6. The stationary branch: background field of the effective action

Part: III, Ultraviolet construction (renormalization group). **Lean:** the work items F01–F05 are stated in `YM/Bid/Station.lean` and F06–F09 in `YM/Bid/Craft.lean`. Their proofs are in `YM/Work/F01–F09`. The other packages are: the uniform branch `YM/Push/STATIONARY_UNIFORM/`; the jet port `YM/Planned/STATIONARY_JET_PORT/`; and the iteration step `YM/Extra/ITERATION_NO_LOSS/`. Also included are `YM/Planned/PRIMARY_INTEGRAL_295/`, `YM/Further/PHYSICAL_PERIOD_LIMIT_F933/`, `YM/Planned/ATOM_REGULAR_ESTIMATE/`, `YM/Next/ROTATIONS/` and `YM/Next/WARD_BIN/`. **Size:** 21,329 lines in the main-theorem closure. Of these:

- the F work items F01–F09 with their auxiliary files: about 3,800;
- the uniform branch: 3,274;
- the jet port: 1,123;
- the iteration step: 1,700;
- the primary-integral package: 3,662;
- the period-limit comparison: 1,262;
- the atom regularity estimate: 474;
- the rotation and Ward material: about 5,900, including the chain of results leading to the rotation theorem.

3.9.1 1. What this unit proves

Setting. The definitions are those of III.1. All lattice sums use counting measure.

Lattices, group, distance.

- One renormalization step goes from a fine torus Λ_f to a coarse torus Λ_c, with an odd block factor $L \geq 3$ and $f = Lc$, in dimension 4. The patch order satisfies $J \geq 4$.
- G is the gauge group and $\mathfrak{g}$ its Lie algebra, normed in physical colour coordinates.
- $\mathrm{chord}(U) = \|U - 1\|/c_G$ and $\mathrm{price}(U) = \frac{1}{2}\mathrm{chord}(U)^2$. With this normalization $\mathrm{price}(e^{tX}) = \frac{1}{2}t^2|X|^2 + o(t^2)$; the price is the Wilson plaquette function up to a constant.

The step datum S. It consists of:

- a block-spin kernel, with penalty $\kappa > 0$, from the kernel choices K;
- a patch, with order J and macro-radius $R_S = (J + 7)L$;
- a finite set of tags (frame and path choices; `StepData.Choices`);
- for each tag k, a predictor $U \mapsto Q_k(U)$ from fine configurations to coarse configurations.

P denotes the derivative of the predictor at the identity. It acts on fine $\mathfrak{g}$-valued 1-forms; it is the linearized block average plus a gradient correction at the block roots. $\bar{Q}$ denotes the linearized

block average alone.

"Larger". S is *larger* (`Larger`) if three conditions hold:

- the minimizing frame and predictor are unique whenever their minimizing branches are selected, and the fallback set of eligible axial frames is nonempty; see `Branches`;
- $100R_S \le f$;
- $c > 1$.

Forests and the axis chart.

- A *forest* H is a spanning forest of the fine torus, made of one tree in each block, rooted at the block centre, with tree height at most $4L$ (`Forest`). Its bonds are the *tree bonds*, all internal to blocks; the remaining bonds are the *free bonds* H^{fr}.
- The *replica* $\mathcal{R}(V)$ of a coarse configuration V puts $V_{(i,\mu)}$ on every fine bond that leaves block i in direction μ, and 1 on bonds internal to a block. Its linearization is ι.
- The *axis chart* has coordinates $x \in \mathfrak{g}^{\text{coarse bonds}}$ and $a \in \mathfrak{g}^{H^{\mathrm{fr}}}$. It describes the fine field

$$U^H(x,a)_e = e^{a_e}\,\mathcal{R}(e^x)_e \quad (e \in H^{\mathrm{fr}}), \qquad U^H(x,a)_e = 1 \quad (e \text{ a tree bond}).$$

Clip and path datum.

- A *clip* p carries the Wilson coefficient $c_p > 0$.
- A *path datum* W (`PathData`) consists of:
  - a series $\ell_W(U)_\pi = \sum_i w_i \,\mathrm{slot}_i(U)$ of parallel-transported plaquette variables, attached to each plaquette π;
  - loop atoms, cubic atoms and quartic atoms: Ad-invariant polynomials of degree $2, 3, 4$ in such slot values.
- *Bounds* (p, σ, C, Λ) for W (`Bounds`) mean four things:
  - the coefficients are summable with weight $e^{\sigma\cdot\text{length}}$, with sums at most C (cubic, quartic, ℓ-rows) and Λ (loop atoms);
  - the atoms are Ad-invariant;
  - if r_{ij} is the sum of row-i coefficients whose plaquette type is j, then $c_p\delta_{ij} + \sum_k r_{ki}r_{kj} = \delta_{ij}$;
  - for each pair of plaquette types and colour indices, the loop coefficients have zero sum and zero first moment in every component of the relative slot offset; these moment series are summable.

The tested action.

- The *main term* (`mainTerm`) is the average over the finite group $\mathcal{S}$ of signed coordinate permutations, acting by pullback s^*U:

$$\mathcal{M}_p(W;U) = \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}}\Big[\sum_\pi \big(c_p\,\mathrm{price}((s^*U)_\pi) + \tfrac{1}{2}|\ell_W(s^*U)_\pi|^2\big) + \sum_x\sum_i (\mathrm{cub}_i + \mathrm{quart}_i)(s^*U)_x\Big].$$

- The *tested action* for tag k (`tested`) is

$$\mathcal{T}_k^H(x,a) = \mathcal{M}_p(W;U^H(x,a)) + \kappa\sum_b \mathrm{price}(e^{x_b}\,Q_k(U^H(x,a))_b^{-1}),$$

  where b runs over the coarse bonds. The second term is the block-spin penalty.

Beds, towers and the quadratic identities.

- A *bed* B (`Bed`) is a family of translation-invariant kernels on plaquette 2-forms: a metric M_B, an inverse kernel, a diagonal kernel D_B, a dual diagonal kernel, and roots R_B. Set $K_B = d^\top M_B d$, which acts on 1-forms.
- A *floor* (a, A) for B (`Floor`) requires:
  - the root identity $M_B = c_p\,1 + R_B^\top R_B$;
  - inverse identities for the metric and diagonal kernels;
  - the Fourier-symbol normalization $\widehat{M}_B(0) = I$ and $M_B d = D_B d$ on the specified torus;
  - symmetry of M_B and D_B;
  - the bounds $a|\omega|^2 \le \langle \omega, M_B \omega\rangle \le A|\omega|^2$, and the same bounds for D_B.
- W is *attached* to B if the coefficients of ℓ_W, summed by offset and plaquette type, are the entries of R_B. B is *sworn* if $\langle dA, M_B dA\rangle$ is $\mathcal{S}$-invariant.
- *Towers* T for the pair of beds (B, C) (`Towers.Diagonal`) provide a linear map $\mathcal{W}$ from coarse 1-forms to fine 1-forms. Its first clause, `Inputs`, is

$$K_B \mathcal{W} = \bar{Q}^\top K_C, \qquad \bar{Q}\mathcal{W} = 1 - \kappa^{-1} K_C.$$

- $\widehat{\mathcal{W}}_H = \mathcal{W} + d_0 \chi_H$ is the linearized gauge transform of $\mathcal{W}$ that vanishes on tree bonds (`extracted`).
- $\mathrm{rel}_H(x) = (\widehat{\mathcal{W}}_H x - \iota x)|_{H^{\mathrm{fr}}}$ (`relative`).

Critical branch. A *critical branch* for (H, p, W) (`Critical`) is a map ρ from coarse fields to $\mathfrak{g}^{H^{\mathrm{fr}}}$ such that:

- $\rho(0) = 0$ and ρ is real-analytic at 0;
- for every tag k, $\mathcal{T}_k^H$ is jointly analytic at 0;
- for every tag k, $D_a \mathcal{T}_k^H(x, \rho(x)) = 0$ for x near 0.

Chart ratio and thermal function.

- A *local chart ratio* $\mathcal{J}$ (`LocalRatio`) is a density $J_{\mathcal{J}}$ on a ball $\{|X| < r_{\mathcal{J}}\} \subset \mathfrak{g}$, where $r_{\mathcal{J}} > 0$ and $s_{\mathcal{J}} > 0$ is a scale. It satisfies:
  - $J_{\mathcal{J}}$ is measurable, positive and real-analytic on the ball, with $J_{\mathcal{J}}(0) = 1$;
  - exp is injective on the ball and its image is open;
  - the pushforward of $s_{\mathcal{J}} J_{\mathcal{J}}(X)\,dX$ restricted to the ball is probability Haar measure restricted to its exponential image.

  Only these density properties enter the forest comparison.
- The *thermal function* of a branch (`thermal`) is

$$\Theta_k^H(x) = \mathrm{loop}_W(U^H(x, \rho(x))) + \tfrac{1}{2} \log\det \mathrm{Hess}_a \mathcal{T}_k^H(x, \rho(x)) - \sum_{e \in H^{\mathrm{fr}}} \log J_{\mathcal{J}}(\rho(x)_e).$$

  The Hessian is taken in the orthonormal colour basis. For a branch with a positive vertical gap, the determinant and chart densities are positive near 0.

Proposition III.6.1 (second-order jet of the tested action on the tree slice; F03). Let S be larger, H a forest and B a bed with a floor (a, A) on Λ_f for the clip p. Let W be a path datum with bounds (p, σ, C, Λ), attached to B, with B sworn on Λ_f. Assume that the main term of W is analytic along analytic families through the identity (true by III.6.2). Then for every tag k, $\mathcal{T}_k^H$ is real-analytic at 0, with $\mathcal{T}_k^H(0) = 0$ and $D\mathcal{T}_k^H(0) = 0$. Moreover, for all $(x, a), (x', a')$,

$$D^2\mathcal{T}_k^H(0)[(x,a),(x',a')] = \langle ds,\, M_B\, ds'\rangle + \kappa\,\langle x - Ps,\; x' - Ps'\rangle, \qquad s = \iota x + a,\;\; s' = \iota x' + a'.$$

Here a is extended by 0 to tree bonds, and the pairings are summed over plaquettes or coarse bonds and over colours. *Status:* theorem. Only the well-posedness clause of "larger" is used. *Lean:* `Bid.F03`, with conclusions `Prismatic` and `Comptroller`; proof `Work.F03.answer`.

Proposition III.6.2 (analyticity of the main term; F02). Let W have bounds (p, σ, C, Λ). Let X be a finite-dimensional real normed space and $x_0 \in X$. Let $y \mapsto A(y)$ be a family of fine configurations with $A(x_0) = 1$ whose link matrices are real-analytic at x_0. Then $y \mapsto \mathcal{M}_p(W; A(y))$ is real-analytic at x_0. The proof does not use $A(x_0) = 1$: the main term is analytic along analytic families at every point. *Status:* theorem. It supplies the hypothesis `Fires` of III.6.1. *Lean:* `Bid.F02`, `Work.F02.answer`.

Proposition III.6.3 (gap of the vertical Hessian; F01). Assume the shell comparison E02 (§2). Fix G, L, J, kernel choices K with penalty κ, and $a > 0$. Then there is $\eta > 0$ with the following property. Take sides c, f; a larger step datum S with kernel choices K, block factor L and order J; a forest H; a clip p; a bed B with floor (a, A) for some A; and a path datum W such that every $\mathcal{T}_k^H$ is analytic at 0 with the value, gradient and Hessian specified in III.6.1. Then for every tag r and every $v \in \mathfrak{g}^{H^{\mathrm{fr}}}$,

$$\eta \sum_{e \in H^{\mathrm{fr}}} |v_e|^2 \le D_a^2 \mathcal{T}_r^H(0,0)[v,v].$$

One may take

$$\eta = \frac{\min(a,\kappa)}{1+\beta+\gamma}, \qquad \beta = 2C_{\mathrm{E02}}^2\, L^4 \cdot 4 \cdot 6\, |X|^2, \qquad \gamma = 2L^4,$$

where:

- $C_{\mathrm{E02}} = C_{\mathrm{E02}}(G, L, J, K)$ is the constant of E02;
- $X = \{-R_S, \dots, R_S\}^4 \subset \mathbb{Z}^4$, where $R_S = (J+7)L$, so $|X| = (2R_S+1)^4$;
- the factors L^4, 4 and 6 are the numbers of fine sites per block, of axes and of plaquette orientations.

So η depends only on (G, L, J, K, a). It does not depend on c, f, H, B beyond its floor, or W. *Status:* theorem, conditional on E02 in its statement. E02 is proved in III.10 (`Work.E02.answer`), and the combination is unconditional in III.6.6. *Lean:* `Bid.F01`, with conclusion `PriceGap`; proof `Work.F01.answer`.

Proposition III.6.4 (existence, tangent and value of the critical branch; F04). Let S be larger, H a forest, B, C beds, p a clip, W a path datum, $a > 0$, and T towers for (S, B, C) satisfying the diagonal identities. Assume:

- every $\mathcal{T}_k^H$ is analytic at 0;
- every $\mathcal{T}_k^H$ has the value, gradient and Hessian specified in III.6.1 with bed B;
- the gap of III.6.3 holds with constant a;

- there are a clip q and real constants v, u for which C has floor (v, u) on Λ_c.

Then there is a critical branch ρ for (H, p, W) such that:

- (ii) $D\rho(0) = \mathrm{rel}_H$, built from the tower map $\mathcal{W}$;
- (iii) for every tag k, $x \mapsto \mathcal{T}_k^H(x, \rho(x))$ is analytic at 0, and its second derivative at 0 is $\langle dx, M_C\, dx' \rangle$. Equivalently, its 2-jet is that of $\frac{1}{2}\langle dx, M_C dx \rangle$.

The floor of C is unused in the proof. Of the diagonal identities, only the clause $K_B \mathcal{W} = \bar{Q}^\top K_C$, $\bar{Q}\mathcal{W} = 1 - \kappa^{-1} K_C$ is used. *Status:* theorem. It gives a germ only: no control of the size of the neighbourhood. *Lean:* `Bid.F04`, with conclusions `Critical`, `Guessed` and `HeatG`; proof `Work.F04.answer`.

Proposition III.6.5 (independence of the gauge-fixing forest; F05). Let S be larger, H, H' forests, and W a path datum with bounds (p, σ, C, Λ). Let ρ, ρ' be critical branches for H and H'. Assume positive gap constants a, b for H, H', respectively, and let $\mathcal{J}$ be a local chart ratio. Then for every tag k, for x near 0:

- (i) $\mathcal{T}_k^H(x, \rho(x)) = \mathcal{T}_k^{H'}(x, \rho'(x))$;
- (ii) $\Theta_k^H(x) = \Theta_k^{H'}(x)$.

The gap for H is unused in the proof. The statement of (ii) allows an additive constant; the proof gives 0. *Status:* theorem. *Lean:* `Bid.F05`, `Work.F05.answer`, `Work.F05.same_thermal`.

Proposition III.6.6 (the stationary branch from F01–F04). For every G, L, J, kernel choices K and $a > 0$ there is $\eta > 0$ with the following property. Take a larger S with these parameters; a forest H; beds B, C; clips p, q; a path datum W; and towers T. Assume:

- T satisfies the diagonal identities for (S, B, C);
- B has floor (a, A) for clip p on Λ_f, and C has floor (v, u) for clip q on Λ_c, for some real A, v, u;
- W has bounds $(p, \sigma, C_0, \Lambda)$ and is attached to B;
- B is sworn.

Then the gap of III.6.3 holds with constant η, and there is a critical branch with properties (ii)–(iii) of III.6.4. *Status:* theorem. `Bid.cospons` derives it from E02 and F01–F04. `Integrated.stationary_scene` instantiates it with the proofs of E02 and F01–F04, so no work-item assumption remains. The geometric, floor and path hypotheses above are retained. *Lean:* `Bid.Scene`.

Proposition III.6.7 (the branch on a volume-independent ball).

(a) Uniform implicit function theorem. Let ι, τ be finite sets and let $\ell : \iota \sqcup \tau \to \Lambda$ assign their coordinates to sites of a four-dimensional torus, with at most K_0 coordinates per site. Equip the coordinate spaces with the sup norm. Fix $C_0, a_0, r, m > 0$ and an integer $n \geq 3$. Suppose $f : \mathbb{R}^{\iota \sqcup \tau} \to \mathbb{R}$ is real-analytic at every point z of $\{\|z\|_\infty < r\}$ and, for $1 \leq k \leq n$, every coordinate tuple $q_1, \dots, q_k$ and every $j \leq k$,

$$\left|\partial_{q_1} \cdots \partial_{q_k} f(z)\right| \leq C_0 \exp\left(-a_0 \sum_{i=1}^{k} \mathrm{dist}(\ell(q_j), \ell(q_i))\right).$$

Here distance is the periodic max-distance. Assume $\nabla_\tau f(0) = 0$ and $D^2_{\tau\tau} f(0) \geq mI$ in counting coordinates. Then there are positive radii $r_{\mathrm{hor}} \leq r_{\mathrm{zone}} \leq r_{\mathrm{safe}} < r$, depending only on

(K_0, C_0, a_0, r, m), and a map $g : \mathbb{R}^\iota \to \mathbb{R}^\tau$ with $g(0) = 0$, such that whenever $\|x\|_\infty < r_{\mathrm{hor}}$:

- g is analytic at x;
- $\|g(x)\|_\infty \le r_{\mathrm{zone}}/2$;
- $g(x)$ is the unique critical point of $f(x, \cdot)$ in $\{\|v\|_\infty \le r_{\mathrm{zone}}\}$.

(b) Application to the tested action. Fix G, real constants μ, C, a patch order J, kernel choices K, and an integer $n \ge 3$. There is a choice of auxiliary constants, made before L, with the following property. For every clip p, real Λ, larger step S of order J with kernel choices K, forest H and path datum W with bounds (p, μ, C, Λ), part (a) applies to the tested action in individual real colour components. It gives a critical branch g common to all tags: for each tag t and $\|x\|_\infty < r_{\mathrm{hor}}$, $g(x)$ is the unique critical point of $\mathcal{T}_t^H(x, \cdot)$ in the closed sup-ball of radius r_{zone}. The radii depend on the auxiliary constants, p and L, but not on c, f, H, Λ or W beyond its bounds. Coarse coordinates are located at their fine block roots when measuring decay. *Status:* theorem. *Lean:* (a) `Receipt.throughout` with hypotheses `Receipt` and radii `reserveX`. (b) `Before.graph`, `Before.Service`, `pre_order`.

Proposition III.6.8 (derivatives along the branch). Fix an integer $N \ge 0$, the data and hypotheses of III.6.7(b), auxiliary constants chosen at order $N + 3$, and any branch g satisfying its analytic, size and uniqueness conclusions. Then for every tag the following hold, with constants depending only on the corresponding bounds and radii.

- (a) At every point of the closed joint sup-ball of radius $r_{\mathrm{safe}} \ge r_{\mathrm{zone}}$, the vertical Hessian, as a matrix function of all variables, has a *block package* of order N:
  - it is analytic and $\ge m/2$;
  - its entries and their derivatives up to order $N + 1$ are localized;
  - for every principal sub-block and every shift $t \ge 0$, the shifted sub-block is invertible;
  - the inverse of the shifted sub-block is exponentially localized with constant $O((1+t)^{-1})$, has localized derivatives up to order $N + 1$, and has ℓ^2 norm $\le (m/2 + t)^{-1}$;
  - $\log\det$ of the shifted sub-block is analytic and its derivatives of orders $1, \dots, N + 1$ are localized, with bounds proportional to $(1 + t)^{-1}$.
- (b) On the coarse sup-ball of radius r_{hor}, the map g and the graph $x \mapsto (x, g(x))$ have localized derivatives of orders $1, \dots, N + 2$.
- (c) The classical action $x \mapsto \mathcal{T}_t^H(x, g(x))$ has localized derivatives of orders $1, \dots, N + 3$.
- (d) The Hessian along the branch, $x \mapsto \mathrm{Hess}_a \mathcal{T}_t^H(x, g(x))$, has a block package of order N.

For a matrix function $A(z)$, localization of its coefficients means a bound

$$|\partial_{q_1} \cdots \partial_{q_k} A_{ij}(z)| \le C_k \exp\left[-a_k \left(\mathrm{dist}(i, j) + \sum_\nu \mathrm{dist}(i, q_\nu)\right)\right],$$

with coordinates replaced by their assigned sites. For positive-order derivatives of a map such as g, omit j; for a scalar function use the bound in III.6.7(a). No volume-independent bound on the value of the classical action or of $\log\det$ is asserted.

Status: theorem under the hypotheses of III.6.7(b). *Lean:* `Counter.service`, `Counter`, `BlockPack`.

Proposition III.6.9 (one level of the iteration). Assume the following.

- The schedule hypotheses of III.2: constants issued before L, warm-up, and a paid schedule.

- A horizon of beds $M_{N,m}$ and towers with the prescribed strip and derivative bounds, floors on every torus, diagonal and stack identities, and identity diagonal at the cutoff (`Suffices`).
- All beds are sworn at the periods used. The towers are harmonic: the zero-momentum coefficient of their 2-form component is independent of the fine residue within a block.
- A gap quote, i.e. III.6.6 at every level, which the Lean supplies from `Iteration.post`.
- A local chart ratio.
- The fresh-data hypotheses. For each admissible input datum and each family of branches satisfying (i), these hypotheses supply the fresh cubic, quartic and loop atoms and their bounds, with the classical and thermal matchings each verified on a chosen forest and tag per period. These choices may differ between the two matchings. It also supplies the cutoff cubic and quartic atoms.

Let $m < N$ and let T be a path datum *funded* at level $m+1$ (`Work item`). That is, the rows of T enumerate the root-kernel coefficients of $M_{N,m+1}$ using the fixed coordinate paths to each offset; T has the level-$(m+1)$ bounds; and the sums of absolute cubic and quartic coefficients weighted by exponential path length are each at most the level cap. This row convention is stronger than attachment alone. Then there exist a path datum R funded at level m, branches $\rho_{p,F}$ for every period p and every forest F, and a number α such that:

- (i) each $\rho_{p,F}$ satisfies the conclusions of III.6.6 with the level-$(m+1)$ gap, its tower map, and the bed $M_{N,m}$;
- (ii) $|\alpha + \kappa_G \log L_m| + 1 \le \mathrm{drift}_{m+1}$, where κ_G is the one-loop slope constant of G;
- (iii) for every p and F, $x \mapsto \mathcal{T}_k^F(x, \rho_{p,F}(x))$ (built with T and the clip of level $m+1$) has the same jets of orders $0, \dots, 4$ at 0 as $x \mapsto \mathcal{M}_q(R; e^x)$ (clip q of level m);
- (iv) for every p and F, the thermal function is analytic and the Hessian determinant is positive near 0. Moreover, for any two real plane waves w_1, w_2, each the real part of a complex lattice plane wave whose coefficient in every axis and colour has modulus at most 1,

$$\left|D^2\Theta_k^F(0)[w_1,w_2] - D^2(\alpha\,\mathcal{M}_q^{(2)}(R) + \mathrm{loop}_R)(0)[w_1,w_2]\right| \le |\Lambda_c|\,\Gamma\,\lambda_{m+1}\,e^{-\gamma c}.$$

Here $\mathcal{M}_q^{(2)}(R;x)$ denotes the symmetrized Wilson-plus-row-square term evaluated on e^x:

$$\mathcal{M}_q^{(2)}(R;x) = \frac{1}{|\mathcal{S}|}\sum_{s\in\mathcal{S}}\sum_{\pi}\left[c_q\,\mathrm{price}((s^*e^x)_\pi) + \tfrac{1}{2}|\ell_R(s^*e^x)_\pi|^2\right].$$

It includes all higher-order dependence on x in these terms; it is not their degree-two Taylor polynomial. The constants $\Gamma, \gamma > 0$ are chosen after L. All conclusions hold for every tag k, and the same R, α serve every period and forest.

A companion statement, `Iteration.peak`, gives a funded path datum at the cutoff level N whose main term has the jets of order ≤ 4 of the Wilson energy. *Status:* theorem under the stated schedule, horizon and fresh-data hypotheses. Their discharge is discussed in §7. *Lean:* `Iteration.extend`, `Move`, `ClassicalMatch`, `FiniteBound`.

Further results attached to this unit (statements only; see §4.7).

- *Beds and towers (F06–F09).* These are combined in `Bid.pinning` (III.10) into a horizon meeting the prescribed bounds. The construction retains the supplied beds and the first two tower components; the floors and original diagonal identities are inputs, and the stack identities are outputs.

  - `F06.answer`: a 3-form kernel that is positive on exact forms can be replaced by one that is positive everywhere, without changing $d^{\top}(\cdot)d$.
  - `F07.answer`: square roots of positive kernels, with exponential decay.
  - `F08.answer`: towers with uniform widths.
  - `F09.answer`: the stack identities.
- *Comparison of two tori* (`Loom.spinning`).
- *Primary-integral comparisons* (`Ramble.billFor`, `PPX.toll`, `Oval.sticker`).
- *Rotation and Ward material* (`Outer.outer`). Mathematically it belongs to VI.1.

3.9.2 2. What it uses

- **III.1 (block-spin step and symmetric gauge fixing).**
  - The step data, tags and predictors, and the well-posedness condition.
  - Spanning forests rooted at block centres.
  - The definitions of the tested action, critical branch, thermal function, beds, floors and towers.
  - The identities $P\widehat{\mathcal{W}}_H = 1 - \kappa^{-1}K_C$ and $K_B\widehat{\mathcal{W}}_H = P^{\top}K_C$, which follow from the tower identities (`stationed`).
  - The change of forest slice$(H) \to$ slice(H'), a gauge transformation trivial at the roots, preserves product Haar measure (`Profiles.spending`).
  - The predictor is invariant under gauge transformations trivial at the block roots.
- **III.10 (one nonlinear step).** E02, the shell comparison (`Bid.E02`, `Cops`). For given (G, L, J, K) there is $C_{\mathrm{E02}} > 0$ with the following property. For every larger S, forest H, fine configuration U in the slice of H, coarse V, fine bond e (in block i, direction μ) and tag k, assume the selected frames have positive mass at all coarse sites within periodic offset $[-2, 2]^4$ of i. Then:

$$\operatorname{chord}\bigl(U_e \mathcal{R}(V)_e^{-1}\bigr) \le C_{\mathrm{E02}} \operatorname{port}_i(U) + \mathbf{1}_{e \text{ crossing}} \operatorname{wire}(Q_k(U), V)_{(i,\mu)}.$$

  Here $\operatorname{wire}(Q, V)_b = \operatorname{chord}(V_b Q_b^{-1})$, and $\operatorname{port}_i(U) \le \sum_{z \in X} \sum_{\pi} \operatorname{chord}(U_\pi)$, summed over plaquettes π at $\mathrm{root}_i + z$. The estimate is proved in `Work.E02.answer`. The combination `Bid.cospons` and the horizon construction `Bid.pinning` are filed under III.10.
- **III.3 (free kernels).** The beds are free lattice kernels in Fourier (Bloch/Laurent) form. F06–F09 complete the towers while retaining the supplied beds and floors.
- **III.2 (order of constants).** The constants issued before L, the block sizes L_m, the post-L costs, and the drift window, all used in III.6.9. In particular the gap constant is post-L and the constants of III.6.7(b) are chosen before L.
- **I.1, II.1, II.2.** The lattice, lattice forms and d; the group G and its exponential, chord, price and Haar measure; the path data and their bounds.
- **III.16 (analytic estimates).** Real-analytic functions, Taylor jets and iterated derivatives.
- **Standard mathematics.**
  - The analytic implicit and inverse function theorems (Krantz–Parks, *The Implicit Function Theorem*, Birkhäuser 2002).
  - Weierstrass's theorem on normally convergent series of holomorphic functions.

- The second-order chain rule at a critical point, so that det of the Hessian changes by the squared Jacobian.
- The change-of-variables formula.
- The Banach fixed-point theorem.
- The Combes–Thomas estimate (Combes–Thomas, *Comm. Math. Phys.* 34 (1973) 251–270).
- The background-field method in lattice gauge theory: T. Bałaban, *Propagators and renormalization transformations for lattice gauge theories I, II*, Comm. Math. Phys. 95 (1984), 96 (1984); *The variational problem and background fields in renormalization group method for lattice gauge theories*, Comm. Math. Phys. 102 (1985); *Renormalization group approach to lattice gauge field theories I*, Comm. Math. Phys. 109 (1987); J. Dimock, *The renormalization group according to Balaban I*, Rev. Math. Phys. 25 (2013).

Imported constants and assumptions.

- The constant $C_{\mathrm{E02}}(G, L, J, K)$.
- The penalty κ of the kernel choices.
- The floors (a, A) of the beds.
- The one-loop slope κ_G, which enters only through the drift window of III.6.9 (see III.9).
- The fresh-data, schedule and horizon hypotheses of III.6.9.

3.9.3 3. Strategy

At fixed coarse field, the renormalization step expands the fine-field integral around a stationary background. Gauge fixing removes the blockwise gauge directions. A positive vertical Hessian then gives an analytic branch, and the tower identities identify its tangent and the quadratic part of its classical value. The one-loop contribution is the Hessian determinant corrected by the Haar density and the loop atoms.

The construction uses the penalty $\kappa \sum_b \mathrm{price}(e^{x_b} Q_k(U)_b^{-1})$ and fixes the fine field on a rooted forest inside each block. The shell comparison bounds the free links by curvature and predictor response. After linearization and summation, it gives a positive vertical Hessian with a lower bound independent of the torus volume. The analytic implicit function theorem produces the branch germ. A change of forest preserves both the stationary value and the thermal function: the squared Jacobian in the Hessian determinant cancels the Haar-density ratio.

For a branch on a ball independent of the volume, exponential decay of the inverse Hessian gives a uniform sup-norm bound. A Newton contraction produces the branch on that ball; differentiating the stationarity and resolvent identities controls its localized derivatives. Finally, the iteration combines inherited atoms with fresh classical and one-loop data supplied by the hypotheses of III.6.9. This is a penalized background-field construction in the spirit of Bałaban's method. It establishes local stationarity and quantitative control near the identity.

3.9.4 4. Steps

Lemmas 4.1–4.4 use the hypotheses of III.6.1, Lemmas 4.5–4.7 those of III.6.3, Lemmas 4.8–4.12 those of III.6.4, and Lemmas 4.13–4.17 those of III.6.5. Lemmas 4.19–4.20 use III.6.7(a)–(b), respectively, and Lemmas 4.22–4.24 use III.6.9.

3.9.4.1 4.1 The second-order jet (F02, F03) Lemma 4.1 (holomorphic extension of the main term). Let W have bounds (p, σ, C, Λ) and let U be any configuration on a finite torus. The ℓ_W-series and the sums of cubic and quartic atoms have holomorphic extensions near the matrix pair (U, U^*). Their summands are uniformly summable on the ambient ball of radius $R_\sigma = e^{\sigma/2} - 1$, using the maximum of the linkwise Frobenius norms. *Sketch:* Treat the link and inverse-link matrices as independent complex variables. A word of length l has norm at most $(1 + R_\sigma)^l \|1\|$. A slot of length l uses words of length at most $2l$, so its norm is at most $C_G e^{\sigma l}$, with C_G independent of l. The coefficient bounds therefore give normal convergence; a Cauchy estimate on a smaller ball controls the derivative series. The Wilson term is polynomial and symmetrization is finite. Composition with $y \mapsto (A(y), A(y)^*)$, followed by restriction to real scalars, proves III.6.2.

Lean: `F02.analytic_series`, `rowExt_series`, `atomExt_series`, `mainTerm_contDiffAt`.

Lemma 4.2 (the chart and its tangent). The map $(x, a) \mapsto U^H(x, a)$ is analytic near 0, with $U^H(0, 0) = 1$. The derivative of $t \mapsto U^H(t(x, a))_e$ at 0 is s_e, where $s = \iota x + a$. On tree bonds $s_e = 0$, because tree bonds are internal and ιx vanishes there. Consequently $\mathrm{incl}(x, a) = (x, \log U^H(x, a))$ is analytic near 0, with $\exp \circ \log U^H = U^H$ near 0 and $D\,\mathrm{incl}(0)(x, a) = (x, s)$. *Lean:* `F03.live_rate`, `inclusion_works`, `inclusion_deriv`.

Lemma 4.3 (flat jets). Let $\mathcal{F}_k(x, A) = \mathcal{M}_p(W; e^A) + \kappa \sum_b \mathrm{price}(e^{x_b} Q_k(e^A)_b^{-1})$ be the action in logarithmic coordinates on all fine bonds. Under the hypotheses of III.6.1:

- $\mathcal{F}_k(0) = 0$ and $D\mathcal{F}_k(0) = 0$;
- $D^2\mathcal{F}_k(0)[(x, A), (x', A')] = \langle dA, M_B dA' \rangle + \kappa \langle x - PA, x' - PA' \rangle$.

Sketch: The computation rests on an abstract principle: if f is C^2 at 0 and, for every z, $f(tz)/t^2 \to \frac{1}{2} Q(z, z)$ for a symmetric bilinear Q, then $f(0) = 0$, $Df(0) = 0$ and $D^2 f(0) = Q$. The limits are:

- $\mathrm{price}((e^{tA})_\pi)/t^2 \to \frac{1}{2}|dA_\pi|^2$;
- $\ell_W(e^{tA})/t \to R_B\, dA$, because W is attached to B;
- so the quadratic part tends to $\frac{1}{2}\langle dA, (c_p + R_B^\top R_B) dA \rangle = \frac{1}{2}\langle dA, M_B dA \rangle$, by the floor identity;
- the cubic and quartic sums are $o(t^2)$; each slot vanishes at the identity and has a linear tangent;
- averaging over $\mathcal{S}$ does not change the limit, because B is sworn;
- $Q_k(e^{tA})$ has derivative PA at $t = 0$ (this uses well-posedness), so the penalty tends to $\frac{\kappa}{2}|x - PA|^2$.

Lean: `F03.hessian_of_limit`, `quadratic_limit`, `main_limit`, `noise_limit`, `flat_jets`, `metric_pair`. *(proof read; the metric identity and cubic/quartic limit: statement only)*

Lemma 4.4 (transport to the tree slice). Near 0, $\mathcal{T}_k^H = \mathcal{F}_k \circ \mathrm{incl}$. Since $D\mathcal{F}_k(0) = 0$, $D^2\mathcal{T}_k^H(0)[u, u'] = D^2\mathcal{F}_k(0)[D\mathrm{incl}(0)u, D\mathrm{incl}(0)u']$; no term with $D^2\mathrm{incl}$ survives. With Lemmas 4.2–4.3 this gives III.6.1. *Lean:* `F03.from_flat`, `hessian_comp`.

3.9.4.2 4.2 The gap (F01) Lemma 4.5 (link bound). Let $v \in \mathfrak{g}^{H^{\mathrm{fr}}}$ and $y = s(0, v)$, i.e. v on free bonds and 0 on tree bonds. Assume E02. Then for every fine bond e, in block i and direction μ,

$$|y_e| \le C_{\mathrm{E02}} \sum_{z \in X} \sum_{\pi} \left| dy_{(\mathrm{root}_i + z, \pi)} \right| + \left| (Py)_{(i,\mu)} \right|.$$

Sketch: Apply E02 with $V = 1$ to $U_t = e^{ty}$. The configuration lies in the slice, and for small t the selected frames have positive mass. One obtains $\mathrm{chord}(U_{t,e}) \le C\,\mathrm{port}_i(U_t) + \mathrm{chord}(Q_k(U_t)_{(i,\mu)})$.

Divide by $|t|$ and let $t \to 0$, using $\mathrm{chord}(\gamma(t))/|t| \to |\gamma'(0)|$ along curves through 1. *Lean:* `F01.link_bound`, `chord_limit`.

Lemma 4.6 (summation). For the extension y of Lemma 4.5, $\sum_e |y_e|^2 \le \beta \sum |dy|^2 + \gamma \sum |Py|^2$, with β, γ as in III.6.3. *Sketch:*

- Start from $(CZ + Y)^2 \le 2C^2 Z^2 + 2Y^2$.
- Cauchy–Schwarz over $X \times \{\text{plaquette orientations}\}$ costs $6|X|$.
- Each coarse site carries L^4 fine sites and 4 directions.
- In the re-summation over coarse sites, $i \mapsto \mathrm{root}_i + z$ is injective for fixed z, which costs one more factor $|X|$.

Lean: `F01.sum_estimate`, `sum_ports_sq`.

Lemma 4.7 (the gap). By III.6.1 on vertical vectors (where $x = 0$, so the penalty argument is $-Py$) and by the floor,

$$D_a^2 \mathcal{T}_r^H(0)[v, v] = \langle dy, M_B dy \rangle + \kappa |Py|^2 \ge a|dy|^2 + \kappa |Py|^2.$$

Hence $\eta|v|^2 \le \eta(\beta|dy|^2 + \gamma|Py|^2) \le a|dy|^2 + \kappa|Py|^2$. This proves III.6.3. *Lean:* `F01.hess_lower`, `curl_floor`, `vertical_hessian`.

3.9.4.3 4.3 The branch (F04) Lemma 4.8 (tags agree near the identity). If S is well posed, the predictors Q_k for different tags coincide on a neighbourhood of the identity configuration. Hence the functions $\mathcal{T}_k^H$ coincide near 0. Near the identity, the frame and connector rules select their unique minimizers, so the tags do not affect the predictor. Local logarithms transfer this equality to arbitrary nearby configurations and then to the axis chart. *Lean:* `F04.prediction_germ`, `tested_germ`.

Lemma 4.9 (existence). The gap makes the vertical Hessian at 0 injective, hence invertible in these finite-dimensional spaces. The analytic implicit function theorem, applied to $D_a \mathcal{T}_{k_0}^H(x, a) = 0$ for one tag k_0, gives an analytic ρ with $\rho(0) = 0$. By Lemma 4.8 it is critical for every tag. *Lean:* `F04.invertible`, `exists_branch`.

Lemma 4.10 (test-pair identity). Assume the tower identities and the second jet of III.6.1. Then for every $z = (y, b)$ and coarse x,

$$D^2 \mathcal{T}_k^H(0)[z, (x, \mathrm{rel}_H x)] = \langle dy, M_C\, dx \rangle.$$

Sketch:

- $s(x, \mathrm{rel}_H x) = \widehat{\mathcal{W}}_H x$, since both sides vanish on tree bonds.
- The identities of §2 (III.1) give $x - P\widehat{\mathcal{W}}_H x = \kappa^{-1} K_C x$ and $K_B \widehat{\mathcal{W}}_H = P^\top K_C$.
- So the left side equals $\langle s(z), P^\top K_C x \rangle + \langle y - Ps(z), K_C x \rangle = \langle y, K_C x \rangle$.

When M_B is positive, the tower identities also imply that $A = \mathcal{W}x$ minimizes $\frac{1}{2}\langle dA, M_B dA \rangle + \frac{\kappa}{2}|x - \bar{Q}A|^2$, with minimum value $\frac{1}{2}\langle dx, M_C dx \rangle$. *Lean:* `F04.test_pair`.

Lemma 4.11 (tangent). Differentiating $D_a \mathcal{T}(x, \rho(x)) = 0$ at 0 gives $D^2 \mathcal{T}(0)[(x, D\rho(0)x), (0, v)] = 0$. Lemma 4.10 with $z = (0, v)$, together with the symmetry of D^2, gives the same identity with $\mathrm{rel}_H x$ in place of $D\rho(0)x$. Subtracting and inverting the vertical Hessian gives $D\rho(0) = \mathrm{rel}_H$. *Lean:* `F04.tangent`.

Lemma 4.12 (value). Let $\beta(x) = (x, \rho(x))$. Since $D\mathcal{T}(0) = 0$,

$$D^2(\mathcal{T} \circ \beta)(0)[x, x'] = D^2\mathcal{T}(0)[(x, \mathrm{rel}_H x), (x', \mathrm{rel}_H x')] = \langle dx, M_C dx' \rangle,$$

by Lemma 4.10. This is III.6.4(iii). *Lean:* `F04.stationary_value`.

3.9.4.4 4.4 Independence of the forest (F05) Lemma 4.13 (change of forest is a gauge transformation). For (x, a) near 0 define $\mathrm{sw}(x, a)_{e'} = \log(\widetilde{U}_{e'}\, \mathcal{R}(e^x)^{-1}_{e'})$ for $e' \in H'^{\mathrm{fr}}$. Here $\widetilde{U}$ is the gauge transform of $U^H(x, a)$ into the gauge of H', by a gauge function equal to 1 at the block roots. Then near 0:

- $U^{H'}(x, \mathrm{sw}(x, a)) = \widetilde{U}$;
- $\mathcal{T}^H_k(x, a) = \mathcal{T}^{H'}_k(x, \mathrm{sw}(x, a))$;
- $\mathrm{loop}_W(U^H(x, a)) = \mathrm{loop}_W(U^{H'}(x, \mathrm{sw}(x, a)))$.

Gauge invariance holds for each term:

- the atoms are Ad-invariant;
- ℓ_W is Ad-covariant;
- the price is a class function;
- the predictor is invariant under gauge transformations trivial at the roots.

Lean: `F05.tested_switch`, `loop_switch`, `mainTerm_gauge`, `root_noise`, `switch`.

Lemma 4.14 (the two graphs correspond). Near 0, $\mathrm{sw}(x, \rho(x)) = \rho'(x)$. Indeed, $\mathrm{sw}(x, \rho(x))$ is a critical point of $\mathcal{T}^{H'}_k(x, \cdot)$, by the chain rule through the local inverse of $\mathrm{sw}(x, \cdot)$, and both branches tend to 0. Critical branches are unique near 0 when the vertical Hessian at 0 is non-degenerate: this is the inverse function theorem for $(x, a) \mapsto (x, D_a\mathcal{T}(x, a))$, using the gap of H'. With Lemma 4.13 this gives III.6.5(i). *Lean:* `F05.graph_switch`, `station_unique`.

Lemma 4.15 (Hessian determinants). Write $\mathrm{sw}(x, \cdot) = \pi_\epsilon \circ \mathrm{mv}(x, \cdot)$, where π_ϵ relabels coordinates along a bijection $\epsilon : H^{\mathrm{fr}} \to H'^{\mathrm{fr}}$; the two forests have the same number of free bonds. At a critical point the Hessian pulls back by $D_a\mathrm{mv}$, so near 0

$$\det \mathrm{Hess}_a \mathcal{T}^H_k(x, \rho(x)) = \left(\det D_a \mathrm{mv}(x, \rho(x))\right)^2 \det \mathrm{Hess}_a \mathcal{T}^{H'}_k(x, \rho'(x)),$$

and the right-hand determinant is positive (by the gap). *Lean:* `F05.hessian_ratio`, `hessian_pos`, `move`.

Lemma 4.16 (Haar Jacobian identity). Near 0,

$$\prod_{e \in H^{\mathrm{fr}}} J_{\mathcal{J}}(a_e) = \left|\det D_a \mathrm{mv}(x, a)\right| \prod_{e' \in H'^{\mathrm{fr}}} J_{\mathcal{J}}(\mathrm{sw}(x, a)_{e'}).$$

Sketch:

- By the defining property of $J_{\mathcal{J}}$ and right invariance of Haar measure, $a \mapsto (e^{a_e}\mathcal{R}(e^x)_e)_e$ maps the density $\prod_e s_{\mathcal{J}} J_{\mathcal{J}}(a_e)\, da_e$, restricted to the chart, to product Haar measure restricted to the chart image in the slice of H.
- The change of forest preserves product Haar measure (III.1).
- By the change-of-variables formula, the two sides therefore have equal integrals over every small compact set.

- The forests have the same number of free bonds, so the factors of the Haar scale cancel. Continuity and comparison over small closed balls give pointwise equality.

Lean: `F05.toll_switch`.

Lemma 4.17 (cancellation). If $B, j, k > 0$, $A = d^2 B$ and $j = |d|k$, then $d \neq 0$, $A > 0$, and $\frac{1}{2}\log A - \log j = \frac{1}{2}\log B - \log k$. With Lemmas 4.13–4.16 and the equality of the loop terms, this gives III.6.5(ii) with constant 0. *Lean:* `F05.same_thermal`.

3.9.4.5 4.5 The volume-independent branch and its jets Lemma 4.18 (Combes–Thomas bound). Let S be a real symmetric matrix on a finite index set assigned to sites of a four-dimensional torus, with multiplicity at most K_0. Suppose $a_0, M > 0$, $C_0 \geq 0$, $|S_{ij}| \leq C_0 e^{-a_0 \mathrm{dist}(i,j)}$, and $S \geq MI$. Then

$$|(S^{-1})_{ij}| \leq \frac{2}{M} e^{-r_{\mathrm{CT}} \mathrm{dist}(i,j)}, \qquad r_{\mathrm{CT}} = \min\left(\frac{a_0}{4}, \frac{M}{2(1 + 4C_0 K_0 c(a_0/2)/a_0)}\right).$$

Here the site labels are understood in the distances, and

$$c(\alpha) = \frac{16 \cdot 6!\, e^{2\alpha}}{\alpha^6}, \qquad \sum_y e^{-\alpha \mathrm{dist}(x,y)} \leq c(\alpha) \quad (\alpha > 0),$$

uniformly over the torus side. *Sketch:* Conjugate S by the diagonal weight $e^{r_{\mathrm{CT}} \mathrm{dist}(\cdot, j)}$. The difference has both absolute row and column sums at most $M/2$. Its quadratic form is therefore bounded by $(M/2)\|v\|_2^2$, so the conjugated matrix retains coercivity $M/2$, although it need not be symmetric. Applying this inequality to the tilted j-th inverse column gives the displayed entry bound.

Lean: `tiltedCharge`, `inverse_sealed`.

Lemma 4.19 (uniform implicit function theorem; III.6.7(a)). In the notation of III.6.7(a), set $D = K_0 c(a_0)$ and

$$\Pi = 1 + \frac{4}{m} K_0 c(r_{\mathrm{CT}}),$$

where r_{CT} is the rate of Lemma 4.18 with gap $M = m/2$. The radii may be chosen as

$$r_{\mathrm{safe}} = \min\left(\frac{r}{2}, \frac{m}{2(1 + C_0 D^2)}\right),$$
$$r_{\mathrm{zone}} = \min\left(r_{\mathrm{safe}}, \frac{1}{4\Pi(1 + C_0 D^2)}\right),$$
$$r_{\mathrm{hor}} = \min\left(r_{\mathrm{zone}}, \frac{r_{\mathrm{zone}}}{4\Pi(1 + C_0 D)}\right).$$

Sketch: The third-derivative estimate bounds the change of a Hessian entry by $r_{\mathrm{safe}} C_0 D$ times its spatial decay factor. Summing once more gives absolute row and column sums at most $r_{\mathrm{safe}} C_0 D^2 \leq m/2$. Thus the vertical Hessian stays $\geq (m/2)I$, and its inverse has sup-norm operator bound Π throughout the closed safety ball. For $\|x\|_\infty < r_{\mathrm{hor}}$, set $H_0 = D^2_{\tau\tau} f(0)$ and

$$Y(v) = v - H_0^{-1} \nabla_\tau f(x, v).$$

On the closed ball of radius r_{zone}, $\|DY\|_{\infty\to\infty} \le \Pi r_{\mathrm{zone}} C_0 D^2 \le 1/4$, while $\|Y(0)\|_\infty \le \Pi \|x\|_\infty C_0 D \le r_{\mathrm{zone}}/4$. Hence Y maps this ball into the ball of half the radius and has a unique fixed point. The analytic implicit function theorem at each fixed point, together with uniqueness, makes the resulting branch analytic at every allowed x. *Lean:* `Receipt.adjusted`, `Receipt.snaps`, `Receipt.priced_inverse`, `Receipt.throughout`.

Lemma 4.20 (the tested action satisfies III.6.7(a)). Take the auxiliary constants of III.6.7(b). For every clip p, larger S with order J and kernel choices K, forest H and W with bounds (p, μ, C, Λ), the tested action in sup-norm colour coordinates satisfies:

- analyticity and localized derivatives of orders $1, \dots, n$, with width $\mu/(8(n+1)^2(1+L))$, on a ball whose radius is independent of c, f and H;
- the multiplicity bound $8 \dim \mathfrak{g}$;
- tag-independence on the whole ball and zero vertical gradient at 0;
- the vertical gap

$$m = \frac{\min(c_p/|\mathcal{S}|,\ \kappa)}{1 + \beta(L, \ell_L) + 2L^4}.$$

This is a coercivity estimate for the same vertical Hessian, with no bed or attachment hypothesis. In it:

- the constant of E02 is replaced by an explicit L-dependent constant ℓ_L, obtained by proving the shell comparison directly (`link_charge`);
- the floor a is replaced by $c_p/|\mathcal{S}|$, since the vertical Hessian is $\ge (c_p/|\mathcal{S}|)|dy|^2 + \kappa|Py|^2$ (`wallet`).

Lean: `Before.receipt`, `symbolic_gap`, `zeroGap`, `Before.full_pack`. *(proof read for the gap and the assembly; the derivative bounds of the action and the direct shell estimate: statement only)*

Lemma 4.21 (derivatives of the inverse and of $\log\det$; III.6.8). Let $M(y)$ be a finite matrix function analytic at y_0, with $M(y_0)$ invertible. Assign its matrix and variable indices to lattice sites, with matrix-index multiplicity at most K_0. Suppose its coefficient derivatives through order N have the localization bounds of III.6.8 at rate $a_0 > 0$, and $M(y_0)^{-1}$ has exponentially decaying entries at that rate. Then the derivatives of M^{-1} through order N are localized, with rates and constants defined recursively from these bounds and K_0. *Sketch:* Differentiate

$$D(M^{-1}) = -M^{-1}(DM)M^{-1}.$$

The product rule reduces each new order to lower inverse derivatives and derivatives of M. Exponential convolution bounds control the internal index sums; each multiplication reduces the available rate. For a positive determinant, use $D \log\det M = \mathrm{tr}(M^{-1}DM)$ and repeat the same estimates. Under the coercivity hypotheses of III.6.8 this applies to every principal submatrix with every nonnegative shift. Composing with the stationary graph gives the estimates along the branch; the joint action has order $N+3$, leaving order $N+1$ after taking two Hessian derivatives.

Lean: `inverse_lines`, `establish`, `along_block`, `Counter.background`. *(proof read for the inverse induction and the assembly; the convolution and composition estimates: statement only)*

3.9.4.6 4.6 One level of the iteration

Lemma 4.22 (gap quote and branch selection). Fix a stage index j. Apply III.6.6 with the block factor and floor bound assigned to that stage. This gives post-L gap constants η_j (`Iteration.post`). At level m, use $\eta_{J(m+1)}$. Every funded T at level $m+1$ then has, for each period and forest, a branch satisfying III.6.9(i) (`select`).

Lemma 4.23 (the glued datum is funded). The fresh-data hypotheses supply fresh cubic and quartic atoms t, v and fresh loop atoms l. Let R be the bundle on the bed $M_{N,m}$ with:

- cubic atoms: t joined with the transport of the cubic atoms of T through the towers ("inherited");
- quartic atoms: v joined with the inherited quartic atoms;
- loop atoms: l joined with the inherited loop atoms.

Then R is funded at level m. The inherited cubic and quartic atoms are charged with relative factors L^{-2} and L^{-4}, and the resulting charges stay within 6 times the level price. *Lean:* `resume`. *(proof read for the assembly; the inherited-atom estimates: statement only)*

Lemma 4.24 (the move). The fresh one-loop data supply $|\alpha + \kappa_G \log L| \le s_{j,k}$, where $s_{j,k}$ is the prescribed slope bound for the two stage indices. The schedule gives $s_{j,k} + 1 \le \mathrm{drift}_{m+1}$, proving (ii). The matching (iii) is proved in three steps:

- On one reference forest per period, orders ≤ 2 follow from III.6.4(iii) and the attachment of R to $M_{N,m}$ (`Calibration.lower`).
- Orders $3, 4$ split into the inherited part, matched through the towers, and the fresh part, supplied by the fresh-data hypotheses.
- Tag-independence and III.6.5 transfer the match to every tag and forest (`all_forests`).

The bound (iv) is proved on the reference forest from the fresh determinant hypothesis, and transferred to every forest by the equality of thermal functions (`transplant`). *Lean:* `extend`. *(proof read for the assembly; inherited-atom matching: statement only)*

3.9.4.7 4.7 Remaining packages (statements only)

- *F06–F09.*
  - F06: fix a strip bound A, a positive margin v and strip width $\eta > 0$. A real translation-invariant 3-form kernel Q, Hermitian on real Fourier phases and bounded below by v on the image of the 2-form symbol, can be replaced by P bounded above and below on all 3-forms. The replacement preserves $d_2^{\top}(\cdot)d_2$ on every torus. Its positive lower bound, upper bound and strip width depend only on (A, v, η).
  - F07: with analogous uniform positivity and strip bounds, a real kernel has a symmetric positive square root on every torus, with strip width and bounds depending only on the input bounds and the finite component space.
  - F08: fix the two clips, penalty, bed strip bounds, a 3-form kernel strip bound and a derivative order. Towers whose first two components have the prescribed derivative bounds can be completed by replacing the remaining components. The completed towers satisfy the replacement identities and uniform strip bounds chosen before the odd factor L.
  - F09: the replacement identities, floors of the two beds, diagonal identities for the original towers and a square-root factorization of the defect imply the stack identities for the completed towers.
- *Comparison of two tori* (`Loom.spinning`). Let $\pi : \beta \to \alpha$ have exactly M elements in every fibre. Assign the indices in β to torus sites with multiplicity at most K_0, and suppose distinct elements of a fibre are separated by at least N. Let F, F' be matrix functions on a real

normed space, analytic at 0, with invertible values there and

$$F(x)E = EF'(x) \quad \text{near } 0, \qquad E_{i,j} = \mathbf{1}_{\pi(i)=j}.$$

Let y, z be analytic scalar functions with $y = Mz$ near 0. Fix two directions. Suppose the entries of $F(0)^{-1}$, the two first variations and the mixed second variation are bounded by $B_0 e^{-d\,\mathrm{dist}}$, $T_0 e^{-d\,\mathrm{dist}}$ and $Q_0 e^{-d\,\mathrm{dist}}$, respectively, where $d > 0$ and $B_0, T_0, Q_0 \geq 0$. The mixed second derivatives of $\frac{1}{2}\log|\det F| - y$ and $M(\frac{1}{2}\log|\det F'| - z)$ then differ by at most

$$|\beta|\, C(K_0, T_0, Q_0, B_0, d) e^{-dN/8}.$$

For positive determinants these are the thermal determinant terms. The proof expands the second derivative into two traces. Compatibility identifies the small-system trace with sums over covering fibres; the error consists of off-diagonal entries between distinct elements of a fibre, whose separation is at least N.

- *Primary integral.* This package contains:
  - Newton remainder bounds from bounded finite differences;
  - vanishing of a symmetric tensor on lists of pure directions of a given total engineering degree when its polynomial component of that degree vanishes;
  - under strip bounds for the tower kernel, comparable probe amplitudes, equality of the logarithmic probe germs within radius χ, and derivative bounds B on the true path, a bound $D(1+B)e^{-\sigma\chi/2}$ for the difference of the true and forecast probe jets (`Ramble.billFor`);
  - triangular structure of the primary transition matrix.

  These comparisons enter the repeatable nonlinear step in III.10.

- *Atom regularity.* Suppose $t \geq 0$, every plaquette incident to Y has chord at most t, the halo of radius $q+1$ about o lies in Y, and $2(q+1) < n$. There is a gauge g such that every link b with $\mathrm{dist}(o, b.1) \leq q$ satisfies $\mathrm{chord}((g \cdot V)_b) \leq 4t(1 + \mathrm{dist}(o, b.1))$.

- *ROTATIONS, WARD_BIN.* This material is the conclusion of the rotation theorem (line-sum property and vanishing of the local rotation charge on collinear points) and its chain: moment bounds for reflection-positive spectral measures, decay of the Ward integrand, and the flux identity. It belongs to VI.1.

3.9.5 5. Where the difficulty lies

The main estimate is the volume-independent lower bound on the vertical Hessian. Curvature alone leaves gauge and flat directions; the forest condition and predictor penalty together control every free link. The denominator of the gap contains a term proportional to $C_{\mathrm{E02}}^2 L^4 |X|^2$, and $|X| = (2(J+7)L+1)^4$. The gap is chosen after L; no lower bound uniform in L is asserted.

A spectral gap gives an ℓ^2 inverse bound. To obtain a sup-norm radius independent of the volume, one also needs the exponential entry bounds and their summability. The uniform construction uses a direct shell estimate and the positive Wilson coefficient, so it needs no bed or attachment hypothesis. It reuses the linearization and counting estimates of F01, while supplying its own gap and radii.

Forest independence requires both the stationary change-of-coordinates formula for the Hessian and preservation of product Haar measure. Their Jacobians cancel exactly, including the Haar

normalization scales. The proof needs only the gap for the target forest. Other unused hypotheses are the coarse floor in F04, the identity base value in F02, and the size clauses of "larger" in F03–F05.

The conclusions concern a branch near the identity and uniqueness in a specified sup-ball. Global constrained minimization is not asserted. The one-loop coefficient and the classical remainders remain inputs to the iteration through the fresh-data hypotheses in III.6.9.

3.9.6 6. Notation

Written proof	Lean	Meaning
S, κ, K, J, L	`StepData`, `KernelChoices`	step datum; penalty; kernel choices; order; block factor
larger	`Bid.Larger`	well posed, $100R_S \le f$, $c > 1$
Q_k, P, $\bar{Q}$	`prediction`, `probe`, `mean`	predictor; its linearization; linearized block average
H, H^{fr}	`Forest`, `Free`	forest gauge; free bonds
$\mathcal{R}$, ι	`replica`, `linearCopy`	coarse field copied onto crossing fine bonds
$U^H(x,a)$, a	`lifting`, `AxisChart`	fine field in the axis chart
s, $x - Ps$	`stretch`, `crop`, `bills`	linearized fine field; linearized penalty argument
W, p, c_p	`PathData`, `Clip`	path datum (row series, loop/cubic/quartic atoms); clip; Wilson coefficient
$\mathcal{M}_p(W;\cdot)$, $\mathcal{M}^{(2)}$	`mainTerm`, `secondTerm`	symmetrized main term; Wilson-plus-row-square term
$\mathcal{T}_k^H$	`tested`, `noiseTaylor`	tested action; penalty term
B, M_B, R_B, K_B	`Bed`, `K`	bed; metric; roots; $d^\top M_B d$
$\langle dA, M_B\, dA' \rangle$	`gaugeFiscal`	quadratic form of the bed
floor, attached, sworn	`Floor`, `Attached`, `Sworn`	ellipticity of the bed; ℓ_W linearizes to $R_B\, d$; symmetry invariance
T, $\mathcal{W}$, $\widehat{\mathcal{W}}_H$	`Towers.Diagonal`, `Inputs`, `extracted`	towers; linear background map; its forest-gauge version
rel_H	`relative`, `Guessed`	linear background field in chart coordinates
critical branch ρ	`Critical`	background field
gap η	`PriceGap`	lower bound of the vertical Hessian at 0
second jet; value jet	`Comptroller`, `HeatG`, `SameJet`	III.6.1; III.6.4(iii); equality of j-th derivatives
$J_{\mathcal{J}}$	`ChartRatio`, `LocalRatio`	Haar density in exponential coordinates
Θ_k^H	`thermal`, `hessian`	thermal function; Hessian matrix in the colour basis
sw, mv	`switch`, `move`	chart change between forests

Written proof	Lean	Meaning
localized derivatives; localized matrix	`Scatter`, `Sealed`, `Cable`	exponential decay of entries and derivatives
hypotheses of III.6.7(a)	`Receipt`	analytic, localized, gapped, critical at 0
auxiliary constants chosen before L	`Before`	radii and costs of the uniform layer
block package; jet service	`BlockPack`, `Counter`	III.6.8
fresh-data hypotheses	`Rates`	cutoff atoms, fresh classical remainders and one-loop matching
funded datum; move	`Work item`, `Move`, `OnShell`	III.6.9

"Price", "bill", "work item", "receipt", "toll" and "cost" are numerical bounds and their bookkeeping; they have no economic meaning.

3.9.7 7. What remains unexplained

1. **Fresh classical and one-loop inputs.** Proposition III.6.9 is conditional on `Rates`. The adapter `IterationRecipe.stocked` supplies the cutoff atoms and retains the classical and quantum matching assumptions from `Heating`; it does not discharge those assumptions. Their discharge in the main construction remains unverified here.
2. **Small-field domains.** The comparison between the componentwise radii in III.6.7–III.6.8 and the small-field thresholds used downstream remains unverified. The relevant branch and derivative interfaces are linked in III.6.7–III.6.8.
3. **Iteration over all levels.** The induction beyond one step is in `Iteration.sweep` and `Luna.uniformZero` (III.12). Their proofs are outside this verification.
4. **Supporting proofs not checked in full.** The remaining unread supporting proofs are the metric/path-limit estimates identified after Lemma 4.3; the direct shell and action-derivative estimates used in Lemma 4.20; the convolution and graph-composition estimates used in Lemma 4.21; and the inherited-atom matching and bounds in Lemmas 4.23–4.24. F06–F09 and the primary-integral, atom-regularity, rotation and Ward packages are represented by their statements only. The period-comparison trace argument was checked; its geometric realization for the actual Hessians was not.
5. **Attached packages.** The primary-integral package feeds the repeatable step in III.10; the rotation and Ward material belongs with VI.1. The application of the period comparison in `Fresco.subscribed` remains unverified here. The size figures in the header are the inherited dependency-closure counts; they have not been recounted.

3.10 III.7. Matching the Wilson energy to fourth order

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/Next/STATIONARY/`, `YM/WilsonBulk/Spectral/`, `YM/UVExtension/Projection/`, and a few lines of `YM/UVExtension/Reward/`. **Size:** 19,945 lines in the main-theorem closure: 19,391 in `Next/STATIONARY`, 286 in `UVExtension/Projection`, 215 in `WilsonBulk/Spectral`, and about 50 elsewhere.

Conventions. Levels $m = 0, \dots, N$ run from the coarsest level ($m = 0$) to the Wilson lattice ($m = N$). Blocking from fine level $m+1$ to coarse level m uses an odd factor L_m. The stage J_m is the index of the constants used at level m; it satisfies $J_{m+1} \in \{J_m, J_m + 1\}$. In one step the fine torus Λ_f has side $f = Lc$ and the coarse torus Λ_c has side c; in general $|\Lambda_n| = n^4$. Fields are Lie-algebra link variables x, and configurations are $U = e^x$. $E_W(U) = \sum_p (1 - d_{\text{rep}}^{-1} \text{Re}\,\text{tr}\, U_p)$ is the Wilson energy in the fixed matrix representation of dimension d_{rep}, C_G is the adjoint Casimir constant of I.1, and $\kappa = 11C_G/(24\pi^2)$.

3.10.1 1. What this unit proves

At level m the effective action has the form $\beta\,\text{main}_{\mathbf{c},R}(U) + \text{loop}_R(U)$, plus polymer activities (III.10). Here R is a *path datum*: an exponentially weighted, explicit sum of gauge-covariant curvature monomials along lattice paths. This unit constructs the path data and the coupling shifts.

- **(A)** At the finest level there is a path datum whose leading part has the same Taylor jet as E_W through order four (III.7.2).
- **(B)** For one block-spin step, the coarse leading part can be chosen as the classical (tree-level) transform of the fine one through order four. The second-order Taylor coefficient of the one-loop term equals α times the coarse quadratic form plus coarse two-leg atoms, up to an error exponentially small in the side of the torus. Moreover $|\alpha + \kappa \log L|$ is bounded independently of L (III.7.4).
- **(C)** The value $\kappa = 11C_G/(24\pi^2)$ is computed, not assumed. The one-loop determinant of the background-field operator in Feynman gauge, regularized in the infrared at scale $1/L$, has F^2-coefficient $-\kappa \log L + O(1)$ (III.7.3). Two results connect this determinant to the blocked one: an exact Faddeev–Popov identity (III.7.6) and uniform invertibility of the completed fluctuation operators (III.7.5, §4.6).

Iterated over all levels, (A)–(C) give the kinetic data satisfying the conditions required in III.9. The slope κ, a defined constant in I.1 and III.9.1, is identified here with the one-loop coefficient of the blocked action.

Proposition III.7.1 (path data, beds, Fourier kernels).

- *Legs and atoms.* A leg is a lattice walk γ from the origin of $\mathbb{Z}^4$ together with a plaquette orientation P. At base point x it evaluates U to $\text{Ad}(U_\gamma)\,\mathcal{F}(U_{\partial p}) \in \mathfrak{g}$. Here p is the plaquette of orientation P at the end of γ, U_γ is the transport along γ, and $\mathcal{F}(u)$ is the $\mathfrak{g}$-component of $(u - u^{-1})/2$. The length of a leg is $|\gamma| + 4$.
- A k-atom is a k-tuple of legs with a coefficient tensor $a : \{1, \dots, \dim \mathfrak{g}\}^k \to \mathbb{R}$. It evaluates to $\sum_b a(b) \prod_i (\text{leg}_i)_{b_i}$. It is *invariant* if this polynomial is Ad-invariant. Its exponential weight at width σ is $\|a\|_1 e^{\sigma \cdot \text{length}}$.
- *Path datum R.* For each of the six plaquette orientations P, a sequence of rows $(\lambda_{P,k}, \text{leg}_{P,k})$; and sequences of 2-atoms (*loop atoms*), 3-atoms and 4-atoms. The rows define $\ell_R(U)_{(x,P)} = \sum_k \lambda_{P,k} \text{leg}_{P,k}(U, x)$.
- *Main term.* For a clip $\mathbf{c}$ (only $c_{\text{dir}} > 0$ enters here), whenever the row and atom series

converge absolutely:

$$\mathrm{main}_{\mathbf{c},R}(U) = \frac{1}{384}\sum_s \Big[\sum_p \Big(c_{\mathrm{dir}}(1-d_{\mathrm{rep}}^{-1}\mathrm{Re}\,\mathrm{tr}\,U_p)+\tfrac{1}{2}|\ell_R(U)_p|^2\Big)+\sum_x\sum_k (\mathcal{A}_k^{(3)}+\mathcal{A}_k^{(4)})(U,x)\Big]_{U\to sU}.$$

This is an average over the 384 signed permutations s of the axes. loop_R is the same average of the sum of all loop atoms. These are the unclipped analytic germs; the small-field clipping of the full action does not enter them.

- *Weight clauses* $R \in \mathrm{Bounds}(\mathbf{c};\sigma,C,\Lambda)$:
  - $\sigma > 0$, and $\sum_k |\lambda_{P,k}| e^{\sigma\,\mathrm{len}} \le C$ for each P;
  - the sums of exponential weights are $\le C$ for the cubic and quartic atoms and $\le \Lambda$ for the loop atoms;
  - all atoms are invariant;
  - normalization at zero momentum: $c_{\mathrm{dir}}\delta_{PQ} + \sum_{P'} \hat{\ell}_{P'P}(0)\hat{\ell}_{P'Q}(0) = \delta_{PQ}$;
  - the *moment conditions*: for each pair of leg orientations and colours, the loop coefficients have total weight 0 and first moment 0 in the displacement between the two legs (with the required summability).
- *Bed* $B = (D,\tilde{D},H,\tilde{H},\rho)$: five translation-invariant kernels on plaquette fields (diagonal, its inverse, metric, its inverse, roots).
  - R is *attached* to B if its rows, summed by end plaquette and displacement, give ρ.
  - $\mathrm{Floor}(B,\mathbf{c},n;\varepsilon,C)$ on Λ_n means: $D\tilde{D} = 1$; $H\tilde{H} = 1$; $\Phi^{\mathsf{T}}\Phi = H$ with $\Phi = (\sqrt{c_{\mathrm{dir}}}\,\mathbf{1};\rho)$; $\hat{H}(0) = 1$; $H\,d_1 = D\,d_1$ on link fields; H, D symmetric with $\varepsilon I \le H, D \le CI$ as quadratic forms.
  - A kernel is in $\mathrm{strip}(\sigma,C)$ if $\sum_y e^{\sigma|y|_1}|K(y)| \le C$ entrywise.
- *Fourier kernels.* A $(t{+}1)$-leg kernel assigns a number $\hat{K}_{b,a}(p_1,\dots,p_t)$ to colours b, directions a and momenta, with $p_0 = -\sum p_j$.
  - $\hat{K}$ is *strip-analytic of width σ and size K* if it is holomorphic, 2π-periodic in each scalar momentum coordinate, and bounded by K on $|\mathrm{Im}\,p_{j,a}| < \sigma$.
  - A function F on link fields on Λ_n is *recorded* by $\hat{K}$ with error ϵ if its complexified Taylor coefficient $D^{t+1}F(0)/(t+1)!$, evaluated on canonical unit plane waves, differs from $|\Lambda_n|\hat{K}_{b,a}(p)$ when the momenta sum to zero, and from 0 otherwise, by at most ϵ. All tuples of lattice momenta are tested.
  - *F vanishes to order s along rays* if $F(tv) = O(t^s)$ as $t \to 0$ for every v, with the constant allowed to depend on v.
- *F^2-coefficient.* For a two-leg kernel, $\alpha_b(\hat{K})$ is twice the real part of the coefficient of s^2 in $\hat{K}_{(b,b),(1,1)}(s\,e_0)$. For the quadratic kernel of a bed whose metric has positive strip width and $\hat{H}(0) = 1$, it equals 1, so α is measured in units of the classical F^2 term.

Status: definitions. Path data, weight clauses, main term, beds and Floor are specifications of III.1; the Fourier notions are defined in this unit. *Lean:* `RG.PathData`, `RG.Slot`, `RG.Atom`, `PathData.Bounds`, `metric_zero`, `loop_jets`, `RG.Clip`, `Precision.mainTerm`, `Precision.Bed`, `Attached`, `Floor`, `Firm`, `Rays`, `Recorded`, `alphaPay`, `tip_fuel`.

Proposition III.7.2 (the Wilson jet at the finest level). Let G be a gauge group, $\sigma_0 > 0$ and $A, Z \ge 0$. There are $\sigma \in (0,\sigma_0]$ and $C \ge 0$ with the following property.

Hypotheses: sides $n_i \to \infty$; a clip with $c_{\mathrm{dir}} \le Z$; a bed B such that

- $\rho \in \mathrm{strip}(\sigma_0, A)$;
- $\mathrm{Floor}(B, \mathbf{c}, n; a, b)$ holds for every side n and some a, b;
- the quadratic gauge form of B is invariant under signed permutations on each Λ_{n_i};
- $D = 1$ on each Λ_{n_i}.

Conclusion: there is a path datum Q such that:

1. the rows of Q are the entries of ρ placed on straight lattice routes, Q is attached to B, and all loop coefficients vanish;
2. $Q \in \mathrm{Bounds}(\mathbf{c}; \tau, k, \Lambda)$ for all $0 < \tau \le \sigma$, all $k \ge \max(C, 6Ae^{4\sigma_0})$ and all $\Lambda \ge 0$;
3. for every i and every $j \le 4$, the functions $x \mapsto \mathrm{main}_{\mathbf{c},Q}(e^x)$ and $x \mapsto E_W(e^x)$ on Λ_{n_i} are real-analytic at 0 and have equal j-th derivatives there.

Status: theorem. *Lean:* `Stationary.launch`, `rowPath`, `rowPrice`, `SameJet`.

All stages and all volumes.

- With stage widths and amplitude bounds given, the statement holds at every stage with stage constants (σ_j, A_j) (`beckon`, `BasesFine`).
- Take a compatible history satisfying the thresholds of III.2 and a horizon satisfying the free-field conditions of §2, and apply the statement to the sequence of periodic sides at level N. The result is a top datum (`budding`, `BranchRow`) that:
  - satisfies the level-N weight clauses with loop amplitude $\mathrm{loopCap}_N$;
  - matches E_W through order four on every volume;
  - has vanishing loop coefficients.

  This is the top condition used in III.9.

Proposition III.7.3 (the one-loop slope). Let G be simple. For $\sigma, \varepsilon > 0$, $C, Z \ge 0$, $\tau > 0$ and $A \ge 0$ there is $D \ge 0$ with the following property.

Hypotheses:

- $L \ge 1$ is odd, $\mu = 1/L$, and $n_i \to \infty$;
- $\mathbf{c}$ is a clip with $c_{\mathrm{dir}} \le Z$;
- B is a bed with $H \in \mathrm{strip}(\sigma, C)$, $\mathrm{Floor}(B, \mathbf{c}, n; \varepsilon, C)$ for every n, and gauge form invariant under signed permutations on each Λ_{n_i};
- for any $\Lambda \in \mathbb{R}$, a datum $R \in \mathrm{Bounds}(\mathbf{c}; \tau, A, \Lambda)$ is attached to B.

For a link field x on Λ_{n_i} and $U = e^x$ put

$$\Pi_\mu(x) = \tfrac{1}{2} \log\det\left(\mathcal{H}_{\mathbf{c},R}(U) + d_U d_U^* + \mathcal{M}_\mu(U)\right) - \log\det\left(d_U^* d_U + \mathfrak{m}_\mu(U)\right).$$

Here:

- $\mathcal{H}_{\mathbf{c},R}(U)$ is the Hessian at $y = 0$ of $y \mapsto \mathrm{main}_{\mathbf{c},R}(e^y U)$;
- d_U is the covariant difference $(d_U\phi)_e = \phi(e_-) - \mathrm{Ad}(U_e)\phi(e_+)$ from adjoint site fields to link fields;
- $\mathfrak{m}_\mu(U) = \mu^2(\mu^2(d_U^* d_U + \mu^2)^{-1})^9$;
- $\mathcal{M}_\mu$ applies $\mathfrak{m}_\mu$ separately to the four link components, each regarded as a site field at the initial endpoint.

Conclusion: there are a strip-analytic two-leg kernel $\hat{Y}$ and numbers $e_i \to 0$ such that:

- (a) $\hat{Y}$ records the second-order Taylor coefficient of Π_μ at 0 on Λ_{n_i} with error $|\Lambda_{n_i}|\, e_i$, for every i;
- (b) $|\alpha_b(\hat{Y}) + \kappa \log L| \le D$ for every colour b.

Π_μ is the one-loop effective action of the fine leading part in the background U, in Feynman background gauge: $d_U d_U^*$ is the gauge-fixing term and $d_U^* d_U$ the ghost operator. The infrared regulator of mass $\mu = 1/L$ stands in for the block-spin constraint. D does not depend on L. The error e_i decays exponentially in n_i, at an L-dependent rate.

Status: theorem, for all odd $L \ge 1$: the minimal block factor produced is 1. *Lean:* `pouring` (proves `Fluent`, i.e. `HorizCalc` for all parameters), `PoleTest`, `fineMag`, `rankAux`, `grenade`, `Gauge.slopeConstant`.

Proposition III.7.4 (one renormalization step of the leading part). Fix a simple G, kernel parameters K (penalty $a > 0$), a stage J, and constants $\varepsilon, w, \delta > 0$ and $D, Z, C, Z' \ge 0$ for the fine and coarse beds and clips. There are $A_* \ge 0$ and $s > 0$ (output amplitude and width) such that, for every input width $\tau > 0$ and amplitude $A \ge 0$, there are:

- $d \ge 0$ (the drift) and $\gamma > 0$ (a rate);
- a minimal block factor u;
- functions $\Theta_0(L)$ and $\Lambda_0(L)$,

with the following property.

Hypotheses.

- *Steps.* Coarse sides $m_i \to \infty$; block-spin steps $\Lambda_{Lm_i} \to \Lambda_{m_i}$ with one admissible patch of stage J and block factor $L \ge u$, and kernel K; each coarse side is greater than 1, and each fine side is at least 100 times the patch extent.
- *Clips and beds.*
  - A fine clip $\mathbf{p}$ with $p_{\mathrm{dir}} \le Z$ and a coarse clip $\mathbf{q}$ with $q_{\mathrm{dir}} \le Z'$.
  - A fine bed B with $H, \rho \in \mathrm{strip}(w, D)$, $\mathrm{Floor}(B, \mathbf{p}, n; \varepsilon, D)$ for every n, and signed-permutation-invariant gauge form on the fine tori.
  - A coarse bed T with $H_T, \rho_T \in \mathrm{strip}(\delta, C)$, $\mathrm{Floor}(T, \mathbf{q}, n; \varepsilon', C)$ for every n and some $\varepsilon' \in \mathbb{R}$, and signed-permutation-invariant gauge form on the coarse tori.
  - A tower with interpolation kernel A_0, satisfying the quadratic-form interpolation, curl-transport and diagonal-transport identities of III.1 on each step. For every list ν of at most two fine-coordinate differences, $\nabla^\nu A_0$ has strip width w and entrywise weighted sum at most $D/L^{1+|\nu|}$.
- *Fine datum.* $R \in \mathrm{Bounds}(\mathbf{p}; \tau, A, \Lambda)$ attached to B, with $\Lambda \ge 1$.
- *Stationary branches.* A common $\eta > 0$ and, for each i, an axial forest F_i and a branch g_i (coarse fields $\to$ free fine variables) such that:
  - for every choice of block-kernel data, the second variation in the free variables at $(0, 0)$ is at least $\eta \sum_e |v_e|^2$;
  - $g_i(0) = 0$, g_i and the fine action are real-analytic at 0, and g_i is stationary for every choice of block-kernel data;
  - on every free edge, $Dg_i(0)v$ is the free interpolant in the chosen axial gauge minus the derivative of the copied coarse configuration;
  - the second jet of the tree-level coarse action is the quadratic form of T.
- *Chart.* A local Haar chart ratio J.

Conclusion. There are a coarse path datum Y and $\alpha \in \mathbb{R}$ such that:

1. the rows of Y are the straight-route rows of T, and Y is attached to T;
2. $Y \in \mathrm{Bounds}(\mathbf{q}; \eta', A', \Lambda')$ whenever $0 < \eta' \le s$, $A' \ge A_*$ and $\Lambda' \ge \max(\Lambda_0(L), \Lambda/2)$;
3. $|\alpha + \kappa \log L| + 1 \le d$;
4. *classical match:* for every i, every choice t of block-kernel data and every $j \le 4$, the j-th jets at 0 of
$$x \mapsto S_t(x, g_i(x)) \quad \text{and} \quad x \mapsto \mathrm{main}_{\mathbf{q},Y}(e^x)$$
coincide. Here $S_t(x,z) = \mathrm{main}_{\mathbf{p},R}(U_{x,z}) + a\sum_e (1 - d_{\mathrm{rep}}^{-1}\mathrm{Re}\,\mathrm{tr}(V_e\,\Pi_t(U_{x,z})_e^{-1}))$ is the fine leading part plus the block-spin penalty. $U_{x,z}$ is the fine configuration lifted from $V = e^x$ through the forest with free variables z, and Π_t is the block predictor;
5. *finite bound:* for every i, every $\Theta \ge \Theta_0(L)$ and every $c' \le \gamma$, consider the one-loop function
$$\mathrm{th}_{i,t}(x) = \mathrm{loop}_R(U_{x,g_i(x)}) + \tfrac{1}{2}\log\det \mathrm{Hess}_z S_t(x, g_i(x)) - \log\textstyle\prod_e \mathsf{J}(g_i(x)_e).$$
It is analytic at 0, the Hessian determinant is positive near 0, and for all coarse lattice momenta p, p' and real plane waves $w_{p,\xi}(x) = \mathrm{Re}(e^{ip\cdot x}\xi)$ and $w_{p',\zeta}$, with every direction-colour component of ξ, ζ of modulus at most 1,
$$\left| D^2\mathrm{th}_{i,t}(0)[w_{p,\xi}, w_{p',\zeta}] - D^2(\alpha\,\mathrm{quad}_{\mathbf{q},Y} + \mathrm{loop}_Y)(0)[w_{p,\xi}, w_{p',\zeta}] \right| \le |\Lambda_{m_i}|\,\Theta\, e^{-c'm_i}.$$
Here $\mathrm{quad}_{\mathbf{q},Y}$ is the signed-permutation average of the Wilson-plaquette and row-square terms in $\mathrm{main}_{\mathbf{q},Y}$, with the cubic and quartic atoms omitted. Its Hessian is the coarse quadratic form. The comparison function is also analytic at 0. The estimate holds for every t.

On the constants. A_* and s are fixed before (τ, A). The drift is $d = d_{\mathrm{cmp}} + D_{\mathrm{slope}} + 1$, where d_{cmp} is the comparison constant of §4.8 and D_{slope} is the constant of III.7.3. Both may depend on (τ, A) but are independent of L. The construction gives $\Lambda_0(L) = a_0L^4 + b_0\log L + c_0$ and $\Theta_0(L) = b_1L^4$, with nonnegative constants depending on the fixed input parameters; both are $O(L^4)$.

Iteration. Fix positive stage widths and lower bounds for the free quadratic forms, stage amplitude bounds, kernel parameters K and clip bounds. The constants are then chosen in the following order:

1. maximal path widths σ_j and minimal path amplitudes A_j;
2. for each positive sequence $\tau_j \le \sigma_j$ and sequence $A'_j \ge A_j$: drifts $d_j \ge 0$, rates $\gamma_j > 0$ and minimal block factors u_j;
3. for every compatible choice of stage parameters, with direct clip coefficients below their prescribed bounds, drifts at least d_j and block-factor lower bounds at least u_j, and then every admissible choice of block factors: volume thresholds and positive Hessian lower bounds η_j;
4. for every admissible protection sequence θ: entry thresholds Δ_j for the reference coupling.

For each such choice, take post-step constants $\Gamma_j \ge 1$ and $0 < c_j \le \gamma_j$ and the Hessian lower bounds η_j just chosen. The conclusion holds for every history satisfying the compatibility, size and coupling thresholds of III.2; every free-field horizon satisfying the strip, Floor, symmetry, top-identity, interpolation and three-form coercivity conditions of III.10; every second horizon with the same beds and the same interpolation and curl-transport kernels; and every local Haar

chart ratio. For such data there are kinetic data $(\alpha_{N,m}, P_{N,m}, g_{N,m})$ satisfying all the following conditions:

- every $P_{N,m}$ is attached to its bed and satisfies the level-m clauses with loop amplitude $\mathrm{loopCap}_m$;
- $P_{N,N}$ is the Wilson top datum of III.7.2;
- for every $m < N$, every volume index and every axial forest: the second-variation lower bound $\eta_{J_{m+1}}$, stationarity, the derivative condition and the classical match; also $|\alpha_{N,m} + \kappa \log L_m| + 1 \le \mathrm{drift}_{m+1}$, and the finite bound with constants $(\Gamma_{J_{m+1}}, c_{J_{m+1}})$ at loop amplitude $\mathrm{loopCap}_{m+1}$ and also at amplitude 1.

Status:

- The step and iteration are theorems; their output conditions are the definitions used in III.9.
- `transition_cap` proves the step property at every stage using III.7.3 for the input constants. `claiming`, `under_contract` and `geared` iterate it.
- The iteration supplies the one-loop slope internally; its remaining hypotheses are the free-field, history and Haar-chart conditions just listed.
- The coarse-datum construction and induction are also expressed by `color_pipe`, `energized` and `plowed`, together with the preceding step theorem.

Lean: `Transition`, `Scud`, `Hull`, `Bubble`, `FullPaths`, `Kin.Planned`, `ClassicalMatch`, `tested`, `thermal`, `FiniteBound`, `draft`.

Proposition III.7.5 (uniform inverse of the completed free operator). Let j be a stage, K kernel parameters, $\sigma, \varepsilon > 0$ and $\Omega \ge 0$. There is $\eta > 0$, and for every block factor L a constant $A_L \ge 0$, with the following property.

For every admissible patch of stage j and block factor L, and every bed B with $H \in \mathrm{strip}(\sigma/L, \Omega)$ and $\mathrm{Floor}(B, \mathbf{c}, n; \varepsilon, D)$ for all n:

- the inverse $\mathcal{G}_B$ of the block-Bloch symbol of the completed free operator lies in $\mathrm{strip}(\eta, A_L)$ on the coarse lattice;
- for every step with this patch (any coarse side c, fine side cL, any choice of block-kernel data), $\mathcal{C}_B\, \mathcal{G}_B = 1$.

In the rooted coordinates, the joint variables are a fine link field v and a coarse site field h. Before gauge completion, the quadratic form is

$$\langle dv, H\, dv \rangle + a \| -Qv + d_c h \|^2,$$

where Q is the linear block average and d_c the coarse gradient. The completed operator adds $F_{1/L}^{\mathsf{T}} F_{1/L}$, where $F_\mu = d_1^* P_f + \mathfrak{m}_\mu X$, P_f projects onto fine links, and X is minus the derivative of the logarithm of the fine rooted frame. Here d_1 is d_U at $U = 1$. All matrices use physical counting bases. The decay rate η is uniform in L; only the amplitude depends on L.

Status: theorem; used in III.4. *Lean:* `WilsonBulk.uniform_backing`, `completedM`, `jointInverse`.

Proposition III.7.6 (Faddeev–Popov identity for the one-loop term). Fix an admissible step, a clip $\mathbf{p}$ and a datum $R \in \mathrm{Bounds}(\mathbf{p}; \sigma, A, \Lambda)$ for some $\sigma > 0$ and $A, \Lambda \in \mathbb{R}$. Let $g(0) = 0$ be a real-analytic stationary branch for an axial forest, stationary for every choice of block-kernel data, with second variation $\ge \eta \sum_e |v_e|^2$ $(\eta > 0)$. Let $\mu > 0$, let t be a choice of block-kernel data, and let J be a local Haar chart ratio. Then for coarse fields x near 0:

- the completed Hessian $\mathcal{P}_\mu(x)$ has positive determinant;
- $\mathrm{th}(x) = \mathrm{loop}_R(U_{x,g(x)}) + \frac{1}{2}\log\det\mathcal{P}_\mu(x) - \log\det\mathcal{G}_\mu(x)$.

$\mathcal{P}_\mu(x)$ acts on all fine-link and coarse-frame directions. It is the Hessian in left exponential coordinates of the joint action

$$\mathcal{S}_t(V;U,h) = \mathrm{main}_{\mathbf{p},R}(U) + a\sum_e \Big(1 - d_{\mathrm{rep}}^{-1}\mathrm{Re}\,\mathrm{tr}(V_e[\mathrm{gauge}_h\Pi_t(U)]_e^{-1})\Big),$$

at $V = e^x$ and the lifted joint stationary configuration, plus $F_\mu^{\mathsf{T}}F_\mu$. Here $F_\mu = d_U^*P_f + \mathfrak{m}_\mu(U)X$, with the rooted-frame derivative X as in III.7.5. The joint stationary configuration combines the lifted fine field $U_{x,g(x)}$ with its coarse frame. $\mathcal{G}_\mu = d_U^*d_U + \mathfrak{m}_\mu(U)$ is the ghost operator at the lifted configuration. The underlying algebraic identity is

$$T^2\det\mathcal{P}_\mu = \det M\,(\det\mathcal{G}_\mu)^2,$$

with M the axial Hessian and T the Haar Jacobian of the chart.

Status: theorem; used in the renormalization and remainder comparisons of III.4, III.2 and VII.2. *Lean:* `tribute`, `bounty`, `medals`.

3.10.2 2. What it uses

- **I.1 (gauge groups, Wilson action).** E_W, the exponential chart, Ad, the Casimir C_G, and κ with $\kappa > 0$ (`slopeConstant_pos`, `casimir`). Simplicity enters as $\mathrm{tr}(\mathrm{ad}_b\mathrm{ad}_b) = -C_G$ for every unit vector b, and through Schur's lemma.
- **III.1 (block-spin step).** The specifications of III.7.1. Also: steps, patches (block factor L, stage J), axial forests (tree gauge), the fine action plus block penalty, and the stationarity notions (stationarity, second-variation lower bound, derivative condition, equality of jets, classical match). Further: the one-loop function, the finite bound, the kinetic data and the iteration conditions, towers, horizons, and local Haar chart ratios.
- **III.6 (stationary branch).** For every admissible step there is a Hessian lower bound $\eta_j > 0$ (chosen after L) such that every admissible fine datum has a stationary branch with:
  - second variation $\geq \eta_j$;
  - derivative given by the free interpolation in the chosen axial gauge, after subtracting the copied coarse field;
  - quadratic part equal to the coarse bed.

  This is `Integrated.stationary_scene`, packaged here by `scene_gate`. Also from III.6: positivity of the Hessian near 0 (`hessian_pos`) and independence of the one-loop function from the forest (`same_thermal`).
- **III.10 (conditions on beds and steps).**
  - `Bid.Sworn`: signed-permutation invariance of the bed's gauge form.
  - `Bid.HeatG`: quadratic jet of the tree-level action.
  - `Bid.Larger`: size of the torus.
  - `Craft.Rack` and `Bid.Retained`: uniform strip and Floor bounds, symmetry and three-form coercivity for the free-field horizon; and equality of the beds and the two interpolation kernels for the second horizon.

  - `Bid.one_for_all`: the classical match does not depend on the forest.

- **III.2 (order of constants).** The admissible stage parameters, block factors, protection exponents, entry and volume thresholds, and the history compatibility conditions `Supply.Paid`, free-field bounds and constants chosen after the block factors. Also the reference coupling `β̂m` $= \bar{\beta} + \kappa s_m$ and the loop cap `loopCap_m` $= 2\exp\left(\theta_{J_m}(\log\log\hat{\beta}_m)^2\right)$.
- **II.1.** The local Haar chart ratio of the exponential chart (`chart_ok`), used in the iteration.
- **III.16, III.3.** Laurent/Bloch calculus on strips (products, inverses, Cauchy estimates, Riemann sums versus integrals); the block-Bloch ("slab") representation of fine translation-invariant operators on the coarse lattice.
- **Standard mathematics.**
  - Taylor expansion of the plaquette action; polarization (a jet is determined by its restrictions to lines).
  - The Koszul complex of the regular sequence $q_j(p) = e^{ip_j} - 1$ with Hadamard division in a strip, i.e. the lattice Poincaré lemma in momentum space.
  - Schur's lemma for invariant forms on a simple Lie algebra.
  - The Faddeev–Popov procedure.
  - The background-field method ('t Hooft and Abbott; Peskin–Schroeder, §16.6).
  - The one-loop β-function (Gross–Wilczek and Politzer) and its orbital/spin split $-\frac{11}{3} = \frac{1}{3} - 4$.
  - Balaban's propagator bounds and background-field renormalization group; see also Dimock, *The renormalization group according to Balaban*.
  - Rotation-invariant integrals over shells in $\mathbb{R}^4$.
- **Imported constants and assumptions.** No one-loop coefficient is assumed. κ is defined in I.1, and III.7.3 proves that it is the one-loop slope. The hypotheses carried are those displayed in the propositions; the free-field and history assumptions are supplied by III.4.

3.10.3 3. Strategy

Classical part. At every level the quadratic part of the leading action is fixed by the free-field bed of that level. The cubic and quartic parts are curvature atoms, determined by Taylor agreement through order four with a given gauge-invariant function E. At the top, E is the Wilson energy. At each step, E is the tree-level coarse action $x \mapsto S_t(x, g(x))$. The atom-emission lemma (§4.3, Lemma 4.4) serves both cases. Its mechanism has four parts:

1. The Taylor coefficients of a translation sum of local analytic functionals have Fourier kernels on $\mathbb{Z}^4$ that are strip-analytic. Finite-volume coefficients agree with $|\Lambda_n|$ times these kernels up to $o(|\Lambda_n|)$ along any sequence of sides; this is how volume-uniformity enters.
2. Gauge invariance makes the kernels transversal in each leg (linearized Ward identity).
3. Transversal kernels with at least three legs factor through the curvature. This is the lattice Poincaré lemma, proved by contracting the Koszul complex of the differences $e^{ip_j} - 1$, one axis at a time, with Hadamard division. A one-leg solve reduces the width from σ to $\sigma/4$ once, regardless of the four coordinate divisions within it. Successive leg solves cause further width losses, depending only on the fixed number of legs.
4. The real, colour-invariant Laurent coefficients of the factored kernels are the coefficients of atoms, and exponential decay is the strip bound.

One-loop part.

1. In the axial chart the one-loop term is loop $+\frac{1}{2}\log\det - \log$(Haar Jacobian). An exact Faddeev–Popov identity turns it into $\frac{1}{2}\log\det$ of a covariantly completed Hessian minus log det of a ghost operator.
2. Its second-order Taylor coefficient is a two-leg kernel. Its F^2-coefficient α is compared, with an L-independent error, to that of the regulated Feynman-gauge background-field determinant on the fine lattice.
3. The latter is computed. The lattice integrand is the continuum bubble integrand $h(k)$ of the vector loop (weight $\frac{1}{2}$) and the ghost loop (weight -1), up to errors $O(|k|^{-3}+\mu^2|k|^{-6})$; its angular average is $-\frac{11}{3}C_G|k|^{-4}$. Integration over the shell $1/L<|k|\le 1$ gives $-\kappa\log L$, and the rest is bounded by absolute values.
4. After subtracting α times the bed's quadratic kernel, the remaining kernel vanishes to fourth order at zero momentum. It is therefore represented by loop atoms satisfying the two moment conditions, i.e. by irrelevant terms. Inherited fine loop atoms contribute with a factor L^{-2}.

Constants. The output width and cubic/quartic amplitude of a step are fixed before the input amplitude. The input enters the output only with factors L^{-1} or L^{-2}, which a large L makes small. The strip widths and inverse bounds for the completed operators are uniform in L in block-scaled norms. The newly generated loop-atom amplitude is $O(L^4)$. It is absorbed by $\mathrm{loopCap}_m$, which grows with $\hat{\beta}_m$.

Relation to Balaban. The construction uses background-field renormalization with block-spin constraints. Here:

- the effective action is carried by explicit, exponentially weighted path data;
- the F^2-coefficient is obtained by comparison with a regulated covariant determinant plus a Fourier computation, with $O(1)$ error, for an arbitrary simple compact gauge group.

3.10.4 4. Steps

3.10.4.1 4.1 Fourier kernels of local functionals Lemma 4.1 (translation sums of local functionals; limits of kernels). (a) Let F_k be functions of configurations on $\mathbb{Z}^4$. Suppose F_k depends only on links in a box of size d_k, has a holomorphic extension with bound w_k on a polydisc of radius r around the identity, uniformly after translation and periodic restriction, and $\sum_k w_k \exp(4(t+1)^2\sigma d_k)\le A$, with $r,\sigma>0$ and $w_k\ge 0$. Then for every side n the translation sum on Λ_n is analytic at 1. Its $(t{+}1)$-th Taylor coefficient is recorded *exactly* by a kernel that is strip-analytic of width σ and size $A\,c(t,r)$, independent of n. For the Wilson energy, kernels of every order exist with width 1. (b) Suppose a strip-analytic kernel records F_i on Λ_{n_i} with error $|\Lambda_{n_i}|\epsilon_i$, where $n_i\to\infty$ and $\epsilon_i\to 0$. If the F_i are analytic, invariant near 0 under global colour rotations and lattice gauge transformations, and their Taylor coefficients of positive orders $1,\dots,t$ vanish, then the kernel is transversal in each leg, real and colour invariant. *Sketch:* (a) Cauchy estimates on the polydisc bound each local Taylor coefficient; its Fourier support has size at most $4(t+1)^2 d_k$, which gives the stated exponential factor. (b) Each property holds exactly for the finite-volume coefficients; divide by $|\Lambda_{n_i}|$ and pass to the limit along lattice momenta approaching any prescribed real momenta. Strip analyticity gives the required continuity. *Lean:* `gathered`, `Wilson_stream`, `withdrew`, `wedlock`. *(proof read at top level; Ward identity input statement only)*

3.10.4.2 4.2 The lattice Poincaré lemma in momentum space

Lemma 4.2 (one-leg solve). Let the momentum space have d scalar coordinates, including four selected coordinates, and let δ contract differential forms with $q_j(p) = e^{ip_j} - 1$ in those four coordinates. Let $\sigma > 0$, $A \geq 0$, and let f be a form of degree $r \geq 0$ with the following properties:

- each component is strip-analytic of width σ and size A;
- $\delta f = 0$;
- f vanishes to order s along rays;
- on real momenta, f takes values in subspaces unchanged when any of the four selected momentum coordinates varies;
- if $r = 0$, f vanishes where the projected momentum is 0.

Then $f = \delta B$, where B has degree $r + 1$, is strip-analytic of width $\sigma/4$ and size $4A\,\pi_d(\sigma)$ with $\pi_d(\sigma) = (4/\sigma)(2(1 + \sigma/4)/(\sigma/4))^d$, vanishes to order $\max(s - 1, 0)$, and takes values in the same subspaces on real momenta. *Sketch:* Comb one axis at a time. $f(p) - f(p)|_{p_j=0} = q_j(p)D_j(p)$ by Hadamard division (`div_var`). D_j has width $\sigma/4$ and size $A\pi_d(\sigma)$ by Cauchy estimates, while the restriction $f|_{p_j=0}$ keeps width σ. After all four axes, no positive-degree component remains. In degree zero the remainder is f with all selected momenta set to zero, which vanishes by hypothesis. The factor 4 counts the axes; the width loss is a single one. For positive vanishing order, division by $q_j \sim ip_j$ lowers it by one; order zero remains order zero. *Lean:* `wall_lift`.

Lemma 4.3 (transversal kernels are curvature kernels; atoms). For every $t \geq 2$ and $\sigma > 0$, there are $\eta \in (0, \sigma]$ and $K \geq 0$ such that every $(t{+}1)$-leg kernel f, strip-analytic of width σ and size $A \geq 0$ and transversal in each leg, admits the factorization

$$f(p) = \sum_b \prod_i \hat{s}(p_i)_{b_i a_i}\, g_b(p),$$

for real momenta, where $\hat{s}$ is the symbol of the lattice curl and g is a plaquette kernel, strip-analytic of width η and size AK, with η, K depending on t, σ. If f is also real and colour invariant, the Laurent coefficients of g can be taken real, colour invariant and exponentially decaying. Placed on straight routes, they define atoms. If f records an analytic functional invariant under signed permutations, the signed-permutation average of their translation sum reproduces that Taylor coefficient up to the recording error. *Sketch:* Solve the dependent-momentum leg first, then the independent legs by Lemma 4.2, preserving the subspace constraints from the earlier solves. Symmetrize by conjugate reflection to obtain real Laurent coefficients and project the colour tensors onto the invariant subspace. *Lean:* `allleg`, `strokes`, `colors_align`. *(proof read at top level)*

3.10.4.3 4.3 Classical emission and the Wilson jet

Lemma 4.4 (emission of cubic and quartic atoms). Let $\sigma, \sigma_0 > 0$ and $A, Z \geq 0$. There are $\tau \in (0, \sigma]$ and $C \geq 0$ with the following property.

Hypotheses:

- sides $n_i \to \infty$;
- a bed B with $\rho \in \mathrm{strip}(\sigma_0, A)$ and Floor on the one-site torus, and a clip with $c_{\mathrm{dir}} \leq Z$;
- real-analytic functions E_i of link fields on Λ_{n_i}, invariant near 0 under lattice gauge transformations, global colour rotations and signed permutations, such that E_i minus the bare row action of B vanishes to order 3 along rays; the bare row action has the straight-route rows of ρ and no atoms;

- $M \geq 1$ such that the Taylor coefficients of orders 3 and 4 of E_i are recorded exactly by kernels strip-analytic of width σ and size M.

Conclusion: there are invariant cubic atoms X and quartic atoms V with sums of exponential weights at most MC at width τ such that, for every choice f of loop atoms and every i, $x \mapsto E_i(x) - \mathrm{main}_{\mathbf{c},(\rho\text{-rows},f,X,V)}(e^x)$ vanishes to order 5 along rays. *Sketch:* At order 3, subtract the full bare row action, whose quadratic jet agrees with that of E. The residual kernel is transversal (Lemma 4.1(b)) and is matched by cubic atoms (Lemma 4.3). At order 4, subtract the same bare row action and the cubic atoms; the remainder is transversal and is matched by quartic atoms. *Lean:* `assembly`.

Lemma 4.5 (proof of III.7.2).

1. Take $E_i = E_W$ on Λ_{n_i}. The kernels of orders 3 and 4 come from Lemma 4.1(a).
2. Under $D = 1$, Floor and invariance, H agrees with the identity on exact 2-forms. Hence E_W minus the quadratic term of the bare datum (rows of ρ, no atoms) vanishes to order 3 (`top_Wilson`).
3. Lemma 4.4 gives X, V, and Q is formed from the rows of ρ, zero loop atoms, X and V.
4. Order-5 vanishing along every ray of the difference of two analytic functions gives equality of all derivatives of order ≤ 4 (`matched`).
5. The weight clauses come from the row prices $6Ae^{4\sigma_0}$, the zero-momentum normalization (Floor on the one-site torus, `symbol_match`), and the common atom bound $(1 + C_3 + C_4)C$, where C_3, C_4 bound the Wilson Taylor kernels and C is the emission constant.

Lean: `launch`, `beckon`, `budding`.

Lemma 4.6 (every admissible leading part is Wilson up to irrelevant terms). Fix $\sigma, \tau > 0$, $\varepsilon \in \mathbb{R}$ and $C, Z, A \geq 0$. Let $n_i \to \infty$, $c_{\mathrm{dir}} \leq Z$, and let $R \in \mathrm{Bounds}(\mathbf{c}; \tau, A, \Lambda)$ be attached to a bed with $\mathrm{Floor}(B, \mathbf{c}, n; \varepsilon, C)$ for every n, invariant gauge form on each Λ_{n_i} and $H \in \mathrm{strip}(\sigma, C)$. Then the Taylor kernels of orders 2, 3 and 4 of $\mathrm{main}_{\mathbf{c},R} - E_W$ vanish at zero momentum to orders 3, 2 and 1 respectively. Their strip bounds depend only on $(\sigma, \varepsilon, C, Z, \tau, A)$. In words: all admissible leading parts share the marginal (dimension-four) part of E_W. *Sketch:* $\hat{H}(0) = 1$ makes the quadratic difference vanish to order 3. Ward identities then force the cubic and quartic differences to vanish to orders 2 and 1. Apply the same Ward argument to the invariant cubic and quartic atom sums, whose quadratic jets vanish. These statements concern their full nonlinear Taylor coefficients, including transport and curvature expansions. *Lean:* `greyson`, `veneer`, `extra_curves`, `saunter`, `tideMark`. *(proof read at top level)*

3.10.4.4 4.4 Two-leg kernels: the F^2-coefficient and irrelevance Lemma 4.7 (a two-leg kernel without F^2-term vanishes to fourth order). Let G be simple. Let f be a two-leg kernel that is strip-analytic, transversal, real, colour invariant, symmetric under exchange of the legs, and invariant under signed permutations. If the p_q^2-coefficient of $f_{(b,b),(w,w)}(p)$, for p along an axis $q \neq w$, vanishes for one colour b, then f vanishes to order 4 at zero momentum. Consequently, for every bed with positive metric strip width, $\hat{H}(0) = 1$ and signed-permutation-invariant quadratic form, $f - \alpha_b(f)\,\hat{K}_B$ vanishes to order 4, where $\hat{K}_B$ is the bed's quadratic kernel. *Sketch:*

- Transversality gives vanishing to order 2.
- The third-order coefficient is odd under $p \mapsto -p$, hence 0 by reflection symmetry.
- By transversality, hypercubic symmetry and Schur's lemma, the second-order coefficient is a multiple of $(p^2\delta_{aa'} - p_a p_{a'})\delta_{bb'}$, and the multiple is the given coefficient. *Lean:*

`TributeColor.vanish`, `subtract_fee`. *(proof read at top level)*

Lemma 4.8 (loop atoms are irrelevant). Let $R \in \mathrm{Bounds}(\mathbf{c}; \sigma, C, \Lambda)$. The two-leg kernel of the loop atoms, symmetrized over legs and lattice symmetries:

- records loop_R exactly on every torus;
- is strip-analytic of width σ and size $\Lambda\, r(\sigma)$;
- vanishes to order 4 at zero momentum.

Sketch: The kernel is $\hat{s}(-p)^{\mathsf{T}} \hat{K}(p) \hat{s}(p)$ with $\hat{K}(p) = \sum \mathrm{coeff}\, e^{ip\cdot y}$. Here $\hat{s} = O(p)$, and the two moment conditions give $\hat{K}(0) = 0$ and $\nabla \hat{K}(0) = 0$. *Lean:* `filing_ready`.

3.10.4.5 4.5 The one-loop term in covariant form Lemma 4.9 (proof of III.7.6). Let H_{joint} be the Hessian of the joint action before gauge completion, and let R select its axial directions. Three determinant identities combine:

- the completed Hessian factors as $\det \mathcal{P}_\mu = (\det \mathcal{G}_\mu)^2 \det(\mathsf{R}^{\mathsf{T}} H_{\mathrm{joint}} \mathsf{R})$ (`root_determines`);
- the axial Hessian satisfies $\det M = (\det \partial s)^2 \det(\mathsf{R}^{\mathsf{T}} H_{\mathrm{joint}} \mathsf{R})$, where s is the left-trivializing chart change (`certified`);
- the Haar Jacobian is $T = |\det \partial s|$ (`treasury`, by comparison of densities on compact sets).

Together they give $T^2 \det \mathcal{P}_\mu = \det M (\det \mathcal{G}_\mu)^2$. $\det M > 0$ by the second-variation lower bound and $\det \mathcal{G}_\mu > 0$ since $\mathcal{G}_\mu \geq \mathfrak{m}_\mu > 0$; take logarithms.

In addition, the one-loop function is analytic at 0 (`born`), and a finite bound proved for one forest transfers to any other (`transfer_bound`). *Lean:* `tribute`. *(proof read; determinant factorization inputs statement only)*

3.10.4.6 4.6 Uniform invertibility of the fluctuation operators Lemma 4.10 (real stability and complex invertibility). Fix a stage j, a penalty $a > 0$, a lower bound $\varepsilon > 0$ and a bound $A_F \geq 0$ for the gauge-completing map F. Let

$$\mathcal{C}_F(p) = \mathcal{Q}_B(p) + F(-p)^{\mathsf{T}} F(p),$$

where $\mathcal{Q}_B$ is the block-Bloch symbol of the joint quadratic form in III.7.5. Suppose B has metric strip bound $(\sigma/L, \Omega)$ and Floor lower bound ε on every torus, and F is holomorphic and periodic in width $\sigma > 0$, with block-scaled norm at most A_F. On real momenta require

$$F(p) R_{\mathrm{gauge}}(p) = \mathcal{G}_{1/L}(p), \qquad F(-p)^{\mathsf{T}} = F(p)^*,$$

where R_{gauge} inserts a fine gauge direction into the joint variables and $\mathcal{G}_{1/L}$ is the free regulated ghost symbol.

(a) There is $y > 0$, depending only on $(G, j, a, A_F, \varepsilon)$, such that $\|\mathcal{C}_F(p) v\| \geq y \|v\|$ in the block-scaled coordinates for every real p, every L and every patch of stage j.

(b) If $0 < \sigma \leq 1$, $\Omega \geq 0$, the patch is admissible, and F represents the rooted gauge-completing map on a sequence of tori with coarse sides tending to infinity, then there are $\eta \in (0, \sigma]$ and $C \geq 0$, depending only on $(G, j, K, \varepsilon, \sigma, \Omega, A_F)$, such that $\mathcal{C}_F^{-1}$ is holomorphic in width η and has block-scaled norm at most C, uniformly in L.

Sketch: The real estimate controls the quotient by gauge directions using the quadratic form and block penalty; the displayed identity controls the remaining gauge directions. Strip bounds give a uniform perturbation estimate from real to nearby complex momenta. Choose the width so that this perturbation is smaller than the real lower bound.

Lean: `stadium`, `post_cert`. *(proof read at top level; real quotient and strip perturbation estimates not expanded)*

Lemma 4.11 (all five operators). Fix the numerical parameters of III.7.4. There are $0 < \eta \le w_0 \le \min(w, \delta, 1/128)$ and $C \ge 0$, depending only on the free-field bounds, G, K and J, with the following property. Under the geometric, fine-bed, fine-datum and interpolation hypotheses of III.7.4, form the actual fluctuation symbols using the free interpolation A_0 and the block predictor. At zero background, the following five operators are invertible in width η, with inverse norm at most C in their respective block-scaled coordinates, uniformly in L and in the input path width, cubic/quartic amplitude and loop amplitude:

- the auxiliary operator;
- its block-aliased copy;
- the completed Hessian;
- the completed Hessian without the contact term;
- the ghost operator.

The unaliased free auxiliary operator and the free regulated ghost operator satisfy corresponding inverse bounds. *Sketch:* Bound the fine auxiliary and ghost inverses directly in Fourier space and transfer the auxiliary bound to its aliased copy. Apply Lemma 4.10 to the joint completion, then use finite-volume compatibility for the completed and contact-removed symbols. Take minima of widths and maxima of constants. *Lean:* `honored`, `GrantF`, `completion_choice`. *(proof read at top level)*

Lemma 4.12 (proof of III.7.5).

1. Choose (η, C) by Lemma 4.10(b) for the regulator family of the completed operator.
2. For each L, a constant $W(L)$ converts block-scaled norms into the plain strip norm.
3. The bound on the inverse gives the strip bound $(\eta/2,\ \mathrm{cost}(\eta, CW(L)))$.
4. The inverse Fourier coefficients are real by finite-volume compatibility. On every torus their Fourier multiplier is the inverse symbol, which gives the matrix identity.

Lean: `uniform_backing`.

3.10.4.7 4.7 The slope of the regulated determinant Lemma 4.13 (the continuum bubble). Assume G is simple. Take an external field of colour b, polarization e_1 and momentum $t\,e_0$, and loop momentum $k \neq 0$.

- The vector bubble has orbital vertex $-2ik_1$ and spin vertex $\pm 2it$ in the $(0,1)$ plane.
- Both bubbles carry the colour trace $\mathrm{tr}(\mathrm{ad}_b^2) = -C_G$, which uses simplicity.
- The propagators are $|k|^{-2}$ and $|k + te_0|^{-2}$.

Writing $c_{\mathrm{gh}}(t)$ for the ghost bubble and $c_{\mathrm{vec}}(t)$ for the vector bubble,

$$\partial_t^2 c_{\mathrm{gh}}(0) - \tfrac{1}{2}\,\partial_t^2 c_{\mathrm{vec}}(0) = 2h(k), \qquad h(k) = 4C_G\Big(\frac{k_1^2}{|k|^6} - \frac{4k_1^2k_0^2}{|k|^8} - \frac{1}{|k|^4}\Big).$$

The terms in k_1^2 come from the orbital coupling: one adjoint scalar's worth, $\frac{1}{2}\cdot 4$ vector components minus one ghost. The term $-|k|^{-4}$ comes from the spin coupling of the vector field. *Lean:* `haloRate`, `poleVec`.

Lemma 4.14 (the shell integral). For $0 < \mu \le 1$,

$$\frac{1}{(2\pi)^4}\int_{\mu<|k|\le 1} h(k)\,d^4k = -\kappa \log \mu^{-1}.$$

Sketch: By rotation invariance $\int k_1^2|k|^{-6} = \frac{1}{4}\int |k|^{-4}$, and $\int k_1^2 k_0^2|k|^{-8} = \frac{1}{24}\int |k|^{-4}$, the latter from averaging over $e_i \pm e_j$. Also $\int_{\mu<|k|\le 1} |k|^{-4} = 2\pi^2 \log \mu^{-1}$. So h averages to $4C_G(\frac{1}{4} - \frac{1}{6} - 1)|k|^{-4} = -\frac{11}{3}C_G|k|^{-4}$: orbital $+\frac{1}{3}C_G$, spin $-4C_G$. Finally $\frac{11}{3}C_G \cdot 2\pi^2/(2\pi)^4 = \kappa$. *Lean:* `poleSpin`, `joy_two`, `joy_heights`, `polar_m`.

Lemma 4.15 (the lattice integrand is close to h). Under the hypotheses of III.7.3, let Φ_μ be the unintegrated second-derivative kernel of Π_μ, built from the vertex kernels of Lemma 4.6:

$$\Phi_\mu = \tfrac{1}{2}\,(\text{vector film}) - (\text{ghost film}).$$

Let $k_b(k) = \frac{1}{2}\partial_s^2 \Phi_{\mu,(1,b),(1,b)}(s\,e_0; k)|_{s=0}$ be its integrand at loop momentum k. The recorded quadratic Taylor kernel is

$$\hat{Y}_\mu = \frac{1}{2(2\pi)^4}\int_{[-\pi,\pi]^4} \Phi_\mu(\,\cdot\,; k)\,d^4k,$$

and $\alpha_b(\hat{Y}_\mu) = (2\pi)^{-4}\int_{[-\pi,\pi]^4} \operatorname{Re} k_b(k)\,d^4k$. There are $a, b' \ge 0$, uniform over all data of III.7.3 and all $\mu \in (0,1]$, such that:

$$|k_b(k)| \le \frac{a}{(|k|+\mu)^4} \text{ on the Brillouin cube,} \qquad |k_b(k) - h(k)| \le b'(|k|^{-3} + \mu^2|k|^{-6}) \text{ for } |k| \ge \mu.$$

Sketch: Each film is (tadpole) − (bubble). The tadpoles are $O(|k|^{-2} + \mu^2|k|^{-6})$. The lattice bubbles differ from the continuum ones by $O(|k|^{-3} + \mu^2|k|^{-6})$, using the softness of Lemma 4.6. Lemma 4.13 identifies the continuum combination. *Lean:* `astronomy`, `Rally.coin`, `Rally.shell`, `polar_clock`, `amena`. *(proof read at top level; analytic tadpole and bubble estimates not expanded)*

Lemma 4.16 (from the bounds to the slope). For $0 < \mu \le 1$, under the strip regularity and two bounds of Lemma 4.15 with $a, b' \ge 0$,

$$\left|\alpha_b + \kappa \log \mu^{-1}\right| \le (2\pi)^{-4}\left(a\tfrac{\pi^2}{2} + a(2\pi)^4 + 3\pi^2 b'\right).$$

Sketch: Split the cube into three regions.

- On $|k| \le \mu$: the integrand is $\le a\mu^{-4}$ and the volume is $\pi^2\mu^4/2$.
- On the shell: compare with h and use Lemma 4.14; the error is $\int b'(|k|^{-3} + \mu^2|k|^{-6}) \le 3\pi^2 b'$.
- On $|k| > 1$: the integrand is $\le a$.

Lean: `absoluteGeode`.

Lemma 4.17 (proof of III.7.3).

1. Lemma 4.6 and Lemma 4.1 give the vertex kernels of $\text{main}_{\mathbf{c},R} = E_W + (\text{main} - E_W)$.

2. The film $\Phi_{1/L}$ represents $\Pi_{1/L}$ (`gemF`). For fixed L it is strip-analytic, since the vector and ghost symbols are invertible at real momenta, by Floor and the regulator, and hence in a strip (`HouseWord.done`).
3. One half of the normalized loop-momentum integral of the second-derivative film is the Taylor kernel $\widehat{Y}$. Comparing the torus Riemann sum with the integral gives an error $|\Lambda_{n_i}|K_L e^{-z_L n_i/4}$ (`profit`).
4. Lemmas 4.15–4.16 with $\mu = 1/L$ give (b), with $D = (2\pi)^{-4}(\dots)$ independent of L.

Lean: `pouring`.

3.10.4.8 4.8 From the regulated determinant to the blocked determinant Lemma 4.18 (comparison of F^2-coefficients). Assume G is simple and the hypotheses of III.7.4, together with the interpolation, vertex and inverse bounds of Lemma 4.11. Consider the two-leg films of:

- the medal film: $\frac{1}{2}\log\det$ of the completed Hessian minus $\log\det$ of the ghost operator, with regulator $1/L$;
- the core film: the same with the contact term removed;
- the auxiliary film: built from the block-aliased auxiliary operator.

Then:

- the medal film is strip-analytic of width $\eta/2$ and size $L^4 a$;
- one half of its normalized loop-momentum integral records the determinant part of the one-loop function along the branch on Λ_{m_j}, with error $|\Lambda_{m_j}|\, c_L e^{-\eta m_j/8}$;
- $|\alpha_b(\text{medal}) - \alpha_b(\text{auxiliary})| \le d_{\text{cmp}}$, with d_{cmp} depending on the stage and vertex bounds, including the input width and amplitude, but not on L.

Sketch:

- For the contact contribution, the tested diagonal entry of the integrated kernel vanishes to order 3 along the external momentum ray. Hence $\alpha(\text{medal}) = \alpha(\text{core})$ (`drift_tip`).
- The difference core − auxiliary is strip-analytic with an L-independent bound. Cauchy estimates bound its F^2-coefficient (`boundedTip`).

Lean: `quoting`, `medalCue`, `coreCue`, `rankCue`. *(proof read at top level; contact identities not expanded)*

Lemma 4.19 (aliasing preserves α). Under the hypotheses of Lemma 4.18, the F^2-coefficient of the auxiliary film on the coarse lattice equals $\alpha_b(\widehat{Y})$, where $\widehat{Y}$ is the kernel of III.7.3 for $\Pi_{1/L}$ on the fine tori Λ_{Lm_i}. *Sketch:* The coarse symbol at momentum p is the sum over the L^4 aliases of the fine symbol. The nonzero aliases vanish to order 4. The zero alias agrees through order two with $L^4 L^{-2}$ times the fine kernel at p/L. The additional L^{-2} from momentum rescaling gives $L^4 L^{-2} L^{-2} = 1$, so the F^2 coefficient is unchanged. *Lean:* `pole_reduced`, `carry_plan`, `fine_alpha`. *(proof read at top level; high-alias and zero-alias estimates statement only)*

3.10.4.9 4.9 The step and the iteration Lemma 4.20 (construction of the coarse datum). Assume the geometric, bed, Haar-chart and stationary-branch hypotheses of III.7.4. Fix a common Fourier width $\sigma > 0$, coarse-bed width $w > 0$, coarse metric and root bound $M \ge 0$, and a bound on the coarse direct clip coefficient. Require $H_T, \rho_T \in \text{strip}(w, M)$. There

are $s \in (0, \min(\sigma, w)]$ and $K, D \geq 0$, independent of the following kernel amplitudes, with this property.

Suppose the tree-level action has exact Taylor kernels of orders 3 and 4, of width σ and size $X \geq 1$. Suppose a two-leg kernel $\widehat{A}$ of width σ and size Φ records the determinant part with errors $|\Lambda_{m_i}|e_i$, where $e_i \geq 0$ and $e_i \to 0$. Suppose a two-leg kernel $\widehat{L}$ of the same width and size R_0 records the inherited loop functional exactly and vanishes to order 4 at zero momentum. Set $\alpha = \alpha_b(\widehat{A})$ for one fixed colour b.

There is a coarse datum Y, with the straight-route rows of T and attached to T, whose classical jets agree through order four. It satisfies the weight clauses at every $0 < s' \leq s$, every row/cubic/quartic bound

$$A' \geq \max\{6Me^{4w}, XK\},$$

and every loop bound

$$\Lambda' \geq (\Phi + |\alpha|C_T + R_0)D,$$

where C_T is the uniform strip bound for the quadratic kernel of T at width $\min(\sigma, w)$.

For every i, Θ, c', Ω, the finite bound of III.7.4 holds with right-hand side $|\Lambda_{m_i}|\Theta\Omega e^{-c'm_i}$ whenever $B_G e_i \leq \Theta\Omega e^{-c'm_i}$, where $B_G = 2(4 \dim \mathfrak{g})^2$.

Sketch:

- Apply Lemma 4.4 to the tree-level action. Its quadratic jet agrees with the bare row action by the stationary-branch hypotheses.
- The determinant and inherited-loop functionals are analytic and invariant under gauge, colour and signed-permutation symmetries. Their recorded kernels inherit these properties. Thus $\widehat{A} - \alpha\widehat{K}_T + \widehat{L}$ vanishes to order 4 by Lemma 4.7 and the hypothesis on $\widehat{L}$.
- Represent this residual kernel by two-leg curvature atoms with zero zeroth and first displacement moments (`reach_symbol`). The weighted Laurent bound gives the stated loop amplitude.
- The Faddeev–Popov identity converts the determinant recording error to the error for the one-loop function. Expanding real waves into complex basis waves gives the factor B_G (`draft_pipe`). Classical jet equality follows from order-5 ray vanishing (`achieved`, `violet`).

Lean: `color_pipe`. *(proof read at top level)*

Lemma 4.21 (proof of III.7.4).

- *Coercivity.* Lemma 4.11 at widths $\min(w, \delta)$.
- *Tree-level vertices.* Cubic and quartic vertex bounds have the form $a + L^{-1}\ell$ and $c + L^{-1}r$ (`setWest`). By `exchangeYield`, the order-3 and order-4 kernels of the tree-level action, the latter including the one-propagator exchange of two cubic vertices, are bounded by

  $$M = 1 + \tfrac{a+1}{6} + \tfrac{(c+1)+3(a+1)^2E}{24}.$$

  Here E bounds the completed free inverse. The term $3(a+1)^2E$ bounds the three pairings of two cubic vertices joined by one propagator.
- *Inherited loops.* The inherited loop kernel has size $L^{-2}\Lambda C_0$ (`bequest`).

- *The slope.* Lemma 4.18 and Lemma 4.19 give $\alpha = \alpha_b(\text{medal film})$ with $|\alpha + \kappa \log L| \le d_{\text{cmp}} + D_{\text{slope}}$.
- *Assembly.* Lemma 4.20 builds Y.
- *Choice of L.* $L \ge u$ is chosen so that L^{-1} times the input-dependent constants lies below the output thresholds, and $L^{-2} C_0 D_{\text{lift}} \le \frac{1}{4}$, where D_{lift} is the loop-representation constant of Lemma 4.20 (`age_control`). The loop amplitude is the sum of a newly generated term and an inherited term. Their bounds are $\le \Lambda'/2$ and $\le \Lambda'/2$, respectively, when $\Lambda' \ge \Lambda_0(L)$ and $\Lambda' \ge \Lambda/2$ (`cap_invoice`).

Lean: `transition_cap`. *(proof read at top level)*

Lemma 4.22 (loop caps and the iteration).

- *Loop caps.* For every W_j there are thresholds Δ_j such that, once $\hat{\beta}$ at the entry exceeds them, $W_{J_{m+1}} \le \text{loopCap}_m$, $\text{loopCap}_{m+1} \le 2\,\text{loopCap}_m$ and $\text{loopCap}_m \ge 1$. The history must satisfy the stage, block-factor and entry conditions of III.2, with positive protection exponents nonincreasing with stage. Use $\hat{\beta}_{m+1} = \hat{\beta}_m + \kappa \log L_m$, and $(\log\log(s+k))^2 - (\log\log s)^2 \le 3k/s$ for each fixed $k \ge 0$ and all sufficiently large s (`bump`).
- *Iteration.*
  - `claiming` applies Lemma 4.21 to the stage pairs (j, j) and $(j+1, j)$ and takes minima and maxima of the constants.
  - `energized` produces one step's record at level m.
  - `plowed` assembles the kinetic data by downward induction from $m = N$, starting from the top datum.
  - `under_contract` chooses the volume thresholds so that $F(L)e^{-\gamma n} \le \Gamma e^{-(\gamma/2)n}$ (`waits_up`).
  - The final iteration uses III.7.3.

Lean: `clocks`, `geared`.

3.10.5 5. Where the difficulty lies

- **The coefficient and its identification.** The continuum bubble yields the factor 11/3 from the orbital and spin vertices and $\text{tr}(\text{ad}_b^2) = -C_G$. Relating this calculation to the blocked determinant also requires the exact Faddeev–Popov identity, cancellation of the contact contribution to α, a comparison uniform in L, and control of every Fourier alias.
- **Quantitative curvature factorization.** The lattice Poincaré lemma must preserve exponential decay, colour invariance and the constraints already imposed on other legs. Each one-leg solve loses a factor four in strip width; the number of legs is fixed.
- **Uniformity in the volume.** A single infinite-volume kernel or path datum works on every torus in the chosen sequence. Limits along increasingly fine momentum meshes transfer Ward identities and colour symmetries to that kernel. In the iteration, the sequence consists of the sides at one fixed level as the volume index varies.
- **Order of constants.** The output width and cubic/quartic amplitude precede the input path amplitude. Contributions depending on the latter carry a factor L^{-1}; inherited loop atoms carry L^{-2}. The new loop amplitude is $O(L^4)$, and the reference-coupling thresholds make the allowed loop amplitude large enough to absorb it.
- **Strength of the conclusion.** The slope estimate has an $O(1)$ error uniform in L. This

suffices for III.9 when $\log L$ dominates that error. Polymer activities are absent from the argument.

- **The regulator.** The scalar regulator is $\mu^{20}(\Delta_U+\mu^2)^{-9}$. Its componentwise extension to link fields respects gauge transformations and unsigned axis permutations; full signed-permutation symmetry is not asserted for this auxiliary determinant. The blocked action has the full lattice symmetry.

3.10.6 6. Notation

A determinant film records the second derivative before loop-momentum integration. Its quadratic Taylor kernel is one half of its normalized loop-momentum integral.

Written proof	Lean	Meaning
$R = (\ell, \mathcal{A}^{(2)}, \mathcal{A}^{(3)}, \mathcal{A}^{(4)})$	`RG.PathData`	path datum (rows, loop/cubic/quartic atoms)
leg, atom	`RG.Slot`, `RG.Atom`	transported curvature; curvature monomial
Bounds$(\mathbf{c}; \sigma, C, \Lambda)$	`PathData.Bounds`	exponential summability, invariance and moment clauses
$\mathrm{main}_{\mathbf{c},R}$, loop_R	`mainTerm`, `loop`	leading part; loop functional
clip $\mathbf{c}$, c_{dir}	`RG.Clip`	clipping parameters; `direct`
bed B, Floor, attached	`Bed`, `Floor`, `Attached`	free quadratic-form data of a level
straight-route rows of ρ	`rowPath`, `emitBase`	rows realizing the bed roots on straight lattice routes
strip-analytic (σ, K)	`Firm`	bounded holomorphic periodic kernel
vanishing to order s along rays	`Rays`	raywise $O(t^s)$ at $t = 0$
$(t{+}1)$-leg kernel; records	`Picture`, `Recorded`	Fourier kernel of a Taylor coefficient
$\alpha_b(\widehat{K})$	`alphaPay`	F^2-coefficient, normalized by the bed
Π_μ	`PoleTest`, `fineMag`	regulated background-field one-loop action
$\mathcal{G}_\mu$, $\mathcal{P}_\mu$	`grenade`, `medals`	ghost operator; completed Hessian
$\mu = 1/L$	`inch`	regulator mass
$h(k)$; $k_b(k)$	`poleVec`; `keelRay`	continuum bubble integrand; lattice integrand
bounds of Lemma 4.15	`Waves1`, `Waves2`	global infrared bound and shell comparison bound
$(2\pi)^{-4}$	`cartWeight`	normalization of the momentum integral
$S_t(x, z)$; th	`tested`; `thermal`	fine action plus penalty; one-loop function
classical match; finite bound	`ClassicalMatch`; `FiniteBound`	items 4 and 5 of III.7.4

Written proof	Lean	Meaning
slope statement	`HorizCalc`, `Fluent`	III.7.3 for given / all parameters
step property; step record	`Transition`; `Bubble`	III.7.4
top datum	`BranchRow`	III.7.2 at level N
kinetic data, planned conditions	`Kin`, `Kin.Planned`, `FullPaths`	output to III.9
invertibility bounds	`GrantF`, `Freezer`	strip invertibility with inverse bound
medal, core, auxiliary films	`medalCue`, `coreCue`, `rankCue`	§4.8
$\hat{\beta}_m$, $\mathrm{loopCap}_m$	`hatBeta`, `loopCap`	reference coupling; loop-amplitude cap
κ	`Gauge.slopeConstant`	$11C_G/(24\pi^2)$
$D^r F(0)/r!$	`noteAt`	complexified Taylor coefficient
analytic gauge and lattice symmetry	`Pointwise`	hypotheses of atom emission
$\pi_d(\sigma)$	`priceDivision`	one-coordinate division constant in d momentum coordinates
joint free form; gauge completion	`frontR`, `postroom`, `cannonade`	fine-link and coarse-frame operators

3.10.7 7. What remains unexplained

1. **Analytic estimates below the stated lemmas.** The real quotient estimate and complex-strip perturbation estimate used in Lemma 4.10, and the analytic tadpole and bubble bounds used in Lemma 4.15, have not been checked through every subsidiary proof. Their statements support the estimates given here. The determinant-factorization and Haar-density lemmas of Lemma 4.9 remain inputs at statement level. The contact identities used in Lemma 4.18 and the high-alias and zero-alias estimates used in Lemma 4.19 likewise remain inputs at statement level.
2. **Choice of regulator power.** The ninth power is part of the explicit definition, and the cited estimates use it. A minimal power, or a conceptual reason for choosing exactly nine, has not been established.
3. **Supply of the global hypotheses.** III.4 supplies the free-field horizons and history conditions. The `free/stationary connection` explicitly constructs the two horizons with identical beds and interpolation kernels before applying the iteration. The discharge of these conditions in every downstream application has not been checked.
4. **Limits of path data.** The projection results support the compactness argument in III.13: exponential weights give Hilbert-space parameters for curvature coefficients (`fitting`); coefficientwise convergence with uniform row bounds gives convergence of row-action jets (`insideLine`); compatible source jets and a common positive Hessian lower bound give continuity of stationary, tree-level and one-loop jets (`interview`). The complete compactness construction and passage through all its quantifiers belong to III.13 and remain outside this unit.

3.11 III.8. Connected activities and backward dependence cones

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/RG08/Steam/`, `YM/RG08/Forms/`, `YM/RG08/Skiff/`, `YM/RG08/Jetty/`, `YM/Push/GRID_APPROXIMATION/`. The support packages are listed in §7. **Size:** 4,147 lines in the main-theorem closure:

- 1,093 lines for the backward cones (`GRID_APPROXIMATION`);
- 1,083 lines for the connected heat expansion (`Steam` 138, `Forms` 412, `Skiff` 306, `Jetty` 227);
- about 1,970 lines in the support packages listed in §7.

The unit has two independent halves.

- **Part A, connected activities.** Bounds at fixed order for a recursive heat expansion, organised by powers of the coupling g, the heat time t and the field. For finite index sets and symmetric inputs, its unmarked coefficients satisfy the formal equation for $\log \mathbb{E}_{t\Gamma} e^{V}$. A second recursion treats one marked insertion. Both admit polynomial bounds in an input size W and Lipschitz bounds for differences, uniform in the index set and geometry.
- **Part B, backward dependence cones.** The pull-back of a coarse observable through the block-spin kernel of III.1 is its conditional expectation given the fine field. It is local in an explicit fine window, and pull-backs of observables on disjoint coarse sets multiply exactly. Along a history the radius of the window grows by an explicit affine map whose increments, in physical units, decay geometrically.

3.11.1 1. What this unit proves

3.11.1.1 Part A *Conventions.*

- I is an arbitrary set (of cells), d is a pseudometric on I, and $\sigma \geq 0$.
- A rank-n tensor is a function $F : I^n \to \mathbb{R}$ with ordered slots and no symmetry factors. For a tuple $x \in I^n$, $\operatorname{diam}_d(x) = \max(0, \max_{i,j} d(x_i, x_j))$, with value 0 for the empty tuple.
- The *rooted norm* is

$$\|F\|_\sigma = \max_{1\leq i\leq n} \sup_{z\in I} \sum_{x\in I^n,\ x_i=z} |F(x)|\ e^{\sigma\ \operatorname{diam}_d(x)} \ \in [0,\infty].$$

  For $n = 0$ set $\|F\|_\sigma = 0$: a rooted bound imposes no condition on the scalar coefficient. Suprema over empty index sets also have value 0.
- A *propagator* is any $\Gamma : I \times I \to \mathbb{R}$; symmetry is not assumed. It is measured by its weighted row and column sums,

$$[\Gamma]_\sigma = \max\Big(\sup_u \sum_v |\Gamma(u,v)| e^{\sigma d(u,v)},\ \sup_v \sum_u |\Gamma(u,v)| e^{\sigma d(u,v)} \Big).$$

- A pair (F, F') is (W,δ)-*controlled* if $\|F\|_\sigma \leq W$, $\|F'\|_\sigma \leq W$ and $\|F - F'\|_\sigma \leq W\delta$. The same definition applies to a pair of propagators, with $[\cdot]_\sigma$ in place of $\|\cdot\|_\sigma$.
- A *polynomial majorant* is a function P with $0 \leq P(x) \leq Cx^t$ for all $x \geq 1$, for some $C \geq 0$ and $t \in \mathbb{N}$. The class is closed under sums and products.

Formal potentials. A formal potential is a family $V = (V_{e,n})_{e,n\geq 0}$ of rank-n tensors; $V_{e,n}$ is the coefficient of g^e with n field variables. In Gaussian applications the variables may include

spectators, on which Γ vanishes. V is *eligible* if $V_{0,n} = 0$ for all n, and $V_{e,n} \neq 0$ implies $n \leq e+2$ and $e+n$ even. Eligibility does not require symmetry.

Heat coefficients. The heat coefficients $V^{(j)} = V^{(j)}(\Gamma, V)$ are defined by $V^{(0)} = V$ and

$$V^{(j+1)}_{e,n} = \frac{1}{j+1}\Big[\binom{n+2}{2}\Delta_\Gamma V^{(j)}_{e,n+2} + \sum_{c=0}^{n}\sum_{i=0}^{j}\sum_{a=0}^{e}\frac{(c+1)(n-c+1)}{2}\, V^{(i)}_{a,c+1} \star_\Gamma V^{(j-i)}_{e-a,n-c+1}\Big]. \tag{A1}$$

Here Δ_Γ (contraction) and $\star_\Gamma$ (join) are defined in Step A1. For infinite I, the recursions assign value 0 to a sum that is not summable. The bounds below justify the required sums absolutely for positive-rank unmarked outputs and for all marked outputs; they make no convergence assertion for unmarked scalar contractions. The formal heat-equation interpretation for finite I and symmetric V, Γ is given in A1.

Proposition III.8.1 (bounds at fixed order for the connected heat coefficients). Fix $j, e, n \in \mathbb{N}$ and put

$$T_j(e,n) = \{(a,r) : 1 \leq a < e,\ 1 \leq r \leq a+2\} \cup \{(e,r) : 1 \leq e,\ n < r \leq e+2\},$$

enlarged by (e,n) when $j = 0$, $e \geq 1$ and $1 \leq n \leq e+2$. There is a polynomial majorant P with the following property. Let I be a set, d a pseudometric on it, $\sigma \geq 0$, $\delta \geq 0$, $W \geq 1$, V, V' eligible and Γ, Γ' propagators. Suppose (Γ, Γ') is (W,δ)-controlled, and so is $(V_{a,r}, V'_{a,r})$ for every $(a,r) \in T_j(e,n)$. Then $(V^{(j)}_{e,n}(\Gamma, V),\ V^{(j)}_{e,n}(\Gamma', V'))$ is $(P(W),\delta)$-controlled.

- P depends on (j,e,n) only. It does not depend on I, d, σ or δ, hence not on the number of cells, the period, the scale or the stage at which it is applied.
- For $j \geq 1$, inputs at the same order e have strictly higher rank.
- For $n = 0$ the conclusion is empty.

Status: theorem. *Lean:* `YM.RG08.Steam.navigation`, input range `taps`, norms `Held`, `Wire`, `Cot`, `WireCot`, majorants `Wage`.

Proposition III.8.2 (support; the heat-time series is finite). Let V be eligible and Γ arbitrary.

- Every $V^{(j)}$ is eligible.
- $V^{(j)}_{e,n} \neq 0$ implies $n + 2j \leq 3e$.
- For $j \geq 1$, $V^{(j)}$ is symmetric under permutations of its slots.

Hence, at fixed (e,n), the heat-time series is zero if $n > 3e$ and otherwise is a polynomial in t of degree at most $\lfloor (3e-n)/2 \rfloor$. For every real t the complete series is

$$\mathcal{E}_t(V)_{e,n} = \sum_{j=0}^{3e} t^j V^{(j)}_{e,n} = V_{e,n} + \mathcal{R}_t(V)_{e,n}, \qquad \mathcal{R}_t(V)_{e,n} = \sum_{j=1}^{3e} t^j V^{(j)}_{e,n},$$

and $\mathcal{E}_t(V)$ is eligible. *Status:* theorem. *Lean:* `preserves`, `live`, `off`, `coal`, `ember`, `cinder`, `smoulder`, `appropriate`.

Proposition III.8.3 (expansion with one marked insertion).

Test functions and their norms. Let $\mathcal{S}$ be the real vector space of smooth real-valued functions on $\mathbb{R}^4$ whose derivatives of every order have polynomial growth. For $k \in \mathbb{N}$ and $\sigma > 0$, using the Euclidean norm $|y|$, put

$$\|f\|_{k,\sigma} = \max_{|\alpha|\le k} \sup_{y\in\mathbb{R}^4} |\partial^\alpha f(y)| e^{-\sigma|y|},$$

and, for a real-linear functional λ on $\mathcal{S}$, $\|\lambda\|^*_{k,\sigma} = \sup\{|\lambda(f)| : \|f\|_{k,\sigma} \le 1\}$.

Marked tensors and the marked norm. A marked rank-n tensor assigns to each $x \in I^n$ a real-linear functional $M(x)$ on $\mathcal{S}$. Let $h : I \to [0,\infty)$ be 1-Lipschitz for d (a height, for instance a distance to the insertion), and set $h(x) = \max(0, \max_i h(x_i))$ for a tuple x. The marked norm is the unrooted weighted ℓ^1 norm

$$\|M\|_h = \sum_{x\in I^n} \|M(x)\|^*_{k,\sigma}\, e^{\sigma h(x)}.$$

Unlike the rooted norm, it gives a genuine bound for $n = 0$. A pair (M, M') is (U,δ)-controlled if $\|M\|_h \le U$, $\|M'\|_h \le U$ and $\|M - M'\|_h \le U\delta$.

Marked heat coefficients. For a marked series $F = (F_{e,n})$ they are $M^{(0)} = F$ and

$$M^{(j+1)}_{e,n} = \frac{1}{j+1}\Big[\binom{n+2}{2}\Delta_\Gamma M^{(j)}_{e,n+2} + \sum_{c=0}^{n}\sum_{i=0}^{j}\sum_{a=0}^{e}(c+1)(n-c+1)\; M^{(i)}_{a,c+1} \star_\Gamma V^{(j-i)}_{e-a,n-c+1}\Big].$$

The cross term has twice the coefficient of its unmarked counterpart. The recursive outputs are initially maps on test functions; the bounds below prove that they are real-linear. Their interpretation as a normalised Gaussian expectation is discussed in §7.

(i) *Fixed-order bound.* Fix j, e, n. There is a polynomial majorant P with the following property, for all I, d, h as above, $\sigma > 0$, $k \in \mathbb{N}$, $\delta \ge 0$, $W \ge 1$, $U \ge 0$, eligible V, V', propagators Γ, Γ' and marked series F, F'. Assume:

- (Γ, Γ') is (W,δ)-controlled;
- the unmarked inputs $(V_{a,r}, V'_{a,r})$ with $1 \le a \le e$ and $1 \le r \le a+2$ are (W,δ)-controlled;
- the marked inputs $(F_{a,r}, F'_{a,r})$ are (U,δ)-controlled for: $a < e$ (any r); or $a = e$ and $r > n$; or $j = 0$ and $(a,r) = (e,n)$.

Then $\big(M^{(j)}_{e,n}(\Gamma, V, F),\ M^{(j)}_{e,n}(\Gamma', V', F')\big)$ is $(P(W)U,\delta)$-controlled, with real-linear output values. The majorant depends only on (j,e,n); in particular it is independent of k, σ and the height h.

(ii) *Support.* Let $l \in \mathbb{N}$. If V is eligible and $F_{a,r} \ne 0$ implies $r \le l + a$ for every a, r, then $M^{(j)}_{e,n} \ne 0$ implies $n + 2j \le l + 3e$ and $n \le l + e$.

(iii) *Full positive-time part.* Fix l, e, n. There is a polynomial majorant P such that the pair

$$\sum_{j=1}^{l+3e} t^j M^{(j)}_{e,n}(\Gamma, V, F), \qquad \sum_{j=1}^{l+3e} t^j M^{(j)}_{e,n}(\Gamma', V', F')$$

is $(P(W)U,\delta)$-controlled for every real t with $|t| \le W$. The hypotheses are those of (i), with marked inputs required only for $a < e$, or $a = e$ and $r > n$. By (ii), this is the whole positive-time part when both $F_{a,r}$ and $F'_{a,r}$ vanish for $r > l + a$, for every a, r.

Status: theorem. This is the form in which the main theorem uses Part A (in III.14). *Lean:* `journey`, `full`, `registry`. Test space: `Pearl`. Norms: `Sings`, `Dyad`, `purse`, `beam`, `Loom`, `fits`.

3.11.1.2 Part B *Setting.*

- The gauge group is as in I.1, with its Haar probability measure.
- A block-spin step (III.1) maps a fine periodic lattice to a coarse one with odd block factor $L \geq 3$. The coarse site x sits at the fine site ιx.
- The kernel is built from patches: boxes of radius $\rho = (J+3)L$ around the points ιx, where the stage satisfies $J \geq 4$.
- $k_\beta(U, V)$ is the conditional density of the coarse configuration V given the fine configuration U, with respect to the product Haar probability measure.
- The pull-back of a bounded measurable complex function F of the coarse configuration is

$$(\mathcal{B}F)(U) = \int k_\beta(U, V) F(V)\, dV.$$

- For $r \in \mathbb{N}$, $\Lambda_r(x)$ is the periodic ℓ^∞ box of radius r about x; periodic images are allowed. All lattice radii below are nonnegative integers.
- F is *local on* X if $F(V)$ depends only on the links of V with both endpoints in X.
- $\mathcal{I}(X)$ is the class of observables that are gauge invariant, measurable, bounded and local on X (IV.1).

Patch hypothesis (III.1). The minimising frame and the best predictor are unique whenever their respective branches are used. The set of admissible fallback frame tags is nonempty for every cover configuration.

Proposition III.8.4 (one step: the pull-back is local). Assume the patch hypothesis, and let $\beta \in \mathbb{R}$.

(i) Let F be a bounded measurable complex function local on a coarse set X, and let Y be a set of fine sites containing $\Lambda_{10\rho}(\iota x)$ for every $x \in X$. Then $\mathcal{B}F$ is local on Y.

(ii) Assume moreover that the density is gauge covariant, $k_\beta(U^g, V^{g\circ\iota}) = k_\beta(U, V)$ for every fine gauge transformation g. If $F \in \mathcal{I}(\Lambda_R(o))$, then $\mathcal{B}F \in \mathcal{I}(\Lambda_{LR+10\rho}(\iota o))$, with the same sup bound.

Status: theorem. III.1 supplies these kernel properties for its chosen kernel parameters: the patch hypothesis is `Next.Foundations.branches`. Gauge covariance follows from `StepData.covary`. For each step of a history, both are fields of the block-control data built in `Calendar.blocks_go`. *Lean:* `back_local`, `back_cylinder`.

Proposition III.8.5 (exact factorisation). Assume the patch hypothesis.

- If F, G are bounded, measurable and local on disjoint coarse sets X, Y, then $\mathcal{B}(FG) = \mathcal{B}F \cdot \mathcal{B}G$ for every β.
- In particular $\mathcal{B}\prod_{i\in l} F_i = \prod_{i\in l} \mathcal{B}F_i$ when $F_i \in \mathcal{I}(C_i)$ for distinct cells C_i of a tiling.
- Along a history with the kernel properties stated below, fix N, p and let $h \leq k \leq m$. Let l be a finite set of level-h sites and $F_i \in \mathcal{I}(\Lambda_r(\iota_{h\to k} i))$ at level k for $i \in l$. If $\mathrm{room}(h, k, r)$ holds, then $\mathcal{B}_{k\to m}\prod_{i\in l} F_i = \prod_{i\in l} \mathcal{B}_{k\to m} F_i$.
- The same holds for the product of the tile copies of one $F \in \mathcal{I}(\Lambda_r(0))$ (translated, reflected, and complex-conjugated on odd tiles), the copies on the right being taken at level m.

Status: theorem. It is an identity, not an estimate, and the fine windows of $\mathcal{B}F$ and $\mathcal{B}G$ may overlap. *Lean:* `back_factors`, `back_cells`, `backTo_cells`, `backTo_copies`.

History.

- Levels $m = 0, 1, 2, \ldots$ get finer as m grows. Level m has spacing a_m.
- The step from level $m+1$ to level m has block factor L_m (odd, ≥ 3), so $a_{m+1} = a_m / L_m$. Its patch radius is $\rho_m = (J_{m+1} + 3)L_m$, where $J_m \geq 4$ and $J_{m+1} \leq J_m + 1$. The inverse couplings are $\beta_{N,p,m}$.
- Every step satisfies the patch hypothesis, gauge covariance and covariance under lattice translations and signed permutations of the axes. These are part of the history data supplied by III.1.
- $\mathcal{B}_m$ is the pull-back through this step at $\beta_{N,p,m+1}$. Put $\mathcal{B}_{k\to m} = \mathcal{B}_{m-1} \circ \ldots \circ \mathcal{B}_k$ (with $\mathcal{B}_{k\to k} = \mathrm{id}$), and let $\mathcal{B}^*_{k\to m}$ be the composite of the push-forwards of III.1 from level m to level k.
- For a level-h site i and $m \geq h$, $\iota_{h\to m} i$ is its image at level m.
- The cone radius is $r_{k\to k}(r) = r$, $r_{k\to m+1}(r) = L_m \; r_{k\to m}(r) + 10(J_{m+1} + 3)L_m$.
- The physical budget is $\tau_m = 30 a_m (J_m + 5)$.
- $\mathrm{room}(h, k, r)$ is the condition $2(a_k r + \tau_k) < a_h$.

Proposition III.8.6 (pull-back along the history). For all N, p and $k \leq m$:

(i) $\mathcal{B}_{k\to m}$ maps bounded measurable functions to bounded measurable functions, with $\sup |\mathcal{B}_{k\to m} F| \leq \sup |F|$. It is linear, fixes constants and commutes with complex conjugation.

(ii) For every bounded measurable complex F at level k and every integrable complex g at level m, $\int (\mathcal{B}^*_{k\to m} g) F = \int g \; \mathcal{B}_{k\to m} F$ and $\int \mathcal{B}^*_{k\to m} g = \int g$. All integrals here use the Haar probability measures at their respective levels; the push-forward is integrable.

(iii) Let $h \leq k$ and i a level-h site. If $F \in \mathcal{J}(\Lambda_r(\iota_{h\to k} i))$, then $\mathcal{B}_{k\to m} F \in \mathcal{J}(\Lambda_{r_{k\to m}(r)}(\iota_{h\to m} i))$.

(iv) On bounded measurable functions, $\mathcal{B}_{k\to m}$ commutes with signed permutations of the axes followed by translation by $\iota_{h\to \cdot} i$, for $h \leq k$ and each level-h site i, and with the copy maps of the centred level-h tiling.

Status: theorem. The hypotheses on the kernels are supplied by the history. *Lean:* `backTo`, `pushTo`, `backTo_bound`, `backTo_env`, `backTo_add`, `backTo_const`, `backTo_star`, `pushTo_dual`, `pushTo_integral`, `backTo_local`, `backTo_orient`, `backTo_copy`.

Proposition III.8.7 (size of the cone in physical units). For the history above and all nonnegative integer levels and radii:

(i) $\tau_{m+1} \leq \frac{2}{3}\tau_m$; hence $\tau_m \leq (2/3)^m \tau_0 \to 0$.

(ii) $a_m \; r_{k\to m}(r) + \tau_m \leq a_k r + \tau_k$ for $k \leq m$. Thus an observable of lattice radius r at level k has, at every finer level, a backward cone of physical radius at most $a_k r + \tau_k$, uniformly in m, N and p.

(iii) If $h \leq k$ and $\mathrm{room}(h, k, r)$ holds, then for every $m \geq k$ and every level-h site i, the box $\Lambda_{r_{k\to m}(r)}(\iota_{h\to m} i)$ lies in the cell of i of the centred level-h tiling at level m. That cell has $L_h \cdots L_{m-1}$ sites per side, i.e. physical side a_h.

(iv) For fixed h and r, $\mathrm{room}(h, k, r)$ holds for all sufficiently large $k \geq h$.

Status: theorem. *Lean:* `tail`, `tail_next`, `tail_zero`, `scales_next`, `scales_budget`, `room`, `room_fits`, `plenty`.

3.11.2 2. What it uses

- **III.1 (block-spin step, history).**
  - *The kernel.* k_β (`StepData.density`) is a sum over site tags and bond tags of a `likelihood` times a product over bonds of normalised penalty densities. The likelihood is a product over sites of frame masses times a product over bonds of predictor masses.
  - *Frames and patches.* `Patch.frame`, `Patch.frameMass`, `StepData.connector`, and the patch radius `Patch.radius` $= (J+3)L$.
  - *Patch hypothesis and kernel properties.* The patch hypothesis `Patch.Branches` is proved in `Next.Foundations.branches`. Also used: `frameMass_sum`; `StepData.isKernel` (positivity, measurability, boundedness, mass one); the push-forward `StepData.push`.
  - *The history.* `History`, `Levels`, `Calendar.Fits` ($J_{m+1} \le J_m + 1$), `Levels.transition_mass`. The per-step `Calendar.BlockControl` holds the patch hypothesis and the lattice-symmetry and gauge covariance of each density; it is constructed in `Calendar.blocks_go`.
- **I.1.**
  - Periodic lattices, links, Haar measure, gauge transformations, transport along routes, and locality `LocalOn`.
  - The spacing schedule `Schedule`: L_m odd and ≥ 3; `a_step` ($a_{m+1} = a_m/L_m$); `shrinks` ($a_m \to 0$); `embeds_scale` ($a_m\, L_h \cdots L_{m-1} = a_h$).
- **IV.1.** Tilings, cells and copies; the class `Interior` $= \mathcal{J}$; and, in Step B7 only, the chessboard root `rootTest` and the chessboard bound `Tiling.planted`. Its reflection-positivity hypotheses are packaged in `Seam` and proved for $\beta \ge 0$ in `laws`.
- **Part A is an abstract estimate.** The consumer, III.14, supplies V, Γ and the bounds W, δ. The weighted ℓ^1 estimates behind the rooted norm come from the support packages listed in §7. The covariance used by III.14 is discussed there.
- **Standard mathematics.**
  - *Part A.* The formal Hamilton–Jacobi heat equation, often called the Polchinski equation, generates connected diagrams by contractions and joins. For finite I and symmetric inputs, put $W_t(\phi) = \log \mathbb{E}_{t\Gamma} e^{V(g,\phi+\zeta)}$, with expectation interpreted by Wick's rule. Then $\partial_t W = \frac{1}{2}\sum_{u,v} \Gamma(u,v)(\partial_u\partial_v W + \partial_u W \partial_v W)$. Rooted coefficient bounds give a direct inductive estimate for this equation. Background references include Polchinski, Nucl. Phys. B 231 (1984) 269; Brydges–Kennedy, J. Stat. Phys. 48 (1987) 19; Salmhofer, *Renormalization: an introduction*, Springer 1999; and Brydges, *Lectures on the renormalisation group*, IAS/Park City Math. Ser. 16, 2009.
  - *Part B.* Conditional expectation for a Markov kernel; Fubini for product measures and for sums over products of probability vectors. The model is Bałaban's block averaging (Comm. Math. Phys. 98 (1985) 17); the kernel here is the probabilistic one of III.1.
  - *Step B7.* The chessboard estimate (Fröhlich–Israel–Lieb–Simon, Comm. Math. Phys. 62 (1978) 1).
- **Imported constants and assumptions.**
  - The patch radius $(J+3)L$ (III.1).
  - $L_m \ge 3$ and $J_{m+1} \le J_m + 1$ (I.1, III.1).

– The patch hypothesis is assumed in III.8.4–III.8.5; gauge covariance is also assumed in III.8.4(ii). III.1 supplies these properties for its chosen kernels, together with lattice-symmetry covariance for iteration.
– The constants 10 (read radius), 8 (route length), 30 and 5 (in τ_m) are internal.
– Part A imports no constants.

3.11.3 3. Strategy

Part A. Each step of (A1) either contracts two slots of one coefficient or joins two coefficients by a propagator line. The rooted norm absorbs the internal sums without changing σ: the output diameter is bounded by the input diameters plus the line length, so the exponential weights multiply. For a marked coefficient, the height inequality plays the same role, with the marked factor summed over all its slots.

Induction on heat order gives a polynomial majorant in W. Telescoping the inputs gives the corresponding difference bound with the same δ. Every primitive vertex has positive coupling order and at most three times that order in slots; every propagator removes two slots. This gives $n + 2j \le 3e$ and finiteness in heat time at each coupling order. Positive heat order uses only lower coupling orders and higher ranks at the same coupling order.

Part B. Given the fine field, the kernel first draws independent site tags, then independent predictor tags on bonds, and finally independent coarse links conditional on these tags. Integrating an observable local on X removes all unused bonds and site tags. The remaining factors read the fine field within radius 10ρ of ιX. Observables on disjoint coarse site sets use disjoint tags and bonds, so their pull-backs multiply. Under iteration the radius grows by $R \mapsto L_m R + 10\rho_m$. In physical units the added radius is at most $\tau_m/3$, while $\tau_{m+1} \le 2\tau_m/3$. The total radius therefore stays within $a_k r + \tau_k$. When this fits in one level-h cell, different cells remain separate at every intermediate step, allowing the product formula to iterate.

3.11.4 4. Steps

3.11.4.1 A. Connected heat coefficients A1. Operations and the recursion.

Definitions. For F of rank $m + 2$,

$$(\Delta_\Gamma F)(x) = \mathrm{Sym} \sum_{u,v \in I} \Gamma(u, v) F(u, v, x).$$

For F of rank $p + 1$ and G of rank $q + 1$, the join has rank $p + q$:

$$(F \star_\Gamma G)(x, y) = \mathrm{Sym} \sum_{u,v \in I} F(u, x) \Gamma(u, v) G(v, y).$$

Sym averages over permutations of the slots.

Combinatorial factors. On a finite index set, write $W = \sum_n \langle W_n, \phi^{\otimes n} \rangle$ with symmetric coefficient tensors and no slot factorials. For symmetric Γ, the differential operators $\frac{1}{2} \sum_{u,v} \Gamma(u, v) \partial_u \partial_v W$ and $\frac{1}{2} \sum_{u,v} \Gamma(u, v) \partial_u W \partial_v W$ give the factors $\binom{n+2}{2}$ and $(c + 1)(n - c + 1)/2$. For nonsymmetric tensors (A1) remains a definition, without this identification.

Vanishing terms and symmetry. For eligible V the pair terms with $a = 0$ or $a = e$ vanish, because $V^{(i)}_{0,\cdot} = 0$. Every $V^{(j)}$ with $j \ge 1$ is symmetric, since every term is symmetrised.

Formal heat equation. For finite I, eligible V symmetric in every rank (`waxed`), and symmetric Γ, III.14 proves:

- the generating series solves the formal Hamilton–Jacobi equation (`Glyph.cloudy`);
- its exponential solves the linear heat equation (`Cloud.Clockwise.linear`);
- summation at time t gives $\mathcal{E}_t(V)$ (`Glyph.breaktime`).

These identities support the formal cumulant interpretation. They do not assert convergence of a Gaussian integral or of the series in g. The norm estimates use only the recursion.

Lean: `lap`, `bridge`, `bleach`, `unilateral`, `pair`, `wheels`, `bank`, `burn`, `blaze`, `coal`.

A2. Support (Proposition III.8.2). Since $\Delta_\Gamma 0 = 0$ and $0 \star_\Gamma G = F \star_\Gamma 0 = 0$, a nonzero $V^{(j+1)}_{e,n}$ has a nonzero predecessor: either $V^{(j)}_{e,n+2} \neq 0$, or both $V^{(i)}_{a,c+1}$ and $V^{(j-i)}_{e-a,n-c+1}$ are nonzero for some $c \leq n$, $i \leq j$, $a \leq e$. Strong induction on j gives $n + 2j \leq 3e$:

- at $j = 0$, $n \leq e + 2 \leq 3e$ because $e \geq 1$;
- the contraction step uses $(n + 2) + 2j \leq 3e$;
- the pair step adds $(c + 1) + 2i \leq 3a$ and $(n - c + 1) + 2(j - i) \leq 3(e - a)$.

Parity and $n \leq e + 2$ propagate the same way. Splitting off $j = 0$ gives $\mathcal{E}_t = V + \mathcal{R}_t$. *Lean:* `predecessors`, `preserves`, `live`, `off`, `smoulder`, `appropriate`.

A3. The two contraction estimates. Let $\sigma \geq 0$, let the input rooted norms and $[\Gamma]_\sigma$ be finite, and require positive output rank. Then

$$\|\Delta_\Gamma F\|_\sigma \leq \|F\|_\sigma \sup|\Gamma| \leq \|F\|_\sigma [\Gamma]_\sigma, \qquad \|F \star_\Gamma G\|_\sigma \leq \|F\|_\sigma [\Gamma]_\sigma \|G\|_\sigma.$$

Symmetrisation. Sym does not increase the rooted norm: it permutes the slots and preserves diameters.

Contraction. Fix a slot of x. Then $\operatorname{diam}(x) \leq \operatorname{diam}(u, v, x)$ and $|\Gamma(u, v)| \leq \sup|\Gamma|$, and the sum over (u, v, x) with that slot fixed is a rooted sum of F.

Join. The key inequality is

$$\operatorname{diam}(x, y) \leq \operatorname{diam}(u, x) + d(u, v) + \operatorname{diam}(v, y).$$

With a slot of x fixed, sum in three stages:

1. over y at fixed v: at most $\|G\|$, rooted at v;
2. over v: at most the row sum of Γ;
3. over (u, x): at most $\|F\|$.

A fixed slot of y is treated symmetrically, with column sums.

Differences. Absolute convergence permits telescoping: $\Delta_\Gamma F - \Delta_{\Gamma'} F' = \Delta_\Gamma(F - F') + \Delta_{\Gamma - \Gamma'} F'$, and the join gives three terms. If the tensor pairs are (P, δ)- and (Q, δ)-controlled and (Γ, Γ') is (B, δ)-controlled, with $P, Q, B, \delta \geq 0$, the contraction pair is $(2PB, \delta)$-controlled and the join pair is $(3PBQ, \delta)$-controlled. At output rank 0 these controlled-pair conclusions are vacuous; telescoping need not be valid for divergent scalar sums.

Lean: `Ground`, `Held` with `ensign`, `Wire`, `unary`, `Held.lap`, `arc`, `root_sum`, `binary_EN`, `Held.bridge`, `Held.bleach`, `lap_compare`, `bridge_compare`, `Held.lap_diff`, `Held.bridge_diff`, `Cot.lap'`, `Cot.bridge'`.

A4. Proof of Proposition III.8.1. Strong induction on j, for all (e, n) at once.

Base, $j = 0$. For $n = 0$ the claim is vacuous. For $e = 0$, or $n > e + 2$, the coefficient is zero by eligibility. Otherwise $(e, n) \in T_0(e, n)$ and the bound is W.

Step, $j \to j + 1$. Use (A1).

- The contraction term needs the case $(j; e, n + 2)$. Its inputs ($a < e$, or $a = e$ with $r \geq n + 2$) lie in $T_{j+1}(e, n)$.
- Only pair terms with $0 < a < e$ survive. Both factors are of order less than e, so they need only inputs of order less than e.

With $B = W$ for the propagator, A3 gives the polynomial majorant

$$P_{j+1,e,n}(x) = \frac{1}{j+1}\Big[\binom{n+2}{2}\, 2x\, P_{j,e,n+2}(x) + \sum_{c=0}^{n}\sum_{i=0}^{j}\sum_{a=0}^{e} \frac{(c+1)(n-c+1)}{2}\, 3x\, P_{i,a,c+1}(x) P_{j-i,e-a,n-c+1}(x)\Big].$$

Lean: `navigation`, `taps`, `Wage`.

A5. Full correction and change of index set (outside the main-theorem closure).

- *Full correction.* For fixed e, n, there is a polynomial majorant P such that, under III.8.1's hypotheses on the input range $T_1(e, n)$, the pair $(\mathcal{R}_t(V)_{e,n}, \mathcal{R}_t(V')_{e,n})$ is $(P(W), \delta)$-controlled whenever $|t| \leq W$. Sum A4 over $1 \leq j \leq 3e$, using $|t|^j \leq W^j$.
- *Transport.* Let $n \geq 1$, let (I, d) and (J, D) be pseudometric spaces, and let $\sigma, \sigma' \geq 0$. For each slot i, choose a kernel $K_i : I \times J \to \mathbb{R}$, a nonnegative weight E_i, and $b_i \geq 0$ bounding both weighted row and column sums of $|K_i| e^{E_i}$. Define $(KF)(y) = \sum_{x \in I^n} F(x) \prod_i K_i(x_i, y_i)$. Suppose, for distinct slots i, j, that $\sigma' D(y_i, y_j) \leq \sigma d(x_i, x_j) + E_i(x_i, y_i) + E_j(x_j, y_j)$ whenever the two kernel factors are nonzero. Then $\|KF\|_{D,\sigma'} \leq \|F\|_{d,\sigma} \prod_i b_i$ for every F with finite rooted norm. Fix an output slot: use its column bound to sum its input position, and use row bounds for the remaining slots. The geometric inequality bounds the output weight by the input weight times all slot weights.
- *Differences.* Suppose (F, F') is (C, δ)-controlled, with $C, \delta, B \geq 0$. If every K_i, K_i' has weighted row and column bounds B, every $K_i - K_i'$ has bounds $B\delta$, and the same geometric inequality holds for every combination of their supports, then $(KF, K'F')$ is $((n+1)CB^n, \delta)$-controlled. Telescope once in F and once in each of the n kernels.
- *Combination.* At $t = 1$, take $K_i = \mu$ in every slot and $B = W$. For eligible V, V' and the input bounds above, the transported correction has a polynomial bound in W. A bound for the transported full potential additionally requires $(V_{e,n}, V'_{e,n})$ to be (W, δ)-controlled.
- *Rank* 0. The unmarked rooted bounds leave the scalar term uncontrolled. On an infinite lattice its sums may diverge.

Lean: `postage`, `raft`, `lighthouse`, `street`, `Alley`, `Cot.raft`, `paving`, `strict`, `allG`.

A6. Marked expansion (Proposition III.8.3).

Operations. For M marked and G unmarked, the contraction $\Delta_\Gamma M$ is as in A1, and the join is $(M \star_\Gamma G)(x, y) = \mathrm{Sym} \sum_{u,v} \Gamma(u, v) G(v, y) M(u, x)$.

Estimates.

$$\|\Delta_\Gamma M\|_h \leq \sup|\Gamma|\ \|M\|_h, \qquad \|M \star_\Gamma G\|_h \leq \|M\|_h [\Gamma]_\sigma \|G\|_\sigma.$$

The first holds because $h(x) \le h(u,v,x)$. The second follows from $h(x,y) \le h(u,x) + d(u,v) + \mathrm{diam}_d(v,y)$, which uses that h is 1-Lipschitz. Sum the marked factor over all its slots, the propagator by its row bound, and the unmarked factor by its rooted bound at v. With finite input bounds, the sum of the dual norms is finite. Evaluation at each test function is therefore absolutely summable and preserves linearity, including at output rank 0. The difference constants are 2 and 3, as in A3.

Induction for (i). Induct on j as in A4.

- The contraction term uses the case $(j; e, n+2)$.
- Pair terms with $a = e$ vanish, because the unmarked factor then has order 0.
- For $a < e$, the marked factor $M^{(i)}_{a,c+1}$ is bounded by the induction hypothesis, which needs marked inputs of order at most $a < e$. The unmarked factor $V^{(j-i)}_{e-a,n-c+1}$ is bounded by III.8.1, whose inputs lie in $\{1 \le a' \le e,\ 1 \le r \le a'+2\}$.

This gives (i) with output $P(W)U$.

(ii) and (iii). For (ii), a nonzero contraction has a marked predecessor of rank $n+2$. A nonzero join has marked and unmarked predecessors whose ranks sum to $n+2$ and whose coupling orders sum to e. Add the marked induction bounds to the unmarked bounds of III.8.2 to obtain $n+2j \le l+3e$ and $n \le l+e$. For (iii), sum (i) over $1 \le j \le l+3e$, using $|t|^j \le W^j$; no bound for the initial rank-n, order-e marked coefficient is needed.

Consistency. Evaluating a marked tensor at a fixed test function turns its contraction and join into those of A1.

Lean: `Loom`, `Sings`, `purse`, `beam`, `Lyrics`, `Dyad`, `fits`, `pocket`, `cloak`, `Dyad.pocket`, `Dyad.cloak`, `shim`, `shift`, `rungs`, `skate`, `climbs`, `across`, `journey`, `Lit`, `full`, `shaker`, `cruising`, `registry`, `clocked`, `reel_stamp`.

3.11.4.2 B. Backward dependence cones B1. Product structure of the kernel. Summing the bond tags out of the definition in III.1 gives

$$k_\beta(U,V) = \sum_t \prod_x q_x(U,t_x) \prod_b p_b(U,t;V_b).$$

- $t = (t_x)$ runs over the site (frame) tags, and $q_x(U,\cdot)$ is the frame mass at x.
- $p_b(U,t;\cdot)$ is a mixture, over predictor tags r with weights summing to one, of normalised penalty densities centred at the predictions $Q_r(Y_b)$, which are built from the connector Y_b of the bond:
$$p_b(U,t;v) = \sum_r w_r(Y_b)\ Z_\beta^{-1} \exp\big(-\beta\kappa\ \mathrm{price}(v\ Q_r(Y_b)^{-1})\big).$$
Here $\kappa > 0$ is the kernel penalty of III.1, called b in the main theorem.

Each p_b has Haar integral 1. The identity $\sum_r q_x(U,r) = 1$ uses the patch hypothesis (the fallback set is nonempty). Hence

$$\mathcal{B}F(U) = \sum_t \Big(\prod_x q_x(U,t_x)\Big) S_F(U,t), \qquad S_F(U,t) = \int \prod_b p_b(U,t;V_b) F(V)\ dV.$$

Given U, the site tags are independent; given U and the tags, the coarse links are independent. Adjacent bonds share a tag, so the links are not independent given U alone. *Lean:* `density_parts`, `noiseLink`, `perLink`, `perSite`, `noiseLink_mass`, `perLink_mass`, `site_unit`, `parts_back`.

B2. Read sets.

(a) *Frames.* The frame and the frame mass at x depend only on U on links inside $\Lambda_{10\rho}(\iota x)$. There are two branches:

- if a small-field test on transports in the patch passes, the frame is the minimiser, normalised at the centre, of an averaged price of transports between patch points, and the mass is uniform over all tags;
- otherwise the frame is the axial gauge built from the tag, normalised at the centre, and the mass is uniform over admissible tags (those whose axial cost is bounded by a multiple of the plaquette cost in the patch).

All transports run along coordinate routes between points of Λ_ρ, of length at most 8ρ, so they stay in $\Lambda_{9\rho}$. The minimiser is selected by a global choice. Its uniqueness (patch hypothesis) is what forces two configurations that agree on the window to select the same frame.

(b) *Connectors.* The connector of the bond from x to $x + e_\mu$ depends only on the tags at both ends and on U in $\Lambda_{10\rho}(\iota x) \cup \Lambda_{10\rho}(\iota(x + e_\mu))$. It is built from the two frames and from straight transports of length $L \le \rho$.

Lean: `patch_dep`, `route_bound`, `visits_bound`, `site_dep`, `connector_dep`, `reach`.

B3. Locality (Proposition III.8.4). Two elementary facts:

- (P1) if a function reads only the coordinates in a set s, its integral against a product of probability densities does not change when the densities outside s are changed;
- (P2) the same holds for sums over products of probability vectors.

Proof of (i). If F is local on X, it reads only bonds with both ends in X. By (P1) and B2(b), $S_F(U, t)$ depends only on $t|_X$ and on U in the windows around ιX. By (P2), the tags off X sum out. By B2(a), q_x for $x \in X$ reads U only in $\Lambda_{10\rho}(\iota x)$.

Proof of (ii). If $y \in \Lambda_R(o)$ then $\iota y \in \Lambda_{LR}(\iota o)$, and $\Lambda_{10\rho}$ of a point of $\Lambda_{LR}(\iota o)$ lies in $\Lambda_{LR+10\rho}(\iota o)$. Gauge invariance follows from the covariance of k_β and the invariance of Haar measure. The sup bound and measurability hold because k_β is a probability kernel.

Lean: `Reads`, `pi_ignores`, `sum_ignores`, `actualEdges`, `state_reads`, `back_local`, `window_embed`, `window_add`, `back_natural`, `back_bound`, `back_env`, `back_cylinder`.

B4. Factorisation (Proposition III.8.5).

- If $X \cap Y = \emptyset$, the bond sets read by F and G are disjoint, so $S_{FG}(U, t) = S_F(U, t) S_G(U, t)$ by Fubini.
- S_F depends on $t|_X$ and S_G on $t|_Y$, and the $q_x(U, \cdot)$ are probability vectors, so the sum over tags factorises.
- Induction over the cells of a tiling gives the finite-product version.
- Along the history, induct on m. By III.8.7(iii), each $\mathcal{B}_{k\to m} F_i$ lies in $\mathcal{J}$ of its own level-h cell, so the one-step product formula applies at every step.
- The version for copies follows with III.8.6(iv).

Lean: `pi_independent`, `sum_independent`, `back_factors`, `back_cells`, `backTo_cells`, `copy_window`, `backTo_copies`.

B5. Iteration (Proposition III.8.6). Induct on $m - k$, with $\mathcal{B}_{k\to m+1} = \mathcal{B}_m \circ \mathcal{B}_{k\to m}$.

- (i) and (ii) hold for one step because k_β is a real probability kernel; (ii) is Fubini.
- (iii) is III.8.4(ii) at each step. The kernel hypotheses come from the block-control data, and $\iota(\iota_{h\to m}i) = \iota_{h\to m+1}i$.
- (iv) uses the lattice-symmetry covariance of each density.

Lean: `backTo_next`, `backTo_env`, `backTo_bound`, `backTo_add`, `backTo_const`, `backTo_star`, `back_dual`, `pushTo_integral`, `pushTo_dual`, `anchor_next`, `backTo_local`, `backTo_orient`, `backTo_copy`.

B6. Physical size of the cone (Proposition III.8.7).

- (i) $\tau_{m+1} \le 30\frac{a_m}{3}(J_m+6) \le \frac{2}{3}\tau_m$.
- (ii) Put $r' = L_m r + 10(J_{m+1}+3)L_m$. Then $a_{m+1}r' = a_m r + 10a_m(J_{m+1}+3) \le a_m r + \tau_m/3$. Adding $\tau_{m+1} \le \frac{2}{3}\tau_m$ gives $a_{m+1}r' + \tau_{m+1} \le a_m r + \tau_m$, and one iterates.
- (iii) The cell of i at level m is exactly $\Lambda_{(D-1)/2}(\iota_{h\to m}i)$, with $D = L_h \cdots L_{m-1}$ odd and $a_m D = a_h$. By (ii) and room, $2a_m r_{k\to m}(r) < a_h - 2\tau_m < a_m D$, so $2r_{k\to m}(r) < D$.
- (iv) $a_k \to 0$ and $\tau_k \to 0$.

Lean: `expansion`, `grows`, `grows_next`, `tail_next`, `tail_zero`, `scales_next`, `scales_budget`, `tiling_window`, `window_fits`, `room_fits`, `plenty`.

B7. Two consequences used downstream.

(a) *Replicated differences (used by III.11).* Fix N, p and $h \le k \le N$. Let O be a bounded measurable complex function at level N, let $F \in \mathcal{J}(\Lambda_w(0))$ at level k with $\mathrm{room}(h,k,w)$, and let s be a finite set of level-h sites. Then

$$Z_N\langle \prod_{i\in s} \mathrm{copy}_i(\mathcal{B}_{k\to N}F - O)\rangle_N = \sum_{l\subseteq s}\int \mathcal{B}^*_{k\to N}\Big[e^{-S_N}\prod_{i\in s\setminus l}(-\mathrm{copy}_i O)\Big]\prod_{i\in l}\mathrm{copy}_i F.$$

Here $S_N = \beta_{N,p,N}E_N$ is the Wilson action, $Z_N = \int e^{-S_N}\,dU$, and $\langle Q\rangle_N = Z_N^{-1}\int e^{-S_N}Q\,dU$. The copies on each side use the level-h tiling at the level of their argument; the integrals on the right are at level k. Expand the product, commute the copies with subtraction, then apply III.8.5 and the duality in III.8.6(ii). The old factors stay together inside one push-forward; no independence of O is required.

(b) *Chessboard telescoping (used by III.13, VII.2).* Fix $h \le N$, p and a real coupling β for which the chessboard inequality of IV.1 holds; the Wilson law has this property for $\beta \ge 0$. Let T be the centred level-h tiling at level N. For a central-cell observable Q, define

$$\mathrm{root}_T(Q) = \Big|\Big\langle \prod_{a\in T}\mathrm{copy}_a Q\Big\rangle_\beta\Big|^{1/(\sharp T)},$$

where $\sharp T$ is the number of cells. Let the index set be finite, let a_i be distinct level-h sites, and let $O_i, Z_i \in \mathcal{J}(T_0)$, where T_0 is the central cell. Put $b_i = \iota_{h\to N}a_i$ and let τ_{b_i} act on configurations by lattice translation. If real bounds c_i, e_i satisfy $\mathrm{root}_T(O_i) \le c_i$ and $\mathrm{root}_T(Z_i - O_i) \le e_i$, then

$$\Big|\langle \prod_i (Z_i\circ\tau_{b_i})\rangle_\beta - \langle \prod_i (O_i\circ\tau_{b_i})\rangle_\beta\Big| \le \prod_i(e_i+c_i) - \prod_i c_i.$$

Expand $\prod_i(O_i + (Z_i - O_i))$ over subsets. The empty-subset term cancels, and the chessboard estimate bounds each remaining term by the corresponding product of c_i and e_i. To apply it

to literal translates, undo the reflection in each copy and, on odd cells, conjugate its source. These operations preserve the central-cell class and its root. Summing the nonempty-subset bounds gives the right-hand side.

Lean: `Joint.restored`, `tariff`; `rooted_translates`.

3.11.5 5. Where the difficulty lies

- **Input ranges.** At positive heat order, inputs of coupling order e have strictly higher rank. The purely contracting contribution to $V^{(j)}_{e,n}$ starts at $V_{e,n+2j}$. This permits downward induction in rank at fixed coupling order. The bound is uniform in geometry, but its polynomial may depend on (j, e, n); no resummation in g is proved.
- **Scalar terms.** The unmarked rooted norm gives no scalar estimate. The marked norm sums all slots and does control rank 0, including absolute convergence of its contractions. The same $\sigma > 0$ occurs in the line, diameter, height and test-function weights.
- **Locality of a selected minimiser.** Agreement on the read window makes the minimisation problems identical. Uniqueness then makes their selected frames equal. The read radius 10ρ also includes fallback tests and connectors; the coordinate routes used by frames lie in $\Lambda_{9\rho}$.
- **Conditional factorisation.** Disjoint coarse site sets use disjoint tags and bonds. Their pullbacks may nevertheless read overlapping fine windows. Factorisation therefore concerns the conditional kernel, not the fine Gibbs law. Iteration requires separation at every intermediate level, supplied by the cell-containment bound.
- **Physical size.** The lattice radius grows by $R \mapsto L_m R + 10\rho_m$, while its physical radius stays below $a_k r + 30 a_k (J_k + 5)$. For fixed h, r, this bound eventually fits inside a level-h cell.

3.11.6 6. Notation

Written proof	Lean	Meaning
I, d, diam_d	`Ground`, `Ground.radius`	cells with a pseudometric; largest distance within a tuple
rank-n tensor	`Tape`, `Flake`; file	function on I^n, ordered slots
$\|F\|_\sigma \le p$	`Held`	rooted weighted ℓ^1 bound (empty at rank 0)
Γ, $[\Gamma]_\sigma \le B$	`Line`, `Wire`	propagator; weighted row and column sums
(W, δ)-controlled	`Cot`, `WireCot`	both at most W, difference at most $W\delta$
polynomial majorant	`Wage`	$0 \le P(x) \le Cx^t$ for $x \ge 1$
$V = (V_{e,n})$, eligible	`Grain`, `Eligible`	formal potential
Δ_Γ, $\star_\Gamma$, Sym	`lap`, `bridge`, `bleach`	contraction, join, symmetrisation
$V^{(j)}$	`burn`	heat coefficient
$T_j(e, n)$	`taps` (`Residual` is T_1)	admissible inputs
$\mathcal{E}_t$, $\mathcal{R}_t$	`ember`, `cinder`	full series; positive-time part
h, $h(x)$	`Loom`, `Loom.cage`	height; its maximum over a tuple

Written proof	Lean	Meaning
$\|f\|_{k,\sigma}$, $\|\lambda\|^*_{k,\sigma}$	`beam`, `purse`	weighted C^k norm of test functions; dual norm
$\|M\|_h$; (U,δ)-controlled	`Sings`, `Dyad`	marked norm; marked pairs
marked Δ_Γ, $\star_\Gamma$	`pocket`, `cloak`	marked contraction and join
$M^{(j)}$; $\sum_{j\geq 1} t^j M^{(j)}$	`skate`; `shaker`	marked heat coefficients; positive-time part
k_β	`StepData.density`	conditional density of the coarse field given the fine field
$\mathcal{B}$	`back`	one-step pull-back
q_x, p_b	`perSite`, `perLink`	tag law at a site; link law at a bond
$\Lambda_r(x)$	`window`	periodic ℓ^∞ box
10ρ; $\rho = (J+3)L$	`reach`; `Patch.radius`	read radius; patch radius
$\mathcal{J}(X)$	`Interior`	gauge invariant, measurable, bounded, local on X
$\mathcal{B}_{k\to m}$, $\mathcal{B}^*_{k\to m}$	`backTo`, `pushTo`	iterated pull-back; push-forward
$\iota_{h\to m}$	`anchor`	image of a level-h site at level m
$r_{k\to m}(r)$	`grows`, `expansion`	cone radius in lattice units
τ_m	`tail`	physical cone budget $30a_m(J_m+5)$
$\mathrm{room}(h,k,r)$	`room`	$2(a_k r + \tau_k) < a_h$

3.11.7 7. What remains unexplained

1. **Application of the bounds.** The marked bound III.8.3 feeds III.14 through `Necks.slushy_uniform`. The support and finite-sum identities of III.8.2 also enter III.14 directly. The choices of W, δ, σ, h and the covariance line bounds must be explained there; their derivation from earlier covariance estimates has not been traced here. The paired covariance and potential bounds are explicit hypotheses of that application.
2. **Gaussian interpretation.** A1 verifies the unmarked formal heat equations for finite index sets and symmetric input tensors and propagators. For symmetric marked inputs, the intended identity is

$$M_t(\phi) = \frac{\mathbb{E}_{t\Gamma}[F(g,\phi+\zeta)e^{V(g,\phi+\zeta)}]}{\mathbb{E}_{t\Gamma}[e^{V(g,\phi+\zeta)}]} \qquad \text{(probable)},$$

   coefficientwise in g, with expectation interpreted by Wick's rule. The marked source announces this identification for later work, but a proof was not located. In particular the marked rank-0 coefficient should not yet be described as a proved normalised expectation. No analytic convergence of the series in g is asserted.
3. **Infinite-lattice scalars.** The unmarked norm leaves rank 0 uncontrolled. The scalar recursions are total-sum definitions, whose values need not represent convergent infinite-volume sums. A use of these scalars requires a separate finite-volume or vacuum-density argument.

4. **Scope beyond the stated results.** Step A5 gives bounds outside the main-theorem closure; III.14 uses the transported-potential definitions without those bounds. The marked transport in `Skiff/Pave.lean` and the further applications in `GRID_APPROXIMATION/Birth`, `Nil`, `Candidates`, `Picture`, `Estimate` and `Attach*` are not developed in this outline. The relation between these formal connected coefficients and the polymer activities of III.5 also remains to be explained in III.14 (tentative).
5. **Support packages.** About 1,970 assigned lines lie outside the two arguments above. Their mathematical roles have only been classified from documentation (tentative): `Carrel` treats path templates and their leading curl; `Camera` treats jets of chart germs; `Room` treats field pull-backs, Jacobians and centred Wick operations; `Rudder` treats symmetric lifts; and `Pert/Bulk` and `Packets` supply weighted sums. `AnalyticNext/QUARTIC_MATCH` belongs with the quartic analysis of III.7 and III.10. Other assigned packages include `Isles`, `Anchor`, parts of `Cloud` and `Glyph`, `WilsonBulk/Web` and `ProofPorts/…`. These packages need separate outlines if their results are to be included here.

3.12 III.9. The marginal flow: the running coupling and its reset

Part: III, Ultraviolet construction (renormalization group). **Lean:** chapter 21: `YM/UVNext/COUPLING_RESET/`, `YM/AnalyticBridge/VERIFIED_RESET_STEP/`, `YM/Extra/WILSON_STARTUP/`, `YM/Push/BETA_TRANSPORT/`, `YM/Further/STRONG_STARTUP_CREDIT/`, `YM/Push/PRICE_COMPILER/`, `YM/Extra/MULTISOURCE_BOUNDS/`. It also includes five small packages: `PhysicalNext/UNMASKED_THROUGH_LAUNCH`, `Planned/RECIPE_STARTUP_PORT`, `R2I/Wind`, `Further/THRESHOLD_SOURCE_RESTART` and `PhysicalNext/SERVICE`. The interfaces this unit fills are in `YM/RG/History.lean` and `YM/PCalc/Band.lean`, both in unit III.1. **Size:** 7,215 lines in the main-theorem closure:

package	lines
verified reset step	2,659
Wilson start-up	2,422
transport of the coupling	652
coupling reset	592
multi-source bounds	344
each remaining package	at most 213

Conventions.

- **Levels.** Levels $m = 0, 1, \dots, N$ run from the coarsest to the finest:
  - $m = 0$ is the torus of side L_p at unit spacing; the sides L_p are multiples of 4 and strictly increase with p;
  - $m = N$ is the Wilson lattice.
- **Blocking.** The step $m + 1 \to m$ blocks by an odd factor $L_m \geq 3$. The spacing at level m is $a_m = (L_0 \cdots L_{m-1})^{-1}$, and the scale is $s_m := \log(1/a_m) = \sum_{i<m} \log L_i$. We write Λ_m for the level-m torus (at volume index p) and $|\Lambda_m|$ for its number of sites.
- **Gauge group.** $G \subset U(n)$ is closed, and its Lie algebra $\mathfrak{g}$ carries the inner product $\langle X, Y \rangle = -\mathrm{Retr}(XY)/n$.
- **Plaquettes.** $s(u) = 1 - \mathrm{Retr}u/n$ is the plaquette function. $F_p(U) \in \mathfrak{g}$ is the orthogonal projection of $\frac{1}{2}(U_p - U_p^{-1})$, and $S_W = \sum_p s(U_p)$ is the Wilson action. Since $s(e^X) =$

$|X|^2/2 + O(|X|^4)$, the inverse coupling β equals $1/g_0^2$ in this normalization.

Effective actions. An effective action at level m is a *representation* $\rho = (P, c, E, K, b, \beta, v)$. It evaluates to

$$\rho(U) = \exp\left(v|\Lambda_m| - \beta\,\mathcal{K}_P(U) - \mathcal{L}_P(U) - \mathcal{E}(U)\right)\mathcal{G}(U).$$

- **The kinetic functional $\mathcal{K}_P$.** This is the β-coefficient of the leading terms of the path data P:
  - a clipped plaquette term;
  - smeared curvature-squared terms;
  - cubic and quartic local polynomials in curvatures along lattice paths.

  These are averaged over the point group of the lattice, i.e. signed permutations of the axes.
- **The other factors.** $\mathcal{L}_P$ is the β-independent loop functional. $\mathcal{E}$ is the sum of the regular activities multiplied by their small-field indicators. Their weighted norms are summed at each anchor. $\mathcal{G}$ is the gas of forced activities, whose weighted uniform bounds are summed over supports containing each site. $b = (k, q, a, B, t)$ records the derivative order $k \geq 6$, spatial power $q \geq 2$, support exponent $a > 0$, cover radius $B \geq 1$ and threshold $0 < t \leq 1$ of the polymer norms.
- **Normalization.** The zero-momentum quadratic symbol of the β-term is the identity on plaquette orientations. The loop kernel has vanishing zeroth and first moments. These conditions fix the normalization of β within the representation.

The block-spin step. $T^{(m)}_{\beta'}$ denotes one step from level $m+1$ to level m at inverse coupling β':

$$(T^{(m)}_{\beta'}\rho')(V) = \int k_{\beta'}(V, U)\,\rho'(U)\,dU.$$

The kernel $k_{\beta'}$ is a probability kernel (unit III.1). Densities are taken relative to product Haar probability measure and are not normalized.

3.12.1 1. What this unit proves

This unit tracks the scalar coefficient β of the normalized curvature-squared term through the blocking history. Its three results are:

- **(a) Start-up.** With the start-up bounds, Wilson top path data and recipe acceptance, the Wilson weight at the finest level is admissible for every β in the band.
- **(b) One step.** Given the early presentation, local bounds, numerical conditions and recipe acceptance stated below, one block-spin step followed by a *reset* of β maps a continuous admissible family at level $m+1$ to one at level m. The coupling moves by $\alpha_{N,m} + d$, where $\alpha_{N,m}$ is the one-loop coefficient of the step and $|d| \leq r_m \leq 1$.
- **(c) Thresholds.** The delay-dependent numerical conditions hold beyond entry thresholds, provided the earlier constants satisfy the compatibility inequalities of III.9.7. These choices respect the order of constants in III.2.

Combined with the shooting argument of III.2 and the constructions of III.10 and III.13, this yields the selected history of III.9.3 and the law $\beta_{N,p}/\log(1/a_N) \to \kappa$ of the main theorem.

Proposition III.9.1 (the slope constant). Let $\mathfrak{g}$ be simple and nonabelian, and let (b_i) be an orthonormal basis of $(\mathfrak{g}, \langle\cdot,\cdot\rangle)$. Put

$$C_G = -\frac{1}{\dim\mathfrak{g}}\sum_i \mathrm{tr}_{\mathfrak{g}}((\mathrm{ad}b_i)^2), \qquad \kappa = \frac{11\,C_G}{24\pi^2}.$$

Then $\kappa > 0$.

Here C_G is the adjoint Casimir in the chosen inner product, and $\kappa = 2b_0$ in the conventional one-loop notation $b_0 = 11C_G/(48\pi^2)$. For $SU(N)$ in the defining representation, this normalization gives $C_G = 2N^2$ and $\kappa = 11N^2/(12\pi^2)$.

The invariant form $-\mathrm{tr}(\mathrm{ad}X\,\mathrm{ad}Y)$ is $C_G\langle X, Y\rangle$. Its value at (X, X) is $\sum_i |[X, b_i]|^2$, and nonabelianness makes this positive for some X. Thus $C_G > 0$.

Status: definition; positivity is a theorem and uses simplicity. Inside this unit only two things are used: $\kappa > 0$, and the identity $\hat{\beta}_{m+1} = \hat{\beta}_m + \kappa \log L_m$. The fact that the one-loop coefficient of the actual blocked action is $-\kappa \log L + O(1)$ is proved in III.7, where $11/(24\pi^2)$ is the value of an explicit momentum integral (§2).

Lean: `YM.Gauge.slopeConstant`, `YM.Gauge.casimir`, `YM.Gauge.slopeConstant_pos`, `YM.Gauge.trace_color`.

Proposition III.9.2 (reference coupling and scale functions). Fix $\kappa > 0$, $\bar{\beta} > 1$ and an infinite sequence of odd block factors $L_m \geq 3$. Define

$$\hat{\beta}_m = \bar{\beta} + \kappa s_m, \quad M_m = e^{(\log\log\hat{\beta}_m)^2}, \quad t_m = \hat{\beta}_m^{-1/2} M_m, \quad \varepsilon_m = \hat{\beta}_m^{-11/10}, \quad \tilde{\varepsilon}_m = e^{-M_m^{1/5}},$$

and set $B_m = \lceil (\log\hat{\beta}_m)^2 \rceil$ and $\mathfrak{B}_m = [\hat{\beta}_m/2,\, 2\hat{\beta}_m]$. These are, in order: the reference coupling, the threshold multiplier, the small-field threshold, the regular and forced allowances, the cover size and the band.

Then $\hat{\beta}_{m+1} = \hat{\beta}_m + \kappa \log L_m$, $\hat{\beta}$ is nondecreasing, and $s_m \geq m \log 3$, so $\hat{\beta}_m \to \infty$.

The scale conditions are *hypotheses* on the scale sequence, not consequences of the definitions. They are:

- **Size conditions.** $\hat{\beta}_m > 1$ and $t_m \leq 1$. For every volume index p, the level-m side is at least $100(B_m + 1)$, and the level-$(m+1)$ side is at least $100(R_m + 1)$, where R_m is the block-patch radius.
- **Stage index.** The stage index satisfies $J_m \geq 4$, is nondecreasing, $J_{m+1} \leq J_m + 1$, and $J_m \to \infty$. The derivative orders, spatial powers and support exponents are nondecreasing along the levels. There are nondecreasing stage block factors ℓ_j with $L_m = \ell_{J_{m+1}}$.
- **Smallness exponent.** $0 < \theta_{J_m} \leq 1/J_m$, and θ is antitone.
- **Gentle scales.** $\log L_m / s_m \to 0$.
- **Drift and logarithmic costs.** With ω_j the drift allowance of stage j,

$$\frac{\omega_{J_{m+1}}}{\log L_m} \leq J_{m+1}^{-2}, \qquad \frac{\log(1 + \mathrm{paths} + \mathrm{width}^{-1} + \mathrm{columns} + \mathrm{relaxation} + \mathrm{support\ exponent})_{J_{m+1}}}{\log L_m} \leq J_{m+1}^{-2}.$$

Status: definitions (III.1) and elementary theorems. The scale conditions are part of the scale sequence produced in III.2.

Lean: `Calendar.hatBeta`, `threshold`, `allowance`, `forcedAllowance`, `coverSize`, `Calendar.Fits`, `Shoot.band`, `Deployment.hat`, `Timing.hat_mono`, `Schedule.divergent`.

Proposition III.9.3 (selected histories). A *history* assigns to all N, p and $m \leq N$ a level-m representation $\rho_{N,p,m}$ with coupling $\beta_{N,p,m}$. It is *selected* if the following hold.

1. **Common path data.** The path data $P_{N,m}$ of $\rho_{N,p,m}$ do not depend on p.
2. **Shape.** $\rho_{N,p,m}$ carries the level-m norm parameters and clip. It is lawful at level m:
   - its path data obey the stage bounds, with loop cap $2M_m^{\theta_{J_m}}$;
   - its regular activities are local, gauge invariant, vanish at the flat field, and have zero Hessian on affine link directions whenever their supports do not wrap; their norm bound is ε_m;
   - its forced activities satisfy the locality, eligibility and symmetry conditions of II.2, with norm bound $\tilde{\varepsilon}_m$.

   The whole activity family transforms under translations and signed permutations of the axes.

   Moreover $\beta_{N,p,m} \in \mathfrak{B}_m$.
3. **Start.** $v_{N,p,N} = 0$ and $\rho_{N,p,N}(U) = e^{-\beta_{N,p,N} S_W(U)}$ for all U.
4. **Transfer.** For $m < N$, $\rho_{N,p,m} = T^{(m)}_{\beta_{N,p,m+1}} \rho_{N,p,m+1}$ at every configuration, and
$$|\beta_{N,p,m+1} - \beta_{N,p,m} - \kappa \log L_m| \le \omega_{J_{m+1}}.$$
5. **End point.** $\beta_{N,p,0} = \bar{\beta}$.
6. **Running.** For every $\varepsilon > 0$ there is m_0 with $|\beta_{N,p,m} - \hat{\beta}_m| \le \varepsilon s_m$ for all $m \ge m_0$, $N \ge m$ and all p.

Consequences:

- **Partition function.** For all $m \le N$, $\int \rho_{N,p,m}\, d\mathrm{Haar} = Z_W(\Lambda_{N,p}; \beta_{N,p,N})$, the Wilson partition function.
- **Running law.** $\beta_{N,p,m}/s_m \to \kappa$ uniformly in $N \ge m$ and p, because $s_m \to \infty$.
- **Bare coupling.** The bare coupling of the Wilson family, $\beta_{N,p} := \beta_{N,p,N}$, satisfies $\sup_p |\beta_{N,p}/\log(1/a_N) - \kappa| \to 0$. This is (Reg1) of the main theorem with $b_* = \kappa = 2b_0$. The continuum running-coupling comparison belongs to VII.5.

Status: (1)–(6) are a specification (III.1). It is proved for the history of the main theorem in III.13 (`Spire.Uses.exists_bolt`), using the reset steps of III.9.5, the step program from III.10, and the finite-cutoff construction of Lemma 4.22. The interval shooting of Lemma 4.20 also gives a selected history from the step program. In both constructions, Lemma 4.21 derives (6) from (4) and (5).

The partition identity follows by induction from probability-kernel normalization and integrability of lawful representations. The calibration condition of VIII.1 requires one datum (scales, kinetic data and recipe) and one family of trails realizing this history, with every step a reset step for all N, p and $m < N$.

Lean: `History.Selected`, `Conv`, `Transfer`, `Running`, `History.partition_eq`, `Spec.Calibrated`.

Proposition III.9.4 (one reset step along a curve). Fix a datum $\mathfrak{D}$ consisting of three things:

- levels satisfying the scale and kernel conditions of III.9.2 and §2, and meeting the entry thresholds of the fixed numerical choices;

- a compatible horizon, a local Haar chart, positive post-block constants, and kinetic data $(\alpha_{N,m}, P_{N,m})$ satisfying the planned one-loop conditions (§2);
- a recipe, with initial, retention and continuation tests.

Fix N, p and $m < N$, let $I = [l, u]$ with $l \le u$, and let $x \mapsto \rho'_x$ $(x \in I)$ be a continuous curve of admissible level-$(m+1)$ representations.

- *Admissible* means: path data $P_{N,m+1}$, the level-$(m+1)$ norm parameters and clip, lawful with the level-$(m+1)$ constants, and accepted by the recipe.
- *Continuous* means: β and v are continuous; the regular and forced addresses, including supports, inventories and eligibility tests, are fixed; and the coefficients vary continuously in the weighted sum of regular sup-jet norms and forced uniform norms. Regular jets are measured only where their supports are small-field, at the prescribed derivative order and spatial power.

Write $\beta'_x = \beta(\rho'_x)$ and $H_x = T^{(m)}_{\beta'_x}\rho'_x$. Let (c_m, r_m) be the reset constants of the stage pair (J_{m+1}, J_m), with $c_m > 0$ and $0 < r_m \le 1$. Let $g = g_{J_{m+1}} > 0$ be the prescribed fine-stage forced-decay constant. Assume:

- **(P) Priced.** $\eta = \eta_{J_m}$ is a sensor constant (Lemma 4.7), and $2(c_m+1)\varepsilon_m \le \eta\, t_m^2\, r_m$.
- **(V) Probe.** V is a small-field configuration with $\mathcal{K}_{P_{N,m}}(V) \ge |\Lambda_m| \eta t_m^2$. Such a V exists by (P).
- **(I) Income.** There is a family $F(x,d)$, $x \in I$, $|d| \le r_m$, of level-m activities with fixed addresses, valid with $(\varepsilon_m, \tilde{\varepsilon}_m)$. Assume the catalogue conditions: each coefficient is continuous in its regular sup-jet norm or forced uniform norm, and every convergent parameter sequence admits a subsequence with common summable majorants in those weighted norms. Moreover $H_x(1), H_x(V) > 0$, and

$$\Big|(\beta'_x + \alpha_{N,m})\mathcal{K}(V) + \mathcal{L}(V) - \log\frac{H_x(1)}{H_x(V)}\Big| \le |\Lambda_m| c_m \varepsilon_m,$$

$$|\mathcal{E}_{F(x,b)}(V) - \mathcal{E}_{F(x,a)}(V)| \le |\Lambda_m| c_m \varepsilon_m |b-a|.$$

- **(R) Remainder.** Put

$$\mathrm{shape}_{x,d}(U) = e^{-(\beta'_x+\alpha_{N,m}+d)\mathcal{K}(U)-\mathcal{L}(U)-\mathcal{E}_{F(x,d)}(U)}\,\mathcal{G}_{F(x,d)}(U).$$

  Require:
  - *Shape control:* $|\mathrm{shape}_{x,a}(U) - H_x(U)/H_x(1)| \le C_{x,U}\,|f(x,a)|$ for all U and $|a| \le r_m$, where f is the scalar residual of Lemma 4.9 and, for each x, U, a finite $C_{x,U} \ge 0$ works for every $|a| \le r_m$;
  - *Forced bound:* the forced activities are bounded by $e^{-g\,M_{m+1}^2}$;
  - *Recipe:* the recipe accepts every exact trial.

Conclusion. There is a continuous $d : I \to [-r_m, r_m]$ with the following properties. Let ρ_x have path data $P_{N,m}$, activities $F(x, d(x))$, coupling $\beta'_x + \alpha_{N,m} + d(x)$ and $v_x = |\Lambda_m|^{-1} \log H_x(1)$. Then:

- $x \mapsto \rho_x$ is a continuous curve;
- $\rho_x = H_x$ at every configuration;
- ρ_x is admissible, its forced activities are bounded by $e^{-gM_{m+1}^2}$, and $|\beta(\rho_x) - \beta'_x - \alpha_{N,m}| = |d(x)| \le r_m \le 1$.

Exact density transfer, recipe continuation, admissibility, the stronger forced bound and the error bound $|\beta - \beta' - \alpha_{N,m}| \le 1$ define a *reset step*. Without (R), the same d is the unique zero in $[-r_m, r_m]$ of the scalar residual. In that case $\rho_x(1) = H_x(1)$ and $\rho_x(V) = H_x(V)$.

Status: theorem.

Lean: `CouplingReset.advance`, `correct`, `Income`, `Remainder`, `PricedReset`, `Usable`, `Directed.Move`.

Proposition III.9.5 (explicit reset and step program).

(a) *Explicit correction.* Fix the datum, input curve and reset constants of III.9.4, and assume G is simple and the input couplings lie in $\mathfrak{B}_{m+1}$. Suppose that, for every $x \in I$, the blocked density has the early presentation

$$H_x(U) = \exp\Big(\tau_x|\Lambda_m| - \beta'_x\mathcal{K}(U) - \mathcal{L}(U) - \sum_i \chi_{X_i}(U)E_{x,i}(U)\Big)\,\mathcal{G}_x(U)\,e^{-\alpha_{N,m}\mathcal{S}(U)},$$

where χ_X indicates that X is small-field and $\mathcal{S}$ is the masked quadratic receipt of the coarse path data. Assume:

- The early regular functions are local, gauge invariant and smooth through the required derivative order. Their sup-jet bounds $h_{x,i} \ge 0$ satisfy, at every anchor o,

$$\sum_{i:o_i=o} e^{(a+3)\,\mathrm{cover}_{B_m}(X_i)}h_{x,i} \le H_x^E,$$

where a is the level-m support exponent. The early forced functions satisfy the forced-family conditions and a site-pinned bound w_x at support exponent $a+1$. The full family transforms under translations and the point group.
- The addresses are fixed along I, and the early functions satisfy the coefficient-continuity and summable-tail conditions used in III.9.4(I).
- The reset is priced; the clip is compatible with the coarse norm parameters; and the width and cubic conditions hold. There are numerical output bounds e_x, k_x such that $H_x^E \le c_m\varepsilon_m/4$, the regular and forced reorganization estimates fit within e_x, k_x, and these bounds imply both validity with $(\varepsilon_m, \tilde{\varepsilon}_m)$ and the forced bound $e^{-gM_{m+1}^2}$. These are the fee and closure conditions of Lemma 4.17.
- The recipe accepts the reorganized representation and its continuation from ρ'_x.

Then the reset correction is the isotropic Hessian coefficient δ_x of the early regular functions (Lemma 4.12):

$$d(x) = \delta_x, \qquad \beta(\rho_x) = \beta'_x + \alpha_{N,m} + \delta_x, \qquad t_m^2|\delta_x| \le H_x^E \le c_m\varepsilon_m/4, \qquad v_x = \tau_x - |\Lambda_m|^{-1}\sum_i E_{x,i}(1).$$

The output is continuous and has all the properties of III.9.4. The Gaussian-sector formulation additionally requires its integration licenses and the identity between that integral and the push; a direct early presentation of the push supplies the equality used in the reset.

(b) *Program.* Assume G is simple. Suppose the numerical conditions above hold at every level, the early presentation and its local bounds and continuity are available along every continuous admissible input curve with couplings in $\mathfrak{B}_{m+1}$, and the recipe accepts these outputs. Suppose also that the start-up curves below exist and pass the recipe's initial and retention tests. Then:

- For all N, p, there is a continuous admissible curve $\beta \mapsto F(\beta)$ on $\mathfrak{B}_N$ with coupling β, vacuum constant 0 and density $e^{-\beta S_W}$, passing the initial tests.
- For all $N, p, m < N$, every continuous admissible input curve on a closed interval has, when its couplings lie in $\mathfrak{B}_{m+1}$, a continuous output curve related to it pointwise by reset steps. The finer bound $|\beta - \beta' - \alpha_{N,m}| \le r_m$ also holds.

The output coupling may lie outside $\mathfrak{B}_m$; shooting restricts the parameter interval to restore the band (Lemma 4.20).

Status: conditional theorems. III.10 supplies the physical inputs, including the Gaussian-sector identity and the early presentation. Its Roost.Factor.worksNow concludes the step program from these results.

Lean: `Return.reset`, `Return.result`, `Collection.earlyValue`, `Use.supplier`, `CouplingReset.program`, `Supplier.ports`, `Directed.Program`, `Return.Prices`, `Collection.Fees`, `Collection.Compared`.

Proposition III.9.6 (Wilson start-up). Let c be a clip and b a set of norm parameters with $0 < t \le 1$. They are *compatible* if $c_{\mathrm{dir}} \le 1$, $B \ge 2$, and $t\, e^{(B-2)/c_{\mathrm{rel}}} \ge 2$.

1. **Exact start.** For every $\beta \in \mathbb{R}$ and every configuration U, $\rho_\beta^{\mathrm{start}}(U) = e^{-\beta S_W(U)}$. Here $\rho_\beta^{\mathrm{start}}$ has:
   - the *Wilson top* path data P_c^{top} (Lemma 4.2);
   - $v = 0$;
   - regular activities $\beta(1 - c_{\mathrm{dir}})\times$ (the one-plaquette residuals of small-field plaquettes), averaged over the point group;
   - forced activities $e^{-\beta\, \mathrm{charge}(D,U)}$, one for each connected component D of the large-field region for the padded adjacency relation.
2. **Residual bound.** For every gauge group G and all orders k and spatial powers q there are $C > 0$ and $\delta \in (0, 1]$ with the following property. For every set of norm parameters with order k, power q and threshold $t \le \delta$, every torus, every plaquette p and every site o of p, the residual R_p has weighted sup-jet norm at most Ct^5 at o.
3. **Eventual bounds and continuity.** For each fixed stage with $c_{\mathrm{dir}} \le 1$, there is β^* such that, for every realized choice of levels and every N at that stage with $\widehat{\beta}_N \ge \beta^*$, the start-up conditions hold: compatibility, small threshold, and the regular and forced norm bounds $\varepsilon_N, \tilde{\varepsilon}_N$. The start-up curve is continuous on $\mathfrak{B}_N$. It is admissible for a given recipe if its prescribed top paths are P^{top}, these paths satisfy the stage bounds, and its initial and retention tests accept the displayed curve.
4. **Installation.** Suppose the old kinetic data satisfy the planned conditions, the Wilson top paths satisfy the stage bounds, and a revised horizon is supplied whose top quadratic datum is the Wilson one, whose lower data agree with the old horizon, and whose initial stationary guesses agree. Then the kinetic data can be replaced by data for that horizon with $P_{N,N} = P^{\mathrm{top}}$, the same α, and every planned condition preserved.

Status: theorem.

Lean: `WilsonStartup.Start.evaluates`, `residual_price`, `Start.Allows`, `Timing.enough`, `Planned.Port.card`, `Change.engine`, `WilsonStartup.Decent`, `Design.Suitable`, `Change.Overlay`.

Proposition III.9.7 (entry thresholds). Fix a pre-block stage profile and assume G is simple.

There are sensor constants $\eta_j > 0$, chosen before the block sizes, with the following property. Fix the block factors ℓ_j, exponents $0 < \theta_j \le 1$, reset constants, forced-decay forecasts and post-block costs. For each allowed stage transition $i \le j \le i+1$, let f_{ji} be the inherited coefficient and assume

$$8 f_{ji} \le q_i \ell_j^{\min}, \qquad u_{ji}(x) \longrightarrow 0 \quad (x \to \infty),$$

where $q_i = 1/[64(1+A_i)] > 0$, $A_i \ge 0$ is the coefficient of the early size in the regular reorganization bound (Lemma 4.17), and $\ell_j^{\min}$ is the preassigned block-size floor.

Every entry-threshold record V has a componentwise increase U such that every further increase $W \ge U$ satisfies the following conditions beyond its thresholds:

- the reset pricing inequality;
- the cubic inequality of Lemma 4.10;
- clip compatibility when $c_{\mathrm{dir}} \le 1$, the width condition and the regular and forced reorganization inequalities;
- the inherited inequality of Lemma 4.17.

For a realized choice of levels, these conclusions require the same stage profile, block factors and exponents, and payment of the thresholds: $W^{\mathrm{entry}}_{J_{m+1}} \le \widehat{\beta}_m$ for every m. To obtain III.9.5's physical bounds, also assume $4q_i \le c_{ji}$ and $4g_j \le \mathrm{rate}_{ji}$; raising thresholds does not supply these inequalities or the early presentation. Start-up bounds admit the same monotone choice of thresholds when $c_{\mathrm{dir}} \le 1$ at every used stage.

Status: theorem about numerical conditions, conditional on the compatibility inequalities stated above.

Lean: `Time.beforeWait`, `StrongStartup.prices`, `Timing.choose`, `Time.credit`.

Proposition III.9.8 (marker-rooted to site-pinned polymer sums). Fix $M, \mu \in \mathbb{N}$, $a \in \mathbb{R}$ and $c \ge 0$. Let (X_i) be polymers on a four-dimensional torus with weights $f_i \ge 0$, indexed by an arbitrary set. Fix a finite set of marker names; each polymer carries a subset of these names, the markers sitting at sites $o(v)$, with at most μ markers per site. Suppose that every polymer with $f_i > 0$ contains the site of one of its markers, and that for every marker v

$$\sum_{i:\, v \text{ is a marker of } i} e^{(a+1)\,\mathrm{cover}_M(X_i)} f_i \le c.$$

All sums of nonnegative terms here are required to be summable, including when the index set is uncountable. Then for every site x

$$\sum_{i:\, x \in X_i} e^{a\,\mathrm{cover}_M(X_i)} f_i \le \mu\, e\, (2(16M+9))^4\, c,$$

uniformly in the size of the torus.

Status: theorem. It is used for observable insertions in III.12 and III.15, not for the coupling flow.

Lean: `Extra.Multi.root_hit`.

What the rest of the construction uses.

- The shape, band and transfer clauses of III.9.3.
- The reference coupling $\widehat{\beta}$ used in the later estimates.
- The step program III.9.5(b), the input of the shooting.
- The finer accuracy $|\beta_{N,p,m} - \beta_{N,p,m+1} - \alpha_{N,m}| \le r_m$.
- III.9.8.

3.12.2 2. What it uses

- **III.1 (block-spin step and interfaces).** This unit uses:
  - the step $T_\beta^{(m)}$ with its symmetry, gauge covariance and probability normalization (`Calendar.op`, `BlockControl`, `StepData.push_integral`);
  - the interface records for histories and selection, the datum $\mathfrak{D}$ (`Directed`), and reset step / program;
  - the conversion of a reset step into the transfer inequality (`Directed.links`). This uses $|\beta - \beta' - \alpha| \le 1$ and $|\alpha + \kappa \log L| + 1 \le \omega$.
- **II.2 (polymer activities).** Representations, activities, their bounds and the large-field masks: `Rep`, `Activities.Valid`, `EBound`, `KBound`, `PathData`, `PathData.Bounds`.
- **III.2 (order of constants and shooting).** This unit uses:
  - the constants chosen before the block size, and the block sizes (`Supply.Sheet`, `Supply.Large`). At each used stage, the block factor ℓ_j satisfies $\omega_j / \log \ell_j \le \kappa/4$ and $\omega_j / \log \ell_j \le j^{-2}$.
  - the record of the scale sequence having met every entry threshold (`Supply.Paid`);
  - the shooting construction and the running estimate (Lemmas 4.20–4.21).
- **III.7 (one-loop coefficient).** The kinetic data satisfy the planned conditions (`Kin.Planned`). Among them:
  - $|\alpha_{N,m} + \kappa \log L_m| + 1 \le \omega_{J_{m+1}}$;
  - the Hessian of the fluctuation term (fine loop functional plus $\frac{1}{2} \log \det$ of the fluctuation Hessian minus the log Haar Jacobian) agrees with that of $\alpha_{N,m}$ times the coarse quadratic action plus the coarse loop functional, with an error bounded by $|\Lambda_m| \Gamma \Lambda e^{-c \operatorname{side}(\Lambda_m)}$ on every pair of bounded plane-wave directions (`FiniteBound`);
  - $P_{N,N}$ matches the Wilson energy to fourth order.

  The bound on α comes from `Stationary.pouring`, which, for fixed admissible analytic input parameters, constructs one-loop coefficients and constants $D \ge 0, L_{\min}$ with $|\alpha + \kappa \log L| \le D$ for every odd $L \ge L_{\min}$ (`HorizCalc`). The constant is produced by two steps:
  - the scalar-minus-half-vector bubble equals $2\,\mathrm{pole}(p)$, where $\mathrm{pole}(p) = 4C_G(p_1^2|p|^{-6} - 4p_1^2 p_0^2 |p|^{-8} - |p|^{-4})$ (`haloRate`, `poleVec`);
  - $(2\pi)^{-4} \int_{\mu \le |p| \le 1} \mathrm{pole} = -\kappa \log \mu^{-1}$ (`poleSpin`), with the remaining errors bounded (`absoluteGeode`).

  The angular average of pole is $-\frac{11}{3} C_G |p|^{-4}$, the classical split $\frac{1}{3} - 4$ into orbital and spin parts.
- **III.13 (kinetic data and history).** Finite-cutoff kinetic data are built by downward recursion in m. The limiting one-step relation used to compare such data contracts with factor $\frac{1}{2}$ (`Reward.fieldArray`, `Reward.step_contracted`, `Reward.ties`). The selected history of the main theorem is also built there.
- **III.10 and III.4 (the nonlinear step).** These provide the hypotheses of III.9.5:
  - the identification of the blocked density with a Gaussian-sector integral (`Return.earnings`);
  - its early presentation (`Return.atlas`);
  - the local activity bounds (`Collection.Local`), continuity and recipe acceptance.

  These are delivered by `Rain.glazing` and `Roost.Factor`.
- **I.1 and III.16 (analytic toolkit).** Lie-group facts and Taylor jets of the plaquette function.

Standard mathematics.

- Intermediate value theorem, and continuity of the inverse of a uniformly increasing family.
- Taylor's formula with uniform constants on compact sets.
- Schur's lemma: an Ad-invariant bilinear form on a simple $\mathfrak{g}$ is a multiple of the inner product.
- The representation theory of the hyperoctahedral group on plaquette orientations, used through explicit reflections and permutations.
- Fourier analysis on tori, analytic matrix inversion and exponential decay of Fourier coefficients.

References:

- One-loop β-function: Gross–Wilczek, *Phys. Rev. Lett.* 30 (1973) 1343; Politzer, *Phys. Rev. Lett.* 30 (1973) 1346.
- The orbital/spin split $\frac{1}{3} - 4$: Nielsen, *Am. J. Phys.* 49 (1981) 1171; Hughes, *Nucl. Phys. B* 186 (1981) 376.
- Lattice gauge theory: Wilson, *Phys. Rev. D* 10 (1974) 2445; Seiler, *Lecture Notes in Physics* 159 (1982).
- Block-spin renormalization of lattice gauge fields with coupling-constant renormalization in four dimensions: Bałaban, *Commun. Math. Phys.* 109 (1987) 249; 116 (1988) 1; 119 (1988) 243; 122 (1989) 175 and 355.
- Exposition of Bałaban's method: Dimock, "The renormalization group according to Balaban" I–III (*Rev. Math. Phys.* 25 (2013); *J. Math. Phys.* 54 (2013); *Ann. Henri Poincaré* 15 (2014)).
- Tracking a marginal coupling and choosing the bare parameters so that the flow ends at a prescribed value: Bauerschmidt–Brydges–Slade, *Introduction to a Renormalisation Group Method*, LNM 2242 (2019).

Imported constants and assumptions.

- **The drift allowances ω_j.** These are constants fixed before the block size, with $\omega_j \geq 1 + D$. The one-loop error D is fixed in III.7 and III.13, and ω is then made small relative to $\log L$ by choosing L large (III.2).
- **The exponents.** The exponents $-1/2$, $-11/10$, $(\log\log)^2$ and $1/5$ in III.9.2 are part of the definitions.
- **The hypotheses of III.9.5(a).** These are carried here as assumptions and discharged in III.10.

3.12.3 3. Strategy

The nonlinear step gives an early presentation of the blocked density. Its marginal coefficient has three contributions: the input coupling β', the one-loop shift $\alpha_{N,m}$, and the isotropic Hessian coefficient δ of the early regular functions. The reset sets $\beta = \beta' + \alpha_{N,m} + \delta$. Subtracting flat values and quadratic projections, adding compensating path terms, and absorbing mask discrepancies into the forced gas preserves the full density. The output activity bounds and $|\delta| \leq r_m \leq 1$, together with $|\alpha_{N,m} + \kappa \log L_m| + 1 \leq \omega$, give the transfer inequality.

Two configurations identify the reset: the flat field and a small-field stripe V with $\mathcal{K}(V) \geq |\Lambda|\eta t^2$. Matching at the flat field fixes $v = |\Lambda|^{-1} \log H(1)$. Matching the log-ratio at V gives a scalar equation $f(x, d) = 0$. The pricing and activity-motion bounds make f increasing with slope at least $\mathcal{K}(V)/2$, and give $|f(x, 0)| \leq \mathcal{K}(V)r/2$. Its unique root lies in $[-r, r]$ and depends continuously on x. Shape control gives equality at every configuration. For the reorganized activities, $f(x, d) = (d - \delta_x)\mathcal{K}(V)$, so the correction is explicit.

The interval shooting chains these steps from the Wilson lattice down to level 0. At each level, the transfer inequality forces the two endpoint couplings to lie on opposite sides of $[\bar{\beta}+\kappa s_m/2, \bar{\beta}+2\kappa s_m]$. Continuity selects a subinterval whose couplings stay within that window and attain its endpoints. At level 0, a surviving bare parameter gives $\beta_0 = \bar{\beta}$. Summing the transfer errors gives the running law, uniformly in cutoff and volume. III.13 uses compatible finite-cutoff fixed points to obtain the centered history and its analytic continuations.

3.12.4 4. Steps

3.12.4.1 4.1 Reference scales Lemma 4.1 (recursion of the reference coupling). $s_{m+1} = s_m + \log L_m$, so $\hat{\beta}_{m+1} = \hat{\beta}_m + \kappa \log L_m$. $\hat{\beta}$ is nondecreasing if $\kappa \geq 0$. Since $L_i \geq 3$, $s_m \geq m \log 3 \to \infty$, and $\hat{\beta}_m \to \infty$ when $\kappa > 0$.

Sketch: $a_m = (L_0 \cdots L_{m-1})^{-1}$ and the definitions.

Lean: `Schedule.scale_succ`, `Deployment.hat`, `Timing.hat_mono`, `Schedule.divergent`.

3.12.4.2 4.2 The Wilson start-up Lemma 4.2 (splitting the plaquette action). Write $s(u) = |F(u)|^2/2 + Q_4(F(u)) + R(u)$. Near the identity, curvature gives local coordinates on G. Here Q_4 is the diagonal fourth Taylor coefficient of the plaquette function in those coordinates. The odd coefficients vanish by inversion symmetry. The residual R, defined by the displayed identity for every $u \in G$, has vanishing derivatives of orders $0, \ldots, 4$ at the identity.

For $c_{\mathrm{dir}} \leq 1$, the top path data P_c^{top} consist of:

- the clipped plaquette term, which equals $c_{\mathrm{dir}}\, s(U_p)$ in the small-field region;
- the row term $\sqrt{1 - c_{\mathrm{dir}}}\, F_p$, contributing $(1 - c_{\mathrm{dir}})|F_p|^2/2$;
- the quartic atoms $(1 - c_{\mathrm{dir}})Q_4(F_p)$;
- no cubic and no loop terms.

They satisfy the zero-momentum normalization ($c_{\mathrm{dir}} + (1 - c_{\mathrm{dir}}) = 1$). On the small-field region, $\beta S_W = \beta\, \mathcal{K}_{P^{\mathrm{top}}} + \beta(1 - c_{\mathrm{dir}}) \sum_p R_p$.

Sketch: The derivative of the curvature map at the identity is the identity, so the analytic inverse function theorem gives curvature coordinates. Inversion sends F to $-F$ and preserves s. Taylor expansion through order 4 gives the quadratic and quartic terms and the residual. The identity $s = \frac{1}{2}\mathrm{chord}^2$ fixes the normalization.

Lean: `convert_taylor`, `surplus`, `surplus_jet`, `Design.top`, `Design.principal_top`.

Lemma 4.3 (exact start). For a compatible pair (c, b), every β and every U: the path-data channels of P^{top} at β, plus the regular activities (residuals of small-field plaquettes), plus the large-field remainders, add up to $\beta S_W(U)$, the sum being averaged over the point group. Moreover the gas equals $\exp(-\beta \cdot (\text{large-field remainder}))$. Hence $\rho_\beta^{\mathrm{start}} = e^{-\beta S_W}$, which is III.9.6(1).

Sketch: Fix U. The only nonzero gas term is the family of connected components of the large-field region for the padded adjacency relation (cover size $B \geq 2$ and the compatibility condition separate the padded components). So the gas collapses to a single product $\prod_D e^{-\beta\, \mathrm{charge}(D,U)}$. The charges sum to the large-field remainder, and the channel identity accounts for the rest.

Lean: `channel_split`, `Start.gas_raw`, `Start.issues`, `Start.evaluates`.

Lemma 4.4 (one-plaquette residual). This is III.9.6(2).

Sketch: R_p is a real-analytic function of the four link matrices of p. On a small-field plaquette (chord $\le t$) its derivatives of order $l \le k$ along directions of size $\le t(1+4)^q$ are bounded by $C_l t^5$. This is Taylor's formula with the flatness of Lemma 4.2 and compactness of G; the factor $(1+4)^q$ is the spatial weight of sites within distance 4. Take $C = 1 + \sum_l C_l$ and the smallest δ.

Lean: `residual_price`, `residual_jet`, `healthy_chord`, `Quad.small_jet`.

Lemma 4.5 (eventual start-up bounds). Under the hypotheses of III.9.6(3), the start-up bounds hold beyond a stage-dependent threshold, uniformly in N, p and $\beta \in \mathfrak{B}_N$. With the top-path bounds and recipe approval specified there, the start-up curve is admissible.

Sketch: Put $x = \widehat{\beta}_N$ and $M = e^{(\log\log x)^2}$. The regular bound is a stage-dependent constant times $2xt_N^5 = 2x^{-3/2}M^5 = o(x^{-11/10})$. For the forced terms, the proof uses a decay exponent M and an additional polymer-counting factor $e^\gamma - 1$, with $\gamma = e^{-M/2}$. The connected-polymer estimate bounds the forced norm by $e^{-M+1} \le e^{-M^{1/5}}$ for large x. Compatibility follows from $B_N \asymp (\log x)^2$ and $t_N = x^{-1/2}M$. These estimates give the displayed continuous start-up family; admissibility also uses its path bounds and the recipe tests.

Lean: `Timing.enough`, `Start.ink`, `Start.Loan.kernel`, `Planned.Port.card`.

Lemma 4.6 (installing the Wilson top). Assume the old fine paths satisfy the stationary, positive-Hessian, initial-guess, classical-matching and finite-period one-loop conditions for a step. Let the new fine paths have the same main-action jets through order 4, and suppose both fine loop kernels vanish. With the same local Haar chart and block branches, there are new stationary data satisfying those same conditions, with the same coarse paths and one-loop coefficient. With the revised horizon and top-path bounds of III.9.6(4), this replacement gives planned kinetic data at every cutoff, preserving the stationary data for $m + 1 < N$.

Sketch: Equality of fourth jets gives equality of the fluctuation Hessians at the flat field and of their determinant jets through order 2. The analytic implicit function theorem constructs the new stationary branch; finite-order agreement transfers the initial guess, classical match and one-loop bound. The old and Wilson top actions both match the Wilson energy through order 4. Only the step $N \to N-1$ changes.

Lean: `Change.top_step`, `Change.engine`.

3.12.4.3 4.3 The scalar response **Lemma 4.7 (sensor).** Assume $\mathfrak{g} \neq \{0\}$, fix a clip with $A = c_{\mathrm{dir}} > 0$, and let $C \ge 0$. Then there is $\eta > 0$, depending only on these data, with the following property. For every even torus side, every path datum satisfying the path bounds with cap C, and every set of norm parameters with $0 < t \le 1$, there is a configuration V that is small-field everywhere and satisfies $\mathcal{K}(V) \ge |\Lambda|\,\eta\, t^2$.

Sketch:

- Choose $u \in G$ with chord $h = \tau t$, where $\tau = \min(\varepsilon_G, \frac{1}{4}, A/(512(1+C)))$; such u exists by the intermediate value theorem along a one-parameter subgroup.
- Put u on the axis-0 links at sites whose coordinate 1 is even, and 1 elsewhere. Every $(0,1)$-plaquette has chord h, and every plaquette has chord $\le 4h \le t$.
- The quadratic term contributes $\ge |\Lambda| A h^2/2$. The cubic and quartic atoms contribute at least $-|\Lambda|(C(4h)^3 + C(4h)^4) \ge -|\Lambda| A h^2/4$.

- The point-group average permutes plaquettes and preserves these bounds. Hence $\eta = A\tau^2/4$.

Lean: `CouplingReset.sensor`, `reachable_chord`.

Lemma 4.8 (monotone response, continuous root). Let $I = [l, u]$ with $l \le u$, let $r \ge 0$, and let $f : I \times [-r, r] \to \mathbb{R}$ satisfy:

- there is $\mu > 0$ such that $f(x, b) - f(x, a) \ge \mu(b - a)$ for every $x \in I$ and $-r \le a \le b \le r$;
- $f(x, \cdot)$ is continuous for every $x \in I$, and $f(\cdot, a)$ is continuous for every $a \in [-r, r]$;
- $|f(x, 0)| \le \mu r$ for every $x \in I$.

Then there is a unique $d(x) \in [-r, r]$ with $f(x, d(x)) = 0$. The function d is continuous, and $|d(x) - a| \le |f(x, a)|/\mu$ for every $x \in I$ and $a \in [-r, r]$.

Sketch: $f(x, -r) \le 0 \le f(x, r)$, so the intermediate value theorem gives a zero, unique by strict monotonicity. Moreover $|d(y) - d(x)| \le |f(y, d(x)) - f(x, d(x))|/\mu \to 0$.

Lean: `Response`, `Response.balance_solve`, `Response.continuous`, `solve`.

Lemma 4.9 (two-point readout). Fix path data, a clip, norm parameters and activities $F(x, d)$ whose regular functions vanish at the flat configuration. Let V be small-field everywhere and suppose $H_x(1), H_x(V) > 0$. Define

$$f(x, d) = (\beta'_x + \alpha + d)\,\mathcal{K}(V) + \mathcal{L}(V) + \mathcal{E}_{F(x,d)}(V) - \log \frac{H_x(1)}{H_x(V)}.$$

Install the trial with coupling $\beta'_x + \alpha + d(x)$ and vacuum constant $v_x = |\Lambda|^{-1} \log H_x(1)$. If $f(x, d(x)) = 0$, the trial equals H_x at 1 and at V. If moreover the template satisfies shape control (III.9.4(R)), it equals H_x at every configuration.

Sketch: At $U = 1$ the kinetic, loop and activity terms vanish and the gas is 1. At small-field V the gas is 1. Shape control at a zero of f gives shape $= H_x/H_x(1)$.

Lean: `Template.residual`, `Template.passes`, `Template.realizes`, `Template.response`.

Lemma 4.10 (surplus from a pre-reset presentation). Suppose that on $V = 1$ and on every small-field V

$$H_x(V) = \exp\left(u_x|\Lambda_m| - \beta'_x\,\mathcal{M}(V) - \alpha_{N,m}\,\mathcal{S}(V) - \mathcal{L}(V) - \mathcal{E}_x(V)\right),$$

where $\mathcal{M}$ is the unmasked main term (quadratic, cubic and quartic), $\mathcal{S}$ its quadratic part, and $\mathcal{E}_x$ has size $\le c_m\varepsilon_m/4$. Assume the coarse path bounds with cap C and the cubic condition $|\alpha_{N,m}|\,2Ct_m^3 \le c_m\varepsilon_m/2$. Then the surplus inequality of III.9.4(I) holds. If, in addition, the output activities satisfy the catalogue and slow-variation assumptions of III.9.4(I), then all of (I) holds.

Sketch:

- On small-field V, $\mathcal{K}(V) = \mathcal{M}(V)$ and the loop functional is unmasked.
- $|\mathcal{K}(V) - \mathcal{S}(V)| \le 2C|\Lambda|t^3$, since the cubic and quartic atoms are $O(t^3)$ per site.
- The regular activities at 1 and at V cost $2 \cdot |\Lambda|c\varepsilon/4$.
- The cubic condition absorbs $|\alpha|\,|\mathcal{K} - \mathcal{S}|$. It holds eventually because $|\alpha| \le \omega + \kappa \log L$ and $t^3 \ll \widehat{\beta}^{-11/10}$.

Lean: `Prereset.income`, `Prereset.value_income`, `pays_surplus`, `higher_estimate`, `CubicGate`, `within_alpha`.

3.12.4.4 4.4 The reset step Lemma 4.11 (one step). This is III.9.4.

Sketch:

- (P) and (V) give $|\Lambda_m|(c_m+1)\varepsilon_m \le \frac{1}{2}\mathcal{K}(V)r_m$.
- Slow motion in d gives slope $\ge \mu := \frac{1}{2}\mathcal{K}(V)$.
- The surplus inequality and $|\mathcal{E}(V)| \le |\Lambda|\varepsilon_m$ give $|f(x,0)| \le \mu r_m$.
- f is continuous in x by Lemma 4.15.
- Lemma 4.8 gives d; Lemma 4.9 installs the trial; (R) gives the identity at all configurations, the forced bound and recipe acceptance.
- $|\beta - \beta' - \alpha| = |d| \le r_m \le 1$.

Lean: `correct`, `advance`, `Income.paidResponse`, `Income.balance`.

Lemma 4.12 (the marginal coefficient of the activities). Let f be a local, gauge-invariant regular function anchored at o, smooth through the prescribed derivative order, and let $\mathfrak{g}$ be simple. The second derivative of f at the trivial configuration, along the canonical affine link test fields centered at o, with curls X in orientation b and Y in orientation b', equals $\mathrm{coeff}_f(b,b')\langle X, Y\rangle$ (Schur). For the projection used below, retain this Hessian coefficient when the support enlarged by the radius-2 halo of o does not wrap around the torus, and set the projected coefficient to 0 otherwise.

For a family A of such functions with an anchored regular norm bound, whose addresses and values transform under translations and the point group, assume the torus side is at least 4 and put

$$\delta(A) = \sum_{f \text{ anchored at } 0} q_f(01,01).$$

Here q_f denotes the projected coefficient just defined. Then:

- for every site o, $\sum_{f \text{ anchored at } o} q_f(b,b') = \delta(A)\,\mathbf{1}_{b=b'}$;
- the sum of the projected Hessian forms at o is $\delta(A)\sum_b |w_b|^2$. Its quadratic energy is $\frac{1}{2}\delta(A)\sum_b |w_b|^2$;
- if A has regular size c, then $t^2|\delta(A)| \le c$.

Sketch: Gauge invariance makes each color bilinear form invariant under the adjoint action, hence scalar by simplicity. For each pair of distinct plaquette orientations, reflecting an axis belonging to exactly one of them reverses the corresponding off-diagonal coefficient; its sum therefore vanishes. Signed permutations carry each diagonal coefficient to $(01,01)$. The bound follows from the second derivative in a direction of size t.

Lean: `Averaging.delta`, `Averaging.isotropic`, `Averaging.identity`, `Averaging.delta_bound`, `label_color`, `projection`.

Lemma 4.13 (exact reorganization). Assume the early presentation, local bounds, compatible clip, width, reorganization inequalities and recipe acceptance of III.9.5(a). Let δ_x be the coefficient of Lemma 4.12. The reorganized representation has the same paths, coupling $\beta'_x + \alpha_{N,m} + \delta_x$, and vacuum constant $\tau_x - |\Lambda_m|^{-1}\sum_i E_{x,i}(1)$. Its density equals H_x at every configuration, and it obeys the level-m allowances and the forced bound $e^{-gM^2_{m+1}}$.

Sketch: For an early regular function E_i with support X_i and anchor o_i, put $Y_i = X_i \cup \mathrm{halo}_2\{o_i\}$. Let Q_i be one half of its projected Hessian applied to the local clover curvatures, and set

$E_i^0 = E_i - E_i(1) - Q_i$. Define D_i to vanish when Y_i is small-field, and otherwise by

$$D_i = E_i(1) + \chi_{\{o_i\}} Q_i - \chi_{X_i} E_i.$$

Then, at every configuration,

$$\chi_{X_i} E_i = E_i(1) + \chi_{\{o_i\}} Q_i - D_i + \chi_{Y_i} E_i^0.$$

The projected terms sum to $\delta_x/2$ times the small-field clover-curvature square. The regular terms also acquire δ_x times the difference between each quadratic clover proxy and its original quadratic path term, and $-(\alpha_{N,m} + \delta_x)$ times each masked cubic or quartic path term. The remaining mask discrepancies become logarithmic factors in the forced gas.

More precisely, let $\mathcal{E}_x^{\mathrm{new}}$ denote the resulting regular sum and ℓ_x the sum of these logarithmic factors before they are absorbed into the gas. The activity and path splitting identities give

$$(\alpha_{N,m} + \delta_x)\mathcal{K} + \mathcal{E}_x^{\mathrm{new}} - \ell_x = \alpha_{N,m}\mathcal{S} + \mathcal{E}_x^{\mathrm{early}} - \sum_i E_{x,i}(1).$$

Absorbing e^{ℓ_x} into the original forced gas proves equality of the full densities, including at configurations where masks fail. The reorganization estimates bound the regular terms and the added forced factors; the numerical closure conditions give the claimed allowances. Recipe acceptance is a separate hypothesis.

The trial activities are independent of the correction parameter d. Thus exact equality at $d = \delta_x$ gives $f(x, d) = (d - \delta_x)\mathcal{K}(V)$ and shape control on $[-r_m, r_m]$. The unique root of Lemma 4.11 is therefore δ_x.

Lean: `Collection.Local.actual`, `Local.exponent`, `Return.settled`, `Return.grants`, `Return.earns`, `Return.remainders`, `Return.physical_shape`, `Return.reset`, `Residual.splits`, `Local.first_split`, `Collection.path_split`. *(proof read, including the reorganized activities in `Local.renew`)*

Lemma 4.14 (from per-step inputs to the program). This is III.9.5(b). The numerical conditions (`Use.Machine`), the existence of early presentations along every admissible input curve (`Use.Offering`) and the start-up curves (`Use.Origins`) together give the step program, assuming G is simple and the machine uses the compatible recipe and physical bounds of III.9.5.

Sketch: For each input curve, Lemma 4.7 provides a probe. Lemma 4.13 provides (I) and (R). Lemma 4.11 then gives the output curve and the bound $|\beta - \beta' - \alpha| \le r_m$.

Lean: `Use.supplier`, `Machine.packet`, `Supplier`, `program`.

3.12.4.5 4.5 Continuity of the transport

Lemma 4.15 (joint continuity of the blocked density). Let $x \mapsto \rho'_x$ be a continuous curve of lawful representations with fixed path data, clip and norm parameters, and assume the block kernel satisfies the branch conditions of III.1. Then $(x, V) \mapsto (T^{(m)}_{\beta'_x} \rho'_x)(V)$ is jointly continuous, and $x \mapsto \rho'_x(U)$ is continuous for every U.

Sketch: Continuity of the activities in the polymer norm gives uniform convergence of the additive part and of the gas in U. This yields a local uniform bound on ρ'_y, and dominated convergence applies under the kernel. The kernel itself is continuous in β and in the coarse configuration.

Lean: `FixedRepData.continuous_push`, `FixedRepData.continuous_eval`, `Catalog.stable`.

3.12.4.6 4.6 Entry thresholds Lemma 4.16 (monotone thresholds). Let $T_j(x)$ be properties, one for each stage j, each holding for all large x. Then any record u_j can be raised to a nondecreasing record $m_j \geq u_j$ such that $T_j(x)$ holds for all $x \geq m_j$. For each fine stage j, combine the two possible coarse targets $i = j$ and $i = \max(j-1, 0)$ first.

Sketch: Take m_j to be the prefix maximum of $\max(\text{threshold}_i, u_i)$ over $i \leq j$.

Lean: `collect_upper`, `eventually_transitions`, `StepStages`, `DelayLe`.

Lemma 4.17 (eventual reset inequalities). Fix an allowed transition from fine stage j to coarse stage i, its block factor L, and all the constants in III.9.7. Write $x = \hat{\beta}_m$, $y = x + \kappa \log L$, $M = e^{(\log\log x)^2}$ and q_i for the early-activity fraction. The conclusions are uniform for $|\alpha| \leq \omega_j + \kappa \log L$ and $\beta \in [y/2, 2y]$. Then all the delay-dependent reset conditions hold for sufficiently large x.

Sketch: The sufficient pricing condition $2(c+1) \leq \eta r M^2$ holds because $M \to \infty$. Multiplication by x^{-1} gives $2(c+1)x^{-1} \leq \eta r t^2$; since $\varepsilon = x^{-11/10} \leq x^{-1}$ for $x \geq 1$, this implies the required pricing inequality. The cubic condition follows from $t^3/\varepsilon = x^{-2/5} M^3 \to 0$.

For the local reorganization, its regular bound has the form $A_i H + B_i |\alpha| t^3$, with A_i, B_i fixed before L. The choice $H = q_i \varepsilon$, with $A_i q_i \leq 1/64$, leaves room for an output bound $\varepsilon/32$. The forced estimate controls both the early forced terms and the added logarithmic factors, at support exponent $a+1$, where a is the coarse support exponent. The width condition is $4(a+1) \leq \sigma B$, where σ is the path-decay width. It holds since $B \to \infty$, as does clip compatibility. The corresponding forced output bound is $e^{-R_{ji}(y,\beta)}$. If $4g_j \leq \text{rate}_{ji}$ and $\beta \in [y/2, 2y]$, its definition gives $R_{ji}(y, \beta) \geq g_j M(y)^2$; for large x it also gives the ordinary forced allowance at x.

The inherited condition is

$$(f_{ji}/L + u_{ji}(x))\varepsilon(y) + \mathcal{R}_{ji}(x, y, \beta) \leq q_i \varepsilon(x), \qquad \beta \in [y/2, 2y].$$

Here $\mathcal{R}_{ji}$ is the fixed step-estimator residual at exponent θ_j; it is $o(\varepsilon(x))$ uniformly in this band when $\theta_j \leq 1$, and $u_{ji}(x) \to 0$. The pre-block inequality gives $f_{ji}/L \leq q_i/8$. Together with $\varepsilon(y)/\varepsilon(x) \to 1$, these bounds prove the inequality. Lemma 4.16 combines the conditions in the order: inherited estimates, local reorganization, cubic estimate, reset pricing; the sensors were fixed before L.

Lean: `can_delay_reset`, `buys_reset`, `sensors`, `delay_cubic`, `Time.cash`, `Time.late`, `Time.open_shop`, `Time.now`, `Time.rawGate`, `Time.beforeWait`.

Lemma 4.18 (joint start-up thresholds). Assume G is simple and $c_{\text{dir}} \leq 1$ at every used stage. Fix the observable sections, marker bounds and nonnegative additional start-up requests. Before choosing block sizes, enlarge the path and column bounds so that the Wilson top paths satisfy the stage bounds and the column bound is at least the factorial of the derivative order. Then the block-size floors and the coupling and scale thresholds can be enlarged so that every realized choice of levels meeting them has the start-up bounds of Lemma 4.5 and the following additional bounds. Write $x = \hat{\beta}_N$, $t = t_N$, $M = M_N$, $B = B_N$ and a for the support exponent at stage $j = J_N$.

- For every prescribed observable section of degree $d \leq j$, its stencil diameter is at most B, and

  $$\tilde{\varepsilon}_N C_d t^{-s_d} \leq \sqrt{\tilde{\varepsilon}_N},$$

  where s_d is the total slot order of its section family and $C_d \geq 1$ bounds its evaluation on every configuration by C_d times the coefficient ℓ^1 norm.

- For $\beta \in [x/4, 2x]$, the start-up regular bound remains ε_N after increasing the support exponent by any preassigned $\delta_j \geq 0$. The connected-polymer estimate retains a forced-decay factor $e^{-\nu M^2}$, where $\nu = (128 h_4)^{-1} > 0$ and $h_4 = 2^4\, 4!$ is the order of the lattice point group; its final forced bound is at most $\tilde{\varepsilon}_N$.
- The mixture estimates have $B \geq 8$, $x^{-1/2}(k+1)e^{a5^4} \leq 1$, and a connected-polymer bound at most $1/2$ with support exponent increased by 8.
- For each preassigned integer observation radius r_j, the first-step numerical bound is

$$(4B+1)^4(2r_jB+1)^4\, 8\left[\frac{b_K}{2} + D_G \log\left(4(1+2x)\right)\right] \leq \frac{x\, t^2}{64 h_4},$$

  where $b_K > 0$ is the kernel penalty and $D_G = \dim_{\mathbb{R}} M_n(\mathbb{C})$.

These conclusions persist when the thresholds are increased.

Sketch: At a fixed stage only finitely many observable degrees are requested. At fixed stage, the stencil and cylinder constants are fixed, while $B \to \infty$ and $e^{-M^{1/5}}$ absorbs every fixed power of t^{-1}. The additional support weights and the first-step cost grow polynomially in $\log x$, whereas $xt^2 = M^2$ grows faster than every such polynomial. Take common thresholds and enlarge the block-size floors for the cross-level support estimates. The joint pricing theorem also increases the scale thresholds, while preserving the prescribed volume and growth data. This establishes numerical preparation; subsequent nonlinear steps still require their own activity estimates.

Lean: `StrongStartup.prices`, `StrongStartup.before`, `Extra.Multi.pricing`, `Timing.receive`, `ThroughLaunch.developer`, `Mark.Ready`, `StrongStartup.Escrow`, `StrongStartup.BondFuel`, `Mixture.Extras`.

3.12.4.7 4.7 Assembly and the running law Lemma 4.19 (bounds at the edges of the shooting window). Assume $\kappa > 0$ and $\bar{\beta} > 0$. Put $\phi_m = \bar{\beta} + \frac{\kappa}{2} s_m$ and $\psi_m = \bar{\beta} + 2\kappa s_m$. Then $[\phi_m, \psi_m] \subset \mathfrak{B}_m$, and $\phi_0 = \psi_0 = \bar{\beta}$. Assume $\omega_{J_{m+1}} \leq \frac{\kappa}{4} \log L_m$ (guaranteed by the choice of block sizes). If a step satisfies the transfer inequality, then:

- input coupling ϕ_{m+1} forces output coupling $< \phi_m$;
- input coupling ψ_{m+1} forces output coupling $> \psi_m$.

Sketch: $\beta_m \leq \phi_{m+1} - \kappa \log L_m + \omega \leq \phi_m - \frac{\kappa}{4} \log L_m$, and symmetrically for ψ.

Lean: `Shoot.gates`, `Shoot.limits`, `Shoot.Drifting.pay`.

Lemma 4.20 (shooting in the bare coupling). Suppose the step program holds and G is simple. Then for all N, p there is a trail $(\rho_{N,p,m})_{m \leq N}$ with the following properties:

- it starts with a Wilson start-up $e^{-\beta S_W}$;
- each level is admissible, with coupling in $[\phi_m, \psi_m]$;
- consecutive levels satisfy the transfer clause;
- $\beta_{N,p,0} = \bar{\beta}$.

Choosing these trails for all N, p gives a history satisfying (1)–(5) of III.9.3. The common-path clause follows because every trail uses the same prescribed $P_{N,m}$.

Sketch: Decreasing induction on m carries an interval $[l, u]$ of bare couplings and a continuous curve of trails.

- At $m = N$, restrict the start-up curve to $[\phi_N, \psi_N]$.
- At each step, apply the step program, giving an output curve over the same interval. By Lemma 4.19 its couplings at l and u lie below ϕ_m and above ψ_m. A compactness and intermediate-value argument gives $[l', u'] \subset [l, u]$ whose image is $[\phi_m, \psi_m]$, with ϕ_m, ψ_m attained at the ends.
- At $m = 0$ the left end gives $\beta_0 = \bar{\beta}$.

Lean: `Shoot.steps`, `Shoot.narrow`, `Shoot.successful`, `Recipe.races`, `Graces.winnow`.

Lemma 4.21 (running). For a history on levels satisfying the drift-smallness and stage-divergence conditions of III.9.2, clauses (4) and (5) of III.9.3 imply clause (6).

Sketch:

- By induction from $\beta_{N,p,0} = \bar{\beta} = \hat{\beta}_0$: $|\beta_{N,p,i} - \hat{\beta}_i| \le \sum_{k=1}^{i} \omega_{J_k}$.
- Given ε, pick j with $J_{k+1}^{-2} \le \varepsilon/2$ for $k \ge j$, using $J \to \infty$.
- The drift condition gives $\sum_{k \le i} \omega_{J_k} \le \sum_{k \le j} \omega_{J_k} + \frac{\varepsilon}{2}(s_i - s_j)$.
- This is $\le \varepsilon s_i$ for i large, since $s_i \to \infty$. The bound is uniform in $N \ge i$ and p.

Lean: `Shoot.trajectory`, `Orchard.traject`. The second version replaces (5) by the band at level 0.

Lemma 4.22 (the history of the main theorem). Fix the compatible data of III.13, including its analytic continuation bounds and finite-cutoff solutions. For every N, p and every real b with $|b| < r_*$, there is a trail in that construction with level-0 coupling $\bar{\beta} + b$. Its bare coupling is unique within the specified class of finite-cutoff solutions. The coupling and activity arrays admit analytic continuation to the complex disc $|b| < r_*$, with $r_* > 0$ independent of N, p. At $b = 0$, these trails form a selected history, and every step is a reset step.

Sketch: III.13 solves a coupled finite-cutoff fixed-point problem for all the level couplings. For real offsets its fixed point gives real trails and the stated endpoint; the comparison theorem gives uniqueness in this class. At $b = 0$, their admissibility gives the common paths and level-wise bounds; their reset steps give exact transfer. Lemma 4.21 supplies running. This is the history used in the Wilson family. The calibration condition records its realized trails and reset steps; it does not assert uniqueness among all representations of the same density.

Lean: `Spire.Uses.exists_bolt`, `Spire.Uses.hits`, `Upland.Arc.exists_trial`, `Mast.calibrated`, `Upland.Arc.finiteCut`, `Spire.Fitted.acts`, `Spire.Fitted.unions`. *(proof read for the displayed assembly and finite-cutoff uniqueness theorem; inputs from III.13)*

3.12.4.8 4.8 Auxiliary results

Lemma 4.23 (marker-rooted sums). This is III.9.8.

Sketch:

- If $x \in X_i$ and a marker v of X_i sits in X_i, then $\operatorname{dist}(o(v), x) \le \operatorname{cover}_M(X_i)(4M + 2) + 2M$. So $e^{a \operatorname{cover}} \le e^{(a+1)\operatorname{cover}}\, e^{1 - \operatorname{dist}/(4M+2)}$.
- Summing over i marker by marker gives at most c per marker.
- Summing over markers gives $\le \mu \sum_y e^{1 - \operatorname{dist}(y,x)/(4M+2)} \le \mu e (2(16M + 9))^4$, uniformly in the torus.

Lean: `Extra.Multi.root_hit`, `torus_kernel`.

Lemma 4.24 (decay of a shifted Gram inverse). Fix a finite channel set I, $r, \eta > 0$ and $A \geq 0$. There are $C \geq 0$ and $\tau > 0$, depending only on I, r, η, A, such that every real convolution kernel P on $\mathbb{Z}^4$ with

$$\sum_{z \in \mathbb{Z}^4} e^{\eta |z|_1} |P_{ab}(z)| \leq A \qquad (a, b \in I)$$

has a real convolution kernel S with the corresponding bound C at exponent τ, and, on every torus,

$$S_n(rI + P_n^{\mathsf{T}} P_n) = I.$$

Here P_n, S_n are periodizations. Thus $S_n = (rI + P_n^{\mathsf{T}} P_n)^{-1}$, uniformly in the torus side.

Sketch: On real momentum, $rI + \hat{P}(k)^* \hat{P}(k)$ is positive definite. Compactness in the fixed finite channel space and the weighted kernel bound give invertibility and a uniform inverse bound in a narrower complex strip. Fourier coefficient estimates give an exponentially summable inverse kernel. The discrete Fourier transform identifies its periodization with the inverse matrix on every torus.

Lean: `Polar.deliver`, `Polar.invert_tori`, `Polar.writes`.

3.12.5 5. Where the difficulty lies

The early presentation with local bounds comes from III.10, the one-loop coefficient from III.7, and the compatible kinetic data from III.13. Given those inputs, the reset must preserve the full masked density, including failed-mask discrepancies and the forced gas. Two-point matching alone does not prove that identity. Lemma 4.13 supplies it and proves the output activity bounds.

The marginal coefficient is isolated by gauge symmetry, simplicity and lattice isotropy. It is a Hessian coefficient, so its quadratic energy carries a factor 1/2. The Wilson start-up has a uniform $O(t^5)$ residual even when individual links are far from the identity. Continuity in the bare parameter must hold in the activity norms despite the hard configuration masks; shooting depends on this continuity.

The block factors grow. The planned one-loop condition implies $\omega_{J_{m+1}} \geq 1$, while the drift condition gives $\log L_m \geq J_{m+1}^2 \to \infty$. The gentle-scale condition requires $\log L_m = o(s_m)$. The reset correction is small:

$$|\delta_x| \leq \frac{c_m}{4} \hat{\beta}_m^{-1/10} e^{-2(\log \log \hat{\beta}_m)^2}.$$

The constants c_m depend on the stage pair, so this is a fixed-stage estimate. The separate one-loop error is bounded independently of L once its earlier parameters are fixed; no decay of that error in L is required. The argument gives $\beta_m = \bar{\beta} + \kappa s_m + o(s_m)$ uniformly in the cutoff and volume. It does not give a bounded remainder or a two-loop expansion.

The coarsest coupling is prescribed exactly. Shooting chooses the bare coupling separately for each cutoff and volume and restricts the surviving parameter intervals to keep every level in its band. The uniqueness theorem in III.13 applies to its specified continuation class.

3.12.6 6. Notation

Written proof	Lean (glossary)	Meaning
$\kappa = \frac{11C_G}{24\pi^2} = 2b_0$	`Gauge.slopeConstant`, `Gauge.casimir`	slope of β against $\log(1/a)$
s_m, a_m, L_m	`Schedule.scale`, `Schedule.a`, `block`	scale $\log(1/a_m)$, spacing, block factor (odd, ≥ 3)
$\bar{\beta}$	`Calendar.bar` (“barrier”)	coupling at the coarsest level
$\hat{\beta}_m$, $\mathfrak{B}_m$	`Calendar.hatBeta`, `band`	reference coupling, band $[\hat{\beta}/2, 2\hat{\beta}]$
t_m, M_m, B_m	`threshold`, `monitor`, `coverSize`	small-field threshold, its multiplier, cover size
ε_m, $\tilde{\varepsilon}_m$	`allowance`, `forcedAllowance`	sizes of regular and forced activities
J_m, ω_j, θ_j	`Calendar.J`, `Stage` (`drift`), `theta`	stage index, drift allowance, smallness exponent
$\rho = (P, c, E, K, b, \beta, v)$	`Rep`	effective action (“representation”)
$\mathcal{K}_P$	`BetaTransport.kinetic`	β-coefficient of the leading terms
$T_\beta^{(m)}$	`StepData.push`	one block-spin step
$\beta_{N,p,m}$, $\rho_{N,p,m}$	`History.β`, `History.rep`	selected history
$\alpha_{N,m}$, $P_{N,m}$	`Kin` (`alpha`, `path`)	one-loop shift of the step, path data
$\mathfrak{D}$	`Directed`	scale sequence, kinetic data and recipe
reset step	`Directed.Move`	one step with exact transfer and $\lvert\beta - \beta' - \alpha\rvert \leq 1$
step program	`Directed.Program`	start-up curves and reset along all admissible curves
(c_m, r_m)	`ResetPrices` (`cost`, `radius`)	income multiplier, reset window (≤ 1)
η_j, probe V	`Sensor`, `Usable`	lower bound $\mathcal{K}(V) \geq \lvert\Lambda\rvert\eta t^2$
$f(x,d)$, d	`Template.residual`	scalar residual, coupling correction
χ_X, Q_i, E_i^0, D_i	`Residual.splits`	small-field indicator, local quadratic projection, subtracted regular function, mask discrepancy
$\delta(A)$	`Averaging.delta`	isotropic curvature-squared coefficient of activities
early presentation	`Collection.earlyValue`, `Return.atlas`	blocked density before the reset
P^{top}, R_p, Q_4	`Design.top`, `residual`, `fourth`	Wilson top path data, one-plaquette residual, quartic coefficient
$\rho_\beta^{\mathrm{start}}$	`Bid.Cake` with `payload`, `dictionary`	Wilson start-up representation

Written proof	Lean (glossary)	Meaning
compatible (c, b)	`WilsonStartup.Decent`	$c_{\mathrm{dir}} \le 1$, $B \ge 2$, $te^{(B-2)/c_{\mathrm{rel}}} \ge 2$
ϕ_m, ψ_m	`floorBeta`, `ceilBeta`	shooting window
numerical bounds and entry thresholds	glossary item 1	constants in the polymer norm estimates and lower bounds on $\hat{\beta}$

3.12.7 7. What remains unexplained

1. **Nonlinear inputs.** The reset is conditional on the early presentation, stronger local activity bounds and continuity. Their supply is the conclusion of Rain.glazing and Roost.Factor.offering. Their assembly proofs identify the required inputs; the underlying nonlinear and Gaussian-sector estimates belong to III.10 and III.4 and are not independently verified here.
2. **Activity reorganization.** The explicit splitting and the proof of Local.renew give Lemma 4.13. A full written proof should expand the local reorganization constants and the connected-polymer estimate that absorbs the logarithmic factors. Their detailed norm estimates are inputs from II.2 and III.10.
3. **Finite-cutoff uniqueness.** The uniqueness in Lemma 4.22 concerns the class of solutions specified in III.13, with its fixed data and continuation bounds. This construction solves coupled fixed-point equations rather than choosing a subinterval by Lemma 4.20. The displayed assembly and uniqueness theorem have been checked; the analytic existence and comparison estimates underneath them remain for III.13. Real offsets satisfy $|b| < r_*$; complex offsets parametrize analytic continuations, not real positive densities.
4. **Relaxed histories.** The intermediate history predicate with no endpoint restriction is distinct from the selected-history specification. Its running estimate uses the band at level 0. The final centered history satisfies the exact endpoint clause.
5. **Numerical and bibliographic scope.** Lemma 4.18 gives the start-up numerical bounds; their use for source polynomials and first-step observations requires the norm definitions in III.12 and III.15. The cited numerical preparation does not itself construct those observables or their nonlinear continuations. The literature citations in §2 are background references; their exact bibliographic details and comparison with this construction remain unchecked here.
6. **Verification scope.** The correspondence checks use the supplied Lean source and link table. No Lean build or independent axiom-closure check was performed. The package sizes and main-theorem dependency counts are retained from the unit map (tentative).

3.13 III.10. One full nonlinear renormalization step: bounds on irrelevant terms

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/Work/` (items E01–E09, H01–H04, I01–I04), `YM/Bid/`, `YM/Next/FLUCTUATIONS/`, `YM/AnalyticNext/CONNECTED_ACTIVITY_JETS_E6/`, `YM/PhysicalNext/RG_NEXT_COMPLETE_REPEATABLE_NONLINEAR_STEP/`, `YM/PhysicalNext/V9_JOINT_ENDPOINT_LOAD/`, `YM/Pressure/` (definitions). **Size:** 49,448 lines in the main-theorem closure.

Conventions. Levels are indexed from coarse to fine, as in III.9. - *Step and couplings.* One step

S goes from a fine torus of side $f = cL$ to a coarse torus of side c, with odd block factor $L \geq 3$. Write x for the coarse reference coupling $\hat{\beta}_m$. The fine coupling is $\beta_F = x + b_G \log L = \hat{\beta}_{m+1}$ (`clock_next`), where $b_G \geq 0$ is the one-loop slope of III.9; $b_G > 0$ when the Lie algebra of G is simple. - *Thresholds, radius and cutoff.*

$$t_c(x) = x^{-1/2} e^{(\log\log x)^2} \text{ (coarse field)}, \qquad t = t_c(\beta_F) \text{ (fine field)}, \qquad T = \beta_F^{-49/100} \text{ (large field proper)},$$

with cluster radius $B = \lceil (\log x)^2 \rceil$ and path-length cutoff $R = \lfloor B/10^5 \rfloor$. - *Sizes.* Since $e^{(\log\log\beta)^2} = \beta^{o(1)}$, we have $t = \beta_F^{-1/2+o(1)} \ll T$. On the other hand $\beta_F t^2 = e^{2(\log\log\beta_F)^2}$ grows faster than every power of $\log\beta_F$. - *Field sizes.* $\chi(g)$ is the chordal distance from $g \in G$ to 1. For a plaquette p, $|U_p| = \chi(U_{\partial p})$. For a coarse bond b, $w_b = \chi(V_b Q_b^{-1})$ measures how far the coarse field V is from the block-spin predictor $Q = Q(U,k)$ of the fine field U with gauge-fixing tags k (III.1). - "For all large x" means $x \geq x_0$, with x_0 independent of the period and volume. Other dependencies are specified in each statement.

3.13.1 1. What this unit proves

The unit does not construct the effective actions; that is III.9, with III.5–III.8. It proves the estimates on one nonlinear block-spin step that the construction consumes: - a deterministic lower bound on the action gap of the step near large-field regions (III.10.1); - bounds on the activities produced when the fluctuation field is integrated out, and on all their derivatives in the coarse field (III.10.2, III.10.3); - bounds on the Gaussian sector of the step (III.10.4); - the numerical schedule under which these estimates hold at every level (III.10.5).

III.10.6 collects the start-up fusion and marker estimates. In the energy argument, constants are chosen before L, after L, or after W; this order determines which losses they can absorb. The derivative and Gaussian estimates have their own quantifier orders, stated below.

Data of a step. - *Request.* A request P (`Teller`) consists of: - kernel choices K, with penalty $a > 0$ and cuts; - a fine stage and a coarse stage, each with clip constants $\alpha, \lambda, \rho > 0$ satisfying $\lambda < \alpha/16$ (`Clip`), a path width and a path budget; - a step index j and constants $\eta > 0$, C. - *Tube.* A tube $\mathcal{T}$ (`Tube`) carries the towers Φ, Φ', N and the fine and coarse path data. Φ maps coarse plaquette coordinates to fine plaquette coordinates; Φ' maps them to coarse-bond ("interior") coordinates; N maps them to gauge coordinates. - *Admissibility.* A step with tube $(S, \mathcal{T})$ is admissible for P (`Teller.Funds`) if: - S uses K and index j; - the patch is well posed (`Branches`: unique minimizers whenever the frame or predictor selector uses minimization, and a nonempty set of eligible axial tags for every configuration); - the scalar tower coefficients have exponentially weighted absolute sums at rate η bounded by C/L^2 for the fine kernel and C for the interior and gauge kernels. Finite differences of order $\ell \leq j-2$ of the one-form and two-form cross kernels have bounds $C/L^{1+\ell}$ and $C/L^{2+\ell}$, respectively (`YM.Precision.Towers.Deep`); - the clips and path budgets are those of P; - the flat, uncut tower matrices satisfy the exact identities `Towers.Roll`, among them the resolution of the identity

$$\Phi_0^* \Phi_0 + \Phi_0'^* \Phi_0' + N_0^* N_0 = 1.$$

The transported kernels cut at length R need not satisfy this identity; their defects are estimated in §4.5.

Seeds. A store $A = (A_F, A_I, A_\flat)$ (`Stores`) consists of finite sets of fine plaquettes ("costs"), coarse bonds ("aisles") and coarse sites ("flags"). $\sharp A$ is the sum of their cardinalities, including coincident

seeds of different types. $D_r(A)$ is the coarse r-neighbourhood of their positions: all parent sites touched by a fine plaquette, both endpoints of a coarse bond, and each flag site. A is *consistent* with (U,k,V) at (x,W) (`Plot`) if the following hold: - whenever a parent site touched by p meets $D_{5B}(A)$, $p\in A_F$ implies $|U_p|\ge t/4$, and $p\notin A_F$ implies $|U_p|\le t/2$; - whenever an endpoint of b meets $D_{5B}(A)$, $b\in A_I$ implies $w_b\ge Wt$, and $b\notin A_I$ implies $w_b\le 2Wt$; - $A_\flat$ agrees with the flags of V at threshold $t_c(x)$ on $D_{5B}(A)$; - each site tag based in $D_{5B}(A)$ has positive frame weight.

Nothing is assumed outside $D_{5B}(A)$. The *spent* seeds A^{sp} are the costs with $|U_p|>T$ and the aisles with $w_b>T$.

Action gap. - *Clipped density.* For a stage with clip (α,λ,ρ) and threshold s, the clipped density of a plaquette is $\phi_s(|U_p|)+\mathbf{1}_{\mathrm{kept}}\frac12|\ell_p(U)|^2$ (`crown`). Here: - $\phi_s(r)=\alpha r^2/2$ for $r\le s$ and $\phi_s(r)=\lambda s r$ for $r>s$ (`clipped`); - $\ell_p(U)$ is the path series of the stage, cut at length R; - "kept" means $|U_q|\le s\,e^{\mathrm{dist}(p,q)/\rho}$ for all q (`kept`). - *Actions.* $\mathcal{A}_F(U)$ is the fine clipped action (fine stage, threshold t) and $\mathcal{A}_c(V)$ the coarse one (coarse stage, threshold $t_c(x)$). - *Step functional.* $\mathcal{E}_A(U,V)$ (`assessed`) is the sum of the row contributions of Lemma 4.1 over the rows located in $D_{2B}(A)$. Summed over all rows of the torus, the same contributions give exactly

$$\mathcal{A}_F(U)+\frac{a}{2}\sum_b w_b^2-\mathcal{A}_c(V).$$

So $\mathcal{E}_A$ is the action gap of the step (fine action, plus block-spin penalty, minus coarse action), localized near the seeds.

Proposition III.10.1 (energy per seed). Let P be a request. There is L_0 such that for every block factor $L\ge L_0$ there is $W_0(L)$ such that for every $W\ge W_0$ there is x_0 with the following property. Take $x\ge x_0$ and: - a step S of block factor L with coarse side $c\ge 100(1+B)$, and a tube $\mathcal{T}$ admissible for P; - a spanning forest F of the block structure, and a fine field U equal to 1 on the trunks of F; - a store A, tags k and a coarse field V such that A is consistent with (U,k,V) at (x,W).

Then

$$\kappa\, t^2\,\sharp A\ \le\ \mathcal{E}_A(U,V),\qquad \kappa=\tfrac12\min\{\lambda_F/8,\ a/32,\ \alpha_c/64\},$$

where λ_F is the linear clip constant of the fine stage and α_c the quadratic clip constant of the coarse stage.

Status: theorem under the stated admissibility, tree-gauge and consistency hypotheses. - The Lean statement (`Teller.cash`, `Teller.Grants`) asserts only some $\kappa>0$ chosen after L and W. The value above is the one the proof produces (Lemma 4.14), and it is independent of L and W. - This deterministic estimate precedes the Gaussian tail and activity bounds of III.10.2.

Lean: proved by `Fluctuations.grants`. This is `Integrated.pressure_grants` applied to `Fluctuations.bare`, through `Bid.issues`.

Proposition III.10.2 (activities of one step). Let P be a request and $\mathcal{T}$ target data (`Target`): constants $\varepsilon,\delta>0$, $e\ge 0$, a rate $r\in(-\frac12,0)$, a halo $h\ge 500$ and a local exponential chart of G, with 4ε below the support exponent of the fine stage and 8ε below both stage widths. Put $a_*=e+2\varepsilon+\delta$.

There is L_0 such that for every $L\ge L_0$ and every large W there is $\mu>0$ with the following property. For every patch D of block factor L satisfying the branch condition there are window

constants o (`Target.Order`: a decay rate, a spectral gap, a shell width K, a Lipschitz constant γ, a drift cap, a plateau a_o and a trigger level ξ) and a constant $C \geq 0$ such that the following holds for all large x, every step S on D with $c \geq 100(1+B)$ and every tube admissible for P (`Works`).

- (a) The conclusions `GrantBodies` hold:

  1. the bound of III.10.1 holds at this x with constant μ (`PointCharge`);
  2. if U is in tree gauge and, within coarse distance $B/4$ of the bond's parent site, site tags have positive frame weight, fine plaquettes have size at most $t/2$, and coarse mismatches are at most $2Wt$, that link lies within Kt of the replicated coarse link; the Lie logarithm is reliable at that radius (`ShellAccess`);
  3. in every sector whose flags are the actual flags of V (`manifest`), the Gaussian precision matrix is symmetric, has exponentially decaying entries and a positive spectral lower bound, and the covariance formed from its local inverses is positive definite (`Rotator`);
  4. the Gaussian centring drift has norm at most $\mathrm{cap} \cdot t$ at each fine bond. If the tube also satisfies the compatibility condition in (b), then, for every $\beta > 0$, a fluctuation satisfying $|z_j| \leq \xi t\sqrt{\beta}$ on all coordinates within distance B of a site farther than $30B$ from the seeds has site and bond tag masses one, $|U_p| < t/4$ for plaquettes based there, and $w_b < Wt$ for bonds based there;
  5. numerical margins hold: $4a_o t$ is below the chart radius, and, for the window's positive inverse-kernel constants τ_0, C_0 and window bound $Q_0 \geq 1$, the quantity $\epsilon_B = e^{-\tau_0 B}(Q_0 C_0 + 1)$ satisfies $\epsilon_B < 1$ and $Q_0 C_0 \epsilon_B < 1$.

- (b) The conclusions `Invoices` hold for every tube satisfying the free-kernel compatibility condition `Stamp`, and for every input representation K of the fine-level density as in `Intake`. The input has the fine-stage support exponent, threshold t, coupling $\beta \in [\beta_F/2, 2\beta_F]$, the path data and clips of $\mathcal{T}$, cluster radius $B(\beta_F)$, and activities obeying the incoming bounds of III.9: E-norm $\beta_F^{-11/10}$, K-norm $e^{-M^{1/5}}$ and loop budget $2M$, where M is the fine monitor. The conclusions are:

  1. the cost record with gain $\beta\mu t^2$ per seed and spend $C\,\mathrm{bill}(x)^5$, where $\mathrm{bill} = 1 + B + (\log \beta_F)^2$, passes the tree-graph summability test `Buy`;
  2. the letters of the new gas carry weights k_i, y_i, E satisfying a rooted Kotecký–Preiss condition with exponent ε and total β_F^r (`Sales`). For a letter with support X_i and extra length d_i, this means nonnegative weights u_i with

$$\sum_{i : o \in X_i} u_i \exp\left(a_* N_B(X_i) + 2\varepsilon d_i\right) \leq \beta_F^r \qquad \text{for every coarse site } o.$$

     Here $N_B(H)$ is the minimum number of radius-B balls whose connected union contains H. The letters are the forced activities of K, the logarithmic letters, and the warning letters. Their weights are respectively k_i, y_i and E times a fixed patch-dependent warning-amplitude bound; warning letters have $d_i = 0$;
  3. in every manifest sector, the following bounds hold:
     - the old activities are bounded by k_i;
     - the logarithmic letters ℓ_i and the differences $e^{\ell_i} - 1$ are bounded by y_i;
     - every packet weight satisfies $|\mathrm{Pack}| \leq \exp\left((C\,\mathrm{bill}^5 - \beta\mu t^2)\,\sharp A\right)$ (`Pack`);
     - for every set v of warning addresses, the Gaussian measure of the catastrophe set is at most $E^{\sharp v}$. The catastrophe set consists of the fluctuations z for which,

near the anchor of each address in v, some coordinate within distance $2B$ satisfies $|z_j| \geq \xi t\sqrt{\beta}$. Every fluctuation that triggers the warnings on v lies in that set.

Status: theorem. The target data, fine input bounds and free-kernel compatibility are hypotheses. The input bounds are supplied along the RG trajectory (III.9); the compatibility construction is discussed in §7. - *Quantifier order:* L_0, then W, then μ (uniform in D), then (o, C) for each patch, then x. - *Size of the per-seed factor.* Since $\beta\mu t^2 \geq \frac{\mu}{2} e^{2(\log\log\beta_F)^2}$ and $\mathrm{bill}^5 = O((\log\beta_F)^{10})$, the factor $e^{C\mathrm{bill}^5 - \beta\mu t^2}$ is eventually smaller than every power of β_F.

Lean: `Target.issue`. `Target.quotesTogether` repackages the output as the bundle `Cash` (the activity estimates together with witnesses for the letter weights).

Proposition III.10.3 (derivatives of the new activities). Let P and $\mathcal{T}$ be as in III.10.2. There is L_0 with the following property. Fix $L \geq L_0$, a patch D satisfying the branch condition, an order $i \in \mathbb{N}$, a sector cutoff χ, $W, \mu > 0$, window constants o, and chart constants (A, f) satisfying `YM.ConnectedActivityE6.Blaze.Eider.Canary`. The latter require $A, f > 0$, $\log(\exp u) = u$ for $|u| \leq 2A$, $\exp(\log(\exp u\, g)) = \exp u\, g$ for $|u| \leq A$ and $\|g - 1\| \leq f$, and analyticity of $M \mapsto \log(\exp u\, M)$ for $|u| \leq 2A$, $\|M - 1\| \leq f$. Fix a function $\mathcal{V}$ of rate $-1/2$ that is eventually nonnegative (Lemma 4.15).

There are $C \geq 0$, functions q, ρ, p of rates $-1/2, 0, 0$, respectively, and $l \in \mathbb{N}$ such that, for all large x, the following holds for every step on D satisfying the compatibility condition, every fine representation R, and every bound bundle H of III.10.2 at (W, x, μ). One has $8a_o t \leq A/2$ and a derivative-bound structure Ξ (`Utopia`) with the following properties in every sector at x, of width $o.K$, cutoff χ, and with the actual coarse flags.

- Its nonpacket weights satisfy the rooted bound of III.10.2(b) with total β_F^r. Its catastrophe probabilities, with trigger $\xi/2$, are at most $(\mathrm{rad}^2)^{\sharp v}$.
- Let $V(y)$ be a family of coarse fields on a finite-dimensional real normed parameter space, based at the sector field $V(y_0) = V_0$. Suppose each link matrix is C^i at y_0 and

$$\|D^m V_e(y_0)\| \leq c_G \mathcal{V}(x)^m \qquad (0 \leq m \leq i),$$

  where c_G is the fixed matrix-norm bound in `Mode`. Transport the fluctuation coordinates compatibly with this family: the physical and Gaussian coordinates represent the same field, their physical displacement has all derivatives through order i bounded by $q(x)$, and, near y_0, the resulting fine field agrees with the held fine field on every cold bond and on every hot bond whose physical coordinate has norm at most $8a_o t$ (`YM.ConnectedActivityE6.Blaze.Silica.Clutch`). In a region disjoint from the coarse flags, each letter a_j is C^i at y_0 and satisfies

$$\|D^m a_j(y_0)\| \leq \mathrm{cost}_j(\rho(x)\mathrm{span}_j)^m \qquad (0 \leq m \leq i)$$

  (`Brush`). Packet spans are one; other spans include their support and path lengths. The costs use the packet estimate and the nonpacket weights before extracting the catastrophe-probability factor.

The same Ξ satisfies an integrated estimate (`Herd`). Let $\mathfrak{p}$ be a finite molecule with connected support-overlap graph and total support disjoint from the flags. Require that its packet stores are connected with pairwise disjoint radius-hB neighbourhoods, that its old forced supports are pairwise disjoint, and that every site in an old forced inventory touches a fine-plaquette seed in the

union of the packet stores. Along a family $V(y)$ as above, the prescribed transported integrand is C^i at y_0 and has every derivative of order at most i bounded by an integrable function g. With $d\nu = d\lambda_F\, d\gamma$, the product of tree-slice Haar measure and Gaussian fluctuation measure,

$$\int g\, d\nu \le |p(x)| \prod_{j\in\mathfrak{p}} \mathrm{fare}_j\, (1+\mathrm{shape}(\mathfrak{p}))^l.$$

The fares include a factor rad for each distinct warning address. For one measurable, gauge-invariant source letter of amplitude at most w_M, local with its eligibility tests included, the same estimate replaces its fare by w_M. A forced source must vanish outside the original eligibility set; a regular source occupies an unused regular-letter address. The remaining warning and shape factors are unchanged.

Status: theorem, conditional on the input bound bundle and compatibility. The constants and functions are uniform in the volume. `Captain` and `rocketsC` also control iterated left derivatives in spatially weighted directions.

Lean: `Utopia.whittle` (letter jets), `timing` (integrable majorants).

Proposition III.10.4 (Gaussian sector of the step). Fix a patch D, a request Q for which D satisfies the branch condition, an aperture $B_0 > 0$, a sector cutoff χ, and $k, p \in \mathbb{N}$. There are $\sigma, d > 0$ and $\pi \ge 1$ (`Charges`) such that the following holds for all large x.

Take $0 < \beta \le 2\beta_F$, a budget point of power p and threshold $t_c(x)$, a step on D with $c \ge 100(1+B)$ admissible for Q, a forest, tags, and a store A whose flags equal the flags of V on the whole coarse torus. Freeze A, the forest and tags while differentiating. Let Γ be the sector precision matrix, $\tilde\Gamma$ its local approximate inverse, m the smooth sector mean at aperture $B_0 t$, and $h = -g^{-1}$ times its stationarity residual, with $g = \beta^{-1/2}$. These are the objects in `Value`. Then Γ and $\tilde\Gamma$ are positive definite, and m equals the stationary logarithmic mean for every frozen-link input.

For a coarse site o and $0 \le m \le k$, let $\mathcal{D}^m$ be an iterated left derivative in m directions satisfying $|A_e| \le t_c(x)(1+\mathrm{dist}(e,o))^p$. If x_i is the coarse location of coordinate i, put

$$G_i = (1+\mathrm{dist}(x_i, o))^{pk}.$$

The bounds are

$$|\mathcal{D}^m \Gamma_{ij}| \le \pi e^{-\sigma \mathrm{dist}(x_i,x_j)} G_i^m, \qquad |\mathcal{D}^m(\Gamma\tilde\Gamma - 1)_{ij}| \le e^{-dB}\pi e^{-\sigma \mathrm{dist}(x_i,x_j)} G_i^m,$$

$$|\mathcal{D}^m m_i| \le \pi t G_i^m, \qquad |\mathcal{D}^m h_i| \le e^{-dB}\pi G_i^m.$$

Thus the spatial exponent at derivative order m is pkm.

Status: theorem. `JointLoad.stage` also supplies a covariance cap and a trigger-tail estimate; its scales are given in §7.

Lean: `JointLoad.capital`.

Proposition III.10.5 (schedule). - (a) Let P be a sheet of pre-L data (III.2) and K a price map. There are floors k_i with the following property. Take block factors $L_i \ge k_i$, any protection, any delays V and any $m \ge 0$. Then there are delays W, differing from V only by larger entry couplings β_i^*, and numbers $F_{ij} \ge m$, such that for every pair of stages $P.\mathrm{first} \le j \le i \le j+1$ the bound of III.10.1 holds at every coarse coupling $x \ge \beta_i^*$, for the request with fine stage i, coarse

stage j and step index i, with block factor L_i, aisle multiplier F_{ij} and some $\mu_{ij} > 0$ uniform in x, period and volume (`Ringing`). - (b) If G is simple, then for every window datum of III.2 (entry and plateau requirements) there is a level schedule that meets all requirements of III.2 (`Paid`) and whose periods are divisible: for every side c and every z there is a period index $p \geq z$ with $c \mid \mathrm{period}(p)$. - (c) Along a directed trajectory satisfying (a), fix an adjacent stage pair $i = J(m)$, $j = J(m+1)$ with $P.\mathrm{first} \leq i$. There is $\mu > 0$, uniform in the cutoff N, level m and period, such that for $m < N$ at this pair and $x = \hat{\beta}_m$, one has $\mu\, t_c(\hat{\beta}_{m+1})^2 \sharp A \leq \mathcal{E}_A$ whenever the fine field is in the chosen tree gauge and the store is consistent at multiplier F_{ji}.

Status: theorems. Part (a) is unconditional (`pressure_pricing` applies `Bid.pricing_order` to III.10.1). Divisibility is what lets torus tilings fit in III.12–III.13.

Lean: `Wait.engine`, `notation_ticket`.

Proposition III.10.6 (start-up fusion H01–H04; markers I01–I04).

(H) *Setting.* Fix a budget point and:

- a *pipe* X: regular and forced addresses, and supports of logarithmic letters;
- a *tap* d: regular activities, forced activities K_i and logarithmic letters ℓ_i.

The *fused* gas (`fused`) groups the forced activities and the factors $e^{\ell_i} - 1$ into connected clusters (pellets). Each pellet contains at least one forced activity, and the forced supports in a pellet are pairwise disjoint.

Hypotheses. Fix a bijective enumeration of the pellets. Let $\kappa, \gamma \geq 0$. - d is *digested* (`Digests`): the gas is valid with specified regular and forced bounds; the logarithmic family has a summable uniform majorant, and each letter is measurable, gauge invariant, local with the flag tests included, vanishes on healthy configurations, and is translation and lattice-symmetry covariant. - d is *bought* with constants κ, γ (`Buys`): $|K_i| \leq e^{-\kappa} f_i$ and $|e^{\ell_i} - 1| \leq f_i$, where $f \geq 0$ satisfies the rooted Kotecký–Preiss condition at support exponent one larger than the budget exponent, with total γ, and $(2B+1)^4 \gamma \leq 1$.

Conclusions. - (H01) For every configuration V, the fused gas equals the original gas times $\exp(\sum_{i=0}^{\infty} \ell_i(V))$. - (H02) The fused activities satisfy the K-bound with any price $v \geq e^{-\kappa}(e^{\gamma} - 1)$. - (H03) The fused system is a forced family and admits the lattice-symmetry moves. - (H04) Fix the paths, clip, budget point, addresses and pellet enumeration. Suppose the original representations form a continuous curve on a subset of $\mathbb{R}$, each forced or exponentiated logarithmic letter varies continuously in the supremum norm over configurations, and a single nonnegative rooted majorant as above works throughout the parameter set. Then the fused representations form a continuous curve in the same representation norm. This is a fixed-volume assertion.

(I)

- (I01) Let $B \geq 8$, let F be a set of at least two markers with pairwise separation at least B^2, and let $F \subset H$. Then $B \sharp F \leq 64\, N_B(H)$, where $N_B(H)$ is the minimum number of radius-B balls with connected union covering H. The markers lie on a finite periodic lattice with its max-distance; no nonwrapping hypothesis is required.
- (I02) Fix a representation with budget order n. Suppose marker bands satisfy `YM.Mix.Stubs.Holds` at $r > 0$: valid local addresses, r times every unit amplitude at most one, $B \geq 8$ and separation at least B^2 for each pair of distinct markers, factorially weighted derivative majorants, and separate rooted budgets at most one for single-marker

and multiple-marker terms. For the multiple-marker budget one may instead use positive weights t_v and an extra exponent $\delta \geq 0$, with $r \leq t_v e^{\delta B/64}$ for every marker v and a rooted budget at most one using support exponent increased by δ. Then, at weight $r/n!$, the bands satisfy `YM.Overlay.Bands.Radial`: positive weights, unit-amplitude bound one, and rooted ordinary derivative and forced-entry bounds at most two each.

- (I03) Let a family satisfy the marker schedule condition: $B_{r+1}^2 \leq 2B_r^2$ for every r, and $(L_r B_r^2)^4 \mathcal{F}_{r+1} = o(\mathrm{scale}_{r+1})$. Here a_r is the lattice spacing, $L_r = a_r/a_{r+1}$, and the vacuum bound is
$$\mathcal{F}_r = 2(2 + 4P_r) + 1 + 6(\alpha_r + P_r^2) + 4d_G(\log 16 + \tfrac{1}{2}\log \hat{\beta}_r),$$
with P_r the stage's path budget, α_r its direct clip coefficient and $d_G = \dim_{\mathbb{R}} M_m(\mathbb{C})$ for the defining matrix representation. This is the hypothesis of `YM.Stock.Tabs`. Fix $k \geq 1$ and levels $m \leq i$ such that $kB_i^2 \leq a_m/a_i$ (an open stop). For every $N \geq i$ and every period, the centred copies at level N are the embedded sites from level m; their parents agree at consecutive levels from m to N; and distinct embedded sites are separated by at least kB_r^2 for every $i \leq r \leq N$.
- (I04) Assume G is simple. For every family, every $p > 0$ and $r \in \mathbb{N}$, if $\mathrm{order}_i! \leq \mathrm{columns}_i$ for all i, then the widths
$$w_i = \frac{\hat{\beta}_i^{-r/2}}{p\,\mathrm{columns}_i \prod_{m<i} \mathrm{columns}_{m+1}}$$
are positive, and for every $\varepsilon > 0$ eventually $5 - \log(w_i/\mathrm{order}_i!) \leq \varepsilon\,\mathrm{scale}_i$.

Status: theorems. In the main-theorem closure: - H01–H04 are used by the start-up representation at the finest level; - I01, I03 and I04 are used by the replicated-marker constructions; - the assemblies `Oust.begins` and `Bid.installs` are not in the closure.

Lean: - Statements: `H01`, `H02`, `H03`, `H04`, `I01`–`I04`. - Proofs: `H01.gathered`, `H02.answer`, `H03.fused_forcedFamily`, `H04.answer`, `I01.answer`, `I02.answer`, `I03.answer`, `I04.answer`.

3.13.2 2. What it uses

From other units. - *III.1 (block-spin step and gauge fixing).* - The step data: patch, kernel choices, tree gauge, predictor Q and tags. - The definitions of request, admissible tube, store, consistency and action gap, including the identity `Tube.locates`. - The branch condition. It is proved for the kernel choices of the construction by `Foundations.branches` (from `Axial.viable`) and is carried by every sheet (`Sheet.has`). - With usable tags at the base site, the curvature bound for the predictor `qdc`: - $|Q_p| \leq C_Q L^{-2}\,\mathrm{port}(x,U)$; - $|V_p| \leq \sum_{b \subset \partial p} w_b + C_Q L^{-2}\,\mathrm{port}(x,U)$.

```
Here ${\rm port}(x,U)$ is the sum of the fine $|U_q|$ over the macro box of the coarse site $x$.
2}$.
```

- *III.2 (constants and schedule):* sheets, block factors, protection, delays, bar and waits. The target data precede the block choice; the per-step window constants are chosen in III.10.2.
- *III.3 and III.4 (free kernels, Gaussian integration):* the towers, their flat identities and exponential decay (III.3 attribution probable); the Gaussian tail bound `rectangle`, which for covariance at most M reads $\Pr(|\xi_i| > s\ \forall i \in X) \leq 2^{|X|} e^{(-\nu s + M\nu^2/2)|X|}$ for every $\nu > 0$.
- *III.5 (cluster expansion):* polymer norms, budget points, representations of the effective density, Kotecký–Preiss sums.
- *III.6–III.8 (stationary branch, matching, connected activities):* the stationary mean and its jets, the quartic matching of the path data, and the locality of connected activities.

- *III.9:* the slope b_G, the thresholds, the monitor and the incoming allowances. III.9 supplies the fine-level density representation and consumes the activity bounds; the norm conversion remains to be written out (§7).
- *I.1, II.2, III.16:* lattice, Haar measure, chordal distance, polymer activities, Taylor jets.

Standard mathematics. The analytic method is Balaban's lattice-gauge renormalization group. Relevant references are *Renormalization group approach to lattice gauge field theories*, I, CMP 109 (1987), 249–301, II, CMP 116 (1988), 1–22, and *Large field renormalization*, I, CMP 122 (1989), 175–202, II, CMP 122 (1989), 355–392. The summability estimates use rooted polymer bounds of Kotecký–Preiss, CMP 103 (1986), 491–498, tree-graph estimates, Schur's test, Gaussian exponential moments, and Taylor estimates on compact charts. Exact correspondences with Balaban's lemmas remain open (§7).

Imported constants and assumptions. - The request and target data precede L. The window constants in III.10.2 are chosen after L, W and the patch, before the threshold in x. Simplicity of G is required by III.10.5(b) and I04. - The following are internal choices of the proof: the exponent $49/100$ in T; the factor 10^{-5} in R; the hysteresis thresholds $t/4$, $t/2$, Wt, $2Wt$; and the radii $5B$, $2B$, $4B/5$, $2B/5$, $30B$. - The incoming bounds on the fine activities ($\beta_F^{-11/10}$, $e^{-M^{1/5}}$, $2M$) are hypotheses supplied by III.9.

3.13.3 3. Strategy

The method combines large-field suppression with Gaussian integration in the small-field region. Seeds record the fine plaquettes, mismatches and coarse flags that exceed their thresholds. The deterministic estimate gives an action cost κt^2 per seed. Since $\beta \geq \beta_F/2$, its Boltzmann factor is at most $e^{-(\kappa/2)e^{2(\log\log\beta_F)^2}}$. This pays every fixed polylogarithmic entropy loss and is smaller than every inverse power of β_F, while $t = \beta_F^{-1/2+o(1)}$ remains perturbatively small.

An exact completed-square decomposition separates the fine, interior, coarse and gauge contributions from two kernel defects. Large L makes the fine prediction small; large W separates mismatch seeds from the interior prediction. Coarse plaquettes above a fixed threshold lie near spent seeds, whose larger cost tT pays the affected-cell losses. Away from those seeds, local gauge fixing and the flat tower identities reduce the defects to chart errors and exponentially small kernel tails. Summation over a radius-B neighbourhood still leaves an error $o(t^2)$ per seed.

The resulting gas has packet, old forced, logarithmic and warning letters. The energy bound and annulus estimate control packets; Gaussian tails control warnings. Rooted support bounds and tree-graph estimates sum connected molecules. Derivatives are taken with sector data fixed and fluctuation coordinates transported compatibly. The Leibniz rule controls products, and Cauchy–Schwarz retains the square root of the simultaneous-warning probability when integrating their jets.

3.13.4 4. Steps

Unless a lemma gives different hypotheses, the energy estimates in §§4.2–4.6 use the step, tube, forest, fields, tags and consistent store of III.10.1. A sufficiently large threshold in x may depend on the fixed request, L, W, and any additional parameters already chosen, but never on the period. Row and column bounds refer to sums of operator norms; vector bounds are pointwise in the coordinate index. Distances are native torus max-distances.

3.13.4.1 4.1 The algebraic identity Lemma 4.1 (completed squares). Notation: - $v = v(V)$ are the coarse coordinates (`vision`). For each coarse plaquette q they are the pair $(\sqrt{\alpha_c}\,\mathrm{curv}(V_q),\ \ell_q(V))$, with the first component multiplied by $\min(1, t_c/|V_q|)$, interpreted as one when $|V_q| = 0$, and the second set to zero unless q is kept. Let d_F, d_c denote the clipped densities of §1. - $X(U)$ and $X^c(V)$ are the unthinned fine and coarse coordinates. - $Z_b = \sqrt{a}\,\mathrm{curv}(V_b Q_b^{-1})$.

The rows are: - *fine plaquette p:* $d_F(U,p) + \frac{1}{2}|(\Phi v)_p|^2 - \langle (\Phi v)_p, X_p\rangle$; - *coarse bond b:* $\frac{a}{2}w_b^2 + \frac{1}{2}|(\Phi' v)_b|^2 - \langle (\Phi' v)_b, Z_b\rangle$; - *coarse plaquette q:* $\langle v_q, X_q^c\rangle - \frac{1}{2}|v_q|^2 - d_c(V,q)$; - *gauge coordinate n:* $\frac{1}{2}|(Nv)_n|^2$; - *defects at q:* - the linear-defect contribution $\langle v_q, (\Phi^* X + \Phi'^* Z - X^c)_q\rangle$; - the completeness-defect contribution $\frac{1}{2}\langle v_q, (v - (\Phi^*\Phi + \Phi'^*\Phi' + N^*N)v)_q\rangle$.

Their sum over the torus is $\mathcal{A}_F(U) + \frac{a}{2}\sum_b w_b^2 - \mathcal{A}_c(V)$. The functional $\mathcal{E}_A$ keeps only the rows located in $D_{2B}(A)$.

Proof: adjointness for the three kernels; all cross terms cancel. The cancellation is algebraic and works for any coarse coordinate vector v if the rows are defined with that vector.

Lean: `Tube.locates`; rows `workF`, `workI`, `workC`, `workN`, `workL`, `workD`.

3.13.4.2 4.2 Packing and geometry Lemma 4.2 (packing; E01). For every $\delta > 0$ there is N, depending on P and δ but not on L or W, such that for all L, all $W > 0$, all large x and all admissible data with A consistent: - (i) for every coarse site y, $\mathrm{port}(y,U) = \sum_q m(y,q)|U_q|$ with nonnegative integer charges m satisfying $\sum_y m(y,q) \le N$, and $m(y,q) > 0 \Rightarrow \mathrm{dist}(y,\bar{q}) \le N$; - (ii) every coarse plaquette of $D_{2B}(A)$ with $|V_q| > \delta$ (a tide) lies within $4B/5$ of A^{sp} (the hit set); - (iii) $\sharp A_\flat \le 4\,\sharp\{q \in D_{2B} : |V_q| > t_c\}$; - (iv) $\sharp\{\text{tides}\} \le N\sharp A^{\mathrm{sp}}$; - (v) the coarse plaquettes of $D_{2B}(A)$ number at most $NB^4\sharp A$, and the hit set at most $NB^4\sharp A^{\mathrm{sp}}$.

Proof: the macro box has radius $(j+7)L$. Counting possible parent sites gives a bounded total multiplicity for each fine plaquette. If no spent seed is within distance $j+8$ of a coarse plaquette, consistency bounds every relevant input by T; the predictor-curvature bound then gives $|V_q| \le C(L,W)T \le \delta$ for large x. Thus every tide has a nearby spent seed, and $j + 8 \le 4B/5$ eventually. A flag is incident to a coarse plaquette above t_c; each plaquette has at most four incident sites. Four-dimensional ball counts give the remaining bounds.

Lean: `Bid.E01` (`Payments`, `MacroStore`), `Work.E01.answer`.

Lemma 4.3 (shell comparison; E02). For every gauge group G, block factor L, index j and kernel choices K, there is $C > 0$ with the following property. Take a step using K, block factor L and index j, whose fine side is at least 100 times the macro radius, with $c > 1$; any U in the tree gauge; any V; and any fine link e with usable tags. Then

$$\chi(U_e \tilde{V}_e^{-1}) \le C\,\mathrm{port}(\bar{e}, U) + w_{(\bar{e},\mu)},$$

where $\tilde{V}$ is the replica of V on fine links and $\bar{e}$ the parent block. The last term is present only when the ends of e lie in different blocks.

Sketch: close e by the two tree paths, lift the loop to $\mathbb{Z}^4$, and bound the chord of the product by the sum of chords along a walk whose length is bounded by the macro budget. For a crossing link, the predictor estimate controls the discrepancy between the two block roots; the triangle inequality then adds the coarse mismatch w_b.

Lean: `Bid.E02`, `Work.E02.answer`.

3.13.4.3 4.3 Kernel bounds Lemma 4.4 (rows and columns; E03). There is $A > 0$, independent of L and W, such that: - Φ has row sums at most A/L^2 and column sums at most AL^2; - Φ' and N have row and column sums at most A; - all entries vanish beyond distance $B/50$.

Take absolute values of the cover coefficients. Adjoint transport has norm one, periodization charges each cover term once per row or column, and the fine residue multiplicity is L^4 (`anchored_of_terms`).

Lean: `Bid.E03` (`Allowance`), `Work.E03.answer`.

Lemma 4.5 (path series and predictions; E04, given E03). There is $A > 0$, independent of L and W, such that, with $s = t$ at the fine stage and $s = t_c$ at the coarse stage: - the fine and coarse path series have absolute row and column sums at most A and range at most $B/50$; - $|\ell_p(U)| \le \sum_q c_{pq}|U_q|$ for the nonnegative absolute path coefficients c_{pq}, and $|\ell_p| \le As$ on kept plaquettes; - $|v| \le At$, $|\Phi v| \le At/L^2$ and $|\Phi' v| \le At$.

Proof: sum the absolute path coefficients with their exponential length weights. Periodization gives the column count; the kept test bounds each contributing plaquette by the threshold times the same exponential weight. Thinning bounds the coarse coordinates, and the row estimates of Lemma 4.4 bound their predictions. The comparison $t_c(x) \le 2t$ holds eventually at fixed L.

Lean: `Bid.E04` (`LaneCounts`, `Maximum`), `Work.E04.answer`.

3.13.4.4 4.4 Row-by-row lower bounds Lemma 4.6 (coarse rows; E05). There are $\delta, B_1 > 0$, independent of L and W, such that for every coarse plaquette with $\tau = |V_q|$: - $t_c < \tau \le \delta$ implies row $\ge \frac{\alpha_c}{4} t_c \tau$; - $\tau \le t_c$ implies row $\ge -B_1 t_c^4$; - $\tau > \delta$ implies row $\ge -\lambda_c t_c \tau$.

Sketch: in matrix space the curvature is the tangential projection of $V_q - 1$, so $\chi^2 - |\mathrm{curv}|^2$ is the squared normal displacement divided by the fixed matrix-to-colour normalization, which is $O(\chi^4)$ in a tubular neighbourhood of G. - The path component cancels whether kept or dropped. For $\tau \le t_c$ the direct component is unthinned, and the row equals $-\frac{\alpha_c}{2}(\chi^2 - |\mathrm{curv}|^2)$. - For $t_c < \tau \le \delta$ the row is $\ge \alpha_c t_c \tau(1 - O(\delta^2)) - \frac{\alpha_c}{2} t_c^2 - \lambda_c t_c \tau \ge \frac{\alpha_c}{4} t_c \tau$, using $\lambda_c < \alpha_c/16$.

Lean: `Bid.E05` (`Crowds`), `Work.E05.answer`.

Lemma 4.7 (fine and interior rows; E06, given E03, E04). There is L_1 such that for every $L \ge L_1$ and every $W \ge 2A/\sqrt{a}$ (with A from E04), for large x, with $d = \alpha_F/128$:

$$\tfrac{\lambda_F}{2} t \sum_{p \in A_F,\, |U_p| > t} |U_p| + d\, t^2\, \sharp\{p \in A_F : |U_p| \le t\} \le \sum \text{fine rows}, \qquad \tfrac{a}{8} \sum_{b \in A_I} w_b^2 \le \sum \text{interior rows}.$$

Proof: - *Costs with* $|U_p| \le t$. The direct part of the fine row is at least $\frac{1}{2}(\sqrt{\alpha_F}|U_p| - |(\Phi v)_p|)^2$. For a cost, $|U_p| \ge t/4$, while $|(\Phi v)_p| \le At/L^2 \le \sqrt{\alpha_F} t/8$ for $L \ge L_1$; this gives $\alpha_F t^2/128$. - *Costs with* $|U_p| > t$. The clipped action is $\lambda_F t|U_p|$, and the cross term is at most $AL^{-2}\sqrt{\alpha_F} t|U_p| \le \frac{\lambda_F}{4} t|U_p|$. - *Path-series part.* It is a complete square on kept plaquettes. On dropped plaquettes the cross term is paid by the column bounds of E04, at total cost $\frac{\lambda_F}{4} t \sum |U_p|$; this works only after summation, not row by row. - *Aisles.* On an aisle $w_b \ge Wt$ and $|(\Phi' v)_b| \le At \le \frac{\sqrt{a}}{2} w_b$, so the row is at least $\frac{1}{2}(\frac{\sqrt{a}}{2} w_b)^2 = \frac{a}{8} w_b^2$. - *Unlisted rows.* Interior rows are nonnegative by the same square lower bound. Unlisted fine direct parts and kept path parts are nonnegative; the dropped path parts are included in the summed loss above.

Lean: `Bid.E06` (`Seals`), `Work.E06.answer`.

Lemma 4.8 (affected cells; E07, given E01, E03, E04). For every fixed $\delta > 0$ there is a pre-L constant $C_7 > 0$, and for every L and $W > 0$ a constant Z, such that for large x:

$$\text{defects(hit)} + \frac{C_7}{L^2} t \sum_{p \in A_F, |U_p| > t} |U_p| + C_7 t \sum_{b \in A_I} w_b + Z t^2 B^4 \sharp A^{\text{sp}} \;\geq\; \lambda_c t_c \sum_{\text{tides}} |V_q|.$$

Proof: 1. By qdc, on the hit set $|V_q| \leq r_0 t + H_q$. The macro excess H is bounded, through the charges of E01, by a pre-L constant times $L^{-2} \sum_{\text{swell}} |U_p| + \sum_{A_I} w_b$. 2. The defect rows on hit cells are bounded below by Cauchy–Schwarz, using $|v| \leq At$ and the kernel bounds of E03–E04. The inputs X, Z, X^c split into a uniform part, $O(t)$ per cell, and the excess. 3. The uniform part is paid by $Z t^2 \sharp\text{hit} \leq Z N t^2 B^4 \sharp A^{\text{sp}}$; the excess is paid by the two middle terms.

No cancellation of linear terms is used.

Lean: `Bid.E07` (`Lapse`), `Work.E07.answer`.

Lemma 4.9 (unaffected cells; E08, given E01, E02). For every L and $W > 0$ there is C such that for large x the following holds. For every coarse plaquette q of $D_{2B}(A)$ outside the hit set there is a gauge transformation z with: - $\chi(V_b^z) \leq h := CB^{10}T$ for all coarse bonds within $2B/5$ of q; - $\chi(U_e^{z \circ \text{parent}}) \leq h$ for all fine bonds whose parent is within $2B/5$.

Sketch: there is no spent seed within $4B/5$ of q, so all tolls and wires near q are at most T. An axial gauge on a box of the coarse cover, with E02 transferring the bound to fine links, loses a polynomial in B.

Lean: `Bid.E08` (`Stitches`), `Work.E08.answer`.

3.13.4.5 4.5 Linearization defects on unaffected cells Lemma 4.10 (flat identities). Consider the linearized uncut kernels Φ_0, Φ_0', N_0 (scalar matrices built from the towers), and the linearized inputs for log-fields u (fine) and v (coarse): $X_0(u)$ (face curls and path series of u) and $Z_0(u, v) = \sqrt{a_{\text{pen}}}\,(v - \text{probe}\, u)$, where a_{pen} is the penalty. Under the admissibility identities and agreement of the beds with the fine and coarse path data, for every u, v and every coarse two-layer vector w,

$$\Phi_0^* X_0(u) + \Phi_0'^* Z_0(u, v) - X_0^c(v) = 0, \qquad w - (\Phi_0^* \Phi_0 + \Phi_0'^* \Phi_0' + N_0^* N_0) w = 0.$$

Proof: the flat tower identities, applied componentwise to the scalar matrices and the attached path data.

Lean: `roll_l`, `roll_d`.

Lemma 4.11 (chart estimates on the torus). For a patch D satisfying the branch condition there are $C, \tau, r > 0$ with the following property. For every admissible step on D, all tags, all R, and all log-fields a, b with $\|a\|, \|b\| \leq h \leq r$, set $U = \exp a$ and $V = \exp b$. Define the defect vectors $\mathfrak{l} = \Phi^* X + (\Phi')^* Z - X^c$ and $\mathfrak{d}(w) = w - (\Phi^* \Phi + (\Phi')^* \Phi' + N^* N) w$. At every coarse plaquette, componentwise in its two coordinates, - $\|\mathfrak{l}\| \leq Ch\,\omega$; - $\|\mathfrak{d}(w)\| \leq Cq\,\omega$ whenever $q \geq 0$ and $\|w\| \leq q$.

Here $\omega = \omega(R, h, \tau) = (1 + R)h + e^{-\tau R}$ (`widen`).

Proof: compare every actual object with its linearization, then subtract Lemma 4.10. - *Kernels:* they are close to their flat versions (`kernel_chart_cost`). - *Path-series coordinates* (`x_chart_cost`):

- transporting a curvature of size $O(h)$ along a path of length at most R costs $O(Rh \cdot h)$; - the paths longer than R are paid by the weight $e^{\sigma \,\mathrm{length}}$ in the path budget, which gives $e^{-(\sigma/d)R}$. - *Predictor coordinates:* quadratic error (`z_chart_cost`).

Lean: `global_chart_cost`.

Lemma 4.12 (interior cost). For the same D there are $C, \tau, r > 0$ with the following property. Suppose that in some gauge z both U and V are within $h \le r$ of the identity on all bonds within distance $3R + 4 +$ reach of a coarse plaquette q, where reach is the predictor reach and $h \ge 0$. The step must be on D and admissible for P; the tags and cut length $R \in \mathbb{N}$ are arbitrary. Then the bounds of Lemma 4.11 hold at q for every coarse coordinate vector bounded by a nonnegative constant.

Proof: 1. *Gauge covariance:* the defect vectors transform by $\mathrm{Ad}(z_q)$ (`gauge_gaps`). 2. *Locality:* the defect vectors depend only on nearby bonds (`local_gaps`). 3. *Extension:* a configuration that is h-close to 1 on a ball agrees there with exp of a global log-field of norm $O(h)$ (`extend_interior`). 4. Apply Lemma 4.11.

Lean: `interior_cost`.

Lemma 4.13 (unaffected defects; EBare). Assume E03, E04 and the conclusion of E08 at (P, L, W). Then for every $\varepsilon > 0$ and all large x, the sum of the defect rows over the unhit coarse plaquettes of $D_{2B}(A)$ is at least $-\varepsilon t^2 \sharp A$.

Proof: 1. Take $h = CB^{10}T$ from E08 and $\omega = \omega(R, h, \tau)$. Lemma 4.12 applies, since $3R{+}4{+}\text{reach} \le 2B/5$. 2. Cauchy–Schwarz with $|v| \le At$ gives, on every unhit cell, defects at least $-\ell$, where $\ell = 2(At)(Ch\omega) + (At)(CAt\,\omega)$. 3. There are at most $NB^4 \sharp A$ such cells (E01 with $\delta = 1$). 4. $NB^4\ell \le \varepsilon t^2$ for large x, because $B^n\omega \to 0$ and $B^n h\omega/t \to 0$ for every n (`radius_stitch_errors`). Indeed $h = \beta_F^{-0.49+o(1)}$, $h^2/t = \beta_F^{-0.48+o(1)}$, and $e^{-\tau R} \le \beta_F^{-1}$ because $R \asymp (\log x)^2$. 5. If no patch with the request's data satisfies the branch condition, the statement is vacuous.

Lean: `Bid.EBare` (`Belly`), `Fluctuations.bare`.

3.13.4.6 4.6 Assembly of III.10.1 Lemma 4.14 (assembly; E09). For a fixed request, the packing estimate of Lemma 4.2, fine/interior bounds of Lemma 4.7, coarse bounds of Lemma 4.6 at some $\delta > 0$, affected-cell bound of Lemma 4.8 at that δ, and unaffected-cell bound of Lemma 4.13 for every $L, W > 0$, imply III.10.1.

Choice of constants: - $L \ge \max(L_1, L_7)$, where $C_7/L_7^2 \le \lambda_F/4$; - $W \ge \max(1,\ 2A/\sqrt{a},\ 16C_7/a)$; - $m = \min(\lambda_F/8,\ \alpha_F/128,\ a/32,\ \alpha_c/64)$ and $\kappa = m/2$; since $\lambda_F < \alpha_F/16$, this is the value in III.10.1; - x so large that: $t_c \le \delta$; $t \le T$; $t \le 2t_c$; $(4mN + ZB^4)t^2 \le m\,tT$; $B_1 t_c^4 NB^4 \le \frac{m}{4}t^2$; and the unaffected-cell bound holds with $\varepsilon = m/4$.

Proof: add the rows. 1. *Swell costs:* $\frac{\lambda_F}{2} t|U_p|$. Half of it pays the $C_7 L^{-2}$ term of Lemma 4.8; the remainder is at least $2mt^2$, and at least $m\,tT$ when the cost is spent. 2. *Other costs:* at least $\alpha_F t^2/128 \ge mt^2$. 3. *Aisles:* $\frac{a}{8}w^2 - C_7 tw \ge \frac{a}{16}w^2 \ge mt^2$; spent aisles carry $m\,tT$. 4. *Tides:* paid by Lemma 4.8. 5. *Coarse plaquettes in the intermediate range:* $\frac{\alpha_c}{4} t_c |V_q| \ge \frac{\alpha_c}{16} t^2$, which pays $4mt^2$ per intermediate plaquette, since $\sharp A_\flat \le 4(\sharp\text{intermediate} + N\sharp A^{\mathrm{sp}})$. 6. *Small coarse plaquettes:* in total at least $-\frac{m}{4}t^2\sharp A$. 7. *Gauge rows:* ≥ 0. 8. *Defects:* on hit cells they are inside Lemma 4.8; on unhit cells they total at least $-\frac{m}{4}t^2\sharp A$. 9. *Spent seeds:* their $m\,tT$ covers the flag charge against A^{sp} and the Z term.

The total is at least $\frac{m}{2}t^2\sharp A$.

Lean: `Bid.E09`, `Work.E09.answer` (with `assemble` and `bank`), `Bid.issues`.

3.13.4.7 4.7 Activities of one step The next estimates use the hypotheses of III.10.2. In statements about packets, logarithmic letters or warnings, the tube satisfies the compatibility condition, the input satisfies the stated fine-level bounds, and sectors have the actual coarse flags. Auxiliary window constants are fixed before the threshold in x.

Lemma 4.15 (rate calculus). A function w has rate a if for every $v > a$, $|w(x)| \le \beta_F^v$ eventually. Products of functions with rates a, b have rate $a + b$; their sums have rate $\max(a, b)$. Eventual absolute domination preserves a rate, and constants have rate 0. Thus t has rate $-\frac{1}{2}$, and B and bill have rate 0.

Lean: `Rate`.

Lemma 4.16 (from the action gap to packet weights). - (a) There is L_0 such that for all $L \ge L_0$ and all large W there is $\mu > 0$ with two properties. First, the bound of III.10.1 holds with constant μ at all large x (`Grant.order`; here μ is the κ of III.10.1). Second, fix any halo and nonnegative support exponent, $d \in \mathbb{N}$, $D \ge 0$, $z < 2$ and $u > 0$. For all large x and every $\beta \ge u\beta_F$, the gain $\beta\mu t^2$ passes the packet summability test for any spend at most $D\,\mathrm{bill}^d M^z$ and any nonnegative prefactor at most its exponential. The total packet charge is exponentially small in the retained gain. The threshold may depend on these additional parameters. - (b) Suppose the local sector and tag tests hold through radius $300B$, and the physical fluctuation on hot bonds is bounded by a nonnegative function of rate $-1/2$ through radius $304B$. On the annulus $2B < \mathrm{dist} \le 200B$, every remainder row is at least $-E(x)$ for some nonnegative E of rate $-3/2$ (`Coat.annulus`). After multiplying E by the annulus volume and fixed coordinate multiplicity, it still has rate $-3/2$. Consequently the assigned energy is at least $(\mu t^2 - E)\sharp A$ (`Coat.earns`). Since βE has rate at most $-1/2$, the loss is absorbed in bill^5. - (c) In every manifest sector, $|\mathrm{Pack}| \le \exp\left((C\,\mathrm{bill}^5 - \beta\mu t^2)\sharp A\right)$ (`steam_prices`).

((a) and (c) proof read; (b) statement only)

Lemma 4.17 (old and logarithmic letters). Fix $-1/2 < s < 0$ and any $u \in \mathbb{R}$. Assume the sector drift and residual estimates, with a residual envelope of rate -1, as well as the common hypotheses of this subsection. For sufficiently large L and then all large x, the old forced letters have a rooted bound with total β_F^u, and all logarithmic letters have one with total β_F^s. The latter include inherited regular activities, local Jacobians and Gaussian determinant corrections. The weights bound both $|\ell_i|$ and $|e^{\ell_i} - 1|$. Taking $u = s$ gives the two totals used in III.10.2.

Proof: transport the incoming forced bounds to coarse supports, and split the logarithmic alphabet into inherited activities, Jacobians and Gaussian corrections. Their envelopes have rate $-1/2$ or smaller. Choose an intermediate rate strictly between $-1/2$ and s to absorb their finite sum, and use $|e^z - 1| \le 2|z|$ once the individual logarithmic weights are at most one.

Lean: `recommends`.

Lemma 4.18 (window, drift, no new seeds). For each patch there are window constants such that, for large x: - every manifest sector satisfies the Hessian conditions of III.10.2(a)(iii) (`shoots`); - the Gaussian centring drift is $O(t)$ and the residual of its linear stationarity equation has an envelope of any prescribed rate, in particular -1 (`YM.ConnectedActivityE6.Coat.tends_val`); -

there is $\xi > 0$ with $\sharp\{\text{colours}\} \cdot \xi \leq 1$ such that fluctuations satisfying the radius-B window test create no seeds at sites farther than $30B$ from the given ones (`Coat.ready`).

Proof: exponential row bounds and coercivity give the precision estimate; the local-inverse error tends to zero and yields a positive covariance. Applying the approximate inverse to the linear term gives the centring drift and a small linear residual. For the no-new-seed assertion, choose ξ so the Gaussian fluctuation contributes at most one third of the available margin. Linear cancellation from the compatibility identities and the small nonlinear remainder pay the other two thirds. The fine and mismatch prediction bounds then give the strict thresholds $t/4$ and Wt.

Lemma 4.19 (catastrophes). For every set v of warning addresses, the Gaussian measure of the catastrophe set is at most $E^{\sharp v}$, with E eventually smaller than every power of β_F. The threshold $\xi t\sqrt{\beta}$ equals $\xi e^{(\log\log\beta_F)^2}(\beta/\beta_F)^{1/2}$ in Gaussian units.

Proof: the covariance has a volume-independent Schur bound. Apply the Gaussian rectangle estimate to simultaneous coordinate excursions and sum the possible witnesses in each radius-$2B$ neighbourhood. The resulting envelope can be taken as $CB^4e^{-cM^{1/2}}$, where $M = e^{(\log\log\beta_F)^2}$ and $c > 0$. It is smaller than every inverse power of β_F. Repeated warning locations are paid through the fixed multiplicity bound.

Lean: `Coat.alarm_bills`.

Lemma 4.20 (assembly of III.10.2). The preceding estimates imply III.10.2 with the following parameter choices. 1. The floor on L is the sum of the floors from Lemma 4.16(a), the no-new-seed margin, and Lemma 4.17. 2. W is taken large for the energy estimate and the no-new-seed margin; μ comes from Lemma 4.16(a) and does not depend on the patch. 3. For each patch: - K, γ come from the shell comparison; - the window constants from the precision estimate; - the drift constant C_d from the centring estimate, and $a_o = \max(C_d + 1, \gamma K)$; - ξ from the no-new-seed estimate; - C from the packet estimate; - s is chosen with $-\frac{1}{2} < s < r$. 4. All conditions holding for large x are merged into one threshold. The three letter totals have rate $s < r$, so their sum is eventually at most β_F^r.

Lean: `Target.issue`.

3.13.4.8 4.8 Derivatives Lemma 4.21 (letter jets). Under the hypotheses of III.10.3, the letter-jet structure exists before integration. Choose $-1/2 < s < r$ and set $\mathrm{rad} = \beta_F^s$. The warning amplitude before probability extraction is a patch constant. The jet scale is a sum $\rho(x) = B_0(x) + A_0(x)$ of two rate-zero envelopes, chosen independently of the input representation and volume.

Proof: combine the packet-jet estimate, the old and logarithmic letter estimates, and the differentiated warning cutoffs, all along compatible transported fluctuation coordinates. Choose the catastrophe envelope of rate $2s - 1$; its eventual bound by β_F^{2s} gives the probability factor rad^2. The three rooted letter totals have rate $s < r$.

Lean: `Utopia.whittle`.

Lemma 4.22 (molecule integrals). The letter bounds of Lemma 4.21 imply the integrated majorants of III.10.3, for the compatible connected molecules specified there.

Proof: the Leibniz rule for `Brush` adds the letter jet scales. Their sum is bounded by twice the molecule's shape, giving, for rate-zero transport envelopes U, J' and fixed integers $k, K \geq 0$, a

pointwise factor

$$(1+|U(x)|+2(\rho(x)+J'(x)))^{i}(1+\mathrm{shape})^{i}\,\mathrm{facescale}^{ki}.$$

The bound vanishes outside the catastrophe set for the molecule's distinct warning addresses. If that set is $\mathcal{C}$ and its address set is v, Cauchy–Schwarz gives

$$\int \mathbf{1}_{\mathcal{C}}\,\mathrm{facescale}^{ki}\,d\gamma \le \left(\int \mathrm{facescale}^{2ki}\,d\gamma\right)^{1/2}\gamma(\mathcal{C})^{1/2} \le e(x)(1+\mathrm{shape})^{K}\mathrm{rad}^{\sharp v},$$

where e has rate zero. Integration over the probability measure on the tree slice adds no factor. Multiplying by the pointwise coefficient gives a rate-zero prefactor p and shape power $l = K + i$. For a marked source, remove its ordinary fare and insert its amplitude bound.

Lean: `timing`.

3.13.4.9 4.9 Start-up fusion and markers Lemma 4.23 (H01–H04). Under the respective hypotheses in III.10.6(H), the four fusion conclusions hold. - *H01* is the Mayer identity $e^{\sum \ell_i} = \prod_i (1 + (e^{\ell_i} - 1))$, regrouped with the forced activities into pellets. Absolute domination of the original and logarithmic series justifies the product expansion and regrouping; the argument also proves summability of the fused series. - *H02:* retain one factor $e^{-\kappa}$ from a forced letter in every pellet and use the rooted tree sum $e^{\gamma} - 1$. - *H03:* transport every pellet support, inventory and eligibility test under each lattice symmetry. The local flag tests and the permutations of the original addresses induce those of the fused family. - *H04:* each pellet is a fixed finite product; a common summable envelope then gives continuity of the weighted series by dominated convergence.

(proof read for H01, H02 and H04; H03 forced-family proof read, symmetry moves statement only)

Lean: `H01.gathered`, `H02.answer`, `H03.fused_forcedFamily`, `H04.answer`.

Lemma 4.24 (I01–I04). Under the respective hypotheses in III.10.6(I), the four marker conclusions hold. - *I01:* traverse the centres of a minimizing connected B-ball cover. Distinct markers force gaps of at least $B^2 - 2B$ between the traversal times of their covering centres. Comparing total traversal length with the number of balls gives the factor 64. - *I02:* I01 converts extra support decay to an increase of multiple-marker radii. Splitting single-marker and multiple-marker sums gives total two. Shrinking the radius by the order factorial converts factorially weighted derivative bounds to ordinary derivative bounds. - *I03:* centred embeddings compose and preserve the parent relation. The open-stop inequality persists to later levels because $B_{r+1}^2 \le 2B_r^2$ and each block factor is at least three. - *I04* follows from `columns_receipt`: for every level schedule, $\sum_{m<i} \log \mathrm{columns}_{m+1} = o(\mathrm{scale}_i)$. Combine this with $\log \mathrm{order}_i! \le \log \mathrm{columns}_i$ and $\log \hat{\beta}_i = o(\mathrm{scale}_i)$.

Lean: `I01.answer`, `I02.answer`, `I03.answer`, `I04.answer`.

3.13.4.10 4.10 The schedule Lemma 4.25 (pricing order). Under the hypotheses of III.10.5(a), the following choices produce its floors and raised entry couplings. 1. For each pair of adjacent stages, III.10.1 gives a floor b_{ij}; set $k_i = \max(b_{ii}, b_{i,i-1}, \text{sheet floor}_i)$. 2. For $L_i \ge k_i$, choose $W \ge m$ in the admissible range and a threshold x_{ij}. 3. Raise β_i^* to $\max(\beta_i^*, x_{ii}, x_{i,i-1})$. 4. The conclusion persists under further raising (`recheck`).

Lean: `Bid.pricing_order`.

Lemma 4.26 (divisible schedule). Under the hypotheses of III.10.5(b), the following construction gives a schedule satisfying all entry, plateau and divisibility requirements. 1. Entry couplings come from the bar (using G simple), leads from the wait data, and a monotone stage-index function J with steps at most 1 and $J \to \infty$ by the finite-request scheduling lemma of III.2. 2. The scales are s_i; the initial length M is the minimal length plus the largest bill over the first stages; the periods are $4(M + p + 1)$.

The checks: - $\hat{\beta}_m \geq$ the bar value, and $\hat{\beta}_m \geq \beta^*_{J(m+1)}$; - the scale entries; - the volume inequalities: shells $\cdot\, (1 + B_i + B_{i+1})^{\text{growth}} \leq$ side, $100(B_i + 1) \leq$ side, and $100($patch radius $+ 1) \leq$ next side; - the mesh condition $\log L_{J(i+1)}/s_i \to 0$; - divisibility: $p = c(M + z + 1) - (M + 1)$ gives the period $4c(M + z + 1)$.

Lean: `Wait.engine`.

3.13.5 5. Where the difficulty lies

The energy estimate must preserve L-independent coefficients on the large fine inputs and W-independent coefficients on the mismatches. The fine prediction is $O(t/L^2)$, so increasing L absorbs its cross terms. The interior prediction is $O(t)$, so increasing W absorbs the mismatch cross terms. Losses depending on both L, W are charged to spent seeds and absorbed only by increasing x.

On unaffected cells, the transported, truncated kernels have no exact completeness identity. The flat identity cancels the linear part, while local gauge fixing, chart estimates and exponential tails control the remaining defects. This is the unaffected-defect estimate `Bid.EBare`. The constant κ is independent of L, W; the coupling threshold is not.

The activity and derivative estimates additionally require compatible quadratic forms, controlled fluctuation transport and probability bounds for simultaneous warnings. Their product estimates retain one gain for each distinct warning address before the connected-molecule sum. The schedule is numerical; start-up fusion and marker estimates serve separate parts of the construction.

3.13.6 6. Notation

Written proof	Lean (glossary)	Meaning
$x = \hat{\beta}_m$, $\beta_F = x + b_G \log L$	`betaF`	coarse and fine reference couplings
$t_c(x)$, $t = t_c(\beta_F)$	`temp`, `fiery`	small-field thresholds; `fiery` is at the **fine** coupling
$T = \beta_F^{-49/100}$	`largeT`	spent-seed threshold
B, R	`radiusB`, `till`	cluster radius $\lceil (\log x)^2 \rceil$, path cutoff $\lfloor B/10^5 \rfloor$
$\lvert U_p \rvert$, w_b	`toll`, `wire`	plaquette size; mismatch $\chi(V_b Q_b^{-1})$
request P; admissible	`Teller`, `Teller.Funds`	data of one call of the step estimator
$A = (A_F, A_I, A_\flat)$, consistent	`Stores`, `Stores.Plot`	seeds; two-sided thresholds on D_{5B}

Written proof	Lean (glossary)	Meaning
A^{sp}, tides, hit	`spend`, `tides`, `hit`	seeds above T; coarse plaquettes above δ; cells near A^{sp}
$\mathcal{E}_A$	`Stores.assessed`	localized action gap
v, Φ, Φ', N	`vision`, `Φ`, `Φ'`, `N`	coarse coordinates; transported kernels cut at R
Φ_0, Φ_0', N_0	`flatF`, `flatI`, `flatN`	flat, uncut scalar tower operators
$\omega(R, h, \tau)$	`widen`	$(1+R)h + e^{-\tau R}$
rate a	`Rate`	$\le \beta_F^{a+o(1)}$
target data $\mathcal{T}$	`Target`	ε, δ, e, rate r, halo, chart
letter weights, KP condition	`Sales`, `Buy`	rooted Kotecký–Preiss and tree-graph tests
new-gas bounds	`Cash`, `Utopia`, `Herd`, `Captain`	III.10.2 bundle; letter jets; molecule integrals
$N_B(H)$	`coverNumber`	minimum connected ball-cover size
$\rho(x)$, rad	`Utopia.active`, `Utopia.rad`	jet scale of rate zero; square root of a warning-probability bound
Gaussian sector	`Value`, `Charges`	precision, approximate inverse, mean, residual

3.13.7 7. What remains unexplained

1. **Conversion to the norms of III.9.** The rooted optional-letter bound and the packet/tree test are explicit here. Their complete conversion to the outgoing E- and K-norms of II.2 and III.5, including the reset, remains to be written out. The relevant per-step assemblies are the activity-bound bundle, its jet version, and the molecule-integral bounds.
2. **Free-kernel compatibility along the trajectory.** The condition requires fine and coarse beds satisfying the diagonal and flat tower identities, invariance under lattice symmetries, attachment to the chosen path data, and positive invertibility bounds. `YM.PhysicalNext.Repeat.Pane.ovation` constructs it on the auxiliary torus from the trajectory's beds, assuming their lattice-symmetry property at every period. The proof of that symmetry property for the final selected trajectory, and the precise III.3 attribution of all tower inputs, have not been checked here.
3. **Depth of the analytic proof checks.** The assembly proofs of III.10.2–III.10.4 and their defining estimates have been read. The underlying stationary coercivity, local-inverse, packet-jet and individual letter-jet estimates have not all been read through; their detailed analytic proofs remain to be checked. The proof of H03's forced-family structure was read; its full lift of lattice-symmetry permutations was not checked through. These limitations do not alter the conditional statements above.
4. **Additional reset estimates.** `meritBar` and `rimfire` give summable jet bounds proportional to the input difference, under matched-input and derivative-control hypotheses. `erudite` gives positivity of a residual density and a $\beta_F^{-23/20}$ jet bound on the off-diagonal remainder

under matching, finite-bound and small-chart hypotheses. Their complete hypotheses and role in III.9's reset are not included in this unit's numbered outputs.

5. **Gaussian trigger scales.** In `JointLoad.stage`, the scales are $b(x) = 1 + B(x)$ and $p(x) = e^{(\log\log\beta_F)^2}$; the trigger estimate absorbs fixed powers of b into e^{-cp^2} per distinct endpoint. Its full use in the preconstruction has not been traced here.
6. **Comparison with Balaban.** The cited papers describe the corresponding small-field, cluster and large-field operations. A lemma-by-lemma identification, including threshold and normalization choices, remains open. The bibliographic dates and volumes above do not establish such an identification.

3.14 III.11. Comparison with the free Gaussian measure: interacting rank and determinant budgets

Part: III, Ultraviolet construction (renormalization group). **Lean:** chapter 24: `YM/Next/RANK/`, `YM/PhysicalNext/AFTERNOON_INTERACTING_RANK_COMPARISON/`, `YM/PhysicalNext/RANK_COVARIANCE_ATLAS/`, `YM/MergeProof/` (with `West/`, `North/`, `East/`, `South/`). **Size:** 42,556 lines (about 5,300 declarations).

Scope. The unit compares the interacting renormalization-group (RG) history with the free Maxwell field. It also contains the stress-flux argument and the label algebra of the Maxwell equation ideal.

3.14.1 1. What this unit proves

The unit has two outputs. The first is the **rank comparison**. In the continuum limit, the space F_d of local fields of degree d is isomorphic to the corresponding space of the free Maxwell field, compatibly with complex conjugation and with the lattice symmetry group. Here F_d is the space of labels of degree d modulo labels whose correlations vanish. The second output is the **stress tensor of the limit**: a nonzero conserved stress tensor that satisfies the translation and rotation Ward identities and is normalized against the free Maxwell stress tensor along a scaling sequence. Both outputs rest on quantitative estimates proved here: a per-site determinant budget, chessboard bounds for connected parts of corrected observables, and comparisons of covariances with the free curvature kernel. The trial systems test a finite separating family of free one- and two-point correlations.

Notation. Set $\mu_k = \exp((\log\log\widehat{\beta}_k)^2)$. The centered estimate and the trial systems use this fixed schedule quantity. G is a compact matrix group, assumed simple only where stated; $\dim G$ is the dimension of its Lie algebra; h is normalized Haar measure; $\mathrm{chord}(u) \geq 0$ is the chordal distance from u to 1; and $\mathrm{shell}(r) = \{u \in G : \mathrm{chord}(u) \leq r\}$. The torus of side $n \geq 2$ is $\Lambda_n = (\mathbb{Z}/n)^4$, with $c = n^4$ sites, $4c$ bonds and $6c$ plaquettes. The prior μ_{prior} is product Haar measure on bonds. An RG representation R on Λ_n has an inverse coupling $\beta = \beta(R)$, an exponent $\mathrm{exponent}_R$ with constant part $\mathrm{scalar}(R)$, and an unnormalized density $e^R = e^{\mathrm{exponent}_R} \cdot (\text{polymer gas})$.

Proposition III.11.1 (determinant budget on the torus, and its per-site price). (a) Let R be a lawful representation on Λ_n, $n \geq 2$, with lawful-class parameters $(\sigma, A, \Lambda, \delta, w)$, and assume $e^R \geq 0$ pointwise. Let $0 < s \leq 1 \leq b$ be a Haar budget (Lemma 4.1). Let $g > 0$ with $g \leq \mathrm{threshold}(R)$ and $\beta g^2 = 1$. Let F be a μ_{prior}-integrable complex function and $z \geq 0$, $X \in \mathbb{R}$

such that $z|F(U)| \le \exp(\text{exponent}_R(U) + cX)$ for every configuration U. Then

$$\frac{z\left|\int F\, d\mu_{\text{prior}}\right|}{\int e^R\, d\mu_{\text{prior}}} \;\le\; \exp\Big(c\,\pi(R,X,g) - 4\dim G\,\log g\Big),$$

where the per-site price is

$$\pi = 2\,\text{exCost}(A,\Lambda,\delta) + 6 + \beta\,\text{baseBudget}(A,g) + X + 3\,\text{ent} - 4\log s, \qquad \text{ent} = \log b - \dim G\,\log\sigma_R,$$

with $\sigma_R = \min(1,\ell_R)$, where q_R and ℓ_R are the quadratic and linear clip coefficients. Here exCost $= \delta +$ loopsPrice bounds the loop and additive activity terms per site, and baseBudget $= 3(q_R + A^2)g^2$ bounds the remaining action on configurations whose bonds all lie in shell$(g/4)$. (b) Let $(\mathcal{A}_m)$ be the levels of an RG schedule for simple G, and let $O = (s,b)$ be a Haar budget. For all sufficiently large levels m, every period p, every representation P on the level-m torus (with $c_{m,p}$ sites), every $k \in \mathbb{N}$, every $X \le 1 + k$, and every (g,Λ,δ) admissible at level m, the following holds:

$$\pi(P,X,g) - \frac{4\dim G\,\log g}{c_{m,p}} \;\le\; \text{workitem}(O,m,k).$$

Admissible at level m means: same clip as the stage of m, $\beta(P) \le 4\hat{\beta}_m$, $\beta(P)g^2 = 1$, $0 < g \le \text{threshold}(P) \le \text{threshold}_m$, $\delta \le 1$, $\Lambda \le \text{loopCap}_m$. The work item is

$$\text{workitem} = 2(8\,\text{paths}_m + 2) + 6 + 3(\text{direct}_m + \text{paths}_m^2) + (1+k) + 3\,\text{ent} - 4\log s + 1.$$

In (b), π uses $A = \text{paths}_m$ and the clip of stage m. The work item depends on that stage, O and k, uniformly over the admissible couplings and periods. *Status:* theorems. *Lean:* `determinant`, `protection`, `HaarBudget`; (b) `affordable`, `Meets`, `work item`.

Proposition III.11.2 (free pair terms). Fix G, $d, n \in \mathbb{N}$, and two colored plaquette derivative stencils L, S based at the origin, with scaling degrees $\Delta_L = 2 + \text{ord}(L) \le d$ and $\Delta_S = 2 + \text{ord}(S) \le d$. There is $C \ge 0$, depending only on these data, such that the following holds. Let $x \in (\mathbb{R}^4)^n$ have distinct entries and let $i \ne j$. Suppose $D \ge 0$ satisfies $|\text{Re}\,\Gamma_{(i,a),(j,b)}| \le D$ for every pair of free-current keys a, b of scaling degree at most d. Then, for every $r \ge 1$,

$$r\left|r^{\Delta_L+\Delta_S}\,\langle \text{ev}_L \otimes \text{ev}_S, B_r(i,j)\rangle - \text{light}(L,S)(x_i,x_j)\right| \le CD.$$

Here

$$B_r(i,j) = \sum_{a,b} \text{Re}\,\Gamma_{(i,a),(j,b)}\; r^{-\text{gen}(a)} t_a \otimes r^{-\text{gen}(b)} t_b,$$

where t_a is the polynomial jet sample of key a, $\text{gen}(a)$ is its scaling degree, and ev_L applies the stencil L. The kernel light(L,S) is the corresponding derivative of the free Maxwell curvature two-point function. *Status:* theorem. *Lean:* `Prayer.fence`, `deck`.

Proposition III.11.3 (window estimate for the free covariance, uniform in the regulator). Let $\alpha, \eta > 0$, $\Omega \in \mathbb{R}$ and $n \in \mathbb{N}$. Then there is $C \ge 0$ with the following property. For every regulator f in the family $\text{Firm}(\alpha,\Omega,\eta)$ of III.3, every G, and every pair of colored plaquette stencils a, b, the frequency chart $K = K_f(a,b)$ is measurable, vanishes for $|\xi| > 2$, is C^n away from 0, and satisfies:

- $|K(0)| \le C$;

- $|\partial^i K(\xi)|\,|\xi|^i \le C$ for $0 < |\xi| \le 2$ and $i \le n$;
- $|\partial^i (K - H_{a,b})(\xi)|\,|\xi|^i \le C|\xi|$ for $0 < |\xi| \le 1$ and $i \le n$.

Here ∂^i denotes the i-th Fréchet derivative, measured in operator norm, and $H_{a,b}$ is the degree-zero Maxwell curvature symbol for the plaquettes and colors of a, b, with heat parameter $(2\pi)^2$. If a, b have different colors, then $K = H_{a,b} = 0$. *Status:* theorem. *Lean:* `window_prices`, `Window`, `Cast`.

Proposition III.11.4 (frequent good scales produce corrected trial systems). Let G be simple and C a Wilson family satisfying the construction record. Let d be a degree and $\mathcal{F}$ a stage package for degree d (fixed band $b_0 \ge \max(6,\, s_d + 3)$, with s_d the label's scaling saving, and Taylor order $\ge d$). Let X consist of a real basis of F_d^{free}, starting with the unit, and finitely many generic one- and two-point tests that separate this space. Choose representatives of that basis in the label space. Let O be a Haar budget, J a chart ratio, and B a binding of the RG history of C to its price lists. Fix $w \in \mathbb{R}$. Assume that at infinitely many levels k the following conditions hold simultaneously:

- the four price conditions: tender, metering with (O, w), and, for some common auxiliary parameters l, r, hours with (J, B, l, r, w) and vents with (J, B, l, r);
- the lantern-mark condition for X.

Then there is a trial system Z for X that travels with the station and satisfies the purchase conditions:

- depths $k_1 < k_2 < \dots$ and test scales $s_i = a_{k_i} \mu_{k_i}^{b'}$ (fixed $b' > 0$) strictly decreasing to 0, with strictly increasing width levels $\mathrm{width}_i \le k_i$ and the stage index constant on $[\mathrm{width}_i, k_i]$;
- for each trial i, coefficient vectors $v_i(N, p)$ of corrected observables that are real, bounded uniformly in (N, p), and have first entry the reference unit;
- generic test configurations lying strictly inside the centred width cell, whose rescalings converge to the prescribed configuration of each row;
- for each trial i, eventually in N and for every period p, on the centred tiling of the cutoff torus: for every row, every basis column and every nonempty set s of insertion modes, the chessboard root, at coupling $\beta_{N,p}$, of the s-connected part of the corrected observable (taken relative to its free Maxwell one- and two-point profile) is at most $\gamma_i = \mu_{k_i}^{-1/32} \to 0$.

Status: theorem, with the frequent-scale conditions carried as hypotheses. In the main chain they are discharged as follows. In `Accent.rated`, the lantern marks hold eventually in k by `bursts`. The four price conditions are supplied to the rank assembly by `franchise` (this unit) inside the assembly `jointOffers`. That assembly is closed by `uv_log` (III.13), whose only hypotheses are that G is simple and $b > 0$. *Lean:* `cast_off`, `wheeled`, `v_culminates`, `v_obtain_purchases`, `Purchases`.

Proposition III.11.5 (the field spaces of the limit are those of the free Maxwell field). Let G be simple. Write $\mathcal{L}_d$ for abstract labels and $\mathcal{V}_d$ for their fixed coordinate space. Let C be a Wilson family satisfying the construction record and the centred estimate: for each d and $R, \varepsilon > 0$ there is $A \ge 0$ such that, at every level $m \ge \mathrm{ref}(d)$, eventually in N and for all p, every label of degree d and norm $\le R$ has centred budget $A\, a_m^{-d-\varepsilon}$. Assume:

- (i) a projector presentation of the labels: for each d and cutoff (N, p), a projector $\Pi_{d,N,p}$ on the coordinate label space $\mathcal{V}_d$ whose kernel is the image of the Maxwell equation ideal $\mathcal{E}_d \subset \mathcal{L}_d$ under an invertible corrected label map $\mathcal{L}_d \to \mathcal{V}_d$. For each d, both are

real and equivariant for all sufficiently large N, uniformly in p. Assume also affordability and relay-cost bounds along every extraction;

- (ii) the one-step cost package along the gauge-door maps between successive RG levels;
- (iii) panels Q_d, the mark exponents $c_d = 2(w_d + 1)$, a chart ratio and a binding such that, for each d, the four price conditions of III.11.4 hold simultaneously at infinitely many levels, with $w = -3$. The panels are specialized to the family using c_d.

Then for every extraction order f there are a refinement g, a smooth limit table T with a differentiation plan E, and the extraction record of IV.1 along $f \circ g$ (raw limits, bins, spans, cleaning to T, Euclidean axioms), together with a seed record for $(C, f \circ g, T)$. The seed record consists of: cutoff, ordered and generic point limits, Euclidean axioms, cell hypothesis, and corrected rank widths. Moreover the following holds: there is a real structure on the free Maxwell table and, for every d, a linear isomorphism

$$\Phi_d : F_d(T) \xrightarrow{\sim} F_d^{\mathrm{free}} = \mathcal{L}_d/\mathcal{E}_d$$

that intertwines complex conjugation and the action of the signed permutations of the four axes. More precisely, there is a real, equivariant linear isomorphism $L_d : \mathcal{L}_d \to \mathcal{V}_d$ carrying $\mathcal{E}_d$ onto $\mathrm{null}_d(T)$. Thus $\dim F_d(T) = \dim F_d^{\mathrm{free}}$; the identification of null spaces uses this corrected map. *Status:* theorem with hypotheses (i)–(iii). All three are discharged in the main chain: (i) by `reception` through `chaletrail` (this unit), (iii) by `franchise`, and the binding by `castle`. The one-step package (ii) and the centred estimate are also included in `Chateau` and consumed by `uv_log` (III.13). *Lean:* `Accent.rated`, `Maxwell.Ranks`, `Spec.Seed`, `Scotia`, `Wcmain`.

Proposition III.11.6 (the stress tensor of the limit, normalized against free Maxwell). (a) Let C satisfy the construction record and f be an extraction order with ordered and generic point limits, giving a table S. Assume:

- S satisfies the Euclidean axioms (with spin structure H) and the cell hypothesis;
- W is a differentiation plan for S with its inclusion and derivative laws;
- $\dim F_d(S) \le \dim F_d^{\mathrm{free}}$ for every d;
- $\mathcal{T}$ is a free Maxwell comparison model, with Euclidean axioms a, inclusion, derivative and orientation laws, and a divergence-free stress;
- the station hypothesis linking C, f and a holds;
- the shell estimate of C holds.

Then W satisfies the orientation law, and there are a stress record I (stress tensor $\mathcal{J}$, translation parts, spin rows) and a scaling X of S such that:

- $\mathcal{J} \neq 0$ and $\mathcal{J}$ lies in the kernel space of H;
- the scaled tables approach the isotropic panel, and the scaling is stable and compatible between the spin structures of S and $\mathcal{T}$;
- the scaled $\mathcal{J}$ is metered against the free stress tensor $\mathcal{T}.\mathcal{J}$;
- $\mathcal{J}$ is conserved and symmetric and satisfies the translation and rotation Ward identities (III.11.7);
- the action law relating $\mathcal{J}$ to the free model holds.

(b) Let G be simple, let C carry the shell estimate, and assume the cofinal grid approximation condition and weak completeness of the corrected comparison tests for f, together with the seed record of III.11.5 for (C, f, S). Then for every differentiation plan W with inclusion and derivative laws there are Maxwell data M and a stress record I with $\mathcal{J} \neq 0$ satisfying the

UV interface record $\mathrm{UV}(C, f, S, W, M, I)$. *Status:* theorems. In (b), the free model comes from `maxwell_supplies` (VI.2). The station hypothesis is built from the seed and weak completeness, and the hypothesis `DimAbove` from the seed's corrected rank widths; this is how III.11.5 feeds III.11.6. The shell budget is proved by `laurelE` and `fugacious` (this unit) inside `uv_log`. *Lean:* `Acme.readyA`, `Acme.emanate`, `Aroma`.

Proposition III.11.7 (Ward identity: the flux of the stress against a Killing field is a local field). Let C satisfy the construction record, let f be an extraction order with the cell hypothesis and generic point limits S, and suppose S satisfies the Euclidean axioms with real structure H. Let W satisfy the derivative laws, and assume the shell estimate of III.11.6. Let:

- $\mathcal{J}$ be a stress tensor of S that is conserved with respect to W, and $v \in F_d(S)$;
- U be a smooth Killing field on $\mathbb{R}^4$, homogeneous of degree p ($U(ry) = r^p U(y)$ for $r > 0$);
- $d, D, p \in \mathbb{N}$ and $D + 3 + p = d + 4$.

Then there is $B \in F_D(S)$ with the following property. For every n, all fields $q_1, \dots, q_n$, every configuration of distinct points $x_0, \dots, x_n$, and every window χ adapted to x (smooth, compactly supported, equal to 1 near x_0, with support avoiding $x_1, \dots, x_n$), the flux of $\mathcal{J}$ against $U(\cdot - x_0)$ and χ, inserted with v at x_0 and q_k at x_k, equals the correlation of B at x_0 with the q_k at x_k. The sign convention is

$$\text{flux} = -\sum_{i,j} \int_{\mathbb{R}^4} \partial_i \chi(y)\, U_j(y - x_0)\, \langle \mathcal{J}_{ij}(y)\, v(x_0)\, q_1(x_1) \cdots q_n(x_n) \rangle \, dy.$$

Nonzero homogeneous Killing fields have $p = 0$ (translations, $D = d + 1$) or $p = 1$ (rotations, $D = d$). *Status:* theorem. *Lean:* `annular_fieldA`, `tenet`, `shells_law`.

Further inputs.

- **One-step and root-test estimates for polynomial and coefficient data.** `single` and `asideP` apply the coefficient bound to already packed coefficients across one gauge-door step, with parameters scaled by ηt. Also in this group: `grant`, `y_ready` and `console`.
- **Eventual estimates and uniform choices.** These include bounds at sufficiently large levels, uniform versions on compact sets of distinct-point configurations, and choices of enlarged waiting windows: `pedestals`, `chamber`, `herald`, `scopes`, `on_hold`, `on_demand`, `chamber_now`, `mural_cast`, `earnings`, `lava_v`, `sponsored`, `boot`, `forge`. The weak-completeness theorem in this list enlarges the waiting window. The order of the choices is part of each statement.
- **Polynomial and integral identities.** Wick expansion, Taylor polynomials in link variables, pushforward of densities as sums over sectors, and factorization on disjoint supports; see §4.3.

3.14.2 2. What it uses

Inputs from other units.

- I.1 (lattice, Wilson measure): the periodic torus, plaquette values, the product Haar prior, gauge transformations, the chordal distance on G (`prior`, `chord`).
- II.1 (field labels): the label spaces $\mathcal{L}_d$ with their coordinates and real structure; the free Wick realization of labels, whose kernel is the Maxwell equation ideal $\mathcal{E}_d$; the weave and portfolio data used for the corrected label maps (`equationIdeal`).

- II.2 (polymer activities, Wilson family, construction record, cells): lawful representations and their exponent and gas; the Wilson family and its construction record; the tilings and the chessboard root (`Rep.Lawful`).
- III.2 (choice of constants): the levels and stages ($\hat{\beta}_m$, thresholds, loop caps, clip coefficients, stage index J_m), the pre-L and post-L constants, delays and windows; the order of choice supplies the eventual estimates and waiting windows listed above, including the stage construction `ruese`.
- III.3, III.4 (free covariances, Gaussian integration): the free covariance and its frequency chart, the regulator families with their symbol bounds (used in Lemma 4.10), the window-to-kernel transfer `intended`.
- III.9, III.10 (marginal flow, one nonlinear step): growth of $\hat{\beta}_m$, the RG history ("course") from cutoff N to level k with its exact vacuum identity, the one-step bounds along paths, and the waiting-window engine `Wait.engine`.
- III.13 (common Wilson family): `uv_log`, the theorem that closes the chain and supplies every hypothesis of III.11.4–III.11.6 on the main path.
- IV.1 (extraction): extraction orders and refinements, raw limits, projection inclusion, spans, cleaning to a smooth table, and the answers to the work items D01–D06 (`Base`).
- VI.1, VI.2 (Ward identities, free Maxwell stress): windows, flux, Killing fields, conservation, finite faithfulness of testers (A04), the free lighthouse and its stress tensor, the canonical free Maxwell data, and the production of the current.
- Consumers: VI.1 (`Climb.comparison` also derives Maxwell ranks from the action law), VII.4 (OPE columns), and VIII.1 (the record of local field laws contains the Maxwell ranks; `Org.UV` is built by III.11.6(b)).

Standard mathematics. The chessboard estimate for reflection-positive measures (Fröhlich–Israel–Lieb–Simon, Comm. Math. Phys. 62 (1978); Glimm–Jaffe, *Quantum Physics*, 2nd ed.), together with reflection positivity of the Wilson action (Osterwalder–Seiler, Ann. Phys. 110 (1978)). Axial (tree) gauge fixing on a torus and invariance of Haar measure (Seiler, *Gauge Theories as a Problem of Constructive Quantum Field Theory and Statistical Mechanics*, Lecture Notes in Physics 159 (1982)). Haar volume of small balls in a compact Lie group. Wick (Isserlis) expansion. Schur complements for Gaussian conditioning. Symbol estimates in frequency space. Finite-dimensional duality (a vector lies in the range of a linear map if and only if it is annihilated by every functional that annihilates the range). Ward identities for a conserved symmetric stress tensor and Killing fields of $\mathbb{R}^4$.

Imported constants and assumptions.

- The Haar-budget constants (s, b) exist by `integral_budget` and `exists_budget`. They are fixed once for all stages and volumes.
- Hypotheses carried by the outputs: the projector presentation, the one-step cost package, the centred budget, the frequent-scale price conditions, the binding, the shell budget, Grill and weak completeness. On the main path all are discharged before `uv_log` (III.13), whose only hypotheses are G simple and $b > 0$. The producers in this unit are `reception`/`chaletrail`, `franchise`, `castle`, `laurelE`, `fugacious` and `grilled`.
- The structures in `LedgerGate.lean` (one-step "Estimate" packages: ignition, local output labellings, non-leading terms) are definitions, not theorems. Their docstring calls them "physics inputs, not consequences of the shape/lawful structure". They are conditional one-step interfaces. The restarted comparison constructs principal integration data from

the binding and the one-step estimates; the older interfaces need not all occur in that route. Their complete correspondence with the restarted interfaces remains open in §7.

3.14.3 3. Strategy

The field-space comparison has two parts. Algebraic equations give an upper bound on the dimension of each interacting field space. Short-distance one- and two-point tests give the reverse bound. A conjugation of projectors then identifies the real representations of the signed permutation group.

At each cutoff, the projector kernel is the image of $\mathcal{E}_d$ under an invertible corrected label map. Free probes separate the on-shell sector, so $\dim F_d^{\mathrm{free}} = \dim \mathcal{V}_d - \dim\ker \Pi_{d,N,p}$. The relay estimates carry these equations to the limit, giving the upper bound. Along a sequence of good levels, the corrected test moments approach their free values. Since the free tests separate a basis, the corrected vectors are eventually linearly independent modulo the limit null space. This gives the lower bound.

The two bounds imply $\mathrm{null}_d(T) = \ker \Pi_\infty$ and $\dim F_d(T) = \dim F_d^{\mathrm{free}}$. An invertible operator close to the identity conjugates a cutoff projector to Π_∞. Composing it with the corrected label map identifies the Maxwell ideal with the interacting null space, preserving conjugation and signed permutations.

The quantitative engine is the chessboard estimate. The product of reflected copies of a connected part, integrated against the Wilson measure at cutoff N, is carried by the exact RG history to a coarse level m. There the number of cells equals the number of sites $c_{m,p}$. The flowed observable is bounded pointwise by $r^{-|s|q}\exp(\text{exponent} + qX)$ for a set s of marked modes, and Proposition III.11.1 bounds the resulting ratio of integrals by $e^{c_{m,p}\pi} g^{-4\dim G}$.

The determinant estimate compares two Haar integrals:

- **Numerator.** Extract the factor e^{6c} from the plaquette estimate. A triangular choice of $3c-3$ plaquettes, each owning a bond, gives an integral equal to $(\int e^{-\mathrm{chord}(u)/(g/\sigma_R)}\, dh(u))^{3c-3}$, bounded by $(b(g/\sigma_R)^{\dim G})^{3c-3}$.
- **Vacuum.** Tree gauge leaves $3c+1$ free bonds. Restricting each to a Haar shell of radius $g/4$ gives the lower bound $(sg^{\dim G})^{3c+1}$.

The powers of g cancel up to $g^{-4\dim G} = \beta^{2\dim G}$, which does not depend on the volume. After taking the $c_{m,p}$-th root, its logarithmic contribution is at most 1, because $2\dim G \log\beta \le 8\dim G\,\widehat{\beta}_m \le a_m^{-1} \le c_{m,p}$ eventually. So the root test costs a per-site bound depending only on the stage parameters. The constants are chosen so that the allowance is e^{2I_k} and the error bound is e^{-I_k}, where $I_k = \log\mu_k/32 \ge$ workitem for the one- and two-point tests; the root of every nonempty connected part is then $\le e^{I_k}/e^{2I_k} = e^{-I_k} \to 0$.

All conditions are needed only at **infinitely many** good levels k. A diagonal choice of depths then produces the trial sequence (III.11.4). The covariance splitting identifies the fluctuation law used in the one-step comparison. The regulator estimates supply the frequency windows, while the uniform train constants control their propagation; the tender condition includes the numerical clearance needed for both.

The stress tensor (III.11.6, III.11.7) is built by comparing the limit with a free Maxwell model along a scaling sequence. The Ward identities follow from a flux argument on shrinking annuli of

radii $r_l \to 0$:

- by conservation the flux does not depend on the window;
- after scaling, the flux is an integral over the annulus $r_l/4 < |y - x_0| < 2r_l$;
- a bound on correlations in annuli (Lemma 4.23) shows that combinations of fluxes annihilating all degree-D testers tend to 0;
- finite-dimensional duality and finite faithfulness of the testers then give one local field B representing the flux.

3.14.4 4. Steps

3.14.4.1 4.1 Haar estimates and the determinant budget **Lemma 4.1 (Haar budget).** There are $0 < s \le 1 \le b$ such that $h(\mathrm{shell}(r/4)) \ge s\, r^{\dim G}$ for $0 < r \le 1$, and $\int e^{-\mathrm{chord}(u)/t}\, dh(u) \le b\, t^{\dim G}$ for $t > 0$. *Sketch:* in exponential coordinates near 1, the chord is comparable to the Lie-algebra norm and the Haar density is bounded above and below. Compactness extends the upper volume bound $h(\mathrm{shell}(r)) \le C r^{\dim G}$ to all $r > 0$. Majorize the exponential by $\sum_{k\ge 0} e^{-k} 1_{\mathrm{shell}((k+1)t)}$ and sum $C t^{\dim G} \sum_{k\ge 0} (k+1)^{\dim G} e^{-k}$. *Lean:* `integral_budget`.

Lemma 4.2 (plaquette elimination). Let $n \ge 2$ and let $\varphi : G \to [0,1]$ be continuous. Then $\int \prod_p \varphi(U_p)\, d\mu_{\mathrm{prior}}(U) \le \left(\int \varphi\, dh\right)^{3c-3}$, the product running over all $6c$ plaquettes of Λ_n. *Sketch:* fix the spanning tree in which each site $x \ne 0$ is joined to $x - e_{i(x)}$, where $i(x)$ is the highest axis with $x_i \ne 0$. The $3c+1$ non-tree bonds are grouped into tiers by axis. Within each tier, take the wrap bonds first, except one "handle" per tier, then the transverse bonds from higher to lower height. This gives $3c-3$ swept bonds e, each assigned a plaquette $p(e) \ni e$. Triangularity: if $e \in p(e')$ with $e' \ne e$, then e' has strictly higher priority than e. Integrate the bond of highest priority first. By left and right invariance of Haar measure, $U_{p(e)}$ is Haar-distributed as a function of U_e, and no other selected plaquette contains e. So $\int \varphi(U_{p(e)})\, \Psi\, d\mu = \left(\int \varphi\, dh\right) \int \Psi\, d\mu$. By induction, the product over the selected plaquettes integrates to exactly $\left(\int \varphi\right)^{3c-3}$; the other plaquette factors are bounded by 1. *Lean:* `elimination`, `charged_integral`, `pull_link`, `independent`. *(proof read; the combinatorics of the sweep, i.e. the count and triangularity lemmas, statement only)*

Lemma 4.3 (vacuum lower bound). Under the hypotheses of III.11.1(a), $\int e^R\, d\mu_{\mathrm{prior}} \ge \exp\left(\mathrm{scalar}(R) - c(\mathrm{exCost} + \beta\, \mathrm{baseBudget})\right) (s\, g^{\dim G})^{3c+1}$. *Sketch:* e^R is gauge invariant (`gauge_eval`), so its integral equals the integral over the $3c+1$ free bonds with tree bonds set to 1 (`free_count`). Restrict to the set where every free bond lies in $\mathrm{shell}(g/4)$. Every bond then has chord $\le g/4$, so every plaquette has chord $\le g$. On calm configurations the polymer gas factor is 1, and the exponent is $\ge \mathrm{scalar} - c(\mathrm{exCost} + \beta\, \mathrm{baseBudget})$. Lemma 4.1 with $r = g \le 1$ bounds the Haar volume of the restriction set from below. Positivity of e^R handles the complement. *Lean:* `vacuum_bound`.

Lemma 4.4 (exponent bound and numerator). Under the hypotheses of III.11.1(a), with $E = g/\sigma_R$: for every U, $\mathrm{exponent}_R(U) \le \mathrm{scalar} + c(\mathrm{exCost} + 6) - \sum_p \mathrm{chord}(U_p)/E$. Consequently $z|\int F| \le \exp(\mathrm{scalar} + c(\mathrm{exCost} + 6 + X))\, (b\, E^{\dim G})^{3c-3}$. *Sketch:* for the clipped plaquette action at $\beta = g^{-2}$, one has $\mathrm{chord}/E - 1 \le \beta \cdot (\text{base term})$:

- in the quadratic regime, with $x = \mathrm{chord}/g$, use $\sigma_R x - 1 \le \sigma_R x^2/2$ and $\sigma_R \le_R$;
- above the clipping threshold t, use $g \le t$ and $\sigma_R \le \ell_R$ to obtain $\sigma_R\, \mathrm{chord}/g \le \beta\, \ell_R\, t\, \mathrm{chord}$; the constant term of the linear branch is nonnegative.

The principal part is an average over signed reflections of the pulled-back configuration. The chord sum is invariant under these, and the loop and activity corrections are bounded by c times their per-site prices. This gives the first bound, with $6 =$ the number of plaquette orientations accounting for the -1 per plaquette. Then $z|F| \le e^{\text{scalar}+c(\dots)} \prod_p e^{-\text{chord}(U_p)/E}$, and Lemma 4.2 with $\varphi = e^{-\text{chord}/E}$ together with Lemma 4.1 conclude. *Lean:* `exponent_retains`, `numerator`.

Lemma 4.5 (the ratio; III.11.1(a)). Divide Lemma 4.4 by Lemma 4.3. With $u = 3c - 3$ and $l = 3c + 1 = u + 4$, the exponent is

$$c(2\,\text{exCost} + 6 + \beta\,\text{baseBudget} + X) + u\,\text{ent} - l\log s - 4\dim G\log g.$$

Since $\text{ent} \ge 0$, $\log s \le 0$, $u \le 3c$ and $l \le 4c$, this is at most $c\pi - 4\dim G\log g$. *Lean:* `determinant`.

Lemma 4.6 (per-site work item; III.11.1(b)). Under the assumptions and uniformly over the parameter ranges of III.11.1(b), for all sufficiently large m:

- (i) $\pi \le \text{workitem} - 1$;
- (ii) $-4\dim G\log g/c_{m,p} \le 1$.

Sketch: (i) uses $\text{exCost} \le 1 + (8\,\text{paths} + 1)$, from $\delta \le 1$, $\Lambda t^2 \le 1$ and $\hat\beta_m t^3, \hat\beta_m t^4 \le 1$ eventually; and $\beta\,\text{baseBudget} = 6(\text{direct} + \text{paths}^2)/2$ by $\beta g^2 = 1$. (ii) Since $-4\log g = 2\log\beta \le 2\beta \le 8\hat\beta_m$, and $\hat\beta_m$ grows linearly in $\log a_m^{-1}$, one gets $8\dim G\,\hat\beta_m \le a_m^{-1} \le c_{m,p}$ eventually. *Lean:* `affordable`, `rolls_law`.

Lemma 4.7 (chessboard root). Let T tile the cutoff torus into q cells, where q is the number of sites of the level-m torus. Suppose the exact RG history has terminal density P. Assume that deleting the additive activities of P gives a lawful representation with activity parameter 0, and that the deleted additive term has absolute value at most $q\delta$. Let V be a finite set of modes and let every component Φ_s be continuous. Assume $r > 0$, $e \ge 0$, $g > 0$, $g \le \text{threshold}(P)$ and $\beta(P)g^2 = 1$. Suppose the local activity bound for the flowed stamped polynomial, at radius r and exponents (e, l, L), gives

$$r^{q\,\text{card}\,s}|F_s(U)| \le \exp\left(\text{exponent}_P(U) + q(L + \text{card}\,Vl)\right)$$

for every terminal configuration U and every $s \subseteq V$. Here F_s is the flow of the unnormalized Wilson density multiplied by the product of the reflected copies of Φ_s. Then

$$\rho_T(\Phi_s) \le \frac{\exp\left(\pi(P, L + \text{card}\,Vl, g) - 4\dim G\log g/q\right)}{r^{\text{card}\,s}},$$

where ρ_T is the absolute value of the stamped Wilson expectation to the power $1/q$.

For the station at level k, assume allocation, per-site affordability at the width level, $\text{card}\,V \le M$, $l, L \le 1$, and $r = e^{2I}$ with $I \ge \text{workitem}(O, k, M)$. Then, for every nonempty s, $\rho_T(\Phi_s) \le e^{-I}$. *Sketch:* the exact vacuum identity supplies the Wilson partition function as denominator. Apply III.11.1(a) to the representation with its additive activities deleted. Comparing both numerator and denominator costs $e^{q\delta}$ twice, supplying the 2δ already included in $2\,\text{exCost}$ in π. Take the q-th root. For the last assertion, allocation keeps the stage constant between width and k; $L + \text{card}\,Vl \le 1 + M$, and $r^{\text{card}\,s} \ge r$. *Lean:* `s_postage`, `denture`, `coupon`. *(proof read; the flow identity and the cell-stamp bookkeeping, statement only)*

3.14.4.2 4.2 Free covariances Lemma 4.8 (III.11.2). *Sketch:* expand the pair term as $\sum_{a,b} \operatorname{Re}\Gamma_{ab}\,(r^{\Delta_L}\mathrm{ev}_L(r^{-\mathrm{gen}\,a}t_a))(r^{\Delta_S}\mathrm{ev}_S(r^{-\mathrm{gen}\,b}t_b))$. Keys of generation below the scaling degree of the stencil evaluate to 0. Keys of generation equal to the scaling degree give exactly the free curvature kernel (`deck`). Every other term carries an extra factor r^{-1} or smaller. So r times the remainder is at most $D\sum|\mathrm{ev}_L(t_a)||\mathrm{ev}_S(t_b)|$, summed over the non-leading pairs. *Lean:* `fence`. Its eventual form is the pair penalty `Prayer.penalty`.

Lemma 4.9 (measured Gaussian data). Let Q be the symmetric covariance array and m the mean vector obtained by applying the normalized stencil and remainder functionals to measured pair and one-point data. Let F be the free-current covariance on the stencil keys, extended by zero on the remainder indices. Suppose $B \geq 1$, $T, P, \varepsilon \geq 0$, and, entrywise,

$$|m_i| \leq PT, \qquad |Q_{ij} - F_{ij}| \leq PT^2.$$

On the stencil keys assume the sharper bounds $|m_i| \leq \varepsilon T$ and $|Q_{ij} - F_{ij}| \leq \varepsilon T^2$. If

$$PT \leq B, \qquad |F_{ij}| + PT^2 \leq B, \qquad \varepsilon T \leq e, \qquad \varepsilon T^2 \leq e,$$

then all measured means and covariances are bounded by B, and their restriction to the stencil keys differs from $(0, F)$ by at most e in both means and covariances. *Sketch:* use the triangle inequality for Q. In the application, the measured horizons and bridges satisfy the discrete Bianchi identities and scaled finite-difference bounds. The stencil normalization T bounds the sampling coefficients and the remainder factor M^{d-1} for addresses within radius M. These give the displayed broad and sharp bounds. Positivity of Q is not required. *Lean:* `resolution`.

Lemma 4.10 (III.11.3). *Sketch:* write $\lambda = 2\pi$ and let $T(\zeta)$ be exterior multiplication by $i\zeta$ from one-forms to two-forms. In the chart $\zeta = \lambda\xi$, the heat symbol is

$$e^{-|\zeta|^2}|\zeta|^{-2}(T(\zeta)T(\zeta)^*)_{a,b},$$

with the color Kronecker factor. The regulator estimates in III.3 bound the chart as a symbol of order 0 and its difference from this heat symbol as a symbol of order 1. The chain rule under $\xi \mapsto \lambda\xi$ preserves these orders and changes the constants by fixed powers of λ. Sum the constants over the finitely many plaquette pairs to obtain a bound independent of G and the colors. *Lean:* `window_prices`, `heat_window`.

Lemma 4.11 (covariance splitting and propagation). (i) Let an atlas be compatible with the chosen schedule and its retained unmasked quadratic forms, and assume the atlas symmetry, coercivity, inverse and crossing laws. Let $i < N$, choose a tree forest for the step, and assume the step torus satisfies the size condition for tree inversion. Then

$$\mathcal{C}_{i+1} - W_i\mathcal{C}_iW_i^{\mathsf{T}} = d_f\Gamma_i d_f^{\mathsf{T}}, \qquad \Gamma_i \succeq 0,$$

where $\mathcal{C}_i$ is the complete curvature covariance, W_i the passage map, d_f the fine exterior derivative, and Γ_i the uncut tree covariance, extended by zero on tree bonds. (ii) Fix kernel choices and stage indices $j, t \geq 4$. There is $R \geq 1$ such that, for every G, every schedule with these kernel choices, and every N, i, p, d satisfying $J_{i+1} = j$, $J_i = t$ and $d - 1 \leq j - 2$ (natural-number subtraction), the difference-motion propagation bound holds with constant R. *Sketch:* (i) is a Schur-complement identity on the curl image; the restricted tree inverse gives positivity. For (ii), use the uniform resolvent estimate for the two stage indices and sum its six plaquette components.

The table `prepared` chooses, for each kernel, the uniform window, covariance and propagation constants; their required smallness is a separate clearance condition. *Lean:* `one_fold`, `train_issue`.

Lemma 4.12 (Wick expansion). Let A be a finite index set, let $\mathrm{loc} : A \to S$ take values in an ordered location set, and let Z_a be formal Gaussian generators with zero mean and symmetric covariance Q. Assume $Q(Z_a, Z_b) = 0$ whenever $\mathrm{loc}(a) = \mathrm{loc}(b)$. For every $p \subseteq A$, the formal Wick moment of $\prod_{a \in p} Z_a$ is the sum, over perfect matchings of p joining different locations, of the products of their covariances. No positivity assumption is imposed on Q. *Sketch:* remove one index and use the Gaussian moment recursion. The zero mean removes the singleton term, and the covariance hypothesis removes pairs at a common location. The remaining choices are in bijection with the admissible matchings. *Lean:* `headed`.

3.14.4.3 4.3 Polynomial calculus and identities

Lemma 4.13 (polynomial and integral calculus). Write $\|P\|_1$ for the sum of the absolute values of the coefficients of a polynomial P.

- (i) Fix derivative rules for G and an order d. There is $C \geq 0$ such that, whenever a local observable A is regular for a parameter set of order at least $d+1$ and has the prescribed derivative bounds c, its pullback $A \circ \mathcal{N}$ to a link-coordinate chart has a Taylor polynomial P of degree at most d satisfying $\|P\|_1 \leq c(\mathrm{card\,links} + 1)^d C$ and $|A(\mathcal{N}(x)) - P(x)| \leq c\|x\|^{d+1}$ for $\|x\|$ within the fixed chart radius. `minted`
- (ii) For a family A_i satisfying (i) with bounds c_i, and coefficients f_i such that $S = \sum_i |f_i| c_i < \infty$, there is a degree-$d$ polynomial P with $\|P\|_1 \leq S(\mathrm{card\,links} + 1)^d C$ and remainder at most $S\|x\|^{d+1}$. `till`
- (iii) Fix real normed spaces E, V, with V a normed algebra, and $D \in \mathbb{N}$. There are $C \geq 0$ and $q \in \mathbb{N}$ such that, for every finite list $f_1, \ldots, f_l : E \to V$, point x, and $B \geq 1$, if each f_i is C^D at x, $f_i(x) = 1$, and $\|\partial^j f_i(x)\| \leq B$ for $j \leq D$, then $\|\partial^j \prod_i f_i(x)\| \leq C((1+l)B)^q$ for $j \leq D$. `conquer`
- (iv) For one- or two-insertion polynomial data in the staged comparison, assume the common-mask condition, the comparison bounds on means and covariances, $0 < \varepsilon \leq 1$, coefficient bound w, degree bound D, the prescribed variable supports, and vanishing of inadmissible pair terms. The difference between the resulting connected component and its free polynomial comparison is at most $t + z + (u+t)t + tu$, where u, t, z are respectively the polynomial size, one-factor error and two-factor error bounds of that comparison. `console`
- (v) For a bounded measurable gauge-invariant density g, a branching step and an allocation satisfying its chart and integrability conditions, the pushforward is exactly the sum over matching-flag sectors and tree/tag choices of their weighted conditional integrals. If these integrands have a common factor $\Lambda(U)$ and a finite local decomposition with the required integrability, the result is $\Lambda(U)$ times the sum of products over pairwise disjoint compatible local components. `arriving`, `rendered`
- (vi) Fix a valid coarse configuration and a tree/tag choice. If finitely many integrable observables satisfy the locality and compatibility conditions of the conditional fluctuation law, and their coarse supports are pairwise disjoint, their product is integrable and its conditional integral equals the product of their conditional integrals. `peer_beams`

Sketch: Taylor's theorem gives (i), and absolute coefficient summability gives (ii). For (iii), split each factor as $1 + (f_i - 1)$ and induct on the derivative order; the highest derivative terms add rather than multiply. For (iv), the two-point cumulant is the second moment minus the product of the means; expand the difference and use the three stated bounds. For (v), apply the allocation

identity before discarding sectors with the wrong flags, then group the local components. For (vi), the joint local-coordinate marginal is a product measure, so Fubini gives factorization. *(proof read; local comparison and allocation inputs statement only)*

3.14.4.4 4.4 From frequent good scales to trial systems (III.11.4) Lemma 4.14 (multi-step propagation). Fix a level $k \le N$, a band b, a fitted configuration x with v insertions, and an exact RG history. Let m_0 and m_1 be its width and middle levels. Assume allocation at k, admissible representations along $[m_0, m_1]$, a numerical propagation bound for arity M, and

$$\hat{\beta}_u \in [\hat{\beta}_k/2, \hat{\beta}_k] \quad (m_0 \le u \le m_1), \qquad v \le M.$$

Assume the instance decomposition for the tested labels: at the middle level, the marked sources approximate the negative free-profile terms within the prescribed error, with the required background and source bounds, and the subsequent steps satisfy the exhibited bounds. Finally assume each free-profile term has absolute value at most $\exp((\log\log\hat{\beta}_k)^2/4)$.

Then the flow of the corrected stamped observable has a local activity decomposition at m_0 with radius allowance(M, k) and exponents $(e, 1, 1)$, where e is the input source exponent. *Sketch:* use the middle-level decomposition and the free-profile bound to initialize the source estimate. The propagation bound controls its iteration through $[m_0, m_1]$; the coupling window keeps every intermediate estimate in the same numerical range. The correction cancels the leading free-profile terms. This is the local decomposition used in Lemma 4.7. *Lean:* `v_conditioned`. *(proof read; helper estimates statement only)*

Lemma 4.15 (the chain). Under the hypotheses of III.11.4:

1. At infinitely many k, the simultaneous price and insertion conditions give covariance data, an exact RG history and compatible restarted representations, eventually in N and for every p. The uniform covariance and test-point estimates supply the local Gaussian comparison. (`cast_off` via `backstage`.)
2. Construct a common correction map and polynomial approximations for all rows. The one-step comparison supplies the coefficient, support and remainder bounds on the same path of representations. (`v_paid_factory`, *proof read; analytic inputs statement only.*)
3. Compare the resulting Gaussian polynomial moments with the free profiles. The eventual finite-difference bounds transfer this comparison from the chosen representatives to the actual corrected columns, giving the instance estimates of Lemma 4.14. (`v_culminates`, *proof read.*)
4. Apply Lemma 4.14 to propagate the corrected estimates to the width level, retaining the terminal density identity and admissibility needed for the root estimate. (`v_pay_all`, *proof read.*)
5. Offers give a trial system. Choose depths k_i inductively among the good levels, with $k_{i+1} > k_i$, strictly increasing widths, and strictly decreasing test scales tending to 0. The lifts come from `ticketed` (bounded, real, with first entry the reference unit). The chessboard bounds come from Lemma 4.7, using the per-site affordability `banking`, which follows from Lemma 4.6. (`v_obtain_purchases`, *proof read.*)

Lean: `wheeled` composes items 2–5.

Lemma 4.16 (eventual insertion estimates). Assume G is simple, C satisfies the construction record, and the one-step cost package uses the prescribed weight sequence and multiplicities

at least 1. Fix a degree-d column family X and stage package whose mark exponent satisfies $c \geq 2(w(G,d)+1)$, where $w(G,d) = w_d$ is the prescribed coefficient exponent. Then, for all sufficiently large k, metering at k implies the lantern-mark condition for X at k. *Sketch:* write $x = \widehat{\beta}_k$ and $b_a(x) = \exp(a(\log\log x)^2)$. For every fixed a, the inequality $b_a(x)s \leq b_{-8}(x)$ with $s \geq 0$ gives

$$s \leq \exp(-(8+a)(\log\log x)^2) \leq x$$

for sufficiently large x. Hence $x^{-c/2}s \leq x^{-w(G,d)}$. The one-step estimate and spacing of the rounded test points turn this coefficient bound into the required insertion estimate. The threshold is uniform over the tested coefficients, cutoffs and periods. *Lean:* `bursts`, `quaking`. *(proof read; analytic inputs statement only)*

Lemma 4.17 (discharge of the frequent-scale conditions). Given compatible pre-L price data and one-step bounds for a sheet, with the stage thresholds below the chosen block lengths and each degree index eventually present in the stage index sets, a Haar budget, nonnegative mark exponents and simple G, there are panels Q_d and a waiting window extending any given one with the following property. For every Wilson family C paid on that window and every binding with these prices, for every d, the four price conditions hold at infinitely many levels. *Sketch:* the restarted comparison and the scale-selection construction produce the panels and the window; frequency follows because each requested degree index is eventually present in those stage sets. *Lean:* `franchise`. *(proof structure read; helpers statement only)*

3.14.4.5 4.5 The rank comparison (III.11.5) Lemma 4.18 (equation count). Under the projector presentation of III.11.5(i), at every cutoff, $\dim \mathcal{L}_d \leq \dim\ker \Pi_{d,N,p} + \dim F_d^{\mathrm{free}}$. *Sketch:* $\ker \Pi_{d,N,p}$ is the image of $\mathcal{E}_d$ under an injective corrected label map, by the projector presentation. The free null space is the image of $\mathcal{E}_d$ under the label coordinates. This uses `projectionExact` (a label is null in the free table if and only if its on-shell image lies in the residual) and `no_residual` (the residual is 0: free probes separate the on-shell sector). Then $\dim F_d^{\mathrm{free}} + \dim \mathcal{E}_d = \dim \mathcal{L}_d$. *Lean:* `savings`, `EquationCount`, `projectionExact`, `no_residual`. *(proof read; free positivity input statement only)*

Lemma 4.19 (short-distance comparison and independence). Assume C satisfies the construction record, f gives generic point limits S, and the correlation kernels of S are continuous on generic configurations and integrable against every separated test function. Let Z satisfy the purchase conditions, and suppose all its coefficient vectors have iterated limits along f. Then:

- (i) for every row and every basis column, the correlation of its limiting corrected vectors at the trial's physical points differs from the free correlation at the rescaled points by at most $P_k\gamma_k$, where P_k is the polynomial coefficient weight of the free profile. The numbers P_k converge and $\gamma_k \to 0$, so these correlations converge to their prescribed free test values;
- (ii) for all sufficiently large k, the limiting vectors of trial k are linearly independent modulo the subspace of labels annihilated by every integrated correlation test of S.

Sketch: expand the ordinary moment using the relative connected components and apply the chessboard inequality to every nonempty component. This gives the error $P_k\gamma_k$ before the cutoff limits, and the bound passes through those limits. Each row functional descends to the quotient by the integrated-test null space. The resulting finite matrix converges to the injective free detector matrix. Injectivity is open in finite dimensions, giving (ii). *Lean:* `measured`; the comparison is `Purchases.samples/converge` in `Bought.lean`. *(proof read; chessboard input statement only)*

Lemma 4.20 (seed and null space). Assume C satisfies the construction record. Under a centred port for C along f, raw limits S along that same extraction, relay bins with convergent projectors, the equation count for every d, and trial systems with the purchase conditions for every d, there is a refinement along which the following hold:

- the arrival and spans records;
- a cleaned smooth table T;
- the seed record for T;
- for every d: $\operatorname{null}_d(T) = \ker \Pi_\infty$, where Π_∞ is the limit projector, and $\dim F_d(T) = \dim F_d^{\mathrm{free}}$.

Sketch: the relay estimates give $\ker \Pi_\infty \subseteq \operatorname{null}_d(T)$. Lemma 4.19(ii) gives the reverse dimension inequality, using the equation count and eventual stability of the projector kernel dimension. Hence the subspaces coincide, and the field dimensions agree. *Lean:* `arriving_seed`, `equation_retains`. *(proof read; analytic inputs statement only)*

Lemma 4.21 (equivariant identification). Under the projector presentation of III.11.5(i) and the extraction record for its projectors and cleaned table T, there are the real, signed-permutation-equivariant isomorphisms Φ_d of III.11.5. The free table has its canonical real structure and zero residual null space. *Sketch:* let $A_i \to \Pi_\infty$ be the projectors along the extraction, and set $Y_i = \Pi_\infty A_i + (1 - \Pi_\infty)(1 - A_i) \to 1$. For large i, Y_i is invertible and $Y_i A_i = \Pi_\infty Y_i$. Since A_i and Π_∞ are real and equivariant, so is Y_i. Then $L = Y_i \circ$ (corrected label map) maps $\mathcal{E}_d$ onto $\ker \Pi_\infty = \operatorname{null}_d(T)$ (`drumroll`). The inverse of the induced quotient map, followed by the fixed free coordinate identification, is Φ_d. *Lean:* `thrush`, `drumroll`.

Lemma 4.22 (III.11.5). Under the hypotheses of III.11.5, its conclusion follows by the following steps:

1. extract raw point limits (IV.1) and include the label projections;
2. build, for each d, the free detector columns X_d, which form a basis of F_d^{free};
3. combine assumption (iii) with Lemma 4.16 and apply III.11.4 to get trial systems with the purchase conditions;
4. apply Lemma 4.20 with Lemma 4.18;
5. assemble the extraction record (using D05);
6. conclude with Lemma 4.21.

Lean: `rated`.

3.14.4.6 4.6 The stress tensor (III.11.6, III.11.7) Lemma 4.23 (annihilated flux combinations). Under the hypotheses of III.11.7, let I be a finite family of generic tester configurations on the schedule grids, each with $x_0 = 0$, and let (g_a) be coefficients with $\sum_a g_a \operatorname{test}_a(z) = 0$ for all $z \in F_D$. Here test_a inserts z at the root with the remaining fields of tester a; flux_a inserts the flux built from $(\mathcal{J}, U, v)$ there. Then $\sum_a g_a \operatorname{flux}_a = 0$. *Sketch:* let $r_l \to 0$ be the radii of the shell budget. Choose a smooth radial bump χ, equal to 1 on the ball of radius $1/2$ and supported in the unit ball. Conservation makes the flux independent of the window, so for large l use $\chi_l(y) = \chi(y/r_l)$. By homogeneity of U, the flux becomes $-\sum_{ij} \int M_{ij}(y)\, r_l^{3+p} \langle \mathcal{J}_{ij}(r_l y) \cdots \rangle \, dy$, where $M_{ij} = \partial_i \chi \cdot U_j$ is integrable and supported in $\frac{1}{2} \le |y| \le 1$. The annulus bound `zone` says that, for annihilating combinations, $\sup_{|y| \in (r_l/4, 2r_l)} \left| \sum_a g_a\, r_l^{3+p} \langle \mathcal{J}(y) \cdots \rangle_a \right| \to 0$. So the combination is a constant sequence tending to 0. *Lean:* `tenet`, `zone`. *(proof read; annulus comparison inputs statement only)*

Lemma 4.24 (III.11.7). *Sketch:*

- for each finite set of tester configurations, Lemma 4.23 and finite-dimensional duality (`obtainable`) give B matching all their fluxes;
- fix a finite separating family of testers (A04), choose B matching it, and compare with the field matching its union with any additional tester; separation forces the two fields to agree;
- translation covariance of flux and correlations moves any grid configuration to $x_0 = 0$;
- continuity of the flux and of the limit kernels, plus density of grid configurations, extends the identity to all distinct points.

Lean: `annular_fieldA`, `obtainable`.

Lemma 4.25 (stress laws). Under the hypotheses of III.11.7, there are linear translation maps $F_d(S) \to F_{d+1}(S)$ and rotation maps $F_d(S) \to F_d(S)$ satisfying the respective flux identities. The rotation maps also depend real-linearly on the skew matrix. *Sketch:* apply Lemma 4.24 with constant fields $U = e_i$ ($p = 0$, $D = d + 1$) and with the rotation fields $U(y) = ky$, k skew ($p = 1$, $D = d$). Linearity follows from linearity of the flux and separation of fields by correlations. *Lean:* `shells_law`.

Lemma 4.26 (III.11.6). Under the respective hypotheses of III.11.6(a) and (b), the stated stress and ultraviolet records exist. *Sketch of (a):*

1. D06 gives the orientation law.
2. The station equivalences identify the field spaces of S with those of the free model; this uses the hypothesis `DimAbove`.
3. `produce_current` (VI.2) gives $(\mathcal{J}, X)$ with $\mathcal{J} \neq 0$, the approach, stability, the meter and zero current divergence.
4. `linear_cons` converts zero current divergence to conservation. The weak comparison along the scaling sequence supplies the action law.
5. Lemma 4.25 gives the stress laws.

Sketch of (b): take free Maxwell data for simple G and replace them by the canonical data (`FreeStress.fire`). Build the station hypothesis from the seed and weak completeness, and the hypothesis `DimAbove` from the corrected rank widths. Then apply (a). *Lean:* `readyA`, `emanate`. *(proof read; helper estimates statement only)*

3.14.5 5. Where the difficulty lies

- **The volume-uniform root test.** A per-site loss of $\log \beta$ in the chessboard root would destroy the comparison, because the budget must tend to 0 while $\beta \to \infty$. The tree-gauge count of III.11.1 avoids this: it costs only $\beta^{2 \dim G}$ in total, not per site. The price for this is a crude, but uniform, per-site constant. This estimate uses Haar integration rather than a Hessian determinant.
- **The direction of the rank argument.** The upper bound (equation count) is algebraic and holds at every cutoff. The lower bound needs the short-distance comparison, at infinitely many scales only, through trial sequences. The equivariant isomorphism comes from conjugating projectors, not from an explicit basis. The conclusion identifies complex representations with compatible conjugations under signed permutations.
- **Frequent rather than eventual scales.** Every price condition is required only at infinitely many levels; a diagonal extraction then selects the trials. This is weaker than an "eventually

in k" statement, and it is all the rank argument needs.

- **Uniformity.** The constants must be chosen before the block lengths and must control every period. Eventual estimates cannot replace bounds on a specified comparison window without the corresponding delay.
- **Connections.** The stress-flux argument supplies the local charge fields of VI.1. The equation-ideal calculation uses the free label algebra of II.1.

3.14.6 6. Notation

Written proof	Lean (glossary)	Meaning
schedule factor μ_k	`monitor`	$\exp((\log\log\widehat{\beta}_k)^2)$
coefficient exponent w_d	`winter`	degree of the input coefficient family plus 4
Haar budget (s, b)	`HaarBudget` (`small`, `big`)	lower constant for shells, upper constant for $\int e^{-\mathrm{chord}/t}$
σ_R, ent	`strength`, `ent`	$\min(1, \text{linear clip})$; $\log b - \dim G \log\sigma$
per-site price π, work item	`protection`, `work item`	III.11.1
admissible at level m	`Meets`	conditions on (P, g, Λ, δ) in III.11.1(b)
chessboard root ρ_T	`Cells.Tiling.rootTest`	$\lvert E\prod_{\text{cells}}\text{copy}\rvert^{1/\text{card cells}}$
free curvature kernel	`light`	derivative of the free Maxwell curvature two-point kernel between colored stencils
scale-r pair term	`collect`, `arc`, `arbor`, `Beamer.amount`	$\sum \Gamma_{ab}\, r^{-\mathrm{gen}} t_a \otimes r^{-\mathrm{gen}} t_b$; $r^{2+\mathrm{ord}}$
Fourier chart rate	`circleRate`	2π; heat parameter $(2\pi)^2$
window estimate	`Window`, `Cast`	symbol bounds of III.11.3
stage package, metering	`Cohesion`, `Metering`	per-degree stage data; charge and launch at k
price conditions	`Tender`, `Metering`, `ShowHours`, `Vents`, `LanternMarks`	hypotheses of III.11.4 at a level k
provisions, aggregate, turning, offers	`s_provisions`, `s_empire`, `Turning_`, `Offers_`	successive frequent-scale conditions of Lemma 4.15
trial system, purchase conditions	`Columns.Trials`, `Purchases`, `Travels_`	III.11.4
compatible quadratic atlas	`Binding`	shape, coercivity, size and path compatibility on the same Wilson family
launch conditions	`Launch`	local comparison and propagation conditions at a specified level
projector presentation Π	`Scotia`	projectors whose kernels are corrected images of $\mathcal{E}_d$
one-step cost package	`Scrim`	step costs along gauge doors

Written proof	Lean (glossary)	Meaning
Maxwell equation ideal $\mathcal{E}_d$, F_d^{free}	`Maxwell.equationIdeal`, `Maxwell.freeTable`	labels vanishing on the free field; free field spaces
Maxwell ranks	`Maxwell.Ranks`	equivariant isomorphisms $F_d(T) \cong F_d^{\mathrm{free}}$
numerical upper dimension bound	`DimAbove`	$\dim F_d(S) \le \dim F_d^{\mathrm{free}}$
equation count	`EquationCount`	Lemma 4.18
seed record	`Spec.Seed`	cutoff, limits, axioms, cell hypothesis, corrected rank widths
shell budget	`Aroma`, `Economics`	shell estimates, open tabs, radii $r_l \to 0$
flux, window	`flux J χ U x q v`, `Window` (VI.1)	flux of the stress against a vector field
arbitrarily fine comparison systems	`Stations`	for every order, size and positive tolerances, an admissible station at an arbitrarily small scale
cofinal grid approximation	`Grill`	approximation systems at cofinally many grid levels
weak completeness	`CompleteWeak`	comparison systems for every fixed arity, degree and positive size bound
free lighthouse	`Lighthouse`, `MaxwellData`	free Maxwell comparison model

3.14.7 7. What remains unexplained

1. **Analytic inputs to the annulus estimate.** The reduction in `zone` and the flux argument are checked. The discrete stencil bound, interpolation estimate and shell estimates invoked there have not been independently checked through their analytic dependencies.
2. **Detailed one-step interfaces.** The principal-data construction and the proof assembling the aggregate comparison were checked at their application points. The underlying nonlinear integration and remainder estimates were not checked in full. The relation between the older conditional interfaces in `LedgerGate.lean` and every restarted replacement remains to be documented; their shared names do not identify their hypotheses.
3. **Record definitions.** The stage, launch, hours, vents, relay, station and shell hypotheses in the numbered statements refer to the linked records. Their complete translation into ordinary mathematical estimates remains to be written. Lemmas 4.11, 4.13(iv)–(vi), 4.14 and 4.17 retain these structural hypotheses; the displayed inequalities alone are not their full assumptions.
4. **Auxiliary results and upstream estimates.** The additional estimates listed after III.11.7 were checked at statement level, including their parameter order. The sweep triangularity in Lemma 4.2 was checked at statement level. The proofs of the lower-level producers of the centred estimate and one-step package, and of the schedule-existence theorem in III.10, were

not checked in full. The source links identify these dependencies; the present outline does not independently establish them.

3.15 III.12. Bounds uniform in the volume and convergence as the spacing tends to zero

Part: III, Ultraviolet construction (renormalization group). **Lean:** chapter 30, `YM/FullVolume/`. **Size:** 18,973 lines in the main-theorem closure.

Scope. This unit proves volume-uniform bounds and takes the ultraviolet limit at each fixed volume. It has three outputs:

- (a) Bounds on lattice correlation functions of composite observables and of their finite differences at separated points. The bounds are uniform in the volume and eventually uniform in the cutoff.
- (b) Convergence of these correlation functions as $N \to \infty$ at each fixed volume, with smooth limits.
- (c) Power-series bounds on their dependence on the renormalized coupling. The assembly uses these.

Notation.

- N is the mesh index. The lattice spacing is $a_N = 1/D_N$, where $D_N = \ell_0 \cdots \ell_{N-1}$ is a product of odd block sizes.
- p is the volume index. $\Lambda_{N,p}$ is the torus of physical side L_p, with $L_p D_N$ sites per axis. The block size ℓ_j and the physical period L_p are different quantities.
- m, j, k are renormalization-group levels. $\beta_{N,p}$ is the bare inverse coupling, $\mathbb{E}_{N,p}$ the Wilson expectation, and $\hat{\beta}_j$ the renormalized inverse coupling at level j.
- Lab_d is the finite-dimensional space of labels of degree d. $O_{N,p}(v; o)$ is the lattice representative of $v \in \mathrm{Lab}_d$ at the site o (II.1).
- $\mathcal{N}_d \subset \mathrm{Lab}_d$ is the equation ideal of the free Maxwell field, and $F_d = \mathrm{Lab}_d / \mathcal{N}_d$.
- *Eventually in the mesh* means: there is N_0 such that the statement holds for all $N \geq N_0$ and all p (the filter `mesh`). In statements of the form "for all large m, eventually in the mesh," N_0 may depend on m.
- A sequence $w_{N,p}$ in a normed space has *sub-power growth* if for every $\varepsilon > 0$, eventually in the mesh, $\|w_{N,p}\| \leq a_N^{-\varepsilon}$ (`Sluggish`). The quantity `flight` (ε) equals $a_N^{-\varepsilon}$, because `scale` $(N) = \log a_N^{-1}$.

3.15.1 1. What this unit proves

Definitions.

- A *trial* F is a family of complex functions $F_{N,p}(o)$ of the gauge field on $\Lambda_{N,p}$, indexed by a site o. It is *local* if, eventually in the mesh, it is translation covariant and each $F_{N,p}(o)$ is gauge invariant, measurable, bounded and depending only on links within $o + X$, for one fixed finite $X \subset \mathbb{Z}^4$ (`Local`, `Stencil`).
- *Chessboard root at scale m.* For $m \leq N$, $T_m(o)$ is the tiling of $\Lambda_{N,p}$ by cubes of physical side a_m with one cell centred at o (`center`). Let G be bounded, measurable and gauge invariant, localized in the central cell. Then $R_m(G) = \left|\mathbb{E}_{N,p} \prod_c G^{(c)}\right|^{1/\sharp\text{cells}}$. Here $G^{(c)}$ is obtained by

reflecting G into the cell c through the cell walls, and is complex conjugated after an odd number of reflections. The number of cells is $\sharp\text{cells} = (L_p/a_m)^4$ (`rootTest`).

- F is *affordable at cost* $u(m, N, p)$ if F is local and the following holds for all large m, eventually in the mesh: $F_{N,p}(0)$ is localized in the central cell of $T_m(0)$ and $R_m(F_{N,p}(0)) \le u(m, N, p)$.
  - F is *owed at cost* $v(N, p)$ if it is affordable at a cost independent of m.
  - F is *funded* if it is affordable at a cost $u(m)$ independent of (N, p).

  Lean: `Affords`, `Owed`, `Funded`.
- *Normalized differences:* $(\Delta_\mu F)_{N,p}(o) = a_N^{-1}(F_{N,p}(o + e_\mu) - F_{N,p}(o))$, and $\Delta_l = \Delta_{\mu_1} \cdots \Delta_{\mu_k}$ for a list $l = (\mu_1, \dots, \mu_k)$ of axes (`farther`, `hops`).
- A *coin* of degree d is a sequence $x = (x_{N,p})$ in Lab_d whose inverse terminal coordinates have sub-power growth (`Coin`). Thus, if $\tau_{d;N,p}$ is the terminal coordinate isomorphism, $\|\tau_{d;N,p}^{-1} x_{N,p}\| \le a_N^{-\varepsilon}$ eventually for every $\varepsilon > 0$. Under the consistency conditions on the family, bounded label sequences are coins. $O(x)$ denotes the trial $O_{N,p}(x_{N,p}; \cdot)$.
- A *presentation* is, for each d and each (N, p), an idempotent linear map $\pi_{N,p}$ on Lab_d (`Presentation`, IV.1). The kernel condition is an additional hypothesis.
- For a real configuration x, write $\alpha_N(x)_{i\mu} = \lceil x_{i\mu}/a_N \rceil$ for its coordinatewise ceiling address (`address`). This is the right endpoint convention. Interpolation means multilinear interpolation on the corresponding cubes (`interpolated`).

Proposition III.12.1 (kernel bounds uniform in the volume). Let R be a run (§2) for a simple gauge group G, with Wilson family C. Fix n, labels $v_i \in \mathrm{Lab}_{d_i}$ $(1 \le i \le n)$, a finite list λ of pairs (slot i, axis μ), and $r > 0$. Then there are $B \ge 0$ and N_0 with the following property. For all $N \ge N_0$, p and integer configurations x satisfying $a_N \operatorname{dist}(x_i, x_j) \ge r$ for $i \ne j$,

$$\left|\Delta_\lambda \, \mathbb{E}_{N,p}[O_{N,p}(v_1; x_1) \cdots O_{N,p}(v_n; x_n)]\right| \le B.$$

Here $\Delta_{(i,\mu)}$ is the normalized forward difference in x_i along e_μ, and dist is the periodic ℓ^∞ lattice distance. N_0 and B may depend on $R, n, (v_i), \lambda, r$, but not on p. *Status:* theorem, for every run; the run supplies all inputs (§2). *Lean:* `FixedBounds` (statement), `gKernel` (the differenced kernel), `genuine` (theorem), `funding` (reduction to III.12.2).

Proposition III.12.2 (derivative closure of the chessboard budget). Let R be a run for a simple G. Let $w = (w_{N,p})$ be a sequence in Lab_d with $\|w_{N,p}\| \le K$ eventually in the mesh, for some $K \ge 0$, and let l be a list of axes. Then $\Delta_l O(w)$ is funded: for some $u(m)$ and all large m, eventually in the mesh it is localized in the central cell and $R_m(\Delta_l O(w)_{N,p}(0)) \le u(m)$. *Status:* theorem. *Lean:* `Run.financed`.

Proposition III.12.3 (closure lemma). Fix a degree d. Let C be a Wilson family satisfying its consistency conditions and the following hypotheses:

1. the centred budget (§2);
2. the scale-selection data of §2;
3. sound section data V_e and linear automorphisms W_e of the abstract labels, for every degree e, satisfying the section, symmetry and exact local identities described in §2;
4. a presentation π and the full section-comparison data of §2, including the quantitative comparison with V_e, identification of $\ker \pi_{N,p}$ with a dressed equation ideal, and a bound $\sup_{N,p} \|\pi_{N,p}\| < \infty$ for each degree;
5. degree-d detector rows satisfying the placement and bounded-lift conditions of Lemma 4.6 and the short-distance comparison of Lemma 4.8 (`VitalRank`).

Let F be a funded trial whose values $F_{N,p}(o)$ are continuous functions of the gauge field, and let l be a list of axes. Let x be a coin of degree d such that $O(x) - \Delta_l F$ is owed at a cost $u_{N,p} \to 0$ in the mesh. Then $\Delta_l F$ is funded. *Status:* theorem under the stated hypotheses. For a run, these follow from its ties, with the detector comparison obtained from its short-distance estimates (§2). *Lean:* `improved`, `cap`.

Proposition III.12.4 (uniform rank detection). Assume hypotheses 1–5 of III.12.3 and fix a list l of axes. For all large k and all sufficiently small $b > 0$ there is $Z \geq 0$ such that, eventually in the mesh,

$$\|\pi_{N,p} x\| \leq Z \big\| \mathcal{P}_{k,b,l;N,p}(\pi_{N,p} x) \big\| \qquad \text{for all } x \in \mathrm{Lab}_d.$$

Here $\mathcal{P}_{k,b,l;N,p} : \mathrm{Lab}_d \to \mathbb{C}^J$ is the linear detector of Lemma 4.6: correlations with fixed partners, averaged successively in the directions of l over lengths between b and $2b$. Repeated axes give repeated averages. *Status:* theorem (conditional on 1–5). *Lean:* `trained`, `probe`.

Proposition III.12.5 (labels in the equation ideal give negligible observables). Assume the consistency conditions on C and hypotheses 2–4 of III.12.3. Let x be a coin of degree d with $\pi_{N,p} x_{N,p} = 0$ for all (N,p). Then $O(x)$ is owed at cost $a_N^{1/32}$: it is local, and for all large m, eventually in the mesh, $R_m(O(x)_{N,p}(0)) \leq a_N^{1/32}$. *Status:* theorem. *Lean:* `sweep`, `banked`.

Proposition III.12.6 (polynomial bounds on the dependence on the coupling). Let R be a run, T a campaign for R with pillars W, and assume the bounds `Sky`. Let $\bar\beta$ be the target level-0 coupling and $w > 0$ the fan width (`ask`). Fix p, labels V_0 (an n-tuple) and $r > 0$. Then there are $b, m \in \mathbb{N}$ with $m \geq 1$, $B \geq 0$ and $c \in (0,1]$ with the following property.

Take $m \leq j < N$ and $a \in [-1,1]$. Take a family $s \mapsto F(s)$, $s \in [-1,1]$, of admissible campaign members, with the same block sequence, sections and labels, whose level-0 inverse coupling at (N,p) is $\bar\beta + w\,a\,s$. Use the expectation and representatives of the attached auxiliary transfer family; their correlations agree with those of the actual successful trajectory at these positive cutoffs (Lemma 4.18). Take an r-separated configuration o of sites of $\Lambda_{j+1,p}$, embedded into $\Lambda_{N,p}$. Then the correlation

$$s \longmapsto \mathbb{E}_{N,p}^{F(cs/2)} \Big[\prod_i O(V_{0,i}; o_i) \Big], \qquad s \in [-1,1],$$

equals $\sum_{i \geq 0} P_i s^i$ with $\sum_i |P_i| \leq B\, a_{j+1}^{-b}$. The constants depend only on the campaign data, p, the labels and r. *Status:* theorem, conditional on the campaign records. These are supplied on the main-theorem route by the assembly (VIII.1). *Lean:* `Face.financed`, `PaidBeam`, `Ray.sun` ($= [-1,1]$), `intervalClock` ($= a_{j+1}^{-1}$ at level $j+1$), `slash`, `verso`.

Proposition III.12.7 (Cauchy property in the mesh at fixed volume). Fix a simple gauge group and one renormalization-group construction, with its stationary-path bounds, free amplification, observable-transfer bounds and common reference sections. Assume the uniform perturbation estimates for that construction (§2). Fix p. Let (C_n) be families formed from histories in this construction at cutoffs $f(n) \to \infty$. Assume:

- (i) `Rising`: each history is a selected branch from a level k on, and agrees with a fixed kinetic path. The stationary kinetic paths converge with a strict reserve in their admissible bounds. For every fixed level $i \geq k$, the inverse couplings are Cauchy and the polymer activities admit Cauchy representatives in the common weighted space (`Helmet`).

- (ii) For each label degree used below, its reference level $t(d)$ satisfies $k \le t(d)$ and $d+3 \le J(t(d))$. The transfer data are based at the reference section at $t(d)$ and are obtained by composing the branch transfers of Lemma 4.16.
- (iii) For every sufficiently large base level r, the multiscale remainder bounds hold for all sufficiently large sequence indices n. Their label allowances $\zeta(d,r)$ are eventually positive for each degree used, and their occupancy allowances are at least 1 at every level.

Fix a finite label list $(v_i)_{i=1}^{s}$ and a configuration $x \in (\mathbb{R}^4)^s$ with $x_i - x_j \notin L_p\mathbb{Z}^4$ for $i \neq j$. Assume that x lies exactly on a common lattice of some level r: $x = a_r z$ for an integer configuration z. Then the correlations of C_n at cutoff $f(n)$, evaluated at the coordinatewise ceiling lattice points of x, form a Cauchy sequence in n. *Status:* theorem under these hypotheses. For a run equipped with a campaign, pillars, fixed-level convergence data and the uniform perturbation estimates (§2), it applies with $C_n = C$ and $f(n) = n$. *Lean:* `Rising.cinema`, `Rising.arrives`; run instance `Campaign.sees`.

Proposition III.12.8 (positional ultraviolet limit). Let C be a Wilson family and e a strictly increasing sequence of mesh indices. Assume:

- the conclusion of III.12.1 (`FixedBounds`);
- the lattice correlations at coordinatewise ceiling lattice points are Cauchy along e at every configuration satisfying $x_{i\mu} - x_{j\mu} \notin L_p\mathbb{Z}$ for every $i \neq j$ and every axis μ (`Purchased`).

Then for every p, n and $v_1, \dots, v_n$ there is a function f on $(\mathbb{R}^4)^n$, C^∞ on $\mathrm{NoHit}_p = \{x : x_i - x_j \notin L_p\mathbb{Z}^4 \ (i \neq j)\}$ (`NoHit`), such that:

- (a) for every smooth compactly supported test function t whose support avoids collisions and lies strictly inside the centred period cell in every coordinate, the smeared lattice sums converge along e to $\int tf$;
- (b) fix any finite list of coordinate directions (slot, axis), and in each direction choose integers $q_-, q_+ \ge 0$ with $q_- + q_+ > 0$. Use the stencil $(F(x + q_+e) - F(x - q_-e))/((q_- + q_+)a_N)$. If g is the corresponding iterated coordinate derivative of f, then:
  - along $e(s_i)$ for every sequence $s_i \to \infty$, the difference quotients at lattice configurations whose physical positions converge to $x \in \mathrm{NoHit}_p$ converge to $g(x)$ (`Cluster`);
  - their multilinear interpolations converge locally uniformly on NoHit_p;
  - tg is integrable and the smeared sums of the differences converge to $\int tg$ for every test t as in (a).

Status: theorem under the two displayed hypotheses. For a run with a campaign, pillars, fixed-level convergence and uniform perturbation estimates, the full-sequence instance $e = (0, 1, 2, \dots)$ is `Campaign.realUV`. Its pointwise Cauchy hypothesis is derived from III.12.7 in `Campaign.purchased`. *Lean:* `PositionsUV`, `positioned`. `printed` is a variant along an extraction order and is not on the main-theorem route.

Use in the assembly. The record `Final.Jewel` is produced by `jewelOf17`. Its fields take the outputs of this unit:

- The kernel-bound conclusion is III.12.1.
- The full-mesh conclusion is III.12.8 on the full sequence. This is (D5); through `from_spec` it gives the ordered limit with the full ultraviolet sequence, (D4).
- The analytic-family conclusion uses III.12.6, through the chain of §4.4 ending in `Fan.standing` and `Travel`.

3.15.2 2. What it uses

The run and its ties. A `Run` (II.2, III.14) fixes the block sizes and numerical bounds, bare history, interlevel gauge maps and observable transfers. Its degree-d reference level satisfies $J(t(d)) \geq d+3$, and its Wilson family satisfies the consistency conditions.

`Run.again`, proved through `ahead` gives an extracted hierarchy and reconstruction after refining any extraction order, together with `Run.Ties`, which has these fields:

- The centred budget `Wcmain` for the family. For every d, every $R > 0$ and every $\varepsilon > 0$ there is $A \geq 0$ such that, for every $m \geq t(d)$, eventually in N, for all p, all $v \in \mathrm{Lab}_d$ with $\|v\| \leq R$ and all sites o: $O_{N,p}(v; o)$ is localized in the cell of $T_m(o)$ centred at o, and its chessboard root is at most $A\, a_m^{-d-\varepsilon}$ (`centeredBudget`).
- The scale-selection conditions.
- Remainder and regularity estimates for observables (III.11).
- Section data V_d, dressings W_d satisfying `Portfolio`, a presentation, and `Hearthside`.
- For every limit table and every degree d, detector rows (`Columns.Trials`) with the short-distance data `Purchases`.

Under the section hypotheses of III.12.3, the section is sound, has the prescribed polynomial realization and finite regularity bounds, and its dressing commutes with signed lattice symmetries and conjugation. Dressed equation labels satisfy the distant-spectator identity of Lemma 4.10.

The section-comparison estimates include cutoff-dependent automorphisms $A_{d;N,p}$ on terminal coordinates, auxiliary comparison columns, and a sub-power sequence $E_d(N)$. Eventually, uniformly in p, $A_{d;N,p}$ is the symmetry-averaged map determined by these columns and its inverse increases coefficient size by at most a factor 2. For every comparison column and every signed or conjugated version, the centred root of the difference between the auxiliary section evaluation and the chosen section evaluation is bounded by

$$C_d(m,k) E_d(N) a_N^{1/8}.$$

This holds for every admissible auxiliary level $k < N$ and centred scale $m \leq N$; $C_d(m,k)$ is the comparison constant from III.11. The scale-selection data choose such a k for each sufficiently large m. The kernel of $\pi_{N,p}$ is the image of $\mathcal{N}_d$ under the dressing, threshold rescaling, terminal map and $A_{d;N,p}$. The data also retain the projection symmetries and the finite-cutoff equation and difference comparisons needed by extraction.

Other existence formulations are `Run.ramp`, `realized`, `dagger`, `rampant`, `Final.july`.

From other units.

- **III.11.**
  - The centred budget for the run's family is `gardena`, via `coastwise` and `tent`. The centred logarithmic root bound is eventually at most $(d+\varepsilon)\log a_i^{-1}$.
  - The short-distance estimates for detector lifts compare their cumulant polynomials with the free Maxwell profile in chessboard norm. The lifts are bounded and admissibly placed, and the rescaled test configurations converge.
  - The remainder comparison for normalized local observables and their regularity constants is the input of `prepared_rows`.
- **II.1.** Labels and representatives. The normalized observables $a_N^d F$ (`unpriced`) and their regularity class (`clothed`, `section_bounds`, `offmask`). `offset`: the root on a coarse tiling at

any interior offset is bounded by the centred root at a finer scale. The matching rate `gamma` $= 1/8$.

- **II.2.** The Wilson family, its consistency conditions (translation covariance, locality of the representatives, continuity) and the run record.
- **IV.1.** Tilings, chessboard roots and localized observables. `Seam` holds for every $\beta \geq 0$ (`laws`) and gives two inequalities:
  - the chessboard inequality $|\mathbb{E} \prod_c G_c^{(c)}| \leq \prod_c R(G_c)$;
  - the triangle inequality $R(G + G') \leq R(G) + R(G')$.

  `seamless` gives $\beta_{N,p} \geq 0$. Also: `stepTotal` (Lipschitz bound in the positions for correlations at separated points, derived from centred budgets); `growFrom` (Cauchy on a dense set plus asymptotic equicontinuity gives locally uniform convergence).
- **III.2.** Existence of the section-comparison data and the numerical parameter sequence.
- **III.6, III.8, III.9, III.10.** Histories and selected branches; stationary-branch records (used in `Rising.flooded`). The running coupling satisfies $\hat{\beta}_j = \text{const} + \kappa_G \log a_j^{-1}$ with $\kappa_G > 0$ (`slopeConstant_pos`). The large-field radius grows without bound in $\hat{\beta}$ (`radiusB_tendsto`).
- **VIII.1.** `Winter`, proved by `offers`, gives fixed-level convergence of the inverse couplings and normalized activities for sequences of successful trajectories whose level-0 couplings converge. It includes the kinetic-path reserve, free amplification and analytic versions of the limits. The activity norm includes regular derivatives, forced terms and support weights; the scalar part of the effective action is excluded. The assembly supplies its stationary-path, free-amplification and uniform perturbation inputs; the pillars identify the run's transfers with the branch transfers. The bare coupling's analytic extension is `bareTrial`.

Standard mathematics.

- Reflection positivity of the Wilson action: Osterwalder–Seiler, *Ann. Phys.* 110 (1978).
- Chessboard estimates: Fröhlich–Israel–Lieb–Simon, *Comm. Math. Phys.* 62 (1978) and *J. Stat. Phys.* 22 (1980); Shlosman, *Russian Math. Surveys* 41 (1986).
- The chessboard seminorm (triangle inequality for R): Fröhlich–Lieb, *Comm. Math. Phys.* 60 (1978); Biskup, *Reflection positivity and phase transitions in lattice spin models* (2009).
- Summation by parts. Stability of injectivity of a linear map under small perturbations. The dimension count for a linear map of known rank.
- Arzelà–Ascoli type arguments. Taylor's theorem with remainder.
- Absolutely convergent power series, uniqueness from their values on a real interval, polynomial coefficient estimates and dominated convergence for series.
- The renormalization-group control of effective actions uniformly in the volume: Balaban, *Comm. Math. Phys.* 1984–1989 (references as in III.9).

Imported constants and assumptions.

- The exponent $-d - \varepsilon$ in the centred budget. III.12.2 uses it with $\varepsilon = 1$.
- The matching chain: rate $\gamma = 1/8$, giving an owed cost $a_N^{1/16}$ in Lemma 4.14; the equation-ideal estimate gives $a_N^{1/32}$ (III.12.5).
- The tail factor $e^{-B_j/64}$ (`brisk`), where B_j is the budget radius at level j. Positions must have mutual lattice distance at least B_j^2 at level j.
- All numerical thresholds of the run (III.2).

3.15.3 3. Strategy

The centred budget applies to primaries with bounded physical labels. A normalized difference carries the divergent factor a_N^{-1}, so it needs a separate argument. The local regularity estimates and the matching theorem of III.11 write $\Delta_l O(w)$ as a degree-$(d+|l|)$ primary $O(x)$ with an error of root cost $a_N^{1/16}$. The inverse terminal coordinates of x have only sub-power control. To bound πx, test against fixed partner observables and average in the differenced directions. Summation by parts bounds these tests by $(2/b)^{|l|}$ times an undifferenced correlation bound. At short distances the detector approaches a free Maxwell matrix that separates the quotient by the equation ideal. Finite-dimensional stability then bounds πx. The auxiliary equation sections have zero chessboard root by their distant-spectator identity; section comparison makes $O(x-\pi x)$ negligible. Thus $\Delta_l O(w)$ is funded, and the chessboard inequality bounds products at separated points (III.12.1–III.12.5).

For convergence, fix the volume and an intermediate level j. Fixed-level convergence of the couplings, normalized activities and effective insertions makes the scalar-normalized level-j correlation Cauchy as the cutoff grows. The full correlation differs from it by at most $2F_0 e^{-B_j/64}$ for all sufficiently large cutoffs, where F_0 is independent of the cutoff and j. Since $B_j = \lceil (\log \hat{\beta}_j)^2 \rceil \to \infty$, the full correlations are Cauchy at configurations lying on a common lattice. Positional Lipschitz bounds extend convergence from this dense set to every non-collision configuration. Taylor bounds identify limits of differences with derivatives, giving smoothness and the smeared limits (III.12.7–III.12.8).

For dependence on the coupling, vary the level-0 value in a family parametrized by $\bar{\beta} + ws$. After reducing the parameter interval, the lattice correlations have absolutely convergent power series whose coefficient norms grow polynomially in a_j^{-1}, uniformly in the cutoff (III.12.6). Their real level-to-level differences are $O(a_j)$. A polynomial coefficient estimate after a further interval reduction makes the differences summable in coefficient norm. Fixed-level Cauchy convergence of the series then gives analytic continuum correlations on a complex disc. This uses the same fixed-level convergence input as the preceding paragraph, together with the additional analytic estimates.

3.15.4 4. Steps

3.15.4.1 4.1 Calculus of the chessboard budget Throughout §§4.1–4.3, the family satisfies its consistency conditions. In §4.2, assume hypotheses 1–5 of III.12.3, with the same detector rows and presentation throughout.

Lemma 4.1 (closure properties). For trials F, H and costs u, v:

- $\mathrm{Aff}(F,u)$ and $\mathrm{Aff}(H,v)$ imply $\mathrm{Aff}(F \pm H, u+v)$.
- $\mathrm{Aff}(F,u)$ implies $\mathrm{Aff}(cF, |c|u)$ for any $c = c_{N,p} \in \mathbb{C}$.
- $\mathrm{Aff}(0,0)$, and finite sums behave the same way.
- Affordability is monotone in the cost.
- It is unchanged if F is modified off a tail of the mesh.

Hence funded trials form a vector space. *Sketch:* linear combinations preserve locality; the root is homogeneous and satisfies the chessboard triangle inequality. *Lean:* `Affords.sub`, `Affords.scalar`, `Affords.enlarged`, `Affords.recurses`, `Funded.sub`; negation, addition, the zero trial and finite sums are treated in the same file.

Lemma 4.2 (all offsets). Let F be affordable at cost u. For every m and every integer $J \geq 1$

there is $k \geq m$ with the following property, eventually in the mesh. For every tiling of physical cell side a_m and every site x in its central cell at least one J-th of the cell side from each wall, $F_{N,p}(x)$ is localized in that cell and its root on the tiling is at most $u(k, N, p)$. *Sketch:* choose k with $3Ja_k \leq a_m$ and apply the offset comparison (II.1), using translation covariance. *Lean:* `Affords.every`. *(proof read; offset comparison imported from II.1)*

Lemma 4.3 (bounded primaries are funded). Under the centred budget, let $v_{N,p} \in \mathrm{Lab}_d$ satisfy $\|v_{N,p}\| \leq Y$ eventually in the mesh for some fixed Y. Then $O(v)$ is funded, with $u(m) = A\, a_m^{-d-1}$. *Sketch:* apply the centred budget with $\varepsilon = 1$ and $R = \max(1, Y)$, at the offset 0. *Lean:* `Funded.primaries`.

Lemma 4.4 (products at separated points). Let $F_1, \dots, F_n$ be affordable at costs $u_1, \dots, u_n$, and let $r > 0$. For all large k, eventually in the mesh, every configuration z with $a_N \operatorname{dist}(z_i, z_j) \geq r$ satisfies

$$\left|\mathbb{E}_{N,p} \prod_i F_i(z_i)\right| \leq \prod_i u_i(k, N, p).$$

If all F_i are funded, the left side is bounded uniformly, eventually in the mesh. *Sketch:*

- Choose l with $a_l < r$, and k with $6n\, a_k < a_l$.
- Translate the tiling of side a_l so that each z_i lies in the interior margin of width $3a_k$ of its cell. Distinct points lie in distinct cells, since their distance exceeds the cell side.
- Write each $F_i(z_i)$ as the reflected copy of an observable localized in the central cell. The centred estimate at scale k and the offset comparison bound its root by $u_i(k, N, p)$.
- Apply the chessboard inequality. The remaining cells carry the constant 1, whose root is 1.

Lean: `cohort`, `FullVolume.financed`.

Lemma 4.5 (III.12.1 from III.12.2). Suppose $\Delta_l O(v)$ is funded for every fixed label v, in every degree, and every list l. Then the bounds of III.12.1 hold for C. *Sketch:* induction on the list λ of pairs (slot, axis). Differencing slot i differences its insertion (`actuate`), by linearity of expectation. Thus the differenced kernel is a correlation of $\Delta_{l_i} O(v_i)$, each funded. Conclude by Lemma 4.4. Then `genuine` applies III.12.2 to constant sequences. *Lean:* `funding`.

3.15.4.2 4.2 Detectors, rank detection and the closure lemma

Lemma 4.6 (detectors; summation by parts). A detector row j at test scale k consists of:

- a test configuration $y_{k,j} \in (\mathbb{R}^4)^{1+e_j}$, with distinct projections on every coordinate axis and contained in a box of side $a_{\mathrm{width}(k)}$;
- partner labels $w_{k,j,u}$ (lifted basis labels whose norms, for fixed k, are bounded uniformly in N, p and the finite indices).

For a trial F, the *monitor* is the correlation $M_j(F)(x) = \mathbb{E}_{N,p}[F(x_0) \prod_u O(w_{k,j,u}; x_u)]$ (`monitor`). The *assay* averages the first slot over $n_b = \lceil b/a_N \rceil$ consecutive sites in each direction of l, starting at the prescribed lattice approximation of $y_{k,j}$ (`assay`). The detector is $\mathcal{P}_{k,b,l}(x)_j = \mathrm{assay}_j(O(x))$. For each fixed k, l, there is $b_0 > 0$ such that, for every $0 < b < b_0$, the following hold:

- (a) if F is funded with continuous values, then $\|\mathrm{assay}(\Delta_l F)\| \leq v$ eventually in the mesh, for some constant v depending on F, k, b, l;
- (b) if F is owed at a cost $u_{N,p} \to 0$, then $\mathrm{assay}(F) \to 0$.

Sketch:

- `scores_hops`: $M_j(\Delta_l F)$ is the a_N-normalized lattice difference of $M_j(F)$ in the first slot.
- `same_delta`: averaging n consecutive normalized differences telescopes to $(na_N)^{-1}(f(x + ne_\mu) - f(x))$, with $na_N \in [b, 2b]$. Iterating gives `reserve_abs`: $|\mathrm{assay}(\Delta_l F)| \le (2/b)^{|l|} \sup |M_j(F)|$ near the test point.
- `warded`: near a generic test point, $|M_j(F)| \le u(w) \cdot A$. This is Lemma 4.4 applied to F and the partners, which are funded by Lemma 4.3 (`siblingsPrice`).
- For (b), averaging does not increase the bound (`rounds`).

Lean: `invites`.

Lemma 4.7 (averaging is harmless on primaries). For every $\varepsilon > 0$ and k, for small b and eventually in the mesh, $|\mathcal{P}_{k,b,l}(\beta_c)_j - M_j(O(\beta_c))(y_{k,j})| \le \varepsilon$ for all rows j and all lifted basis labels β_c. *Sketch:* `lobby` gives a radius $t > 0$ such that correlations of labels of bounded norm vary by less than ε within distance t of the test configuration. This follows from the positional Lipschitz bound derived from the centred budget (IV.1), uniformly in p. Then choose $2|l|b \le t$. *Lean:* `probesUp`. *(proof read; the single-step Lipschitz bound is an input from IV.1)*

Lemma 4.8 (short distances: the free Maxwell values). For every $\varepsilon > 0$, eventually in k, eventually in the mesh, simultaneously for all rows j and basis indices c: $|M_j(O(\beta_c))(y_{k,j}) - \phi_{jc}| < \varepsilon$. Here ϕ_{jc} is the value of the j-th test of the free Maxwell table on the c-th basis vector of F_d (`focus`). *Sketch:* the short-distance comparison of III.11 makes the lattice correlation approach the free Maxwell correlation at the rescaled test configuration. The latter configurations converge to the prescribed free tests, whose correlations are continuous there. *Lean:* `gaugesFull`, packaged as `VitalRank`. *(proof read; short-distance estimates imported from III.11)*

Lemma 4.9 (rank detection; III.12.4). The matrix (ϕ_{jc}) is injective on F_d (`true_lower`): the free tests separate the quotient. *Sketch:*

- `firm` gives $\delta, L > 0$: every linear map whose matrix is δ-close to (ϕ_{jc}) satisfies $\|y\| \le L\|\cdot\, y\|$.
- Take $\varepsilon = \delta/4$. The entries of $\mathcal{P}(\pi\beta_c)$ are within δ of $\phi_{\cdot c}$, by three estimates: Lemma 4.7; Lemma 4.8; and $\mathcal{P}(\beta_c - \pi\beta_c) \to 0$, which holds by Lemma 4.10 and Lemma 4.6(b).
- $\operatorname{rank} \pi_{N,p} \le \dim F_d$ (`weighed`). Indeed $\ker \pi_{N,p}$ is the dressed image of $\mathcal{N}_d$, and $\mathcal{N}_d$ is the null space of the free table by the exact free-field projection.
- `rank_upper`: the map $y \mapsto \pi(\sum_c y_c \beta_c)$ is injective by the previous steps, hence onto $\operatorname{ran} \pi$ by dimension. This gives $\|\pi x\| \le Z\|\mathcal{P}(\pi x)\|$ with $Z = A\,(\dim F_d)\,VL$. Here A bounds $\|\pi\|$ and V bounds $\|\beta_c\|$.

Lean: `trained`.

Lemma 4.10 (equation terms; III.12.5). Let x be a coin with $\pi x = 0$. Then $O(x)$ is owed at cost $a_N^{1/32}$. *Sketch:* undo the section-comparison automorphism and terminal map to obtain a terminal-coordinate row y of sub-power growth whose associated abstract labels belong to the dressed equation ideal. The inverse coefficient-size bound in §2 controls this change of coordinates.

- `ocean`: $O(x) - H(y)$ is owed at $a_N^{1/32}$. Here $H(y)_{N,p}(o)$ is the bare lattice evaluation of the section data at o, scaled by a_N^{-d}. The proof goes through `studded`, a bound by a constant times a sub-power factor times $a_N^{1/8}$.
- `hole`: $H(y)$ is owed at cost 0. By the section hypotheses, the evaluation of a dressed element of $\mathcal{N}_d$ is `MicroAt`: for every $\beta \ne 0$ and every large torus, its Wilson expectation against any bounded measurable observable depending only on links at lattice distance $> s$ vanishes. The spectator need not be gauge invariant. Once the cell side exceeds $2s + 2$ lattice units,

the copies in the other cells are such observables. So the chessboard product has expectation 0 and $R_m = 0$.

The auxiliary equation sections are an input from II.1 (§7). *Lean:* `sweep`, `banked`.

Lemma 4.11 (a priori bound on πx). Under the hypotheses of III.12.3, there is a constant A such that $\|\pi_{N,p} x_{N,p}\| \le A$ eventually in the mesh. *Sketch:*

- By Lemma 4.6, $\|\mathcal{P}(x)\| = \|\mathrm{assay}(O(x))\| \le \|\mathrm{assay}(\Delta_l F)\| + \|\mathrm{assay}(O(x) - \Delta_l F)\| \le c$.
- Put $X = \pi x$, $\theta = (1 + \|X\|)^{-1}$ and $z = \theta(x - X)$. Since $|\theta| \le 1$, θx is a coin. Since $\|\theta X\| \le 1$, θX is a coin too. Thus z is a coin and $\pi z = 0$, so $Z\|\mathcal{P}(z)\| \le \frac{1}{2}$ eventually by Lemmas 4.10 and 4.6(b).
- Hence $\|X\| \le Z\|\mathcal{P}(X)\| \le Z\|\mathcal{P}(x)\| + Z\|\mathcal{P}(x - X)\| \le Zc + \frac{1}{2}(1 + \|X\|)$.

Rearrange to obtain $A = 2Zc + 1$. *Lean:* `cap`.

Lemma 4.12 (closure lemma; III.12.3). $O(\pi x)$ is funded by Lemmas 4.11 and 4.3. $O(x - \pi x)$ is owed at $a_N^{1/32} \to 0$ by Lemma 4.10, hence funded: a cost tending to 0 is eventually below 1 (`fixed_cost`). So $O(x)$ is funded. Finally $\Delta_l F = O(x) - (O(x) - \Delta_l F)$ is funded, by Lemma 4.1. *Lean:* `improved`.

3.15.4.3 4.3 Differences of primaries (III.12.2)

Lemma 4.13 (regularity under normalized differences). Call F *tucked of degree* d if the following holds. There are a stencil containing 0 and sequences $c_{N,p}, b_{N,p}$ of sub-power growth. Eventually in the mesh, for every site o, $a_N^d F_{N,p}(o)$ is regular on the stencil around o, its field-derivative bounds are at most $c_{N,p}$, and its sup norm is at most $b_{N,p}$. Then:

- for a simple gauge group, $O(w)$ is tucked of degree d for every coin w;
- if F is tucked of degree d and the constants for shifting the field-derivative bounds by one lattice site have sub-power growth, then $\Delta_\mu F$ is tucked of degree $d + 1$. The run's remainder data supply these shift bounds.

Sketch: $a_N^d O(w)$ is the section evaluation at the threshold-scaled coefficients (`charged`, II.1); the bounds come from the regularity and supremum estimates for section evaluations. For the difference, $a_N^{d+1} \Delta_\mu F(o) = a_N^d F(o + e_\mu) - a_N^d F(o)$. The stencil grows by one step, and the derivative bounds pick up the sub-power shift cost. *Lean:* `Tucked`, `Tucked.up`, `Tucked.farther`, `Tucked.hops`.

Lemma 4.14 (matching a local observable by a primary). Assume a simple gauge group, the scale-selection data and the observable remainder and regularity bounds of III.11 (§2). Let F be tucked of degree d, local, and continuous as a function of the gauge field at the insertion site 0, eventually in the mesh. Then there is a coin v of degree d such that $O(v) - F$ is owed at cost $a_N^{1/16}$. *Sketch:* apply the local regularity comparison of III.11 to $a_N^d F$. After undoing this normalization, the centred root of $O(v) - F$ at scale m is at most a constant (depending on m and an auxiliary level chosen by the scale-selection conditions) times a sub-power factor times a_N^γ, with $\gamma = 1/8$. `readyCoin` and the sub-power absorption estimate absorb the sub-power factor into $a_N^{1/8 - 1/16}$. *Lean:* `prepared_rows`. *(statement and reduction read; the III.11 inputs statement only)*

Lemma 4.15 (proof of III.12.2). In a run for a simple gauge group, let w be eventually bounded in the mesh and set $F = O(w)$. Lemma 4.3 funds F. By Lemma 4.13, $\Delta_l F$ is tucked of degree $d + |l|$; it is also local and continuous in the gauge field. Lemma 4.14 gives a coin x of that degree

with $O(x) - \Delta_l F$ owed at $a_N^{1/16} \to 0$. The run's ties supply hypotheses 1–5 at degree $d + |l|$, so III.12.3 funds $\Delta_l F$. *Lean:* `Run.financed`.

3.15.4.4 4.4 Observable transfer and coupling dependence The analytic transfer bounds use III.12.6's campaign data; analytic continuum convergence also uses III.12.7's fixed-level convergence and perturbation inputs.

Lemma 4.16 (composition of observable transfers). Fix a simple gauge group, a branch from level k to cutoff N, the observable-transfer bounds and compatible intermediate representations of §2. Fix $k \le r \le N$ and $d + 3 \le J(r)$. There are degree-d transfer data based at the reference section at r, obtained by composing the one-step transfers along the branch. *Sketch:* a one-step transfer tool with its capacity estimate acts on an observable at level $m + 1$ and produces an observable at level m whose density is the exact renormalization-group pushforward with the degree-dependent block normalization. It preserves linearity and symmetry and supplies coefficient, remainder and response bounds (`motor`). These steps compose with the prescribed top observable (`composeT`). The branch includes selectedness, available coefficient spaces and the one-step transfer data (`Branch`); its estimates provide the required capacities. *Lean:* `Branch.composing`. Fixed-level convergence follows from `Branch.stays`, using the stability of the step maps under two cutoff subsequences (`Branch.tiers`) (Lemma 4.19). *(proof read for composition and fixed-level reduction; one-step capacity estimate statement only)*

Lemma 4.17 (power series in the tuning parameter, level by level). Assume the campaign data and estimates of III.12.6. Write $f \in \mathrm{PB}(t)$ if $f(s) = \sum_i P_i s^i$ on $[-1, 1]$ with $\sum \|P_i\| \le t$. This class is closed under sums ($\mathrm{PB}(t) + \mathrm{PB}(t') \subset \mathrm{PB}(t + t')$) and under rescaling $s \mapsto as$ with $|a| \le 1$.

- The one-step analytic capacity estimates bound the remainder map, applied to a coefficient vector x, by $\mathrm{PB}(\nu\|x\|)$. Here ν is the prescribed one-step bound in the completed remainder norm; the parameter interval has already been reduced by a factor 2.
- A bound $\mathrm{PB}(t)$ at level m propagates to level $m+1$ as $\mathrm{PB}(\ell_m t)$ (`linked`), or as $\mathrm{PB}(2v_T\chi_d t)$ (`destination`).
- Here v_T is the prescribed transfer bound and χ_d the finite-dimensional coefficient constant. Since $\prod_{m=0}^{j} \ell_m = a_{j+1}^{-1}$, iterating gives bounds polynomial in a_{j+1}^{-1}.

`cruise` gives per-label bounds. `polynomial` bounds a correlation item at $s \mapsto cs$ by $B_0 a_{j+1}^{-b_0}$ for fixed B_0, b_0. Here c is fixed by `aperture` so that $|c|\, Q/Z_* \le \frac{1}{2}$, where Q bounds the source contribution and $Z_* > 0$ bounds the scalar-normalized partition function from below; see also `storms`. `Face.financed` identifies the item for the full mode with the correlation (`absolute`), which gives III.12.6. The weighted analytic bounds for the parameter-dependent effective densities (`attic_close`, `many`, `program`) belong to this fan analysis. `Amp.give` and `Amp.stage` extend a multilinear series from a dense isometric image, with geometric bounds; this builds the series on real jets without complexifying the field. *(proof read for the power-series calculus, parameter rescaling, interval selection and final correlation bound; intermediate analytic estimates statement only)*

Lemma 4.18 (analytic continuum correlations). Assume a campaign and pillars, the analytic estimates of III.12.6, fixed-level convergence and the uniform perturbation estimates of §2. Fix p, finitely many labels, a compact $K \subset \mathrm{NoHit}_p$ and $q \in \mathbb{N}$. There is a positive parameter width, independent of the cutoff, and an open neighbourhood Ω of K within NoHit_p, on which the limiting kernels are C^q in position and analytic in the complex parameter. For each spatial stencil

list of length at most q, the lattice kernels, their multilinear interpolations and their admissible smeared sums have analytic continuations whose parameter derivatives of every fixed order converge uniformly on compact subdiscs and compact spatial subsets of Ω. *Sketch:* `pricedB` gives coefficient-norm bounds Ba_j^{-b} for differenced kernels on a fixed compact set. `traceS` chooses series representing these kernels. Their real values at consecutive levels differ by $O(a_j)$. A polynomial coefficient estimate after a fixed reduction of the parameter interval turns this into a coefficient-norm bound $C\sqrt{a_j}$, summable in j. For each fixed j, Cauchy convergence of the real values gives Cauchy convergence of the series in the cutoff after another interval reduction (`astronomer`). Thus the series converge uniformly in position (`charted`). Their absolutely convergent complex power series give analytic limits and convergence of all parameter derivatives on smaller discs. The width may depend on p, K, q. The final identification uses the actual successful Wilson trajectories, whose correlations agree with those of the auxiliary transfer families at every positive cutoff. This is the content of the analytic spatial statement `Spending` and its theorem `Fan.standing`, used by the assembly's analytic family. *(proof read for the series-convergence and final assembly reductions; spatial derivative identification statement only)*

Additional estimates in this folder.

- Under the connected-activity hypotheses of III.8, the large-field step estimate applies when its rate parameter is at most $-7/16$ (`projects`, namespace of the connected-activity bounds, III.8).
- Under the one-step source, degree, threshold and release hypotheses of III.10, the primary insertion satisfies its step estimate (`primaryJG`, III.10).
- Fix $g < e$, $w < w'$, $w \le 1$, a torus with at least four sites per axis, and a budget point and clipping parameter satisfying the admissibility and kinetic-width conditions. There is $t > 0$ such that every $|a| \le t$ sends a representation with bounds $(\sigma, C, \Lambda, g, w)$ to one with bounds $(\sigma, C, \Lambda, e, w')$, preserving its evaluation (`earned`).
- Comparisons of rank data between two cutoff sequences (`cams`, `linters`).
- The connected-activity and stationary-branch results used by these estimates.

These are inputs to the transfer and analytic estimates, not additional convergence assertions. Their full hypotheses and proofs belong to III.8–III.10; see §7.

3.15.4.5 4.5 Convergence in the mesh Lemma 4.19 (fixed-level convergence). In the setting of III.12.7, fix $m \ge k$ and above the reference levels of all labels used. Then:

- the inverse couplings and weighted activities are Cauchy (`flooded`, by the fixed-level convergence input);
- the level-m coefficients and coins of each label converge (`grow`, by stability of the composed transfers);
- the effective observable and its polymer parts converge uniformly (`hail`). The property `Confluence` means eventual uniform boundedness and uniform Cauchy convergence.

Sketch: the kinetic-path reserve and activity convergence control the normalized density. Stability of the composed transfers gives Cauchy coefficients and remainders; evaluation preserves their uniform Cauchy bounds. *(proof read for these reductions; transfer stability estimates statement only)*

Lemma 4.20 (correlations computed at a fixed level converge). Let the level-j data be as in Lemma 4.19. Assume the effective insertions and their polymer components are measurable,

and that the scalar-normalized partition function is eventually bounded below by $a > 0$. Then the level-j value

$$C_j^I(n) = Z_{j,n}^{-1} \int Q_{j,n}^I(U)\, d\mu_{\mathrm{Haar}}(U)$$

is Cauchy in n for every fixed label subset I. Here $Q_{j,n}^I$ is the coefficient indexed by I in the scalar-normalized effective polynomial obtained by retaining the vacuum and singleton source data; $Z_{j,n}$ is its vacuum normalization, and μ_{Haar} is the product Haar probability measure. *Sketch:* uniform convergence passes through the Haar integral. The normalizations have a positive limit, so the quotients converge. The additive action scalar cancels. *Lean:* `Confluence.laven`, `Rising.latter`, with the uniform bounds `ships` and `limitVault`.

Lemma 4.21 (tail bound uniform in the cutoff). In the setting of III.12.7, fix a non-collision configuration lying exactly on a lattice of some level. Choose a sufficiently large base level r, above the reference levels and with distinct insertion sites. Fix positive label weights so that their weighted coefficient costs are within the allowances at r; these weights exist by `creditP`. For every fixed subset I of the labels, there is $F_0 \geq 0$, independent of j and the cutoff, such that for each sufficiently large $j > r$, eventually in n,

$$|p_n^I - C_j^I(n)| \leq 2F_0 e^{-B_j/64}.$$

Here p_n^I is the product correlation at cutoff $f(n)$ and $C_j^I(n)$ is its scalar-normalized level-j approximation. The lower threshold on j ensures separation by B_j^2 lattice units, support exponent at least 2, and $e^{B_j/64} > 2$. The threshold on n may depend on j. *Sketch:* propagate the exact polynomial density of the inserted observables down the renormalization group. The weighted remainder estimate controls its part of degree at least two. Separation forces a nonzero such term to occupy enough blocks to gain the factor $e^{-B_j/64}$. The remaining omission estimate divides by a uniformly positive scalar-normalized partition function. Uniform bounds on the effective insertions at r and j follow from Lemma 4.19; the final numerator depends only on the data at r. *Lean:* `Sever.echoes`. *(proof read for the exact-density, remainder and omission reduction; underlying multiscale estimates are inputs)*

Lemma 4.22 (Cauchy property). Under the hypotheses of Lemma 4.21, p_n^I is Cauchy for every fixed subset I of the labels. *Sketch:* $B_j = \lceil (\log \hat{\beta}_j)^2 \rceil$, so $g(j) = e^{B_j/64} \to \infty$ because $\hat{\beta}_j = \mathrm{const} + \kappa_G \log a_j^{-1} \to \infty$. Choose j with $g(j) > 2$ and $2F_0/g(j) \leq \varepsilon/4$. Lemma 4.21 bounds $|p_n^I - C_j^I(n)|$ by $\varepsilon/4$ for large n. Lemma 4.20 gives the Cauchy property of $C_j^I(n)$. Conclude with the triangle inequality. *Lean:* `Rising.arrives`.

Lemma 4.23 (III.12.7). Under the hypotheses of III.12.7, take the subset containing all labels. The full-subset quantity is the correlation itself (`absolute`). The positions are $o_N = \mathrm{wrap}(\alpha_N(x))$. For $x \in \mathrm{NoHit}_p$ lying exactly on a lattice of some level, these satisfy the separation B_j^2 for all large j by the lattice-ancestry estimates, and they are injective at the base level. *Lean:* `Rising.cinema`.

Lemma 4.24 (all non-collision points, full sequence). For a run with a campaign, pillars, fixed-level convergence and uniform perturbation estimates, III.12.7 holds on the full sequence of cutoffs (`Campaign.sees`, with the same cutoff-dependent history family repeated for every sequence index and $f(n) = n$). Configurations lying exactly on some lattice are dense. III.12.1 gives a local Lipschitz bound (`localLip`), hence asymptotic equicontinuity on compact subsets of NoHit_p (`circulating`). The dense-set convergence theorem (IV.1) then gives locally uniform

convergence on NoHit_p, in particular the Cauchy property on the subset with no coordinate ties. *Lean:* `Campaign.purchased`.

Lemma 4.25 (positional limits; III.12.8). Assume the two hypotheses of III.12.8.

- `established`: the kernels converge locally uniformly on NoHit_p to a continuous f. The argument is the one of Lemma 4.24.
- `from_previous`: if the l-differences converge, so do the differences with one additional coordinate direction a. Fix $q = \min(t/2, r/3, \varepsilon/(8(B+1))) > 0$, where t is a neighbourhood radius and r, B are the Taylor constants. Choose integers n_N with $k_N = n_N a_N \in [q, 2q]$ and $k_N \to q$. If $a_N x_N \to x$, compare the unit difference D_N with $W_N = k_N^{-1}(F_N(x_N + n_N e_a) - F_N(x_N))$. Both endpoint values converge (`returns`), so W_N converges. The Taylor estimate $|F_N(x_N + n_N e_a) - F_N(x_N) - k_N D_N| \le B k_N^2$ (`taylor`) gives $|W_N - D_N| \le \varepsilon/4$ eventually. Hence D_N is Cauchy.
- Limits of differences are derivatives of f, so $f \in C^\infty(\mathrm{NoHit}_p)$ (`reachable`).
- Every fixed forward/backward interval stencil is a finite average of translates of a unit forward difference. These translates approach the same physical point, so they have the same derivative limit. Multilinear interpolation preserves this limit. For admissible tests, the corresponding piecewise constant Riemann sums have an integrable bound on a fixed compact set, and dominated convergence gives the smeared limits.

Under the hypotheses of Lemma 4.24, the run instance is `Campaign.realUV`: III.12.8 applied to III.12.1 and Lemma 4.24. *Lean:* `positioned`. *(proof read for equicontinuity, Taylor comparison, derivative convergence, stencil averaging, interpolation and smearing)*

3.15.5 5. Where the difficulty lies

Matching controls only inverse terminal coordinates. Rank detection must bound the projected physical labels, while the equation-section comparison controls the projection kernel. These estimates precede every limit.

The estimates of III.12.1–III.12.5 are uniform in every admitted volume. The convergence results III.12.7–III.12.8 fix the volume first. Their thresholds may depend on the period, labels, compact spatial set, derivative order and test function. The parameter width in Lemma 4.18 may also depend on these data.

All positional limits are off collisions on the torus, including configurations with coordinate ties and configurations crossing the boundary of a period cell. Extension across collisions belongs to VII.4. Every derivative order or stencil is fixed before the cutoff threshold is chosen.

Sub-power factors are absorbed by the positive error exponents. Fixed-level convergence uses a positive lower bound on the scalar-normalized partition function; the additive action scalar need not converge.

3.15.6 6. Notation

Written proof	Lean (glossary)	Meaning
eventually in the mesh	`mesh`	for $N \ge N_0$ and all p
sub-power growth	`Sluggish`, `flight`	$\|w\| \le a_N^{-\varepsilon}$ for all $\varepsilon > 0$; flight $(\varepsilon) = a_N^{-\varepsilon}$

Written proof	Lean (glossary)	Meaning
trial F, local	`Trial`, `Stencil`	site-indexed lattice observables on a fixed stencil
$T_m(o)$, R_m	`center`, `rootTest`	centred tiling of side a_m; chessboard root
affordable, owed, funded	`Affords`, `Owed`, `Funded`	centred root bound by $u(m,N,p)$, $v(N,p)$, $u(m)$
Δ_μ, Δ_l	`farther`, `hops`	a_N-normalized forward differences
tucked of degree d	`Tucked`	$a_N^d F$ regular with sub-power bounds
coin; $O(x)$	`Coin`, `Bid.upper`	label sequence with sub-power inverse terminal coordinates; its representative
$\pi_{N,p}$	`Presentation`, `display`	idempotent with kernel the dressed equation ideal
M_j, detector $\mathcal{P}_{k,b,l}$	`monitor`, `assay`, `probe`	test correlation; iterated directional average
ϕ_{jc}	`focus`	free Maxwell value of test j on basis vector c
centred budget	`Wcmain`	root $\le A\, a_m^{-d-\varepsilon}$
kernel bounds	`FixedBounds`	III.12.1
run, ties	`Run`, `Run.Ties`	record of one construction; its inputs for this unit
fan parameter $s \in [-1,1]$	`Ray.sun`, `ask`	level-0 coupling $\bar\beta + ws$
PB(t)	`PaidBeam`	power series on $[-1,1]$ with ℓ^1 coefficient norm $\le t$
a_j^{-1}	`intervalClock`	$D_j = \ell_0 \cdots \ell_{j-1}$
campaign, pillars, sky	`Campaign`, `Pillars`, `Sky`	records carrying the fan and its bounds
fixed-level convergence; function convergence	`Helmet`, `Confluence`	Cauchy couplings and weighted activities; eventual uniform boundedness and uniform Cauchy convergence of functions
point-Cauchy property	`Purchased`	Cauchy correlations at ceiling addresses when no coordinate projections tie
positional limit	`PositionsUV`	III.12.8; (D5) of the main statement

3.15.7 7. What remains unexplained

1. **Local matching and equation sections.** Lemma 4.14 imports the local-to-primary estimate of III.11; only its reduction to a coin and an $a_N^{1/16}$ root bound is checked here. The construction of the auxiliary sections from the Maxwell equation ideal, and its relation

to a lattice Schwinger–Dyson identity, remain unverified. Lemma 4.10 uses their exact distant-spectator identity, without identifying it with the Wilson action's naive equation of motion.

2. **One-step observable transfer.** The capacity estimate used by Lemma 4.16 remains at statement level. Its coefficient and remainder constants require the one-step analysis.
3. **Analytic estimates.** The intermediate weighted estimates in Lemma 4.17 and the identification of spatial derivatives of the analytic limit in Lemma 4.18 remain at statement level. The outer series-convergence argument is checked.
4. **Additional estimates in §4.4.** The large-field and primary-step results and the two transfer comparisons retain their full hypotheses in III.8–III.10. Their proofs, and the detailed small-shift lawfulness estimate, are not checked here.
5. **Fixed-level convergence and tails.** The analytic trajectory-convergence theorem and the multiscale remainder and omission estimates remain imported inputs; their applications are checked. The tail argument needs only $B_j = \lceil (\log \hat{\beta}_j)^2 \rceil \to \infty$, not geometric decay in j.

3.16 III.13. Extension to all labels: one common Wilson family

Part: III, Ultraviolet construction (renormalization group). **Lean:** `YM/SharpGram/Common.lean`, `YM/SharpGram/Measured.lean`, `YM/SharpGram/Assembly/`, `YM/UVExtension/`. **Size:** about 22,100 lines in the unit inventory. Principal assembly results: `multitude`, `uv_log`, `multitude_foot`.

Place in the proof. This unit supplies the ultraviolet construction for `habitat` (Proposition III.13.4). The main theorem `YangMills.constructed` does not use it.

- The main theorem gets its Wilson family from the construction run of III.14 (`constructionRun`, `Run.extractible17`).
- The two constructions rest on the same per-step estimates (III.1–III.12) and package them differently.

3.16.1 1. What this unit proves

- **The constants are chosen once.** One choice of renormalization-group (RG) constants supplies a certified schedule, chessboard bounds, the logarithmic OPE package, coverage, weak comparison systems and extraction data (§4.1).
- **Fixed-volume ultraviolet limits.** On every fixed, sufficiently large torus, the correlation functions of the renormalized observables at separated points converge along the *whole* sequence of lattice spacings, together with their lattice derivatives. Couplings and normalized effective-action readings depend holomorphically on a terminal-coupling shift in a disc independent of the cutoff and period. Transfer data and correlations have the analytic extensions specified in III.13.7 (§4.2–4.4).

Conventions.

- G is a compact matrix group with simple nonabelian real Lie algebra ("G simple").
- $b > 0$ is the *penalty* of the block-spin kernel. It is the Gaussian block-spin coefficient in the linearized kernel identity. Its comparison with Balaban's notation is left to §7.
- **Levels.** RG levels are $m = 0, 1, \dots, N$. Level N (the *cutoff*) is the finest lattice, the Wilson end. Level 0 is the coarsest. The spacings are $a_0 = 1$ and $a_{m+1} = a_m / L_m$ with odd block sizes L_m, and $\sigma_m = \log(1/a_m)$.
- **Tori.** The period index is p. The torus $\Lambda_{N,p}$ has spacing a_N and side ℓ_p.

- **Couplings.** A Wilson family C has a selected RG trajectory with inverse couplings $\beta_m(N,p)$. Its Wilson coupling is $\beta_{N,p} = \beta_N(N,p)$, its value at level 0 at the centre of the shooting family is the *barrier* $\bar\beta$, and $\hat\beta_m$ is the reference running coupling.
- **Labels and observables.**
  - $\mathcal{L}_d$ is the space of labels of degree d, and $\mathrm{ref}(d)$ is the reference level of degree d.
  - $O_v^{N,p}(x)$ is the renormalized lattice observable of a label v at a site x (II.1), and $\langle\cdot\rangle_{N,p}$ is the Wilson expectation at $\beta_{N,p}$.
- **Separated configurations.** $\Omega_p^n = \{x \in (\mathbb{R}^4)^n : x_i \not\equiv x_j \bmod \ell_p\mathbb{Z}^4 \text{ for } i \neq j\}$. A test function is a complex-valued C^∞ function with compact support in the set of pairwise distinct configurations in $(\mathbb{R}^4)^n$. It is *admissible at period* p if its closed support lies in $(-\ell_p/2, \ell_p/2)^{4n}$; this support is then a compact subset of Ω_p^n.
- **Extraction orders.** An extraction order f is a pair of subsequences (mesh indices, period indices) (II.2), and $f \circ g$ is its refinement by g.

Lean conventions: `Test`, `Extraction.suitable`.

Definition (chessboard bound in the schedule). For $v \in \mathcal{L}_d$, levels $m \le N$ and $A \ge 0$, the *centered budget* holds if:

- $\mathrm{ref}(d) \le N$;
- for every site o of $\Lambda_{N,p}$, the observable $O_v^{N,p}(o)$ is supported in the reference cell of the level-m reflection-positivity tiling centred at o;
- its chessboard root $\left|\langle \prod_{\text{cells}} \text{reflected copies}\rangle_{N,p}\right|^{1/\sharp\text{cells}}$ is at most A.

The family has the **chessboard bound in the schedule** if for every d and all $R, \varepsilon > 0$ there is $A \ge 0$ such that, for every level $m \ge \mathrm{ref}(d)$ and all sufficiently large N, the centered budget holds with constant $A\, a_m^{-d-\varepsilon}$ for every period p and every $v \in \mathcal{L}_d$ with $\|v\| \le R$.

Lean: `Extraction.Wcmain`, `centeredBudget`.

Proposition III.13.1 (one family for the ultraviolet and logarithmic packages). Let G be simple and $b > 0$. There is a Wilson family C with the following properties.

1. Its kernel penalty is b.
2. C **is calibrated.** There are a full schedule of constants (sheet, block sizes, protection, delays, bar, wait, chart ratio, prices, post-cost) and a directed history whose trails generate the history of C. Every step from level $m+1$ to level m satisfies the per-step estimates of III.9–III.12.
3. C **is consistent.**
   - The trajectory is the selected one.
   - The local sections are sound.
   - For every d: $d+3 \le J(\mathrm{ref}(d))$, and $d+1 \le \mathrm{order}_m$ and $d \le \mathrm{power}_m$ for every $m \ge \mathrm{ref}(d)$.
   - Stencils fit: $20(|x_i|+1) \le \mathrm{side}(N,p)$ for every stencil point x of the degree-d local section, every coordinate i, and all N, p with $\mathrm{ref}(d) \le N$.
   - For each basis label, the placed local section is regular on its footprint and has the prescribed derivative bounds with size $\mathrm{columns}_N\, \theta_N^q$, where q is its factor degree and θ_N is the level-N threshold.
   - The transfer data are calibrated from $\mathrm{ref}(d)$ on.
4. C has the chessboard bound in the schedule.
5. C is *floral*: it is the centre of an analytic shooting family (Proposition III.13.7).

6. C satisfies the three entrance conditions of the OPE:
   - the shell estimates of III.11 for every d, D, h with $D + h = d + 4$: there are $t > 1$ and $j(i) \to \infty$ with $\sigma_{j(i)}/\sigma_i \to t$ for which the prescribed shell remainder bounds hold, for every choice of labels and extraction order;
   - slow growth of the cover sizes, $\mathrm{cs}_{i+1}^2 \le 2\,\mathrm{cs}_i^2$, with $(L_i\,\mathrm{cs}_i^2)^4 \cdot (\text{vacuum cost}) \le \varepsilon\,\sigma_{i+1}$ eventually;
   - a positive auxiliary radius sequence ρ_i satisfying $\log \rho_i/\sigma_i \to -1$.
7. There are coefficient-map bound data W, allowance indices $r : \mathbb{N} \to \mathbb{N}$ and one-step transfer data D for which two properties hold. Write K for the bound function supplied by W.
   - **The logarithmic OPE package.** This contains the transfer and remainder estimates of III.11, quantitative triangular label steps, auxiliary label comparisons, geometric separation bounds, and the sup bounds for products and the additional arity bounds of VII.4. It also gives $L_m \le \widehat{\beta}_m$ and $\mathrm{columns}_m$, $\mathrm{columns}_{m+1} \le \widehat{\beta}_m$; $r \to \infty$; and support exponent 100 at every level.
   - **Calibrated label steps.** For each d, N, p, m with $\mathrm{ref}(d) \le m < N$, there is a linear isomorphism T such that, writing x_{m+1}, x_m for the coefficient maps,

     $$\|x_m(v) - T(V_{L_m} x_{m+1}(v))\|_1 \le \mathrm{barunit}(L_m)\,\|\mathrm{core}(x_{m+1}(v))\|_1 \qquad (v \in \mathcal{L}_d),$$

     where V_L multiplies a component of engineering grade e by L^{d-e}. For an input basis vector of engineering grade e and factor degree q, its only output component of engineering grade $\le e$ is itself, multiplied by $(\theta_m/\theta_{m+1})^q$. No component of lower factor degree occurs. The map commutes with the core projection and lattice symmetries, preserves the unit, and satisfies $\|T\|_1, \|T^{-1}\|_1 \le K(J(m+1))$.
8. *Coverage* holds. There is $g(k) \to \infty$ with $g(k) \le k$ and $a_{g(k)} \le \widehat{\beta}_k\, a_k$ for large k such that, for every d and all large k, three probe panels of depth d are poised at level $g(k)$ (VII.1).
9. For every extraction order F satisfying the cell hypotheses (IV.1), the grid approximation property holds at arbitrarily fine fixed levels, for bounded label degrees and the reflection pairs allowed in III.11. Complete weak comparison systems exist for every finite demand and degree horizon, every positive radius bound, and every real-spin structure on the smooth free comparison table.
10. For every extraction order f there are:
    - a refinement g;
    - tables A, (L_k) and T, with a bin R and a derivative plan E, forming extraction base data along $f \circ g$ (IV.1);
    - a seed s for $(C, f \circ g, T)$ such that the Maxwell rank condition holds for T.

Status: theorem. The only hypotheses are G simple and $b > 0$.

Lean: `uv_log`. The predicates, in the order of the items: `Spec.Calibrated`, `Family.Makes`, `Oasis.Floral`, `Acme.Aroma`, `LogPackage`, `TrueSteps` with `CalibratedStep`, `Dense.Coverage`, `Org.Grill`, `East.CompleteWeak`, `Next.Extract.Base`, `Spec.Seed`, `Maxwell.Ranks`. The engineering rescaling and leading map are `Bid.vowel` and `Heading.leads`.

Proposition III.13.2 (the coordinated family). Let G be simple and $b > 0$. There is one Wilson family C with the following properties:

- it has kernel penalty b and is consistent, calibrated and floral;
- it has the chessboard bound in the schedule;
- it has the fixed-volume limits of III.13.5;

- it has the logarithmic OPE package and the calibrated label steps;
- it has coverage;
- it has complete weak comparison systems along every extraction order satisfying the cell hypotheses.

In addition, for **every** extraction order F there are a refinement g and a table T such that:

- (a) the ordered limit holds for T along the *full* mesh sequence and the period subsequence of $F \circ g$;
- (b) the OPE services of VII.4 hold for $(C, F \circ g, T)$;
- (c) a construction record of II.2 for $(C, F \circ g, T)$ exists.

Status: theorem. The family is chosen before F.

Lean: `coordination`, `Extraction.Order.Ordered`, `OPE.Facilities`, `Spec.Final`.

Proposition III.13.3 (the common family, all phases). Let G be simple, $b > 0$, and let f be an extraction order. There are a family C, a refinement g, a table T, a construction record F for $(C, f \circ g, T)$ and a reconstruction record $\mathcal{A}$ for every phase of F (V.3) such that:

- everything listed in III.13.2 holds for $(C, f \circ g, T)$;
- the reconstruction is *completed* for every phase: Wightman realization, surjective comparison maps to the Osterwalder–Schrader Hilbert spaces, unique vacuum, light-cone and heat-kernel bounds uniform over the phases, trivial commutant, the continued stress tensor, continuous dependence of boundary correlations on the phase, and the barycentre identities;
- the derivative sector of F has a prescription P giving the OPE estimates for all fixed finite cluster trees and the corresponding estimates with separated spectators (VII.4). These estimates allow an arbitrary positive power loss.

Status: theorem. In the statement C is quantified after f. The proof takes C from III.13.2, so in fact one family serves all f.

Lean: `common`, `EveryPhase.Completed`, `EveryPhase.Almanac`, `OPE.AllShapes`.

Proposition III.13.4 (headline: moving-frame Gram limits on the common family). Let G be simple, $b > 0$, and let f be an extraction order. There are $C, g, T, F, \mathcal{A}, P$ as in III.13.3 with all of its conclusions. In addition, coverage holds at the cutoff of F, and the following holds for every ultrafilter $\mathcal{U}$ on $\mathbb{N} \times \mathbb{N}$ finer than the filter of $f \circ g$ ("period index large, then mesh index large") and every depth D:

1. *Moving-frame portfolios* of depth D along $\mathcal{U}$ exist. A portfolio consists of:
   - levels $k' = k'(k) \to \infty$ with $k' \le k$ and $a_k \le a_{k'} \le \widehat{\beta}_k\, a_k$ for all sufficiently large k;
   - three probe panels poised at level k' for all sufficiently large k;
   - label matrices having limits along $\mathcal{U}$ for all sufficiently large k.
2. For every portfolio X there are linear isomorphisms $Y_k(d)$ from the degree-d field space of the free Maxwell table onto that of T ($d \le D$, $k \in \mathbb{N}$), with the following properties.
   - For all sufficiently large k, they respect the grading through depth D.
   - For each $d \le D$, the Gram forms expressed in these bases are bounded and coercive uniformly for all sufficiently large k.
   - For each $d \le D$, there is $q_d \in \mathbb{N}$ such that, for every sufficiently large k and every field x, there are representatives v and $v_{N,p} \to v$ along $\mathcal{U}$, projecting to x, whose source costs at level k are eventually at most $\widehat{\beta}_k^{q_d} a_k^d \|Y_k(d)^{-1} x\|$. Conversely, there is $K_d \ge 0$ such

that every convergent family of representatives with source cost at most z along any nontrivial filter finer than $\mathcal{U}$ satisfies $\|Y_k(d)^{-1}x\| \le K_d a_k^{-d} z$. Both assertions hold for all sufficiently large k.

- For every $d \le D$, the Gram form of X at step k, read in the basis $Y_k(d)$, converges as $k \to \infty$ to the Gram form of the free Maxwell model in degree d.

Status: theorem; headline declaration. In the statement C is quantified after f, as in III.13.3.

Lean: `habitat`, `Weak.Portfolio`, `PricedIn`. *(proof of the wrapper read)*

Proposition III.13.5 (fixed-volume limit with k derivatives). Let C be as in III.13.1, and fix $d, k \in \mathbb{N}$. For all sufficiently large periods p (the threshold depends on d and k), every n, and all labels $v_1, \dots, v_n$ of degrees $\le d$, there is a function g on $(\mathbb{R}^4)^n$ such that, as $N \to \infty$ along the whole sequence:

1. g is continuous on Ω_p^n. The lattice correlation functions $x \mapsto \langle \prod_i O_{v_i}^{N,p}(x_i) \rangle_{N,p}$, extended periodically and then as step functions using coordinatewise ceiling projection onto $(a_N \mathbb{Z}^4)^n$, converge to g uniformly on compact subsets of Ω_p^n.
2. For every word u of at most k coordinate directions, the step-function extensions of the normalized forward differences converge to $\partial_u g$ uniformly on compact subsets of Ω_p^n. The same limit is obtained by multilinear interpolation of any fixed word of normalized coordinate differences with positive integer step lengths and fixed integer stencil offsets, including forward, backward and centred differences.
3. On Ω_p^n these iterated directional derivatives exist: $\partial_a(\partial_u g) = \partial_{(a,u)} g$ whenever the word u has length $< k$.
4. For every test function t admissible at period p, tg is integrable and $a_N^{4n} \sum_r t(r) \langle \prod_i O_{v_i}^{N,p}(r_i) \rangle_{N,p} \to \int tg$, where the sum is over the centred representatives of the torus lattice.

Status: theorem.

Lean: `Poppy.Margins`, `Poppy.bravo`, `Poppy.blossoms`.

Proposition III.13.6 (convergence at separated lattice points). Let $C^{\mathbf{s}}$ be a member of the shooting family of III.13.7. At a fixed period p, assume:

- its shifts converge, $s_{N,p} \to s$ as $N \to \infty$, with $|s| < r_*$;
- $C^{\mathbf{s}}$ has the chessboard bound in the schedule;
- the slow-growth, label-transfer and remainder bounds hold (these are supplied by the bundle of III.13.7 for every assignment of shifts).

Let $J \in \mathbb{N}$, let $u_1, \dots, u_I$ be labels with $\mathrm{ref}(\deg u_i) \le J$, and let $o_1, \dots, o_I$ be pairwise distinct sites of the level-J lattice of the period-p torus. Then the sequence

$$\Psi_N = \Big\langle \prod_{i=1}^{I} O_{u_i}^{N,p}(o_i) \Big\rangle_{N,p},$$

with o_i read as sites of $\Lambda_{N,p}$, converges as $N \to \infty$. *Status:* theorem. For the family of III.13.1, $\mathbf{s} \equiv 0$.

Lean: `Hearth.food`.

Proposition III.13.7 (the family is the centre of an analytic shooting family). For the family C of III.13.1 there are:

- a schedule with its per-step data;
- a shooting structure with an arc;
- a label-transfer bundle;
- reference levels ref;

such that the following holds.

1. *Shooting.* There is $r_* > 0$, depending only on the schedule, such that for all N, p and every real s with $|s| < r_*$ there is exactly one Wilson coupling whose trajectory under the fixed recursion stays in the prescribed scalar coupling bands and reaches $\bar{\beta} + s$ at level 0. For an assignment $\mathbf{s} = (s_{N,p})$ of real shifts with $|s_{N,p}| < r_*$, let $C^{\mathbf{s}}$ be the Wilson family with these trajectories and the labels, sections and reference levels of the bundle. Then $C = C^{\mathbf{0}}$.
2. *Analyticity, uniform in the cutoff.* For $m \le N$, the level-m coupling $\beta_m(N, p; s)$ and the normalized activity readings of the level-m effective action extend holomorphically to $|s| < r_*$. They are bounded there by $3\hat{\beta}_m$ and by soft $=$ rim/8 respectively. For fixed m, p they converge as $N \to \infty$, with all derivatives in s, locally uniformly on the disc.
3. *Bounds.* C satisfies the calibration package (with constant allowance) and the priced sup bounds. For every assignment $\mathbf{s}$, $C^{\mathbf{s}}$ satisfies the label-transfer and remainder bounds and has the chessboard bound in the schedule. Extraction base data exist at the centre.
4. *Analytic transfer data.* There is $a > 0$, with its disc contained in the shooting domain, on which the bundle has holomorphic coefficient and transfer data for all d, N, p with $\mathrm{ref}(d) \le N$. In particular, the inverse terminal label maps extend holomorphically there.
5. *Analytic correlations.* Fix d, m. For all large p, every n and labels of degree $\le d$, there are $b' > 0$ and functions $W_N(z, x)$, holomorphic in $|z| < b'$, and $f(z, \cdot)$ on $(\mathbb{R}^4)^n$ such that:
   - $W_N(s, \cdot)$ is the lattice correlation function of $C^{\mathbf{s}}$ whenever $s_{N,p} = s$ (for every N with $\mathrm{ref}(\deg v_i) \le N$ for all i);
   - for every complex z in the disc, $f(z, \cdot)$ has continuous iterated coordinate derivatives through order m on Ω_p^n. The step-function differences and interpolated differences converge to these derivatives, with all derivatives in z, uniformly on products of compact subsets of the disc and Ω_p^n. At every real shift these limits agree with the fixed-volume limit of the constant-shift family;
   - for admissible t, $z \mapsto \int t\, f(z, \cdot)$ is holomorphic, and the smeared lattice correlations converge to it, with all derivatives in z, locally uniformly.

Status: theorem (part of III.13.1). The full-mesh ordered limits in §4.5 use the real shooting trajectories and the bounds of item 3; they do not use the analytic correlation limits of item 5.

Lean: `Oasis.Floral`, `Oasis.Spring`, `Upland.Arc`, `Mast.Cellar`, `Cellar.yard`, `Harvest.Barn`, `Gale.Circuit`, `Beach.Shoal`, `Beach.Realizes`. *(definitions and producing proofs read)*

3.16.2 2. What it uses

Inputs from other units.

- **I.1.** The Wilson measure and partition function, and the one-loop slope $\kappa_G = 11\, C_G/(24\pi^2)$ (C_G the Casimir).
- **II.1, II.2.** Labels, local sections, renormalized observables; tables, extraction orders, seeds and construction records.
- **III.1.** Kernel choices with penalty b for which every patch branches (unique minimizing frames and predictors, a non-empty axial fallback): `Axial.viable`, `KernelChoices`.

- **III.2.** The order of choice. The pre-L constants form the sheet; then come the block sizes, the wages, the protection θ, the delays, the bar and the waits, in this order.
- **III.4 and III.8.** The relay supply `posters`, and the chessboard product estimate `GridApprox.tariff`.
- **III.7.** The one-loop annulus calculation. `HorizCalc` holds for every simple G by `Stationary.pouring` through `Fluent`. This is where the coefficient of $\log L$ in the marginal flow comes from (Lemma 4.3).
- **III.9–III.12.** The per-step estimates, the selected trajectory and the band, and the directed histories, alloys and factors. They enter through the step package `uv_commerce` of this unit.
- **III.11.**
  - The calibration package `Chateau`. It includes a certified schedule, consistency, transfer and remainder estimates, the chessboard bound in the schedule, and extraction base data, seeds and Maxwell ranks after refinement of every extraction order.
  - The priced sup bounds `Velour`.
  - The OPE entrance `Acme.emanate`.
  - The weak comparison systems `sponsored`.
  - The cost bound `Ribbons.laurelE`.
- **III.15, III.16.** Cost accounting for labels, and analytic tools.
- **IV.1.** The chessboard bound in the schedule (as a definition), the cell hypotheses, extraction base data, the Maxwell ranks, and the Arzelà–Ascoli-type lemma `Extraction.growFrom`.
- **V.3.** Reconstruction for every phase: `all_characters`.
- **VII.1.**
  - Coverage, through `Dense.covered`.
  - The slim-window port `slim_wait_for`.
  - The calibrated label steps.
  - The portfolios `fromCoverage`.
- **VII.4.**
  - The logarithmic package and the OPE services.
  - The prescriptions `Prescription.chosen` and the shapes.
  - The delay raises `OPE.halo_delay` and `OPE.Log.aux_delay`.
- **VIII.1.** The construction record and `Acme.glistens`, which turns the UV package and the calibration into a record.

Standard mathematics used.

- **Balaban's block-spin renormalization** for lattice gauge theory (Balaban, CMP 95–122, 1984–89; Dimock, *The renormalization group according to Balaban I–III*). The convergence statements of §4.2–4.4 use the per-step estimates together with stability of the chosen recursion.
- **Shooting by contraction** (Banach fixed point), uniform in the cutoff.
- **Vitali's convergence theorem** (Vitali–Porter). Locally bounded holomorphic functions that converge on a real segment converge locally uniformly, with all derivatives. The Lean proof maps into $\ell^\infty(\mathbb{N}; Y)$ and applies the identity principle to the closed subspace of convergent sequences.
- **Arzelà–Ascoli**, in the form: asymptotic equicontinuity on compacts together with convergence on a dense set gives locally uniform convergence.
- **Dominated convergence.**
- **Chessboard estimates** for reflection-positive measures (Fröhlich–Israel–Lieb–Simon 1978;

Glimm–Jaffe, ch. 10).
- **Ultrafilters.**

Imported constants and assumptions. The statements of III.13.1–III.13.4 have no hypotheses beyond G simple and $b > 0$. The one-loop slope κ_G enters the coupling-shift estimate (Lemma 4.3) through the hypothesis the annulus calculation, which III.7 proves for every simple G. The construction fixes the following constants:

- the stage data: first stage 4, Taylor order $6 + i$, power $2 + i$, and support exponent 100 at level i;
- the allowances $Q \equiv 1$ (for the floral property) and $Q_2(j) = j + 1$ (for the OPE iteration);
- auxiliary data $(40, 1, 1)$;
- a bar parameter 2;
- the target rate $-7/16$ of the factor.

The analytic argument uses soft = rim/8 < 1, rad = rim/4 and the gain $e^{(\sigma_k - \sigma_m)/8}$.

3.16.3 3. Strategy

The order of choice. Each estimate of III.9–III.12, and each requirement of the later parts (OPE with logarithmic corrections, coverage, weak comparison), has the form "there is a threshold beyond which the property holds". It is also stable under further raising: in the preorders on delays, waits and wages, and under decreasing the protection. Successive raises meet the finitely many requirements while preserving this order: sheet → block sizes → wages → protection → delays → bar → wait.

Increasing stage parameters (Taylor order $6 + i$, power $2 + i$) accommodate the countably many label degrees: degree d is required only from its fixed reference level ref(d) onwards.

Shooting and convergence at a fixed level.

1. The family is the member $s = 0$ of a one-parameter family in which the terminal coupling at level 0 is $\bar{\beta} + s$. For each cutoff N the trajectory is found by shooting: it is the fixed point of a contraction with rate $1/2$, uniformly in N.
2. This uniformity, the convergence of the background fields and coupling shifts at each level (Lemma 4.7), and an soft^v decay of the influence of levels far above m together give convergence, as $N \to \infty$ along the whole sequence, of the coupling and the effective action at any fixed level m, for every real s in the shooting disc (Lemma 4.8).
3. Holomorphic extension in s to a disc independent of N and p, with uniform bounds, and Vitali's theorem upgrade this to locally uniform convergence of all derivatives in s on the disc (Lemma 4.9).

From the effective action to correlations. A correlation of renormalized observables at distinct points is handled in three steps.

1. **RG identity.** For a function of the level-k variables, its conditional average through the block-spin kernels has Wilson expectation equal to its expectation in the level-k effective measure. That measure converges as $N \to \infty$ (Lemmas 4.11–4.12).
2. **Coarse approximation.** Each fine renormalized observable is close to the conditional average of a level-k observable whose coefficients converge as $N \to \infty$. The error is measured

in the chessboard norm and is small for k large, because the gain $e^{(\sigma_k-\sigma_m)/8}$ beats the subexponential growth of the costs (Lemma 4.13).

3. **Product estimate.** The chessboard product inequality turns these one-point approximations into an approximation of the product (Lemma 4.14).

A Cauchy argument then gives Proposition III.13.6. Proposition III.13.5 follows by Arzelà–Ascoli: there is convergence on the dense set of grid points of all levels, and there is equicontinuity from uniform bounds on lattice differences of order $k+1$. Those bounds need a non-degeneracy ("slicing") of the label snapshots, which is known only for large periods, so the fixed-volume statement holds for p large. The reports then feed this family to the OPE entrance, reconstruction and Gram machinery defined in other units.

Balaban's estimates are used only through the per-step package; what this unit adds is the analysis just described and the order of choice.

3.16.4 4. Steps

3.16.4.1 4.1 Choosing the constants once Lemma 4.1 (the step package with all later constants raised). Let G be simple. Fix the purchased sheet, prices, numerical bounds, scope, tariffs and block-size floor of Lemma 4.2, and assume that, for every stage pair $i \le j \le i+1$ with i beyond the first stage and every $d+3 \le i$, the local-section and nonconstant-label expansion costs are bounded by the supplied budget at stage j. Let $PB, P_2 : \mathbb{N} \to \mathbb{N}$ be arbitrary allowance sequences. There is a floor $f \ge v_0$ on the block sizes such that, for every block-size sequence $L \ge f$, the following holds.

- Block-size rolls D exist.
- For every roll D there is a wage threshold U. For every wage $w \ge U$, postage data, exchange data and insertion tariff, there is a post-cost P.
- For every sequence $\theta_0 : \mathbb{N} \to \mathbb{R}$ with $\theta_0(i) > 0$ for all i, there is a protection sequence $\theta \le \theta_0$.
- For all menus, auxiliary data and delays T_0 there is an ordered delay $V \ge T_0$ such that the following holds for every delay $Y \ge V$, every bar, every wait I and every level system a paid by I whose spacing schedule has the divisor property. There are a directed history B on the given level system a and a set of per-step data R for this schedule, with the prescribed wages, exchange, menu and auxiliary data, postage rates bounded by the requested rates, the prepared insertion data, and:
  - label-transfer bounds and connected-activity bounds for iteration;
  - target rates $\le -7/16$;
  - repeated-insertion bounds for both allowances;
  - startup and smallness data for shooting;
  - continuity bounds for a single RG step;
  - fixed-level limits of the kinematic coupling shifts and paths;
  - covariance comparison data.

Sketch: The per-step estimates are supplied in the required order of choice. Successive delay raises add the later requirements. Each raise is a lemma "there is $V' \ge V$ such that every $Y \ge V'$ has the property". Raise the entry-coupling threshold pointwise by taking its maximum with the shooting threshold. Every subsequent delay increase preserves the preceding estimates.

Lean: `multitude`, `uv_commerce`. *(proof of* `multitude` *read;* `uv_commerce` *read only as far as its call to* `Roof.uv_timberT`*)*

Lemma 4.2 (the purchased sheet). Let G be simple, K kernel choices for which every patch branches, and v stage data satisfying the following conditions:

- first stage ≥ 4;
- Taylor orders monotone and $\geq \max(6, i)$;
- powers monotone and $\geq \max(2, i)$;
- support exponents monotone and positive.

For every two allowance sequences there are a sheet with kernel K and stage data v, prices, a budget sequence and a bound W, chosen before the tariffs and scopes. For every tariff and scope there is then a block-size floor above the sheet floor such that Lemma 4.1 applies to every larger block-size sequence.

Sketch: Choose the sheet using three inputs:

- the kinematic constant issuer: pre-L constants chosen so that, once L is revealed, only waiting is needed and the target estimates hold;
- the horizon theorem: horizon data for every sheet with kernel K;
- the kinematic convergence of Lemma 4.7.

Then apply Lemma 4.1.

Lean: `multitude_foot`, `sailsV`, `office`, `profiles`. *(proof of* `multitude_foot` *read; the others statement only)*

Lemma 4.3 (the one-loop running of the marginal coupling). Let G be simple. Fix kernel choices, a stage J, a colour direction, positive tolerances ε, w, δ, and nonnegative D, Z, C, Z'. There are $\kappa \geq 0$ and $s > 0$ with the following property. For all $\tau, \mathrm{pace} > 0$ and $A \geq 0$ satisfying the annulus calculation of III.7 with parameters $(G, w, \varepsilon, D, Z, \tau, A)$, there are:

- $d \geq 0$ and $\gamma > 0$;
- an integer u;
- functions $\beta(L)$ and $\Lambda_0(L)$;

such that every sequence of step data with stage J, block size $L \geq u$, diverging coarse side lengths, and the comparison hypotheses of III.7 admits the following data. Those hypotheses include input and output covariance strip bounds, input path bounds, admissible forests and charts, spectral gap and criticality estimates, heat bounds, direct clip bounds Z, Z', a positive gap parameter and amplitude $\Lambda \geq 1$. The output consists of:

- background path data Y;
- a number α, the shift of the marginal coupling read from the actual quadratic form of the step in a colour direction, with

$$|\alpha + \kappa_G \log L| + 1 \leq d;$$

- equality of the classical Taylor coefficients through order 4, and a finite-volume thermal comparison estimate. For every $\Theta \geq \beta(L)$ and $c' \leq \gamma$, on a coarse torus with ν sites in each coordinate, the second derivatives differ by at most $\nu^4 \Theta e^{-c'\nu}$ on the prescribed unit-bounded wave probes. The output path has its announced bounds whenever $0 < \eta' \leq s$, $A' \geq \kappa$ and $\Lambda' \geq \max(\Lambda_0(L), \mathrm{pace}\,\Lambda)$.

The annulus calculation supplies a comparison quadratic form whose coupling shift differs from $-\kappa_G \log L$ by a constant independent of L. Its conclusion applies to all odd L beyond a threshold under the input path, covariance, floor and compatibility assumptions of III.7.

Sketch: α is compared with the coupling shift of the annulus comparison form, and the two bounds are added. The constant d is chosen before L, so the error is bounded uniformly in L.

Lean: `ceilings`, `colorReady`, `HorizCalc`. *(the part of the proof of* `ceilings` *that bounds* α *read;* `colorReady` *statement only)*

Lemma 4.4 (the bundle and the joint offers). Let G be simple, K kernel choices for which every patch branches, v satisfying the stage conditions of Lemma 4.2 and having support exponents ≥ 100, and $Q, Q_2 : \mathbb{N} \to \mathbb{N}$ monotone with values ≥ 1. There are:

- a sheet with kernel K and stage data v;
- prices, a budget sequence and a bound K';
- a floor $\tau \geq \mathrm{floor}_L$;

such that for every block-size sequence $k \geq \tau$ rolls exist, and the following holds through the quantifier chain of Lemma 4.1. In that chain the waits receive one more raise Z, of readiness, minimal length and window. Every level system paid by this raised wait and satisfying the divisor property then carries:

- a directed history and a factor;
- a shooting structure $\mathcal{S}$ with an arc;
- a label-transfer bundle over $\mathcal{S}$;

and the family $C^{\mathbf{0}}$ built from the bundle has the following properties:

- the calibration package for both Q and Q_2;
- slow growth of the cover sizes;
- the priced sup bounds (three variants);
- the bundle conditions of III.13.7.3;
- calibrated label steps;
- the leases, and target rates $\leq -7/16$.

In addition:

- there is a radius $r > 0$ such that the disc of radius r lies in the analytic domain and the transfer data extend holomorphically to it for all d, N, p with $\mathrm{ref}(d) \leq N$;
- complete weak comparison systems exist for every extraction order.

Sketch: Lemma 4.2 supplies the schedule. Increase the block-size floor to ensure $\mathrm{rate}(L)(1 + 2\,\mathrm{columns})\,\mathrm{orbitCost} \leq \mathrm{barunit}(L)$, then construct the bundle. The relay supply and a further wait increase give weak comparison systems. The leases and target rates give the common disc for holomorphic transfer.

Lean: `havens_log`, `jointOffers`, `Gale.sheaf`. *(proof of* `jointOffers` *read; statement and opening of* `havens_log` *read)*

Lemma 4.5 (instantiation). Proposition III.13.1 follows from Lemma 4.4 with the following choices.

- **Kernel.** The kernel choices have penalty b (III.1). The stage data are $(4,\ 6+i,\ 2+i,\ 100)$, and the allowances are $Q \equiv 1$, $Q_2(j) = j+1$.
- **Block sizes.** Take the maximum of the floor of Lemma 4.4 and the relay floor.
- **Post-L data.** The menu is canonical, the auxiliary data are $(40, 1, 1)$, the wage is the threshold itself, and $\theta_0 \equiv 1$.
- **Delays.** Zero delays are raised by the geometric delay bound and the logarithmic delay bound; the latter gives L_m, columns $\leq \widehat{\beta}_m$.
- **Bar and wait.** A bar above the delays with parameter 2 is chosen. A wait with slim window is thickened by the slim-window theorem so that it carries a certified station port, then joined with the wait of Lemma 4.4. The schedule existence theorem gives a paid level system.

Then:

- coverage follows from the certified station port;
- floral from Lemma 4.18;
- the OPE entrance conditions from the shell estimate, the slow growth of cover sizes and the radius asymptotics;
- the logarithmic package is assembled from the label-transfer and remainder bounds, the priced bounds, the calibrated steps, the budget, $r \to \infty$ (from the depth of the fits) and support exponent 100.

Lean: `uv_log`, `Axial.viable`.

3.16.4.2 4.2 Shooting, and the limit at a fixed level **Lemma 4.6 (shooting).** Let $r_* > 0$ be the radius of the arc; it depends only on the schedule. For all N, p and real $|s| < r_*$:

- there is exactly one $\beta \in \mathbb{R}$ for which the fixed startup and step recursion has a trajectory staying in all prescribed scalar bands and satisfying $\beta_N = \beta$, $\beta_0 = \bar{\beta} + s$;
- the realized trajectory satisfies the band and step conditions at every level, and its level-0 coupling is $\bar{\beta} + s$;
- β is the value at s of the holomorphic function $\beta_N(N, p; \cdot)$.

Lean: `Arc.exists_trial`, `Oasis.spring`, `Spire.Uses`. *(the wrappers read; existence and uniqueness come from the fitted structure, statement only)*

Lemma 4.7 (kinematic convergence at each level). Fix a paid level system, the horizon and chart data, a colour direction, the small positive pulse parameter, the one-step kinematic estimates and the limiting covariance data supplied by Lemma 4.2. There are background paths and coupling shifts $\alpha_N(m)$ satisfying all kinematic step estimates such that, for every level m:

- $\alpha_N(m)$ converges as $N \to \infty$;
- there are a limiting background path and w_m strictly larger than the level-m width such that, for every $u < w_m$, the path coefficients converge in the exponentially weighted sum norm of width u. Their path supports agree with those of the limit for all sufficiently large N.

Lean: `Reward.work item`, `Reward.turning`. *(proof of `work item` read; `turning` statement only)*

Lemma 4.8 (ultraviolet convergence at a fixed level, real shifts). Fix the shooting structure and assume its one-step continuity property and the kinematic convergence of Lemma

4.7; both are supplied by the arc. For every real s in the analytic domain, every level m and every period p, two quantities converge as $N \to \infty$ along the whole sequence: the effective activities at level m, and the level-m coupling data of the trajectory ending at $\bar{\beta} + s$.

Sketch:

1. **Contraction.** The trajectory with cutoff N is the fixed point of a map on coupling sequences. Its k-th iterate from 0 lies within $\mathrm{rad} \cdot 2^{1-k}$ of the fixed point, uniformly in N.
2. **Stability.** If the level-$(m+1)$ data converge as $N \to \infty$, so do the level-m data and the claimed coupling shift of the step. By induction on k, every iterate therefore converges level by level. The coupling shifts converge by Lemma 4.7.
3. **Decay of far-ultraviolet influence.** The level-m activities depend on the levels above $m + v$ only up to soft^v with $\mathrm{soft} < 1$, uniformly in N.
4. **Conclusion.** An $\varepsilon/4$ argument closes each step: in a complete space, a sequence that can be approximated uniformly by convergent sequences converges.

Lean: `Camp.artistry`, `Camp.climb`, `Camp.income`, `Camp.Sign`, `Camp.trim`. *(proof read; the contraction rate and the* soft^v *decay are taken from statements)*

Lemma 4.9 (Vitali, and analyticity uniform in the cutoff). Let $r > 0$ and let F_N be holomorphic on $|z| < r$ with values in a complex Banach space, uniformly bounded, and convergent at every real $z \in (-r, r)$. Then there is a holomorphic f on the disc such that every iterated derivative of F_N converges to that of f, locally uniformly. The lemma applies to $\beta_m(N, p; \cdot)$ (bounded by $3\widehat{\beta}_m$) and to the normalized level-m activity readings (bounded by soft). With Lemma 4.8 this gives III.13.7.2.

Lean: `Camp.viva`, `Camp.SummitUV`, `Arc.uv`.

3.16.4.3 4.3 Renormalized observables Lemma 4.10 (label transfer). Fix the shooting structure, coefficient bundle and reference levels of Lemma 4.4.

- **Coefficient data.** For each degree d, period p, level i, cutoff N and shift s, the bundle provides the coefficient data of the degree-d renormalized observables at level i.
- **Bounds.** For every assignment of shifts, degree d and level $m \geq \mathrm{ref}(d)$, the coefficient histories satisfy the supplied transfer and remainder bounds. Also $d + 3 \leq J(\mathrm{ref}(d))$.
- **Continuity in the shift.** If the shifts at period p converge to s, the level-i data converge for $i \geq \mathrm{ref}(d)$.
- **Transfer data.** The transfer data of $C^{\mathbf{s}}$ are read from these coefficient data and are calibrated from $\mathrm{ref}(d)$ on.
- **Composition.** In a selected history with sound local sections, bounds on the coefficient maps and their inverses, and compatible one-step transfer estimates at every $r \leq m < N$, a valid top coefficient at level N extends to a multiscale history down to r. It retains that top coefficient and satisfies the one-step transfer, remainder and matching identities. The one-step construction transports a prescribed coefficient; downward induction composes it across the required levels.

Lean: `Mast.Cellar`, `Cellar.gate`, `Lacing.compose`, `repriseQ`, `acclaim`, `antilles`. *(`Cellar` definition read; the Lacing lemmas statement only)*

3.16.4.4 4.4 Correlations at fixed volume Lemma 4.11 (the RG identity). Let the history be selected, $m \le N$, and F a bounded measurable function of the level-m configuration. Let $B_{N\to m}F$ be its conditional average obtained by iterating the block-spin kernels from level m to level N. Then

$$\langle B_{N\to m}F\rangle_{N,p} = \frac{\int \rho_{m,N,p}(U)F(U)\,d\mathrm{Haar}(U)}{\int \rho_{m,N,p}(U)\,d\mathrm{Haar}(U)},$$

where $\rho_{m,N,p}$ is the effective density with its field-independent scalar factor removed.

Sketch: Iterating the one-step density identity gives the effective measure at level m. Duality with the conditional averaging kernels and equality of partition functions give the formula; the removed scalar factor cancels in the ratio.

Lean: `Hearth.scaledK`, `Hearth.tasting`, `GridApprox.back`, `GridApprox.backTo`.

Lemma 4.12 (convergence of the effective measure at a fixed level). Fix the arc and its coefficient bundle. At a fixed period p, let the shifts satisfy $s_{N,p} \to s$ with s real in the disc. If F_N are bounded measurable functions of the level-m configuration and form a uniformly Cauchy sequence, then the level-m effective expectations of F_N converge as $N \to \infty$.

Sketch: The effective densities with their scalar factors removed converge uniformly, by stability of the trajectory family. Their integrals have a uniform positive lower bound. The effective expectations are ratios of convergent integrals.

Lean: `Hearth.arrival`, `Hearth.seating`, `strippedOn`, `zPositive`, `Mast.signs`. *(proof of* `arrival` *read; the three inputs statement only)*

Lemma 4.13 (coarse approximation of renormalized observables). Fix the arc and coefficient bundle. At period p, assume the shifts converge to a real shift in the shooting disc, and assume the transfer and remainder estimates and slow-growth bounds of III.13.6. Let $J \le m \le k$, with $\mathrm{cs}_m^2 \le a_J/a_m$. Require enough separation for the supports of the degree-d coarse observables: $2(a_k r_d + \mathrm{tail}_k) < a_J$, where r_d is their fixed support radius and $\mathrm{tail}_k = 30a_k(J(k)+5)$ bounds the reach of the later kernels. Let u be a degree-d label with $\mathrm{ref}(d) \le m$, and let $w > 0$ satisfy

$$w\,a_m^{-d}\cdot 4(1+\mathrm{columns}_k)\,e^{-(\sigma_k-\sigma_m)/8} \le \tfrac{1}{2}\,\mathrm{fly}_{d,m}.$$

Then there are level-k observables $g_N = \sum_j c_j(N)\,\Lambda_j$ with the following properties:

- the Λ_j are fixed level-k observables, and the coefficients $c_j(N)$ converge as $N \to \infty$;
- for large N, the chessboard root, at level J, of $B_{N\to k}g_N - O_u^{N,p}(0)$ is at most $\mathrm{stock}_k(u)\cdot \mathrm{stakes}(J,m)/w$.

For fixed d, m, J, u and every $\zeta, \varepsilon > 0$, all sufficiently large k admit $w > 0$ with $wa_m^{-d}4(1+\mathrm{columns}_k)e^{-(\sigma_k-\sigma_m)/8} \le \zeta/2$ and $\mathrm{stock}_k(u)\mathrm{stakes}(J,m)/w \le \varepsilon$. Apply this with $\zeta = \mathrm{fly}_{d,m}$. The product of the two requirements tends to 0, because columns and stock grow subexponentially in σ_k while $e^{(\sigma_k-\sigma_m)/8}$ grows exponentially.

Lean: `Hearth.peeling`, `Hearth.seatingM`. *(proofs read; the level-k representation itself is statement only)*

Lemma 4.14 (chessboard product estimate). Assume the Wilson measure satisfies the reflection-positivity and translation properties for the level-h tiling. Let O_i, Z_i be bounded

measurable observables supported in its reference cell, with chessboard roots $\|O_i\| \le c_i$ and $\|Z_i - O_i\| \le e_i$, and let a_i be distinct sites of the level-h lattice. Then

$$\Big|\Big\langle \prod_i Z_i(\cdot + a_i)\Big\rangle - \Big\langle \prod_i O_i(\cdot + a_i)\Big\rangle\Big| \le \prod_i (e_i + c_i) - \prod_i c_i.$$

Sketch: Expand $Z_i = O_i + (Z_i - O_i)$ and subtract the term containing only the O_i. Apply the chessboard inequality to each remaining product on distinct cells; summing gives the displayed bound.

Lean: `GridApprox.tariff`, applied in `Hearth.banquet`. *(`tariff` statement only; `banquet` proof read)*

Proof of Proposition III.13.6.

1. The chessboard bound in the schedule gives, for each i, a bound t_i on the chessboard root of O_{u_i} at level J.
2. Fix $q > 0$. Choose k large and weights w_i by Lemma 4.13, so that each error is $\le q$. The support separation makes conditional averaging commute with the product of translated coarse observables. The RG identity (Lemma 4.11) and the chessboard product estimate (Lemma 4.14) then give a sequence F_N such that, for large N, $|F_N - \Psi_N| \le \Phi(q) := \prod(q + t_i) - \prod t_i$. Here F_N is the level-k effective expectation of the product of the translated $g_N^{(i)}$, and it converges by Lemma 4.12.
3. Since Φ is continuous with $\Phi(0) = 0$, one can choose q with $\Phi(q) < \varepsilon/4$. The triangle inequality through two terms of the convergent approximating sequence shows that (Ψ_N) is Cauchy.

Lean: `Hearth.food`, `Hearth.closet`.

Lemma 4.15 (slicing for large periods).

Definition. *Slicing*(d, p) holds if there are:

- finitely many test shots, admissible at period p;
- a constant $a > 0$;
- a subspace $U \subset \mathcal{L}_d$ of codimension charge(G, d), the dimension of the degree-d Maxwell equation ideal;

such that, for large N, $a\|x\| \le \sum_i \|\mathrm{snapshot}_i(N, p)x\|$ for $x \in U$.

Statement. Assume a fixed presentation identifying the finite-cutoff kernels of the label projections with the transported Maxwell equation ideals. Suppose every admissible snapshot at every fixed period converges, and every extraction order has a refinement with base data whose projections agree with this presentation. Then Slicing(d, p) holds for all large p. For the bundle of III.13.7 the conclusion takes this form: for each d and all large p there is $\delta > 0$ such that Slicing holds for every assignment of shifts with $|s_{N,p}| \le \delta$ eventually.

Sketch: By contradiction. A sequence of bad periods defines an extraction order. Along a refinement, finite detection on the range of the limiting projection gives Slicing at all large indices.

Lean: `Harvest.rootwards`, `Harvest.Slicing`, `Barn.ripe`.

Lemma 4.16 (proof of Proposition III.13.5). Fix d, k and p beyond the slicing threshold for the degrees up to $d + k + 1$. Use the arc and bundle hypotheses of Lemma 4.4, zero shifts, and labels of degree $\leq d$. Then:

1. **Dense set.** On the points of Ω_p^n whose coordinates lie on the level-m grid for some $m \geq \max \mathrm{ref}$ (a dense set), the lattice correlations converge by III.13.6.
2. **Uniform difference bounds.** Slicing and the label-transfer bounds give bounds on lattice differences of order $\leq k + 1$, uniform in N on compact subsets of Ω_p^n. Hence the correlations and their differences of order $\leq k$ are asymptotically equicontinuous there.
3. **Arzelà–Ascoli.** It gives locally uniform convergence to a continuous g, and to continuous limits of the difference quotients. These limits are the iterated derivatives of g.
4. **Smeared limits.** For admissible tests, the smeared correlations converge by dominated convergence, with domination by the uniform period-p bound.

Lean: `Poppy.blossoms`, `Poppy.coveringR`, `Extraction.growFrom`, `Poppy.nourishing`, `Poppy.mature`; difference bounds: `Fallow.reaped`. *(proofs read through the stated transfer estimates)*

Lemma 4.17 (analytic dependence of the correlations on the shift). Assume the arc, bundle bounds, paid schedule, analytic transfer disc, leases and target rate bound of Lemma 4.18. Fix d, m, a period p with slicing stable for all degrees through $d + m + 1$ for sufficiently small shifts, and finitely many labels of degree $\leq d$. There is $b' > 0$ such that, for all sufficiently large N, the lattice correlations extend holomorphically to $|z| < b'$. On every compact subset of Ω_p^n, their lattice differences through order m have bounds uniform in N and z throughout this disc. At arbitrary lattice configurations the bound supplied at each cutoff may depend on N.

Sketch: Slicing gives uniform real difference bounds through one higher order. The analytic transfer estimates and complex correlation bounds permit the Banach-valued Vitali argument on each separated compact set. The resulting limits have the spatial derivatives and smeared limits of III.13.7.5. Shrinking the disc once more incorporates every cutoff at which all reference levels fit. At each fixed level and period, the normalized effective densities have bounded holomorphic extensions; after shrinking the disc, the moduli of their partition integrals are bounded away from zero uniformly over the allowed cutoffs.

Lean: `Beach.inlet`, `Beach.sparksP`, `Beach.keep_prefix`, `Pier.roomR`, `Pier.mounted`. *(proofs of `sparksP` and `keep_prefix` read; `inlet`, `roomR` and `mounted` statement only)*

3.16.4.5 4.5 Reports Lemma 4.18 (the centre is floral).

Assume:

- an arc and the bundle conditions;
- a paid family;
- a disc in the analytic domain, with holomorphic transfer data for all d, N, p with $\mathrm{ref}(d) \leq N$;
- the leases and target rates $\leq -7/16$;
- the calibration package with $Q \equiv 1$, and the priced sup bounds.

Then the member $s = 0$ is floral.

Sketch: The shooting and analyticity properties come from the arc (Lemmas 4.6, 4.9). For large p, the analytic correlation data come from Lemma 4.17.

Lean: `Oasis.survives`.

Lemma 4.19 (mesh limits along the full sequence). For a floral family C, an extraction order f and a table T with a construction record for (C, f, T), the order (full mesh sequence, periods of f) is ordered for T.

Sketch: At each fixed period there are tables whose smeared limits are taken along the full mesh sequence; they come from Proposition III.13.6 with shift 0, through the fixed-period smearing theorem. The record supplies the limit along the mesh subsequence of f, and uniqueness of limits identifies the two.

Lean: `Reports.full_tests`, `Reports.useFull`.

Lemma 4.20 (assembly of the reports).

- **III.13.2.** For each extraction order F:
  - item 10 of III.13.1 gives g, the base data and a seed;
  - the seed gives the cell hypotheses, hence grid approximants and weak comparison systems;
  - the OPE entrance theorem gives Maxwell data, infrared data with non-zero J, and the UV package;
  - the reconstruction assembly theorem gives a construction record;
  - Lemma 4.19 gives the full-mesh ordering;
  - the OPE services are assembled from the seed, the logarithmic package and the base data.

  Proposition III.13.5 supplies the fixed-volume limits.
- **III.13.3.** Apply the every-phase reconstruction theorem to the record, and the coherent-prescription theorem to the derivative sector.
- **III.13.4.**
  - Coverage gives portfolios along any finer ultrafilter: take the levels $g(k)$ of item 8 of III.13.1, and the limits along $\mathcal{U}$.
  - The OPE services give the bases Y_k and the convergence of the Gram forms (the Gram comparison theorem, through the grading theorem and the Gram limit theorem).

Lean: `coordination`, `common`, `habitat`, `fromCoverage`, `attains`. *(proofs of the three reports and of* `fromCoverage` *read;* `attains` *read to the level of its two inputs)*

Lemma 4.21 (further estimates used in choosing the step constants).

- **Connected-activity targets.** Take a target with an expense bound. There is M such that for every side $L \geq M$, $W, \mu > 0$, $a \geq 0$, indices j, k, N, $0 \leq \xi <$ exp and $e > 0$ with rate $+ e < \eta < 0$, the prescribed connected-activity estimate holds for all sufficiently large x, with weight $\beta_F(x, L)^e$ and every target order with parameter W. This bound supplies the steepness requirement in Lemma 4.1.
- **Ursell sum.** Assume the support compatibility tests, nonnegative weights, a nonnegative support exponent, the cluster-expansion readiness bound with $0 \leq q < 1$, and a rooted input budget B_0. The averaged marked Ursell sum over clusters, weighted by the cost of the union of their supports, has budget $B_0/(1-q)$ (III.5 in the form of III.8).
- **Uniform rate.** For a connected-activity target there is a side threshold such that, after fixing a larger side, a branching patch, cutoff function, positive W, μ, arity and order

bounds, $0 < c < \exp$ and $e > 0$ with rate $+ e < \gamma < 0$, there is a rate function $\vartheta(x)$ with $|\vartheta(x)| \le \beta_F(x, L)^{\gamma+\varepsilon}$ eventually for every $\varepsilon > 0$. Eventually in x, it bounds the prescribed analytic costs uniformly over admissible tubes, representatives and budget points with exponent at most c and the specified order, power, threshold and radius bounds.

- **Jet realization.** Let the real variable space be finite dimensional, and fix a point x and jet order j. Suppose a determining parameter set carries C^j germs at x, and the prescribed jet coefficients extend holomorphically along every holomorphic curve in the parameter domain, agreeing with those germs on the determining set. There is a family of C^j germs at x realizing these coefficients throughout that domain, holomorphic along such curves together with its jets through order j.

Lean: `Linden.lives`, `Copse.tangled`, `Concert.basisE`, `Lamppost.forge`.

3.16.5 5. Where the difficulty lies

Fixed-level convergence requires control of arbitrarily many ultraviolet levels. The contraction estimate and the soft^v bound give two uniform approximations by convergent sequences. For correlations, Lemma 4.13 balances exponentially improving coarse approximation against subexponential coefficient costs; the chessboard inequality then controls products.

The derivative limit holds only for sufficiently large periods, with threshold depending on the degree and derivative order. Slicing gives a lower bound on a subspace of the required dimension. The plain smeared limit, used in Lemma 4.19, holds at every fixed period.

The coupling and activity analyticity holds in one disc for all cutoffs and periods. The analytic correlation disc may depend on the finite label family and the period. The construction fixes support exponent 100 and target rate $-7/16$; no alternative rate is asserted here.

The family in III.13.2 is chosen before the extraction order. III.13.3 and III.13.4 state a weaker quantifier order, but their proofs use that same choice. The main theorem follows the separate construction route of III.14; equality of the families produced by the two routes is not asserted.

3.16.6 6. Notation

Written proof	Lean	Meaning
C	`Family`	Wilson family: history, holonomy data, label frames, local sections, reference levels, transfer data
$\beta_{N,p}$	`Family.beta`	Wilson coupling at cutoff N, period p
consistent	`Family.Makes`	selected trajectory, sound sections, stage and stencil conditions, priced bounds, calibrated transfer
calibrated	`Spec.Calibrated`	history generated by a full schedule with per-step estimates
ref(d)	`Family.ref`	reference level of degree d

Written proof	Lean	Meaning
penalty b	`KernelChoices.penalty`	coefficient of the Gaussian block-spin term
sheet, rolls, wages, θ, delays, bar, wait, paid	`Sheet`, `Rolls`, `Return.Wages`, `Protection`, `Delays`, `Bar`, `Wait`, `Paid`	constants in the order of choice
alloy, factor, directed history	`Alloy`, `Roost.Factor`, `Directed`	a schedule of RG steps with their bounds
shooting structure, arc, r_*	`Spire.Uses`, `Upland.Arc`, `Arc.radius`	trajectories ending at $\bar{\beta}+s$; disc of analyticity
shift s, assignment $\mathbf{s}$	`Orchard.Rose`, `Orchard.Keys`, `Mast.roseA` $(\mathbf{s}=0)$	real shift of the terminal coupling
label-transfer bundle, $C^{\mathbf{s}}$	`Mast.Cellar`, `Cellar.yard`	coefficient and transfer data for every shift assignment; the associated family
floral	`Oasis.Floral`, `Oasis.Spring`	III.13.7
bundle conditions	`Harvest.Barn`	Scrim bounds and scheduled chessboard bound for all shifts; base data at the centre
chessboard bound in the schedule	`Extraction.Wcmain`	§1 definition
fixed-volume limit with k derivatives	`Poppy.Margins`	III.13.5
slicing	`Harvest.Slicing`	Lemma 4.15
logarithmic OPE package	`OPE.Log.LogPackage`	III.13.1.7
calibrated label steps	`SharpGram.TrueSteps`	III.13.1.7
coverage	`Dense.Coverage`	III.13.1.8
OPE entrance conditions	`Acme.Aroma`	III.13.1.6
grid approximants, weak comparison systems	`Org.Grill`, `East.CompleteWeak`	III.13.1.9
OPE services	`OPE.Facilities`	services of VII.4 with a log package
construction record, reconstruction for every phase	`Spec.Final`, `EveryPhase.Almanac`, `EveryPhase.Completed`	II.2, V.3
portfolio	`Weak.Portfolio`	moving-frame probe data along an ultrafilter
effective expectation	`Hearth.tasting`	normalized expectation against the effective density with its scalar factor removed
$B_{N\to m}$	`GridApprox.backTo`	iterated conditional averaging through the block-spin kernels
V_L	`Bid.vowel`	engineering rescaling by L^{d-e} on grade e
$\mathrm{stock}_k(u)$	`stockK`	coefficient bound with subexponential growth in σ_k

Written proof	Lean	Meaning
stakes(J, m)	`stakes`	fixed-scale chessboard comparison cost
$\mathrm{fly}_{d,m}$	`Bandage.fly`	positive allowance in the label-transfer estimate
soft, rad, gain	`Camp.soft`, `Camp.rad`, `Bid.gainChart`	rim/8, rim/4, $e^{(\sigma_k-\sigma_m)/8}$

3.16.7 7. What remains unexplained

1. **Quantitative packages.** The shell estimates in the OPE entrance condition, the full transfer, remainder, sup and weak comparison bounds, and the one-step comparison assumptions in Lemma 4.3 retain their contracts from III.7, III.11 and VII.4. Their complete expansion into conventional notation remains to be written there. The summaries here do not replace those contracts.
2. **Proofs not fully checked.** The detailed constant issuers underlying Lemmas 4.1–4.4, the fitted recursion behind Lemma 4.6, the contraction and far-level decay estimates in Lemma 4.8, the one-step label estimates behind Lemma 4.10, and the auxiliary estimates in Lemma 4.21 are used as stated. Their internal proofs have not all been checked. The difference estimate in Lemma 4.16 is traced through slicing and the transfer bounds; its preceding one-step estimates remain inputs.
3. **Two construction routes.** No identification of the families from III.13 and III.14, or transfer of III.13.5 to the III.14 family, is established here. A common witness satisfying both lists would require a separate argument.
4. **Moving-frame estimates.** The basis construction and its comparison with the free Gram form belong to VII.1/VII.3. Lemma 4.20 uses those estimates; their proofs are not reproduced here.
5. **The penalty.** Its role as the Gaussian block-spin coefficient is fixed by the kernel choices. Its precise normalization relative to Balaban's parameter a has not been verified.
6. **Other reports.** `Reports.ultraviolet`, `Oasis.fruition` give additional reports. They are not needed for the assembly of III.13.1–III.13.4. The same applies to the offset and analytic-tuning test wrappers and the separate fixed-volume existence theorem.

3.17 III.14. The construction run: flat representations and the ultraviolet rate

Part: III, Ultraviolet construction (renormalization group). **Lean:**

- the endpoint `YM/Independent/ConstructionRun.lean`;
- the native component recipe `YM/Integration/NativeComponentRecipe.lean` and `ActualRunNativeComponentRecipe.lean`;
- the flat-representation chain `YM/RG08/Flat/`, `YM/RG08/Packets/` and `YM/RG08/Rate/`;
- the local-estimate package of `YM/RG12/Burst/Wardrobe.lean`;
- the packages that define the numerical floors of the recipe: `YM/RG08/Redo/`, `Yamen/`, `Sideshow/`, `Necks/`, `Rudder/`, `Room/`, `Weald/`, `Wisp/`, `Trill/`, `Halo/`, `Line/`, … (§4.6).

Size: 14,290 lines in the main-theorem closure (2,509 declarations).

The unit selects the construction run and proves a quantitative limit theorem for its Gaussian

quadratic part.

- **The endpoint.** For a compact matrix gauge group G with simple nonabelian Lie algebra and a penalty $b > 0$, it selects a *run*: the Wilson family of the main theorem with its scale and extraction data. The run satisfies the *funded* condition of III.15 and carries the *native component recipe*.
- **The Gaussian limit.** At each fixed stage, the order-zero quadratic kernel converges as the ultraviolet cutoff is removed, with a power rate in the ratio of lattice spacings (Proposition III.14.3). This is one of three components of the recipe.

Use in the main theorem. VIII.1 (`continuum_with_position_tests`) uses the run, its penalty, the pillars and the funded property. It discards the attached-recipe conjunct.

The selection theorem of III.2 accepts any nonempty family of demands. Here the demand is a recipe, with lower bounds on the block factors and numerical requests on the running coupling. The Gaussian limit is therefore an additional property of the selected run; VIII.1 does not invoke it separately. Replacing this selection by one with fewer demands is a separate question (§7).

3.17.1 1. What this unit proves

Conventions. We use the level systems and sufficiency condition of III.1. Throughout, G is a compact matrix gauge group; "simple" means that its real Lie algebra is simple and nonabelian.

- **Level system.** A level system $\mathcal{L}$ fixes stages $i \in \mathbb{N}$ (finer as i grows), odd block factors $L_i \geq 3$ for the step $i+1 \to i$, and spacings $a_0 = 1$, $a_{i+1} = a_i/L_i$, with log-scales $s_i = \log(1/a_i)$. Its smoothing indices satisfy $J_i \leq J_{i+1} \leq J_i + 1$. Write ℓ_j for the block factor assigned to smoothing index j, so $L_i = \ell_{J_{i+1}}$. The level system also fixes the block-spin kernel with penalty $b > 0$ and the reference inverse coupling $\hat{\beta}_i = \bar{\beta} + s_G s_i > 1$, where $s_G = 11C_G/(24\pi^2)$ (`hatBeta`, `slopeConstant`). The remaining compatibility conditions are those of III.1.
- **Cutoff.** "Cutoff N" means the lattice of spacing a_N, renormalized through stages $N, N-1, \ldots$.
- **Horizon.** A horizon H assigns plaquette-metric and transfer data to each cutoff and stage. Write $M^{\infty}_{N,i}$ for its translation-invariant metric kernel on $\mathbb{Z}^4$, $\widehat{M}_{N,i}(\xi)$ for its Fourier symbol, and $M^{\mathbb{T}}_{N,i}$ for its periodization. On a torus $\mathbb{T}$, the Gaussian quadratic form on 1-forms has matrix $\mathcal{K}^{\mathbb{T}}_{N,i} = d^{\top} M^{\mathbb{T}}_{N,i} d$.
- **Price map and sufficiency.** A price map $P = (\alpha_j, \Omega_j, \eta_j)_j$ specifies positive strip widths α_j, upper bounds Ω_j and positive coercivity bounds η_j. The full sufficiency condition of III.1 includes strip and derivative bounds for all metric and transfer data, inverse and factorization identities, symmetry and coercivity on every torus, the background-field and fluctuation identities between consecutive stages, and the bare terminal condition. In particular, a sufficient horizon satisfies $\widehat{M}_{N,i}(0) = 1$ and $\mathcal{K}^{\mathbb{T}}_{N,N} = d^{\top} d$.
- **Position-space kernel.** $\mathcal{O}^{H}_{N,i} = d^{\top} M^{\infty}_{N,i} d$ is the quadratic kernel on the links of $\mathbb{Z}^4$ (`output`). This is the Hessian convention: no factor $1/2$ is included in $\mathcal{O}$.
- **Kernel norms.** Let $x(u)$ be the base point of a link or plaquette u. For link kernels, put $\|F\|_{\sigma} = \sup_u \sup_{\phi} \sum_{y \in \mathbb{Z}^4} |F(u,(y,\phi))| e^{\sigma |x(u) - y|_1}$. For plaquette kernels, put $\|B\|^{\mathrm{pl}}_{\sigma} = \sup_u \sum_{v \in \mathbb{Z}^4 \times \mathcal{F}} |B(u,v)| e^{\sigma |x(u) - x(v)|_1}$, where $\mathcal{F}$ is the set of plaquette orientations. The latter sum includes all target orientations.
- **Age.** $\Lambda_{k,i} = 5 + \log \hat{\beta}_k + |s_i - s_k|$ is the age of stage i seen from a reporting scale k.

Proposition III.14.1 (the construction run). Let G be simple and $b > 0$. There exist a run

R, a campaign T for R and pillars W for (R, T) such that:

1. the penalty of the block-spin kernel of R equals b;
2. R satisfies the full *funded* condition of III.15 relative to the local-estimate package Δ_{10}. Its thirteen clauses (§6) include the prescribed local estimates, $Q(i) \to \infty$, support exponent 100 at every stage, and the extraction, OPE and thermal conditions. The selection uses $Q(i) = i + 2$;
3. the native component recipe is *attached* to R. There are a sheet s, a price map q, a recipe C for s (Proposition III.14.2) and floors $n \geq \max(n_s, E_C)$, where n_s is the sheet's floor and E_C the combined recipe floor of Lemma 4.14, such that:
   - R is built on s with floors raised to n, with the primary price map $P_{s,q}$ of III.12;
   - both countable request families, before and after the block factors are chosen, satisfy the eventual bounds of Lemma 4.16; the recipe's post-length bill B_C also satisfies $B_C(\ell, J_{m+1}) \leq \hat{\beta}_m$ for every m;
   - the realized recipe (Lemma 4.14) holds on R's own levels and horizon.

Status: theorem. A run is fixed by choice as `fundedRun`.

Lean: `YM.Independent.constructionRun`; `R2I.Funded`; `garment10`; `Campaign.Pillars`; `ActualRunNativeComponentRecipe.Attached`.

Proposition III.14.2 (existence of the native component recipe). Let G be simple and let s be a sheet (the kernel choice and stage profile, fixed before any block factor). There exist a price map X (the *secondary* price map) and floors $E^{\flat}, E^{\mathrm{sw}}, E^{\mathrm{du}} : \mathbb{N} \to \mathbb{N}$ with the following properties, for every level system $\mathcal{L}$ with the kernel and stage profile of s.

- **(flat representation)** If $L_i \geq E^{\flat}(J_{i+1})$ for all i, then for every e satisfying $\Omega^X_{J_i} \leq \log \hat{\beta}_i$ for $i \geq e$, there are kernels and constants satisfying clauses (a)–(c) of Proposition III.14.3 with this X, simultaneously for every sufficient horizon.
- **(point-group invariance)** If $L_i \geq E^{\mathrm{sw}}(J_{i+1})$ for all i, then for every horizon H sufficient for some price map, every $i \leq N$ and every torus, the colour-summed bilinear form $\sum_a \langle A^a, \mathcal{K}^{\mathbb{T}}_{N,i} B^a \rangle$ is invariant under simultaneous pullback of A, B by signed permutations of the axes fixing the origin. Pullback reverses the sign when a link orientation is reversed.
- **(background-field identities)** If $L_i \geq E^{\mathrm{du}}(J_{i+1})$ for all i, then for every such H, every $m < N$ and every torus in the level system, the canonical free background-field map of III.3 satisfies the relations of Proposition III.14.5 for the fine and coarse metrics $(M^{\mathbb{T}_f}_{N,m+1}, M^{\mathbb{T}_c}_{N,m})$. It therefore gives a minimizer of the Gaussian variational problem for the actual quadratic forms.

Status: theorem.

Lean: `NativeComponentRecipe.Choice`, `available`; the three components follow from `Flat.entrance_gate`, `Cast.assured` (with `Bid.Sworn`) and `Aura.dual_support` (with `Aura.licensed`).

Proposition III.14.3 (ultraviolet rate for the Gaussian quadratic part). Let G be simple and s a sheet. There exist a price map X and a floor $E : \mathbb{N} \to \mathbb{N}$ with the following property. Let $\mathcal{L}$ be a level system with the kernel and stage profile of s and with $L_i \geq E(J_{i+1})$ for all $i \geq 0$. Let $e \in \mathbb{N}$ satisfy $\Omega^X_{J_i} \leq \log \hat{\beta}_i$ for $i \geq e$. Then there are kernels F_i $(i \geq e)$ on the links of $\mathbb{Z}^4$ and constants $A, B \geq 0$, $c > 0$, $d, l \in \mathbb{N}$ such that the following holds for every horizon H of $\mathcal{L}$ sufficient for some price map, every reporting scale $k \in \mathbb{N}$ and every $i \geq e$. Write $w_j = \min(1, \alpha^X_j, \alpha^X_{j+1})$.

- (a) $\|F_i\|_{w_{J_i}} \le B\,\Lambda_{k,i}^{l}$;
- (b) $\mathcal{O}^{H}_{n,i}(u,v) \to F_i(u,v)$ as $n \to \infty$, for all links u, v;
- (c) $\|\mathcal{O}^{H}_{N,i} - F_i\|_{w_{J_i}} \le A\,\Lambda_{k,i}^{d}\,(a_N/a_i)^{c}$ for all $N \ge i$.

One may take $c = 1/4$, $d = l = 1$, $A = 34\sigma_0|\mathcal{F}|(1+s_G)/(1-3^{-1/8})$ and $B = \sigma_0|\mathcal{F}|(1+s_G)$, where $\sigma_0 = |\mathcal{F}|^2\lambda_d^2$ and λ_d is a width-one strip bound for the lattice exterior derivative. These constants depend only on G and the lattice, independently of the sheet, level system, entry stage and horizon.

Status: theorem. This is an order-zero estimate (see `Mesh.lean`).

Lean: `Flat.Quorum` (the three clauses), `Flat.entrance_gate` (existence), `Flat.supplies`, `Flat.meshed`, `Flat.clock`, `Flat.curve`, `Flat.Delay`.

Proposition III.14.4 (flat representations at all stages, abstract form). Let P be a price map, $\theta > 0$, and $w : \mathbb{N} \to (0,\infty)$. Let $(\mathcal{L}_x, H_x, r_x, N_x, n_x, k_x, Z_x)_{x\in X}$ be a family of level systems with horizons, integers $r_x \le N_x \le n_x$, reporting scales $k_x \in \mathbb{N}$ and offsets $Z_x \ge 0$. Put $\Lambda_x(i) = 2 + Z_x + |s_i - s_{k_x}|$, with the scale data understood to belong to $\mathcal{L}_x$. Fix $A_0 \ge 0$ and $d_0 \in \mathbb{N}$, independent of x. Assume:

- each H_x is sufficient for P, and its plaquette-transfer kernels at cutoff N_x and stages $r_x \le i < N_x$ are the canonical free kernels of III.3;
- $w_j \le \alpha_j$ for all j, and for $r_x \le i < N_x$:
  - $w_{J_i} \le \min(1, \alpha_{J_i}, \alpha_{J_{i+1}})$;
  - $2w_{J_i} \le L_i\, w_{J_{i+1}}$;
  - $q_i(w_{J_{i+1}}) \le L_i^{-\theta}$, where q_i is the one-step factor of Lemma 4.3;
- $2|\mathcal{F}|\Omega_{J_{N_x}} \le A_0\Lambda_x(N_x)^{d_0}$ for every x.

Then there are $A \ge 0$, $c > 0$, $d \in \mathbb{N}$ and kernels $K_{x,i}$ such that, for all x and $r_x \le i \le N_x$, $K_{x,i}$ is a flat representation (Definition 4.1) of $\mathcal{K}_{n_x,i} - \mathcal{K}_{N_x,i}$ at stage i, of width w_{J_i} and cost $A\,\Lambda_x(i)^d(a_{N_x}/a_i)^c$. One may take $c = \min(1, \theta/2)$ and $d = d_0$, with A depending only on A_0, d_0, θ.

Status: theorem.

Lean: `Flat.World.overlap`, with `World`, `World.Funding`, `World.Provides`.

Proposition III.14.5 (Schur identity for two cutoffs, and the flat step).

Identity. Fix one RG step: a fine torus $\mathbb{T}_f$ with L times the side of the coarse torus $\mathbb{T}_c$, the block average Q and penalty $b > 0$. Say that $(W; M, M')$ satisfies the background-field relations when

$$\mathcal{K}(M)\,W = Q^{\top}\mathcal{K}(M'), \qquad QW = 1 - b^{-1}\mathcal{K}(M').$$

If M is symmetric and positive semidefinite, these relations make WA' a minimizer of $\langle A, \mathcal{K}(M)A\rangle + b|A' - QA|^2$, with minimum $\langle A', \mathcal{K}(M')A'\rangle$. For the algebraic identity below, positivity is unnecessary: if $(W_1; M_1, M_1')$ and $(W_2; M_2, M_2')$ satisfy the relations and M_2, M_2' are symmetric, then

$$\mathcal{K}(M_2') - \mathcal{K}(M_1') = W_2^{\top}(\mathcal{K}(M_2) - \mathcal{K}(M_1))W_1.$$

Flat step. Let $\tau > 0$, $r \ge 0$, and let B be a neutral, translation-invariant plaquette kernel with $\|B\|_{\tau}^{\mathrm{pl}} \le r$, whose periodization represents $\mathcal{K}(M_2) - \mathcal{K}(M_1)$ on $\mathbb{T}_f$. Suppose T, U are order-two strip-bounded curvature-transport kernels with constants $q, p \ge 0$ in width ω. Assume

$T(x+Lv, y+v) = T(x,y)$ and the same identity for U, that their periodizations C_2, C_1 satisfy $dW_2 = C_2 d$, $dW_1 = C_1 d$, and that $\sum_z U((x,\varphi),(z,\psi))$ is independent of x for each pair of orientations. If $0 \le \sigma \le \omega$ and $2\sigma \le L\tau$, then

$$B'(y,z) = \sum_{x,w} T(x,y) B(x,w) U(w,z), \qquad \|B'\|_\sigma^{\mathrm{pl}} \le \frac{2|\mathcal{F}|^2 qp}{\tau L} r.$$

The kernel B' is neutral and translation invariant, and its periodization represents $\mathcal{K}(M_2') - \mathcal{K}(M_1')$ on $\mathbb{T}_c$.

Sketch: transpose the second pair of background-field relations and substitute the first pair; the mixed $b^{-1}\mathcal{K}(M_2')\mathcal{K}(M_1')$ terms cancel. The intertwining identities then give the asserted curvature representative. Neutrality permits subtraction of the second transport kernel at the first base point, yielding the norm estimate in Lemma 4.3.

Status: theorem. No inverse of the curvature form is used. Preservation of neutrality requires the separate constant-row-sum condition.

Lean: `Flat.poised` (identity), `Flat.toward` (step), `Packets.push_null`, `neutralCoin`.

Proposition III.14.6 (rate lemma for fine-to-coarse recursions). For each $x \in X$, fix a scale system with $L_i \ge 3$, stages $r_x \le i \le t_x$ and age $\Lambda_x(i) = 2 + Z_x + |s_i - s_{k_x}|$, where $Z_x \ge 0$. Fix $\theta, c_0 > 0$, $A_0 \ge 0$ and $d_0 \in \mathbb{N}$, uniformly in x. Suppose $f_x(i), g_x(i) \ge 0$ on these stage ranges and

- $f_x(i) \le L_i^{-\theta} f_x(i+1) + g_x(i)$ for $r_x \le i < t_x$, and $f_x(t_x) \le g_x(t_x)$;
- $g_x(i) \le A_0 \Lambda_x(i)^{d_0} (a_{t_x}/a_i)^{c_0}$ for $r_x \le i \le t_x$.

Then $f_x(i) \le A \Lambda_x(i)^{d_0} (a_{t_x}/a_i)^{\mu}$ throughout each range, where $\mu = \min(c_0, \theta/2) > 0$. One may take $A = A_0 (K+1)^{d_0}/(1 - 3^{-\theta'/2})$, with $\theta' = \theta - \mu > 0$ and $K = 2(d_0+1)/\theta'$.

Status: theorem.

Lean: `Rate.Stair.fade_solve`, `poly_solve`, `slope`, `Fade.cap`.

Further content. The auxiliary estimates in §4.6 supply numerical requirements for the recipe. They include the one-step transit estimate `harbor`, curvature-coefficient lifts `arc` and `warehouse`, and formal Gaussian identities `bottle`, `still` and `absorbed`.

3.17.2 2. What it uses

From other units.

- **III.1 (block-spin step and its linearization).**
  - Level systems: stages, spacings, block factors, the smoothing indices J, and $\hat{\beta}_i$.
  - Horizons and their sufficiency for a price map, and the metric data with kinetic normalization $\widehat{M}(0) = 1$.
  - The background-field relations, and the identity $\mathcal{K}(M') = W^\top \mathcal{K}(M) W + b(1 - QW)^\top (1 - QW)$.
  - Links: `Levels`, `Horizon`, `Horizon.Suffices`, `MapPrices`, `Floor`, `Precision.K`, `Inputs`, `action`.
- **III.2 (choosing the constants).**
  - Sheets: `Sheet`.

  - Selection with a preselected demand: `selectRunWithTreasuryPeriod_preselected_quote_orders_pos` and `PostnumericPreselection`. The quantified conclusion used here is Lemma 4.16.
- **III.3 (free kernels).** `retained_support` gives, for each kernel choice, the following:
  - a clip profile γ and a floor n;
  - for every sheet with that kernel, a price map X such that every level system with clip parameters $\le \gamma$ and $L_i \ge n(J_{i+1})$ (the "credit", `Credit`) carries a horizon sufficient for X, whose transfer data are the canonical free data (`destination`) and whose forms are point-group invariant.
- **III.8 (kernel norms).** Weighted row and column norms, and order-m strip bounds for kernels from fine to coarse lattices: `RowCost`, `Atlas`, the carrier map `carrier`.
- **III.12.** The structures `Run` and `Pillars`, and the primary price map $P_{s,q}$ (`raging`).
- **III.15.**
  - The funded property `Funded`.
  - The predicate `EventuallyPaid`: a property of level systems is eventually paid if it can be enforced by raising the entry thresholds of β. Here it is applied to the trivial property.
- **I.1.** The group data and the one-loop slope s_G.
- **Use in VIII.1.** `continuum_with_position_tests` applies `R2IGoals.fromFunded` to the pillars and funded property.

Standard mathematics.

- Gaussian block-spin transformation with background field (Balaban, *Propagators and renormalization transformations for lattice gauge theories I, II*, CMP 95 (1984), CMP 96 (1984); *Averaging operations for lattice gauge theories*, CMP 98 (1985); Gawędzki–Kupiainen, *Block spin renormalization group for dipole gas and* $(\nabla\varphi)^4$, Ann. Phys. 147 (1983)). The coarse quadratic form is the Schur complement obtained by minimizing over the fine field.
- Exponentially weighted ℓ^1 kernel norms and their behaviour under composition and blocking, as in Balaban's propagator estimates.
- A discrete Grönwall argument for $f_i \le \lambda_i f_{i+1} + g_i$.
- Uniqueness of a translation-invariant kernel of finite weighted row norm from its periodizations on tori of diverging size (dominated convergence).

Imported constants and assumptions.

- The endpoint assumes only simplicity of G and $b > 0$.
- The local-estimate package Δ_{10} is defined in this unit, in `Wardrobe.lean`. It is the pair of the predicate `Attic` and the lemma `annex`:
  - the predicate combines one-step connected-activity bounds, tree-sum smallness and bounds through any fixed finite range of marks and degrees;
  - for a target and expense with expense exponent at most $-7/16$, there is a block-factor threshold such that, for every larger block factor, every $W, \mu > 0$, admissible order, stage profile and finite range, the predicate holds for all sufficiently large x. The threshold for x may depend on those fixed data.
- Inside the flat chain every hypothesis is discharged in this unit or in III.3:
  - sufficiency, free transfer data and point-group invariance come from III.3 via the shadow (Lemma 4.11);
  - the width conditions follow from the floors (Lemma 4.6);
  - the delay condition follows from the post-length bill (Lemma 4.14).

3.17.3 3. Strategy

The endpoint applies the selection theorem of III.2 to the demand for a recipe. First choose the sheet, primary price map, recipe and combined floor E_C. Next choose the block factors ℓ_j above that floor. Finally raise the coupling thresholds to pay the bill $B_C(\ell, j)$ and the countable numerical requests. The selection retains the resulting inequalities on the run itself, so Lemma 4.15 realizes the recipe on that run's levels and horizon.

For the Gaussian estimate, compare two cutoffs $N \le n$ at a common stage i. At stage N, their metric difference $M^{\infty}_{n,N} - M^{\infty}_{N,N}$ represents the quadratic-form difference and has zero row sums in each orientation, since both Fourier symbols equal the identity at zero momentum. The mixed Schur identity transports this representative through the two background-field maps. A transport kernel has row scale L^{-2}, column scale L^2 and first-difference scale L^{-3}. Neutrality permits subtraction in one leg, gaining a factor L^{-1}; the distance introduced by subtraction is absorbed using half of the input exponential weight. The block-factor floor makes the one-step factor at most $L_i^{-1/2}$. A discrete Grönwall estimate absorbs polynomial growth in the age and gives the rate $(a_N/a_i)^{1/4}$. Passing from curvature representatives to link kernels and then taking a Cauchy limit gives Proposition III.14.3.

The argument carries representatives modulo kernels annihilated by $d^{\top}(\cdot)d$, without inverting the degenerate curvature form. The plaquette metrics do have inverses, used in the transport kernels. The Gaussian quadratic forms are independent of the sufficient horizon, because the terminal form and the Schur recursion determine them. This permits the estimate to be proved on a shadow level system with altered clip parameters, where III.3 supplies canonical free transfer data, and then transferred to the original horizon.

3.17.4 4. Steps

3.17.4.1 4.1 Flat representations and their propagation Definition 4.1 (flat representation). Let H be a horizon of $\mathcal{L}$, $N \le n$, $i \le N$, and $\sigma, p \ge 0$. A kernel K on plaquettes $\mathbb{Z}^4 \times \mathcal{F}$ is a *flat representation* of $\mathcal{K}_{n,i} - \mathcal{K}_{N,i}$ at stage i, of width σ and cost p, if:

- $\|K\|^{\mathrm{pl}}_{\sigma} \le p$;
- $\sum_{u \in \mathbb{Z}^4} K(x, (u, \varphi)) = 0$ for every x and every orientation φ (*neutral*);
- $K(x + u, y + u) = K(x, y)$;
- for every torus $\mathbb{T}$ of the level system at stage i, the periodization $K^{\mathbb{T}}$ satisfies $d^{\top} K^{\mathbb{T}} d = \mathcal{K}^{\mathbb{T}}_{n,i} - \mathcal{K}^{\mathbb{T}}_{N,i}$.

Lean: `Flat.FlatRep`, `Sigil`, `Folded`, `neutral`, `town`.

Lemma 4.2 (the finest stage). Let H be sufficient for P, $N \le n$, and $0 < \sigma \le \alpha_{J_N}$. Then $M^{\infty}_{n,N} - M^{\infty}_{N,N}$ is a flat representation at stage N of width σ and cost $2|\mathcal{F}|\Omega_{J_N}$.

Sketch: each metric has plaquette norm $\le |\mathcal{F}|\Omega_{J_N}$ by its strip bound. The difference is neutral because $\widehat{M}(0) = 1$ for both metrics. The torus identity is $\mathcal{K}(M_{n,N}) - \mathcal{K}(M_{N,N}) = d^{\top}(M_{n,N} - M_{N,N})d$.

Lean: finest-stage lemma (private, `World.lean`), via `turn_class`; cost `topTab`.

Lemma 4.3 (one step to a coarser stage). Let H be sufficient for P, with $i < N \le n$. Assume the plaquette-transfer kernel of H at (N, i) is the canonical free kernel. Let K be a flat

representation at stage $i+1$ of width $u>0$ and cost $p\ge 0$, and let $0<\sigma\le \min(1,\alpha_{J_i},\alpha_{J_{i+1}})$ with $2\sigma\le L_i u$. Then

$$K'(y,z)=\sum_{x,w} T_{n,i}(x,y)\,K(x,w)\,T_{N,i}(w,z)$$

is a flat representation at stage i of width σ and cost $q_i(u)\,p$, where:

- $T_{m,i}=(M^{\infty}_{m,i+1})^{-1}\mathcal{T}_{m,i}M^{\infty}_{m,i}$ is the curvature-transport kernel at cutoff m, with $\mathcal{T}_{m,i}$ the installed plaquette-transfer kernel and products denoting kernel composition;
- $q_i(u)=2|\mathcal{F}|^2h_i^2/(uL_i)$;
- $h_i=\exp(4)|\mathcal{F}|^2\Omega^2_{J_{i+1}}\Omega_{J_i}$.

Sketch: the four properties come from four inputs.

- *Algebra.* The torus identity follows from the Schur identity and from the intertwining $dW_m=C_m d$ (`inherited_precision`). These give $\mathcal{K}_{n,i}-\mathcal{K}_{N,i}=d^{\top}C_n^{\top}K^{\mathbb{T}}C_N d$, and $C_n^{\top}K^{\mathbb{T}}C_N$ is the periodization of K' (`town_hall`).
- *Bounds on the transport kernels.* Both are order-2 strip-bounded with constant h_i (`response`, `allotment`).
- *Neutrality.* The free plaquette-transfer kernel has a zero-momentum symbol independent of the fine residue class, and therefore position-independent row sums. Composition with the translation-invariant metric kernels preserves this property for $T_{N,i}$ (`travels`). Thus constants of each orientation go to constants and neutrality passes to K' (`invariant`).
- *Norm.* Neutrality replaces $T_{N,i}(w,z)$ by $T_{N,i}(w,z)-T_{N,i}(\bar w,z)$, where $\bar w$ has the base point of x and the orientation of w. In the weighted row estimate this costs rh_i/L_i^3, where r is the ℓ^1 distance between the base points of x,w. The inequality $re^{\sigma r/L_i}\le (2/u)e^{ur}$ for $r\ge 0$ follows from $2\sigma\le L_i u$. The column estimate for $T_{n,i}$ contributes $h_iL_i^2$; summing orientations gives the factor $2|\mathcal{F}|^2h_i^2/(uL_i)$.

Lean: one-step lemma (private, `World.lean`); `toward`, `poised`, `push_null`, `quote`.

Lemma 4.4 (induction over stages). Under the hypotheses of Proposition III.14.4 there are flat representations $K_{x,i}$ $(r_x\le i\le N_x)$ of width w_{J_i} and cost $c_x(i)=2|\mathcal{F}|\Omega_{J_{N_x}}(a_{N_x}/a_i)^{\theta}$.

Sketch: the costs are defined explicitly, so that $c_x(i)=L_i^{-\theta}c_x(i+1)$. The proof is downward induction on N_x-i:

- start with Lemma 4.2;
- apply Lemma 4.3 with $u=w_{J_{i+1}}$, $\sigma=w_{J_i}$;
- use $q_i(w_{J_{i+1}})\le L_i^{-\theta}$ and monotonicity of flat representations in the cost.

The representatives for all (x,i) are then fixed by choice.

Lean: downward-induction lemma (private, `World.lean`).

Lemma 4.5 (the rate; proof of Proposition III.14.4). Write $g_x(i)=2|\mathcal{F}|\Omega_{J_{N_x}}\mathbf{1}_{i=N_x}$. The costs of Lemma 4.4 satisfy the recursion of Proposition III.14.6 with forcing g. By the last hypothesis of III.14.4, g has the required form with $c_0=1$ (it vanishes below the top). Proposition III.14.6 then gives cost $\le A\Lambda_x(i)^{d_0}(a_{N_x}/a_i)^{\min(1,\theta/2)}$.

Proof of III.14.6.

1. Multiply by $(a_i/a_t)^{\mu}$; the recursion keeps its form with $\theta'=\theta-\mu\ge\theta/2$.

2. Prove $\tilde{f}_x(i) \le \tilde{B}\,(K+\Lambda_x(i))^{d_0}$ by downward induction, with $\tilde{B} = A_0/(1-3^{-\theta'/2})$.
3. The step uses the polynomial estimate: since $\Lambda_x(i+1) \le \Lambda_x(i) + \log L_i$, one has $(K+\Lambda_x(i+1))^d L_i^{-\theta'} \le (K+\Lambda_x(i))^d L_i^{-\theta'/2}$ when $d \le K\theta'/2$. Then $L_i^{-\theta'/2} \le 3^{-\theta'/2}$, and $\tilde{B}3^{-\theta'/2} + A_0 = \tilde{B}$.

Finally, $\Lambda_x(i) \ge 2$ gives $(K+\Lambda_x(i))^{d_0} \le (K+1)^{d_0}\Lambda_x(i)^{d_0}$. Undoing the tilt gives the stated value of A.

Lean: `World.overlap`, `fade_solve`, `poly_solve`, `slope`, `Fade.of_top`, `drop_length`.

3.17.4.2 4.2 Choice of widths and passage to position space Lemma 4.6 (choice of widths). Let G be simple, X a price map and $w_j = \min(1, \alpha_j^X, \alpha_{j+1}^X)$. Put

$$\beta_{j,j'} = 2|\mathcal{F}|^2 h_{j,j'}^2 / w_j, \qquad h_{j,j'} = \exp(4)|\mathcal{F}|^2 (\Omega_j^X)^2 \Omega_{j'}^X, \qquad E^\flat(j) = \lceil 2/w_j + \beta_{j,j}^2 + \beta_{j,j-1}^2 \rceil.$$

Here $j-1$ is truncated at zero. Suppose $L_i \ge E^\flat(J_{i+1})$ and $\Omega_{J_i}^X \le \log \hat{\beta}_i$ for every $i \ge e$. With q_i computed from X, the following hold for $i \ge e$:

- $w_{J_i} \le \min(1, \alpha_{J_i}^X, \alpha_{J_{i+1}}^X)$;
- $2w_{J_i} \le L_i w_{J_{i+1}}$;
- $q_i(w_{J_{i+1}}) \le L_i^{-1/2}$;
- $\Omega_{J_j}^X \le (1+s_G)\Lambda_{k,j}$ for all k and all $j \ge e$.

Sketch: since $J_i \in \{J_{i+1}, J_{i+1}-1\}$, the constant h_i is one of $h_{j,j}, h_{j,j-1}$ with $j = J_{i+1}$. Then $q_i(w_j) = \beta/L_i \le L_i^{-1/2}$ because $\beta^2 \le L_i$. Next, $2w_{J_i} \le 2 \le L_i w_j$. For the age cap: $\hat{\beta}$ is affine in s, so $\log \hat{\beta}_j \le \log \hat{\beta}_k + s_G|s_j - s_k|$.

Lean: `spacing_choice`, `universeClock`, `work items`, `entrance`, `before`, `Spacing`.

Lemma 4.7 (the family estimate). Assume the conclusions of Lemma 4.6 and let H be sufficient for X. Index the family by $x = (k, N, n)$ with $k \in \mathbb{N}$ and $e \le N \le n$, and put $r_x = e$, $\theta = 1/2$ and $Z_x = 3 + \log \hat{\beta}_k$. Then the numerical hypotheses of Proposition III.14.4 hold with $A_0 = 2|\mathcal{F}|(1+s_G)$ and $d_0 = 1$. If the plaquette-transfer data at every (N,i) with $e \le i < N$ are the canonical free data, all its hypotheses hold.

Sketch: the first three width estimates are exactly the required one-step inequalities. The age bound gives $2|\mathcal{F}|\Omega_{J_N} \le A_0 \Lambda_{k,N}$ uniformly in k, N, n.

Lean: `berth`, `Spacing.funded`.

Lemma 4.8 (from a representative to the position-space kernel). Let $0 \le \sigma, \varepsilon \le 1$ and $p, C \ge 0$. Suppose both metric kernels are strip-bounded in width σ with constant p, and that K represents $\mathcal{K}_{n,i} - \mathcal{K}_{N,i}$ on tori whose sides tend to infinity, with $\|K\|_\varepsilon^{\mathrm{pl}} \le C$. Then

$$\mathcal{O}_{n,i} - \mathcal{O}_{N,i} = d^\top K\, d \quad \text{on the links of } \mathbb{Z}^4, \qquad \|\mathcal{O}_{n,i} - \mathcal{O}_{N,i}\|_\varepsilon \le \sigma_0 C, \qquad \sigma_0 = |\mathcal{F}|^2 \lambda_d^2,$$

where λ_d is a width-one strip bound for d.

Sketch: $\mathcal{O}_{n,i} - \mathcal{O}_{N,i}$ and $d^\top K d$ have finite row norms and the same periodizations on the given tori. Dominated convergence as the torus side tends to infinity identifies the kernels. Composition with the two exterior derivatives gives the factor σ_0.

Lean: `resolve`, `output_cost`, `sumToken`.

Lemma 4.9 (rate for pairs of cutoffs). Under the conclusions of Lemma 4.6, with H sufficient for X and with canonical free plaquette-transfer data at all (N, i), $e \le i < N$, the constant A of Proposition III.14.3 satisfies

$$\|\mathcal{O}_{n,i} - \mathcal{O}_{N,i}\|_{w_{J_i}} \le A\,\Lambda_{k,i}\,(a_N/a_i)^{1/4} \qquad \text{for all } k \text{ and } e \le i \le N \le n.$$

Sketch: apply Proposition III.14.4 to the family of Lemma 4.7, then Lemma 4.8.

Lean: `meshed`.

Lemma 4.10 (the limit kernel). Under the same hypotheses there are kernels F_i satisfying (a)–(c) of Proposition III.14.3, in the same width w_{J_i}.

Sketch: for fixed i, Lemma 4.9 gives a Cauchy modulus $A\Lambda_{k,i}(a_n/a_i)^{1/4} \to 0$ in the weighted row norm. Completeness of the scalar entries gives a pointwise limit, and finite-sum bounds followed by monotone convergence preserve the row estimates (`solvent`). This gives norm convergence and (c). For $n \ge i$, $\|\mathcal{O}_{n,i}\|_{w_{J_i}} \le \sigma_0|\mathcal{F}|\Omega_{J_i} \le \sigma_0|\mathcal{F}|(1+s_G)\Lambda_{k,i}$, which passes to the limit and gives (a). Uniqueness of pointwise limits makes F_i independent of k.

Lean: `supplies`.

3.17.4.3 4.3 Uniqueness of the Gaussian part; the three recipe components Lemma 4.11 (the Gaussian quadratic forms do not depend on the horizon). Let H, H' be horizons, each sufficient for some price map, on level systems $\mathcal{L}, \mathcal{L}'$ that differ at most in the clip parameters of their stages; for example $\mathcal{L}'$ may be the *shadow* of $\mathcal{L}$, with clip profile replaced by γ. Then $\mathcal{K}^{\mathbb{T}}_{N,i}(H) = \mathcal{K}^{\mathbb{T}}_{N,i}(H')$ for all $i \le N$ and all tori, and $\mathcal{O}^{H}_{N,i} = \mathcal{O}^{H'}_{N,i}$.

Sketch: downward induction from $i = N$, where both forms equal $d^\top d$. If the forms agree at stage $i+1$, the Schur identity gives $\mathcal{K}_{N,i}(H) - \mathcal{K}_{N,i}(H') = W^\top \cdot 0 \cdot W' = 0$; the clip parameters do not enter the step data. This proves equality on the tori of the level system. Lemma 4.8 then identifies the position-space kernels; periodizing that equality gives equality on every torus.

Lean: `same_path`, `same_output`, `same_terminal`, `shadow`, `same_shadow`.

Lemma 4.12 (the flat component; proof of Proposition III.14.3). For a simple G and a sheet s, take the clip profile γ and floor E_{free} from III.3, and obtain X by applying its sufficiency construction to the sheet with clip profile γ. Set $E = \max(E^\flat, E_{\text{free}})$. Given $\mathcal{L}$ and e as in Proposition III.14.3, the shadow $\mathcal{L}'$ meets the assumptions of III.3, so it carries a horizon Y sufficient for X with canonical free transfer data. Apply Lemmas 4.6–4.10 to $(\mathcal{L}', Y)$; the delay condition is unchanged, since $\widehat{\beta}$ and the scale data are unchanged. Transfer (b) and (c) to every sufficient horizon H of $\mathcal{L}$ by Lemma 4.11. The floor E is required at every stage because III.3 requires its floor everywhere.

Lean: `entrance_gate`, `rearrange`.

Lemma 4.13 (the other two components of the recipe). For each sheet, the point-group and background-field conclusions of Proposition III.14.2 hold above suitable floors. Both use the shadow construction of Lemma 4.12. Their floors may be chosen to be E_{free}; the flat component also requires $E^\flat$.

- *Point-group invariance.* III.3 supplies point-group invariance for the shadow horizon, and the forms coincide by Lemma 4.11.
- *Background-field identities.* On the shadow the free background-field map satisfies the relations of Proposition III.14.5, because its transfer data are the free data. The shadow's quadratic forms equal those of H (Lemma 4.11), and the free data of $\mathcal{L}$ and of its shadow coincide. Hence the same relations hold for H.

Lean: `Cast.assured`, `Aura.dual_support`, `available`.

3.17.4.4 4.4 Realizing the recipe on a run Lemma 4.14 (realization). Let s be a sheet, q a price map, C a recipe for s with secondary price map X, and $\mathcal{L}$ a level system with the kernel and profile of s. Let H be sufficient for the primary price map $P_{s,q}$. Let E_C be the maximum of $E^{\flat}, E^{\mathrm{sw}}, E^{\mathrm{du}}$ and the two numerical floors in §4.6, and let $B_C(\ell, j)$ be the maximum of the two post-length bills there. Assume

- $\ell_j \geq E_C(j)$ for every smoothing index j;
- $\hat{\beta}_m \geq B_C(\ell, J_{m+1})$ for every stage m.

Then on $(\mathcal{L}, H)$:

- both numerical floor and bill conditions hold;
- the background-field identities and point-group invariance hold;
- Proposition III.14.3 holds with $e = 0$.

Sketch: the floor bounds apply to $L_i = \ell_{J_{i+1}}$ and to all smoothing-index requirements. For the delay, the second bill is the exponential of a finite bound S_i for the actual patch, with $\Omega_q^X \leq S_i$ for every $q \leq J_{i+1}$. Its payment gives $S_i \leq \log \hat{\beta}_i$ (`Yamen.hatch`). Since $J_i \leq J_{i+1}$, the delay holds with $e = 0$. Apply the three components of C; the fixed chart of III.3 supplies the remaining local-chart condition.

Lean: `realize`, `Realized`, `floor`, `post`, `numeric`.

Lemma 4.15 (binding to the run). Let s, q, C be as in Lemma 4.14 and $n : \mathbb{N} \to \mathbb{N}$ with $n \geq E_C$. Let R be a run whose sheet is s with floors raised to n, whose price map is $P_{s,q}$, and whose levels pay B_C. Then the recipe is realized on R's own levels and horizon.

Sketch: the sheet and price-map identities identify the kernel, profile and sufficient horizon required in Lemma 4.14. The run's block-factor inequalities give $\ell_j \geq n(j) \geq E_C(j)$. Apply that lemma. The retained selection data of III.2 contain exactly these identities and inequalities.

Lean: `on_run`, `of_preselection`.

3.17.4.5 4.5 The run Lemma 4.16 (selection with a preselected demand; III.2). Fix the comparison and custody data used in III.2. Let G be simple, $b > 0$, Λ a property of level systems that can be enforced by raising coupling-entry thresholds, and Δ a local-estimate package. For every sheet s and price map q, suppose a nonempty demand set $\mathcal{D}(s, q)$ is given. For each $C \in \mathcal{D}(s, q)$, prescribe an integer floor $E_C(j)$, real requests $U_C(r, j)$, and real post-length requests $V_C(\ell, r, j)$ depending on the block-factor sequence ℓ.

Then there are a run R, a campaign T, a sheet s, a price map q, a choice $C \in \mathcal{D}(s, q)$ and an integer floor $n \geq \max(n_s, E_C)$ such that:

- the penalty is b, Λ holds on the levels of R, pillars exist, and R is funded for Δ;
- R has the sheet s with floor raised to n, and price map $P_{s,q}$;
- $\log\log\hat{\beta}_m \geq 0$ for every m;
- for each of the request families $F = U_C$ and $F(r,j) = V_C(\ell,r,j)$, and every fixed r, there is m_0 such that $|F(r,j)| \leq 1 + \log\log\hat{\beta}_m$ whenever $m \geq m_0$ and $j \leq J_{m+1}$;
- $V_C(\ell,0,J_{m+1}) \leq \hat{\beta}_m$ for every m.

The comparison and custody data have canonical choices depending only on G, used in Lemma 4.17. The additional conclusions about orders, periods and extraction in III.2 are not needed here.

Lean: `selectRunWithTreasuryPeriod_preselected_quote_orders_postnumeric`; the plain version is `selectRun`. *(statement only; the construction behind it belongs to III.2)*

Lemma 4.17 (assembly of the construction run). Proposition III.14.1 holds.

Sketch: use the canonical comparison and custody data and apply Lemma 4.16 with Λ identically true, $\Delta = \Delta_{10}$, and $\mathcal{D}(s,q)$ the set of recipes for s. Proposition III.14.2 makes this set nonempty. Use floor E_C, reciprocal-width requests $U_C(r,j) = \rho_C(r,j)^{-1}$, where $\rho_C(r,j) > 0$ is the recipe width for demand r at smoothing index j, and $V_C(\ell,r,j) = B_C(\ell,j)$, independent of r. Lemma 4.15 realizes the selected recipe on the run.

Lean: `constructionRun`, `postnumeric`.

3.17.4.6 4.6 The packages behind the numerical floors The combined recipe floor includes `rail`, and its post-length bill uses `receipt`/`after`. The second numerical component uses `Yamen.need` and `Yamen.stamp`.

The supporting estimates below supply finite widths, numerical floors and post-length bounds. The widths also enter the reciprocal-width requests of Lemma 4.17. These are summaries under the hypotheses of the linked results; the complete auxiliary statements are not reproduced here.

- **One-step transit and graded differences.** `harbor`: for admissible parameters there are a constant $A \geq 1$ and a growth function J of controlled degree bounding a one-step transit of paired marked kernel families. Supporting results: `DocketBill`, `slushy_uniform`, `conquer`, `drive` (graded differences of two kernel families with scale-independent constants).
- **Curvature-coefficient lifts with linear cost.** Closed coefficients satisfying the required grading, symmetry and vanishing conditions admit curvature lifts, with reduced width and a bound linear in the input size. The bilinear case needs an additional zero-momentum condition: `line_legs`, `doubleLift`, `arc`, `warehouse`, `arm`, `multi`.
- **Formal Gaussian change of variables.** Near-identity substitutions preserve the formal Gaussian functional after Jacobian correction. The supporting results include marked-variable vanishing and compatibility of connected logarithms: `bottle`, `still`, `absorbed`, `impress`.
- **Formal Ward identities and gauge variations.** Change of variables with an insertion, a finite-torus Ward identity, and bounds on the response to adding gradients: `stirred`, `welcomed`, `spark`.
- **Representatives and evaluation.** Gauge-equivalent parametrizations give equal block-integral germs and rescaled coefficients under the stated regularity conditions; the block-integral comparison also assumes equality of the quadratic germs. A related formula evaluates

completed coefficients by averaging over signed lattice symmetries: `blossom`, `switch_rep`, `hatch`.
- **Marked-kernel receipts.** Bounds on marked kernels including the exact leading term, with constants announced before the block factor: `podium`, `baton`.
- **Line inverse in cell coordinates.** An algebraic inverse on the constrained subspace, followed by strip bounds and the finite-torus inverse identity: `jointProjection`, `preheat`, `transcribe`.

3.17.5 5. Where the difficulty lies

- **Degenerate quadratic forms.** Gauge zero modes prevent inversion of $d^\top M d$. The mixed Schur identity propagates curvature representatives without this inversion.
- **Neutrality.** Zero row sums are required separately in every target orientation. Kinetic normalization gives the initial cancellation; the constant-row-sum identity for the free transfer kernel preserves it. Scaled translation covariance alone does not suffice.
- **The exponent.** The width floor reduces the L^{-1} gain to $L^{-1/2}$, and the recursion estimate uses half of the remaining exponent to absorb age growth. The resulting exponent $1/4$ is not claimed to be optimal.
- **Order of choices.** The flat floor is fixed from the price map before the block factors. It involves the constants at indices $j, j-1$ and widths at $j, j+1$. The post-length bill then enforces the delay condition through the coupling thresholds.
- **Scope.** The estimate controls the Gaussian link kernel $d^\top M^\infty d$. It gives no convergence rate for the undetermined part of the plaquette metric, positive-order coefficients or source kernels. The formal recipe is not an identification with physical marked observables.

3.17.6 6. Notation

Written proof	Lean	Meaning
run R, campaign T, pillars W	`Run`, `Campaign`, `Pillars`	Wilson family with its scale and extraction data; campaign comparison data; selected histories from stage 0 for all (N, p), with local step data and accounting identities
funded	`R2I.Funded`	13 clauses: `localBudget`, `attentive`, `logCopy`, `locrian`, `wild`, `silk`, `poly`, `fixed`, `tic`, `halo`, `paced`, `stoves`, `hasSky`
Δ_{10}	`garment10`, `Decal`	local one-step estimate package with its eventual-validity lemma
sheet s	`Sheet`	kernel choice, stage profile, floors; fixed before L
price map $P = (\alpha_j, \Omega_j, \eta_j)$	`MapPrices`	strip width, upper bound and coercivity bound for each smoothing index

Written proof	Lean	Meaning
horizon H, sufficient for P	`Horizon`, `Suffices`	metrics $M_{N,i}$ and transfer data, with strip bounds and Schur relations
$\mathcal{K}^{\mathbb{T}}_{N,i} = d^{\top} M^{\mathbb{T}}_{N,i} d$	`Precision.K`, `quadratic`	Gaussian quadratic form on a torus
$\mathcal{O}_{N,i}$	`output`, `writ`	position-space kernel $d^{\top} M^{\infty} d$ on links
background-field relations	`Inputs`	$\mathcal{K} W = Q^{\top} \mathcal{K}'$, $Q W = 1 - b^{-1} \mathcal{K}'$
curvature transport $T_{m,i}$	`fine_share`, `response`	$(M^{\infty}_{m,i+1})^{-1} \mathcal{T}_{m,i} M^{\infty}_{m,i}$
h_i, $q_i(u)$	`header`, `quote`, `neutralCoin`	$\exp(4)\lvert\mathcal{F}\rvert^2 \Omega^2_{J_{i+1}} \Omega_{J_i}$; $2\lvert\mathcal{F}\rvert^2 h_i^2/(u L_i)$
flat representation	`FlatRep`, `Sigil`	Definition 4.1
weighted norms	`Folded`, `RowCost`, `Atlas`	plaquette norm $\lVert K\rVert^{\mathrm{pl}}_{\sigma}$; link row norm $\lVert F\rVert_{\sigma}$; row, column and first-difference bounds of order m
neutral, translation invariant	`neutral`, `town`	zero symbol at zero momentum; $K(x+u, y+u) = K(x, y)$
w_j, $\Lambda_{k,i}$	`curve`, `clock`, `Lambda'`	$\min(1, \alpha_j, \alpha_{j+1})$; $5 + \log \widehat{\beta}_k + \lvert s_i - s_k\rvert$
delay condition	`Delay`	$\Omega_{J_i} \le \log \widehat{\beta}_i$ for $i \ge e$
flat rate (III.14.3)	`Quorum`	clauses (a)–(c)
family and numerical assumptions	`World`, `Funding`, `Spacing`	hypotheses of III.14.4; their form on a common level system
stairs, drop, weight	`Stair`, `drop`, `Fade`	index family; $L_i^{-\theta}$; age polynomial times $(a_t/a_i)^c$
shadow	`shadow`	same level system with clip profile replaced
recipe C, floor E_C, bill B_C	`Choice`, `floor`, `post`	three components with pre-length floors; combined floor E_C and post-length bill B_C
point-group invariance (“sworn”)	`Bid.Sworn`	colour-summed bilinear form invariant under signed axis permutations
background-field identities (“licensed”)	`Aura.licensed`	free background map satisfies the background-field identities for the actual forms
“prices”, “bill”, “floor”, “numeric”, “post”	—	numerical bounds on block factors and the reference coupling

3.17.7 7. What remains unexplained

1. **Selection in III.2.** Lemma 4.16 is used as an input; the underlying construction of the scale and extraction data is not audited here. Its two auxiliary inputs have canonical choices. Removing the recipe demand would give a separate existence statement, not an identification of the resulting run with the fixed run of Proposition III.14.1.
2. **The funded condition.** Its thirteen clauses are retained in full, rather than replaced by their summary in Proposition III.14.1. The route through `fromFunded` uses the clauses for extraction, comparison, OPE and asymptotic control; their detailed mathematical meanings and proofs belong to III.15 and later units. In particular, that route does not project the local-estimate clause involving Δ_{10}.
3. **Auxiliary estimates in §4.6.** Only their statements and numerical uses are summarized. Their full admissibility conditions and proofs, including the curvature-lift, marked-kernel and formal Ward estimates, remain outside this verification. The summaries are not substitutes for those hypotheses.
4. **Inputs from III.3 and III.8.** The free-kernel existence theorem and the underlying lattice kernel estimates are used at their stated interfaces. The proofs of those upstream results are not audited here.

3.18 III.15. Cost accounting for observables and the remainder comparison

Part: III, Ultraviolet construction (renormalization group). **Lean:** chapter 32: `YM/RemainderComparison/` (packages Chron, Cadenza, Halo, Muse, Solo, Lyric, Herbal, Cascade, Sleet, Access, Portal, Aux and about twenty smaller ones), `YM/Progress/DIRECTED_RECIPE/`, `YM/Next/SCHEDULE/`, `YM/RG05Readout/`, `YM/Integration/ChronPreconstruction/`, `YM/Integration/RG05Preconstruction/`, and small parts of `YM/R2I/`, `YM/RG12/`, `YM/OPE/Therm/`. **Size:** 7,382 lines in 1,053 declarations used by the main theorem. About 4,200 of these lines are the launch theorem and its prerequisites.

Overview. The unit has three independent parts, plus some record types.

- **(A) Pair-step pricing** (`DIRECTED_RECIPE`). Two numerical facts used by the RG step of III.9 and III.10. A hard family of polymer activities is merged with the Mayer expansion of the exponential of soft local terms. The merged family keeps a weighted polymer bound and carries an explicit Gaussian large-field factor (III.15.1–2).
- **(B) Derivative bounds for local observables** (`SCHEDULE/Wages`). On the small-field region, a transported curvature slot has left derivatives of every order $\leq K$ bounded by a constant times t. A k-leg monomial in slots has them bounded by a constant times t^k. Here t is the small-field threshold (III.15.3–4).
- **(C) Launch data for the remainder comparison** (`RemainderComparison`). For connected-activity targets with rates below $-\frac{3}{8}$, a sufficiently large block side permits the numerical conditions of a comparison step. After the auxiliary data are fixed, one chooses the comparison functions and then an entry coupling above which those conditions hold (III.15.5–7).
- **(D) Records and numerical definitions.** These include the construction-run bounds, the additional-expense and readout records, and the RG05 readout prices. Their scope is described in §4.6.

Status of the parts for the main theorem.

- (A) and (B) are consumed by III.2, III.9 and III.10.
- (C) supplies additional comparison data in the main-theorem dependency chain:
  - III.2's assembly lemma `YM.PhysicalNext.Repeat.Legato.R2I.RG05Joint.joinExpensesShipWithPrec` calls the launch and delay-enlargement theorems to build the additional conjunct `YM.PhysicalNext.Repeat.Legato.Stagehands.News`, which records a family of quarries and their stagewise bounds.
  - That conjunct is passed unchanged up to the selected-run theorem `YM.PhysicalNext.Repeat.Legato.R2I.RG05Joint.selectRunWithTreasuryPeriod_preselected_quo` of III.14.
  - In the main-theorem dependency chain, `YM.Independent.constructionRun` discards it. The retained construction-run conditions form a separate record.
  - Part C imposes additional lower bounds on block sides and entry couplings. Its comparison conclusions are absent from the final construction-run statement.

Conventions.

- x denotes the coarse renormalized inverse coupling $\hat{\beta}$.
- $\beta_F(x, L) = x + b_G \log L$ is the inverse coupling at the next finer scale (`YM.Pressure.betaF`); b_G is the slope constant of G, positive when G is simple.

For $\beta > 0$ and $x > 1$, use the scale functions

$$g(\beta) = \beta^{-1/2}, \qquad p(x) = e^{(\log\log x)^2}, \qquad t(x) = g(x)\,p(x), \qquad \mathcal{B}(x) = \lceil (\log x)^2 \rceil.$$

- t is the small-field threshold (`YM.Pressure.temp`, equal to `YM.Next.Scheduling.Instrument.t` for the plain instrument).
- $\mathcal{B}$ is the cluster radius (`YM.Pressure.radiusB`, the same formula as `YM.Next.Scheduling.Clock.cover`).
- The band of y is $[y/4, 4y]$ (`YM.Next.Scheduling.inBand`). On it, $g(y)/2 \le g(\beta) \le 2g(y)$ (`YM.Next.Scheduling.band_gap`).

Growth classes, at fixed L, as $x \to \infty$:

- **Rate** $\le a$ (`YM.ConnectedActivityE6.Rate`): for every $v > a$, eventually $|w(x)| \le \beta_F(x, L)^v$.
- **Reserve** (`YM.RemainderComparison.Reserve`): $|w(x)| \le M(x)^k$ eventually for some $k \in \mathbb{N}$, where $M(x) = p(\beta_F(x, L))$ (`YM.Push.AbsoluteSums.Delay.fineMonitor`). Every reserve has rate at most 0.
- **Gear** a (`YM.RemainderComparison.Gear`): $\beta_F^{-a} w$ is a reserve, equivalently $|w(x)| \le \beta_F(x, L)^a M(x)^k$ eventually for some $k \in \mathbb{N}$. This implies rate at most a.

Weighted hit bound (`YM.Ac.Hit`): for supports X_i, weights $f_i \ge 0$, scale M and exponent a, write $\mathrm{Hit}_M(f; a) \le \gamma$ if

$$\sum_{i:\, z \in X_i} f_i\, e^{a N_M(X_i)} \le \gamma \quad \text{for every site } z,$$

where $N_M(X)$ is the minimal number of M-balls with connected union covering X.

The quote is $\mathrm{quote}(\kappa, \gamma) = e^{-\kappa}(e^{\gamma} - 1)$ (`YM.Bid.quote`).

In the code, a price or bill is a numerical bound; "financed" means that the bound satisfies the specified growth conditions.

3.18.1 1. What this unit proves

3.18.1.1 Part A: pair-step pricing Proposition III.15.1 (pair-step rates). Let $\rho > 0$, $C \geq 0$, $r > \rho$, $d \in \mathbb{N}$, $e \in \mathbb{R}$ and $q > 0$. For $y > 1$ and $\beta > 0$ put

$$s(y,\beta) = \frac{t(y)}{g(\beta)} = p(y)\sqrt{\beta/y}, \quad \mathrm{mild}(y) = C(1+\mathcal{B}(y))^d p(y)^e g(y)^q,$$

$$\mathrm{strong}(y,\beta) = Ce^{-rs^2}, \quad \kappa(y,\beta) = \rho s^2.$$

For all sufficiently large $y > 1$, the following holds for every $\beta \in [y/4, 4y]$ and every integer $0 \leq B \leq \mathcal{B}(y)$. With $a = \mathrm{strong}(y,\beta)$, $b = \mathrm{mild}(y)$ and $\gamma = e^{\kappa}a + 2b$:

$$\kappa \geq 0, \quad a \geq 0, \quad 0 \leq b \leq 1, \quad (2B+1)^4\gamma \leq 1, \quad \mathrm{quote}(\kappa,\gamma) \leq e^{-\kappa}.$$

These are the rate conditions with target $w = e^{-\kappa}$. The threshold in y depends only on (ρ, C, r, d, e, q). *Status:* theorem. *Lean:* `YM.Progress.DirectedRecipe.Payments.Pairing.currency`; data `YM.Progress.DirectedRecipe.Payments.Pairing`; conditions `YM.Progress.DirectedRecipe.Payments.Rates` and `YM.Progress.DirectedRecipe.Payments.inflation`. *Used in:*

- III.9, in `YM.AnalyticBridge.VerifiedReset.Time.cash`, with $d = e = 0$, $q = \frac{1}{2}$, $y = x + b_G \log L$ and $B = \mathcal{B}(x)$.
- III.2, in `YM.PhysicalNext.Repeat.Jams.reunite`.

Proposition III.15.2 (merging a hard and a soft family). Fix a parameter point P with derivative order $K \geq 6$, direction power $P_0 \geq 2$, cluster scale $B_P \geq 1$, support exponent $\sigma > 0$ and threshold $0 < t_P \leq 1$. Consider a countably indexed regular family E_i, a **hard family** of forced activities F_i with supports X_i, and a **soft family** of real local exponential corrections ℓ_i with nonempty supports X_i'. Assume:

1. The regular terms satisfy the regular-family conditions at P: regularity on their supports, vanishing at the flat field, and vanishing affine Hessian on nonwrapping supports. The forced terms and their eligibility tests are gauge invariant, measurable and determined by the links and final flags on their supports; a nonzero forced term satisfies its eligibility test. The regular and forced families admit the prescribed translation/reflection permutations, including the support and inventory transformations. Each ℓ_i is gauge invariant, measurable and determined by the links and final flags on X_i'; it vanishes on configurations small-field on X_i'. The soft family is also covariant under translations and reflections.
2. The regular family has bound δ: it admits nonnegative bounds on all left derivatives through order K on its small-field supports, in the directions specified by P, whose weighted sum at each anchor is at most δ.
3. For every i and every configuration U, $|F_i(U)| \leq p_i$ and $|\ell_i(U)| \leq q_i$, with $p_i, q_i \geq 0$.
4. $\mathrm{Hit}_{B_P}(p; \sigma+1) \leq a$ on (X_i) and $\mathrm{Hit}_{B_P}(q; \sigma+1) \leq b$ on (X_i').
5. The rate inequalities of III.15.1 hold with $B = B_P$, with the final inequality replaced by $\mathrm{quote}(\kappa, e^{\kappa}a + 2b) \leq w$.

Then the fusion input condition with (δ, w) holds. Explicitly:

- the original regular and forced families retain bounds δ and a, respectively; the soft corrections admit a summable uniform majorant;
- the merged family $(F_i) \sqcup (e^{\ell_i} - 1)$, with weights $e^{\kappa} p_i$ and $2q_i$, satisfies $|F_i| \le e^{-\kappa} \cdot e^{\kappa} p_i$ and $|e^{\ell_i} - 1| \le 2q_i$;
- that family has $\mathrm{Hit}_{B_P}(\cdot\,; \sigma + 1) \le \gamma := e^{\kappa} a + 2b$, with $(2B_P + 1)^4 \gamma \le 1$;
- $\mathrm{quote}(\kappa, \gamma) \le w$.

Status: theorem. The only analytic input is $|e^z - 1| \le 2|z|$ for $|z| \le 1$. Through the fusion lemmas of III.10 (`Work.H01`–`H03`, used in `YM.Progress.DirectedRecipe.Ingredients.gates`), the conclusion gives the fused polymer family with norm bound $\le w$. With $w = e^{-\kappa}$ from III.15.1, this is the Gaussian large-field factor $\exp(-\rho\,(\beta/y)\,p(y)^2)$. *Lean:* `YM.Progress.DirectedRecipe.Payments.tender`; `YM.Progress.DirectedRecipe.Payments.hardCost`, `YM.Progress.DirectedRecipe.Payments.softCost`, `YM.Progress.DirectedRecipe.Payments.rooted`, `YM.Progress.DirectedRecipe.Payments.Fittings`, `YM.Progress.DirectedRecipe.Ingredients`. *Used in:* III.9 (`YM.AnalyticBridge.VerifiedReset.Collection.Local.renew`) and III.10 (`YM.PhysicalNext.Repeat.Marlin.cannon`).

3.18.1.2 Part B: derivative bounds for local observables A *slot* s is a lattice walk ω from the origin, followed by a plaquette π (two axes) at the endpoint of ω (`YM.RG.Slot`). Its length is $|s| = |\omega| + 4$, and its value at a site o is the transported curvature

$$s(U, o) = \mathrm{Ad}(U_\omega)\, \mathcal{C}(U_\pi) \in \mathfrak{g},$$

where $\mathcal{C}$ is the colour projection of the anti-Hermitian part.

A *k-leg atom* a has slots $s_1, \dots, s_k$ and a real coefficient tensor c_b indexed by colour multi-indices $b \in \{1, \dots, \dim \mathfrak{g}\}^k$ (`YM.RG.Atom`). Its value is $a(U, o) = \sum_b c_b \prod_i s_i(U, o)^{b_i}$, its size is $|a| = \sum_b |c_b|$, and its length is $\mathrm{len}(a) = \sum_i |s_i|$.

Fix a parameter point with derivative order $K \ge 6$, power $P \ge 2$ and threshold $t \in (0, 1]$. Directions are colour-valued increments A of the left derivative ∂_A. A direction is *admissible at o* if

$$\|A_e\| \le t\,(1 + \mathrm{dist}(e, o))^P \quad \text{for every link } e.$$

Here $\mathrm{dist}(e, o)$ is the periodic maximum distance from the basepoint of e to o (`YM.RG.BudgetPoint.dirBound`).

A configuration is *small-field on X* (healthy, `YM.RG.Healthy`) if every plaquette touching a site of X has chord distance $\le t$ from the identity. Let $\kappa_G > 0$ be the colour scale of G and $\|1\|$ the Frobenius norm of the identity matrix.

Proposition III.15.3 (slot derivative bound). Let $\ell \in \mathbb{N}$ and s be a slot with $|s| \le \ell$, let S be a finite set of points of $\mathbb{Z}^4$ containing the stencil of s, and let o be a site. Then $U \mapsto s(U, o)$ is K times left-differentiable. Let U be small-field on $o + S$ (wrapped on the torus), and let $A_1, \dots, A_m$ be admissible directions at o with $m \le K$. Then

$$\left\| \partial_{A_1} \cdots \partial_{A_m} s(U, o) \right\| \le \mathrm{slotCost}(P, K, \ell)\, t,$$

$$\mathrm{slotCost}(P, K, \ell) = (1 + \kappa_G^{-1} \|1\|)\, \mathrm{margin}(P, \ell)^K, \qquad \mathrm{margin}(P, \ell) = 2(\kappa_G + 1)\, 2^P (\ell + 1)^{P+1}.$$

The constant depends on P, K, ℓ, G only, not on t, o or the volume. At $m = 0$ the stronger order-zero bound is $\|s(U,o)\| \le t$. *Status:* theorem. *Lean:* `YM.Next.Scheduling.Wages.slotPays`; `YM.Next.Scheduling.Wages.slotCost`, `YM.Next.Scheduling.Wages.margin`, `YM.Work.G07.LeftBound`. *Used in:* III.9 (`YM.AnalyticBridge.VerifiedReset.Low.plain_bill`) and III.15.4.

Proposition III.15.4 (bound for k-leg atoms). Let $k \in \mathbb{N}$, let a be a k-leg atom, o a site, and U small-field on the footprint $o + \mathrm{stencil}(a)$. Then $U \mapsto a(U,o)$ is K times left-differentiable, and every left derivative of order $m \le K$ in admissible directions at o is bounded by

$$\mathrm{price}(a) = |a|\, 2^K \big(2^K\, \mathrm{slotCost}(P, K, \mathrm{len}(a))\, t\big)^k = \mathrm{start}(G,k,P,K)\; t^k\; |a|\; (\mathrm{len}(a)+1)^{(P+1)Kk},$$

where $\mathrm{start}(G,k,P,K) = 2^K \big(2^K \mathrm{slotCost}(P,K,0)\big)^k$. *Status:* theorem. *Lean:* `YM.Next.Scheduling.Wages.fin`, `YM.Next.Scheduling.Wages.price`, `YM.Next.Scheduling.Wages.scan`, `YM.Next.Scheduling.Wages.startup`. *Used in:* III.9 (`YM.AnalyticBridge.VerifiedReset.High.paid`) and III.10.

3.18.1.3 Part C: launch data for the remainder comparison

A *connected-activity target* Q (`YM.ConnectedActivityE6.Target`, III.10) carries:

- a rate $\rho_Q \in (-\frac{1}{2}, 0)$;
- an exponent $\mathrm{e}_Q \ge 0$;
- small parameters $\varepsilon_Q, \delta_Q > 0$ with $4\varepsilon_Q$ below the top support exponent;
- a halo $h_Q \ge 500$;
- a chart table with its local-ratio property. The target also requires $8\varepsilon_Q$ below both input chart widths.

Its relative exponent is $\mathrm{rel}_Q = \mathrm{e}_Q + 2\varepsilon_Q + \delta_Q$.

Stage pairs (j,i) are those with $i \in \{j, j-1\}$ (`YM.Push.PriceCompiler.StepStages`). The stage profile P_j has support exponent σ_j.

Proposition III.15.5 (launch data). Let G be simple. Fix:

- a sheet with stage profiles P_j and a kernel choice;
- a compatible price table and targets R_{ji} with $\frac{3}{8} + \rho_{R_{ji}} < 0$ for all $j, i \in \mathbb{N}$;
- tolerances $h_{ji} > 0$;
- libraries C_{ji} (of the order of P_i);
- mark numbers r_{ji}.

Then there are thresholds $f : \mathbb{N} \to \mathbb{N}$ and constants $C'_{ji} \ge 1$ such that:

1. For all j, i, the *quotation bound* holds for $(C_{ji}, \mathrm{power}(P_i), r_{ji})$ with fine exponent σ_j, coarse exponent $\sigma_i + 2$ and constant C'_{ji}. More precisely, take an RG step of side L, fine and coarse parameter points B, P, and supports X, Y such that Y lies in the $r_{ji}\mathcal{B}(x)$-neighbourhood of the coarse image of X. Assume P has the power of P_i, B has support exponent σ_j, $B_P = \mathcal{B}(x)$, $B_B \le 2B_P$, $t_P \le 2t_B$, and the acceptance condition in (c) below. For every $u \ge 0$ satisfying the local analytic smallness condition for the return fee and every $\tau \ge 0$,

$$e^{(\sigma_i+2)N_{B_P}(Y)}\, \tau\, \mathrm{returnFee} \le C'_{ji} L^{-5} e^{\sigma_j N_{B_B}(X)} \tau.$$

2. For every stage pair (j,i) and every block side $L \ge f_j$:
   - (a) the quay condition (III.15.6) holds for R_{ji} at side L;
   - (b) $(\mathrm{rel}_{R_{ji}}+1)\,10\,c_{h+5}/L + 2\varepsilon_{R_{ji}} \le \sigma_j$, where c_{h+5} is the pre-cost of halo $h_{R_{ji}}+5$;
   - (c) the acceptance condition $20(\sigma_i+2)\,\mathrm{toll}(r_{ji}) \le L\sigma_j$;
   - (d) $C'_{ji}/L \le h_{ji}$.

Status: theorem. The construction run calls it with rates $< -\frac{7}{16}$; these are supplied by III.2's company selection, and $-\frac{7}{16}$ implies the hypothesis. *Lean:* `YM.RemainderComparison.Chron.sails`, `YM.RemainderComparison.Chron.Pledge`, `YM.PhysicalNext.JointLoad.Heir.Release.Quotation`, `YM.PhysicalNext.JointLoad.Heir.Release.accepts`, `YM.RemainderComparison.Lyric.radiusCost`, `YM.RemainderComparison.Chron.Contemple` (tolerances).

The *heart condition* on rays for a stage E' and exponent e_Q consists of the support inequalities

$$0 \le \sigma_*, \quad \sigma_{E'} \le \sigma_*, \quad \sigma_* + 1 \le a_*, \quad \varepsilon_* > 0, \quad a_* + 3\varepsilon_* \le \mathrm{e}_Q,$$

and the order inequalities

$$\mathrm{ord}(E') \le J_*, \quad \mathrm{ord}(E') + (p_*+1) + p_* + 2 \le n_*, \quad p_* + 1 \le s_*, \quad k_* + 1 \le s_*.$$

The starred quantities are the support exponents, slack and requested orders carried by the rays.

Proposition III.15.6 (the quay for large sides). Let G be simple, K a kernel choice, and Q a target with $\frac{3}{8} + \rho_Q < 0$. There is n such that for every block side $L \ge n$ the following holds. Take any:

- patch p of side L satisfying the branching condition;
- $W, \mu > 0$, and order datum at W;
- songbook and scrollwork at side L, and χ;
- stages E, E', and rays with the heart property for (E', e_Q);
- library C of the order of E';
- record with parameter $\varepsilon \le \frac{1}{4}$;
- $j \in \mathbb{N}$, reserve f, and $\eta, \delta > 0$.

Then there exist a quarry D (a record of bounding functions of x, with $d(D) > 0$ and paint supply) and a threshold P_1 with:

- D is financed at rate ρ_Q;
- for every $x \ge P_1$, the poster comparison and chart conditions and the instructions hold. For every $\beta \in [\beta_F(x,L)/4, 4\beta_F(x,L)]$ and every parameter point Z tempered to E' at x, the active, resonance and tolerance conditions hold.

Here *tempered* means that Z has cluster radius $\mathcal{B}(x)$, threshold $t(x)$, the order, power and support exponent of E', and threshold at most $2t(\beta_F(x,L))$. The active conditions comprise the local analytic estimates, readiness of the spatial restrictions, compatibility estimates and the prescribed tail bounds. The block-side threshold n is independent of the auxiliary data; the quarry and P_1 may depend on all of them. The quarry is a fixed record of functions of x. *Status:* theorem. *Lean:* `YM.RemainderComparison.Cadenza.buoys`, `YM.RemainderComparison.Cadenza.Quay`, `YM.RemainderComparison.Cadenza.Allot`, `YM.RemainderComparison.Cadenza.Coaxes`, `YM.RemainderComparison.Cadenza.Quarry.Financed`.

Proposition III.15.7 (using the launch data). Fix a sheet, compatible target and expense tables R, X, chosen block sizes L_j, a protection, a register and a firmament. The register supplies W_{ji}, μ_{ji} and order data, with positivity for stage pairs. The firmament gives, for each stage pair, finitely many *request slots*; each carries a scrollwork, a songbook, grade and order data, a reserve, positive constants, and an extra condition that holds eventually for every financed quarry. Fix also tolerances, libraries C_{ji}, mark numbers r_{ji} and records u_{ji}. Assume:

- launch data as in III.15.5 with thresholds $k_j \le L_j$;
- $\varepsilon(u_{ji}) \le \frac{1}{4}$ for stage pairs with i at least the first stage;
- $\sigma_i + 3 < \mathrm{e}_{R_{ji}}$ for all j, i.

Then for every initial delay datum T_0 there are a delay datum $W \ge T_0$ and a quarry V_{jig} for every request slot g, indexed at all (j, i), with the following properties for every delay datum $U \ge W$:

- (a) launch data with $f_j \le L_j$;
- (b) for every stage pair (j, i) with $j \ge 4$, i at least the first stage, and the patch branching, and for every slot g: the slot's rays have the heart property; $d(V_{jig}) > 0$; paint supply holds; the allotment of III.15.6 holds with threshold $\beta_j^*(U)$; and the slot's extra condition holds for $x \ge \beta_j^*(U)$;
- (c) for every stage pair with i at least the first stage, every request slot g, every $x \ge \beta_j^*(U)$ and every torus side $c \ge \mathrm{shells}_j(U)\,(1 + \mathcal{B}(x))^{\mathrm{growth}_j(U)}$, the slot's scope is tiled at (x, c). This means $2z_g(x) < c$ and $4\,\mathrm{halo}_g(x) + 16 \le c$, where z_g and halo_g belong to its scrollwork.

The order on delay data increases entry couplings, scale-entry functions, shell counts and growth exponents componentwise. The quarries are chosen before U.

Status: theorem. *Lean:* `YM.RemainderComparison.Chron.grows`, `YM.RemainderComparison.Chron.Pillar`, `YM.RemainderComparison.Chron.Firmament`, `YM.RemainderComparison.Chron.Glint`. *Use of III.15.5 and III.15.7:* III.2 (`YM.PhysicalNext.Repeat.Legato.R2I.RG05Joint.joinExpensesShipWithPrediction`), whose comparison conjunct reaches the selected-run theorem of III.14 and is discarded by the final construction-run theorem (see the overview).

3.18.2 2. What it uses

From other units.

- III.1: the scale functions t, $\mathcal{B}$ and β_F; slots and atoms.
- II.1: ordered left-derivative calculus; the order-zero slot bound `YM.Work.G07.slot_bound`.
- II.2: weighted hit bounds.
- III.5: the logarithmic tail bound of the cluster expansion, used in Lemma 4.15.
- III.10, a large set of inputs:
  - the growth classes, connected-activity targets and step records;
  - the three fusion theorems, which turn the fusion input condition into a fused activity bound;
  - the quotation constants `YM.PhysicalNext.JointLoad.Heir.Release.quoting` (including the factor L^{-5});
  - the per-vertex weight `YM.PhysicalNext.Repeat.Height.grow` $= \beta_F^{3/8}$ and the criterion `YM.PhysicalNext.Repeat.Height.pressureYoung`;

  - the local derivative and connected-activity estimates, including the rate-ρ_Q bound on the tree sum.
- III.2: the targets R_{ji} with rates $< -\frac{7}{16}$ and $\sigma_i + 3 < \mathrm{e}_{R_{ji}}$ (the expense records of the company selection). Also the block-size and delay order into which III.15.5 and III.15.7 are inserted.

Standard mathematics.

- Cluster and polymer expansions with weighted per-site bounds (Kotecký–Preiss, Commun. Math. Phys. 103 (1986); Brydges, *A short course on cluster expansions*, Les Houches 1984).
- The small-field/large-field decomposition and Mayer expansion used in the RG step. The L^{-5} estimate is an input from III.10, with its explicit support and analytic hypotheses.
- The Leibniz rule and the chain rule for products of group elements along a path.
- $|e^z - 1| \le 2|z|$ for $|z| \le 1$.
- Elementary asymptotics: $e^{a(\log\log y)^d}$ grows faster than every power of $\log y$ when $a > 0$ and $d > 1$, and $e^{-c\,p(y)^a}$ decays faster than every inverse power of y when $a, c > 0$.

Imported constants and assumptions.

- Simplicity of G. In Part C it enters through the local single-occurrence estimate used in Lemma 4.16. Parts (A) and (B) do not use it.
- $\frac{3}{8} + \rho < 0$ for the targets. The value $\frac{3}{8}$ is the exponent of the per-vertex tree weight $\beta_F^{3/8}$ of III.10. In Lemma 4.17 it is used to make that weight times the activity summable. Since targets have $\rho_Q > -\frac{1}{2}$, the admissible window is $(-\frac{1}{2}, -\frac{3}{8})$; the run uses $(-\frac{1}{2}, -\frac{7}{16})$.
- The tolerance table h_{ji} and the firmament are inputs. In the run they come from a fixed default (`YM.Integration.RiderResourceDynamics.defaultDynamics`).

3.18.3 3. Strategy

Part A separates the numerical loss from the fusion argument. On the coupling band, the hard bound is Ce^{-rs^2} with $s \ge p(y)/2$. Retaining $e^{-\rho s^2}$ leaves $Ce^{-(r-\rho)s^2}$ to absorb the support-counting factor. The soft bound decays because its negative power of y dominates its powers of $p(y)$ and $\mathcal{B}(y)$. The Mayer expansion then combines the hard activities with $e^{\ell_i} - 1$; their weighted bounds add to $e^{\kappa}a + 2b$. Both families may be supported on large fields. The terminology "hard" and "soft" distinguishes their estimates.

Part B represents a transported curvature slot as the projection of the single group word $\omega \cdot \partial\pi \cdot \omega^{-1}$. Unitarity bounds each differentiated product without an exponential loss in path length. Each derivative contributes one direction norm, at most $t(1 + 2\ell)^P$, and has at most 2ℓ possible link occurrences. For positive derivative order, $t^m \le t$; at order zero, the small-field plaquette bound gives the factor t. The Leibniz rule gives t^k for a product of k slots, with a polynomial loss in total path length.

Part C first chooses a block-side threshold and then, for each permitted side and each fixed set of auxiliary data, chooses comparison functions and an entry coupling. The connected-activity bounds have rate ρ_Q; the extra tree weight $\beta_F^{3/8}$ still leaves negative rate. A marked input family of size $\delta \le \beta_F^{-11/10}$ gives an anchored output bound $(C'/L + o(1))\delta$, subject to the support and local analytic conditions in Lemma 4.16. Clusters with at least one marked occurrence and m further occurrences outside the inserted leaves gain an additional factor $S'(x)^m$ of negative rate. At each fixed expansion order, the tail bounds are small enough to absorb any prescribed power of

$g(\beta)$. Finally, finitely many requests at each stage are met by enlarging the entry coupling and spatial-size thresholds.

3.18.4 4. Steps

3.18.4.1 4.1 Elementary asymptotics Lemma 4.1. For every $k \in \mathbb{N}$, as $x \to \infty$:

1. $p(x)^a g(x)^q \to 0$ for all $a \in \mathbb{R}$ and $q > 0$;
2. $(1 + \mathcal{B}(x))^k / p_d(x)^a \to 0$ for $d > 1$ and $a > 0$, where $p_d(x) = e^{(\log\log x)^d}$;
3. $g(x)^q e^{-c\, p_d(x)^a} \to 0$ for every real q, $c, a > 0$ and $d > 1$.

Sketch: substitute $u = \log\log x$, so that $\log x = e^u$; each claim reduces to a polynomial or exponential in u against e^u or e^{u^d}. *Lean:* `Clocks.lean`; `YM.Next.Scheduling.Clock.light`, `YM.Next.Scheduling.Clock.ceiling_cheaper`, `YM.Next.Scheduling.Clock.burnt`.

3.18.4.2 4.2 Pair-step rates Lemma 4.2 (III.15.1). Under the hypotheses of III.15.1, the rate conditions hold eventually, uniformly over the coupling band and $B \le \mathcal{B}(y)$.

Proof sketch:

1. Put $\eta = (r-\rho)/4$. On the band, $g(\beta) \le 2g(y)$, so $s \ge p(y)/2$. Hence e^κ strong $= Ce^{-(r-\rho)s^2} \le h(y) := Ce^{-\eta p(y)^2}$.
2. Put $v = h + 2\,\mathrm{mild}$. For every k, $(1+\mathcal{B})^k v \to 0$:
   - for h, write $(1+\mathcal{B})^k e^{-\eta p^2} = \big[(1+\mathcal{B})^k g\big] \cdot \big[g^{-1} e^{-\eta p^2}\big]$ and use Lemma 4.1(1, 2, 3);
   - for mild, write $(1+\mathcal{B})^{d+k} p^e g^q = \big[(1+\mathcal{B})^{d+k}/p\big] \cdot \big[p^{1+e} g^q\big]$.
3. With $k = 4$: $(2B+1)^4 \gamma \le 2^4 (1+\mathcal{B})^4 v \le 1$ eventually.
4. $e^v - 1 \le 1$ eventually gives quote$(\kappa, \gamma) \le e^{-\kappa}$, and mild $\to 0$ gives mild ≤ 1.

All thresholds are independent of β and of $B \le \mathcal{B}(y)$. *Lean:* `YM.Progress.DirectedRecipe.Payments.Pairing.currency`, the two private asymptotic estimates in `Economy.lean`. The same file also defines the record of these rate conditions along the chosen sequence of scales.

3.18.4.3 4.3 Merging hard and soft families Lemma 4.3 (blend). Let $p, q \ge 0$ satisfy $\mathrm{Hit}_B(p; \sigma+1) \le a$ on (X_i) and $\mathrm{Hit}_B(q; \sigma+1) \le b$ on (X'_i). Then the weights $(e^\kappa p_i) \sqcup (2q_i)$ on the disjoint union satisfy $\mathrm{Hit}_B(\cdot\,; \sigma+1) \le e^\kappa a + 2b$.

Sketch: multiply a hit bound by a constant, then add two summable families. *Lean:* `YM.Progress.DirectedRecipe.Payments.give`, `YM.Progress.DirectedRecipe.Payments.blend`, `YM.Progress.DirectedRecipe.Payments.merges`.

Lemma 4.4 (activity bounds of the merged family). Under the nonempty-support and positive-exponent conventions of III.15.2, assume $p_i, q_i \ge 0$, $|F_i| \le p_i$, $|\ell_i| \le q_i$, the soft hit bound b, and $b \le 1$. Then $\|F_i\| \le e^{-\kappa}(e^\kappa p_i)$ and $|e^{\ell_i} - 1| \le 2q_i$.

Sketch:

1. Dropping one unit of support exponent gives a sum bound for the weights $e^{\sigma N(X'_i)} q_i$.
2. A single term is at most the sum, so $|\ell_i| \le q_i \le e^{\sigma N(X'_i)} q_i \le b \le 1$.
3. Then $|e^z - 1| \le 2|z|$ for $|z| \le 1$.

Lean: `YM.Progress.DirectedRecipe.Payments.shares`, `YM.Progress.DirectedRecipe.Payments.drop_reserve`

Lemma 4.5 (III.15.2). Under the hypotheses of III.15.2, the fusion input condition holds, with witness $(\kappa,\ e^{\kappa}a + 2b,\ a)$.

Sketch: the original-family estimate combines the structural assumptions, the regular bound δ, the hard sup bounds and the reduction of the support exponent by one. The merged-family estimate uses $\gamma \geq 0$ and $\kappa \geq 0$, the capital inequality, Lemma 4.3 and Lemma 4.4. The quote inequality is the last rate condition. *Lean:* `YM.Progress.DirectedRecipe.Payments.tender`.

3.18.4.4 4.4 Slots and atoms **Lemma 4.6 (a slot is one group word).** For every slot s, site o and configuration U,

$$s(U,o) = \Pi(U_{\omega\cdot\partial\pi\cdot\omega^{-1}}), \qquad \Pi(A) = \mathcal{P}_G(\tfrac{1}{2}(A - A^*)), \qquad \|\Pi(A)\| \leq \kappa_G^{-1}\|A\|.$$

Here $\mathcal{P}_G$ is the colour-coordinate projection with the normalization of G.

Sketch: $\mathcal{C}(hgh^{-1}) = \mathrm{Ad}(h)\,\mathcal{C}(g)$. The detour has length $2|\omega| + 4 \leq 2|s|$. *Lean:* `YM.Next.Scheduling.Wages.detour`, `YM.Next.Scheduling.Wages.projector`, `YM.Next.Scheduling.Wages.hangs`, `YM.Next.Scheduling.Wages.norm_projector`.

Lemma 4.7 (derivatives of a word). Let w be a product of link variables and inverses with n link uses, and let $T \geq 0$. Write ι_G for the colour-to-matrix map. Suppose the matrix representative of every increment satisfies $\|\iota_G(A_e)\| = \kappa_G\|A_e\| \leq T$ on every used link e, for every direction in the list $\vec{A}$. Then

$$\|\partial_{\vec{A}} U_w\| \leq \|1\|\,(Tn)^{|\vec{A}|}.$$

Sketch: the Leibniz rule; each derivative replaces one factor by its left increment. *Lean:* `YM.Next.Scheduling.Wages.Path.bill`.

Lemma 4.8 (III.15.3). Under the hypotheses of III.15.3, the slot derivative bound holds.

Sketch:

1. Every used link basepoint of the detour is within distance 2ℓ of o. Admissible directions therefore give $T = \kappa_G t(1 + 2\ell)^P$, and $T \cdot 2\ell \leq \mathrm{margin} \cdot t$.
2. Lemma 4.7, then Π, give $\kappa_G^{-1}\|1\|(\mathrm{margin}\,t)^m$.
3. Since $\mathrm{margin} \geq 1$, $t \leq 1$ and $1 \leq m \leq K$, this is at most $\kappa_G^{-1}\|1\|\,\mathrm{margin}^K t$.
4. For $m = 0$, use the order-zero small-field slot bound.

Lean: `YM.Next.Scheduling.Wages.slotPays`.

Lemma 4.9 (III.15.4). Under the hypotheses of III.15.4, the atom derivative bound and its closed form hold.

Sketch:

1. Extract colour components by an orthonormal projection.
2. Apply the Leibniz rule to the product of the k leg components: two derivative bounds c_1, c_2 up to order K give $2^K c_1 c_2$.
3. Multiply by c_b and sum over b.
4. The closed form follows from $\mathrm{slotCost}(P,K,\ell) = \mathrm{slotCost}(P,K,0)(\ell+1)^{(P+1)K}$.

Lean: YM.Next.Scheduling.Wages.fin, YM.Next.Scheduling.Wages.scan.

3.18.4.5 4.5 The Chron chain (Part C) The following conventions specify the conditional estimates in this chain. An *admissible step record* consists of step data with the prescribed patch, a stamped tube, an input representation R, and its connected-activity hypotheses at (W, x, μ, L, Q). A *marked input* (E, w, δ) is a set E of regular input indices with derivative bounds $w_i \geq 0$, zero outside E, satisfying

$$\sum_{i:\, o_i = z} e^{\sigma_R N_{B_R}(X_i)} w_i \leq \delta \qquad \text{for every fine site } z.$$

Here o_i, X_i are the input anchors and supports, and the derivative bounds use the input parameter point. In particular, $\delta \geq 0$.

For a finite family of inserted leaves with supports Z_α, nonnegative coefficients l_α, attached root sets A_α and leaf weights ω_α, the *leaf-scale condition* with parameters (n, σ, Y) requires $Y \geq 0$, at most n roots over each coarse site, and

$$\sum_{\alpha:\, v \in A_\alpha} e^{\sigma N_{B_R}(Z_\alpha)} \omega_\alpha \leq Y \qquad \text{for every root } v.$$

The weights ω_α include the leaf norm and l_α, and each positive weight must have an attached root lying in its leaf support. The *radius condition at* σ is $(\mathrm{rel}_Q + 1)10c_{h_Q+5}/L + 2\varepsilon_Q \leq \sigma$.

A cluster together with chosen leaf insertions has N *remaining occurrences*: the sum, over its vertices, of the numbers of input occurrences outside those insertions. It is *marked* if at least one remaining occurrence comes from E. Thus $N \geq 1$ for a marked cluster, but the other remaining occurrences need not be marked. The insertion count is the total number of leaves, counted with multiplicity. These are the two counts used below.

Lemma 4.10 (quotation and acceptance). Let t, u be stage profiles, C a library of order $\mathrm{ord}(u)$, $r' \in \mathbb{N}$ and $h > 0$. There are $E \geq 1$ and k such that:

- the quotation bound holds with exponents $(\sigma_t, \sigma_u + 2)$ and constant E under the support and analytic hypotheses of III.15.5;
- for every $L \geq k$: $20(\sigma_u + 2)\mathrm{toll}(r') \leq L\sigma_t$ and $E/L \leq h$.

Sketch: the quotation theorem of III.10 gives an L-independent E. Then $E/L \to 0$ and $L\sigma_t \to \infty$. The return-fee estimate gives the factor L^{-5}. Summing over the L^4 fine sites of a block leaves E/L, by the anchored-sum estimate. *Lean:* inner step of YM.RemainderComparison.Chron.sails; YM.PhysicalNext.JointLoad.Heir.Release.quoting.

Lemma 4.11 (radius cost). If $b > 0$ and $b + 2\varepsilon_Q \leq v$, then $(\mathrm{rel}_Q + 1)10\, c_{h_Q+5}/L + 2\varepsilon_Q \leq v$ for all large L. In III.15.5, take $b = \varepsilon_Q$: the target condition gives $3\varepsilon_Q < 4\varepsilon_Q < \sigma_j$.

Sketch: multiply the desired inequality by $L > 0$ and choose L so large that $(\mathrm{rel}_Q + 1)10c_{h_Q+5} \leq bL$. *Lean:* YM.RemainderComparison.Lyric.bidLift.

Lemma 4.12 (comparison functions). Fix a target Q, a patch, stages E, E', a kernel choice, an order datum at W, a songbook, a scrollwork, $\mu > 0$, and rays satisfying the heart condition for (E', e_Q). There is a poster D such that $d(D) > 0$, its paint-supply condition holds, and its radius, charge and four comparison functions are reserves. Eventually in x, it satisfies the conditional

comparison statement, the chart-radius estimates and the axis-chart existence statement required by the poster condition.

Sketch: obtain the radius and chart estimates from the chart construction, and choose the paint constant. The comparison-function theorem applies when the chart parameter is positive; otherwise the chart hypotheses are inconsistent. It requires positive comparison parameters a, d, a subpower bound on the volume function, reserves for the radius, charge and derivative control, paint supply, and orders satisfying $j + (r + 1) + r + 2 \le n$ and $n + N \le k$. The ray heart condition supplies these order inequalities. It produces four reserve functions, uniformly over the subsequent step and chart choices. Its conclusion is conditional on local derivative bounds, the chart estimates and coefficient matching: derivatives through order j of the differences of the expansion coefficients through degree r have the prescribed bound. The proof combines finite-order exponential bounds, analytic expansion estimates and the assumed coefficient match. *Lean:* `YM.RemainderComparison.Cadenza.Poster.quorum`, `YM.RemainderComparison.Herbal.compile`, `YM.RemainderComparison.Herbal.Airstrip`, `YM.RemainderComparison.Portal.JMatch` *(proof read; proofs of the analytic expansion estimates not expanded).*

Lemma 4.13 (connected-activity structures). Fix an input specification and target Q. There is l such that, for every $L \ge l$, branching patch, order j, auxiliary choice χ, $W, \mu > 0$ and order datum at W, one can choose constants and functions before x as follows: d, v have gear $-\frac{1}{2}$; B, r, T' have rate at most 0; and p is a reserve, with $T', p \ge 0$. Eventually in x, every admissible step record admits a connected-activity structure with the heavy, derivative and marked-input estimates specified by its three accompanying conditions.

Sketch: combine the connected-activity existence theorem with the derivative estimates for its local factors and the marked-input estimate. The growth-class rules keep d, v at gear $-\frac{1}{2}$ and the remaining stated functions at rate at most 0. The structure is chosen separately for each step record; the bounding functions are common to all those records. *Lean:* `YM.RemainderComparison.Halo.seeds` *(proof read; analytic inputs from III.10).*

Lemma 4.14 (remaining-occurrence bound). There is a block-side threshold depending only on the input specification and Q such that, for every L above it and every branching patch, the following holds. Fix the remaining data of Lemma 4.13 and $a \ge 0$, $\varepsilon > 0$ with $a + 2\varepsilon \le \mathrm{e}_Q$. One can choose bounding functions such that, for every fixed pair of orders I, J, reserve Y and nonnegative integer-valued function n whose real-valued cast is a reserve, there are a reserve V and a function S' of gear $q < 0$ with the following property. Eventually in x, every admissible step record admits a connected-activity structure for which the following holds. For every finite leaf family satisfying the leaf-scale condition $(n(x), \sigma, Y(x))$ and the radius condition at σ, every marked input with $\delta \le \beta_F^{-11/10}$, and every $m \in \mathbb{N}$, the bills restricted to marked clusters with $N \ge 1 + m$ remaining occurrences and insertion count at most I satisfy

$$\mathrm{Hit}_{\mathcal{B}(x)}(\mathrm{bill}; a) \le V(x) S'(x)^m \delta.$$

The derivative order is j and the support-polynomial order in the bill is J. The bill is formed from the local majorant supplied by the chosen structure.

Sketch: increase the weights of the remaining occurrences by a factor whose inverse has negative gear. On the indicated clusters the inflated product contains at least one marked factor and m further factors. Apply the weighted cluster bound and divide by those factors. The leaf-scale and radius conditions control the inserted leaves; the insertion cap I controls the accompanying polynomial

losses. *Lean:* YM.RemainderComparison.Halo.ramp *(proof read; weighted-cluster estimates used as inputs).*

Lemma 4.15 (restriction to a cluster predicate). Fix an admissible step record, a connected-activity structure, a nonnegative local majorant satisfying its derivative condition, nonnegative leaf coefficients and nonnegative vertex weights γ. Let $a \geq 0$, $e_1, \delta > 0$ and $q' < 1$. Suppose the polymer majorant has hit bound e_1 at exponent $a + 2\delta$, and its degree price with slack δ is at most q'. If a cluster predicate V and $v \geq 0$ satisfy $1 \leq v \prod \gamma$ on V, then the restricted bills have hit bound at exponent a at most

$$vA_\delta(x; j, m)e_1(1 - q')^{-1}, \qquad A_\delta(x; j, m) = [4(2\mathcal{B}(x) + 1)]^{mj}(mj)!\,\delta^{-mj}e^{\delta}.$$

The bill uses derivative order j and support-polynomial order m.

Sketch: the logarithmic cluster-tail estimate gives the geometric sum $e_1/(1 - q')$. Absorb $(1 + N)^{mj}$ into $e^{\delta N}$, using $N^{mj}e^{-\delta N} \leq (mj)!\delta^{-mj}$ with the corresponding shift by 1. The spatial factor is $[4(2\mathcal{B}(x) + 1)]^{mj}$. Finally insert $v \prod \gamma \geq 1$ on the retained clusters. *Lean:* YM.RemainderComparison.Lyric.gathers, YM.RemainderComparison.Lyric.run.

Lemma 4.16 (anchored bound for marked outputs). There is a block-side threshold depending only on the input specification and Q such that the following holds for every L above it and every branching patch. Fix the remaining data of Lemma 4.13 and $a \geq 0$, $\varepsilon > 0$ with $a + 3\varepsilon \leq \mathrm{e}_Q$. The connected-activity functions can be chosen as follows. Fix orders J, r, I, n, P_0 with $2 \leq n \leq j$, assume G simple, and fix an auxiliary choice, a reserve Y and a nonnegative integer-valued function i whose cast is a reserve, a record with parameter at most $\frac{1}{4}$, and a library of order n. There are a reserve V and a function V' of some rate $q' < 0$ such that, eventually in x, every admissible step record has:

- the marked hit bound $V(x)\delta$ at exponent $a + \varepsilon$, for leaf families with scale $(i(x), \sigma, Y(x))$ satisfying the radius condition, marked inputs $\delta \leq \beta_F^{-11/10}$, and insertion count at most I;
- an anchored output bound $(C_0/L + V'(x))\delta$ at exponent a, provided the output parameter point has radius $\mathcal{B}(x)$, threshold $t(x)$ and power $P_0 \leq J$, the library's compatibility condition holds, $r \geq h_Q + 5$, and the acceptance, quotation and local analytic smallness conditions hold with quotation constant C_0.

The anchored bound includes regularity and all left derivatives through the output order on the small-field supports. The proof gives $q' = \max(q, -\frac{1}{2})$, where $q < 0$ is the gear exponent in Lemma 4.14.

Sketch: separate clusters with a single remaining occurrence from those with at least two, and separate supports whose covering number exceeds $\mathcal{B}(x)$. The single-occurrence contribution uses the quotation bound and the sum over L^4 sites. Lemma 4.14 bounds the other contributions by VS' and $e^{-\varepsilon\mathcal{B}}V$. A further local comparison error has rate at most $-\frac{1}{2}$. Their sum defines V'. *Lean:* YM.RemainderComparison.Solo.atlas *(proof read; local singleton estimate used as an input).*

Lemma 4.17 (tail bounds). Fix L, $\mu > 0$, $a \geq 0$, $b > 0$ with $a + 2b \leq \mathrm{e}_Q$, and $\frac{3}{8} + \rho_Q < 0$. Fix orders j, n, k, s, J, functions B, p_0 of rate at most 0, and a nonnegative integer-valued function m whose cast has rate at most 0. Put $G_*(x) = \beta_F(x, L)^{3/8}$ and $\tau(x) = G_*(x)^5$. Let $I(x)$ be the additional root-activity bound formed from m, p_0, and let $\Sigma(x)$ be the resulting tree-sum majorant with derivative factor $1 + 2^j B(x)$. Eventually in x:

- the degree prices of $G_*((\text{root activity}) + I)$ and of Σ are at most $\frac{1}{2}$, with slacks δ_Q and b, respectively;
- $G_* \geq 1$, I is below the prescribed root allowance, and $G_*^{5k+2s} \leq e^{\mathrm{mark}}$ and $G_*^{5k+2s} \leq e^{\varepsilon_Q \lfloor \mathcal{B}/100000 \rfloor}$;
- for every β in the band of β_F,

$$\tau^k G_*^{-(5k+2s)} A_b(x; j, J)\, 2\Sigma(x) \leq g(\beta)^s,$$

$$\tau^k e^{-\mathrm{mark}}\, 2\mathrm{pledge}(x) \leq e^{-\mathrm{mark}/2} g(\beta)^s;$$

- $e^{-\mathrm{mark}/2} \leq e^{-4p(x)^{1/5}}$.

Sketch: the additional root term has rate at most $-\frac{3}{2}$. Thus Σ has rate at most $\frac{3}{8} + \rho_Q < 0$, and the polynomial spatial factors have rate at most 0. These facts give the two degree-price inequalities. In the first displayed tail bound the factors τ^k and G_*^{-5k} cancel; G_*^{-2s} supplies $\beta_F^{-3s/4}$, sufficient for $g(\beta)^s$ uniformly on the band. The large marking exponent absorbs the remaining powers in the second bound. *Lean:* `YM.RemainderComparison.Lyric.epochs`, `YM.RemainderComparison.Lyric.Premium`.

Lemma 4.18 (simultaneous derivative and tail estimates). Suppose $\frac{3}{8} + \rho_Q < 0$. There is ℓ such that, for every $L \geq \ell$, branching patch, $W, \mu > 0$, order datum, auxiliary choice, orders j, J, k, s, and $a \geq 0$, $\varepsilon, \Delta > 0$ with $a + 2\varepsilon \leq \mathrm{e}_Q$, the following holds. Fix a reserve Y, a nonnegative integer-valued function n whose cast is a reserve, and a function p_0 of any fixed finite rate. There are reserves A, B, C and a function E of rate $q < 0$ such that, eventually in x, each admissible step record admits three connected-activity structures with these properties:

- the first has the derivative estimates, mass condition and tail bounds of Lemma 4.17, and the required depth bound for G_*^{5k+2s};
- $A(x) \geq 1$, and the full bill from the second has hit bound $B(x)$ at exponent a for leaves whose coefficients have been divided by $A(x)$, whenever the original leaf-scale condition $(n(x), \sigma, Y(x))$ and radius condition hold;
- the third gives hit bound $C(x)E(x)^i\delta$ at exponent a for marked clusters with $N \geq 1 + i$, insertion count at most k, and $\delta \leq \beta_F^{-11/10}$, under the same leaf-scale and radius conditions;
- writing $\pi_\Delta(x) = e^{-\Delta \lfloor \mathcal{B}(x)/100000 \rfloor}$, the crop inequalities hold:

$$\pi_\Delta p_0 \leq \beta_F^{-11/10}, \qquad C\pi_\Delta p_0 \leq g(\beta_H)^s, \qquad \pi_\Delta A^k B \leq g(\beta_H)^s.$$

The full and marked majorants also satisfy their local derivative conditions. Here β_H is the coupling of the step record; the last bounds are for each fixed s.

Sketch: use separate connected-activity structures for the mass bound, the full bill and the marked bill. Combine the full-bill estimate, Lemma 4.14, Lemma 4.17 and the decay of π_Δ against every function of finite rate. The three structures need not coincide. *Lean:* `YM.RemainderComparison.Muse.generation`, `YM.RemainderComparison.Muse.Roles` *(proof read; connected-activity and full-bill estimates used as inputs).*

Lemma 4.19 (extending the comparison functions). Suppose G is simple and $\frac{3}{8} + \rho_Q < 0$. There is ℓ such that, for every branching patch of side at least ℓ, $W, \mu > 0$, order datum, songbook, scrollwork, auxiliary choice, mark number r, stage E', rays with the heart property, library of order $\mathrm{ord}(E')$, record with parameter at most $\frac{1}{4}$, and financed poster D, there is a quarry F extending D, chosen before x, such that:

- its new functions Y, spin and e are reserves, its connected-activity bound R_b has rate at most ρ_Q, and its marked-output correction h has some negative rate;
- eventually in x, every parameter point tempered to E' and every compatible admissible step record has the required local analytic bounds and a connected-activity structure with tree-sum bound at most $R_b(x)$;
- its marked-output bound is $(C_0/L + h(x))\delta$, under all the marked-input, compatibility, acceptance, quotation and local analytic hypotheses of Lemma 4.16, with $r \geq h_Q + 5$.

Sketch: take the new reserve functions and the finite-rate tail function from Lemma 4.18. The tree-sum bound is twice the polynomial spatial factor times the connected-activity majorant, whose rate is at most ρ_Q. Take $h = V'$ from Lemma 4.16 and apply that lemma with an empty leaf family. The additional tail function is a field of the quarry but carries no separate growth assertion in its financing condition. *Lean:* `YM.RemainderComparison.Cadenza.rollBefore`, `YM.RemainderComparison.Cadenza.Instructions`, `YM.RemainderComparison.Cadenza.Quarry.Staked`, `YM.RemainderComparison.Cadenza.Quarry.Bequest`.

Lemma 4.20 (III.15.6). Under the hypotheses of III.15.6, the quay condition holds for all sufficiently large block sides.

Sketch: take n from 4.19. For given data, 4.12 gives a poster, and 4.19 extends it to a quarry. The remaining step conditions hold eventually in x by their local analytic estimates and the scrollwork growth condition. Take P_1 beyond all of these thresholds. *Lean:* `YM.RemainderComparison.Cadenza.buoys` *(proof read; individual local analytic estimates not expanded).*

Lemma 4.21 (III.15.5). Under the hypotheses of III.15.5, the launch data exist.

Sketch: for each (j, i), take k_{ji} (4.10), g_{ji} (4.20) and n_{ji} (4.11). Set

$$f_j = \max_{i \in \{j, j-1\}} g_{ji} + \max_{i \in \{j, j-1\}} k_{ji} + \max_{i \in \{j, j-1\}} n_{ji}$$

and take $C'_{ji} = E_{ji}$ from 4.10. *Lean:* `YM.RemainderComparison.Chron.sails`.

Lemma 4.22 (III.15.7). Under the hypotheses of III.15.7, the quarries and a common enlargement of the delay data exist.

Sketch:

1. For each stage pair and slot, apply the pledge at $L = L_j$ (allowed since $k_j \leq L_j$) to the slot's data. The chosen rays have the heart property because $\sigma_i + 3 < \mathrm{e}_{R_{ji}}$. This gives a quarry V_{jig} and a threshold. The slot's extra condition holds eventually.
2. Set $\mathrm{shells}_j = \max(\mathrm{shells}_j(T_0), \max_{i \in \{j, j-1\}, g}(4\ell_g + 50))$ and $\mathrm{growth}_j = \max(\mathrm{growth}_j(T_0), 1)$. The inner maximum is 0 when there are no request slots. Tiling follows from the scrollwork growth conditions.
3. Since there are finitely many slots and two stage pairs per j, choose β^*_j beyond all thresholds.

Lean: `YM.RemainderComparison.Chron.grows`.

3.18.4.6 4.6 Records and other fragments (Part D) The construction-run record collects local estimates, growth of the copy count, bounds for the selected representations, and the conditions required for later OPE and comparison arguments. Its fields are established when

the run is assembled. The additional-expense records collect the hypotheses needed on the same selected run. The readout record extends three primitive conditions and adds a growth condition, a bound on block sizes by an iterated logarithm of the coupling, and cross-stage comparison inequalities.

The unit also contains numerical definitions and lemmas for the RG05 readout, the scale functions, and the instrument and postage records. These are not all mere projections: some prove asymptotic and transport estimates. The readout formulas and the remaining RG12 and R2I fragments are outside the detailed statements above; see §7.

Lean: `YM.PhysicalNext.Repeat.Legato.R2I.Funded`, `YM.PhysicalNext.Repeat.Legato.R2I.Extras`, `YM.Integration.RG05Preconstruction.PrimitiveExtras`, `YM.Integration.RG05Preconstruction.ReadoutPur`

3.18.5 5. Where the difficulty lies

The active outputs III.15.1–4 require uniform estimates in the coupling band and the volume. In III.15.1, the threshold must work for every β in the band and every $B \le \mathcal{B}(y)$. In III.15.3–4, the constants depend on derivative order and path length but not on t or the torus side. The single-word representation and unitarity give that independence.

Part C adds the conditional comparison bounds to the selected run, but those bounds are discarded in the final construction-run theorem. Its main estimates concern an inherited marked family: the anchored bound $(C'/L + o(1))\delta$ and the geometric gain per additional remaining occurrence. Their application requires the leaf-scale, radius, quotation and local analytic conditions. The gain counts all remaining occurrences, with at least one marked occurrence.

The exponent $\frac{3}{8}$ is the tree-weight exponent inherited from III.10. Requiring $\rho_Q < -\frac{3}{8}$ makes the relevant tree-sum majorant decay; the run's choice $\rho_Q < -\frac{7}{16}$ leaves a margin greater than $\frac{1}{16}$. The large-side threshold is independent of the later auxiliary data. The comparison functions and entry-coupling threshold may depend on those data, and the estimates then hold for all larger x.

3.18.6 6. Notation

Written proof	Lean (glossary)	Meaning
$g(\beta)$	`YM.Next.Scheduling.Clock.gap`, `gStep`	$\beta^{-1/2}$
$p(x)$	`YM.Next.Scheduling.Instrument.P` (plain), `YM.Next.Scheduling.Clock.watch` 2	$t^{(\log\log x)^2}$
$t(x)$	`YM.Pressure.temp`, `YM.Next.Scheduling.Instrument.t`	small-field threshold $g(x)p(x)$
$\mathcal{B}(x)$	`YM.Pressure.radiusB`, `YM.Next.Scheduling.Clock.cover`	cluster radius $\lceil(\log x)^2\rceil$
$\beta_F(x, L)$	`YM.Pressure.betaF`	$x + b_G \log L$
band of y	`YM.Next.Scheduling.inBand`	$[y/4, 4y]$
$\mathrm{Rate}_L(a)$	`YM.ConnectedActivityE6.Rate`	$\lvert w\rvert \le \beta_F^v$ eventually, for every $v > a$

Written proof	Lean (glossary)	Meaning
reserve; $M(x)$	`YM.RemainderComparison.Reserve`, `YM.Push.AbsoluteSums.Delay.fineMonitor`	$\lvert e_k\rvert \le M^k$ with $M = p(\beta_F)$
gear a	`YM.RemainderComparison.Gear`	$\beta_F^{-a} w$ is a reserve
$\mathrm{Hit}_M(f;a) \le \gamma$	`YM.Ac.Hit`	weighted per-site polymer bound
quote(κ,γ)	`YM.Bid.quote`	$e^{-\kappa}(e^{\gamma}-1)$
rate conditions (B, w, a, b, κ)	`YM.Progress.DirectedRecipe.Payments.Rates`	the five inequalities of III.15.1
mild, strong, κ	`YM.Progress.DirectedRecipe.Payments.Paying` (`mild`, `strong`, `kGoal`)	power-decaying and Gaussian pair-step costs
hard and soft family	`YM.Progress.DirectedRecipe.Payments.hardCost`, `YM.Progress.DirectedRecipe.Payments.softCost`	forward hard cost and local exponential correction
Ingredients	`YM.Progress.DirectedRecipe.Ingredients`	input to the fusion of III.10
slot s, atom a	`Slot`, `Atom` (III.1)	transported curvature; k-leg monomial
small-field on X	`YM.RG.Healthy`	plaquettes touching X within chord t of 1
admissible direction	`YM.RG.BudgetPoint.dirBound`	$\lVert A_e\rVert \le t(1+\mathrm{dist}(e,o))^P$
slotCost, margin, price	`YM.Next.Scheduling.Wages.slotCost`, `YM.Next.Scheduling.Wages.margin`, `YM.Next.Scheduling.Wages.price`	decorative constants of III.15.3–4
target Q; ρ_Q, e_Q, ε_Q	`YM.ConnectedActivityE6.Target`	connected-activity target (rate, exponent, …)
$\beta_F^{3/8}$	`YM.PhysicalNext.Repeat.Height.over`	per-vertex tree weight (III.10)
pledge	`YM.RemainderComparison.Chronicle.Pledge`	large-field conditions (1)–(2) of III.15.5
quay	`YM.RemainderComparison.Cadenza.Quay`	all quay conditions available at side L
quarry, financed	`YM.RemainderComparison.Cadenza.Quarry`, `YM.RemainderComparison.Cadenza.Quarry.Financed`	bounding functions of x with reserve/rate finances
Pillar	`YM.RemainderComparison.Chronicle.Pillar`	pledge plus per-slot allotments and tilings
radius cost	`YM.RemainderComparison.Lyric.radiusCost`	$r_Q = C_0 c_{h+5}/L + 2\varepsilon_Q \le v$
comparison estimate	`YM.RemainderComparison.Herbal.Arresting`, `YM.RemainderComparison.Portal.Match`	interacting vs. model, with jets matched to order r
heart condition	`YM.RemainderComparison.Cadenza.Bars.Heart`	supports and order inequalities stated before III.15.6
B_P, t_P	`YM.RG.BudgetPoint`	cluster radius and threshold of a parameter point
support enlargement	`YM.PhysicalNext.JointLoad.Heir.Valid`	Y lies within distance $r\mathcal{B}(x)$ of the coarse image of X
local analytic smallness	`YM.PhysicalNext.JointLoad.Heir.Plented`	numerical hypothesis on the return fee
library compatibility	`YM.PhysicalNext.JointLoad.Heir.Bays`	registered local estimates for the library

Written proof	Lean (glossary)	Meaning
marked input	`YM.RemainderComparison.Halo.presents`	decisive bounds on an inherited regular subfamily
leaf-scale condition	`YM.RemainderComparison.Lyric.Scale`	nonnegative weights, attached roots, rooted bound and occupancy
N; marked cluster	`YM.RemainderComparison.Halo.grows`	remaining occurrences after removing leaf insertions
insertion count	`YM.RemainderComparison.Lyric.number`	number of inserted leaves, with multiplicity
$A_b(x; j, J)$	`YM.ConnectedActivityE6.Blaze.polynomial`	polynomial spatial factor in Lemma 4.15
$\pi_\Delta(x)$	`YM.RemainderComparison.Sleet.pigment`	$e^{-\Delta\lvert\mathcal{B}(x)\rvert/100000}$
tempered point	`YM.RemainderComparison.Cadenza.Temper`	agreement of scale, threshold and stage parameters
comparison structure	`YM.RemainderComparison.Portal.Ray`	analytic expansion data and coefficient match
ι_G; $\mathcal{P}_G$	`YM.Gauge.colorMat`, `YM.Gauge.projectColor`	colour-to-matrix map; normalized colour projection
$I(x)$; $\Sigma(x)$	`YM.RemainderComparison.Lyric.unit`; `YM.RemainderComparison.Lyric.sum`	additional root term; tree-sum majorant in Lemma 4.17
pledge(x)	`YM.PhysicalNext.Repeat.Daisy.tail`	tail coefficient in Lemma 4.17
marked cluster	`YM.RemainderComparison.Halo.dyed`	remaining occurrence belongs to the inherited family

3.18.7 7. What remains unexplained

1. **Scope of Part C.** The selected-run theorem retains its comparison data as an additional conjunct; the final construction-run theorem discards that conjunct. A written proof using only the final run properties can omit these comparison conclusions. An all-order expansion using them would also have to supply the coefficient-matching hypotheses; existence of the comparison functions alone does not supply that match.
2. **Local analytic conditions.** The quotation bound and Lemmas 4.12, 4.16 and 4.19 retain the source's support-enlargement, library-compatibility and analytic-smallness conditions. Their full numerical definitions, and the local analytic, readiness, compatibility, resonance and tolerance conditions in III.15.6, remain to be written out. In particular, the local analytic condition attached to a funded step has not been translated into an independent mathematical statement.
3. **Provider proofs.** The assembly proofs in §4 have been read. The deeper proofs of the local singleton estimate, weighted-cluster bounds, full-bill estimates and analytic expansion estimates have not all been checked here. The contraction and occurrence-count statements are consequences of the cited conditional estimates, not unconditional bounds for arbitrary local terms.
4. **Historical interpretation.** The estimate L^{-5} per fine site and its conversion to L^{-1} per coarse block are explicit. Identifying it with a particular dimension-five operator is not justified by those inequalities alone. The reason for the choice $\frac{3}{8}$ belongs to III.10's tree

estimates.

5. **Other fragments.** The construction-run, additional-expense and readout record statements have been read. The RG05 readout price formulas and their proofs, and the remaining RG12 and R2I fragments, are not covered by the detailed verification. The thermodynamic fragment also includes eventual numerical estimates and a delay-enlargement lemma; it is not solely a collection of definitions.

3.19 III.16. Analytic toolkit

Part: III, Ultraviolet construction (renormalization group). **Lean:** eight folders, none a numbered chapter of its own: `YM/Complex/`, `YM/TimeKernel/`, `YM/Analysis/`, `YM/Laurent/`, `YM/CellFourier/`, `YM/Work/F07/`, `YM/Maxwell/`, `YM/Labels/`. **Size:** 5,318 physical source lines in 41 files, plus elementary helper lemmas in other folders.

The results fall into three groups.

- **(A) Traces of kernel products.** A Hölder inequality and Cauchy's estimate give a dimension-free Lipschitz bound for moving insertions between time slices. Quadrature passes the bound from matrices to kernels; IV.1 uses it for equicontinuity.
- **(B) Analytic and Fourier estimates.** Holomorphic symbols correspond to exponentially summable Laurent coefficients. Fourier and Bloch identities diagonalize periodized kernels, and a binomial series supplies a matrix square root. Taylor coefficient identities and several-variable analyticity support these arguments.
- **(C) Maxwell fields and labels.** Radial-gauge potentials correspond to closed curvature jets. Restriction to the Maxwell shell defines the free contractions and Wick hierarchy. Gauge-invariant polynomial labels admit coherent complex bases with real coefficients.

3.19.1 1. What this unit proves

The contraction kernel and Wick hierarchy are definitions whose properties are proved here.

Notation for (A). - Matrices are complex $\iota \times \iota$ matrices with ι finite, and $\|\cdot\|$ is the operator norm. - $\|b\|_{2q} := (\mathrm{tr}((b^*b)^q))^{1/(2q)}$ is the Schatten $2q$-norm. - For $A \succeq 0$ with spectral decomposition $A = \sum_\lambda \lambda P_\lambda$, put $A^z := \sum_{\lambda>0} e^{z\log\lambda} P_\lambda$ for $z \in \mathbb{C}$. This is entire in z. It vanishes on $\ker A$ for every z, so A^0 is the projection onto the range, and it equals the ordinary power A^m for every integer $m \geq 1$. - Products $\prod_{k=1}^n$ are ordered by k.

Proposition III.16.1 (integrating a positive-semidefinite kernel).

Hypotheses. - X is a compact metric space with a finite Borel measure μ, and $\mathcal{A}, \mathcal{B}$ are sets. - $K : \mathcal{A} \times \mathcal{A} \to \mathbb{C}$ is positive semidefinite: for every m and all $a_1, \dots, a_m \in \mathcal{A}$ (repetitions allowed), the matrix $(K(a_i, a_j))_{i,j}$ is Hermitian and positive semidefinite. - $g : \mathcal{B} \times X \to \mathcal{A}$ is such that $(x, y) \mapsto K(g(b, x), g(c, y))$ is continuous on $X \times X$ for all $b, c \in \mathcal{B}$.

Conclusion. The kernel

$$\tilde{K}(b, c) = \int_X \int_X K(g(b, x), g(c, y))\, d\mu(y)\, d\mu(x)$$

is positive semidefinite on $\mathcal{B}$.

Status: theorem. *Lean:* `YM.Analysis.NNKernel.accumulate`; the notion of positive semidefinite kernel is `NNKernel`.

Proposition III.16.2 (Hölder inequality for traces with complex exponents).

Hypotheses. - $A \succeq 0$ and $b_1, \dots, b_n$ are complex $\iota \times \iota$ matrices. - $q, M \geq 1$ are integers, and $Z_A := \operatorname{tr} A^M > 0$. - $v \in \mathbb{C}^n$ satisfies $\operatorname{Re} v_k \geq 0$ for all k and $\frac{n}{2q} + \frac{1}{M} \sum_k \operatorname{Re} v_k = 1$.

Conclusion.

$$\left| \operatorname{tr} \prod_{k=1}^{n} b_k A^{v_k} \right| \leq \prod_k \|b_k\|_{2q} \cdot Z_A^{1-n/(2q)}.$$

For real $v_k > 0$, the corresponding Schatten exponent is M/v_k and $\|A^{v_k}\|_{M/v_k} = Z_A^{v_k/M}$. At $v_k = 0$, the factor is the range projection, whose operator norm is 1 since $Z_A > 0$. The bound has no dimension factor.

Status: theorem. *Lean:* `YM.Holder.estimation`. The left side is `total` and the right side is `cost`.

Proposition III.16.3 (Lipschitz bound in the gaps, matrices; key result).

Hypotheses. - $A \succeq 0$ and $b_1, \dots, b_n$ are as in III.16.2. - $q, w \geq 1$ are integers, $M = 2qw$, and $\operatorname{tr} A^M > 0$. - $Y \in \mathbb{N}^n$ satisfies $nw + \sum_k Y_k = M$. - For some $i \neq j$ and some $\ell \in \mathbb{N}$, the vector $Y' \in \mathbb{N}^n$ moves ℓ units from gap j to gap i: $Y'_i = Y_i + \ell$, $Y'_j + \ell = Y_j$, and $Y'_k = Y_k$ otherwise. - $r > 0$, $Y_k > 2r$ for every k, and $\ell \leq r$.

Conclusion.

$$\left| \operatorname{tr} \prod_k b_k A^{Y'_k} - \operatorname{tr} \prod_k b_k A^{Y_k} \right| \leq \frac{\ell}{r} \prod_k \|b_k\|_{2q} \left(\operatorname{tr} A^M\right)^{1-n/(2q)}.$$

The condition $i \neq j$ forces $n \geq 2$.

Status: theorem. *Lean:* `YM.Holder.finite_lip`.

Proposition III.16.4 (Lipschitz bound in the gaps, kernels; key result).

Setting. X is a compact metric space with a finite Borel measure μ. For $L \geq 1$ and continuous $k_1, \dots, k_L : X \times X \to \mathbb{C}$, write

$$\operatorname{Tr}(k_1 \cdots k_L) = \int_{X^L} \prod_{l=1}^{L} k_l(x_l, x_{l+1}) \, d\mu(x_1) \cdots d\mu(x_L), \qquad x_{L+1} = x_1,$$

and write $k^*(x, y) = \overline{k(y, x)}$ for the adjoint kernel. Powers inside Tr denote repetitions in the cyclic word.

Hypotheses. - K is a continuous positive-semidefinite kernel, and $B_1, \dots, B_n$ are continuous kernels. - $q, w \geq 1$, $M = 2qw$, and $\operatorname{Tr}(K^M) = Z$ with $Z > 0$ real. - Y, Y', ℓ, r are as in III.16.3. - $c_1, \dots, c_n$ are real numbers with $\left|\operatorname{Tr}((B_k^* B_k)^q)\right|^{1/(2q)} \leq c_k \, Z^{1/(2q)}$ for every k.

Conclusion.

$$\left|\operatorname{Tr}(B_1 K^{Y'_1} \cdots B_n K^{Y'_n}) - \operatorname{Tr}(B_1 K^{Y_1} \cdots B_n K^{Y_n})\right| \leq Z \Big(\prod_k c_k \Big) \frac{\ell}{r}.$$

Status: theorem. *Lean:* `YM.TimeKernel.passage`, with `closed` (the cyclic integral), `along` (the word with gaps) and `multiply` (the column word $(B^* B)^q$).

Notation for (B). - α is a finite index set; in the applications it is the four axes. - $|x|_1 = \sum_a |x_a|$ for $x \in \mathbb{Z}^\alpha$, and $e^{ip\cdot x} = \exp(i\sum_a p_a x_a)$ for $p \in \mathbb{C}^\alpha$. - $S_\eta = \{p \in \mathbb{C}^\alpha : |\operatorname{Im} p_a| < \eta \ \forall a\}$. - For $f : \mathbb{Z}^\alpha \to \mathbb{C}$, the symbol is $\hat f(p) = \sum_x f(x) e^{ip\cdot x}$.

Proposition III.16.5 (holomorphy in a strip and exponentially weighted Laurent coefficients). Let $\eta > 0$.

(a) Let u be holomorphic on S_η and 2π-periodic there: $u(p+2\pi x) = u(p)$ for $p \in S_\eta$ and $x \in \mathbb{Z}^\alpha$. For $p \in S_\eta$ let $\hat u_p(x) = \int_{[0,1]^\alpha} u(p+2\pi\theta) e^{-2\pi i\theta\cdot x} d\theta$, and put $\hat u = \hat u_0$. Then:

- the integral $\int u(p+2\pi\theta)\, d\theta$ does not depend on $p \in S_\eta$;
- $\hat u_p(x) = e^{ip\cdot x}\hat u(x)$.

If moreover $|u| \le K$ on S_η, then: - $|\hat u_p(x)| \le K$; - for $H \ge 0$ and $b > 0$ with $H + b < \eta$, $\sum_x e^{H|x|_1}|\hat u(x)| \le K(2(1+b)/b)^{\operatorname{card}\alpha}$; - $u(p) = \sum_x \hat u(x) e^{ip\cdot x}$ on S_η, with absolute convergence.

(b) Let $\sum_x e^{\eta|x|_1}|f(x)| \le W$. Then:

- $|\hat f(p)| \le W$ whenever $|\operatorname{Im} p_a| \le \eta$ for all a;
- for $0 < \sigma < \eta$, $\hat f$ is holomorphic on S_σ with $\|D\hat f(p)\| \le W/(\eta - \sigma)$ (operator norm for the sup norm on $\mathbb{C}^\alpha$), so $|\hat f(p) - \hat f(p')| \le W\|p - p'\|_\infty/(\eta - \sigma)$ on S_σ;
- for real p, the symbol of $\operatorname{Re} f$ equals $\frac12(\hat f(p) + \overline{\hat f(-p)})$.

Status: theorems. *Lean:* - for (a): `market_cap`, `full_turn`, `coat_price`, `Vane.issues`, `Vane.paints`; - for (b): `Width.estimate`, `readDeriv`, `readDiff`, `readRealPart`.

Proposition III.16.6 (Fourier and Bloch decomposition on tori).

Setting. - All torus side lengths n, c, f, L are positive integers, and $\Lambda_n = (\mathbb{Z}/n\mathbb{Z})^4$. For $p \in \Lambda_n$, represented in $\{0, \dots, n-1\}^4$, let $k_p = 2\pi p/n$, and let $F_n = (e^{ik_p\cdot x})_{x, p\in\Lambda_n}$. - α, β are finite index sets, and the real kernel $R = (R_{ab}(x))$, $x \in \mathbb{Z}^4$, satisfies $\sum_x |R_{ab}(x)| < \infty$ for every $a \in \alpha$, $b \in \beta$. - Its periodization on Λ_n is $R_n((y,a),(y',b)) = \sum_{x:\, y+x\equiv y'} R_{ab}(x)$, and its symbol is $\hat R(k)_{ab} = \sum_x R_{ab}(x) e^{ik\cdot x}$.

(a) $F_n F_n^* = F_n^* F_n = n^4 I$, and $R_n(F_n \otimes I_\beta) = (F_n \otimes I_\alpha)\operatorname{diag}_{p\in\Lambda_n} \hat R(k_p)$. This holds for every p, including $p = 0$.

(b) Let $f = cL$. For $p \in \Lambda_c$ and $l \in \Lambda_L$ let $k_{p,l} = 2\pi(p + cl)/f$. These are the L^4 fine momenta k with $Lk \equiv 2\pi p/c$ modulo 2π, i.e. the aliases of the coarse momentum $2\pi p/c$. The fine Fourier matrix $\Phi = (e^{ik_{p,l}\cdot y})_{y\in\Lambda_f,(p,l)}$ satisfies:

- $\Phi\Phi^* = f^4 I$;
- $R_f(\Phi \otimes I_\beta) = (\Phi \otimes I_\alpha)\operatorname{diag}_{p\in\Lambda_c}\operatorname{diag}_{l\in\Lambda_L} \hat R(k_{p,l})$.

(c) For every $p \in \mathbb{C}^4$, the alias block $E_L(p) = (e^{i(p+2\pi l)\cdot x/L})_{x, l\in\Lambda_L}$ satisfies $E_L(\bar p)^* E_L(p) = L^4 I$.

Status: theorems. *Lean:* `gram`, `coGram`, `circulant`, `readingsGram`, `aligns`, `straightens`.

Proposition III.16.7 (analytic square root near a spectral band). Let $m > 0$ and let the finite matrix index set be nonempty. For complex matrices X with $\|X/m - 1\| < 1$, put $\Phi_m(X) := \sqrt{m}\sum_{k\ge0}\binom{1/2}{k}(X/m - 1)^k$. - Φ_m is holomorphic on this ball. - Φ_m is bounded on

each smaller ball $\|X/m - 1\| \le r < 1$. - Φ_m commutes with entrywise complex conjugation. - If X is Hermitian and $v \le X \le m$ as quadratic forms, with $v > 0$, then $\Phi_m(X)$ is Hermitian, $\Phi_m(X)^2 = X$, and $\sqrt{v} \le \Phi_m(X) \le \sqrt{m}$.

Status: theorem. *Lean:* `binRoot_on_band`, `binRoot_deriv`, `binSum_bound`, `binRoot_conj`.

Notation for (C). - Fix a compact matrix gauge group G. Its colour space $\mathfrak{g}$ is the real Lie algebra in coordinates orthonormal for $-(\dim \rho)^{-1}\mathrm{Re}\,\mathrm{tr}(XY)$, where ρ is the fixed defining representation. - A homogeneous polynomial on spacetime is identified with its symmetric multilinear Taylor coefficient. - For $j \ge 0$: - $\mathcal{A}_j$ is the space of 4-tuples (A_l) of $\mathfrak{g}$-valued homogeneous polynomials of degree $j+1$ on $\mathbb{R}^4$ with $\sum_l x_l A_l(x) = 0$ (radial, or Fock–Schwinger, gauge); - $\mathcal{C}_j$ is the space of $\mathfrak{g}$-valued homogeneous polynomials F_{il} of degree j with $F_{il} = -F_{li}$ and $\partial_a F_{bc} + \partial_b F_{ca} + \partial_c F_{ab} = 0$; - $\mathcal{S}_j \subset \mathcal{C}_j$, the Maxwell shell, adds the condition $\sum_a \partial_a F_{ab} = 0$.

Proposition III.16.8 (free Maxwell field).

(a) *(Poincaré lemma in radial gauge.)* The curl $A \mapsto F_{il} = \partial_i A_l - \partial_l A_i$ is a linear isomorphism $\mathcal{A}_j \to \mathcal{C}_j$. Its inverse is $A_i(x) = (j+2)^{-1} \sum_k x_k F_{ki}(x)$.

(b) *(Taylor coefficients of solutions.)* Let $U \subset \mathbb{R}^4$ be open, and let the $\mathfrak{g}$-valued field $f = (f_{ab})$ be smooth and antisymmetric on U and satisfy the Bianchi and Maxwell equations there. Then for every $x \in U$ and $j \ge 0$, the j-th Taylor coefficient of f at x lies in $\mathcal{S}_j$.

(c) *(Free two-point function.)* The function $G_0(x) = 1/(4\pi^2|x|^2)$ is harmonic on $\mathbb{R}^4 \setminus \{0\}$. Put

$$C_{ab,km} = \partial_a \psi_{b,km} - \partial_b \psi_{a,km}, \qquad \psi_{b,km} = \delta_{kb}\partial_m G_0 - \delta_{mb}\partial_k G_0.$$

This is the field-strength two-point function $\langle F_{ab}(x) F_{km}(0)\rangle$ of one colour component of the free field in Feynman gauge. Then:

- for fixed (k,m), $(a,b) \mapsto C_{ab,km}$ is smooth and antisymmetric on $x \ne 0$ and solves the Bianchi and Maxwell equations there;
- the same holds in (k,m) for fixed (a,b);
- $C_{ab,km}(x) = C_{km,ab}(x) = C_{ab,km}(-x)$.

(d) *(One contraction between jets.)* Let $j, l \ge 0$, and let p and q be real linear forms on $\mathcal{S}_j$ and $\mathcal{S}_l$ (probes). Write $c_j h(z) = D^j h(z)/j!$, and let (e_c) be the orthonormal colour basis. Choose linear extensions P, Q of p, q to the unrestricted two-index multilinear tensors. Define the colour-valued field

$$R_{P,km}(z) = \sum_c P((c_j(C_{ab,km} e_c)(z))_{a,b}) e_c,$$

and, for $z \ne 0$, put

$$\kappa(p,q)(z) = (-1)^l Q((c_l R_{P,km}(z))_{k,m}).$$

This definition is independent of both extensions. It is bilinear in (p,q), smooth for $z \ne 0$, and satisfies $\kappa(p,q)(z) = \kappa(q,p)(-z)$.

For the coordinate probes $p(F) = F_{ab}[u]_c$ and $q(F) = F_{km}[v]_d$, with $u \in (\mathbb{R}^4)^j$ and $v \in (\mathbb{R}^4)^l$,

$$\kappa(p,q)(z) = \delta_{cd}(-1)^l c_l(t \mapsto c_j C_{ab,km}(t)[u])(z)[v].$$

Equivalently, at $z = x - y$ this is the mixed Taylor coefficient at (x,y) of $\delta_{cd} C_{ab,km}(x-y)$. Thus κ is the free covariance of the corresponding jet observables, with Taylor coefficients used at both endpoints.

Status: theorems in (a)–(c); definition and theorems in (d). *Lean:* - (a): `potential`, `closedCurl`, `curl_inverse`, `potential_recover`; - (b): `WaveOn.jets`; - (c): `harmonic`, `Bare.equation`, `Bare.equation_right`, `Bare.exchange`, `Bare.even`; - (d): `legKernel`, `descends`, `pairing`, `exchange`, `raw_mixed`, `reads_mixed`.

Proposition III.16.9 (field labels).

Setting. Let $d \geq 0$ be an integer.

- Fix real bases of the spaces $\mathcal{A}_j$.
- Pol is the algebra of complex polynomials in the resulting coordinates of a jet sequence $(A_j)_{j\geq 0}$.
- A coordinate from $\mathcal{A}_j$ has weight $j+2$, and a monomial has the sum of the weights of its factors.
- $\mathrm{Label}_d \subset$ Pol consists of the polynomials of weight $\leq d$ that are invariant under the simultaneous adjoint action of G on all A_j. Invariance is imposed on the whole polynomial.

(a) A polynomial that vanishes on all real jet sequences is zero.

(b) There are finitely many monomials of weight $\leq d$, so Label_d is finite-dimensional.

(c) Every element of Label_d is the sum of its bihomogeneous components (weight, polynomial degree), both indices $\leq d$.

- Each component is again a label.
- Taking components commutes with complex conjugation of coefficients and with the action of $O(4)$ on jets.
- For $s \in O(4)$, set $(T_s A_j)_i(x) = \sum_k s_{ik}(A_j)_k(s^{-1}x)$. The induced label action is $(s \cdot P)(A) = P(T_{s^{-1}}A)$; it satisfies $(st)\cdot P = s\cdot(t\cdot P)$.

(d) For every integer $d \geq 0$, Label_d has a complex basis (a frame) $e_0 = 1, e_1, \dots, e_{N_d}$ of bihomogeneous labels with real coefficients. These bases can be chosen simultaneously so that, under the natural inclusion, every basis vector at level d occurs at level $d+1$. No ordering beyond $e_0 = 1$ is prescribed.

Status: theorems. *Lean:* `eval_ext`, `cap_finite`, `labels_finite`, `decompose`, `gradePart_bar`, `gradePart_turn`, `rotation_law`, `Frame`, `frames`.

Proposition III.16.10 (free Wick hierarchy on labels).

Definitions. Let $n \geq 0$, let $d_1, \dots, d_n \geq 0$ be integers, and let $x_1, \dots, x_n \in \mathbb{R}^4$ be pairwise distinct. - For Wick monomials in the coordinates of shell jets, the *free moment* is the sum, over all pairings of the factor occurrences in which partners sit at different points, of the products of the kernels $\kappa(\cdot,\cdot)(x_i - x_{i'})$ of the paired coordinate forms. It is extended multilinearly to Wick polynomials. - The *free correlation* of labels $P_k \in \mathrm{Label}_{d_k}$ is the moment of their restrictions to on-shell jets, $A_j = \mathrm{curl}^{-1} F_j$ with $F_j \in \mathcal{S}_j$.

Properties. - The free correlation is smooth on the configuration space. - It vanishes if one label vanishes on all on-shell jets, i.e. lies in the equation ideal. - It is invariant under permuting the points together with the labels. More generally, embedding the index set into a larger configuration of distinct points preserves the correlation when each added point carries the label 1. - The moment with every insertion equal to 1 is 1, including at $n = 0$. For constant insertions $a_1, \dots, a_n \in \mathbb{C}$, it is $\prod_i a_i$.

Status: definitions and theorems. *Lean:* `Wick.moment`, `Wick.correlation`, `Wick.smooth`, `Wick.kills_equation`, `ideal_meaning`, `Wick.exchangeSites`, `Wick.monExtension`, `Wick.adjoin`, `Wick.constants`.

Other units also quote, by name, the general lemmas of §4: the algebra of positive kernels (4.1), quadrature (4.2), the calculus of Taylor coefficients (4.19), and analyticity in several variables and weak analyticity (4.20).

3.19.2 2. What it uses

- **I.1 (lattice, symmetries, gauge groups).**
  - The torus Λ_n, its cover $\mathbb{Z}^4$, and the coarse/fine subdivision maps: `Lattice.Site`, `Lattice.wrap`, `Lattice.Ratio`.
  - $\mathbb{R}^4$ with its axes, and $O(4)$: `E4`, `Ortho`.
  - The gauge-group data, namely the colour space and the adjoint action: `Gauge`, `Gauge.Color`, `Gauge.adjoint`.
  - The configuration space of distinct points: `conf`.
  - All of these are definitions.
- **III.1:** the summability condition "nonnegative, summable, sum $\le W$", `RG.SumBudget`, used to define the weighted norms of (B). It is a definition.
- **Mathlib:**
  - Hadamard's three-lines theorem;
  - Cauchy's estimate for the derivative on a circle, and the mean value inequality on convex sets;
  - the spectral theorem for Hermitian matrices;
  - the Schur product theorem;
  - dominated convergence;
  - Cauchy's theorem for rectangles and the Cauchy integral formula;
  - multivariable Fourier inversion for continuous torus functions with absolutely summable Fourier coefficients;
  - the binomial series in Banach algebras;
  - symmetry of second derivatives;
  - the identity theorem for polynomials on infinite sets.
- **Standard references:** B. Simon, *Trace Ideals and Their Applications*, for Schatten inequalities; Berg–Christensen–Ressel, *Harmonic Analysis on Semigroups*, for positive kernels; Katznelson, *An Introduction to Harmonic Analysis*, for torus Fourier series; Reed–Simon, *Methods of Modern Mathematical Physics*, vol. I, for operator functional calculus; Hörmander, *An Introduction to Complex Analysis in Several Variables*, and Rudin, *Functional Analysis*, for holomorphy; Glimm–Jaffe, *Quantum Physics*, for Wick products and reflection positivity.
- **Imported constants and assumptions.** All analytic assumptions appear in the statements above. The numerical factors are explicit. The parameters q, w, M, r in (A), the strip widths and torus sizes in (B), and the label cutoff d are chosen in the applications.

Uses in other units. The principal applications are listed below; the scope of the longer dependency lists is noted in §7. - **III.16.1–III.16.4:** the external applications of this trace argument are in IV.1. - III.16.1 gives the positivity of the Wilson transfer kernel: `Slice.step_positive`, Lemma IV.1.f. - III.16.4 gives the time-direction Lipschitz bound: `Slice.Time.powering`, Lemmas IV.1.e–f. - Lemma 4.1 (the algebra of positive kernels) is also used in III.2, III.4, III.6, III.10,

III.11, III.13 and VIII.1. - **III.16.5:** III.7, III.3, III.6, III.2, III.4, III.9, III.10, III.12–III.15, VII.1, VII.2. - **III.16.6:** III.7, III.3, III.4, III.2, III.6, III.14, III.10, III.13, III.9, III.11, III.1, III.5, VII.1, VII.2. - **III.16.7:** III.6, in the square-root theorem described in §3. - **III.16.8 and III.16.10:** VI.2, VII.1, VII.3, VI.1, III.11, II.1, VII.2, IV.1. - **III.16.9:** II.1, III.11, VII.2, III.12, III.13, III.6, VI.2, VI.1, III.10, III.2, VII.3, VII.1, VII.4, VIII.1. - **Lemma 4.20:** III.6 and V.2 use `finiteD`; III.13, V.1 and V.2 use `lift_weak`.

3.19.3 3. Strategy

(A) Moving an insertion. Transfer-kernel correlations have the form $\mathrm{Tr}(B_1 K^{Y_1} \cdots B_n K^{Y_n})/Z$. Moving one insertion transfers ℓ units between two gaps. For finite matrices, continue the exponents along $Y + z(e_i - e_j)$. The trace is entire, and the Hölder balance is unchanged. III.16.2 bounds it by a constant C on $|z| \le 2r$; Cauchy's estimate gives $|F'| \le C/r$ on $|z| \le r$, hence a change of at most $C\ell/r$. The dimension-free bound passes to kernels by sampling every factor and adjoint with the same partition. In IV.1, for nonnegative Wilson coupling, positivity follows from III.16.1 and the positive-kernel algebra; Z is the partition function. The condition $M = 2qw$ makes the reflected column $(B^*B)^q$ fill the time period, so chessboard estimates bound its normalized trace. The normalized difference is at most $(\prod_k c_k)\ell/r$; uniformity in volume and spacing requires corresponding control of c_k and ℓ/r.

(B) Kernels and symbols. Exponentially weighted summability gives bounded holomorphic symbols. Moving the Fourier integration torus in the imaginary directions gives the converse, with a narrower exponential weight. Discrete Fourier identities then transfer symbol identities to every finite torus, including the zero momentum. For the square-root construction in III.6, take a real square matrix kernel Q with $\sum_x e^{\eta|x|_1}|Q_{ab}(x)| \le A$ for every entry, $\eta > 0$, and Hermitian real-momentum symbols satisfying $vI \le \hat{Q}(p) \le AI$, where $0 < v \le A$. The derivative bound, together with the norm conversion from entrywise bounds to the matrix operator norm, keeps $\hat{Q}/A - I$ in a ball of radius less than 1 on a thinner strip. III.16.7 gives a bounded holomorphic root there. Its Laurent coefficients have an exponential bound; taking their real parts preserves the positive square-root symbol at real momenta because $\hat{Q}(-p) = \overline{\hat{Q}(p)}$. Fourier diagonalization yields $P_n^{\mathsf{T}} P_n = Q_n$ with P_n symmetric on every torus. The parameter choices and assembly belong to III.6.

(C) Free fields and labels. For homogeneous closed curvature jets, the radial-gauge formula $A_i = (j+2)^{-1} \sum_k x_k F_{ki}$ inverts the curl. Imposing the Maxwell equation gives the shell. The explicit two-curl kernel derived from $G_0(x) = (4\pi^2|x|^2)^{-1}$ solves both equations in each leg away from the origin, so its Taylor coefficients define pairings on shell probes. Wick sums of these pairings give the free hierarchy on polynomial labels. Bihomogeneous decomposition and conjugation provide coherent complex bases with real coefficients. This hierarchy is the comparison object for short-distance limits, stress-tensor identities, the trace anomaly and nontriviality in VI.1–VI.2 and VII.1–VII.3; its kernels are used at distinct points.

3.19.4 4. Steps

3.19.4.1 4.1 Positive kernels and quadrature Lemma 4.1 (algebra of positive-semidefinite kernels).

Statement. - Generators: the class contains the Gram kernels $\overline{f(x)} f(y)$ and the constant 1. - Closure: it is closed under multiplication by constants $c \ge 0$, finite sums, pointwise (Schur)

products, nonnegative integer powers, pullbacks $K(\varphi(x),\varphi(y))$, pointwise limits, and $K \mapsto e^K$. - Criterion: a kernel is positive semidefinite if and only if it is Hermitian and $\sum_{i,j} \bar{c}_i K(x_i,x_j)c_j \geq 0$ for all finite families. - Integrated form: on a compact metric space with a finite Borel measure, for continuous positive-semidefinite K and bounded measurable f, $\iint K(x,y)\overline{f(x)}f(y)\,d\mu\,d\mu \geq 0$.

Sketch. - Products follow from the Schur product theorem. - e^K is the limit of the partial sums $\sum_{m<N} K^m/m!$. - Limits are handled by the quadratic-form criterion. Hermitian symmetry passes to the limit through the 2×2 Gram matrices. - For the integrated form, replace K by its values at the tags of a fine partition (Lemma 4.2). The integral becomes $\sum_{i,j} \overline{\int_{\text{cell } i} f}\, K(x_i,x_j) \int_{\text{cell } j} f \geq 0$, and dominated convergence finishes the proof.

Lean: `gram`, `scaled`, `plus`, `times`, `pow`, `titled`, `criterion`, `closed`, `exp`, `integrated`.

Lemma 4.2 (quadrature).

Statement. Let X be a compact metric space with a finite Borel measure μ. - For each $k \geq 0$ there is a finite measurable partition of X with assigned tags such that every point lies within $1/(k+1)$ of its cell's tag. Empty cells and zero masses are allowed; a tag need not lie in its cell. - Let $m_i = \mu(\text{cell } i)$. For every finite index set I and every continuous $F : X^I \to E$ with E a Banach space, the Riemann sums $\sum_{i\in \text{cells}^I} \prod_{j\in I} m_{i_j} F((x_{i_j})_{j\in I})$ converge to $\int F\, d\mu^{\otimes I}$.

Sketch. - The partitions: cover X by finitely many balls of radius $1/(k+1)$ and disjointify. - The convergence: the Riemann sum equals $\int F\circ \pi_k\, d\mu^{\otimes I}$, where π_k sends every point to the tag of its cell. Since $\pi_k \to \mathrm{id}$ pointwise and F is bounded, dominated convergence applies. - The statement concerns each fixed F and finite product; it asserts no uniform rate over families of integrands.

Lean: `Quad`, `refine_`, `shrink`, `counting`, `approximation`, `approx`.

Lemma 4.3 (proof of III.16.1).

Sketch. - For a partition, define $G_Q(b,c) = \sum_{i,j} m_i m_j K(g(b,x_i), g(c,x_j))$. - G_Q is Hermitian. For each finite family of elements $b \in \mathcal{B}$ and coefficients r_b, its quadratic form $\sum_{b,c} \bar{r}_b G_Q(b,c) r_c$ is the quadratic form of K at the points $g(b,x_i)$, indexed by (b,i), with coefficients $r_b m_i$. Hence G_Q is positive semidefinite. - By Lemma 4.2 in two variables, $G_Q \to \tilde{K}$ pointwise. - Positive-semidefinite kernels are closed under pointwise limits (Lemma 4.1).

Lean: `accumulate`.

3.19.4.2 4.2 The finite trace inequality Lemma 4.4 (three-lines Hölder lemma).

Definition. A *strip path* is an entire map $V : \mathbb{C} \to M_\iota(\mathbb{C})$ with: - $\|V(t)\| \leq 1$ for $0 \leq \operatorname{Re} t \leq 1$; - trace norm $\sup_{\|C\|\leq 1} |\operatorname{tr} V(t)C| \leq 1$ for $\operatorname{Re} t = 1$.

Statement. If $V_1,\dots,V_n$ are strip paths and $h_k \in \mathbb{C}$ satisfy $\operatorname{Re} h_k \in [0,1]$ and $\sum_k \operatorname{Re} h_k = 1$, then $|\operatorname{tr} V_1(h_1)\cdots V_n(h_n)| \leq 1$.

Sketch. Prove $|\operatorname{tr}(c\,V_1(h_1)\cdots V_n(h_n))| \leq 1$ for every contraction c, by induction on n. The case $n=0$ is excluded by the balance condition. In the induction step let $t = \operatorname{Re} h_1$. - *Case* $t = 1$. The other factors are contractions. Apply the trace-norm bound of V_1 after a cyclic permutation. - *Case* $t < 1$. Put $E(x) = \operatorname{tr}(c\,V_1(x + i\operatorname{Im} h_1)\prod_{k\geq 2} V_k(x_k(x)))$ with $x_k(x) = \operatorname{Re} h_k \frac{1-x}{1-t} + i\operatorname{Im} h_k$. - E is entire and bounded on $0 \leq \operatorname{Re} x \leq 1$. The bound depends on $\operatorname{card} \iota$, but it is used only qualitatively. - On $\operatorname{Re} x = 0$, the first factor is a contraction absorbed into c, and the remaining

real parts sum to 1; the induction hypothesis gives $|E| \le 1$. - On $\operatorname{Re} x = 1$, the factors $V_k(x_k(x))$ with $k \ge 2$ are contractions, and the trace-norm bound of V_1 gives $|E| \le 1$. - The three-lines theorem gives $|E(t)| \le 1$, and $E(t)$ is the trace to be bounded.

Lean: `Flex`, `holder_core`, `convexStrip`.

Lemma 4.5 (Schatten factorization). For every matrix b and integer $q \ge 1$ there is a strip path V with $b = \|b\|_{2q}\, V(1/(2q))$.

Sketch. - Write $b^*b = U \operatorname{diag}(s_i^2)\, U^*$. Then $b = W \operatorname{diag}(s)\, U^*$ with $\|W\| \le 1$, where $W = bU \operatorname{diag}(s)^{-1}$ on the support of s. - If $b = 0$, take $V = 0$. Otherwise let $f_i = s_i^{2q} / \sum_j s_j^{2q}$ and $V(t) = W \operatorname{diag}(f_i^t)\, U^*$, with the convention $0^t = 0$. - Since $f_i \le 1$, $\|V(t)\| \le 1$ on the closed strip. - For $\operatorname{Re} t = 1$, $|\operatorname{tr} V(t)C| \le \sum_i f_i |(U^*CW)_{ii}| \le 1$. - At $t = 1/(2q)$, $f_i^{1/(2q)} = s_i / \|b\|_{2q}$.

Lean: `even_cost`, `imageForm`, `weighted`, `specPow`.

Lemma 4.6 (complex powers). For $A \succeq 0$ and integer $M \ge 1$ there is a strip path P with $A^z = Z_A^{z/M} P(z/M)$ for all $z \in \mathbb{C}$. Here $Z_A^w = e^{w \log Z_A}$ if $Z_A > 0$, and $Z_A^w = 0$ otherwise. Moreover A^z equals the ordinary power A^m at every integer $z = m \ge 1$.

Sketch. If $Z_A = 0$, positivity gives $A = 0$ and one may take $P = 0$. Otherwise take $P(w) = U \operatorname{diag}((\lambda_i^M / Z_A)^w) U^*$. The normalized eigenvalues sum to 1, so the strip and trace-norm bounds follow as in Lemma 4.5.

Lean: `leap`, `leap_at`, `flexPower`.

Lemma 4.7 (proof of III.16.2).

Sketch. - Write $b_k = \|b_k\|_{2q} V_k(1/(2q))$ (Lemma 4.5) and $A^{v_k} = Z_A^{v_k/M} P(v_k/M)$ (Lemma 4.6). - The scalar prefactors have modulus $\prod_k \|b_k\|_{2q}\, Z_A^{\sum \operatorname{Re} v_k/M} = \prod_k \|b_k\|_{2q}\, Z_A^{1-n/(2q)}$. - The $2n$ strip paths $V_1, P, V_2, P, \dots$ are evaluated at exponents whose real parts, $1/(2q)$ and $\operatorname{Re} v_k/M$, lie in $[0,1]$ and sum to 1. - Lemma 4.4 bounds the remaining trace by 1.

Lean: `estimation`.

Lemma 4.8 (proof of III.16.3).

Sketch. - Let $F(z) = \operatorname{tr} \prod_k b_k A^{W_k(z)}$ with $W(z) = Y + z(e_i - e_j)$, and put $C = (\prod_k \|b_k\|_{2q})(\operatorname{tr} A^M)^{1-n/(2q)}$. The function F is entire. - The relation $nw + \sum Y_k = M$ is the balance $n/(2q) + \sum_k Y_k/M = 1$, and $\sum_k \operatorname{Re} W_k(z) = \sum_k Y_k$. - On $|z| \le 2r$, $\operatorname{Re} W_k(z) \ge Y_k - 2r > 0$, so III.16.2 gives $|F| \le C$ there. - For $|x| \le r$, the circle of radius r about x lies in this disc, and Cauchy's estimate gives $|F'(x)| \le C/r$. - The mean value inequality on the convex disc $|z| \le r$, between 0 and ℓ, gives the claim. - $F(0)$ and $F(\ell)$ are the two traces, because all $Y_k, Y_k' \ge 1$.

Lean: `finite_lip`.

3.19.4.3 4.3 From matrices to kernels Lemma 4.9 (sampled traces). Let $L \ge 1$. For a partition with masses m_i and tags x_i, sample a kernel as $k^Q = (\sqrt{m_i}\, k(x_i, x_j) \sqrt{m_j})_{ij}$. Then: - $(k^*)^Q = (k^Q)^*$; - k^Q is positive semidefinite whenever k is; - $\operatorname{tr}(k_1^Q \cdots k_L^Q) = \sum_{i_1,\dots,i_L} \prod_l m_{i_l} \prod_l k_l(x_{i_l}, x_{i_{l+1}})$, because every vertex contributes two square-root weights; - this Riemann sum converges to $\operatorname{Tr}(k_1 \cdots k_L)$ for continuous k_l along the partitions of Lemma 4.2.

Lean: `sample`, `safe`, `measureWeight`, `learns`.

Lemma 4.10 (proof of III.16.4).

Sketch. Apply III.16.3, for each partition, to $A = K^Q$ and $b_l = B_l^Q$. - By Lemma 4.9, $\mathrm{tr} A^M \to Z > 0$, so the hypothesis $\mathrm{tr} A^M > 0$ holds for fine partitions. - $\|b_l\|_{2q} \to |\mathrm{Tr}((B_l^* B_l)^q)|^{1/(2q)}$, because the column word is sampled with the same partition. - The two gap traces converge to the cyclic integrals. - Passing to the limit at fixed (q, w, Y, Y', ℓ, r) gives the bound $\frac{\ell}{r} \prod_l |\mathrm{Tr}((B_l^* B_l)^q)|^{1/(2q)}\, Z^{1-n/(2q)}$. - The column hypothesis turns this into $Z \prod_l c_l\, \ell/r$.

Lean: `passage`, `goes`, `totality`.

3.19.4.4 4.4 Holomorphy in a strip and Laurent coefficients

Lemma 4.11 (torus integrals inside the strip). These are the first three claims of III.16.5(a).

Sketch. - The coefficient bound: $\hat{u}_p(x)$ is an average of u over the torus $p + 2\pi[0,1]^\alpha$ against a unimodular character. - Independence of p: change one coordinate at a time. A real shift leaves a full-period integral unchanged. For an imaginary shift in coordinate i, apply Cauchy's theorem to a rectangle of horizontal width 2π between heights $\mathrm{Im}\, p_i$ and $\mathrm{Im}\, p_i'$. Periodicity cancels the vertical sides, so the horizontal integrals agree. - The identity $\hat{u}_p(x) = e^{ip\cdot x}\hat{u}(x)$ is independence applied to $u(p)e^{-ip\cdot x}$.

Lean: `market_cap`, `full_turn`, `coat_price`.

Lemma 4.12 (weighted ℓ^1 bound). Under the holomorphy and periodicity hypotheses of III.16.5(a), if $|u| \le K$ on S_η, $H \ge 0$, $b > 0$, and $H + b < \eta$, then $\sum_x e^{H|x|_1} |\hat{u}(x)| \le K(2(1+b)/b)^{\mathrm{card}\,\alpha}$.

Sketch. - For given x, take p with $\mathrm{Im}\, p_a = -(H+b)\,\mathrm{sign}\, x_a$. Then $|e^{ip\cdot x}| = e^{(H+b)|x|_1}$, and Lemma 4.11 gives $|\hat{u}(x)| \le K e^{-(H+b)|x|_1}$. - Then $\sum_{x\in\mathbb{Z}^\alpha} e^{-b|x|_1} \le (2\sum_{k\ge 0} e^{-bk})^{\mathrm{card}\,\alpha} \le (2(1+b)/b)^{\mathrm{card}\,\alpha}$.

Lean: `Vane.issues`, `Width.compactPay`, `pins`.

Lemma 4.13 (Laurent representation). Under the same holomorphy and periodicity hypotheses, if $|u| \le K$ on S_η, then $u(p) = \sum_x \hat{u}(x) e^{ip\cdot x}$ on S_η.

Sketch. - By Lemma 4.12, with $H = \max(\{0\} \cup \{|\mathrm{Im}\, p_a| : a \in \alpha\})$ and $b = (\eta - H)/2$, the Fourier series of $\theta \mapsto u(p + 2\pi\theta)$ is absolutely summable. - Fourier inversion for the continuous shifted function identifies this series with the function. - Evaluate at $\theta = 0$ and use $\hat{u}_p(x) = e^{ip\cdot x}\hat{u}(x)$.

Lean: `Vane.paints`.

Lemma 4.14 (Laurent sums). This is III.16.5(b).

Sketch. - Differentiate term by term, using $\|D_p e^{ip\cdot x}\| \le |x|_1 e^{\sigma|x|_1} \le (\eta-\sigma)^{-1} e^{\eta|x|_1}$ on S_σ. - The derivative series is uniformly summable on the open convex strip, which gives holomorphy and the derivative bound. - The Lipschitz bound follows from the mean value inequality. - The real-part identity follows from $\overline{e^{-ip\cdot x}} = e^{ip\cdot x}$ for real p.

Lean: `Width.estimate`, `reward`, `readDeriv`, `readDiff`, `readRealPart`.

3.19.4.5 4.5 Fourier analysis on tori

Lemma 4.15 (orthogonality). $\sum_{x\in\Lambda_n} e^{i(k_p - k_{p'})\cdot x} = n^4 \delta_{pp'}$, and hence $F_n F_n^* = F_n^* F_n = n^4 I$.

Sketch. For $p \neq 0$ some unit step e_a has $e^{ik_p \cdot e_a} \neq 1$. Translation invariance of the sum then forces it to vanish.

Lean: `sum_cell`, `orthogonal`, `gram`, `coGram`.

Lemma 4.16 (circulant diagonalization; III.16.6(a)). Under the hypotheses of III.16.6, periodization is diagonalized by the Fourier matrix. For real p, the symbol is 2π-periodic and satisfies $\hat{R}(-\bar{p}) = \overline{\hat{R}(p)}$ entrywise. These identities also hold at complex momenta wherever the series converge absolutely.

Sketch. For fixed a, b, p, summing the periodized kernel against the Fourier column gives $\sum_z R_{ab}(z) e^{ik_p \cdot (y+z)} = e^{ik_p \cdot y} \hat{R}(k_p)_{ab}$. Absolute convergence permits regrouping by residues. Periodicity follows from $e^{2\pi i m \cdot z} = 1$, and conjugation follows term by term because R is real.

Lean: `circulant`, `Row.periods`, `Row.conjugate`.

Lemma 4.17 (aliases; III.16.6(b)–(c)).

Sketch. - The fine Fourier matrix factors as the coarse Fourier matrix composed with the alias blocks $E_L(2\pi p/c)$. - $E_L(\bar{p})^* E_L(p) = L^4 I$ is Lemma 4.15 on Λ_L. The common factor $e^{ip \cdot x/L}$ cancels against its conjugate taken at $\bar{p}$. - The intertwining relation is Lemma 4.16 on the fine torus, using $e^{ik_{p,l} \cdot fz} = 1$.

Lean: `aliasAt`, `aliasBlock`, `straightens`, `energy`, `readings`, `readingsGram`, `aligns`.

3.19.4.6 4.6 Matrix square root Lemma 4.18 (= III.16.7).

Sketch. - Diagonalize $X = U \operatorname{diag}(s) U^*$ with $v \leq s_i \leq m$. Then $X/m - 1$ has eigenvalues of modulus $\leq 1 - v/m < 1$. - The scalar identity $\sum_k \binom{1/2}{k} t^k = \sqrt{1+t}$ for $|t| < 1$, applied to the eigenvalues, gives $\Phi_m(X) = U \operatorname{diag}(\sqrt{s_i})\, U^*$. - The binomial series in the Banach algebra of matrices has radius of convergence at least 1, giving holomorphy. Continuity and compactness give boundedness on each smaller closed ball. Its real scalar coefficients imply commutation with entrywise conjugation.

Lean: `binRoot_on_band`, `binSum_deriv`, `Band`.

3.19.4.7 4.7 Taylor coefficients and analyticity Lemma 4.19 (Taylor coefficients).

Let E, F be real normed spaces, let $j, l \geq 0$ be integers, let $f : E \to F$, and put $c_j f(x) = D^j f(x)/j!$.

- If f is C^j at x, then $D^j f(x)$ is symmetric.
- Two symmetric continuous j-linear forms with the same diagonal are equal.
- If $U \subset E$ is open, f is smooth on U, $x \in U$, $u \in E^j$, and $v \in E^l$, then

$$c_j(y \mapsto c_l f(y)[v])(x)[u] = c_l(y \mapsto c_j f(y)[u])(x)[v].$$

- If f is C^j at $-x$, then $c_j(f(-\cdot))(x) = (-1)^j c_j f(-x)$. If f is C^j at $x - y$, then $c_j(y' \mapsto f(x - y'))(y) = (-1)^j c_j f(x - y)$.
- If f is C^{j+1} at x, then, for $u, z \in E$,

$$D_u(t \mapsto c_{j+1} f(x)[t, \dots, t])(z) = c_j(D_u f)(x)[z, \dots, z].$$

Sketch. Induct on j for symmetry, using symmetry of the second derivative to exchange the first two slots and the induction hypothesis for the remaining slots. Equality on diagonals then

implies equality of the forms by differentiating the diagonal and inducting on the arity. For commutation, peel off one directional derivative at a time. The chain rule gives the reflection identities. Differentiating a symmetric diagonal contributes $j+1$ equal terms; the factorials give the last identity.

Lean: `sym_Jet`, `commuteD`, `ext_diag`, `coeff`, `exchangeJet`, `reflectionJet`, `oppositeCoeff`, `slot_lower`.

Lemma 4.20 (several-variable and weak analyticity). - A map between finite-dimensional complex normed spaces that is complex-differentiable on a neighbourhood of x is analytic at x. - Let E be a complex Banach space and $f : \mathbb{C} \to E$ continuous on an open set U. If $L \circ f$ is holomorphic on U for every L in a family of continuous complex linear functionals that separates points, then f is analytic on U.

Sketch. - First claim: choose coordinates and a closed polydisc contained in the differentiability neighbourhood. Successive applications of the one-variable Cauchy formula express each scalar component as a continuous linear functional applied to the product of the Cauchy kernels and the fixed boundary values. Each kernel is analytic in the interior variables, since its denominator is invertible in the Banach algebra of continuous functions on the angle torus. Their finite product and the resulting integral are analytic. Coordinate changes recover the assertion for the original spaces. - Second claim: the Cauchy integral formula holds after applying each L, the separating family lifts it to E, and the Cauchy power series converges.

Lean: `finiteD`, `lift_weak`.

3.19.4.8 4.8 Free Maxwell field Lemma 4.21 (Poincaré lemma in radial gauge; III.16.8(a)).

Sketch. - The curl is closed by symmetry of second derivatives. - Inverse on $\mathcal{C}_j$: for $F \in \mathcal{C}_j$ and $A_i = (j+2)^{-1} x_k F_{ki}$, compute $\partial_i A_l - \partial_l A_i = (j+2)^{-1}(2F_{il} + x_k(\partial_i F_{kl} - \partial_l F_{ki}))$. By Bianchi this equals $(j+2)^{-1}(2F_{il} + x_k \partial_k F_{il})$, and by Euler's identity this is F_{il}. - The map lands in radial gauge, since $x_i x_k F_{ki} = 0$ by antisymmetry. - Inverse on $\mathcal{A}_j$: differentiating $x_k A_k = 0$ gives $x_k \partial_i A_k = -A_i$. Hence $x_k F_{ki} = (j+1)A_i + A_i = (j+2)A_i$.

Lean: `takeCurl`, `closedCurl`, `contracted`, `curl_inverse`, `potential_recover`, `potential`.

Lemma 4.22 (the shell; III.16.8(b)). - $\mathcal{S}_j$ is a subspace of $\mathcal{C}_j$. - The inverse curl restricted to the shell is injective: `gaugeJet`, `leg_injective`.

Sketch for III.16.8(b). By Lemma 4.19, for $j \geq 1$ the derivatives of the j-th homogeneous Taylor polynomial are the $(j-1)$-th Taylor coefficients of the derivatives; the degree-zero case is constant. The Bianchi and Maxwell combinations of f vanish near x, so the corresponding combinations of Taylor coefficients vanish.

Lean: `onShell`, `WaveOn`, `WaveOn.jets`.

Lemma 4.23 (the free kernel; III.16.8(c)).

Sketch. - $\partial_i \partial_j G_0 = -2N\delta_{ij}|x|^{-4} + 8N x_i x_j |x|^{-6}$ with $N = (4\pi^2)^{-1}$. The trace vanishes in dimension four. - $\psi_{\cdot,km}$ is divergence-free and harmonic, so $\sum_a \partial_a C_{ab,km} = \Delta\psi_b - \partial_b \operatorname{div}\psi = 0$. - Bianchi holds because C is a curl. - Exchange and evenness follow from the explicit Hessian.

Lean: `Green`, `second_Green`, `harmonic`, `curlOpp`, `bareCurl`, `Bare.equation`, `Bare.exchange`.

Lemma 4.24 (jet pairing; III.16.8(d)).

Sketch. - By III.16.8(b)–(c), the Taylor coefficients of C in either leg lie on the shell. Hence only the restrictions of p and q matter. - Symmetry: it suffices to check probes that read single mixed Taylor coefficients, because these span the dual of the shell. For those, the claim follows from: - commutation of Taylor coefficients of different orders (Lemma 4.19); - the reflection rule, with sign $(-1)^l$; - $C_{ab,km}(x) = C_{km,ab}(-x)$.

Lean: `Probe`, `contraction`, `descends`, `kernel_smooth`, `exchange`.

Lemma 4.25 (Wick hierarchy; III.16.10).

Definitions. - *Pairings* are fixed-point-free involutions of the factor occurrences whose partners sit at different points; no self-contractions are allowed. - The *moment* is defined on the monomial basis and extended multilinearly. - Labels enter through the substitution $A_j = \mathrm{curl}^{-1} F_j$, $F_j \in \mathcal{S}_j$. - The *equation ideal* is the kernel of this substitution. It consists of the labels that vanish on every on-shell jet sequence.

Sketch of the properties. - Vanishing on the equation ideal: multilinearity. - Relabelling and insertion of 1: added unit monomials have no occurrences. An injection of the remaining indices induces bijections of occurrences and pairings. If the orientation of an edge reverses, the exchange identity for κ preserves its value. Extend from monomials by multilinearity. - Smoothness: each polynomial has finitely many monomials and each monomial finitely many pairings, so the result is a finite sum of products of smooth kernels off the diagonal. When all insertions are 1, there are no occurrences and exactly one empty pairing, whose product is 1.

Lean: `Wick.entry`, `Wick.Occ.Matching`, `Wick.monomial`, `reduceEq`, `observable`, `equationIdeal`, and the links of III.16.10.

3.19.4.9 4.9 Labels Lemma 4.26 (= III.16.9).

Sketch. - (a) A complex polynomial that vanishes on $\mathbb{R}^N$ is zero. - (b) A coordinate's exponent in a monomial of weight $\le d$ is at most d, and only finitely many coordinates have weight $\le d$. - (c) Bihomogeneous components of an invariant polynomial are invariant, because the adjoint action is linear in each A_j and preserves bidegree. They are real if the label is real. - (d) Real bihomogeneous labels span Label_d over $\mathbb{C}$, by taking real and imaginary parts of its bihomogeneous components. A basis containing 1 can be extracted, and the bases are extended inductively from level d to level $d+1$. - Each frame gives coordinates on Label_d. For every real t, rescaling the coordinate of e_i by $t^{\deg e_i}$ (polynomial degree) commutes with the action of $O(4)$, since that action preserves each polynomial degree: `Frame.scaled`, `Frame.scaling`.

Lean: `radial`, `eval_ext`, `cap_finite`, `component_mem`, `decompose`, `frame_exists`, `frames`, `rotation_law`.

3.19.5 5. Where the difficulty lies

The analytic issue in III.16.3–III.16.4 is to retain a bound independent of matrix dimension while varying the exponents and passing to integral kernels. The three-lines argument uses dimension only for qualitative boundedness on the strip. Arbitrary insertions are allowed: their Schatten norms supply the entire insertion cost. Positivity and a positive cyclic trace are required of the transfer kernel, and quadrature is taken at fixed exponents and period.

The complex-power convention matters at zero: A^0 is the range projection. The strict inequalities $Y_k > 2r$ ensure that the integer endpoints agree with ordinary positive powers. Fourier inversion loses strip width and the explicit factor $(2(1+b)/b)^4$; differentiating a Laurent sum costs $1/(\eta-\sigma)$. The Bloch identities retain all momenta, and at complex momentum the alias identity uses the adjoint evaluated at $\bar{p}$.

For Maxwell jets, the conventions are Taylor coefficients $D^j/j!$, real orthonormal colour coordinates, the normalization $(4\pi^2)^{-1}$, and the sign $(-1)^l$ at the second endpoint. The equation subspace is defined by restriction to shell jets; no presentation by invariant generators is needed. The chosen label bases are coherent as sets of polynomials, rather than as prescribed ordered lists.

3.19.6 6. Notation

Written proof	Lean	Meaning
positive-semidefinite kernel	`NNKernel`	all finite Gram matrices Hermitian and positive semidefinite
tagged partitions Q_k, masses m_i	`Quad`, `Quad.shrink`	finite measurable partition with tags; mesh $1/(k+1)$
$\|b\|_{2q}$	`paysNorm`, `rate` $= 1/(2q)$	Schatten norm via $\mathrm{tr}((b^*b)^q)$
strip path	`Flex`	entire; contraction on the strip; trace norm ≤ 1 on $\mathrm{Re} = 1$
A^z	`leap`, `specPow`	complex power, 0 on $\ker A$
$Z_A = \mathrm{tr} A^M$	`Holder.Q`	absolute trace, equal to the trace for $A \succeq 0$
sides of III.16.2	`total`, `cost`	trace product and Hölder bound
$q, w, M, \ell, K, Z, c_k, Y'$	`p`, `width`, `N`, `m`, `T`, `R`, `K`, `Z`	Lean parameters of `finite_lip`, `passage`
$\mathrm{Tr}(k_1 \cdots k_L)$	`TimeKernel.closed`	cyclic integral
k^Q, k^*	`sample`, `op`	sampled matrix; adjoint kernel
$B_1 K^{Y_1} \cdots$; $(B^*B)^q$	`along`, `multiply`	gap word; column word
S_η; $e^{ip\cdot x}$; $\|x\|_1$	`strip`, `mode`, `size`	strip, Fourier character and lattice one-norm
$\sum e^{\eta\|x\|_1}\|f(x)\| \le W$	`Width`	weighted norm bound
$\hat{f}$	`Laurent.read`, `Row.read`	symbol (Laurent series)
periodic holomorphic u on S_η; $\hat{u}_p$	`Vane`, `market`	strip hypotheses and shifted Fourier coefficients
kernel R; periodization R_n	`Bloch.Row`, `Row.acts`, `Row.strip`	real kernel over $\mathbb{Z}^4$ with internal indices
k_p; F_n	`mon`, `cell`, `F`	torus momentum, character and Fourier matrix
alias momenta; $E_L(p)$	`aliasAt`, `aliasBlock`	fine momenta over one coarse momentum and their Fourier block
Φ_m; $v \le X \le m$	`binRoot`, `Band`	analytic root and quadratic-form bounds

Written proof	Lean	Meaning
$c_j f(x) = D^j f(x)/j!$	`Tensor.coeff`	Taylor coefficient
∂_i	`Maxwell.grad`	partial derivative along axis i
$\mathcal{A}_j$	`Labels.Taylor` (`radial`)	radial-gauge potentials, degree $j+1$, weight $j+2$
$\mathcal{C}_j$; $\mathcal{S}_j$	`formSpace` (`ClosedJet`); `Shell`	closed curvature jets; Maxwell shell
curl and its inverse	`takeCurl`, `contracted`	homogeneous radial-gauge identification
G_0; $C_{ab,km}$	`Green`, `bareCurl`	$G_0 = (4\pi^2\|x\|^2)^{-1}$; free field-strength two-point function
probe; $\kappa(p,q)$	`Probe`, `legKernel`	linear form on $\mathcal{S}_j$; one-contraction kernel
free moment; free correlation	`Wick.moment`, `Wick.correlation`	Wick pairing sum and its restriction to labels
equation ideal	`equationIdeal`	labels vanishing on all on-shell jets
Pol; Label_d; weight	`Labels.Pol`, `Labels.Label`, `height`	polynomial algebra, invariant filtered subspace and engineering weight
frame	`Frame`, `frames`	complex basis of real bihomogeneous polynomials, with $e_0 = 1$, coherent in d

3.19.7 7. What remains unexplained

1. **Additional lemmas and dependency lists.** The elementary helper lemmas assigned to this unit outside the eight folders have not been checked individually. The longer unit lists in §2 have not been independently verified entry by entry and should be treated as guides to placement.
2. **Norms and parameter choices.** The square-root argument uses the matrix operator norm; its entrywise Laurent bounds require a finite-dimensional norm conversion. In III.6 the strip is narrowed to $\tau = \min(\eta/2, v/(2CD))$, with $D = 2A/\eta$ and C a norm-conversion constant, and then to $\tau/3$. IV.1 chooses q, w, r and sums successive insertion moves. Those choices must accompany any claim of uniformity in volume or spacing.
3. **Placement.** The trace estimate belongs with IV.1, Laurent and Bloch analysis with III.2–III.7, labels with II.1, and Maxwell jets and Wick correlations with VI.1–VI.2 and VII.1–VII.3. The full square-root theorem is assembled in III.6.

3.20 IV.1. Infinite volume: extraction of a subsequence and the Euclidean axioms of the limit

Part: IV, infinite volume (chapter 10). **Lean:** `YM/Extraction`, `YM/Cells`, `YM/Slice`, `YM/Euclid`, `YM/Dash`, `YM/Next/EXTRACTION`, `YM/Lane`, and the proofs of the work items D01, D02, D03, D04, D05, D06. Their statements are in `YM/Bid/Drops.lean`, which is filed with III.10. **Size:** 7,444 lines

in 1,049 declarations used by the main theorem. Other material in these folders concerns stress tensors, Maxwell comparison and label quotients (Parts V–VII), or corrected fields (III.11–III.12).

3.20.1 1. What this unit proves

Setting and notation. Let G be a compact matrix gauge group and C a Wilson family (II.2, III.13). It has:

- lattice spacings $a_N \downarrow 0$, with $a_0 = 1$, $a_{N+1} = a_N/b_N$ and $b_N \geq 3$ odd;
- periods $L_p \uparrow \infty$, each divisible by 4;
- couplings $\beta_{N,p}$;
- the periodic lattice $\Lambda_{N,p} = (\mathbb{Z}/(L_p/a_N)\mathbb{Z})^4$, whose sites are placed in the box $(-L_p/2, L_p/2]^4$;
- the normalized Wilson expectation $\langle\cdot\rangle_{N,p}$.

Labels. Labels at degree d (polynomial weight at most d) form a finite-dimensional complex space Label_d, given by coefficients in a real basis, so that $v \mapsto \bar{v}$ is defined. We use the supremum norm on coefficient vectors. Label_d carries:

- a representation ρ_d of the group H_4 of signed coordinate permutations;
- a unit label 1_d;
- for each axis a the reflected label $\theta_a v = \rho_d(s_a)\bar{v}$, where s_a reverses axis a.

Observables and correlations. A label $v \in \mathrm{Label}_d$ has a renormalized lattice observable $I_{N,p}(v, r)$ at each site r, linear in v. Covariance, reality and the unit identity are required for $N \geq \mathrm{ref}(d)$. Write $\hat{r}$ for the physical position of a site in $(-L_p/2, L_p/2]^4$. For a label list $v = (v_1, \dots, v_n)$,

$$\mathcal{C}_{N,p}(v; r) = \Big\langle \prod_{i=1}^{n} I_{N,p}(v_i, r_i) \Big\rangle_{N,p}, \qquad \mathcal{C}_{N,p}(v; t) = a_N^{4n} \sum_{r \in \Lambda_{N,p}^n} t(\hat{r})\, \mathcal{C}_{N,p}(v; r).$$

For a lifted tuple $r \in (\mathbb{Z}^4)^n$, $\mathcal{C}_{N,p}(v; r)$ means evaluation after reduction modulo L_p/a_N; its lifted physical position is $a_N r$.

Configuration sets.

- Conf_n: configurations with $x_i \neq x_j$ for $i \neq j$.
- Gen_n: configurations with $x_i^a \neq x_j^a$ for all $i \neq j$ and all axes a. This set is open and dense, and its complement is Lebesgue-null.
- On the torus: $\mathrm{Conf}_n^{(p)}$ requires that for each $i \neq j$ some $x_i^a - x_j^a \notin L_p\mathbb{Z}$; $\mathrm{Gen}_n^{(p)}$ requires all $x_i^a - x_j^a \notin L_p\mathbb{Z}$.

Test functions. $\mathcal{D}_n = C_c^\infty(\mathrm{Conf}_n)$. A test function t is *admissible at volume p* if $\mathrm{supp}\, t \subset (-L_p/2, L_p/2)^{4n}$.

Extractions. An *extraction* f is a pair of strictly increasing sequences (N_j) and (p_k); a refinement $f \circ g$ passes to subsequences of both. A property of (N_j, p_k) holds *eventually along ordered tails* if it holds for $k \geq k_0$ and then $j \geq j_0(k)$. Convergence along ordered tails means that, for each

neighbourhood of the limit, membership holds eventually in this sense. The iterated limits in IV.1.1 (c) also require an actual spacing limit at every admissible fixed volume.

Tables. A *table* S assigns to each arity n, degrees $d_1, \dots, d_n$ and point x a multilinear form $S(\cdot\,; x)$ on $\mathrm{Label}_{d_1} \times \cdots \times \mathrm{Label}_{d_n}$. Its values off Conf_n play no role.

Standing hypotheses.

- **(H1)** The family satisfies its specification (§2, input 1).
- **(H2)** The fixed-volume centered budgets, the power budgets along ordered tails, reflection positivity and the chessboard inequalities hold along f, with the quantifiers in §2, input 2.

Proposition IV.1.1 (subsequential raw limits). Assume (H1) and (H2) for an extraction f. Then there are a refinement $f \circ g = ((N_j), (p_k))$, a table S and tables P_p ($p \in \mathbb{N}$) such that the following hold for every n and every list of labels v.

(a) *(Spacing limit at fixed volume.)* For every p, $P_p(v;\cdot)$ is continuous on $\mathrm{Gen}_n^{(p)}$. For every compact $K \subset \mathrm{Gen}_n^{(p)}$ and $\varepsilon > 0$, for all sufficiently large j and all lifted tuples $r \in (\mathbb{Z}^4)^n$ with $a_{N_j} r \in K$,

$$|\mathcal{C}_{N_j,p}(v;r) - P_p(v; a_{N_j} r)| < \varepsilon.$$

(b) *(Volume limit.)* $S(v;\cdot)$ is continuous on Gen_n, and $P_{p_k}(v;\cdot) \to S(v;\cdot)$ uniformly on compact subsets of Gen_n.

(c) *(Ordered smeared limit.)* For every $t \in \mathcal{D}_n$:

- t is admissible at p_k for k large;
- for every admissible p the limit $V_p = \lim_j \mathcal{C}_{N_j,p}(v;t)$ exists (in the proof, $V_p = \int t P_p(v;\cdot)$);
- $V_{p_k} \to \int t\, S(v;\cdot)$.

(d) $t\, S(v;\cdot)$ is integrable for every $t \in \mathcal{D}_n$, i.e. S is locally integrable on Conf_n.

Status: theorem. (H2) holds for every f as soon as the volume-uniform budget of §2, input 2, holds (`Wcmain.projects`). Part (d) uses the uniform bound of Lemma IV.1.d.

Lean:

- existence: `Extraction.crops`;
- conclusion: `RawLimits`, with fields `Order.Meshes`, `Order.VolumePoints`, `Order.Ordered` and `Order.limits`;
- part (d): `Order.ordinary`;
- hypothesis (H2): `CenterPort`.

Proposition IV.1.2 (limits of label derivatives). Let S be an ordered limit along f (IV.1.1 (c)) that is locally integrable (IV.1.1 (d)), with (H1) and (H2) along f. Let $\mathrm{Null}_d(S)$ be the space of $u \in \mathrm{Label}_d$ with $\int t S(\dots, u, \dots) = 0$ for every arity n, slot, spectator label list and $t \in \mathcal{D}_n$. Assume the relay data of §2, input 3:

- projections $\pi_d(N,p)$ on Label_d converging along ordered tails in operator norm to a projection π_d;
- the relay properties for every d;
- the dimension count $\dim \mathrm{Null}_d(S) \le \dim\ker \pi_d$.

Then $\ker \pi_d = \mathrm{Null}_d(S)$. Moreover there are linear maps $\kappa_{d,e} : \mathrm{Label}_d \to \mathrm{Label}_e$ $(d \le e)$ and $\tau_{d,a} : \mathrm{Label}_d \to \mathrm{Label}_{d+1}$ (a an axis) such that, for all n, slots i, labels v and $t \in \mathcal{D}_n$,

$$\int t\, S(v_{i\to\kappa v_i};\cdot) = \int t\, S(v;\cdot), \qquad \int t\, S(v_{i\to\tau_a v_i};\cdot) = -\int (\partial_{x_i^a} t)\, S(v;\cdot).$$

Here $v_{i\to w}$ is v with the i-th label replaced by w. In other words, $\partial_{x_i^a} S(v;\cdot) = S(v_{i\to\tau_a v_i};\cdot)$ in $\mathcal{D}'(\mathrm{Conf}_n)$.

Status: theorem, conditional on the relay data and the dimension count. Their suppliers in III.10–III.11 are listed in §2, input 3.

Lean: `Work.D04.answer` (conclusion `Spans`), `Bin.exactC`, `lifts_model` (maps `MonitorOps`, identities `MonitorOps.Wakes`), `Wakes.through_ties`.

Proposition IV.1.3 (smoothness of the limit). Let S be as in IV.1.1, along an extraction for which IV.1.2 applies. Then there is a table T with the following properties.

- $T = S$ on Gen_n, and $T(v;\cdot) \in C^\infty(\mathrm{Conf}_n)$ for every n and v.
- IV.1.1 (a)–(d) hold for T along the same extraction and with the same P_p.
- T satisfies the basic identities (E1) below.
- The maps κ and τ descend to linear maps $\mathrm{rise}_{d,e} : F_d \to F_e$ and $\partial_{d,a} : F_d \to F_{d+1}$ on the quotients $F_d = \mathrm{Label}_d/\mathrm{Null}_d(T)$. They satisfy, pointwise on Conf_n,

$$T(\dots, \mathrm{rise}\, u, \dots) = T(\dots, u, \dots), \qquad T(\dots, \partial_a u_i, \dots) = \partial_{x_i^a} T(\dots, u_i, \dots).$$

Such a pair of maps is a *derivative plan* with its inclusion and derivative identities.

On Conf_n, the smooth ordered limit along a given extraction is unique (`Order.determined`).

Status: theorem.

Lean: `Org.Arrives.serves` (filed with VIII.1; hypotheses `Org.Arrives`, conclusion `Org.Cleaner`), `Dash.awake`, `RawLimits.ready`, `DiffPlan` with `Inclusions` and `Derivatives`.

Proposition IV.1.4 (the Euclidean hierarchy). Let T be a table and let f be an extraction. Assume (H1), IV.1.1 (a)–(c) hold for T and $T(v;\cdot) \in C^\infty(\mathrm{Conf}_n)$ for every arity n and label list v. Then T satisfies (E1)–(E5), with all points in Conf_n:

- **(E1) Basic identities.** T is invariant under simultaneous permutation of points and labels; $T = 1$ for $n = 0$; and $T((1_d, v); x) = T(v; (x_2, \dots, x_{n+1}))$.
- **(E2) Real spin.** $T(\rho(s)v; sx) = T(v; x)$ for $s \in H_4$, and $T(\bar v; x) = \overline{T(v; x)}$.
- **(E3) Homogeneity.** $T(v; x + s) = T(v; x)$ for every $s \in \mathbb{R}^4$.

- **(E4) Reflection positivity at points.** Fix an axis a and words $w_1, \dots, w_m$, each a finite list of (point, label) pairs with distinct points and all $x^a > 0$. Let $\Theta_a w$ be the word in reverse order with $x^a \mapsto -x^a$ and labels θ_a. Then for all $c \in \mathbb{C}^m$ the following expression is real and nonnegative:

$$\sum_{i,j} \bar{c}_i c_j \, T(\Theta_a w_i \, w_j) \geq 0.$$

- **(E5) Smeared reflection positivity.** The same holds for tests: replace w_i by pairs (labels $v^{(i)}$, test $t_i \in \mathcal{D}_{n_i}$ supported in $\{x_k^a > 0 \ \forall k\}$) and $T(\Theta_a w_i w_j)$ by $\iint \overline{t_i(y)}\, t_j(z)\, T(\Theta_a(v^{(i)}, y)\,(v^{(j)}, z))$. These integrals converge absolutely, and the Gram matrix is positive semidefinite.

If in addition **(E7)** (polynomial bounds on derivatives) holds, then T satisfies the hierarchy (E1)–(E7), where:

- **(E6)** is smoothness on Conf_n;
- **(E7)** says: for every arity n, degree list, label list v and order $j \geq 0$, there are $A \in \mathbb{N}$ and $C \geq 0$ such that $\|D_x^j T(v; x)\| \leq C r^{-A}$ whenever $0 < r \leq 1$ and $|x_i - x_\ell|_\infty \geq r$ for every $i \neq \ell$. Here D^j is the iterated real Fréchet derivative with its operator norm on $(\mathbb{R}^4)^n$. The constants are independent of r and x.

Status: theorem. (E6) and (E7) are hypotheses of the assembly. (E6) is supplied by IV.1.3; (E7) is supplied by IV.1.5 together with the promotion lemma B10 of V.1.

Lean:

- assembly: `Order.assembles`, producing `Table.Hierarchy`;
- fields: `Basic`, `RealSpin`, `Homogeneous`, `RP`, `RPSmear`, `Smooth`, `Capped`;
- call site: `Org.Cleaner.stay`.

Proposition IV.1.5 (polynomial bound at short distance; D05). Assume (H1) and (H2) along f. Let $T(v; \cdot)$ be smooth on Conf_n for every arity and label list, and suppose T satisfies IV.1.1 (a)–(b) along f. Then for every arity n, degree list (d_i) and label list v there is $C \geq 0$, independent of r and x, such that

$$|T(v; x)| \leq C\, r^{-A} \quad \text{whenever } 0 < r \leq 1 \text{ and } |x_i - x_\ell|_\infty \geq r \text{ for every } i \neq \ell, \qquad A = 4n(1 + \max_i d_i).$$

The maximum of an empty degree list is taken to be 0. The displayed exponent follows by choosing the largest degree in the proof.

Status: theorem. Together with the derivative plan of IV.1.3, the lemma B10 of V.1 upgrades this to (E7).

Lean: `Work.D05.answer` (statement `Bid.D05`, conclusion `Bid.OneCap`); promotion `Bid.B10`, proved by `Work.B10.answer`.

Proposition IV.1.6 (the derivative plan respects the symmetries; D06). Let T be smooth on Conf_n for every arity and label list, permutation symmetric and with real spin, and let $(\mathrm{rise}, \partial)$

be a derivative plan for T satisfying its inclusion and derivative identities. Use ρ_d also for the induced action on F_d. For every $d \le e$, axis and $s \in H_4$:

- $\mathrm{rise}_{d,e}$ and every ∂_a commute with complex conjugation;
- rise commutes with $\rho(s)$ for every $s \in H_4$;
- $\rho_{d+1}(s)\,\partial_b = \sum_a s_{ab}\,\partial_a\,\rho_d(s)$, where (s_{ab}) is the signed permutation matrix of s.

Status: theorem.

Lean: `Work.D06.compat`, `Work.D06.answer`, `DiffPlan.Orientation`.

Remarks.

1. **The package handed downstream** is the bundle `Next.Extract.Base`. It contains (H1), (H2), the relay data, the raw limit S with local integrability and the spanning property, the smooth table T with its plan, the bound of IV.1.5 and the hierarchy of IV.1.4. It also records $\ker \pi_d = \mathrm{Null}_d(T)$ (`Base.null`). On the main theorem's path it is assembled in III.11 (`MergeProof.Accent.rated`) as follows:
   - extract with `crops`;
   - refine further so that the projections converge (`include_projections`);
   - convert the relay estimates (`Projected.ready`).
   - include the rank-detector limits along a further common refinement and obtain the dimension comparison (`Beam.arriving_seed`), then apply IV.1.2–IV.1.5. All earlier limits are retained.
2. **The extraction is subsequential in both indices.** It gives no independence from the volume subsequence. The public form of the theorem (`WilsonFamily.Converges`, via `wilsonLimit`) needs the ordered limit along the *full* sequence of spacings. The upgrade is proved in III.13 by `UVExtension.Reports.useFull`, using `full_tests`. At each admissible fixed volume, uniqueness identifies the full-sequence smeared limit with its subsequential limit; the volume limit and the table T are retained.

3.20.2 2. What it uses

Inputs from other units.

1. **Specification of the family.** The definition is in II.1–II.2. It is proved for the constructed family in III.13 (`UVExtension.Mast.calibrated`). Its consequences used here are the following; covariance, reality and the unit identity require $N \ge \mathrm{ref}(d)$:
   - $\beta_{N,p} \ge 0$;
   - $I_{N,p}(v,x)$ is continuous, gauge invariant and linear in v;
   - $I_{N,p}(v,x)$ depends only on links inside a fixed finite stencil translated to x; for each stencil offset z and axis a, $20(|z^a|+1) \le L_p/a_N$ whenever $N \ge \mathrm{ref}(d)$;
   - for $N \ge \mathrm{ref}(d)$, the terminal transfer $B_{d,N,p}$ from cutoff coordinates to reference coordinates is normalized as follows: the coefficient trajectory of v has $x_{\mathrm{ref}(d)} = v$, $x_N = B_{d,N,p}^{-1} v$, and $\|x_{m+1}\|_1 \le \mathrm{column}_{m+1}\|x_m\|_1$ for $\mathrm{ref}(d) \le m < N$;

- translation covariance;
- H_4-covariance: $I(\rho(s)v, sx+b)$ is $I(v,x)$ composed with the pull-back of the lattice symmetry;
- reality: $I(\bar{v},x) = \overline{I(v,x)}$;
- $I(1_d, x) = 1$ for $N \geq \mathrm{ref}(d)$. *Lean:* `Family.Makes`, `Family.O`, `rotated`, `reality`, `constant`.

2. **Centered chessboard budgets** (interface item 2). A sufficient volume-uniform input is the following: for every degree d, every $R > 0$ and every $\varepsilon > 0$ there is $A \geq 0$ such that for every $m \geq \mathrm{ref}(d)$ there is $N_0 \geq m$ with the following property. For all $N \geq N_0$, all p, all $v \in \mathrm{Label}_d$ with $\|v\| \leq R$, and all sites x:
   - $I_{N,p}(v,x)$ is an interior function of the base cell of the centered tiling $\mathcal{T}_m(x)$;
   - $\|I_{N,p}(v,x)\|_{\mathcal{T}_m(x)} \leq A\, a_m^{-d-\varepsilon}$.

   This stronger volume-uniform input implies **(H2)** along every extraction f. Precisely, (H2) requires:
   - for every fixed p and label v, at every sufficiently fine scale m there is $A \geq 0$ such that the centered interior condition and root bound $\|I_{N_j,p}(v,x)\|_{\mathcal{T}_m(x)} \leq A$ hold for all x and all large j, with $N_j \geq \max(m, \mathrm{ref}(d))$ for $v \in \mathrm{Label}_d$;
   - reflection positivity, the chessboard inequality and subadditivity of roots at every (N,p); and, for every d, $R > 0$ and $\varepsilon > 0$, numbers $m_0 \geq \mathrm{ref}(d)$ and $A \geq 0$ such that for every $m \geq m_0$ and $\eta > 0$, eventually along ordered tails, the centered interior condition and $\|I_{N,p}(v,x)\|_{\mathcal{T}_m(x)} \leq A\, a_m^{-d-\varepsilon} + \eta$ hold for all $\|v\| \leq R$ and all sites x, with $N \geq m$.

   On the main theorem's path it is supplied by `MergeProof.Accent.gardena` (III.11). The variant `Org.Assembly.wcmain` is not on that path. *Lean:* `Wcmain`, `centeredBudget`, `Cells.center`, `CellHyp`, `CenterPort.book`, `CenteredBook`.

3. **Relay data** (interface item 3, in the form used here; III.10–III.11). A family $Q_{N,p}(x)$ is *chessboard-small* if there is a real rate $y(N,p) \to 0$ along ordered tails with the following property. For every integer $q \geq 0$ and lower scale m, there is $k \geq m$ such that, eventually along ordered tails with $N \geq k$, the bound $\|Q_{N,p}(x)\|_{\mathcal{T}} \leq y(N,p)$ and the interior condition hold for every translated bond tiling $\mathcal{T}$ of side a_k and every x in its base cell with lattice coordinates z^a relative to the base corner satisfying

$$M \leq z^a, \qquad z^a + M + 1 \leq a_k/a_N, \qquad M = \left\lfloor \frac{a_k/a_N}{8(q+1)} \right\rfloor.$$

   Thus the error bound must hold for translated cells with prescribed relative interior margins. Centered cells alone do not suffice.

   A label family $w(N,p)$ has *subpolynomial growth* if, for every $\varepsilon > 0$,

$$\|B_{d,N,p}^{-1} w(N,p)\| \leq a_N^{-\varepsilon}$$

   eventually along ordered tails. For each degree d the relay data are:

- projections $\pi_d(N,p)$ on Label_d, bounded at each fixed p and along ordered tails;
- *pace*: for $v \in \mathrm{Label}_d$ and each axis a there are degree-$(d+1)$ labels $w(N,p)$ of subpolynomial growth such that $I_{N,p}(w(N,p),\cdot) - a_N^{-1}\left[I_{N,p}(v,\cdot + e_a) - I_{N,p}(v,\cdot)\right]$ is chessboard-small;

Here e_a is one lattice step, of physical length a_N.

- *transit*: the same, with a label of degree $d' \le d$ approximated at degree d;
- *mark*: if $w(N,p) \in \ker \pi_d(N,p)$ has subpolynomial growth, then $I_{N,p}(w(N,p),\cdot)$ is chessboard-close to measurable observables, bounded at each cutoff and site, whose correlations with arbitrary spectator labels vanish exactly at every fixed positive separation, eventually along ordered tails;
- the dimension count $\dim \mathrm{Null}_d \le \dim \ker \pi_d$.

Supplied in III.11 by `MergeProof.Scotia`, `savings` and `Beam.arriving_seed`. *Lean:* `Presentation`, `Affordable`, `RelayCosts`, `Bid.RelayBin`, `vet`, `close`, `dollar`, `heighten`, `Bin.shortage`.

4. **Analytic toolkit (III.16).** The Lipschitz bound for traces of products of a positive kernel (Lemma IV.1.e). *Lean:* `TimeKernel.passage`, `Holder.finite_lip`, `Holder.estimation`, `Holder.convexStrip`, `TimeKernel.learns`.
5. **Scale schedule (I.1, III.2, V.1).**
   - Gentleness, $\log b_m / \log a_m^{-1} \to 0$, gives $a_{m+1} \ge c\, a_m^2$ (`Schedule.Gentle`, `mesh_inequality`, `OS.thickness`). This is used in D05.
   - The column constants of the renormalization group have $\sum_{m \le N} \log(\mathrm{column}_m) = o(\log a_N^{-1})$. This is used in D01.
6. **Promotion of derivative bounds (V.1).** The lemma B10 (see IV.1.5).

Interface item 4 (whole-sequence convergence at fixed volume) enters only in Remark 2: the extraction itself passes to a spacing subsequence. Interface item 5 (corrected fields) is not used either: `CellHyp` consists only of the chessboard inequality and the budget for $I_{N,p}$. The corrected-field definitions serve III.11–III.12.

Standard mathematics.

- *Reflection positivity of the Wilson action, site and bond reflections.* K. Osterwalder, E. Seiler, Ann. Phys. 110 (1978) 440–471. Axioms: Osterwalder–Schrader, CMP 31 (1973), 42 (1975).
- *Chessboard estimates.*
  - J. Glimm, A. Jaffe, T. Spencer, CMP 45 (1975) 203–216;
  - J. Fröhlich, R. Israel, E. Lieb, B. Simon, CMP 62 (1978) 1–34 and J. Stat. Phys. 22 (1980) 297–347;
  - M. Biskup, *Reflection positivity and phase transitions in lattice spin models*, LNM 1970 (2009), §5.
- *Transfer matrices of reflection-positive lattice theories.* Glimm–Jaffe, *Quantum Physics*, 2nd ed.

- *Hölder's inequality for Schatten norms and complex interpolation (Hadamard three lines).* B. Simon, *Trace Ideals and Their Applications*; Reed–Simon II, §IX.4. Cauchy's estimate for holomorphic functions.
- *Compactness and integration.* Arzelà–Ascoli; Tychonoff for countable products; dominated convergence.
- *Regularity.* A distribution whose localizations have rapidly decreasing Fourier transforms is C^∞ (Hörmander, *The Analysis of Linear Partial Differential Operators I*, Ch. 7).

Imported constants and hypotheses.

- Numbers: the budget constants $A(d, R, \varepsilon)$ and thresholds $N_0(m, d, R, \varepsilon)$ (input 2); the reference levels $\mathrm{ref}(d)$; the gentleness constant c of input 5. The exponent $4n(1 + \max d_i)$ of IV.1.5 comes from choosing $\varepsilon = 1$ and from applying $a_{m+1} \geq c a_m^2$ twice.
- Hypotheses: (H1), (H2), and the relay data and dimension count (input 3). The promotion theorem B10 is proved in V.1.

3.20.3 3. Strategy

Chessboard estimates bound correlations at distinct points uniformly in the cutoff and volume. To obtain equicontinuity, fix an axis and express a correlation as a cyclic trace of transfer kernels and observable kernels supported in slabs. Moving one insertion transfers length between two gaps. Finite-dimensional complex interpolation bounds the trace for complex gap shifts; Cauchy's estimate gives a Lipschitz bound. The required column moments are bounded by the same chessboard budgets. This argument needs separation in every coordinate. Two applications of Arzelà–Ascoli, with countable basis indices, give spacing limits at every fixed volume and then a volume limit on Gen_n. Bounds on all of Conf_n supply domination for smeared limits.

Smoothness uses the label calculus. The relay estimates approximate each lattice forward difference quotient by a degree-raised observable. Summation by parts computes its weak limit, and chessboard-small errors have vanishing smeared limits. The projections and dimension count identify the null labels. A finite-dimensional compactness argument then replaces the varying approximating labels by fixed linear maps, giving distributional derivatives as kernels with raised labels. Iteration yields rapid Fourier decay after localization, hence a smooth representative on all distinct-point configurations.

The exact lattice symmetries pass first at generic grid points and then by continuity. Reflection positivity passes for joint tests supported in a half-space; normalized bump functions give its pointwise form. A two-scale choice of cells yields a polynomial bound at short distances, and the derivative identities promote it to bounds of every order. These arguments assemble the Euclidean hierarchy.

3.20.4 4. Steps

3.20.4.1 4.1 Reflection positivity and chessboard estimates Lemma IV.1.a (reflection positivity of the Wilson measure). Let $\Lambda = (\mathbb{Z}/n\mathbb{Z})^4$ with n even. Let a be an axis, $b \in \mathbb{Z}/n$, and Θ either the site reflection $x^a \mapsto 2b - x^a$ or the bond reflection $x^a \mapsto 2b - 1 - x^a$ (other coordinates fixed), acting on configurations by pull-back. Let Λ_+ be the half-torus $0 < (x^a - b) \bmod n < n/2$ (site case) or $0 \leq (x^a - b) \bmod n < n/2$ (bond case), with residues in

$\{0, \dots, n-1\}$. For every $\beta \geq 0$ and every bounded, measurable, gauge-invariant F depending only on links in Λ_+,

$$\langle F \overline{F \circ \Theta} \rangle_\beta \geq 0.$$

Proof sketch. This is the Osterwalder–Seiler argument. Reparametrize configurations by the links in the two fixed planes, the links crossing them, and the two open halves; this map preserves the measure.

- *Site reflections.* Given the plane links, the halves are independent and exchanged by Θ, so the expectation is a weighted integral of $|\mathbb{E}(F \mid \text{plane links})|^2$ with nonnegative weights.
- *Bond reflections.* Integrating the crossing links gives for each plane the one-step kernel $K_\beta(X, Y)$. It is positive semidefinite because $\operatorname{Re}\operatorname{tr}(XY^*)$ is a sum of Gram kernels, and exponentials, products and averages of positive semidefinite kernels stay positive semidefinite (Schur). The product of the two boundary kernels is positive semidefinite, so its quadratic integral is nonnegative.

Lean: `Slice.reflected_positive`, `Cells.Positive`, `Cells.Interior`, `Cells.Cut`, `Slice.portPos`, `Slice.step_positive`.

Lemma IV.1.b (chessboard estimate). Let $\mathcal{T}$ be a tiling of Λ into M^4 congruent cells (M even), with site or bond seams. For a cell c let θ_c be the composition of seam reflections carrying the base cell to c. Put $\operatorname{copy}_c F = F \circ \theta_c$, or its complex conjugate when θ_c reverses an odd number of axes. For F interior to the base cell (bounded, measurable, gauge invariant, depending only on links inside it) put $\|F\|_{\mathcal{T}} = |\langle \prod_c \operatorname{copy}_c F \rangle_\beta|^{1/M^4}$. If $\beta \geq 0$, then for interior F_c

$$\Big| \Big\langle \prod_c \operatorname{copy}_c F_c \Big\rangle_\beta \Big| \leq \prod_c \|F_c\|_{\mathcal{T}},$$

and $\|\cdot\|_{\mathcal{T}}$ is subadditive. By (H1) this holds for every (N, p) of the family.

Proof sketch. This is the Fröhlich–Israel–Lieb–Simon argument.

- *Fold.* For an assignment A of functions to cells, put $\Phi(A) = |\langle \prod_c \operatorname{copy}_c A_c \rangle_\beta|$. Cauchy–Schwarz for a band cut gives $\Phi(\operatorname{assign}(f, g))^2 \leq \Phi(\operatorname{assign}(f, f))\Phi(\operatorname{assign}(g, g))$, where the two assignments supply the two halves.
- *Maximization.* Among assignments of finitely many functions to cells, the maximum of $|\langle \prod \rangle|$ is attained by a constant assignment, because a fold-closed property satisfied by some assignment is satisfied by a constant one.
- *Normalization.* Apply this to $F_c/(\|F_c\| + \epsilon)$ and let $\epsilon \to 0$.
- *Subadditivity.* Expand $\prod_c (F + F')$ into 2^{M^4} assignments and apply the estimate to each.

Lean: `Cells.Seam`, `Cells.laws`, `Tiling.tiles`, `Tiling.maximize`, `Tiling.fold_both`, `Tiling.existence`, `Tiling.adds`, `Tiling.rootTest`, `Tiling.copy`, `Tiling.planted`, `Cells.seamless`.

Lemma IV.1.c (roots of uncentered observables). Assume (H1), $m \leq k \leq N$ and $N \geq \operatorname{ref}(d)$. Let $\mathcal{X}$ be a bond tiling of $\Lambda_{N,p}$ with cells of side a_m, and let $u \in \operatorname{Label}_d$ have all stencil offsets of

coordinate size $< a_k/a_N$, and assume that for every image gu, $g \in H_4 \times \{1, \mathrm{conj}\}$, the centered roots satisfy $\|I(gu, y)\|_{\mathcal{T}_k(y)} \le A$ for all y, with the centered interior conditions. If o is at distance $\ge 3a_k$ from the boundary of the base cell of $\mathcal{X}$, then $\|I(u, o)\|_{\mathcal{X}} \le A$.

Proof sketch.

- *Centred case.* If o is the centre of a fine cell aligned with $\mathcal{X}$, the $\mathcal{X}$-copies of $I(u, o)$ are copies, in the fine centered tiling, of observables $I(gu, y)$ (by H_4-covariance and reality) sitting in distinct fine cells. The chessboard estimate in $\mathcal{T}_k$, with 1 in the remaining cells, gives $\|I(u, o)\|_{\mathcal{X}}^{\sharp\mathcal{X}} \le A^{\sharp\mathcal{X}}$.
- *Log-convexity.* Under unit translations, $\|F\|_{\mathcal{X}}^2 \le \|F \circ \tau_{-e}\|_{\mathcal{X}} \|F \circ \tau_e\|_{\mathcal{X}}$. This follows from translation invariance and the chessboard estimate with alternating translates, since reversed cells see the opposite shift. A nonnegative log-convex sequence is bounded by its values at the endpoints.
- *Interpolation.* Move between fine-cell centres one axis at a time.

Lean: `Around.interior_from`, `Around.swept`, `Tiling.pathway`, `Tiling.trip`, `Spin.uses`.

3.20.4.2 4.2 Boundedness at separated points Lemma IV.1.d (uniform bound). Assume (H1). Let $x_1, \dots, x_n \in \Lambda_{N,p}$ be such that every pair has some coordinate at cyclic physical distance $\ge r$. Let $m \le k \le N$ with $a_m < r$ and $6n\, a_k < a_m$, and let $A \ge 0$ bound the centered roots at scale k of all images gv_i, $g \in H_4 \times \{1, \mathrm{conj}\}$, with the centered interior conditions. Assume $N \ge \max_i \mathrm{ref}(d_i)$ and that all stencil offsets of these labels have coordinate size $< a_k/a_N$. Then $|\mathcal{C}_{N,p}(v; x)| \le A^n$. Under (H1)–(H2) this yields the bound **(B)**:

- **(B1)** for each p and compact $K \subset \mathrm{Conf}_n^{(p)}$, $|\mathcal{C}_{N_j,p}|$ is bounded at lattice points of K for j large;
- **(B2)** for compact $K \subset \mathrm{Conf}_n$, it is bounded eventually along ordered tails.

Proof sketch.

- By pigeonhole, some translate of the tiling with cells of side a_m puts every x_i at distance $\ge 3a_k$ from all cell boundaries (possible because $n \cdot 2 \cdot 3a_k < a_m$).
- Distinct points lie in distinct cells because $a_m < r$.
- Each $I(v_i, x_i)$ is the copy of some $I(g_i v_i, y_i)$ with y_i in the base cell.
- Apply the chessboard estimate with 1 elsewhere, then Lemma IV.1.c.
- For (B): by compactness a single $r > 0$ works on K, and for large p there is no wrap-around. Choose m, then k, then take the budgets from (H2).

Lean: `Around.fits_budget`, `Around.size_done` (conclusion `Bound`), `Tiling.fits_together`, `Gaps.some`, `Gaps.someGrows`.

3.20.4.3 4.3 Equicontinuity Lemma IV.1.e (Lipschitz bound for traces; proved in III.16). Let X be a compact metric space with a finite Borel measure, K a continuous positive semidefinite kernel, and $B_1, \dots, B_n$ continuous kernels, with $n \ge 2$. Let $q, w \ge 1$ be integers, put $M = 2qw$, and assume $Z = \mathrm{Tr}\, K^M > 0$. Take gaps $Y, Y' \in \mathbb{N}^n$ with $nw + \sum_k Y_k = M$. Suppose

Y' transfers an integer $\ell \geq 0$ units from one gap to a different gap, $Y_k > 2r$ for every k, and $0 \leq \ell \leq r$ with $r > 0$. If

$$\|B_k\|_{2q} := \left(\operatorname{Tr}(B_k^* B_k)^q\right)^{1/(2q)} \leq c_k Z^{1/(2q)},$$

then

$$\Big|\operatorname{Tr} \prod_k B_k K^{Y'_k} - \operatorname{Tr} \prod_k B_k K^{Y_k}\Big| \leq Z\Big(\prod_k c_k\Big)\frac{\ell}{r}.$$

Here traces of kernel products are their cyclic integrals.

Proof sketch. Approximate the cyclic integrals by finite matrices (A, b_k) using one quadrature for all kernels and their adjoints. For the positive semidefinite matrix A, use the entire spectral family equal to $e^{z \log \lambda}$ on positive eigenvalues and zero on $\ker A$ for every z. It agrees with A^m at positive integers m. The trace with gap vector $W(z) = Y + z(e_i - e_j)$ is entire. On $|z| \leq 2r$, all real gap parts are nonnegative and

$$\frac{n}{2q} + \sum_k \frac{\operatorname{Re} W_k(z)}{M} = 1.$$

Writing $Z_A = \operatorname{Tr} A^M$, complex interpolation gives the bound $\prod_k \|b_k\|_{2q} Z_A^{1-n/(2q)}$. Cauchy's estimate bounds the derivative on $|z| \leq r$ by this quantity divided by r. Integrate from 0 to ℓ and pass to the kernel limit; the partition traces and column moments converge under the common quadrature.

Lean: `TimeKernel.passage`, `Holder.finite_lip`, `Holder.estimation`, `Holder.leap`.

Lemma IV.1.f (the Wilson transfer matrix). Fix $\beta \geq 0$, an axis a and $\Lambda_{N,p}$. The one-step kernel K_β on slice configurations, with the links in direction a integrated out and half of each slice action assigned to either side, is continuous and positive semidefinite. With $M = L_p/a_N$, $\operatorname{Tr} K_\beta^M = Z$ is the partition function. Every continuous observable local to a slab of width $1 \leq w < M$ lattice steps has a slab representation, whose weighted integration gives a continuous kernel B. Products of such observables, all of width w and separated by gaps Y, satisfy

$$\Big\langle \prod_k F_k \Big\rangle_\beta = \frac{\operatorname{Tr} \prod_k B_k K_\beta^{Y_k}}{Z}, \qquad nw + \sum_k Y_k = M.$$

If $M = 2qw$ with $q \geq 1$, the normalized column bounds $\left(\operatorname{Tr}(B_k^* B_k)^q / Z\right)^{1/(2q)} \leq c_k$ therefore give the difference bound $(\prod_k c_k)\ell/r$ under the gap and shift hypotheses of Lemma IV.1.e.

Proof sketch. Fill the complement of the slab with a fixed configuration to express the observable as a function of its slab links. Integrate the interior links with the Wilson weights. Fubini's theorem identifies the product with the cyclic kernel integral. Apply Lemma IV.1.e and divide by $Z > 0$.

Lean: `Slice.step_positive`, `Time.partition_seen`, `Time.powering`, `Time.selects`, `Time.organized`.

Lemma IV.1.g (chessboard bound for the column norm). Assume (H1), $m \le N$ and $N \ge \mathrm{ref}(d_v)$ for a label v of degree d_v. Put $d = a_m/a_N$. Suppose $F = I_{N,p}(v, x)$ and $J = I_{N,p}(\theta_a v, x)$ are interior to the base cell of $\mathcal{T}_m(x)$ with roots $\le c$. Then F has a slab representation of width $2d$ lattice steps. Its kernel B and the kernel of J are adjoints. Writing $L_p/a_N = 4qd$, with $q \ge 1$, and letting Z be the partition function, one has

$$\left(\frac{\mathrm{Tr}(B^*B)^q}{Z}\right)^{1/(2q)} \le c.$$

Proof sketch. The observable J equals $\overline{F \circ \Theta}$ for the site reflection through x^a. Reversing the slab gives the adjoint kernel. The column consists of $2q$ alternating copies of J and F, placed in every other cell of the row of $\mathcal{T}_m(x)$ in direction a. Apply the chessboard estimate with 1 in all other cells and take the $1/(2q)$-th power.

Lean: `Tiling.enough`, `Difference.reading`, `Difference.stars`.

Lemma IV.1.h (Lipschitz bound in one coordinate). Assume (H1), $m \le N$ and $N \ge \max_i \mathrm{ref}(d_i)$. Let $n \ge 2$ lattice points have pairwise cyclic physical a-gaps $> 2a_m + 2\rho$, with $\rho > 0$. Assume the centered interior conditions and let c_i bound the roots of $I(v_i, x_i)$ and $I(\theta_a v_i, x_i)$ on $\mathcal{T}_m(x_i)$. Moving one point by $h \in a_N\mathbb{N}$, $0 \le h \le \rho$, in direction $+e_a$ changes $\mathcal{C}_{N,p}(v; x)$ by at most $(\prod_i c_i)\, h/\rho$.

Proof sketch. Put $d = a_m/a_N$ and reorder the points cyclically in the a-direction (permutation symmetry). Represent each observable by a brick of width $2d$; the gaps are then $> 2\rho/a_N$ in lattice units. Combine Lemmas IV.1.f and IV.1.g.

Lean: `Difference.primitive`, `Difference.chrono`.

Lemma IV.1.i (equicontinuity). Under (H1)–(H2), for every arity n:

- **(Q1)** for every p, compact $K \subset \mathrm{Gen}_n^{(p)}$, label list v and $\epsilon > 0$, there is $\delta > 0$ such that, for all large j, the lifted lattice correlations at any $x, y \in K$ with $|x - y| < \delta$ differ by less than ϵ;
- **(Q2)** for every $\epsilon > 0$, the same assertion holds for every compact $K \subset \mathrm{Gen}_n$, eventually along ordered tails, with one δ independent of the volume.

Quantitatively, let $R > 0$ bound below all cyclic coordinate gaps on a closed neighbourhood of K, and let A bound the centered roots of the v_i and all $\theta_a v_i$ at a scale $a_m < R/4$. Then $|\Delta\mathcal{C}| \le 16nA^n|x - y|/R$ for $|x - y|$ sufficiently small.

Proof sketch. Telescope through at most $4n$ coordinate moves inside the neighbourhood and apply Lemma IV.1.h with $\rho = R/4$. For a negative move, interchange its endpoints. For $n = 1$ use translation invariance; $n = 0$ is constant.

Lean: `Difference.completed` (conclusion `TransferLip`), `comparable`, `stepTotal`, `permits`, `Gaps.all`, `Gaps.allGrows`.

3.20.4.4 4.4 Compactness and ordered limits Lemma IV.1.j (Arzelà–Ascoli for countably many families). Let (Y_i) be a countable family of locally compact separable metric spaces, $U_i \subset Y_i$ open, and $T_i^{(j)} : Y_i \to \mathbb{C}$. Assume that $T_i^{(j)}(y)$ is eventually bounded for each $y \in U_i$, and that the family is eventually equicontinuous on compact subsets of U_i. Then one

strictly increasing sequence (t_j) makes every $T_i^{(t_j)}$ converge locally uniformly on U_i to a continuous limit.

Proof sketch. Take countable dense sets and replace the finitely many out-of-bound values at each dense point by a point in its bounding ball. By Tychonoff, the countable product of closed balls is sequentially compact, which gives a single subsequence converging at all dense points. Equicontinuity upgrades this to locally uniform convergence.

Lean: `localAscoli`, `simultaneous`, `oscillation`.

Lemma IV.1.k (two-stage extraction). The bound (B) and equicontinuity (Q) along f imply IV.1.1 (a)–(b) along a refinement.

Proof sketch.

- *Indexing.* Index by $(p, n, \text{degrees}, \text{basis vectors})$, a countable set. Limits are taken on basis labels and extended multilinearly.
- *Spacing.* Use the step extensions of the lattice correlators on $\mathrm{Gen}_n^{(p)}$; the extension moves points by at most $4na_N$. Equicontinuity comes from (Q1) and the bound from (B1). Lemma IV.1.j gives one subsequence of spacings for all p.
- *Volume.* Use the functions P_{p_k} on Gen_n. Equicontinuity and the bound come from (Q2) and (B2) passed to $j \to \infty$; this uses that a compact subset of Gen_n lies in $\mathrm{Gen}_n^{(p)}$ for large p. Apply Lemma IV.1.j again.

Lean: `first_surface`, `setMeshes`, `each_oscillates`, `volume_oscillates`, `fitted`.

Lemma IV.1.l (ordered smeared limits and local integrability). The bound (B) together with IV.1.1 (a)–(b) implies IV.1.1 (c)–(d).

Proof sketch. Dominated convergence, applied twice.

- *Fixed volume.* At an admissible p, the lattice sum is the integral of a step function. It converges almost everywhere (on $\mathrm{Gen}_n^{(p)}$) to tP_p and is dominated by $\sup|t| \cdot D$ on the box, using (B1) on $\mathrm{supp} t \subset \mathrm{Conf}_n^{(p)}$.
- *Volume limit.* $|P_{p_k}| \le D$ on $\mathrm{supp} t \cap \mathrm{Gen}_n$ for large k, by (B2) in the limit $j \to \infty$, and $P_{p_k} \to S$ on the full-measure set Gen_n.

IV.1.1 is Lemmas IV.1.d, IV.1.i, IV.1.k and IV.1.l combined.

Lean: `Order.poses`, `Order.ordinary`, `Order.intraperiod`, `Order.ring_bound`, `almostGeneric`; `crops`.

3.20.4.5 4.5 Label derivatives

Lemma IV.1.m (one-slot difference quotients; D03). Assume (H1), IV.1.1 (c)–(d) and (B) along f. For every arity n, label list v, slot i, axis a and $t \in \mathcal{D}_n$, put $u = v_i$ and replace $I(u, x_i)$ by the forward difference quotient $a_N^{-1}[I_{N,p}(u, x_i + e_a) - I_{N,p}(u, x_i)]$. The resulting smeared lattice correlation converges along ordered tails to $-\int (\partial_{x_i^a} t)\, S(v; \cdot)$.

Proof sketch.

- Summation by parts on the torus, which is exact, moves the difference onto the test as a backward difference quotient.

- For p large and then N large the shifted test is evaluated without wrap-around.
- The backward quotient $[t(x-a_N e_{i,a})-t(x)]/a_N$ converges uniformly to $-\partial_{x_i^a} t$, where $e_{i,a}$ is the coordinate vector for slot i and axis a. The error is at most $L_p^{4n}\cdot\sup|\text{residual}|\cdot D$, with D from (B2) on a compact neighbourhood of $\operatorname{supp} t$.
- The main term converges by (c) applied to $\partial_{x_i^a} t$.

Lean: `Work.D03.answer` (statement `Bid.D03`), `on_lattice`, `error_sum`, `heighten`.

Lemma IV.1.n (chessboard-small observables are invisible; D02). Assume (H1)–(H2). Let $Q_{N,p}$ be chessboard-small with rate $y(N,p)\to 0$. For every $\delta>0$, arity, slot and fixed spectator label list, there is $A_\delta\geq 0$ such that, eventually along ordered tails, the correlation with Q in that slot has modulus at most $A_\delta|y(N,p)|$ at every configuration separated by δ in the periodic physical supremum norm. Consequently every smeared lattice correlation with Q in one slot tends to 0 along ordered tails.

Proof sketch. The first claim is the chessboard argument of Lemma IV.1.d with Q in one cell. For the second, write the smeared sum as the integral of a step function supported in a fixed ball and bounded by $\sup|t|\,A_\delta|y|$.

Lean: `Lease.take`, `Lease.rents`, `Lease.gain`, `Work.D02.answer` (statement `Bid.D02`).

Lemma IV.1.o (bounded labels have subpolynomial cutoff coordinates; D01). Under (H1), every degree-d label family $w(N,p)$ bounded along ordered tails has subpolynomial growth in the sense of §2, input 3.

Proof sketch. The inverse terminal transfer is bounded, up to a fixed degree-dependent norm factor, by the product of the column constants up to level N, and their logarithms sum to $o(\log a_N^{-1})$.

Lean: `Work.D01.answer` (statement `Bid.D01`), `terminal_ball`, `columnLogs_le`.

Lemma IV.1.p (spanning; D04). Assume (H1), (H2), IV.1.1 (c)–(d), and the relay data and dimension count for every d. Then:

1. for each fixed $u\in\mathrm{Label}_d$, the smeared lattice correlations of $u-\pi_d(N,p)u$ (in any slot, with any spectators and tests) tend to 0 along ordered tails; together with the dimension count this gives $\ker\pi_d=\mathrm{Null}_d(S)$;
2. for $d\leq e$ and $u\in\mathrm{Label}_d$ there are degree-e labels $w(N,p)$ whose smeared correlations converge along ordered tails to those of u in the limit;
3. for $u\in\mathrm{Label}_d$ and each axis a there are degree-$(d+1)$ labels $w(N,p)$ whose smeared correlations converge along ordered tails to $-\int(\partial_{x_i^a}t)\,S(v;\cdot)$.

In 2 and 3, one family $w(N,p)$ works for every arity, slot, spectator list and test. It may be chosen with subpolynomial growth, and the normalized remainders $(w-\pi w)/(1+\|\pi w\|)$ also have vanishing smeared limits.

Proof sketch.

- (1) $u-\pi(N,p)u$ is bounded and lies in $\ker\pi(N,p)$. By Lemma IV.1.o it has subpolynomial growth, so *mark* applies: it is chessboard-close to an observable with exactly vanishing separated correlations. Lemma IV.1.n then shows the smeared correlations vanish.

The one-slot smear maps converge in operator norm, since their domains are finite-dimensional. This allows passage to varying labels and gives $\ker \pi_d \subset \mathrm{Null}_d(S)$; the dimension count gives equality.

- (2)–(3) *Transit*, resp. *pace*, gives labels whose observables are chessboard-close to the embedded label, resp. to the forward difference quotient. Lemma IV.1.n transfers the smeared limits, which Lemma IV.1.m computes for the difference quotient.
- For the normalized remainder, dividing w by $1+\|\pi w\|$ preserves subpolynomial growth, while $\pi w/(1+\|\pi w\|)$ is bounded and Lemma IV.1.o applies. Their difference lies in the projection kernel, so *mark* and Lemma IV.1.n apply.

Lean: `Work.D04.answer` (statement `Bid.D04`), `Spans`, `Bin.Upper`, `Bin.exactC`.

Lemma IV.1.q (fixed derivative labels). The conclusions of Lemma IV.1.p yield the maps κ and τ of IV.1.2.

Proof sketch. This is a finite-dimensional compactness argument.

- Take a cluster point (c, z) of $((1+\|\pi w\|)^{-1},\ \pi w/(1+\|\pi w\|))$ along the ordered-tail filter. Use one refining ultrafilter for this finite-dimensional pair; all spectator tests then retain their limits.
- Idempotence and convergence of the projections give $\pi z = z$. For every spectator test, the limits give $\langle S, z\rangle = c \cdot (\text{target})$.
- $c \neq 0$: otherwise $z \in \mathrm{ran}\pi \cap \ker \pi = \{0\}$ (since $\ker \pi = \mathrm{Null}$), contradicting $\|z\| + c = 1$. So z/c is a fixed label with the target kernels.
- Define κ and τ this way on a basis and extend linearly.

The identity for τ is the distributional relation $\partial_{x_i^a} S_v = S_{v_{i\to\tau_a v_i}}$.

Lean: `lifts_model`, `Dash.brace`, `Wakes.through_ties`, `Dash.monitor`.

3.20.4.6 4.6 Smoothness Lemma IV.1.r (distributional regularity). Let K_w, with w ranging over label lists of arity n, be locally integrable on Conf_n and continuous on Gen_n. Assume that for every w, slot i and axis a there is w' with $\partial_{x_i^a} K_w = K_{w'}$ in $\mathcal{D}'(\mathrm{Conf}_n)$. Then every K_w agrees almost everywhere on Conf_n, and everywhere on Gen_n, with a function in $C^\infty(\mathrm{Conf}_n)$.

Proof sketch.

- For $\varphi \in \mathcal{D}_n$: $\xi_{i,a}\,\widehat{\varphi K_w}(\xi)$ is a constant times $\widehat{\varphi K_{w'}} + (\widehat{\partial_{i,a}\varphi})K_w$.
- By induction on m, $|\xi_{i,a}|^m|\widehat{\varphi K_w}|$ is bounded for all m and all coordinates. So $\widehat{\varphi K_w}$ decays rapidly, and φK_w equals a smooth function almost everywhere.
- Take $\varphi = 1$ near an arbitrary point of Conf_n, generic or not.
- The local smooth representatives agree on Gen_n, which has full measure, hence on overlaps; glue them. Multilinearity in the labels follows from the same density argument.

Lean: `Dash.awake`, `aggregate`, `repair`, `arises`, `Hidden`, `currents`, `closeBy`.

Lemma IV.1.s (the smooth version is a raw limit with a derivative plan). Let T be given by Lemma IV.1.r applied to S and the maps of IV.1.2. Then T satisfies the conclusions of IV.1.3.

Proof sketch.

- T satisfies IV.1.1 (a)–(d) with the same P_p, because the volume limit only sees Gen_n, and (c) follows from Lemma IV.1.l;
- (E1) holds by Lemma IV.1.t;
- the kernel identities pass from S to T, since the smeared values are the same;
- κ and τ preserve null labels: their smeared kernels are respectively the original kernel and its distributional derivative. Smoothness turns these into pointwise identities, so the maps descend to the quotients F_d.

Lean: `Org.Arrives.serves`, `RawLimits.ready`, `Table.Field`.

3.20.4.7 4.7 Euclidean axioms, bounds and assembly Lemma IV.1.t (symmetries and basic identities). If T is smooth on Conf_n and is a point limit (IV.1.1 (a)–(b)) along f, and (H1) holds, then T satisfies (E1)–(E3).

Proof sketch.

- *Grid convergence.* For a point x of the level-k grid $a_k\mathbb{Z}^{4n}$ with $x \in \mathrm{Gen}_n$, use its reduction in $\Lambda_{N,p}$ for $N \geq k$ and sufficiently large p. The lattice correlations there converge to $T(v;x)$ along ordered tails.
- *Lattice identities.* Permutation symmetry, H_4-covariance about the origin, translations by grid vectors, reality and the unit label hold exactly on the lattice, by (H1) and the invariance of the Wilson measure.
- *Extension.* Equality on the dense set of generic grid points extends to Gen_n by continuity, and then to Conf_n, since both sides are continuous there and Gen_n has full measure. Real translations are approximated by grid translations.

Lean: `Order.limits_basic`, `Order.spin`, `Order.homogeneous`, `Order.inPatch`, `Order.equal`.

Lemma IV.1.u (reflection positivity of the limit). If T is smooth, symmetric and an ordered limit along f, and (H1) holds, then (E5) holds; (E5) implies (E4).

Proof sketch.

- *Lattice side.* For a test supported in $\{x^a > 0\}$, take p large enough that it is admissible. The smeared lattice observable $a_N^{4n}\sum_r t(\hat{r})\prod I(v_k, r_k)$ is then, for N large, interior to the half-torus $0 < x^a < L_p/2$. Its reflection through the site plane $x^a = 0$ carries labels θ_a, because $I(\theta_a v, \Theta x) = \overline{I(v,x)\circ\Theta}$. The lattice Gram matrix of such observables is the smeared lattice correlation of $\overline{t_i}\otimes t_j$ pulled back by reflection of the first configuration. Symmetry permits reversal of its entries. The resulting joint test has compact support in $\mathrm{Conf}_{n_i+n_j}$, since the two configurations lie strictly on opposite sides of the plane. Lemma IV.1.a makes the lattice Gram matrix positive semidefinite.
- *Limit.* Smoothness gives absolute integrability on the compact joint supports. Pass to the ordered limit using (c).
- *(E4) from (E5).* Use bump functions concentrating at the points, together with continuity of T.

Lean: `Order.realPositive`, `Banks.at_cut`, `Banks.kept`, `Banks.plane`, `Pairing.at_point`, `Pairing.nearGram`.

Lemma IV.1.v (the polynomial bound). Under the hypotheses of IV.1.5, its stated exponent $4n(1 + \max_i d_i)$ is valid.

Proof sketch. Set $D = 1 + \max_i d_i$. The centered power budgets for all transformed labels can be bounded by Ba_k^{-D} for every $k \geq M$, with fixed $B \geq 0$ and M.

- Choose the first $m \geq M$ with $a_m < r/2$, and the first $k \geq M$ with $a_k < a_m/[6(n+1)]$. The scale inequality, including the possible endpoint $m = M$ or $k = M$, gives $a_m \geq c(r/2)^2$ and then $a_k \geq c'r^4$.
- Lemma IV.1.d bounds the lattice correlations by $(Ba_k^{-D})^n$. Pass first to the spacing limit and then to the volume limit at generic points with separation $> r/2$.
- Continuity and density extend the bound to every r-separated configuration. Since $a_k \geq c'r^4$, the bound is at most Cr^{-4Dn}, with C independent of positions and r.

Lean: `Work.D05.answer`, `image_power`, `scales`, `separated_bound`.

Lemma IV.1.w (orientation). Under the hypotheses of IV.1.6, its four symmetry identities hold on the quotient spaces F_d.

Proof sketch. A class in F_d is determined by its kernels against all spectators. Compare the kernels of both sides using (E2) and the chain rule for $x \mapsto sx$ and for complex conjugation.

Lean: `Work.D06.compat`.

Lemma IV.1.x (assembly). If T is an ordered limit and a point limit along f, (H1) holds, T is smooth and (E7) holds, then T satisfies the hierarchy (E1)–(E7).

Proof sketch. Lemmas IV.1.t and IV.1.u give (E1)–(E5); (E6)–(E7) are assumed. When (H2) and the derivative plan of IV.1.3 are also available, IV.1.5 and B10 supply (E7). B10 inducts on the derivative order: each coordinate derivative is a kernel with one raised label, and the operator norm is bounded by a finite sum of coordinate norms. The base case is IV.1.5. On the main path, compactness of the bounded projection matrices gives a common refinement before forming the bundle of Remark 1.

Lean: `Order.assembles`, `Org.Cleaner.stay`, `Bid.healthy`, `Work.B10.capped_tags`, `include_projections`, `Projected.ready`, `Next.Extract.Base`.

3.20.5 5. Where the difficulty lies

The main estimates are the transfer-kernel Lipschitz bound and the control of uncentered chessboard roots. Complex interpolation and Cauchy's estimate turn reflection positivity into equicontinuity without prior derivative bounds. Translation log-convexity allows the centered budgets to control observables at arbitrary interior positions.

The regularity argument depends on more than compactness: the relay estimates supply degree-raised approximants to differences, and the rank comparison identifies precisely which labels vanish. The normalized remainder estimate in Lemma IV.1.p is needed to obtain fixed derivative labels. Fourier regularity, passage of symmetries by density, and positivity of limiting Gram matrices then use standard arguments.

Three distinctions matter:

- Equicontinuity is proved on Gen_n, whereas bounds and the smooth representative extend to Conf_n.
- The maps $\tau_{d,a}$ are chosen by compactness. Their kernel identities determine them only modulo Null_{d+1}; the induced maps on F_d are intrinsic.
- The hierarchy here has signed coordinate symmetries and reflection positivity at coordinate planes. Translation invariance moves those planes. Continuous rotation symmetry is a later result.

3.20.6 6. Notation

object	code name	proposed notation
Wilson family; spacing, period, coupling	`Family` (`C.S.a`, `C.S.period`, `C.beta`)	C; a_N, L_p, $\beta_{N,p}$
specification of the family	`Family.Makes`	(H1)
label, register of labels	`Register.Token`, `Family.book`	$v \in \mathrm{Label}_d$
unit label; reflected label	`Register.one`, `Register.reflectedToken`	1_d; $\theta_a v = \rho_d(s_a)\bar{v}$
renormalized lattice observable	`Family.O`	$I_{N,p}(v,x)$
lattice correlation; smeared	`Family.numeric`, `Order.smears`	$\mathcal{C}_{N,p}(v;x)$, $\mathcal{C}_{N,p}(v;t)$
configuration sets	`conf`, `generic`, `NoHit`, `Cyclic`	Conf_n, Gen_n, $\mathrm{Conf}_n^{(p)}$, $\mathrm{Gen}_n^{(p)}$
test functions; admissible at p	`Test`, `suitable`	$\mathcal{D}_n$; $\mathrm{supp} t \subset (-L_p/2, L_p/2)^{4n}$
extraction; refinement; ordered sense	`Order`, `refine`, `late`, `fuse`	$f = ((N_j),(p_k))$, $f \circ g$, "k large, then j large"
table	`Table`	$S(v;x)$, $T(v;x)$
raw limits	`RawLimits`, `PointLimits`	IV.1.1 (a)–(c); P_p
tiling, copy, chessboard root	`Tiling`, `copy`, `rootTest`	$\mathcal{T}$, copy_c, $\|F\|_{\mathcal{T}}$
centered tiling at x, cells of side a_m	`Cells.center`	$\mathcal{T}_m(x)$
interior function of a cell	`Interior`	$F \in \mathcal{A}(\mathrm{cell})$
reflection positivity and chessboard inequality	`Seam`	(RP), (CB)
budgets	`Wcmain`, `CenterPort`	input 2; (H2)
uniform bound; equicontinuity	`Bound`, `TransferLip`	(B1)–(B2), (E1)–(E2)
forward difference quotient	`heighten`	$a_N^{-1}[I(v,\cdot+e_a) - I(v,\cdot)]$
chessboard-small; subpolynomial growth	`close`, `dollar`	—
projections; relay data	`Dash.Bin`, `RelayBin`	$\pi_d(N,p) \to \pi_d$; pace, transit, mark
null labels; label quotient	`Dash.monitor`, `Table.null`, `Table.Field`	Null_d, $F_d = \mathrm{Label}_d/\mathrm{Null}_d$
derivative labels; derivative plan	`MonitorOps`, `DiffPlan`	$\kappa_{d,e}$, $\tau_{d,a}$; $\mathrm{rise}_{d,e}$, $\partial_{d,a}$

object	code name	proposed notation
Euclidean hierarchy	`Table.Hierarchy`	(E1)–(E7)
zeroth-order polynomial bound	`Bid.OneCap`	IV.1.5
bundle handed downstream	`Next.Extract.Base`	—

3.20.7 7. What remains unexplained

1. **Ultraviolet relay and rank estimates.** The interfaces in `MergeProof.Scotia` and `savings` were checked, together with their use in the dimension argument. The upstream proofs of the matching estimates and interacting rank comparison in III.10–III.11 were not audited here. In particular, the dimension inequality is a separate input; it does not follow from the centered root budgets.

2. **Volume-uniform budget.** The supplier is `gardena` (III.11). Its proof reduces to earlier estimates on the same family, but their relation to the volume-uniform bounds assigned to III.12 was not traced in full.

3. **Fixed-volume ultraviolet limit.** The full-spacing upgrade required by `wilsonLimit` is located and explained in Remark 2. The upstream construction of the full fixed-volume kernels used by that upgrade was not audited here.

4. **Alternative assembly.** `Next.Extract.finish` and `common` give another conditional route to the same bundle. They require relay costs and detector estimates in addition to the centered budgets. Their hypotheses were checked; they are not additional proofs of the ultraviolet inputs.

3.21 V.1. Decomposition of the limit into phases

Part: V, phases and reconstruction. **Lean:** `YM/OS`, `YM/Phase`, `YM/Ground`, `YM/Heat`, `YM/SNAG`, `YM/Paley`, `YM/Spectral`, the work items B01–B18 in `YM/Work`, and the assembly files `Bid/Phase.lean`, `Bid/Fetch.lean`, `Bid/Market.lean`, `Org/Phase.lean`. **Size:** about 5,700 lines in declarations used by the main theorem, including about 2,700 in B01–B18, plus the assembly files.

3.21.1 1. What this unit proves

Setting and notation.

- **Tables and words.** S is the limit table of IV.1 (the mixed state S^{mix} of the main theorem). A *label* is a pair (d, v): its degree $d \in \mathbb{N}$ and a vector v in the finite-dimensional reference space V_d. A *word* $w = ((x_1, v_1), \dots, (x_n, v_n))$ is a finite list of points of $\mathbb{R}^4$ with labels; $|w| = n$, and $S(w) = S(x; v)$ is its value, multilinear in the labels.
- **Time, slabs, buffers.** Time is the last coordinate $t(x)$, with unit vector e_t; $y \perp e_t$ denotes a spatial vector. For a different coordinate axis a, replace $t(x)$ by x^a in the following definitions.
  - A word is *separated* if its points are distinct, and *positive* if in addition all $t(x_i) > 0$.
  - It lies in the *slab* $(0, d)$ if $0 < t(x_i) < d$ for all i.
  - It is r-*buffered* in that slab if its points are at mutual sup-distance $\geq r$ and $r \leq t(x_i) \leq d - r$.

- **Operations on words.** $w + a$ is the translate, $\tau_d w = w + de_t$, $w \cdot w'$ is concatenation, and $\bar{w}$ conjugates only the labels.
- **Two reflections.**
  - Θw, the OS reflection: reverse the list, reflect the points in $\{t = 0\}$, and apply to the labels the antilinear map (conjugation followed by the time-mirror action).
  - ϑw, the linear time reflection: reflect the points and apply the time-mirror action to the labels, with neither conjugation nor reversal.
  - For real labels, Θw is a permutation of ϑw.
- **Null spaces.** $N_d(S) \subset V_d$ consists of labels whose correlations with every separated family of spectators vanish. Write $F_d(S) = V_d/N_d(S)$ and $q_d \colon V_d \to F_d(S)$ for the quotient map. A table S' *respects* S if $N_d(S) \subset N_d(S')$ for all d.
- **Field algebra (VI.1).** It consists of label derivatives $\partial_j \colon F_d(S) \to F_{d+1}(S)$, inclusions $F_d(S) \to F_e(S)$ for $d \le e$, an action R of $O(4)$ on these spaces, and a stress tensor $J_{ab} \in F_4(S)$. In a word, $\partial_j v$ and $R(K)v$ mean representatives of $\partial_j q_d(v)$ and $R(K)q_d(v)$. All tables and operators to which this notation is applied factor through the relevant quotients. Covariance means $S(Kx_1, \dots, Kx_n; R(K)v_1, \dots, R(K)v_n) = S(x; v)$ for every $K \in O(4)$ and separated configuration $x \in \mathrm{Conf}_n$.
- **Caps.** For each fixed arity, degree list, label list and derivative order j, there are $A \in \mathbb{N}$ and $C \ge 0$ such that $\|D^j S(x; v)\| \le C r^{-A}$ whenever $0 < r \le 1$ and the points have mutual sup-distance at least r.
- **The OS space.** The OS space $\mathcal{H}$ of S for an axis a is the completion of the span of a-positive words modulo null vectors, with $\langle b, b' \rangle = S(\Theta b \cdot b')$. $|b\rangle$ denotes the vector of b, and $\Omega = |\emptyset\rangle$, with $\|\Omega\| = 1$.

Nowhere does the unit assume clustering, a unique vacuum or a spectral gap.

Proposition V.1.1 (chambers and ground space). Let S satisfy the Euclidean hierarchy and let a be a coordinate axis. Then:

- On the OS space $\mathcal{H}$ there is a strongly continuous semigroup $(E_t)_{t \ge 0}$ of self-adjoint contractions with $E_t|b\rangle = |b + te_a\rangle$.
- There is a strongly continuous unitary group $(U_y)_{y \perp e_a}$ commuting with it, with $U_y|b\rangle = |b+y\rangle$.
- The triple is a *chamber*. Let $\mathcal{H}_0 = \{\psi : E_t\psi = \psi \ \forall t\}$ be its *ground space* and P_0 the orthogonal projection onto it. Then $\Omega \in \mathcal{H}_0$, and $E_t\psi \to P_0\psi$ as $t \to \infty$ for every ψ.

Status: theorem. *Lean:* `OS.realization`, `OS.implement`, `OS.Semigroup.emerges`; definitions `OS.Chamber`, `OS.Semigroup`, `Semigroup.ground`.

Proposition V.1.2 (spectral calculus in a chamber). In a chamber, write formally $E_t = e^{-tH}$.

- (a) There is a strongly continuous unitary group $s \mapsto e^{isH}$, determined by $e^{isH}E_1\psi = E_1^{1-is}\psi$. Here E_1^z, for $\mathrm{Re}\, z > 0$, is the continuous functional calculus of the positive contraction E_1. The map $V(x) = e^{i\,x^a H}U_{x_\perp}$, with $x_\perp = x - x^a e_a$, is a strongly continuous unitary representation of $\mathbb{R}^4$.
- (b) There is a projection-valued measure P on $\mathbb{R}^4$ with $\langle \psi, V(x)\psi \rangle = \int e^{ip\cdot x}\, d\mu_\psi(p)$, where $\mu_\psi(B) = \langle \psi, P(B)\psi \rangle$ for Borel B.
- (c) Every μ_ψ is supported in $\{p^a \ge 0\}$.
- (d) For $\mathrm{Im}\, z \ge 0$:

$$\langle \psi, V(x)\, e^{i(\mathrm{Re}\, z)H} E_{\mathrm{Im}\, z}\, \psi \rangle = \int e^{ip\cdot x}\, e^{ip^a z}\, d\mu_\psi(p).$$

Status: theorem. It holds for every semigroup with the properties of V.1.1. No gap is assumed or proved. *Lean:* `Semigroup.edge`, `Semigroup.unit`, `Spectral.Orbit.total`, `Orbit.P`, `Semigroup.nonneg`, `Semigroup.heat`.

Proposition V.1.3 (slab operators). Assume the hierarchy and the bench (V.1.7). Then every chamber of S for the time axis carries a unique family of bounded operators $T_d(w)$, one for each $d > 0$ and each word w in the slab $(0, d)$, with

$$T_d(w)\,|b\rangle = |w \cdot \tau_d b\rangle \qquad \text{for every positive word } b.$$

- **All-copy bound.** For every $D \in \mathbb{N}$ and $\rho > 0$ there are $C > 0$ and $\kappa \in \mathbb{N}$ such that

$$\|T_d(w)\| \le C^{|w|}(|w|/r)^{\kappa|w|}$$

for every $d > 0$, every $r \in (0, 1]$, and every word w that is r-buffered in $(0, d)$ with degrees $\le D$ and label norms $\le \rho$. The right side is 1 for the empty word.

- **Smoothness.** If moreover S satisfies the derivative identities of the plan ∂, then for fixed d and labels the map $x \mapsto T_d(w(x))$ is C^∞ in operator norm on the open set where $w(x)$ lies in the slab. Its derivatives are $\partial_{x_p^j} T_d(w(x)) = T_d(w'(x))$, where w' replaces the label v_p by $\partial_j v_p$.

Status: theorem, given the bench. More generally, smoothness follows for any slab family with the all-copy bound on a time-axis chamber of a table S' that satisfies the hierarchy, respects S, and satisfies the derivative identities of the plan on S. *Lean:* `Work.B01.answer`, `Work.B02.answer`; definitions `Spread`, `Spread.AllCopy`, `Spread.Laws`.

Proposition V.1.4 (the compressed vacuum algebra). Assume the hierarchy and $O(4)$-covariance of S. Fix a time-axis chamber and any family $T_d(w)$ with the defining identity of V.1.3. Then:

- (a) $U_y\psi = \psi$ for every $\psi \in \mathcal{H}_0$ and every spatial y.
- (b) For a separated word w, the operator $M(w) := P_0\,T_d(w+s)|_{\mathcal{H}_0}$ does not depend on the choice of $s \in \mathbb{R}^4$ and $d > 0$ with $w + s$ in the slab $(0, d)$. For positive b, $M(b)\Omega = P_0|b\rangle$. For every separated w, $\langle\Omega, M(w)\Omega\rangle = S(w)$.
- (c) Cluster limit. Let $q_1, \dots, q_m$ be separated words and $a_1(n), \dots, a_m(n) \in \mathbb{R}^4$ with $t(a_j(n) - a_i(n)) \to +\infty$ for $i < j$; the spatial components are arbitrary. Then for large n the word $(q_1 + a_1(n)) \cdots (q_m + a_m(n))$ is separated, and its value under S tends to $\langle\Omega, M(q_1) \cdots M(q_m)\Omega\rangle$.
- (d) The norm-closed $*$-algebra $\mathcal{A} \subset B(\mathcal{H}_0)$ generated by the $M(w)$ is commutative, $\mathcal{A}\Omega$ is dense in $\mathcal{H}_0$, and $p \in \mathcal{A}$ with $p\Omega = 0$ implies $p = 0$.
- (e) Suppose in addition that the family satisfies the smoothness part of V.1.3. Then:
  - M is invariant under permutations of the word and under all translations; $M(\emptyset) = 1$; a unit label may be dropped together with its point; and $M(w)^* = M(\Theta w)$.
  - For $x \in \mathrm{Conf}_n$, the map $v \mapsto M(x; v)$ is multilinear and factors through $F_{d_1}(S) \times \cdots \times F_{d_n}(S)$.
  - $x \mapsto M(x; v)$ is C^∞ in norm on Conf_n.
  - $\mathcal{A}$ is norm-separable.

Status: theorem. *Lean:* `Ground.inert`, `Ground.overline`, `Ground.almost`, `Spread.insertValue`, `Ground.arcs`, `Ground.adjunction`, `Ground.full`, `Ground.attains`, `Work.B03.answer`, `Work.B12.answer`; definitions `Ground.M`, `Spread.algebra`, `Bid.Theme`.

Proposition V.1.5 (decomposition into phases). Assume the setting of V.1.4(a)–(e). Then:

- There are a compact metrizable space X (the Gelfand spectrum of $\mathcal{A}$), a Borel probability measure ν on X with $\nu(O) > 0$ for every nonempty open O, and characters χ_ξ ($\xi \in X$).
- $\xi \mapsto \chi_\xi$ is a bijection onto the characters of $\mathcal{A}$; each $\xi \mapsto \chi_\xi(p)$ is continuous; and $\int_X \chi_\xi(p)\, d\nu(\xi) = \langle \Omega, p\Omega \rangle$ for every $p \in \mathcal{A}$.
- For each ξ, the formula $S^\xi(x; v) = \chi_\xi(M(x; v))$ on Conf_n (and 0 off it) defines a table.
- $S^\xi(w) = \chi_\xi(M(w))$ for every separated w; $\xi \mapsto S^\xi(w)$ is continuous; and the barycenter identity holds:

$$\int_X S^\xi(w)\, d\nu(\xi) = S(w) \qquad \text{for every separated word } w.$$

Status: theorem. *Lean:* `Ground.resolution`, `Work.B07.state`, `Resolution.barycenter`; definitions `Spread.Resolution`, `Resolution.Out`, `Resolution.receives`.

Proposition V.1.6 (every phase satisfies the hierarchy and the field laws). In addition to V.1.3–V.1.5 (with the smoothness laws for the plan of the field algebra), assume:

- the Seed of S (IV.1);
- the grid hypothesis (§2);
- the laws of the field algebra for S (VI.1).

Then two operator identities hold before any character is applied:

- (e) *phase reflection*: $M(w) = M(\vartheta w)$ for every separated word with arbitrary complex labels;
- (f) $M(w) = M(Kw)$ for every separated w and every $K \in O(4)$, where K acts on points and, through $R(K)$, on labels.

As a consequence, for **every** $\xi \in X$:

- (a) S^ξ satisfies the Euclidean hierarchy:
  - permutation symmetry, $S^\xi(\emptyset) = 1$, and unit labels;
  - invariance under signed permutations of the axes, and reality $S^\xi(x; \bar{v}) = \overline{S^\xi(x; v)}$;
  - translation invariance;
  - reflection positivity for every coordinate axis, in point and smeared form;
  - smoothness on Conf_n, and caps on all derivatives.
- (b) S^ξ respects S and is $O(4)$-covariant for the same action R. It satisfies the inclusion and derivative identities of the plan and the stress-tensor Ward identities of VI.1.
- (c) Uniform cap: for every $D \in \mathbb{N}$ and $\rho > 0$ there are $K > 0$ and $\kappa \in \mathbb{N}$ with

$$|S^\xi(w)| \le K^{|w|} \big(|w|/r\big)^{\kappa|w|}$$

  for all ξ, all $r \in (0, 1]$, and all words with degrees $\le D$, label norms $\le \rho$ and points at mutual sup-distance $\ge r$.
- (d) $N_d(S^\xi) = N_d(S)$ for every d, hence $F_d(S^\xi) = F_d(S)$.

Status: theorem. *Lean:* `Work.B05.answer`, `B06.answer`, `B07.answer`, `B08.answer`, `B09.answer`, `B10.answer`, `B11.answer`, `Retain.lift_detect`; definitions `Phase.reflects`, `Bid.Garland`, `Phase.Scripts`.

Proposition V.1.7 (bench). For every $D \in \mathbb{N}$ and $\rho > 0$ there are $C_0 > 0$ and $\kappa \in \mathbb{N}$ with the following property. Let $d > 0$, $r \in (0, 1]$, and let $w \neq \emptyset$ be r-buffered in $(0, d)$, with degrees $\leq D$ and label norms $\leq \rho$. Then there are a lattice level b and a width $s \leq d$ with $s \in a_b\mathbb{N}$ and w in the slab $(0, s)$, such that every positive word v has a constant B_v with

$$\big|S(\Theta v_n \cdot v_n)\big| \leq B_v\, A^{4n} \ \ (n \geq 0), \qquad A = C_0^{|w|}(|w|/r)^{\kappa|w|}, \qquad v_0 = v, \ \ v_{n+1} = \tau_s(\Theta w)\cdot\tau_s(w\cdot\tau_s v_n).$$

Here v_n is v preceded by $2n$ consecutive slabs of width s, holding alternately w and its reflection. Thus $|v_n\rangle = (T_s(w)^*T_s(w))^n|v\rangle$ once $T_s(w)$ exists, and $S(\Theta v_n \cdot v_n)$ is a reflection-symmetric train of $4n$ copies of w.

Status: a hypothesis of this unit, required of every limit table with a Seed. It is proved in VII.4 (`Next.Phase.seats`) from the cutoff family, the hierarchy, the point limits and the cell hypothesis of the Seed.

- **Chessboard estimate.** The cell hypothesis bounds every signed or reflected field of degree $\leq D$ by $J a_k^{-(D+1)}$ in chessboard norm. The scale can be chosen with $a_k \geq c\,(r/|w|)^4$, where $c > 0$ depends only on the fixed family and (D, ρ). A finer scale handles the exterior word. The resulting train bound is $B^{2|v|}(Ja_k^{-(D+1)})^{4n|w|}$ (`Next.Phase.travel_price`). Thus one may take $\kappa = 4(D + 1)$; the exterior constant is independent of n.

Lean: `Bid.Bench`, `Bid.Seats`, `OS.QRead`, `Next.Phase.replicas`, `Next.Phase.rates`.

Proposition V.1.8 (vacuum uniqueness in almost every phase). Let (X, ν, χ) and S^ξ be as in V.1.5, with every S^ξ satisfying the hierarchy. For each ξ fix a time-axis chamber of S^ξ, with vacuum Ω_ξ and ground space $\mathcal{H}_0^\xi$. Then:

- $\mathcal{H}_0^\xi = \mathbb{C}\,\Omega_\xi$ for ν-almost every ξ; in particular such phases exist.
- Every time-axis chamber of every S^ξ carries a slab family $T_d^\xi(w)$ with $\|T_d^\xi(w)\| \leq \|T_d(w)\|$ for each valid slab word. If the original family has the all-copy bound, so does this family. Under V.1.6 it also has the smoothness laws.

Status: theorem. Uniqueness for every phase is proved later, in V.3 (`EveryPhase.every`), by an argument outside this unit. *Lean:* `Cluster.exp_real`, `Cluster.lifts`, `Cluster.selects`, `Work.B13.answer`, `Work.B14.answer`; definition `Chamber.UniqueHeat`.

Proposition V.1.9 (selected-phase certificate). Let C be a Wilson family with extraction order f and limit table S. Assume:

- the Seed of S;
- the grid hypothesis;
- a field algebra on S with its laws.

Then there are:

- a time-axis chamber of S whose ground space is fixed pointwise by spatial translations;
- a slab family on it with the all-copy bound and the smoothness laws;
- a decomposition $(X, \nu, \chi_\xi, S^\xi)$ as in V.1.5, with every S^ξ satisfying V.1.6(a), (b), (d);
- a phase ξ_*, with a time-axis chamber of S^{ξ_*} whose ground space is $\mathbb{C}\Omega_{\xi_*}$, and a slab family on that chamber with the smoothness laws.

Status: theorem.

- **Assembly.** The bench theorem and B01–B18 give the slab and phase constructions (`Bid.Looms`); the selection argument produces the certificate (`Phase.Loom.cleanse`).
- **Use in the main theorem.** The main theorem applies it in `Pert.withFinal` and `MergeProof.Acme.glistens`. The result is the `phase` field of `Spec.Final`.

Lean: definitions `Phase.Loom`, `Phase.Stall`, `Phase.Vendor`, `OS.Selection`, `Selection.Certificate`, `Spec.Materializes`.

3.21.2 2. What it uses

- **IV.1 (limit table and Seed).**
  - The limit table S with the Euclidean hierarchy: Lean `Table.Hierarchy`, with fields basic laws, real-spin, translations, point and smeared reflection positivity, smoothness, caps.
  - The Seed, Lean `Spec.Seed`. Its components:
    - (i) the cutoff family;
    - (ii) the ordered limit;
    - (iii) point limits at generic points (`PointLimits`);
    - (iv) the hierarchy;
    - (v) the *cell hypothesis* (`CellHyp`): reflection-positivity seams at every cutoff; and, for every degree d and $\varepsilon, \rho > 0$, a threshold m_0 and $A \geq 0$ such that for every $m \geq m_0$ and $\eta > 0$, late in the ordered cutoff, every degree-d label of norm at most ρ is supported inside its centered cell and has chessboard root at most $A\, a_m^{-d-\varepsilon} + \eta$, uniformly over cell centers;
    - (vi) the *corrected-field data* (`RankWidths`, `Columns`, `Trials.Valid`). For each degree d: finitely many one- or two-point test rows and a basis of the free-field space of degree d, for which the free test map is injective; and lattice "corrected" fields at levels k whose classes span $F_d(S)$ for all sufficiently large k, whose one-field differences from free values and centered two-field products minus free cumulants have chessboard bounds $\mathrm{gap}_k + \eta$ late in the cutoff for every $\eta > 0$, with $\mathrm{gap}_k \to 0$. The free values use the rescaled test points.
  - The free table (`Cells.standard`).
  - Where these are used: (i)–(iv) in B04, B16 and the bench; (v) in the bench, B15 and the grid hypothesis; (vi) in B18 and `lift_detect`.
- **VII.4 (the bench).** V.1.7, i.e. `Next.Phase.seats`.
- **III.11 (the grid hypothesis)** `Org.Grill`. Proved from bounded ultraviolet approximants and the cell hypothesis by `MergeProof.Accent.grilled`.
  - Setup: take a degree bound d and a level m. There is a level $k \geq m$ such that the following holds for every word w with real labels of degree $\leq d$ at generic points of the level-k grid.
  - Let F be the lattice product of the fields of w, and ϑF the product of the mirrored fields at the reflected points.
  - The lattice variables f_j are the translates of $F - \vartheta F$ by $t\, j$ grid steps in every coordinate.
  - Conclusion: for each approximation index $\varkappa$ the f_j admit real measurable approximants, bounded by $R_\varkappa$ uniformly in the cutoff, whose pair integrals, for $i < j$, differ from $\mathbb{E}[\mathrm{Re}(f_i f_j)]$ by at most $\delta_\varkappa \to 0$. For a fixed grid word, $R_\varkappa$ and $\delta_\varkappa$ are independent of the translation, copy indices and cutoffs. The pair estimates hold for all sufficiently large t, then late in the cutoff (`Grid.Covering`). It is used only in B05.

- **VI.1 (field algebra and its laws on S)** (`Spec.Algebra`, `Spec.Locals`).
  - $O(4)$-covariance of S (`Rotation.Oriented`) is used in V.1.4(a), (d), in `Ground.resolution`, and in B06, B07.
  - The rotation laws (`Rotation.Laws`): group law, and $R(s)$ equal to the mirror action of S for signed permutations s. Used in B06, B07.
  - The derivative identities are used in B02 and B09; the inclusion identities and stress-tensor laws (`StressLaws`) in B09.
  - **Order of dependence.** VI.1 uses the chamber construction and the bounded heat functional calculus from the beginning of this unit, as well as elementary geometry and norm estimates. It does not use the compressed algebra, phase reflection or phase selection. Those constructions then use the $O(4)$ action supplied by VI.1.
- **V.2.** The Gram/completion construction of the OS space (`Kernel.Gram`, with `by_bound`, `by_bud`).
- **I.1, II.1.** Lattice, generic points, label registers, the Wilson family.

Standard mathematics.

- **Osterwalder–Schrader reconstruction** and the doubling bound for the semigroup: Osterwalder–Schrader, CMP 31 (1973), 42 (1975); Glimm–Jaffe, *Quantum Physics*, Ch. 19 (probable).
- **Chessboard estimates:** Fröhlich–Israel–Lieb–Simon, CMP 62 (1978).
- **Pure phases.** The classical analogue is the decomposition of a Nelson–Symanzik positive theory into pure phases, Fröhlich, Ann. Phys. 97 (1976) (probable), and the decomposition of states along abelian algebras, Bratteli–Robinson I, Ch. 4.
- **Operator algebras and measure theory:**
  - Gelfand–Naimark for commutative C*-algebras and spectral permanence (Bratteli–Robinson I, §2.1–2.2);
  - Riesz–Markov (Reed–Simon I, Thm. IV.14);
  - continuous functional calculus;
  - Hahn–Banach;
  - dominated convergence.
- **Harmonic analysis:** the Stone–Naimark–Ambrose–Godement theorem (Folland, *A Course in Abstract Harmonic Analysis*, §4.4) and a Paley–Wiener support theorem for bounded holomorphic functions in a half-plane (Reed–Simon II, §IX.3) (probable).
- **Geometry:** Cartan–Dieudonné (reflections generate $O(4)$).

Imported constants and assumptions.

- No numerical constant is fixed in the unit.
- The all-copy bound uses the constants of the bench; one may choose $\kappa = 4(D+1)$. These constants may depend on the fixed family and (D, ρ), but not on the slab width, buffer, word length or number of copies.
- The hypotheses are: the hierarchy and the Seed (IV.1), the grid hypothesis (III.11), the bench (VII.4), and the field-algebra laws (VI.1).

3.21.3 3. Strategy

The limit table as a mixture. The table S is obtained by ordered limits of finite-volume Wilson correlations. Its OS semigroup may have a multidimensional ground space $\mathcal{H}_0$. The unit

decomposes the correlations through an algebra acting on that space.

- **The algebra.** Inserting a separated word into a time slab gives a bounded operator; compressing it to $\mathcal{H}_0$ gives $M(w)$. These operators generate a commutative C*-algebra $\mathcal{A}$, for which Ω is cyclic and separating.
- **Phases.** The Gelfand spectrum X of $\mathcal{A}$, with the measure ν representing $p \mapsto \langle \Omega, p\Omega \rangle$, is the space of phases. Each character gives a table, $S^\xi = \chi_\xi \circ M$, and $S = \int S^\xi \, d\nu$.
- **Comparison with the classical case.** The algebra describes observables at infinite time separation. It replaces the invariant observables used in the ergodic decomposition of a Euclidean measure. Here the construction uses only the OS space of the limit correlations. Operator identities give the hierarchy and field laws for every ξ; vacuum uniqueness is first obtained almost everywhere.

Four inputs. Four inputs make this work, and each is used where the classical argument would use the measure.

- **(i) Boundedness.** A chessboard estimate controls alternating trains of $4n$ copies with constants independent of n. A power argument converts this estimate into the slab-operator bound $C^{|w|}(|w|/r)^{\kappa|w|}$.
- **(ii) $O(4)$-covariance** (VI.1). It shows that ground states are invariant under spatial translations, which makes $M(w)$ well defined. Spatial displacements are rotated into time displacements, which are asymptotically invisible. It also makes $\mathcal{A}$ commutative: exchanging the time axis with a spatial axis reverses the time order of two clusters inside a four-cluster limit.
- **(iii) Phase reflection** $M(w) = M(\vartheta w)$. Together with $M(w)^* = M(\Theta w)$, this makes every phase value on real labels real. Put $D = M(w) - M(\vartheta w)$ for such a word. Then $D^* = -D$, so $\langle \Omega, D^2 \Omega \rangle = -\|D\Omega\|^2$. Bounded real lattice approximants give the reverse inequality by positivity of the variance of many far-separated copies. Thus $D = 0$. Conjugating time reflection gives invariance of M under $O(4)$.
- **(iv) Chessboard detectors.** For the corrected fields they show that the one- and two-point functions of every phase, on a spanning family of fields, are close to free-field values for which the test map is injective. Hence every phase has the null spaces of S.

Selecting a phase. For a positive word b, the quantity $\|P_0^\xi |b\rangle\|^2 - |S^\xi(b)|^2 \geq 0$ has ν-integral zero. This follows by dominated convergence along E_s and the barycenter identity. One countable family of words is total in every phase chamber, so a single null set suffices, and outside it every phase has a one-dimensional ground space. The spectral calculus of V.1.2 (SNAG, positivity of the energy from analyticity of the heat semigroup) is built without assuming a gap and is used downstream (V.2, V.3).

3.21.4 4. Steps

3.21.4.1 4.1 OS space, chambers, spectral calculus Lemma V.1.a (chamber). Under the hierarchy, V.1.1 holds except for the convergence $E_t \to P_0$. *Proof sketch.*

- **Contraction (OS doubling trick).** For p in the span of positive words, $\|E_t p\|^2 = \langle p, E_{2t} p \rangle \leq \|p\| \sup_s \|E_s p\|$. The supremum c is finite, because $s \mapsto S(\Theta b \cdot \tau_s b')$ is bounded for $s \geq 0$: the points stay at a fixed positive distance, and the zeroth cap applies. Hence $c^2 \leq \|p\| c$, so $\|E_t p\| \leq \|p\|$.

- **Self-adjointness and unitarity.** Self-adjointness of E_t and unitarity of U_y follow from translation invariance.
- **Strong continuity.** It follows from continuity of the Gram functions (smoothness) together with the uniform bounds.

Lean: `OS.contract`, `OS.implement`.

Lemma V.1.b (ground projection). For a semigroup of self-adjoint contractions, $E_t\psi \to P_0\psi$ for every ψ. *Proof sketch.*

- For $u \le v$, $\langle E_u\psi, E_v\psi\rangle = \|E_{(u+v)/2}\psi\|^2 \ge \|E_v\psi\|^2$. Hence $\|E_u\psi - E_v\psi\|^2 \le \|E_u\psi\|^2 - \|E_v\psi\|^2$, and the decreasing limit of $\|E_t\psi\|^2$ makes $E_t\psi$ Cauchy.
- The limit is fixed by every E_s, and ψ minus the limit is orthogonal to $\mathcal{H}_0$, because $\langle E_t\psi, u\rangle = \langle\psi, u\rangle$ for $u \in \mathcal{H}_0$.
- No spectral theorem is used.

Lean: `Semigroup.emerges`.

Lemma V.1.c (real-time group and SNAG). V.1.2(a), (b) hold. *Proof sketch.*

- **Real-time group.** $E_1 = E_{1/2}^2$ is positive, with spectrum in $[0,1]$. The functional calculus gives E_1^z, holomorphic for $\operatorname{Re} z > 0$, and $E_1^t = E_t$ for $t > 0$ by positive roots and continuity. If $E_t\psi = 0$, analyticity makes $E_s\psi = 0$ for every $s > 0$, and continuity at zero gives $\psi = 0$. Self-adjointness then makes the range of E_1 dense. Since $\|E_1^{1-is}\psi\| = \|E_1\psi\|$, the map $E_1\psi \mapsto E_1^{1-is}\psi$ extends to an isometry. The power laws give the group law and inverse at $-s$; continuity on the dense range gives strong continuity.
- **SNAG.** Integrate V against Fourier transforms of Schwartz functions to obtain a $*$-representation of continuous functions on the one-point compactification of $\mathbb{R}^4$. Riesz–Markov gives a projection-valued measure there. The point at infinity carries no mass: evaluate at $x = 0$ along an approximate identity.

Lean: `Semigroup.roots_heat`, `Semigroup.pillar`, `Semigroup.edge`, `Semigroup.unit`, `Spectral.Orbit.total`, `Spectral.PVM`.

Lemma V.1.d (spectrum condition and Laplace formula). V.1.2(c), (d) hold. *Proof sketch.*

- **The function J.** Fix ψ and x. The function $J(z) = \langle\psi, V(x)e^{i(\operatorname{Re} z)H}E_{\operatorname{Im} z}\psi\rangle$ is continuous on $\operatorname{Im} z \ge 0$. In the interior it equals $\langle\psi, V(x)E_1^{-iz}\psi\rangle$, so it is holomorphic there. It is bounded by $\|\psi\|^2$. Its boundary values are the Fourier transform of $e^{ip\cdot x}d\mu_\psi(p)$ in the variable p^a.
- **Support theorem.** Let μ be a finite positive Borel measure on $\mathbb{R}^4$, let $\ell\colon \mathbb{R}^4 \to \mathbb{R}$ be continuous, and let F be bounded and holomorphic in the upper half-plane, continuous up to the boundary, with $F(b) = \int e^{i\ell(y)b}d\mu(y)$. Then $\operatorname{supp}\mu \subset \{\ell \ge 0\}$. To see this, integrate F against the Gaussian $e^{-ibz-\varepsilon z^2}$ on horizontal lines and shift the contour upward. For $b < 0$, choosing height $-b/(2\varepsilon)$ bounds the mass where $|\ell(y) - b| < |b|/2$ by $Ce^{-3b^2/(16\varepsilon)}$. Letting $\varepsilon \downarrow 0$ excludes negative ℓ from the support. Apply this with $x = 0$ in J and $\ell(p) = p^a$.
- **Laplace formula.** Both sides of (d) are bounded and holomorphic with the same boundary values, so they agree.

Lean: `Semigroup.altitude`, `Semigroup.nonneg`, `Semigroup.heat`, `Paley.supports`, `Paley.heat`, `Paley.Air`.

3.21.4.2 4.2 Slab operators Lemma V.1.e (power argument).

- **Setting.** Let $\Delta > 0$, let A, B be linear maps on a complex vector space, and let Y be a linear map into a Hilbert space, with $\langle YAv, Yw\rangle = \langle Yv, YBw\rangle$ for all v, w.
- **Statement.** If every v has a constant K_v with $\|Y(BA)^n v\| \le K_v \Delta^{2n}$ for all n, then $\|YAv\| \le \Delta \|Yv\|$.

Proof sketch.

- Put $\delta_n = \|Y(BA)^n v\|/\Delta^{2n}$. Then $\delta_n^2 \le \|Yv\|\,\delta_{2n}$, so $\sup_n \delta_n \le \|Yv\|$.
- Hence $\|YAv\|^2 = \langle Yv, YBAv\rangle \le \Delta^2 \|Yv\|^2$.
- Apply this with $A =$ insertion of w, $B =$ insertion of $\tau_s \Theta w$, and $Y =$ the OS map. The bench gives $K_v = \sqrt{B_v}$ for single words; for finite sums use the triangle inequality. Extend by continuity.

Lean: `Drive.against`, `OS.size`, `OS.producer`.

Lemma V.1.f (existence and all-copy bound; B01). The first part of V.1.3 holds. *Proof sketch.*

- Let w be r-buffered in $(0, d)$. The bench gives $s \le d$ with the train bound at width s. V.1.e then gives a bounded operator T' with $\|T'\| \le A$ and $T'|b\rangle = |w \cdot \tau_s b\rangle$.
- Put $T_d(w) = T' E_{d-s}$, so $\|T_d(w)\| \le A$; for the empty word, $T_d(\emptyset) = E_d$.
- Every word in a slab is buffered for some (D, ρ, r). Operators agreeing on all positive-word vectors coincide, so the family is unique on valid slab words.

Lean: `Work.B01.answer`.

Lemma V.1.g (smoothness laws; B02).

- **Hypotheses.** S' is a table respecting S, both satisfying the hierarchy, and S' satisfies the derivative identities of the plan of S. The slab family on a chamber of S' has the all-copy bound.
- **Conclusion.** The family is C^∞ in operator norm, with derivatives given by differentiated labels.

Proof sketch.

- **Local bound on the candidate derivative.** The candidate derivative is the operator with the label differentiated. It is locally bounded: an r-buffered configuration is $r/2$-buffered on a neighbourhood, so the all-copy bound applies there.
- **Matrix elements.** Between positive-word vectors, the derivative identity of S' gives the candidate's matrix elements as derivatives.
- **From matrix elements to norm.** A mean-value estimate with constants uniform in the observing vectors gives first Lipschitz continuity in norm, then norm continuity of the candidate (apply the same to the next derivative), then Fréchet differentiability. Induction on the order gives C^∞.

Lean: `Work.B02.answer`, `B02.norm_remainder`, `B02/Slab`.

3.21.4.3 4.3 The compressed algebra Lemma V.1.h (ground states are spatially invariant). Under the hierarchy and $O(4)$-covariance, $U_y \psi = \psi$ for all $\psi \in \mathcal{H}_0$. *Proof sketch.* Let p, q be positive words, y spatial, and $u = y + e_t$.

- **Pure time shifts are invisible.** For any $d(s)$ with $t(d(s)) \to \infty$ and any fixed λ, $S(\Theta p \cdot (q + d(s) + \lambda e_t)) - S(\Theta p \cdot (q + d(s))) \to 0$. Both terms are matrix elements $\langle p', U_{z(s)} E_{\tau(s)+\lambda} q' \rangle$ and $\langle p', U_{z(s)} E_{\tau(s)} q' \rangle$, with $\tau(s) \to \infty$ and arbitrary spatial $z(s)$, and $E_\tau q'$ converges (V.1.b).
- **Rotate the spatial displacement into time.** Choose a hyperplane reflection K with $Ku = |u| e_t$ and $t(Ke_t) > 0$. Covariance turns $S(\Theta p \cdot (q + u + s e_t)) - S(\Theta p \cdot (q + s e_t))$ into a pure time shift by $|u|$ at displacement sKe_t, so it tends to 0.
- **Conclusion.** Combining the two, $\langle p, U_y E_s q \rangle - \langle p, E_s q \rangle \to 0$. Hence $U_y P_0 |q\rangle = P_0 |q\rangle$, and this extends to all of $\mathcal{H}_0$ by density.

Lean: `Ground.inert`, `Ground.rays`, `Ground.across_axis`, `Ground.overtops`.

Lemma V.1.i (the compression and its identities; B03). V.1.4(b) and (e), except separability, hold. *Proof sketch.*

- **Independence of the shift.** For $u \geq 0$ the defining identity gives $T_{d+u}(w) = T_d(w) E_u$ and $T_{u+d}(\tau_u w) = E_u T_d(w)$; spatial translations give $U_y T_d(w) = T_d(w + y) U_y$. Together with $P_0 E_u = P_0$ and V.1.h, these identities make $M(w)$ independent of the shift and width.
- **Values.** $\langle \Omega, T_d(w) \Omega \rangle = S(w)$ because $\Omega = |\emptyset\rangle$; translation invariance removes the shift.
- **Identities.** Pairing against positive-word vectors gives $\langle p, T_d(w) q \rangle = S(\Theta p \cdot w \cdot \tau_d q)$ with separated exterior words. Permutation symmetry, unit labels, multilinearity and dependence on $F_d(S)$ therefore hold before compression.
- **Adjoint and smoothness.** $T_d(w)^* = T_d(\tau_d \Theta w)$ by translation invariance. Smoothness: near a configuration one shift and width serve (the slab condition is open), compression is bounded linear, and V.1.g applies.

Lean: `Ground.at_any`, `Ground.overline`, `Ground.pins`, `Ground.slim`, `Work.B03.answer`, `Bid.depot`.

Lemma V.1.j (cluster limits). V.1.4(c) holds. *Proof sketch.*

- **Reduction to positive clusters.** Shift each cluster by a fixed vector to make it positive; this does not change M.
- **Induction on m.** Shift the union so that q_1 sits in its slab. Its vector is then $T_d(q_1) U_{y_n} E_{u_n} |\mathrm{rest}_n\rangle$ with $u_n \to \infty$.
- **Key convergence.** If $\psi_n \to \psi$ and $u_n \to \infty$, then $U_{y_n} E_{u_n} \psi_n \to P_0 \psi$ for arbitrary spatial y_n, by V.1.b and V.1.h. Inductively the vector converges to $T_d(q_1) M(q_2) \cdots M(q_m) \Omega$, and pairing with Ω inserts P_0.

Lean: `Ground.arcs`, `Ground.future_train`, `Ground.receives`.

Lemma V.1.k (commutativity, cyclicity, separation). V.1.4(d) holds. *Proof sketch.* Write $\langle q_1, \ldots, q_m \rangle = \langle \Omega, M(q_1) \cdots M(q_m) \Omega \rangle$.

- **Simultaneous rotation.** If $K \in O(4)$ maps some u with $t(u) > 0$ to a vector with positive time, then $\langle q_1, \ldots, q_m \rangle = \langle Kq_1, \ldots, Kq_m \rangle$. Place cluster j at $n\, j\, u$, apply V.1.j to both orders, and use covariance of S at finite n.
- **Exchange of two clusters.** Let k exchange the time axis with the first spatial axis. Place four clusters at 0, $n(e_t + 2e_1)$, $n(2e_t + e_1)$ and $n(3e_t + 3e_1)$, and apply k: the two middle clusters exchange their time order. With the simultaneous-rotation step for $u = e_t + e_1$, this gives $\langle x, y, w, z \rangle = \langle x, w, y, z \rangle$.
- **Commutativity of generators.** The vectors $P_0 |f\rangle = M(f) \Omega$ for positive f have dense

linear span in $\mathcal{H}_0$, and $M(f)^* = M(\Theta f)$. Thus $\langle M(f)\Omega, M(u)M(v)M(p)\Omega\rangle = \langle \Theta f, u, v, p\rangle$ is symmetric in u, v. Testing on both dense spans proves $M(u)M(v) = M(v)M(u)$.

- **Extension to $\mathcal{A}$.** The generators are closed under adjoints, so $\mathcal{A}$ is commutative.
- **Cyclicity and separation.** $\mathcal{A}\Omega \supset P_0\{|b\rangle\}$ is dense. If $p\Omega = 0$, then $p\,a\Omega = a\,p\Omega = 0$ for all $a \in \mathcal{A}$, so $p = 0$.

Lean: `Ground.same_up`, `Ground.exchanges`, `Ground.gens`, `Ground.adjunction`, `Ground.full`, `Ground.attains`.

Lemma V.1.l (separability; Gelfand–Riesz; B12). $\mathcal{A}$ is norm-separable, and the first part of V.1.5 holds. *Proof sketch.*

- **Separability.** By multilinearity each generator is a finite combination of $M(x; e)$ with basis labels e. For fixed arity, degrees and basis indices, $x \mapsto M(x; e)$ is continuous on the separable set Conf_n. A countable union of separable images, finite products, adjoints and closure stay separable.
- **Gelfand–Riesz.** Gelfand gives $\mathcal{A} \cong C(X)$, and a countable dense sequence makes X metrizable. The vacuum state is represented by a probability measure ν.
- **Full support.** If $\nu(O) = 0$ for some open $O \ni \xi_0$, a continuous bump supported in O corresponds to an element $p \neq 0$ with $\|p\Omega\|^2 = \int |\widehat{p}|^2 d\nu = 0$, contradicting separation.

Lean: `Work.B12.answer`, `Ground.resolution`, `Ground.tiny`, `Ground.supports`.

3.21.4.4 4.4 Phase reflection and rotation invariance Lemma V.1.m (grid passage; B04).

- **Setting.** Assume the cutoff laws and generic point limits of the Seed. Let w have real labels at generic points of the level-k grid, and let w_j be its translate by $t\,j$ grid steps in every coordinate.
- **Statement.** For $i < j$ and all large t, the lattice pair expectation $\mathbb{E}_{N,p}[\mathrm{Re}(f_i f_j)]$ of §2 converges, along the ordered limit, to

$$e_{ij}(t) = \mathrm{Re}\big[S(w_i \cdot w_j) - S(w_i \cdot (\vartheta w)_j) - S((\vartheta w)_i \cdot w_j) + S((\vartheta w)_i \cdot (\vartheta w)_j)\big].$$

Proof sketch. Expand the product into four point moments of the Wilson family. For large t the union of two translated grid words is generic, and the point limits of the Seed apply term by term. The reflection acts before the shift, so the shift stays outside it. *Lean:* `Work.B04.answer`, `Grid.Expects`, `Grid.event`.

Lemma V.1.n (phase reflection; B05).

- **Hypotheses.** The Seed, the grid hypothesis, $O(4)$-covariance, V.1.h and V.1.i.
- **Statement.** $M(w) = M(\vartheta w)$ for every separated word with arbitrary labels.

Proof sketch.

- **(1) Real labels at generic grid points.** Here Θw is a permutation of ϑw, so $M(w)^* = M(\vartheta w)$. Hence $D := M(w) - M(\vartheta w)$ satisfies $D^* = -D$ and $a := \langle \Omega, D^2\Omega\rangle = -\|D\Omega\|^2 \leq 0$.
- **(2) The variance bound.** Fix q and $\varkappa$. For all large t, the approximants $g_0, \dots, g_q$ satisfy, late in the cutoff,

$$0 \leq \int \Big(\sum_{j=0}^{q} g_j\Big)^2 \leq (q+1)R_\varkappa^2 + 2 \sum_{0 \leq i < j \leq q} (\mathbb{E}_{N,p}[\mathrm{Re}(f_i f_j)] + \delta_\varkappa).$$

First take the ordered cutoff limit using V.1.m. Then let $t \to \infty$; V.1.j gives $e_{ij}(t) \to a$, so

$$0 \le (q+1)(R_\varkappa^2 + q(a + \delta_\varkappa)).$$

Let $q \to \infty$, then $\varkappa \to \infty$. This gives $a \ge 0$, hence $D\Omega = 0$ and $D = 0$ by V.1.k.

- **(3) Extension to all configurations.** Approximate a generic configuration by grid configurations of finer levels; the grid hypothesis provides a cofinal set of levels. Both sides are continuous (V.1.i), so equality holds on the co-null set of generic configurations, and then on Conf_n.
- **(4) Complex labels.** Extend by multilinearity, since real basis vectors span.

Lean: `Work.B05.answer`, `B05.real_depot`, `Grid.horizon`, `Grid.estimates`, `Grid.Covering.square` in `Phase/Square.lean`, `Grid.anti`.

Lemma V.1.o (invariance under $O(4)$; B06). Given phase reflection and the rotation laws, $M(w) = M(Kw)$ for all $K \in O(4)$. *Proof sketch.*

- **Products.** The set of K with $M(\cdot) = M(K\cdot)$ is closed under products. It contains the time reflection, because by the rotation law $R(\vartheta)$ is the mirror action on $F_d(S)$ and V.1.n applies.
- **Two-point functions.** $\langle p, x\rangle = \langle Kp, Kx\rangle$ for every K. This is V.1.k's simultaneous-rotation step. If $t(Ke_t) > 0$ use $u = e_t$; if $t(Ke_t) = 0$ use $u = e_t + K^{-1}e_t$; if $t(Ke_t) < 0$ compose with the time reflection.
- **Conjugates of the time reflection.** Rotating a pair back by J^{-1}, reflecting the second factor and rotating forward gives $\langle p, x\rangle = \langle p, J\vartheta J^{-1}x\rangle$ for all p. Since $M(\Theta p) = M(p)^*$, this gives $M(x)\Omega = M(J\vartheta J^{-1}x)\Omega$, hence equality by separation.
- **Generation.** Every hyperplane reflection is such a conjugate, and reflections generate $O(4)$ (Cartan–Dieudonné). No separate treatment of time–space rotations is needed.

Lean: `Work.B06.answer`, auxiliary lemmas in `Work/B06.lean`.

3.21.4.5 4.5 The phase tables Lemma V.1.p (phase tables, symmetries, barycenter; B07).

- **Definition.** $S^\xi(x; v) = \chi_\xi(M(x; v))$ is a table.
- **Properties.**
  - It is basic: permutation symmetry, $S^\xi(\emptyset) = 1$, unit labels.
  - It is translation invariant, respects S (labels in $N_d(S)$ give $M = 0$), and is $O(4)$-covariant (V.1.o).
  - It is real-spin. Signed permutations act through R by the rotation law. Reality: $S^\xi(x; \bar v) = \chi_\xi(M(\bar w)) = \chi_\xi(M(\vartheta\bar w)) = \chi_\xi(M(\Theta w)) = \overline{\chi_\xi(M(w))}$, using V.1.n and a permutation.
  - It is smooth: χ_ξ is extended by Hahn–Banach to a bounded functional on $B(\mathcal{H}_0)$ and composed with the smooth map $x \mapsto M(x; v)$.
- **Barycenter.** $\int S^\xi(w)\, d\nu = \langle \Omega, M(w)\Omega\rangle = S(w)$, by V.1.l and V.1.i.

Lean: `Work.B07.answer`, `B07.state_spin`, `B07.state_smooth`, `Resolution.barycenter`.

Lemma V.1.q (reflection positivity in every phase; B08). Every S^ξ is reflection positive for every axis, in point and smeared form. *Proof sketch.*

- **Time axis.** Given finitely many positive words w_i, choose d so they all lie in $(0, d)$. By V.1.i, $M(\Theta w_i \cdot w_j) = Y_i^* Y_j$ with $Y_i = T_d(w_i)|_{\mathcal{H}_0} : \mathcal{H}_0 \to \mathcal{H}$. Therefore $\sum \bar{c}_i c_j M(\Theta w_i \cdot w_j) = Y^* Y \geq 0$, where $Y = \sum c_i Y_i$. Spectral permanence makes its character value nonnegative.
- **Other axes.** Real-spin exchanges the axis with time.
- **Smeared form.** Realize the positive point kernel in a Hilbert space. A compactly supported test gives the integral of its word vectors against the test function. Smoothness makes these vector-valued functions continuous and integrable; the smeared quadratic form is the squared norm of a finite sum of these integrals.

Lean: `Work.B08.answer`, `B08.char_nonneg`, `B08.smear`.

Lemma V.1.r (derivative, inclusion and stress laws in every phase; B09). Every S^ξ satisfies the inclusion and derivative identities of the plan and the stress-tensor Ward identities. *Proof sketch.*

- **The evaluator.** There is a bounded functional φ_ξ on $B(\mathcal{H})$ with $S^\xi(x; v) = \varphi_\xi(T_D(w(x)+s))$ on a padded neighbourhood of each configuration; it is χ_ξ extended by Hahn–Banach, composed with compression.
- **Derivatives and inclusions.** The identities hold among slab operators. Derivatives come from V.1.g; pairing with positive-word vectors reduces inclusions to the identities of S.
- **Stress laws.** Choose one shift s and width D that put the configuration and the support of a smooth compactly supported window χ in $(0, D)$, with $\chi = 1$ near the distinguished point x_0 and all spectators outside its support. The operators $-\sum_a \int \partial_a \chi(y)\, T_D((J_{ab}(y) B(x_0) \cdots) + s)\, dy$ are Bochner integrals. The shell weight vanishes near every possible collision, which gives integrability. Pairing with positive-word vectors adds only exterior points outside the window, so the Ward identities of S apply. The angular law uses the additional weight $(g(y - x_0))_b$ for a skew matrix g. Apply φ_ξ to both operator identities.

Lean: `Work.B09.answer`, `B09.has_evaluator`, `B09.Evaluates.flux`.

Lemma V.1.s (caps in every phase; B10, B11).

- (i) Uniform zeroth cap: V.1.6(c).
- (ii) A smooth basic table that respects S, satisfies the inclusion and derivative identities, and has the zeroth cap for every fixed list of labels, has caps on all derivatives.

Proof sketch.

- (i) Translate w into a slab with buffer r and unit padding. Then $|\chi_\xi(M(w))| \leq \|M(w)\| \leq \|T_d(w)\|$, and the all-copy bound applies; the constants depend only on (D, ρ).
- (ii) A coordinate derivative replaces one label by its derivative label, raising its degree by one. Induct on the order, and bound each multilinear derivative norm by the sum over its finitely many coordinate directions. The constants may depend on the arity, labels and derivative order, but not on the positions or r.

Lean: `Work.B11.answer`, `B11.phase_cap`, `Work.B10.answer`, `B10.capped_tags`, `Bid.OneCap`.

3.21.4.6 4.6 Null spaces of the phases Lemma V.1.t (moment bounds give bounds in every phase). Let $\tau \geq 0$ and $p \in \mathcal{A}$. If $\langle \Omega, (p^* p)^n \Omega \rangle \leq \tau^{2n}$ for every $n \in \mathbb{N}$, then $|\chi_\xi(p)| \leq \tau$ for every ξ. *Proof sketch.*

- The hypothesis says $\int |\chi_\xi(p)|^{2n} d\nu \leq \tau^{2n}$ for all n.

- If $|\chi_{\xi_0}(p)| > \tau' > \tau$, then continuity gives an open $O \ni \xi_0$ on which $|\chi_\xi(p)| > \tau'$.
- Full support gives $\nu(O) > 0$, so $\nu(O)\tau'^{2n} \le \tau^{2n}$ for all n, a contradiction.

Lean: `Retain.bit`, `Retain.Tall`.

Lemma V.1.u (detectors; B15–B18).

- **The detectors.** Assume the Seed and phase reflection. Fix a degree d, an index k of its corrected-field data, and a test row with a basis index. Let F be a one-point word at a prescribed grid position, with free value s at the rescaled position. For a two-point row, let G be its second one-point word, s' its free value and γ the free cumulant. There are detectors in $\mathcal{A}$:
  - the one-point detector $\Delta_1 = M(F) - s \cdot 1$;
  - the two-point detector $\Delta_2 = M(F \cdot G) - s\,M(G) - s'M(F) + (ss' - \gamma) \cdot 1$.
- **Statement.** For every slot there is a one-point detector; for a two-point row there is also the two-point detector. Each satisfies $\langle \Omega, (\Delta^*\Delta)^n \Omega \rangle \le \mathrm{gap}_k^{2n}$ for every $n \in \mathbb{N}$, where $\mathrm{gap}_k \ge 0$ and $\mathrm{gap}_k \to 0$.

Proof sketch. Each detector is the cluster element $c_E = \sum_i \lambda_i M(\mathrm{word}_i)$ of a finite lattice *episode* E, whose chessboard budget is gap_k (B18, from the corrected-field data). Then:

- **B15 (cutoff level).** The absolute lattice expectation of the product of $2n$ diagonal translates of the episode function, alternately conjugated, is at most $(\mathrm{gap}_k + \varepsilon)^{2n}$ late in the cutoff. This is the chessboard estimate.
- **B16 (limit table).** For fixed n and all sufficiently large translations, expand the product into finitely many monomials. Their points are generic: within each cluster this is part of the data, and between clusters each coordinate tie excludes at most one translation parameter. Point limits apply to each monomial, including the conjugated cutoff labels.
- **B17 (vacuum moments).** Let the translation tend to infinity and apply V.1.j to each ordered monomial, using $M(\bar{w}) = M(w)^*$ from V.1.i and V.1.n. The limit is $\langle \Omega, (c_E^* c_E)^n \Omega \rangle$. Finally let $\varepsilon \downarrow 0$.

Lean: `Work.B15.answer`, `B16.answer`, `B17.answer`, `B18.answer`, `Bid.episode`; definitions `Bid.Episode`, `Episode.Sold`, `Retain.Δone`, `Retain.Δtwo`, `Retain.Funds`.

Lemma V.1.v (retention of null spaces). Assume the corrected-field data and detector bounds of V.1.u. If a table S' is symmetric, respects S, and satisfies $S'(w) = \chi_\xi(M(w))$ for every separated word and some fixed ξ, then $N_d(S') = N_d(S)$ for every d. *Proof sketch.*

- **Comparison maps.** Let F_d^{free} be the free-field space in the corrected-field data. Define $U_k \colon F_d^{\mathrm{free}} \to V_d$ by sending its chosen basis to the corrected labels. Compose U_k, the quotient map for S', and the finite family of S'-probes to obtain $f_k \colon F_d^{\mathrm{free}} \to \mathbb{C}^m$.
- **Injectivity.** V.1.t and V.1.u give one-point errors at most gap_k and two-point errors at most $\mathrm{gap}_k(1 + |s| + |s'|)$. The free singleton values converge and are bounded, and the rescaled test points converge. Thus f_k converges in norm to the injective free test map. Injectivity is open in this finite-dimensional operator space, so f_k is injective for large k.
- **Conclusion.** For all large k, $q_d U_k$ is also surjective by the spanning hypothesis. If $v \in N_d(S')$, choose $z \in F_d^{\mathrm{free}}$ with $q_d U_k z = q_d v$. Since S' respects S, $U_k z \in N_d(S')$, hence $f_k z = 0$. Injectivity gives $z = 0$, so $q_d v = 0$. The reverse inclusion is the respect hypothesis.

Lean: `Retain.lift_detect`, `Retain.result`, `Retain.pair_near`.

3.21.4.7 4.7 Choosing a phase Lemma V.1.w (slab operators in a phase chamber; B13). For every ξ, every time-axis chamber of S^ξ carries a slab family with $\|T_d^\xi(w)\| \le \|T_d(w)\|$. *Proof sketch.*

- **The train in the phase.** In the phase chamber the train Gram is $S^\xi(\Theta v_n \cdot v_n) = \chi_\xi(Y(v_n)^* Y(v_n))$, where $Y(v_n) = (T_d(w)^* T_d(w))^n Y(v)$.
- **Bench in the phase.** Hence $|S^\xi(\Theta v_n \cdot v_n)| \le \|\chi_\xi\| \, \|Y(v)\|^2 \, \Delta^{4n}$ for every $\Delta \ge \|T_d(w)\|$. This is the bench in the phase, with a constant depending on v.
- **Conclusion.** V.1.e gives norm $\le \Delta$ for every $\Delta > \|T_d(w)\|$; the constant depending on v drops out. No operator inequality between the chambers is used.

Lean: `Work.B13.answer`, `B13.buys`.

Lemma V.1.x (one countable family; B14). There is a single sequence (b_k) of positive words, depending only on the label register and the time axis, whose vectors have dense linear span in every OS Hilbert space realizing the positive-word Gram form of a symmetric smooth table. *Proof sketch.*

- **Parametrization.** For fixed arity and degrees, positive words are parametrized by a separable space: positive separated configurations times label coefficients. Choose countable dense subsets, together with the empty word.
- **Continuity.** Inner products of the vectors are values of the table at combined configurations, which are continuous in the points (smoothness) and in the coefficients (finite-dimensional multilinearity). So the vector map is continuous, and the closed span of the countable family contains all vectors.

Lean: `Work.B14.universal`, `B14.answer`.

Lemma V.1.y (almost every phase has a unique ground state).

- **Statement.** In the setting of V.1.8, for each positive b, $P_0^\xi |b\rangle_\xi = S^\xi(b)\,\Omega_\xi$ for ν-almost every ξ. This proves the uniqueness assertion of V.1.8.

Proof sketch.

- **The variance identity.** Let $\omega_\xi = \|P_0^\xi |b\rangle_\xi\|^2 = \lim_{n\to\infty} S^\xi(\Theta\tau_n b \cdot \tau_n b)$. For $n \in \mathbb{N}$ these functions of ξ are continuous, nonnegative, decreasing in n and dominated by their value at $n = 0$, which is integrable on compact X. Their integrals are the corresponding quantities for S. Dominated convergence gives $\int \omega_\xi \, d\nu = \|P_0|b\rangle\|^2 = \|M(b)\Omega\|^2 = \int |\chi_\xi(M(b))|^2 d\nu$.
- **Almost-everywhere vanishing.** Since $\|P_0^\xi|b\rangle - S^\xi(b)\Omega_\xi\|^2 = \omega_\xi - |S^\xi(b)|^2 \ge 0$ has integral zero, it vanishes a.e.
- **Uniqueness.** Apply this to the family (b_k) of V.1.x; the union of the countably many null sets is null. Outside it, P_0^ξ maps a dense set into $\mathbb{C}\Omega_\xi$, so $\mathcal{H}_0^\xi = \mathbb{C}\Omega_\xi$. Since $\nu(X) = 1$, such ξ exist; the choice is not canonical.

Lean: `Cluster.exp_real`, `Cluster.champ`, `Cluster.fits`, `Cluster.lifts`, `Cluster.selects`.

Lemma V.1.z (assembly). V.1.9 holds. *Proof sketch.*

- **Slab and phase constructions.** B01 and B02 give the original slab family with its laws. V.1.h gives spatial invariance; B03 gives the compression identities; B04 and B05 give phase reflection; B15–B18 give the detector bounds; and B12 gives separability. For any decomposition, B06 and B07 construct the tables, and B08–B11 give their hierarchy and field

laws. B13 and B02 give slab families in the phase chambers, and B14 supplies a common countable total family.
- **The certificate.** Choose a chamber for every phase, select ξ_* by V.1.y, and apply V.1.v to obtain $N_d(S^\xi) = N_d(S)$ for every ξ, d.

Lean: `Bid.Looms`, `Phase.Loom.cleanse`, `Pert.withFinal`.

3.21.5 5. Where the difficulty lies

Distinctive arguments.

- **The decomposition uses the OS space of the limit correlations.** Three properties require separate arguments:
  - spatial invariance of the ground states (V.1.h);
  - commutativity of $\mathcal{A}$ (V.1.k);
  - reality of phase values (V.1.n).
- **Lattice input after the limit.** It enters at three places only: the bench (V.1.7), phase reflection (V.1.n, through the grid hypothesis), and the detectors (V.1.u).
- **Phase reflection is the least standard argument.** It is a variance-positivity statement for far-separated copies of a real lattice variable. It is equivalent to the claim that the abstract characters assign real values to real observables.
- **Detectors.** They require quantitative UV input: one- and two-point chessboard deviations $\mathrm{gap}_k \to 0$ of corrected fields from free values. These estimates preserve the field null spaces in every phase.

Routine but long.

- B02, B03, B09, B12, B14, B16, B18: smooth dependence on positions, label quotients, finite polynomial limits, separability.
- `SNAG`, `Paley`, `Heat` (about 1,100 lines) reprove textbook theorems in an explicit form adapted to the library.

Unusual or worth checking. 1. **Selection.** The selected phase comes from a measure-theoretic existence argument and is not canonical. No finite bound on the number of phases is asserted. V.3 later proves unique ground states for every phase. 2. **Sub-width in the bench.** The bench gives the train bound only at a lattice-compatible width $s \le d$. The remaining width is absorbed by the contraction E_{d-s} (V.1.f). 3. **Which symmetries are used where.** - Commutativity uses covariance only under the exchange of the time axis with one spatial axis. This is a lattice symmetry, already contained in the real-spin part of the hierarchy. - Spatial invariance of $\mathcal{H}_0$ needs arbitrary hyperplane reflections, i.e. the continuum $O(4)$ action of VI.1. In the Lean, both steps draw on that action. - Both use improper elements of $O(4)$ (reflections), not only rotations. 4. **Scope of the certificate.** Its unique-ground-state assertion concerns the time-axis chamber of the selected phase. It also records the hierarchy, field laws and null-space equality for every phase. The continuation linked to the selected Wightman realization uses that selected table. 5. **Smoothness hypothesis in B02.** B02 needs the derivative identities of the table on which the slab family lives. For phases these come from B09, which uses the slab laws of S, so the two applications of B02 are ordered.

3.21.6 6. Notation

object	code name	proposed notation
limit table, value on a word	`Table`, `S.expect w`	$S = S^{\mathrm{mix}}$, $S(w)$
OS reflection; linear time reflection of a word	`reflect w timeAxis`; `Grid.anti`	Θw; ϑw
OS space, vector of a positive word	`GNS.H`, `GNS.plant`	$\mathcal{H}$, $\lvert b\rangle$
chamber	`Chamber`	$(\mathcal{H}, E_t, U_y)$
ground space, projection, vacuum	`Semigroup.ground`, `orth`, `omega0`	$\mathcal{H}_0$, P_0, Ω
real-time group; 4-parameter group; spectral measure	`edge`; `unit`; `Orbit.P`	e^{isH}; $V(x)$; $P(dp)$
slab, buffer	`Slab`, `buffer`	slab $(0, d)$, r-buffered
slab operators ("Spread") and all-copy bound	`Spread`, `place`, `AllCopy`, `Laws`	$T_d(w)$; $C^n(n/r)^{\kappa n}$
bench, train bound	`Bench`, `Seats`, `QRead`	bench (V.1.7)
spatial invariance of $\mathcal{H}_0$	`Steady`	V.1.4(a)
compression; its multilinear form; algebra	`M`, `atom`; `depot`; `Spread.algebra`	$M(w)$; $M(x; v)$; $\mathcal{A}$
identities of M ("Theme")	`Theme`	V.1.4(e)
phase reflection	`Phase.reflects`	$M(w) = M(\vartheta w)$
decomposition ("Resolution")	`Spread.Resolution`, `X`, `μ`, `character`	X, ν, χ_ξ
phase tables; their properties	`ensemble`, `Garland`, `Scripts`	S^ξ
S^ξ receives χ_ξ	`Resolution.receives`	$S^\xi(w) = \chi_\xi(M(w))$
grid hypothesis, bounded approximants	`Org.Grill`, `Grid.Covering`	grid hypothesis
lattice episode, cluster element, budget	`Episode`, `cluster`, `Sold`	E, c_E
detectors; moment bound	`Δone`, `Δtwo`; `Tall`, `Funds`	Δ_1, Δ_2
countable dense family	`Cluster.horizon`	(b_k)
unique ground state	`UniqueHeat`	$\mathcal{H}_0^\xi = \mathbb{C}\Omega_\xi$
selected phase and certificate	`Selection`, `atPoint`, `Certificate`, `Materializes`	ξ_*; V.1.9
bundles of obligations	`Loom`, `Stall`, `Vendor`	assembly assumptions

The measure ν corresponds to μ in the Lean.

3.21.7 7. What remains unexplained

1. **Ultraviolet inputs.** The chessboard geometry below `Next.Phase.travel_price` belongs to VII.4. Its train estimate and the subsequent scale choice give the bench used here; the supporting packing argument remains outside this unit. The bounded-approximation input to `Accent.grilled` belongs to III.11. Its output has the cutoff-independent bounds required in V.1.n, but the underlying ultraviolet estimates and the production of the corrected-field data remain inputs.
2. **Order of limits.** In V.1.n, fix the number of copies and approximation index, then fix a

sufficiently large translation, take the ordered cutoff limit, and let the translation tend to infinity. Only then let the number of copies tend to infinity and the approximation error tend to zero. The detector argument similarly takes the cutoff limit before the large-translation limit and sends its budget slack to zero last.

3. **Later vacuum uniqueness.** V.1.8 gives uniqueness almost everywhere, which suffices to select ξ_*. The stronger result for every phase, `EveryPhase.every`, uses reconstruction in V.3; its proof is outside this unit.
4. **Placement of the spectral calculus.** V.1.2 supplies V.2 and V.3 and can be placed at the start of reconstruction in the written proof. The $O(4)$ action from VI.1 is needed only after the preliminary chamber and heat constructions.
5. **Bibliography.** The reference locations marked "probable" in §2 need bibliographic verification.

3.22 V.2. Osterwalder–Schrader reconstruction of a Wightman theory

Part: V, phases and reconstruction.

Lean:

- The work items A16 and C01–C29.
  - Statements: `Bid/Corners`, `Bid/Tar`, `Bid/Grow`.
  - Proofs: `YM/Work`, files C01–C29, subfolders C07/ and C13/.
  - Collected by `Integrated.reconstructionArcs`.
- The folders `YM/Boundary`, `YM/Fields`, `YM/Hilbert`, `YM/Compact`, `YM/Distribution` and `YM/Spark`. Spark is the Schwartz-space calculus used by C05, C08, C12, C14 and C25.
- Material filed elsewhere but described here because it completes the reconstruction (§4.3, §4.6, §4.8):
  - The boundary-value and germ lemmas of `YM/Shift`, filed partly under VI.1 by the declaration table.
  - The assembly in `Next/BOUNDARY/Stage.lean` and `Org/Theatre.lean`, filed under VIII.1.

Size: 6,456 lines in 705 declarations.

part	lines
proofs collected by reconstructionArcs	4,279
Fields	438
Compact	218
Distribution	189
Spark	120
Hilbert	85
Boundary	66
Overlay and Vac (ultraviolet material)	558

Three bundles in Integrated.lean are also ultraviolet material placed here by the dependency cut; they are listed at the end of §1.

3.22.1 1. What this unit proves

Conventions.

- **Points.** A point of $\mathbb{R}^4$ is $x = (\vec{x}, x^0)$; time is the last coordinate, as in the code. A configuration is $x = (x_1, \dots, x_n) \in \mathbb{R}^{4n}$.
- **Pairing.** $\langle \cdot, \cdot \rangle$ is the *Euclidean* inner product of $\mathbb{R}^{4n}$, extended complex-bilinearly. Dual cones are taken with respect to this pairing. Hilbert-space inner products are antilinear in the first argument.
- **Cones and groups.** $V_+ = \{|\vec{x}| < x^0\}$, with closure $\bar{V}_+$. $\mathcal{L}_+^\uparrow$ is the proper orthochronous Lorentz group. $\Lambda_a(t)$ is the boost of rapidity t along the spatial axis a. The Poincaré group $\mathcal{P}$ acts on tests by $(g_* f)(x) = f(g^{-1}x_1, \dots, g^{-1}x_n)$.
- **Heights.** Heights $y \in \mathbb{R}^n$ may be negative. $E_n = \{y_1 < \cdots < y_n\}$ is the set of ordered heights, and $\Sigma_n = \{z \in \mathbb{C}^n : \operatorname{Im} z \in E_n\}$ is the strict region. We write $\underline{y} = ((\vec{0}, y_1), \dots, (\vec{0}, y_n)) \in \mathbb{R}^{4n}$.
- **Fourier transform.** $\hat{f}(p) = \int e^{-i\langle x,p\rangle} f(x)\, dx$, $\check{f}(p) = (2\pi)^{-4n} \int e^{i\langle p,x\rangle} f(x)\, dx$, and $(\mathcal{F}W)(f) = W(\hat{f})$.

Unless stated otherwise, arities n, m range over $\mathbb{N}$, including 0, and label tuples range over $\mathcal{F}^n$. Supports of functions are closed supports.

Tables and the OS space.

- S and T are tables on one label register (glossary). $N_d(Q)$ denotes the raw degree-d labels whose correlations in Q vanish with arbitrary other insertions. The quotient label space of S is $F_d = \mathrm{Label}_d / N_d(S)$, and $\mathcal{F}$ is their union, into which F_d embeds by pin_d.
- $T(x; v)$ is the Euclidean n-point function of T at a non-coincident configuration x, read on tags $v_k = (d_k, v_k)$ of S. For a tuple of tags, $\mathrm{pin}\, v$ means $(\mathrm{pin}_{d_k} v_k)_k$; the insertion into $\mathcal{F}$ is implicit in ψ_v. It is meaningful when T *respects* S, i.e. $N_d(S) \subseteq N_d(T)$.
- $\mathcal{H}$ is the OS Hilbert space of T for the time axis. It carries the vacuum Ω, a Hamiltonian $H \geq 0$ with holomorphic contraction semigroup $e^{-\zeta H}$ ($\operatorname{Re}\zeta > 0$), the spatial translations $U(\vec{a})$, and a unitary space-time translation group.
- A *spread* A gives slab operators $A_d(w)$ for words w in a time slab of width d (V.1).
- A differential plan P consists of linear maps $\iota_{d,e} : F_d \to F_e$ for $d \leq e$ and derivative maps $\partial_{d,a} : F_d \to F_{d+1}$. Its inclusions are *compatible with a table* Q if replacing any tag (d, v) by $(e, \iota_{d,e} v)$ leaves $Q(x; v)$ unchanged at every non-coincident configuration, with arbitrary other tags. Its derivatives are compatible with Q if inserting $\partial_{d,a} v$ differentiates the corresponding correlation in that slot and coordinate. Writing C_d for label conjugation and $M_d(s)$ for a signed coordinate symmetry, P is *oriented* if its inclusions commute with $C_d, M_d(s)$, its derivatives commute with conjugation, and

$$M_{d+1}(s)\partial_{d,b} = \sum_a s_{ab}\, \partial_{d,a} M_d(s).$$

  The induced star satisfies $J\mathrm{pin}_d v = \mathrm{pin}_d(M_d(r_0) C_d v)$, where r_0 is time reflection.

Continuations and boundary families.

- A **continuation** F is a family of maps $F_n(z; \vec{x})$, $z \in \mathbb{C}^n$, $\vec{x} \in (\mathbb{R}^3)^n$, each multilinear on $\mathcal{F}^n$. We write
  - $F_n^y(x) = F_n(x^0 + iy; \vec{x})$, with real times x_k^0 and heights y_k;
  - $\psi_v(y; f) = \int F_n^y(x)(v) f(x)\, dx$.

- A **boundary family** W is a family of multilinear maps $W_n : \mathcal{F}^n \to \mathcal{S}'(\mathbb{R}^{4n})$.
  - For a word $b = ((u_1, f_1), \dots, (u_n, f_n))$, with $u_k \in \mathcal{F}$ and $f_k \in \mathcal{S}(\mathbb{R}^4)$, we set $W(b) = W_n(u)(f_1 \otimes \cdots \otimes f_n)$.
  - For an antilinear map J on $\mathcal{F}$ we set $b^* = ((Ju_n, \bar{f}_n), \dots, (Ju_1, \bar{f}_1))$ and $\langle b, c\rangle_W = W(b^* c)$.

The thirty reconstruction statements are conditional results for arbitrary tables, OS data and continuations. VIII.1 applies them to the mixed limit table and to a selected phase (§4.8).

Proposition V.2.1 (the reconstruction bundle). The statements A16 and C01–C29 (V.2.2–V.2.8) hold simultaneously. Each retains its own hypotheses. *Status:* theorem. *Lean:* `Integrated.reconstructionArcs`, `Bid.Arcs`.

Proposition V.2.2 (strict-time continuation; C01–C03). For the tables S, T, the OS chamber V with Hilbert space $\mathcal{H}$, a differential plan P and a spread A, assume the slab hypotheses (H):

- S and T satisfy the Euclidean hierarchy;
- T respects S;
- the inclusions of P are compatible with S and T, and its derivatives with T;
- the spread A obeys the slab bound (SB) (§2), is smooth in the insertion positions on every valid slab, and differentiating an insertion replaces its label by the corresponding derivative under P.

Then:

(a) (C01) There are linear maps $B_p : \mathcal{F} \to \mathcal{B}(\mathcal{H})$, $p \geq 0$, such that:

- for $p > 0$, $B_p(\mathrm{pin}_d v)$ is the slab operator of width $2p$ of the one-point word made of the representative of v placed at $(\vec{0}, p)$;
- for every $u \in \mathcal{F}$ there are $C > 0$ and $a \in \mathbb{N}$ with $\|B_p(u)\| \leq C p^{-a}$ for $0 < p < 1$.

(b) (C02) Let B be such a family. Put

$$G_p(w, \vec{a}) = e^{-(\mathrm{Im}\, w - 2p)H}\, e^{i\,\mathrm{Re}\, w\, H}\, U(\vec{a}),$$

$$M_p(z; \vec{x})(v) = \langle \Omega,\, B_p(v_1)\, G_p(z_2 - z_1, \vec{x}_2 - \vec{x}_1)\, B_p(v_2) \cdots G_p(z_n - z_{n-1}, \vec{x}_n - \vec{x}_{n-1})\, B_p(v_n)\Omega \rangle.$$

For $n = 0$ set $M_p = \|\Omega\|^2$. Call p *admissible* for z if $0 < p < 1$ and $\mathrm{Im}(z_j - z_i) > 4p$ for $i < j$. Then:

- $M_p = M_q$ whenever p and q are both admissible for z;
- $M_p(iy; \vec{x})(\mathrm{pin}\, v) = T((\vec{x}_k, y_k)_k; v)$ for $y \in E_n$ and p admissible.

(c) (C03, from (b) alone) There is a continuation F with $F_n(z; \vec{x}) = M_p(z; \vec{x})$ for $z \in \Sigma_n$ and every admissible p, satisfying:

- (SC) for each fixed label tuple v, $(z, \vec{x}) \mapsto F_n(z; \vec{x})(v)$ is continuous on $\Sigma_n \times (\mathbb{R}^3)^n$ and holomorphic in $z \in \Sigma_n$, and $F_n(iy; \vec{x})(\mathrm{pin}\, v) = T((\vec{x}_k, y_k)_k; v)$ for $y \in E_n$;
- (PG) for all n and v there are $m \in \mathbb{N}$ and $C \geq 0$ with $|F_n^y(x)(v)| \leq C k^{-m}$ for all x, all $k \in (0, 1]$, and all y with $y_j - y_i \geq k$ for $i < j$.

Status: theorem. The derivative and spatial regularity hypotheses remain part of the stated input package. *Lean:*

- `C01.answer`, `C02.answer`, `C03.answer`.

- Statements `Bid.C01`, `Bid.C02`, `Bid.C03`.
- Hypothesis and conclusion structures: `Pantry`, `Piece`, `Slice`, `Direct`.
- Compatibility definitions: `DiffPlan.Inclusions`, `DiffPlan.Orientation`.
- Definitions `Small`, `Small.matrix`, `Small.rated`, `Small.Defined`, `Waves`, `Waves.Region`, `Waves.Breathes`.

Proposition V.2.3 (Cauchy–Riemann identities and tempered boundary values; C04). Let F satisfy (SC) and (PG).

(a) (C04) Three properties hold:

- (I) $x \mapsto F_n^y(x)(v)f(x)$ is integrable for $y \in E_n$ and $f \in \mathcal{S}(\mathbb{R}^{4n})$.
- (HC) $y \mapsto \psi_v(y; f)$ is continuous on E_n.
- (CR) For $a \in \mathbb{R}^n$ and $y + sa \in E_n$,

$$\tfrac{d}{dt}\psi_v(y + ta; f)\big|_{t=s} = \psi_v(y + sa; -i\,\partial_{\underline{a}} f), \qquad \partial_{\underline{a}} = \textstyle\sum_k a_k\,\partial/\partial x_k^0.$$

(b) If F satisfies (I), (HC), (CR) and (PG), there is a boundary family W with (BV): the integrands defining $\psi_v(y; f)$ are integrable for $y \in E_n$, and $\psi_v(y; f) \to W_n(v)(f)$ as $y \to 0$ within E_n, for all n, v and f. In particular, the radial boundary limit ($\mathrm{BV}_{\mathrm{ray}}$) holds: $\psi_v(\varepsilon u; f) \to W_n(v)(f)$ as $\varepsilon \downarrow 0$ for each $u \in E_n$.

Status: theorem. Part (b) lives in Shift/Breath and Shift/Osc, which are filed under VI.1. It is used through `Show.Rhythm.opens` (VIII.1). The boundary family is an output of the reconstruction. *Lean:*

- `C04.answer`, `Bid.C04`.
- `Waves.Bounds`, `Waves.Lines`, `Waves.Along`.
- `Waves.jointly`, `Waves.NonTangentialFree`, `Waves.Opens`, `Waves.exposed`.

Proposition V.2.4 (Laplace identity and spectral support; C05, C06). Let F satisfy (SC), (PG), (CR) and (HC), and have boundary family W (BV).

(a) (C05; Laplace identity (L)) For $y \in E_n$ and compactly supported $f \in \mathcal{S}(\mathbb{R}^{4n})$,

$$W_n(v)(\hat{f}) = \psi_v(y; \widehat{e^{\langle \underline{y}, \cdot \rangle} f}).$$

(b) (C06, from (SC), (PG), (I) and (a)) $\mathcal{F}W_n(v)$ is supported in

$$\mathcal{P}_n = \Big\{p : \textstyle\sum_k y_k p_k^0 \geq 0 \text{ for every nondecreasing } y\Big\} = \Big\{p : \textstyle\sum_k p_k^0 = 0,\ \sum_{k \geq j} p_k^0 \geq 0\ \forall j\Big\}.$$

(c) (Lorentz covariance) If W also satisfies (Cov) (V.2.5), then $\mathcal{F}W_n(v)$ is supported in

$$\mathcal{V}_n = \bigcap_{\Lambda \in \mathcal{L}_+^{\uparrow}} \tilde{\Lambda}^{-1} \mathcal{P}_n,$$

where $\tilde{\Lambda} = \mathsf{r}\Lambda\mathsf{r}$ is the dual action (r reverses the spatial components). Every $p \in \mathcal{V}_n$ has $\sum_k p_k = 0$ and $\sum_{k \geq j} p_k \in \bar{V}_+$ for all j. This is the cone-support hypothesis of C14–C16 and C19.

Status: theorem. (SC) is not used in C05 or C06, and (I) is not used in C06. *Lean:*

- `C05.answer`, `C06.answer`.
- Statements `Bid.C05`, `Bid.C06`.
- `Waves.during`, `Surface.cone_tests`, `half`, `coneM`, `tents`, `Bid.cones`.

Proposition V.2.5 (Poincaré covariance; A16, C07, C08, C09, C28).

Data.

- A rotation structure R on S (VI.1): an action of $O(4)$ on the F_d, with Lie action ρ of $\mathfrak{so}(4)$. It is subject to Euclidean covariance (the functions $T(x;v)$ are $O(4)$-covariant) and the rotation laws of VI.1.
- A spin structure K: representations $K(\Lambda)$ of $\mathcal{L}_+^\uparrow$ on $\mathcal{F}$ and on each F_d, with spin compatibility with R, relative to the real structure of S, which consists of:
  - compatibility with pin_d;
  - the representation property and continuity on each finite-dimensional level;
  - $\frac{d}{dt}K_d(\Lambda_a(t))|_{t=0} = -i\rho(e_0 \wedge e_a)$;
  - $K = R$ on spatial rotations;
  - commutation with the reflection adjoint.

(a) (A16) $\mathcal{L}_+^\uparrow$ is generated, as an abstract group, by the spatial rotations and the coordinate boosts.

(b) (C08, no hypotheses) For all n and $f \in \mathcal{S}(\mathbb{R}^{4n})$:

- $g \mapsto g_* f$ is continuous from $\mathcal{P}$ into $\mathcal{S}(\mathbb{R}^{4n})$;
- $t^{-1}(\Lambda_a(t)_* f - f) \to -D_a f$ in $\mathcal{S}$ as $t \to 0$, where $D_a = \sum_k (x_k^a \partial_{x_k^0} + x_k^0 \partial_{x_k^a})$.

(c) (C07) Assume S and T satisfy the hierarchy, the inclusions of P are compatible with S, T respects S, and Euclidean covariance, the rotation laws of VI.1 and spin compatibility hold. Then every F with (SC), (PG), (I), (CR) and (HC) satisfies the boost identity (B_y): for all $y \in E_n$, tags v, spatial axes a and $f \in \mathcal{S}(\mathbb{R}^{4n})$,

$$\sum_{k=1}^{n} \psi_{v^{(k)}}(y; f) = \psi_v(y; D_{a,y} f), \qquad D_{a,y} = \sum_k \Big(x_k^a \, \partial_{x_k^0} + (x_k^0 + i y_k) \, \partial_{x_k^a} \Big).$$

Here $v^{(k)}$ is v with v_k replaced by $-i\rho(e_0 \wedge e_a) v_k$. Among the table, plan and spin inputs, the boost argument uses the Euclidean hierarchy and rotation data.

(d) (C09) Assume A16, C08 and the same table, plan, rotation and spin hypotheses as in (c). If F satisfies (SC), (PG) and (B_y) and has boundary family W (BV), then W satisfies (Cov):

- $W_n(v)(f(\cdot - a)) = W_n(v)(f)$ for $a \in \mathbb{R}^4$;
- $W_n(K(\Lambda)v)(\Lambda_* f) = W_n(v)(f)$ for all $\Lambda \in \mathcal{L}_+^\uparrow$.

(e) (C28) Assume C08, S symmetric, the inclusions of P compatible with S, spin compatibility, and J the induced star. Let $g = (a, \Lambda)$ act on words letterwise by $(u, f) \mapsto (K(\Lambda)u, g_* f)$. Then (Cov) implies:

- (invariance of the word Gram form) $\langle g \cdot b, g \cdot c \rangle_W = \langle b, c \rangle_W$;
- (continuity of word matrix elements) $g \mapsto \langle b, g \cdot c \rangle_W$ is continuous.

Status: theorem. *Lean:*

- `A16.answer` (statement `Bid.A16`, `beamGens`).

- `C07.answer`, `C08.answer`, `C09.answer`, `C28.answer`.
- Statements `Bid.C07`, `Bid.C08`, `Bid.C09`, `Bid.C28`.
- Definitions `Boosty`, `inclined`, `kick`, `Urs`, `Spins.Descends`, `Surface.Transforms`, `Surface.Strong`.
- Rotation laws: `Table.Rotation.Laws`.

Proposition V.2.6 (Fourier–Laplace continuation in the forward tube; C10–C16).

Definitions.

- Tube and cones:
  - $V_n^+ = \{\eta \in \mathbb{R}^{4n} : \eta_{k+1} - \eta_k \in V_+,\ 1 \le k < n\}$ (the open imaginary-part cone);
  - $\bar V_n^+$ is the analogue with $\bar V_+$ (the closed imaginary-part cone);
  - $\mathcal{T}_n = \{z \in \mathbb{C}^{4n} : \operatorname{Im} z \in V_n^+\}$ (the tube). This is the forward tube in absolute coordinates; there is no condition on $\operatorname{Im} z_1$.
- A *cutoff* χ_n is a real smooth function whose derivatives are all polynomially bounded, equal to 1 on the 1-neighbourhood of $\mathcal{V}_n$, with closed support contained in its open 3-neighbourhood.
- *Fourier–Laplace continuation:*
$$\mathcal{W}_n(v)(z) = (2\pi)^{-4n}\,(\mathcal{F}W_n(v))(e^{i\langle\cdot,z\rangle}\chi_n), \qquad z \in \mathcal{T}_n.$$
- *Partial boundary values:* $W_n(v)[\eta](https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/f) = (\mathcal{F}W_n(v))(e^{-\langle\eta,\cdot\rangle}\chi_n\,\check f)$, formally $W_n(v)(\cdot + i\eta)$.

Statements.

(a) (C10, cone duality) For $p \in \mathcal{V}_n$ and $\eta \in \bar V_n^+$, $\langle p, \eta\rangle \ge 0$. For $\eta \in V_n^+$ there is $c > 0$ with $\langle p, u\rangle \ge c|p|$ for all $p \in \mathcal{V}_n$ and all u near η.

(b) (C11) Cutoffs exist.

(c) (C12) Given (a), every cutoff satisfies the damping properties (D):

- $e^{-\langle\eta,\cdot\rangle}\chi_n$ has temperate growth for $\eta \in \bar V_n^+$;
- $\eta \mapsto e^{-\langle\eta,\cdot\rangle}\chi_n f$ is continuous from $\bar V_n^+$ into $\mathcal{S}$.

(d) (C13) Given (a), $\mathcal{W}_n(v)$ is holomorphic on $\mathcal{T}_n$.

(e) (C14, the integral representation) Given (c), for $\eta \in V_n^+$ the identity $W_n(v)[\eta](https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/f) = \int \mathcal{W}_n(v)(x + i\eta) f(x)\,dx$ holds, with integrable integrand.

(f) (C15) Suppose:

- (c) holds and $\operatorname{supp}\mathcal{F}W \subseteq \mathcal{V}_n$;
- (d) and (e) hold;
- F satisfies (SC), (PG), (I) and the Laplace identity V.2.4(a).

Then $\mathcal{W}_n(v)(x + i\underline{y}) = F_n^y(x)(v)$ for all $y \in E_n$ and $x \in \mathbb{R}^{4n}$. Together with (d) and (e) this is (TC).

(g) (C16) Given $\operatorname{supp}\mathcal{F}W \subseteq \mathcal{V}_n$ and (Cov), $\mathcal{W}_n(v)(z) = \mathcal{W}_n(K(\Lambda)v)(\Lambda z)$ for $z \in \mathcal{T}_n$ and $\Lambda \in \mathcal{L}_+^{\uparrow}$, with Λ acting on real and imaginary parts.

Status: theorem. Several listed hypotheses are not used (§5). *Lean:*

- `C10`, `C11`, `C12`, `C13`, `C14`, `C15`, `C16` (all answer theorems).
- Statements `Bid.C10`, `C11`, `C12`, `C13`, `C14`, `C15`, `C16`.
- Definitions `Wide`, `Income`, `wave_turn`, `Blanket`, `Blanket.Modified`, `Blanket.partialD`, `Surface.Aurora`, `Aurora.emit`, `Aurora.FullBody`, `tube`, `shores`, `fields`.

Proposition V.2.7 (locality; C17–C21, C27).

Setting.

- Let $n \geq 2$ and let $Q = (i, i+1)$ be an adjacent pair. s is the transposition of i and $i+1$, and σ swaps slots i and $i+1$ of real, imaginary or complex configurations.
- $\Gamma_Q = \{x : x_{i+1} - x_i \text{ spacelike}\}$.
- A height profile y is a *spectator profile* for Q if $y_i = y_{i+1}$ and $y_k < y_l$ for all other pairs $k < l$. Then $\underline{y}$ and $\sigma\underline{y}$ lie in $\bar{V}_n^+$, and for every real x the point $x + i\underline{y}$ lies in the closures of $\mathcal{T}_n$ and of $\sigma^{-1}\mathcal{T}_n$.
- $\Lambda_e(\zeta)$ is the complexified boost of rapidity $\zeta \in \mathbb{C}$ along a unit spatial direction e.

Statements.

(a) (C17) There are spectator profiles $y_m \to 0$ with the following property. For every m and every $x \in \Gamma_Q$ there are a unit direction e, $t > 0$, and neighbourhoods $U_{\mathrm{o}}, U_{\mathrm{t}}$ of $x + i\underline{y_m}$ such that, for every $b \in [0, t]$:

- $\Lambda_e(ib)w \in \mathcal{T}_n$ for all $w \in U_{\mathrm{o}} \cap \mathcal{T}_n$;
- $\sigma\Lambda_e(-ib)w \in \mathcal{T}_n$ for all $w \in U_{\mathrm{t}} \cap \sigma^{-1}\mathcal{T}_n$;
- the same holds at $b = t$ for $w = x + i\underline{y_m}$ itself.

The profiles are $y_m = (m+1)^{-1}d$, where $d_k = k - 1$ except that $d_{i+1} = i - 1$. Thus the entries at i and $i+1$ agree; all later entries keep their original values.

(b) (C18) Under spin compatibility, $t \mapsto K_d(\Lambda_e(t))$ extends to an entire function of $t \in \mathbb{C}$, for each d and e.

(c) (C19) Assume (TC), cone support V.2.4(c), spin compatibility, S symmetric, the inclusions of P compatible with S, (b), and the real covariance V.2.6(g). Let v be labels, Q a pair, and y a spectator profile admitting the paths of (a). Then every $x \in \Gamma_Q$ carries two germs:

- g_x^{o}, holomorphic near $x + i\underline{y}$ and equal to $\mathcal{W}_n(v)$ on $\mathcal{T}_n$ near that point;
- g_x^{t}, equal to $\mathcal{W}_n(v \circ s) \circ \sigma$ on $\sigma^{-1}\mathcal{T}_n$ near that point.

(d) (C20) Assume S and T symmetric, the inclusions of P compatible with S, T smooth, T respects S, (SC) and (TC). For any such germs and any $x \in \Gamma_Q$ with all $x_k^0 = 0$, $g_x^{\mathrm{o}} = g_x^{\mathrm{t}}$ near $x + i\underline{y}$.

(e) (C21) Assume (D) and (TC). Each $x \in \Gamma_Q$ has a radius $r > 0$ such that, for compactly supported f with support in $B(x, r)$:

$W_n(v)[\underline{y}](\mathit{https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units}$

(f) (Boundary exchange) Suppose:

- $\mathrm{supp}\mathcal{F}W_n(u) \subseteq \mathcal{V}_n$ for all u;
- the cutoff satisfies (D);
- for a sequence of spectator profiles $y_m \to 0$ there are germs satisfying (d) and (e).

Then $W_n(v)(f) = W_n(v \circ s)(f \circ \sigma)$ for every $f \in \mathcal{S}$ with $\mathrm{supp} f \subset \Gamma_Q$. When this holds for every arity, label tuple and adjacent pair, W satisfies *joint locality*.

(g) (C27) Joint locality implies *word locality*: $W(a\,(u,f)(u',g)\,c) = W(a\,(u',g)(u,f)\,c)$ for all words a, c, labels u, u', and tests f, g such that $\mathrm{supp} f - \mathrm{supp} g$ consists of spacelike vectors.

Status: theorem. *Lean:*

- `C17.answer`, `C18.answer`, `C19.answer`, `C20.answer`, `C21.answer`, `C27.answer`.
- Statements `Bid.C17`, `C18`, `C19`, `C20`, `C21`, `C27`.
- Path definitions: `augment`, `travelable`, `Weighty`.
- Pair definitions: `NearPair`, `gap`, `spectator`, `Fleck`.
- Germ definitions: `Panes`, `sameGround`, `Forms`.
- Locality lemmas: `Panes.reaches`, `Panes.forgets`, `NearPair.actor`, `Actor`, `Surface.Regional`.

Proposition V.2.8 (real-time vectors, Gram limits, nulls; C22–C26, C29).

Heated vectors. For $n \geq 1$, take $v \in \mathcal{F}^n$, heights $y \in E_n$ with positive entries, $p \in (0,1)$ satisfying $4p < y_k$ and $4p < y_j - y_i$ for $i < j$, and real points $r \in \mathbb{R}^{4n}$. Set $z_k = r_k^0 + iy_k$. The heated vector is

$$\Psi_p(v;y;r) = e^{-(y_1-p)H} e^{ir_1^0 H} U(\vec{r}_1)\, B_p(v_1)\, G_p(z_2 - z_1, \vec{r}_2 - \vec{r}_1)\, B_p(v_2) \cdots G_p(z_n - z_{n-1}, \vec{r}_n - \vec{r}_{n-1})\, B_p(v_n)\Omega.$$

For $n = 0$ set $\Psi_p = \Omega$. For a word $b = ((u_1, f_1), \dots, (u_n, f_n))$ and $t > 0$, the heat-cup vector is

$$\Phi_t(b) = \int f_1(x_1) \cdots f_n(x_n)\, \Psi_{p(t)}(u; (t, 2t, \dots, nt); x)\, dx, \qquad p(t) = \min(t,1)/8.$$

(a) (C22, the pointwise Gram identity) Assume (H), J the induced star, the outputs of V.2.2(a)–(c), the orientation of P, and T respects S. Take $x \in E_n$ and $y \in E_m$ with positive entries, and widths t, u admissible with all heights exceeding $4t$ (resp. $4u$). Then

$$\langle \Psi_t(v;x;r), \Psi_u(w;y;q) \rangle = F^h_{n+m}(\varrho)\,(Jv_n, \dots, Jv_1, w_1, \dots, w_m),$$

with $h = (-x_n, \dots, -x_1, y_1, \dots, y_m)$ and $\varrho = (r_n, \dots, r_1, q_1, \dots, q_m)$.

(b) (C23, integrability and translation covariance of heat cups) Assume (H), the slab-insertion properties V.2.2(a), and spin compatibility. For every word b and $t > 0$, the integrand of $\Phi_t(b)$ is Bochner integrable, and $\Phi_t(g \cdot b) = U(a)\Phi_t(b)$ for translations $g = (a, 1)$, where $U(a)$ is the unitary translation of $\mathcal{H}$.

(c) (C24, the heat-cup Gram limit) Assume (H), (SC), (BV), integrability and translation covariance of heat cups and the pointwise Gram identity. Then $\langle \Phi_s(b), \Phi_{s'}(c) \rangle \to \langle b, c \rangle_W$ as $(s, s') \to (0^+, 0^+)$ independently.

(d) (C25) For every $n \geq 0$, tensor products span a dense subspace of $\mathcal{S}(\mathbb{R}^{4n})$.

(e) (C26, label faithfulness) Assume C25, $N_d(T) = N_d(S)$ for all d, the hierarchies for S, T, compatibility of the inclusions of P with both tables, (SC) and (TC). A label $u \in \mathcal{F}$ with $W(a\,(u, f)\,c) = 0$ for all words a, c and all f is zero. This concerns single labels, not words.

(f) (C29, no hypotheses) For every boundary family W, every antilinear map J, all words a, c and all labels u, u', the following hold:

- (word bilinearity) $(u, f) \mapsto W(a\,(u, f)\,c)$ is bilinear;
- (continuity in the two tests) $(f, g) \mapsto \langle (u, f)a, (u', g)c\rangle_W$ is continuous on $\mathcal{S}(\mathbb{R}^4)^2$.

Status: theorem. *Lean:*

- `C22.answer`, `C23.answer`, `C24.answer`, `C25.answer`, `C26.answer`, `C29.answer`.
- Statements `Bid.C22`, `C23`, `C24`, `C25`, `C26`, `C29`.
- Definitions `Hull`, `Roof`, `Small.shot`, `Small.heatCup`, `Small.Lamp`, `Surface.kernel`, `Persists`, `Symbol`, `Cam`.

Proposition V.2.9 (the target: a Wightman structure realizing W).

A *Wightman structure* consists of:

- families $K(\Lambda)$ of linear automorphisms of $\mathcal{F}$ and of each F_d;
- a field system: a Hilbert space $\mathcal{H}_W$, a subspace $\mathcal{D}$, a vacuum $\Omega_W \in \mathcal{D}$, and fields $\varphi(u, f) \in \mathrm{End}(\mathcal{D})$, bilinear in $(u, f) \in \mathcal{F} \times \mathcal{S}(\mathbb{R}^4)$;
- an antilinear map $J : \mathcal{F} \to \mathcal{F}$;
- a unitary, strongly continuous representation U of $\mathcal{P}$;
- a projection-valued measure E on $\mathbb{R}^4$.

It *realizes* W if:

- J is the induced star and K satisfies spin compatibility;
- totality and temperedness: $\|\Omega_W\| = 1$; $\mathcal{D}$ is dense and equals the span of the word vectors $\varphi(b)\Omega_W$; $f \mapsto \varphi(u, f)\psi$ is continuous;
- the adjoint identity: $\langle \varphi(u, f)\psi, \chi\rangle = \langle \psi, \varphi(Ju, \bar{f})\chi\rangle$ on $\mathcal{D}$;
- vacuum expectations: $\langle \Omega_W, \varphi(b)\Omega_W\rangle = W(b)$;
- local commutativity: $[\varphi(u, f), \varphi(u', g)] = 0$ on $\mathcal{D}$ when $\mathrm{supp} f - \mathrm{supp} g$ is spacelike;
- field faithfulness: for $u \neq 0$ some $\varphi(u, f) \neq 0$;
- covariance: $U(g)$ preserves $\mathcal{D}$ and fixes Ω_W, and $U(g)\varphi(u, f)\psi = \varphi(K(\Lambda)u, g_* f)U(g)\psi$;
- E is the spectral measure of the translations and vanishes off $\bar{V}_+$;
- every translation-invariant vector is a multiple of Ω_W.

The *field axioms without vacuum uniqueness* (VIII.1) are the same list without the last clause. *Status:* definition (a specification). It is produced in VIII.1 by the construction of V.2.10. Consequence: `Fields.general` and `Fields.sees` show that a realization has Gram form $\langle \varphi(b)\Omega_W, \varphi(c)\Omega_W\rangle = \langle b, c\rangle_W$, so W is positive on words. *Lean:* `Wightman`, `Wightman.Realizes`, `Fields`, `Fields.Total`, `Symmetries`, `Action`, `FieldCore`.

Proposition V.2.10 (field realization of W). Fix rotation data R and a real structure on S. Let J be the induced involutive star and assume spin compatibility. Let W satisfy word bilinearity, continuity in the two tests, word locality, label faithfulness, continuity of word matrix elements and invariance of the word Gram form. Suppose vectors $\Phi_s(b) \in \mathcal{H}$, $s > 0$, satisfy the Gram limit V.2.8(c), $\Phi_s(\emptyset) \to \Omega$ as $s \downarrow 0$, and, for every word b and translation a, $\Phi_s(a \cdot b) = U(a)\Phi_s(b)$ for

all sufficiently small $s > 0$. Assume T satisfies the normalized, unital, permutation-symmetric Euclidean laws. Then there is a Wightman structure satisfying every clause of V.2.9 except possibly vacuum uniqueness. If the common fixed vectors of all e^{-tH}, $t > 0$, are multiples of Ω, it realizes W. *Status:* theorem. The route used by the main theorem is `Next.BOUNDARY.establishing` followed by `Compatible.vacuum` (VIII.1). `Boundary.harvest` states the same in one lemma and is used only by an alternative route. The ingredients are in §4.8.

Not reconstruction. Integrated.lean also proves three ultraviolet bundles, and the folders Overlay and Vac are ultraviolet material; they belong to Part III (III.6, III.10–III.13):

- `source_sections` (G01–G07);
- `pressure_grants` (E01–E09, given `EBare`). The energy hypothesis is proved by `Next.Fluctuations.bare`;
- `stationary_scene` (E02, F01–F04).

3.22.2 2. What it uses

Inputs from other units.

1. **The Euclidean hierarchy** (IV.1; for each phase, V.1): `Table.Hierarchy`. It consists of permutation symmetry, normalization, the unit-insertion law, compatibility with signed symmetries and conjugation, translation homogeneity, reflection positivity on words and on smeared joint tests, smoothness off coincidences, and capped polynomial bounds near collisions. Its uses include:
   - permutation symmetry of S and T (C02, C20, C22, C26);
   - smoothness of T off coincidences (C07, C20, C26);
   - translation homogeneity of T (C02, C09);
   - the reflection adjoint on labels (the star J);
   - the null spaces (null-space compatibility, C26).
2. **The OS space and the slab operators** (V.1, the B-items; files YM/OS, YM/Heat).
   - `OS.Chamber` provides:
     - the GNS space with vacuum Ω;
     - the semigroup e^{-tH} and its holomorphic extension $e^{-\zeta H}$, a contraction for $\operatorname{Re}\zeta > 0$, with the semigroup law and $e^{-tH}\Omega = \Omega$ (Heat/Rows, Heat/Extend);
     - spatial translations;
     - a unitary space-time translation group whose spectral measure lies in $\{p^0 \geq 0\}$;
     - the ground space, i.e. the invariant vectors.
   - `OS.Spread`: slab operators $A_d(w)$ acting on GNS vectors by concatenation of words.
   - The laws `Spread.Laws`. Among them is the slab bound (SB), `AllCopy`: for each degree bound D and representative bound R there are $C > 0$ and $a \in \mathbb{N}$ such that

     $$\|A_d(w)\| \leq C^{|w|}(|w|/r)^{a|w|}$$

     for every word w whose labels have degree at most D and representative norm at most R, in a slab of width d, with pairwise distances in the maximum norm and distances from both time boundaries at least $r \in (0, 1]$. This is the only quantitative input of the unit; it gives the bound p^{-a} of C01.
   - VIII.1 obtains (H) from the phase data, both for the mixed table $S = T$ and for the selected phase.

3. **Rotation, spin and star data** (VI.1).
   - R with Euclidean covariance and rotation laws (A10–A15).
   - K with spin compatibility (A16–A17).
   - A16 itself, whose proof `Work/A16` is a field of the bundle.
   - The induced star J (the induced-star identity, from the orientation of the differential plan).
4. **The label structures** of I.1 and II.1: the gauge group, the register, the quotient labels, pin, and the common level of finitely many labels.

Standard mathematics.

- *Osterwalder–Schrader reconstruction:* Osterwalder–Schrader, *Axioms for Euclidean Green's functions* I, II, CMP 31 (1973), 42 (1975); Glimm–Jaffe, *Quantum Physics*, Ch. 6 and 19.
  - Here the continuation is built from operators on the OS space, so the Laplace-transform estimates of OS II are not needed.
- *Boundary values of holomorphic functions of polynomial growth as distributions of finite order:* Hörmander, *ALPDO I*, Thm. 3.1.15; Vladimirov, *Methods of the Theory of Generalized Functions*; Streater–Wightman, Thm. 2-10.
- *Fourier–Laplace transforms of distributions with support in a cone, and holomorphy in tubes:* Reed–Simon II, Thm. IX.16; Streater–Wightman, Thm. 2-6.
- *Locality:* Jost points and the complex Lorentz group (Jost, *The General Theory of Quantized Fields*; Streater–Wightman, Thms. 2-11, 2-12). Locality from the permutation symmetry of the Schwinger functions follows OS I.
  - A holomorphic germ is determined by its restriction to a maximal totally real subspace (elementary, by power series).
- *Hilbert-space tools:* the Kolmogorov (GNS) construction; Riesz–Markov; Stone–Weierstrass; the spectral measure of a continuous unitary representation of $\mathbb{R}^4$ (Stone–Naimark–Ambrose–Godement); Reed–Simon I, Ch. VII–VIII.
- *Other standard facts:*
  - density of tensor products in $\mathcal{S}(\mathbb{R}^{4n})$ (Reed–Simon I, §V.3);
  - continuous one-parameter groups in finite dimension are exponentials;
  - generation of $\mathcal{L}_+^{\uparrow}$ by rotations and boosts.

Imported constants and assumptions.

- There are no numerical constants beyond conventions: the width $p < 1$, gaps $> 4p$, and the choice $p = \min(k, 1)/8$. The exponents a and constants C of C01 are those of (SB) (V.1).
- The hypotheses not discharged here are:
  - (H);
  - the data R, K, J with their properties;
  - $N_d(T) = N_d(S)$ for every d;
  - one-dimensionality of the heat-fixed space.

  VIII.1 supplies the table, slab and symmetry data from V.1 and VI.1. One-dimensionality of the heat-fixed space is supplied for the selected phase.
- The boundary family W, (BV) and cone support are produced by V.2.3(b) and V.2.4(c) from the outputs of C03, C04, C06 and C09.

3.22.3 3. Strategy

The slab bound controls a field insertion $B_p(v)$ between two heat factors. The relation $B_{p+u} = e^{-uH} B_p e^{-uH}$ makes products of these insertions independent of the admissible width. Their vacuum matrix elements give a continuation to strictly ordered imaginary times, with polynomial growth as the time differences shrink. Cauchy–Riemann identities and Taylor's formula along $(0, 1, \dots, n-1)$ then give tempered boundary values for arbitrary approaches through ordered heights. A Laplace identity yields total energy zero and nonnegative suffix energies.

Euclidean covariance gives Poincaré covariance of the boundary family: translations and spatial rotations pass through a one-variable identity theorem; infinitesimal rotations in a time–space plane give the boost identity, which is integrated along the real boost groups. Lorentz covariance improves the spectral support to $\mathcal{V}_n$. Fourier–Laplace transformation extends the continuation to the forward tube. Locality follows from two holomorphic germs near each spacelike adjacent pair: imaginary boosts construct the germs, Euclidean permutation symmetry identifies their jets on the Wick slice, and continuation along a rescaling of real times propagates their agreement. Partial boundary values then give exchange of adjacent spacelike fields.

Finally, inner products of heated OS vectors equal the continuation at a glued, reflected configuration. Their limit gives positivity of the word kernel and an isometric embedding of its Kolmogorov space into the OS Hilbert space. Left multiplication defines the fields on the common word domain. The embedding intertwines translations, transferring positive energy; Lorentz covariance gives the forward-cone spectrum condition. A one-dimensional OS heat-fixed space gives vacuum uniqueness. These are the operator and local-germ forms of Osterwalder–Schrader reconstruction used here.

3.22.4 4. Steps

3.22.4.1 4.1 Tools Lemma V.2.a (Kolmogorov). Let K be a positive semidefinite kernel on a set X. There are a Hilbert space and vectors $(\psi_x)_{x\in X}$ with $\langle \psi_x, \psi_y\rangle = K(x,y)$ and dense span. *Proof sketch.* Complete the finitely supported functions modulo the null space of $\sum \bar{f}(x) g(y) K(x,y)$. *Lean:* `Kernel.host`, `Kernel.Gram`.

Lemma V.2.b (projection-valued measure). Let X be a compact Hausdorff space and $A : C(X) \to \mathcal{B}(\mathcal{H})$ a unital complex $*$-algebra homomorphism. There is a strongly countably additive projection-valued measure on the Borel sets of X representing A. *Proof sketch.*

1. For each ψ, the positive functional $f \mapsto \langle \psi, A(f)\psi\rangle$ gives a Riesz–Markov measure μ_ψ.
2. $f \mapsto A(f)\psi$ extends to an isometry $L^2(\mu_\psi) \to \mathcal{H}$.
3. $E(s)\psi$ is the image of 1_s; for disjoint Borel sets, the corresponding indicator sums converge in $L^2(\mu_\psi)$, giving strong countable additivity.

It is used (V.1, SNAG/Escape) on the one-point compactification of $\mathbb{R}^4$ to build the spectral measure of the translations. *Lean:* `Spectral.Rep.ships` *(proof read for additivity; Compact/Borel statement only).*

Lemma V.2.c (Schwartz calculus; tensor density, C25).

1. Let $f_\lambda, g \in \mathcal{S}(\mathbb{R}^d)$. If every jet of f_λ converges locally uniformly to the corresponding jet of g, and every Schwartz seminorm of f_λ is eventually bounded uniformly in λ, then $f_\lambda \to g$ in $\mathcal{S}$.
2. Let K be compact metric, μ a finite Borel measure, and $u \mapsto f_u$ continuous from K into $\mathcal{S}(\mathbb{R}^d)$. If $g \in \mathcal{S}$ satisfies $g(x) = \int_K f_u(x)\, d\mu(u)$, then the sums $\sum_j \mu(E_j) f_{u_j}$ over finite measurable

partitions, with $u_j \in E_j$ and $\max_j \operatorname{diam}(E_j) \to 0$, converge to g in $\mathcal{S}$.
3. The linear span of tensor products is dense in $\mathcal{S}(\mathbb{R}^{4n})$.

Proof sketch. For 3:

- reduce to compactly supported v;
- choose a product cutoff equal to 1 on the support;
- replace $\hat{v}$ by compactly supported cutoffs $\hat{v}_N \to \hat{v}$ in $\mathcal{S}$;
- the fixed product cutoff times $(2\pi)^{-4n} \int \hat{v}_N(\xi) e^{i\langle \xi, \cdot \rangle}\, d\xi$ converges to v by Fourier inversion;
- approximate each compact integral by quadrature sums of products of cutoff plane waves, using 2. *Lean:* `Spark.ready`, `Spark.records`; `C25.compact_mem`, `C25.cut_port_mem`.

3.22.4.2 4.2 The strict-time continuation Lemma V.2.d (slab insertions; C01). As in V.2.2(a). *Proof sketch.*

- **Construction.** For each degree, B_p is defined on representatives by the one-point slab operator.
- **Linearity and descent to quotient labels.** An operator on $\mathcal{H}$ is determined by its matrix elements between GNS vectors. These matrix elements are Schwinger functions of T on the word *reflected word, centre insertion, raised word*; they are linear in the inserted label and vanish on S-null labels because T respects S. The degrees are glued using compatibility with the inclusions. For $p = 0$ the map is set to 0.
- **Bound.** (SB) applies to a word of length 1 in a slab of width $2p$ with buffer $r = p \le 1$ and gives $\|B_p(\mathrm{pin}_d w)\| \le C(1/p)^a$. The constants depend on the degree and on the norm of the representative.

Lean: `C01.single`, `C01.inner_raw`, `C01.at_width`, `C01.answer`.

Lemma V.2.e (Euclidean values and width independence; C02). As in V.2.2(b). *Proof sketch.*

- **Euclidean values.** At imaginary times each factor G_p is the heat semigroup at a positive real time. By induction from the right, the vector $B_p(v_k)G_p \cdots B_p(v_n)\Omega$ is the GNS vector of a positive-time Euclidean word. At the end, $\langle \Omega, \text{vector} \rangle$ is the Schwinger function, and translation invariance removes the offset.
- **Width independence.**
  - First, $B_{p+u} = e^{-uH} B_p e^{-uH}$.
  - Second, $G_p(w) = e^{-uH} G_{p+u}(w) e^{-uH}$.
  - The factors e^{-uH} telescope, and $e^{-uH}\Omega = \Omega$ removes the last one.

Only gaps $> 2p$ are needed; the factor 4 in "admissible" is slack. *Lean:* `C02.train_pure`, `C02.pad_bar`, `C02.matrix_pad`.

Lemma V.2.f (holomorphy and polynomial growth; C03). As in V.2.2(c). *Proof sketch.*

- **Definition.** $F_n(z) = M_p(z)$ with $p = \min(k, 1)/8$, where $k > 0$ is any lower bound for the imaginary-time differences when $n \ge 2$; take $k = 1$ for $n = 0, 1$. By V.2.e this does not depend on the admissible p, and admissibility is an open condition.
- **Holomorphy and continuity.** These follow from those of $\zeta \mapsto e^{-\zeta H}$ in norm, combined with strong continuity of U, through the finite product.

- **Growth.** For $0 < k \le 1$, take $p = k/8$. The factors G_p are contractions, so $|F| \le \|\Omega\|^2 \prod_j \|B_p(v_j)\| \le \|\Omega\|^2 (\prod_j C_j 8^{a_j}) k^{-\sum_j a_j}$. This gives (PG) with $m = \sum a_k$.

Lean: `C03.defined_exists`, `C03.defined_region`, `C03.defined_power`.

3.22.4.3 4.3 Boundary values and spectral support Lemma V.2.g (C04). As in V.2.3(a). *Proof sketch.*

- (PG) gives a bound on F^y that is uniform in x and locally uniform in $y \in E_n$; this yields integrability and continuity by dominated convergence.
- On each one-variable slice $w \mapsto F(\cdot + w\underline{a})$, Cauchy–Riemann gives $\partial_t F(y + ta) = i\,\partial_{\underline{a}} F$.
- Cauchy estimates on discs of radius uniform in x allow differentiation under the integral.
- Integration by parts moves $\partial_{\underline{a}}$ onto f.

Lean: `C04.cr_height`, `C04.deriv_bound_near`, `C04.lines`.

Lemma V.2.h (boundary values). As in V.2.3(b). For every nondecreasing y, including $y = 0$, the boundary value is given by an explicit formula. Let $\varrho = (0, 1, \dots, n-1)$ and $D = -i\partial_{\underline{\varrho}}$, and let m be the exponent of (PG). Then

$$W_n^y(v)(f) = \sum_{k=0}^{m} \frac{(-1)^k}{k!}\, \psi_v(y + \varrho; D^k f) - \int_0^1 \frac{(-s)^m}{m!}\, \psi_v(y + s\varrho; D^{m+1} f)\, ds.$$

Proof sketch.

- For $y \in E_n$, Taylor's formula with integral remainder along $t \mapsto y + t\varrho$ (by (CR)) shows $W^y = \psi(y; \cdot)$.
- For nondecreasing y the profile $y + s\varrho$ has all gaps $\ge s$. So (PG) bounds the remainder integrand by $C\|D^{m+1} f\|_{L^1}/m!$, uniformly in y.
- Hence:
  - $y \mapsto W^y(f)$ is continuous on the closed set of nondecreasing heights, by dominated convergence;
  - $|W^y(f)|$ is bounded by finitely many L^1 norms of $D^k f$, so W^y is tempered.
- Multilinearity in v follows from uniqueness of limits. At $y = 0$ this gives (BV).

Lean: `Waves.desired`, `Waves.meant`, `Waves.maintains`, `Waves.cost`, `Waves.seeks`.

Lemma V.2.i (Laplace identity; C05). As in V.2.4(a). *Proof sketch.* Put $A(s) = \psi_v(sy; \widehat{e^{s\langle \underline{y}, \cdot \rangle} f})$ for $s > 0$.

1. A **is constant.** By (CR) along y, and since $\frac{d}{ds} \widehat{e^{s\langle \underline{y}, \cdot \rangle} f} = i\,\partial_{\underline{y}} \widehat{e^{s\langle \underline{y}, \cdot \rangle} f}$ in $\mathcal{S}$ (here compact support is needed), $A'(s) = 0$. The product rule is justified by the uniform bound $|\psi_{y'}(g)| \le \mathrm{toll}(g)$ of V.2.h.
2. **The limit.** As $s \to 0^+$, $A(s) \to W_n(v)(\hat{f})$, by the radial limit and the same bound.

Only the radial limit is used. *Lean:* `C05.hasDerivAt_zero_psi`, `C05.tendsto_psi`.

Lemma V.2.j (support in $\mathcal{P}_n$; C06). As in V.2.4(b). *Proof sketch.* Let y be nondecreasing and f compactly supported in $\{p : \langle \underline{y}, p \rangle < -1\}$.

1. Apply V.2.i at the ordered heights $Y(t) = ty + \varrho$, whose gaps are all ≥ 1.
2. (PG) with $k = 1$ gives $|\psi_{Y(t)}(\hat{g})| \le C\|\hat{g}\|_{L^1}$.

3. Absorb $e^{\langle \underline{\varrho}, \cdot \rangle}$ into the compact test. Every seminorm of the remaining factor $e^{t\langle \underline{y}, \cdot \rangle}$ times that test is bounded by a polynomial in t times e^{-t}, and tends to zero.
4. Hence $W(\hat{f}) = 0$. Rescaling y removes the -1.

Constant profiles of both signs force $\sum_k p_k^0 = 0$. *Lean:* `C06.patch`, `C06.lift_falls`.

3.22.4.4 4.4 Poincaré covariance Lemma V.2.k (Poincaré action on $\mathcal{S}$; C08). As in V.2.5(b). *Proof sketch.*

- **Continuity.** The seminorm estimate $p_{i,k}(f \circ (A(\cdot - b))) \leq (\|A^{-1}\| + |b|)^i \|A\|^k 2^i \max_{(i',k') \leq (i,k)} p_{i',k'}(f)$ combines with locally uniform convergence of jets (V.2.c, item 1).
- **Derivative.** The pointwise derivative in t of $\Lambda_a(t)_* f$ is $-\Lambda_a(t)_* D_a f$. The slope error is written as $\int_0^1 (\cdot)\, du$ (fundamental theorem of calculus) and controlled in $\mathcal{S}$ by the continuity just proved.

Lean: `C08.size_affine_le`, `C08.continuous_gpush`, `C08.probe_slope_of_pointwise`.

Lemma V.2.l (boost identity at strict heights; C07). As in V.2.5(c). *Proof sketch.*

1. **Euclidean input.** Differentiating the $O(4)$ covariance of T along $\exp(s\, e_0 \wedge e_a)$ at a non-coincident configuration gives the rotation Ward identity: the derivative of $T(x; v)$ along the rotation field equals $-\sum_k T(x; \dots, \rho(e_0 \wedge e_a) v_k, \dots)$.
2. **The analytic function $E(z)$.** Fix f with compact support and introduce a complex parameter z and the times $z\, x_k^0 + i y_k$. Let $E(z)$ be the integral of

$$F \cdot \Big(\sum_k [x_k^a \partial_{x_k^0} + z(z x_k^0 + i y_k) \partial_{x_k^a}] f \Big) - z \sum_k F(v^{(k)}) f.$$

   The times stay strictly ordered on $\operatorname{supp} f$ for z in a convex open set containing $\mathbb{R}$, and E is holomorphic there.
3. **E vanishes on a segment of the imaginary axis.** At $z = iw$, $w > 0$ small, the configuration is Euclidean, with times $y_k + w x_k^0$. There the vector field is w times the pull-back of the $(0, a)$-rotation field, and F is the Schwinger function. By step 1 the integrand is the divergence of the compactly supported vector field fT times the divergence-free rotation field, so $E(iw) = 0$.
4. **Conclusion.** The identity theorem gives $E(1) = 0$, which is (B_y) for compactly supported f. Density (cutoffs) and the continuity of $f \mapsto \psi(y; f)$ (V.2.h at fixed y) give all f.

The term $i y_k \partial_{x_k^a}$ is the imaginary part of the complex time multiplying the spatial derivative. *Lean:* `C07.oriented_differential`, `C07.integral_orbit`, `C07.error_imag`, `C07.error_one`, `C07.compact_boost`, `C07.answer`.

Lemma V.2.m (real Poincaré covariance; C09, A16). As in V.2.5(a), (d). *Proof sketch.*

- **Translations by $a = (\vec{a}, a^0)$.** For fixed $y \in E_n$, $F_n^y(x + a) = F_n^y(x)$. This follows from a one-variable identity theorem in c for the times $i \operatorname{Im} z_k + c(\operatorname{Re} z_k + a^0)$: at $c = ib$ the times are imaginary, and Euclidean translation invariance by $(\vec{a}, b a^0)$ applies. Change variables and let $y \to 0$ (BV).
- **Spatial rotations.** The same argument applies, using $O(3) \subset O(4)$ covariance and $K = R$ on rotations.
- **Boosts.**

  - Passing (B_y) to $y \to 0$ removes the term $\sum iy_k\partial_{x_k^a}$ and gives $\sum_k W(v^{(k)})(f) = W(v)(D_a f)$.
  - The function $h(t) = W_n(K(\Lambda_a(t))v)(\Lambda_a(t)_* f)$ has derivative 0 at $t = 0$, by C08(b), the boost derivative, and finite-dimensional expansion of the labels.
  - By the group law h' vanishes everywhere, so h is constant.
- **The Lorentz group.** The Λ for which covariance holds form a subgroup containing the rotations and the coordinate boosts. A16 makes it all of $\mathcal{L}_+^\uparrow$. The proof of A16 is purely group-theoretic: up to three boosts bring a future unit vector to rest, and its stabilizer consists of rotations.

Lean: `C09.time_compare`, `C09.restricted_translation`, `C09.restricted_spatial`, `C09.boost_corner`, `C09.boost_invariant`, `A16.answer`.

Lemma V.2.n (word covariance and cone support; C28). As in V.2.5(e) and V.2.4(c). *Proof sketch.*

- **Gram invariance.** The induced star commutes with the letterwise action by spin compatibility. Pushed tensor tests are g_* of the tensor test; then apply (Cov).
- **Continuity.** Use C08(b, first part), continuity of $\Lambda \mapsto K(\Lambda)$ on a finite-dimensional level, and continuity of $W_n(v)$.
- **Lorentz improvement of spectral support.** The Fourier transform intertwines Λ with $\tilde{\Lambda}$, so covariance carries the support condition $\mathcal{P}_n$ to $\tilde{\Lambda}^{-1}\mathcal{P}_n$ for every Λ. The properties of $\mathcal{V}_n$ follow from half-space support applied to step profiles and to constant profiles.

Lean: `C28.answer`, `tents`.

3.22.4.5 4.5 Continuation to the full tube

Lemma V.2.o (cone geometry, cutoff, damping; C10–C12). As in V.2.6(a)–(c). *Proof sketch.*

- **C10.** Summation by parts gives $\langle p, \eta\rangle = \sum_k \langle P_{k+1}, \eta_{k+1} - \eta_k\rangle$, where $P_j = \sum_{l\geq j} p_l$ and $P_1 = 0$. Each term is ≥ 0 because $\bar{V}_+$ is self-dual for the Euclidean pairing. For the local bound: if $B(\eta, r) \subset V_n^+$ and $p \neq 0$, then for $u \in B(\eta, r/2)$ the point $u - \frac{r}{2}p/|p|$ lies in $\bar{V}_n^+$, so $\langle p, u\rangle \geq \frac{r}{2}|p|$.
- **C11.** Take χ_n = (a bump of integral 1 and outer radius 1/2) $*$ (indicator of the open 2-neighbourhood of $\mathcal{V}_n$).
- **C12.** By C10, $-\langle \eta, p\rangle \leq 3|\eta|$ on the 3-neighbourhood. Leibniz bounds then hold uniformly for bounded $\eta \in \bar{V}_n^+$, and continuity into $\mathcal{S}$ follows by V.2.c, item 1.

Lean: `C10.telescope`, `C10.cone_bound`, `C11.answer`, `C12.cone_estimate`, `C12.attenuate_estimate`.

Lemma V.2.p (holomorphy in the tube; C13). As in V.2.6(d). *Proof sketch.*

1. **Exponential reserve.** For $z \in \mathcal{T}_n$, on $\mathrm{supp}\chi_n$ one has $\mathrm{Re}\,(i\langle p, z\rangle) = -\langle p, \mathrm{Im}\, z\rangle \leq A - c|p|$, by C10 and distance ≤ 3. So $e^{i\langle\cdot, z\rangle}\chi_n$ has finite seminorms weighted by $e^{c|p|/2}$.
2. **Differentiability.** Write $L_z(p) = e^{i\langle p, z\rangle}\chi_n(p)$. For each Schwartz seminorm q, exponential decay gives $q(L_{z+h} - L_z - i\langle\cdot, h\rangle L_z) \leq C_q|h|^2$ for all sufficiently small h. Thus $z \mapsto L_z$ is complex Fréchet differentiable into $\mathcal{S}$.
3. **Analyticity.** Compose with the continuous functional $\mathcal{F}W$. Complex differentiability on an open set gives analyticity.

The argument applies to every Schwartz test with the prescribed exponential form; C13 asserts the existence of such tests and the resulting holomorphy. *Lean:* `C13.lens_reserve`, `C13.analytic_interior`; `ExpReserve.of_cut`, `hasFDerivAt_shift`.

Lemma V.2.q (partial boundary values as integrals; agreement with F; C14, C15). As in V.2.6(e), (f). *Proof sketch.*

- **C14.** For real x the Fourier–Laplace test at $x+i\eta$ is the modulation $e^{i\langle\cdot,x\rangle}g$, with $g = e^{-\langle\eta,\cdot\rangle}\chi_n$. Hence $\int \mathcal{W}_n(v)(x+i\eta)f(x)\,dx = (\mathcal{F}W)(g\check{f})$. The distribution is moved inside the x-integral via Riemann sums in $\mathcal{S}$, first for compactly supported f, then by density. Integrability comes from the polynomial growth of $x \mapsto \mathcal{W}(x+i\eta)$. Cone support is not needed.
- **C15.**
  1. Since $\underline{y} \in V_n^+$ and $\operatorname{supp}\mathcal{F}W \subseteq \mathcal{V}_n$, the multipliers $e^{-\langle \underline{y},\cdot\rangle}\chi_n$ and $e^{-\langle \underline{y},\cdot\rangle}$ agree near the support.
  2. So V.2.i gives $W_n(v)[\underline{y}] = \psi_v(y;\cdot)$ on compactly supported tests, hence on all tests ($\psi(y;\cdot)$ is tempered by (I) and (PG)).
  3. With C14, $\int \mathcal{W}(x+i\underline{y})f = \int F^y(x)f$ for all f. Both integrands are continuous, so they are equal.

Lean: `C14.modulation_eq`, `C14.emit_integral`, `C15.blanketed_eq`, `C15.partialD_psi`.

Lemma V.2.r (Lorentz covariance in the tube; C16). As in V.2.6(g). *Proof sketch.*

- $\mathcal{V}_n$ is invariant under the dual action.
- The Fourier–Laplace test at Λz, composed with $\tilde{\Lambda}$, differs from the lens at z only away from $\mathcal{V}_n$: both cutoffs equal 1 there, and the phases agree by duality.
- Cone support, the intertwining of $\mathcal{F}$ with Λ and $\tilde{\Lambda}$, and (Cov) conclude.

Lean: `C16.lens_support`, `C16.answer`.

3.22.4.6 4.6 Locality

Lemma V.2.s (imaginary boosts at a spectator profile; C17). As in V.2.7(a). *Proof sketch.*

1. **Spectator profiles.** The profiles $y_m = d/(m+1)$ are spectator profiles. Monotone heights give $\underline{y_m} \in \bar{V}_n^+$, and σ fixes $\underline{y_m}$. The closure statements follow by approaching through strictly ordered heights.
2. **Choice of direction.** Let $\xi = x_{i+1} - x_i$ be spacelike, so $\vec{\xi} \neq 0$, and take $e = \vec{\xi}/|\vec{\xi}|$. Then
$$\operatorname{Im}(\Lambda_e(ib)w)_k = \operatorname{Im} w_k + \sin b \cdot c_e(\operatorname{Re} w_k) + (\cos b - 1)\, s_e(\operatorname{Im} w_k),$$
with $c_e(\vec{X},t) = (te, e\cdot\vec{X})$ and $s_e(\vec{X},t) = ((e\cdot\vec{X})e, t)$.
3. **The pair difference.** Its imaginary part is 0 at $b=0$, and $c_e(\xi) = (\xi^0 e, |\vec{\xi}|) \in V_+$ exactly because $|\xi^0| < |\vec{\xi}|$ (the Jost condition). So the difference enters V_+ for small $b > 0$.
4. **The other consecutive differences.** They are already in V_+ and stay there by continuity, uniformly near the base point.
5. **The swapped tube.** After the swap the pair difference is $-\xi$, and the opposite rapidity $-ib$ compensates.

For $n = 2$ the profile is identically 0, so the germs sit at real points. *Lean:* `C17.height_path`, `C17.fleck_of_spectator`, `C17.choose_direction`, `C17.enters`, `C17.local_paths`.

Lemma V.2.t (complexified labels; the two germs; C18, C19). As in V.2.7(b), (c). *Proof sketch.*

- **C18.** A continuous one-parameter group on a finite-dimensional space is e^{tX} (via $\int_0^t$ and invertibility for small t), and $e^{\zeta X}$ is entire.
- **C19.**
  1. Put all labels at one level d and write $K_{d,e}(\zeta)$ for the entire extension from C18. Define the germ at $x + i\underline{y}$ by $w \mapsto \mathcal{W}_n(\mathrm{pin}_d K_{d,e}(it)v)(\Lambda_e(it)w)$. It is holomorphic near $x + i\underline{y}$ because $\Lambda_e(it)(x + i\underline{y}) \in \mathcal{T}_n$ and $\Lambda_e(it)$ is complex linear.
  2. Fix $w \in \mathcal{T}_n$ near the point. The function $\zeta \mapsto \mathcal{W}_n(\mathrm{pin}_d K_{d,e}(\zeta)v)(\Lambda_e(\zeta)w)$ is holomorphic near the segment $i[0,t]$, by the path conditions, and constant for real ζ, by V.2.6(g).
  3. The identity theorem makes it constant on the segment, so the germ equals $\mathcal{W}_n(v)$ on $\mathcal{T}_n$ near $x + i\underline{y}$.
  4. The tilted germ uses $-it$, the labels $v \circ s$, and σ.

Lean: `C18.eq_exponential`, `C18.entire`, `C19.paneAlong`.

Lemma V.2.u (agreement at Euclidean points; C20). As in V.2.7(d). *Proof sketch.*

1. **The Wick slice.** Let $M(\vec{x}, \tau) = (\vec{x}, i\tau)$. Its complex span is $\mathbb{C}^{4n}$. If x has zero times, then $M(\vec{x}, y) = x + i\underline{y}$.
2. **The original germ.** On the open set of strictly ordered τ, M maps into $\mathcal{T}_n$, and $g^{\mathrm{o}} \circ M = \mathcal{W} \circ M = F$ at imaginary times. That equals the Schwinger function $T((\vec{x}_k, \tau_k); v)$ (V.2.6(f), (SC)).
3. **The tilted germ.** On the swapped-ordered set, $g^{\mathrm{t}} \circ M$ equals the same Schwinger function, by permutation symmetry of T.
4. **Smoothness at the base point.** The point $(\vec{x}, y)$ is non-coincident: the pair has equal times but $\vec{x}_i \neq \vec{x}_{i+1}$ by spacelikeness. So T is smooth there.
5. **Jets.** Both open sets accumulate at $(\vec{x}, y)$. Hence all real jets of $g^{\mathrm{o}} \circ M$ and $g^{\mathrm{t}} \circ M$ at that point equal those of T.
6. **Conclusion.** Real jets on a slice that spans over $\mathbb{C}$ determine all complex derivatives, so $g^{\mathrm{o}} = g^{\mathrm{t}}$ near $x + i\underline{y}$.

No Schwarz reflection and no edge-of-the-wedge theorem are used. *Lean:* `C20.jets_on_open_side`, `C20.complex_germ`, `C20.emit_table`.

Lemma V.2.v (local realization and exchange; C21). As in V.2.7(e), (f). *Proof sketch.*

- **C21.** As $\eta \to \underline{y}$ within V_n^+, $\int \mathcal{W}(\xi + i\eta) f(\xi)\, d\xi \to W[\underline{y}]$(*https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/f*), by C14 and continuity of $\eta \mapsto W[\eta]$ on $\bar{V}_n^+$. Near $x + i\underline{y}$ the germ is bounded and equals $\mathcal{W}$ on $\mathcal{T}_n$, so dominated convergence gives the integral of the germ. The swapped case is the same after a change of variables.
- **Propagation.** For $x \in \Gamma_Q$, the path $x \mapsto (\vec{x}, tx^0)$, $t \in [0,1]$, stays in Γ_Q. Agreement of the two germs is an open and closed condition along this path, so it passes from the zero-time point to x.
- **Local equality.** On a small ball around each point of Γ_Q both partial boundary values are integrals of the same germ, so their difference vanishes on Γ_Q.
- **Exchange.** Let $m \to \infty$. Then $W[\underline{y_m}] \to W[0] = W$, by continuity in η and because $\chi_n = 1$

near $\mathcal{V}_n$, and similarly on the swapped side.

Lean: `C21.realize`, `Panes.reaches`, `Panes.forgets`, `NearPair.actor`; `Pane.propagate`, `Blanket.flatD`.

Lemma V.2.w (locality for words; C27). As in V.2.7(g). *Proof sketch.* If $\mathrm{supp} f - \mathrm{supp} g$ is spacelike, the tensor test of the word $a\,(u,f)(u',g)\,c$ is supported in Γ_Q, with Q the slots of (u,f) and (u',g). Composing with σ swaps the two factors. Joint locality applies to all Schwartz tests supported in Γ_Q, so no density argument is needed. The pair may be at any position in a word of any length ≥ 2. *Lean:* `C27.coordinate_tsupport`, `C27.exchange`.

3.22.4.7 4.7 Real-time vectors, Gram limits, nulls

Lemma V.2.x (Gram form of heated vectors; C22). As in V.2.8(a). *Proof sketch.*

1. **Imaginary times.** At imaginary times each heated vector is the GNS vector of a positive-time Euclidean word.
   - Their inner product is the reflection-positive form, i.e. the Schwinger function of *reflected first word, second word*: times negated, order reversed, labels starred.
   - By (SC) this is F at $i \cdot h$. Here the star J appears because it is the induced reflection adjoint, and null-space compatibility is used.
2. **Continuation in real times.** Introduce one complex parameter c and the times $iy_k + c\,r_k^0$. Both sides are holomorphic on a strip $|\mathrm{Im}\, c| < \delta$; the left side is antiholomorphic in the first slot, handled by conjugation.
3. **Conclusion.** The two sides agree at $c = i\tau$ for small τ, hence at $c = 1$.

The width formula of C03 is not used. *Lean:* `C22.euclideanRef`, `C22.onImag`, `C22.pointJoined`.

Lemma V.2.y (heat cups and their Gram limit; C23, C24). As in V.2.8(b), (c). *Proof sketch.*

- **C23.**
  - $x \mapsto \Psi(\cdot; x)$ is continuous and bounded by $\|\Omega\| \prod \|B_p(v_k)\|$, since the propagators are contractions. So the integrand is Bochner integrable.
  - Translation covariance comes from the difference structure of Ψ and $U(a)\Omega = \Omega$.
- **C24.**
  1. By the pointwise Gram identity and Fubini, $\langle \Phi_s(b), \Phi_{s'}(c)\rangle$ is the integral of F at heights $h = (-ns, \dots, -s, s', \dots, ms')$ against the test $f_b^* \otimes f_c$.
  2. As $(s, s') \to 0$, $h \to 0$ within E_{n+m} along a two-parameter family with no fixed direction. The unrestricted boundary limit applies to this family, whose two scales vary independently.
  3. (BV) gives the limit $W(b^* c)$.

Lean: `C23.shot_control`, `C23.heatCup_shift`, `C24.toward`, `C24.integrated`.

Lemma V.2.z (nulls; word insertion; C26, C29). As in V.2.8(e), (f). *Proof sketch.*

- **C26.** Let $u = \mathrm{pin}_d[w]$ be a marker.
  1. By C25, every $W_n(\dots, u, \dots)$ vanishes as a distribution.
  2. By V.2.6(f), $F^y = 0$ for ordered y, so the Schwinger functions with w in some slot vanish at time-ordered non-coincident points.

3. Symmetry and continuity give vanishing at all non-coincident points. So $w \in N_d(T) = N_d(S)$, and $u = 0$.

- **C29.**
  - word bilinearity is the multilinearity of W_n composed with that of $\otimes$.
  - continuity in the two tests follows from continuity of $W_n(v)$, of $(f_1, \dots, f_n) \mapsto \otimes f_k$, and of conjugation.

Lean: `C26.marker_on_joint`, `C26.strict_zero`, `C26.read_zero`, `C29.answer`.

3.22.4.8 4.8 Assembly (VIII.1) and the field realization The reconstruction is assembled in `Next/BOUNDARY/Stage.lean`, in this order.

1. `rhythm` chains C01 → C02 → C03 → C04.
   - `Rhythm.opens` (V.2.h) produces W.
   - `at_contour` shows that any prescribed continuation–boundary pair for T satisfies (BV).
2. `illuminated` applies C07, C09 (with A16 and C08), C05 and C06. `spectra` uses Lorentz improvement of spectral support to get cone support.
3. `blankets` (C10, C12), `visible` (C13) and `neon_any` (C13–C15) give (TC). `emit_eq` shows that the continuation on $\mathcal{T}_n$ does not depend on the cutoff.
4. `windows` combines C17, C19 (with C18 and C16), C20 and C21. `Show.uses` then gets joint locality from the boundary exchange lemma and Lorentz improvement of spectral support.
5. `revealing` applies C22, C26 with C25, and C24 with C23.
6. `establishing` applies C28, C29 and C27, then builds the field system (step 7).
7. The field system is built (V.2.10) in the following order.
   1. The heat cups converge: the two-variable Gram limit makes $s \mapsto \Phi_s(b)$ Cauchy (`norms_kernel`, `Cups.out`). The limit vectors $\Phi_0(b)$ satisfy $\langle \Phi_0(b), \Phi_0(c) \rangle = \langle b, c \rangle_W$ and $\Phi_0(\emptyset) = \Omega$.
   2. Hence $\langle \cdot, \cdot \rangle_W$ is positive (`Outfield.safe`), and V.2.a gives its Kolmogorov space.
   3. Fields act by left multiplication of words. The identity $\langle (u, f)b, c \rangle_W = \langle b, (Ju, \bar{f})c \rangle_W$ shows that a null linear combination of words remains null after left multiplication, so the fields are well defined (`Soil.starter`).
   4. The same identity gives the formal adjoint. The other properties of W give the remaining clauses of V.2.9: temperedness (continuity in the two tests); local commutativity (word locality); field faithfulness (label faithfulness).
   5. A unitary representation U of $\mathcal{P}$ is defined by $U(g)[b] = [g \cdot b]$, using invariance of the word Gram form and continuity of word matrix elements (`Soil.mobilize`).
   6. $[b] \mapsto \Phi_0(b)$ is an isometry into $\mathcal{H}$ that intertwines translations and sends the vacuum to Ω (`universal_embed`, `Outfield.daylight`).
   7. **Spectrum condition** (`Action.spectrum`). The isometric intertwiner identifies each scalar spectral measure with that of its image in $\mathcal{H}$, by equality of characteristic functions. These measures are supported in $\{p^0 \geq 0\}$ by positivity of H. Lorentz covariance of U upgrades this to support in $\bar{V}_+$.
   8. **Vacuum uniqueness** (`Action.only`). Under one-dimensionality of the heat-fixed space, an invariant vector is mapped to the ground space of $\mathcal{H}$, which is $\mathbb{C}\Omega$.
8. `Compatible.vacuum` adds uniqueness for the selected phase. `after_materializes` runs the chain for the mixed table ($S = T$, without vacuum uniqueness) and for the selected phase.

3.22.5 5. Where the difficulty lies

The boost argument must control moving labels and tests without imposing a topology on the full label union. All labels are first placed in one finite-dimensional level; Schwartz estimates then justify the derivative and its boundary limit. The locality argument needs equality of complete jets on the Wick slice. Equality of values at zero real times would not determine the two germs. Smoothness of the Euclidean correlations at non-coincident configurations, including equal pair times, supplies those jets.

The Taylor formula in V.2.h gives a bound uniform over all nondecreasing heights. This uniformity permits varying tests in the Laplace identity and independent heating scales in the Gram limit. Positivity of the word kernel then suffices for the field construction; surjectivity of the embedding into the OS space is not required.

Several formal hypotheses are stronger than the corresponding proof needs. The slab-insertion construction uses the slab bound but not the derivative laws of the spread. C05 and C06 use the tested identities and growth bound without Euclidean root values. C14 uses the damping properties without spectral support; C16 uses spectral support and Lorentz covariance without agreement with the strict-time continuation. These distinctions do not remove the hypotheses from the statements above.

The pairing is Euclidean, so momenta transform by $\tilde{\Lambda} = \mathsf{r}\Lambda\mathsf{r}$. The tube uses absolute coordinates and imposes no condition on its first imaginary point. For $n = 1$, it is $\mathbb{C}^4$ and $\mathcal{V}_1 = \{0\}$. The polynomial bound is per label tuple, with no uniformity in the arity. The chosen value $B_0 = 0$ is outside the admissible widths. The Laplace identity is stated on compactly supported tests.

3.22.6 6. Notation

object	code name	proposed notation
OS Hilbert space of T, time axis	`OS.Chamber`	$\mathcal{H}$, Ω, H, $U(\vec{a})$
slab operators of a spread; slab bound	`OS.Spread`, `AllCopy`	$A_d(w)$; (SB)
hypothesis package of C01	`Bid.Pantry`	(H)
labels of S; tags; insertion	`S.Union`, `S.Tag`, `pin`	$\mathcal{F}$, (d, v), pin_d
Euclidean function of T on labels of S	`S.useAt T`	$T(x; v)$
single insertions between heat factors	`Small` (`bar`)	$B_p(v)$
propagator between insertions	`Small.rip`	$G_p(w, \vec{a})$
strict-time matrix element	`Small.matrix`	$M_p(z; \vec{x})$
ordered heights; strict region	`east`, `strict`	E_n, Σ_n
heights placed as times	`dawn`	$\underline{y}$
continuation; restriction; smearing	`Waves`, `restricted`, `psi`	$\overline{F}_n$, F_n^y, $\psi_v(y; f)$
strict continuation; integrability; height continuity; Cauchy–Riemann	`Region`, `Bounds`, `Along`, `Lines`	(SC), (I), (HC), (CR)

object	code name	proposed notation
polynomial growth in the inverse time separation	`Breathes`	(PG)
boundary family (Wightman functions)	`Surface`	$W_n(v)$; $W(b)$
boundary-value hypotheses	`Opens`, `NonTangentialFree`	$(\mathrm{BV}_{\mathrm{ray}})$, (BV)
Laplace identity; half-space support	`during`, `cone_tests`, `half`	(L); $\mathcal{P}_n$
Wightman spectral set	`coneM`, `cones`	$\mathcal{V}_n$
imaginary-part cones; tube	`fields`, `shores`, `tube`	V_n^+, $\bar{V}_n^+$, $\mathcal{T}_n$
cutoff; damping properties; partial boundary value	`Blanket`, `Modified`, `partialD`	χ_n; (D); $W_n(v)[\eta]$
Fourier–Laplace continuation	`Aurora.emit`, `FullBody`	$\mathcal{W}_n(v)(z)$; (TC)
rotation structure; spin structure	`Rotation` (R, `lie`), `Spins` (`rail`, `shell`)	R, ρ; $K(\Lambda)$
boost identity; Poincaré covariance	`Boosty`; `Urs`	(B_y); (Cov)
induced star; word star; kernel	`StarOp`, `rev`, `kernel`	J; b^*; $\langle b, c\rangle_W$
adjacent pair; spacelike set; swap	`NearPair`, `gap`, `switch/name`	$Q = (i, i+1)$, Γ_Q, σ
spectator profile; complex boost	`spectator/Fleck`; `augment`	y; $\Lambda_e(\zeta)$
germs at a spacelike pair	`Panes`	g_x^{o}, g_x^{t}
heated vector; heat cup	`shot`, `heatCup`	$\Psi_p(v; y; r)$; $\Phi_t(b)$
reconstructed theory	`Boundary.Wightman`	$(\mathcal{H}_W, \mathcal{D}, \Omega_W, \varphi, J, U, \mathsf{E})$
Kolmogorov space of $\langle\cdot,\cdot\rangle_W$	`Soil`	$\mathcal{H}_W$
the bundle	`Bid.Arcs`	(A16), (C01)–(C29)

3.22.7 7. What remains unexplained

1. **Upstream slab estimate.** The constants in (SB) are independent of the slab width and word length for fixed degree bound D and representative bound R; they may depend on D, R. VIII.1 obtains the mixed and selected slab laws from the phase-data package. The upstream proof of those laws, beyond their use in reconstruction, is outside this check.
2. **Imported analytic results.** The holomorphic heat-semigroup construction and the OS translation spectral measure are used by their stated properties. The construction of the compact-model Borel functional calculus, and the estimates underlying complex differentiation of the Fourier–Laplace tests, have not been checked in full. Their exact uses are identified in V.2.b, V.2.f and V.2.p.
3. **Location of the arguments.** The boundary-value construction, the spectral-support improvement and the germ exchange lemmas are filed under VI.1; the field realization and application to phase data are filed under VIII.1. They belong with the reconstruction argument in a written proof.
4. **Vacuum uniqueness.** The construction for the mixed table supplies the field axioms without vacuum uniqueness. The selected phase has uniqueness because its OS heat-fixed space is one-dimensional. No uniqueness assertion for the mixed table follows here.

5. **References.** The bibliographic pointers in §2 identify the relevant standard methods; their theorem numbering has not been verified.

3.23 V.3. Reconstruction in every phase

Part: V, phases and reconstruction. **Lean:** `YM/EveryPhase` (34 files). Its hypotheses are recorded in `YM/Spec/Closed.lean` and `YM/Phase/Mixture.lean`. **Size:** 5,268 lines.

- The 436 declarations occupy about 4,180 lines; 419 of them (about 4,070 lines) occur in the main theorem's dependency graph.
- About half the folder (15 files, about 2,600 lines) continues the stress tensor (§4.7).

3.23.1 1. What this unit proves

Setting. Fix a Wilson family, an extraction and the limit table $S = S^{\mathrm{mix}}$, together with a record `Final` for them. The unit uses the following parts of it.

- The Euclidean axioms of S^{mix} (IV.1).
- The phase decomposition of V.1:
  - a compact metric space X with a Borel probability measure ν of full support;
  - the commutative C*-algebra $\mathfrak{A}$ generated by the time-slab operators of S^{mix}, compressed to the ground space $\mathcal{D}$ of the Osterwalder–Schrader (OS) space $\mathcal{H}^{\mathrm{OS}}$ of S^{mix};
  - continuous characters χ_ξ of $\mathfrak{A}$, indexed bijectively by X, with $\xi \mapsto \chi_\xi(M)$ continuous for every $M \in \mathfrak{A}$ and $\int_X \chi_\xi(M)\, d\nu = \langle \omega, M\omega \rangle$;
  - spatial translations acting trivially on $\mathcal{D}$;
  - tables S^ξ satisfying the Euclidean axioms, with $S^\xi(w) = \chi_\xi(M_w)$ for every word w of pairwise distinct points. Here $M_w \in \mathfrak{A}$ is the compressed slab operator of a time translate of w.
- The relativistic theory $\mathfrak{W}_{\xi_*}$ of V.2 at the selected phase ξ_*, and the analytic continuations of S^{mix} and S^{ξ_*}.
- The field algebra of VI.1 (derivatives, rotations, the Euclidean stress tensor $T^{\mathrm{E}}_{ab} \in F_4$), and the Euclidean Ward identities in every phase.

Notation.

- **OS side.** For each ξ, $\mathcal{H}^{\mathrm{OS}}_\xi$ is an OS space of S^ξ for the time axis. It carries the transfer semigroup $e^{-sH^{\mathrm{OS}}_\xi}$ and the Euclidean vacuum ω_ξ. Its ground space is $\mathcal{D}_\xi = \{v : \ e^{-sH^{\mathrm{OS}}_\xi} v = v \text{ for all } s \geq 0\}$.
- **Wightman theory.** $\mathfrak{W}_\xi = (\mathcal{H}_\xi, \mathfrak{D}_\xi, \Omega_\xi, \phi_\xi, U_\xi, P_\xi)$ consists of:
  - a Hilbert space $\mathcal{H}_\xi$ and a dense common invariant domain $\mathfrak{D}_\xi$ spanned by word vectors;
  - a unit vacuum Ω_ξ;
  - fields $\phi_\xi(a, f)$ for labels $a \in \mathbb{F}$ (the common label space generated by the F_d) and $f \in \mathcal{S}(\mathbb{R}^4)$;
  - a strongly continuous unitary Poincaré representation U_ξ and an energy–momentum spectral measure P_ξ.
- **Time and energy.** Time is the fourth coordinate. H_ξ generates the time translations, and $P^0_\xi = P_\xi(\{p^4 = 0\})$.
- **Words.** A *finite sum of words* is $A = \sum_k c_k\, \phi(a_{k1}, f_{k1}) \cdots \phi(a_{kn_k}, f_{kn_k})$. It is *compactly supported* if every $f_{kj} \in C^\infty_c(\mathbb{R}^4)$. $A^\dagger$ is its formal adjoint (conjugate coefficients and tests,

apply the star operation to labels, and reverse each word). The spectral measure of $A\Omega_\xi$ is $\mu_A^\xi(B) = \langle A\Omega_\xi, P_\xi(B)A\Omega_\xi\rangle$.

- **Continuations.** $\mathsf{F}_n^\xi(a; z, \vec{x})$ is the analytic continuation of S^ξ to complex times with $\mathrm{Im}z_1 < \cdots < \mathrm{Im}z_n$. $W_n^\xi(a; g)$, for $g \in \mathcal{S}((\mathbb{R}^4)^n)$, are its boundary distributions. W_n^{mix} are those of the continuation of S^{mix} (V.2).

Proposition V.3.1 (reconstruction in every phase). There is a family indexed by X with the following data for each $\xi \in X$:

- an OS space $\mathcal{H}_\xi^{\mathrm{OS}}$ of S^ξ with its time-slab operators;
- a continuation F^ξ of S^ξ with boundary distributions W^ξ;
- a Wightman theory $\mathfrak{W}_\xi$ on the label space $\mathbb{F}$, with the same star operation and the same Poincaré action on labels as $\mathfrak{W}_{\xi_*}$;
- an isometry $J_\xi : \mathcal{H}_\xi \to \mathcal{H}_\xi^{\mathrm{OS}}$ with $J_\xi\Omega_\xi = \omega_\xi$. It intertwines the space–time translations and sends word vectors to the boundary values of the corresponding Euclidean vectors at imaginary times (Lemma V.3.b).

At $\xi = \xi_*$ these are the data of V.1 and V.2. For **every** $\xi \in X$, with no exceptional set, the following hold.

1. *Wightman axioms.*
   - The vacuum expectation values of products of fields are W^ξ.
   - The Hilbert-space adjoint of $\phi_\xi(a, f)$ extends $\phi_\xi(a^*, \bar{f})$.
   - Fields with spacelike separated supports commute on $\mathfrak{D}_\xi$.
   - U_ξ implements Poincaré covariance.
   - The joint spectrum of the translations lies in the closed forward cone $\overline{V}_+$.
   - For every nonzero label a, there is a test function f for which $\phi_\xi(a, f) \neq 0$ on $\mathfrak{D}_\xi$.
   - Every translation-invariant vector is a multiple of Ω_ξ.
2. *Surjectivity.* J_ξ is unitary.
3. *Euclidean vacuum.* $\mathcal{D}_\xi = \mathbb{C}\omega_\xi$.
4. *Zero energy.* For every $x \in \mathcal{H}_\xi$, $P_\xi^0 x = x$ if and only if $x \in \mathbb{C}\Omega_\xi$.
5. *Low-energy bound.* For every compactly supported finite sum of words A there is $M \geq 0$ such that $\mu_A^\xi(\{0 < p^4 \leq u\}) \leq Mu^2$ for all $\xi \in X$ and $u \in (0, 1]$.
6. *Heat decay.* For every such A there is $M \geq 0$ such that, for all $\xi \in X$ and $s \geq 1$,
$$0 \leq \langle A\Omega_\xi, (e^{-sH_\xi} - P_\xi^0)A\Omega_\xi\rangle \leq Ms^{-2}.$$
7. *Irreducibility.* For a bounded complex-linear operator B on $\mathcal{H}_\xi$, the following are equivalent: $B \in \mathbb{C}\,1$; and, for every a, f and $u, v \in \mathfrak{D}_\xi$, one has $u \in \mathrm{dom}\,\phi_\xi(a, f)^*$ and $\langle \phi_\xi(a, f)^* u, Bv\rangle = \langle u, B\phi_\xi(a, f)v\rangle$.
8. *Stress tensor.* The conclusions (T0)–(T6) of Proposition V.3.y hold for $\mathfrak{W}_\xi$.
9. *Dependence on the phase.*
   - For all n, $a \in \mathbb{F}^n$ and $g \in \mathcal{S}((\mathbb{R}^4)^n)$, the map $\xi \mapsto W_n^\xi(a; g)$ is continuous and $\int_X W_n^\xi(a; g)\, d\nu(\xi) = W_n^{\mathrm{mix}}(a; g)$.

- For every word w of pairwise distinct points, $\int_X S^\xi(w)\,d\nu(\xi) = S^{\mathrm{mix}}(w)$.

In item 6, e^{-sH_ξ} is defined as $J_\xi^* e^{-sH_\xi^{\mathrm{OS}}} J_\xi$. Since J_ξ intertwines the spectral projections (V.3.b), this equals $\int e^{-sp^4}\,dP_\xi(p)$. The projection subtracted in item 6 is the full zero-energy projection. Items 5 and 6 are therefore proved before item 4 and independently of it.

Status: theorem, for an arbitrary record `Final`. The record is constructed in the proof of the main theorem (VIII.1), so the statement is unconditional there.

- The detailed main theorem states V.3.1 as its clause (D7), via `Independent.Core`.
- The public theorem uses the Euclidean barycenter formula of item 9 for clause (R12), via `publicReconstruction`.
- Item 1 and the averaged Wightman distributions of item 9 enter `Headline.Reconstruction`.
- No unit uses items 2–8 as hypotheses.

Lean: `all_characters`, `Completed` (the statement; see §6 for the fields), `member` (weak commutant), `Almanac` (the data), `almanac` (existence).

Proposition V.3.2 (no exceptional phase). For every $\xi \in X$ and $x \in \mathcal{H}_\xi$,

$$P_\xi^0 x = \langle \Omega_\xi, x\rangle\, \Omega_\xi.$$

Status: theorem (Lemma V.3.r). Items 3, 4 and 7 of V.3.1 are derived from it. *Lean:* `all_plant`, `boundary_zero_energy`.

Remark V.3.3 (recorded consequences, not used by the main theorem). Under V.3.1, for every $\xi \in X$ and $x \in \mathcal{H}_\xi$:

- (a) $e^{-sH_\xi} x = x$ for all $s \geq 0$ if and only if $x \in \mathbb{C}\Omega_\xi$;
- (b) x is invariant under all translations if and only if $x \in \mathbb{C}\Omega_\xi$;
- (c) for every degree d and degree-d token coefficient vector v, v is null for S^ξ if and only if the field associated to its class in F_d vanishes on every test function;
- (d) reflection positivity holds for every affine hyperplane. More precisely, let $O \in O(4)$, $b \in \mathbb{R}^4$ and $a \in \{1,2,3,4\}$. For any finite family of words w_i with distinct points in $\{x : (O^{-1}(x-b))^a > 0\}$,

$$\sum_{i,j} \overline{c_i} c_j\, S^\xi(\theta_{O,b,a} w_i\, w_j) \geq 0 \qquad (c_i \in \mathbb{C}).$$

Here $\theta_{O,b,a}$ reverses the word, reflects its points across the hyperplane, and acts on labels by transporting the a-axis reflection through O.

Lean: `fixedHeat`, `exactly_fixed`, `recognizes`, `allPlanes`.

3.23.2 2. What it uses

From other units.

- **IV.1 (infinite volume).** The Euclidean axioms of S^{mix}: translation homogeneity, reflection positivity, smoothness. Lean: `Seed`.
- **V.1 (decomposition into phases).** The data:
  - the triple (X, ν, χ) with the properties of §1: `Resolution`, `Resolution.Out`, `Resolution.receives`;

  - for every ξ: the Euclidean axioms of S^{ξ}, null spaces equal to those of S^{mix}, the inclusion, derivative and rotation laws, and the Euclidean Ward identities of the stress tensor;
  - the slab laws of S^{mix}, including its copy bounds (below): `Selection`, `Selection.Certificate`.
- **V.1, tools.**
  - The OS space of a table: `OS.realization`.
  - The multiple-reflection bound for slab operators: `OS.producer`.
  - A countable family of positive-time words whose OS vectors are total in the OS space of every symmetric smooth table: `Work.B14.universal`.
  - Continuity of OS vectors in the points: `Work.B14.plant_cont`.
  - The full slab laws from the copy bounds: `Work.B02.answer`.
  - Almost-everywhere uniqueness of the Euclidean vacuum (Lemma V.3.c): `Cluster.lifts`.
  - The Euclidean barycenter formula: `Resolution.barycenter`.
- **V.2 (OS reconstruction).**
  - The continuations of S^{mix} and of S^{ξ_*}, and the theory $\mathfrak{W}_{\xi_*}$ with all Wightman axioms: `Relativistic`.
  - Compatibility of both continuations with the slab operators, which includes the polynomial bound on one-point slab operators used in V.3.d: `onFinal`.
  - The estimates of work items C01–C29 on the continuation and its partial boundary values (used in V.3.d, V.3.g, V.3.i).
- **Reconstruction machinery from V.2:**
  - Wightman data lacking only vacuum uniqueness: `FieldCore`.
  - The boundary isometry and its properties: `Attachment`, `Attachment.measures`, `Attachment.projections`, `Attachment.with_heat`.
  - The constructions run at each phase: `contour`, `compatible`, `establishing`.
- **VI.1 (Ward identities, rotations, stress tensor).** Derivative plan, rotations and Euclidean stress tensor: `Algebra`. Their laws (conservation and symmetry of T^{E}, tensor covariance, rotation laws): `Locals`.

Standard mathematics.

- OS reconstruction: Osterwalder–Schrader, CMP 31 (1973) and CMP 42 (1975); Glimm–Jaffe, *Quantum Physics*, Ch. 6 and 19.
- Boundedness of transfer operators by iterated reflection and the Cauchy–Schwarz inequality: Glimm–Jaffe, Ch. 10; Fröhlich–Israel–Lieb–Simon, CMP 62 (1978).
- Decomposition of a state along an abelian algebra of the vacuum sector, used in V.1 and in V.3.c: Bratteli–Robinson, vol. 1, Ch. 4.
- Spectral theorem for unitary representations of $\mathbb{R}^4$, Bochner's theorem, Laplace transforms of positive measures: Reed–Simon I–II.
- Boundary values of holomorphic functions of polynomial growth and their Taylor representation: Streater–Wightman, Ch. 2; Reed–Simon II, §IX.3.
- Complex analysis: Vitali–Montel theorem, identity theorem, Morera's theorem, Liouville's theorem.
- Uniqueness of the vacuum and irreducibility of the field algebra: Streater–Wightman, Ch. 4; Jost, *The General Theory of Quantized Fields*.

Imported constants and assumptions. The record `Final` is the only hypothesis. Its quantita-

tive inputs are as follows.

- **(i) Copy bounds of S^{mix}** (V.1, from the chessboard estimates). For every degree bound D and coefficient bound R there are $C > 0$ and $Q \in \mathbb{N}$ with

$$\|\Pi(d, w)\| \le C^{|w|}(|w|/r)^{Q|w|}.$$

  This holds for every word w of $|w|$ points with degrees $\le D$ and coefficients $\le R$, lying in the slab $[r, d-r]$, with points r-separated and $0 < r \le 1$.

- **(ii) One-point slab operators** (V.2). For each label $a \in \mathbb{F}$ there are $Q_a > 0$ and $N_a \in \mathbb{N}$ such that the one-point slab operator of S^{mix}, of thickness $2p$ with a at $(\vec{0}, p)$, has norm at most $Q_a p^{-N_a}$ for $0 < p < 1$. The exponent N_a is a bound, not an asserted scaling dimension.

The one-point bound is `Boundary.Small.rated`.

The unit's own constants are explicit:

- $R = 4r + 1$ in V.3.m;
- $c(R, M) = 16\pi\,|B_1|\,|B_R|\,M/c_3$ in V.3.k, where $|B_r|$ is the volume of the ball of radius r in $\mathbb{R}^3$ and $c_3 = (2\pi)^3$ for the Fourier convention in V.3.k;
- the factor 2 in V.3.l;
- $\Gamma(3) = 2$ in V.3.o.

The Fourier convention and constant are `Boundary.Probe.hat_eq`, `Boundary.Probe.pref`.

3.23.3 3. Strategy

V.2 reconstructs a relativistic theory at a selected phase ξ_*. The same construction applies to every S^ξ before vacuum uniqueness is known. The norm of each phase's slab operator is bounded by the corresponding norm for S^{mix} (V.3.a): repeated reflection gives a power bound, and the fixed norm of the character disappears on taking roots. This gives polynomial continuation bounds uniform in ξ. Character continuity at imaginary times propagates to complex times by Vitali's theorem, then to boundary distributions by a Taylor formula with an integrable remainder. Integration over phases commutes with both passages (V.3.d–V.3.h).

Vacuum uniqueness holds almost everywhere by the phase decomposition (V.3.c). To extend it to every phase, fix a compactly supported finite sum of words A and set

$$r_\xi(s) = \langle A\Omega_\xi, e^{-sH_\xi} A\Omega_\xi\rangle - |\langle \Omega_\xi, A\Omega_\xi\rangle|^2.$$

Locality makes the vacuum commutator vanish at large spatial separation for $|t| \le 1$. Its time-averaged odd part is the spatial Fourier transform of a positive measure, weighted by $(1-\cos E)/E$. A momentum bump of radius u bounds its shell mass by $O(u^2)$: the bump contributes u^3, while the weight contributes u. Dyadic summation and a Laplace transform give decay Ms^{-2}, uniformly in ξ, after subtracting the full zero-energy projection (V.3.k–V.3.o). Almost everywhere this is the vacuum projection. Since $r_\xi(s)$ is continuous, the same estimate holds everywhere by full support of ν. Letting $s \to \infty$ identifies the zero-energy projection with the vacuum projection at every phase (V.3.q–V.3.r). This needs no continuity of the zero-energy projection itself.

Separately, positive-height Euclidean vectors are images of Wightman vectors with exponentially weighted frequency tests, so J_ξ is onto (V.3.i–V.3.j). Vacuum uniqueness therefore holds on both sides. A Liouville argument gives irreducibility. Finally, continuation of the Euclidean Ward

identities gives the local stress-charge commutator (V.3.u–V.3.y). The removal of the exceptional set uses uniform low-energy control from locality, rather than a mass gap.

3.23.4 4. Steps

3.23.4.1 4.1 Slab bounds and the family of reconstructions Lemma V.3.a (slab operators of a phase are dominated). Fix $\xi \in X$. Let $\mathcal{K}$ be any OS space of S^ξ for the time axis, and Π_ξ any family of slab operators on it. Then for every thickness $d > 0$ and every word w of pairwise distinct points in the slab $(0, d)$,

$$\|\Pi_\xi(d, w)\| \le \|\Pi(d, w)\|,$$

where Π are the slab operators of S^{mix} on $\mathcal{H}^{\mathrm{OS}}$. Here $\Pi(d, w)$ maps the OS vector of a positive-time word v to that of w followed by v translated by d.

Sketch.

1. Let $u = \Pi(d, w)$ and $D > \|u\|$. For a positive-time word v, let $X(v)$ be the slab operator of v restricted to the ground space $\mathcal{D}$.
2. The compression of the doubled word $\theta v\, v$ is $X(v)^* X(v) \in \mathfrak{A}$. Let r_n be v with n reflected pairs of copies of w inserted. Then $X(r_n) = (u^* u)^n X(v)$.
3. Hence the diagonal Gram entry of r_n in S^ξ satisfies
$$|S^\xi(\theta r_n\, r_n)| = |\chi_\xi(X(r_n)^* X(r_n))| \le \|\chi_\xi\|\, \|X(v)\|^2 D^{4n}.$$
4. The power argument (iterated Cauchy–Schwarz on the span of OS vectors) turns this into an operator on $\mathcal{K}$ of norm at most D that inserts w. The fixed factor $\|\chi_\xi\|\, \|X(v)\|^2$ disappears in the limit.
5. Slab operators are determined by their action on OS vectors, so this operator is $\Pi_\xi(d, w)$. Finally let $D \downarrow \|u\|$.

Lean: `sameBound`, with `x_train`, `buys`, `actual`, and `OS.producer`.

Lemma V.3.b (the family of reconstructions). For every ξ there are:

- (i) an OS space $\mathcal{H}^{\mathrm{OS}}_\xi$ of S^ξ with slab operators Π_ξ satisfying the slab laws (copy bounds, smoothness in the positions, derivative rule), with the copy constants of S^{mix}, hence uniformly in ξ;
- (ii) a continuation F^ξ of S^ξ with non-tangential boundary values W^ξ;
- (iii) a Wightman theory $\mathfrak{W}_\xi$ with all properties of V.3.1(1) except uniqueness of the vacuum, with the star operation and label action of $\mathfrak{W}_{\xi_*}$;
- (iv) an isometry J_ξ with $J_\xi \Omega_\xi = \omega_\xi$ and $J_\xi U_\xi(a) = U^{\mathrm{OS}}_\xi(a) J_\xi$ for $a \in \mathbb{R}^4$, where U^{OS}_ξ is the OS translation group. It is uniquely determined by the following boundary property: Euclidean word vectors at positive imaginary heights, smeared with a joint Schwartz test g, converge in Hilbert norm to $J_\xi \Phi_\xi(a; g)$ as the heights and slab widths tend to zero with the positive margins of V.3.i. Here Φ_ξ is the continuous extension of word vectors to joint tests.

At ξ_* these are the data of V.1–V.2. Moreover:

- $J_\xi P_\xi(B) = P^{\mathrm{OS}}_\xi(B) J_\xi$ for every Borel set $B \subset \mathbb{R}^4$, so μ^ξ_x is the spectral measure of $J_\xi x$;

- if $\mathcal{D}_\xi = \mathbb{C}\omega_\xi$, then $\mathfrak{W}_\xi$ satisfies all of V.3.1(1).

Sketch.

- For $\xi \neq \xi_*$, the OS space is built from S^ξ and the slab operators are those of V.3.a.
- Their copy bounds are those of S^{mix}, by V.3.a; the remaining slab laws follow from the copy bounds.
- The continuation, the compatibility data and a theory without vacuum uniqueness come from the reconstruction machinery of V.2. The boundary property in (iv) determines the isometry on a dense span of word vectors.
- At ξ_* the existing data are taken, so the construction does not alter V.2.
- The intertwining of the spectral projections follows from intertwining the translations, by comparing characteristic functions.
- The last assertion holds because a translation-invariant x has $J_\xi x \in \mathcal{D}_\xi$ and J_ξ is injective.

Lean: `storeys`, `almanac`, `Almanac.priced`, `Attachment.projections`, `Attachment.with_heat`. *(proof read for these;* `contour`, `compatible`, `establishing` *and* `Work.B02.answer` *also read; underlying analytic inputs from V.2)*

Lemma V.3.c (unique Euclidean vacuum for almost every phase; from V.1). For ν-almost every ξ, $\mathcal{D}_\xi = \mathbb{C}\omega_\xi$.

Sketch (a variance identity). For a positive-time word b, let $\Psi_\xi(b) \in \mathcal{H}_\xi^{\mathrm{OS}}$ be its OS vector and Q_ξ the projection onto $\mathcal{D}_\xi$.

1. $\|Q_\xi \Psi_\xi(b)\|^2 = \lim_{s\to\infty} S^\xi(\theta b_s\, b_s)$, where b_s is b translated by s in time. The terms are character values, continuous in ξ, nonincreasing in s and dominated by the value at $s = 0$.
2. By dominated convergence and the barycenter identity, $\int \|Q_\xi \Psi_\xi(b)\|^2 d\nu = \|Q\Psi(b)\|^2$.
3. On the other hand $S^\xi(b) = \chi_\xi(M_b)$ and $M_b\omega = Q\Psi(b)$. Multiplicativity of characters gives $\int |S^\xi(b)|^2 d\nu = \int \chi_\xi(M_b^* M_b)\, d\nu = \|Q\Psi(b)\|^2$.
4. Hence the nonnegative function $\|Q_\xi \Psi_\xi(b)\|^2 - |S^\xi(b)|^2 = \|Q_\xi \Psi_\xi(b) - S^\xi(b)\omega_\xi\|^2$ has integral zero, so $Q_\xi \Psi_\xi(b) \in \mathbb{C}\omega_\xi$ almost everywhere.
5. Let b run over a countable family of words whose OS vectors are total in every OS space (`Work.B14.universal`).

Lean: `Almanac.almostEvery`, `Cluster.lifts`, `Cluster.exp_real`.

3.23.4.2 4.2 Continuity in the phase and barycenter Lemma V.3.d (polynomial bound, uniform in the phase). For every n and $a \in \mathbb{F}^n$ there are $m \in \mathbb{N}$ and $C_0 \geq 0$ with the following property. For all ξ, all real configurations x, and all heights y with $y_j - y_i \geq k$ for $i < j$, where $k \in (0, 1]$,

$$|\mathsf{F}_n^\xi(a;\, t(x) + iy,\, \vec{x})| \leq C_0\, k^{-m}.$$

Sketch.

- F_n^ξ at such points is a matrix element, between ω_ξ and ω_ξ, of a product of contractions (complex-time evolutions, spatial translations) and one-point slab operators of thickness of order k.
- Each one-point slab operator is bounded by the corresponding one for S^{mix} (V.3.a), which is polynomial in $1/k$ (input (ii) of §2).

- Choose slab half-width $p = k/8$. With the constants of §2(ii), take $m = \sum_j N_{a_j}$ and $C_0 = \prod_j (Q_{a_j} 8^{N_{a_j}})$.

Lean: `power`. *(proof read; the matrix-element representation of the continuation comes from V.2)*

Lemma V.3.e (strict heights). For all n, a, real x and $y_1 < \cdots < y_n$, the map $\xi \mapsto \mathsf{F}_n^\xi(a; t(x) + iy, \vec{x})$ is continuous and its ν-integral is $\mathsf{F}_n^{\mathrm{mix}}(a; t(x) + iy, \vec{x})$.

Sketch.

1. At purely imaginary times $z_j = iy_j$, F_n^ξ is S^ξ of a word of distinct points, hence a character value. It is therefore continuous in ξ, with barycenter S^{mix}.
2. Along the complex line $c \mapsto iy + c\,t(x)$, for c in a strip $|\mathrm{Im} c| < \delta$ (where the heights stay ordered with gaps $\geq k/2$), the functions are holomorphic and bounded uniformly in ξ by V.3.d.
3. By Vitali's theorem, convergence along nets $\xi' \to \xi$ on the imaginary segment through 0 propagates to the whole strip.
4. The integral over X is holomorphic in c. It agrees with the unselected function on that segment, hence on the strip by the identity theorem. Evaluate at $c = 1$.

Lean: `imaginary_evaluation`, `strict_evaluation`, `integral_holomorphic`, `propagate_limit`.

Lemma V.3.f (tested strict heights). For $y_1 < \cdots < y_n$ and $g \in \mathcal{S}((\mathbb{R}^4)^n)$, define $\psi^\xi(a, y, g) = \int \mathsf{F}_n^\xi(a; t(x) + iy, \vec{x})\, g(x)\, dx$. Then $\xi \mapsto \psi^\xi(a, y, g)$ is continuous, and $\int \psi^\xi \, d\nu = \psi^{\mathrm{mix}}$.

Sketch. Dominated convergence with dominator $C_0 k^{-m} |g|$ (V.3.d), and Fubini.

Lean: `eager`.

Lemma V.3.g (continuity and barycenter of the Wightman distributions). Item 9 of V.3.1, first part.

Sketch. Let $\rho = (0, 1, \ldots, n-1)$ and $D = -i \sum_{j=1}^n (j-1) \partial_{t_j}$ on joint test functions. The Cauchy–Riemann equations and integration by parts give $\frac{d}{dt} \psi(a, y + t\rho, g) = \psi(a, y + t\rho, Dg)$.

1. With m as in V.3.d, the boundary value has the Taylor representation

$$W_n^\xi(a; g) = \sum_{k=0}^{m} \frac{(-1)^k}{k!}\, \psi^\xi(a, \rho, D^k g) - \int_0^1 \frac{(-t)^m}{m!}\, \psi^\xi(a, t\rho, D^{m+1} g)\, dt.$$

2. By V.3.d the integrand is bounded by $C_0 \| D^{m+1} g \|_{L^1}$, uniformly in t and ξ.
3. Continuity then follows from V.3.f and dominated convergence in t.
4. For the barycenter: $\psi^\xi(a, y, g) \to W_n^\xi(a; g)$ as $y \to 0$ among ordered heights, with a bound uniform in ξ and y given by the same representation. Integrate over X by dominated convergence, and use V.3.f at each y.

Lean: `word_continuity`, `barycenter`, `Waves.desired`, `Waves.meant`. The derivative convention is `Boundary.Waves.clock`.

Lemma V.3.h (inner products). For finite sums of words A, A':

- $\xi \mapsto \langle A\Omega_\xi, A'\Omega_\xi \rangle$ is continuous on X;

- $M_A := \sup_\xi \|A\Omega_\xi\|^2 = \sup_\xi \mu_A^\xi(\mathbb{R}^4) < \infty$.

Sketch. $\langle \phi_\xi(w)\Omega_\xi, \phi_\xi(w')\Omega_\xi\rangle = W^\xi(w^\dagger w')$, with the same star operation for all ξ. Apply V.3.g, then use compactness of X.

Lean: `word_pair`, `sum_pair`, `word_bound`.

3.23.4.3 4.3 Surjectivity of the boundary isometry Lemma V.3.i (Euclidean vectors at positive heights are in the range). Fix $n \geq 1$, $\xi \in X$, labels $a \in \mathbb{F}^n$, heights $0 < y_1 < \cdots < y_n$ and $p \in (0,1)$ with $4p < y_1$ and $4p < y_{j+1} - y_j$ for $1 \leq j < n$.

- Let $B_{\xi,p}(a)$ be the centered one-point slab operator of thickness $2p$. For $\mathrm{Im}z > \ell$ put

$$R_{\xi,\ell}(z,\vec{x}) = e^{-(\mathrm{Im}z-\ell)H_\xi^{\mathrm{OS}}} e^{i\mathrm{Re}z H_\xi^{\mathrm{OS}}} U_\xi^{\mathrm{OS}}(\vec{x}).$$

For a real configuration q, write $z_j = q_j^4 + iy_j$ and define

$$\Psi_p^\xi(a; q+iy) = R_{\xi,p}(z_1,\vec{q}_1)B_{\xi,p}(a_1)\left(\prod_{j=2}^{n} R_{\xi,2p}(z_j - z_{j-1}, \vec{q}_j - \vec{q}_{j-1})B_{\xi,p}(a_j)\right)\omega_\xi,$$

where the factors are ordered by increasing j from left to right; the product is empty when $n = 1$.

- Let $h \in C_c^\infty((\mathbb{R}^4)^n)$ and use the convention $g(q) = \widehat{h}(q) = \int e^{-i\sum_j p_j\cdot q_j} h(p)\,dp$. Set $g_y = e^{-\widehat{\sum_j y_j p_j^4}} h$. Then

$$\int g(q)\,\Psi_p^\xi(a; q+iy)\,dq = J_\xi\,\Phi_\xi(a; g_y).$$

  Here $\Phi_\xi(a;\cdot)$ is the joint-test extension from V.3.b. Compact support of h ensures that the exponential multiplier preserves the Schwartz class.

- Consequently every $\Psi_p^\xi(a; q+iy)$ lies in $\mathrm{ran}\,J_\xi$.

Sketch.

1. Call the two sides v and $J_\xi\Phi$. Their squared norms and their inner product, $\|v\|^2$, $\langle v, J_\xi\Phi\rangle$ and $\|J_\xi\Phi\|^2$, are each expressed through partial boundary values of W^ξ at heights y.
2. On tests of compact frequency support these partial boundary values equal W^ξ tested against the damped test functions, so the three numbers coincide and $\|v - J_\xi\Phi\| = 0$.
3. Density of compact-frequency tests and continuity in g extend the membership in $\mathrm{ran}\,J_\xi$ to all Schwartz g.
4. Testing against all g, and using continuity in q, gives the pointwise statement. The range is closed since J_ξ is an isometry.

Lean: `vector_cooled`, `vector_in`, `shot_in`, `boundary_tilt`, `chilled`. *(proof read; V.2's partial-boundary identities statement only)*

Lemma V.3.j (surjectivity). Item 2 of V.3.1.

Sketch.

1. The empty word gives $\omega_\xi = J_\xi\Omega_\xi$. Take a nonempty positive-time Euclidean word with strictly increasing times. Its OS vector is a vector Ψ_p^ξ of V.3.i (real times 0, heights equal to the Euclidean times), so it lies in the range.

2. Take a general positive-time word. By symmetry its points may be reordered so that the times are nondecreasing, say r. Then $r + u\rho$ is strictly increasing for $u > 0$.
3. Its OS vector is the limit, as $u \downarrow 0$, of OS vectors in the range, by continuity of OS vectors in the points. It therefore lies in the range, because the range is closed.
4. These vectors span a dense subspace, so $\operatorname{ran} J_\xi$ is dense and closed.

Vacuum uniqueness is not used.

Lean: `onto`, `ordered_covered`, `bright_shot`, `reaches`.

3.23.4.4 4.4 Low-energy bound and heat decay Lemma V.3.k (time-averaged commutator). Let ρ, σ be finite positive measures on $\mathbb{R}^4$ supported in $\overline{V}_+$, with $\rho(\mathbb{R}^4) + \sigma(\mathbb{R}^4) \le M$. Put

$$\hat\rho(\vec x, t) = \int e^{i(\vec p\cdot\vec x + p^4 t)}\, d\rho(p), \qquad a = \hat\rho(\cdot) - \hat\sigma(-\cdot), \qquad D(\vec x) = \int_0^1 \frac{a(\vec x, t) - a(\vec x, -t)}{2i}\, dt.$$

Then $|D| \le M$. Write $\mathcal{F}f(x) = \int e^{-i\vec p\cdot x} f(\vec p)\, d\vec p$, so $c_3 = (2\pi)^3$. If moreover $D(\vec x) = 0$ for $|\vec x| > R$, then for $0 < u \le 1$

$$\rho(\{u/2 < p^4 \le u\}) \le c(R, M)\, u^2, \qquad c(R, M) = 16\pi\, |B_1|\, |B_R|\, M / c_3.$$

Sketch.

1. By Fubini, D is the spatial Fourier transform of $\rho^\sharp + \sigma^\sharp$. Here $\rho^\sharp$ is the image under $p \mapsto \vec p$ of $\pi(p^4)\, d\rho(p)$, with $\pi(E) = \int_0^1 \sin(Et)\, dt$, equal to $(1 - \cos E)/E$ for $E > 0$ and to 0 at $E = 0$, and $\sigma^\sharp$ is the analogous image of σ under $p \mapsto -\vec p$.
2. On $\overline{V}_+$ one has $\pi(p^4) \ge 0$ at all energies, and $\pi(E) \ge E/\pi$ for $0 \le E \le 1$.
3. Test D against $\mathcal{F}f$, where f is a smooth spatial-momentum bump with $0 \le f \le 1$, equal to 1 on the ball of radius u and supported in the ball of radius $2u$. Then $|\mathcal{F}f| \le |B_{2u}| = 8|B_1|u^3$. Since D vanishes outside B_R, the pairing is at most $8|B_1||B_R|Mu^3$.
4. The pairing equals $c_3(\int f\, d\rho^\sharp + \int f\, d\sigma^\sharp)$, which is at least $c_3\, \frac{u}{2\pi}\, \rho(\{u/2 < p^4 \le u\})$. On that shell $|\vec p| \le p^4 \le u$, so $f = 1$ there, and $\pi(p^4) \ge u/(2\pi)$.
5. Thus the exponent 2 is 3 (spatial dimension, from the volume of the momentum ball) minus 1 (first-order vanishing of the weight).

Lean: `annulus`, `averaged_bound`, `fee`, `two_average`, `one_average`, `pulse_positive`, `pulse_lower`.

Lemma V.3.l (dyadic summation). Let ρ be a finite measure and $C \ge 0$ with $\rho(\{u/2 < p^4 \le u\}) \le Cu^2$ for all $u \in (0, 1]$. Then $\rho(\{0 < p^4 \le u\}) \le 2Cu^2$ for $u \in (0, 1]$.

Sketch. Cover $(0, u]$ by the shells with $u_n = u2^{-n}$. Then $\sum_n Cu^2\, 4^{-n} \le \frac{4}{3}Cu^2$; the bound 2 follows already from the coarser majorant $\sum_n 2^{-n} = 2$.

Lean: `dyadic`.

Lemma V.3.m (locality at large distance, uniformly in the phase). Let A be a compactly supported finite sum of words, and $r \ge 0$ a bound for $|t|$ and $|\vec x|$ on the supports of all its test functions. Put $R = 4r + 1$. Then for all ξ, $|t| \le 1$ and $|\vec x| > R$:

$$\hat\mu_A^\xi(\vec x, t) - \hat\mu_{A^\dagger}^\xi(-\vec x, -t) = \langle \Omega_\xi, [A^\dagger, \alpha_{(\vec x, t)} A]\, \Omega_\xi \rangle = 0,$$

where α denotes translation of the test functions. Hence the function D of V.3.k, built from $\rho = \mu_A^\xi$ and $\sigma = \mu_{A^\dagger}^\xi$, vanishes for $|\vec{x}| > R$.

Sketch.

- The first identity follows from covariance and the adjoint relation.
- The supports of $A^\dagger$ and $\alpha_{(\vec{x},t)}A$ are spacelike separated, so the commutator vanishes by locality.
- R depends only on the supports, and translating test functions is the same operation for every ξ.

Lean: `late`, `distant`, `comm_expect`, `op_comm`.

Proposition V.3.n (low-energy bound). Item 5 of V.3.1, with $M = 2c(R, M_A + M_{A^\dagger})$.

Sketch. Apply V.3.k to $\rho = \mu_A^\xi$ and $\sigma = \mu_{A^\dagger}^\xi$. These are supported in $\overline{V}_+$ by the spectrum condition, with mass bounds from V.3.h and R from V.3.m. Then apply V.3.l.

Lean: `light_bound`.

Proposition V.3.o (heat decay with the full zero-energy projection). Item 6 of V.3.1, with $M = 2\max(M_A, 2c(R, M_A + M_{A^\dagger}))$.

Sketch.

1. By the spectral theorem and $\operatorname{supp}\mu_A^\xi \subset \{p^4 \geq 0\}$:

$$\langle A\Omega_\xi, (e^{-sH_\xi} - P_\xi^0)A\Omega_\xi\rangle = \int_{\{p^4>0\}} e^{-sp^4}\, d\mu_A^\xi \;\geq 0.$$

2. Let ρ be a finite measure with $\rho(\{0 < p^4 \leq t\}) \leq Ct^2$ for all $t > 0$; for $t > 1$ this follows from the total-mass bound. Write $e^{-sE} = \int_E^\infty s e^{-st}\, dt$. Fubini gives

$$\int_{\{p^4>0\}} e^{-sp^4} d\rho = \int_0^\infty s e^{-st} \rho(\{0 < p^4 \leq t\})\, dt \leq Cs \int_0^\infty t^2 e^{-st} dt = \Gamma(3)\, C\, s^{-2}.$$

Lean: `residual_pair`, `decay`, `inverse_square`, `bHeat_pair`.

Lemma V.3.p (continuity of the heat form). For every finite sum of words A and $s > 0$, $\xi \mapsto \langle A\Omega_\xi, e^{-sH_\xi} A\Omega_\xi\rangle = \int e^{-sp^4}\, d\mu_A^\xi$ is continuous.

Sketch.

1. The Fourier transform $\hat{\mu}_A^\xi(p) = \langle A\Omega_\xi, (\alpha_p A)\Omega_\xi\rangle$ is continuous in ξ by V.3.h, and bounded by M_A.
2. Hence $\int h\, d\mu_A^\xi$ is continuous in ξ for every Schwartz h, by Parseval and dominated convergence.
3. On $\overline{V}_+$ outside the ball of radius $2N$ one has $p^4 \geq N$. Multiplying e^{-sp^4} by a bump equal to 1 on that ball changes the integral by at most $M_A e^{-sN}$, uniformly in ξ.
4. A uniform limit of continuous functions is continuous.

Lean: `uniform_heat`, `char_cont`, `test_cont`.

3.23.4.5 4.5 Every phase

Lemma V.3.q (vacuum-subtracted decay at every phase). For every compactly supported finite sum of words A there is $M \geq 0$ such that $0 \leq r_\xi(s) \leq Ms^{-2}$ for all $\xi \in X$ and $s \geq 1$, with $r_\xi(s)$ as in §3.

Sketch.

1. $r_\xi(s)$ is continuous in ξ by V.3.p and V.3.h; Ω_ξ is the vector of the empty word.
2. Where $\mathcal{D}_\xi = \mathbb{C}\omega_\xi$, which is ν-almost everywhere by V.3.c, J_ξ intertwines the zero-energy projections. Then $P^0_\xi x = \langle \Omega_\xi, x\rangle \Omega_\xi$, so $r_\xi(s) = \langle A\Omega_\xi, (e^{-sH_\xi} - P^0_\xi)A\Omega_\xi\rangle \in [0, Ms^{-2}]$ by V.3.o.
3. The set $\{\xi : r_\xi(s) \in [0, Ms^{-2}]\}$ is closed and of full measure. Since ν charges every nonempty open set, it is X.

Lean: `proj_eq`, `univ_of_closed`, `extends_decay`.

Lemma V.3.r (zero-energy projection = vacuum projection). Proposition V.3.2.

Sketch.

1. $e^{-sH_\xi} \to P^0_\xi$ strongly as $s \to \infty$, so V.3.q gives $\langle x, P^0_\xi x\rangle = |\langle \Omega_\xi, x\rangle|^2$ for $x = A\Omega_\xi$.
2. Since P^0_ξ is an orthogonal projection with $P^0_\xi \Omega_\xi = \Omega_\xi$, this gives $\|P^0_\xi x - \langle \Omega_\xi, x\rangle \Omega_\xi\|^2 = \|P^0_\xi x\|^2 - |\langle \Omega_\xi, x\rangle|^2 = 0$.
3. The identity extends from compactly supported to Schwartz test functions by cutoffs converging in $\mathcal{S}$ and continuity of the word vectors in the joint test function.
4. Bounded operators that agree on all word vectors are equal.

Lean: `groundvac`, `bHeat_limit`, `cluster_zero`, `all_plant`.

Lemma V.3.s (Euclidean vacuum and Wightman axioms at every phase). Items 1, 3 and 4 of V.3.1.

Sketch.

1. Let $v \in \mathcal{D}_\xi$. By V.3.j, $v = J_\xi x$ for some x.
2. The projection onto $\mathcal{D}_\xi$ equals the zero-energy spectral projection of the OS translation group: both are the strong limit of $e^{-sH^{\mathrm{OS}}_\xi}$.
3. Intertwining with V.3.2 gives $v = J_\xi P^0_\xi x = \langle \Omega_\xi, x\rangle\, \omega_\xi$. This is item 3.
4. Item 1 then follows from V.3.b, and item 4 from V.3.2.

Lean: `every`, `reflection_projection`, `boundary_realizes`, `boundary_zero_energy`.

3.23.4.6 4.6 Irreducibility

Proposition V.3.t (trivial weak commutant). Item 7 of V.3.1.

Sketch. The weak commutant is first rewritten with $\phi_\xi(a^*, \bar f)$ in place of the Hilbert adjoint, since the former is the restriction of the latter to $\mathfrak{D}_\xi$. The converse direction is the adjoint relation. Now let B be weakly central and write $\Omega = \Omega_\xi$.

1. For word vectors $u = \phi(w)\Omega$ and $v = \phi(w^\dagger)\Omega$, and $\mathrm{Im} z \geq 0$, set

$$p(z) = \langle J_\xi u, e^{izH^{\mathrm{OS}}_\xi} J_\xi B\Omega\rangle, \qquad q(z) = \langle J_\xi B^*\Omega, e^{izH^{\mathrm{OS}}_\xi} J_\xi v\rangle.$$

   Both are bounded, continuous, and holomorphic for $\mathrm{Im} z > 0$. On the real axis $p(t) = \langle u, U_\xi(t)B\Omega\rangle$, and at $z = is$ one has $p(is) = \langle u, e^{-sH_\xi} B\Omega\rangle$.

2. Covariance and weak centrality give $p(t) = q(-t)$ for real t.
3. Gluing p on the upper half-plane to $q(-\cdot)$ on the lower one gives a bounded continuous function, holomorphic off $\mathbb{R}$. It is entire by Morera's theorem and constant by Liouville's theorem.
4. Hence $\langle u, e^{-sH_\xi} B\Omega\rangle = \langle u, B\Omega\rangle$ for all $s \geq 0$. Letting $s \to \infty$ and using V.3.2 gives $B\Omega = a\Omega$ with $a = \langle \Omega, B\Omega\rangle$.
5. Weak centrality then gives $\langle \phi(c)\Omega, B\phi(b)\Omega\rangle = \langle \phi(b^\dagger)\phi(c)\Omega, B\Omega\rangle = a\langle \phi(c)\Omega, \phi(b)\Omega\rangle$ for all words, so $B = a\,1$.

The proof uses V.3.2 and the reconstruction data of V.3.b. It requires neither surjectivity of J_ξ nor invariance of $\mathfrak{D}_\xi$ under B.

Lean: `commutant`, `saturated`, `paste`, `precise`, `irreducible`.

3.23.4.7 4.7 The continued stress tensor Conventions.

- Indices $a, b \in \{1, 2, 3, 4\}$, with 4 the time index.
- $\eta = \mathrm{diag}(-1, -1, -1, 1)$, $\zeta_4 = -i$, $\zeta_j = 1$.
- The Minkowski stress labels are $T^{\mathrm{M}}_{ab} = -\zeta_a\zeta_b T^{\mathrm{E}}_{ab}$, that is $T^{\mathrm{M}}_{44} = T^{\mathrm{E}}_{44}$, $T^{\mathrm{M}}_{4j} = iT^{\mathrm{E}}_{4j}$, $T^{\mathrm{M}}_{jk} = -T^{\mathrm{E}}_{jk}$. Write $T_{ab}(f) = \phi_\xi(T^{\mathrm{M}}_{ab}, f)$.
- A Killing field is $K(x) = k + Vx$, where $k \in \mathbb{R}^4$ and $V^{\mathsf{T}}\eta + \eta V = 0$. Write Σ_K for the infinitesimal label action of V (VI.1). Its Euclidean continuation is $K^b_{\mathrm{E}}(y) = \zeta_b(k^b + \sum_c V_{bc}\kappa_c y^c)$, where $\kappa_4 = i$ and $\kappa_j = 1$ for $j \leq 3$. The Euclidean derivative label is denoted by $\partial^{\mathrm{E}}_b B$ (`EveryPhase.Stress.Killing`).
- The charge is
$$Q_K(f, c) = \sum_b T_{4b}(K^b\,(c \otimes f)),$$
for a time profile f and a spatial cutoff c.

Lemma V.3.u (Euclidean contact identity and current jump). Fix $\xi \in X$, distinct Euclidean points $x_1, \dots, x_{n+1}$, a distinguished label $B \in F_d$ at x_k, arbitrary spectator labels, and a Killing field K. Let $\chi \in C^\infty_c(\mathbb{R}^4; \mathbb{R})$ equal 1 near x_k, with every other x_j outside $\operatorname{supp}\chi$. Then the following integral is absolutely convergent and

$$-\int \sum_{a,b} \partial_a\chi(y)\, K^b_{\mathrm{E}}(y)\, S^\xi(\dots, T^{\mathrm{E}}_{ab}(y), \dots)\, dy = S^\xi(\dots, \Sigma_K B(x_k), \dots) + \sum_b K^b_{\mathrm{E}}(x_k)\, S^\xi(\dots, \partial^{\mathrm{E}}_b B(x_k), \dots).$$

If the exterior Euclidean times are strictly ordered and $c \in C^\infty_c(\mathbb{R}^3; \mathbb{R})$ equals 1 near $\vec{x}_k$, the difference of the currents $\sum_b \int c(\vec{y}) K^b_{\mathrm{E}}(\vec{y}, t) S^\xi(\dots, T^{\mathrm{E}}_{4b}(\vec{y}, t), \dots)\, d\vec{y}$, with the stress insertion immediately after B at $t = x^4_k + \varepsilon$ and immediately before B at $t = x^4_k - \varepsilon$, tends to the right side as $\varepsilon \downarrow 0$.

Sketch. Decompose V into a spatial rotation and a boost. The translation and rotation Ward identities give the displayed formula, including the Wick factors. Choose $\chi(y) = c(\vec{y})h(y^4 - x^4_k)$ with a fixed smooth time cutoff supported between the neighboring exterior times. Conservation identifies the time derivative of the integrated current with the flux through ∇c. The fundamental theorem of calculus, applied on both sides of x^4_k, gives the jump. For complex exterior times, the spacelike continuation germs in (T0) give equal limits for the two spatial fluxes; bounded

derivatives give joint current limits. Vitali's theorem then continues the jump identity from imaginary exterior times. Only spatially integrated currents have a jump limit here.

Lean: `euclidean_shell`, `Stress.pure_clock`, `Stress.through_jump`. The cutoff condition is `Table.Window`.

Lemma V.3.v (Minkowski jump identity). Fix ξ, a Killing field K, labels at $n+1$ points and a distinguished index k. Let $M \in C_c^\infty((\mathbb{R}^4)^{n+1})$, $f \in C_c^\infty(\mathbb{R})$ and $c \in C_c^\infty(\mathbb{R}^3;\mathbb{R})$. For every $x \in \operatorname{supp} M$, assume that $c = 1$ near $\vec{x}_k$ and

$$|s - x_k^4| < |\vec{y} - \vec{x}_k| \qquad (s \in \operatorname{supp} f,\ \vec{y} \in \operatorname{supp} \partial_j c,\ j = 1,2,3).$$

Then

$$\sum_b \zeta_b \Big[W^\xi(\dots, a_k, \mathcal{T}_b, \dots) - W^\xi(\dots, \mathcal{T}_b, a_k, \dots) \Big] = \Big(\int f \Big) \Big[W^\xi(\dots, \Sigma_K a_k, \dots; M) + \sum_b \zeta_b W^\xi(\dots, \partial_b^{\mathrm{E}} a_k, \dots; K_k^b M) \Big].$$

The left side is tested jointly against M in the original variables and $K^b(y)c(\vec{y})f(y^4)$ in the inserted stress variable y. Thus $\mathcal{T}_b$ denotes the Euclidean label T_{4b}^{E} inserted immediately after, respectively before, the k-th field. On the right, $K_k^b M(x) = K^b(x_k)M(x)$.

Sketch.

- Use the common sequence of tied heights from (T0). At each height, integrate the continued current jump of V.3.u against $M(x)f(t)$. Joint convergence and compactness of the test supports give a common integrable bound; the right side is independent of t, producing $\int f$.
- Pass to the limit with the continuity of the partial boundary values.

Lean: `Stress.verdict`, `Stress.partial_jump`, `Stress.Jury`.

Lemma V.3.w (Ward identity with arbitrary spectators). In V.3.v take $\int f = 1$ and a product test with a compactly supported factor m at the distinguished field. The resulting identity holds with arbitrary Schwartz spectator words before and after the inserted pair. The cutoff conditions need hold only for points of $\operatorname{supp} m$.

Sketch. Cut off the spectator test functions. Apply V.3.v. Pass to the limit by continuity of W^ξ on joint test functions.

Lean: `Stress.Schwartz_shells`, `Stress.compact_shells`.

Lemma V.3.x (the charge generates the transformation). Fix $\xi \in X$, $B \in F_d$ and a Killing field K. Let $\mathcal{O} \subset \mathbb{R}^4$ be compact, $m \in C_c^\infty(\mathbb{R}^4)$ with $\operatorname{supp} m \subset \mathcal{O}$, and $f \in C_c^\infty(\mathbb{R};\mathbb{R})$ with $\int f = 1$. Let $c \in C_c^\infty(\mathbb{R}^3;\mathbb{C})$ equal 1 on an open ball such that, for every $s \in \operatorname{supp} f$, every point $(\vec{y}, s)$ over its complement is strictly spacelike to every point of $\mathcal{O}$. Then, on $\mathfrak{D}_\xi$,

$$i[Q_K(f,c), \phi_\xi(B,m)] = \phi_\xi(\Sigma_K B, m) - \sum_{a=1}^{4} \phi_\xi(B, \partial_a(K^a m)).$$

The derivative includes the affine coefficient K^a. Equivalently, the last term is $\sum_a \zeta_a \phi_\xi(\partial_a^{\mathrm{E}} B, K^a m)$.

Sketch.

- Replace c by its real part. The difference of the two charges commutes with $\phi_\xi(B,m)$ by locality, since it is supported off the ball.

- Then take matrix elements between word vectors and apply V.3.w.

Lean: `Stress.local_action`, `Stress.measure_identity`, `Stress.Charge`, `Stress.Season`, `Stress.Reservation`.

Proposition V.3.y (the stress tensor of $\mathfrak{W}_\xi$). For every $\xi \in X$ the following hold. Statements (T1)–(T4) hold for all $f \in \mathcal{S}(\mathbb{R}^4)$, $a, b \in \{1,2,3,4\}$ and $u, v, x \in \mathfrak{D}_\xi$.

- **(T0)** There are holomorphic extensions of F^ξ to the full forward tube and continuous tested partial boundary values, equal to W^ξ at zero height. For each adjacent pair there is a sequence of tied imaginary heights tending to zero, independent of the labels, at which the two field orders have equal continuation germs at spacelike separation. The tube consists of complex configurations with consecutive imaginary differences in the open forward cone; partial boundary values are taken on the corresponding closed cone. These data are used in V.3.u–V.3.v.
- **(T1)** $U_\xi(g)\mathfrak{D}_\xi \subset \mathfrak{D}_\xi$ for every proper orthochronous Poincaré transformation g, and $U_\xi(g)T_{ab}(f)x = \sum_{i,j} \Lambda(g)_{ia}\Lambda(g)_{jb}\, T_{ij}(f \circ g^{-1})\, U_\xi(g)x$.
- **(T2)** $T_{ab} = T_{ba}$.
- **(T3)** $\langle T_{ab}(f)u, v\rangle = \langle u, T_{ab}(\bar{f})v\rangle$.
- **(T4)** $\sum_a \eta_{aa}\, T_{ab}(\partial_a f) = 0$.
- **(T5)** The commutator identity of V.3.x.
- **(T6)** Under the cutoff and profile conditions of V.3.x, for every finite sum of words A whose test functions are supported in $\mathcal{O}$, $i[Q_K(f,c), A]$ is the sum obtained by replacing one factor at a time by the right side of V.3.x and retaining the original order of the other factors. The formula holds for any choice of representatives of the labels in the spaces F_d.

Sketch.

- (T0) is supplied by the full-tube and partial-boundary reconstruction of V.2, with one common sequence of exchange heights for all labels (`EveryPhase.Stress.Prospect`, `EveryPhase.Stress.Gallery`).
- (T1)–(T4) are the tensor covariance, symmetry, reality and conservation of T^{E} (VI.1), read through the Wick factors and the derivative rule for fields.
- (T5) is V.3.x; (T6) follows from (T5) by the derivation property of commutators.

Lean: `Stress.Force`, `Stress.force`, `Stress.Generated`, `Stress.tensor_lorentz`, `Stress.tensor_herm`, `Stress.tensor_conserved`.

3.23.4.8 4.8 Assembly Proof of V.3.1.

Take the family of V.3.b. Then:

- item 1 is V.3.s; item 2 is V.3.j; item 3 is V.3.s; item 4 is V.3.2;
- item 5 is V.3.n; item 6 is V.3.o; item 7 is V.3.t; item 8 is V.3.y;
- item 9: the Wightman part is V.3.g; the Euclidean part is V.1's barycenter formula, which translates a word of distinct points into a slab using the homogeneity of S^{mix}.

Lean: `all_characters`, `Resolution.barycenter`.

3.23.5 5. Where the difficulty lies

The delicate points are the common bounds near the analytic boundary, the recovery of positive-height Euclidean vectors from joint frequency tests, and the joint current limits in the stress-tensor

argument. The last passage uses equal continuation germs only at spacelike separation; the current itself is spatially integrated. It gives a charge commutator for every admissible time profile, without defining a stress field at a fixed real time.

The low-energy estimate needs only locality, the spectrum condition and the two vector-norm bounds. Its positive weight avoids differentiating spectral integrals or imposing generator-domain bounds. The exponent 2 is compatible with the massless scalar measure, since $\int_{|\vec{p}|\le u} d^3p/|\vec{p}| \sim u^2$. Thus this estimate does not imply a mass gap. For removal of the exceptional set, any decay rate tending to zero uniformly in ξ would suffice.

The remaining integrations, dyadic summation and assembly are routine once their uniform bounds have been established.

What looks unusual.

- Item 8 quantifies over all admissible time profiles and spatial cutoffs. Admissible profiles and cutoffs exist (`Stress.accepts`, `Stress.reservable`); this existence assertion is separate from the commutator conclusion. The main theorem does not use it or the auxiliary results of Remark V.3.3.
- The reconstruction agrees with V.2 at ξ_*. The subsequent proof of vacuum uniqueness applies to every phase.
- The Euclidean half of item 9 restates V.1, and the public clause (R12) is derived from it.
- Nothing here or in the main theorem decides whether X is a single point. Each $\mathfrak{W}_\xi$ is irreducible, while S^{mix} is the ν-mixture of the S^ξ.

3.23.6 6. Notation

object	code name	proposed notation
phases, measure, selected phase	`F.α.space.X`, `F.α.space.μ`, `F.α.atPoint`	X, ν, ξ_*
characters of the ground algebra	`F.α.space.character`, `Spread.algebra`	χ_ξ on $\mathfrak{A}$
unselected table; table of phase ξ	`S`; `F.α.ensemble ξ`	S^{mix}; S^ξ
OS space and slab operators of S^{mix}	`F.α.room`, `F.α.unselected`	$\mathcal{H}^{\mathrm{OS}}$, $\Pi(d,w)$
ground space, Euclidean vacuum	`Chamber.D`, `Chamber.omega0`	$\mathcal{D}_\xi$, ω_ξ
one-dimensional ground space	`Chamber.UniqueHeat`	$\mathcal{D}_\xi = \mathbb{C}\omega_\xi$
family of reconstructions	`Almanac`	$(\mathcal{H}^{\mathrm{OS}}_\xi, \Pi_\xi, \mathsf{F}^\xi, \mathfrak{W}_\xi, J_\xi)_{\xi\in X}$
OS space, slab operators of S^ξ	`L.chambers ξ`, `L.spreads ξ`	$\mathcal{H}^{\mathrm{OS}}_\xi$, $\Pi_\xi(d,w)$
continuation and boundary values	`L.contours ξ` (`waves`, `world`)	F^ξ_n, W^ξ_n
Wightman theory of phase ξ	`L.actors ξ`	$\mathfrak{W}_\xi = (\mathcal{H}_\xi, \mathfrak{D}_\xi, \Omega_\xi, \phi_\xi, U_\xi, P_\xi)$
boundary isometry	`L.J ξ`, `L.links ξ` (`Attachment`)	J_ξ
Wightman data without vacuum uniqueness	`FieldCore`	—

object	code name	proposed notation
heat semigroup on $\mathcal{H}_\xi$	`bHeat`	$e^{-sH_\xi} = J_\xi^* e^{-sH_\xi^{\mathrm{OS}}} J_\xi$
zero-energy projection	`momentum.region zeroE`	P_ξ^0
finite sum of words; compact supports; adjoint; vector	`SumOfWords`, `finite`, `reverseP`, `stands`	A, $A^\dagger$, $A\Omega_\xi$
spectral measure of $A\Omega_\xi$	`vletter`	μ_A^ξ
low-energy region; shell	`lowE`, `ringE`	$\{0 < p^4 \le u\}$; $\{u/2 < p^4 \le u\}$
vacuum commutator function	`delta`	$\langle \Omega_\xi, [A^\dagger, \alpha_p A]\Omega_\xi \rangle$
time-averaged odd part; weight	`averaged`, `pulse`	$D(\vec{x})$; $\pi(0) = 0$, $\pi(E) = (1 - \cos E)/E$ for $E > 0$
strict-height values; tested values	`waves.restricted`, `waves.psi`	$\mathsf{F}_n^\xi(a; t(x) + iy, \vec{x})$; $\psi^\xi(a, y, g)$
weak commutant	`member`, `weaklyCentral`	—
Minkowski stress labels and fields	`Stress.M`, `Stress.fieldM`	T_{ab}^{M}, $T_{ab}(f)$
Killing field; charge; Lie derivative of a field	`Stress.Killing`, `Stress.Charge`, `Stress.action`	K, $Q_K(f, c)$, $\phi(\Sigma_K B) + K^a \partial_a \phi(B)$
stress record	`Stress.Force` (fields appointment, orbital, symmetric, hermitian, conserve, commutator, rule)	(T0)–(T6)
complete record	`Completed` (fields isTheory, into_onto, euclid, mink, light, heat, commutant, continued_stress, depends, at_boundary, at_words)	items 1, 2, 3, 4, 5, 6, 7, 8, 9, 9, 9 of V.3.1

3.23.7 7. What remains unexplained

1. **Analytic inputs from V.2.** The constructions `contour`, `compatible`, `establishing` and their use here were checked in source. The underlying C01–C29 analytic estimates, including the partial-boundary identities used in V.3.i and the exchange continuations in (T0), remain inputs from V.2; their proofs are not audited here. The local statements and proof sketches have been checked against the declarations that consume them.
2. **Partial-boundary data.** (T0) states the existence and compatibility needed for the current-jump argument. A detailed proof must specify the admissible tube faces and the spacelike neighborhoods of the tied pair; it must not replace integrated current limits by pointwise limits of the stress insertion.

3.24 VI.1. Ward identities, the rotation action and the conserved stress tensor

Part: VI, symmetry: stress tensor and Ward identities. **Lean:**

- the work items A01–A17, stated in `YM/Bid/Angular.lean`, combined in `YM/Bid/Town.lean` and proved in `YM/Work/A01`–`A17`;

- the collinear flux estimate, in YM/Next/ROTATIONS with YM/Axis;
- the analytic continuation, in YM/Ward;
- the scaling comparison, in YM/Climb;
- the descent of the rotation action, in YM/Spin;
- the assembled record, in YM/Org/Twist.lean;
- in part, YM/Push/STRESS_FLUX, YM/Shift and YM/Next/BOUNDARY.

Size (approximate):

- 7,704 lines are assigned to this unit and used by the main theorem. Of these, about 900 (Shift and most of Next/BOUNDARY) are reconstruction material for V.2, and about 1,000 (STRESS_FLUX) are free-field algebra used in VII.2 (Remark in §4.10).
- About 6,100 further lines of Next/ROTATIONS carry the collinear flux estimate. The declaration table files them under III.6, but they belong here, and the III.6 note says so too.

Conventions.

- *Table.* S is the limit table of IV.1. For labels $v_i \in F_{d_i}$ and $x \in \mathrm{Conf}_n$ (pairwise distinct points of $\mathbb{R}^4$), $S(v_1, \dots, v_n; x_1, \dots, x_n)$ is a correlation kernel, smooth on Conf_n.
- *Label spaces.* F_d is the finite-dimensional complex space of labels of degree $\le d$ modulo null labels. It has inclusions $\iota = \iota_{d\to e}$, a unit $\mathbf{1}_d$, the action M_σ of the signed permutation matrices $\sigma \in B_4 \subset O(4)$, and an antilinear conjugation Γ.
- *Euclidean axioms*:
  - permutation symmetry, normalisation of the empty correlation, deletion of unit insertions, and translation invariance;
  - covariance under B_4 and Γ;
  - reflection positivity in the coordinate hyperplanes, for point words and for smeared tests;
  - smoothness on configuration space;
  - for each fixed arity, labels and derivative order j, constants $C \ge 0$ and $A \in \mathbb{N}$ such that $\|D^j S(x)\| \le C r^{-A}$ whenever $0 < r \le 1$ and the points are separated by at least r in the maximum norm. The constants are uniform in the positions.
- *Differential plan.* Complex linear maps $\partial_a : F_d \to F_{d+1}$ satisfying three laws:
  - the *derivative law* $S(\dots, \partial_a v_k, \dots; x) = \partial_{x_k^a} S(\dots, v_k, \dots; x)$;
  - the *inclusion law* $S(\dots, \iota v_k, \dots) = S(\dots, v_k, \dots)$;
  - *orientation*: Γ and M_σ commute with the inclusions, ∂ commutes with Γ, and $M_\sigma \partial_b = \sum_a \sigma_{ab}\, \partial_a M_\sigma$. Commutation of ∂ with the inclusions follows from the inclusion and derivative laws.
- *Rotations.* $\mathfrak{so}(4)$ is the space of real skew 4×4 matrices with $[K, K'] = KK' - K'K$. $K^{(ba)} = E_{ba} - E_{ab}$ generates the rotation field $x_a e_b - x_b e_a$.
- *Spin rows.* A system of spin rows is a family of $\mathbb{R}$-linear maps $T : \mathfrak{so}(4) \to \mathrm{End}_{\mathbb{C}} F_d$.
- *Angular sum.* For spin rows T and maps $P_b : F_d \to F_{d+1}$,

$\mathcal{W}_K[T, P](https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q; x$

By the derivative law, $\mathcal{W}_K[T] = 0$ says $\frac{d}{dt}\big|_0 S(q; e^{tK}x) = -\sum_i S(\dots, T_K q_i, \dots; x)$. This is infinitesimal rotation covariance with spin part T. - *Lorentz group.* Axes are numbered $0, 1, 2, 3$; axis 3 is

time and $0, 1, 2$ are spatial. Write $e_t = e_3$. The axis 0 used for the continuation in §4.4 has no distinguished physical role. - All identities hold at separated points. No contact term is evaluated.

3.24.1 1. What this unit proves

Inputs.

- The table and its differential plan.
- An unnormalised stress $J = (J_{ab})$, $J_{ab} \in F_4$, with complex linear *raw translation maps* $p_a : F_d \to F_{d+1}$ and real linear *raw spin rows* $Y : \mathfrak{so}(4) \to \operatorname{End} F_d$. These satisfy the shell identities below with unknown right-hand sides.
- Free-Maxwell comparison data: a free table with its differential plan ∂^{free}, rotation action R^{free} with generators $\mathcal{L}_K^{\text{free}}$, stress J^{free} and Ward constant $\kappa \neq 0$.
- A weak scaling comparison of S with the free table.

The table and differential plan come from IV.1. The stress, shell maps and comparison data come from VI.2, using ultraviolet inputs as specified in §2. Every result below assumes the Euclidean axioms for S, unless it is a statement solely about groups or operators.

Outputs. Given the differential plan and the comparison inputs, the unit constructs:

- an action of $O(4)$ on every F_d whose generators are $c^{-1}Y$;
- $O(4)$-covariance of S;
- a conserved, symmetric, real, $O(4)$-tensorial stress J' satisfying the Ward identity (13) of the assessment and its rotation analogue;
- a Lorentz action on the labels, compatible with the rotation action.

Shells and stress laws. A *window* for $x = (x_0, \dots, x_n) \in \operatorname{Conf}_{n+1}$ is a real function $\chi \in C_c^\infty(\mathbb{R}^4)$ that equals 1 near x_0 and whose closed support excludes $x_1, \dots, x_n$. For $B \in F_d$ and tags q at $x_1, \dots, x_n$, the two shells are

$$\Phi_\chi^{\text{tr}}(b) = -\sum_a \int \partial_a \chi(y)\, S(J_{ab}(y), B(x_0), q)\, dy, \qquad \Phi_\chi^{\text{rot}}(K) = -\sum_{a,b} \int \partial_a \chi(y)\, \big(K(y - x_0)\big)_b\, S(J_{ab}(y), B(x_0), q)\, dy.$$

The integrands vanish near all insertions. Given maps $P_b : F_d \to F_{d+1}$ and $A : \mathfrak{so}(4) \to \operatorname{End} F_d$, the triple $(J; P, A)$ satisfies the *stress laws* if

$$\Phi_\chi^{\text{tr}}(b) = S(P_b B(x_0), q) \qquad \text{and} \qquad \Phi_\chi^{\text{rot}}(K) = S(A_K B(x_0), q)$$

for all windows, all B and all q (`Table.StressLaws`, `shellGrad`, `shellSpin`, `Window`).

Raw hypotheses (R) (`Climb.Pre`):

- J is *conserved*: symmetric, Γ-real, and $\sum_a \partial_a J_{ab} = 0$ in F_5 (`Conserved`);
- J is *raw-tensorial*: $M_\sigma J_{ab} = \sum_{ij} \sigma_{ia}\sigma_{jb} J_{ij}$ for $\sigma \in B_4$ (`RawTensor`);
- $(J; p, Y)$ satisfies the stress laws.

Proposition VI.1.1 (collinear rotation Ward identity and line sums).

Hypotheses.

- S satisfies the Euclidean axioms.
- ∂ satisfies the derivative law and orientation.
- (J, p, Y) satisfies (R).

Conclusions.

- (i) Let $x \in \mathrm{Conf}_n$ have all its points on a coordinate axis $\mathbb{R}e_l$. Then for every $K \in \mathfrak{so}(4)$ and all tags q: $\mathcal{W}_K[Y, p](\mathit{https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q}; x) = 0$.
- (ii) (Line sums.) Let $x \in \mathrm{Conf}_n$ lie on $\mathbb{R}e_l$, let $j \neq l$, and let $v_1, \dots, v_n \in F_d$. Then $\sum_i S(v_1, \dots, p_j v_i, \dots, v_n; x) = 0$.

Status: theorem.

- It answers the request `Bid.Miles` through `Outer.miles`.
- The analytic input is the exterior flux estimate (Lemma VI.1.e). It uses reflection positivity and the polynomial bounds, but no mass gap or clustering.

Lean: `Outer.outer`, `Bid.LineSum`, `Climb.localTurn`.

Proposition VI.1.2 (normalisation of the shell maps: $p = c\,\partial$).

Hypotheses.

- ∂ satisfies all three laws.
- The ultraviolet record holds, in particular (R) and the weak comparison law against the free data (input 4 of §2).
- The free-Maxwell data hold (input 3 of §2).

Conclusion. There is a real $c \neq 0$, independent of the degree and coordinate axis, such that:

- $p_a = c\,\partial_a$ on every F_d, for every a;
- the rescaled rows $D := c^{-1} Y$ satisfy
  - (C) $\partial_b D_K + \sum_i K_{ib}\,\partial_i = D_K\,\partial_b$ on every F_d;
  - (L) $\mathcal{W}_K[D](\mathit{https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q}; x) = 0$ for every configuration x on a coordinate axis.

Status: theorem.

Lean: `Bid.pressured`, `Bid.formed`; the record is `Climb.Crystal`.

Proposition VI.1.3 (rotation Ward identity at all separated configurations).

Hypotheses.

- S satisfies the Euclidean axioms.
- ∂ satisfies the derivative law.
- The spin rows T satisfy (C) and (L).

Conclusion. $\mathcal{W}_K[T](\mathit{https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q}; x) = 0$ for all n, all $x \in \mathrm{Conf}_n$, all tags q and all K.

Status: theorem.

Lean: `Ward.global`; the vanishing is `Ward.Still`; the hypotheses are `Ward.commutes` and `Ward.Collinear`; the sum is `Ward.angular`.

Proposition VI.1.4 ($\mathfrak{so}(4)$ brackets; A08).

Hypotheses.

- ∂ satisfies the inclusion and derivative laws.
- T satisfies the conclusion of VI.1.3.
- T is *compatible* (`Bid.Hyper`): it commutes with Γ and with ι, kills $\mathbf{1}$, and $M_\sigma T_K = T_{\sigma K \sigma^{-1}} M_\sigma$.
- *Internal row erasure* holds (`Org.erases`). That is, for some N and all $d \geq N$, let $\Delta : \mathfrak{so}(4) \times \mathfrak{so}(4) \to \operatorname{End} F_d$ be a real bilinear family of complex linear endomorphisms with these properties (`Org.Winds`):
  - it is covariant under signed permutations of determinant 1;
  - it is Γ-real;
  - it kills $\mathbf{1}$;
  - it is alternating;
  - for every $e \leq d$ and u, v, $\Delta(u, v)$ preserves $\iota_{e \to d}(F_e)$.

  If moreover $\sum_i S(w_1, \dots, \Delta(u, v) w_i, \dots, w_m; x) = 0$ for all m, all $x \in \mathrm{Conf}_m$, all $u, v \in \mathfrak{so}(4)$ and all $w \in F_d^m$, then $\Delta = 0$.

Conclusion. $[T_u, T_v] = T_{[u,v]}$ on every F_d.

Status: theorem. Erasure for S is itself proved (A07, Lemma VI.1.q).

Lean: `Work.A08.answer` for `Bid.A08`; the conclusion is `Bid.Brackets`.

Proposition VI.1.5 (descent of the free rotation action). Under the hypotheses of VI.1.2, let c be its normalisation constant and put $D = c^{-1} Y$.

Conclusion. There are complex linear isomorphisms $V_d : F_d^{\mathrm{free}} \to F_d$ with these properties:

- $V_d M_\sigma^{\mathrm{free}} = M_\sigma V_d$ and $V_d \Gamma^{\mathrm{free}} = \Gamma V_d$;
- $D_K V_d = V_d \, \mathcal{L}_K^{\mathrm{free}}$ for all $K \in \mathfrak{so}(4)$.

Hence $R_d(k) := V_d \, R_d^{\mathrm{free}}(k) \, V_d^{-1}$, $k \in O(4)$, is a continuous representation of $O(4)$ on F_d. It equals M_σ on B_4, commutes with Γ, and has infinitesimal generator D. Call such an action, with its specified generators, a *rotation port*. The Ward identity of VI.1.3 holds for D.

Status: theorem. The V_d are chosen independently for each d. Compatibility with the inclusions comes later (VI.1.6).

Lean: `Climb.descends`, `Spin.produces`, `Spin.meet`, `Bid.services`; the transported action is `Spin.morph` and a port is `Spin.Port`.

Proposition VI.1.6 (covariance and the rotation laws; A14).

Hypotheses.

- ∂ satisfies the inclusion and derivative laws.
- R is a port with generators T.
- $\mathcal{W}_K[T] = 0$ on all of Conf.

Conclusions.

- R is *oriented*: $S(R(k)v_1, \dots, R(k)v_n; kx_1, \dots, kx_n) = S(v; x)$ for all $k \in O(4)$ and all $x \in \mathrm{Conf}_n$.
- R satisfies the *rotation laws*: it fixes $\mathbf{1}$; it commutes with ι; it acts on derivatives as a vector, $R_{d+1}(k)\,\partial_a v = \sum_i k_{ia}\,\partial_i R_d(k) v$; and it keeps the port properties.

Status: theorem. It uses A13: $O(4)$ is generated by B_4 and the one-parameter groups e^{tK}.

Lean: `Work.A14.answer` for `Bid.A14`; `Rotation.Laws`, `Rotation.Oriented`; `Work.A13.answer`.

Proposition VI.1.7 (the averaged stress tensor and the Ward identity (13); A15).

Hypotheses.

- ∂ satisfies the derivative law.
- R is a rotation action satisfying the rotation laws, and R is oriented.
- $p = c\,\partial$ with $c \neq 0$, and (J, p, Y) satisfies (R).
- The generators of R are $c^{-1}Y$.

Conclusion. There is a stress J' with these properties:

- it is conserved, i.e. symmetric, real and divergence-free;
- it is *tensorial*: $R_4(k)J'_{ab} = \sum_{ij} k_{ia}k_{jb}J'_{ij}$ for all $k \in O(4)$;
- $(J'; \partial, c^{-1}Y)$ satisfies the stress laws.

The translation law is the Ward identity (13):

$$-\sum_a \int \partial_a \chi(y)\, S(J'_{ab}(y), B(x_0), q)\, dy = \frac{\partial}{\partial x_0^b} S(B(x_0), q).$$

Status: theorem. It uses A10.

Lean: `Work.A15.answer` for `Bid.A15`; `Table.Tensorial`.

Proposition VI.1.8 (Lorentz action on the labels). For (b)–(c), assume the hypotheses of VI.1.2 and choose V_d, R and J' as in VI.1.5–7.

(a) $SO^+(1,3)$ is generated by the spatial rotations and the coordinate boosts (A16).

(b) Let B^{free} be the Lorentz action on the free labels, which is part of the free data. Let $B_d(\Lambda) = V_d B_d^{\mathrm{free}}(\Lambda) V_d^{-1}$. Then $B_e(\Lambda)\,\iota = \iota\, B_d(\Lambda)$ for all $d \le e$ and $\Lambda \in SO^+(1,3)$ (A17).

(c) The compatible actions B_d define a representation on the union of the label spaces. For every degree d and spatial axis i:

- the action on F_d is continuous;
- $\frac{d}{ds}\big|_{s=0} B_d(\mathrm{boost}_i(s))v = -i\, D_{K^{(3i)}} v$ for every $v \in F_d$;
- B_d equals R_d on spatial rotations;
- B_d commutes with the time-reflection adjoint $M_{\theta_t}\Gamma$, where θ_t reverses the time coordinate.

Status: theorems.

Lean: `Work.A16.answer` for `Bid.A16`; `Work.A17.answer` for `Bid.A17`; `Spin.wand`, `Spin.reach`, `Boundary.Spins.Descends`.

Proposition VI.1.9 (the assembled record). Let the free data, the infrared triple (J, p, Y) and the ultraviolet record be as in §2 (inputs 3–4). Then the six assembly clauses hold: compatibility of the raw maps, their normalisation, descent of rotations, covariance, normalisation and averaging of the stress, and compatibility of the Lorentz action with inclusions.

Hence, if ∂ satisfies the inclusion and derivative laws, there exist c, (V_d) and J' such that the following local field laws hold:

- the normalisation of VI.1.2 for c;
- V_d intertwines M and Γ;
- $R = V R^{\mathrm{free}} V^{-1}$ with generators $c^{-1}Y$ is a port;
- the three plan laws; the rotation laws and orientation of R; each F_d is isomorphic to F_d^{free} compatibly with M and Γ; and J' is conserved, tensorial and satisfies the stress laws with $(\partial, c^{-1}Y)$.

There is also a Lorentz action descending from R (VI.1.8(c)).

Status: theorem conditional on the inputs in §2. Their production uses VI.2 and the ultraviolet units; VIII.1 assembles the resulting local field laws with the phase and reconstruction data.

Lean: `Outer.commission`, `Bid.station`, `Org.Commission`, `Commission.picks`, `Org.Tab`, `Spec.Locals`.

3.24.2 2. What it uses

Inputs from other units.

1. **IV.1 (limit table and differential plan).**
   - The Euclidean axioms `Table.Hierarchy`.
   - The plan laws `DiffPlan.Derivatives`, `Inclusions` and `Orientation`.
   - Also: $\partial_a \partial_b = \partial_b \partial_a$; injectivity of ι; the unique lift of compatible maps to the union of label spaces.
2. **V.1 (OS chambers).**
   - For each coordinate axis there is the Osterwalder–Schrader Hilbert space of reflection in $\{x_a = 0\}$ (`OS.Chamber`). It carries a self-adjoint contraction semigroup e^{-tH}, the unitary translations parallel to the hyperplane, and their joint spectral measure.
   - It is built from the axioms by `OS.realization`, so it is not an assumption.
3. **VI.2 (free-Maxwell data)** `Org.MaxwellData`:
   - a free table with its axioms;
   - free comparison data (`Climb.Lighthouse`), i.e. $(\partial^{\mathrm{free}}, R^{\mathrm{free}}, J^{\mathrm{free}}, \kappa \neq 0)$, with laws `Goes`:
     - plan laws;
     - rotation laws and orientation;
     - stress laws for $(J^{\mathrm{free}}; \kappa\partial^{\mathrm{free}}, \kappa\mathcal{L}^{\mathrm{free}})$;
     - ∂^{free} is nonzero on F_4^{free};
     - at every sufficiently large degree, finitely many collinear readouts whose common kernel within the free vector-map space is $\mathbb{R}\partial^{\mathrm{free}}$;
   - a free Lorentz action B^{free} descending from R^{free};
   - the perfect commutator relations among the free generators;
   - Casimir projection polynomials;
   - free row erasure.

The free-data producer `Maxwell.Free.maxwell_supplies` and its stress replacement `FreeStress.fire` are supplied in VI.2. A12 and A10+A09 give the Casimir polynomials and the generator relations. These free-data producers assume G simple.

4. **VI.2 and III.11 (raw stress, shell maps and weak comparison).**
   - `Org.IR` is the triple (J, p, Y).
   - `Org.UV` contains the lattice and extraction data, corrected-field bounds and Euclidean axioms for S; orientation of ∂; (R); the *weak action law*; and grid approximations.
   - The weak action law `Climb.ActLaw` says: for every arity bound N, degree bound d, derivative bound m and $R \geq 1$, there is a *scaling system* $X = (r_k \downarrow 0,\ \xi_k > 0,\ U_{k,e} : F_e^{\mathrm{free}} \to F_e)$ (`Climb.Scaling`) with a *track* (`Climb.Track`):
     - every $U_{k,e}$ is a complex linear isomorphism; for $e \leq \max(d, 4)$ it intertwines M, Γ and the unit, and these maps respect inclusions within that degree range;
     - for $n \leq N$, labels of degree at most d, and $j \leq m$, $D_y^j S(U_k v; r_k y) \to D^j S^{\mathrm{free}}(v; y)$ uniformly on the compact set $\|y\| \leq R$ with pairwise maximum-norm separation at least R^{-1};
     - $\xi_k^{-1} r_k^4 U_{k,4}^{-1} J_{ab} \to J_{ab}^{\mathrm{free}}$ for every a, b.
   - The scaling system may depend on all four bounds. Any finite collection of requests can be combined by increasing them.
   - The conserved-current construction produces the raw stress; the annular-field construction supplies its shell maps. The immediate suppliers are `Acme.readyA` and `Acme.shells_law`. They provide the weak comparison and shell laws for this same stress.
5. **VIII.1 (assembly).** Combines the local field laws obtained here with the phase decomposition and reconstruction.

Standard mathematics.

- Ward identities for a conserved symmetric stress tensor: the flux of $J_{ab}V_b$ through a closed shell, for a Killing field V (Noether; e.g. Itzykson–Zuber §2-4).
- Osterwalder–Schrader reconstruction and the transfer operator along a coordinate axis (Osterwalder–Schrader 1973/1975; Glimm–Jaffe, *Quantum Physics*, ch. 6 and 19). Spectral theorem and Laplace representations.
- Analytic vectors (Nelson, Ann. of Math. 1959). Holomorphic contraction semigroups (Reed–Simon II). Formal uniqueness at a noncharacteristic line (as proved in VI.1.m). Identity theorem for real-analytic functions.
- Representation theory:
  - Haar measure and Weyl's unitarian trick;
  - the Casimir element;
  - McWeeny purification $x \mapsto 3x^2 - 2x^3$ of approximate idempotents (Rev. Mod. Phys. 1960).
- Groups and ODEs:
  - Cartan–Dieudonné ($O(n)$ is generated by reflections; Mathlib);
  - Givens rotations;
  - generation of $SO^+(1, 3)$ by rotations and boosts;
  - Picard–Lindelöf uniqueness.

Imported constants and assumptions.

- $\kappa \neq 0$ comes from the chosen free data. The initial free-data constructor has $\kappa = -\tau$; the

stress replacement used in assembly supplies a common nonzero Ward constant without identifying its numerical value.
- c is determined by $p = c\,\partial$. Along scaling systems satisfying the degree-4 requests in VI.1.g, $c/\xi_k \to \kappa$ (Lemma VI.1.u).
- The flux estimate uses $\lambda_k = e^k$ and the exponents $100, 150, 300, 3000$. These are internal choices.
- Inputs 3–4 are hypotheses of the assembly theorem; their existence is supplied by the units cited above.

3.24.3 3. Strategy

Conservation and the raw shell laws make the maps p and Y compatible with inclusions, conjugation and signed permutations. Moving the center inside a fixed window gives their commutators with derivatives. To obtain a global rotation identity on a coordinate line, test the stress against a compact vector field that equals the rotation field near the insertions. Its cutoff has widths λ^{100} in the rotation plane, λ^{150} in the line direction if that direction is transverse to the plane, and λ^{300} in the remaining directions. Two derivative terms cancel by symmetry of the stress. Reflection positivity and spectral estimates make the remaining exterior flux $O(\lambda^{-1})$, so the collinear angular sum vanishes. Translation along the line then gives the transverse translation sums.

The free comparison identifies $p = c\partial$: finitely many collinear tests have a one-dimensional kernel in the free vector-map space, and this rank condition persists along a sufficiently accurate scaling system. Dividing by c gives the collinear Ward identity for $D = c^{-1}Y$. Its derivatives determine the transverse Taylor series of each two-point function from an explicit analytic comparison function built from radial correlations. Positivity turns the resulting bounds on Gram-kernel jets into factorial bounds for derivatives of the corresponding OS vectors. Correlations are therefore analytic when one insertion moves while remaining extreme in some coordinate. The Ward identity propagates along finite chains of these moves; density gives it at every separated configuration.

Differentiating the global Ward identity twice gives the $\mathfrak{so}(4)$ brackets, after row erasure is transported from the free theory. A relative-Casimir polynomial and McWeeny iteration produce exact intertwiners with the free rotation generators. Transporting the free action through them gives $O(4)$ actions on each degree. The Ward identity implies covariance, which in turn gives compatibility with inclusions, the unit and derivatives. Haar averaging of $c^{-1}J$ gives the tensorial stress J' without changing its Ward laws. The free Lorentz action is transported through the same intertwiners; uniqueness for the boost ODEs makes it compatible across degrees.

3.24.4 4. Steps

3.24.4.1 4.1 Local structure of the raw shell maps (A01–A03) Lemma VI.1.a (A01: compatibility). Assume (R), the derivative law and the inclusion law. Then:

- p lies in the *colour space* (`Climb.ColorLaw`): $p_i\mathbf{1} = 0$; p commutes with ι and Γ; $M_\sigma p_i = \sum_j \sigma_{ji}\, p_j M_\sigma$.
- Y is compatible (`Bid.Hyper`).

Proof sketch. Each identity is tested against all words in a fixed window. The shells are fluxes of J against $\partial\chi \otimes V$, with $V \equiv e_i$ for p and $V(y) = K(y - x_0)$ for Y.

- *Unit.* If the label at the centre is $\mathbf{1}$, the unit axiom deletes that insertion. Integrating by

parts against χ gives $\int \chi \,\mathrm{div}(\sum_b V_b S(J_{ab}(y),\dots))$, and this vanishes. Conservation handles one term. The Killing property ($\partial_a V_b$ antisymmetric) together with $J_{ab} = J_{ba}$ handles the other.

- *Mirrors.* Substitute $y \mapsto \sigma y$ and use the window $\chi \circ \sigma^{-1}$ at σx. Then use B_4-covariance of S, raw tensoriality of J, and $\sigma K(\sigma^{-1}\cdot) = (\sigma K \sigma^{-1})\cdot$.
- *Reality.* Reality of S and J, and the weights are real.
- *Inclusions.* The inclusion law inside the integrand.
- The orientation hypothesis is not used.

Lean: `Work.A01.answer`, `translation_color`, `angular_hyper`, `cutoff_zero`, `flux_unit`, `flux_mirror`.

Lemma VI.1.b (A02: moving the centre). Assume (R) and the derivative law. Then

$$\partial_b Y_K u + \sum_c K_{cb}\, p_c u = Y_K\, \partial_b u \qquad \text{and} \qquad \partial_b\, p_i = p_i\, \partial_b$$

on every F_d (`Bid.ShellCom`).

Proof sketch. Keep χ fixed and move the centre to $x_0 + te_b$; for small t, χ is still a window. Differentiate both shell laws at $t = 0$.

- The right-hand sides give $S(\partial_b Y_K B)$ and $S(\partial_b p_i B)$, by the derivative law.
- On the left, the kernel's t-derivative replaces B by $\partial_b B$. The weight $K(y - x_0 - te_b)$ contributes $-\sum_c K_{cb}$ times the translation shell.
- Differentiation under the integral is dominated on $\operatorname{supp}\nabla\chi \times [-r, r]$, so no near-diagonal behaviour is needed.

Lean: `Work.A02.answer`, `deriv_shellSpin`, `stable`.

Lemma VI.1.c (A03: transversality). Assume the plan laws, the colour law for p, the second identity of VI.1.b, and the line sums of VI.1.1(ii). Then p and ∂ are *transverse* (`Climb.transverse`). That is: for $x \in \mathrm{Conf}_n$ on $\mathbb{R}e_a$, $b \neq a$ and $v \in F_d^n$, the function $y \mapsto \sum_k S(v_1, \dots, P_b v_k, \dots, v_n; y)$ has vanishing jet of every order at x, for $P = p$ and for $P = \partial$.

Proof sketch.

- The inclusion law extends the line sums to labels of mixed degree.
- By $\partial p = p\,\partial$ and the derivative law, a derivative of such a row sum in a direction $e_{k,c}$ is the same kind of row sum with ∂_c inserted at slot k. Induction on the order, quantified over all words, reduces every jet to a line sum at x.
- For ∂, the line sum is translation invariance, and the colour law is `Climb.native`, which uses orientation.

Lean: `Work.A03.answer`, `all_line`, `row_D`, `full_jet`.

3.24.4.2 4.2 The collinear Ward identity

Lemma VI.1.d (flux identity). Assume (R) and the derivative law. Let $x \in \mathrm{Conf}_n$ and $K \in \mathfrak{so}(4)$. Let F be a smooth compactly supported vector field that equals $y \mapsto Ky$ near every x_i. Then

$\mathcal{W}_K[Y,p](https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q;x)$

Proof sketch.

- Choose disjoint windows χ_i around the x_i, and split $F = (F-u)+u$ with $u = \sum_i \chi_i\, K(\cdot)$.
- The flux of $F-u$ vanishes: it is a sum of constant-field fluxes against a cutoff supported away from all points, which are zero by conservation (as in VI.1.a).
- For u, write $\partial_a(\chi_i K y)_b = \partial_a \chi_i\,(Ky)_b + \chi_i K_{ba}$. The second term pairs to zero with the symmetric J.
- In the first term, write $Ky = K(y-x_i) + Kx_i$. These produce, at x_i, the rotation shell and $(Kx_i)_b$ times the translation shell. Use permutation symmetry to place x_i in the first slot.
- The shell laws turn these into the two terms of $\mathcal{W}$.

Lean: `Outer.flux_value`, `sands`, `paid_zero`.

Lemma VI.1.e (exterior flux estimate). Fix data and cutoff field as follows.

- Fix axes $a \neq b$ and a line direction l. Put $\lambda = \lambda_k = e^k$, with $k \to \infty$.
- Let η be an even smooth bump equal to 1 on $|s| \le 1$ and zero for $|s| \ge 2$, and put $\ell(s) = s\eta(s)$. Then ℓ is odd and equals s on $|s| < 1$. Its derivative ℓ' is even, compactly supported and equal to 1 on $|s| < 1$.
- Put $\mathrm{petal}_{ab}(y) = \lambda^{100}\,\ell(\lambda^{-100} y_a) \prod_{j \neq a} \ell'(\lambda^{c_j} y_j)$, with $c_j = -100$ for $j \in \{a,b\}$, $c_l = -150$ if $l \notin \{a,b\}$, and $c_j = -300$ otherwise.

Hypotheses.

- J is conserved and raw-tensorial.
- ∂ satisfies the derivative law and orientation.
- $x \in \mathrm{Conf}_n$ lies on $\mathbb{R}e_l$.
- $d \neq a$.

Conclusion. For every fixed tuple of tags q, the integrand is absolutely integrable for all sufficiently large k, and $\int \partial_d\, \mathrm{petal}_{ab}(y)\, S(J_{db}(y), q; x)\, dy = O(\lambda^{-1})$. The constant may depend on the tags, configuration, axes and stress, but not on k. The integrand is supported in the exterior shell $\lambda^{-c_d} \le |y_d| \le 2\lambda^{-c_d}$.

Proof sketch.

- *Cluster form.* Off the insertions, $S(J_{db}(y), q; x) = \langle \bar{B}; J_{db}(y) \rangle$. Here B is the reflected word of the collinear insertions. Its insertion points have transverse coordinates 0 (`collimated`).
- *Parity.* Decompose B into components of definite reflection parity in each axis. Only the component odd in a and b survives integration against the test (`Yard.lean`: `pick_price`).
- *Laplace representation.* Represent the correlation through the OS space of axis d. Because the test vanishes for $|y_d| < \lambda^{-c_d}$, the y_d-integral becomes a Laplace transform. Its time factor satisfies $|\mathrm{Lap}(E)| \le C\lambda^{c_d}/(E + \lambda^{c_d})$, paired with the polarised spectral measure $\nu_d(B, J_{db})$ of the transfer operator (`YM/Axis`; `exterior_size`).
- *Outside the low box.* Where $E > 1$ or some $|p_j| > \lambda^{1+c_j}$, the contribution is $O(\lambda^{-\infty})$. This uses only Schwartz decay of the profiles and the zeroth moment.

- *Inside the box.* A weighted Cauchy–Schwarz (Young) inequality splits the bound between $\nu_d(B)$ and $\nu_d(J_{db})$.
- *Shear case $d \notin \{a, b\}$.*
  - The line cost of the cluster is $\int_{E\le 1}(E+\lambda^{-3000})^{-1}\prod_j h_j\, d\nu_d(B) = O(\lambda^1)$ for $d = l$, and $O(\lambda^{150})$ otherwise.
  - The point cost $\int_{E\le 1}(E+\lambda^{c_d})^{-1}\prod_j h_j\, d\nu_d(J) = O(\lambda^\delta)$ holds for every $\delta > 0$.
  - Here h_j are fixed nonnegative even smooth bumps, rescaled in L^1 at widths λ^{1+c_j}; products run over $j \neq d$.
  - Before multiplication by the prefactor λ^{100} in the cutoff, the exponent is at most -146.
- *Diagonal case $d = b$.*
  - J_{bb} is even in every axis, so the mixed-parity point estimate does not apply. Split at $E = \lambda^{-3000}$.
  - Above the split, the cluster's line cost is paired with the *mass estimate* $\int_{E\le 1} EW^4(E+W)^{-4}\prod h_j\, d\nu_b(J_{bb}) = O(\lambda^\delta W^4)$ for every $\delta > 0$, at $W = \lambda^{-99}$.
  - Below the split, the packet estimate gains λ^{-1500}; the line estimate then suffices with a polynomial bound on the stress measure.
- *Point cost.* Proved by a *sign alternation.* The same Euclidean integral is a positive spectral pairing along one axis and minus a positive pairing along an axis of opposite parity. Moment-subtracted profiles have Laplace responses bounded below by $c/(E+w)$ down to $E = 0$. Walking through the axes, with orders chosen backwards, moves every profile to scale ≥ 1 at subpower cost. These estimates also control the contribution at zero energy and require no spectral gap.
- *Moment bounds.* For an axis-pointlike weight, $\int e^{-tE}|p_j|^{2m}\, d\nu \le Ct^{-D}$ for $0 < t \le 1$ and $j \neq d$. The polynomial kernel bounds give this estimate by finite differences. Shifted cluster weights have analogous damped moment estimates.
- *Mass estimate.* Uses conservation in tested form. Tested conservation relates the contact-normalised two-point symbols by $B = p_i^2 A + D_B$ and $C = p_i^2 B + D_C$. Contact terms vanish on tests with enough zero moments; the remaining error estimates enter after normalisation by the band widths. Positivity gives alternating signs for the three residual band differences. Comparing the bands W and $2W$ bounds the positive residual, which dominates the displayed mass integral.
- The mass estimate uses conservation, the derivative law and orientation. Raw tensoriality supplies the parities. Combining the two spectral regions gives $O(\lambda^{-101})$ before the prefactor λ^{100}, and hence the stated $O(\lambda^{-1})$ bound.

Lean: `Outer.blows`, `point_moment`, `alternate`, `budget_lock`; the cutoff profile is defined in `Patch.lean`; the line and point estimates are in `Tunnel.lean` and the mass estimate is in `Mass.lean`. *(proof read for the reduction and combination of the spectral estimates; supporting analytic inputs remain as listed in §7)*

Lemma VI.1.f (collinear identity and line sums). This is Proposition VI.1.1.

Proof sketch.

- Take $F = \mathrm{vane}_{ab}$, with $F_b = \mathrm{petal}_{ab}$, $F_a = -\mathrm{petal}_{ba}$ and $F_d = 0$ otherwise. For sufficiently large cutoff index it equals the Killing field of $K^{(ba)}$ near every x_i.
- In the flux of VI.1.d, the diagonal terms $\partial_a \mathrm{petal}_{ab}\, J_{ab}$ and $\partial_b \mathrm{petal}_{ba}\, J_{ba}$ cancel identically, since $\partial_a \mathrm{petal}_{ab} = \partial_b \mathrm{petal}_{ba}$ and J is symmetric. What remains is a finite sum of integrals of

VI.1.e.

- Hence $\mathcal{W}_{K^{(ba)}}[Y,p]$(*https : //github.com/scottnarmstrong/yang − mills − informalization/blob/main/outline/units/q; x*), which does not depend on n, is $O(e^{-n})$, so it is 0.
- $\mathcal{W}_K[Y,p]$ is $\mathbb{R}$-linear in K, so it vanishes for all K.
- Translating the configuration by e_l along its line changes $\mathcal{W}_{K^{(jl)}}[Y,p]$ by exactly $\sum_i S(\dots, p_j q_i, \dots)$. Both values vanish, which gives the line sums.

Lean: `draught_river`, `drained`, `axisTurn`, `fullTurn`, `stagger`, `vane_home`.

3.24.4.3 4.3 Normalisation $p = c\,\partial$ (A04–A06) Lemma VI.1.g (scaling limits; A04, A05). Assume the inclusion and derivative laws, (R), and the free data of §2. For each degree d, there are bounds d_0, N_0, m_0, R_0, with $R_0 \geq 1$, such that every track whose arity, degree, derivative and size bounds are at least N_0, d_0, m_0, R_0, respectively has the following limits. Write X for its scaling system and define $\mathrm{aisle}_{k,d}(P)_a = U_{k,d+1}^{-1} P_a U_{k,d}$. At this fixed degree:

- $r_k\,\mathrm{aisle}_{k,d}(\partial) \to \partial^{\mathrm{free}}$;
- $(r_k/\xi_k)\,\mathrm{aisle}_{k,d}(p) \to \kappa\,\partial^{\mathrm{free}}$;
- $\xi_k^{-1} U_{k,d}^{-1} Y_K U_{k,d} \to \kappa\,\mathcal{L}_K^{\mathrm{free}}$.

Proof sketch.

- *First limit.* Test against a finite family of free correlations that is faithful on F_{d+1}^{free}. The chain rule gives the factor r_k. Then use first-order jet convergence, and invert by finite-dimensional linear algebra.
- *Second and third limits.* Write the shell law at $r_k x$ with the window $\chi(\cdot/r_k)$. Substituting $y = r_k z$ produces $r_k^4 \cdot r_k^{-1}$. For the translation shell, multiplication by r_k/ξ_k gives the stress factor $\xi_k^{-1} r_k^4 J$. For the rotation shell, the weight $K(y - r_k x_0) = r_k K(z - x_0)$ supplies the additional factor, so the multiplier is ξ_k^{-1}. The stress convergence and uniform kernel convergence on the compact shell supports give the free shell laws. Finite faithful tests recover the two operator limits.

Lean: `Work.A04.answer`, `Work.A05.answer`, `integrate_sheet_scale`, `recover_on`; the requests are `Bid.Watches` and `Bid.ConLap`.

Lemma VI.1.h (A06 and the identification $p = c\,\partial$). Assume the hypotheses of VI.1.2 and the transversality conclusions of VI.1.c.

Setting.

- The free data supply d_* such that, for every $d \geq d_*$, there is a finite *selection* of collinear transverse readouts on F^{free}: m-th jets of row sums at configurations on a coordinate axis.
- These readouts vanish, within the colour space of the free table, exactly on $\mathbb{R}\,\partial^{\mathrm{free}}$ (`Climb.Ens`, `Select`).

Claims.

- (A06) Along any scaling system meeting a sufficiently large finite request for these readouts, the actual rescaled readouts converge to the free ones. They annihilate $\mathrm{aisle}_{k,d}(p)$ and $\mathrm{aisle}_{k,d}(\partial)$ exactly at every k. The reason is that a readout of $\mathrm{aisle}_{k,d}(P)$ equals r_k^m times an actual collinear jet of P, which vanishes by VI.1.c.
- Consequently there is a real $c \neq 0$ with $p_a = c\,\partial_a$ on all F_d.

Proof sketch.

- *Persistence lemma.* Let $\mathcal{C}$ be the fixed real space of free vector maps and suppose $Z_k \to Z$ with $\ker Z \cap \mathcal{C} = \mathbb{R}x$, $x \neq 0$. Choose a real linear functional ℓ with $\ell(x) \neq 0$. The map (Z, ℓ) is injective on $\mathcal{C}$; injectivity persists under small perturbations in finite dimension. Thus $\ker Z_k \cap \mathcal{C}$ has dimension at most one for large k. This gives $\operatorname{aisle}_{k,d}(p) = c_d \operatorname{aisle}_{k,d}(\partial)$, hence $p = c_d \partial$ on F_d for large d.
- The constants agree across d, and lower degrees follow, by injectivity of ι and the colour law.
- $p \neq 0$ and $\partial \neq 0$ by VI.1.g and $\kappa \partial^{\mathrm{free}} \neq 0$.

Lean: `Work.A06.answer`, `Climb.extends_form`, `Bid.pressured`.

Lemma VI.1.i (the normalised Ward data). With c as in VI.1.h, $D = c^{-1}Y$ satisfies (C) and (L). This proves Proposition VI.1.2.

Proof sketch. Divide the first identity of VI.1.b, and the identity of VI.1.1(i), by c, using $p = c\,\partial$.

Lean: `Bid.formed`, `Bid.witnessing`.

3.24.4.4 4.4 Continuation off the line Throughout this subsection S satisfies the Euclidean axioms, T satisfies (C) and (L), and ∂ satisfies the derivative law.

Lemma VI.1.j (derivatives of angular sums).

- $\partial_{x_k^a} \mathcal{W}_K[T](\mathit{https} : \mathit{//github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q}; x) = \mathcal{W}_K[T](\mathit{https} : \mathit{//github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q'}; x)$, where q' is q with q_k replaced by $\partial_a q_k$. In particular $\mathcal{W}_K[T](\mathit{https} : \mathit{//github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q}; \cdot)$ is C^∞ on Conf.
- Hence, if $\mathcal{W}_K[T](\mathit{https} : \mathit{//github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/\cdot}\,; y) = 0$ for all tags at one point y, then every derivative of $\mathcal{W}_K[T](\mathit{https} : \mathit{//github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/q}; \cdot)$ at y vanishes, for every q.

Proof sketch. Product rule. The derivative of $(Kx_k)_b$ in x_k^a is K_{ba}. Use $\partial_a \partial_b = \partial_b \partial_a$, and (C) absorbs the extra term. The second statement is induction on the order, with q generalised.

Lean: `Ward.traverses`, `Ward.ac`, `Ward.jet_line`.

Lemma VI.1.k (OS vectors of one insertion). Let $\mathcal{H}_a$ be the OS space for reflection in $\{x_a = 0\}$. Write Θ_a for the corresponding reflection of words, including the label adjoint $M_{\theta_a}\Gamma$. For $x_a > 0$ let $\operatorname{spot}(A, x) \in \mathcal{H}_a$ be the vector of the one-point word $A(x)$.

- (i) For any word w supported in $\{y_a > 0\}$ with distinct insertion points, its OS vector $[w]$ satisfies $\langle [w], \operatorname{spot}(A, x)\rangle = S(A(x), \Theta_a w)$.
- (ii) $\operatorname{spot}(A, x + s e_a) = e^{-sH}\operatorname{spot}(A, x)$ for $s \geq 0$, and $\operatorname{spot}(A, x + v) = U(v)\operatorname{spot}(A, x)$ for $v \perp e_a$.
- (iii) For $b \neq a$, $s \mapsto \operatorname{spot}(A, x + s e_b)$ is C^∞ with derivative $\operatorname{spot}(\partial_b A, \cdot)$. The weak derivative comes from the derivative law and is upgraded to a strong one.
- (iv) $t \mapsto \operatorname{spot}(A, x + t e_a)$ is real-analytic for $x_a + t > 0$, because e^{-zH} is holomorphic on $\operatorname{Re} z > 0$.

- (v) $t \mapsto S(A(te_a), B(0))$ is real-analytic for $t > 0$.

Lean: `Ward.spot`, `link`, `responds`, `blooms`, `advancing`, `Analysis.represents`, `Ward.radial`.

Lemma VI.1.l (transverse jets of the two-point function). Fix $a \neq b$ and $K = K^{(ba)}$. For a pair of labels $\pi = (\pi_1, \pi_2)$ put $\Phi_\pi(x_1, x_2) = S(\pi_1(x_1 e_a + x_2 e_b), \pi_2(0))$ on $\{x_1 > 0\}$. Define

$$(\mathcal{L}\Phi)_\pi = (x_1\partial_2 - x_2\partial_1)\Phi_\pi + \Phi_{(T_K\pi_1, \pi_2)} + \Phi_{(\pi_1, T_K\pi_2)}.$$

Then:

- $(\mathcal{L}\Phi)_\pi(x) = \mathcal{W}_K[T](\pi; (x_1 e_a + x_2 e_b, 0))$;
- $\partial_2^n(\mathcal{L}\Phi)_\pi(t, 0) = 0$ for all n, all $t > 0$ and all π.

Proof sketch. The first identity is the definition of $\mathcal{W}$, by the derivative law. At $x_2 = 0$ both points lie on $\mathbb{R}e_a$, so (L) gives $\mathcal{W} = 0$ for all labels. VI.1.j then gives all x_2-derivatives. This supplies the collinear jets needed for the analytic comparison.

Lean: `Ward.sigma`, `Ward.realized`.

Lemma VI.1.m (comparison function and formal uniqueness). Let (r, θ) be polar coordinates of (x_1, x_2) with $x_1 > 0$. Put

$$Q_\pi(x) = S((e^{-\theta T_K}\pi_1)(re_a),\ (e^{-\theta T_K}\pi_2)(0)).$$

- Q is real-analytic on $\{x_1 > 0\}$, by VI.1.k(v), finite-dimensionality of the label spaces, and the matrix exponential.
- $\mathcal{L}Q = 0$ identically, since θ has derivative 1 and r has derivative 0 along the rotation field.
- $Q = \Phi$ on $\{x_2 = 0\}$.
- (Formal uniqueness.) Let L be smooth on the half-plane with $L_\pi(t, 0) = 0$ and $\partial_2^n(\mathcal{L}L)_\pi(t, 0) = 0$ for all n, π and $t > 0$. Then $\partial_2^n L_\pi(t, 0) = 0$ for all n.

  Applied to $L = \Phi - Q$: the x_2-Taylor series of Φ at $(t, 0)$ is that of the analytic function Q.

Proof sketch. Strong induction on n, for all π at once. For $n \geq 1$, at $(t, 0)$,

$$t\,\partial_2^{n+1} L_\pi = -\partial_2^n(L_{(T_K\pi_1, \pi_2)} + L_{(\pi_1, T_K\pi_2)}) + n\,\partial_1\partial_2^{n-1} L_\pi.$$

For $n = 0$ the transport equation directly gives $t\partial_2 L_\pi = -L_{(T_K\pi_1, \pi_2)} - L_{(\pi_1, T_K\pi_2)} = 0$. In the displayed recurrence, the first term vanishes by induction, and the last term vanishes because $\partial_2^{n-1} L_\pi$ vanishes on the whole half-line. Since $t \neq 0$, the line is noncharacteristic.

Lean: `Ward.Q`, `htQ`, `solves`, `PDE.sure`.

Lemma VI.1.n (analytic Gram kernel).

- (Abstract.) Let $f : \mathbb{R} \to \mathcal{H}$ be C^∞ with $\langle f(s), f(s')\rangle = A(s' - s)$, where A has at 0 the Taylor series of a real-analytic germ. Then f is real-analytic on $\mathbb{R}$.
- (Application.) Hence, for $x_a > 0$ and $b \neq a$, $s \mapsto \mathrm{spot}(A, x + se_b)$ is real-analytic on $\mathbb{R}$.

Proof sketch.

- $\|f^{(m)}(s)\|^2 = |A^{(2m)}(0)| \le C\rho^{-2m}(2m)!$ uniformly in s. With $(2m)! \le 4^m(m!)^2$ this gives $\|f^{(m)}\| \le \sqrt{C}(\rho/2)^{-m}m!$ everywhere, and the Taylor remainder finishes.
- For the application: by reflection and translation invariance, $\langle f(s), f(s')\rangle = \Phi_{(A,\Theta_a A)}(2x_a, s'-s)$. VI.1.m identifies its Taylor series at 0 with that of $s \mapsto Q_{(A,\Theta_a A)}(2x_a, s)$.
- Uniformity in s, which comes from positivity and translation invariance, is why a jet at a single point suffices.

Lean: `Analysis.blossoms`, `bud`, `builds`, `Ward.fires`, `swing`, `Ward.pure`.

Lemma VI.1.o (analyticity at an extreme point). Suppose that after moving x_k to $x_k + se_b$, the point k is strictly latest, or strictly earliest, in coordinate a. Then $u \mapsto S(q; x \text{ with } x_k + ue_b)$ is real-analytic at s, for every b (including $b = a$). Hence $\mathcal{W}_K[T]$ is real-analytic along such a move.

Proof sketch.

- Translate in e_a so that the other points lie in $\{y_a < 0\}$ and $x_k + se_b$ in $\{y_a > 0\}$.
- Then $S = \langle \Theta(\text{word of the others}), \mathrm{spot}(q_k, \cdot)\rangle$, a bounded functional of one vector. This is analytic by VI.1.n for $b \ne a$, and by VI.1.k(iv) for $b = a$.
- The "earliest" case reduces to the "latest" case by the reflection Θ_a with mirrored labels.

Lean: `Ward.higher`, `last_adv`, `first_adv`, `Ward.channel`.

Lemma VI.1.p (walks; proof of VI.1.3). A *walk* is a finite chain of segments. Each segment moves one point x_k along e_b, and on the whole segment x_k is strictly earliest or latest in some coordinate a.

- If $\mathcal{W}_K[T](\mathit{https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/\cdot\,; y}) = 0$ (all tags) at the end y of a walk, then the same holds at its start.
- Every configuration with pairwise distinct 0-coordinates is joined by a walk to a configuration on $\mathbb{R}e_0$.
- Such configurations are dense in Conf_n.

Proof sketch.

- *Propagation.* On a segment, $\mathcal{W}$ is analytic (VI.1.o), and all its derivatives vanish at the endpoint (VI.1.j). By the identity theorem it vanishes on the segment.
- *Geometry.* Take the earliest point in coordinate 0. Move it beyond all others in coordinate 1, then beyond all others in coordinate 0, then set its coordinates $1, 2, 3$ to 0, keeping it latest in coordinate 0. Induct on the remaining points.
- *Density.* Add $t(0, 1, \dots, n-1)$ to the 0-coordinates.
- *Conclusion.* (L) holds on $\mathbb{R}e_0$, so $\mathcal{W} = 0$ on a dense set. $\mathcal{W}$ is continuous on Conf (VI.1.j).

Lean: `Ward.stood`, `Walk`, `along`, `park1`, `pushes`, `line`, `global`.

3.24.4.5 4.5 Brackets (A07, A08) Lemma VI.1.q (A07: erasure for the limit table).

The weak action law, with its alignment of M, Γ, units and inclusions, transports free row erasure to S and ∂.

Proof sketch.

- For $d \ge N$, the free bilinear families of the kind in VI.1.4 form a finite-dimensional real space $\mathcal{M}$. Free erasure means that the free row tests separate $\mathcal{M}$, so finitely many of them do.

- Along a scaling system, the transported actual rows converge to these, so they are injective on $\mathcal{M}$ at some scale k.
- An actual family Δ with vanishing row sums, pulled back by U_k, lies in $\mathcal{M}$, because U_k respects M, Γ, $\mathbf{1}$ and ι. Its rows are actual row sums at $r_k x$, which vanish. So $\Delta = 0$.

Lean: `Work.A07.answer`, `transfer`.

Lemma VI.1.r (A08: brackets). This is Proposition VI.1.4.

Proof sketch.

- Let $\mathcal{O}_u$ be the orbital derivative along $x_i \mapsto u x_i$ for all points at once. VI.1.3 says $\mathcal{O}_u S(q) = -\sum_i S(\dots, T_u q_i, \dots)$ on the open set Conf.
- Apply it twice: $\mathcal{O}_u \mathcal{O}_v S = \sum_{i,j} S(\dots T_u q_j \dots T_v q_i \dots)$. Linear vector fields satisfy $[\mathcal{O}_u, \mathcal{O}_v] = -\mathcal{O}_{[u,v]}$.
- Subtracting, the off-diagonal terms cancel. So the defect $\Delta(u,v) = [T_u, T_v] - T_{[u,v]}$ has vanishing row sums.
- By compatibility of T and $\sigma[u,v]\sigma^{-1} = [\sigma u \sigma^{-1}, \sigma v \sigma^{-1}]$, Δ is a family of the kind in VI.1.4. Erasure gives $\Delta = 0$ for $d \geq N$.
- Lower degrees follow by injecting into degree $\max(d, N)$.

Lean: `Work.A08.answer`, `orbital_comm`, `orbit_twice`, `defect_winds`, `defect_ward`.

3.24.4.6 4.6 Descent of the free rotation action (A09–A12) Fix d. Index the generators by ordered pairs $\pi = (i,j)$, with $K_\pi = E_{ij} - E_{ji}$ and $C_\pi = \mathcal{L}^{\mathrm{free}}(K_\pi)$ on F_d^{free}. For tuples B, C of operators on F_d^{free} define

- $\mathrm{ad}_{B,C}(X) = BX - XC$ on $\operatorname{End} F_d^{\mathrm{free}}$;
- $\Omega(B,C) = \sum_\pi \mathrm{ad}^2_{B_\pi, C_\pi}$, the relative Casimir.

A tuple satisfies the *perfect relations* if $X_{ij} = [X_{ik}, X_{kj}]$ for a fixed $k \notin \{i,j\}$. Any $\mathfrak{so}(4)$ representation does (A09).

Lemma VI.1.s (A12: the intertwiner projection). There is a real polynomial z such that $P = z(\Omega(C,C))$ satisfies:

- $P^2 = P$;
- $P(\mathrm{id}) = \mathrm{id}$;
- $\mathrm{ad}_{C_\pi, C_\pi} P = 0$ for all π.

So P projects onto the commutant of the C_π, along $\operatorname{range}\Omega$.

Proof sketch.

- Haar averaging over $O(4)$ gives an inner product on $\operatorname{End} F_d^{\mathrm{free}}$ in which every $\operatorname{ad} C_\pi$ is skew (unitarian trick). So Ω is self-adjoint, $\ker\Omega = \bigcap_\pi \ker \operatorname{ad} C_\pi$, and $\ker \Omega^2 = \ker \Omega$.
- Cayley–Hamilton, with the powers of X removed, gives a real z with $z(0) = 1$ and $\Omega z(\Omega) = 0$.

Lean: `Work.A12.answer`, `Spin.Works`, `wave`.

Lemma VI.1.t (approximate to exact intertwiners).

- (i) *Purification.* In a Banach algebra, let $Z_n \to z$ with $z^2 = z$. There are $C > 0$ and n_0 such that, for $n \geq n_0$, the iterates of $\varphi(x) = 3x^2 - 2x^3$ starting at Z_n converge to idempotents

$P_n \to z$, with $\|P_n - Z_n\| \le 2C\|Z_n^2 - Z_n\|$. This uses $\varphi(x) - x = (1-2x)(x^2-x)$ and $\operatorname{def}(\varphi x) = (4\operatorname{def} x - 3)(\operatorname{def} x)^2$, where $\operatorname{def} x = x^2 - x$.

- (ii) *Norm gap.* If operators D_π satisfy the perfect relations and all $\|D_\pi\| < \frac{1}{2}$, then all $D_\pi = 0$, since $\max\|D\| \le 2\max\|D\|^2$.
- (iii) *Exact intertwiners.* Assume:
  - tuples $B_n \to C$, with B_n (eventually) and C satisfying the perfect relations;
  - $P = z(\Omega(C,C))$ is as in VI.1.s;
  - there is a closed additive subgroup $\mathcal{F} \subset \operatorname{End}_{\mathbb{C}} F_d^{\mathrm{free}}$ containing id such that, eventually, $z(\Omega(B_n, C))$ preserves $\mathcal{F}$ and commutes with every $\operatorname{ad}_{B_{n,\pi}, C_\pi}$.

  Then for large n there is an invertible $f \in \mathcal{F}$ with $B_{n,\pi} f = f C_\pi$ for all π.

Proof sketch of (iii).

- Work in $\operatorname{End}(\operatorname{End} F_d^{\mathrm{free}})$ with $D_\pi(n) = \operatorname{ad}_{B_{n,\pi}, C_\pi}$. These satisfy the perfect relations and converge to $\operatorname{ad}_{C_\pi, C_\pi}$.
- $z(\Omega(B_n, C)) \to P$. Purify it to idempotents Q_n. Every iterate preserves $\mathcal{F}$ and commutes with $D_\pi(n)$. The preserving operators form a closed subring of $\operatorname{End}(\operatorname{End} F_d^{\mathrm{free}})$, so these properties pass to Q_n.
- The compressions $D_\pi(n) Q_n$ satisfy the perfect relations and tend in norm to $\operatorname{ad}_{C_\pi, C_\pi} P = 0$. By (ii) they vanish for large n.
- Then $f_n = Q_n(\mathrm{id}) \to P(\mathrm{id}) = \mathrm{id}$, so f_n is invertible. It lies in $\mathcal{F}$, and $\operatorname{ad}_{B_{n,\pi}, C_\pi} f_n = 0$.
- No complete reducibility of the representation $D = c^{-1}Y$ on F_d, and no Lie integration for it, is used.

Lean: `Spin.rectify`, `stable`, `enough`, `onto`, `realize`, `Tracks`.

Lemma VI.1.u (descent of rotations; proof of VI.1.5). Assume the hypotheses of VI.1.2. The normalised rows $D = c^{-1}Y$ satisfy the global Ward identity by VI.1.3.

- **Choice of U_n.** Choose one scaling system meeting the requests of VI.1.g at degrees d and 4, and aligned through degree d. Put $U_n = U_{n,d}$.
- **Convergence.** At degree 4, $p = c\,\partial$ and VI.1.g give $c/\xi_k \to \kappa$. Hence $U_n^{-1} D U_n \to \mathcal{L}^{\mathrm{free}}$.
- **Inputs of VI.1.t(iii).** These are the *tilt* (`Spin.Tilt`):
  - U_n intertwines M and Γ (track);
  - D satisfies the perfect relations, by A08 (with erasure from A07) and A09;
  - the free data supply these relations and the Casimir polynomial of VI.1.s for $\mathcal{L}^{\mathrm{free}}$;
  - let $\mathcal{F} \subset \operatorname{End}_{\mathbb{C}} F_d^{\mathrm{free}}$ be the closed real subspace of maps commuting with the free M and Γ. The operators $z(\Omega(U_n^{-1} D U_n, C))$ preserve $\mathcal{F}$ (A11: real coefficients and cancellation of the signed-permutation factors in the sum of squares) and commute with the relative generators (A09).
- **Exact intertwiner.** VI.1.t gives f_n. Put $V_d = U_n f_n$. Then V_d intertwines M and Γ, and $D_{K_\pi} V_d = V_d C_\pi$. By linearity this holds for all K.
- **Port.** $R_d = V_d R_d^{\mathrm{free}} V_d^{-1}$ inherits continuity, the group law, $R(\sigma) = M_\sigma$, reality, and the generator D.

Lean: `Bid.services`, `Bid.currencies`, `Bid.imaginary`, `Work.A09.answer`, `Work.A11.answer`, `Spin.fine`, `Spin.meet`, `Spin.produces`, `Climb.descends`.

3.24.4.7 4.7 Integration to $O(4)$ (A13, A14) Lemma VI.1.v (A13). $O(4)$ is generated by B_4 and the one-parameter groups e^{tK}, $K \in \mathfrak{so}(4)$.

Proof sketch. $O(4)$ is generated by hyperplane reflections (Cartan–Dieudonné). The reflection in $v^{\perp}$ equals $g^{-1}R_0 g$, where R_0 is a coordinate mirror and g is a product of plane rotations $e^{t(E_{ij}-E_{ji})}$ that turn v onto e_0 (Givens rotations).

Lean: `Work.A13.answer`, `reflection_conjugate`.

Lemma VI.1.w (A14: covariance and laws). This is Proposition VI.1.6.

Proof sketch.

- The set of k with $S(R(k)v; kx) = S(v; x)$ for all v, x is a subgroup of $O(4)$.
- It contains B_4, since $R(\sigma) = M_\sigma$ and S is B_4-covariant.
- It contains each e^{tK}. The function $f(t) = S(R(e^{tK})v; e^{tK}x)$ is differentiable, and by the group law $f'(t)$ is $\mathcal{W}_K[T]$ at the moved data, which is 0. So f is constant.
- By A13 the subgroup is all of $O(4)$.
- The rotation laws:
  - $R(k)\mathbf{1} = \mathbf{1}$: apply covariance, delete the unit insertion, apply covariance again;
  - commutation with ι: covariance plus the inclusion law (`Spin.team`);
  - the vector law: differentiate the covariance identity and use the chain rule for $x \mapsto kx$;
  - the remaining fields are those of the port.

Lean: `Work.A14.oriented`, `cov_parameter`, `unit_fixed`, `vectorLaw`, `laws_of_oriented`.

3.24.4.8 4.8 The averaged stress tensor (A15)

Lemma VI.1.x (A15: Haar averaging). Under the hypotheses of VI.1.7, put $J_0 = c^{-1}J$, so that $(J_0; \partial, c^{-1}Y)$ satisfies the stress laws. Define

$$J' = \int_{O(4)} k \cdot J_0 \, d\mu(k), \qquad (k \cdot J)_{ab} = \sum_{ij} k_{ai} k_{bj} \, R_4(k) J_{ij},$$

with μ normalised Haar measure on $O(4)$. Then J' satisfies Proposition VI.1.7.

Proof sketch.

- $k\cdot$ is a linear group action on the finite-dimensional space of stresses. Fixed points are exactly the tensorial stresses, and invariance of μ makes J' fixed.
- Each $k \cdot J_0$ is conserved: symmetry by the index structure, reality by the reality law, divergence by the vector law.
- Each $k \cdot J_0$ satisfies the same stress laws. Substitute $y \mapsto ky$ and use the window $\chi \circ k$ at $k^{-1}x$; pull the labels back by $R(k^{-1})$; use covariance of S.
  - The translation weight e_b becomes $k^{-1}e_b$. The right side is independent of k by the vector law.
  - The weight $K(y - x_0)$ becomes $k^{-1}Kk$ centred at $k^{-1}x_0$. The right side is independent of k by A10: $R(k)\,\mathcal{L}_K = \mathcal{L}_{kKk^{-1}}\,R(k)$.
- All the conditions are linear, so they pass to the integral.

Lean: `Work.A15.answer`, `normalize_flux`, `averageStress`, `tensorial_average`, `conserved_average`, `stressLaws_average`, `rotate_flux`, `Work.A10.answer`.

3.24.4.9 4.9 Lorentz group, and assembly Lemma VI.1.y (A16, A17 and the descended Lorentz action). This is Proposition VI.1.8.

Proof sketch.

- *(a) Generation.* Let $\Lambda \in SO^+(1,3)$. Then Λe_t is a future unit timelike vector. At each step, apply a coordinate boost with rapidity $-\operatorname{artanh}(y_i/y_t)$ to the current vector y. This kills its i-th spatial component and preserves the other spatial components. After three steps the vector is e_t. The resulting Lorentz transformation fixes e_t, so it is a spatial rotation.
- *(b) Compatibility with inclusions.* Induct over the generators.
  - Spatial rotations: here B is the transported rotation action, which commutes with ι by VI.1.6.
  - Boosts: $t \mapsto B_e(\mathrm{boost}_i(t))\iota v$ and $t \mapsto \iota B_d(\mathrm{boost}_i(t))v$ both solve $f' = Gf$ with $G = -i\,D_{K^{(3i)}}$, and they agree at $t = 0$. Here G commutes with ι, which follows by differentiating the inclusion law of R. Uniqueness for the linear ODE finishes.
- *(c) Descent.* Compatibility with inclusions and the unique lift to the union of label spaces give the representation. Its descent properties transfer from the free action through V (`Spin.transmits`).

Lean: `exists_rest`, `run_boost`, `rise_generator`, `Spin.run`, `Spin.reach`.

Lemma VI.1.z (assembly). Under the hypotheses of VI.1.9, the preceding constructions give its local field laws and Lorentz action.

Proof sketch. A01–A03 and the collinear flux argument give the raw compatibility laws. VI.1.h–i give c and the normalised Ward data. VI.1.3 and VI.1.u give the intertwiners, rotation port and global Ward identity. A13–A14 give covariance and the rotation laws; A10 and A15 give the conserved tensorial stress with the required shells. Each F_d has the free comparison isomorphism by any aligned scaling system at that degree. A16–A17 give compatibility of the Lorentz actions, and the unique lift to the union gives the final representation.

Lean: `Bid.station`, `Commission.picks`, `Climb.comparison`.

3.24.4.10 4.10 Free-field stress algebra

Assume G is simple. The free stress satisfies the following finite-dimensional algebraic facts:

- adjoint-invariant linear forms vanish, and invariant bilinear forms are multiples of the inner product.
- Gauge-invariant composites of weight ≤ 5 have explicit normal forms.
- The real symmetric B_4-covariant 2-tensors of weight at most 4 form a space of dimension ≤ 4, spanned by the four-parameter family

$$T_{ab} = \delta_{ab}\Big(w_0 + w_1 \sum_i \langle F_{ai}, F_{ai}\rangle + w_2 \sum_{ij} \langle F_{ij}, F_{ij}\rangle\Big) + (1-\delta_{ab})\, w_3 \sum_i \langle F_{ai}, F_{bi}\rangle.$$

Here F_{ab} is the free curvature and $w \in \mathbb{R}^4$. The Maxwell stress is $w = (0, 1, -\frac{1}{4}, 1)$. - Wick's theorem computes the free one- and two-point tests. Centring and conservation force $w = w_3 \cdot (0, 1, -\frac{1}{4}, 1)$.

These facts feed VII.2's statement that, in the free table, the divergence map from centred, real, symmetric, B_4-tensorial stresses in $(F_4^{\mathrm{free}})^{4\times 4}$ to real B_4-vector fields in $(F_5^{\mathrm{free}})^4$ is onto and has a one-dimensional kernel (`free_rank`). They are not used in the Ward chain above.

Lean: `bilinear_invariant`, `finrank_two`, `finrank_one`, `stress_tensor`, `div_kernel`, `stress_bijective`. *(proof of the divergence test read; polynomial classification and dimension bounds: statement only)*

3.24.5 5. Where the difficulty lies

The exterior flux estimate (VI.1.e) bears most of the analytic burden. Reflection positivity along axes of opposite parity gives the subpower estimate without a gap. Diagonal stress components require a separate conservation argument, since they have no odd reflection parity. The widths $\lambda^{100}, \lambda^{150}, \lambda^{300}$ and the split at λ^{-3000} make the powers in these estimates compatible.

Continuation off the line (VI.1.j–p) requires more than a smooth function with vanishing jets. The collinear Ward identities determine the transverse jets of the two-point function; Hilbert-space positivity turns those jets into uniform factorial derivative bounds and hence analyticity.

The descent (VI.1.s–u) constructs exact intertwiners from the free comparison. Lie brackets alone would give an action of a covering group after integration; they would not by themselves identify the required $O(4)$ action or its signed-permutation subgroup. Covariance subsequently forces compatibility between degrees.

The remaining steps use compactly supported integration by parts, finite-dimensional tests, group generation, Haar averaging and uniqueness for linear ODEs.

3.24.6 6. Notation

object	code name	proposed notation
label spaces, tags, inclusions, unit	`Field d`, `Tag`, `rise`, `unit`	F_d, (d, v), ι, $\mathbf{1}$
signed permutations, conjugation	`Signed`, `mirror`, `conj`	B_4, M_σ, Γ
differential plan and its laws	`DiffPlan`, `Derivatives`, `Inclusions`, `Orientation`	∂_a; derivative, inclusion, orientation laws
stress, window, shells	`Table.Stress`, `Window`, `shellGrad`, `shellSpin`	$J_{ab} \in F_4$, χ, Φ^{tr}_χ, Φ^{rot}_χ
raw shell maps; raw hypotheses	`Climb.Parts` (p), `Ward.SpinRows` (Y); `Climb.Pre`	p_a, Y_K; (R)
angular sum; its vanishing	`localTurn`, `Ward.angular`; `Still`, `Collinear`, `commutes`	$\mathcal{W}_K[Y, p]$, $\mathcal{W}_K[T]$; (L), (C)
line sums, transversality	`LineSum`, `Climb.transverse`	line sums, transverse
colour space and law	`Colors`, `ColorLaw`	colour space
normalised data	`Climb.Crystal`	$p = c\,\partial$, $D = c^{-1}Y$
free data	`MaxwellData`, `Climb.Bel`, `Lighthouse`, `Goes`	$F_d^{\mathrm{free}}, \partial^{\mathrm{free}}, R^{\mathrm{free}}, \mathcal{L}^{\mathrm{free}}, J^{\mathrm{free}}, \kappa$
scaling system, track, weak law	`Climb.Scaling`, `Track`, `ActLaw`, `aisle`	$(r_k, \xi_k, U_{k,d})$, $\mathrm{aisle}_{k,d}$
cutoff field	`petal`, `vane`, `fuel`	petal_{ab}, $\lambda = e^k$
OS vector of one insertion	`spot`, `OS.Chamber`	$\mathrm{spot}(A, x) \in \mathcal{H}_a$

object	code name	proposed notation
row erasure	`erases`, `Winds`	erasure
compatibility of spin rows	`Bid.Hyper`	compatible
perfect relations, Casimir projection	`Spin.rel`, `wave`, `Works`, `effect`	perfect relations, $z(\Omega(B,C))$, $\mathrm{ad}_{B,C}$
tilt, intertwiners	`Spin.Tilt`, `realH`, `change`	U_n, V_d
port, transported action	`Spin.Port`, `Spin.morph`, `Org.mill`	$R_d(k) = V_d R_d^{\mathrm{free}}(k) V_d^{-1}$
rotation laws, covariance	`Rotation.Laws`, `Rotation.Oriented`	rotation laws; $S(R(k)v; kx) = S(v; x)$
conserved, tensorial; average	`Conserved`, `Tensorial`, `averageStress`	$J' = \int k \cdot c^{-1} J \, d\mu$
Lorentz action on labels	`Spin.run`, `wand`, `Boundary.Spins`, `Descends`	$B_d(\Lambda)$
record	`Org.Commission`, `Org.Tab`, `Spec.Locals`	(c, V, J'); local laws

3.24.7 7. What remains unexplained

1. **Supporting analysis in the exterior flux estimate (probable).** The reductions to spectral integrals, the point and line estimates, the diagonal mass estimate and their combination in VI.1.e were checked against the cited proofs. The proofs of the remote-gap profiles, the odd temporal variant, the contact extension on product tests, and the finite-difference and interpolation estimates were not checked in full. These are the files `GapTest`, `Nodal/Mode/Odd`, `Collision/Fold/Journal`, `MomentFD` and `Nodes`. In particular, the construction and normalisation of the contact extension remain inputs to the mass-estimate sketch.
2. **Free-field stress algebra (probable).** The statements in §4.10 and the proof that centring and the two divergence tests select the Maxwell line were checked. The polynomial normal-form and dimension-count proofs were not checked in full. The map in the cited rank theorem has centred stresses as its domain; centring cannot be omitted from the one-dimensional-kernel claim.
3. **Production of the inputs.** The immediate producers of the raw stress, shell maps, weak comparison and free data are identified in §2. Their ultraviolet hypotheses, including the annular estimates and finite comparison systems, are not discharged in this unit. The unused lattice shell-law interfaces in `STRESS_FLUX` are not the source of (R).
4. **Normalisation and centring.** The Ward coefficient of J' is exactly 1. Centring is not part of the conclusion of VI.1.7; the centred trace is formed in VII.2. The stress replacement used before the weak comparison supplies $\kappa \neq 0$ without the numerical identity $\kappa = -\tau$. Along the scaling systems used in VI.1.u, positivity of ξ_k and $c/\xi_k \to \kappa \neq 0$ imply that c and κ have the same sign. Neither sign is specified by this unit's hypotheses.
5. **Scope.** The folder ownership and line counts in the header have not been independently recounted. The assignment of `Shift` and most of `Next/BOUNDARY` to V.2 is tentative. The statements above are source comparisons; Lean elaboration and the transitive axiom dependencies were not checked.

3.25 VI.2. The free Maxwell stress tensor and the local Ward charge

Part: VI, symmetry: stress tensor and Ward identities. **Lean:** `YM/Next/MAXWELL`, `YM/UVNext/FREE_STRESS_CERTIFICATE`, `YM/Next/UV_STRESS`, `YM/Pert`. **Size:** 14,179 lines used by the main theorem (Maxwell 8.2k, free-stress classification 3.1k, UV stress 2.6k, Pert 0.2k).

3.25.1 1. What this unit proves

The free Maxwell theory supplies the comparison model for the stress tensor of the Wilson limit. Its local Ward identities fix the sign and normalization of the free stress. For a simple gauge group, a finite-dimensional calculation identifies its centred conserved tensors of degree at most four as a line. Short-distance comparison transfers this line to the interacting limit; the translation Ward identity then fixes the normalization of the interacting stress.

Throughout, G is a gauge group as in I.1, $\mathfrak{g}$ is its Lie algebra, and $c = 1, \dots, \dim \mathfrak{g}$ indexes an orthonormal colour basis for the invariant inner product. Spacetime indices range over $0, 1, 2, 3$. Inner products of $\mathfrak{g}$-valued forms include the colour sum.

A *Maxwell wave* on an open set $U \subset \mathbb{R}^4$ is a smooth $\mathfrak{g}$-valued antisymmetric field f_{ab} satisfying

$$\partial_a f_{bc} + \partial_b f_{ca} + \partial_c f_{ab} = 0, \qquad (\mathrm{Div}\, f)_b := \sum_a \partial_a f_{ab} = 0.$$

Here $\mathrm{Div} = -d^*$ with the usual Euclidean codifferential convention. The space Sh_j consists of the homogeneous Taylor coefficients of order j satisfying these linear equations, not the cumulative jets through order j. The algebra $\mathcal{W}$ consists of complex polynomials in coordinates on these spaces; a coordinate from Sh_j has weight $j+2$. Its *shell sector* $\mathcal{P}_d$ is the image of labels of degree at most d under substitution of on-shell Taylor coefficients. Equivalently, it consists of the $\mathrm{Ad}(G)$-invariant polynomials of weight at most d (`sector`). A *probe* of order j is a real linear functional on Sh_j; $p(f)(x)$ denotes its value on the order-j Taylor coefficient of f at x.

The curvature propagator is

$$\langle F^c_{im}(x) F^{c'}_{ab}(y)\rangle_0 = \delta^{cc'} \mathcal{C}_{im,ab}(x-y), \qquad G_0(x) = \frac{1}{4\pi^2 |x|^2}, \qquad -\Delta G_0 = \delta,$$

where $\mathcal{C}$ is the curl–curl of G_0. The generated field $f^{p,x}(y) = \langle p(F)(x) F(y)\rangle_0$ is a Maxwell wave off x. Correlations of Wick polynomials at distinct sites sum over complete pairings of legs between different sites; there are no contractions within a site.

Write

$$T^{\mathrm{free}}_{ab} = \sum_{c,i} F^c_{ai} F^c_{bi} - \frac{1}{4}\delta_{ab} \sum_{c,i,k} F^c_{ik} F^c_{ik} \in \mathcal{P}_4.$$

Its polarization and the current associated with a vector field v are

$$\Theta_{ab}(f,g) = \sum_i \big(\langle f_{ai}, g_{bi}\rangle + \langle f_{bi}, g_{ai}\rangle\big) - \frac{1}{2}\delta_{ab} \sum_{i,k} \langle f_{ik}, g_{ik}\rangle, \qquad j^v_a(f,g) = \sum_b v_b \Theta_{ab}(f,g).$$

Thus $\Theta(f,f) = 2T^{\mathrm{free}}(f)$. Both tensors are symmetric and traceless. For a real cutoff $\chi \in C^\infty_c(\mathbb{R}^4)$, the outward flux is $\Phi_\chi(j) = -\sum_a \int \partial_a \chi \, j_a$.

The free table on a register L is $\mathcal{M} = \mathrm{Bel}(L)$, with filtered field spaces $\mathcal{F}_d^{\mathrm{free}}$. The Wilson limit table S has spaces $\mathcal{F}_d^S$. For a lighthouse $\mathcal{T}$, write T_0 for its free stress and κ for its Ward constant. Let Conf_n denote the distinct-point configurations. For $R > 0$, set

$$K_{R,n} = \{y \in (\mathbb{R}^4)^n : \ |y_i - y_j|_\infty \geq R^{-1} \ (i \neq j), \ \|y\| \leq R\},$$

where $\|y\|$ is the Euclidean norm of the full configuration.

Proposition VI.2.1 (the free Maxwell table). For every register L, the free table $\mathcal{M} = \mathrm{Bel}(L)$ has the following properties.

(a) For every $d \geq 0$, substitution of on-shell Taylor coefficients induces a complex linear isomorphism $\iota_d : \mathcal{F}_d^{\mathrm{free}} \to \mathcal{P}_d$. A label is null exactly when its polynomial vanishes on every on-shell Taylor tower. At distinct points, field correlations equal the Wick moments of their images under ι.

(b) The table satisfies the Euclidean hierarchy of IV.1: permutation symmetry, the unit laws, covariance under signed coordinate permutations, reality, translation invariance, point and smeared reflection positivity, and smoothness on Conf_n. The correlations are in fact real-analytic there. For each fixed tuple of fields and derivative order, there are constants $A \geq 0$ and $C \geq 0$ bounding that derivative by Cr^{-A} on configurations with mutual sup-distances at least r, for $0 < r \leq 1$.

(c) Differentiation of Taylor coefficients induces maps $\partial_a^{\mathrm{free}} : \mathcal{F}_d^{\mathrm{free}} \to \mathcal{F}_{d+1}^{\mathrm{free}}$ satisfying the inclusion, derivative and orientation laws. In particular,

$$\langle (\partial_a^{\mathrm{free}} B)(x_0) \cdots \rangle^{\mathrm{free}} = \partial_{x_0^a} \langle B(x_0) \cdots \rangle^{\mathrm{free}}.$$

The action of $O(4)$ on coefficients induces actions R_g on all field spaces and generators $\rho(\omega)$ for $\omega \in \mathfrak{so}(4)$. These satisfy the rotation laws of VI.1: continuity, the group law, extension of signed permutations, reality, preservation of the unit and inclusions, vector covariance of derivatives, and the infinitesimal-generator identity. All free correlations are rotation invariant.

(d) The fields $\iota_4^{-1}(T_{ab}^{\mathrm{free}})$, also denoted T_{ab}^{free}, are symmetric, real, traceless and centred, and satisfy $\sum_a \partial_a^{\mathrm{free}} T_{ab}^{\mathrm{free}} = 0$ in $\mathcal{F}_5^{\mathrm{free}}$.

Status: theorem for every gauge group in the setting of I.1; simplicity is not required. *Lean:* `Maxwell.Free.hierarchy`, `fields`, `no_residual`, `scheme`, `gear`, `gear_laws`, `gear_motion`, `original`, `realStress`.

Proposition VI.2.2 (local Ward charge). Let $U \subset \mathbb{R}^4$ be open, $x \in U$, and f a Maxwell wave on U. Let v be a smooth conformal Killing field on $\mathbb{R}^4$:

$$\partial_a v_b = -\partial_b v_a \quad (a \neq b), \qquad \partial_0 v_0 = \partial_1 v_1 = \partial_2 v_2 = \partial_3 v_3.$$

Let $\chi \in C_c^\infty(U; \mathbb{R})$ equal one near x. Since $df = 0$, its Lie derivative is

$$(\mathcal{L}_v f)_{il} = \sum_b [\partial_i (v_b f_{bl}) - \partial_l (v_b f_{bi})] = (d\iota_v f)_{il}.$$

(a) For every zero-order component probe F_{il}^c,

$$\Phi_\chi(j^v(f^{F_{il}^c, x}, f)) = -(\mathcal{L}_v f)_{il}^c(x).$$

(b) If $\mathcal{L}_v f$ is also a Maxwell wave on U, then, for every order $j \geq 0$ and every probe p of that order,

$$\Phi_\chi(j^v(f^{p,x}, f)) = -p(\mathcal{L}_v f)(x).$$

For translations $v = e_a$, this is $-\partial_a[p(f)](https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/x)$. For rotations $v(y) = \omega(y - x)$, it is $-(\omega \cdot p)(f)(x)$, where $\omega \cdot p$ is the infinitesimal label action. For dilation $v(y) = y - x$, it is $-(j + 2)p(f)(x)$.

(c) For any two Maxwell waves f, g on a neighbourhood of $\operatorname{supp}\chi$, the flux $\Phi_\chi(j^v(f, g))$ vanishes. It also vanishes for two fields $f^{p,x}$, $f^{q,x}$ generated at the same point, provided v is homogeneous about x of degree $e \in \{0, 1\}$.

Status: theorem, with the propagator, polarization and outward-flux conventions above. *Lean:* `Maxwell.Free.local_force`, `jet_force`, `drift_flux`, `angular_flux`, `radial_flux`, `sourcefree_flux`, `two_flux_zero`; definitions `Killing`, `variation`, `pulse`, `singleFlux`.

Proposition VI.2.3 (Ward identities of the free table). Let $n \geq 0$, $x \in \mathrm{Conf}_{n+1}$, and $p_0, \dots, p_n \in \mathcal{W}$. Let $\chi \in C_c^\infty(\mathbb{R}^4; \mathbb{R})$ equal one near x_0, with $x_1, \dots, x_n \notin \operatorname{supp}\chi$. Take one of the pairs

$$(v, D) = (e_a, \partial_a), \qquad (\omega(y - x_0), \rho(\omega)), \qquad (y - x_0, \varepsilon),$$

where a is an axis, $\omega \in \mathfrak{so}(4)$, and ε multiplies each monomial by its weight. The operators act on $\mathcal{W}$ by derivations. Then

$$-\sum_{a,b} \int \partial_a \chi(y)\, v_b(y)\, \langle T^{\mathrm{free}}_{ab}(y) p_0(x_0) \cdots p_n(x_n) \rangle_0 \, dy = -\langle (Dp_0)(x_0) p_1(x_1) \cdots p_n(x_n) \rangle_0.$$

Consequently, for every $\tau \in \mathbb{R}$, the stress τT^{free} satisfies the translation and rotation Ward identities with rows $-\tau\partial^{\mathrm{free}}$ and $-\tau\rho$. The stress normalized to give $+\partial^{\mathrm{free}}$ in the translation identity is $-T^{\mathrm{free}}$.

Status: theorem. *Lean:* `Maxwell.Free.moment_force`, `straight_motion`, `rotational_motion`, `radial_motion`, `beacon_signal`, `Table.StressLaws`.

Proposition VI.2.4 (the free stress line). Suppose G is simple. Let W be a derivative plan on the free table satisfying the derivative and orientation laws. The derivative law determines its derivative maps uniquely as ∂^{free}. Define the following real vector spaces:

- $\mathcal{Z}$ consists of symmetric real tensors $J \in (\mathcal{F}_4^{\mathrm{free}})^{4\times 4}$ that are covariant under signed coordinate permutations and centred: $\langle J_{ab} \rangle = 0$.
- $\mathcal{V}$ consists of real vectors in $(\mathcal{F}_5^{\mathrm{free}})^4$ covariant under signed coordinate permutations.

The divergence $\mathrm{div} : \mathcal{Z} \to \mathcal{V}$, $(\mathrm{div} J)_b = \sum_a W_a J_{ab}$, is surjective, and

$$\dim_{\mathbb{R}} \mathcal{Z} = 3, \qquad \dim_{\mathbb{R}} \mathcal{V} = 2, \qquad \ker \mathrm{div} = \mathbb{R} T^{\mathrm{free}}.$$

Thus a lighthouse with such a derivative plan and stress τT^{free}, $\tau \neq 0$, satisfies (FD): its divergence is onto, its kernel has real dimension one, and its stress is a nonzero centred conserved tensor.

Status: theorem. *Lean:* `FreeStress.rank_on`, `line_dim`, `lighting`, `certificate`, `FreeDiv`, `KernelSpace`, `Vectors`, `currentDiv`.

Proposition VI.2.5 (Maxwell data). Suppose G is simple, let C be a Wilson family for G, and let $\tau \in \mathbb{R} \setminus \{0\}$.

(a) The lighthouse $\mathcal{T}_\tau = (\partial^{\mathrm{free}}, R, \tau T^{\mathrm{free}}, \kappa = -\tau)$ satisfies all lighthouse laws of VI.1. These include the derivative, rotation and Ward laws of VI.2.1–3, a nonzero derivative in degree four, and the following selection property. There is N such that for every $d \geq N$, finitely many derivatives of Ward sums at collinear configurations detect the line $\mathbb{R}\partial^{\mathrm{free}}$ among complex linear maps $t_a : \mathcal{F}_d^{\mathrm{free}} \to \mathcal{F}_{d+1}^{\mathrm{free}}$ satisfying:

- they kill the unit, are real, and transform as a vector under signed permutations;
- for every $e \leq d$, t_a maps the image of $\mathcal{F}_e^{\mathrm{free}}$ into the image of $\mathcal{F}_{e+1}^{\mathrm{free}}$.

For component t_b, a diagram is an iterated directional derivative of

$$x \longmapsto \sum_i \langle w_1(x_1) \cdots (t_b w_i)(x_i) \cdots w_n(x_n) \rangle^{\mathrm{free}}, \qquad w_i \in \mathcal{F}_d^{\mathrm{free}},$$

evaluated at a distinct-point configuration on a coordinate axis perpendicular to e_b. The selected diagrams all vanish exactly when t is a real multiple of the derivative tuple.

(b) There exists a Maxwell datum for C with lighthouse $\mathcal{T}_\tau$. It also contains the free Euclidean hierarchy, the Casimir polynomial data of VI.1, compatible representations of the proper orthochronous Lorentz group on the field spaces and their union, continuous in each degree, and the following Ward-vanishing property. For every $d \geq 4$, let $T(u, v)$ be a real-bilinear family of complex linear endomorphisms of $\mathcal{F}_d^{\mathrm{free}}$, with $u, v \in \mathfrak{so}(4)$. Suppose it is alternating, real, equivariant under orientation-preserving signed permutations, kills the unit, and preserves every lower-degree image. If

$$\sum_i \langle w_1(x_1) \cdots (T(u, v) w_i)(x_i) \cdots w_m(x_m) \rangle = 0$$

for every $m \geq 0$, every $x \in \mathrm{Conf}_m$, every $w_i \in \mathcal{F}_d^{\mathrm{free}}$ and every u, v, then $T = 0$. The Lorentz representation agrees with R on spatial rotations, commutes with time-reflected conjugation, and has boost generator $-i\rho$ in each time–space plane.

(c) Every Maxwell datum can be replaced by one with the same derivative plan, rotations, hierarchy and Lorentz representation, stress τT^{free}, full rotation covariance of that stress, and (FD). The Ward constant of any Maxwell datum with stress τT^{free} is $-\tau$.

Status: theorem. The construction used in III.11 chooses $\tau = 1$ and hence $\kappa = -1$. *Lean:* `Maxwell.Free.herald`, `shines`, `maxwell_supplies`, `erasing`, `scanner`, `launched`, `FreeStress.fire`, `names`, `required`; records `Org.MaxwellData`, `Org.erases`, `Climb.Lighthouse`, `Lighthouse.Goes`.

Proposition VI.2.6 (strict reflection positivity). Fix a coordinate axis a, and let θ_a reflect in $y_a = 0$. Let ϑ_a act on polynomials by the corresponding jet reflection and coefficient conjugation.

(a) For every x with $x_a > 0$ and every nonzero $p \in \mathcal{W}$,

$$\langle (\vartheta_a p)(\theta_a x) p(x) \rangle_0 > 0.$$

(b) For every finite set of distinct coordinate probes of the spaces Sh_j, the reflected propagator Gram matrix at $(\theta_a x, x)$ is positive definite.

(c) The free table is reflection positive for finite families of words in the positive half-space, with distinct sites within each word.

(d) For every $d \geq 0$, a polynomial $p \in \mathcal{P}_d$ whose correlations with every finite family of spectators from $\mathcal{P}_d$ vanish at every distinct-point configuration is zero.

Status: theorem. *Lean:* `detects`, `coordGram`, `positive_table`, `ghosts_out`.

Proposition VI.2.7 (a conserved stress on the limit table). Let C be a Wilson family satisfying the construction conditions of II.2. Let F be an extraction order with ordered smeared convergence and generic point limits to a table S with Euclidean hierarchy H. Let W be a derivative plan on S satisfying the inclusion and derivative laws; its orientation law then follows from IV.1. Let H_0 be the free Euclidean hierarchy, and $\mathcal{T}$ a lighthouse whose plan satisfies the inclusion, derivative and orientation laws and for which (FD) holds. Assume the station property of VI.2.8 and

$$\dim_{\mathbb{C}} \mathcal{F}_d^S \leq \dim_{\mathbb{C}} \mathcal{F}_d^{\mathrm{free}} \qquad (d \geq 0).$$

(a) There exists a scaling $X = (r_k, \xi_k, U_{k,d})$, where $r_k > 0$ decreases strictly to zero, $\xi_k > 0$, and $U_{k,d} : \mathcal{F}_d^{\mathrm{free}} \to \mathcal{F}_d^S$ is a complex linear isomorphism for every k, d. It is *approaching*: for each fixed tuple $v = (v_i)$ of free fields, the difference

$$K_k(y) = \langle U_k v_1(r_k y_1) \cdots U_k v_n(r_k y_n) \rangle^S - \langle v_1(y_1) \cdots v_n(y_n) \rangle^{\mathrm{free}}$$

satisfies, for every compact $K \subset \mathrm{Conf}_n$,

$$\sup_{\substack{t \in C_c^\infty(\mathrm{Conf}_n;\mathbb{C}) \\ \operatorname{supp} t \subset K,\ \|t\|_\infty \leq 1}} \left| \int t(y) K_k(y)\, dy \right| \longrightarrow 0.$$

It is *stable*: the maps eventually commute with conjugation and signed permutations in each degree, preserve the unit in each degree, and intertwine each fixed inclusion of degrees. The initial choice may have $\xi_k = 1$.

(b) For every approaching stable scaling, the divergence $\mathrm{div}^S : \mathcal{Z}^S \to \mathcal{V}^S$ is onto and has a one-dimensional real kernel. These spaces are defined as in VI.2.4, using S and W.

(c) There exist a nonzero $J \in \mathcal{Z}^S$ with $\mathrm{div}^S J = 0$ and an approaching stable refinement Y of X, obtained by taking a subsequence and changing ξ_k, such that

$$\xi_k^{-1} U_{k,4}^{-1}(r_k^4 J_{ab}) \longrightarrow T_{0,ab} \qquad \text{for all } a, b.$$

(d) For every finite arity bound N, degree bound d, derivative order m, and $R \geq 1$, a tail of Y is a track: it intertwines symmetries, units and inclusions through degree $\max(d, 4)$, satisfies (c), and has convergence of all correlation derivatives of orders $0 \leq j \leq m$ uniformly on $K_{R,n}$ for $n \leq N$. The convergence is for each fixed tuple of fields of degrees at most d; its tail may depend on that tuple and the error tolerance. In particular, J satisfies the comparison law of VI.1.

Status: theorem under these hypotheses. III.11 supplies the station estimates; the rank-corrected widths of the construction give the dimension bound; VI.2.4 gives (FD) for the canonical datum. To obtain the infrared and ultraviolet data used later, III.11 also supplies the shell estimates needed for the Ward rows of J. *Lean:* `Stations.equivalences`, `differential_rank`, `produce_current`, `focused`, `cap_given`; records `Climb.Scaling`, `Scaling.meter`, `Approaches`, `Stable`, `ActLaw`, `Org.IR`, `Org.UV`.

Proposition VI.2.8 (weak comparison stations). Fix C, F, S, H, H_0 as in VI.2.7. A *station* of depth D consists of $r > 0$ and complex linear lifts $A_{N,p,d} : \mathcal{F}_d^{\mathrm{free}} \to \mathrm{Ref}_d$ for $d \le D$ at every lattice cutoff (N, p).

It is *admissible* for arity bound $n_{\max}$, radius $R \ge 1$ and tolerance $\varepsilon > 0$ if:

- the lifts are eventually uniformly bounded along the extraction order, real and signed-covariant, and preserve the degree-zero unit;
- for every $n \le n_{\max}$, every tuple v_i of degrees at most D, every smooth complex test t supported in $K_{R,n}$, every $M > 0$ with $\|t\|_\infty \le M$, and every $\eta > 0$, eventually along the ordered cutoff tails,

$$\left| \mathcal{S}_{N,p}(Av; r^{-4n} t(\cdot/r)) - \int t(y) \langle v_1(y_1) \cdots v_n(y_n) \rangle^{\mathrm{free}} \, dy \right| \le \varepsilon M \prod_i \|v_i\| + \eta.$$

  Here $\mathcal{S}_{N,p}$ is the lattice-smeared correlation of IV.1. The cutoff tail may depend on v, t, M, η.

The *station property* requires an admissible station with $r \le \delta$ for every $D, n_{\max}$, $R \ge 1$ and $\varepsilon, \delta > 0$.

(a) An admissible station yields a panel $U_d : \mathcal{F}_d^{\mathrm{free}} \to \mathcal{F}_d^S$ for $d \le D$. It is real, signed-covariant, preserves the degree-zero unit, and satisfies

$$\left| \int t(y) [\langle U v_1(r y_1) \cdots U v_n(r y_n) \rangle^S - \langle v_1(y_1) \cdots v_n(y_n) \rangle^{\mathrm{free}}] dy \right| \le \varepsilon M \prod_i \|v_i\|$$

for the same tuples and tests.

(b) Quantitative grid, strip and finite-difference estimates satisfying the balance conditions in VI.2.s, for every finite arity bound, degree bound and $R \ge 1$, imply the station property.

Status: the station and balance conditions are definitions; (a) and (b) are theorems. III.11 supplies the balanced estimates through its complete weak-comparison result. *Lean:* `Station`, `Station.Admissible`, `Stations`, `harvested`, `Granular`, `realizes_stations`, `MergeProof.East.CompleteWeak`.

Proposition VI.2.9 (the normalized stress and its short-distance law). Let C be a Wilson family, F an extraction order, S a table with derivative plan W, and M a Maxwell datum with lighthouse $\mathcal{T}$ and Ward constant $\kappa \neq 0$. Let $I = (J_0, p^I, Y^I)$ and let X be a scaling. Assume the ultraviolet-data conditions of VIII.1 for (C, F, S, W, M, I). In particular, they give a Euclidean hierarchy on S, the orientation law for W, a symmetric real conserved signed-covariant J_0 with Ward rows (p^I, Y^I), and tracks comparing J_0 to T_0. Assume also the inclusion and derivative laws for W, and the following additional properties: J_0 is a nonzero member of $\mathcal{Z}^S$; $\mathcal{T}$ satisfies (FD); and X is approaching and stable, with

$$\xi_k^{-1} U_{k,4}^{-1}(r_k^4 J_{0,ab}) \longrightarrow T_{0,ab}.$$

Let $J \in (\mathcal{F}_4^S)^{4\times 4}$ be symmetric, real, conserved and covariant under signed permutations. Suppose it satisfies the translation Ward identity with rows W and the rotation Ward identity with some rotation rows. Then:

(a) $U_{k,4}^{-1}(r_k^4 J_{ab}) \to \kappa^{-1} T_{0,ab}$ for every a, b.

(b) For every finite tuple of free fields q_i, every assignment of slots to either $U_k q_i$ or $r_k^4 J_{a_i b_i}$, and every $j \geq 0$, the resulting correlations at $r_k y_i$ converge with their order-j derivatives uniformly on compact subsets of Conf_n. The limit is the free correlation with each stress slot replaced by $\kappa^{-1} T_{0,a_i b_i}$.

(c) For the canonical datum with plan ∂^{free} and stress T^{free}, $\kappa = -1$. Thus the Ward-normalized physical stress converges to $-T^{\mathrm{free}}$ without the factor ξ_k^{-1}.

Status: theorem. VI.1 supplies a stress J satisfying these Ward laws. The mixed-correlation conclusion follows from (a) and VI.2.u; the linked mixed-correlation theorem states its specialization to that stress. *Lean:* `Pert.meter_physical`, `Pert.mixed_leading`, `kappa_eq`, `common_stress`, `Pert.Supplier`, `Org.Tab`.

3.25.2 2. What it uses

From other units.

- III.16 supplies on-shell Taylor spaces, label frames, substitution into curvature polynomials, and the cross-site Wick pairing formula: `YM.Maxwell.Wick.moment`; `YM.Maxwell.observable`.
- IV.1 supplies tables, quotient fields, ordered cutoff limits and lattice smearing. The derivative and inclusion laws give the orientation law: `YM.Work.D06.answer`. Its rank-corrected widths give the upper dimension bound used here: `YM.Next.UVStress.cap_given`.
- V.1 gives smeared reflection positivity from point reflection positivity and derivative bounds from order-zero bounds: `YM.Work.B08.smear`; `YM.Work.B10.answer`.
- VI.1 supplies local windows, finite separating correlation tests, collinear Ward diagrams, and the derivative, rotation and track laws. Rotation laws follow from invariance: `YM.Work.A14.laws_of_oriented`. It also supplies the Casimir relations and annihilating polynomials: `YM.Work.A09.answer`; `YM.Work.A10.answer`; `YM.Work.A12.answer`.
- III.10 assembles Maxwell data from these components: `YM.Bid.masts`.
- III.11 supplies balanced weak-comparison estimates and the shell estimates for the interacting Ward identities. It combines these with the present stress-line construction to obtain the infrared and ultraviolet data: `YM.MergeProof.East.CompleteWeak`; `YM.MergeProof.Acme.readyA`; `YM.MergeProof.Acme.emanate`.
- VII.5 supplies the comparison of stress and derivative normalizations along tracks and the nontriviality test for a Ward-normalized stress: `YM.Extra.WeakScaling.tracks_ratio`; `YM.Extra.WeakScaling.flux_nontrivial`.
- VIII.1 defines the Maxwell, infrared and ultraviolet data: `YM.Org.MaxwellData`; `YM.Org.IR`; `YM.Org.UV`.

Standard mathematics. Wick's theorem and the bosonic Fock inner product (Glimm–Jaffe, ch. 6); reflection positivity of Gaussian fields (Osterwalder–Schrader, 1973/75); the Euclidean Green function and Cartan's formula; integration by parts and approximate identities; homogeneous harmonic functions and the Kelvin transform; finite-dimensional rank–nullity, openness of injectivity and surjectivity, compactness of spheres, and uniqueness for linear ODEs. The passage from uniform order-zero test estimates to smooth convergence uses closure under gradients, localization and Fourier inversion (VI.2.u).

Imported constants and assumptions. Simplicity of G enters the stress classification and the rigidity arguments for the Maxwell data. It is essential for the one-dimensional stress line: for

$\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$, the two factors have separately invariant conserved Maxwell stresses. The propagator normalization is $c_0 = (4\pi^2)^{-1}$. The explicit correlation tests give the integers $-256, -512, 128, -128$ in VI.2.l and the factor $256 \dim \mathfrak{g}\, c_0^2$ in VI.2.n. The station estimates and dimension bound enter as hypotheses of the transfer argument, with their discharge stated in VI.2.7–8.

3.25.3 3. Strategy

The free table is realized by invariant polynomials in on-shell curvature coefficients with Wick correlations. Translation and rotation covariance, analyticity and polynomial bounds follow from the propagator. Reflection positivity uses a Fock-space factorization. For two polynomials at mirror points, the reflected Gram matrix of their finitely many coordinate probes is positive definite. Factoring it as L^*L identifies their correlation with a Fock inner product after an invertible linear substitution. Strict positivity eliminates further null fields and identifies the free quotient with $\mathcal{P}_d$.

For the Ward identity, replace the singular propagator by a smooth curl source. Its divergence is a derivative of an approximate identity. Two integrations by parts and removal of the regulator give the local charge $-\mathcal{L}_v f$. Differentiation in the source point extends the charge to every Taylor order. Wick expansion then leaves three possibilities: both stress legs meet the marked site, giving zero by homogeneity; one meets that site and one a spectator, giving the desired derivation; or both meet spectators, giving a source-free current inside the cutoff.

The free stress line is computed in finite-dimensional spaces. Signed covariance allows three centred symmetric tensors in degree four and at most two vectors in degree five. Two explicit correlation tests prove that divergence has rank two. Its kernel is therefore the line generated by T^{free}. A translation Ward test fixes the constant $\kappa = -\tau$ for stress τT^{free}.

Stations give comparison maps at scales tending to zero. Correct them recursively to intertwine inclusions. Finite separating tests make the corrected maps eventually injective; the dimension bound makes them isomorphisms. Integration by parts and the derivative law show that $r_k U_k^{-1} W_a U_k \to \partial_a^{\mathrm{free}}$. The correlation differences, including those with convergent field coefficients, are closed under gradients, so the uniform test estimates improve to smooth local convergence.

Transport the interacting divergence to the free spaces, recentering tensors after transport. It converges to free divergence, so surjectivity and rank–nullity transfer the one-dimensional kernel to S. Normalize a nonzero vector in this kernel to a sphere in the free stress space and pass to a subsequence. Its limit is a nonzero centred conserved free tensor, hence a multiple of T^{free}; a single sign choice fixes the limit on the entire comparison sequence.

Finally, a Ward-normalized stress J on S has centred part proportional to the constructed stress: $c(J - \langle J \rangle \mathbf{1}) = J_0$. Comparing the derivative and stress Ward identities along tracks gives $c/\xi_k \to \kappa$. Thus $U_k^{-1}(r_k^4 J) \to \kappa^{-1} T_0 = -T^{\mathrm{free}}$ in the canonical normalization. This obtains the interacting stress from the finite-dimensional conservation law, then fixes its scale by the Ward identity.

3.25.4 4. Steps

3.25.4.1 4.1 The free table Lemma VI.2.a (field spaces and correlations). For every d, the null labels are exactly the kernel of substitution into $\mathcal{P}_d$, and $\iota_d : \mathcal{F}_d^{\mathrm{free}} \to \mathcal{P}_d$ is an isomorphism carrying correlations to Wick moments.

Sketch. The free table is defined by substitution followed by the Wick moment. A residual polynomial has zero correlations with every family of shell-sector spectators. A nonzero polynomial

has positive two-point correlation with its own reflected conjugate by VI.2.b, which remains in the same sector. Thus the residual subspace is zero. *Lean:* `fields`, `no_residual`, `freeTable`, `Wick.moment`.

Lemma VI.2.b (strict reflection positivity). For $x_a > 0$, the pairing $\langle(\vartheta_a p)(\theta_a x)q(x)\rangle_0$ is a positive-definite Hermitian form on $\mathcal{W}$.

Sketch. Let s be the finite union of coordinate variables appearing in p and q. The reflected propagator Gram matrix M_s is positive definite; factor it as $M_s = L^*L$. The Wick pairing between the two sites then equals $\langle T_L p, T_L q\rangle_{\mathrm{Fock}}$, where T_L substitutes $e_i \mapsto \sum_j L_{ji} e_j$. The Fock form makes monomials orthogonal with squared norms $\alpha!$. Since L is invertible, the form is definite. Taking $q = p$ gives strict positivity; testing a residual polynomial against its reflected conjugate gives VI.2.6(d). *Lean:* `detects`, `core`, `Fock.halves`, `coordGram`, `ghosts_out`. *(`detects`, `core` and `ghosts_out`: proof read; reflected Gram positivity and Fock pairing factorization: statement checked)*

Lemma VI.2.c (the Euclidean hierarchy). The free table is reflection positive, real-analytic on distinct-point configurations, translation invariant, and has polynomial bounds there, including bounds for all derivatives.

Sketch. For words with distinct sites in a positive half-space, the multi-site Fock factorization makes the matrix of reflected correlations positive semidefinite. A leg kernel between probes of orders j, l is analytic and harmonic off zero, homogeneous of degree $-(4 + j + l)$, and becomes analytic on all of $\mathbb{R}^4$ after multiplication by $|x|^{2(3+j+l)}$. Its bound on a sphere and homogeneity give a polynomial bound away from zero. Finite Wick sums inherit analyticity and bounds. The derivative law gives bounds for all derivatives; V.1 gives smeared positivity. The propagator depends only on point differences, which gives translation invariance. *Lean:* `positive_table`, `cabinets_positive`, `kernel_pole`, `cap_moment`, `hierarchy`. *(proof read; multi-site Fock factorization used as an input)*

Lemma VI.2.d (derivatives, rotations and conservation). The Taylor derivations and rotation action give VI.2.1(c), and T^{free} satisfies VI.2.1(d).

Sketch. The directional derivative raises the Taylor order by one and the weight by one. Applying it at one insertion differentiates precisely the propagator legs incident to that insertion. Rotation covariance follows from the natural jet action and invariance of the propagator; VI.1 then gives the rotation laws.

To prove conservation as a polynomial identity, evaluate on an arbitrary Taylor tower. Only its orders zero and one occur, so realize them by a wave $f = A_0 + A_1$ with homogeneous components of degrees zero and one. The evaluated divergence is $\frac{1}{2}\sum_a \partial_a \Theta_{ab}(f, f) = \sum_i \langle(\mathrm{Div}\, f)_i, f_{bi}\rangle = 0$. Evaluation on all towers separates polynomials. Symmetry, reality, tracelessness and centring follow from the displayed stress polynomial and the absence of one-site contractions. *Lean:* `scheme`, `gear`, `realStress`, `pol_conservation`, `polStress`.

3.25.4.2 4.2 The local Ward charge

Lemma VI.2.e (divergence of the current). For smooth closed two-forms f, g on an open set U and a smooth conformal Killing field v,

$$\sum_a \partial_a j^v_a(f, g) = \sum_b v_b \sum_i (\langle(\mathrm{Div}\, f)_i, g_{bi}\rangle + \langle f_{bi}, (\mathrm{Div}\, g)_i\rangle).$$

In particular, the flux vanishes when both fields are Maxwell waves near the cutoff support.

Sketch. Apply the product rule. The contraction of ∇v with Θ vanishes because Θ is symmetric and traceless and the symmetric part of ∇v is scalar. The remaining identity follows by the Bianchi equations. Integrating against the cutoff proves the flux statement. *Lean:* `current_div`, `sourcefree_flux`.

Lemma VI.2.f (regulated source). Under the hypotheses of VI.2.2, fix i, l, c and $t > 0$. Set

$$G_t(y) = c_0(t^2 + |y|^2)^{-1}, \qquad \rho_t(y) = -\Delta G_t(y) = 8c_0 t^2 (t^2 + |y|^2)^{-3},$$

and let $R^t = d\alpha^t\, e_c$, with $\alpha^t_b = \delta_{ib}\partial_l G_t(\cdot - x) - \delta_{lb}\partial_i G_t(\cdot - x)$. Then R^t is closed and

$$(\operatorname{Div} R^t)_b = (\delta_{lb}\partial_i \rho_t - \delta_{ib}\partial_l \rho_t)(\cdot - x) e_c.$$

Writing $h_a = (\iota_v f)^c_a$, its flux is

$$\Phi_\chi(j^v(R^t, f)) = \int \rho_t(y-x)[\partial_l(\chi h_i) - \partial_i(\chi h_l)](https://github.com/scottnarmstrong/yang-mills-informaliza$$

Sketch. Apply VI.2.e and integrate $\chi \operatorname{div} j^v$ by parts. Insert the displayed divergence of R^t and integrate by parts again to move derivatives from ρ_t to χh. *Lean:* the regulated-flux identity in `Contact.lean`, `regulated`.

Lemma VI.2.g (removing the regulator). The regulated flux extends continuously to $t = 0$ and gives VI.2.2(a).

Sketch. On the compact support of $\nabla\chi$, the source point is excluded and R^t is jointly continuous down to zero, where it equals $f^{F^c_{il},x}$. The functions ρ_t form an approximate identity. Hence VI.2.f tends to $(\partial_l h_i - \partial_i h_l)(x) = -(\mathcal{L}_v f)^c_{il}(x)$, since $\chi = 1$ near x. *Lean:* `local_force`.

Lemma VI.2.h (charges of higher-order probes). Under the hypotheses of VI.2.2(b), the flux identity holds in every order. Translations, rotations and dilation give its three stated specializations.

Sketch. Probes of order $j+1$ are spanned by directional derivatives of probes of order j. Keep v and χ fixed while differentiating in the source point. The generated fields satisfy $\frac{d}{ds} f^{p,x+su} = f^{\partial_u p, x+su}$, where $\partial_u p$ is the probe representing the derivative of $p(f)$. Compact support away from the source justifies differentiation under the flux integral. The right side differentiates as a probe evaluation because $\mathcal{L}_v f$ is a Maxwell wave.

For a centred affine motion, the order-j coefficient of $\mathcal{L}_v f$ at the centre depends only on the order-j coefficient of f. For rotations it is the derivative of the inverse action on solution coefficients: if that action is Q_g, then $(\omega \cdot p)(A) = \frac{d}{dt}|_{t=0} p(Q_{\exp(-t\omega)}A)$. For dilation, Euler's identity on the homogeneous wave gives the factor $j + 2$. *Lean:* `jet_force`, `drift_flux`, `angular_flux`, `radial_flux`.

Lemma VI.2.i (two sources at one point). With χ, v as in VI.2.2, probes p, q of orders j, l, and v homogeneous about x of degree $e \in \{0, 1\}$, the flux $Z = \Phi_\chi(j^v(f^{p,x}, f^{q,x}))$ vanishes.

Sketch. The current is divergence-free off x, so changing the cutoff while keeping it equal to one near x preserves its flux. Replace it by $\chi(x + (\cdot - x)/2)$. The two generated fields have homogeneities $-(4 + j)$ and $-(4 + l)$, so a change of variables gives $Z = 2^{3+e-(8+j+l)} Z$. The factor is strictly less than one. *Lean:* `two_flux_zero`.

Lemma VI.2.j (Ward identities for Wick polynomials). The three charges of VI.2.h imply VI.2.3 for every tuple of Wick polynomials.

Sketch. More generally, take a conformal Killing field v and a polynomial derivation D for which the two-source flux at the marked site vanishes and

$$\Phi_\chi(j^v(f^{p,x_0}, f^{q,y})) = -\langle (Dp)(x_0)q(y)\rangle_0 \qquad (y \notin \operatorname{supp}\chi).$$

The three motions satisfy these conditions by VI.2.h–i. For monomials, expand the two stress legs in the Wick moment. The integrand is one half the sum over ordered pairs of extracted legs of their polarized current times the remaining Wick moment. Two marked legs give zero by VI.2.i; two spectator legs give zero by VI.2.e. The mixed pairs give D on each marked leg, with the two orders cancelling the factor one half. The Leibniz rule reassembles the result as Dp_0. Multilinearity gives the identity for all polynomials, and ι carries it to field-space Ward laws. *Lean:* `fees`, `moment_force`, `Motion`, `beacon_signal`.

3.25.4.3 4.3 The free stress line and the Ward constant Lemma VI.2.k (the tensor and vector spaces). Assume G is simple. Put

$$Q_{ak,bj} = \sum_c F^c_{ak}F^c_{bj}, \qquad P_{ab} = \sum_k Q_{ak,bk}, \qquad N = \sum_{b,k} Q_{bk,bk},$$

$$A_{ab} = \delta_{ab}P_{aa}, \qquad B_{ab} = \delta_{ab}(N - 2P_{aa}), \qquad C_{ab} = (1-\delta_{ab})P_{ab}.$$

Then A, B, C form a real basis of $\mathcal{Z}$, and $\dim_{\mathbb{R}} \mathcal{V} \le 2$.

Sketch. The simple-group invariant calculation through weight four, followed by sign changes and coordinate permutations, leaves two diagonal coefficients and one off-diagonal coefficient. Centring removes the scalar unit term, and reality makes the coefficients real. The tensors above have zero one-point functions. Their independence follows from the tests in VI.2.l together with a normalized two-point test of the 01 component against $Q_{02,12}$, whose values on (A, B, C) are $(0, 0, -16)$.

For the vector bound, the Bianchi and Maxwell equations reduce the zeroth component to the real span of two fixed curvature–first-derivative combinations. Signed permutations determine every other component from it. Thus projection to the zeroth component embeds $\mathcal{V}$ into a space of dimension at most two. *Lean:* `exhaust`, `mixed_space`, `capped_v`. *(proof read; invariant spanning calculations used as inputs)*

Lemma VI.2.l (rank of free divergence). Under the hypotheses of VI.2.4, let s_1, s_2 test the zeroth component of a vector at e_0 against $Q_{01,01}(0)$ and $Q_{12,12}(0)$, respectively, taking real parts and dividing by $\dim \mathfrak{g}\, c_0^2$. Then

$$(s_1, s_2)(\mathrm{div}A) = (-256, 0), \qquad (s_1, s_2)(\mathrm{div}B) = (0, -512), \qquad (s_1, s_2)(\mathrm{div}C) = (128, -128).$$

Sketch. Differentiate the explicit two-point curvature correlations and evaluate the finite index sums. The first two columns make $s \circ \mathrm{div}$ onto $\mathbb{R}^2$. By VI.2.k, s is therefore an isomorphism and divergence is onto. Rank–nullity gives a one-dimensional kernel. The columns also show that $\frac{1}{2}A - \frac{1}{4}B + C$ lies in it, and this tensor is $P - \frac{1}{4}\delta N = T^{\mathrm{free}}$. *Lean:* `east_column`, `west_column`, `cross_column`, `scope_exact`, `rank_on`, `line_dim`, `lighting`, `certificate`. *(proof read; underlying differentiated propagator table used as an input)*

Lemma VI.2.m (canonical choice of stress). For any Maxwell datum for a simple G and any $\tau \neq 0$, one can retain its derivative plan, rotations, hierarchy and Lorentz representation while replacing its stress by τT^{free} and obtaining (FD) and full rotation covariance.

Sketch. Divide the original stress by its Ward constant to obtain Ward rows (∂, ρ). Average over $O(4)$ with Haar measure and take the compatible real part. The Ward identities imply symmetry and conservation of the resulting tensor. Centre it by subtracting its expectation times the unit; this preserves its Ward laws and covariance. The centred tensor j_0 is nonzero, since otherwise its translation Ward law would make the derivative vanish in degree four. By VI.2.l, $j_0 = cT^{\mathrm{free}}$ with $c \neq 0$. Scaling it by τ/c gives the desired stress with Ward constant τ/c; VI.2.n identifies this constant as $-\tau$. *Lean:* `ignite`, `every_light`, `fire`. *(proof read; Haar averaging and Ward-implied conservation and symmetry: statement checked)*

Lemma VI.2.n (the Ward constant). If a Maxwell datum for a simple G has stress τT^{free}, its Ward constant is $\kappa = -\tau$.

Sketch. Test its translation Ward identity in direction zero with $Q_{01,01}$ at 0 and at $-e_0$. Wick expansion leaves one stress leg at each insertion. The single-contraction flux is $-\partial_0 u_{01}(0)$ for a smooth Maxwell wave u on the cutoff support. Evaluating the remaining propagators gives flux $256\tau \dim \mathfrak{g}\, c_0^2$. The derivative side is $-256\kappa \dim \mathfrak{g}\, c_0^2$. Since simplicity implies $\dim \mathfrak{g} > 0$, the constants agree only if $\kappa = -\tau$.

There is also a structural proof in the canonical case. If the plan is ∂^{free} and the stress is T^{free}, VI.2.3 gives Ward constant -1. Comparing translation Ward identities for two constants, using separating correlations, makes their difference times the derivative tuple in degree four zero. That tuple is nonzero, so the constant is unique. *Lean:* `negated`, `required`, `single_cross`, `battery`, `at_tower`, `kappa_eq`, `same_translation`.

3.25.4.4 4.4 The remaining Maxwell data

Lemma VI.2.o (selection of the derivative). For simple G, there is d_0 such that spectators in $\mathcal{P}_{d_0}$ detect every shell-sector polynomial, of any weight. For every $d \geq \max(d_0, 4)$, the admissible vector maps in VI.2.5(a) whose Ward diagrams vanish form the line $\mathbb{R}\partial^{\mathrm{free}}$. Finitely many collinear diagrams suffice.

Sketch. For detection, induct on the weight. Mixed contractions lower the weight and constrain derivatives in the highest-order variables. An explicit colour matrix has nonzero determinant as a matrix of polynomials, so these constraints eliminate the high-order variables. The remaining bounded Taylor orders are detected by a fixed finite exterior degree.

For selection, the low-degree classification first gives a scalar multiple of the derivative on degree four. Subtract that multiple. The remainder kills the free stress; the dilation Ward identity and the detection result make it commute with the weight derivation. Vector covariance makes it anticommute with total coordinate inversion. Since inversion acts on weight w by $(-1)^w$, the remainder is zero. Analyticity transports vanishing of all derivatives at collinear configurations to arbitrary configurations. Finite-dimensionality reduces the intersection of all diagram kernels to a finite intersection. *Lean:* `exposed`, `matrix_meal`, `charges_quiet`, `finite_lenses`, `scanner`, `born`. *(proof read; low-degree vector-map classification and bounded-order detection used as inputs)*

Lemma VI.2.p (the Ward-vanishing test). The endomorphism family in VI.2.5(b) vanishes for every $d \geq 4$.

Sketch. Write the family on polynomials. Exact degree-four two-point correlations imply that each

endomorphism commutes with coordinate quarter turns on the degree-four image. Equivariance then makes its alternating bilinear dependence on $u, v \in \mathfrak{so}(4)$ invariant under simultaneous conjugation by those turns. The finite quarter-turn calculation forces this alternating form to vanish. In particular, the family kills T^{free}.

Apply the local rotational Ward identity to its vanishing Ward sums. The commutator with every $\rho(\omega)$ has zero correlations with degree-d spectators and still lies in $\mathcal{P}_d$; VI.2.6(d) makes it zero. Uniqueness for the linear rotation ODE gives commutation with quarter turns on all of $\mathcal{F}_d^{\mathrm{free}}$. The same alternating-form calculation now gives $T = 0$. *Lean:* `erasing`, `commute_ghost`, `complete_turn`, `low_motion`.

Lemma VI.2.q (assembly of Maxwell data). For simple G and $\tau \neq 0$, the free lighthouse and the remaining data satisfy VI.2.5(a),(b).

Sketch. VI.2.d gives the derivative and rotation laws; VI.2.j gives the Ward laws. The derivative is nonzero in every degree $d \geq 4$: differentiate a nonconstant invariant quadratic curvature polynomial and use faithfulness of the associated colour pairing. VI.2.o supplies the selection property. The Lorentz action on the free polynomial fields descends compatibly to their union, with boost generator $-i\rho$ and the prescribed spatial rotations. Combine these results with the Casimir statements of VI.1 and the Ward-vanishing test VI.2.p. *Lean:* `shines`, `maxwell_supplies`, `Bid.masts`, `voyage`. *(assembly and nonzero derivative: proof read; Lorentz and Casimir data: statement checked)*

3.25.4.5 4.5 Comparison with the Wilson limit

Lemma VI.2.r (panels from a station). An admissible station under the hypotheses of VI.2.8 yields the panel and estimate of VI.2.8(a).

Sketch. The bounded lift maps lie in a fixed finite-dimensional space. Pass to a convergent subfilter and project their limits to the quotient fields of S. Reality, signed covariance and the degree-zero unit identity pass to the limit. Ordered smeared convergence also holds for convergent label coefficients, by finite-dimensional multilinearity. The change of variables in the dilated test then turns the admissibility estimate into the panel estimate. The subfilter proves an estimate for the same continuum table; it does not change the physical extraction order. *Lean:* `harvested`, `calibrated_smear`.

Lemma VI.2.s (balanced estimates give stations). Fix the extraction hypotheses of VI.2.8 and finite bounds $N, D \geq 0$ and $R \geq 1$. Suppose stations satisfy $r_k \to 0$ and the eventual lift bounds and symmetries of VI.2.8. Let their normalized grid spacings be $s_k > 0$ and their strip widths $\sigma_k > 0$. For each basis word w of arity $n \leq N$ and degrees at most D, suppose there are errors $e_k \to 0$ and eventually nonnegative costs A_k, B_k with

$$\sigma_k \to 0, \qquad 4ns_k \leq \sigma_k \text{ eventually}, \qquad A_k^n \sigma_k \to 0, \qquad B_k^n s_k/\sigma_k \to 0.$$

For every sufficiently large fixed k and every convergent subfilter of the lifts along the extraction order, let Q_k be the rescaled continuum correlation of their projected limits. Write Q_0 for the free correlation of w and

$$E_\sigma = \{y : \ |y_i^a - y_j^a| > \sigma \text{ for every } i \neq j, a\}.$$

Assume the following bounds for every such limit:

$$|Q_k| \leq A_k^n \quad \text{on } K_{R,n}, \qquad \|DQ_k\| \leq 4nB_k^n/\sigma_k \quad \text{on } K_{2R,n} \cap E_{4\sigma_k},$$

$$|Q_k - Q_0| \le e_k \quad \text{on } (s_k\mathbb{Z})^{4n} \cap K_{2R,n} \cap E_0.$$

These bounds for all N, D, R imply the station property. The grid, bulk and finite-difference assumptions of the balanced estimate in III.11 imply precisely these bounds for every cluster limit.

Sketch. The excluded band $K_{R,n} \setminus E_{8\sigma_k}$ has volume $O_{n,R}(\sigma_k)$, so its contribution is $O_{n,R}((A_k^n + \|Q_0\|_\infty)\sigma_k)$. Outside the band, rounding to the grid stays in $E_{4\sigma_k}$ and $K_{2R,n}$ for large k. Interpolation costs $O_n(B_k^n s_k/\sigma_k)$ plus the free correlation's modulus of continuity. All these terms and e_k tend to zero. A contradiction subfilter with convergent lifts converts the bound into the ordered cutoff estimate. Expansion in the finite product basis, with coefficient sum bounded by a constant times $\prod_i \|v_i\|$, gives admissibility for arbitrary tuples. Costs may depend on the basis word; only finitely many words occur for the fixed bounds. *Lean:* `cutoff_balance`, `pays_film`, `Balances`, `Wheel.Spots`, `supply_wheel`, `realizes_stations`.

Lemma VI.2.t (comparison isomorphisms). Under the hypotheses of VI.2.7, there is an approaching stable scaling, initially with $\xi_k = 1$.

Sketch. Choose stations of depth and arity bound k, radius bound $k+1$, tolerance $(k+1)^{-1}$, and strictly decreasing radii $r_k \le (k+1)^{-1}$. Their panels approach in the test sense of VI.2.7(a). For each consecutive free inclusion choose a real, signed-covariant left inverse R_d. Starting with $Q_{k,0} = U_{k,0}$, set

$$Q_{k,d+1} = \mathrm{inc}^S Q_{k,d} R_d + U_{k,d+1}(1 - \mathrm{inc}^{\mathrm{free}} R_d).$$

These maps intertwine inclusions exactly. Expansion into mixed tuples of lower-degree fields and complementary fields preserves the comparison estimates. Finitely many separating joint tests make $Q_{k,d}$ eventually injective. The dimension bound then makes it bijective. In each degree, fill the finitely many earlier indices with any one of these isomorphisms. This preserves eventual comparison, symmetry and inclusion identities and yields isomorphisms for every k, d. *Lean:* `Stations.ranging`, `Stations.build_curve`, `signed_section`, `corrected`, `MeterTest.injects`, `Stations.equivalences`, `DimAbove`.

Lemma VI.2.u (smooth convergence). With the hierarchy and derivative laws of VI.2.7, every approaching scaling satisfies

$$r_k U_{k,d+1}^{-1} W_a U_{k,d} \longrightarrow \partial_a^{\mathrm{free}}$$

in each finite-dimensional operator space. Its rescaled correlations converge with all derivatives uniformly on compact subsets of Conf_n. This also holds for free field tuples $v_k \to v$ with fixed degrees.

Sketch. The chain rule identifies a position derivative of a rescaled correlation with the correlation obtained by inserting the displayed transported derivative. Integrate against tests and move the derivative to the test. Approaching convergence and finitely many separating tests then determine the limit of the operator.

Finite-dimensional multilinearity extends the test estimates to convergent field tuples. The resulting correlation differences are closed under gradients, by operator convergence. After localization, their Fourier transforms tend uniformly to zero: multiplication of a fixed test by any Fourier phase preserves its sup norm. Repeated integration by parts gives arbitrarily high decay in frequency, using the same estimates for gradient descendants. Fourier inversion gives uniform convergence; repeating this for descendants gives every derivative order. *Lean:* `scaled_identity`, `bend_limit`, `closed_path`, `travels_jet`, `GradientFamily.jets`.

Lemma VI.2.v (rank transfer). Every approaching stable scaling under VI.2.7 transfers surjectivity and the one-dimensional divergence kernel to S.

Sketch. For large k, transport a free centred tensor by $U_{k,4}$ and then recenter it in S; this defines a real isomorphism $C_k : \mathcal{Z}^{\mathrm{free}} \to \mathcal{Z}^S$. The inverse transports back and recentres; preservation of the unit makes the two composites identities. Let $V_k : \mathcal{V}^{\mathrm{free}} \to \mathcal{V}^S$ be transport by $U_{k,5}$. Then

$$Z_k = V_k^{-1} \circ (r_k \operatorname{div}^S) \circ C_k \longrightarrow \operatorname{div}^{\mathrm{free}}.$$

The centring term disappears on differentiation because the derivative kills the unit. Compose Z_k with a right inverse of free divergence. The result tends to the identity on $\mathcal{V}^{\mathrm{free}}$, hence is eventually invertible. Thus div^S is onto. The two isomorphisms identify the dimensions of domain and target with the free dimensions, so rank–nullity gives kernel dimension one. *Lean:* `differential_rank`, `centerEquiv`.

Lemma VI.2.w (one current along one comparison sequence). Under VI.2.7, there are a nonzero centred conserved J and a refinement Y satisfying VI.2.7(c),(d).

Sketch. Choose $B \neq 0$ in $\ker \operatorname{div}^S$ and normalize its entire transported stress tensor:

$$\xi_k = \frac{\|U_{k,4}^{-1}(r_k^4 B)\|}{\|T_0\|}, \qquad \beta_k = \xi_k^{-1} U_{k,4}^{-1}(r_k^4 B).$$

A subsequence converges on the sphere of radius $\|T_0\|$. The limit is symmetric, real and signed-covariant by closedness. Convergence of one-point functions makes it centred. VI.2.u and the equation $\sum_a (r_k U_{k,5}^{-1} W_a U_{k,4}) \beta_{k,ab} = 0$ make it conserved. By (FD), it equals cT_0 for a real c with $|c| = 1$. Choose $J = B$ or $-B$ once so the limit is T_0. For each finite set of track requirements, stability holds on a tail and VI.2.u supplies the required derivatives. This tail is a track for the same J. *Lean:* `produce_current`, `subsector`, `on_axis`, `center_expect`, `focused_track`, `focused`.

Lemma VI.2.x (physical normalization). Under VI.2.9, there is $c \neq 0$ such that

$$c(J - \langle J \rangle \mathbf{1}) = J_0, \qquad c/\xi_k \longrightarrow \kappa.$$

Consequently VI.2.9(a)–(c) hold and $\xi_k \to c/\kappa \neq 0$.

Sketch. The centred stress J_c lies in $\ker \operatorname{div}^S$. It is nonzero: otherwise the translation Ward identity would make the derivative tuple in degree four zero. Its transported limit would then contradict the nonzero free derivative. The kernel is a real line, so $cJ_c = J_0$ for some $c \neq 0$. Comparing their translation Ward identities gives $p^I = cW$ in degree four.

Along tracks, the transported derivatives tend to ∂^{free}, whereas the transported Ward rows of J_0, divided by ξ_k, tend to $\kappa \partial^{\mathrm{free}}$. Since the free derivative is nonzero, a separating linear functional gives $c/\xi_k \to \kappa$. Therefore

$$U_{k,4}^{-1}(r_k^4 J_c) = \frac{\xi_k}{c}\left[\xi_k^{-1} U_{k,4}^{-1}(r_k^4 J_0)\right] \longrightarrow \kappa^{-1} T_0.$$

The unit part tends to zero because $r_k^4 \to 0$ and the maps eventually preserve the unit. VI.2.u applies to these convergent preimages in every combination of slots, giving the mixed-correlation limit. VI.2.n and the equality of the two displayed free-stress constructions give the canonical limit $-T^{\mathrm{free}}$. *Lean:* `centered_line`, `ratio`, `meter_physical`, `mixed_leading`, `tracks_ratio`.

3.25.5 5. Where the difficulty lies

The transfer to S depends on exact field-space isomorphisms, not merely comparison maps. Finite separating tests and the upper dimension bound provide them. The derivative law then turns the uniform test estimates into convergence of transported derivatives and smooth correlations. Recentring is necessary before transferring divergence: a field-space isomorphism need not preserve the expectation functional.

The normalization is a separate assertion. Conservation determines a line, while the Ward identity determines a point on it. Comparing that identity along tracks removes the initially unknown factor ξ_k and fixes the negative free-stress limit.

Two algebraic rigidity results support VI.1: detection of arbitrary-weight fields by bounded-degree spectators, and vanishing of alternating internal Ward rows. They require more than reflection positivity alone. The three-dimensional tensor space also reflects the chosen symmetry group: signed coordinate permutations permit three coefficients, and conservation reduces them to one. The local current calculation admits conformal Killing fields, although the polynomial Ward argument uses only translations, rotations and dilation.

3.25.6 6. Notation

object	code name	proposed notation
free Maxwell table	`Maxwell.freeTable`, `Climb.Bel`, `Maxwell.Free.Model`	$\mathcal{M}$, fields $\mathcal{F}_d^{\mathrm{free}}$
homogeneous on-shell Taylor coefficients, polynomial algebra	`Shell`, `WickPol`	Sh_j, $\mathcal{W}$
shell sector	`shellSector`	$\mathcal{P}_d$
field ↔ polynomial	`fields`, `address`	ι_d
Wick moment	`Wick.moment`	$\langle p_1(x_1)\cdots p_n(x_n)\rangle_0$
probe evaluation, generated field, leg kernel	`sample`, `pulse`, `K`	$p(f)(x)$, $f^{p,x}$, $\langle p(x)q(y)\rangle_0$
derivative plan, rotations	`scheme`, `gear`, `rewired` (`glide`, `wheel`, `metric`)	$\partial_a^{\mathrm{free}}$; R_g, $\rho(\omega)$; ε
free Maxwell stress	`polStress`, `original`, `canonicalStress`, `torch`	T^{free}, τT^{free}
polarized stress, current, flux	`cross`, `current`, `exportFlux`	Θ_{ab}, j^v, Φ_χ
conformal Killing field, Lie derivative	`Killing`, `variation`	v, $\mathcal{L}_v f = d\iota_v f$
centred stresses, vectors, divergence	`KernelSpace`, `Vectors`, `currentDiv`	$\mathcal{Z}$, $\mathcal{V}$, div
free divergence property	`FreeDiv`	(FD)
lighthouse, Maxwell data	`Climb.Lighthouse`, `herald`, `Org.MaxwellData`	$\mathcal{T} = (\partial^0, R^0, T_0, \kappa)$, M
null-test space, Ward vanishing	`ghost`, `silence`	$\mathcal{N}_d$; $\sum_i \langle \cdots T p_i \cdots \rangle = 0$
station, admissibility, estimates	`Station`, `Stations`, `Granular`, `Balances`	$(r, A_{N,p})$

object	code name	proposed notation
panel, scaling, rescaled derivative	`Panel`, `Climb.Scaling`, `tiltScale`	U; (r_k, ξ_k, U_k); tilt_k
uniform test convergence, stability, stress limit	`Approaches`, `Fades`, `Stable`, `Scaling.meter`	—

3.25.7 7. What remains unexplained

1. **Inputs not checked through all subsidiary proofs.** The reflected Gram positivity and Fock pairing factorization in VI.2.b were checked at statement level. The Haar-averaging and Ward-implied conservation and symmetry results used in VI.2.m were also checked at statement level. The invariant spanning results behind VI.2.k, the differentiated propagator table behind VI.2.l, and the low-degree vector-map classification and bounded-order detection behind VI.2.o remain inputs to the sketches. Their direct applications and hypotheses were checked, but their full derivations were not independently reconstructed.
2. **Lorentz and Casimir data.** The representation, descent and generator statements were checked, together with their use in assembly. The complexified-rotation construction of the representation and the Casimir proofs A09–A12 were not checked in full.
3. **Renormalization-group estimates.** The conversion of balanced grid, bulk and finite-difference estimates into stations was checked here. Their production in III.11, including all parameter choices in the complete weak-comparison result and the shell estimates needed for the interacting Ward rows, remains to be verified in that unit. The excluded region here is precisely a band around coordinate-projection ties; it is not a neighbourhood of the full collision diagonal.
4. **Short-distance normalization.** The comparison of derivative and stress rows uses the track estimates of VI.1. The scalar-ratio argument and its application here were checked; the underlying track Ward estimates A04–A05 remain inputs. The mixed-correlation theorem is stated in Lean for the stress assembled in VI.1; VI.2.9(b) gives the general consequence of the same convergent-preimage argument.

3.26 VII.1. Short distances: convergence to free Maxwell fields

Part: VII, identification as Yang–Mills. **Lean:** the chain of `YM/Pert` (folders `Fourier`, `Align`, `Warp`, `Rig`, the jet stencils of `Lattice`, the statement `Position/Crest.finite_leading`, and `Dense`), `YM/Push/FINITE_DETECTION`, `YM/SharpGram/Weak`, `YM/SharpGram/Dense`, `YM/RG08` (about thirty small sub-folders), `YM/PertExtension`, `YM/OPE/Levy`. **Size:** 8,117 lines. They divide as follows:

- 3,275 lines in the chain ending in `finite_leading`;
- 1,006 in `FINITE_DETECTION`;
- 616 in `SharpGram/Weak`;
- about 2,250 in `RG08` (`Dart` 538, `Ballast` 338, `Vale` 315, the rest under 160 each);
- 316 in `SharpGram/Dense`, 310 in `PertExtension`, 108 in `SharpGram/Matrix`, 95 in `Pert/Fourier` and 63 in `OPE/Levy`;
- the remainder in packages of fewer than 40 lines.

The main result identifies the order-zero lattice correlations with the free Maxwell hierarchy. The short-distance limit of the interacting correlations is stated in `FirstExtension` (VI.2), established

by `firstTarget` in VIII.1. The results are:

- the order-zero term of lattice perturbation theory converges, as the spacing tends to zero, to the Wick-ordered free Maxwell hierarchy evaluated on the on-shell images of the labels (VII.1.1–VII.1.3);
- the free hierarchy detects every label outside the Maxwell equation ideal through a single two-point function at a fixed pair of points (VII.1.4–VII.1.5);
- using this detection, the measuring charts of the construction can be corrected to charts that are exactly compatible with the filtration, the symmetries and the unit (VII.1.6);
- estimates on backward windows and local expansions used in the rank comparison (VII.1.7).

3.26.1 1. What this unit proves

Setting.

Gauge group and continuum objects.

- G is a compact matrix gauge group whose Lie algebra $\mathfrak{g}$ is real simple and nonabelian, with an invariant inner product and an orthonormal basis of colours.
- A *key* $a = (j, \alpha)$ is a coordinate of the real radial-gauge potential space T_j: homogeneous $\mathfrak{g}$-valued 1-forms of polynomial degree $j+1$ with $x \cdot A(x) = 0$. Curl identifies T_j with the closed homogeneous curvature jets of degree j. Each key has engineering weight $j+2$.
- Pol is the complex polynomial algebra in the keys. V_d consists of its $\mathrm{Ad}(G)$-invariant polynomials of total engineering weight at most d. The free field space F_d is the quotient of V_d by the free null space (VI.2).
- Sh_j is the space of curvature tensors, symmetric in the j Taylor slots and antisymmetric in the two form indices, whose homogeneous fields solve $dF = d{*}F = 0$. Taylor coefficients are normalized by $1/j!$. Write $\kappa : \mathrm{Sh}_j \to T_j$ for the inverse radial-gauge curl map (`gaugeJet`).
- $\rho : \mathrm{Pol} \to \mathrm{WickPol} = \mathbb{C}[\text{shell coordinates}]$ restricts a polynomial to on-shell jets (`reduceEq`).
- The *equation ideal* $\mathcal{I}_d = V_d \cap \ker \rho$ consists of labels that vanish on every on-shell Taylor tower (`equationIdeal`).
- $\omega_d = \rho \circ \mathrm{sec}_d : F_d \to \mathrm{WickPol}$ is the on-shell image of a label. Here sec_d is the representative polynomial of VII.3, obtained by projecting radial curvature coordinates onto the Maxwell shell (`sections`, `fixedPolynomial`).

The free Maxwell hierarchy (III.16).

- $\Gamma(x) = 1/(4\pi^2|x|^2)$ (`Green`).
- The kernel `bareCurl` is $\delta_{bk}\partial_a\partial_m\Gamma - \delta_{bm}\partial_a\partial_k\Gamma - \delta_{ak}\partial_b\partial_m\Gamma + \delta_{am}\partial_b\partial_k\Gamma$. This is $\langle F_{ab}(x)F_{km}(y)\rangle$ at $x-y$ for $F = dA$, where A is the Feynman-gauge free field with $\langle A^c_\mu(x)A^{c'}_\nu(y)\rangle = \delta^{cc'}\delta_{\mu\nu}\Gamma(x-y)$.
- For linear forms P on Sh_j and Q on Sh_l, $K(P,Q)(x-y)$ is P applied to the j-jet in x and Q to the l-jet in y of this kernel, colour-diagonal (`legKernel`). The value is independent of extensions of P, Q to full curvature tensors, because the kernel solves the free equations in each variable.
- For a key a, write x_a for its real coordinate functional on T_j and $\kappa_a = x_a \circ \kappa$, and write $K_{ab} = K(\kappa_a, \kappa_b)$. The Taylor coefficients in both legs include their factorial denominators; differentiation at the second endpoint contributes $(-1)^l$.
- The *Wick moment* $\mathcal{W}_x(q_1, \dots, q_r)$, for $x \in \mathrm{Conf}_r$ and $q_i \in \mathrm{WickPol}$, is multilinear (`Wick.moment`). On monomials it is the sum, over perfect matchings of the variable oc-

currences that never pair two occurrences at the same point, of the product of the entries $K(\cdot,\cdot)(x_i - x_j)$. It is the correlation of the Wick-ordered products $:q_i:(x_i)$ of the Gaussian jet field with covariance K.

- The free hierarchy of the labels is $x \mapsto \mathcal{W}_x(\omega_d p_1, \dots, \omega_d p_r)$ (`Ward.disc` of the free table).

The lattice prescription at order zero (definitions from VII.3).

- $\Lambda_N = (\mathbb{Z}/N)^4$ with N even, $N \geq 4$.
- The field space $\mathcal{A}_N$ consists of $\mathfrak{g}$-valued link variables that are odd under the point reflection: $A(-x - e_i, i) = -A(x, i)$, i.e. A is invariant under $x \mapsto -x$ as a 1-form (`LinkP`).
- The quadratic form $\mathsf{Q}_N(A) = \frac{1}{2}(\sum_{x,i<j} |\mathrm{curl}_{ij} A(x)|^2 + \sum_x |\mathrm{div} A(x)|^2)$ is the Hessian at the identity of the Wilson action plus the Feynman gauge-fixing term $\frac{1}{4}\sum |\mathrm{div} A|^2$ (`form`, `action`).
- Q_N is positive definite on $\mathcal{A}_N$: a closed and divergence-free field is constant, and a reflection-odd constant field is zero. μ_N is the centred Gaussian with density proportional to $\exp(-\mathsf{Q}_N(A)/2)$ (`data`).
- Expectations are formal power series in the coupling. The interaction exponent has order-n coefficient equal to minus the order-$(n+2)$ action coefficient for $n \geq 1$, and zero constant coefficient. Its exponential is multiplied by the Taylor series of the Haar Jacobian and Faddeev–Popov density. Each jet-coordinate series uses its order-$(n+1)$ Taylor coefficient at order n (`weight`, `seriesLex`). Order zero is therefore a μ_N-expectation; $\langle\cdot\rangle^{(0)}$ denotes the order-zero coefficient.
- $\ell^h_x : \mathcal{A}_N \to T_j$ is the radial-gauge image of the linear lattice curvature jet at site x (`slope`). It is obtained from the j-fold forward difference of the plaquette curl divided by $j!$ by symmetrization and the radial-gauge map, scaled by $h^{-(j+2)}$ and averaged over the hyperoctahedral group W_4 of signed axis permutations. $\ell^h_{x,a}$ is its a-th coordinate.
- The order-zero lattice representative of a polynomial $p \in \mathrm{Pol}$ using only keys of rank at most d, at o is

$$R_h(p, o)(A) = \mathbb{E}_\xi\, p(\ell^h_o(A) + i\xi),$$

where ξ is an independent centred Gaussian on the keys of rank $\leq d$ with covariance

$$S_h(a, b) = (2\pi)^{-4} \int_{[-\pi,\pi]^4} \hat{\Delta}(k)^{-1} \sum_c \lambda_a(c, k)\lambda_b(c, k)\, dk.$$

Here $\hat{\Delta}(k) = \sum_i 4\sin^2(k_i/2)$, c runs over cosine/sine, axis and colour channels, and $\lambda_a(c,\cdot)$ is $\ell^h_{0,a}$ applied to the corresponding plane wave on $\mathbb{Z}^4$. So S_h is the covariance of the jet stencils at one site for the Feynman-gauge Gaussian on the infinite lattice $\mathbb{Z}^4$, and $R_h(p, o) =: p(\ell^h_o) :_{S_h}$ is Wick-ordered with respect to the infinite-lattice self-contraction (`column`, `link`).

- **Regulators.** For an integer $L \geq 2$ set $h_m = L^{-m}$ and $N_m = 4L^{m^2+m+1}$. Lattice points are reported relative to the centre $c_m = (N_m/4, \dots, N_m/4)$, through $o \mapsto h_m(o - c_m)$ with representatives in $(-N_m/2, N_m/2]^4$.

- **Profile.** For $f : (\mathbb{R}^4)^r \to \mathbb{C}$ and $p \in F_d^r$,

$$\Pi_m(f;p) = h_m^{4r} \sum_{o \in \Lambda_{N_m}^r} f(h_m(o - c_m)) \left\langle \prod_j R_{h_m}(\sec_d p_j, o_j) \right\rangle_{N_m}^{(0)}$$

  (`profile`, `row`).

- **Itinerary.** An itinerary is a sequence (N_i, h_i) with N_i even, $h_i > 0$, $N_i h_i$ a positive integer tending to ∞, $h_i \to 0$, and $N_i h_i^k \to \infty$ for every $k \in \mathbb{N}$ (`Itinerary`). Sites $o_i \in \Lambda_{N_i}$ are *homing* to z if $h_i(o_i - c_{N_i}) \to z$ (`Home`).

Proposition VII.1.1 (regulator removal at order zero). Fix an integer $L \geq 2$, $d, r \in \mathbb{N}$, label frames, and $p_1, \dots, p_r \in F_d$. Let f be continuous with compact support and $\operatorname{supp} f \subset \mathrm{Conf}_r$, where support is closed. Then

$$\lim_{m \to \infty} \Pi_m(f;p) = \int f(x)\, \mathcal{W}_x(\omega_d p_1, \dots, \omega_d p_r)\, dx.$$

In board form: let Q be a position board whose profiles are smooth test functions compactly supported in Conf ("clean profiles"). Then the order-zero board readings converge to the free readings, slot by slot and for every tuple of basis labels. *Status:* theorem. There is no hypothesis on the construction; it concerns lattice Gaussian integrals and the continuum kernel only. *Lean:* `Rig.zerothProfile`. Board form: `Position.finite_leading`, using the definitions `finite`, `white` and `CleanProfiles` (all VII.3).

Proposition VII.1.2 (pointwise convergence of order-zero correlations). Let (N_i, h_i) be an itinerary with $4 \mid N_i$. Let $z \in \mathrm{Conf}_r$, and let $o_{i,j} \in \Lambda_{N_i}$ be homing to z_j for each j. Then for all $p_1, \dots, p_r \in F_d$,

$$\left\langle \prod_j R_{h_i}(\sec_d p_j, o_{i,j}) \right\rangle_{N_i}^{(0)} \longrightarrow \mathcal{W}_z(\omega_d p_1, \dots, \omega_d p_r).$$

Status: theorem. *Lean:* `Warp.row_limit`, through `poly_limit` and `earth`.

Proposition VII.1.3 (corrected two-leg covariance). Let (N_i, h_i) be an itinerary with $4 \mid N_i$. Let ι be any index set, let $z : \iota \to \mathbb{R}^4$ be injective, and for every $j \in \iota$ let $o_{i,j}$ be sites homing to z_j. For indices $j, l \in \iota$ and arbitrary keys a, b, define

$$\kappa_h(j,a;l,b) = \mathrm{Cov}_{\mu_N}(\ell^h_{o_j,a}, \ell^h_{o_l,b}) - \delta_{jl}\, S_h(a,b).$$

Then

$$\kappa_{h_i}(j,a;l,b) \to \begin{cases} K_{ab}(z_j - z_l), & j \neq l, \\ 0, & j = l. \end{cases}$$

Status: theorem. *Lean:* `Align.corrected_limit`, from `orbit_limit`, `orbit_contact` and `shell_exact`.

Proposition VII.1.4 (two-point detection). Let e_{t} be the unit vector of the time axis, σ the time reflection $x_{\mathrm{t}} \mapsto -x_{\mathrm{t}}$ acting on labels, and $\bar{v}$ the complex conjugate label. If $v \in V_d \setminus \mathcal{J}_d$, then

$$\operatorname{Re} \mathcal{W}_{(e_{\mathrm{t}},0)}(\rho v,\ \rho(\sigma \bar{v})) > 0.$$

More generally, let Φ and Ψ be the algebra maps sending shell coordinates to the diagonalizing and reflected probe variables of Lemma VII.1.m. For every nonzero $q \in \mathrm{WickPol}$, $\operatorname{Re} \mathcal{W}_{(e_{\mathrm{t}},0)}(\Phi q, \Psi \bar{q}) >$

0. Here Φ is surjective, and $\rho(\sigma\bar{v}) = \Psi(\bar{q})$ whenever $\rho v = \Phi q$. *Status:* theorem, about the free kernel only. No Wilson family and no reflection positivity of a measure is involved. *Lean:* `Free.two_detect`, `Free.positive`, `detectors`.

Proposition VII.1.5 (Kelvin formulas for the free kernel).

(a) Let A be a real symmetric j-tensor on $\mathbb{R}^4$ whose diagonal $x \mapsto A(x, \dots, x)$ is harmonic. Then for $x \neq 0$

$$\frac{1}{j!} A(\partial, \dots, \partial)\, \Gamma(x) = \frac{(-2)^j}{4\pi^2}\, |x|^{-2(j+1)} A(x, \dots, x).$$

(b) Let $Y \in \mathrm{Sh}_j$ be $\mathfrak{g}$-valued, with homogeneous Maxwell solution $Y^\natural(x) = Y(x, \dots, x)$. Contract the first leg of the curvature kernel at x with Y, using its rank-j Taylor coefficient and summing over colours, ordered form indices and jet indices, without a factor $1/2$. For $x \neq 0$ the result equals

$$\gamma_j\, (\iota^* Y^\natural)(x), \qquad \gamma_j = \frac{2^j (j+1)(j+2)}{\pi^2}.$$

Here $\iota(x) = -x/|x|^2$ and ι^* is the pullback of 2-forms:

$$(\iota^* F)_{km}(x) = |x|^{-4} \left[B_{km} - 2|x|^{-2} \left(x_k \sum_a x_a B_{am} - x_m \sum_a x_a B_{ak} \right) \right], \qquad B_{ab} = F_{ab}(\iota x).$$

Status: theorem. *Lean:* `Kelvin.harmonic_Green` (with `R` $= |x|^{-2}$), and `Kelvin.response_eq` (with `inverted` and `gain`).

Proposition VII.1.6 (corrected measuring charts). Fix a Wilson family C with a realization of its defining conditions, a construction record U with limit table S and extraction order f, and OPE table data as in VII.4. Let $D \in \mathbb{N}$, and let u be a nontrivial filter on $\mathbb{N} \times \mathbb{N}$ finer than the extraction filter of f.

Assume a weak measuring portfolio X through degree D. Its data include the observable bounds, Haar bounds, a Haar chart ratio, three price panels, and a constant dominating the label costs through D. It supplies an index $g(k) \to \infty$, finite-regulator linear label maps, and limiting maps $\Phi_{k,d}$ on the reference label-coordinate spaces. Eventually in k:

- $g(k) \leq k$ and $\mathrm{scale}(k) \leq \mathrm{scale}(g(k)) + \log \hat{\beta}_k$;
- the finite-regulator maps satisfy the simultaneous rank-comparison conditions through D: every degree is available at step $g(k)$, the block ratio is below the monitor bound, the maps are tested and equivariant, and all three panels supply the required observable and remainder estimates;
- their rescaled read-out maps converge to $\Phi_{k,d}$ along the same filter u, for every $d \leq D$ and every reference label.

Let λ_d be the fixed equivariant representative of the free quotient in reference label coordinates, and let π_d^S be projection to S_d. Set $D_{k,d} = \pi_d^S \Phi_{k,d} \lambda_d$. Define $\chi_k(d) = D_{k,d}$ when this map is bijective, and choose a fixed isomorphism $F_d \to S_d$ otherwise. Then there are complex linear isomorphisms $p_k(d) : F_d \to S_d$, for every $k, d \in \mathbb{N}$, such that:

1. For all sufficiently large k, simultaneously for $d \leq D$, $p_k(d)$ commutes with signed permutations and conjugation and preserves the unit. For all $d \leq e \leq D$,

$$p_k(e) \circ \iota_{d,e}^F = \iota_{d,e}^S \circ p_k(d).$$

2. For every $d \le D$ and $v \in F_d$,
$$\chi_k(d)^{-1} p_k(d) v \longrightarrow v.$$

The simultaneous finite-regulator conditions are `Poised`, and the representative is `ladder`.

Status: theorem under the stated construction and portfolio hypotheses. Lemma VII.1.q supplies the required two-point convergence. *Lean:* `Weak.Portfolio.tune_grades`, with `Portfolio` and `chart`.

Proposition VII.1.7 (backward windows and local expansions).

(a) Fix the RG constants of III.2, Haar bounds, a Haar chart ratio satisfying the local ratio bounds, and a relay of admissible price panels with activation bounds $u_j \le L_j$. Each degree is eventually active; its three panels satisfy the comparison inequalities and the analytic bounds required for the rank estimates. Let U be a wait with window
$$w_j(t) = \left\lfloor \frac{\log(\bar{\beta} + c_G t)}{\log L_j} \right\rfloor,$$
where $c_G > 0$ is the coupling slope. There is a wait W with no smaller minimum stage length or entry thresholds and with the same window, together with panels and cost bounds for every degree, having the following property. For every realized Wilson family satisfying W, the prescribed observable bounds and the polynomial-representative conditions, and every D, all sufficiently late steps k whose scale exceeds the stage entry threshold and whose backward window has length at most k and lies in one stage admit simultaneous finite-regulator rank-comparison data through D, as in VII.1.6. The returned panels satisfy the three comparison inequalities at every degree, and each cost bound dominates the label costs through that degree.

(b) Let U have the window in (a), and let V have no smaller stage-length bounds, entry thresholds or windows. Assume, for every stage j, that the entry threshold of V at j is at least that of U at $j+1$, and that $\bar{\beta} + c_G t \ge c_G$ above the V threshold at j. Every RG trajectory satisfying V admits $g(k) \to \infty$ such that eventually $g(k) \le k$, the step $g(k)$ is above its U entry threshold, its backward U window has length at most $g(k)$ and lies in one stage, and
$$\operatorname{scale}(k) \le \operatorname{scale}(g(k)) + \log \hat{\beta}_k.$$
These threshold conditions are `LateEnough`.

(c) Fix a finite index set I, a marker subset $T \subset I$, an anchor $p \in T$, $D, k, E, q \in \mathbb{N}$, a support parameter $R \in \mathbb{R}$, and $v > 0$, with $q \le k + E$. There are $0 < w \le v$ and $C_0 \ge 0$, independent of the layout and its bound B, with the following property. Suppose a layout has marker weights $a_i \in \mathbb{N}$ with $\sum_{i \in T} a_i = k$, depends only on the occupied markers, vanishes on the empty marker set, and satisfies the prescribed tensor symmetry, rank and analytic realization conditions (`Sense`). Suppose also that every coefficient of expansion order $i \le E$, at every leg rank, has the anchored v-weighted absolute-sum bound B. Translate the root to x_p and shift the order by q, setting coefficients with $q + n < k$ to zero. Through leg rank $k + E - q$, the resulting coefficients are exactly the Taylor coefficients of a finite sum of curvature expressions of ranks
$$k \dot{-} q, \dots, k + E - q,$$

where $k \dot{-} q = \max(k - q, 0)$. Each expression depends only on T, retains the required symmetries and analytic realization, and has the anchored absolute-sum bound BC_0 with weight $\exp(w\,\mathrm{girth})$, including the exponential weight on relative marker positions.

Status: theorems. *Lean:* `Dense.slim_wait_for`, `Dense.nearby`, `Dart.coin_pre`.

3.26.2 2. What it uses

From other units.

- *VII.3 (the lattice prescription and position boards):*
  - the reflection-odd field space, the Feynman-gauge form, the Gaussian μ_N and the coupling series;
  - the jet stencils (`wideLinear`), the Wick-subtraction covariance S_h (`shellCov`) and the on-shell representatives `physicalLabel`;
  - rows, profiles and the board objects `Conditions`, `finite` and `white`.
- *III.16 (free Maxwell analysis):* the shells, `reduceEq`, `equationIdeal`, Γ, the curvature kernel `legKernel` and the Wick moment `Wick.moment`.
- *VI.2 and VI.1:* the free table and its label spaces F_d (VI.2); the hierarchy as a function of positions, `Ward.disc` (VI.1).
- *II.1 and I.1:* labels, keys and Taylor jets (II.1); the lattice and the signed permutations W_4 (I.1).
- *For VII.1.6:*
  - the construction record (VIII.1) and the limit table S;
  - the OPE table work items, of which only the system part is used (VII.4);
  - the chart-correction machinery `retraction`, `adjusted`, `rotateChart` and `Graded` (VII.5);
  - the rank-comparison error bound and its decay `Ship.diminishes` (III.11), and the chessboard-to-moment estimate for observables `Read.full_cost` (III.11);
  - the lemma `tendsto_of_fitted` (VI.1): if $f_k, g : E \to F$ are linear maps between finite-dimensional complex normed spaces, $f_k \to g$ pointwise, g is injective and $f_k(w_k) \to g(z)$, then $w_k \to z$.
- *For VII.1.7:* stage-length bounds, entry thresholds, backward windows, and the RG scale sequence (III.2); price panels and the linear checks on their exponents (II.2).

Standard mathematics.

- Gaussian moments and Wick's theorem in recursive form; Wick ordering as $:p(\varphi):= \mathbb{E}_\xi\, p(\varphi + i\xi)$. References: Glimm–Jaffe, *Quantum Physics*, §6.3 and §8.6; B. Simon, *The $P(\varphi)_2$ Euclidean (Quantum) Field Theory*, Ch. I.
- Fourier analysis on $(\mathbb{Z}/N)^4$; asymptotics of the lattice Green function and its finite differences (Lawler–Limic, *Random Walk: A Modern Introduction*, Ch. 4); power counting for lattice Feynman integrals (T. Reisz, *Commun. Math. Phys.* 116 (1988) 81).
- Riemann sums for continuous compactly supported functions; dominated convergence.
- Derivatives of $|x|^{2-n}$ against harmonic polynomials (Hobson's formula; E. W. Hobson, *The Theory of Spherical and Ellipsoidal Harmonics*, 1931); the Kelvin transform (Axler–Bourdon–Ramey, *Harmonic Function Theory*, Ch. 4); invariance of the source-free Maxwell equations under inversion (Bateman, Cunningham, 1910).

Imported constants and assumptions.

- VII.1.1–VII.1.5 use the stated gauge, label and regulator data and no assumptions on an interacting construction.
- VII.1.6 imports the construction record, the portfolio estimates at step $g(k)$, the error bound of III.11, which tends to 0, and the OPE work items.
- VII.1.7 imports the RG scale sequence (stage lengths L_j, the scale and $\widehat{\beta}_k$) from III.2.

3.26.3 3. Strategy

At order zero, the normalized perturbative expectation is Gaussian. Wick ordering introduces an independent imaginary Gaussian for each insertion, so the pair contraction is the torus stencil covariance minus the infinite-lattice self-covariance at a common insertion. Fourier analysis identifies the contractions at separated points with the Maxwell kernel. At one insertion, the remaining finite-volume error tends to zero because the period grows faster than every inverse power of the spacing. The lattice moments and the continuum Wick moments satisfy the same deletion recursion; induction on the number of distinguishable legs gives VII.1.2. Compactness and dominated convergence then give the tested limit VII.1.1.

For detection, inversion and translation turn a homogeneous Maxwell solution of degree j into a field whose Taylor expansion at the detector begins at order j. Adding the corresponding lower-rank probes gives a triangular change of variables. In an orthogonal shell basis, the reflected one-leg pairing becomes diagonal with positive entries. Wick's rule then makes distinct monomials orthogonal and gives every monomial a positive squared norm, proving VII.1.4.

The interacting comparison uses the rank estimates of III.11. They imply convergence of all mixed two-point readings in one portfolio. Free detection makes the raw charts eventually invertible and forces their filtration defects to tend to zero. Fixed equivariant retractions of the free inclusions remove these defects recursively, giving the exactly compatible charts of VII.1.6. Proposition VII.1.7 supplies the backward-window and local-expansion estimates used in this construction.

3.26.4 4. Steps

3.26.4.1 4.1 The free lattice covariance Lemma VII.1.a (covariance of link forms). For links $a = (x, i)$, $b = (y, j)$ and colours u, t,

$$\mathrm{Cov}_{\mu_N}(\langle A(a), u\rangle, \langle A(b), t\rangle) = \langle u, t\rangle\, \mathbf{1}_{i=j}\,[g_N(y-x) - g_N(y+x+e_i)].$$

Here g_N is the mean-zero torus Green function, $-\Delta g_N = \delta_0 - N^{-4}$ and $\sum_x g_N = 0$, written as a Fourier sum over $(\mathbb{Z}/N)^4$ with the zero mode omitted (`toroid`, `lap_eq`). *Proof sketch.* Summation by parts gives $\mathsf{Q}_N(A, B) = -\frac{1}{2}\sum\langle A(x,i), \Delta B_i(x)\rangle$, the Feynman-gauge identity $\mathrm{curl}^2 + \mathrm{div}^2 = -A \cdot \Delta A$ (`total`). The response to a unit forcing at a is $g_N(\cdot - x)e_i$ minus the same function at the image link. The constants N^{-4} cancel between the two images, and oddness of the test field doubles the pairing (`inverse_eq`). *Lean:* `Orb.covar_link`.

Lemma VII.1.b (separated legs). Let (N_i, h_i) be an itinerary. For a torus site x, write $[x]_N$ for its centred lift in $(-N/2, N/2]^4$. Let x_i, y_i satisfy $h_i[x_i - y_i + p]_{N_i} \to z \neq 0$ and $\|h_i[x_i + y_i + p]_{N_i}\| \to \infty$ for every fixed $p \in \mathbb{Z}^4$. Then for real linear forms P on T_j and Q on T_k,

$$\mathrm{Cov}_{\mu_{N_i}}(P \circ \ell^{h_i}_{x_i}, Q \circ \ell^{h_i}_{y_i}) \to K(P \circ \kappa, Q \circ \kappa)(z).$$

The image condition holds automatically when both families are homing and $4 \mid N_i$, since then each coordinate of $x_i + y_i$ is $N_i/2 + O(h_i^{-1})$ modulo N_i (`finale_home`). *Proof sketch.*

1. Average over W_4 and expand the forms in component functionals. By Lemma VII.1.a each component pairing becomes $h^{-(|D|+2)}$ times a forward difference ∇_D^+ of g_N, evaluated at $x-y$ (direct term) and at $x+y+e$ (image term). The Fourier symbol of ∇_D^+ is $\prod_{k\in D}(e^{ip_k}-1)$.
2. *Direct term* (`joint_casual`). At physical momentum scale s, set $t=hs$ and write lattice momentum as tk. The unscaled Fourier multiplier becomes $N^{-4}t^{|D|-2}$ times a rescaled symbol; after multiplication by $h^{-(|D|+2)}$, the prefactor is $(Nh)^{-4}s^{|D|-2}$. The rescaled symbol is jointly smooth near $(0,k)$ for $k\neq 0$ and equals $\prod_{l\in D}(ik_l)/|k|^2$ at $t=0$. Split momentum space with a smooth bump into a low-momentum piece, dyadic annuli and a tail.
   - Each annulus is a Riemann sum of a compactly supported rescaled integrand and converges to its continuum integral.
   - For separations bounded below by $R>0$, the tail is bounded by $C(D,R)/s$ uniformly in N once the mesh is fine enough relative to hs (`upper_tail`). The proof uses $|D|+3$ summations by parts in momentum and the lower bound $|e^{2\pi i[x]_f/N}-1|\gtrsim \|h[x]_N\|/(Nh)$ for a suitable coordinate f.
   - The low piece is $O(s^{|D|+2})$.
   - The same bounds hold for the continuum integral after heat regularization. A 3ε argument concludes (`wrap_tube`).
   - Besides $h\to 0$, this part requires only $Nh\to\infty$, rather than faster volume growth.
3. *Image term.* The same decomposition gives 0 when the norm of the centred image displacement tends to infinity in physical units (`joint_far`).
4. The limit is identified with K on rotated on-shell jets, and W_4-averaging returns $K(P\circ\kappa, Q\circ\kappa)$.

Lean: `Align.chart_limit`, `Align.orbit_limit`. *(proof read through the Fourier bounds; remaining analytic inputs in §7)*

Lemma VII.1.c (coincident legs). Let (N_i,h_i) be an itinerary and let $x_i\in\Lambda_{N_i}$ satisfy $\|h_i[2x_i+p]_{N_i}\|\to\infty$ for every fixed $p\in\mathbb{Z}^4$. For real linear forms P on T_j and Q on T_k,

$$\mathrm{Cov}_{\mu_{N_i}}(P\circ\ell_{x_i}^{h_i}, Q\circ\ell_{x_i}^{h_i})-\mathcal{S}_{h_i}(P,Q)\to 0,$$

where $\mathcal{S}_h$ is the $\mathbb{Z}^4$ covariance of the same stencils. In particular $S_h(a,b)=\mathcal{S}_h(x_a,x_b)$ (`shell_exact`). *Proof sketch.* The image term vanishes as in Lemma VII.1.b. The direct term is a finite combination of values $g_N(a)$ at fixed lattice offsets a, compared with the $\mathbb{Z}^4$ Green function $G_{\mathbb{Z}^4}(a)$ and multiplied by $h^{-(j+k+4)}$. The estimate for each fixed offset a (`toroid_cost`) gives constants $C,A\geq 0$, independent of N and r, such that for N even and $0<r\leq 1$, $8/N\leq r$,

$$|g_N(a)-G_{\mathbb{Z}^4}(a)|\leq C\,r^2+A/(Nr^4).$$

It is proved by viewing $g_N(a)$ as a Riemann sum of $\cos(2\pi a\cdot q)/\hat{\Delta}(2\pi q)$:

- near $q=0$ both terms are $O(|q|^{-2})$, whose integral over $\{|q|\leq r\}$ is $O(r^2)$;
- elsewhere the integrand is $O(r^{-4})$-Lipschitz and the grid points move by $O(1/N)$.

With $r=h^{n+1}$, where $n=j+k+4$ is the power of h^{-1} to be beaten, the error times h^{-n} is at most $Ch^{n+2}+A/(Nh^{5n+4})\to 0$. This comparison uses the condition $N_ih_i^k\to\infty$ for every k (`strike_two`). *Lean:* `Align.chart_return`, `Align.orbit_contact`.

Lemma VII.1.d (VII.1.3). This is Lemma VII.1.b for $j\neq l$ (with `scene_home` and `finale_home`) and Lemma VII.1.c for $j=l$. The covariance matrix Q_N^{-1} evaluated on the stencil vectors equals the orbital pairing (`key_orb`). *Lean:* `corrected_limit`.

3.26.4.2 4.2 Order zero as a Gaussian recursion Lemma VII.1.e (order zero is a joint Gaussian expectation). Fix $h > 0$, an even period $N \geq 4$, insertion sites $o_1, \dots, o_r$, and $d \in \mathbb{N}$. Let s be a finite set of distinguishable occurrences, with insertion map $\tau : s \to \{1, \dots, r\}$ and key map $a : s \to \{\text{keys of rank at most } d\}$. Repeated keys are allowed. Set $\mathrm{root}(s)_j = \prod_{\nu:\tau(\nu)=j} x_{a(\nu)}$. Then

$$\Big\langle \prod_j R_h(\mathrm{root}(s)_j, o_j) \Big\rangle^{(0)} = E_h(s) := \mathbb{E}\Big[\prod_{\nu \in s} (\ell^h_{o_{\tau(\nu)}, a(\nu)} + i\, \xi^{(\tau(\nu))}_{a(\nu)}) \Big],$$

under $\mu_N \otimes \bigotimes_{j=1}^{r} \mathcal{N}(0, S_h)$. *Proof sketch.* The weight and the vacuum both have constant coefficient 1, so the order-zero coefficient of the normalized expectation is a μ_N-integral. The constant coefficient of each lattice jet is its linear part ℓ^h. Fubini over the independent auxiliary copies finishes the proof. *Lean:* `Warp.amplitude_zero`. The pair covariance of two legs is κ_h (`node_wage`): the auxiliary copies contribute $i^2 S_h = -S_h$ for legs assigned to the same insertion and 0 otherwise, even if two insertion sites happen to coincide.

Lemma VII.1.f (symmetric deletion identity). Let μ be a centred Gaussian on a finite-dimensional real space (possibly degenerate) and P_a, $a \in s$, real linear functionals with values in $\mathbb{C}$. Then

$$|s|\, \mathbb{E} \prod_{a \in s} P_a = \sum_{a \in s} \sum_{b \in s \setminus \{a\}} \mathbb{E}[P_a P_b]\, \mathbb{E} \prod_{c \in s \setminus \{a,b\}} P_c.$$

The products are bilinear, without complex conjugation. For the legs of Lemma VII.1.e, write $\kappa_h(\nu, \mu) = \kappa_h(\tau(\nu), a(\nu); \tau(\mu), a(\mu))$; then $|s| E_h(s) = \sum_{\nu \neq \mu} \kappa_h(\nu, \mu) E_h(s \setminus \{\nu, \mu\})$. *Proof sketch.* For powers of a single real functional L of variance v, $\mathbb{E}L^{n+1} = nv\, \mathbb{E}L^{n-1}$ for $n \geq 1$, with $\mathbb{E}L = 0$, read off from the moment generating function (`ev_next`). The multilinear case follows by polarization (`packet_cancel`). The complex case follows by splitting into real and imaginary parts. *Lean:* `exchange_s`, `Warp.eq_cancel`.

Lemma VII.1.g (the continuum recursion). For $z \in \mathrm{Conf}_r$ set $c_z(s) = \mathcal{W}_z(\rho \circ \mathrm{root}(s))$. Then for every $\nu \in s$,

$$c_z(s) = \sum_{\mu \in s \setminus \{\nu\}} \mathbf{1}_{\tau(\mu) \neq \tau(\nu)}\, K_{a(\nu), a(\mu)}(z_{\tau(\nu)} - z_{\tau(\mu)})\, c_z(s \setminus \{\nu, \mu\}),$$

and summing over $\nu \in s$ gives the analogue of Lemma VII.1.f. *Proof sketch.* Removing one on-shell variable $\rho(x_{a(\nu)})$ at the point $z_{\tau(\nu)}$ from a Wick moment amounts to applying, at every other point, the derivation that replaces one variable by its pairing with $a(\nu)$ (`take_line`). On a monomial the derivation produces the sum over its variables (`D_kill`). There are no same-point pairings. *Lean:* `cut_one`, `cut_cancel`.

Lemma VII.1.h (convergence of leg products). In the setting of VII.1.2, $E_{h_i}(s) \to c_z(s)$ for every finite set of legs. *Proof sketch.* The proof is by strong induction on $|s|$.

- For $s = \emptyset$ both sides equal 1 (probability measures).
- Otherwise divide the identities of Lemmas VII.1.f and VII.1.g by $|s|$. Use VII.1.3 for the coefficients and the induction hypothesis for the shorter products.

Lean: `Warp.earth`.

Lemma VII.1.i (monomials, polynomials, labels; VII.1.2).

- Every tuple of monomials involving only keys of rank at most d is $\mathrm{root}(s)$ for a suitable set of distinguishable occurrences, so Lemma VII.1.h applies.

- Multilinearity extends the limit to arbitrary polynomials involving only keys of rank at most d; no bound on their number of factors is needed.
- The representatives sec_d have engineering weight at most d, hence use only keys of rank at most d. Composing gives the label limit, with $\rho \circ \mathrm{sec}_d = \omega_d$.

Lean: `mono_single`, `mono_limit`, `poly_limit`, `row_limit`.

3.26.4.3 4.3 Tested sums

Lemma VII.1.j (Riemann sums with continuous convergence). Hypotheses:

- centred torus grids of positive integer side N_m and spacing $t_m = 1/n_m$, $n_m \in \mathbb{N}_{>0}$, with $t_m \to 0$ and $t_m N_m \to \infty$;
- kernels $\mathcal{K}_m$ on r-tuples of grid points;
- f continuous with compact support in an open set ω;
- along every cofinal reindexing $\sigma : \mathbb{N} \to \mathbb{N}$, and every sequence of tuples on the reindexed grids whose embeddings converge to $z \in \omega$, the corresponding kernels converge to $\mathcal{G}(z)$.

Writing $\jmath_m(o) = t_m[o - c_{N_m}]_{N_m}$ for the embedding, one has $t_m^{4r} \sum_o f(\jmath_m(o))\mathcal{K}_m(o) \to \int f\mathcal{G}$. *Proof sketch.*

- The kernels are eventually bounded uniformly on $\operatorname{supp} f$ (`bounded_fee`). Otherwise there are points of $\operatorname{supp} f$ along which they blow up, and a convergent subsequence of these points contradicts the hypothesis.
- Write the sum as the integral of a step function supported in a fixed ball. Apply dominated convergence; the box eventually contains the support.

Lean: `Rig.grid_limit`, with `Arrives`.

Lemma VII.1.k (VII.1.1). Apply Lemma VII.1.j with $\omega = \mathrm{Conf}_r$, $\mathcal{K}_m$ the order-zero correlations and $\mathcal{G} = \mathcal{W}(\omega_d p)$, along the regulator sequence $h_m = L^{-m}$, $N_m = 4L^{m^2+m+1}$ (`routine`).

- The arrival property is VII.1.2, applied to each cofinal reindexing of the regulator sequence, which is again an itinerary with $4 \mid N$.
- The tested sum is the order-zero coefficient of the profile.
- The identity $\rho \circ \mathrm{sec}_d = \omega_d$ identifies the continuum moment with the free hierarchy.
- The board form applies this slot by slot to basis labels; clean profiles are smooth test functions with compact support in Conf (`Test`).

Lean: `Rig.zerothProfile`, `Position.finite_leading`.

Downstream use (not part of this unit). The assembly in VIII.1 uses the tested order-zero limit in two ways:

- `credible` subtracts it from `measured_tends` (which assumes the interacting short-distance comparison of VI.2). This shows that measured scaled means minus order-zero lattice readings tend to 0.
- `calibrate` solves the finite position equations when the leading Jacobian is invertible.

The position-comparison data are described in VII.3.

3.26.4.4 4.4 Detection on the free side

Lemma VII.1.l (Wick rule under linear substitution). Fix $x \in \mathrm{Conf}_r$ and a finite list of variable occurrences assigned to its insertions.

Replace each occurrence a by a finite complex linear combination u_a of shell coordinates, allowing different Taylor ranks. Extend K to these combinations by complex bilinearity. The Wick moment of the resulting products equals

$$\sum_{\text{matchings}} \prod_{\text{pairs}} K(u_a, u_{a'})(x_{i(a)} - x_{i(a')}).$$

Only matchings between distinct insertions are included. *Proof sketch.* Expand the linear forms in the coordinate basis. Multilinearity reduces the formula to the defining Wick sum on coordinate monomials. *Lean:* `Free.linear_substitution`.

Lemma VII.1.m (diagonal one-leg kernel). With the bilinear extension of K from Lemma VII.1.l, choose in each Sh_j a basis (Y_a) orthogonal for full component contraction, and set $\lambda_a = \gamma_j \langle Y_a, Y_a \rangle > 0$. For $Y \in \mathrm{Sh}_j$, let Y_i be the order-i Taylor coefficient of $Y^\natural$ at e_t and let $\langle Y_i, \cdot \rangle$ denote the full-contraction probe. Define

$$\epsilon_a = \sum_{i=0}^{j} \frac{\gamma_j}{\gamma_i} \langle (Y_a)_i, \cdot \rangle, \qquad \epsilon_a^\sigma = \epsilon_a \circ \sigma.$$

Then

$$K(\epsilon_a, \epsilon_b^\sigma)(e_\mathrm{t}) = \delta_{ab} \lambda_a.$$

Proof sketch. Expand the translated field $x \mapsto Y^\natural(e_\mathrm{t} + x)$ into its homogeneous Taylor pieces and apply the Kelvin formula to each piece. Since $e_\mathrm{t} + \iota(e_\mathrm{t}) = 0$, its inverted field has no Taylor coefficient at e_t below order j; its order-j coefficient is $(-1)^j$ times the time-reflected tensor Y. The weighted sum defining ϵ_a produces exactly γ_j times this inverted translated field. The second-endpoint sign cancels $(-1)^j$ on the diagonal. Thus pairings with lower ranks vanish and equal-rank pairings are γ_j times full contraction. Symmetry gives vanishing for the opposite rank ordering, and the chosen orthogonal bases give the displayed matrix.

Lean: `Free.pairing_var`, with `center_lower` and `center_diag`.

Lemma VII.1.n (orthogonality and positivity; VII.1.4). For finitely supported exponent vectors α, β in the probe variables,

$$\mathcal{W}_{(e_\mathrm{t},0)}(\Phi x^\alpha, \Psi x^\beta) = \delta_{\alpha\beta} s_\alpha, \qquad s_\alpha = \prod_a \alpha_a! \lambda_a^{\alpha_a} > 0.$$

The empty monomial has $s_0 = 1$. *Proof sketch.*

- By Lemmas VII.1.l and VII.1.m only matchings that pair each variable at e_t with the same variable at 0 survive.
- For equal exponent vectors, there are $\prod_a \alpha_a!$ surviving matchings, each of amplitude $\prod_a \lambda_a^{\alpha_a}$.
- For $q \neq 0$, expanding $q = \sum c_\alpha x^\alpha$ gives $\mathrm{Re} = \sum_\alpha |c_\alpha|^2 s_\alpha > 0$.
- For a label v:
  - Φ is onto (`forward_full`), so $\rho v = \Phi q$ for some q;
  - $\rho(\sigma \bar{v}) = \Psi \bar{q}$ (`reversed_observable`);
  - $q \neq 0$ because $v \notin \mathcal{J}_d = V_d \cap \ker \rho$.

Lean: `pair_test`, `wick_orth`, `positive`, `two_detect`.

Lemma VII.1.o (Kelvin formulas; VII.1.5). *Proof sketch for (a).* The case $j = 0$ is the definition of Γ. For $j \geq 1$, use

$$\tfrac{1}{j!} A(\partial) = \tfrac{1}{j} \sum_i \partial_i \tfrac{1}{(j-1)!} A(e_i, \cdot)(\partial).$$

Each contraction $A(e_i,\cdot)$ is again symmetric with harmonic diagonal. Then use $\partial_i|x|^{-2j} = -2j\,x_i|x|^{-2(j+1)}$ and the vanishing of $\sum_i \partial_i A(e_i,x,\dots,x)$ (harmonicity). The contraction identity $\sum_i x_i A(e_i,x,\dots) = A(x,\dots,x)$ closes the step. *Proof sketch for (b).* Work colour by colour. Each component $Y^\natural_{ab}$ is harmonic (closed and coclosed), so (a) applies to every term of the curvature kernel. The terms are recombined using the divergence equation and $\partial_k(x\lrcorner Y^\natural)_m - \partial_m(x\lrcorner Y^\natural)_k = (j+2)Y^\natural_{km}$ (Bianchi plus Euler). *Lean:* `harmonic_Green`, `response_eq`.

Lemma VII.1.p (finite detection). Let β be a basis of F_d. The map $v \mapsto \big(\mathcal{W}_{(e_t,0)}(\omega_d v, \omega_d \beta_i)\big)_i$ is injective on F_d. A label whose two-point functions with all labels vanish at the detector pair is null in the free table. *Proof sketch.* If all basis probes vanish, linearity gives vanishing against every second label. For a representative of v, take its time-reflected conjugate as the second label and apply VII.1.4. Conversely, $\rho v = 0$ makes every free correlation containing that label vanish. Thus the free null space in V_d is $\mathcal{J}_d$. *Lean:* `free_criterion`, `free_null`, `free_injective`. In the same spirit, `OPE.Levy.free_positive` shows that the reflected free two-point form at the detector pair is positive on $F_d \setminus 0$.

For bounded complex linear maps $P : E \to E$, $X : V \to E$, $A : E \to Z$ and $H : V \to Z$ between complex normed spaces, the perturbation lemma `test_projected` is not used by the main theorem. It states that a lower bound $a\|x\| \le \|Hx\|$ for every $x \in V$ and $\|H - AX\| + \|(A - AP)X\| \le \delta$ imply $(a-\delta)\|x\| \le \|APXx\|$.

3.26.4.5 4.5 Corrected charts and bookkeeping **Lemma VII.1.q (two-point convergence at moving detectors).** Under the hypotheses of VII.1.6, let $\delta_k = r_k y_k$ be the portfolio detector pair, where $r_k > 0$, $r_k \to 0$, and $y_k \to (e_t, 0)$. For fixed reference labels v_0, v_1 of respective weights at most $d_0, d_1 \le D$, put $\hat{v}_{k,j} = \pi^S_{d_j} \Phi_{k,d_j} v_j$. Then

$$S_{d_0,d_1}(\delta_k; \hat{v}_{k,0}, \hat{v}_{k,1}) \longrightarrow \mathcal{W}_{(e_t,0)}(\rho v_0, \rho v_1),$$

where ρv_j means the on-shell image of the label polynomial represented by v_j. *Proof sketch.* The finite-regulator estimate `Read.full_cost` converts bounds on all nonempty marker components into a moment bound. The portfolio supplies these component bounds, and passage to the common extraction subfilter gives `actualRead`. The resulting comparison `compareC` bounds the difference between the interacting moment at δ_k and the free moment at y_k by $C\varepsilon_{g(k)}$, with $\varepsilon_{g(k)} \to 0$ by III.11. Continuity of the free kernel at the detector pair gives the limit. *Lean:* `Portfolio.twoC`. *(proof read through the rank-comparison input)*

Lemma VII.1.r (raw charts are eventually bijective and nearly compatible). Under the hypotheses of VII.1.6:

1. The read-out maps $D_{k,d}$ are eventually bijective for every $d \le D$, so $\chi_k(d) = D_{k,d}$. The real part of the reflected measured form converges in operator norm to a free form bounded below by $\delta\|v\|^2$ for some $\delta > 0$. It is therefore bounded below by $(\delta/2)\|v\|^2$ eventually. This proves injectivity; the comparator isomorphism in the construction record gives equality of dimensions and hence surjectivity (`freely`, `certified`, `chartDraw`).
2. For every $d < D$ and $v \in F_d$, $\chi_k(d+1)^{-1}\iota^S\chi_k(d)v \to \iota^F v$ (`stepFits`). Both sides are probed by two-point functions against $\chi_k(d+1)\beta_i$, where (β_i) is a basis of F_{d+1}; Lemma VII.1.q gives convergence of the probes, and injectivity of the free probe (Lemma VII.1.p), through the finite-dimensional fitting lemma of §2, gives convergence of the arguments.

3. Eventually each $\chi_k(d)$ commutes with conjugation and signed permutations (`symmetrical`), and it preserves the unit.

Lemma VII.1.s (VII.1.6). Under the hypotheses of VII.1.6:

1. Take a retraction $P_d : F_{d+1} \to F_d$ of the free inclusion that is equivariant and commutes with conjugation (averaging over W_4 and symmetrizing with the conjugation).
2. For fixed k, define recursively $A_0 = \chi_k(0)$ and
$$A_{d+1} = \iota^S A_d P_d + \chi_k(d+1)(1 - \iota^F P_d).$$
3. For each $d \le D$, the operators $\chi_k(d)^{-1} A_d$ tend to the identity by induction using Lemma VII.1.r(2). They are eventually invertible. Set $p_k(d) = A_d$ when it is bijective and $p_k(d) = \chi_k(d)$ otherwise.
4. Exact compatibility follows from the construction together with VII.1.r(3), and $\chi_k^{-1} p_k \to 1$ from VII.1.r(2).

Lean: `tune_grades`.

Lemma VII.1.t (VII.1.7).

- (a) `slim_wait_for`. The required minimum window is $o(\log \widehat{\beta})$ at each fixed stage, while the prescribed window is $\lfloor \log \widehat{\beta} / \log L_j \rfloor$. Raise entry thresholds to fit the finitely many active estimates inside it. Repeat for the three panels and raise the thresholds further to ensure that the block ratio is below the monitor bound. The station estimates then give the simultaneous data in VII.1.7(a).
- (b) `nearby`. If the backward window at k stays in its stage, take $g(k) = k$. Otherwise take the last step of the preceding stage. Within one stage the logarithmic window grows by at most one per level. The threshold conditions ensure that this last step is usable; the number of intervening levels times $\log L_j$ is at most $\log \widehat{\beta}_k$. The selected stage tends to infinity, so $g(k) \to \infty$.
- (c) `coin_pre`. Put $z = k + E - q$ and first reduce the weight to $\sigma = v/(z+2)$ to pay for re-rooting. Apply `arose` from rank $\max(k - q, 0)$ through z. Inductively lift the lowest remaining coefficient to a curvature expression, subtract its Taylor expansion, and continue. At each of these finitely many steps, reduce the exponential weight and enlarge the constant to control the subtraction. The analytic realization hypothesis permits the lift at every step.

Parts (a) and (b) supply the common portfolio used in VII.1.6; the assembly is in III.13. Part (c) supplies local curvature expansions for the RG estimates.

(proof read through the stated station and curvature-lifting inputs)

3.26.5 5. Where the difficulty lies

The main analytic work is the control of finite differences of the torus Green function, uniformly as the spacing shrinks and the volume grows. The Fourier argument must retain oscillatory cancellation; an absolute bound on the full momentum integral would not give the required derivative limits.

Wick ordering uses the infinite-lattice covariance. Consequently the same-insertion contraction is not exactly zero at finite volume. For each fixed stencil offset, the available estimate is

$Cr^2 + A/(Nr^4)$, multiplied by h^{-n}. Taking $r = h^{n+1}$ gives $Ch^{n+2} + A/(Nh^{5n+4})$. The chosen period handles every fixed n at once. This is a sufficient growth condition; no optimal volume rate is asserted.

The reflection-odd field space removes constant zero modes but introduces an image term. Reporting around the quarter period sends that image to infinite physical distance. Detection requires a separate algebraic argument: the translated inversion formula and the triangular probe substitution establish strict positivity on every nonzero on-shell polynomial, including constants. The chart correction then uses finite-dimensional perturbation theory and the interacting estimates of III.11.

3.26.6 6. Notation

object	code name	proposed notation
reflection-odd link fields, Feynman-gauge form, Gaussian	`LinkP`, `form`, `data`	$\mathcal{A}_N$, Q_N, μ_N
torus Green function (mean zero)	`toroid`	g_N
linear lattice curvature jet	`slope`	ℓ_x^h
$\mathbb{Z}^4$ self-covariance of the jets (Wick subtraction)	`link`, `shellCov`	S_h
Wick-ordered representative	`column`	$R_h(p, o)$
corrected and limit two-leg covariances	`corrected`, `indirect`	κ_h, K_{ab}
itinerary; homing sites	`Itinerary`, `Home`	(N_i, h_i); $h_i(o_i - c_{N_i}) \to z$
regulator sequence, centre, profile	`topSpacing`, `period`, `center`, `profile`	h_m, N_m, c_m, $\Pi_m(f; p)$
shell, on-shell reduction, equation ideal	`Shell`, `reduceEq`, `equationIdeal`	Sh_j, ρ, $\mathcal{I}_d$
Green function, curvature jet kernel	`Green`, `legKernel`	Γ, $K(P, Q)$
Wick moment	`Wick.moment`	$\mathcal{W}_x$
detector pair, time reflection	`detectors`, `Kelvin.σ`	$(e_{\mathrm{t}}, 0)$, σ
inversion pullback, gain	`inverted`, `gain`	ι^*, γ_j
weak measuring portfolio, raw chart, corrected chart	`Portfolio`, `chart`, `tune_grades`	X, $\chi_k(d)$, $p_k(d)$

3.26.7 7. What remains unexplained

1. **Analytic inputs below the Fourier limits.** The symbol smoothness, higher-difference estimate, continuum annular decay, tail estimate and fixed-offset quadrature estimate support Lemmas VII.1.b–c. The remaining lower-level estimates for periodic annular sums, the low-momentum bound and the annular Riemann-sum comparison have not been checked line by line. Their uniformity in the period and spacing remains an input to the sketches. (probable)

2. **Imported interacting estimates.** The two-point limit follows from the simultaneous portfolio conditions and the rank-comparison error bound. The full analytic proofs supplying

those conditions, including the station estimates and curvature-lifting estimates used in VII.1.7, are outside this verification. (probable)

3. **Other assigned packages.** The remaining RG local-expansion packages and the perturbative-extension packages are not described here; no proof outline for their internal estimates is supplied. Their placement relative to III.13, III.15 and VII.6 remains to be settled. (tentative)

4. **Scope of the comparison.** The Gaussian μ_N is defined independently of the RG Gaussian measures. Proposition VII.1.1 requires no identification with them. The interacting short-distance theorem and its combination with the tested order-zero limit are the linked results in VIII.1; no all-orders perturbative comparison follows from this unit.

5. **References and size estimates.** The literature citations and attributed source-line counts have not been independently verified. (tentative)

3.27 VII.2. The trace of the stress tensor and the trace anomaly

Part: VII, identification as Yang–Mills. **Lean:** `YM/TraceExtension` (namespaces `YM.TraceExtension` and `YM.Trace`). The folder splits as follows:

- the centered trace and its spectral measure: the top-level files `Centered`, `WilsonTrace`, `LogScaling`, `ConeLaplace`, `Dyadic`, `Belfry`, `Agora`, `Endpoint`;
- the renormalization-group proof of the anomaly: `Chart`, `Slate`, `Shelves`, `Terrace`, `Margin`, `Loft`, `Lamp`, `Det`, `Perfect`, `Anchor`, `Cask`, `Shepherd`, with the auxiliary folders `Zn`, `Weir`, `Pills`, `Turret`, `Wicket`, `Fen`, `Swift`, `Filch`, `Quince`, `Pi`, `Xo`;
- the short-distance law: `Verse` and `Foundation`;
- the exclusions: `Wick` and `Fock`;
- a position-space variant: `Atlas`, `Pith`, `Sundial`, `Link`.

The unit also contains `FreeRanks.lean`, the specification `HeadlineSpec.lean`, `ManuscriptTraceTests.lean` and small pieces of `YM/Integration` and `YM/RG08/Calendar`.

Size: 20,681 lines. The argument headed by `Store_l` and the argument headed by `agenda` are the two main RG arguments.

Conventions. G is a compact matrix group with simple nonabelian Lie algebra $\mathfrak{g}$; $\dim \mathfrak{g}$ is the real dimension. C_A is the adjoint Casimir of I.1, defined by $-\mathrm{tr}(\mathrm{ad}_x \mathrm{ad}_y) = C_A\langle x, y\rangle$. The constants are

$$\kappa_G = \frac{11\,C_A}{24\pi^2}, \qquad b_0 = \frac{\kappa_G}{2} = \frac{11\,C_A}{48\pi^2}, \qquad d_2 = -\frac{\kappa_G}{4} = -\frac{b_0}{2}, \qquad A_G = 12\Big(\frac{d_2}{b_0}\Big)^2 \frac{\dim \mathfrak{g}}{\pi^4} = \frac{3\dim \mathfrak{g}}{\pi^4}.$$

κ_G is the slope of the coupling schedule $\hat{\beta}_k = \bar{\beta} + \kappa_G \log(1/a_k)$ of the renormalization group (RG) at level k (II.2, III.9). Write $g(r)^2 = \beta(r)^{-1}$ for the running coupling in free-field normalization.

- Unless other hypotheses are stated, fix the reconstructed theory supplied by a final record F (VIII.1), over a Wilson family C, an extraction order f and a limit table S.
- F_d is the space of degree-d field labels; $\langle\cdot\rangle_{\xi_*}$ is the one-point function of the selected phase.
- $T = (T_{\mu\nu})$ is the Ward-normalized stress tensor of VI.1.
- F is the free Maxwell table (the two constructions listed in §6 coincide by definition): $\dim \mathfrak{g}$ independent free Maxwell fields, with propagator $\langle AA\rangle = 1/(4\pi^2|x|^2)$.

- $\mathcal{O} \in \mathsf{F}_4$ is the Wick-ordered label $\frac{1}{4}\sum_{a,\mu,\nu} F^a_{\mu\nu}F^a_{\mu\nu}$, and T^{free} is the traceless free Maxwell stress tensor.
- $G_\Theta(r) = S^{\xi_*}(\Theta(-\frac{r}{2}e_0)\,\Theta(\frac{r}{2}e_0))$ is the time-axis two-point function of the selected phase.

3.27.1 1. What this unit proves

The unit defines the centered trace Θ and proves four results about it:

- the trace anomaly at the level of fields: at scale r, Θ equals $-\kappa_G\beta(r)^{-1}\mathcal{O} + o(\beta(r)^{-1})$, where $\beta(r) \sim \kappa_G \log(1/r)$ is the running inverse coupling;
- the resulting short-distance law $r^8 \log^2(1/r)\, G_\Theta(r) \to 3\dim\mathfrak{g}/\pi^4$, and the corresponding law for the spectral measure;
- the uniqueness (rank one) of the conserved stress tensor;
- two exclusions of free-field descriptions.

Most of the lines prove the field-level anomaly, Proposition VII.2.4. That proof is a renormalization-group argument: a lattice Ward identity is transported through the block-spin steps, and a one-loop computation fixes the coefficient.

Proposition VII.2.1 (the centered trace). Let

$$\Theta := \sum_\mu T_{\mu\mu} - \Big\langle \sum_\mu T_{\mu\mu} \Big\rangle_{\xi_*} \mathbf{1} \in F_4.$$

(a) $\langle\Theta\rangle_{\xi_*} = 0$. Θ is real, invariant under every $K \in O(4)$ for the rotation action of VI.1, and fixed by the time-reflection adjoint. The same holds for the trace centered in any phase table respecting S.

(b) In the reconstructed Wightman theory of the selected phase, the pinned field label $\iota(\Theta)$ is fixed by the conjugation J and by the Lorentz group. Moreover $\langle\Omega, \Phi_\Theta(f)\Omega\rangle = 0$ for every Schwartz f.

(c) For every Schwartz test f, as an identity on the common invariant field domain, $\Phi_\Theta(f) = \Phi_{\sum_\mu T_{\mu\mu}}(f) - \langle\sum_\mu T_{\mu\mu}\rangle_{\xi_*}(\int f)\,\mathrm{id}$.

(d) If $r^8\log^2(1/r)\,G_\Theta(r) \to A > 0$, then $\iota(\Theta) \neq 0$.

Status: theorem. It uses the conservation, reality and tensoriality of T and the rotation laws (VI.1), and the reconstruction (V.2–V.3). *Lean:*

- definitions: `stressTrace`, `centerIn`, `theta`, `center`;
- (a): `centerIn_vac`, `trace_real`, `trace_turn`, `trace_adj`;
- (b): `self_adjoint`, `pinned_scalar` with `scalar_rail`, `pinned_centered` with `vev_boundary`;
- (c): `center_field`;
- (d): `nonzero_strength`;
- public form: the zero-expectation and field-identification clauses of `TraceExclusions`, i.e. (T1).

Proposition VII.2.2 (spectral representation). There is a unique positive measure μ_Θ on $\mathbb{R}^4$ with the following properties.

- It has tempered growth: there are $C_\mu \geq 0$ and $k \in \mathbb{N}_0$ such that $\mu_\Theta\{p : p_0 \leq R\} \leq C_\mu R^k$ for every $R \geq 1$.
- It is supported in the closed forward cone $\bar{V}_+$ and is Lorentz invariant.
- $e^{-sp_0}\mu_\Theta$ is finite for $s > 0$, and $G_\Theta(s) = \int e^{-sp_0}\, d\mu_\Theta(p)$ for $s > 0$.
- For every Schwartz φ, the spectral measure of the translation group in $\Phi_\Theta(\varphi)\Omega$ is $|\widehat{\varphi}|^2\mu_\Theta$.
- For every Schwartz test ϕ on $(\mathbb{R}^4)^2$, the Wightman function $W_2(\Theta,\Theta)(\phi)$ equals $\int \widehat{\phi}^{\,\mathrm{wave}}\, d\mu_\Theta$.

In particular G_Θ is real, nonnegative and nonincreasing on $(0,\infty)$.

Status: theorem: the Källén–Lehmann representation, obtained from the spectral theorem for the translations of the reconstruction (V.2–V.3). *Lean:* `TraceSpectrum`, `traceSpectrum_exists`, `monotone_axis`.

Proposition VII.2.3 (one-loop certificates along every filter). Let G be simple and let C be the Wilson family issued by run data satisfying the RG construction hypotheses of §2, item 4, including the chosen bounded block-spin run. Let $\mathcal{L}$ be any filter on $\mathbb{N} \times \mathbb{N}$ along which both coordinates tend to infinity. Then there are work items $t = (t_{k,N,p})$, that is, level-5 lattice data at RG level k for the pair (N,p) of spacing and volume indices, with the following property. For all large k and $\mathcal{L}$-eventually in (N,p), the lattice certificate of §6 holds at (k,N,p). In particular the Ward dart $J = t_{k,N,p}.J$ (Lemma VII.2.d), a lattice stress tensor with its Ward identity at level k, satisfies for every color index q

$$\left|\mathrm{midPay}_q(J) - (-12\kappa_G)\right| \leq \widehat{\beta}_k^{-100}.$$

Here midPay_q is the real coefficient selector applied to the Haar divergence of the displacement field rooted at the origin; its exact definition is `midPay`.

Status: theorem. It is used by III.2 to choose the run and as a hypothesis of VII.2.4. *Lean:* `agenda`, `Grant`, `WorthB`.

Proposition VII.2.4 (the gallery property: the anomaly at the level of fields). Assume the following:

- C satisfies the Wilson consistency conditions of II.1, and G is simple;
- the quantitative RG and scaling-sequence hypotheses of §2, items 4 and 9, hold; the auxiliary iteration depths n_k are eventually at least 2, and a sequence z_d bounds every generator budget of degree at most d and satisfies the degree-d scaling estimates;
- f is an extraction order and $\mathcal{L}$ is an ultrafilter on $\mathbb{N}^2$ finer than the fuse filter of f;
- the work items t of VII.2.3 satisfy the certificate along $\mathcal{L}$.

Then the following property holds. For every limit table S along f with ordered Wilson convergence, a Euclidean hierarchy, point limits, the cell hypothesis and corrected rank estimates, and every field algebra on S satisfying the derivative, rotation, conservation and Ward laws, there exist:

- limits Z of the RG data along $\mathcal{L}$: $Z.\beta(k) = \lim_{\mathcal{L}} \beta_{N,p}(k)$, the renormalized inverse coupling at level k, with $\widehat{\beta}_k/2 \leq Z.\beta(k) \leq 2\widehat{\beta}_k$, together with the limits of the label transfers;
- a free Maxwell hierarchy a on F,

such that the following two statements hold.

(i) Let (r_i, U_i) be a scaling sequence of S that approaches a, with the RG lifts and pointwise lattice limits of VII.6. Write $k_i \to \infty$ for its base levels and ℓ_i for the later readout levels

selected by these data. Here $r_i \downarrow 0$, the $U_i : \mathsf{F}_d \to F_d$ are complex-linear identifications that eventually preserve the unit, reality and signed coordinate actions in each degree, and their rescaled correlations approach the free hierarchy. Set $w_i := Z.\beta(\ell_i)$. Then:

- $w_i \to \infty$ and $\log(1/r_i)/w_i \to \kappa_G^{-1}$;
- $r_i^4\, U_i^{-1} \mathring{T}_{\mu\nu} \to T^{\mathrm{free}}_{\mu\nu}$, where $\mathring{T}$ is T centered in S;
- for every phase table T' with a hierarchy respecting S, there are real sequences d_i, e_i such that the trace $\Theta_{T'}$ centered in T' satisfies, eventually,

$$r_i^4\, U_i^{-1} \Theta_{T'} = e_i\, \mathbf{1} + 4 d_i\, \mathcal{O}, \qquad w_i d_i \to d_2, \quad w_i e_i \to 0,$$

hence $w_i\, r_i^4 U_i^{-1} \Theta_{T'} \to 4 d_2\, \mathcal{O} = -\kappa_G\, \mathcal{O}$.

(ii) Every sequence $s_j > 0$ with $s_j \to 0$ has a subsequence that is the scale sequence of a scaling sequence satisfying (i).

Thus the rescaled trace has leading term $-\kappa_G g(r)^2 \mathcal{O}$. In canonically normalized field-strength notation, this is the one-loop term $\frac{\beta_{\mathrm{RG}}(g)}{2g} F^a_{\mu\nu} F^a_{\mu\nu}$, with $\beta_{\mathrm{RG}}(g) = -b_0 g^3 + o(g^3)$. The precise field statement is the sequence limit above.

Status: theorem. Its hypotheses are data of the construction run. They are discharged in III.2: in `plentifulWithPrediction`, for each f one takes an ultrafilter $\mathcal{L}$ finer than the fuse filter of f, the work items of VII.2.3, and the gallery assembly. The result is recorded as `Funded.gallery` (VIII.1). *Lean:* `gang`, `Gallery`, `Scribe`, `Store_l`.

Proposition VII.2.5 (short-distance law and spectral law). Let G be simple, and suppose the gallery property holds for (C, f). Then:

(a) $\lim_{r \downarrow 0} r^8 \log^2(1/r)\, G_\Theta(r) = A_G = \dfrac{3 \dim \mathfrak{g}}{\pi^4} > 0$, as a limit in $\mathbb{C}$.

(b) As $\rho \to \infty$,

$$\rho^{-8} \log^2 \rho \int \varphi(p/\rho)\, d\mu_\Theta(p) \to \frac{A_G}{1536\pi} \int_{V_+} (p^2)^2\, \varphi(p)\, d^4p,$$

for continuous compactly supported φ and for Schwartz φ. Here $p^2 = p_0^2 - |\vec{p}|^2$; the measure on the right has Laplace transform $A_G s^{-8}$. More explicitly, put $\phi_\rho(x,y) = \rho^8 \phi(\rho x, \rho y)$ and $\hat{\phi}^{\,\mathrm{wave}}(p) = \int e^{ip\cdot(y-x)} \phi(x,y)\, d^4x\, d^4y$, in the Fourier convention of VII.2.2. Then $\rho^{-8} \log^2 \rho\, W_2(\Theta,\Theta)(\phi_\rho)$ tends to the same cone integral with $\varphi = \hat{\phi}^{\,\mathrm{wave}}$.

Status: theorem under the gallery property; (b) uses only (a) and VII.2.2. The final short-distance and spectral conclusions are filed in VIII.1. *Lean:*

- (a): `outside`, `EuclideanAsymptotic`, `award`, `award_pos`;
- (b): `compact_limit`, `spectral_limit`, `relative_limit`, `joint_limit`;
- bundles: `Crest` and `crest_ready`, `HighEnergy`;
- public form: the amplitude identity and `publicTrace`, i.e. (T2), (T3).

Proposition VII.2.6 (the conserved stress tensor is unique). Let $\mathcal{K}(S)$ be the real space of symmetric tensors $(B_{\mu\nu})$ in F_4 that are real, centered with respect to S, and covariant under signed coordinate permutations. For a derivative plan satisfying the derivative laws and the orientation condition, let $\mathrm{div} : \mathcal{K}(S) \to \mathcal{V}(S)$ be the current divergence, where $\mathcal{V}(S)$ consists of real degree-5 vectors covariant under signed coordinate permutations.

(a) (free) For G simple, the free Maxwell hierarchy on F and a derivative plan satisfying the derivative laws and the orientation condition: div is onto and $\dim\ker \mathrm{div} = 1$.

(b) (interacting) In the setting of VII.2.4(i), $\dim\ker \mathrm{div} = 1$ on $\mathcal{K}(S)$; moreover $\dim \mathcal{K}(S) = 3$ and the target of div is two-dimensional.

Status: theorem. (b) is the uniqueness statement of the assessment, §7.1. It is used in VII.2.4 to identify T with the tensor constructed from the RG. *Lean:* `free_rank` (from `free_closure`), `ranking`, the rank clause of `ChoiceK`.

Proposition VII.2.7 (no finite Wick profile). Suppose $r^8 \log^2(1/r)\, G_\Theta(r) \to A$ with $A > 0$.

(a) Let $W(r) = \sum_\alpha c_\alpha \prod_\beta K_{m_{\alpha\beta},q_{\alpha\beta},h_{\alpha\beta}}(r)$ be any finite combination, with $c_\alpha \in \mathbb{C}$, where

$$K_{m,q,h}(r) = \int_{\mathbb{R}^3} \frac{E_m(p)^q\, h(p)}{2E_m(p)}\, e^{-rE_m(p)}\, dp, \qquad E_m = \sqrt{m^2 + |p|^2},$$

with $m \geq 0$, $q \in \mathbb{N}_0$, and $h : \mathbb{R}^3 \to \mathbb{R}$ continuous and positively homogeneous of degree $j \in \mathbb{N}_0$. Then for every $\delta > 0$ there is $r \in (0, \delta)$ with $G_\Theta(r) \neq W(r)$.

(b) No such collection reproduces the Wightman two-point function of Θ through its mass-shell expression on all product tests $\varphi \otimes \psi$ whose wave transform has compact support. The same holds for the Wick contraction of two finite polynomial fields from any palette of free-field modes.

Status: theorem. (a) uses only the law; (b) also uses VII.2.2. The internal class is the public class of Definition 1.15 field by field (`profile_eq`). With VII.2.5 this gives (T4). The proof is also described in VII.3. *Lean:* `euclidean_excludes`, `polynomial_excludes`, `wick_excludes`, `Mode`, `Track`.

Proposition VII.2.8 (no relatively local realization in free-photon Fock space). Suppose the law of VII.2.7 holds with $A > 0$. For every $m \in \mathbb{N}_0$, let $\mathcal{F}_m$ be the Fock space of m species of free photons, realized by symmetric transverse ("physical") wave functions of on-shell momenta. Let $U : \mathcal{H} \to \mathcal{F}_m$ be a complex-linear isometry from the reconstructed space of the selected phase with:

- $U\Omega = \Omega_{\mathcal{F}}$;
- image in physical waves;
- U intertwines the translations with multiplication by $e^{ib\cdot p_{\mathrm{tot}}}$.

Then there are a compactly supported smooth scalar test f and compactly supported smooth tests $h_c : \mathbb{R}^4 \to \mathbb{R}^3$, $1 \leq c \leq m$, each causally separated from f, such that

$$\langle W(-h)\Omega_{\mathcal{F}},\, U\Phi_\Theta(f)\Omega\rangle \neq \langle U\Phi_\Theta(\bar{f})\Omega,\, W(h)\Omega_{\mathcal{F}}\rangle,$$

where $W(h)\Omega_{\mathcal{F}}$ is the coherent state defined in §6. Equality for all such pairs would hold if $\Phi_\Theta(f)$ commuted with the free Weyl operators localized spacelike to f.

The same conclusion holds in two variants:

- with any bounded self-adjoint operator fixing $U\mathcal{H}$ applied to both coherent vectors;
- with an average over a probability family of maps that preserve inner products of physical waves and fix $U\mathcal{H}$; for every $r > 0$, this variant has tests supported in open sets Q, W with $\bar{Q} \subset B(0, r)$ and $\bar{Q} - \bar{W}$ spacelike, with f supported in Q and every component of h in W.

Status: theorem. These are the free-photon and averaged free-photon clauses of the trace-exclusion specification; the final conclusions are filed in VIII.1. *Lean:* `Comparison`, `Nfl`, `excludes`, `saturate`, `excludes_projected`, `smell`, `regions_excludes`.

Proposition VII.2.9 (position-space coefficient; supplementary). For every simple G, every block-spin kernel parameter $b > 0$ and every prescribed extraction order p, there are a consistent Wilson family C with that kernel parameter and a refinement f of p for which final records exist and the following holds for every final record F over (C, f).

There are a free hierarchy, coupling limits Z and an ultrafilter such that admissible degree-4 position-space frames exist. For every such pair of frames $P(s), P'(s) : \mathsf{F}_4 \to F_4$, with a window cutoff χ, there are real functions κ, d, e such that, for all sufficiently small $s > 0$,

$$P'(s)^{-1}(s^4 T_{\mu\nu}) = \kappa(s) T^{\mathrm{free}}_{\mu\nu} + d(s)\delta_{\mu\nu}\mathcal{O} + e(s)\delta_{\mu\nu}\mathbf{1}, \qquad \kappa(s) \to 1.$$

Admissibility means that P has the position-space Gram normalization, branch and angular alignment, and convergence of all prescribed finite off-diagonal jets; P' is obtained from P by positive rescaling of the centered scalar sector and its centered complement, fixing the unit, with the two shell tests normalized to their free values. These conditions are specified by the frame predicate linked below.

For every coupling function g that is eventually positive and satisfies $\log(1/s)g(s)^2 \to \kappa_G^{-1}$,

$$\frac{d(s)}{g(s)^2} \to d_2.$$

The coupling g_X defined by the position-space tests satisfies $g_X/g_W \to 1$, where g_W is the Wilson coupling from Z, and obeys the stronger estimate

$$d(s) - d_2 g_X(s)^2 = O(g_X(s)^4).$$

Here the free scalar is exactly $\mathcal{O}$. Centering gives the exact identity $s^4\Theta = 4d(s)\,\mathrm{center}_{\xi_*}(P'(s)\mathcal{O})$ for sufficiently small $s > 0$.

Proof sketch. Subsequence compactness gives position-space frames at every sufficiently small scale. The two positive sector rescalings tend to 1 and impose the shell normalizations. Put $B_0 \neq 0$ for the common normalized shell value and $Q(s)$ for the remaining shell readout, which has a finite limit. Conservation and the Ward identity give $d = d_2 g_X^2 \kappa$ and $\kappa - 1 = -dQ/B_0$. Since $d/g_X^2 \to d_2$, these identities imply $\kappa - 1 = O(g_X^2)$ and then $d - d_2 g_X^2 = O(g_X^4)$. The Wilson comparison is established before the definition of g_X by computing the same divergence diagnostic in Wilson normalization.

Status: theorem for the family and refinement supplied by the supplementary construction. *Lean:* `meterQ` *(proof read; frame-existence inputs statement only)*; `main`, `XScheme`, `ShellNorm`; the stress clause of `Result`, `FieldComparison`, `odd_limit`, `Measures`.

3.27.2 2. What it uses

Inputs from other units.

1. *Gauge data* (I.1): the Casimir $C_A > 0$ and $\kappa_G > 0$ (`slopeConstant`); simplicity; a nontrivial color space.

2. *Coupling schedule* (II.2, III.9): $\hat{\beta}_k = \bar{\beta} + \kappa_G \log(1/a_k)$, with $\hat{\beta}_k \to \infty$ (`hrate`). The actual couplings follow the schedule: $Z.\beta(k)/\hat{\beta}_k \to 1$ (`gamut`, `Barry`).

3. *The one-loop integral* (III.7): with infrared mass $\mu \in (0, 1]$, the one-loop momentum integral has coefficient $-\kappa_G \log(1/\mu)$ up to a bounded error (`absoluteGeode`).

4. *The RG step* (III.4–III.15):

   - the stationary background, criticality and fourth-order classical matching of actions (III.6–III.7): `encamp`;
   - cluster and polymer bounds (III.5, III.8) and determinant budgets (III.4, III.11);
   - the bounded block-spin run, eventual level estimates, classical and determinant bounds, support conditions, and small-field and large-field estimates. The exact run predicates are listed in §6.

   These hypotheses are discharged when the run is chosen in III.2.

5. *Infinite volume* (IV.1): the extraction order and its fuse filter, point limits, `CellHyp`, and the chessboard root tests on centered chambers.

6. *Phases and reconstruction* (V.1–V.3): the selected phase; the analytic continuation and its boundary values; the spectral measure of the translations; the Lorentz action and the conjugation J. All of this is packaged in `Final`.

7. *Field algebra and stress tensor* (VI.1): the derivative plan, the rotation laws (including that rotations and boosts generate the Lorentz group), and the stress tensor with its conservation, tensoriality and Ward identities (`Locals`).

8. *Free Maxwell model* (VI.2, VII.1, VII.3):

   - the free table F and its Wick calculus;
   - the free stress tensors (`materialB`, `Litpay`);
   - the normalization $\langle \mathcal{O}(x)\mathcal{O}(y)\rangle = 3 \dim \mathfrak{g}/(\pi^4 |x-y|^8)$ (`wick_norm`, proved by an explicit curl computation);
   - convergence of rescaled two-point functions at lattice-rounded points (`rainbow`).

9. *Scaling sequences* (VII.6): scaling sequences approaching the free hierarchy (`StarPath`, `Gondola`) exist along a subsequence of any positive null sequence (`pays_sky`, `gravity`).

Standard mathematics.

- The trace-anomaly mechanism (Adler–Collins–Duncan, Phys. Rev. D 15 (1977); Collins–Duncan–Joglekar, Phys. Rev. D 16 (1977); Nielsen, Nucl. Phys. B 120 (1977)) and the background-field one-loop beta function ('t Hooft 1973; Abbott, Nucl. Phys. B 185 (1981)).
- The Källén–Lehmann representation and the spectral theorem for the translation group (Reed–Simon II, §IX.8; Glimm–Jaffe, ch. 19).
- The identity theorem for holomorphic functions in tube domains.
- Laplace transforms of positive measures, continuity theorems and Tauberian monotonicity (Feller II, §XIII.1–5; Widder, *The Laplace Transform*).
- Fock space, Weyl operators and coherent states (Reed–Simon II, §X.7; Bratteli–Robinson II, §5.2).
- Simultaneous eigenvector decomposition for commuting operators annihilated by polynomials with distinct real roots.

- Lattice difference operators and finite-difference stencils of order 8.

Imported constants and assumptions. κ_G, b_0 and d_2 are defined from C_A (`b0`), so $\kappa_G = 2b_0$ and $d_2 = -b_0/2$ are identities by definition (`beta_eq`, `force_b`). The value κ_G enters the proofs in two independent places:

- as the slope of the coupling schedule (input 2);
- as the logarithmic coefficient of the one-loop integral (input 3).

The amplitude $3\dim\mathfrak{g}/\pi^4$ depends on these two being equal (§5). The gallery property is a hypothesis of VII.2.5; it is discharged for the constructed family in III.2, so the trace data must be carried by the family chosen in the ultraviolet construction.

3.27.3 3. Strategy

Physical picture and the approach taken. Classically the Yang–Mills stress tensor is traceless. The running coupling produces a nonzero trace after renormalization. It characterizes T by two properties: uniqueness (the conserved admissible tensors form a line, VII.2.6) and the Ward normalization. At each scale r_i it builds three lattice tensors $T_i^{(B)} = \mathrm{tub}_i(B)$ from the reference tensors $B = L, \Delta, \Xi$: the free Maxwell stress L, the pure-trace tensor $\Delta_{\mu\nu} = \delta_{\mu\nu}\mathcal{O}$, and a third traceless tensor Ξ.

The candidate is $X_i = w_i T_i^{(L)} + c_{i,0} T_i^{(\Xi)} + c_{i,1} T_i^{(\Delta)}$, where $w_i \sim \beta(r_i)$. The suitably rescaled divergence of the canonical part $w_i T_i^{(L)}$, tested by ρ, tends to $-\kappa_G$. Conservation forces $c_{i,1} \to -\kappa_G/4 = d_2$. Uniqueness makes X_i a multiple of the centered stress tensor; the Ward normalization makes that multiple tend to 1. The weighted trace therefore satisfies $w_i \operatorname{tr}(r_i^4 U_i^{-1} X_i) \to 4d_2\mathcal{O}$. Combining this with $w_i \sim \kappa_G \log(1/r_i)$ gives the leading rescaled trace $-\mathcal{O}/\log(1/r_i)$.

The one-loop input. The value $-\kappa_G$ comes from a lattice Ward identity at a single RG level, a "dart". A dart is a lattice stress tensor whose divergence equals the first variation of the action along a displacement field, through fourth order in the field. Darts at a given level are obtained from finer levels by block-spin transport. The transport preserves the Ward identity, because the first variations of the fine and coarse effective actions agree to fourth order; errors contract by $\frac{1}{2}$ per step. The level-5 "payoff" of the dart is then compared with the infrared-regulated one-loop effective action,

$$\tfrac{1}{2}\log\det(\text{gauge Hessian} + \mu^2) - \log\det(\text{ghost operator} + \mu^2).$$

Its normalized quadratic momentum coefficient has the logarithmic term $-\kappa_G \log(1/\mu)$ per plane, 12 ordered planes in total. A mean-value argument in $\log\mu$ turns this pole into the limit $-12\kappa_G$ for the payoff. This is Balaban's renormalization-group framework (III) together with a background-field one-loop computation; what is specific here is that the anomaly coefficient is extracted from the Ward identity carried along the RG flow.

From fields to the law, and the exclusions. By free-field convergence (VII.1), the two-point function of $w_i r_i^4 U_i^{-1}\Theta$ converges to $(4d_2)^2 \cdot 3\dim\mathfrak{g}/(\pi^4|x|^8)$, and $\log(1/r_i)/w_i \to 1/\kappa_G$. Rotation and translation invariance move the points to the time axis. Monotonicity of G_Θ in the separation removes the lattice rounding of points. Finally, every positive null sequence has a good subsequence, which upgrades the subsequential limit to $r \downarrow 0$. A Tauberian argument transfers the law to the spectral measure. The exclusions use only the law and μ_Θ:

- every finite Wick profile expands in integer powers and nonnegative powers of logarithms, which are incompatible with $r^{-8}\log^{-2}(1/r)$;
- a relatively local Fock realization forces each n-photon amplitude to split into eigencomponents of the commuting momentum-doubling maps, with total scaling degree at most 3; this bounds $\mu_\Theta\{p_0 \le R\}$ by R^7, against $R^8/\log^2 R$.

3.27.4 4. Steps

3.27.4.1 4.1 The centered trace and its spectral measure Lemma VII.2.a (centering and symmetry). The map $v \mapsto \langle v\rangle$ is linear with $\langle \mathbf{1}\rangle = 1$, so $\langle\Theta\rangle = 0$. Reality and $O(4)$ invariance follow from those of T: contracting the tensorial law with $KK^{\mathsf{T}} = 1$ gives $\sum_a K_{ia}K_{ja} = \delta_{ij}$. Time-reflection invariance follows from reality and rotation invariance. For Lorentz invariance of $\iota(\Theta)$, the Lorentz group is generated by spatial rotations and axis boosts. Along a boost, $t \mapsto \mathrm{boost}(t)\iota(\Theta)$ solves the linear ODE $\dot y = -iX_{0a}y$, and the generator annihilates $\iota(\Theta)$ because Θ is rotation invariant. Uniqueness of solutions of the ODE gives invariance. *Lean:* `trace_turn`, `scalar_rail`.

Lemma VII.2.b (zero Wightman vacuum expectation). For a centered label, the one-point function continued to complex times is holomorphic and vanishes on the Euclidean slice, so it vanishes identically by the identity theorem. Its boundary value, the Wightman one-point function, is therefore 0. *Lean:* `vev_boundary`.

Lemma VII.2.c (spectral measure). The boundary realization (V.2) assigns a positive spectral measure to the label $\iota(\Theta)$. Smearing the field with φ gives a Hilbert-space vector whose translation spectral measure is $|\widehat{\varphi}|^2\mu_\Theta$.

- The spectrum condition gives support in $\bar V_+$; the boundary estimates give tempered growth and the polynomial energy budget.
- Lorentz invariance comes from Lemma VII.2.a.
- The Laplace formula comes from continuing $G_\Theta(2s)$ to the boundary pairing at imaginary time (`continuation`).
- Uniqueness holds because μ is determined by the integrals $\int |\widehat{\varphi}|^2 d\mu$.

Monotonicity holds because e^{-sp_0} decreases in s on $\bar V_+$. *Lean:* `traceSpectrum_exists`, `monotone_axis`. *(proof read at the top level; the boundary spectral lemmas are inputs from V.2)*

3.27.4.2 4.2 Ward darts and their transport through the RG Lemma VII.2.d (sound darts and the block-spin step). A dart on a torus of side f consists of site-indexed functions and vector fields: a tensor J_{ab}, a displacement field Y_a on link configurations, scalar entries E_a, Q_a, and scalar entries tw_{abi}. It is *sound* if:

- the scalar entries are analytic and gauge invariant near the identity configuration, with zero constant and linear terms; the displacement fields are analytic and gauge covariant and vanish at the identity;
- $\sum_a J_{aa} = \sum_a \partial_a Q_a$ and $J_{ab} - J_{ba} = \sum_i \partial_i \mathrm{tw}_{abi}$, with lattice differences ∂;
- the Ward relation holds: $\sum_a \partial_a J_{ab} - E_b$ agrees through fourth order at zero field with the first variation of the action in the direction Y_b.

The block-spin map takes a dart at the fine level to one at the coarse level, by block averaging of inherited jets scaled by L^k with $k = 3, 4, 5$. It preserves soundness, under the criticality, branch,

bound, positive price-gap and classical-matching hypotheses of III.6–III.7. The mechanism is fourth-order matching of first variations: along an analytic coarse direction Y with $Y(0) = 0$, the derivative of the coarse action at the stationary background in the direction Y agrees through fourth order with the derivative of the fine action in the lifted direction. This follows by the chain rule through the stationary background map. *Lean:* `Dart`, `Dart.Sound`, `Dart.down`, `Sound.balance`; `aroma_transfer`, `recoup`.

Lemma VII.2.e (darts with prescribed precision). Under the bounded-run and eventual level hypotheses of VII.2.3, for all large levels k, every $\varepsilon > 0$ and every filter along which $N \to \infty$ and the lattice side at each fixed level tends to infinity, the following holds eventually. There is a sound dart at level k for the actual configuration with these properties:

- its jets obey the size bounds;
- its tested second derivatives are within ε of the tree-level values: total differential weight at most 4 for J and at most 5 for Y; the derivatives of E of orders 2–4 have size at most ε.

Proof sketch. Choose i with $2^{-(i+1)}C(\hat{\beta}_k) < \varepsilon$, where $C(\hat{\beta}_k)$ bounds the finitely many relevant derivative orders. Darts at level k are obtained by transporting darts from level $k+i+1$ through $i+1$ steps. Each step multiplies the deviation from tree level by at most $\frac{1}{2}$. The resulting geometric bound is converted into the required estimates. *Lean:* `treasury`, `cash_viz`; `through`, `pay_step`.

3.27.4.3 4.3 The one-loop slope Lemma VII.2.f (logarithmic pole). Let $\Gamma_\mu(U) = \frac{1}{2}\log\det(\mathcal{H}(U) + MM^{\mathsf{T}} + \mu^2) - \log\det(\mathcal{G}(U) + \mu^2)$ be the infrared-regulated one-loop effective action of a single RG level. Here $\mathcal{H}$ is the Hessian of the action, MM^{T} the gauge-fixing term, and $\mathcal{G}$ the ghost operator. Let $R_\mu = \sum_{d\neq i} y_{di}(\Gamma_\mu)$, where y_{di} is the quadratic color-q, polarization-i momentum test in direction d, divided by the number of sites. It applies the centered second difference to the quadratic field derivative at the opposite momenta $\pm h e_d$, with the factor h^{-2}, where $h = 2\pi/f$ for torus side f. For a sequence of lattices of side $\to \infty$ satisfying the uniform covariance, resolvent and action bounds of the one-loop comparison (the predicates in §6), there is C' such that for every $\mu \in (0, 1]$, eventually,

$$\left|\operatorname{Re} R_\mu + 12\,\kappa_G \log(1/\mu)\right| \le C'.$$

Proof sketch. By symmetry all 12 planes contribute equally. The $(0, 1)$ coefficient converges, along the sequence, to a momentum integral whose $\log(1/\mu)$ coefficient is $-\kappa_G$ up to a bounded error (`absoluteGeode`, III.7). *Lean:* `slopeL`, `marke`, `ridge` *(proof read;* `effect`*,* `banana`*,* `payday`*,* `jury` *statement only; the momentum selector is defined in §6).*

Lemma VII.2.g (from the pole to a value: mean-value extraction). The function R satisfies $\mu\,\partial_\mu R_\mu = -\mathrm{seaD}_\mu$, where $\mathrm{seaD}_\mu = \mu^2 \sum_{d\neq i} u_{di}(\mu)$ and u_{di} collects the same normalized momentum tests of $-\operatorname{tr}(\mathcal{H} + MM^{\mathsf{T}} + \mu^2)^{-1} + 2\operatorname{tr}(\mathcal{G} + \mu^2)^{-1}$ (from $\partial_\mu \log\det(F + \mu^2) = 2\mu \operatorname{tr}(F + \mu^2)^{-1}$). By Lemma VII.2.f, $x \mapsto \operatorname{Re} R_{e^x} - 12\kappa_G x$ is bounded by C' eventually for each fixed $x \le 0$. For every $a < b \le 0$, the mean value theorem on $[a, b]$ then gives $\mu_w \in [e^a, e^b]$ with $|\operatorname{Re}\mathrm{seaD}_{\mu_w} - (-12\kappa_G)| \le 2C'/(b-a)$ eventually. *Lean:* `ridge_deriv`, `coastMark`.

Lemma VII.2.h (the payoff converges to $-12\kappa_G$). Under the operator bounds of VII.2.f, take translation-covariant sound darts B_w on sides $g(w) \to \infty$, with a fixed positive decay parameter, uniform jet bounds and precision errors $0 \le \varepsilon_w \le 1$ tending to 0. Let X_w be the summed normalized momentum test of the Haar divergence of their rooted displacement fields. Then $\operatorname{Re} X_w \to -12\kappa_G$; the difference between the finite-momentum test and the Taylor selector, including its quadratic

correction, tends to 0, so $\mathrm{midPay}_q(B_w) \to -12\kappa_G$. *Proof sketch.* For every fixed $0 < b' \le c \le 1$ and every choice $\mu_w \in [b', c]$, the Ward relation gives $X_w = \mathrm{seaD}_{\mu_w} + \mu_w^2 K_w(\mu_w) + o(1)$ with $|\mu_w^2 K_w(\mu_w)| \le B'\mu_w$ eventually, where B' is independent of the fixed regulator interval. Here K_w is a trace of resolvents paired with the linearized displacement and ghost terms. The regulated Ward identity yields the decomposition; its other action-variation and precision terms vanish in the period limit. Choose $u \le 0$ with $B'e^u \le \varepsilon/4$, then C with $2C'/C \le \varepsilon/4$, and apply Lemma VII.2.g on $[u - C, u]$. *Lean:* `awake`, `midAwake`; `saga`, `crest_v` *(proof read; their analytic inputs statement only)*; `fawn`.

Lemma VII.2.i (uniform certificates; proof of VII.2.3).

1. At a fixed level k, fix a positive decay parameter and a uniform jet bound. For every $\delta > 0$ there are a precision tolerance $\eta > 0$ and a volume threshold T such that every translation-covariant dart satisfying those bounds with precision η, with $N > k$ and $p \ge T$, has $|\mathrm{midPay} - (-12\kappa_G)| \le \delta$. The proof is by contradiction: a sequence of counterexamples with $p \to \infty$ contradicts Lemma VII.2.h.
2. Average the darts of Lemma VII.2.e over lattice translations. The bounds are preserved, so this gives darts with payoff accuracy ε for all large k, eventually along every filter for which $N, p \to \infty$.
3. With $\varepsilon = \hat{\beta}_k^{-100}$ and the remaining stage, support, cluster and coefficient bounds, the work items are chosen by choice along $\mathcal{L}$.

Lean: `wire_mid`, `dewWt`, `agenda`. *(proof read; stage and certificate-supply inputs statement only)*

3.27.4.4 4.4 From the slope to the coefficient of the trace Notation. Fix the run data and a scaling sequence as in VII.2.4(i). All statements in §§4.4–4.5 have these hypotheses. Write $\hat{\beta}_i = \hat{\beta}_{\ell_i}$, $E_i = \exp(\frac{1}{5}(\log\log\hat{\beta}_i)^2)$ and $R_i = \exp(\frac{1}{6}(\log\log\hat{\beta}_i)^2)$.

- *Reference tensors.* The centered, real degree-4 free tensors covariant under signed coordinate permutations form a family $\mathrm{st}(w)$, $w \in \mathbb{R}^3$, with $\mathrm{tr}\,\mathrm{st}(w) = 4(w_0+4w_1)\mathcal{O}$ (`mechanical_trace`). The three reference tensors are $L = \mathrm{st}(-(1, -\frac{1}{4}, 1)) = T^{\mathrm{free}}$, $\Delta = \mathrm{st}(0, \frac{1}{4}, 0) = \delta_{\mu\nu}\mathcal{O}$ and $\Xi = \mathrm{st}(0, 0, 1)$.
- *The functional ρ.* On raw degree-5 divergence labels, before quotienting by the Maxwell equations, $\rho(\nabla_{\mathrm{raw}}\mathrm{st}(w)) = -4(w_0 + 4w_1)$ (`rho_vector`). It annihilates the Maxwell equation direction. Below, $\rho(B)$ abbreviates $\rho(\nabla_{\mathrm{raw}}B)$, so $\rho(L) = \rho(\Xi) = 0$ and $\rho(\Delta) = -4$. Explicitly, $\Xi_{\mu\mu} = 0$ and $\Xi_{\mu\nu} = \sum_{a,c} : F^c_{\mu a}F^c_{\nu a} :$ for $\mu \ne \nu$.
- *Lattice-built tensors.* $\mathrm{tub}_i(B) \in \mathcal{K}(S)$, for $B \in \{L, \Delta, \Xi\}$, are built from the work items at the readout level ℓ_i.
- *Rescaled divergence.* Eventually, $K_i := w_i r_i^5\, U_i^{-1} \circ \mathrm{div}$ (`Optics`).
- *Rescaling of tensors.* $\mathrm{Ph}_i := w_i r_i^4 U_i^{-1}$.

Lemma VII.2.j (the divergence vector of the dart). Let Σ_k be the level-5 coefficient vector of the divergence of the dart's stress tensor at level k, normalized by the coupling limits and taken along $\mathcal{L}$. Then eventually $\|\Sigma_k\| \le c\, e^{\frac{1}{6}(\log\log\hat{\beta}_k)^2}$, which grows more slowly than every positive power of $\hat{\beta}_k$, and $\rho(\Sigma_k) \to -\kappa_G$. *Proof sketch.* At each (k, N, p) the revised readout decomposes into the payoff term midPay/12 plus terms bounded by a fixed multiple of $e^{-(\log\log\hat{\beta}_k)^2}$ and $\hat{\beta}_k^{-30}$. Passing to the ultrafilter limit adds a coefficient-transfer error $O(e^{-\frac{1}{4}(\log\log\hat{\beta}_k)^2})$. The certificate of VII.2.3 then gives $|\mathrm{midPay}/12 + \kappa_G| \le \hat{\beta}_k^{-100}/12$. For the three reference tensors, the analogous

raw vectors converge to $\nabla_{\mathrm{raw}}L$, $\nabla_{\mathrm{raw}}\Delta$ and $\nabla_{\mathrm{raw}}\Xi$, at rate $o(e^{-\frac{1}{5}(\log\log\hat{\beta})^2})$ after applying ρ. *Lean:* `penumbra`, `brass`; `tiara`, `diadem`, `article`.

Lemma VII.2.k (divergences at scale r_i). There are linear maps s_i from the fixed raw coefficient space to $\mathcal{V}(\mathsf{F})$, converging to the free projection s, and functionals μ_i on $\mathcal{V}(\mathsf{F})$ with $\|\mu_i\| \le c$ eventually, such that $E_i\|\mu_i \circ s_i - \rho\| \to 0$ in operator norm, where ρ is expressed in the lattice coefficient coordinates. With these:

- $w_i K_i(\mathrm{tub}_i L) - s_i(\Sigma_{\ell_i}) \to 0$, hence $\mu_i(w_i K_i(\mathrm{tub}_i L)) \to -\kappa_G$;
- $\|w_i K_i(\mathrm{tub}_i L)\| \le C R_i$ eventually;
- for $B = \Delta, \Xi$: $K_i(\mathrm{tub}_i B) \to \mathrm{div}^{\mathrm{free}} B$ and $E_i(\mu_i(K_i \mathrm{tub}_i B) - \rho(B)) \to 0$.

Inputs.

- The lattice-to-continuum remainders are $o(\hat{\beta}_i^{-p})$ for every $p \in \mathbb{N}_0$. This rests on the stage estimates and on the cluster bounds, which are the RG estimates of III.5 and III.8 run once more on the trace observables.
- Lattice observables are identified with continuum labels by two tools. Pullback along centered chambers: if a lattice observable differs from the local representative of v_k by a chamber-interior term with fast chessboard root, then its correlations with displaced spectators agree with those of the limit of v_k up to a fast error. Eighth-order difference stencils: the lattice stencil of an observable converges to the continuum derivative of its limit.

Lean: `pairingD`, `sizeQ'`, `transferX`, `beater`, `mere`. The following are statement only: `Prosper`, `rungs`, `legacy`, `hullage`, `gapC`, `crestship`, `touring`, `trod`, `small_means`, `expire`, `lay`, `welfare`, `garnet`, `quaff`, `from_center`, `axis_advance`, `differentiation`.

Lemma VII.2.l (the coefficients of the improvement terms). There are $c_i \in \mathbb{R}^2$ and $C \ge 0$ such that, eventually:

- $\|c_i\| \le C R_i$;
- $f_i : c \mapsto c_0 K_i(\mathrm{tub}_i \Xi) + c_1 K_i(\mathrm{tub}_i \Delta)$ is bijective;
- $w_i K_i(\mathrm{tub}_i L) + f_i(c_i) = 0$.

Moreover $c_{i,1} \to -\kappa_G/4 = d_2$. *Proof sketch.*

1. The maps f_i converge to $c \mapsto c_0 \,\mathrm{div}\,\Xi + c_1 \,\mathrm{div}\,\Delta$, which is bijective for simple G. Hence f_i is eventually bijective with uniformly bounded inverse.
2. Set $c_i = -f_i^{-1}(w_i K_i(\mathrm{tub}_i L))$.
3. Apply μ_i to the balance relation and use Lemma VII.2.k:

$$4c_{i,1} = \mu_i(w_i K_i \mathrm{tub}_i L) + c_{i,0}\,\mu_i(K_i \mathrm{tub}_i \Xi) + c_{i,1}(\mu_i(K_i \mathrm{tub}_i \Delta) + 4) \to -\kappa_G.$$

   The last two terms tend to 0 because $|c_{i,j}| \le C R_i$ and $R_i/E_i \to 0$.

Lean: `formulate`, `wend`, `quill`.

Lemma VII.2.m (the conserved candidate). Let $X_i := w_i \,\mathrm{tub}_i L + c_{i,0}\,\mathrm{tub}_i \Xi + c_{i,1}\,\mathrm{tub}_i \Delta$. Then:

- X_i is conserved eventually, since $K_i X_i = 0$ and K_i vanishes exactly on conserved tensors;
- $r_i^4 U_i^{-1} X_i \to T^{\mathrm{free}}$, because $\mathrm{Ph}_i\,\mathrm{tub}_i L \to T^{\mathrm{free}}$ and $c_i/w_i \to 0$;
- $w_i \,\mathrm{tr}(r_i^4 U_i^{-1} X_i) \to 4 d_2\, \mathcal{O}$, because for $B \in \{L, \Xi\}$ the trace of $\mathrm{Ph}_i\,\mathrm{tub}_i B$ is $o(\hat{\beta}^{-p})$ for every p, while $\mathrm{tr}\,\mathrm{Ph}_i\,\mathrm{tub}_i \Delta \to 4\mathcal{O}$.

Lean: `serene`, `yields`, `blossoms`, `sum_delta`; `aprilA`; `nearing_h`, `muteP` *(proof read at the top; inputs statement only).*

Lemma VII.2.n (identification with T; proof of VII.2.6).

- *Rank.* The free side gives div onto with one-dimensional kernel; this is reduced by linear algebra to the free closure map. Eventually $K_i \circ (c \mapsto c_0 \mathrm{tub}_i \Xi + c_1 \mathrm{tub}_i \Delta)$ is onto the two-dimensional target, and $\ker K_i = \ker \mathrm{div}$. Since $\dim \mathcal{K}(S) = \dim \mathcal{K}(\mathsf{F})$, a dimension count gives $\dim \ker \mathrm{div} = 1$ on $\mathcal{K}(S)$.
- *Identification.* Hence eventually $X_i = v_i \mathring{T}$ for real v_i.
- *Normalization.* Choose a free observer o and a label u with $o(\partial_b u) \neq 0$. The rescaled Ward flux $r_i \, \mathrm{flux}_b(\mathring{T})$ tested against $U_i u$ converges to $o(\partial_b u)$, and the same flux for X_i converges to the free value. Comparing the two limits gives $v_i \to 1$.
- *Conclusion.* $r_i^4 U_i^{-1} \mathring{T} \to T^{\mathrm{free}}$ and $w_i \, \mathrm{tr}(r_i^4 U_i^{-1} \mathring{T}) \to 4 d_2 \mathcal{O}$.

Lean: `ranking`, `reveals`, `year_spec`, `free_rank`; `free_closure`; the Ward-flux comparison inputs *(statement only; VI.1).*

Lemma VII.2.o (the field-level anomaly; proof of VII.2.4).

1. *Centering is invisible at short distance.* For any phase table T', $\Theta_{T'} = \mathrm{tr} \, \mathring{T} - x \, \mathbf{1}$ for some constant x. Since $w_i r_i^4 \to 0$, the term $x\mathbf{1}$ disappears after rescaling. This gives $w_i r_i^4 U_i^{-1} \Theta_{T'} \to 4 d_2 \mathcal{O}$.
2. *Form of* Θ. $\Theta_{T'}$ is a real $O(4)$-scalar. Eventually U_i^{-1} maps it into the real degree-4 free labels fixed by signed coordinate permutations, which are spanned by $\mathbf{1}$ and $\mathcal{O}$. This gives the stated form with coefficients e_i, d_i. The two basis labels are linearly independent; applying their coordinate functionals to the limit gives $w_i e_i \to 0$ and $w_i d_i \to d_2$.
3. *Clock.* $\log(1/r_i)/w_i \to \kappa_G^{-1}$ holds because $r_i = a_{k_i} m_i$ with $\log m_i = o(\hat{\beta}_{k_i})$, $\hat{\beta}_k = \bar{\beta} + \kappa_G \log(1/a_k)$, and $w_i / \hat{\beta}_{k_i} \to 1$.
4. *Gallery property.* The free Maxwell comparison supplies the hierarchy a and its Ward-normalized stress tensor; compactness along the ultrafilter supplies Z. The preceding conclusions apply to every compatible scaling sequence. The subsequence statement comes from VII.6.

Lean: `july`, `shift_unit`, `leaderB`, `zoom_BQ`, `band_cost`, `Store_l`, `gang`; `payload`, `styling`.

3.27.4.5 4.5 The short-distance law Lemma VII.2.p (law along an anomaly sequence at fixed configurations). Let $x = (x_0, x_1)$ have distinct coordinates in every axis and lie in the interior of the fixed scaling window with bound 100. Let $x^{(i)} \to x$ be its lattice roundings at level i. Then

$$r_i^8 \log^2(1/r_i) \, G_\Theta(r_i |x_0^{(i)} - x_1^{(i)}|) \to \frac{A_G}{|x_0 - x_1|^8}.$$

Proof sketch.

1. Write $U_i(a_i) = w_i r_i^4 \Theta$, where $a_i = U_i^{-1}(w_i r_i^4 \Theta)$.
2. The two-point function in degree 4 is quadratic, $\langle (pu)(pu) \rangle = p^2 \langle uu \rangle$.
3. By $O(4)$ and translation invariance, any pair of points is moved to the time axis, so the left side equals $(\log(1/r_i)/w_i)^2 \, S^{\xi_*}(U_i a_i(r_i x_0^{(i)}) \, U_i a_i(r_i x_1^{(i)}))$.
4. Since $a_i \to 4 d_2 \mathcal{O}$, the free-field convergence and the normalization of $\mathcal{O}$ give the limit $\kappa_G^{-2} (4 d_2)^2 \cdot 3 \dim \mathfrak{g} / (\pi^4 |x_0 - x_1|^8) = A_G / |x_0 - x_1|^8$.

Lean: `tagline`, `axis_shape`, `wick_norm`; `rainbow`, `rounds` *(proof read at the top).*

Lemma VII.2.q (removing the rounding; the full limit).

1. *Removing the rounding.* Take $x = (0, (\frac{s}{2}, \frac{s}{2}, \frac{s}{2}, \frac{s}{2}))$ with $s \in (\frac{1}{2}, 2)$, so that $|x_0 - x_1| = s$. Since G_Θ is nonincreasing (VII.2.2) and the rounded separations tend to s:

   - $\liminf_i r_i^8 \log^2(1/r_i)\, G_\Theta(r_i) \geq A_G s^{-8}$ for $1 < s < 2$;
   - $\limsup_i r_i^8 \log^2(1/r_i)\, G_\Theta(r_i) \leq A_G s^{-8}$ for $1/2 < s < 1$.

   Letting $s \to 1$ gives the limit A_G along (r_i).

2. *The full limit.* If the law failed as $r \downarrow 0$, there would be a null sequence avoiding a neighbourhood of A_G. By VII.2.4(ii), a subsequence satisfies the field-level anomaly, which contradicts step 1.

Lean: `border`, `outside`.

Lemma VII.2.r (Tauberian step; proof of VII.2.5(b)). Suppose VII.2.2 holds and $r^8 \log^2(1/r) G_\Theta(r) \to A \geq 0$. Let $\nu_\rho := \rho^{-8} \log^2 \rho\ \mu_\Theta(\rho\,\cdot)$. By Lorentz invariance, the Laplace transforms $\int e^{-u\cdot p} d\nu_\rho$ for u in the open forward cone reduce to time-axis values. The law then gives their convergence to $A(u^2)^{-4}$, the Laplace transform of $\frac{A}{1536\pi}(p^2)^2 \mathbf{1}_{V_+} d^4p$. The law also gives a uniform energy budget $\nu_\rho\{p_0 \leq R\} \leq CR^8$ for all $R \geq 1$ and all sufficiently large ρ, with one constant C. Laplace convergence gives convergence on continuous compactly supported tests. The uniform polynomial growth bound controls the tails of Schwartz tests and extends the limit to that class. *Lean:* `transform_limit`, `rescaled_energy`, `compact_limit`, `joint_limit`; `laplace_schwartz`, `cone_laplace`.

3.27.4.6 4.6 Exclusions Lemma VII.2.s (power–logarithm expansions cannot carry the law). Suppose $f = u + O(1)$ as $s \downarrow 0$, where u is a finite real combination of $s^a \log^k s$ with $a \in \mathbb{Z}$ and $k \in \mathbb{N}_0$, and suppose $s^8 \log^2(1/s)\, f(s) \to A$. Then $A = 0$. *Proof sketch.* The $O(1)$ part contributes $s^8 \log^2(1/s) \cdot O(1) \to 0$. Along $s = 2^{-n}$, $s^8 \log^2(1/s)\, u(s)$ is a combination of $c_t\, \rho_t^n n^{k_t+2}$ with $\rho_t = 2^{-(8+a_t)}$. Every term has a factor n^{k_t+2} with $k_t + 2 \geq 2$, so no term tends to a nonzero constant. Collect equal pairs (a_t, k_t), discard zero coefficients, and order the remaining terms lexicographically by $(\rho_t, k_t + 2)$. If the leading term has $\rho_t < 1$, all terms tend to 0. Otherwise the leading term diverges and dominates the others, so a finite limit is impossible unless its coefficient vanishes. *Lean:* `no_powerlog`.

Lemma VII.2.t (Wick profiles expand; proof of VII.2.7).

- *Expansion.* Each mode kernel $K_{m,q,h}$ has an expansion to all orders in $\{s^a \log^k s\}$. The class is closed under products and sums, so $\operatorname{Re} W$ has the form required in Lemma VII.2.s; part (a) follows. Equality on $(0, \delta)$ is formulated as eventual equality as $s \downarrow 0$.
- *Part (b).* If the Wightman function equals the mass-shell expression on all products with compactly supported wave transform, their wave transforms include every compactly supported Schwartz momentum test, so $\int \psi\, d\mu_\Theta$ equals the mass-shell expression for every compactly supported ψ. Compactly supported Schwartz $w_n \to e^{-sp_0}$, dominated by e^{-sp_0}, together with dominated convergence on both sides, give $G_\Theta(s) = W(s)$ for $s > 0$. The palette version reduces to the collection version.

Lean: `Expands`, `moments_expand`, `real_profile`, `identifies_euclid`, `polynomial_excludes`. *(proof read; the single-mode expansion not read)*

Lemma VII.2.u (Fock sectors; proof of VII.2.8). Assume the positive short-distance law of VII.2.8 and a comparison U satisfying its relative-locality identity.

1. *Sector decomposition.* By translation covariance, $U\Phi_\Theta(\varphi)\Omega = \hat{\varphi}(p_{\mathrm{tot}})\,Y$ for fixed measurable sector amplitudes $Y = (Y_{n,s})$, which need not form a square-integrable vector before smearing. Hence μ_Θ is the image of $\sum_{n,s} |Y_{n,s}|^2\, d(\text{photon measures})$ under total momentum, and $\mu_\Theta\{p_0 \le R\} = \sum_{n,s} \mathrm{mass}(Y_{n,s}; \{\sum |k_j| \le R\})$.
2. *Evaporation.* The mass of $Y_{n,s}$ on the box $\{|k_j| \le 2^J \text{ for all } j\}$ is at most $\mu_\Theta\{p_0 \le (n+1)2^J\}$. By the law, 2^{-8J} times this tends to 0.
3. *Dilation polynomials.* Relative locality gives equality of the forward and adjoint sector pairings when each test is supported spacelike to the origin. Approximate identities in time show that the Cauchy data of their difference are supported at the spatial origin. A distribution supported at one point has finite jet order. If D_j doubles the jth momentum, these jet bounds and mass-shell uniqueness give, for some K, $\prod_{a=0}^{K-1}(D_j - 2^{a-1})Y_{n,s} = 0$ in every slot. The roots are distinct and the D_j commute, so $Y_{n,s}$ is a finite sum of joint eigencomponents.
4. *Finitely many sectors.* A joint eigencomponent with slot degrees a_j has box mass proportional to $R^{2\sum_j a_j}$ at dyadic radii R. Evaporation excludes $\sum_j a_j \ge 4$. A component with any $a_j = 0$ vanishes by local square integrability near zero momentum. Thus every nonzero component has $a_j \ge 1$, so $Y_{n,s} = 0$ for $n \ge 4$. The remaining components have box mass at most $C_{n,s}2^{6i}$ at radius 2^i.
5. *Contradiction.* Summing over the finitely many sectors with $n \le 3$ gives $\mu_\Theta\{p_0 \le 2^k\} \le D\,2^{7k}$. Multiply this bound by $R^{-8}\log^2 R$: it tends to zero. The spectral law instead gives a positive limit against a nonnegative compact test supported in $\{0 < p_0 < 1,\ p^2 > 0\}$, a contradiction.

Lean: `form_exists`, `parts`, `burn`, `consumes`, `finite_particles`, `saturate`, `local_eigen`; `local_position`, `cauchy_point`, `binds`, `splitParts`, `log_high`, `does_not_fit` *(proof read; the point-support and analytic uniqueness inputs are not expanded here).*

3.27.5 5. Where the difficulty lies

Hard. Three pieces carry the mathematics, and all three are needed for the exact constant.

- The transport of the lattice Ward identity through the RG with fourth-order matching (VII.2.d–e).
- The one-loop pole with coefficient exactly κ_G (VII.2.f, using III.7), and its conversion into a value by the mean-value argument (VII.2.g–h).
- The control of lattice-to-continuum remainders to $o(\hat{\beta}^{-p})$ for every p (VII.2.k). The trace must be controlled after multiplication by the inverse coupling. The separate coefficient errors have the slower exponential rates displayed in VII.2.k. The bounds on c_i grow like $e^{\frac{1}{6}(\log\log\hat{\beta})^2}$ and are beaten by decay factors $e^{-\frac{1}{5}(\log\log\hat{\beta})^2}$.

The identification of the explicit conserved tensor with T (VII.2.n) is short once the rank-one statement and the Ward normalization are available.

Routine.

- Centering and the symmetries of Θ.

- The passage from subsequential limits at rounded points to the full limit (monotonicity and the subsequence principle).
- The Tauberian step.
- The Wick exclusion.

The Fock exclusion also needs a distributional argument: relative locality yields point-supported Cauchy data, which force polynomial identities for momentum doubling. The growth comparison alone does not supply this step.

Unusual or worth checking.

1. *Logic of dependence.* The law is derived from the gallery property, which is established while the ultraviolet run is chosen (III.2): the same Wilson family must carry the trace certificates. The exclusions are proved from the law, not independently. Their public form combines the two.

2. *The cancellation of b_0 is a consistency check, not a tautology.* The identities $\kappa_G = 2b_0$ and $d_2 = -\kappa_G/4$ are definitions. The content is:

   - $c_{i,1} \to -\kappa_G/4$, where κ_G comes from the one-loop integral (VII.2.f);
   - $\log(1/r)/w \to 1/\kappa_G$, where κ_G is the slope of the coupling schedule.

   Only because both are κ_G is the amplitude $A_G = (4d_2/\kappa_G)^2 \cdot 3\dim\mathfrak{g}/\pi^4 = 3\dim\mathfrak{g}/\pi^4$, independent of C_A. Had the one-loop slope differed from the schedule, the amplitude would differ by the square of their ratio.

3. *Geometry of the points.* The law is first obtained at the generic configuration $(0, \frac{s}{2}(1,1,1,1))$, as the lattice machinery requires distinct coordinates in each axis. It is then rotated to the time axis, which uses the full $O(4)$ invariance of Θ from VI.1.

4. *Precision.* The payoff certificate is accurate to $\widehat{\beta}_k^{-100}$, much more than the argument needs.

5. *Scope of the exclusions.* The Wick class is exactly the public finite-profile class. The conclusion does not exclude continuous-mass generalized free fields. The Fock comparison is tested through vacuum matrix elements with coherent states; it gives an obstruction to relative locality with free photons under the stated embedding hypotheses.

3.27.6 6. Notation

object	code name	notation
trace of the stress tensor	`stressTrace`	$\sum_\mu T_{\mu\mu}$
one-point function in a table	`euclideanVEV`	$\langle v\rangle_\xi$
centered trace, its pinned label	`theta`, `center`, `pinned`	Θ, $\Theta_{T'}$, $\iota(\Theta)$
time-axis two-point function	`euclidean`	$G_\Theta(r)$
spectral measure	`TraceSpectrum`	μ_Θ
cone measure	`coneMeasure`	$\frac{A}{1536\pi}(p^2)^2\mathbf{1}_{V_+}d^4p$
scale factor	`Zlog`, `clue`	$\rho^{-8}\log^2\rho$; $r^8\log^2(1/r)$
slope, one-loop coefficient, force	`slopeConstant`, `b0`, `d2`	κ_G, b_0, d_2
amplitude; payoff constant	`award`; `landPay`	A_G; $-12\kappa_G$

object	code name	notation
free table; $\frac{1}{4}F^2$ label	`Climb.Bel`, `Cells.standard`; `OScale`, `square4`	F; $\mathcal{O}$
free tensors, divergence functional	`materialB`, `BornZ`, `rho`, `Lease`	$T^{\mathrm{free}} = L$, Δ, Ξ; ρ
scaling sequence; coupling limits	`Climb.Scaling` (r_i, U_i); `Arch`, `BQ`	(r_i, U_i); $Z.\beta(k)$, w_i
rescaled fields	`ZoomValue`, `Year`, `Phase4`, `Optics`	$U_i^{-1}(w_i r_i^4 \cdot)$, $r_i^4 U_i^{-1}$, Ph_i, K_i
lattice-built tensors; candidate	`tub`, `pollen`; `Cloak`	$\mathrm{tub}_i(B)$; X_i
divergence vector of the dart	`Shine5`	Σ_k
lattice Ward dart; precision	`Dart`, `Dart.Sound`; `Viz`	dart $(J, Y, E, Q, \mathrm{tw})$
regulated one-loop action and coefficient	`marke`, `ridge`, `seaD`	Γ_μ, R_μ, seaD_μ
payoff of a dart	`midPay`, `Holt`	midPay_q
work items and certificate	`Work items`, `Grant`, `WorthB`	certificates along $\mathcal{L}$
slowly varying rates	`Wave`, `AgeK`	$e^{a(\log\log\hat{\beta}_k)^2}$
field-level anomaly; its bundle	`Scribe`, `Gallery`	Scribe property; gallery property
short-distance law	`EuclideanAsymptotic`	$r^8 \log^2(1/r) G_\Theta(r) \to A$
Wick modes and tracks	`Mode`, `Track`	$K_{m,q,h}$, $\prod_\beta K$
Fock comparison; relative locality	`Comparison`, `Nfl`, `Ambient`, `coherent`	$U : \mathcal{H} \to \mathcal{F}_m$; $W(h)\Omega_{\mathcal{F}}$
output bundles	`TraceExclusions`, `Crest`, `Result`, `Tests`	(T1)–(T4) and internal variants

For the coefficient selectors, let $M^H_{iq}(p_1, p_2)$ be the quadratic field derivative of H at zero, complexified and evaluated on two plane waves of polarization i and color q, using the rooted trigonometric interpolation. Write $[t^j]$ for the Taylor coefficient, including $1/j!$. For the rooted displacement $Y_b(0)$, set $D_b(z) = \mathrm{tr}(\mathrm{tidal}(Y_b(0))(\exp z))$, with the exponential applied linkwise. Let Λ be the set of lattice sites and let $\mathrm{i} = \sqrt{-1}$. Then the payoff is exactly

$$\mathrm{midPay}_q = \mathrm{Re} \sum_{d \neq i} \left(\mathrm{i} \sum_b [u^2 v] M^{D_b}_{iq}(u e_d, -u e_d + v e_b) - \frac{1}{2}[t^2] M^{D_i}_{iq}(t e_d, -t e_d) \right).$$

The cubic momentum coefficient is extracted from the quadratic field derivative. The one-loop selector is

$$y_{di}(H) = \frac{M^H_{iq}(h e_d, -h e_d) - 2M^H_{iq}(0,0) + M^H_{iq}(-h e_d, h e_d)}{2h^2 \operatorname{card} \Lambda}, \qquad h = 2\pi/f.$$

Lean: `lush`, `molt`, `neap`, `pia`, `Deb`, `cob`, `yard`.

For the Fock tests, $h = (h^a_c)_{1 \le c \le m,\, 1 \le a \le 3}$ is a list of real scalar spacetime tests. Its one-particle wave is

$$\eta^a_c(k) = |k| \sum_b \left(\delta_{ab} - \frac{k_a k_b}{|k|^2} \right) \hat{h}^b_c(|k|, k), \qquad d\nu(k) = \frac{d^3 k}{2|k|}.$$

The coherent vector has n-particle component $e^{-\|\eta\|^2/2}\mathrm{i}^n\eta^{\otimes n}/\sqrt{n!}$. The value at $k=0$ is irrelevant to ν.

The run predicates used in VII.2.3 are `Bolt`, `Bandage`, `Nest`, `Vigil`, `Acclaim` and `supportQ`; those used in VII.2.4 are `PostT`, `Stock.Tabs`, `Scrim`, `Sceptre`, `supportQ`, `Moonlit` and `auraQ`, together with `Binding`, `HaarBudget`, `ChartRatio`, the generator bounds and `FiresSky`. The one-loop hypotheses in VII.2.f–h are `Cred` and `Flare`. Their parameters and dependencies are those of the fixed run and operator data.

3.27.7 7. What remains unexplained

1. **Transport and one-loop estimates.** The statements of the fourth-order first-variation transfer and the quantitative contraction through successive RG steps were checked, but their proofs were not read in full. The same limitation applies to the covariance and resolvent estimates feeding the one-loop pole. The proof sketches use these as inputs from III.4–III.11.
2. **Remainder estimates.** The stage, cluster and chamber-pullback estimates listed after VII.2.k were checked as statements, including their scale, degree, support and separation restrictions. Their detailed proofs, and the construction of the fast bounds behind the trace estimate, remain to be expanded. The slowly decaying coefficient errors and the remainders smaller than every inverse power are distinct estimates.
3. **Wick expansion.** The finite-profile reduction and the incompatibility with the trace law were checked. The all-orders expansion of a single massive mode is used as a theorem; its proof was not read.
4. **Distributional inputs to the Fock exclusion.** The reductions through remote tests, equal-time data, momentum-doubling identities, eigenvector splitting and dyadic growth were read. The finite-order theorem for point-supported distributions and the analytic uniqueness results used in those reductions remain inputs; their full proofs were not audited here.
5. **Position-space frames.** The scalar identification, frame quantifiers, shell normalization and the $O(g_X^4)$ deduction were checked. The full existence proof for the position-space Gram and angular alignment, and the detailed formulas for the two shell observables, remain to be expanded from the frame definitions. Proposition VII.2.9 concerns the family and extraction refinement supplied by the supplementary construction.
6. **Unit boundaries.** The declaration table places the spectral measure, the final short-distance steps and the Fock conclusion in VIII.1, the scaling-sequence existence results in VII.6, and the logarithmic momentum integral in III.7. Their mathematical roles are described here. The remaining integration and RG bookkeeping records are outside the proof sketches in this unit.

3.28 VII.3. The theory is not free: Gram positivity and exclusion of free-field profiles

Part: VII, identification as Yang–Mills. **Lean:** `YM/Extra/FREE_DETECTOR_ALGEBRA`, `YM/Pert/Position`, `YM/Pert/Rigid`, `YM/Pert/Lattice`, and a few declarations of `YM/Push/LOOP_COEFFICIENT`. Proposition VII.3.7 is in `YM/TraceExtension/Wick` and `TraceBridge.lean`. **Size:** about 3,800 declaration lines on the main dependency path in the listed folders, excluding the trace-exclusion files.

The reflected Gram form of the free Maxwell Wick algebra is strictly positive. It identifies the free field spaces with on-shell polynomial labels and supplies injective two-point tests (VII.3.1–VII.3.5).

At degree four, rigidity gives finite position equations with invertible linearization and a leading comparison with lattice perturbation theory (VII.3.6). Independently, the trace anomaly excludes every finite Wick profile (VII.3.7). The lattice series used in the position comparison are defined in §4.7.

3.28.1 1. What this unit proves

Setting.

Write $\mathrm{Conf}_n = \{(x_1, \dots, x_n) \in (\mathbb{R}^4)^n : x_i \neq x_j \text{ for } i \neq j\}$.

Coefficients and jets.

- $\mathfrak{g}$ is the Lie algebra of the gauge group, with its invariant inner product. $\mathfrak{g} = 0$ is allowed in VII.3.1–VII.3.5.
- For $j \geq 0$, Sol_j is the real space of $\mathfrak{g}$-valued 2-forms $S = (S_{\mu\nu})$ on $\mathbb{R}^4$ that satisfy:
  - the coefficients are homogeneous polynomials of degree j;
  - $dS = 0$ and $\delta S = 0$, where δ is the codifferential (`Sol`, `onShell`).
- Such S are harmonic (`Sol.harmonic`).
- Sol_j is the space of on-shell j-jets: the j-th Taylor coefficient of a solution of the free Maxwell equations lies in Sol_j.
- $S(x)$ denotes the polynomial 2-form.
- Write $T_j F = (j!)^{-1} D^j F$ for a Taylor coefficient. The inner product $\langle S, T\rangle$ sums the color inner products over both ordered form indices and all ordered derivative indices; there is no factor $1/2$ for the antisymmetric form indices.
- $\mathrm{Tow}_n := \bigoplus_{j<n} \mathrm{Sol}_j$ (`ProbeTower`).

Wick polynomials.

- Fix a basis of each Sol_j. Put $\mathcal{W} = \mathbb{C}[X_s]_{s \in \mathsf{S}}$ with $\mathsf{S} = \{(j, a)\}$ (`WickPol`, `Slot`).
- $X_{(j,a)}$ is the a-th coordinate of the j-th on-shell Taylor coefficient of the free curvature at a point.
- For $p \in \mathrm{Tow}_n$, $\ell_p \in \mathcal{W}$ is the real linear polynomial $A \mapsto \sum_{j<n} \langle p_j, A_j\rangle$ (`digit`, `code`).

Free moments.

- $C_{st}(x-y)$ is the free covariance of jet coordinate s at x with jet coordinate t at y (`Wick.entry`, `legKernel`). It is smooth off the diagonal and built from derivatives of $\Gamma(z) = (4\pi^2|z|^2)^{-1}$.
- For $x \in \mathrm{Conf}_n$, the free moment $\langle P_1(x_1) \cdots P_n(x_n)\rangle_0$ is multilinear (`Wick.moment`). On monomials it is the sum, over perfect matchings of the variable occurrences with no pair inside one factor, of $\prod_{\text{pairs}} C_{st}(x_i - x_k)$.
- These are the Wick-ordered correlations of $\dim \mathfrak{g}$ independent free Maxwell curvatures. Multiplication in $\mathcal{W}$ therefore represents Wick multiplication at one insertion.

Reflected pairing.

- For $z \neq 0$, $\mathcal{R}_z v = v - 2\langle v, z\rangle |z|^{-2} z$ is the reflection in $z^\perp$ (`face`). Orthogonal transformations act on all tensor indices of jets and on polynomials by $(R \cdot P)(A) = P(R^{-1} \cdot A)$ (`pullSol`).
- $\bar{P}$ conjugates the coefficients of P (`barPol`).
- For $x \neq y$ (`duplex`, `aligned`):

$$\langle P, Q\rangle_{x,y} := \big\langle P(x)\,(\mathcal{R}_{x-y}\bar{Q})(y)\big\rangle_0.$$

- This is the local two-point Osterwalder–Schrader pairing for reflection in the perpendicular bisector of $[x, y]$, with the convention that the first argument is linear and the second conjugate-linear.

Tower Gram form.

- On Tow_n the restriction to linear polynomials is $B_{x-y}(p, q) := \langle \ell_p, \ell_q\rangle_{x,y}$. It is real and depends only on $z = x - y$.
- Explicitly, $B_z(p, q) = \sum_{j,k<n} \mathrm{scan}_z(p_j, q_k)$, where $\mathrm{scan}_z(P, Q)$ is the free covariance of $\langle P, T_j F\rangle(z)$ with $\langle \mathcal{R}_z Q, T_k F\rangle(0)$ (`towerB`, `scan`).
- The identification of the two descriptions is `crash_test` together with `join_crash`.

Proposition VII.3.1 (strict reflected positivity of the free Wick algebra). Let $x \neq y$ in $\mathbb{R}^4$. (a) For every $P \in \mathcal{W}$: $\langle P, P\rangle_{x,y} > 0$ if and only if $P \neq 0$. (b) $\overline{\langle P, Q\rangle_{x,y}} = \langle Q, P\rangle_{x,y}$ for all $P, Q \in \mathcal{W}$. (c) For every finite $\mathbb{C}$-linearly independent family (V_i) in $\mathcal{W}$, the matrix $(\langle V_j, V_i\rangle_{x,y})_{ij}$ is positive definite. No simplicity or positive-dimension assumption is needed, and the polynomial degree is unrestricted. Self-pairings are real, so the inequality in (a) is a real inequality. *Status:* theorem. Only (a) lies on the path to the main theorem; (b) and (c) are proved but not used. *Lean:* (a) `reflected_positive`; (b) `reflected_symm`; (c) `reflected_pd`.

Proposition VII.3.2 (tower Gram form and Fock structure). Let $x \neq y$, $z = x - y$, $n \in \mathbb{N}$. (a) B_z is symmetric and positive definite on Tow_n. More precisely, there is a block-unitriangular automorphism σ_z of Tow_n (Lemma VII.3.f) such that

$$B_z(\sigma_z p, \sigma_z q) = \sum_{j<n} \pi_j(z)\,\langle p_j, q_j\rangle, \qquad \pi_j(z) = \frac{(j+1)(j+2)\,2^j}{\pi^2\,|z|^{2j+4}} > 0.$$

(b) Let $b_1, \dots, b_N \in \mathrm{Tow}_n$ satisfy $B_z(b_i, b_k) = d_i \delta_{ik}$, with any $d_i \in \mathbb{R}$. Let $\Phi_b : \mathbb{C}[Y_1, \dots, Y_N] \to \mathcal{W}$ be the algebra homomorphism with $Y_i \mapsto \ell_{b_i}$ (`encode`). Then

$$\langle \Phi_b p, \Phi_b q\rangle_{x,y} = \sum_\alpha p_\alpha\,\overline{q_\alpha}\;\alpha!\,d^\alpha, \qquad \alpha!\,d^\alpha := \prod_i \alpha_i!\,d_i^{\alpha_i}.$$

Status: theorem. *Lean:* (a) `towerB_sym`, `towerB_pos`, `shiftAll_gram`, `price`, `price_pos`; (b) `duplex_script`, `aligned_script`.

Proposition VII.3.3 (Wick recursion; characterization of the free moments). Let $n \geq 1$, $x \in \mathrm{Conf}_n$, $W \in \mathcal{W}^n$, $1 \leq i \leq n$ and $s \in \mathsf{S}$. Let $\partial_s^{y,y'}$ be the $\mathbb{C}$-linear derivation of $\mathcal{W}$ with $\partial_s^{y,y'} X_t = C_{st}(y - y')$. Then

$$\big\langle W_1(x_1)\cdots(W_i X_s)(x_i)\cdots W_n(x_n)\big\rangle_0 = \sum_{k\neq i} \big\langle W_1(x_1)\cdots(\partial_s^{x_i,x_k} W_k)(x_k)\cdots W_n(x_n)\big\rangle_0,$$

where on the right the factor at x_i is W_i. Conversely, for any $n \geq 0$ and $x \in \mathrm{Conf}_n$, a $\mathbb{C}$-multilinear map $T : \mathcal{W}^n \to \mathbb{C}$ with $T(1, \dots, 1) = 1$ satisfying this recursion for every i, s, W is the free moment at x. Consequences:

- $\langle\cdots\rangle_0$ is invariant under any simultaneous $O(4)$ rotation of points and polynomials.
- $\langle \bar{P}_1(x_1)\cdots\bar{P}_n(x_n)\rangle_0 = \overline{\langle P_1(x_1)\cdots P_n(x_n)\rangle_0}$.

Status: theorem. *Lean:* `Pair.remove`, `Pair.extractVar`; characterization `byAnnihilators`; consequences `wick_rotate`, `wick_bar`.

Proposition VII.3.4 (the free table is faithful on on-shell labels).

Setting:

- Label_d is the space of complex polynomials in radial potential jets that are invariant under the adjoint action of the gauge group and have engineering weight at most d; an order-j slot has weight $j+2$ (II.1, III.16; `Label`).
- $\omega_d : \mathrm{Label}_d \to \mathcal{W}$ evaluates a label on free on-shell jets (`observable`).
- $\mathrm{sh}_d := \omega_d(\mathrm{Label}_d)$ (`shellSector`).
- The free table (`freeTable`, IV.1) has correlations $\langle\omega(a_1)(x_1)\cdots\omega(a_n)(x_n)\rangle_0$.
- Its field space E_d (the glossary's F_d, for the free table) is the space of degree-d reference labels modulo null labels.

Statements: (a) For every $d \geq 0$, let $P \in \mathrm{sh}_d$. Suppose $\langle P(x_0)Q_1(x_1)\cdots Q_n(x_n)\rangle_0 = 0$ for every $n \geq 0$, every choice of degrees $d_1, \ldots, d_n \geq 0$, all $Q_i \in \mathrm{sh}_{d_i}$ and all $x \in \mathrm{Conf}_{n+1}$. Then $P = 0$. Hence:

- the null labels of the free table at degree d are exactly the labels with $\omega_d(a) = 0$;
- ω_d induces a linear isomorphism $E_d \cong \mathrm{sh}_d$. Write $\varpi_d : E_d \hookrightarrow \mathcal{W}$ for the resulting injection.

(b) For every basis $(b_i)_{i\in I}$ of E_d and every $x \neq y$, the map $v \mapsto (\langle\varpi v(x)\,\varpi b_i(y)\rangle_0)_{i\in I}$ is injective.

(c) Fix a Wilson family C, any table S on its register, and a pair satisfying $x_\mu \neq y_\mu$ for each coordinate μ, for example $(x, y) = (0, (1, 1, 1, 1))$. For every $d \geq 0$ there is a column system (IV.1) for E_d. It consists of:

- a complex basis fixed by conjugation, with the unit first, whose elements are images of frame labels with recorded weight and polynomial length;
- the pair readings of (b) against this basis at (x, y), which are injective.

Status: theorem. Part (a) enters the rank comparisons of III.11 and III.12; (c) supplies columns for III.11 and III.2. The embedding ϖ_d is used in VII.1. *Lean:* (a) `no_residual` (with the definitions `residual`, `ProbeFamily`), `free_null`, `exactField`, `writes`; (b) `scanner_inj`; (c) `vane`, `vane_possible`, `RStage`, `recorded`.

Proposition VII.3.5 (covariance of the free table). For every gauge group in the setting above, the free table has the following symmetries:

- its correlations are invariant under the signed coordinate permutations, acting on points and labels;
- conjugating the labels conjugates the correlations.

In addition, $O(4)$ acts on each E_d by complex-linear isomorphisms satisfying $\varpi_d(R\cdot v) = R\cdot(\varpi_d v)$. *Status:* theorem. The proof uses the $O(4)$-covariance of the free kernel, not any reconstruction theorem. *Lean:* `table_spin`, `rideO`, `ride`.

Proposition VII.3.6 (rigidity at degree four; position equations). Parts (a) and (c) assume that $\mathfrak{g}$ is simple and nonabelian. Part (b) holds for any gauge group in the setting above.

Buffers. For a degree d, a *buffer* (`buffers`) is a $\mathbb{C}$-linear map $B : E_d \to E_d$ such that:

- B kills the unit;
- for each $e < d$, B preserves the subspace of fields of degree $\le e$;
- B commutes with conjugation and with the signed coordinate permutations.

Buffers form a finite-dimensional real space Buf_d.

Stationarity. B is *stationary* if

$$\sum_{j=1}^{r} \langle v_1(x_1) \cdots (Bv_j)(x_j) \cdots v_r(x_r) \rangle_0 = 0 \quad \text{for all } r,\ v \in E_d^r,\ x \in \mathrm{Conf}_r.$$

Equivalently, the change of identification $v \mapsto (1 + tB)v$ of degree-d fields leaves all free correlations unchanged to first order in t.

(a) *Rigidity.* Every stationary buffer of degree 4 vanishes. In fact $B \in \mathrm{Buf}_4$ vanishes as soon as both of the following hold, with e the unit vector of coordinate axis 0:

- $\langle Bv \rangle_0 = 0$ for all v;
- $\langle (Bv)(e)\, w(0) \rangle_0 + \langle v(e)\, (Bw)(0) \rangle_0 = 0$ for all v, w.

For $d < 4$, $\mathrm{Buf}_d = 0$.

(b) *Square systems.* Suppose every stationary buffer of degree d vanishes. Then there are finitely many *sections* $(r_i, p_i, f_i)_{1 \le i \le k}$, with $r_i \ge 0$, $p_i \in E_d^{r_i}$ and $f_i \in C_c^\infty(\mathrm{Conf}_{r_i}; \mathbb{C})$, and a real-linear map $g : \mathbb{R}^k \to \mathbb{R}^m$ with $m = \dim_{\mathbb{R}} \mathrm{Buf}_d$, such that the map

$$B \longmapsto g\Big(\mathrm{Re} \int f_i(x) \sum_j \big\langle p_{i1}(x_1) \cdots (Bp_{ij})(x_j) \cdots \big\rangle_0 \, dx \Big)_{i \le k}$$

is an isomorphism $\mathrm{Buf}_d \to \mathbb{R}^m$.

- This map is the derivative in the buffer direction, at the free data and $B = 0$, of the *position chart* $(w, B) \mapsto g(\mathrm{Re}\, w_i((1 + B)p_i))_i$. Here w_i ranges over complex-multilinear forms on $E_d^{r_i}$.
- The system is called *square*, and *clean* (all profiles are test functions).
- For simple nonabelian $\mathfrak{g}$ and $d = 4$ such a system exists.

(c) *Leading position comparison at degree four.* Hypotheses:

- b is a final record over the Wilson family, with limit table S.
- $X = (r_k, U_{k,d}, \xi_k)$ is a scaling sequence: $r_k > 0$ decreases strictly to 0, $\xi_k > 0$, and $U_{k,d} : E_d \to F_d(S)$ is a complex-linear isomorphism for every k, d. Write $U_k = U_{k,4}$.
- Along X, the leading free-Maxwell comparison of VI.2 holds, including eventual compatibility of the frames with the unit, conjugation, signed permutations and degree inclusions.
- Q is a clean square degree-four system. The map φ satisfies the four conditions in VII.3.r: a local analytic implicit solution, convergence of the order-zero lattice data, the formal position identities, and an actual leading comparison along X.

Put $B_k := \varphi(m_Q(k))$, where $m_Q(k)$ are the tested rescaled correlations $\int f(x) \langle U_k v_1(r_k x_1) \cdots \rangle_S \, dx$ on basis fields. Then:

- $B_k \to 0$;
- eventually $1 + B_k$ is invertible, and the corrected frames $U_k(1 + B_k)$ satisfy the position equations of Q exactly;
- eventually $U_k(1 + B_k)$ commutes with conjugation and signed permutations, maps the free unit to the unit of S, and is compatible with the frames of degrees ≤ 3 under the inclusions $E_d \subset E_4$;
- for every clean degree-four system R and every integer $L \ge 2$,

$$m_R^{B_k}(k) - \mathrm{chart}_R(\pi_R^{(0)}(L,k), \varphi(\pi_Q^{(0)}(L,k))) \longrightarrow 0.$$

Here $m_R^{B_k}(k)$ is the tested reading after replacing each field v by $U_k(1 + B_k)v$, and $\pi_R^{(0)}(L,k)$ is the array of order-zero lattice readings on basis fields at spacing L^{-k}.

These conclusions, together with the comparisons for all clean systems described in VII.3.r, give clause (D15) of the detailed main theorem.

Status:

- (a) and (b): theorems.
- (c): theorem under the leading comparison hypothesis (VI.2; discharged by `firstTarget`) and the four conditions on φ (discharged by `calibrate`, VIII.1).
- VIII.1 assembles this with the existence of a clean square system: simplicity of $\mathfrak{g}$ and the leading free comparison imply the full degree-four position conclusion.

Lean: (a) `Rigid.frozen_four`, `Rigid.first_frozen` (unused), definition `Frozen`; (b) `square_of_frozen`, `Conditions`, `chart`, `white`, `square`, `square_four`; (c) `residency`, `Resident`, `FourMeters`, `Crest`, `BoardAttains`, `four_on`, `normalized_four`.

Proposition VII.3.7 (exclusion of finite Wick profiles; assigned to VII.2 and VIII.1).

Definitions:

- $G_\Theta(r)$ is the two-point function of the centered trace Θ of the stress tensor in the selected phase, at the points $\mp\frac{r}{2}e_{\mathrm{t}}$ on the time axis (`traceTwo`).
- A *mode* is (m, j, q, h) with $\mathsf{m} \ge 0$, $j, q \in \mathbb{N}$, and $h : \mathbb{R}^3 \to \mathbb{R}$ continuous with $h(tp) = t^j h(p)$ for every $t > 0$ and $p \in \mathbb{R}^3$.
- For $r > 0$, its integrated kernel is $K_{\mathsf{m},q,h}(r) = \int_{\mathbb{R}^3} E^q\, h(p)\, e^{-rE}(2E)^{-1} dp$, with $E = \sqrt{\mathsf{m}^2 + |p|^2}$.
- A *Wick profile* is a finite sum $W = \sum_\alpha c_\alpha \prod_\beta K_{\alpha\beta}$ with $c_\alpha \in \mathbb{C}$; empty products equal 1 (`Mode`, `Mode.kernel`, `Track`, `wickProfile`; statement.md, Definition 1.15).

(a) Suppose $r^8 \log^2(1/r)\, G_\Theta(r) \to A$ as $r \to 0^+$, with $A > 0$. Then for every Wick profile W and every $\delta > 0$ there is $r \in (0, \delta)$ with $G_\Theta(r) \ne W(r)$. Only this asymptotic law of G_Θ is used.

(b) Under the same asymptotic hypothesis, in the selected reconstructed theory, no finite sum of the corresponding mass-shell contraction integrals reproduces the Wightman two-point function of Θ on every product $\bar f \otimes g$ of Schwartz tests for which $\psi_{f,g} = \overline{\hat f}\,\hat g$ is compactly supported, using the positive-sign Fourier convention. A product of mode kernels corresponds to testing $\psi_{f,g}$ at the sum of their positive-energy on-shell momenta, with the product of their densities $E^q h(p)/(2E)$ (`waveTest_product`). Thus, for every proposed finite sum, some such pair f, g gives different readings.

With $A = 3 \dim \mathfrak{g}/\pi^4$ from VII.2, (a) is clause (T4) of the main theorem. *Status:* theorem. (a) is assigned to VII.2 and (b) to VIII.1. *Lean:* (a) `Wick.euclidean_excludes`, `Wick.no_powerlog`; public form `publicTrace` for `Hierarchy.TraceTest`. (b) `Wick.polynomial_excludes`. The asymptotic law and related polynomial-field exclusions occur in `Headline.TraceExclusions`.

The Euclidean conclusion concerns the single function G_Θ along one axis. The excluded class allows arbitrary complex coefficients and continuous homogeneous numerators, but only finitely many modes and products. It does not include general continuous-mass generalized free fields or infinite Wick expansions.

3.28.2 2. What it uses

Inputs from other units.

1. **III.16 (analytic toolkit).**
   - The free curvature kernel, Wick moments, on-shell jet spaces, label evaluation map and real homogeneous reference frames defined above.
   - The response identity `replyTest`: for $P \in \mathrm{Sol}_j$ and $Q \in \mathrm{Sol}_k$, the covariance of $\langle P, T_j F\rangle(z)$ with $\langle Q, T_k F\rangle(0)$ equals $(-1)^k \langle Q, T_k[R_P](https://github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/z)\rangle$. Here R_P is the curvature field produced by P and T_k is the k-th Taylor coefficient.
   - Used in VII.3.1–VII.3.5.
2. **IV.1.** The free table, its on-shell image and spectator tests; correlation tables, null and quotient field spaces, real conjugation and signed-permutation symmetry, column systems and generic configurations.
3. **VI.2 (free Maxwell stress tensor).** For simple nonabelian $\mathfrak{g}$:
   - $E_d = \mathbb{C} \cdot 1$ for $d < 4$ (`below`, `dim_before`).
   - E_4 has the basis consisting of the unit and the 21 invariant quadratics $\langle F_P, F_Q\rangle$, one for each unordered pair $\{P, Q\}$ of coordinate planes. Its dual functionals are built from the one-point function and the two-point functions at separation e (`stand`, `separates`, `portLine`, `up_to`).
   - The free curvature two-point function at e_t is diagonal in planes, with entries $\kappa_t(P) = -\pi^{-2}$ if $t \in P$ and $+\pi^{-2}$ otherwise (`bare_diagonal`, `coat`).
   - Also from VI.2: `FirstExtension`. Along X:
     - the rescaled correlations in the frames U_k converge in every C^j on compacta of Conf_n to those of the free table, with the stress tensor compared to the Maxwell stress tensor;
     - the centered real symmetric degree-four tensors that transform tensorially under signed permutations and have zero divergence form a one-dimensional real space;
     - derivatives and rotation generators converge.
   - This leading comparison is a hypothesis of VII.3.6(c).
4. **VI.1.** Scaling sequences of radii, stress normalizations and degreewise complex-linear frame changes.
5. **VII.1.** `Rig.zerothProfile`: the order-zero coefficient of the lattice perturbative profile converges to the tested free reading. It supplies the lattice limit in VII.3.r.
6. **VIII.1.**
   - The final construction data, including the selected phase and its reconstructed theory.

- `calibrate` and `credible`: the four implicit-position conditions hold for every clean square system, and the leading lattice comparison holds for every clean system.
- `Reports.of_square` (implicit function theorem).

7. **VII.2.** For VII.3.7:
   - the short-distance law `EuclideanAsymptotic` with constant $3 \dim \mathfrak{g}/\pi^4$;
   - the expansion lemmas `moment_expands`, `Track.moments_expand`, `real_profile`.
8. **V.3 and VII.2.** The selected trace field has a positive-energy spectral measure μ, its two-point function is the Laplace transform of μ, and its Wightman two-point distribution is the corresponding momentum-space integral. Exponential damping is integrable for every positive time (`TraceSpectrum`, `traceSpectrum_exists`). This is used only in VII.3.7(b).
9. **I.1.** The lattice, the Wilson action and the Haar measure (for §4.7).

Standard mathematics.

- Wick's theorem, Gaussian integration by parts, and Fock-space orthogonality of Wick monomials: Glimm–Jaffe, *Quantum Physics*, ch. 6; B. Simon, *The $P(\phi)_2$ Euclidean (Quantum) Field Theory*, ch. I; S. Janson, *Gaussian Hilbert Spaces*, ch. 3.
- Reflection positivity of the free field: Osterwalder–Schrader, CMP 31 (1973) and 42 (1975); Glimm–Jaffe, ch. 6. It is not invoked: strict positivity is proved by explicit diagonalization.
- Hobson's formula for $h(\partial)|x|^{-2}$ in four dimensions, h harmonic and homogeneous: Hobson, *The Theory of Spherical and Ellipsoidal Harmonics*, ch. III; Stein–Weiss, *Introduction to Fourier Analysis on Euclidean Spaces*, ch. IV.
- Invariance of the four-dimensional Maxwell equations under inversion: Bateman (1910), Cunningham (1910).
- Existence of orthogonal bases for symmetric bilinear forms; the real-analytic implicit function theorem; characters of the group $(\mathbb{Z}/2)^4$ of axis reflections.
- Small-r expansions of Laplace integrals with logarithms, as for modified Bessel functions: Watson, *A Treatise on the Theory of Bessel Functions*, ch. III.
- Lattice perturbation theory with Faddeev–Popov gauge fixing: Montvay–Münster, *Quantum Fields on a Lattice*, ch. 3–4.

Imported constants and assumptions.

- Constants used:
  - $\Gamma = 1/(4\pi^2|z|^2)$;
  - $s_j = (j+1)(j+2)(-2)^j/\pi^2$, the code's `strength`;
  - $\pi_j(z) = (-1)^j s_j |z|^{-2j-4}$;
  - $\kappa_t(P) = \mp\pi^{-2}$ (VI.2).
- Hypotheses: the leading free comparison (VI.2) and the four implicit-position conditions (VIII.1) for VII.3.6(c); the short-distance law with $A > 0$ for VII.3.7 (VII.2).
- Simplicity of $\mathfrak{g}$ enters VII.3.6 through VI.2: invariant linear color forms vanish, invariant bilinear color forms are multiples of the inner product, and $\dim \mathfrak{g} > 0$.

3.28.3 3. Strategy

The reflected covariance couples different jet orders. Hobson's formula identifies the field of an order-j multipole with s_j times its Kelvin transform. Translating the polynomial multipole by the negative inverted separation and rescaling its Taylor components produces a triangular change of probes whose response has no Taylor terms below order j at the second point. The reflected

covariance becomes diagonal, with positive pivots π_j. Wick recursion extends this to polynomial monomials, with factorial norms $\alpha! d^\alpha$.

Strict positivity detects every nonzero on-shell label. It identifies the free quotient and gives injective two-point columns. At degree four, the explicit quadratic basis reduces buffer rigidity to a coefficient calculation: two-point stationarity gives weighted antisymmetry, while signed permutations give symmetry or zero. Compactly supported tests convert this rigidity into finitely many equations with invertible derivative. The implicit function theorem solves them along the scaling sequence.

Independently, finite Wick profiles have expansions in integer powers of distance and nonnegative powers of its logarithm. Their real parts cannot have the trace field's leading term $Ar^{-8}(\log(1/r))^{-2}$. Comparison along $r = 2^{-n}$ proves the contradiction. The spectral representation transfers equality of tested Wightman two-point functions to equality of the Euclidean profiles.

3.28.4 4. Steps

3.28.4.1 4.1 The Wick algebra

Lemma VII.3.a (Wick recursion). Proposition VII.3.3. *Proof sketch.*

- By multilinearity it suffices to treat monomials. Occurrences of variables are distinguishable copies.
- Adjoin one occurrence of X_s at x_i. Matchings of the enlarged system correspond bijectively to pairs (a partner occurrence of some X_t at some x_k with $k \neq i$, a matching of the reduced system). Here the reduced system has one copy of X_t removed at x_k.
- The multiplicity of X_t in W_k is the coefficient produced by ∂_s.
- Uniqueness: induct on the number of non-unit entries and on the polynomial structure. The recursion and the normalization $T(1, \dots, 1) = 1$ determine T.
- Rotated and conjugated moments satisfy the same recursion, by covariance and reality of the kernel. This gives $O(4)$-invariance and reality.

Lean: `Pair.remove`, `Pair.remove_mon`, bijections in `Match.lean` and `Occurrences.lean`; `byAnnihilators`, `wick_rotate`, `wick_bar`.

Lemma VII.3.b (two-point adjoint relation). For $p \in \mathrm{Tow}_n$ and $x \neq y$, let $D_p^{x,y}$ be the derivation of $\mathcal{W}$ sending each variable at y to its free covariance with ℓ_p at x (`crash`). For all $P, Q \in \mathcal{W}$ and $q \in \mathrm{Tow}_n$:

- $\langle (P\ell_p)(x)\, Q(y) \rangle_0 = \langle P(x)\, (D_p^{x,y} Q)(y) \rangle_0$;
- $D_p^{x,y}(\mathcal{R}_{x-y}\ell_q) = B_{x-y}(p, q)$, a constant.

Proof sketch. Apply VII.3.a with $n = 2$: the sum over $k \neq i$ has a single term. On linear polynomials, the resulting contraction is exactly $B_{x-y}(p, q)$. *Lean:* `join_crash`, `crash_relation`, `crash_test`.

Lemma VII.3.c (exhaustion by towers). For every $P \in \mathcal{W}$ there is $n_0 \geq 0$ with the following property. For all $n \geq n_0$ and every real basis $(b_i)_{1 \leq i \leq N}$ of Tow_n, one has $P = \Phi_b(p)$ for some $p \in \mathbb{C}[Y_1, \dots, Y_N]$. *Proof sketch.*

- Each variable $X_{(j,a)}$ equals ℓ_r, where r is the Riesz dual of the a-th coordinate functional, placed in order j (`digit_coeff`).
- ℓ_r is a real linear combination of the ℓ_{b_i} once $n > j$.

- Induct on the structure of P: constants, sums, and products with one variable.

Lean: `from_large`, `from_basis`.

3.28.4.2 4.2 The tower Gram form Lemma VII.3.d (Hobson's formula in $\mathbb{R}^4$). Let $S \in \mathrm{Sol}_j$ and $x \neq 0$. Let $S(\partial)$ be the differential operator with symbol S, normalized with Taylor coefficients (a factor $1/j!$). Then

$$S(\partial)\Gamma(x) = (-2)^j\,|x|^{-2j}\,S(x)\,\Gamma(x).$$

Proof sketch. Induct on j. In the step from j to $j+1$, take $S \in \mathrm{Sol}_{j+1}$ and let $S_a \in \mathrm{Sol}_j$ be its partial contraction with e_a. Taylor normalization gives $S(\partial) = \frac{1}{j+1}\sum_a \partial_a S_a(\partial)$ (`Sol.freeze`). Apply the induction hypothesis and differentiate:

- the term with the divergence of the partial contractions vanishes, because S is harmonic (`Sol.freeze_trace`);
- the radial derivative gives $-2(j+1)|x|^{-2(j+2)}\sum_a x_a S_a(x)$, and $\sum_a x_a S_a(x) = S(x)$ (`freeze_diagonal`). The factor $j+1$ cancels, leaving the required factor -2.

Lean: `shell_green`.

Lemma VII.3.e (the field of a multipole is a Kelvin transform). For $P \in \mathrm{Sol}_j$ and $z \neq 0$, the curvature field R_P produced by P satisfies

$$R_P(z) = s_j\,|z|^{-4}\,(\mathcal{R}_z \otimes \mathcal{R}_z)\,P(\check{z}) = s_j\,|z|^{-2j-4}\,(\mathcal{R}_z \otimes \mathcal{R}_z)\,P(z), \qquad \check{z} = z/|z|^2.$$

That is, $R_P = s_j\,I^*P$, the pull-back of the 2-form P under the inversion I, whose differential is $|z|^{-2}\mathcal{R}_z$. *Proof sketch.*

- The curvature kernel is a curl-curl of Γ. Its contraction with P is $(R_P)_{im} = 2(\partial_m(D_P)_i - \partial_i(D_P)_m)$, where $(D_P)_i = \sum_a \partial_a(P_{ai}(\partial)\Gamma)$. By VII.3.d,

$$D_P(x) = -2(j+1)(-2)^j(4\pi^2)^{-1}|x|^{-2(j+2)}\,\iota_x P(x)$$

(`divP_value`).

- Its curl is computed with $\partial_i(\iota_x P)_m - \partial_m(\iota_x P)_i = (j+2)P_{im}$. This uses Bianchi and Euler; see `radial_curl`.
- Collecting terms gives $s_j|x|^{-2(j+2)}(P_{im} - 2|x|^{-2}(x_i(\iota_x P)_m - x_m(\iota_x P)_i))$, which is the reflected tensor.

Lean: `curl_kelvin`, `reply_kelvin`, `reply_radial`, `strength_sign`.

Lemma VII.3.f (shifted towers). For $x \neq 0$, $j < n$ and $P \in \mathrm{Sol}_j$, define $\sigma_x(P) := (Q_0, \dots, Q_j, 0, \dots) \in \mathrm{Tow}_n$ with

$$Q_k := (s_j/s_k)\,T_k[P](-\check{x}).$$

Here $T_k[P](https : //github.com/scottnarmstrong/yang - mills - informalization/blob/main/outline/units/a) \in \mathrm{Sol}_k$ is the order-k Taylor coefficient of P at a. In particular $Q_j = P$. For every $y \neq 0$, the total field of the shifted tower is

$$\sum_k R_{Q_k}(y) = s_j |y|^{-4} (\mathcal{R}_y \otimes \mathcal{R}_y) P(\breve{y} - \breve{x}).$$

It vanishes to order j at $y = x$:

$T_k[\sum R_{Q_i}](https : //github.com/scottnarmstrong/yang-mills-informalization/blob/main/outline/units/x)$

Proof sketch.

- Since $\sum_k T_k[P](https : //github.com/scottnarmstrong/yang - mills - informalization/blob/main/outline/units/a)(w) = P(w + a)$ and R is linear, VII.3.e gives the formula for the total field.
- Along the line $y = x + tv$: $\breve{y} - \breve{x} = t\,|x + tv|^{-2}(\mathcal{R}_x v - t|v|^2 \breve{x})$, which is smooth in t.
- By homogeneity the field is t^j times a smooth function whose value at $t = 0$ is $s_j |x|^{-2(j+2)} (\mathcal{R}_x P)(v)$.
- Taylor coefficients of symmetric jets are determined by their diagonal values.

Lean: `shifts`, `towards`, `field_expands`, `fields_shift`, `signal_kelvin`, `invert_line`, `signal_lower`, `signal_top`, `newLeg`.

Lemma VII.3.g (diagonal Gram form in shifted towers). For $x \neq 0$, $i, m < n$, $P \in \mathrm{Sol}_i$ and $Q \in \mathrm{Sol}_m$:

$$B_x(\sigma_x P, \sigma_x Q) = \begin{cases} \pi_i(x) \langle P, Q \rangle & i = m, \\ 0 & i \neq m. \end{cases}$$

Hence $B_x(\sigma_x p, \sigma_x q) = \sum_i \pi_i(x) \langle p_i, q_i \rangle$, where $\sigma_x p := \sum_i \sigma_x(p_i)$. *Proof sketch.*

- By the response identity (III.16), for $q \in \mathrm{Sol}_k$ placed in order k: $B_x(\sigma_x P, q) = (-1)^k \langle \mathcal{R}_x q, \; T_k[\sum R_{Q_i}](https : //github.com/scottnarmstrong/yang - mills - informalization/blob/main/outline/units/x) \rangle$.
- By VII.3.f this is 0 for $k < i$, and $(-1)^i s_i |x|^{-2(i+2)} \langle \mathcal{R}_x q, \mathcal{R}_x P \rangle = \pi_i(x) \langle P, q \rangle$ for $k = i$, since $\mathcal{R}_x$ is orthogonal.
- If $m < i$, expand $\sigma_x Q$ into components of orders $\leq m$; each gives zero. The case $m > i$ follows by symmetry.
- If $m = i$, only the top component Q of $\sigma_x Q$ contributes.

Lean: `slide_sum`, `slide_below`, `slide_top`, `newLeg_below`, `newLeg_diag`, `shiftAll_gram`.

Lemma VII.3.h (VII.3.2(a)). For every $z \neq 0$ and $n \geq 0$, B_z is symmetric and positive definite on Tow_n. *Proof sketch.*

- Symmetry: the kernel is symmetric under exchanging the two probes and replacing z by $-z$. It is also $O(4)$-covariant (`leg_cov`). Apply this with $\mathcal{R}_z z = -z$ and $\mathcal{R}_z^2 = 1$.
- Positivity:

– σ_z is injective, since $B_z(\sigma_z p, \sigma_z p) > 0$ for $p \neq 0$ by VII.3.g;
– hence it is bijective, by finite dimension (it is also visibly block unitriangular);
– so $B_z(p,p) = \sum \pi_j \|p'_j\|^2 > 0$ for $p = \sigma_z p' \neq 0$.

Lean: `scan_sym`, `towerB_sym`, `shiftAll_pos`, `shiftDual`, `towerB_pos`.

3.28.4.3 4.3 From towers to all Wick polynomials Lemma VII.3.i (Fock orthogonality). Let b be as in VII.3.2(b), and let $\alpha, \beta \in \mathbb{N}^N$. Then, without conjugation,

$$\left\langle \Phi_b(Y^\alpha)(x)\, (\mathcal{R}_{x-y}\Phi_b(Y^\beta))(y) \right\rangle_0 = \delta_{\alpha\beta}\, \alpha!\, d^\alpha.$$

Proof sketch.

- Since the b_k are orthogonal, VII.3.b gives $D^{x,y}_{b_i}\, \mathcal{R}\Phi_b(q) = d_i\, \mathcal{R}\Phi_b(\partial_{Y_i} q)$ (`crash_basis`).
- Thus transferring one factor Y_i from the first insertion differentiates the second polynomial and multiplies by d_i (`aligned_step`).
- Induct on α, one factor at a time (`tally_step`).
- The base cases are the unit-unit reading 1 and the zero reading of the unit against a nonconstant monomial (`aligned_unit`, `aligned_one_right`).

This is the factorial orthogonality of Wick monomials. *Lean:* `aligned_script`.

Lemma VII.3.j (VII.3.2(b)). The monomials $\Phi_b(Y^\alpha)$ are real (`bar_mono`). Sesquilinear extension of VII.3.i then gives $\langle \Phi_b p, \Phi_b q \rangle_{x,y} = \sum_\alpha p_\alpha \overline{q_\alpha} \alpha! d^\alpha$. *Lean:* `duplex_script`.

Lemma VII.3.k (VII.3.1). *Proof sketch.*

- (a) Given $P \neq 0$, take n from VII.3.c. Take a B_{x-y}-orthogonal basis of Tow_n; it exists for a symmetric form, and $d_i > 0$ by VII.3.h (`upright`). Then $P = \Phi_b(p)$ with $p \neq 0$, and $\langle P, P \rangle = \sum |p_\alpha|^2 \alpha! d^\alpha > 0$.
- (b) Write P and Q in a common basis; the formula of VII.3.j is Hermitian.
- (c) Apply (a) to $\sum a_i V_i$.

Lean: `reflected_positive`, `script_pos`, `reflected_symm`, `reflected_pd`.

3.28.4.4 4.4 Faithfulness and covariance of the free table Lemma VII.3.l (VII.3.4(a)). *Proof sketch.*

- sh_d is stable under conjugation and under $O(4)$, because ω_d intertwines the label operations with $\bar{}$ and with substitution (`sector_bar`, `sector_rot`, `observe_rot`, `observe_bar`).
- So for $P \in \mathrm{sh}_d$ the one-element family $Q_1 = \mathcal{R}_{x-y}\bar{P}$ at $(x,y) = (0, e_{\mathrm{t}})$ is admissible. Its test value is $\langle P, P \rangle_{0,e_{\mathrm{t}}}$ (`point_probe`), which is positive by VII.3.1 if $P \neq 0$.
- The description of the null labels then follows from the identity between free-table readings and these tests (IV.1).

Lean: `no_residual`, `free_null`, `exactField`.

Lemma VII.3.m (VII.3.4(b), (c)). *Proof sketch of (b).*

- Suppose $v \neq w$ have the same readings against a basis. By bilinearity, $u = v - w$ has zero reading against every field.
- In particular u has zero reading against $\mathcal{R}_{x-y} \cdot \bar{u} \in E_d$ (`faceTest_eq`).

- That reading is $\langle \varpi u, \varpi u \rangle_{x,y} > 0$, a contradiction.

Proof sketch of (c).

- Extend the unit to a complex basis chosen among the images of frame labels. These images are fixed by conjugation (`labeled`, `frameRows`).
- Its elements carry their weighted degree, with an order-j jet weighted $j+2$, and their length.
- The pair readings are injective by (b).

Lean: `scanner_inj`, `recorded`, `vane`.

Lemma VII.3.n (VII.3.5). *Proof sketch.*

- ω intertwines the label action of $O(4)$ with substitution, via the identity relating rotation of curvature jets to rotation of potential jets (`radial_turn`). It intertwines label conjugation with $^-$.
- The free moments are $O(4)$-invariant and real (VII.3.3).
- The $O(4)$ action on E_d is transported through $E_d \cong \mathrm{sh}_d$.

Lean: `table_spin`, `ride`.

3.28.4.5 4.5 Degree four: rigidity and the position equations Lemma VII.3.o (coordinates on E_4). Assume that $\mathfrak{g}$ is simple and nonabelian. Let e be the unit vector of coordinate axis 0. The basis and dual functionals below are from VI.2.

Basis. Quadratic products below are Wick-ordered. The space E_4 has the basis $\{\star_c\}$, $c \in \{\emptyset\} \cup \{\text{faces}\}$, where:

- $\star_\emptyset = 1$;
- $\star_{\{P,Q\}} = \langle F_P, F_Q \rangle$, the invariant contraction of two curvature components, for $P \le Q$ two of the six coordinate planes;
- so $\dim E_4 = 22$.

Dual functionals. The dual basis is $\lambda_\emptyset(v) = \langle v \rangle_0$ and

$$\lambda_{\{P,Q\}}(v) = \frac{m(P,Q)}{\dim \mathfrak{g}\ \kappa(P)\kappa(Q)} \left\langle v(e)\, \langle F_P, F_Q \rangle(0) \right\rangle_0,$$

with $m(P,Q) = \frac{1}{2}$ if $P = Q$ and 1 otherwise, and $\kappa(P) = \kappa_0(P) = \mp\pi^{-2}$.

Two-point formulas. For a quadratic index $c = \{P,Q\}$, put $\rho(c) = \dim \mathfrak{g}\, \kappa(P)\kappa(Q)/m(P,Q) \neq 0$. For every $v \in E_4$:

- $\langle v(e) \star_c (0) \rangle_0 = \rho(c)\lambda_c(v)$;
- if $\lambda_\emptyset(v) = 0$, then also $\langle \star_c(e) v(0) \rangle_0 = \rho(c)\lambda_c(v)$.

Lean: `price_tag`, `backwards`, `rate_ne`, `big_weight`; inputs `stand`, `separates`.

Lemma VII.3.p (VII.3.6(a)). Let $B \in \mathrm{Buf}_4$ satisfy the one- and two-point conditions of VII.3.6(a). Put $\mathrm{coeff}(c,c') = \lambda_c(B \star_{c'})$. Then every coefficient vanishes, and so $B = 0$. *Proof sketch.*

- **Unit row and column.**
  - $\mathrm{coeff}(\emptyset, \cdot) = 0$ by the one-point condition (`alone`).

  - $\operatorname{coeff}(\cdot, \emptyset) = 0$ because $B1 = 0$.
- **Antisymmetry.** The two-point condition with $v = \star_c$, $w = \star_{c'}$ gives, by VII.3.o,

$$\rho(c')\operatorname{coeff}(c', c) + \rho(c)\operatorname{coeff}(c, c') = 0$$

(`cancel`).

- **Different reflection characters.** Reflection of axis a multiplies $\star_{\{P,Q\}}$ by $(-1)^{[a\in P]+[a\in Q]}$ (`draw_mask`). If two quadratic indices have different characters of $(\mathbb{Z}/2)^4$, some reflection separates them, and equivariance of B gives coeff $= 0$ (`unpaint`).
- **Equal reflection characters.** A finite check over the 21 faces (`havePath`, decided by computation) finds an involutive axis permutation exchanging $\star_c$ and $\star_{c'}$, up to a common sign. It also shows the two faces are both diagonal or both off-diagonal.
  - Equivariance then gives the symmetry $\operatorname{coeff}(c', c) = \operatorname{coeff}(c, c')$ (`reflect_path`).
  - It also gives equal rates, $\rho(c) = \rho(c')$ (`common_rate`).
  - Combined with the antisymmetry: $2\rho(c)\operatorname{coeff}(c, c') = 0$.
- **Degrees $d < 4$:** $E_d = \mathbb{C} \cdot 1$, and buffers kill the unit.

Lean: `coeff_barren`, `frozen_four`, `first_frozen`, `orient_transfer`, `wearer`.

Lemma VII.3.q (VII.3.6(b)). *Proof sketch.*

- The *feedback* of a section (r, p, f) is the B-derivative at 0 of $\operatorname{Re}\int f\langle(1+B)p_1\cdots\rangle_0$ (`feedback`). It is a linear functional on Buf_d.
- If all feedbacks vanish at B, then B is stationary (`detected`):
  - the profiles f and if give the real and imaginary parts (`feedback_twist`);
  - the integrand is continuous on Conf_r (`visible_cont`), so it vanishes pointwise.
- The span of all feedbacks lies in the finite-dimensional dual of Buf_d. Pick k sections whose feedbacks span the same subspace; their joint kernel is 0 (`finite_detectors`).
- Take a left inverse g' of the resulting injection $\mathrm{Buf}_d \to \mathbb{R}^k$, and put $g = H \circ g'$ with $H : \mathrm{Buf}_d \cong \mathbb{R}^m$ a coordinate isomorphism.
- The chart derivative is then H (`chart_deriv_formula`).

Lean: `square_of_frozen`.

Lemma VII.3.r (VII.3.6(c)).

The data.

- $m_Q(k)$ is the tested correlation of the limit theory at scale r_k in the frames U_k, on basis fields (`measured`, `exactMean`).
- The leading free comparison, with derivative order 0 and no stress slots, gives $m_Q(k) \to w_Q$, the free tested readings (`measured_tends`, `exact_tape`).

The hypothesis on φ. It consists of four conditions:

- φ is a real-analytic implicit solution of $\mathrm{chart}_Q(w, \varphi(w)) = \mathrm{chart}_Q(w_Q, 0)$ near w_Q, with $\varphi(w_Q) = 0$. Locally the solution set is the graph of φ (`Reports`).
- For every integer $L \geq 2$, $\pi_Q^{(0)}(L, m) \to w_Q$ as $m \to \infty$.

- For every $L \geq 2$ and $p, j \geq 0$, eventually in m, all order-j coupling derivatives at 0 of the position equations evaluated on the order-p lattice truncation and its image under φ equal those of the constant free value. The threshold may depend on L, p, j.
- Some buffers $A_k \to 0$ have $1 + A_k$ invertible and solve the position equations eventually along X. For every clean degree-four system R and $L \geq 2$, their corrected tested readings differ by $o(1)$ from $\mathrm{chart}_R(\pi_R^{(0)}(L,k), \varphi(\pi_Q^{(0)}(L,k)))$.

Proof sketch.

- $B_k = \varphi(m_Q(k)) \to \varphi(w_Q) = 0$.
- By local uniqueness of the implicit solution, $B_k = A_k$ eventually. This transfers exact solvability and the leading comparison from A_k to B_k.
- The required symmetries and unit identity follow from the eventual compatibility of U_k in VI.2 and the fact that buffers commute with conjugation and signed permutations and kill the unit (`adjusted_own`, `adjusted_unit`).
- Compatibility with degrees ≤ 3: $1 + B$ is the identity on $E_d \subset E_4$ for $d < 4$, because $E_d = \mathbb{C} \cdot 1$ (`motor_small`).

The full degree-four position conclusion consists of:

- for every degree, every clean system and every $L \geq 2$, the uncorrected tested data minus the order-zero lattice data at spacing L^{-k} tend to 0; every clean square system admits a map φ satisfying the four conditions above;
- a clean square degree-four system Q and such a map φ for which VII.3.6(c) holds.

Lean: `residency`, `adjusted`, `testedMean`, `finite`, `readJet`; packaged by `four_on` (VIII.1).

3.28.4.6 4.6 Exclusion of Wick profiles (VII.3.7) Lemma VII.3.s (expansions; VII.2). Let $\mathcal{B}$ be the real span of $r^a(\log r)^k$ for $a \in \mathbb{Z}$, $k \in \mathbb{N}$, on $r > 0$. Every integrated mode kernel, and each of the real and imaginary parts of every Wick profile, has an expansion to all orders in $\mathcal{B}$: for every $N \in \mathbb{Z}$ there is $u \in \mathcal{B}$ with $f - u = O(r^N)$ as $r \to 0^+$. *Proof sketch.*

- **Reduction to radial moments.** In polar coordinates, $K_{\mathsf{m},q,h} = \frac{1}{2}(\int_{S^2} h) M_{j,q}$, where

$$M_{j,k}(s) = \int_0^\infty t^{j+2} E^{k-1} e^{-sE} dt, \qquad E = \sqrt{\mathsf{m}^2 + t^2}.$$

- **Relations among the moments.**
  - $M'_{j,k} = -M_{j,k+1}$;
  - $M_{j,k+2} = \mathsf{m}^2 M_{j,k} + M_{j+2,k}$;
  - the Euler identity $sM_{j+2,0} = (j+3)M_{j,1}$, from integration by parts.
  - Together they give $(s^{j+3} M_{j,1})' = -\mathsf{m}^2 s^{j+3} M_{j,0}$.
- **Bootstrap.**
  - Start from $M_{j,0} = O(s^{-j-2})$.
  - If $M_{j,0}$ has an expansion with remainder $O(s^k)$, $k \geq -j-2$, integrating the weighted equation gives one for $M_{j,1}$ with remainder $O(s^{k+1})$.
  - Integrating $M'_{j,0} = -M_{j,1}$ then improves the known remainder for $M_{j,0}$ by at least one power. When the remainder to be integrated has order s^{-1}, first weaken it to $O(s^{-2})$; this still suffices for the induction.

  - The explicit power-log terms have primitives in $\mathcal{B}$; integrating $s^{-1}(\log s)^k$ raises the logarithmic power to $k+1$.
- Products and sums preserve the class.

Lean: `Expands`, `euler`, `derived`, `basic_expands`, `moment_expands`, `Track.moments_expand`, `real_profile`, `primitives`.

Lemma VII.3.t (no power-log law; VII.2). Let $f : (0,\infty) \to \mathbb{R}$ and $A \in \mathbb{R}$. Suppose $f = u + O(1)$ with $u \in \mathcal{B}$ as $s \to 0^+$, and $s^8 \log^2(1/s) f(s) \to A$. Then $A = 0$. *Proof sketch.*

- The $O(1)$ part contributes $o(1)$.
- Collect equal power-log terms and discard zero coefficients. Along $s = 2^{-n}$, $s^8 \log^2(1/s) u(s) = \sum_t c_t \rho_t^n n^{k_t+2}$, with $c_t \neq 0$, $\rho_t = 2^{-(8+a_t)}$ and the pairs (ρ_t, k_t) distinct. The empty sum has limit 0.
- Let t^* be the lexicographically largest term.
  - If $\rho_{t^*} < 1$, every term tends to 0, so $A = 0$.
  - Otherwise its positive growth factor $\rho_{t^*}^n n^{k_{t^*}+2}$ diverges and dominates those of all remaining terms. Dividing by this factor, the left side tends to 0 while the right side tends to $c_{t^*} \neq 0$, a contradiction.

Lean: `no_powerlog`.

Lemma VII.3.u (VII.3.7). *Proof sketch.*

- (a) If $G_\Theta = W$ on $(0,\delta)$, apply VII.3.t to $\operatorname{Re} W$ and the real part of the hypothesis. This gives $A = 0$, a contradiction.
- (b) Write $G_\Theta(s) = \int e^{-sp^0} d\mu(p)$, using the spectral measure μ of Θ.
  - Every compactly supported momentum test ψ has the form $\overline{\hat{f}}\hat{g}$: choose $|\hat{f}| = 1$ on its support and $\hat{g} = \hat{f}\psi$. Thus the hypothesized product-test identities give equality on all compactly supported momentum tests.
  - Approximate e^{-sp^0} by compactly supported Schwartz tests dominated by it (`approximate`).
  - Exponential damping is integrable for the trace spectral measure and for every mode product. Dominated convergence on both sides identifies G_Θ with the profile of the description on $(0,\infty)$ (`reads`, `identifies_euclid`).
  - Then apply (a).
- The public form uses the same mode kernels and identifies the asymptotic constant with $3 \dim \mathfrak{g}/\pi^4$.

Lean: `euclidean_excludes`, `polynomial_excludes`, `publicTrace`.

3.28.4.7 4.7 The formal lattice perturbation theory Definitions VII.3.v. Fix a finite periodic four-dimensional lattice. The perturbation theory uses inversion-odd link potentials,

$$A(-x - e_\mu, \mu) = -A(x, \mu).$$

Its definitions are:

Gauge-fixed linearization and Gaussian measure.

The links are $U = \exp(A)$. The action S is the finite-regulator Wilson term plus the quadratic Feynman gauge-fixing term $\frac{1}{4}\sum_x |\operatorname{div} A(x)|^2$ (`action`).

The Haar Jacobian J and the normalized Faddeev–Popov determinant FP are smooth at 0, with $J(0) = \mathrm{FP}(0) = 1$. The latter is the determinant of the gauge-condition derivative on inversion-even site parameters fixed to zero at the root, divided by its flat value (`Orb.FP`). The matrix determinant, logarithmic determinant and inverse estimates come from the matrix calculus listed in the header.

The centered Gaussian probability has covariance equal to the inverse Hessian of this gauge-fixed action at 0 (`Orb.data`).

Formal series in the coupling.

Write $S_j(A) = (j!)^{-1} D^j S(0)[A, \dots, A]$. After the rescaling $A \mapsto gA$, the formal weight is

$$w(g, A) = \exp\left(-\sum_{i \geq 1} g^i S_{i+2}(A) \right) (J\mathrm{FP})(gA),$$

where the last factor denotes its formal Taylor series at $g = 0$.

$\mathbb{E}_0$ is the Gaussian expectation, taken coefficientwise.

The normalized expectation is $\langle f \rangle_{\mathrm{pert}} = \mathbb{E}_0(wf)/\mathbb{E}_0(w)$ (`Orb.weight`, `Orb.expect`, `Orb.normalize`).

Every coefficient is continuous with at most polynomial growth, hence Gaussian-integrable. Since $w(0, A) = 1$, the denominator has constant coefficient 1 and is invertible as a formal series.

Lattice readers of labels. Substitute the lattice jet series plus iZ into each polynomial label, where Z is an independent centered real Gaussian jet with the prescribed covariance, and average in Z separately within each insertion. Multiply these averaged insertions and take the normalized formal expectation. The separate auxiliary averages implement Wick subtraction without auxiliary contractions between different insertions (`Full.column`, `Full.amplitude`).

Profiles. The profile is the lattice Riemann sum

$$\sum_o h^{4r} f(\iota_h(o)) \operatorname{amplitude}(o), \qquad h = L^{-m}, \ \text{side } 4L^{m^2+m+1},$$

where o ranges over r lattice sites and ι_h takes the chosen integer lifts relative to the quarter-side origin and multiplies them by h (`Full.profile`).

The order-zero profiles enter VII.3.r through the convergence theorem of VII.1. Their higher coefficients enter the formal position identities and the degree-four parity assertion assembled in VIII.1.

`Orb.finite_agreement` shows that retaining the interaction series, density series and insertions only through coupling order k preserves every normalized coefficient through order k. The required action Taylor coefficients extend through order $k + 2$.

3.28.5 5. Where the difficulty lies

The main calculation is the simultaneous diagonalization of all jet orders (VII.3.d–h). The Taylor normalization and the sum over ordered form indices determine the Kelvin constants and positive pivots; changing either convention changes the formulas. Recentring at the inverted separation eliminates all lower Taylor coefficients, so positivity follows from a triangular change of basis.

At degree four, the two-point form is indefinite before reflection. Rigidity uses its nonzero diagonal rates together with signed-permutation symmetry: equal reflection characters force symmetric coefficients and equal rates, while the two-point condition forces weighted antisymmetry. Full rotational equivariance of the buffer is unnecessary.

The position comparison is leading-order only. Its formal identities at every coupling order follow from the local implicit equation at a finite regulator; they give no all-order comparison with the interacting theory. The free quotient and its injective columns likewise give no interacting rank bound without the comparison estimates of III.11.

The non-freeness argument is independent of Gram positivity. Its analytic input is the power-log expansion of the massive Laplace moments, including the remainder estimates needed for integration. It excludes inverse-logarithmic leading behavior; logarithms with nonnegative powers already occur in free massive kernels.

3.28.6 6. Notation

object	code name	notation
on-shell j-jets (homogeneous Maxwell solutions)	`Sol`	Sol_j
towers of multipoles	`ProbeTower`	Tow_n
Wick polynomials	`WickPol`, `Slot`	$\mathcal{W} = \mathbb{C}[X_s]$
linear polynomial of a tower	`code`, `digit`	ℓ_p
encoding in a basis of a tower	`encode`	Φ_b
free moment	`Wick.moment`, `join`	$\langle P_1(x_1)\cdots\rangle_0$
reflection in $z^\perp$	`face`, `pullSol`, `solO`	$\mathcal{R}_z$
reflected pairing	`duplex`, `aligned`	$\langle P, Q\rangle_{x,y}$
tower Gram form	`towerB`, `scan`	B_z
inversion; inverted point	`inversePoint`, `rr`	$I(z) = \check{z} = z/\lvert z\rvert^2$
field of a multipole	`reply`	$R_P = s_j I^* P$
Kelvin constant; pivot	`strength`, `price`	s_j, $\pi_j(z)$
shifted tower	`shifts`, `newLeg`, `shiftAll`	σ_z
two-point annihilator	`crash`, `extractVar`	$D_p^{x,y}$, $\partial_s^{y,y'}$
Fock weight	`tally`	$\alpha!\, d^\alpha$
free-field map of labels; shell sector	`observable`, `shellSector`	ω_d, sh_d
field space of the free table; embedding	`(freeTable F).Field d`, `Position.E`, `writes`	E_d, ϖ_d
buffers; corrected frame	`buffers`, `motor`, `adjusted`	Buf_d, $U_k(1+B)$
stationary buffers vanish	`Frozen`, `stationery`	rigidity
basis and dual basis of E_4	`stand`, `star`, `portLine`, `coeff`	$\star_c$, λ_c, $\mathrm{coeff}(c, c')$

object	code name	notation
rate	`rate`	$\rho(c)$
test system; free readings; chart	`Conditions`, `white`, `chart`, `square`	Q, w_Q, chart_Q
measured tested readings	`measured`, `exactMean`, `testedMean`	$m_Q(k)$
leading position comparison	`FourMeters`, `Resident`, `Crest`, `BoardAttains`	(D15)
mode, track, Wick profile	`Mode`, `Track`, `wickProfile`, `atPoint`	$K_{\mathsf{m},q,h}$, W
span of $r^a(\log r)^k$; full expansion	`board`, `Expands`, `AtOrder`	$\mathcal{B}$
trace two-point function	`traceTwo`, `euclidean`	$G_\Theta(r)$
perturbative weight, expectation, normalization	`Orb.weight`, `expect`, `normalize`	w, $\mathbb{E}_0$, $\langle\cdot\rangle_{\mathrm{pert}}$
lattice perturbative profile	`Full.profile`, `Position.finite`	$\pi_Q^{(n)}(L,m)$

3.28.7 7. What remains unexplained

1. **Image versus intrinsic invariant algebra.** The statements use $\mathrm{sh}_d = \omega_d(\mathrm{Label}_d)$ exactly. An intrinsic characterization as all gauge-invariant on-shell polynomials of weight at most d would require a separate surjectivity argument; no such characterization is needed here.
2. **Interacting rank estimates.** The free columns enter the construction of determinant trials in III.11, and the free-null description enters the upper-rank comparison. The estimates that turn these into interacting rank bounds and compatible interacting field spaces are inputs from III.11 and III.12; their proofs are outside this unit's verification.
3. **Continuum Wick subtraction.** The auxiliary imaginary Gaussian average in VII.3.v is the exact lattice definition. Its comparison with continuum Wick-ordered correlations uses VII.1's order-zero profile theorem. The underlying regulator-removal and Wick-subtraction estimates are accepted as VII.1 inputs; their proofs are not checked here.
4. **Further non-freeness statements.** The larger trace-exclusion package also contains exclusions involving embeddings into Maxwell Fock spaces and invariant averages. Their proofs are not covered by VII.3.7, which treats finite mass-shell contraction profiles and their tested two-point identities.

3.29 VII.4. Operator product expansion and contact extensions

Part: VII, identification as Yang–Mills. **Lean:** `YM/OPE`. Its folders, by role:

- prescription and Reader: `Splitting`, `Orbits`;
- bounds in the ground algebra: `Eye`, `Cards`, `Exchange`;
- lattice cluster estimate: `Sweep`, `Replicas`, `Grouping`, `Columns`, `Prices`;
- passage to arbitrary configurations: `Schemes`, `Interpolate`, `Height`, `Rate`;
- trees: `Nesting`, `Trees`, `Single`, `Subpower`.

Also `YM/Next/PHASE`, `YM/Contacts` (`Normal`, `Score`, `Layers`, `Harvest`, `Solstice`, `Transport`),

`YM/MultipleContact` and `YM/SharpGram`. The public statements are `publicOPE` and `contactDistribution`.

Size: about 10,200 lines in 2,035 declarations assigned to this unit:

- `OPE` about 4,600, mostly in `Grouping`, `Replicas`, `Columns`, `Trees`, `Sweep`;
- `Next/PHASE` 840;
- `Contacts` about 3,400;
- `MultipleContact` about 290;
- `SharpGram` about 250.

Conventions.

- G is the compact gauge group, with simple Lie algebra, and $\mathcal{W}$ is the Wilson family of the construction. Its lattice levels $k = 0, 1, 2, \ldots$ have spacings $a_k \downarrow 0$. Larger k means finer. The sequence is *gentle*: $\log(a_k/a_{k+1})/\log(1/a_k) \to 0$.
- $I_{N,p}(v, x)$ is the renormalized lattice observable of a lattice label (token) v at the site x. $\|F\|_{\mathcal{T}_k}$ is the chessboard root of a lattice function F in the centred tiling of scale a_k. Both follow IV.1.
- S is the limit table along the refined extraction order. It has field spaces F_d and the inclusions $\iota_{m,n} \colon F_m \to F_n$ $(m \le n)$ of the field algebra. A label list is $a = (a_i)_{i \in I}$ with $a_i \in F_{d_i}$, and its total degree is $w(a) = \sum_i d_i$.
- $\mathcal{A}$ is the ground algebra of V.1: the commutative C*-algebra generated by the compressed slab operators $M(u)$ of separated words u. Its vacuum is Ω, its Gelfand spectrum is the phase space X with measure ν of full support, and its characters are χ_ξ.
  - $S^\xi(u) = \chi_\xi(M(u))$ and $S^{\mathrm{mix}}(u) = \langle \Omega, M(u)\Omega \rangle$.
  - Hence, for $A \in \mathcal{A}$ and $c \ge 0$: $\|A\| \le c$ if and only if $|s(A)| \le c$ for every state $s \in X \sqcup \{\mathrm{mix}\}$ (`glance_contract`, `Cards.pressure`).
- Clusters and clearance follow Definition 1.13 of the statement.
  - (x, o) is an (r, M, η)-*cluster* if $|x_i - o| \le Mr$ and $|x_i - x_{i'}| \ge \eta r$ for $i \ne i'$.
  - $y \in (\mathbb{R}^4)^J$ is *c-clear of* o if $|o - y_j| \ge c$ and $|y_j - y_{j'}| \ge c$ for $j \ne j'$.
  - $S^s(a, x; b, y)$ and $M(a, x; b, y)$ denote the concatenated word.
- "For all sufficiently small r" means: there is $r_0 > 0$, depending only on the data quantified before the phrase, such that the assertion holds for all $r \in (0, r_0)$ and all configurations and states quantified after it.

3.29.1 1. What this unit proves

Proposition VII.4.1 (separated OPE; properties (O1)–(O2) of the main theorem).

Setting.

- $(\mathcal{W}, S, F)$ is a family, limit table and final record satisfying the RG hypotheses of Remark VII.4.7, denoted (RG).
- $\mathcal{P}$ is any coherent prescription (VII.4.6).
- $\mathcal{R}$ is the Reader built from $\mathcal{P}$ and the rotation and reality structures of F (Lemma VII.4.c). It provides strata $E_j \subset F_j$, bases fixed by conjugation $(O_{j,\kappa})_{\kappa \in \mathrm{Ch}_j}$ of E_j, and coefficients $c_{j,\kappa}(a; x, o)$.

Conclusions.

- **(O1)** Fix a finite I, labels a, $M \geq 0$, $\eta > 0$, $\varepsilon > 0$, j and $\kappa \in \mathrm{Ch}_j$. For all sufficiently small r and every (r, M, η)-cluster (x, o): $|c_{j,\kappa}(a; x, o)| \leq r^{\,j-w(a)-\varepsilon}$.
- **(O2)** Fix finite I, J, labels a, b, $M \geq 0$, $\eta, c, \varepsilon > 0$ and $D \in \mathbb{N}$. For all sufficiently small r, every (r, M, η)-cluster (x, o) and every y that is c-clear of o:
  - the points of (x, y) are pairwise distinct, and so are those of (o, y);
  - the remainder satisfies

$$\Big\| M(a, x; b, y) - \sum_{j \leq D} \sum_{\kappa \in \mathrm{Ch}_j} c_{j,\kappa}(a; x, o)\, M((j, O_{j,\kappa}), o;\, b, y) \Big\|_{\mathcal{A}} \leq r^{\,D+1-w(a)-\varepsilon};$$

  - hence $\big| S^s(a, x; b, y) - \sum_{j \leq D} \sum_{\kappa} c_{j,\kappa}(a; x, o)\, S^s((j, O_{j,\kappa}), o; b, y) \big| \leq r^{D+1-w(a)-\varepsilon}$ for every $s \in X \sqcup \{\mathrm{mix}\}$.

The coefficients do not depend on b, y, s or D. The thresholds do not depend on x, o, y or s.

Status: theorem for every Reader on the given final record under (RG). The main construction supplies the coherent prescription and (RG) in VIII.1.

Lean:

- `Orbits.Foundation`: clause `coefficients` is (O1); clause `discrepancy` gives injectivity, the bound in $\mathcal{A}$ and the bound in every state.
- `Orbits.produce`, `coefficient_separated`, `quantitative_separated`.
- The error element is `algebraError`.
- Headline: `SeparatedOPE`, clauses `coefficient` and `remainder`, the latter via `Independent.Core.point`.
- Public form: `publicOPE` into `Hierarchy.OperatorProductExpansion`.

Proposition VII.4.2 (integrated remainder; item (D8)). Fix the data of (O2), a measurable space K and $B \geq 0$. For all sufficiently small r the following holds.

Hypotheses. Let μ be a positive measure on K, $w \in L^1(\mu; \mathbb{C})$ with $\int |w|\, d\mu \leq B$, and $k \mapsto (x(k), o(k), y(k))$ an a.e.-measurable configuration map. Assume that μ-a.e. on $\{w \neq 0\}$, (x, o) is an (r, M, η)-cluster and y is c-clear of o.

Conclusion. Take for $s(k)$ either the mixed state, or $\xi(k)$ for an arbitrary a.e.-measurable phase map $\xi \colon K \to X$. Then:

- $w\, S^s(a, x; b, y)$, w times the truncated sum, and w times their difference are integrable;
- the integrals satisfy

$$\Big| \int_K w(k) \Big[S^{s(k)}(a, x; b, y) - \sum_{j \leq D} \sum_{\kappa} c_{j,\kappa}(a; x, o)\, S^{s(k)}((j, O_{j,\kappa}), o; b, y) \Big] d\mu(k) \Big| \leq B\, r^{D+1-w(a)-\varepsilon}.$$

In particular, for every probability measure ν' on X, the ν'-mixture of the phases satisfies the bound of (O2) (`probability_mixtures`). The same pointwise threshold works for every K and B.

Status: theorem.

Lean: `estimates_weighted`, `JointEstimate`, `FitsOn`; clause `profiles` of `SeparatedOPE`, through `Core.joint`.

Proposition VII.4.3 (nested OPE, i.e. composition of coefficients over trees; item (D10)).

Trees and plots.

- t is a finite rooted tree whose root is an inner node, and v labels its leaves.
- w_p is the total degree of the leaves below the inner node p.
- A *plot* z assigns points to the leaves and an anchor z_p to each inner node.
- Radii $s_p > 0$ are attached to the inner nodes, with $\rho_p = s_p / s_{\mathrm{parent}(p)}$ for non-root p.

Admissibility. (s, z) is (r, M, η)-*admissible* if:

- $s_{\mathrm{root}} = r$ and $s_p \le s_{\mathrm{parent}(p)}$;
- at every inner node p, all positions of the subtree lie within $M s_p$ of z_p;
- positions in different children of p are at distance $\ge \eta s_p$.

Costs. Put

$$\mathrm{toll}_m(s) = r^{\,m-w(v)-\varepsilon} \prod_{p \neq \mathrm{root}} \rho_p^{\,-w_p-\varepsilon}, \qquad \mathrm{nest}_D(s) = \sum_{p \neq \mathrm{root}} \rho_p^{\,D_p+1}.$$

Conclusions. Fix m, $\kappa \in \mathrm{Ch}_m$, truncation degrees D_p at the inner nodes, $M \ge 0$ and $\eta, \varepsilon > 0$. For all sufficiently small r and every admissible (s, z):

- (a) $|c_{m,\kappa}(v; z_{\mathrm{leaves}}, z_{\mathrm{root}})| \le \mathrm{toll}_m(s)$.
- (b) $|c_{m,\kappa}(v; \cdot) - c^{(D)}_{m,\kappa}(v; z)| \le \mathrm{toll}_m(s)\, \mathrm{nest}_D(s)$. Define $c^{(D)}$ recursively: each inner child q contributes a channel (j_q, κ_q) with $j_q \le D_q$, its composed coefficient, and the output field O_{j_q,κ_q} at z_q. Each leaf child contributes its original label and the factor 1. Multiply these factors by the root coefficient for the resulting child labels, and sum over the inner children's channels. The root output is fixed at (m, κ); D_{root} is unused.
- (c) Fix a finite spectator list b and $\delta > 0$ before taking r small. If y is at distance $\ge \delta$ from every point of the plot, with mutual distances $\ge \delta$, then the degree-m remainder of VII.4.1 for the leaves at z_{root} has $\mathcal{A}$-norm, and absolute value in every state, at most $\mathrm{toll}_{m+1}(s)$.
- (d) Fix a measurable space K. For each small r, keep the radii s fixed throughout the integration. Let μ be a positive measure on K and $q \in L^1(\mu; \mathbb{C})$ with $\int |q|\, d\mu \le B$. Let the plot, and in (c) the spectators, be a.e.-measurable maps satisfying the stated geometry a.e. where $q \neq 0$. The direct and composed coefficients are integrable after multiplication by q; their integral and the difference of their integrals satisfy (a)–(b) with an additional factor B. The tested remainder has the integrated conclusion of VII.4.2 with bound $B\,\mathrm{toll}_{m+1}(s)$, for the mixed state or any a.e.-measurable phase map. The thresholds are uniform in s, μ, q, B and the configuration maps.

Status: theorem for every Reader on the final record under (RG).

Lean:

- `Composite.token`, `Composite.nShot` distinguish unchanged leaf inputs from inner-node expansions.
- `NestedOPE` (clauses `direct`, `tested`, `joint`, `jointLaw`).
- `Nesting.Uniform`, `Nesting.uniform`.
- `Accept`, `Tree.toll`, `Tree.nestPrice`, `Composite.stacked`, `CoefIntegral`.

Proposition VII.4.4 (contact extensions; items (D2), (D11)–(D13)).

Setting.

- $\mathcal{M}$ is a system of contact devices on the final record. Its prescription is the Reader's ($\mathcal{M}$.basis $= \mathcal{P}$; this is (D11)) and its rotation action is that of F.
- (RG) holds.
- For labels $a_1, \dots, a_k$ of degrees d_i and a state s:
  - let $\mathrm{cl}(n) > 0$ be the clearance of the devices up to degree n;
  - put $B_0 = \min(1, \mathrm{cl}(w(a)))$ and fix a subtraction length $s_0 \in (0, B_0/2]$; the public form uses $s_0 = B_0/4$;
  - let $\mathcal{D}^s_a(f) = \int \langle \mathrm{full} \rangle_s(f; x)\, dx$, the kernel being given by the forest formula of Lemma VII.4.r.

Conclusions.

- (a) $\mathcal{D}^s_a$ is a distribution of order at most $(2 + w(a))\, k$. For every compact K there is C_K with $\int |\langle \mathrm{full} \rangle_s(f; x)|\, dx \le C_K \max_{|\alpha| \le (2+w(a))k} \|\partial^\alpha f\|_\infty$ for all f supported in K.
- (b) **(D2)** $\mathcal{D}^s_a(f) = \int f(x)\, S^s(a; x)\, dx$ for every $f \in C^\infty_c$ supported in Conf_k.
- (c) The same construction gives distributions on affine slices $o + P(E)$, where P is a linear isometry whose range contains the normal space of the diagonal of a group $g \subset \{1, \dots, k\}$, and o lies on that diagonal.
- (d) **(D12)** For the admissible stage families described below, the pairing equals the sum over subtraction choices and dyadic shells. Every shell kernel is integrable, the sum of their L^1 norms is finite, and the iterated forest sums give the same pairing.
- (e) **(D12)** For the single-block partition $Q = \{g\}$ with $g \neq \emptyset$, use pure dilation in its normal space K_g, with $R \in (0, 1]$. Put $q_g = 4(|g| - 1)$ and $f_R(x) = R^{-q_g} f(R^{-1} x_{K_g} + x_{K_g^\perp})$. If $\mathrm{supp} f$ lies in the rescaled stage and $\|f\|_{C^N} \le H$, where $N = (2 + \sum_{i=1}^k d_i)(|g| - 1)$, then for every $\varepsilon > 0$ the integral of the absolute kernel is at most $CHR^{u - w(a_g) - \varepsilon}$. Here $u = 0$ for the full resolution, $u = n$ for the weight-n coefficient and $u = n + 1$ for the remainder after degree n. For the last two bounds the stage must also lie within clearance $\mathrm{cl}(n)$ of the g-diagonal. For $|g| > 1$, tests supported away from that diagonal give the same pairing as the punctured expression, which already includes proper-subcluster subtractions.
- (f) **(D13)** The device strata have bases fixed by conjugation, and the basis in degree 0 contains the unit field.
- (g) **(D13)** For every positive sequence (f_n) there are thresholds $0 < t_n \le f_n$ with $4t_{n+1} < t_n$, $t_n \to 0$ and $2(n + 1)t_n \le \mathrm{cl}(n)$.

In (d)–(e), an admissible stage family consists of a partition Q of a group g, affine placements $o + P(E)$, positive subtraction lengths s, and block scales $R_i \in (0, 1]$. Each o lies on the block diagonals, and the range of P contains their normal spaces. A common ceiling $B \le \min(1, \mathrm{cl}(\sum_i d_i))$ satisfies $2s \le B$. At each placement a common linear frame acts by R_i on every block normal space and preserves each orthogonal complement. The closed stages lie in a fixed ball in frame coordinates and within normal distance B of each block diagonal; for nonsingleton blocks $R_i \le C_1 s$ with fixed C_1. Original and spectator labels have uniform norm bounds. Spectator positions and labels are continuous on each stage and constant in each block's normal directions. All centroids of nonempty subgroups of different blocks, and all spectators, are uniformly separated by some $\delta > 0$. These conditions are the geometric and field hypotheses for the summation and scaling estimates; their constants may depend on the fixed family.

Uniqueness, reflection positivity on overlapping supports, contact Ward identities for arbitrary subtractions and real-time products are not asserted.

Status: theorem (`provided`). In the main theorem the devices are assembled in `contacts_on` (VIII.1), with detectors from `detector_exists`. (D2) appears in the detailed theorem `constructed`, not in the public joint existence statement.

Lean:

- Stage hypotheses: `Route`, `Route.Authority`, `Backdrop.Eligible`; output clearance: `sees`; derivative order: `stairs`.
- `MultipleLight`, with clauses `slope` = `Sunshine`, `moment` = `Resolution`, and `distribution` = `Charge` of order `totalTestOrder`.
- `SystemBasis`, `Thresholds`.
- Public form: `contactDistribution` for `Hierarchy.Contacts`.

Proposition VII.4.5 (slab-operator bound; the bench V.1.7).

Hypotheses. $(\mathcal{W}, f, S)$ is a seed:

- $\mathcal{W}$ has the construction properties;
- the extraction is ordered and has generic point limits;
- S satisfies the Euclidean hierarchy;
- the corrected field estimates of IV.1 hold;
- the cell hypothesis holds: reflection positivity and the chessboard inequality at every cutoff, and centred bounds $\|I_{N,p}(v,x)\|_{\mathcal{T}_m} \le A\, a_m^{-d-\varepsilon} + \eta$ late in the extraction.

Conclusion. For every $D \in \mathbb{N}$ and $R > 0$ there are $C_0 > 0$ and $P \in \mathbb{N}$ with the following property. Let $d > 0$, $r \in (0,1]$, and let $w \neq \emptyset$ be a word with these properties:

- its tokens have degree $\le D$ and coefficient norm $\le R$;
- it lies in the time slab $(0,d)$ and is r-buffered there.

Then there are a level b and a width $0 < s \le d$ with $s \in a_b\mathbb{N}$ and w in the slab $(0,s)$, such that every word v supported at positive times has a constant B_v with

$$|S(\Theta v_n \cdot v_n)| \le B_v\, \rho^{4n} \quad (n \in \mathbb{N}), \qquad \rho = C_0^{|w|}\,(|w|/r)^{P|w|}.$$

Here v_n is v preceded by $2n$ consecutive slabs of width s that hold alternately w and its reflection. In V.1 an iterated Schwarz inequality turns this into $\|T_s(w)\| \le \rho$. One may take $P = 4(D+1)$.

Status: theorem: `Next.Phase.seats` proves `Bid.Seats`.

- It enters the phase construction of V.1 through `Next.Phase.vacuum`.
- The OPE proofs of this unit do not use it directly; they use the algebra $\mathcal{A}$ it helps to build.

Lean: `Bid.Bench`, `OS.QRead`, `Spread.radius`; steps in Lemmas VII.4.o–p.

Proposition VII.4.6 (coherent prescription).

Setting.

- A *probe* is a finite separated configuration of tokens; its first point is the anchor, and the other points are taken relative to it.
- $\mathrm{sp}_n : F_n \to \mathbb{C}^{\{\text{probes}\}}$ sends ϕ to the correlations of ϕ at the anchor with the probe tokens. This map is injective, by the definition of F_n as a quotient by null fields, and compatible with the inclusions.
- A *mode* is a pair $i = (n, \alpha)$, where α indexes a basis of $F_n/\iota(F_{n-1})$, with $F_{-1} = 0$.

Conclusion. There are, for every mode, a vector $u_i \in F_n$ (needle) and a finitely supported family λ_i of complex weights on probes (sheet), with the following property. Write $\lambda_i^{(m)} = \sum_p \lambda_i(p)\,\mathrm{sp}_m(\cdot)(p)$ for the induced functional (test) on F_m. Then for every n:

- $\lambda_i^{(n)}(\iota u_j) = \delta_{ij}$ for all modes i, j of degree $\le n$;
- $\phi = \sum_{\deg i \le n} \lambda_i^{(n)}(\phi)\,\iota u_i$ for all $\phi \in F_n$.

Tests of degree $> n$ vanish on F_n. One prescription serves every truncation degree.

Status: theorem.

Lean: `Prescription`, `SignedOff`, `Coherent`, `Prescription.chosen`, `highVan`.

Remark VII.4.7 (hypotheses and order of choices).

The order of choices. In the main theorem (`WilsonExistence`), the family is chosen first, for G with simple Lie algebra and any penalty $b > 0$. Then, for every extraction order, a refinement, the table, the final record, the Reader and the devices are chosen together.

The hypotheses (RG) (`Services`) bundle four items:

- the grid-approximant property;
- simplicity of the Lie algebra of G;
- the per-step and cluster RG estimates;
- the continuum extraction record.

For the table of the construction these hypotheses are proved by `Run.extractible17` (VIII.1, from Part III outputs; §2 input 6).

Separate existence theorems. `through_final` and the raw tree theorem allow any penalty $b > 0$. `Orbits.foundation`, the single-cluster existence theorem, the covariant tree existence theorem and `celestial` choose $b = 1$. The main theorem instead applies the OPE estimates to its already chosen family.

Further results. The same construction also gives:

- the OPE in the raw (complex, non-covariant) needle basis (`SingleSubpower`);
- tree versions for every coherent prescription (`AllShapes`, via `Services.everything`).

What is not claimed. The coefficients depend on the prescription and the Reader. Independence of the extraction subsequence is not claimed (Assessment §7.5).

3.29.2 2. What it uses

Inputs from other units

1. *The final record and its seed* (IV.1, V.1–V.3, VI.1; assembled in VIII.1). This comprises:
   - the cutoff family with the construction properties (II.2, III.13);
   - the hierarchy (Euclidean axioms) of S, and generic point limits on grid configurations;
   - the field algebra: inclusions ι, derivative identities, rotation action and its laws, reality structure.

   Lean: `Seed`, `Locals`, `Final`.
2. *Reflection positivity and centred bounds* (IV.1, interface item 2; I.1). Two parts:
   - the chessboard inequality at every cutoff;

- for every degree d and $\varepsilon, R > 0$, there are m_0 and $A \geq 0$ such that, for every $m \geq m_0$ and $\eta > 0$, eventually along the extraction every $I_{N,p}(v, x)$ with $\|v\| \leq R$ is interior to its centred cell at scale a_m, with root $\leq A\, a_m^{-d-\varepsilon} + \eta$, uniformly in v and the site x.

Lean: `CellHyp`.

3. *The ground algebra and phases* (V.1). This comprises:
   - $\mathcal{A}$ commutative, with X its Gelfand spectrum and ν of full support;
   - `B17.answer`: $\langle \Omega, (u^* u)^n \Omega \rangle$ is the limit, as the spacing $\to \infty$, of the continuum correlation of $2n$ time-translates of u, alternately conjugated;
   - `Retain.bit`: if $c \geq 0$ and $\langle \Omega, (u^* u)^n \Omega \rangle \leq c^{2n}$ for all n, then $|\chi_\xi(u)| \leq c$ for all ξ;
   - the selection, the every-phase tables, and their smoothness.
4. *The derivative identity* (VI.1, IV.1): the derivative in one point of a correlation is the correlation with the derivative label, of degree one higher. It is used in Lemma VII.4.i.
5. *Haar measure on $O(4)$ and the rotation laws* (VI.1). They are used for the covariant Reader.
6. *The RG hypotheses (RG).* The estimates come from Part III and are collected in VIII.1:
   - `Org.Grill`: for every degree bound d and level m there is $k \geq m$ such that conjugation-fixed token tuples of degree $\leq d$ on generic level-k grid configurations admit coverings that work along the extraction. It is supplied in III.11 and used in the a priori bounds (Lemma VII.4.e).
   - `Package` = `TableTickets`, together with cluster product cuts for every arity. It supplies per-step RG bounds, coefficient transport and column estimates, halo summability, polymer representations, and per-cell scale choices. These are the Part III inputs to Lemmas VII.4.f–g, collected in VIII.1.
   - The continuum extraction record (IV.1, `Next/EXTRACTION/Base.lean`): the extraction to the smooth continuum table. It is supplied with the extraction in III.11 and used in the grid estimates (Lemma VII.4.h).

Standard mathematics

- Commutative C*-algebras: the Gelfand isometry and norm = sup over characters (Bratteli–Robinson I, §2.3.5).
- Reflection positivity and chessboard estimates (Fröhlich–Israel–Lieb–Simon 1978; Fröhlich–Simon 1977; Glimm–Jaffe, Ch. 6 and 18).
- The iterated Schwarz inequality for transfer operators on an invariant pre-Hilbert domain: exponential bounds on the iterates of an adjoint pair give $\|Av\| \leq D\|v\|$. This is used in V.1 (Glimm–Jaffe, Ch. 6).
- Block-spin renormalization and polymer representations (Bałaban; Brydges, *Lectures on the renormalisation group*; Gawędzki–Kupiainen).
- The Wilson–Zimmermann OPE (Wilson 1969; Zimmermann 1973) and its axiomatic form (Hollands–Wald 2008). The conclusions here are finite-order asymptotic bounds with ε-loss.
- Extension of distributions across diagonals with Taylor subtraction of order given by the scaling degree (Epstein–Glaser 1973; Brunetti–Fredenhagen 2000; Hörmander I, Thm 3.2.3–3.2.4). Forest formulas (Zimmermann 1969).
- Normalized Haar measure on $O(4)$, finite-dimensional duality, Taylor's formula with remainder, and the mean value theorem.

Imported constants and assumptions

- $\varepsilon > 0$ is an arbitrary exponent loss. The estimates do not determine logarithmic corrections.
- Constants of the cluster estimate (Lemma VII.4.f):

  - the gain margin $1/16$, from the RG transport factor $(a_k/a_m)^{1/8}$;
  - the loss $(a_m/a_j)^{4|I|}$, where the exponent 4 is bookkeeping slack ($L_i^4 \geq 6L_i$);
  - subpower factors in the per-cell, halo and insertion bounds, absorbed by ε.
- The constant $P = 4(D+1)$ of VII.4.5 comes from degree-d bounds of the form $a_k^{-(d+1)}$ and $a_k \geq c\,(r/|w|)^4$.

3.29.3 3. Strategy

A coherent system of probes reads every output degree from the mixed correlation table. For mode i, the raw coefficient is $\lambda_i(a; x, o) = \sum_p \lambda_i(p) S(a, x - o; \text{probe } p)$, with each reading set to zero when the combined configuration is not separated (`Prescription.ope`). Biorthogonality makes these readings exact on single fields (`sumAll`). Averaging over $O(4)$, symmetrizing the input and output conjugations, and removing lower-degree components gives the Reader's real covariant strata. These choices are fixed before the spectator labels, state and truncation degree.

The RG estimate replaces a lattice cluster by one label Q of prescribed degree t. Its remainder gains $(a_m/a_k)^{t+1/16}$, while the label and remainder both pay the factor $a_j^{-w}(a_m/a_j)^{4|I|}$. Along the scale induction, nearby groups fuse and separated groups pass through the next block-spin step. After extraction, exact reconstruction of Q by the probes yields the grid OPE. Reflection positivity bounds decks of widely translated copies by products of chessboard roots, treating a cluster remainder as one factor. The deck limit gives all vacuum moments of u^*u in the ground algebra; full support of the phase measure and the Gelfand isometry then give $\|u\|_{\mathcal{A}}$. Each spectator costs only its own one-field bound. The resulting norm estimate controls every phase and the mixed state with the same coefficients.

Exponential rates organize the choices of lattice scales. A downward induction in total degree extends the grid bounds to arbitrary configurations: each derivative raises the degree by one, while the distance to the fine grid recovers one power. At sufficiently high degree the crude bound already suffices to start the induction. Bringing the auxiliary scale heights together gives an arbitrary ε-loss; refinement arguments turn these rates into uniform small-radius estimates. Tree expansions repeat this replacement recursively, leaving leaf labels unchanged. Contacts use a forest recursion: each retained weight receives a normal Taylor subtraction of its own order, and the remainder gains enough radial power for integration. Partial collisions are handled by the same recursion on finer partitions. Separately, the slab-operator bound follows by applying the two-scale chessboard estimate to a reflection-positive train and using iterated Schwarz to remove the fixed exterior cost.

3.29.4 4. Steps

3.29.4.1 4.1 Rates and sequences Lemma VII.4.a (from sequences and rates to uniform small-r bounds).

- (i) Let $P(r, z)$ and $Q(r, z)$ be predicates. Suppose that for every choice $r \mapsto z_r$ and every non-trivial refinement of $r \to 0^+$ along which $P(r, z_r)$ holds eventually, a further refinement makes $Q(r, z_r)$ hold eventually. Then, for all sufficiently small r, $P(r, z)$ implies $Q(r, z)$ for every z.
- (ii) *Rates.* A quantity has upper rate κ along $T \to \infty$ if $|f| \leq e^{\kappa' T}$ eventually, for every $\kappa' > \kappa$. With $T = -\log r$ this means $|f| \leq r^{-\kappa-\varepsilon}$ eventually, for every $\varepsilon > 0$. Below, "rate" means upper rate unless an exact rate is specified.

- (iii) *Clusters and spectators.* Under $T = -\log r$, an (r, M, η)-cluster has extent of upper rate -1 and reciprocal separation of upper rate 1. For c-clear spectators, both the reciprocal mutual separation and the reciprocal distance to the centre have upper rate 0.
- (iv) *Heights.* A sequence of levels $m \to \infty$ has height c if $\log(1/a_m)/T \to c$. Every height $c \geq 0$ is attained for a gentle spacing sequence.

Proof sketch. For (i), choose a bad configuration at each bad radius and refine to a non-trivial filter supported on those radii. The assumed further refinement is a contradiction. Parts (ii)–(iii) follow from $r = e^{-T}$. For (iv), choose the first level whose logarithmic spacing exceeds cT when $c > 0$, and $\sqrt{T}$ when $c = 0$; gentleness makes the relative overshoot tend to zero.

Lean: `refine_geometry`, `RateLE`, `loglife`, `room_regime`, `free_clear`, `Height`, `found`.

3.29.4.2 4.2 Prescription and Reader Lemma VII.4.b (extension one level at a time). A prescription that is biorthogonal and reconstructing on F_n extends to one with the same property on F_{n+1}, without changing the needles and sheets of degree $\leq n$.

Proof sketch.

- Let $B = \sum_{\deg i \leq n} \lambda_i^{(n+1)}(\cdot)\, \iota_{\deg i, n} u_i \colon F_{n+1} \to F_n$, a left inverse of $\iota_{n,n+1}$.
- Take any linear section of $F_{n+1} \to F_{n+1}/\iota F_n$, and subtract from it its projection $\iota \circ B$. This gives a lift Φ whose image lies in the kernel of all old tests.
- The new needles are Φ applied to a basis of the quotient.
- The new sheets realize the dual coordinates composed with the quotient map. Such sheets exist because the probe functionals span the dual of F_{n+1} (`reachReads`, by faithfulness).
- A recursive choice, collected along n, gives a coherent prescription.

Lean: `startRows`, `nextRows`, `Prescription.chosen`.

Lemma VII.4.c (the Reader). Fix a coherent prescription.

Construction.

- The *raw vector* of degree n is $\sigma_n(a; x, o) = \sum_{\deg i \leq n} \lambda_i(a; x, o)\, \iota u_i \in F_n$.
- Define the covariant average

$$A_n(a; x, o) = \int_{O(4)} \rho_n(g)\, \sigma_n(\rho(g^{-1})a; g^{-1}x, g^{-1}o)\, dg,$$

  where $\rho(g^{-1})$ acts on every input label. Set $A_n = 0$ outside the domain where all rotated probes through degree n stay clear. The conjugation-symmetric reading is

$$\bar{\sigma}_n(a; x, o) = \tfrac{1}{2}\big(A_n(a; x, o) + \overline{A_n(\bar{a}; x, o)}\big).$$

- Define recursively $\pi_n = \mathrm{id} - \sum_{i<n} \iota \circ \pi_i \circ \ell_{i,n}$ on F_n, where $\ell_{i,n}(\phi) = \bar{\sigma}_i(\phi; 0, 0)$.
- The *cut tests* are $\tau_{n,q} = \pi_n \circ \ell_{n,q}$, and the *stratum* is $E_n = \bigcap_{i<n} \ker \tau_{i,n}$.
- The *fresh vector* is $\pi_n \bar{\sigma}_n(a; x, o) \in E_n$.
- $(O_{n,\kappa})$ is a basis fixed by conjugation of E_n, and $c_{n,\kappa}(a; x, o)$ are the coordinates of the fresh vector in it.

Properties.

- $\tau_{n,q} = 0$ for $q < n$.
- $\sum_{i \le n} \iota\tau_{i,q} = \iota_{q,n}$ for $q \le n$.
- The fresh vectors and cut tests intertwine rotations and conjugation; simultaneous translations and permutations of the inputs preserve the fresh vectors. Coordinates transform in the corresponding channel representations.
- The coefficients are smooth where the probes are clear.

Proof sketch.

- On a single field of degree $j \le n$ at the origin, the degree-n averaged reading is $\iota_{j,n}$ (`checkReal_fixed`); this uses coherence and the rotation laws.
- The identities for τ then follow by strong induction on n.
- A complex finite-dimensional space with an antilinear involution has a basis of fixed vectors.

Lean: `checkValue`, `Reader`, `averageV`, `checkReal`, `knife`, `cutTest`, `reconstruct`, `kills_lower`, `stratum`, `fresh`, `Reader.basis`, `coef`.

3.29.4.3 4.3 Norms in the ground algebra Lemma VII.4.d (norm from decks). Let $u \in \mathcal{A}$ come from a finite combination of separated grid words. It has lattice versions $U_{N,p}$ at the cutoffs.

Hypothesis. Let $c \ge 0$. The lattice labels converge to those defining u along a non-trivial refinement of the extraction. For every n and all sufficiently large translation spacings t, eventually along that refinement the lattice moment of the deck $\prod_{l<2n} U^{(*)}_{N,p} \circ \tau_{2tl}$ (time translates, even ones conjugated) has modulus $\le c^{2n}$.

Conclusion. $\|u\|_{\mathcal{A}} \le c$.

Proof sketch.

- The deck moments converge to the continuum deck correlation.
- As $t \to \infty$, that correlation tends to $\langle \Omega, (u^*u)^n \Omega \rangle = \int |\chi_\xi(u)|^{2n} d\nu$ (B17, V.1).
- Since ν has full support, the bound for all n gives $|\chi_\xi(u)| \le c$ for every ξ.
- By Gelfand, $\|u\| = \sup_\xi |\chi_\xi(u)|$.

Lean: `Cards.carrying`, `Cards.pressure`. *(proof read; the geometric inputs to* `B17.answer` *are taken from V.1)*

Lemma VII.4.e (a priori and replacement bounds in $\mathcal{A}$). Assume (RG).

Setting. Main points are at mutual distance $> 2a_b$. Exterior points are at mutual distance $> 2a_l$, and at distance $> 2a_b$ from the main points. The levels satisfy $b \le f \le l \le n$ and $f \le q$, with

$$|I| \cdot 10a_f < a_b, \qquad |J| \cdot 6a_n < a_l.$$

Main tokens have centred bounds B_i at level q, and exterior tokens have bounds D_j at level n.

Conclusions.

- (i) $\|M(\text{main} \cup \text{exterior})\|_{\mathcal{A}} \le \prod_i B_i \prod_j D_j$.

- (ii) For a cluster on a fixed grid with distinct coordinates on every axis, suppose lattice labels $Q_{N,p}$ of degree t converge to Q along a non-trivial refinement of the extraction. Suppose $R_{N,p} = \prod_i I_{N,p}(v_i, x_i) - I_{N,p}(Q_{N,p}, 0)$ is interior to its centred cell at level q with root $\le \alpha$ along that refinement. Require b no finer than the grid, the origin to be $2a_b$-clear of the exterior, and both the cluster–exterior and origin–exterior configurations to be separated. Use the cell-fitting inequalities above with one main factor. Then

$$\|M(v, x; \mathrm{ext}) - M((t, Q), 0; \mathrm{ext})\|_{\mathcal{A}} \le \alpha \prod_j D_j.$$

- (iii) In rate form, let $0 \le \alpha \le \beta$. If reciprocal main–main and main–exterior distances have upper rate α, and reciprocal exterior–exterior distances have upper rate β, then M has upper rate $w(\mathrm{main})\alpha + w(\mathrm{ext})\beta$, uniformly in states.

Proof sketch.

- (i)–(ii): the two-scale chessboard inequality is applied to decks of $2n$ translates. Main factors occupy distinct cells of the coarse tiling. Exterior factors, possibly several per coarse cell, are grouped and bounded in a finer aligned tiling. This gives the deck bound $(\prod B \prod D)^{2n}$ or $(\alpha \prod D)^{2n}$.
- Lemma VII.4.d then gives the norm bound at grid configurations. Smoothness of correlations extends it to the limit configuration.
- (iii): take bounds $a_k^{-d-\delta}$ at heights slightly above the separation rates.

Lean: `Cards.plant`, `Cards.mixed`, `Cards.Hall`, `civic`, `viewing_admits`, `mirrorMatch`, `admitted`, `phase_budget`, `viewer_cost`. *(proof read;* `TileDoc.global` *statement only)*

3.29.4.4 4.4 The lattice cluster estimate Lemma VII.4.f (lattice cluster replacement).

Fix the RG representation, product-cut and transport data of §2 for output degree t. Let I be finite and nonempty. Levels $J \le k \le m < m+1 \le j < N$ are ordered from coarse to fine. Require $\mathrm{ref}(t) \le k$ and $\mathrm{ref}(d_i) \le j$ for every input degree d_i.

Hypotheses.

- *Geometry.* Lattice observables $f_i = I_{N,p}(s_i, o_i)$ of degrees d_i sit at level-j sites whose level-$(m+1)$ ancestor is the centre cell. They are pairwise at least B_j^2 level-j units apart, where B_j is the polymer scale. Require

$$(B_{m+1}^2 + 1)a_{m+1} \le a_J, \qquad \mathfrak{c}_k^2 a_k \le a_J, \qquad 2(a_m \ell_t + \tau_m) < a_J.$$

  Here $\mathfrak{c}_k$ is the supplied cover count, ℓ_t is the output stencil radius plus 4, and $\tau_m = 30a_m(\jmath_m + 5)$ is the cone-tail bound with supplied aperture parameter $\jmath_m$. All f_i and all $I_{N,p}(z, 0)$ are interior to the centre chamber.

- *Weights.* Choose $\alpha_i > 0$ with $\alpha_i \,\mathrm{paid}_j(s_i) \le \zeta(d_i, j)$. Here paid_j is the insertion-size bound and ζ the allowed RG weight. Choose $0 < \alpha \le 1$ below every halo allowance from m through $j-1$. With $H = |I|$, require

$$0 < A \le \alpha^H \left(\frac{a_j}{a_{m+1}}\right)^{4H} \left(\frac{\alpha}{8}\right)^{H^2} \left(\frac{\delta_j \varphi_j}{8}\right)^H \prod_i \alpha_i.$$

- *Output weights.* Let $\delta_m, x_m, \varphi_m, \zeta(t,k)$ and c_m be the supplied RG constants, with c_m the column count. Write $\mathfrak{m}_{k,m}(b)$ for the weight obtained by transporting $b>0$ from k to m, and $A_{k,t}$ for the output transport map. Require $e>0$ and

$$\mathfrak{m}_{k,m}(b) \le \frac{\delta_m A}{16 x_m}, \qquad 8(1+c_m)a_{m+1}^{t}\frac{4x_m}{A} \le e,$$

$$32(1+c_m)b\,\|A_{k,t}\|\frac{e}{(a_k/a_m)^{1/8}} \le \frac{\zeta(t,k)}{2}.$$

Conclusion. There is a lattice label z of degree t with these properties:

- its coefficient size at level m is at most $2e$;
- $R = \prod_i f_i - I_{N,p}(z,0)$ is interior to the centre chamber;
- $\|R\|_{\mathcal{T}_J} \le 2\,\mathrm{stakes}(J,k)/b$.

Here $\mathrm{stakes}(J,k) = \exp\left((a_J/a_k)^4 \mathrm{treasury}_k + 5\right)$ is the supplied per-cell chessboard factor. In Lemma VII.4.g the two coarse levels are chosen so that this factor has upper rate 0.

Proof sketch.

- *Transport.* From N down to j, each insertion is transported separately, since the points are separated.
- *Induction from j down to $m+1$.* The carried state is a polymer representation of the RG-pushed generating function, with the points grouped and the representatives spaced. At each step one of two things happens:
  - if the representatives stay spaced, one block-spin step is taken, and the weights lose at most a factor L_i^4 per point;
  - otherwise a maximal spaced set is kept and the remaining representatives are fused into it. Enlarging polymer supports to reach a representative costs a bounded factor in halo summability. Each fusion pays a subpower toll.
- *The weights.* Their surviving product contains $(a_j/a_{m+1})^{4|I|}(\alpha/8)^{|I|(|I|-s)}$ when s groups remain. Its reciprocal contributes to the final error bound.
- *Final step.* At level $m+1$, fuse the remaining groups into one variable per cell copy. The RG step to level m splits the result into a coefficient Q_0 of degree $\le t$, with its coefficient and remainder bounds, and an irrelevant remainder. Transport to the coarse level gives a factor $(a_m/a_k)^{t+1}$ in the remainder bound.
- The constant 2 comes from two halves, each $\le \mathrm{stakes}/b$, joined by the triangle inequality for roots.

Lean: `joins`, `inductionLive`, `Ballet`, `ranch`, `hitB`. *(statement read; proof read in outline, at the level of the calls in* `Ballet`, `inductionLive`, `Live.gathers`, `bequeathed`*)*

Lemma VII.4.g (replacement rates and the limit label). Assume (RG), fix a nonempty finite token list of total degree w, and take heights $0 \le b < a < c$ for the coarse, output and insertion levels. Fix a target degree t, and put $Z' = w\,c + 4|I|(c-a)$.

Conclusion. There are rate functions g and o with these properties:

- g has exact rate $Z' - ta$;
- o has upper rate $Z' - (t+\frac{1}{16})(a-b)$;

- eventually, the replacement of Lemma VII.4.f has coefficient size $\le g$ and remainder root $\le o$, whenever its cluster and interior conditions hold.

Up to subpower factors, the coefficient size is bounded by $a_j^{-w}(a_m/a_j)^{4|I|}a_m^t$, and the remainder root by $a_j^{-w}(a_m/a_j)^{4|I|}(a_m/a_k)^{t+1/16}$. For each fixed configuration satisfying the cluster and interior conditions, bounded coefficient size gives a convergent refinement of the extraction in the finite-dimensional label space. Its limit Q retains the remainder bound.

Proof sketch.

- Apply Lemma VII.4.f with all parameters exponential in T, and check its size conditions as inequalities between exponents.
- The loss $4|I|(c-a)$ is the rate of $(a_{m+1}/a_j)^{4|I|}$, the reciprocal of the surviving weight factor.
- The binding condition is the transport condition, which needs exponent $< (t+\frac{1}{8})(a-b)-Z'$. The value $\frac{1}{16}$ leaves strict slack.
- Subpower factors have rate 0.

Lean: `funding`, `PriceAt`, `price_plot`, `Survival`, `for_plot`.

3.29.4.5 4.5 The OPE at grid configurations Lemma VII.4.h (grid estimates).

Setting. Assume (RG) and fix a coherent prescription and a nonempty finite token list a. Heights satisfy $0 < \beta < a_0 < x < d$, for levels k, m, j at heights β, a_0, d. The cluster consists of level-j grid points with distinct coordinates on every axis, extent of upper rate $-x$ about the origin and reciprocal separation of upper rate x. Spectator distances from the origin have reciprocal upper rate $0 \le c < \beta$, and their mutual distances have reciprocal upper rate $u \ge c$. Put $\mathrm{tax} = w(a)d + 4|I|(d-a_0)$.

Conclusions.

- each fixed raw coefficient $\lambda_i(a;\mathrm{cluster},0)$ has upper rate $\mathrm{tax} - (\deg i)a_0$;
- for every $N \in \mathbb{N}$, the correlation with the spectators minus $\sum_{\deg i \le N} \lambda_i\, S^s(u_i, 0; \mathrm{spect.})$ has upper rate $\mathrm{tax} - (N+1)(a_0-c) + w(b)u$, uniformly in the state s.

Proof sketch. For the remainder choose $q \ge N$ with $(N+1)a_0 \le (q+\frac{1}{16})(a_0-\beta)$; for a coefficient choose $q \ge \deg i$ with the analogous inequality. Let Q be the degree-q label of Lemma VII.4.g. Then:

- (1) by Lemma VII.4.e(ii), replacing the cluster by Q at 0 costs rate $\mathrm{tax} - (q+\frac{1}{16})(a_0-\beta) + w(b)u$ in $\mathcal{A}$;
- (2) the same with the probes in the role of spectators: $|\lambda_i(\mathrm{cluster}) - \lambda_i(Q)|$ has the rate of the remainder root;
- (3) for $\deg i \le q$, the coefficient of Q is bounded by its coefficient size times $a_m^{-(q-\deg i)}$, up to subpower factors, and hence has rate $\mathrm{tax} - (\deg i)a_0$;
- (4) exact reconstruction: $S(Q,0;\cdot) = \sum_{\deg i \le q} \lambda_i(Q)\, S(u_i, 0; \cdot)$;
- (5) the modes of degree in $(N, q]$ contribute at most rate $\mathrm{tax} - (N+1)(a_0-c) + w(b)u$, using the upper rate $(\deg i)c + w(b)u$ for $S^s(u_i, 0; \mathrm{spect.})$ from Lemma VII.4.e(iii).

Lean: `grid_coeff`, `grid_remainder`, `see_cost`, `det_cost`, `at_grade`, `sumAll`.

3.29.4.6 4.6 Arbitrary configurations and exponents Lemma VII.4.i (derivative bootstrap). Fix a finite index set I and levels j of height $d > x_0$. A configuration path $x(T)$ lies in the height-x_0 corridor if its extent about the origin has upper rate $-x_0$ and its reciprocal separation has upper rate x_0. Let $h_T(v; x)$ be functions for every token list v on I. Suppose the following hold for every such list and every configuration path in this corridor, with constants c_0, d_0 independent of the list:

- (rough) $h_T(v; x(T))$ has upper rate $x_0 w(v) + c_0$;
- (grid) on level-j grid paths with distinct coordinates on every axis, $h_T(v; x(T))$ has upper rate $dw(v) + d_0$;
- (derivative) eventually $h_T(v; \cdot)$ is differentiable at $x(T)$, and its derivative in each point and coordinate equals h_T with that token replaced by its derivative token, of degree one higher.

Then h has rate $d\,w(v) + d_0$ on the whole corridor.

Proof sketch.

- Use induction downward on $w(v)$.
- For $w(v)$ large, $(d - x_0)w(v) \geq c_0 - d_0$, so the rough bound suffices.
- Otherwise choose a grid path whose distance from $x(T)$ has upper rate $-d$. All intervening points stay in the same corridor. The mean value theorem and the induction bound for the derivative give rate $-d + d(w(v) + 1) + d_0 = dw(v) + d_0$ for the change.

Lean: `interpolated`, `cloud_jump`, `leap_cost`. *(proof read for* `interpolated`*)*

Lemma VII.4.j (raw coefficient and remainder rates).

Setting. Assume (RG) and fix a coherent prescription and finite token lists a, b. Along $T \to \infty$, the cluster has extent of upper rate $-x$ around o and reciprocal separation of upper rate $x > 0$. Spectator distances from o have reciprocal upper rate $0 \leq y < x$, and their mutual distances have reciprocal upper rate $u \geq y$.

Conclusions.

- For each fixed mode i, $\lambda_i(a; \cdot, o)$ has upper rate $(w(a) - \deg i)x$.
- For every N, the remainder of the raw truncation has upper rate $w(a)x - (N+1)(x-y) + w(b)u$, uniformly in states and in $\mathcal{A}$.
- In powers, with $r = e^{-xT}$, spectator distance ρ and spectator separation ρ': the remainder is $\lesssim r^{-w(a)}(r/\rho)^{N+1}\rho'^{-w(b)}$ up to ε-losses.

Proof sketch.

- Apply Lemma VII.4.i to Lemma VII.4.h, with grid height d and auxiliary height a_0. The rough bound is Lemma VII.4.e(iii).
- This gives the bounds of Lemma VII.4.h with $\text{tax} = wd + 4|I|(d - a_0)$. Let $a_0 \uparrow x$ and $d \downarrow x$; continuity of the exponents and availability of all intermediate heights give the claimed rates. Empty clusters are handled by the exact unit identities.
- Translation invariance moves the centre to o.
- The bound in $\mathcal{A}$ follows from the bounds in every state by the norm characterization in the conventions.

Lean: `coefficient_rough`, `common_rate`, `early_rate`, `group_coeff`, `group_remainder`, `ground_cost`, `groundError`, `arrangement_cost`.

3.29.4.7 4.7 The Reader and the separated OPE Lemma VII.4.k (transfer to the Reader).

In the regime of Lemma VII.4.j:

- the fresh vector of degree n has rate $(w(a) - n)x$, and hence so does each $c_{n,\kappa}$;
- the error element of VII.4.1 has rate $w(a)x - (D+1)(x-y) + w(b)u$ in $\mathcal{A}$.

Proof sketch.

- Compare the cluster with its raw truncation at a horizon H, placed at o. Inside every rotated probe correlation, the difference has rate $(w - H - 1)x$ by Lemma VII.4.j.
- The raw estimate holds uniformly over all rotations, including the conjugate input reading. Integrating over normalized Haar measure and applying the fixed linear projections preserve its upper rate. Thus the fresh vector differs from $\tau_{n,H}$ of the raw vector by upper rate $(w - H - 1)x$.
- In $\tau_{n,n}$ of the raw vector, only modes of degree n survive, with rate $(w - n)x$.
- For the remainder, write the error as the raw error plus $\sum_{j \le D}$ (the correlation of $\tau_{j,D}(\mathrm{raw}) - \mathrm{fresh}_j$), and bound each term.

Lean: `fresh_bridge`, `fresh_price`, `balance_rate`, `coefficient_rate`, `algebra_rate`. *(proof read;* `fit_old`, `discrepancy_rate_old` *statement only)*

Lemma VII.4.l (proof of VII.4.1). For (O1) and (O2), apply Lemma VII.4.a with $T = -\log r$.

- An (r, M, η)-cluster sits at height $x = 1$.
- Fixed clearance c gives $y = u = 0$.
- So Lemma VII.4.k gives rates $w - n$ and $w - (D+1)$. These are the powers $r^{n-w-\varepsilon}$ and $r^{D+1-w-\varepsilon}$, uniformly in configurations and states.

Lean: `coefficient_separated`, `quantitative_separated`, `produce`; assembly in `R2IGoals.fromFunded`.

3.29.4.8 4.8 Integrated and nested forms Lemma VII.4.m (integrated forms; VII.4.2).

Proof sketch.

- For small r, every admissible configuration lies in an open set on which the correlations and the coefficients are continuous. There, the raw correlation is bounded by $r^{-w-\varepsilon}$ and the remainder by $r^{D+1-w-\varepsilon}$, in every state.
- So the integrands are measurable and dominated by $|w|$ times a constant, and the integrals are bounded by B times the pointwise bound.
- For any probability measure on X, continuity of each fixed separated correlation in ξ permits termwise integration of the finite truncation; the same pointwise bound then gives the mixture estimate.

Lean: `estimates_weighted`, `probability_mixtures`.

Lemma VII.4.n (nested OPE; VII.4.3).

Proof sketch.

- Along a sequence, put $T = \sum_p \log(1/s_p)$, summing over the inner nodes. The normalized logarithms lie in a compact simplex, so a refinement gives heights $a_p \ge 0$ with $\sum_p a_p = 1$.

This also covers scales shrinking faster than every power of the root radius.
- Then induct over the tree.
- At a fork, children whose rate equals the parent's are merged into the parent cluster.
- Each other child is replaced by its own expansion at an auxiliary horizon high enough that its remainder is below any prescribed rate.
- The replacements are telescoped in a fixed triangular order, so that each switch sees only outputs already fixed.
- Summing the node costs gives (a). The composition error gains the smallest value of $(D_p + 1)(a_p - a_{\mathrm{parent}(p)})$ over non-root inner nodes. The extra ε-loss converts this rate estimate into (b). If there are no such nodes, the composed coefficient equals the direct coefficient exactly.
- (c) and (d) follow as in Lemmas VII.4.k–m.

Lean: `Fusion.tariff`, `Quantified.lean` (`Limits.coe_power`, `Limits.tested_power`), `Tree.Tapered`, `Tree.Relative`. *(proof read for* `coe_power`*,* `tested_power`*,* `tariff`*; supplies proof read;* `sequential` *and* `high` *statement only)*

3.29.4.9 4.9 The slab bound of V.1 Lemma VII.4.o (point and train bounds). Take levels $m \le k \le l \le b$.

- (a) *Point bound.* Assume:
  - all points are separated; main–main and main–exterior sup-distances exceed a_m, and exterior–exterior sup-distances exceed a_l;
  - main points are congruent mod a_m to one of P fixed points;
  - $P \cdot 10a_k < a_m$ and $\sharp(\mathrm{ext}) \cdot 6a_b < a_l$;
  - $A, B \ge 0$; main tokens have one-field bounds A at level k for all signed and reflected variants, and exterior tokens have bounds B at level b, eventually along the extraction;
  - the cutoff family has the construction properties, S is smooth on separated configurations, and generic point limits hold along this extraction.

  Then $|S(x; v)| \le A^{\sharp\mathrm{main}} B^{\sharp\mathrm{ext}}$.
- (b) *Train bound.* Assume the Euclidean hierarchy and the same cutoff and point-limit hypotheses. Let w be r-buffered in a slab of width $d \in a_m\mathbb{N}$, with $a_m < r$. Let v be a word supported at positive times with buffer larger than a_l. Require $2|w| \cdot 10a_k < a_m$ and $2|v| \cdot 6a_b < a_l$, and the one-field bounds in (a). Then for every $q \in \mathbb{N}$ the Gram pairing satisfies
$$|S(\Theta v_q \cdot v_q)| \le B^{2|v|} A^{4q|w|}.$$

Proof sketch.

- (a) The two-scale chessboard estimate at the cutoff: main factors are in separated coarse cells, and exterior groups in a finer tiling, with no restriction on how many exterior factors share a coarse cell. A translation of less than one exterior cell fits all the margins. Point limits at generic configurations and continuity give the continuum bound.
- (b) The Gram pairing is the correlation of $4q|w|$ main and $2|v|$ exterior insertions at separated points. Apply (a).

Lean: `cutoff_bound`, `point_bound`, `Gapped`, `travel_price`.

Lemma VII.4.p (uniform constants; proof of VII.4.5).

Proof sketch.

- *One-field bounds.* For degrees $\le D$ and norms $\le R$ there are F and $J > 0$ such that, for $k \ge F$, every such token, together with all its signed and reflected variants, has one-field bound $J a_k^{-(D+1)}$.
- *Choice of level.* For $r \in (0,1]$, take the first $m \ge F$ with $a_m < r/32$. Gentleness gives a uniform lower bound $a_m \ge c_1 r^2$. A second level choice gives $k \ge m$ with $2|w| \cdot 10 a_k < a_m$ and $a_k \ge c(r/|w|)^4$. Hence $A^{|w|} \le \rho = C_0^{|w|}(|w|/r)^{4(D+1)|w|}$ with $C_0 = Jc^{-(D+1)}$.
- *Exterior.* For each exterior word v, choose its own levels and bound X_v. Lemma VII.4.o(b) then gives the bound of VII.4.5 with $B_v = X_v^{2|v|}$.
- *Width.* A sub-width $s \le d$ covered at level m is chosen before the exterior.

Lean: `rates`, `replicas`, `seats`. In V.1: `Drive.against`, `producer`.

3.29.4.10 4.10 Contact extensions Lemma VII.4.q (Taylor deviation on shells). Let E be a finite-dimensional Euclidean space, $K \subset E$ a subspace, $m, n \in \mathbb{N}$, $B \ge 1$, and U a compact subset of $\{\frac{1}{2} \le |x_K| \le 2\}$. Write $D_a x = a x_K + x_{K^\perp}$. There is $X \ge 1$, depending only on E, K, U, m, n, B, with the following property. Let $a, r, s > 0$ and let $A\colon E \to E$ be a linear isomorphism satisfying $(Ax)_K = r x_K$ for every x and $Ax = rx$ for $x \in K$. Let f be smooth and assume $ra/s \le B$, $r \le Bs$, and

$$\operatorname{supp}(f \circ A) \subset \{x : |x_K| \le B/2\}, \qquad \|f \circ A\|_{C^{n+m+1}} \le H.$$

Then the cut-off normal Taylor operator satisfies

$$\left\|(f - T^K_{m,s} f) \circ A \circ D_a\right\|_{C^n(U)} \le XH \begin{cases} a^{m+1}, & a \le 1, \\ a^m, & a > 1. \end{cases}$$

Proof sketch. Taylor's formula with remainder, together with jet bounds for the smooth cut-off.

Lean: `compact_deviation`.

Lemma VII.4.r (the subtraction formula).

Notation.

- φ_i is the field a_i at x_i.
- For a group $h \subset \{1, \dots, k\}$, $|h| \ge 2$: $x_h^\perp$ is the component of x normal to the h-diagonal, $\bar{x}_h$ the centroid, and $L_h = \max\left(0,\ \sum_{i \in h} d_i + 1 - 4(|h| - 1)\right)$.
- $T^h_m f(x) = \chi(|x_h^\perp|^2/s_0^2) \sum_{l \le m} \frac{1}{l!} D^l f(x - x_h^\perp)[x_h^\perp, \dots, x_h^\perp]$, with $\chi = 1$ near 0 and $\chi(|x_h^\perp|^2/s_0^2) = 0$ for $|x_h^\perp| \ge 2s_0$.
- R^h_w replaces a field polynomial by the weight-w component of its OPE projection, placed at $\bar{x}_h$. The device detectors use finite probe sets, a linear decoder, an $O(4)$-average and simultaneous input and output conjugation, followed by the device's graded splitting.

Definition. Set $C_{\{i\}} = \varphi_i \otimes 1$ and, for $|h| \ge 2$,

$$C_h = -\sum_{w < L_h} R^h_w(\Pi_h) \cdot T^h_{L_h - 1 - w}, \qquad \Pi_h = \sum_{\pi} \prod_{\beta \in \pi} C_\beta, \qquad \mathrm{full}_h = \Pi_h + C_h,$$

where π runs over partitions of h into at least two blocks. For a singleton the full expression is $\varphi_i \otimes 1$; for the empty group it is the unit. The kernel of $\mathcal{D}_a^s$ at x is the s-expectation of $\mathrm{full}_{\{1..k\}}$ applied to f. Expanding the recursion gives a Zimmermann-type forest sum. The child expressions are constructed recursively first. On the test function, the parent Taylor operator acts first, followed by the operators already present in the child expression; nested operators need not commute.

Per-group form. $[\Pi_h - \sum_{w<L_h} R_w^h \Pi_h] f + \sum_{w<L_h} R_w^h(\Pi_h)\,(1 - T^h_{L_h-1-w}) f$.

- Put $d_h = \sum_{i\in h} d_i$ and $q_h = 4(|h|-1)$. After retaining weights $w < L_h$, the remainder has radial order $O(|x_h^\perp|^{L_h - d_h - \varepsilon})$, with $L_h - d_h \geq 1 - q_h$.
- A retained weight-w term has coefficient order $O(|x_h^\perp|^{w-d_h-\varepsilon})$ and test deviation $O(|x_h^\perp|^{L_h-w})$.
- For $0 < \varepsilon < 1$ these radial powers are integrable in q_h normal dimensions. Angular sectors containing partial collisions require the subcluster induction of Lemma VII.4.s.
- Each weight-w coefficient has scaling degree at most $d_h - w$, and receives its own Taylor subtraction.
- If $\sum_{i\in h} d_i + 1 \leq 4(|h|-1)$, nothing is subtracted.

Lean: `takeRight`, `Score/Formula.lean`, `Normal.taylor`, `majorKernel`, `Devices`, `Devices.clearance`. *(definitions read)*

Lemma VII.4.s (absolute bound; summation and scaling). Fix the devices, labels, state and subtraction length as in VII.4.4. For every compact $U \subset (\mathbb{R}^4)^k$ there is $C > 0$ such that

$$\int |\langle \mathrm{full} \rangle_s(f;x)|\,dx \leq C\,\|f\|_{C^{(2+w(a))k}} \quad \text{whenever } \mathrm{supp} f \subset U.$$

The summation formulas and scaling bounds hold under the stage-family hypotheses of VII.4.4(d)–(e).

Proof sketch.

- *Patching.* Cover U by finitely many patches. On each patch the relevant partition is the set of coincidence classes of its centre. A smooth partition of unity and the Leibniz rule combine the patch bounds.
- *Induction.* On a patch, use strong induction on $|h|$ minus the number of blocks.
  - Base case: the separated correlation times f, bounded pointwise.
  - Inductive step: split into the remainder arm and the weight-w arms. Cut into dyadic normal shells $|x_h^\perp| \sim 2^{-n}$ and angular fans. Rescale each sector to a finer partition, and apply the induction hypothesis.
- *OPE input.* The raw coefficient rate and the remainder rate in $\mathcal{A}$ from Lemma VII.4.j.
- *Shell exponent.* Including normal volume, the effective exponent is $4(|h|-1) + L_h - d_h \geq 1$:
  - the remainder gains L_h;
  - a weight-w arm gains w from the coefficient and $L_h - w$ from Lemma VII.4.q.
- *Summing.* Choose $\gamma > 0$ with $(3^{|g|-|Q|}+1)\gamma \leq 1$, where $|Q|$ is the number of blocks. The accumulated exponent losses leave a factor $2^{-n\gamma}$ in the shell sum. The corresponding geometric series converges.

Lean: `glues_total`, `global_charge`, `commissions`, `Claim`, `Annulus`. *(proof read in outline)*

Lemma VII.4.t (distribution and restriction).

- A kernel with the bound of Lemma VII.4.s defines a continuous linear functional on C_c^∞, of order $\le (2 + w(a))k$.
- With the full group, $P = \mathrm{id}$, $o = 0$, $B_0 = \min(1, \mathrm{cl}(w(a)))$ and $s_0 = B_0/4$, the hypotheses of Lemma VII.4.s hold on all of $(\mathbb{R}^4)^k$.
- If $\mathrm{supp} f \subset \mathrm{Conf}_k$, the Taylor terms evaluate jets of f on diagonals, where f vanishes with all its derivatives. So the kernel equals $f(x)S^s(a; x)$.

Lean: `Charge`, `Transport.harmonic`, `contactDistribution`.

Lemma VII.4.u (real bases and thresholds).

- The device strata have bases fixed by the conjugation, and the basis in degree 0 contains the unit. Proof: real and imaginary parts span, and a real independent set extends to a basis.
- Given $f_n > 0$, a valid choice of thresholds is

$$g_n = \min\left(f_n,\ 2^{-n},\ \mathrm{cl}(n)/(2(n+1))\right), \qquad t_0 = g_0, \qquad t_{n+1} = \min(t_n/8,\ g_{n+1}).$$

  The thresholds define smooth scalar cutoffs. At a length $0 < \ell \le t_n$, the cutoff-weighted projection agrees with the original projection through degree n. These cutoffs do not enter $\mathcal{D}_a^s$.

Lean: `real_basis`, `installs`.

3.29.5 5. Where the difficulty lies

Routine but long.

- The rate bookkeeping (Lemmas VII.4.a, j).
- The $O(4)$-covariant Reader (Lemmas VII.4.c, k).
- The tree combinatorics (Lemma VII.4.n).
- The geometry of normal shells, angular sectors and compact patches.

Main estimates.

1. **The OPE comes from the RG.** The basic estimate (Lemmas VII.4.f–g) is a lattice statement: a cluster of renormalized observables equals one observable of degree t plus a remainder that gains $(a_m/a_k)^{t+1/16}$, at a cost $(a_m/a_j)^{4|I|}$. The margin $\frac{1}{16}$ is set by the transport gain $(a_k/a_m)^{1/8}$ of Part III's RG, not by the contraction $(a_m/a_k)^{t+1}$ of the irrelevant part. This is where Part III enters.
2. **Uniformity in states and spectators comes from the C*-norm.** The norm of $\mathcal{A}$ is computed by chessboard bounds on decks of translated copies, together with Gelfand and the full support of ν (Lemmas VII.4.d–e). The spectators then cost only their own one-field bounds, at their own scale, and the bound holds in every phase at once.
3. **The derivative bootstrap** (Lemma VII.4.i). Each derivative costs one power of the fine spacing, so a first-order comparison with the grid gains nothing. The induction therefore starts from high degree, where the crude bound is already strong enough.
4. **One coefficient system for every truncation degree.** It comes from a coherent prescription (VII.4.6) and the strata of the Reader.

Unusual, weaker or stronger than expected.

- *Exponents.* The estimates have an arbitrary ε-loss and do not determine logarithmic corrections. The degree w is the sum of the label degrees; the statement does not identify it with a scaling dimension.
- *Coefficients.* They depend on the prescription and the Reader, and nothing relates them to perturbation theory (VII.5, VII.6).
- *Spectator distance.* The remainder bound holds with power-separated spectators, $(r/\rho)^{N+1}$. The main theorem uses only a fixed clearance.
- *Contacts.* The contact extension is an existence statement. Its order $(2 + w(a))k$ is crude. Its OPE projection uses the device detectors, with the same prescription as the Reader but separate probe sets.
- *Interface to Part III.* The cluster replacement requires the numerical and geometric hypotheses in Lemma VII.4.f.

3.29.6 6. Notation

object	code name	proposed notation
upper and exact exponential rates	`RateLE`, `RateIs`	$f \preceq e^{\kappa T}$; $\log f/T \to \kappa$ with $f > 0$ eventually
height of a level sequence	`Height`	$m \to \infty$, $\log(1/a_m)/T \to c$
cluster, clearance	`WithinRoom`, `ClearOf`	(r, M, η)-cluster; c-clear
total degree	`weight`, `sigma`	$w(a)$
needles, sheets, tests	`Prescription`	u_i, λ_i, $\lambda_i^{(n)}$
raw coefficient	`Prescription.ope`	$\lambda_i(a; x, o)$
Reader	`Orbits.Reader`	$\mathcal{R}$
lower-degree removal, cut test, stratum	`knife`, `cutTest`, `stratum`	π_n, $\tau_{n,q}$, E_n
channels, actors, coefficients	`channels`, `actor`, `coef`	Ch_j, $O_{j,\kappa}$, $c_{j,\kappa}(a; x, o)$
ground algebra, states	`unselected.algebra`, `glance`, `Viewers`	$\mathcal{A}$; $s \in X \sqcup \{\mathrm{mix}\}$
error element	`algebraError`, `groundError`	$\mathcal{E}_D(a, x; b, y)$
chessboard root with interior condition	`Cards.PaidAt`	$\|F\|_{\mathcal{T}_k} \le A$
deck of translates	`deck`, `Episode`	$\prod_{l<2n} U^{(*)} \circ \tau_{2tl}$
limit label of a cluster	`Survival`	Q
coefficient size of a label	`quota`	$\|z\|_m$
RG insertion, step and column bounds	`paid`, `ζ`, `δ`, `φ`, `x`, `columns`, `alto`, `march`, `gainChart`	paid_j, ζ, δ, φ, x, c_m, $A_{k,t}$, $\mathfrak{m}_{k,m}$, $(a_k/a_m)^{1/8}$
cell and stencil fitting data	`coverSize`, `lensScope`, `GridApprox.tail`, `clock.J`	$\mathfrak{c}_k$, ℓ_t, τ_m, $\jmath_m$
RG package	`Services`	(RG)
tree plot, admissibility	`Accept`, `Plot`	(s, z) admissible
tree costs	`Tree.toll`, `nestPrice`	toll_m, nest_D
one-field bound, separation	`Next.Phase.Single`, `Gapped`	$\|I(v, \cdot)\|_{\mathcal{T}_k} \le A$
train, quantum read	`ride`, `QRead`	v_n; bench bound ρ
contact devices, clearance	`Contacts.Devices`, `clearance`	$\mathcal{M}$, $\mathrm{cl}(n)$

object	code name	proposed notation
cumulant, punctured, full	`cumulant`, `punctured`, `full`	C_h, Π_h, full_h
normal Taylor operator	`Normal.taylor`	T^h_m
distribution of finite order	`Charge`, `totalTestOrder`	$\mathcal{D}^s_a$, order $(2+w)k$
contact thresholds	`Thresholds`	t_n

3.29.7 7. What remains unexplained

1. **RG inputs below the cluster replacement.** Lemmas VII.4.f–g identify the scalar inequalities and the scale induction. The polymer norm estimates inside the one-step representation, fusion and final remainder transport have not been checked line by line. The dependence of their constants, and the complete Part III proofs of these inputs, remain to be reconciled with the stated RG package.
2. **Proofs used through their statements.** The full geometry behind the vacuum deck limit and the two-scale continuum transfer belongs to V.1. The rotated comparison estimates used in Lemma VII.4.k, and the triangular replacement and high-horizon lemmas used in VII.4.n, have not been checked internally. Their uses and exponent arithmetic agree with their stated conclusions.
3. **Contact sector induction.** The global patching argument, subtraction recursion, shell Taylor bound and scaling normalization have been checked. The finer angular decompositions and the full induction through partially coincident strata remain checked only at the level of their theorem calls. Lemma VII.4.r's radial estimate alone does not settle these partial collisions.
4. **Contact projection versus Reader coefficients.** The devices share the Reader's prescription but use separate detectors and a separate graded splitting. An identity between their weight-w rows and $\sum_\kappa c_{w,\kappa} O_{w,\kappa}$ has not been located. No such identity is used in the contact statement above.
5. **Limits of the conclusions.** The results do not assert uniqueness of coefficients or contact extensions, independence of the extraction subsequence, convergence as $D \to \infty$, reflection positivity or Ward identities for the extensions, or agreement with perturbative coefficients.

3.30 VII.5. Asymptotic freedom of the running coupling

Part: VII, identification as Yang–Mills. **Lean:** `YM/Pert/Asymptotics` (the running coupling). Auxiliary declarations are in `YM/Extra/WEAK_SCALING_ACTLAW` (files `Orientation`, `Criteria`, `Adjust`, `Analysis`, `Station`), from `YM/TraceExtension/Foundation` (files `Caps`, `Dilate`), and one definition from `YM/TraceExtension/Soot`. **Size:** 232 lines in 43 declarations: 32 lines in `Pert/Asymptotics` (5 declarations), 162 in `WEAK_SCALING_ACTLAW` (24), 37 in `TraceExtension/Foundation` (13), 1 in `TraceExtension/Soot` (1).

3.30.1 1. What this unit proves

(A) The running coupling (5 declarations). The inverse squared coupling of the constructed renormalization-group trajectory grows logarithmically, uniformly in the lattice level and the volume (VII.5.1). This gives the regularity clause (Reg1) of the main theorem with slope $b_* = 2b_0$ (VII.5.2). It also gives the law $g^2 \log(1/a) \to 1/(2b_0)$ (VII.5.3).

(B) Auxiliary results (38 declarations).

- VII.5.4: the normalisation of the stress tensor along free-field scaling sequences (used by VI.2).
- VII.5.5: exactly compatible frames, built from approximately compatible ones (used by VII.1).
- VII.5.6: two elementary convergence lemmas (used by III.12).
- VII.5.7–VII.5.8: the definitions of the classical stress-tensor algebra and a read-out predicate (used by VI.1, VII.2, VIII.1).

Setting for (A).

- *Gauge group.* $G \subset U(q)$, $q \geq 1$, is a compact matrix group with Lie algebra $\mathfrak{g}$ and inner product $\langle X, Y\rangle_{\mathfrak{g}} = -\frac{1}{q}\mathrm{Re}\,\mathrm{tr}(XY)$. For an orthonormal basis (e_i), define $C_{\mathfrak{g}} := -(\dim \mathfrak{g})^{-1}\sum_i \mathrm{tr}_{\mathfrak{g}}((\mathrm{ad}\, e_i)^2)$, with value 0 when $\mathfrak{g} = 0$. Here and below, simple means real simple and nonabelian. If $\mathfrak{g}$ is simple, its Killing form equals $-C_{\mathfrak{g}}\langle\cdot,\cdot\rangle_{\mathfrak{g}}$. For $SU(q) \subset U(q)$ with $q \geq 2$, this gives $C_{\mathfrak{g}} = 2q^2$. Put

$$\kappa_G := \frac{11\, C_{\mathfrak{g}}}{24\pi^2}, \qquad b_0 := \frac{11\, C_{\mathfrak{g}}}{48\pi^2} = \tfrac{1}{2}\kappa_G.$$

- *Scales.* The block factors $b_k \geq 3$ are odd. Set $D_m = \prod_{k<m} b_k$, $a_m = 1/D_m$ and

$$s_m := \log(1/a_m) = \sum_{k<m} \log b_k.$$

  Then $s_m \to \infty$.

- *Trajectory.* For a lattice level N, a volume index p and a stage $m \leq N$, $\beta_{N,p,m}$ is the coefficient of the normalised kinetic term in the effective action at spacing a_m. The represented density is obtained from the Wilson weight by $N - m$ block-spin steps.
  - $\beta_{N,p} := \beta_{N,p,N}$ is the bare coupling of the main theorem.
  - $\beta_{N,p,0} = \bar{\beta} > 0$ for every N, p. This is the renormalization condition at unit scale.
  - The reference line is $\hat{\beta}_m := \bar{\beta} + \kappa_G s_m$. One has $\beta_{N,p,m} \in [\hat{\beta}_m/2,\, 2\hat{\beta}_m]$.
  - The running coupling is $g_{N,p,m} := \beta_{N,p,m}^{-1/2}$.

The **running clause** (R) of the construction conditions reads: for every $\varepsilon > 0$ there is m_0 such that

$$|\beta_{N,p,m} - \hat{\beta}_m| \leq \varepsilon\, s_m \quad \text{for all } m \geq m_0,\ N \geq m,\ p \in \mathbb{N}.$$

Proposition VII.5.1 (uniform logarithmic growth of the inverse squared coupling). Let the family satisfy its construction conditions (II.2). For every $\varepsilon > 0$ there is $m_0 \geq 1$ such that

$$\Big|\frac{\beta_{N,p,m}}{s_m} - 2b_0\Big| < \varepsilon \qquad \text{for all } m \geq m_0,\ N \geq m,\ p \in \mathbb{N}.$$

Status: theorem. Of the construction conditions only (R) is used. Simplicity of $\mathfrak{g}$ is not needed. *Lean:* `YM.Pert.Asymptotics.zeroUp_bound`, with `b0` and `b0_normalizes`.

Corollary VII.5.2 ((Reg1) for the constructed family). Let $\mathfrak{g}$ be simple and let the family satisfy its construction conditions. Then its Wilson family satisfies (Reg1) with slope $b_* = 2b_0 = 11C_{\mathfrak{g}}/(24\pi^2) > 0$: for every $\varepsilon > 0$ there is $N_0 \geq 1$ with

$$\Big|\frac{\beta_{N,p}}{\log(1/a_N)} - b_*\Big| < \varepsilon \quad \text{for all } N \geq N_0 \text{ and all } p.$$

Status: theorem, proved in VIII.1 as the first clause of the regularity of the constructed family. It is VII.5.1 at $m = N$, together with $C_{\mathfrak{g}} > 0$ (I.1). This is how the main theorem uses part (A). The main theorem asserts the existence of a positive slope without specifying its value. *Lean:* `YangMills.regular` (running clause of `YangMills.WilsonRegularity`), via `zeroUp_bound`.

Proposition VII.5.3 (asymptotic freedom of the running coupling). Let $\mathfrak{g}$ be simple and let the family satisfy its construction conditions. Let (N_i) and (k_i) be sequences of lattice levels and volume indices with $i \leq N_i$ for all large i. Then

$$\lim_{i\to\infty} g_{N_i,k_i,i}\,\sqrt{\log(1/a_i)} = \frac{1}{\sqrt{2b_0}}, \qquad \text{equivalently} \qquad g^2_{N_i,k_i,i}\,\log(1/a_i) \longrightarrow \frac{1}{2b_0}.$$

- For $N_i = i$ this is the law of the bare coupling, $g_0(a)^2 \sim (2b_0 \log(1/a))^{-1}$.
- For $N_i \geq i$ it is the law of the effective coupling at spacing a_i of any finer lattice. No condition is placed on the volumes k_i.

Status: theorem; its square gives the second limit. No other declaration cites it. *Lean:* `YM.Pert.Asymptotics.comparison`, `regulatorRun`.

Setting for (B).

- *Spaces and maps.*
  - $\mathcal{F}_d$ is the finite-dimensional complex space of labels of degree at most d in a correlation table S carrying a Euclidean hierarchy H (IV.1).
  - $\mathcal{F}^\circ_d$ is the corresponding space of the free Maxwell comparison table, with its hierarchy H°, built on the same reference labels (VI.1).
  - A *derivative plan* (IV.1) on either table consists of complex-linear inclusions $\iota_{d\to e}$ $(d \leq e)$ and complex-linear partial derivatives $\partial_{d,a} : \mathcal{F}_d \to \mathcal{F}_{d+1}$, $a = 1, \dots, 4$.
  - The inclusion axiom says that inserting $\iota_{d\to e}v$ in a separated correlation gives the same value as inserting v. It implies injectivity, $\iota_{d\to d} = 1$, transitivity and preservation of the unit. The derivative axiom identifies insertion of $\partial_{d,a}v$ with differentiation of the correlation in that insertion point.
  - The orientation axioms say that inclusions commute with the hyperoctahedral group Γ_4 of signed permutations (384 elements) and with conjugation. Derivatives commute with conjugation and transform as vectors under Γ_4.
- *Stress data.* A *candidate stress tensor* is a family $J = (J_{ab})$ in $\mathcal{F}_4$. Its *local Ward identities* (IV.1, VI.1) involve a translation part $p = (p_{d,a} : \mathcal{F}_d \to \mathcal{F}_{d+1})$ and a rotation part Y.
- *Free Maxwell data* (VI.2, VIII.1): the comparison data specified in §2, including a derivative plan W, a rotation action, a tensor J° and $\kappa \neq 0$. The plan satisfies the inclusion, derivative and orientation axioms; J° satisfies the local Ward identities with translation part $\kappa\,\partial^W$ and rotation part κ times the infinitesimal rotation action; and the family $\partial^W_4 = (\partial^W_{4,a})_a$ is nonzero.

- *Scaling sequences.* A *scaling sequence* (VI.1) consists of frames $U_k(d) : \mathcal{F}^\circ_d \to \mathcal{F}_d$ (complex-linear isomorphisms), positive, strictly decreasing scales $r_k \to 0$ and stress normalisations $\xi_k > 0$.
- *Tracking.* A scaling sequence *tracks* (V, W, J) at level (n, d, j, R) (VI.1) if three conditions hold:
  - the frames are exactly compatible with inclusions, signed permutations, conjugation and units up to degree $\max(d, 4)$;
  - for each fixed list of at most n fields of degrees at most d, the correlations transported by U_k and evaluated at $r_k x$ converge, with all Fréchet derivatives of orders $0, \dots, j$, uniformly on the window $\{x : \|x\| \le R,\ \|x_a - x_b\|_\infty \ge R^{-1} \text{ for all } a \ne b\}$, to the free correlations;
  - $\xi_k^{-1} r_k^4\, U_k(4)^{-1} J_{ab} \to J^\circ_{ab}$.

Proposition VII.5.4 (normalisation of the stress tensor along free-field scaling sequences). Fix the two hierarchies and the free Maxwell comparison data above. Let V be a derivative plan on S satisfying the inclusion and derivative axioms. Let J be symmetric, real and Γ_4-covariant, with $\sum_a \partial^V_{4,a} J_{ab} = 0$ for every b, and let (J, p, Y) satisfy the local Ward identities.

(a) Suppose $p_{4,a} = c\,\partial^V_{4,a}$ for $a = 1, \dots, 4$ and some $c \in \mathbb{R}$. Then there are $D, N, M \in \mathbb{N}$ and $R_0 \ge 1$ such that $c/\xi_k \to \kappa$ for every scaling sequence that tracks (V, W, J) at a level (n, d, j, R) with $n \ge N$, $d \ge D$, $j \ge M$ and $R \ge R_0$.

(b) If $p \ne 0$, then $J \ne 0$.

Status: theorems. Part (a) uses A04 and A05 (VI.1); part (b) requires only the hierarchy and local Ward identities. *Lean:* `YM.Extra.WeakScaling.tracks_ratio`, `scale_limit`, `flux_nontrivial`. Used in VI.2 by `YM.Pert.ratio` and `YM.Pert.centered_line`.

Proposition VII.5.5 (exactly compatible frames). Suppose S and its free comparison table carry Euclidean hierarchies. Let V and W be derivative plans on these tables satisfying the inclusion and orientation axioms.

(a) For every d there is a $\mathbb{C}$-linear map $P_d : \mathcal{F}^\circ_{d+1} \to \mathcal{F}^\circ_d$ with $P_d\, \iota^W_{d\to d+1} = 1$ that commutes with Γ_4 and with conjugation.

(b) Let the P_d be as in (a), fix $D \in \mathbb{N}$, and let $A_d : \mathcal{F}^\circ_d \to \mathcal{F}_d$ be complex-linear isomorphisms for every d. Define recursively

$$X_0 := A_0, \qquad X_{d+1} := \iota^V_{d\to d+1}\, X_d\, P_d + A_{d+1}(1 - \iota^W_{d\to d+1} P_d).$$

Suppose that for every $d \le D$ the map A_d commutes with Γ_4 and conjugation and maps unit to unit. Then:

- $X_e\, \iota^W_{d\to e} = \iota^V_{d\to e}\, X_d$ for all $d \le e \le D$;
- for $d \le D$, each X_d commutes with Γ_4 and conjugation and maps unit to unit.

(c) Let $A^{(k)}_d$ be complex-linear isomorphisms such that, for every $d < D$ and every v,

$$(A^{(k)}_{d+1})^{-1}\, \iota^V_{d\to d+1}\, A^{(k)}_d\, v \longrightarrow \iota^W_{d\to d+1} v \qquad (k \to \infty).$$

Let $X^{(k)}$ be built from $A^{(k)}$ as in (b). Then for every $d \le D$:

- $(A_d^{(k)})^{-1}X_d^{(k)} \to 1$ in operator norm;
- $X_d^{(k)}$ is bijective for all large k;
- the isomorphisms $\widetilde{X}_d^{(k)}$ (equal to $X_d^{(k)}$ when it is bijective and to $A_d^{(k)}$ otherwise) satisfy $(A_d^{(k)})^{-1}\widetilde{X}_d^{(k)}v \to v$ for every v.

If the $A_d^{(k)}$ also preserve the symmetries and units in (b), then, for all sufficiently large k, the frames $\widetilde{X}_d^{(k)}$ are exactly compatible simultaneously in all degrees $d \le D$.

Status: theorem (finite-dimensional algebra and analysis). *Lean:* `retraction`, `adjusted`, `Exact`, `adjusted_exact`, `Near`, `perturb_limit`, `eventually_unit`, `rotateChart`, `relative`, `Graded`. Used in VII.1 by `YM.SharpGram.Weak.Portfolio.tune_grades`. Exact compatibility is also used in III.13 and VIII.1.

Proposition VII.5.6 (finite-dimensional convergence).

(a) Let $f : E \to X$ be an injective linear map between finite-dimensional real normed spaces. If $f(a_y) \to f(e)$ along a filter, then $a_y \to e$.

(b) Let $f_y, g : (\mathbb{R}^4)^n \to \mathbb{C}$ be continuous real-linear maps, and fix a filter on the index set of y. If $f_y(e_{i,a}) \to g(e_{i,a})$ at each of the $4n$ coordinate vectors, then $f_y \to g$ in operator norm along that filter.

Status: theorem. *Lean:* `norm_fit`, `norm_fit_deriv`, `components`. Used in III.12 by `YM.FullVolume.stretching`.

Definition VII.5.7 (the classical stress-tensor algebra). Labels of degree d are gauge-invariant polynomials of weight at most d in radial Taylor coefficients at the base point. In the formulas below, F and ∂F denote the linear curls of the degree-0 and degree-1 Taylor coefficients used in the free Maxwell comparison. The curvature has weight 2 and each derivative adds 1 (II.1). $\langle\cdot,\cdot\rangle$ denotes the invariant inner product on $\mathfrak{g}$. These are polynomial identities, before passing to the quotient by null fields; they do not multiply limiting fields at coincident points.

- *Weight 4.* The square is $\mathfrak{s} := \frac{1}{4}\sum_{i,j}\langle F_{ij}, F_{ij}\rangle$, summed over ordered pairs. The candidate stress tensors of VI.1 form the family

$$T_{ab}^{(t)} = \delta_{ab}\Big(t_0\mathbf{1} + t_1\sum_i\langle F_{ai},F_{ai}\rangle + t_2\sum_{i,j}\langle F_{ij},F_{ij}\rangle\Big) + (1-\delta_{ab})\,t_3\sum_i\langle F_{ai},F_{bi}\rangle.$$

  The *canonical* tensor is $T^{\mathrm{can}} := T^{(0,1,-1/4,1)}$, that is, $T^{\mathrm{can}}_{ab} = \sum_i\langle F_{ai},F_{bi}\rangle - \delta_{ab}\,\mathfrak{s}$.

- *Weight 5.* Define the vectors

$$\mu^1_b := \sum_i\langle F_{bi},\partial_b F_{bi}\rangle = \tfrac{1}{2}\partial_b\sum_i|F_{bi}|^2, \qquad \mu^2_b := \sum_{i,j}\langle F_{ij},\partial_b F_{ij}\rangle = \tfrac{1}{2}\partial_b\sum_{i,j}|F_{ij}|^2,$$

  and the equation-of-motion vector $E_b := \sum_{a,i}\langle F_{bi},\partial_a F_{ai}\rangle$ (VI.1). Let $\mathcal{R}_1$ be the real space of real, Γ_4-covariant vectors of degree at most 5 (VI.1). Define $\beta,\delta : \mathbb{R}^3 \to \mathcal{R}_1$ by

$$\beta(w) := w_0\mu^1 + w_1\mu^2 + w_2E, \qquad \delta(w) := \beta(2(w_0-w_2),\ 2w_1+\tfrac{1}{2}w_2,\ w_2).$$

The definitions have two accompanying lemmas:

- $\mu^1, \mu^2 \in \mathcal{R}_1$;
- every degree-0 radial jet is realised by a constant-curvature free Maxwell field, with an affine potential.

Status: definitions and two lemmas. *Lean:* `square4`, `raw`, `momentum1`, `momentum2`, `momentum1_mem`, `momentum2_mem`, `momentum2_turn`, `basisV`, `bareVector` (with `mat3`, `proj3`), `shell_affine`, `sumPerm`. The facts proved about these objects in VI.1 and VII.2 are listed in §4.5.

Definition VII.5.8 (sharp read-out bound). Fix the family, a nondecreasing control function $K : \mathbb{N} \to \mathbb{R}$ with $K(j) \geq 1$ for every j, a degree d, stages $\ell_d \leq k \leq N$, a volume index p and $u \in \mathbb{R}$. Write $\|c\|_1 := \sum_\alpha |c_\alpha|$ for the coefficient norm, and let P remove the unit coordinate. Here ℓ_d is the reference stage fixed by the family. The predicate quantifies over every $w \geq 0$ and all insertion presentations A_i with coefficient vectors c_i, remainder costs q_i, inserted densities $\mathcal{D}_i$ and insertion points at the origin. They must satisfy:

- At each $\ell_d \leq i \leq k$, A_i has the admissible regular and forced remainders of III.12, with total remainder cost at most q_i, using the square root of the stage-i forced allowance.
- $c_{\ell_d} = 0$. Only this coefficient vector must vanish at the reference stage.
- The full inserted densities obey $\mathcal{D}_i = b_i^d \, \mathcal{T}^{(i)}_{\beta_{N,p,i+1}} \mathcal{D}_{i+1}$ for $\ell_d \leq i < k$.
- Put $(S_i c)_\alpha := b_i^{d-d_\alpha} c_\alpha$, where d_α is the weight of the basis label, and $K_i := K(J(i+1))$. For each $\ell_d \leq i < k$, $q_i, q_{i+1} \geq 0$ and there is a complex-linear automorphism B_i commuting with P such that $\|B_i c\|_1, \|B_i^{-1} c\|_1 \leq K_i \|c\|_1$ for every c, and

$$\|c_i - B_i S_i c_{i+1}\|_1 \leq b_i^{-3/8}(\|P c_{i+1}\|_1 + q_{i+1}), \qquad q_i \leq K_i \|P S_i c_{i+1}\|_1 + b_i^{-3/8} q_{i+1}.$$

- The terminal cost satisfies $q_k \leq w + n_k(w + \|c_k\|_1)$, where $w \geq 0$ and n_k is the stage-k column number.

The sharp read-out predicate is the assertion that every such chain satisfies

$$u \, \|c_k\|_1 \leq 4(1 + n_k) w.$$

Status: definition. The admissibility and remainder conditions are those of `Format`, `remBound`; the complete chain conditions are `Duet.Woke` and `direction`. *Lean:* `YM.MergeProof.Accent.SharpRead`. Used by: `YM.Trace.Terrace.Sceptre` and the cost lemmas of VII.2.

3.30.2 2. What it uses

Inputs from other units.

- *I.1 (gauge group and scales).*
  - $C_{\mathfrak{g}}$ (`YM.Gauge.casimir`) and κ_G, which is *defined* as $11 C_{\mathfrak{g}}/(24\pi^2)$ (`slopeConstant`).
  - For simple $\mathfrak{g}$, the Killing-form identity in §1 is `trace_adjoint`.
  - Positivity $C_{\mathfrak{g}} > 0$ and $\kappa_G > 0$ for simple $\mathfrak{g}$ (`casimir_pos`, `slopeConstant_pos`).

  - $a_m = 1/D_m$ (`Schedule.a`); s_m is *defined* as $\log(1/a_m)$ (`Schedule.scale`); and $s_m \to \infty$ (`Schedule.divergent`).
- *II.2 (construction conditions)* `YM.Family.Makes`. Its first field says that the trajectory is a selected history (III.1). The bare coupling is `Family.beta`, that is, $\beta_{N,p,N}$.
- *III.1 (selected history)* `YM.RG.History.Selected`. It contains:
  - $\beta_{N,p,0} = \bar{\beta}$;
  - the per-step relation $|\beta_{N,p,m+1} - \beta_{N,p,m} - \kappa_G \log b_m| \le \delta_{m+1}$ (`Transfer`);
  - the band $[\hat{\beta}_m/2, 2\hat{\beta}_m]$ (`Conv`);
  - the running clause (R) (`Running`), with $\hat{\beta}_m$ given by `hatBeta`.
- *III.2 (choice of constants, shooting).*
  - (R) is derived in `YM.RG.Shoot.trajectory`. The inputs are $\beta_{N,p,0} = \bar{\beta}$, the per-step relation, and $\delta_{m+1}/\log b_m \le J(m+1)^{-2}$, where $J(m) \to \infty$ is the stage index of the scale sequence (Lemma VII.5.b).
  - The history with $\beta_{N,p,0} = \bar{\beta}$ is produced by shooting (`Recipe.races`, `successful`).
  - The per-step relation follows from the coupling reset (III.9). If the one-loop shift $\alpha_{N,m}$ satisfies $|\alpha_{N,m} + \kappa_G \log b_m| + 1 \le \delta_{m+1}$ and the reset changes β by $\alpha_{N,m}$ with error at most 1, the triangle inequality gives the transfer bound. This is `Directed.links`, using `Kin.Planned`.
- *VIII.1:* uses VII.5.2 in the assembly of the main theorem.
- *For VII.5.4.*
  - The work items A04 (`answer`) and A05 (`answer`), both proved in VI.1. Along tracking scaling sequences of sufficient arity, degree, derivative order and window size, they give: $r_k\, \partial_d^V$ conjugated by the frames converges to ∂_d^W; and $(r_k/\xi_k)\, p_d$ conjugated by the frames converges to $\kappa\, \partial_d^W$.
  - The tracking notion (`Track`) and the hypotheses on J (`Pre`), from VI.1.
  - The Ward identities (`StressLaws`) and derivative plans (`DiffPlan`), from IV.1.
  - The free Maxwell data (`MaxwellData`, `Lighthouse.Goes`). Besides the data in §1, the full package includes finite separating tests and the boundary, polynomial and internal-rotation conditions of VI.2. The ratio argument uses the free hierarchy, derivative and Ward identities, and the nonzero degree-4 derivative.
- *For VII.5.5:* the inclusion and orientation axioms of derivative plans (IV.1), and the free comparison table (`Spin.comparator`, VI.1).
- *For VII.5.7:*
  - the label polynomials $\langle F_{ab}, F_{cd} \rangle$ (`pair`) and $\langle F_{ab}, \partial_i F_{cd} \rangle$ (`pairDeriv`);
  - E (`eom`), the family $T^{(t)}$ (`stress`, `weyl`) and $\mathcal{R}_1$ (`RawOne`), all from VI.1;
  - the ideal of labels vanishing on free Maxwell solutions (`Maxwell.equationIdeal`), from III.16.
- *For VII.5.8:* the comparison-chain definitions of III.2 and the insertion estimates of III.12. For the constructed family, `attain` (III.2) supplies the read-out field of the trace-analysis assumptions under its scale and one-step hypotheses.

Standard mathematics.

- *Asymptotic freedom.* Gross–Wilczek and Politzer (Phys. Rev. Lett. 30, 1973). In the conventional normalisation for $SU(q)$, with $\beta = 2q/g^2$, the one-loop law is $1/g^2 \sim 2b_0^{\mathrm{conv}} \log(1/a)$ with $b_0^{\mathrm{conv}} = 11q/(48\pi^2)$.
  - Here $\beta = 1/g_0^2$ in the normalisation $\langle X, Y \rangle_{\mathfrak{g}} = -\frac{1}{q} \mathrm{Re}\,\mathrm{tr}(XY)$, and $C_{\mathfrak{g}} = 2q^2$. So $b_* = 2b_0 = 11q^2/(12\pi^2) = 2q \cdot 2b_0^{\mathrm{conv}}$, which agrees with the conventional law.

  - For $SU(2)$, $b_* = 11/(3\pi^2)$.
  - For the relation between the bare coupling and the lattice spacing see Creutz, *Quarks, Gluons and Lattices* (1983).
- *Block-spin RG for lattice gauge fields.* Balaban's renormalization group (Comm. Math. Phys. 109 (1987), 116 (1988), and related papers) provides the background for the marginal-flow estimate. The specific input here is the per-step bound of III.9; (R) follows by summation.
- *Exactly compatible frames.* Averaging over a finite group gives equivariant projections (Maschke). In finite dimension, injective linear maps form an open set, and linear maps are continuous.
- *Classical stress tensor.* The symmetric gauge-field stress tensor T^{can} is conserved on the equations of motion and is traceless in four dimensions (classical scale invariance; Callan–Coleman–Jackiw, Ann. Phys. 59, 1970). The breaking of tracelessness in the quantum theory is the trace anomaly (Adler–Collins–Duncan 1977; Collins–Duncan–Joglekar 1977), treated in VII.2.

Imported constants and assumptions.

- The value $\kappa_G = 11C_{\mathfrak{g}}/(24\pi^2)$ is a definition (I.1), and $b_0 := \kappa_G/2$, so $\kappa_G = 2b_0$ holds by unfolding. That κ_G is the actual slope of the trajectory is the content of the per-step relation, proved in Part III.
- $\bar\beta > 0$ is the fixed reference coupling (III.2). The limiting slope is independent of $\bar\beta$; the convergence thresholds may depend on it.
- VII.5.1 assumes the construction conditions. VII.5.2–VII.5.3 additionally assume simplicity. The auxiliary hypotheses are stated in §1.

3.30.3 3. Strategy

Part (A). The coupling law follows from the one-step bound by summation and division by the logarithmic scale.

1. *Marginal flow (III.9).* Blocking from stage $m+1$ to stage m gives $\beta_{N,p,m} = \beta_{N,p,m+1} - \kappa_G \log b_m + e_{N,p,m}$, with $|e_{N,p,m}| \le \delta_{m+1}$ and $\delta_{m+1}/\log b_m \le J(m+1)^{-2} \to 0$. The reset uses the one-loop estimate of III.7.

2. *Shooting and summation (III.2).* For each N, p the bare coupling $\beta_{N,p,N}$ is chosen so that the flow reaches $\beta_{N,p,0} = \bar\beta$; this is the renormalization condition at unit scale. Summing the per-step relation from step 0 to step m gives

$$|\beta_{N,p,m} - \hat\beta_m| \le \sum_{i<m} \delta_{i+1}.$$

   Since $\delta_{i+1} = o(\log b_i)$ and $s_m = \sum_{i<m} \log b_i \to \infty$, the right side is $o(s_m)$ by a Cesàro argument. The bound is uniform in $N \ge m$ and in p, because the per-step bounds are.

3. *Division by s_m (this unit).* Since $\bar\beta/s_m \to 0$, one gets $\beta_{N,p,m}/s_m \to \kappa_G = 2b_0$, uniformly. Taking $m = N$ gives (Reg1); taking square roots gives the coupling law.

The result concerns the actual inverse coupling of the trajectory, including its bare endpoint. It does not identify this quantity with a coupling defined from continuum correlations.

Part (B). The auxiliary arguments use finite-dimensional limits and equivariant linear algebra.

- *VII.5.4* compares two convergent sequences in a finite-dimensional space. Along a free-field scaling sequence, the rescaled derivative $r_k\partial_4^V$ tends to $\partial_4^W \neq 0$. The rescaled flux $(r_k/\xi_k)p_4 = (c/\xi_k)\, r_k\partial_4^V$ tends to $\kappa\,\partial_4^W$. Hence the scalar c/ξ_k tends to κ.
- *VII.5.5* uses equivariant averaging and a recursive correction:
  1. an equivariant retraction of each inclusion, obtained by averaging over Γ_4 and conjugation;
  2. a recursive correction, degree by degree, that turns approximately compatible frames into exactly compatible ones;
  3. a perturbation argument showing the correction is eventually invertible and asymptotically equal to the original frames.
- *VII.5.7* is the algebra of stress tensors quadratic in F under the hyperoctahedral group. On the lattice only Γ_4 acts, so the candidate family has four parameters and its divergences have three coordinates. For simple $\mathfrak{g}$, conservation on free Maxwell fields singles out the canonical tensor up to scale and addition of a constant multiple of $\delta_{ab}\mathbf{1}$. The canonical tensor is traceless.

3.30.4 4. Steps

3.30.4.1 4.1 The running coupling

Lemma VII.5.a (normalisation). $2b_0 = \kappa_G$. *Proof.* Unfold both definitions. *Lean:* `b0_normalizes`.

Lemma VII.5.b (the running clause; input from III.2). Let $\delta : \mathbb{N} \to [0,\infty)$ and $J : \mathbb{N} \to \mathbb{N}$ satisfy $J(m) \geq 1$ and $J(m) \to \infty$. With the scales and reference line of §1, suppose:

- $\beta_{N,p,0} = \bar{\beta}$ for all N, p;
- $|\beta_{N,p,m+1} - \beta_{N,p,m} - \kappa_G \log b_m| \leq \delta_{m+1}$ for $m < N$;
- $\delta_{m+1}/\log b_m \leq J(m+1)^{-2}$ for every m.

Then (R) holds.

Proof sketch.

1. Since $s_{m+1} = s_m + \log b_m$, induction on m gives $|\beta_{N,p,m} - \hat{\beta}_m| \leq \Delta_m := \sum_{i<m} \delta_{i+1}$ for $m \leq N$.
2. Given ε, choose j with $J(i+1)^{-2} \leq \varepsilon/2$ for $i \geq j$. Then $\Delta_m \leq \Delta_j + \frac{\varepsilon}{2}(s_m - s_j)$ for $m \geq j$.
3. For m large, $\Delta_j \leq \frac{\varepsilon}{2} s_m$, hence $\Delta_m \leq \varepsilon s_m$.

The constants and thresholds do not involve N or p. *Lean:* `YM.RG.Shoot.trajectory` (III.2).

Lemma VII.5.c (= Proposition VII.5.1). *Proof sketch.*

1. Apply (R) with $\varepsilon/4$. Because $s_m \to \infty$, one may also require $s_m \geq 4(|\bar{\beta}| + 1)/\varepsilon$.
2. Then $s_m > 0$ and $|\bar{\beta}| < \frac{\varepsilon}{4} s_m$.
3. With $\kappa_G = 2b_0$, the triangle inequality gives

$$|\beta_{N,p,m} - 2b_0 s_m| \leq |\beta_{N,p,m} - \hat{\beta}_m| + |\bar{\beta}| < \varepsilon s_m.$$

4. Divide by s_m.

Lean: `zeroUp_bound`.

Lemma VII.5.d (= Corollary VII.5.2). *Proof sketch.* The slope is $2b_0$, which is positive because $C_{\mathfrak{g}} > 0$ for simple $\mathfrak{g}$. Given ε, take $N_0 := m_0$ from VII.5.1 and specialise to $m = N$. The bare coupling of the public family is $\beta_{N,p,N}$, and its $\log(1/a_N)$ is s_N by definition. *Lean:* `YangMills.regular` (VIII.1).

Lemma VII.5.e (= Proposition VII.5.3). *Proof sketch.*

1. For each ε, eventually both $i \geq m_0(\varepsilon)$ and $i \leq N_i$. So VII.5.1 gives $\beta_{N_i,k_i,i}/s_i \to 2b_0$.
2. Since $\kappa_G > 0$ for simple $\mathfrak{g}$, $2b_0 > 0$. Then $x \mapsto \sqrt{x}$ is continuous at $2b_0$ and $y \mapsto y^{-1}$ is continuous at $\sqrt{2b_0} \neq 0$. Hence $(\beta_{N_i,k_i,i}/s_i)^{-1/2} \to (2b_0)^{-1/2}$.
3. For all sufficiently large i, both s_i and $\beta_{N_i,k_i,i}$ are positive, so $\beta_{N_i,k_i,i}^{-1/2} s_i^{1/2} = (\beta_{N_i,k_i,i}/s_i)^{-1/2}$, and $s_i = \log(1/a_i)$ by definition.

Lean: `comparison`.

3.30.4.2 4.2 Normalisation of the stress tensor (VII.5.4) Lemma VII.5.f (ratio lemma). Let Z be a finite-dimensional real normed space. If $X_k \to x \neq 0$ in Z and $f_k X_k \to c\,x$ with $f_k \in \mathbb{R}$, then $f_k \to c$. *Proof sketch.* Choose a linear functional ℓ with $\ell(x) \neq 0$. Then $\ell(X_k) \neq 0$ eventually, and $f_k = \ell(f_k X_k)/\ell(X_k) \to c$. *Lean:* `scale_limit`.

Lemma VII.5.g (= VII.5.4(a)). *Proof sketch.*

1. A04 at degree 4 gives thresholds beyond which $X_k := r_k\, U_k(5)^{-1} \partial_4^V U_k(4) \to x := \partial_4^W$.
2. A05 at degree 4, applied to (J, p, Y), gives thresholds beyond which $(r_k/\xi_k)\, U_k(5)^{-1} p_4 U_k(4) \to \kappa\, \partial_4^W$.
3. Take the componentwise maximum of the two sets of thresholds.
4. Substituting $p_4 = c\,\partial_4^V$ in step 2 gives $(c/\xi_k) X_k \to \kappa x$, where $x \neq 0$ is part of the free data. Lemma VII.5.f gives $c/\xi_k \to \kappa$.

The transport is $U_k(d+1)^{-1}(\,\cdot\,)U_k(d)$, as defined by `carry` and `aisle`.

Lean: `tracks_ratio`.

Lemma VII.5.h (= VII.5.4(b)). *Proof sketch.* Suppose $J = 0$. The translation Ward identity writes every correlation with an insertion of $p_{d,a} v$ as a flux of J through a shell, which is then 0. Correlations separate the elements of $\mathcal{F}_{d+1}$, so $p = 0$. *Lean:* `flux_nontrivial`.

Remark (sign of $c\kappa$). Suppose in addition that tracking sequences exist at every level. Then $c\kappa > 0$: since $\xi_k > 0$, c/ξ_k has the sign of c, and its limit is $\kappa \neq 0$. This is `kappa_sign`. It is not assigned to this unit and is not used by the main theorem. VI.2 uses only (a), and obtains $c \neq 0$ from $\kappa \neq 0$.

3.30.4.3 4.3 Exactly compatible frames (VII.5.5) Lemma VII.5.i (equivariant retraction, = VII.5.5(a)). *Proof sketch.*

1. ι^W is injective, so it has a linear left inverse R.
2. Average over Γ_4: $Q := |\Gamma_4|^{-1} \sum_{s \in \Gamma_4} s^{-1} R s$. Since ι^W commutes with Γ_4, Q is again a left inverse, and Q is Γ_4-equivariant.
3. Let θ be the conjugation, an antilinear involution. Put $P := \frac{1}{2}(Q + \theta Q \theta)$. Then P is $\mathbb{C}$-linear. It is a left inverse, since ι^W commutes with θ. It commutes with θ. It stays Γ_4-equivariant, since θ commutes with Γ_4.

Lean: `retraction`, `averaged_cov`.

Lemma VII.5.j (exactness of the correction, = VII.5.5(b)). *Proof sketch.*

1. *One step.* $X_{d+1}\iota^W v = \iota^V X_d P_d \iota^W v + A_{d+1}(\iota^W v - \iota^W P_d \iota^W v) = \iota^V X_d v$.
2. *Intertwining for $d \le e$.* Induction on e, using $\iota_{d\to e+1} = \iota_{e\to e+1}\iota_{d\to e}$ and $\iota_{d\to d} = 1$.
3. *Symmetries.* Induction on d. Let g be a signed permutation or the conjugation. Then g commutes with P_d, ι^W, ι^V, A_{d+1} and, by induction, with X_d. Hence it commutes with both terms of X_{d+1}.
4. *Units.* $\mathbf{1}_{d+1} = \iota^W \mathbf{1}_d$, so $X_{d+1}\mathbf{1}_{d+1} = \iota^V X_d \mathbf{1}_d = \iota^V \mathbf{1}_d = \mathbf{1}_{d+1}$.

Lean: `adjusted_succ`, `adjusted_exact`.

Lemma VII.5.k (asymptotic identity). Under the hypotheses of VII.5.5(c), for $d \le D$, $\Pi_d^{(k)} := (A_d^{(k)})^{-1} X_d^{(k)} \to 1$ in operator norm. *Proof sketch.*

1. Put $\Phi_d^{(k)} := (A_{d+1}^{(k)})^{-1}\iota^V A_d^{(k)}$. Then

$$\Pi_{d+1}^{(k)} v = \Phi_d^{(k)}\,\Pi_d^{(k)} P_d v + (v - \iota^W P_d v), \qquad \Pi_0^{(k)} = 1.$$

2. $\Phi_d^{(k)} \to \iota^W$ pointwise by hypothesis, hence in norm (finite dimension).
3. By induction, $\Pi_d^{(k)} P_d v \to P_d v$.
4. Therefore $\Pi_{d+1}^{(k)} v \to \iota^W P_d v + v - \iota^W P_d v = v$. Finite dimensionality converts pointwise convergence to operator-norm convergence.

Lean: `perturb`, `perturb_limit`.

Lemma VII.5.l (bijectivity and the new frames, VII.5.5(c)). *Proof sketch.*

1. $\Pi_d^{(k)} \to 1$ in norm, and injective operators form an open set. So $\Pi_d^{(k)}$, and hence $X_d^{(k)}$, is injective for large k.
2. $\dim \mathcal{F}_d^\circ = \dim \mathcal{F}_d$ (by $A_d^{(k)}$), so $X_d^{(k)}$ is bijective.
3. Hence $\widetilde{X}_d^{(k)} = X_d^{(k)}$ eventually, and $(A_d^{(k)})^{-1}\widetilde{X}_d^{(k)} v \to v$ for every v.
4. Intersect the eventual sets for $d = 0, \dots, D$. If the original frames satisfy the symmetry and unit conditions of (b), all corrected frames then agree with the exactly compatible maps $X_d^{(k)}$ on this degree range.

Lean: `eventually_unit`, `rotate_eq`, `relative`.

Use. In VII.1 the frames of the free-field scaling sequence intertwine the inclusions only asymptotically. VII.5.5 replaces them by frames that are exactly graded for all large k and asymptotically equal to the original ones, as tracking requires (`tune_grades`).

3.30.4.4 4.4 Finite-dimensional convergence (VII.5.6) Lemma VII.5.m (= Proposition VII.5.6). *Proof sketch.*

- (a) A linear left inverse g of f is continuous in finite dimension, so $a_y = gf(a_y) \to gf(e) = e$.
- (b) Every point of $(\mathbb{R}^4)^n$ is $\sum_{i,a} v_{i,a} e_{i,a}$. Hence evaluation at the $4n$ coordinate vectors is an injective real-linear map from the space of complex-valued real-linear maps to $\mathbb{C}^{4n}$, and (a) applies.

Lean: `norm_fit`, `components`, `norm_fit_deriv`.

3.30.4.5 4.5 The classical stress-tensor algebra (VII.5.7) Lemma VII.5.n (covariance of μ^1, μ^2). For a signed permutation t with signs $\varepsilon_t(\cdot)$:

$$t \cdot \mu^2_a = \varepsilon_t(a)\, \mu^2_{t(a)}, \qquad t \cdot \mu^1_a = \varepsilon_t(a)\, \mu^1_{t(a)}.$$

Both are real. *Proof sketch.* t multiplies $\langle F_{ij}, \partial_b F_{ij}\rangle$ by $\varepsilon(i)\varepsilon(j)\,\varepsilon(b)\varepsilon(i)\varepsilon(j) = \varepsilon(b)$ and relabels the indices. The double sum over (i,j) is invariant under reindexing by t. For μ^1_b, the sign is $\varepsilon(b)^3\varepsilon(i)^2 = \varepsilon(b)$, and the remaining index is reindexed in the same way. Reality follows from the real inner products. *Lean:* `momentum2_turn`, `momentum1_mem`, `momentum2_mem`, `sumPerm`.

Lemma VII.5.o (affine shells). Every degree-0 radial jet is realised by a free Maxwell field with constant curvature and an affine potential. *Proof sketch.* The associated potential has constant curvature, so $\sum_a \partial_a F_{ab} = 0$. *Lean:* `shell_affine`. VII.2 uses this, with its degree-1 analogue `shell_quadratic`, to build test jets on Maxwell solutions.

Consequences in VI.1 and VII.2. Write $T^{(0,w)} := T^{(0,w_0,w_1,w_2)}$ for $w \in \mathbb{R}^3$.

1. *Trace (VI.1).* $\sum_a T^{(0,w)}_{aa} = 4(w_0 + 4w_1)\,\mathfrak{s}$. In particular T^{can} is traceless (`mechanical_trace`, `conformal`).
2. *Divergence (VII.2).* $\delta(w)_b = \sum_a \partial_a T^{(0,w)}_{ab}$ (`euler`). The key identity is the Bianchi consequence

$$\sum_{a,i} \langle F_{ai}, \partial_a F_{bi}\rangle = \tfrac{1}{2}\mu^2_b.$$

   With it, the shear part $\sum_{a\neq b} \partial_a \sum_i \langle F_{ai}, F_{bi}\rangle$ equals $E_b + \frac{1}{2}\mu^2_b - 2\mu^1_b$.
3. *Coordinates (VII.2).* For simple $\mathfrak{g}$, β and δ are bijections $\mathbb{R}^3 \to \mathcal{R}_1$ (`coordinates1`, `bareCoordinates`). Under the same simplicity hypothesis, $\beta(w)$ vanishes on free Maxwell solutions if and only if $w_0 = w_1 = 0$ (`through_eq`).
4. *Conservation of the canonical tensor (VII.2).* $\delta(1, -\frac{1}{4}, 1) = E$: T^{can} is conserved on the equations of motion (`eom_vector_raw`).
5. *The functional ρ (VII.2).* For simple $\mathfrak{g}$, define $\rho : \mathcal{R}_1 \to \mathbb{R}$ by $\rho(\delta(w)) := -4(w_0 + 4w_1)$. Then $\sum_a T^{(0,w)}_{aa} = -\rho(\delta(w))\,\mathfrak{s}$. The functional ρ vanishes on vectors whose components vanish on free Maxwell solutions (`rho`, `rho_eom`).

For simple $\mathfrak{g}$, a tensor in this four-parameter family is conserved on free Maxwell fields precisely when it is a multiple of T^{can} plus a constant multiple of $\delta_{ab}\mathbf{1}$. With the unit term removed, its trace is determined by its divergence through ρ. VII.2 uses this functional in `Lease` and its limiting comparison `mere`.

3.30.5 5. Where the difficulty lies

The logarithmic coupling law uses only the uniform running clause and elementary real analysis. The analytic work lies in the upstream one-loop estimate, the coupling reset and the construction

of a trajectory reaching $\bar{\beta}$ at unit scale. The auxiliary results use finite-dimensional linear algebra, equivariant averaging and scalar-ratio limits.

The result is uniform both in the volume index p and in the finer lattice level $N \geq m$. Thus VII.5.3 allows arbitrary volume sequences. If inverse couplings converge as $N, p \to \infty$ at fixed m, their limits inherit the non-strict bound $|\beta/s_m - 2b_0| \leq \varepsilon$ for $m \geq m_0(\varepsilon)$. Existence of those limits and comparison with a coupling defined by continuum correlations require separate results.

Only the leading logarithm is established here, for the coefficients of the lattice effective actions. No two-loop correction or Λ-parameter is asserted. The main theorem uses VII.5.1 through (Reg1), retaining only the existence of a positive slope. Proposition VII.5.3 is not used in the assembly.

The 38 auxiliary declarations serve other parts of the proof: stress normalisation in VI.2, frame correction in VII.1, convergence in III.12, and classical stress algebra and read-out estimates in VI.1 and VII.2.

3.30.6 6. Notation

object	code name	proposed notation
Casimir constant	`YM.Gauge.casimir`	$C_{\mathfrak{g}}$ ($= 2q^2$ for $SU(q)$)
slope constant	`YM.Gauge.slopeConstant`	$\kappa_G = 11C_{\mathfrak{g}}/(24\pi^2)$
one-loop coefficient	`b0`	$b_0 = \kappa_G/2$; slope $b_* = 2b_0$
lattice spacing at stage m	`Schedule.a`	$a_m = 1/D_m$
logarithmic scale	`Schedule.scale`	$s_m = \log(1/a_m)$
coupling at stage m	`C.path.β N p m`	$\beta_{N,p,m}$; bare coupling $\beta_{N,p} = \beta_{N,p,N}$
reference line	`hatBeta`	$\hat{\beta}_m = \bar{\beta} + \kappa_G s_m$
running clause	`Running`	(R)
running coupling	`regulatorRun`	$g_{N,p,m} = \beta_{N,p,m}^{-1/2}$
free comparison table	`Free`	$\mathcal{F}^{\circ}_d$
equivariant retraction	`retraction`	P_d
corrected frames	`adjusted`, `rotateChart`	X_d, $\widetilde{X}^{(k)}_d$
exact compatibility up to degree D	`Exact`, `Graded`	"exactly graded"
asymptotic intertwining	`Near`	hypothesis of VII.5.5(c)
square of the curvature	`square4`	$\mathfrak{s} = \frac{1}{4}\sum_{i,j}\langle F_{ij}, F_{ij}\rangle$
canonical stress tensor	`raw`	T^{can}
weight-5 vectors	`momentum1`, `momentum2`	μ^1, μ^2
coordinates on $\mathcal{R}_1$	`basisV`	$\beta(w)$
divergence of $T^{(0,w)}$	`bareVector`	$\delta(w)$
sharp read-out bound	`SharpRead`	Definition VII.5.8

3.30.7 7. What remains unexplained

1. **The upstream one-loop calculation.** The transfer bound is traced through the coupling reset and the one-loop shift bound in §2. The evaluation of the momentum integral giving $11C_{\mathfrak{g}}/(24\pi^2)$ belongs to III.7; its analytic proof is not verified here. The coupling results in this unit take the construction conditions as hypotheses.

2. **The coupling after the limits.** The assessment's Theorem 7.2 concerns $g_W(r)$ after removal of the initial cutoff and a subsequential infinite-volume limit, together with comparison to a coupling read from continuum correlations. VII.5.3 proves the lattice-stage law. A declaration establishing the additional limits and comparison has not been identified here (tentative). The files `Beacon`, `Hymnal` and `Musicology` in `YM/RemainderComparison` are possible starting points.
3. **Existence of the comparison inputs.** VII.5.4 assumes the free Maxwell data of VI.2 and a Euclidean hierarchy on S; it gives a conclusion for every tracking sequence meeting its finite thresholds. It does not construct such sequences. Their existence belongs to VI.1 and VII.1. Likewise, the supply of the read-out bound in III.2 uses the hypotheses of the theorem linked in §2; the underlying one-step estimates are outside this unit.

3.31 VII.6. Scaling variants and fixed-field estimates

Part: VII, identification as Yang–Mills. **Lean:** `YM/TraceExtension/Prom/` contains the "Sky" files, 133 files with "dynamic stops". Other outputs are `Trace.first_grade` and `FreeStress.dim_before`. Also included are `SharpGram.inletFirst` and its helpers in `YM/SharpGram/Clauses.lean`. **Size:** 4,205 lines in 374 declarations:

- about 3,600 lines in the chain of `pays_sky`;
- about 560 further lines in the dynamic-stop files;
- 43 lines in the chain of the supply theorem.

The unit has two parts.

- **(A) Scaling sequences at prescribed radii.** The ultraviolet construction (III.11) compares the lattice theory with the free field at reading scales $\ell(b,k) = a_k \mu_k^b$. Varying the band b over a unit interval makes these scales cover every sufficiently small radius. The stage estimates remain valid across one change of block factor. They produce an approaching scaling of the limit theory along a subsequence of any prescribed positive null sequence.
- **(B) Degrees below four.** In the limit theory every field of degree at most 3 is a multiple of the unit. It has a constant family of lattice representatives whose stage-k coefficient size is at most $\hat{\beta}_k$ on an ordered cutoff tail, for every sufficiently large k.

3.31.1 1. What this unit proves

Setting.

The family and its stages.

- G is a compact matrix group with simple Lie algebra $\mathfrak{g}$, and C is a Wilson family with its construction record (II.2, III.13).
- Its RG stages k have spacings $a_k \downarrow 0$ and block factors $L_k = a_k/a_{k+1} \geq 2$.
- The running inverse coupling is $\hat{\beta}_k = \bar{\beta} + c_G \log a_k^{-1}$, where $c_G > 0$. Put

$$\mu_k = \exp\left((\log\log \hat{\beta}_k)^2\right),$$

  which grows faster than every positive power of $\log \hat{\beta}_k$ and slower than every positive power of $\hat{\beta}_k$ (III.11).

- For a real band b, the *reading scale* of stage k is $\ell(b,k) = a_k \mu_k^b$. For fixed b it tends to 0.

The limit theory and the free reference.

- F is an extraction order and Φ_F its filter of ordered cutoff tails on pairs (N,p) (IV.1).
- S is a limit table of C along F (ordered smeared limits and generic point limits), with a Euclidean hierarchy H.
- $\mathcal{F}_d^S = \mathrm{Label}_d/\mathrm{Null}_d(S)$ is the filtered field space of degree at most d, $\pi_d : \mathrm{Label}_d \to \mathcal{F}_d^S$ is the projection, and $\mathbf{1}_d$ is the unit.
- The reference is the free Maxwell table of VI.2: gauge-invariant Wick polynomials in a free $\mathfrak{g}$-valued field. Its field spaces are $\mathcal{F}_d^{\mathrm{free}}$, its unit is $\mathbf{1}_d^{\mathrm{free}}$ and its hierarchy is H_0.
- Correlation kernels on Conf_n are written $\langle \phi_1(x_1) \cdots \phi_n(x_n) \rangle^S$ and $\langle \cdots \rangle^{\mathrm{free}}$.

Scalings. A *scaling* (VI.1) consists of

- radii $r_k > 0$ strictly decreasing to 0,
- constants $\xi_k > 0$,
- complex-linear isomorphisms $U_k(d) : \mathcal{F}_d^{\mathrm{free}} \to \mathcal{F}_d^S$.

It is *approaching* (VI.2.7(a)) if the following holds for every n and all $v_i \in \mathcal{F}_{d_i}^{\mathrm{free}}$. Put

$$K_k(x) = \langle U_k v_1(r_k x_1) \cdots U_k v_n(r_k x_n) \rangle^S - \langle v_1(x_1) \cdots v_n(x_n) \rangle^{\mathrm{free}}.$$

The kernels K_k are smooth on Conf_n and tend to zero locally with an order-zero bound: for every compact $K \subset \mathrm{Conf}_n$ and all $M, \varepsilon > 0$, eventually $|\int t K_k| \le \varepsilon$ for every smooth test function t supported in K and satisfying $\|t\|_\infty \le M$.

Fixed data.

- B is a binding of the RG history to its price lists, O a Haar budget and J a chart ratio (III.11).
- (S) is the one-step cost package of III.11.
- $c_d = 2(w_d + 1)$ are the mark exponents of III.11, where w_d is the polynomial degree of the degree-d label frame plus 4.
- Write J_k for the price index of stage k, Λ_j for its block-factor list, and T_d for the reference stage of degree d. Thus $L_k = \Lambda_{J_{k+1}}$. Stage packages, poised stages, the uniform-band property $\mathrm{Fire}_d(z)$, vanes and checked vanes are defined in §4.

Proposition VII.6.1 (approaching scalings along a subsequence of any null sequence). Assume:

- (i) G is simple and (S) holds for C;
- (ii) (z_d) are reals with $z_d \ge c_i$ for all $i \le d$;
- (iii) $\mathrm{Fire}_d(z_d)$ holds for every d (Definition VII.6.d);
- (iv) $\dim \mathcal{F}_d^S \le \dim \mathcal{F}_d^{\mathrm{free}}$ for every d;
- (v) $\mathcal{L}$ is an ultrafilter on $\mathbb{N}^2$ finer than Φ_F.

Let $s_i > 0$ with $s_i \to 0$. Then there are a strictly increasing $f : \mathbb{N} \to \mathbb{N}$ and a scaling with $r_k = s_{f(k)}$ and $\xi_k = 1$ such that:

- (a) the scaling is approaching;
- (b) for every d and all large k, $U_k(d)$ is real and covariant under signed coordinate permutations between H_0 and H, and $U_k(d)\mathbf{1}_d^{\mathrm{free}} = \mathbf{1}_d$;

- (c) there are vanes V_k of depth k with the following properties:
  - their stages tend to infinity;
  - V_k is checked at level k;
  - its radius is $\ell(V_k) = r_k$;
  - for every d and all large k, the full lift tuple of V_k in degrees $\le k$ converges along $\mathcal{L}$ to a tuple Y_k, and $U_k(d) = \pi_d \circ Y_{k,d}$.

Status: theorem, with (iii) and (iv) carried as hypotheses. Both are supplied inside the construction run, where VII.2 applies this result:

- (iii) is produced by the uniform-band construction (III.2);
- (iv) is the rank-width bound (VI.2), applied to the corrected rank widths of the seed (IV.1).

Lean: `pays_sky`. Conclusions (a)–(c) are the structure `StarPath`.

Proposition VII.6.2 (checked vanes at prescribed radii). Assume (i)–(iii) of VII.6.1. Use the ordered and generic point limits from the setting. Only smoothness of S, rather than its full hierarchy, is needed here; the free hierarchy H_0 is still assumed. Let $s_i > 0$ with $s_i \to 0$. Then there are

- a strictly increasing $p : \mathbb{N} \to \mathbb{N}$ with $s \circ p$ strictly decreasing, and
- for each $d \in \mathbb{N}$, a vane V_d of depth d with stage $\ge d$, checked at level d, with $\ell(V_d) = s_{p(d)}$.

Status: theorem.

Lean: `wakesSky`, with `Vane` and `TimberSky`.

Proposition VII.6.3 (stage-k chessboard bound with a moving band). Assume (i) of VII.6.1, let $d \in \mathbb{N}$, and let $z \ge c_i$ for $i \le d$. Fix the following data:

- $\Delta \in \mathbb{N}$, $e, u_0 \in \mathbb{R}$, and stage packages P_k of degree $\le d$ with Taylor order Δ, exponent e and bands $b_k \le u_0$;
- $n \ge 2$ and $m + 1 \le n$;
- labels $g_i \in \mathrm{Label}_{d_i}$ with $d_i \le d$, for $0 \le i \le m$;
- a compact $K \subset \mathrm{Conf}_{m+1}$.

Then the following holds for all large k. Take:

- a stage-k cost record L of P_k with exponent -3, mark exponent z and arity bound n;
- $N > k$, a period p, and a stage-k RG representative V compatible with L along the same RG history;
- x whose rounding to the stage-k lattice, in units of $\ell(b_k, k)$, lies in K and fits the stopping stage;
- stage-k polynomial lift data T_i of Taylor order Δ for V, and labels $f_i \in \mathrm{Label}_{d_i}$, satisfying the rescaling equation: after applying T_i and the coupling dilation of VII.6.n, the result is g_i weighted by $\ell(b_k, k)/a_k$.

Then for every nonempty subset $s \subset \{0, \dots, m\}$,

$$\rho_{\mathcal{T}}(\Phi_s) \;\le\; \mathrm{bud}_n(k) := \mu_k^{-1/(16n)}.$$

To define Φ_s, introduce commuting markers η_i with $\eta_i^2 = 0$. If O_i is the lattice probe with label

f_i, and D is the free cumulant polynomial of the g_i at the rounded, rescaled points, then

$$\Phi_s = [\eta_s]\left(e^{-D}\prod_{i=0}^{m}(1+\eta_i O_i)\right), \qquad \eta_s = \prod_{i\in s}\eta_i.$$

The exponential is a finite polynomial in the marker algebra. $\rho_{\mathcal{T}}$ is the chessboard root at coupling $\beta_{N,p}$ over the tiling by cells of the width stage $m_0(b_k, k)$.

Status: theorem. The threshold in k depends on $(d, z, \Delta, e, u_0, g, K, n)$ and on the bands only through u_0. This uniformity is the difference with the fixed-band `earnings` of III.11.

Lean: `earningsSky`. The form for poised families is `lava_vSky`.

Proposition VII.6.4 (fixed-field supply in degrees ≤ 3). Let Y be a final record for (C, F, S) (VIII.1) and G simple. Let φ be a filter on $\mathbb{N}^2$ finer than Φ_F, let $d \leq 3$ and let $x \in \mathrm{Label}_d$. Then $\pi_d x$ is *supplied along* φ. That is, there is $p \in \mathbb{N}$ such that for all large k there are $y \in \mathrm{Label}_d$ and labels $q_{N,p'} \in \mathrm{Label}_d$ with:

- $\pi_d y = \pi_d x$;
- $q \to y$ along φ;
- $\mathrm{quota}_k^{N,p'}(q_{N,p'}) \leq \widehat{\beta}_k^p$ eventually along φ.

Here $\mathrm{quota}_k^{N,p'}(u)$ is the coefficient size of u at cutoff (N, p'), written in stage-k coordinates (`quota`). One can take $p = 1$ and q constant.

Status: theorem.

Lean: `inletFirst`, `Supplies`.

Proposition VII.6.5 (fields of degree below four). Let G be simple and let $d \in \mathbb{N}$ satisfy $d < 4$.

- (a) $\dim \mathcal{F}_d^{\mathrm{free}} = 1$.
- (b) If Y is a final record for (C, F, S), then every $u \in \mathcal{F}_d^S$ is of the form $t\,\mathbf{1}_d$ with $t \in \mathbb{C}$.

Status: theorem. Used by VII.6.4 and by `vacuum_truncation`.

Lean: (a) `dim_before`; (b) `first_grade`.

3.31.2 2. What it uses

From other units.

- **III.11 (interacting rank).** The stage estimates and numerical bounds:
  - The fixed-band estimates underlying the moving-band argument:
    - `earnings`, `lava_v`;
    - `Poised`, `aspect`, `podium`.
  - The reading-scale arithmetic:
    - `Station.gap`, `mag`;
    - the rounding `tested`;
    - the rates `hat_rate` and `gap_rate`.
  - The budget `Ship.budget` and the pattern of the root bound `Ship.rebate` (III.11, Lemma 4.7).

  - The step from root bounds to a pointwise comparison, `Read.full_cost`.
  - The one-step cost package `Scrim`.
  - The band inequalities `Checks` and the vane lift `wok`.
- **III.2 (choice of constants).** Production of the uniform-band property by `embalms`, `stagingSky` and the package extension of III.2. Also the construction runs `plentifulWithPrediction`, `plentiful` and `merano`. These runs:
  - choose $z_d = \sum_{i \le d} c_i$;
  - choose an ultrafilter $\mathcal{L}$ finer than Φ_F;
  - invoke the uniform-band construction and then the gallery theorem of VII.2.
- **III.9, III.10, III.11.** The RG history, background fields and one-step bounds used in VII.6.g.
- **IV.1.**
  - Extraction orders and limit tables.
  - The chessboard estimate (Lemma IV.1.b; reflection positivity of the Wilson action).
  - The corrected rank widths `Cells.RankWidths` that give hypothesis (iv).
- **VI.1, VI.2.** The scaling record `Climb.Scaling`. The free Maxwell table `freeTable`. The rest of the station machinery:
  - stations and admissibility `Station.Admissible` (VI.2.8);
  - balanced estimates give stations: `Granular`, `supply_wheel` (VI.2.s);
  - `Approaches` and `Fades`;
  - eventual injectivity `MeterTest.injects`;
  - the dimension cap `DimAbove` and its proof `cap_given`;
  - the fixed-radius analogue `Stations.equivalences` (VI.2.t);
  - the free labels of degree < 4: `just_one`, `below`, `span_four`.
- **VIII.1.** The final record `Spec.Final`. Part B uses three of its fields:
  - the construction record;
  - the hierarchy laws, giving $\mathbf{1}_d \neq 0$;
  - the comparator, a linear isomorphism $\mathcal{F}_d^S \cong \mathcal{F}_d^{\mathrm{free}}$ for each d.
- **VII.2 (consumer).** `gang` proves the last clause of `Gallery` from VII.6.1. That clause says: for every null sequence s there are a subsequence and a scaling with radii $s \circ f$ carrying the trace-comparison witness.

Standard mathematics.

- Reflection positivity of the Wilson action (Osterwalder–Seiler, Ann. Phys. 110 (1978)) and chessboard estimates (Fröhlich–Israel–Lieb–Simon, CMP 62 (1978)).
- Bałaban's block-spin renormalization group with background fields (CMP, 1983–1989). It is used only through the stage estimates of III.9–III.11.
- The intermediate value theorem and recursive (diagonal) choice of subsequences.
- Ultrafilter limits in finite-dimensional spaces (bounded sequences converge along any ultrafilter).
- For simple $\mathfrak{g}$: the adjoint representation has no nonzero invariant vectors, and invariant symmetric bilinear forms on $\mathfrak{g}$ are multiples of the Killing form. Both are used only through VI.2.

Imported constants and assumptions.

- Hypotheses carried by VII.6.1: (S), the uniform-band property, the dimension cap and the ultrafilter. Their discharge is described above.

- The budget $\mathrm{bud}_n(k) = \exp(-\log \mu_k/(16n))$. The cost-record exponent -3.
- The mark exponents $c_d = 2(w_d + 1)$ (`generator`, `winter`).
- Bands at least $d + 6$, and the inequalities between three bands in VII.6.d.
- Arity bound $d + 2$ for degree d (`trays`).
- In the stopping rule: the margin μ_k^2 and the depth $\lceil (\log \widehat{\beta}_k)^2 \rceil^{10}$.
- In the checked condition: tolerance $1/(j+1)$ with window radius $j+1$, and window radius 100 for the two-point bound.
- The error $\widehat{\beta}_i^{-1/32}$ in Lemma VII.6.j.

3.31.3 3. Strategy

For fixed b, the reading scales $\ell(b,k)$ form a discrete null sequence. The fixed-band argument of III.11 also requires a plateau of constant block factor across its RG window. These restrictions do not ensure that its radii meet a prescribed null sequence. VII.2 needs such a sequence to test the trace anomaly at arbitrary short distances.

Let the band vary over $[b_0, b_0 + 1]$, with $b_0 \geq 0$. Eventually $\mu_k \leq \mu_{k+1}$ and $L_k \leq \mu_{k+1}$, so

$$\ell(b_0, k) = a_{k+1} L_k \mu_k^{b_0} \leq a_{k+1} \mu_{k+1}^{b_0+1} = \ell(b_0 + 1, k + 1).$$

The reading intervals therefore overlap and cover all sufficiently small positive radii. Continuity in the band gives exact equality with each target radius. The stage estimates must now be uniform over a unit interval of bands, and their stopping rule must allow a change of block factor. The uniform-band input comes from III.2; the modified estimates occupy most of this unit.

The chessboard bound gives a pointwise comparison with free correlations. Two auxiliary bands control the strip near the coincidence sets and the Riemann-sum error on its complement. These bounds yield admissible stations at the target radii. A diagonal choice makes their depth and arity increase while their tolerance tends to zero. Their bounded lift tuples converge along a single ultrafilter; projection to the limit field spaces gives approaching linear maps. Free test functionals make these maps eventually injective, and the dimension cap makes them isomorphisms. For degrees below four, a separate argument transports the one-dimensional free field spaces through the final comparator. Multiples of the unit then have constant lattice representatives with the required coefficient bound.

3.31.4 4. Steps

3.31.4.1 4.1 Reading scales and dynamic stops Lemma VII.6.a (stopping stages and fit). Fix a band b and a stage k. The middle stage $m_1(b,k)$ is the largest $j \leq k$ for which $a_j \geq \mu_k^2 \ell(b,k)$, with value 0 if no such j exists. Put $Q_k = \lceil (\log \widehat{\beta}_k)^2 \rceil$. The width stage $m_0(b,k)$ is the largest $j \leq m_1$ for which $a_j \geq Q_k^{10} a_{m_1}$, again with value 0 if the set is empty.

The pair (b,k) is *allowed* when both defining inequalities hold at the selected stages,

$$\min(\Lambda_{J_k}, \Lambda_{J_k - 1}) \geq 2, \qquad J_m \geq J_k - 1 \quad (m_0 \leq m \leq k).$$

Monotonicity also gives $J_m \leq J_k$. Thus each block factor in this window is at least $\lambda_k = \min(\Lambda_{J_k}, \Lambda_{J_k - 1})$, and

$$\lambda_k^{j-m} \leq a_m / a_j \qquad (m_0 \leq m \leq j \leq k).$$

Round x to the stage-k lattice in units of $\ell(b,k)$, and multiply by $\ell(b,k)$ to obtain its physical configuration. It *fits* if that configuration is generic and every coordinate has absolute value at most $a_{m_1}/8$. For fixed $b > 0$ and generic x, it fits for every sufficiently large allowed k. More uniformly, for each $R \in \mathbb{R}$ and arity n, every generic rescaled lattice configuration of norm at most R fits for sufficiently large allowed k, uniformly in b.

Proof sketch. The stopping indices are maxima of finite sets. The allowed conditions give the two scale margins. Since $J_m \in \{J_k - 1, J_k\}$, multiplying the lower block-factor bound gives the ratio estimate. Rounding converges for a fixed positive band; the margin $\mu_k^2 \to \infty$ then absorbs each fixed configuration. For rescaled lattice configurations, rounding is exact, so the same margin absorbs R uniformly in the band.

On an allowed plateau of III.11, the maximum definitions agree with its ceiling-based stopping stages: $a_j/a_k = L^{k-j}$ turns the defining inequalities into the same ceiling inequalities.

Inputs: `readySky`, `open_readSky`.

Lean: `gainSky`, `middleSky`, `widthSky`, `AllowedSky`, `FitSky`; fixed versions `middle`, `Allowed`. *(proof read; stopping definitions read)*

Lemma VII.6.b (overlapping reading intervals). Let G be simple, $b \geq 0$, and assume $\max(\Lambda_{J_k}, \Lambda_{J_k-1}) \leq \mu_k$ for all sufficiently large k. Then eventually $\ell(b,k) \leq \ell(b+1,k+1)$. For every $J \in \mathbb{N}$ there is $r > 0$ such that each $s \in (0,r]$ equals $\ell(b+c,k)$ for some $k \geq J$ and $c \in [0,1]$.

Proof sketch. Eventually $\widehat{\beta}_k \geq e$ and $\mu_k \leq \mu_{k+1}$. The factor L_k is among the two factors controlled at stage $k+1$, so $L_k \leq \mu_{k+1}$. This gives overlap. Choose a sufficiently large $i \geq J$ and put $r = \ell(b,i)$. For $0 < s \leq r$, take the least $k > i$ with $\ell(b,k) \leq s$. Then

$$\ell(b,k) \leq s \leq \ell(b,k-1) \leq \ell(b+1,k).$$

Continuity of $c \mapsto \ell(b+c,k)$ gives c.

Lean: `tide_mode`, `near_skies`, `gap_curve`, `gap_account`.

Lemma VII.6.c (a null sequence is realized at increasing stages). Under the hypotheses of VII.6.b, let $s_j > 0$ with $s_j \to 0$, and let $J \in \mathbb{N}$. Then there are strictly increasing $K, P : \mathbb{N} \to \mathbb{N}$ and $c_j \in [0,1]$ with $P_j \geq J$ and $s_{K_j} = \ell(b + c_j, P_j)$.

Proof sketch. Recursion: apply VII.6.b with threshold $\max(J, P_{j-1}) + 1$, and pick $K_j > K_{j-1}$ with $s_{K_j} \leq r$.

Lean: `gap_orders`.

3.31.4.2 4.2 The uniform-band property

Definition VII.6.d (packages, poised stages and the uniform-band property). A package of degree $\leq D$ consists of a band $b \geq D + 6$, a Taylor order $\theta \geq D$, an exponent e, and a purse of numerical cost parameters at each price index. Every purse has band b, Taylor order θ, exponent e, and credit at least $bD + D + 2$.

A finite family of packages is *poised at stage* k, with arity bound n, mark exponent z and tables $R_{N,p,d} \in \mathrm{End}_{\mathbb{C}}(\mathrm{Label}_d)$ for $d \leq D$, when:

- $T_d \leq k$ for every $d \leq D$;
- $\max(\Lambda_{J_k}, \Lambda_{J_k-1}) \leq \mu_k$;

- for every $N > k$ and p, the tables send the reference unit label to its frame unit vector, preserve the real structure, commute with signed coordinate permutations, and approximate the actual stage-k coordinate maps to relative coefficient-size error at most $1/2$;
- the sharper relative error bound $\mu_k^{-1/2}$ also holds;
- for each package there is a stage-k cost record with exponent -3, mark exponent z and arity bound n; for every $N > k$ and p, it has a compatible RG representative and polynomial lift data realizing the same tables.

The cost record also contains the allowed stopping stages of VII.6.a, numerical and background-field bounds along the RG window, room bounds for each arity at most n, and the one-step estimates used in VII.6.g.

The *uniform-band property* $\mathrm{Fire}_d(z)$ asserts that there are package families $P_s^{(i)}$, $i = 0, 1, 2$, base bands b_0, b_1, b_2, and $\kappa < 0$, satisfying

$$b_1 + 6 < b_0, \quad b_2 + 5 < b_0 + \kappa, \quad (b_0 - b_1)d(d+2) + \kappa < 0, \quad (b_0 - b_2)d(d+2) - b_0 - \kappa < 0.$$

For $s \in [0, 1]$, the bands are $b_i + s$, and each family's Taylor order and exponent are independent of s. There is a single k_0 such that, for every $k \geq k_0$ and every $s \in [0, 1]$, the triple is poised at stage k with arity bound $d + 2$ and mark exponent z, for some tables.

Production in III.2. Starting from its package and block-size requests, a local chart ratio and nonnegative mark exponents, the construction enlarges a waiting window so that every family paid on the enlarged window, with the required covariance of its coordinate maps and a binding to the same prices, satisfies $\mathrm{Fire}_d(z_d)$ for every d. Packages chosen at bands $b_i + 1$ are lowered to $b_i + s$ without changing their Taylor orders or exponents. The poised-stage threshold is uniform in $s \in [0, 1]$.

Lean: `Tapes.Panel`, `PoisedSky`, `LegacyCostSky`, `FiresSky`, `Checks`, `raises`, `embalms`. *(definitions read; production proof read)*

Lemma VII.6.e (leaps). Let G be simple and assume $\mathrm{Fire}_d(z)$ with base bands b_0, b_1, b_2. Then $b_0 > 5$, and the following holds for every $s_i > 0$ with $s_i \to 0$. There are:

- a strictly increasing f;
- strictly increasing stages $k_i \geq k_0$;
- offsets $e_k \in [0, 1]$;
- tables Y_i;

such that the triple $P_{e_{k_i}}$ is poised at stage k_i with Y_i, and $s_{f(i)} = \ell(b_0 + e_{k_i}, k_i)$.

Proof sketch.

1. $b_0 \geq d + 6$.
2. The poised condition gives $\max(\Lambda_{J_k}, \Lambda_{J_{k-1}}) \leq \mu_k$, which is the hypothesis of VII.6.b.
3. Apply VII.6.c with base b_0 and threshold k_0. Put $e_{P_j} = c_j$ and $e = 0$ elsewhere.
4. Fire_d at $s = c_j$ gives poised data at stage $P_j \geq k_0$. Here the threshold must not depend on s.

Lean: `fliesSky`, `LeapSky`, `ArcSky`.

3.31.4.3 4.3 Stage estimates with a moving band Lemma VII.6.f (= Proposition VII.6.3).

Proof sketch. Follow the fixed-band argument of III.11 with three uniform estimates and the modified stopping stages.

1. The local probe bounds hold uniformly for bands $\le u_0$.
2. The polynomial lifts of the g_i have bounds independent of the band.
3. The mark exponent $z \ge c_{d_i}$ converts normalized coefficient bounds on the f_i into source-expansion bounds (`tremor`).
4. Per-site affordability holds from some stage u on. It is needed at the width stage m_0.
5. $u \le m_0$ for large k. If $m_0 < u$, the bracket condition of VII.6.a would give $J_k - 1 \le J_{m_0} \le J_u$, contradicting $J_k > J_u + 1$. In the fixed version the plateau is used here, with $J_k > J_u$.
6. Lemma VII.6.g then gives the bound.

Lean: `earningsSky`, `ActumSky`.

Lemma VII.6.g (root bound at one stage). Fix a cost record of degree bound D, band $b \ge 0$, purse exponent e, mark exponent z and arity bound $u \ge 2$. Fix $N \ge k$ and a period p, let $d_i \le D$ for $0 \le i \le m$, and assume $m+1 \le u$, $T_D \le k$ and $T_{d_i} \le k$. Take an RG history from the cutoff to $m_0(b,k)$ satisfying the cost record, the per-site affordability bound at m_0, the required local work items, and the implication from the stage-k seal to its covariance model. Let $g_i, f_i \in \mathrm{Label}_{d_i}$ and polynomial lift data satisfy the rescaling equation of VII.6.3. Assume the configuration fits, its rounded position belongs to the specified compact set, and the uniform local probe bounds hold there. Write γ for the purse credit and $M_{k,N,p,d}$ for the normalized stage-coordinate coefficient size multiplied by the stage multiplicity bound. Finally assume:

- $\mu_k^e \, M_{k,N,p,d_i}(f_i) \le \mu_k^{-8}$ for each probe;
- its lifted polynomial has coefficient sum at most μ_k^{γ};
- separation of the probe seats implies the source-expansion bound with weight g_k^z, where $g_k = \hat{\beta}_k^{-1/2}$.

Then $\rho_{\mathcal{T}}(\Phi_s) \le \mathrm{bud}_u(k)$ for every nonempty s.

Proof sketch. The history gives compatible background fields and marker estimates at each stage. The covariance model supplies the free comparison data, and VII.6.h supplies the source expansion at the middle stage. Propagating that expansion to the width stage gives a marker estimate with allowance $A = e^{2I}$, where $I = \log \mu_k/(16u)$. The per-site affordability and allocated cost imply

$$\rho_{\mathcal{T}}(\Phi_s) \le e^I / A^{|s|} \le e^{-I} = \mathrm{bud}_u(k).$$

Root estimate: `couponSky`; fixed-band counterpart: `Ship.rebate`.

Lean: `y_readySky`, `yardSky`, `invoiceSky`, `Ship.budget`.

Lemma VII.6.h (source expansion at the middle stage). Fix the history, polynomial lift data and labels of VII.6.g. Assume its launch and survivor data, the compatible background-field record, the room bound for $m+1$ probes at depth $k - m_1$, and the leading and nonleading marker estimates along the window. If the configuration fits, its probe seats are separated, and the normalized coefficient, polynomial coefficient and source-expansion bounds of VII.6.g hold, then the marked probe product has the required source expansion at m_1.

More precisely, for every admissible middle-stage numerical bound, there are finite source and background families and a distinguished source for each cell and nonempty mode subset. They reproduce the propagated marked product exactly. Each distinguished source has the prescribed

support and mode set; on a healthy background it differs from the corresponding free-cumulant expression by at most $\mu_k^{-3/2}$, after the permitted covariance adjustment.

Proof sketch. The polynomial coefficient bound controls the probe family. The launch and survivor data give the marked-source representation. The room, fit and marker estimates propagate it to the middle stage, where the support and comparison bounds give the stated expansion.

Lean: `attireSky`.

Lemma VII.6.i (assembly of weighted words). In a fixed one-step stage configuration, assume the background is healthy, the room and one-step bounds hold with size constant $B \geq 1$, and $0 < \varepsilon \leq 1$. Let P_u be polynomials in the mode variables invariant under simultaneous gauge rotations, each of total degree at most D and coefficient sum at most $r \geq 0$, with all variables in the permitted probe regions. Let S be a finite list of pairs (c, w), where $c \in \mathbb{R}$ and w is a word of allowed modes, each supported in the selected cell. Then the assembled chart integrands are integrable, and

$$\left| \mathcal{A}\left(\sum_{(c,w)\in S} c \prod_{u\in w} P_u \right) - \mathcal{R}\left(\sum_{(c,w)\in S} c \prod_{u\in w} P_u \right) \right| \leq \sum_{(c,w)\in S} |c|\, E(B, \varepsilon, r, D, |w|).$$

Here $\mathcal{A}$ is the chart assembly, $\mathcal{R}$ is the corresponding reference reading with the same cell-orientation test, and E is the one-word error bound from III.11.

Proof sketch. Induct on the list. Integrability is preserved by addition and scalar multiplication. Linearity and the triangle inequality add the one-word errors, with the factor $|c|$ for each word.

Lean: `sweepSky`.

Lemma VII.6.j (one step at a change of block factor). Fix a step $i+1 \to i$ with $i < N$, a coarse-field pose (o, W, V, v), a width multiplier $w \in \mathbb{N}$, and its auxiliary block-spin choices, forest and fluctuation data. Assume the one-step coupling bounds, with the larger of $\Omega_{J_{i+1}}, \Omega_{J_i}$ and the smaller of $\alpha_{J_{i+1}}, \alpha_{J_i}$. Assume also the small-field and Hessian-control bounds on the patch. Put $x = \hat{\beta}_i$, let $\tilde{x}$ be the auxiliary reference coupling used in the stationary-field data, let y be the fine reference coupling, and set $L = wB(y)$, where B is the block radius of III.11.

Let r be a fine-lattice site. Assume the fluctuation guards on the enlarged patch about r,

$$2(L+1) < \mathrm{side}_{i+1,p}, \qquad 100(1 + R_B(\tilde{x})) \leq \mathrm{side}_{i,p},$$

and, for every guarded fine bond e, require its parent to lie inside the pose with the margin

$$\mathrm{dist}(o, \mathrm{parent}(e)) + B(x) + 2 \leq W.$$

Then the fine configuration U constructed by the background-field map is healthy at the small-field threshold $\tau(y)$ on the radius-L patch. For every f and gauge transformation t with $t \cdot f = U$, and every coloured incidence j with $\mathrm{dist}(r, j) + 2 \leq L$,

$$\left| \mathcal{O}_j(f, t; r, g) - (\mathrm{Re}\, m_j + (DZ)_j) \right| \leq \hat{\beta}_i^{-1/32}.$$

Here $R_B(s) = \lceil (\log s)^2 \rceil$, g is the step coupling, and $\mathcal{O}_j$ is the colour component of g^{-1} times the plaquette curvature, in tree gauge rooted at r and rotated by the adjoint action of $t(r)$. The mean m and linear fluctuation correction DZ use the same coordinates.

Proof sketch. The Hessian-control bound and guards give link estimates throughout the enlarged patch. A four-bond estimate makes every tested plaquette small, proving health. The link estimate and stationary-field curl identity give the displayed error on incidences two lattice units inside the patch.

Lean: `laminateSky`, used by `screensSky`.

Lemma VII.6.k (poised families supply the stage bound). Assume (S), simplicity of G, and $z \geq c_i$ for $i \leq D$. Let P_k be a finite family of degree-D packages whose bands are $q_\alpha + e_k$, with $e_k \in [0,1]$, and whose Taylor orders and exponents are independent of k. Fix a package index α, labels $y_i \in \mathrm{Label}_{d_i}$ with $d_i \leq D$, a compact $K \subset \mathrm{Conf}_{m+1}$, and $2 \leq n$, $m+1 \leq n$. For every sufficiently large k, every table with which P_k is poised, every $N > k$ and p, and every fitted configuration whose rounded position belongs to K, the bound of VII.6.3 holds for the probes obtained by the vane lift from y_i.

Proof sketch. Apply VII.6.3 with band ceiling $q_\alpha + 1$. The poised data supply the cost record, RG representative and polynomial lifts. Inverting the charted table gives the rescaling equation.

Lean: `lava_vSky`.

Lemma VII.6.l (pointwise comparison). Under the hypotheses of VII.6.k, fix an extraction order F, a package index α, a strip exponent χ, $R > 0$, and a free word of arity $m \leq n$ with degrees at most D. Assume the free table is smooth. There is C such that, for every sufficiently large k and every poised table, the lattice m-point function of the lifted probes differs from the free function by at most $C\,\mathrm{bud}_n(k)$ in the ordered upper-limit sense, at each generic stage-k lattice configuration in the rescaled window of radius $2R$.

Thus, for each configuration and each $\eta > 0$, the difference is eventually at most $C\,\mathrm{bud}_n(k) + \eta$ along F; no common cutoff threshold over all configurations is asserted.

Proof sketch. VII.6.k bounds every nonempty recentred part. The chessboard inequality and moment–cumulant identity bound the difference of moments by the budget times the sum of absolute coefficients of the free moment polynomial. Smoothness bounds that finite sum on the compact window. At arity zero both moments equal 1.

Inputs: `Read.full_cost`.

Lean: `sightSky`.

Lemma VII.6.m (balanced estimates). Under the hypotheses of VII.6.k, suppose three base bands q_x, q_y, q_w and $c < 0$ satisfy

$$q_y + 6 < q_x, \quad q_w + 5 < q_x + c, \quad (q_x - q_y)Dn + c < 0, \quad (q_x - q_w)Dn - q_x - c < 0.$$

Assume the free hierarchy is given. Let $f(i) \to \infty$, and let Y_i be tables for which $P_{f(i)}$ is poised. Then the stations at stage $f(i)$, main band $q_x + e_{f(i)}$ and strip exponent c satisfy the balanced estimates of VI.2 for arity at most n and every window radius $R > 0$.

Proof sketch. Write $k = f(i)$. The pointwise error is $O(\mathrm{bud}_n(k)) \to 0$. The auxiliary package at band $q_y + e_k$ supplies the strip bound

$$A_k = O(\mu_k^{(q_x - q_y)D}),$$

and the package at $q_w + e_k$ supplies the bound used for interpolation off the strip,

$$B_k = O(\mu_k^{(q_x - q_w)D}).$$

Each auxiliary package supplies both types of local bound; its designated role is chosen in the balancing argument. Their stopping stages fit inside the main geometric window by the first two inequalities. With $\sigma_k = \mu_k^c$ and normalized spacing $h_k = \mu_k^{-(q_x+e_k)} \le \mu_k^{-q_x}$, the last two inequalities give

$$A_k^n \sigma_k \to 0, \qquad B_k^n h_k/\sigma_k \to 0.$$

The radii tend to zero, and the lift bounds and symmetries hold on ordered tails. VI.2's station theorem gives eventual admissibility with any positive tolerance.

Inputs: `thirdSky`, `w_rightSky`, `supply_wheel`.

Lean: `aspectSky`, `podiumSky`.

3.31.4.4 4.4 Checked vanes Definition VII.6.n (vanes and checked vanes). A *vane of depth D* consists of a stage k, band b, strip exponent c, and complex-linear tables $R_{N,p,d}$ on Label_d for $d \le D$.

Its station has radius $\ell(b,k)$, strip width μ_k^c, and lifts

$$A_{N,p,d} = R_{N,p,d}^{-1} \circ D_{k,N,p,d}^{-1} \circ W_{\mu_k^b,d} \circ \iota_d.$$

Here $\iota_d : \mathcal{F}_d^{\mathrm{free}} \to \mathrm{Label}_d$ is the canonical equivariant section, W applies the position scaling weight, and D is the coupling dilation. On a frame vector of bidegree (h,q), $W_{\rho,d}$ multiplies by ρ^h, where h is the engineering degree, while $D_{k,N,p,d}$ multiplies by $s_{k,N,p}^q$, where q is the number of curvature factors and

$$s_{k,N,p} = \frac{\beta_{N,p;k}^{-1/2}}{\widehat{\beta}_k^{-1/2}\mu_k}.$$

Here $\beta_{N,p;k}$ is the actual coupling at stage k. If $R_{N,p,d}$ is not bijective, the lift is zero. Charting ensures bijectivity on the cutoff tails used below.

The vane is *checked at level j* when:

1. $5 < b \le \log\log \widehat{\beta}_k$;
2. $T_u \le k$ for every $u \le D$;
3. for every $N > k$ and p, its tables obey the charting and sharper coordinate bounds in VII.6.d;
4. every generic rescaled stage-k lattice pair of norm at most 100 fits; for every $w \le D$, every pair of free basis vectors in degree w, every $N > k$ and p, and every fitted pair whose rounded configuration is in the radius-100 window, all nonempty recentred parts have chessboard root at most $1/(j+1)$;
5. its station is admissible for arity at most j, window radius $j+1$, and tolerance $1/(j+1)$.

Lean: `Vane`, `Vane.wheel`, `wok`, `ladd`, `ladder`, `SeqPair`, `TimberSky`, `Vane.checked`. *(definitions read)*

Lemma VII.6.o (a checked vane below any radius). Assume simplicity, (S), $z \ge c_i$ for $i \le d$, $\mathrm{Fire}_d(z)$, and the limit-table hypotheses of VII.6.2. Let $l \le d+2$, $s_i > 0$ with $s_i \to 0$, $Y \in \mathbb{N}$ and

$\theta > 0$. There are an index $i \geq Y$, a stage $k \geq Y$, and a vane of depth d, stage k, checked at level l, with

$$\ell(b, k) = s_i < \theta.$$

Proof sketch. VII.6.e gives increasing stages realizing a subsequence of s at bands in $[b_0, b_0 + 1]$. The two-point condition follows from VII.6.k applied to basis pairs, with arity budget $d + 2$, together with uniform fitting and $\mathrm{bud}_{d+2}(k) \to 0$. There are only finitely many degrees and basis pairs. VII.6.m and the station theorem give admissibility through arity $d + 2$, hence through arity l, at window radius $l + 1$ and tolerance $1/(l + 1)$. Choose a stage beyond the reference stages and band ceiling, with both indices at least Y and radius below θ.

Inputs: `await`.

Lean: `boatsSky`.

Lemma VII.6.p (= Proposition VII.6.2).

Proof sketch.

1. For each d, N and $y > 0$, VII.6.o with $l = d$, $Y = \max(d, N + 1)$ and $\theta = y$ gives $k > N$ with $s_k < y$, and a vane of depth d, stage $\geq d$, checked at level d, with radius s_k.
2. Choose recursively $p(d + 1) > p(d)$ with $s_{p(d+1)} < s_{p(d)}$.
3. The stages are $\geq d$, so they tend to infinity.

Lean: `wakesSky`.

3.31.4.5 4.5 Limits along the ultrafilter Lemma VII.6.q (limits of station lifts). Under the ordered-limit and hierarchy hypotheses of VII.6.1, let a station of depth D, with radius $r > 0$, be admissible through arity $n_{\max}$, on the radius-R window, with tolerance ε. Let $\mathcal{L}$ be an ultrafilter finer than Φ_F. Assume its degree-d lift sends $\mathbf{1}_d^{\mathrm{free}}$ to the scalar label eventually along Φ_F, for every $d \leq D$. Then the lift tuple converges along $\mathcal{L}$ to

$$Y \in \prod_{d \leq D} \mathrm{Hom}_{\mathbb{C}}(\mathcal{F}_d^{\mathrm{free}}, \mathrm{Label}_d).$$

The maps $U_d = \pi_d \circ Y_d$ are real, signed-covariant and unit-preserving for $d \leq D$. For $n \leq n_{\max}$, $d_i \leq D$, $v_i \in \mathcal{F}_{d_i}^{\mathrm{free}}$, $M > 0$, and every smooth test t supported in the radius-R window with $\|t\|_\infty \leq M$,

$$\left| \int t(x) \left(\langle U v_1(r x_1) \cdots U v_n(r x_n) \rangle^S - \langle v_1(x_1) \cdots v_n(x_n) \rangle^{\mathrm{free}} \right) dx \right| \leq \varepsilon M \prod_i \|v_i\|.$$

Proof sketch.

1. The lifts are eventually bounded, so a nontrivial filter $\mathfrak{b}$ finer than $\mathcal{L}$ along which they converge exists (`finite_extract`).
2. Since $\mathcal{L}$ is an ultrafilter, $\mathfrak{b} = \mathcal{L}$.
3. Symmetry and unit identities pass to the limit, as in VI.2's `harvested`, and ordered smeared convergence gives the displayed estimate. The extra unit hypothesis covers every degree through D.

Lean: `harvestedSky`.

Lemma VII.6.r (vanes give approaching maps). Under the ordered-limit and hierarchy hypotheses of VII.6.1, let r_k be a strictly decreasing null sequence. Let V_k be vanes of depth k whose stages tend to infinity, checked at level k, with $\ell(V_k) = r_k$, and let $\mathcal{L}$ be an ultrafilter finer than Φ_F. Then there are complex-linear maps $U_k(d) : \mathcal{F}_d^{\mathrm{free}} \to \mathcal{F}_d^S$ such that:

- (r_k, U_k) is approaching;
- $U_k(d)$ is real, signed-covariant and unit-preserving for $d \le k$;
- $U_k(d) = \pi_d \circ Y_{k,d}$ for $d \le k$, where Y_k is the $\mathcal{L}$-limit of the lifts of V_k.

Proof sketch.

1. Conditions 2–3 of VII.6.n give unit preservation of the lifts in every degree through k. VII.6.q then applies to each V_k.
2. Fix n, v, a compact $K \subset \mathrm{Conf}_n$, and M, ε. For $k \ge n$, $k \ge \max d_i$, K contained in the radius-$(k+1)$ window, and $\prod \|v_i\| M/(k+1) \le \varepsilon$, the estimate at tolerance $1/(k+1)$ bounds $|\int tK_k|$ by ε.

Lean: `breadSky`.

Lemma VII.6.s (isomorphisms). Assume in addition the dimension cap (iv). Then there is a scaling with radii r_k and $\xi_k = 1$ such that:

- $U_k(d)$ coincides with the map of VII.6.r for all large k (for each d);
- (a)–(c) of VII.6.1 hold.

Proof sketch.

1. The maps are eventually injective in each degree by VI.2. Finitely many free test functionals separate $\mathcal{F}_d^{\mathrm{free}}$, their rescaled versions converge by the approaching property, and injectivity is an open condition.
2. Injectivity and $\dim \mathcal{F}_d^S \le \dim \mathcal{F}_d^{\mathrm{free}}$ give bijectivity.
3. Where the map is not yet bijective, one of the later isomorphisms in that degree is substituted. This does not affect any eventual property.

Lean: `dockedSky`.

Lemma VII.6.t (= Proposition VII.6.1).

Proof sketch. VII.6.p gives p and V. The radii $s \circ p$ form a strictly decreasing null sequence. VII.6.s then gives the scaling, with $f = p$.

Lean: `pays_sky`.

Consumer. `gang` (VII.2) applies VII.6.1 with:

- the seed's ordered limits, hierarchy and point limits;
- the free hierarchy supplied by the free-field comparison theorem;
- hypothesis (iv) from the rank-width bound.

It selects the lift limits and applies the trace-comparison theorem of VII.2. The run `merano`, together with the other runs cited in §2, supplies the rest:

- $z_d = \sum_{i \le d} c_i$;

- $\mathrm{Fire}_d(z_d)$ from the uniform-band construction;
- $\mathcal{L}$ from the existence of an ultrafilter finer than Φ_F.

3.31.4.6 4.6 Degrees below four Lemma VII.6.u (= Proposition VII.6.5).

Proof sketch.

- (a) For simple G, every free polynomial label of degree $d < 4$ is $c \cdot 1$. The label included in the degree-4 filtration is $c \cdot 1 + Q$, where Q lies in the span of the quadratic invariants and is homogeneous of weight 4 (by the degree-four classification). A label of degree < 4 has no weight-4 part, so $Q = 0$. Since labels map onto $\mathcal{F}_d^{\mathrm{free}}$ and the unit is nonzero, $\dim \mathcal{F}_d^{\mathrm{free}} = 1$. For comparison, $\dim \mathcal{F}_4^{\mathrm{free}} = 22$ (`four_size`).
- (b) The comparator of the final record is an isomorphism $\mathcal{F}_d^S \cong \mathcal{F}_d^{\mathrm{free}}$, so $\dim \mathcal{F}_d^S = 1$. Since $\mathbf{1}_d \neq 0$, $\mathcal{F}_d^S = \mathbb{C}\mathbf{1}_d$.

Lean: `dim_before`, `first_grade`.

Lemma VII.6.v (the unit is supplied). Assume the construction record for C, let φ be finer than Φ_F, let G be simple, and fix $d \in \mathbb{N}$ and $t \in \mathbb{C}$. Then $t\mathbf{1}_d$ is supplied along φ with $p = 1$.

Proof sketch. Let e_d^{ref} be the reference unit label. Take the constant family $q \equiv y = t\, e_d^{\mathrm{ref}}$, for which $\pi_d y = t\mathbf{1}_d$. For $k \geq T_d$ and $N \geq k$, the quota is homogeneous and the unit has stage-k coordinate $a_k^d e_0$; this follows from the construction record. Hence the quota is $|t|\, a_k^d$. Along φ the cutoff eventually exceeds k, and for large k both $a_k \leq 1$ and $\hat{\beta}_k \geq |t|$ hold.

Lean: `supplyUnit`, `quota_smul`, `quota_unit`.

Lemma VII.6.w (= Proposition VII.6.4).

Proof sketch. By VII.6.u(b), $\pi_d x = t\mathbf{1}_d$. Then VII.6.v applies, with the construction record taken from the final record.

Lean: `inletFirst`.

Companion statements (VII.4). The same file contains the dual "release" bounds:

- `coreRead`;
- `physicalRelease`, which supplies an exponent p such that, for all large k, every nontrivial finer filter along which $q \to x$ and $\mathrm{quota}_k(q) \leq z$ eventually satisfies $|L(\pi_d x)| \leq \hat{\beta}_k^p a_k^{-(d-w)} z$. It assumes construction and work item data, simplicity of G, $w \leq 3$, and $L(\mathbf{1}_d) = 0$ when $w > 0$;
- `outletFirst`, where L vanishes on the images of $\mathcal{F}_e^S$, $e < w$, under the inclusions of the derivative plan.

These results belong to VII.4.

3.31.5 5. Where the difficulty lies

The analytic work extends the estimates of III.11 across a window with two neighbouring price indices. The smaller block factor controls the stopping depth; the larger and smaller neighbouring price parameters control the one-step estimates. At the final root estimate, affordability is charged at the width stage itself. These distinctions replace uses of constant-factor plateaus.

The interval covering argument requires one stage threshold for every band in a unit interval. Without that uniformity, the band selected by the target radius could escape the available estimates. The subsequent diagonal choice preserves exact equality with the prescribed radii.

The scaling maps are real, signed-covariant and unit-preserving eventually in each degree. This result does not assert compatibility with inclusions between degrees or stress-tensor normalization. The dimension cap supplies surjectivity. The ultrafilter keeps all lift limits on the same cutoff filter, and the checked tolerances give local order-zero convergence without an explicit rate in the prescribed radius. The fixed-field supply argument uses the comparator and the unit coefficient identity; it requires none of the moving-band estimates.

3.31.6 6. Notation

object	code name	proposed notation
scaling (radii, constants, isomorphisms)	`Climb.Scaling`; radii only: `Zoom`	(r_k, ξ_k, U_k)
approaching; weak vanishing on compacts	`Approaches`, `Fades`	$K_k \to 0$ locally with an order-zero bound
scaling sequence along $\mathcal{L}$ (VII.6.1 (a)–(c))	`StarPath`	scaling sequence along $\mathcal{L}$
free Maxwell reference table	`freeTable`, `Bel`, `Cells.standard`	$\mathcal{F}_d^{\mathrm{free}}$, H_0
running coupling; growth scale	`hatBeta`, `monitor`, `Perch.bill`	$\hat{\beta}_k$; μ_k; μ_k^a
band; reading scale	`Panel.arc`, `Purse.band`; `Station.gap`	b; $\ell(b,k) = a_k \mu_k^b$
middle and width stages; allowed; fit	`middleSky`, `widthSky`, `AllowedSky`, `FitSky`	$m_1(b,k)$, $m_0(b,k)$
position weight; coupling dilation	`weightTag`, `Frame.scaled`, `shrink`	$W_{\rho,d}$; $D_{k,N,p,d}$; $s_{k,N,p}$
normalized probe size	`moneys`	$M_{k,N,p,d}$
stage package	`Tapes.Panel`	$(b, \theta, e, \text{purses})$
stage-k cost record; poised	`LegacyCostSky`, `PoisedSky`	poised at stage k
uniform-band property	`FiresSky`	$\mathrm{Fire}_d(z)$
leap; packages with moving band	`LeapSky`, `ArcSky`	—
budget; chessboard root	`Ship.budget`, `rootTest`	$\mathrm{bud}_n(k) = \mu_k^{-1/(16n)}$; $\rho_{\mathcal{T}}$
recentred probe product	`Read.cylinder`, `Read.cumulants`	Φ_s
mark exponents	`generator`	$c_d = 2(w_d + 1)$
one-step cost package	`Scrim`	(S)
vane; its station; checked	`Vane`, `wok`, `TimberSky`, `SeqPair`	$V = (k, b, c, R)$; checked at level j
dimension cap	`DimAbove`	$\dim \mathcal{F}_d^S \le \dim \mathcal{F}_d^{\mathrm{free}}$
gallery property (VII.2)	`Trace.Over.Gallery`	—
stage-k coefficient size	`quota`	$\mathrm{quota}_k^{N,p}(u)$

object	code name	proposed notation
supply; release; killing lower degrees	`Supplies`, `Releases`, `Kills`	x supplied along φ; L released at weight w

3.31.7 7. What remains unexplained

1. **Internal analytic packages.** VII.6.g–j retain the room, background-field, source-expansion, small-field and Hessian-control hypotheses from III.9–III.11. Their direct implications and the final root estimate are verified, but the numerical definitions and proofs of all constituent one-step estimates have not been expanded here. In particular, the small-field input explicitly bounds the combined nonlinear errors by $x^{-1/32}$; the eventual feasibility of that input remains part of III.11. The formula and guarded domain of VII.6.j are verified.
2. **Downstream trace argument.** The checked two-point bound is retained when the vanes and their lift limits are passed to VII.2. Its subsequent analytic use in deriving the trace anomaly is outside this unit. No such conclusion is inferred from the weak convergence alone.
3. **Construction inputs.** The passage from an enlarged paid waiting window and covariant coordinate maps to the uniform-band property is verified at the cited interfaces. The full construction that pays every numerical request belongs to III.2; it is not an additional conclusion of VII.6.1.

3.32 VIII.1. Assembly of the main theorem

Part: VIII, Assembly of the main theorem. **Lean:** the top-level files `Solution.lean`, `Construction.lean` and `ConcreteSU2.lean`; the bridges `WilsonBridge.lean`, `HierarchyBridge.lean`, `TraceBridge.lean` and `ContactBridge.lean`; the public definitions `PublicTheory`, `PublicWilson`, `PublicView`, `ChallengeSpec` and `ContactSpec`; the folders `YM/Nonperturbative`, `YM/Independent`, `YM/Spec`, `YM/Org` and `YM/Pert/Parity`; the files `YM/R2I/Campaign/Accepted.lean` and `YM/RG17/Joint.lean`; and parts of YM/FullVolume, YM/EveryPhase, YM/TraceExtension, YM/OPE, YM/Contacts and YM/Pert. **Size:** 12,060 lines in 1,674 declarations.

The construction chooses one Wilson family for each gauge group and block-spin parameter $b > 0$. At each requested extraction order, it assembles the continuum table, phases, fields, reconstruction and OPE on one refinement. Fixed-volume ultraviolet convergence removes the mesh subsequence. The bridges then give the public theorem in statement.md.

3.32.1 1. What this unit proves

Conventions. The notation follows statement.md:

- $(m, G, \mathfrak{g})$ is a gauge group: $m \geq 1$, G is a closed subgroup of $U(m)$, and $\mathfrak{g} = \{X : \exp(tX) \in G \text{ for all } t \in \mathbb{R}\}$. Simplicity includes nonabelianness.
- $\mathcal{W} = (L_p, D_N, \beta_{N,p}, h, \mathcal{T}, I_{N,p})$ is a Wilson family, with $a_N = 1/D_N$, smeared moments $C_{N,p}(v; f)$ and separated configurations Conf_n.
- A hierarchy has a Wightman theory $\mathfrak{W}$, finite-dimensional field spaces F_d, label maps ι_d, token projection π, phases (X, ν, ξ_*), correlations S^s, time continuations F_n, OPE data $(\mathrm{Ch}_j, O_{j,\kappa}, c_{j,\kappa})$, and $\Theta \in F_4$. Write $\mathbb{F} = \bigsqcup_d F_d$. The function $G_\Theta(r)$ is the selected two-point correlation of Θ at time-axis points $-r/2$ and $r/2$.

- (Reg1)–(Reg4), (R1)–(R15), (O1)–(O2), (T1)–(T4), (P1)–(P4) and (D1)–(D16) are as in statement.md.
- $b_0 = 11C_{\mathfrak{g}}/(48\pi^2)$, where $C_{\mathfrak{g}}$ is the adjoint Casimir in the construction normalization. All Lie-algebra dimensions are real.

Internal objects:

- An *extraction order* is a pair $\sigma = (\sigma^{\mathrm{m}}, \sigma^{\mathrm{v}})$ of strictly increasing maps $\mathbb{N} \to \mathbb{N}$: a mesh sequence and a volume sequence.
- Its *refinement* by τ is $\sigma \circ \tau := (\sigma^{\mathrm{m}} \circ \tau^{\mathrm{m}}, \sigma^{\mathrm{v}} \circ \tau^{\mathrm{v}})$. The full sequence is written id.
- An *internal family* $\mathcal{C}$ is the record of a renormalization-group construction: trajectory, couplings $\beta_{N,p}(m)$ at levels $m \le N$, labels and local lattice sections. (D3) is its construction predicate, and $\mathcal{W}(\mathcal{C})$ is its Wilson family, with $\beta_{N,p} := \beta_{N,p}(N)$.
- A *table* S is a family of continuum correlation functions $S(v; x)$ indexed by token lists.

Ordered limit. $\mathrm{Ord}_\sigma(\mathcal{C}, S)$ means the following. For every n, $v \in \mathcal{T}^n$ and $t \in C_c^\infty(\mathrm{Conf}_n)$:

- t is $\sigma^{\mathrm{v}}(k)$-admissible for large k;
- there are numbers R_p with $C_{\sigma^{\mathrm{m}}(j),p}(v; t) \to R_p$ for every admissible p;
- $R_{\sigma^{\mathrm{v}}(k)} \to \int tS(v; \cdot)$.

Final object. A final object $\mathfrak{F}$ on $(\mathcal{C}, \sigma, S)$ consists of five parts:

1. Chosen renormalization data and step witnesses whose trajectory is exactly that of $\mathcal{C}$ (D6).
2. A *seed*: (D3), $\mathrm{Ord}_\sigma(\mathcal{C}, S)$, point limits, the Euclidean hierarchy axioms for S, cell hypotheses and corrected-field widths.
3. A field algebra: derivatives, rotations and a stress tensor.
4. A phase selection with its certificate.
5. Analytic continuations of the mixed and selected correlations, and a Wightman theory realizing the selected one.

Proposition VIII.1.1 (main theorem, public form). Let $(m, G, \mathfrak{g})$ be a gauge group with $\mathfrak{g}$ simple. Then there is a Wilson family $\mathcal{W}$ for G satisfying (Reg1)–(Reg4), with the following property. For every strictly increasing $p : \mathbb{N} \to \mathbb{N}$ there are a strictly increasing q and a hierarchy over $\mathcal{W}$ satisfying (P1)–(P4) along $p \circ q$:

- **(P1)** For every n, every $v \in \mathcal{T}^n$ and every $f \in C_c^\infty(\mathrm{Conf}_n)$, f is $p(q(k))$-admissible for all large k. There are numbers $R_{p'}$ such that $C_{N,p'}(v; f) \to R_{p'}$ along all N for every p' for which f is admissible, and $R_{p(q(k))} \to \int f S^{\mathrm{mix}}(\pi(v); \cdot)$.
- **(P2)** The selected phase carries a Wightman theory satisfying (W0)–(W3), with faithful fields. The mixed correlations are the ν-average of continuous phase correlations, with ν a probability measure of full support. The continuation is holomorphic in the complex times at fixed real spatial positions, and its boundary values on product Schwartz tests are the selected Wightman functions. All clauses (R1)–(R15) hold.
- **(P3)** The OPE bounds (O1)–(O2) hold with one system of coefficients, uniformly in the state $s \in X \sqcup \{\mathrm{mix}\}$.
- **(P4)** Θ has zero vacuum expectation, $A_{\mathfrak{g}} := 3 \dim_{\mathbb{R}} \mathfrak{g}/\pi^4 > 0$, and $r^8(\log 1/r)^2 G_\Theta(r) \to A_{\mathfrak{g}}$ as $r \downarrow 0$. For every finite Wick profile and every $\delta > 0$, the profile differs from G_Θ at some $r \in (0, \delta)$.

Status: theorem. *Lean:* `YangMills.jointExistence`, statement `JointExistence`,

with `WilsonRegularity`, `Hierarchy`, `Properties`, `Converges`, `Reconstructed`, `OperatorProductExpansion` and `TraceTest`.

Proposition VIII.1.2 (detailed construction theorem). Fix a gauge group $(m, G, \mathfrak{g})$ with $\mathfrak{g}$ simple and a real $b > 0$. Then there is an internal family $\mathcal{C}$ such that:

- its block-spin kernel has penalty b (D16);
- (D3) holds;
- $\mathcal{W}(\mathcal{C})$ satisfies (Reg1)–(Reg4).

Moreover, for every extraction order σ there are:

- a refinement τ and a table S;
- a final object $\mathfrak{F}$ on $(\mathcal{C}, \sigma \circ \tau, S)$;
- an every-phase reconstruction $\mathfrak{L}$, an OPE reader Q, contact devices M and scaling data $\Sigma = (r_k, \xi_k, U_k)$, with $r_k \downarrow 0$ and $\xi_k > 0$;

such that the following hold:

1. The public hierarchy $T(\mathfrak{F}, Q)$ over $\mathcal{W}(\mathcal{C})$ has (P1)–(P4) along $(\sigma \circ \tau)^{\mathrm{v}}$, and every correlation $S^s(a; \cdot)$ extends to a distribution on $(\mathbb{R}^4)^n$ (D1, D2).
2. $\mathrm{Ord}_{(\mathrm{id}, (\sigma\circ\tau)^{\mathrm{v}})}(\mathcal{C}, S)$ holds: the ordered limit with the full mesh sequence (D4).
3. Positional ultraviolet convergence at every fixed volume holds along the full mesh sequence (D5). See VIII.1.6.
4. Chosen renormalization data and step witnesses generate exactly the trajectory of $\mathcal{C}$ (D6).
5. The bundle (Core) holds: OPE services, reconstruction in every phase, the trace tests, the OPE foundation, and integrated OPE remainder bounds for integrable complex densities against positive measures, with bounded L^1 mass.
6. The internal Wightman axioms hold for the selected phase and for every phase $\xi \in X$, and the public (W0)–(W3) hold for every phase. The ν-average of the phase Wightman distributions is the boundary distribution of the mixed continuation (D7).
7. The separated OPE and the nested OPE over finite trees hold (D8, D10).
8. The seven headline trace clauses hold, together with the unique high-energy spectral law contained in (Core) (D9; VIII.1.n and VIII.1.o).
9. The contact devices use the reader's prescription and give a full contact resolution, real bases of the OPE strata and contact thresholds (D11)–(D13).
10. The leading scaling comparison with free Maxwell fields along Σ and the normalization at degree four hold (D14, D15).

The family $\mathcal{C}$ depends only on (G, b). Items 3 and 4 are properties of $\mathcal{C}$ alone, although they are stated inside the quantifier over σ. *Status:* theorem. *Lean:* `YangMills.constructed`. Internal form: `Nonperturbative.existence`, statement `Headline.WilsonExistence`.

Proposition VIII.1.3 (continuum theorem for the funded run). Fix a gauge group with $\mathfrak{g}$ simple and $b > 0$. Let $\mathcal{C}_{G,b}$ be the family of the run $\mathsf{R}_{G,b}$ chosen in VIII.1.a. Its penalty is b. For every extraction order σ there are $\tau, S, \mathfrak{F}, \mathfrak{L}, Q, \mathsf{M}$, with $\mathfrak{F}$ on $(\mathcal{C}_{G,b}, \sigma \circ \tau, S)$, satisfying:

- items 2, 3, 5 and 9 of VIII.1.2;
- the uniform nested OPE over admissible finite trees;
- a scaling sequence Σ with (D14) and (D15).

Status: theorem. *Lean:* `continuum_with_position_tests`. The form without Σ is `continuum`.

Proposition VIII.1.4 (a funded run supplies everything). Let R be a run with campaign T and pillars W, let Δ be a choice of local step estimates, and let $\mathfrak{g}$ be simple. Assume R is *funded* (Fd): see §2, item 1. Then the assembly package (Goal) holds:

1. (Fd).
2. The ultraviolet package (UV_∞) of VIII.1.5.
3. The logarithmic OPE package.
4. The dense-coverage property of the sharp Gram family.
5. For every σ, every table S and every final object $\mathfrak{F}$ on $(\mathcal{C}_{\mathsf{R}}, \sigma, S)$: the trace result of VIII.1.h.
6. For every σ there are τ, numbers $(B_m)_{m\in\mathbb{N}}$ and data $S, \mathfrak{F}$ (on $\sigma\circ\tau$), $\Sigma, \mathfrak{L}, Q$ such that:
   - for every level m the couplings $\beta_{N,p}(m)$ have the ordered limit B_m along $\sigma\circ\tau$;
   - $\mathrm{Ord}_{(\mathrm{id},(\sigma\circ\tau)^{\mathrm{v}})}(\mathcal{C}_{\mathsf{R}}, S)$ and (D5) hold;
   - (D14) holds;
   - the reconstruction in every phase holds;
   - the trace result holds;
   - the OPE facilities, the OPE foundation for Q and the uniform nested OPE hold;
   - the all-shapes OPE holds for every coherent prescription;
   - for every ultrafilter on $\mathbb{N}^2$ finer than the extraction filter of $\sigma\circ\tau$ there is a compatible allocation of the sharp Gram data.

Status: theorem. *Lean:* `R2IGoals.fromFunded`, statement `Broad`.

Proposition VIII.1.5 (ultraviolet endpoint). Let $\mathfrak{g}$ be simple, let R be a run with campaign T and pillars W, and assume the stationary-point and amplification condition, the delay condition, and the weight-and-tableau data (ϑ, v) of (Fd). Then the ultraviolet package (UV_∞) holds:

1. (D3) holds for $\mathcal{C}_{\mathsf{R}}$, and the weight data exist.
2. *Sampler.* Every σ has a refinement τ carrying a table S and a final object on $(\mathcal{C}_{\mathsf{R}}, \sigma\circ\tau, S)$, and $\mathrm{Ord}_{(\mathrm{id},(\sigma\circ\tau)^{\mathrm{v}})}$ holds.
3. The trajectory at every (p, N) competes in, and is issued by, the fixed tuning recursion.
4. *Tuning and convergence.* The parameter $s \in [-1, 1]$ specifies the target coupling $\bar\beta + ws$ at level 0, where $w > 0$ is the fixed tuning width. For every (p, N, s) there is a successful trajectory, and its bare coupling at level N is unique within the prescribed recursion. These bare couplings are values of a function analytic on the open unit disc in s. At each fixed (m, p), the couplings and activities of successful trajectories converge as $N \to \infty$ whenever their level-zero targets converge. For complex targets in the disc, convergence of the analytic couplings and activities, with every parameter derivative, is uniform on compact subsets; activities are measured in the fixed weighted norm at (m, p).
5. For every fixed (m, p) there is $c_{m,p} > 0$ such that
$$e^{-\mathrm{scalar}_{N,p,m}} Z_{N,p} \geq c_{m,p} \qquad (N \geq m),$$
where $Z_{N,p}$ is the bare Wilson partition function relative to probability Haar measure and $\mathrm{scalar}_{N,p,m}$ is the field-independent scalar in the level-m action. Its scalar density $v_{N,p,m}$ tends to $-\infty$ as $N \to \infty$.
6. The volume-uniform bounds hold, as stated in VIII.1.c, step 1.

7. (D5) holds along the full mesh sequence.
8. For every fixed p there is a family of successful trajectories with target $\bar{\beta} + \lambda w s$, $s \in [-1, 1]$, for some $\lambda > 0$, whose central member is the chosen family. They obey the same step relations and insertion normalization. For each token list, each compact $K \subset \mathrm{NoHit}_p$ and each finite spatial derivative order h, there are an open neighbourhood Ω of K in NoHit_p and a smaller positive parameter width, independent of N. In the resulting unit-disc coordinate, the limiting correlation is holomorphic in the parameter and C^h in position on Ω. All fixed coordinate differences of order at most h, their interpolants and their admissible smearings converge to the corresponding derivatives, uniformly on compact parameter and position sets, with every parameter derivative.

Status: theorem. It is assembled here; its ingredients are in III.12–III.15. *Lean:* `FullVolume.Final.jewelOf17` and `Jewel`.

Proposition VIII.1.6 (the ultraviolet limit along the full spacing sequence). For $p \in \mathbb{N}$, let NoHit_p be the set of configurations no two of whose points are congruent modulo $L_p\mathbb{Z}^4$.

(a) For the family of a run satisfying VIII.1.5's hypotheses, the following holds for every p, n and $v \in \mathcal{T}^n$. There is $f_{p,v} \in C^\infty(\mathrm{NoHit}_p)$ such that:

- $C_{N,p}(v; t) \to \int t\, f_{p,v}$ as $N \to \infty$ along all N, for every p-admissible t;
- for every fixed finite list of normalized forward, backward or centered coordinate differences, with fixed integer stencil widths, the interpolated lattice differences converge locally uniformly on NoHit_p to the corresponding derivatives of $f_{p,v}$. The same limit holds whenever lattice configurations approach a point of NoHit_p, along any cutoff indices tending to infinity. The smeared differences converge to the integral of the derivative against the test, and that integrand is integrable.

The cutoff thresholds may depend on p, n, v, the compact set, the tests and the difference stencils.

(b) Suppose a final object exists on $(\mathcal{C}, \sigma, S)$ and (a) holds for $\mathcal{C}$. Then $\mathrm{Ord}_{(\mathrm{id},\sigma^{\mathrm{v}})}(\mathcal{C}, S)$ holds.

Status: theorems. *Lean:* (a) `Campaign.realUV`, with the definition `PositionsUV`. (b) `FullVolume.Final.from_spec`.

Proposition VIII.1.7 (public bridges). Let $\mathfrak{F}$ be a final object on $(\mathcal{C}, \sigma, S)$ and Q a reader, and let $T(\mathfrak{F}, Q)$ be the public hierarchy built from them (VIII.1.p).

1. The public Wilson energy, Wilson measure, lattice moments and smeared moments of $\mathcal{W}(\mathcal{C})$ equal the internal ones. $\mathrm{Ord}_{(\mathrm{id},s)}(\mathcal{C}, S)$ implies (P1) along s.
2. If (D3) holds and $\mathfrak{g}$ is simple, then $\mathcal{W}(\mathcal{C})$ satisfies (Reg1)–(Reg4), with $b_* = 2b_0$.
3. The reconstruction in every phase implies (R1)–(R15).
4. The internal separated OPE implies (O1)–(O2).
5. If $\mathfrak{g}$ is simple, the seven internal headline trace clauses imply (T1)–(T4), with $3 \dim \mathfrak{g}/\pi^4$ identified with the internal amplitude.
6. The full contact resolution implies (D2).

Status: theorems. *Lean:* `wilsonFamily` and `fromWilson` (1); `regular` (2); `publicReconstruction` (3); `publicOPE` (4); `publicTrace` (5); `contactDistribution` (6).

Proposition VIII.1.8 (exclusion of Wick descriptions in momentum space). Let $\mathfrak{F}$ be a final object. Assume $s^8(\log 1/s)^2 G_\Theta(s) \to A > 0$ as $s \downarrow 0$. Let W be any finite list of complex

coefficients c_α and products of mass-shell modes, in the class defining Wick profiles in statement.md. There are Schwartz functions f, g on $\mathbb{R}^4$ for which

$$v_{f,g}(p) := \overline{\mathcal{F}_+ f(p)}\, \mathcal{F}_+ g(p)$$

has compact support and

$$\mathcal{W}_2^{\xi_*}(\Theta^*, \Theta)(\bar{f} \otimes g) \neq \sum_\alpha c_\alpha \,\mathrm{read}_\alpha(v_{f,g}).$$

Here $\mathcal{F}_+$ is the unnormalized positive-sign Fourier transform used in the spectral representation. The four-dimensional test $v_{f,g}$ is the transform of the joint test restricted to relative momenta $(-p, p)$. *Status:* theorem. It enters the trace result of VII.3. The public (T4) follows directly from the Euclidean exclusion (VIII.1.r). *Lean:* `TraceExtension.Wick.polynomial_excludes`.

Proposition VIII.1.9 ($SU(2)$). $\mathfrak{su}(2) \subset M_2(\mathbb{C})$ is simple. Hence VIII.1.1 holds for $(2, SU(2), \mathfrak{su}(2))$. *Status:* theorem. *Lean:* `su2Simple` (from `Examples.SU2.simple`) and `su2Theory`.

Proposition VIII.1.10 (definitions filed here and used elsewhere). Several definitions used by III.10–III.15, V.1–V.3, VI.1–VI.2 and VII.1–VII.4 are filed in this unit:

- the structure of a final object and its parts: `Spec.Final`, `Seed`, `Locals`, `Materializes`, `Relativistic` and `Calibrated`;
- the input bundles `Org.Alloy`, `Assembly`, `MaxwellData`, `IR` and `UV`;
- the volume-uniform bounds of YM/FullVolume, for example `FixedBounds` and `genuine`.

Status: definitions, and theorems described in the units that use them.

3.32.2 2. What it uses

From other units.

1. **III.14 and III.2 (selection of the run).** For simple $\mathfrak{g}$ and $b > 0$ there are a run R, a campaign T and pillars W such that the penalty is b and R is funded for the fixed local-estimate choice Δ. A run records levels, schedule of constants, step budgets, the trajectory and its tuning, the reference levels and the transfer data of the labels.
   - (Fd) comprises thirteen conditions: local step budgets; phase-comparison and dense-coverage data; divergence of a schedule Q; stationary-point and amplification data; a delay condition; frame compatibility; polynomial schedule compatibility; support exponent 100 at every level; the logarithmic OPE budget; the halo condition; a pacing condition; thermal OPE estimates; and weight-and-tableau data (ϑ, v). The detailed numerical conditions belong to III.15.
   - The run is produced by a selection theorem for runs with a fixed treasury period, a preselected component recipe and the default dynamics (III.2).
   - *Lean:* `Independent.constructionRun`, `Funded`, and the selection theorem in `NativePreselectedPostnumericPeriodRun.lean`.
2. **III.12–III.15 (ultraviolet bounds and convergence).** Used for VIII.1.5 and (D5):
   - the convergence and alignment of the activities along the tuning recursion;
   - uniqueness and analyticity of the trial couplings;

- the volume-uniform bounds on correlation kernels and their lattice differences at separated points (`FullVolume.genuine`);
- the pointwise Cauchy property of lattice correlations along the full spacing sequence at fixed volume (as used in `Campaign.sees`);
- the property that every extraction order has a refinement carrying a final object (`Run.again`, through `ahead`).

3. **I.1, II.1–II.2 (setting and construction record).** The internal family, (D3) including the running law: for every $\varepsilon > 0$, $|\beta_{N,p}(m) - \widehat{\beta}_m| \le \varepsilon \log(1/a_m)$ for all sufficiently large m and all $N \ge m$ and all p, where $\widehat{\beta}_m = \bar{\beta} + 2b_0 \log(1/a_m)$; the band condition $\beta_{N,p}(m) \in [\widehat{\beta}_m/2, 2\widehat{\beta}_m]$; tokens, local sections and their stencils; tables.
4. **IV.1 (infinite volume).** The seed: a subsequential table S with $\mathrm{Ord}_{\sigma\circ\tau}$, point limits, the Euclidean hierarchy axioms, cell hypotheses and corrected-field widths. It is produced inside the run's extraction property and used in VIII.1.g.
5. **V.1–V.3 (phases and reconstruction).**
   - The phase selection with its certificate (work items B01–B18).
   - The Osterwalder–Schrader–Wightman reconstruction of the selected phase (`after_materializes`).
   - The reconstruction in every phase, `EveryPhase.all_characters` with `Completed` (V.3).
6. **VI.1–VI.2 (symmetry).** The field algebra (derivatives, rotations, conserved stress tensor) and its local laws: the local-law package built by `Bid.station` from work items A01–A17, and `Org.Tab`.
7. **VII.1 (free Maxwell comparison).** The free-Maxwell data `maxwell_supplies` and `FreeStress.fire`; the leading short-distance comparison; the degree-four normalization (`credible`, `square_four`).
8. **VII.2–VII.3 (trace).**
   - The gallery property `Trace.Over.Gallery` and the chain from it to the trace result.
   - The Euclidean Wick exclusion `euclidean_excludes`, with `real_profile` and `no_powerlog`.
9. **VII.4 (OPE and contacts).**
   - The existence of a coherent prescription (`Prescription.chosen`).
   - The OPE services, which accompany the final object in VIII.1.g.
   - The integrated estimates `estimates_weighted`.
   - The contact resolution `Solstice.provided`, `real_basis`, `installs` and `detector_exists`.
10. **VII.5 (asymptotic freedom).** The value of the slope, $2b_0 = 11C_{\mathfrak{g}}/(24\pi^2)$, and $C_{\mathfrak{g}} > 0$ (`casimir_pos`).

Standard mathematics used.

- uniqueness of limits, and stability of limits under passage to subsequences;
- sequential compactness of countable products of compact intervals (Tychonoff), with a diagonal extraction;
- extension of convergence from a dense set by equicontinuity, and discrete Taylor estimates to recover derivatives, for (D5);
- dominated convergence, and the Laplace transform of a positive measure (Källén–Lehmann representation of the trace two-point function: Källén 1952, Lehmann 1954), for VIII.1.8;
- asymptotic scales $r^a(\log r)^k$, for (T4).

The inputs rest on further standard mathematics:

- Bałaban's block-spin renormalization group for lattice gauge fields (Bałaban, CMP 1984–1989);
- cluster expansions (Brydges, Les Houches 1984);
- chessboard estimates (Fröhlich–Israel–Lieb–Simon, CMP 1978);
- Osterwalder–Schrader reconstruction (CMP 1973, 1975) and Glaser's linear-growth argument;
- decomposition of a translation-invariant state through the commutative algebra on the zero-energy subspace (Borchers; Ruelle; see Glimm–Jaffe, *Quantum Physics*, §19.7);
- one-loop asymptotic freedom (Gross–Wilczek, Politzer 1973) and the trace anomaly (Collins–Duncan–Joglekar 1977).

Imported constants and assumptions. The final existence theorem has no additional analytic hypotheses. The run, local-law packages, final object and (Core) are supplied by the preceding construction results.

Parameters and auxiliary choices:

- The penalty $b > 0$ of the block-spin kernel. The coarse link V is drawn around the block prediction Q with weight $\exp(-\beta b\, \varrho(VQ^{-1}))$ (`penaltyWeight`). It plays the role of Bałaban's averaging parameter and is unrelated to b_*.
- The implication VIII.1.4 accepts a local-estimate choice Δ; VIII.1.a uses the fixed choice supplied by III.14.
- The run-selection theorem is instantiated with the fixed dynamics and auxiliary estimates of III.2.

Constants:

- $b_0 = 11C_{\mathfrak{g}}/(48\pi^2)$, and $b_* = 2b_0$ in (Reg1).
- The trace amplitude $A_{\mathfrak{g}} = 12(d_2/b_0)^2 \dim \mathfrak{g}/\pi^4$ with $d_2 = -b_0/2$, so $A_{\mathfrak{g}} = 3 \dim \mathfrak{g}/\pi^4$.
- $\bar{\beta} > 0$, the reference coupling at level 0.

3.32.3 3. Strategy

Choose one run $\mathsf{R}_{G,b}$ before any extraction order, observable or test. Given σ, first extract a refinement $\sigma \circ g$ on which the couplings at every fixed level have ordered limits. Next extract $(\sigma \circ g) \circ w$, carrying a continuum table and a final object. The later reconstruction, trace, OPE and contact results attach to that object. Passing to further subsequences preserves the earlier limits, so $\tau = g \circ w$ serves every conclusion.

At each fixed volume, the renormalization-step estimates make the lattice correlations Cauchy on a dense union of grids. Equicontinuity gives convergence on separated configurations; discrete Taylor estimates recover all derivatives. Dominated convergence gives the smeared limits. Uniqueness of limits identifies these full-sequence limits with the rows of the extracted table, removing the mesh subsequence.

Specialize the detailed theorem to $b = 1$ and $\sigma = (\mathrm{id}, p)$. The volume part of τ is the required q. The public hierarchy is read from the final object and its OPE reader. The bridges identify the Wilson measures and moments, reconstruction axioms, OPE estimates and trace tests. The plaquette identity, equality of Wick profiles and evaluation of the trace amplitude are the required computations.

3.32.4 4. Steps

The internal statements use a fixed gauge group $(m, G, \mathfrak{g})$. When a run is specified, its campaign and pillars are those appearing in (Fd) and VIII.1.5.

3.32.4.1 4.1 The run and the ultraviolet endpoint Lemma VIII.1.a (the chosen run). Fix a gauge group with $\mathfrak{g}$ simple and $b > 0$. There are a run $\mathsf{R}_{G,b}$, a campaign and pillars with penalty b satisfying (Fd). Write $\mathcal{C}_{G,b}$ for its family. *Proof sketch.* Apply the run-selection theorem of III.2 with the fixed dynamics, local estimates and component prescription of III.14. The required schedule predicate is identically true, so it is satisfied after any delay extension. Choose one witness before specifying an extraction order.

Lean: `constructionRun` (III.14) and `fundedRun`. *(proof read; selection theorem statement only)*

Lemma VIII.1.b (ultraviolet endpoint). This is VIII.1.5. *Proof sketch.* The convergence theorem of III.13–III.15 supplies the analytic families of couplings and activities. Its uniqueness clause identifies successful trajectories with the same level-zero target, and its analytic clause gives the bare-coupling interpolation. Removing the field-independent scalar from the partition function gives the positive lower bound; decay of the bare partition function then forces the scalar density to $-\infty$. The volume-uniform bounds and VIII.1.c give (D5). The parameter-family construction gives item 8. Finally, refine to a final object and apply VIII.1.d to obtain the sampler with the full mesh sequence. The full funding hypothesis supplies the three assumptions of VIII.1.5. *Lean:* `clients`; `jewelOf17`, `wins`, `bareTrial` and `Funded.jewel`. Further sources: `YM.PhysicalNext.Repeat.Legato.Campaign.Winter`, `YM.PhysicalNext.Repeat.Legato.Campaign.offers`, `YM.FullVolume.stripped`, `YM.FullVolume.saved`, `YM.FullVolume.field_independent`. *(assembly, analytic convergence, tuning, costs and parameter-family proofs read; their estimates are inputs from III.12–III.15)*

Lemma VIII.1.c (convergence along all spacings at fixed volume). This is VIII.1.6(a). *Proof sketch.* Fix p, n, v.

1. The volume-uniform bounds control every fixed list of lattice differences on separated configurations. The bound and its cutoff threshold may depend on n, v, the differences and the separation, but are uniform in p.
2. The union of the configuration grids is dense. Grid nesting makes each point of this union a grid point at all sufficiently fine spacings. At these configurations, the renormalization-step comparison from III.12 gives a Cauchy sequence along all N. Equicontinuity extends convergence locally uniformly to NoHit_p.
3. A discrete Taylor estimate compares each first difference with a difference quotient over a fixed small physical displacement. The function values already converge; first let $N \to \infty$, then let the displacement shrink. This proves that the first differences are Cauchy. Iterating gives limits of all differences, and the Taylor estimate identifies them as the derivatives of a smooth function $f_{p,v}$. Multilinear interpolation has the same limits.
4. The lattice sums are integrals of piecewise constant functions on a fundamental cell. A uniform bound on the test support permits dominated convergence, giving the smeared limits.

Lean: `realUV`, `purchased`, `positioned` and `FixedBounds`. Further sources: `YM.FullVolume.Eves.dense`, `YM.FullVolume.Eves.lateAnch`, `YM.FullVolume.from_previous`, `YM.FullVolume.honors`, `YM.FullVolume.augments`, `YM.FullVolume.pudding`. *(proof read, includ-*

ing dense-grid convergence, Taylor recovery and smearing; the renormalization-step estimate is an input)

Lemma VIII.1.d (removing the mesh subsequence). This is VIII.1.6(b). *Proof sketch.* The seed of $\mathfrak{F}$ gives, for each admissible p, $C_{\sigma^{\mathrm{m}}(j),p}(v;t) \to R_p$, and also $R_{\sigma^{\mathrm{v}}(k)} \to \int tS(v;\cdot)$. By VIII.1.c, $C_{N,p}(v;t) \to \int t f_{p,v}$ along all N. Since σ^{m} is cofinal, uniqueness of limits gives $R_p = \int t f_{p,v}$. Thus the same rows R_p are limits along all N, and the volume limit is unchanged. *Lean:* `from_spec`.

Lemma VIII.1.e (OPE and logarithmic packages). Let R be a run with campaign T and pillars W. Assume $\mathfrak{g}$ simple and (Fd). Then the family carries the OPE package and its logarithmic extension. *Proof sketch.* The pillars, polynomial schedule compatibility, halo condition and weight data supply the ordinary package. The logarithmic extension adds the logarithmic exit estimate, its budget, the fixed support exponent, and divergence of the composed schedule $m \mapsto Q(J(m))$. The last property follows from $J(m) \to \infty$ and $Q(k) \to \infty$.

Lean: `Funded.logPackage` and `opePackage17`. *(assembly proofs read)*

3.32.4.2 4.2 The data at one extraction order

Lemma VIII.1.f (limits of the couplings at fixed level). Assume (D3) holds for $\mathcal{C}$, and let σ be an extraction order. Then there are g and $(B_m)_m$ such that, for every m, $\beta_{N,p}(m)$ has the ordered limit B_m along $\sigma \circ g$. Moreover $B_m/\log(1/a_m) \to 2b_0$. *Proof sketch.* Set $c_N(p,m) := \beta_{N,p}(m)$ for $m \le N$ and $c_N(p,m) := \hat{\beta}_m$ for $m > N$. By the band condition of (D3), $c_N \in \prod_{(p,m)} [\hat{\beta}_m/2, 2\hat{\beta}_m]$, a compact metrizable space.

1. Extract g^{m} with $c_{\sigma^{\mathrm{m}}(g^{\mathrm{m}}(j))} \to a$ in the product.
2. Extract g^{v} with $a(\sigma^{\mathrm{v}}(g^{\mathrm{v}}(k)),\cdot) \to B$ in $\prod_m [\hat{\beta}_m/2, 2\hat{\beta}_m]$.
3. For fixed m the truncation is eventually inactive.
4. The running law of (D3) gives $|B_m - \hat{\beta}_m| \le \varepsilon \log(1/a_m)$ for large m, hence the slope.

The slope is not used by the main theorem. *Lean:* `Running.extraction` and `Running.discrete`.

Lemma VIII.1.g (a final object on a refinement). Let $\mathfrak{g}$ be simple, let R be a run with the OPE package, and let σ be an extraction order. Then there are a refinement w, a table S, a final object $\mathfrak{F}$ on $(\mathcal{C}_{\mathsf{R}}, \sigma \circ w, S)$ and scaling data Σ, with OPE services and (D14). *Proof sketch.*

1. The run's extraction property supplies a table and seed on a refinement, together with the trajectory witnesses (D6).
2. Add the free Maxwell data, with the translation normalization in which $\kappa^{-1}J$ is the free Maxwell stress tensor.
3. The weak comparison estimates give the infrared data, scaling data Σ and the first comparison.
4. The symmetry results of VI.1–VI.2 produce the derivatives, rotations and stress tensor with their local laws.
5. The phase results B01–B18 produce the selection and its certificate. The reconstruction theorem then supplies the mixed and selected continuations and the selected Wightman theory. These data form $\mathfrak{F}$ with the prescribed field algebra.
6. The leading mixed comparison, the one-dimensional space of conserved tensors, and the limits of derivatives and rotations give (D14). *Lean:* `Pert.withFinal`; `Run.extractible17`. *(assembly and final-object proofs read; analytic and symmetry inputs as stated)*

Lemma VIII.1.h (trace result for every final object). Assume (Fd) and $\mathfrak{g}$ simple. For every extraction order σ, every table S and every final object $\mathfrak{F}$ on $(\mathcal{C}_{\mathsf{R}}, \sigma, S)$:

- $d_2 = -b_0/2$ and $A_{\mathfrak{g}} > 0$;
- Θ is fixed by the adjoint and Lorentz actions, and $\langle \Omega, \Theta(f)\Omega \rangle = 0$ for every Schwartz test f;
- on the common invariant domain,

$$\Theta(f) = \Big(\sum_a T_{aa}\Big)(f) - \Big\langle \sum_a T_{aa} \Big\rangle_{\xi_*} \Big(\int f\Big)\mathrm{id},$$

  and $\Theta \neq 0$;
- $s^8(\log 1/s)^2 G_\Theta(s) \to A_{\mathfrak{g}}$ as $s \downarrow 0$;
- the trace has a unique spectral measure μ, with

$$\rho^{-8}(\log \rho)^2 \int \varphi(p/\rho)\, \mu(dp) \longrightarrow \frac{A_{\mathfrak{g}}}{1536\pi} \int_{\overline{V}_+} \mathsf{m}(p)^2 \varphi(p)\, dp \qquad (\rho \to \infty),$$

  for every compactly supported continuous φ and every Schwartz φ, where $\mathsf{m}(p) = (p^4)^2 - |\mathbf{p}|^2$ and $\overline{V}_+ = \{p : p^4 \geq |\mathbf{p}|\}$;
- there exist a scaling hierarchy, a Wilson coupling and an ultrafilter compatible with σ for which a normalized position-space scheme exists. Every scheme with that hierarchy and ultrafilter satisfying the stated alignment, leading-derivative and scalar normalization conditions admits the stress comparison of VII.3;
- the momentum-space Wick countertest of VIII.1.8 and the Maxwell relative-locality countertests of VIII.1.n hold, including the version localized in any ball.

Proof sketch. The phase-comparison condition in (Fd) supplies the trace comparison data at σ. The trace asymptotic, spectral law and exclusions of VII.2–VII.3 then apply to this final object. Centering and scalar covariance follow from the stress-tensor laws. The stress-comparison theorem first constructs one admissible normalized scheme, then proves the comparison for every scheme satisfying those same conditions.

Lean: `Trace.Target.Result`, `assembled`, `arenaT`, `crest_ready` and `Funded.gallery`. *(assembly proofs read; the trace estimates are inputs from VII.2–VII.3)*

Lemma VIII.1.i (Wick exclusion in momentum space). This is VIII.1.8. *Proof sketch.*

1. Let μ be the trace spectral measure. It is tempered and supported in the forward cone. For a joint Schwartz test φ, the selected two-point function is the integral of its relative-momentum transform against μ, and

$$G_\Theta(s) = \int e^{-sp^4}\, \mu(dp) \qquad (s > 0).$$

2. Suppose the Wick identity held for every allowed pair (f, g). Then $\int v\, d\mu = \sum_\alpha c_\alpha \mathrm{read}_\alpha(v)$ for every compactly supported Schwartz v. To see this, choose u with $|\mathcal{F}_+ u| = 1$ on a ball containing $\operatorname{supp} v$, and use Fourier surjectivity to choose w with $\mathcal{F}_+ w = (\mathcal{F}_+ u)v$. Thus $\overline{\mathcal{F}_+ u}\mathcal{F}_+ w = v$.

3. For each $s > 0$, approximate e^{-sp^4} by compactly supported smooth cutoffs bounded in absolute value by e^{-sp^4}. Exponential integrability for μ and for every mass-shell term permits dominated convergence on both sides. Hence $G_\Theta(s) = \sum_\alpha c_\alpha K_\alpha(s)$ for every $s > 0$.

4. The resulting Wick profile contradicts VIII.1.r(c).

Lean: `polynomial_excludes`, `identifies_euclid` and `traceSpectrum_exists`. Further sources: `YM.TraceExtension.productTest`, `YM.TraceExtension.waveTest_product`.

Lemma VIII.1.j (reconstruction in every phase; input from V.3). For every final object $\mathfrak{F}$ there are reconstruction data $\mathfrak{L}$ with the following properties. The phasewise assertions hold for every $\xi \in X$.

1. For every $\xi \in X$, a Wightman theory $\mathfrak{W}_\xi$ realizes the continuation of that phase, with the same labels. At ξ_* it is the selected theory of $\mathfrak{F}$.

2. The comparison map from $\mathfrak{W}_\xi$ to its Osterwalder–Schrader space is onto.

3. The Euclidean ground state is unique, and the zero-energy subspace is $\mathbb{C}\Omega_\xi$.

4. For every finite sum A of field words there is $M_A \geq 0$, independent of ξ, such that

$$\mu_{A\Omega_\xi}(\{0 < p^4 \leq u\}) \leq M_A u^2 \quad (0 < u \leq 1),$$
$$0 \leq \langle A\Omega_\xi, (e^{-sH_\xi} - P_{0,\xi})A\Omega_\xi\rangle \leq M_A s^{-2} \quad (s \geq 1).$$

5. The bounded weak commutant, defined using the unbounded adjoints on the field-word domain, is $\mathbb{C}\,\mathrm{id}$.

6. The stress tensor continues to every phase, and each Wightman distribution evaluated on a fixed joint Schwartz test is continuous in ξ.

7. $\int S^\xi \, d\nu = S$ on separated words, and $\int \mathcal{W}^\xi \, d\nu = \mathcal{W}^{\mathrm{mix}}$ on every joint Schwartz test.

Proof sketch (V.1, V.3). The phases are characters of the commutative C^*-algebra generated by time-slab word operators compressed to the mixed Osterwalder–Schrader zero-energy subspace. Its vacuum measure is ν. Translate a separated word into a slab to obtain the Euclidean barycentre identity. Apply the reconstruction, low-energy, heat-decay, irreducibility and stress-continuation results of V.3 to every character, retaining the original selected theory at ξ_*. Their continuity and boundary barycentre identities give the last two assertions.

Lean: `all_characters` and `Completed`. *(statement and top-level proof read)*

Lemma VIII.1.k (the OPE reader and its estimates). Let $\mathfrak{F}$ be a final object on $(\mathcal{C}, \sigma, S)$ with OPE services for that family, order and table. Choose a coherent prescription P for the field derivatives, adapted to the symmetric laws of the table and to the transport injection. Let Q be the reader formed by P, its coherence, the Euclidean kernel and the transport data (injection, orientation, local laws). Then:

- the OPE foundation holds for Q;
- the uniform nested OPE holds;
- the all-shapes OPE holds for every coherent prescription;
- under the dense-coverage property of the sharp Gram family, a compatible sharp Gram allocation exists for every ultrafilter finer than the extraction filter of σ.

Proof sketch. Extend the finite-degree prescriptions compatibly and take their common restriction in each degree. The OPE services give the coefficient and remainder estimates, which also supply the nested expansions and the all-shapes expansion. Dense coverage supplies the test portfolios used to construct the sharp Gram allocation.

Lean: `Prescription.chosen` (VII.4), `Orbits.produce`, `Nesting.uniform` and `attains_mantle17`. *(assembly proofs read; OPE estimates as stated)*

Lemma VIII.1.l (assembly of the full package). This is VIII.1.4. *Proof sketch.* The proof has three parts.

- *Global items.* (UV_∞) comes from VIII.1.b and the logarithmic package from VIII.1.e. The trace result for every $(\sigma, S, \mathfrak{F})$ comes from VIII.1.h. Coverage comes from the phase-comparison and coverage clause of (Fd).
- *At an order σ.* Apply VIII.1.f to obtain (g, B). Apply VIII.1.g to $\sigma \circ g$ to obtain $(w, S, \mathfrak{F}, \Sigma)$. Apply VIII.1.j to obtain $\mathfrak{L}$, and VIII.1.k to obtain Q. Then:
  - subsequence stability carries the coupling limits from $\sigma \circ g$ to $(\sigma \circ g) \circ w = \sigma \circ (g \circ w)$;
  - VIII.1.d, with the full-mesh clause of (UV_∞), gives $\mathrm{Ord}_{(\mathrm{id}, (\sigma \circ g \circ w)^{\vee})}$;
  - the OPE facilities consist of the services, the run parameters and the logarithmic package.
- *Output.* The refinement returned is $\tau = g \circ w$.

Lean: `fromFunded` and `Order.retained`.

3.32.4.3 4.3 The internal headline Lemma VIII.1.m (contact devices). Let $\mathfrak{F}$ carry a reader Q and OPE services. Then there are devices M with the following properties:

- the prescription of M is the reader's prescription;
- the full contact resolution holds;
- each OPE stratum has a complex basis of real fields, and the degree-zero basis contains the unit field;
- there are positive thresholds $t_n \to 0$ with $4t_{n+1} < t_n$ and $2(n+1)t_n \le \mathrm{clearance}_{\mathsf{M}}(n)$.

Proof sketch. Use the reader's coherent prescription, the rotations, the Euclidean hierarchy and the phase transport laws, together with a multiple-contact detector in each degree. The contact theorem of VII.4 gives the resolution from the OPE services. Real conjugation permits a real basis in each stratum, containing the unit in degree zero. Successively choose each positive threshold below one eighth of its predecessor, 2^{-n} and $\mathrm{clearance}_{\mathsf{M}}(n)/(2(n+1))$. *Lean:* `contacts_on` and `MultipleLight`. Further sources: `YM.Contacts.Solstice.Thresholds`.

Lemma VIII.1.n (continuum theorem). This is VIII.1.3. *Proof sketch.* Take (T, W) and (Fd) from the specification of $\mathsf{R}_{G,b}$, and apply the refinement clause of VIII.1.4 at σ. The coupling limits, the all-shapes OPE and the sharp Gram allocations are omitted. Contacts come from VIII.1.m. (Core) comprises:

- the OPE services;
- the reconstruction in every phase;
- the eight trace tests from VIII.1.h;
- the OPE foundation;
- the integrated remainder estimates of VII.4: fix cluster fields $a_i \in F_{d_i}$ and spectator fields. For a measurable parameter space, an integrable complex density w against a positive

measure μ, with $\int |w|\, d\mu \le B$ and an almost everywhere measurable configuration map that is admissible at scale r wherever $w \neq 0$, the weighted raw correlation, finite OPE sum and their difference are integrable. The norm of the integrated difference is at most $B\, r^{D+1-\sum_i d_i - \varepsilon}$, for every almost everywhere measurable choice of phases and for the mixed state. The radius threshold depends on the finite field lists, geometric constants, $\varepsilon > 0$, truncation degree D, parameter space and mass bound B, and precedes the choice of μ, w, configurations and phases.

The internal trace-test bundle has eight clauses:

- (T1);
- the operator identity of VIII.1.h;
- the asymptotics;
- $A_{\mathfrak{g}} > 0$;
- the unique spectral law;
- the Wick countertest;
- for every n and every complex-linear isometry U from $\mathcal{H}$ into the n-colour free Maxwell Fock space, preserving the vacuum, intertwining translations and taking values in symmetric transverse waves, there are compactly supported Schwartz tests $f, h_1, \dots, h_n$, with each h_i real and spacelike separated from f, such that

$$\langle W(-h)\Omega, U\Theta(f)\Omega\rangle \neq \langle U\Theta(\bar{f})\Omega, W(h)\Omega\rangle;$$

- for every such U, every probability measure on an index space and every family of maps preserving inner products on physical waves and fixing $U\mathcal{H}$ pointwise, the analogous identity with the coherent vectors averaged under those maps also fails. For every $r > 0$, the test f can be supported in an open set whose closure lies in $B(0, r)$, with the real tests h_i supported in a common open set whose closure is spacelike separated from the closure of the support region for f.

Finally, the degree-four determining tests of VII.1 convert (D14) into (D15). *Lean:* `continuum_with_position_tests`, `Core`, `Core.of_receipts`, `ManuscriptTraceTests.of_result` and `Pert.Position.four_on`. *(assembly and degree-four normalization proofs read)*

Lemma VIII.1.o (internal headline). For a gauge group with simple $\mathfrak{g}$ and $b > 0$, the internal detailed existence statement holds with $\mathcal{C} = \mathcal{C}_{G,b}$. *Proof sketch.* Item by item:

- (D3) comes from the run, and (D6) from the final object.
- *Reconstruction.* It consists of:
  - the mixed and selected continuations;
  - the internal Wightman axioms for the selected theory and for every $\mathfrak{W}_\xi$: normalized cyclic vacuum, dense domain, temperedness, adjoints, the moment identity, positivity, locality, covariance, strong continuity, spectral representation, forward support, and uniqueness of the invariant vector;
  - the ν-average of the phase Wightman distributions.

- *Separated OPE.* It consists of the coefficient bounds from the OPE foundation, the single-centre remainder bound and the integrated bounds of (Core).
- *Trace clauses.* The seven headline clauses are centering, the operator identity, short-distance asymptotics, positive amplitude, and the Wick, Maxwell and averaged Maxwell countertests. The unique spectral law is retained separately in (Core).

Lean: `Nonperturbative.existence`, `Reconstruction`, `WightmanAxioms`, `SeparatedOPE`, `NestedOPE`, `TraceExclusions` and `Core.point`.

3.32.4.4 4.4 Translation to the public statement Lemma VIII.1.p (the public Wilson family and hierarchy). Let $\mathfrak{F}$ be a final object on $(\mathcal{C}, \sigma, S)$ and Q a reader. The public Wilson family $\mathcal{W}(\mathcal{C})$ is defined by:

- L_p and D_N are the period and denominator sequences of $\mathcal{C}$;
- $\beta_{N,p} = \beta_{N,p}(N)$;
- h is the normalized Haar measure;
- $\mathcal{T}$ is the set of internal tokens;
- $I_{N,p}$ is the internal insertion.

The public Wilson energy equals the internal one. The proof matches the plaquette $U(y,\mu)U(y+e_\mu,\nu)U(y+e_\nu,\mu)^{-1}U(y,\nu)^{-1}$ with the internal face value. Hence the measures, the lattice moments and the smeared moments agree.

The hierarchy $T(\mathfrak{F}, Q)$ has the following components:

- $\mathcal{H}$ and $\mathfrak{W}$ are those of the selected reconstruction, with the same Hilbert space, domain, labels and operators;
- $\mathcal{L}$ is the filtered union of the field spaces, with $F_d =$ the quotient field spaces of S, $\iota_d =$ the field-label inclusion and $\pi =$ the class map of tokens;
- (X, ν, ξ_*) are the phase space, its measure and the selected point;
- S^s is the table read in state s;
- F_n is the selected continuation;
- $(\mathrm{Ch}_j, O_{j,\kappa}, c_{j,\kappa})$ come from Q;
- $\Theta = \sum_a T_{aa} - \langle \sum_a T_{aa} \rangle_{\xi_*} \mathbf{1}_4$.

Lean: `energy_eq`, `law_eq`, `moment_eq`, `smear_eq` and `hierarchy`.

Lemma VIII.1.q (regularity and the Wilson limit). Let $\mathfrak{F}$ be a final object on $(\mathcal{C}, \sigma, S)$ and Q a reader.

(a) If (D3) holds and $\mathfrak{g}$ is simple, then (Reg1)–(Reg4) hold:

- (Reg1): the running law of (D3) at $m = N$ gives $|\beta_{N,p}/\log D_N - 2b_0| < \varepsilon$ for every $\varepsilon > 0$, all sufficiently large N and all p, and $2b_0 > 0$ by $C_{\mathfrak{g}} > 0$;
- (Reg2) and (Reg3): continuity and gauge invariance of the local sections;
- (Reg4): the stencil of the local section.

(b) If $\mathrm{Ord}_{(\mathrm{id},s)}(\mathcal{C}, S)$ holds, then (P1) holds along s. The smeared moments agree by VIII.1.p. The mixed correlation of $\pi(v)$ equals $S(v;\cdot)$ on Conf_n, and $t = 0$ off Conf_n. *Lean:* `regular`, `zeroUp_bound`, `fromWilson` and `wilsonLimit`.

Lemma VIII.1.r (reconstruction, OPE and trace clauses). Let $\mathfrak{F}$ be a final object and Q a reader for it. The conclusions concern $T(\mathfrak{F}, Q)$.

(a) The reconstruction in every phase implies (R1)–(R15). Clause by clause:

- (R1): the reconstruction bridge assembles the internal domain, adjoint, spectral, covariance and locality properties into (W0)–(W3).
- (R2): faithfulness of the selected theory.
- (R3): injectivity of the field-label inclusion.
- (R4): the unit field of degree 0 is nonzero, and faithfulness applies to it.
- (R5): the labels are covered by the derivative plan.
- (R6): token representatives.
- (R7), (R8): the phase decomposition.
- (R9), (R10): permutation symmetry and smoothness of each state.
- (R11): continuity of the characters, after moving a separated word into a time slab.
- (R12): the barycentre identity of VIII.1.j.
- (R13), (R14): holomorphy and the values at Euclidean points of the selected continuation.
- (R15): the boundary values, together with the moment identity, which identifies the public vacuum expectations with the internal Wightman distributions.

(b) The internal separated OPE implies (O1)–(O2), after enumerating $I \sqcup J$ once.

(c) Assume $\mathfrak{g}$ simple and the seven internal headline trace clauses. Then (T1)–(T4) hold. Clause (T1) is the centered clause. For (T2) and (T3), the internal amplitude is $12(d_2/b_0)^2 \dim \mathfrak{g}/\pi^4 = 3 \dim \mathfrak{g}/\pi^4$, and G_Θ is by definition the internal Euclidean two-point function. For (T4), public and internal Wick profiles coincide; then apply the Euclidean Wick exclusion. The real part of a finite Wick profile has, for every $k \in \mathbb{Z}$, an expansion modulo $O(r^k)$ in the span of $r^a(\log r)^j$ with $a \in \mathbb{Z}$ and $j \in \mathbb{N}$. No function with such an expansion modulo $O(1)$ satisfies $r^8(\log 1/r)^2 f(r) \to A \neq 0$, since that would need the power $(\log r)^{-2}$. Thus (T3), together with $A_{\mathfrak{g}} > 0$ from (T2), implies (T4). *Lean:* `publicReconstruction`, `view_axioms`, `publicMoment`, `publicOPE`, `publicTrace` and `euclidean_excludes`. *(proofs read, including the power-log exclusion)*

Lemma VIII.1.s (contact distributions). Let $\mathfrak{F}$ be a final object with contact devices M satisfying the full contact resolution. For every state s, every k and every $a \in \mathbb{F}^k$, there is a continuous linear functional $\mathcal{D}$ on $C_c^\infty((\mathbb{R}^4)^k)$ with $\mathcal{D}(f) = \int f S^s(a; \cdot)$ whenever $\operatorname{supp} f \subset \mathrm{Conf}_k$, using the usual test-function topology. No uniqueness is asserted. *Proof sketch.* Apply the distribution clause of the resolution with:

- the identity embedding of $(\mathbb{R}^4)^k$;
- the full collision group, centred at 0;
- $B = \min(1, \mathrm{clearance}_{\mathsf{M}}(\sum_i d_i))$ and $r = B/4$.

The contact theorem supplies $\mathcal{D}$. On a separated test, the subtraction formula reduces to the original field word. Reading that word gives the correlation; off the support of the test both integrands vanish. *Lean:* `contactDistribution`.

3.32.4.5 4.5 The headline theorems Lemma VIII.1.t (detailed theorem). This is VIII.1.2. *Proof sketch.* Apply VIII.1.o to obtain $\mathcal{C}$, the penalty and (D3). Regularity follows from VIII.1.q(a). For each σ, unpack the data and conclude as follows:

- (P1) by VIII.1.q(b);
- (P2) by VIII.1.r(a), from (Core);
- (P3) by VIII.1.r(b);
- (P4) by VIII.1.r(c);
- (D2) by VIII.1.s;
- (W0)–(W3) for every phase by the reconstruction bridge.

Lean: `constructed`.

Lemma VIII.1.u (public theorem). This is VIII.1.1. *Proof sketch.* Apply VIII.1.t with $b = 1$. Given p, let $\sigma = (\mathrm{id}, p)$, and let τ be the refinement produced. Take $q := \tau^{\mathrm{v}}$ and the hierarchy $T(\mathfrak{F}, Q)$. Then $(\sigma \circ \tau)^{\mathrm{v}} = p \circ q$, and the mesh part of τ plays no role because (P1) is proved along the full mesh sequence (VIII.1.d). *Lean:* `jointExistence` and `Order.refine`.

Lemma VIII.1.v ($SU(2)$). This is VIII.1.9. *Proof sketch.* The matrices

$$A = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

span $\mathfrak{su}(2)$ over $\mathbb{R}$ and satisfy $[A, B] = 2C$, $[B, C] = 2A$, $[C, A] = 2B$. If an ideal contains $aA + bB + cC \neq 0$, a suitable operator $1 + \frac{1}{4}(\mathrm{ad}\, A)^2$, or its cyclic counterpart, isolates a nonzero multiple of one basis vector. The bracket identities then put all three in the ideal. Apply VIII.1.u. *Lean:* `su2` and `su2Theory`. *(proof read, including simplicity)*

3.32.4.6 4.6 Endpoints filed here and not on the path of the main theorem

These separate results and alternative existence theorems are not inputs to the assembly above:

- the degree-four normalization `normalized_four`, obtained by applying the normalization result to a separately constructed family;
- the parity results `parityEndpoint` and `low_read`: parity is retained by the degree-four normalization, and every odd perturbative coefficient of the finite-volume ledger vanishes at each field-degree horizon $d \leq 5$ for simple $\mathfrak{g}$ (VII.6);
- the contact endpoint `celestial`, built on `signed_placements` and `OPE.single` (VII.4);
- the alternative ultraviolet endpoint `july`.

The volume-uniform ultraviolet bounds belong mathematically to III.12–III.15; the supporting reconstruction and symmetry lemmas belong to V.2–V.3 and VI.1.

3.32.5 5. Where the difficulty lies

The assembly preserves the order of choices. A family fixed before σ must support the coupling limits, continuum table and all later structures on one refinement. In particular, the fixed-volume ultraviolet convergence must concern this same family. Uniqueness of row limits then removes the mesh subsequence without replacing the final object or its point-limit data.

The momentum-space Wick exclusion requires a further argument: every compactly supported relative-momentum test must be realizable as $\overline{\mathcal{F}_+ u}\mathcal{F}_+ w$. Fourier surjectivity and an auxiliary transform of modulus one on the test support give that realization. Exponential integrability then permits passage to Euclidean Laplace transforms.

The public theorem retains less than the detailed construction. Its Wilson limit concerns the mixed state, while its trace test concerns the selected phase. The Euclidean Wick exclusion follows from (T2)–(T3); the detailed theorem also gives momentum-space and Maxwell relative-locality countertests. The low-energy and heat bounds are polynomial and imply no mass gap. Coupling limits, all-shapes OPE and sharp Gram allocations occur in (Goal) but are omitted from VIII.1.3. Neither independence of the volume subsequence nor agreement between different penalties or run choices is asserted.

3.32.6 6. Notation

object	code name	proposed notation
gauge group	`GaugeGroup`, `IsSimple`	$(m, G, \mathfrak{g})$, $\mathfrak{g}$ simple
public Wilson family	`wilsonFamily`	$\mathcal{W}(\mathcal{C}) = (L_p, D_N, \beta_{N,p}, h, \mathcal{T}, I_{N,p})$
regularity	`WilsonRegularity`	(Reg1)–(Reg4)
public hierarchy	`hierarchy`	$T(\mathfrak{F}, Q)$
extraction order, refinement	`Extraction.Order`, `refine`	$\sigma = (\sigma^{\mathrm{m}}, \sigma^{\mathrm{v}})$, $\sigma \circ \tau$
ordered limit	`Ordered`	$\mathrm{Ord}_{\sigma}(\mathcal{C}, S)$
internal family, construction predicate	`Family`, `Makes`	$\mathcal{C}$, (D3)
run, funding conditions	`Run`, `fundedRun`, `Funded`	$\mathsf{R}_{G,b}$, (Fd)
ultraviolet endpoint	`Jewel`	(UV_{∞})
positional convergence	`PositionsUV`	(D5), $f_{p,v}$
full assembly package	`Broad`	(Goal)
table	`Table`	S, read as S^s
final object	`Spec.Final`	$\mathfrak{F}$ on $(\mathcal{C}, \sigma, S)$
reconstruction in every phase	`Almanac`, `Completed`	$\mathfrak{L} = (\mathfrak{W}_{\xi})_{\xi \in X}$
OPE reader	`Orbits.Reader`	Q; $(\mathrm{Ch}_j, O_{j,\kappa}, c_{j,\kappa})$
contact devices	`Contacts.Devices`	M
scaling sequence	`Climb.Scaling`	$\Sigma = (r_k, \xi_k, U_k)$
combined continuum properties	`Independent.Core`	(Core)
first comparison, degree-four normalization	`FirstExtension`, `FourMeters`	(D14), (D15)
centered trace, its two-point function	`TraceExtension.center`, `euclidean`	Θ, G_{Θ}
trace amplitude	`Trace.Verse.award`	$A_{\mathfrak{g}} = 3 \dim \mathfrak{g}/\pi^4$
one-loop constants	`Pert.Asymptotics.b0`, `Trace.Over.d2`	$b_0 = 11C_{\mathfrak{g}}/(48\pi^2)$, $b_* = 2b_0$, $d_2 = -b_0/2$
reference coupling	`Calendar.hatBeta`	$\hat{\beta}_m = \bar{\beta} + 2b_0 \log(1/a_m)$
penalty of the block-spin kernel	`KernelChoices.penalty`	b

3.32.7 7. What remains unexplained

1. **Numerical run conditions.** The thirteen funding conditions in §2 are specified by their roles. The exact inequalities for the frame and polynomial schedules, delays, pacing, thermal

estimates and weight tableau must be supplied from III.15. The numerical proofs are inputs.

2. **Renormalization comparison.** VIII.1.c uses the fixed-volume Cauchy estimate from III.12. The underlying renormalization-step estimate is an input; the dense-grid, Taylor and smearing arguments are given in VIII.1.c.
3. **Normalized trace schemes.** VIII.1.h retains the existence and universal quantifiers of the stress-comparison theorem. The alignment, leading-derivative and scalar normalization conditions require the definitions and notation of VII.3. No comparison for an arbitrary renormalization prescription follows.
4. **Upstream reconstruction.** The phase selection and selected reconstruction are produced inside the final-object construction. The every-phase theorem is an input from V.3; its low-energy, decay, irreducibility and boundary arguments remain inputs from V.3.